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T153 Quad-core High-performance Industrial Automation Processor User Manual

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Contents

Revision History i
Contents ii
Figures xiii
Tables xxii
About This Document xxvii
1 Product Description 1
1.1 Overview 1
1.2 Block Diagram 2
2 Memory Map 3
2.1 CPUX Memory Map 3
2.2 MCU Memory Map 7
3 System Boot 8
3.1 Boot ROM (BROM) 8
3.1.1 Overview 8
3.1.2 External Signals 8
3.1.3 Boot Medium Select 8
3.1.4 Boot Process Description 10
4 Power, Reset, and Clock Management 14
4.1 Clock Controller Unit (CCU) 14
4.1.1 Overview 14
4.1.2 Block Diagram 14
4.1.3 External Signals 15
4.1.4 Functional Description 15
4.1.5 Programming Guidelines 19
4.1.6 Register List 28
4.1.7 Register Description 33
4.2 Power Reset Clock Management (PRCM) 164
4.2.1 Overview 164
4.2.2 External Signals 164
4.2.3 Functional Description 165

Draft Version
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4.2.4 Programming Guidelines 166
4.2.5 Register List 166
4.2.6 Register Description 167
5 Processors 178
5.1 CPUX System 178
5.1.1 Overview 178
5.1.2 Block Diagram 180
5.1.3 Functional Descriptions 181
5.1.4 Register list 182
5.1.5 CPUX_SYS_CFG Register Description 185
5.1.6 CPUX_TIMESTAMP_STA Register Description 187
5.1.7 CPUX_TIMESTAMP_CTRL Register Description 188
5.1.8 CPUX_PLL_CFG Register Description 189
5.1.9 CPUX_CLU_CFG Register Description 190
5.2 MCU System 203
5.2.1 Overview 203
5.2.2 Block Diagram 204
5.2.3 Register list 205
5.2.4 Register Description 206
6 Interprocessor Communication 213
6.1 Message Box (MSGBOX) 213
6.1.1 Overview 213
6.1.2 Block Diagram 213
6.1.3 Functional Description 214
6.1.4 Programming Guidelines 215
6.1.5 Register List 217
6.1.6 Register Description 218
6.2 Spinlock 225
6.2.1 Overview 225
6.2.2 Block Diagram 225
6.2.3 Functional Description 225
6.2.4 Programming Guidelines 227
6.2.5 Register List 229

Draft Version
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6.2.6 Register Description 230
7 Thermal Sensor 233
7.1 Thermal Sensor Controller (THS) 233
7.1.1 Overview 233
7.1.2 Block Diagram 233
7.1.3 Functional Description 233
7.1.4 Register List 235
7.1.5 Register Description 236
8 Timers 242
8.1 Timer 242
8.1.1 Overview 242
8.1.2 Block Diagram 242
8.1.3 Functional Descriptions 243
8.1.4 Programming Guidelines 244
8.1.5 Register List 245
8.1.6 Register Description 245
8.2 Watchdog Timer (WDT) 249
8.2.1 Overview 249
8.2.2 Block Diagram 249
8.2.3 Functional Descriptions 249
8.2.4 Programming Guidelines 251
8.2.5 Register List 252
8.2.6 Register Description 252
9 RTC 257
9.1 RTC 257
9.1.1 Overview 257
9.1.2 Block Diagram 258
9.1.3 External Signals 258
9.1.4 Functional Descriptions 259
9.1.5 Programming Guidelines 263
9.1.6 Register List 266
9.1.7 Register Description 267
10 Memory Controllers 285

Draft Version
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10.1 SDRAM Controller (DRAMC) 285
10.2 SD/MMC Host Controller (SMHC) 286
10.2.1 Overview 286
10.2.2 Block Diagram 287
10.2.3 External Signals 288
10.2.4 Functional Description 289
10.2.5 Programming Guidelines 300
10.2.6 Register List 307
10.2.7 Register Description 310
10.3 SPI Flash Controller (SPIF) 370
10.3.1 Overview 370
10.3.2 Block Diagram 371
10.3.3 External Signals 372
10.3.4 Functional Description 373
10.3.5 Programming Guidelines 388
10.3.6 Register List 393
10.3.7 Register Description 394
11 Interrupts 420
11.1 GIC 420
11.1.1 Overview 420
11.1.2 Functional Description 421
11.1.3 Register List 430
11.1.4 Register Description 430
11.2 CLIC 431
11.2.1 Overview 431
11.2.2 Functional Description 431
11.2.3 Register List 437
11.2.4 Register description 438
12 DMA 439
12.1 DMA Controller (DMAC) 439
12.1.1 Overview 439
12.1.2 Block Diagram 440
12.1.3 Functional Description 441

Draft Version
ALLWINNER

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12.1.4 Programming Guidelines 451
12.1.5 Register List 455
12.1.6 Register Description 457
13 Video & Graphics Engine 475
13.1 Display Engine (DE) 475
13.2 Graphic 2D (G2D) 476
14 Video Input Interface 477
14.1 CSIC 477
14.1.1 Overview 477
14.1.2 Block Diagram 479
14.1.3 External Signals 480
14.1.4 Functional Description 481
14.1.5 Programming Guidelines 484
14.1.6 Register List 485
14.1.7 CSIC CCU Register Description 496
14.1.8 CSIC TOP Register Description 503
14.1.9 PHY COMMON Register Description 531
14.1.10 PHY DIGITAL LAYER Register Description 532
14.1.11 PHY PORT PAYLOAD LAYER Register Description 544
14.1.12 CSIC_PARSER Register Description 574
14.1.13 CSIC_DMA Register Description 616
15 Video Out Interfaces 664
15.1 TCON LCD 664
15.1.1 Overview 664
15.1.2 Block Diagram 664
15.1.3 External Signals 665
15.1.4 Functional Description 668
15.1.5 Programming Guidelines 673
15.1.6 Register List 678
15.1.7 Register Description 679
15.2 LVDS 698
15.2.1 Overview 698
15.2.2 Block Diagram 699

Draft Version
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15.2.3 External Signals....699
15.2.4 Functional Description....700
15.2.5 Programming Guidelines....703
15.2.6 Register List....704
15.2.7 Register Description....704
15.3 MIPI DSI....714
15.3.1 Overview....714
15.3.2 Block Diagram....714
15.3.3 External Signals....715
15.3.4 Functional Description....715
15.3.5 Programming Guidelines....716
15.3.6 Register List....718
15.3.7 Register Description....719
16 Audio Interfaces....755
16.1 Audio Codec....755
16.1.1 Overview....755
16.1.2 Block Diagram....756
16.1.3 External Signals....756
16.1.4 Functional Description....757
16.1.5 Programming Guidelines....763
16.1.6 Register List....764
16.1.7 Register Description....766
16.2 I2S/PCM....795
16.2.1 Overview....795
16.2.2 Block Diagram....796
16.2.3 External Signals....797
16.2.4 Functional Description....797
16.2.5 Programming Guidelines....799
16.2.6 Register List....800
16.2.7 Register Description....801
16.3 DMIC....838
16.3.1 Overview....838
16.3.2 Block Diagram....839
16.3.3 External Signals....839

Draft Version
ALLWINNER

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16.3.4 Functional Description 840
16.3.5 Programming Guidelines 842
16.3.6 Register List 844
16.3.7 Register Description 844
16.4 One Wire Audio (OWA) 857
16.4.1 Overview 857
16.4.2 Block Diagram 858
16.4.3 External Signals 858
16.4.4 Functional Description 858
16.4.5 Programming Guidelines 863
16.4.6 Register List 864
16.4.7 Register Description 865
17 Peripherals 888
17.1 GMAC 888
17.1.1 Overview 888
17.1.2 Block Diagram 889
17.1.3 External Signals 890
17.1.4 Functional Description 892
17.1.5 Register List 892
17.1.6 GMAC_TOP Register Description 892
17.2 CAN-FD 896
17.2.1 Overview 896
17.2.2 Block Diagram 897
17.2.3 External Signals 897
17.2.4 Functional Description 898
17.2.5 Programming Guidelines 900
17.2.6 Register List 920
17.2.7 CAN_TOP Register Description 923
17.2.8 CAN_DMA Register Description 936
17.2.9 M_CAN Register Description 939
17.3 Local Bus Controller (LBC) 976
17.3.1 Overview 976
17.3.2 Block Diagram 977
17.3.3 External Signals 977

Draft Version
ALLWINNER

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17.3.4 Functional Description 978
17.3.5 Programming Guidelines 990
17.3.6 Cautions and notes 1001
17.3.7 Register List 1002
17.3.8 Register Description 1005
17.4 USB2.0 System (USB2P0_SYS) 1045
17.4.1 Overview 1045
17.4.2 Block Diagram 1045
17.4.3 USB2.0 PHY Mapping 1046
17.4.4 Register List 1046
17.4.5 USB2P0_PHY_APP Register Description 1047
17.4.6 USB2P0_SYS_DIG_APP Register Description 1048
17.5 USB2.0 DRD 1050
17.5.1 Overview 1050
17.5.2 Block Diagram 1051
17.5.3 External signals 1051
17.5.4 Functional Description 1052
17.6 USB2.0 HOST 1053
17.6.1 Overview 1053
17.6.2 Block Diagram 1053
17.6.3 External signals 1053
17.6.4 Functional Description 1054
17.7 General Purpose ADC (GPADC) 1055
17.7.1 Overview 1055
17.7.2 Block Diagram 1056
17.7.3 External Signals 1056
17.7.4 Functional Description 1057
17.7.5 Programming Guidelines 1058
17.7.6 Register List 1061
17.7.7 Register Description 1063
17.8 TPADC 1086
17.8.1 Overview 1086
17.8.2 Block Diagram 1086
17.8.3 External Signals 1087

Draft Version
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17.8.4 Functional Description 1087
17.8.5 Register List 1093
17.8.6 Register Description 1093
17.9 GPIO 1100
17.9.1 Overview 1100
17.9.2 Block Diagram 1101
17.9.3 Functional Description 1101
17.9.4 Programming Guidelines 1112
17.9.5 Register List 1116
17.9.6 GPIO Register Description 1121
17.9.7 RTC_GPIO Register Description 1391
17.10 UART 1397
17.10.1 Overview 1397
17.10.2 Block Diagram 1398
17.10.3 External Signals 1398
17.10.4 Functional Description 1399
17.10.5 Programming Guidelines 1405
17.10.6 Register List 1408
17.10.7 Register Description 1409
17.11 SPI 1436
17.11.1 Overview 1436
17.11.2 Block Diagram 1437
17.11.3 External Signals 1438
17.11.4 Functional Description 1438
17.11.5 Programming Guidelines 1444
17.11.6 Register List 1448
17.11.7 Register Description 1449
17.12 SPI_DBI 1472
17.12.1 Overview 1472
17.12.2 Block Diagram 1473
17.12.3 External Signals 1474
17.12.4 Functional Description 1475
17.12.5 Programming Guidelines 1489
17.12.6 Register List 1493

Draft Version
ALLWINNER

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17.12.7 Register Description 1495
17.13 Two Wire Interface (TWI) 1527
17.13.1 Overview 1527
17.13.2 Block Diagram 1527
17.13.3 External Signals 1528
17.13.4 Functional Description 1529
17.13.5 Programming Guidelines 1533
17.13.6 Register List 1537
17.13.7 Register Description 1537
17.14 PWM 1555
17.14.1 Overview 1555
17.14.2 Block Diagram 1556
17.14.3 External Signals 1556
17.14.4 Functional Description 1557
17.14.5 Programming Guidelines 1563
17.14.6 Register List 1565
17.14.7 Register Description 1567
17.15 PWMCS 1619
17.15.1 Overview 1619
17.15.2 Block Diagram 1620
17.15.3 External Signals 1622
17.15.4 Functional Description 1624
17.16 LEDC 1630
17.16.1 Overview 1630
17.16.2 Block Diagram 1630
17.16.3 External Signals 1631
17.16.4 Functional Description 1631
17.16.5 Programming Guidelines 1633
17.16.6 Register List 1637
17.16.7 Register Description 1637
17.17 IR Receiver (IR_RX) 1647
17.17.1 Overview 1647
17.17.2 Block Diagram 1647
17.17.3 External Signals 1647

Draft Version
ALLWINNER

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17.17.4 Functional Description 1648
17.17.5 Operation Guidelines 1651
17.17.6 Register List 1652
17.17.7 Register Description 1652
17.18 IR Transmitter (IR_TX) 1659
17.18.1 Overview 1659
17.18.2 Block Diagram 1660
17.18.3 External Signals 1660
17.18.4 Functional Description 1661
17.18.5 Operation Guidelines 1663
17.18.6 Register List 1664
17.18.7 Register Description 1664
18 Security 1671
18.1 Crypto Engine (CE) 1671
18.1.1 Overview 1671
18.1.2 Block Diagram 1673
18.1.3 Functional Description 1674
18.1.4 Register List 1694
18.1.5 CE_NS Register Description 1696
18.1.6 CE_S Register Description 1700

Draft Version
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Figures

Figure 1-1 System Block Diagram 2
Figure 3-1 Boot Process in Normal BROM Process 11
Figure 3-2 Boot0 Read Process 12
Figure 3-3 Mandatory Upgrade Process 13
Figure 4-1 CCU Block Diagram 14
Figure 4-2 PLL Distribution 15
Figure 4-3 Bus Clock Gating Diagram 27
Figure 4-4 System Bus Tree 165
Figure 4-5 Bus Clock Tree 165
Figure 5-1 CPUX System Block Diagram 180
Figure 5-2 MCU System Block Diagram 204
Figure 6-1 Message Box Block Diagram 213
Figure 6-2 The Communication Process 216
Figure 6-3 Spinlock Block Diagram 225
Figure 6-4 Take and Free Spinlock 226
Figure 6-5 Spinlock State Machine 226
Figure 6-6 CPUX and MCU Taking/Freeing Spinlock0 Process 227
Figure 7-1 THS Block Diagram 233
Figure 7-2 Thermal Sensor Timing Requirement 234
Figure 7-3 Thermal Sensor Controller Interrupt Source 234
Figure 8-1 Block Diagram for the Timer 242
Figure 8-2 Timer Typical Application 243
Figure 8-3 Watchdog Block Diagram 249
Figure 8-4 Watchdog Application Diagram 250
Figure 9-1 RTC Block Diagram 258
Figure 9-2 RTC Application Diagram 259
Figure 9-3 RTC Clock Tree 260
Figure 9-4 RTC Counter 261
Figure 9-5 RTC 1 kHz Counter Step Structure 261
Figure 9-6 NMI Block Diagram 263
Figure 10-1 SMHC Block Diagram 287

Draft Version
ALLWINNER

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Figure 10-2 Command Token Format 289
Figure 10-3 Response Token Format 290
Figure 10-4 Data Packet Format for SDR 290
Figure 10-5 Data Packet Format for DDR 291
Figure 10-6 Data Packet Format for DDR in HS400 Mode 292
Figure 10-7 Single-Block and Multi-Block Read Operation 293
Figure 10-8 Single-Block and Multi-Block Read Operation 293
Figure 10-9 IDMAC Descriptor Structure Diagram 294
Figure 10-10 Initialization Process of Command Queue 306
Figure 10-11 SPI_Flash Block Diagram 371
Figure 10-12 Typical Application 373
Figure 10-13 SPI Transfer Mode 375
Figure 10-14 SPI STR Transfer 376
Figure 10-15 SPI DTR Transfer Example, 1-4-4 376
Figure 10-16 SPI DTR Transfer Example, 4-4-4 377
Figure 10-17 SPI DTR Transfer with DQS Input Clock Signal 377
Figure 10-18 SPI CS Select Timing Relation 378
Figure 10-19 SPI 3-Wire Mode 379
Figure 10-20 SPI CMD with Single IO 379
Figure 10-21 SPI Write CMD and DATA with One Wire 379
Figure 10-22 SPI Write CMD and Read DATA with One Wire (No Dummy) 380
Figure 10-23 SPI Write CMD and Read DATA with One Wire (with Dummy) 380
Figure 10-24 SPI Dual Input/Dual Output Mode (1-1-2 381
Figure 10-25 SPI Dual Input/Dual Output Mode (1-2-2) 381
Figure 10-26 SPI Quad Input/Quad Output Mode (1-1-4) 382
Figure 10-27 SPI Quad Input/ Quad Output Mode (1-4-4) 382
Figure 10-28 SPI Quad Input/ Quad Output Mode (4-4-4) 382
Figure 10-29 SPI Octal Input/ Octal Output Mode (1-1-8) 383
Figure 10-30 SPI Octal Input/ Octal Output Mode (1-8-8) 383
Figure 10-31 SPI Octal Input/ Octal Output Mode (8-8-8) 383
Figure 10-32 Descriptor Structure Diagram 384
Figure 10-33 SPI Programming flow 389
Figure 10-34 SPI Transfer Phase Flow Diagram 391
Figure 12-1 DMAC Block Diagram 440

Draft Version
ALLWINNER

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Figure 12-2 DMA Descriptor 443
Figure 12-3 DMA Linked List Transfer 445
Figure 12-4 Workflow of the DMAC Handshake Mode 449
Figure 12-5 DMA Channel Partition 451
Figure 12-6 DMAC Transfer Process 453
Figure 14-1 CSIC Block Diagram 479
Figure 14-2 8-bit DC Sensor Interface Timing 481
Figure 14-3 8-bit YCbCr4:2:2 with Embedded Syncs (BT.656) 481
Figure 14-4 BT.601 v2f_mode Timing 482
Figure 14-5 RAW-8 Format 483
Figure 14-6 Y Format 483
Figure 14-7 UV-Combined Format 483
Figure 15-1 TCON_LCD Block Diagram 664
Figure 15-2 HV Interface Vertical Timing 669
Figure 15-3 HV Interface Horizontal Timing 670
Figure 15-4 Single-channel Transmission Timing 671
Figure 15-5 CPU interface timing (left read, right write) 672
Figure 15-6 The Data Timing of MIPI DSI Video Mode 673
Figure 15-7 HV Mode Initial Process 673
Figure 15-8 i8080 Mode Initial Process 675
Figure 15-9 LVDS Block Diagram 699
Figure 15-10 VESA 6-Bit format 700
Figure 15-11 Single LVDS VSEA Mode 701
Figure 15-12 Single LVDS JEDIA Mode 701
Figure 15-13 Dual-link LVDS VSEA Mode 702
Figure 15-14 Interface Mapping 703
Figure 15-15 MIPI DSI Block Diagram 714
Figure 15-16 DISPLL Distribution 715
Figure 16-1 Audio Codec Block Diagram 756
Figure 16-2 Audio Codec Clock Diagram 757
Figure 16-3 Audio Codec Digital Part Reset System 758
Figure 16-4 Audio Codec Analog Part Reset System 758
Figure 16-5 Audio Codec Data Path Diagram 759
Figure 16-6 Audio Codec Interrupt System 759

Draft Version
ALLWINNER

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Figure 16-7 DAP Data Flow 760
Figure 16-8 HPF Logic Structure 760
Figure 16-9 DRC static Curve Parameters 761
Figure 16-10 DRC Block Diagram 761
Figure 16-11 Energy Filter Structure 762
Figure 16-12 Gain Smooth Filter 763
Figure 16-13 I2S/PCM Interface System Block Diagram 796
Figure 16-14 Typical Application of I2S/PCM Interface 796
Figure 16-15 I2S Standard Mode Timing 798
Figure 16-16 Left-Justified Mode Timing 798
Figure 16-17 Right-Justified Mode Timing 798
Figure 16-18 PCM Long Frame Mode Timing 799
Figure 16-19 PCM Short Frame Mode Timing 799
Figure 16-20 I2S/PCM Operation Flow 799
Figure 16-21 DMIC Block Diagram 839
Figure 16-22 DMIC Stereo Microphones Application Diagram 840
Figure 16-23 DMIC Mono Microphones Application Diagram 841
Figure 16-24 PDM Format 841
Figure 16-25 HPF Logic Structure 842
Figure 16-26 DMIC Operation Mode 843
Figure 16-27 OWA Block Diagram 858
Figure 16-28 OWA Biphase-Mark Code 859
Figure 16-29 OWA Sub-Frame Format 859
Figure 16-30 OWA Frame/Block Format 860
Figure 16-31 Data-Burst Format 861
Figure 16-32 Data-burst Preamble 861
Figure 16-33 Data-burst Preamble words 862
Figure 16-34 Fields of Burst-information 862
Figure 16-35 Length of the Burst-Payload Specified by Pd 863
Figure 16-36 OWA Operation Flow 864
Figure 17-1 GMAC Block Diagram 889
Figure 17-2 GMAC Typical Application 892
Figure 17-3 CAN-FD Block Diagram 897
Figure 17-4 I/O Signals of CAN Controllers 898

Draft Version
ALLWINNER

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Figure 17-5 CAN DMA Block Diagram 898
Figure 17-6 Bus Monitor Mode 905
Figure 17-7 Loop Back Mode 908
Figure 17-8 Message Ram Configuration Example 909
Figure 17-9 Rx Buffer/FIFO Element 910
Figure 17-10 TX Buffer element 912
Figure 17-11 TX Event FIFO Element 915
Figure 17-12 Standard Message ID Filter Element 917
Figure 17-13 Extended Message ID Filter Element 919
Figure 17-14 LOCAL_BUS_CTRL Block Diagram 977
Figure 17-15 LocalBus Protocol Timing (Single Transfer, Mode 1) 978
Figure 17-16 LocalBus Protocol Timing (Burst Transfer, Mode 1) 979
Figure 17-17 LBC Command Format 982
Figure 17-18 DMA Descriptor Format 985
Figure 17-19 LBC Reset Domain Diagram 988
Figure 17-20 Direct Access FIFO Structure 989
Figure 17-21 LBC1 Sequence Diagram 996
Figure 17-22 USB2P0_SYS Block Diagram 1045
Figure 17-23 USB2.0 PHY Mapping 1046
Figure 17-24 USB2.0 DRD Controller Block Diagram 1051
Figure 17-25 USB2.0 DRD Controller and PHY Connection Diagram 1052
Figure 17-26 USB2.0 Host Controller Block Diagram 1053
Figure 17-27 USB2.0 Host Controller and PHY Connection Diagram 1054
Figure 17-28 GPADC Block Diagram 1056
Figure 17-29 GPADC Timing Requirement 1058
Figure 17-30 GPADC Initial Process 1058
Figure 17-31 TPADC Block Diagram 1086
Figure 17-32 TPADC Single-Ended Mode for AUX ADC 1087
Figure 17-33 TPADC Differential Mode for Touch Panel 1088
Figure 17-34 Single Touch X-Coordinate Measurement for Touch Panel 1088
Figure 17-35 Dual Touch Detection for Touch Panel 1089
Figure 17-36 Touch Pressure Measurement for Touch Panel 1090
Figure 17-37 Pen Down Detection for Touch Panel 1091
Figure 17-38 Dual Touch and Pressure Measurement 1093

Draft Version

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Figure 17-39 GPIO Block Diagram....1101
Figure 17-40 Pull-up, Pull-down, and High-impedance Input Mode....1102
Figure 17-41 IO Buffer Strength Diagram....1103
Figure 17-42 GPIOB Interrupt Diagram....1104
Figure 17-43 GPIO Partition....1112
Figure 17-44 UART Block Diagram....1398
Figure 17-45 I/O Signals of UART Controllers (2/4-wire)....1400
Figure 17-46 I/O Signals of UART Controllers (9-wire)....1400
Figure 17-47 UART Serial Data Format....1400
Figure 17-48 Application Diagram for RTS/CTS Auto flow Control....1401
Figure 17-49 RTS/CTS Auto Flow Control Data Format....1401
Figure 17-50 Application Diagram for IrDA Transceiver....1402
Figure 17-51 Serial IrDA Data Format....1402
Figure 17-52 Application Diagram for RS-485 Transceiver....1402
Figure 17-53 RS-485 Data Format....1403
Figure 17-54 SPI Block Diagram....1437
Figure 17-55 SPI Application Block Diagram....1439
Figure 17-56 SPI Phase 0 Timing Diagram....1440
Figure 17-57 SPI Phase 1 Timing Diagram....1440
Figure 17-58 SPI 3-Wire Mode....1441
Figure 17-59 SPI Dual-Input/Dual-Output Mode....1441
Figure 17-60 SPI Dual I/O Mode....1442
Figure 17-61 SPI Quad-Input/Quad-Output Mode....1442
Figure 17-62 Data Transfer in Query Mode(including CPU and DMA mode)....1445
Figure 17-63 SPI_DBI Block Diagram....1473
Figure 17-64 SPI Application Block Diagram....1475
Figure 17-65 DBI Application Block Diagram....1476
Figure 17-66 DBI 3-Line Display Bus Serial Interface Write Operation Format....1477
Figure 17-67 DBI 3-Line Display Bus Serial Interface 8-bit Read Operation Format....1477
Figure 17-68 DBI 3-Line Display Bus Serial Interface 24-bit Read Operation Format....1478
Figure 17-69 DBI 3-Line Display Bus Serial Interface 32-bit Read Operation Format....1478
Figure 17-70 DBI 4-Line Display Bus Serial Interface Write Operation Format....1479
Figure 17-71 DBI 4-Line Display Bus Serial Interface 8-bit Read Operation Format....1479
Figure 17-72 DBI 4-Line Display Bus Serial Interface 24-bit Read Operation Format....1480

Draft Version
ALLWINNER

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Figure 17-73 DBI 4-Line Display Bus Serial Interface 32-bit Read Operation Format....1480
Figure 17-74 RGB111 3-Line Interface Transmit Video Format....1481
Figure 17-75 RGB444 3-Line Interface Transmit Video Format....1481
Figure 17-76 RGB565 3-Line Interface Transmit Video Format....1482
Figure 17-77 RGB666 3-Line Interface Transmit Video Format....1482
Figure 17-78 RGB111 4-Line Interface Transmit Video Format....1483
Figure 17-79 RGB444 4-Line Interface Transmit Video Format....1483
Figure 17-80 RGB565 4-Line Interface Transmit Video Format....1484
Figure 17-81 RGB666 4-Line Interface Transmit Video Format....1484
Figure 17-82 RGB444 2 Data Lane Interface Transmit Video Format....1485
Figure 17-83 RGB565 2 Data Lane Interface Transmit Video Format....1486
Figure 17-84 RGB666 2 Data Lane Interface Transmit Video Format 0....1486
Figure 17-85 RGB666 2 Data Lane Interface Transmit Video Format 1 (ilitek)....1487
Figure 17-86 RGB666 2 Data Lane Interface Transmit Video Format 2 (New vision)....1487
Figure 17-87 RGB888 2 Data Lane Interface Transmit Video Format....1487
Figure 17-88 TWI Block Diagram....1527
Figure 17-89 Write Timing in 7-bit Standard Addressing Mode....1529
Figure 17-90 Read Timing in 7-bit Standard Addressing Mode....1530
Figure 17-91 Write Timing in 10-bit Extended Addressing Mode....1530
Figure 17-92 Read Timing in 10-bit Extended Addressing Mode....1530
Figure 17-93 TWI Driver Write Packet Transmission....1531
Figure 17-94 TWI Driver Read Packet Transmission....1532
Figure 17-95 TWI Programming State Diagram....1533
Figure 17-96 PWM Block Diagram....1556
Figure 17-97 PWM01 Clock Source Bypass....1557
Figure 17-98 PWM01 Output Logic Module Diagram....1558
Figure 17-99 PWM01 Complementary Pair Output....1559
Figure 17-100 Output Waveform with Dead-time Insertion....1560
Figure 17-101 Group 0-3 PWM Signal Output....1561
Figure 17-102 PWM01 Capture Logic Module Diagram....1562
Figure 17-103 PWM0 Channel Capture Timing....1562
Figure 17-104 Waveform Capture Filtering....1563
Figure 17-105 PWMCS Block Diagram....1620
Figure 17-106 PWMCS Typical Application....1624

Draft Version
ALLWINNER

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Figure 17-107 A/B Phase Output 1625
Figure 17-108 CW/CCW Output 1625
Figure 17-109 Pulse + Direction Mode Output 1625
Figure 17-110 A/B Phase Capture (Quadruple-fold Frequency) 1626
Figure 17-111 A/B Phase Capture (All Frequencies) 1626
Figure 17-112 CW/CCW Decoding 1627
Figure 17-113 Pulse + Direction Mode Capture 1627
Figure 17-114 CAP Filtering 1628
Figure 17-115 FIFO Command Format 1628
Figure 17-116 Internal FIFO Diagram 1629
Figure 17-117 LEDC Block Diagram 1630
Figure 17-118 LED Input Data Structure 1631
Figure 17-119 LED Typical Circuit 1631
Figure 17-120 LED Input Data Code 1631
Figure 17-121 LED Data Transfer Mode 1632
Figure 17-122 LEDC Timeout Abnormal Processing Flow 1635
Figure 17-123 FIFO Overflow Abnormal Processing Flow 1636
Figure 17-124 IR Receiver Block Diagram 1647
Figure 17-125 CIR Receiver Application Diagram 1648
Figure 17-126 A Typical Pulse Train of the NEC 1648
Figure 17-127 Logical '0' and Logical '1' of NEC Protocol 1649
Figure 17-128 ATHR Definition 1649
Figure 17-129 ITHR Definition 1650
Figure 17-130 NTHR Definition 1650
Figure 17-131 APAM Definition 1650
Figure 17-132 CIR Receiver Process 1651
Figure 17-133 IR_TX Block Diagram 1660
Figure 17-134 IR_TX Clock Description 1661
Figure 17-135 Definitions of Logical "1" and Logical "0" 1662
Figure 17-136 CIR Message Timing Diagram 1662
Figure 17-137 CIR Transmitter Process 1663
Figure 18-1 CE Block Diagram 1673
Figure 18-2 DES Encryption and Decryption 1674
Figure 18-3 3DES Encryption and Decryption of a 3-key Operation and a 2-key Operation 1675

Draft Version
ALLWINNER

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Figure 18-4 ECB Mode Encryption and Decryption 1676
Figure 18-5 CBC Mode Encryption and Decryption 1676
Figure 18-6 CTR Mode Encryption and Decryption 1677
Figure 18-7 CFB Mode Encryption and Decryption 1678
Figure 18-8 OFB Mode Encryption and Decryption 1679
Figure 18-9 CTS Mode Encryption and Decryption 1680
Figure 18-10 Word Address of Message 1681
Figure 18-11 Byte Order 1681
Figure 18-12 Bit Order 1681
Figure 18-13 The Storage Method of 32-bit IV 1682
Figure 18-14 The Storage Method of 64-bit IV 1682
Figure 18-15 Task Chaining of Hash Algorithms and Random Bit Generator Algorithms 1684
Figure 18-16 Task Chaining of Other Algorithms 1688

Draft Version

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Tables

Table 3-1 BROM External Signals 8
Table 3-2 BOOT_MODE Setting 9
Table 3-3 GPIO Boot Select Configuration 9
Table 3-4 Groups of eFuse_Boot_Select 9
Table 3-5 eFuse Boot Select Setting 10
Table 4-1 CCU External Signals 15
Table 4-2 PLL Specifications 16
Table 4-3 PRCM External Signals 164
Table 5-1 CPUX_SYS Components 180
Table 5-2 Reset Signal Description 181
Table 5-3 MCU System Component 204
Table 7-1 Calculation Parameters for THS Temperature Conversion Formula 235
Table 7-2 eFuse Address of THS Calibration Value in FT Stage 235
Table 9-1 RTC External Signals 258
Table 9-2 RTC Counter Changing Range 261
Table 10-1 SMHC Sub-blocks 287
Table 10-2 SMHC External Signals 288
Table 10-3 DES0 Definition 294
Table 10-4 DES1 Definition 295
Table 10-5 DES2 Definition 295
Table 10-6 DES3 Definition 295
Table 10-7 Differences of Task Descriptor Between CMDP Mode and CMDQ Mode 296
Table 10-8 Phase Offset of Command and Data 297
Table 10-9 SPI_Flash Sub-blocks 371
Table 10-10 SPI_Flash External Signals 372
Table 10-11 SPI_Flash Support List 373
Table 10-12 SPI Flash Modes with Clock Polarity and Phase 375
Table 11-1 Interrupt Sources in CPU System 421
Table 11-2 Interrupt Sources in E907 System 431
Table 12-1 DMAC Sub-blocks 440
Table 12-2 DMA DRQ Types 441

Draft Version
ALLWINNER

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Table 12-3 DMA Address Descriptor 445
Table 14-1 CSIC Sub-blocks 479
Table 14-2 CSIC External Signals 480
Table 14-3 BT.656 Header Code 481
Table 14-4 BT.656 Header Data Bit Definition 482
Table 14-5 CSIC FIFO Distribution 483
Table 15-1 TCON_LCD Sub-blocks 664
Table 15-2 TCON_LCD External Signals 665
Table 15-3 The Correspondence between LCD and RGB 665
Table 15-4 The Correspondence among LCD, LVDS, and DSI. 666
Table 15-5 HV Panel Signals 669
Table 15-6 Analysis of XY Signal 671
Table 15-7 CPU Panel Signals 671
Table 15-8 HV Mode Configuration Example 674
Table 15-9 BT.656 Mode Configuration Example 674
Table 15-10 i8080 Mode Configuration Example 676
Table 15-11 MIPI DSI External Signals 715
Table 15-12 DISPLL Specifications 715
Table 15-13 Parameter Configuration 718
Table 16-1 Audio Codec External Signals 756
Table 16-2 I2S/PCM External Signals 797
Table 16-3 DMIC External Signals 839
Table 16-4 DVC Gain Configuration 842
Table 16-5 OWA External Signals 858
Table 16-6 Biphase-Mark Encoder 859
Table 16-7 Preamble Codes 860
Table 16-8 Bit Allocation of Data-Burst in IEC 60958 Subframes 861
Table 16-9 The Relation between Different System Clock and Sample Ratio 863
Table 17-1 GMAC Sub-blocks 889
Table 17-2 GMAC External Signals 890
Table 17-3 CAN-FD External Signals 897
Table 17-4 Descriptor0 Definition 899
Table 17-5 Descriptor1 Definition 899
Table 17-6 Descriptor2 Definition 900

Draft Version
ALLWINNER

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Table 17-7 LOCAL_BUS_CTRL External Signals....977
Table 17-8 Command Phase Format (8-bit) 979
Table 17-9 Command Phase Format (16-bit) 980
Table 17-10 Big-Endian Data Mapping....981
Table 17-11 LBC Command Format 982
Table 17-12 DMA Descriptor0 Definition 985
Table 17-13 DMA Descriptor1 Definition 986
Table 17-14 DMA Descriptor2 Definition 986
Table 17-15 DMA Descriptor3 Definition 987
Table 17-16 LBC_MODE_CTRL_REG Definition 990
Table 17-17 LBC_TIMESCALE_CTRL_REG Definition 991
Table 17-18 LBC_READY_CTRL_REG Definition 991
Table 17-19 LBC_DP_CTRL_REG Definition....992
Table 17-20 LBC_BE_CTRL_REG Definition 992
Table 17-21 LBC_CS_TIMING_CTRL_REG Definition....993
Table 17-22 LBC_ALE_TIMING0_CTRL_REG Definition....993
Table 17-23 LBC_ALE_TIMING1_CTRL_REG Definition....993
Table 17-24 LBC_WE_TIMING_CTRL_REG Definition....993
Table 17-25 LBC_OE_TIMING0_CTRL_REG Definition 994
Table 17-26 LBC_OE_TIMING1_CTRL_REG Definition 994
Table 17-27 LBC_CYCLE_TIMING_CTRL_REG Definition 994
Table 17-28 LBC_ACCESS_TIMING_CTRL_REG Definition....994
Table 17-29 LBC_PAGE_TIMING_CTRL_REG Definition....995
Table 17-30 LBC_DATAMUX_TIMING_CTRL_REG Definition 995
Table 17-31 LBC_CLE_TIMING0_CTRL_REG Definition....995
Table 17-32 LBC_TURNAROUND_CTRL_REG Definition....995
Table 17-33 USB2.0 DRD External Signals 1051
Table 17-34 USB2.0 Host External Signals....1053
Table 17-35 GPADC External Signals 1056
Table 17-36 GP_SMP_TMS Value Corresponding to Each Sampling Frequency 1061
Table 17-37 TPADC External Signals 1087
Table 17-38 Median and Averaging Filter Size....1091
Table 17-39 Median and Averaging Filter Example 1092
Table 17-40 Port Function....1101

Draft Version
ALLWINNER

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Table 17-41 Multi-function Port....1105
Table 17-42 Port A Multiplex Function....1106
Table 17-43 Port B Multiplex Function....1106
Table 17-44 Port C Multiplex Function....1107
Table 17-45 Port D Multiplex Function....1107
Table 17-46 Port E Multiplex Function....1108
Table 17-47 Port F Multiplex Function....1108
Table 17-48 Port G Multiplex Function....1108
Table 17-49 Port J Multiplex Function....1109
Table 17-50 Port K Multiplex Function....1109
Table 17-51 RTC_GPIO Multiplex Function....1110
Table 17-52 UART External Signals....1398
Table 17-53 Errors of Baud Rate in UART Mode....1403
Table 17-54 Errors of Baud Rate in IrDA Mode....1403
Table 17-55 Errors of Baud Rate in RS485 Mode....1404
Table 17-56 Errors of Baud Rate in 19.2M and 26M Clock Source....1404
Table 17-57 SPI Sub-blocks....1437
Table 17-58 SPI External Signals....1438
Table 17-59 SPI Transmit Format....1439
Table 17-60 SPI Old Sample Mode....1446
Table 17-61 SPI New Sample Mode....1446
Table 17-62 SPI_DBI Sub-blocks....1473
Table 17-63 GPIO multiplexing of SPI1 and DBI....1474
Table 17-64 SPI_DBI External Signals....1474
Table 17-65 2 data lane sclk frequency....1488
Table 17-66 3 line sclk frequency....1488
Table 17-67 4 line sclk frequency....1488
Table 17-68 TWI Sub-blocks....1528
Table 17-69 TWI External Signals....1528
Table 17-70 PWM External Signals....1556
Table 17-71 PWMCS Block Description....1621
Table 17-72 PWMCS External Signals....1622
Table 17-73 PWMCS Signal Multiplexing....1623
Table 17-74 FIFO Command Write Step....1628

Draft Version
ALLWINNER

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Table 17-75 LEDC Sub-blocks 1630
Table 17-76 LEDC External Signals 1631
Table 17-77 LED Data Input Time 1631
Table 17-78 CIR Receiver External Signals 1647
Table 17-79 IR_TX External Signals 1660

Draft Version
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About This Document

Purpose and Target Readers

This document details the features, logical structures, functional description, programming guidelines, and related registers of each module about the SoC. A basic knowledge of digital circuits, logical circuits, SoCs, and so on is necessary to use this manual. Particular attention should be paid to the cautions and notes within the body of the text.

Related Documentation

For a complete listing of related documentation and development-support tools for the device, contact the Allwinner FAE or access the Allwinner Customer Service Platform by visiting https://open.allwinnertech.com/.

Revision Number Definition

Revision 0.10-0.1x

This document is released based on the experimental sample(s). Therefore, it may be modified, calibrated and supplemented hereafter.

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If you have any questions about the document, please contact us to confirm and obtain the latest version.

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
CAUTION	A caution means that damage to equipment is possible.
NOTE	Provides additional information to emphasize or supplement important points of the main text.

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Table Content Conventions

The table content conventions that may be found in this document are defined as follows.

Symbol	Description
-	The cell is blank.

Notation of Symbols and Numbers

The expressions of the data capacity, the frequency, and the data rate are described as follows.

Type	Symbol	Value
Data capacity	K	1024
	M	1,048,576
	G	1,073,741,824
Frequency, data rate	k	1000
	M	1,000,000
	G	1,000,000,000

The expressions of addresses and data are described as follows.

Number	Example	Description
0x	0x0200,0x79	Address or data in hexadecimal
0b	0b010,0b00 000 111	Data or sequence in binary (register description is excluded.)
X	00X,XX1	In data expression, X indicates 0 or 1.For example, 00X indicates 000 or 001, XX1 indicates 001,011,101 or 111.

Register Attributes

The register attributes that may be found in this document are defined as follows.

Abbr. & Symbol	Description
R	Read Only
R/W	Read/Write
R/WAC	Read/Write-Automatic-Clear, clear the bit automatically when the operation of complete. Writing 0 has no effect.
R/WC	Read/Write-Clear
R/W0C	Read/Write 0 to Clear. Writing 1 has no effect
R/W1C	Read/Write 1 to Clear. Writing 0 has no effect
R/W1S	Read/Write 1 to Set. Writing 0 has no effect
W	Write Only

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Abbr. & Symbol	Description
/	If the default value in the row of a bit or multiple bits is / , this bit is unused.
UDF	If the default value of a bit or multiple bits is UDF , this default value is undefined.

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1 Product Description

1.1 Overview

T153 is a quad-core architecture solution targeted for automation applications, such as programmable logic controllers (PLCs), human-machine interface (HMI), etc.

Equipped with a quad-core Arm® Cortex®-A7, a single-core RISC-V E907, and support for DDR3/DDR3L/DDR4 memory, the T153 delivers powerful compute performance and fast response, making it ideal for demanding automation tasks.

The processor features three Gigabit Ethernet interfaces, two CAN_FD interfaces, and 16-/32-bit local bus, supporting high-throughput networking for complex data-driven use cases. Its integrated image signal processor and display engine enable clear, real-time visual feedback for managing intricate manufacturing processes.

With extensive peripheral support, the T153 offers 24-channel GPADC, 6 TWIs, 30-channel PWM. These interfaces provide flexibility for diverse applications, enabling easy integration and scalability in automation systems.

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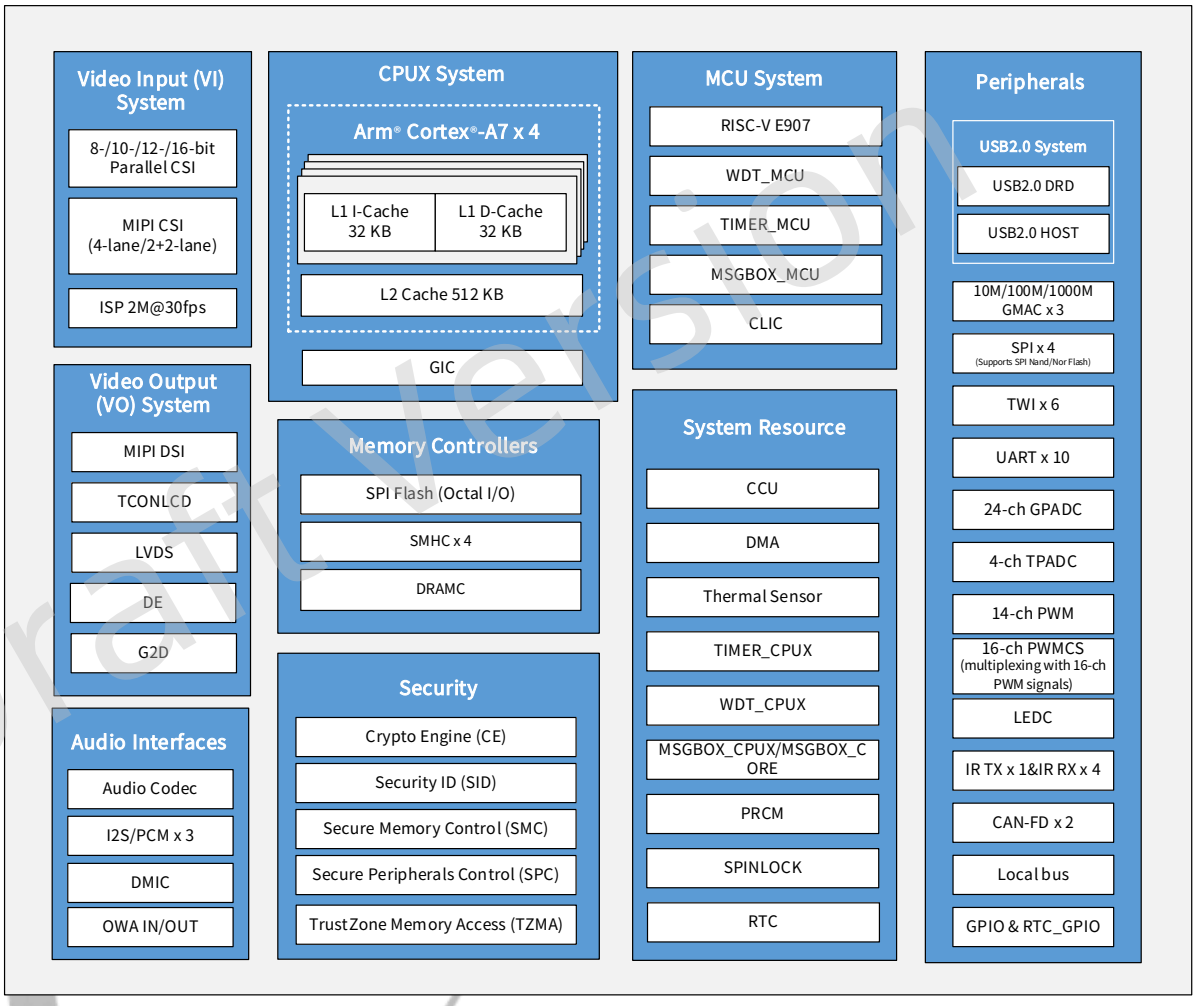
1.2 Block Diagram

The following figure shows the system block diagram.

Figure 1-1 System Block Diagram



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2 Memory Map

2.1 CPUX Memory Map

The following table shows the memory map viewed from Arm® A7.

Region	Address	Size (Bytes)
BROM & SRAM		
FIXS_BROM	0x0000 0000---0x0000 BFFF	48 KB
HS_BROM	0x0001 0000---0x0001 4FFF	20 KB
NS_BROM	0x0001 5000---0x0001 8FFF	16 KB
SRAM C	0x0004 0000---0x0006 7FFF	160 KB
DE_SHARE_SRAM	0x0006 8000---0x0006 BFFF	16 KB
SP0		
CCU	0x0200 2000---0x0200 5FFF	16 KB (OS reserves 8 KB)
SPC0	0x0200 6000---0x0200 6FFF	4 KB
WDT_CPUX	0x0205 0000---0x0205 0FFF	4 KB
IRTX	0x0208 0000---0x0208 0FFF	4 KB
IRRX0	0x0208 1000---0x0208 1FFF	4 KB
IRRX1	0x0208 2000---0x0208 2FFF	4 KB
IRRX2	0x0208 3000---0x0208 3FFF	4 KB
IRRX3	0x0208 4000---0x0208 4FFF	4 KB
LEDC	0x0208 5000---0x0208 5FFF	4 KB
THS	0x0208 6000---0x0208 6FFF	4 KB
GPADC0	0x0208 7000---0x0208 7FFF	4 KB
GPADC1	0x0208 8000---0x0208 8FFF	4 KB
GPADC2	0x0208 9000---0x0208 9FFF	4 KB
TPADC	0x0208 C000---0x0208 CFFF	4 KB
PWMCS0	0x020A 0000---0x020A FFFF	64 KB
PWMCS1	0x020B 0000---0x020B FFFF	64 KB
PWM2	0x020C 0000---0x020C FFFF	64 KB
AXI_MONITOR_RISCV	0x0209 3000---0x0209 3FFF	4 KB
AHB_MONITOR_CPU	0x0209 4000---0x0209 4FFF	4 KB
AHB_MONITOR_DCU	0x0209 5000---0x0209 5FFF	4 KB
AUDIO_SP0		
Audio Codec	0x0203 0000---0x0203 0FFF	4 KB
DMIC	0x0203 1000---0x0203 1FFF	4 KB
I2S0	0x0203 2000---0x0203 2FFF	4 KB
I2S1	0x0203 3000---0x0203 3FFF	4 KB
I2S2	0x0203 4000---0x0203 4FFF	4 KB

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Region	Address	Size (Bytes)
OWA0	0x0203 6000---0x0203 6FFF	4 KB
SP1		
TWI0	0x0251 0000---0x0251 0FFF	4 KB
TWI1	0x0251 1000---0x0251 1FFF	4 KB
TWI2	0x0251 2000---0x0251 2FFF	4 KB
TWI3	0x0251 3000---0x0251 3FFF	4 KB
TWI4	0x0251 4000---0x0251 4FFF	4 KB
TWI5	0x0251 5000---0x0251 5FFF	4 KB
SP_UART		
UART0	0x0260 0000---0x0260 0FFF	4 KB
UART1	0x0260 1000---0x0260 1FFF	4 KB
UART2	0x0260 2000---0x0260 2FFF	4 KB
UART3	0x0260 3000---0x0260 3FFF	4 KB
UART4	0x0260 4000---0x0260 4FFF	4 KB
UART5	0x0260 5000---0x0260 5FFF	4 KB
UART6	0x0260 6000---0x0260 6FFF	4 KB
UART7	0x0260 7000---0x0260 7FFF	4 KB
UART8	0x0260 8000---0x0260 8FFF	4 KB
UART9	0x0260 9000---0x0260 9FFF	4 KB
SH0		
SYSCTRL	0x0300 0000---0x0300 0FFF	4 KB
DMAC0	0x0300 1000---0x0300 1FFF	4 KB
MSGBOX_CPUX	0x0300 3000---0x0300 3FFF	4 KB
SPINLOCK	0x0300 5000---0x0300 5FFF	4 KB
SID	0x0300 6000---0x0300 67FF	2 KB
SECURE_DEBUG_CFG	0x0300 6800---0x0300 6FFF	1 KB
TIMER0_CPUX	0x0300 8000---0x0300 8FFF	4 KB
DCU	0x0301 0000---0x0301 FFFF	64 KB (can only be accessed by external AHB)
CE_NS	0x0304 0000---0x0304 07FF	2 KB
CE_S	0x0304 0800---0x0304 0FFF	2 KB
DRAMC	0x0310 2000---0x0330 1FFF	2 MB
GIC	0x0340 0000---0x034E FFFF	15*64 KB
MSGBOX_CORE0	0x0360 0000---0x0360 0FFF	4 KB
MSGBOX_CORE1	0x0360 1000---0x0360 1FFF	4 KB
MSGBOX_CORE2	0x0360 2000---0x0360 2FFF	4 KB
MSGBOX_CORE3	0x0360 3000---0x0360 3FFF	4 KB
GPIO	0x0360 4000---0x0360 5FFF	8 KB
MSI	0x0360 6000---0x0360 6FFF	4 KB
SMC	0x0360 7000---0x0360 7FFF	4 KB

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Region	Address	Size (Bytes)
SH2		
DMAC1	0x0400 0000---0x0400 0FFF	4 KB
SMHC0	0x0402 0000---0x0402 0FFF	4 KB
SMHC1	0x0402 1000---0x0402 1FFF	4 KB
SMHC2	0x0402 2000---0x0402 2FFF	4 KB
SMHC3	0x0402 3000---0x0402 3FFF	4 KB
SPI0	0x0402 5000---0x0402 5FFF	4 KB
SPI1	0x0402 6000---0x0402 6FFF	4 KB
SPI2	0x0402 7000---0x0402 7FFF	4 KB
SPI3	0x0402 8000---0x0402 8FFF	4 KB
USB2P0_PHY	0x0410 0000---0x0417 FFFF	512 KB
USB2P0_SYS_DIG	0x0418 0000---0x041F FFFF	512 KB
USB0	0x0420 0000---0x042F FFFF	1 MB
USB1	0x0430 0000---0x043F FFFF	1 MB
GMAC0	0x0450 0000---0x0450 FFFF	64 KB
GMAC1	0x0451 0000---0x0451 FFFF	64 KB
GMAC2	0x0452 0000---0x0452 FFFF	64 KB
CAN_SYS	0x0453 0000---0x0454 FFFF	128 KB
DE_SYS		
DE	0x0500 0000---0x053F FFFF	4 MB
G2D	0x0544 0000---0x0547 FFFF	256 KB
VIDEO_OUT_SYS		
DISPLAY_TOP	0x0550 0000---0x0550 0FFF	4 KB
TCON_LCD0	0x0550 1000---0x0550 1FFF	4 KB
MIPI_DSI0	0x0550 6000---0x0550 7FFF	8 KB
VIDEO_IN_SYS		
CSIC_CCU	0x05800000---0x058000FF	256 B
CSIC_TOP	0x05800800---0x058009FF	512 B
CSIC_PARSER0	0x05820000---0x05820FFF	4 KB
CSIC_PARSER1	0x05821000---0x05821FFF	4 KB
CSIC_PARSER2	0x05822000---0x05822FFF	4 KB
CSIC_DMA0	0x05830000---0x05830FFF	4 KB
CSIC_DMA1	0x05831000---0x05831FFF	4 KB
CSIC_DMA2	0x05832000---0x05832FFF	4 KB
ISP	0x0590 0000---0x05CF FFFF	4 MB
SH_LBC		
LOCALBUS_REG	0x0281 0000---0x0281 0FFF	4 KB
APBS0		
SPC1	0x0702 0000---0x0702 0FFF	4 KB
PRCM	0x0701 0000---0x0701 FFFF	64 KB
TWD	0x0702 2000---0x0702 2FFF	4 KB
CPUIDLE	0x0705 0000---0x0705 FFFF	64 KB

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Region	Address	Size (Bytes)
PCK600_QCHANNEL	0x0706 0000---0x0706 7FFF	32 KB
RTC_GPIO	0x0709 6000---0x0709 6FFF	4 KB
AHBS		
RTC	0x0709 0000---0x0709 0FFF	4 KB
CPUX Related		
CPU_CS	0x0800 0000---0x0800 0FFF	4 KB
CPU_SUBSYS_CTRL	0x0810 0000---0x0810 0FFF	4 KB
TIMESTAMP_STA	0x0811 0000---0x0811 0FFF	4 KB
TIMESTAMP_CTRL	0x0812 0000---0x0812 0FFF	4 KB
CPU_IDC_BASE	0x0813 0000---0x0813 0FFF	4 KB
CPUX_PLL_CFG_BASE	0x0814 0000---0x0814 0FFF	4 KB
TZMA	0x0815 0000---0x0815 0FFF	4 KB
C0_CPUX_CFG	0x0901 0000---0x0901 0FFF	4 KB
CPUX_CLU_DEBUG	0x0940 0000---0x970_FFFF	4 MB
SPI_FLASH Direct Running Space		
SPI_FLASH	0x0280 0000---0x0280 0FFF	4 KB
SPI_FLASH_XIP	0x0E00 0000---0x0FFF FFFF	32 MB (SPI Flash memory can be accessed directly from this space and needs to be configured as XIP mode before access.)
SH_MCU		
MCU_CFG	0x01A0 0000---0x01A0 0FFF	4 KB
WDT_MCU	0x01A0 2000---0x01A0 2FFF	4 KB
MCU_LCNT	0x01A0 4000---0x01A0 4FFF	4 KB
MSGBOX_MCU	0x01A0 6000---0x01A0 6FFF	4 KB
TIMER_MCU	0x01A0 8000---0x01A0 8FFF	4 KB
MCU Inner Space (0x2FF0 0000-0x3FFF FFFFcan only be accessed by E907)		
MCU_CLINT	0x3000 0000---0x3000 FFFF	64 KB
MCU_CLIC	0x3080 0000---0x3080 4FFF	20 KB
MCU_SYSMAP	0x3FFF F000---0x3FFF FFFF	4 KB
DRAM Space		
DRAM SPACE	0x4000 0000---0x BFFF FFFF	2 GB
LBC Direct Running Space		
LOCALBUS_DAT	0x2 0000 0000--0x23FFF FFFF	1 GB

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2.2 MCU Memory Map

The following table shows the memory map viewed from the E907.

Region	Address	Size	Description
SRAM			
SRAMC	0x0004 0000---0x0006 7FFF	160 KB	
DE_SHARE_SRAM	0x0006 8000---0x0006 BFFF	16 KB	
Peripheral Space			
MCU_CFG	0x01A0 0000---0x01A0 0FFF	4 KB	
WDT_MCU	0x01A0 2000---0x01A0 2FFF	4 KB	
MCU_LCNT	0x01A0 4000---0x01A0 4FFF	4 KB	
MSGBOX_MCU	0x01A0 6000---0x01A0 6FFF	4 KB	
TIMER_MCU	0x01A0 8000---0x01A0 8FFF	4 KB	
L4_TOP	0x0190 0000---0x0190 0FFF 0x01B0 0000---0x1FFF FFFF	/	system domain, consistent with CPUX memory mapping
MCU_PAHB	0x2FF0 0000---0x2FFF FFFF	1 MB	
E907 TCIP			
MCU_CLINT	0x3000 0000---0x3000 FFFF	64 KB	
MCU_CLIC	0x3080 0000---0x30804 FFFF	20 KB	
MCU_SYSMAP	0x3FFF F000---0x3FFF FFFF	4 KB	
DRAM Space			
DRAM SPACE	0x4000 0000---0x7FFF FFFF	1 GB	Fixed access to the first 1 GB peripheral space and the first 3 GB DDR space by default.
DRAM SPACE	0x8000 0000---0xBFFF FFFF	1 GB	
DRAM SPACE	0xC000 0000---0xFFFF FFFF	1 GB	
E907 accesses DRAM space through AXI. Address mapping is required.			
DRAM SPACE	0x1 0000 0000---0x1 3FFF FFFF	1 GB	When E907_DDR_REMAP=3'b1, map 0xC000 0000---0xFFFF FFFF to 0x1 0000 0000---0x1 3FFF FFFF.

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3 System Boot

3.1 Boot ROM (BROM)

3.1.1 Overview

The BROM system is actually split up into two parts: FEL and Media Boot. The task of FEL is to write the external data to the local NVM, and the task of the Media Boot is to load from NVM to obtain and run an effective and legitimate BOOT0.

The BROM system includes the following features:

- Support the core0 of CPUX boot process
- Support Mandatory Upgrade Process through USB or SMHC0
- Support GPIO to select the kind of boot media to boot
- Support eFuse to select the kind of boot media to boot
- Support Normal Boot and Secure Boot
- Support loads only certified firmware in Secure Boot
- Ensure that the Secure Boot is in a trusted environment

3.1.2 External Signals

Table 3-1 BROM External Signals

Signal Name	Description	Type
FEL	Boot Select Jump to the Try Media Boot process when FEL is high level, or else enter into the mandatory upgrade process.	I
BOOT-SEL0	Boot Media Select 0	I
BOOT-SEL1	Boot Media Select 1	I

3.1.3 Boot Medium Select

The BROM system supports the following boot media:

- SD Card
- eMMC
- SPI NOR Flash (Quad Mode and SIngle Mode)
- SPI NAND Flash

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There are two ways to select the boot medium: GPIO Select and eFuse Select. The BROM will read the state of BOOT_MODE first, and then select the boot medium to boot according to the state of BOOT_MODE. The BOOT_MODE is the BROM_Config in the eFuse mapping.

The following table shows the BOOT_MODE setting:

Table 3-2 BOOT_MODE Setting

BOOT_MODE[0]	Boot Select type
0	GPIO Boot
1	eFuse Boot

NOTE

The BOOT_MODE BIT is bit [0] of the eFuse register 0x03006210.

3.1.3.1 GPIO Boot Select

If the state of the BOOT_MODE is 0, the boot medium is decided by the value of the GPIO pin.

Table 3-3 GPIO Boot Select Configuration

Pin_Boot_Select[1:0]	Boot media
00	SPI0 NAND->SPI0 NOR(4 wire)-> SPI0 NOR(1 wire)->USB0
01	SPI0 NOR(4 wire)-> SPI0 NOR(1 wire)->SPI0 NAND->USB0
10	SD Card->SPI0 NAND->SPI0 NOR(4 wire)-> SPI0 NOR(1 wire)->USB0
11	SD Card -> SPI2 NOR(4 wire)-> SPI2 NOR(1 wire) ->SPI2 NAND -> EMMC2 -> SPI0 NAND->USB

NOTE

- The status of the GPIO boot select pin can be read by the bit[12:11] of the system configuration module (register: 0x03000024).
- The EMMC2 boot process is EMMC_USER-> EMMC_BOOT.

3.1.3.2 eFuse Boot Select

If the state of the BOOT_MODE is 1, select the eFuse Boot.

The eFuse_Boot_Select_Cfg is divided into 3 groups and each group is 4-bit. The following table shows the groups of eFuse_Boot_Select.

Table 3-4 Groups of eFuse_Boot_Select

eFuse_Boot_Select_Cfg [11:0]	Description
eFuse_Boot_Select[3:0]	eFuse_Boot_Select_1

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eFuse_Boot_Select_Cfg [11:0]	Description
eFuse_Boot_Select[7:4]	eFuse_Boot_Select_2
eFuse_Boot_Select[11:8]	eFuse_Boot_Select_3

These three groups take effect with the following priority:

eFuse_Boot_Select_1 -> eFuse_Boot_Select_2 -> eFuse_Boot_Select_3

For example, eFuse_Boot_Select_2 will not take effect unless eFuse_Boot_Select_1 is set as 0b1111(skip), eFuse_Boot_Select_3 will not take effect unless eFuse_Boot_Select_2 is set as 0b1111(skip), etc. If all three groups are set to 0b1111, no other groups can be used for boot select, BROM assume "try" is selected.

In the Try mode, the BROM follows the order below to select the boot medium:

SD Card -> eMMC ->SPI0 NOR -> SPI0 NAND

The following table shows the boot medium priority for the different values of eFuse_Boot_Select_n, where n = [3:1].

Table 3-5 eFuse Boot Select Setting

eFuse_Boot_Select_n	Boot media
0000	Try
0001	SD Card
0010	SD Card->EMMC_USER -> EMMC_BOOT
0011	SD Card->SPI0 NOR
0100	SD Card->SPI0 NAND
0101	SD Card
0110	SD Card->EMMC_BOOT -> EMMC_USER
1011	SD Card->SPI2 NOR in PE
1100	SD Card->SPI2 NAND in PE
1111	Skip (when n=1,2) -- Boot media is decided by eFuse_Boot_Select_(n+1) Try (when n=3)

NOTE

The status of the eFuse boot select pin is the bit [11:0] of the eFuse register 0x03006214.

3.1.4 Boot Process Description

3.1.4.1 Normal BROM Process

In Normal BROM Mode, the system boot starts from CPUX Core0, and then the BROM will read the state of the FEL pin. If the FEL pin is low, the system will jump to the mandatory upgrade process.

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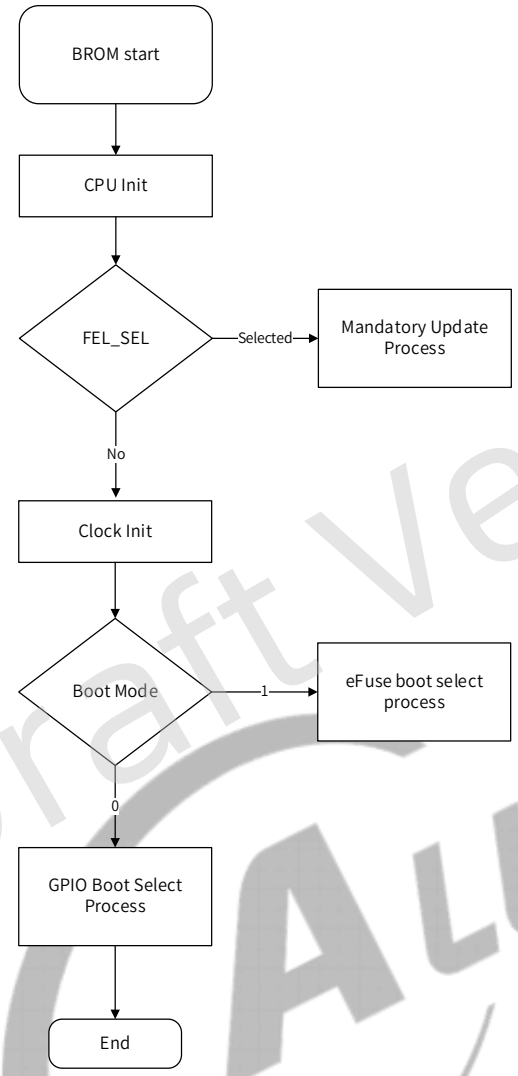
The following figure shows the boot process in Normal BROM Mode.

Figure 3-1 Boot Process in Normal BROM Process



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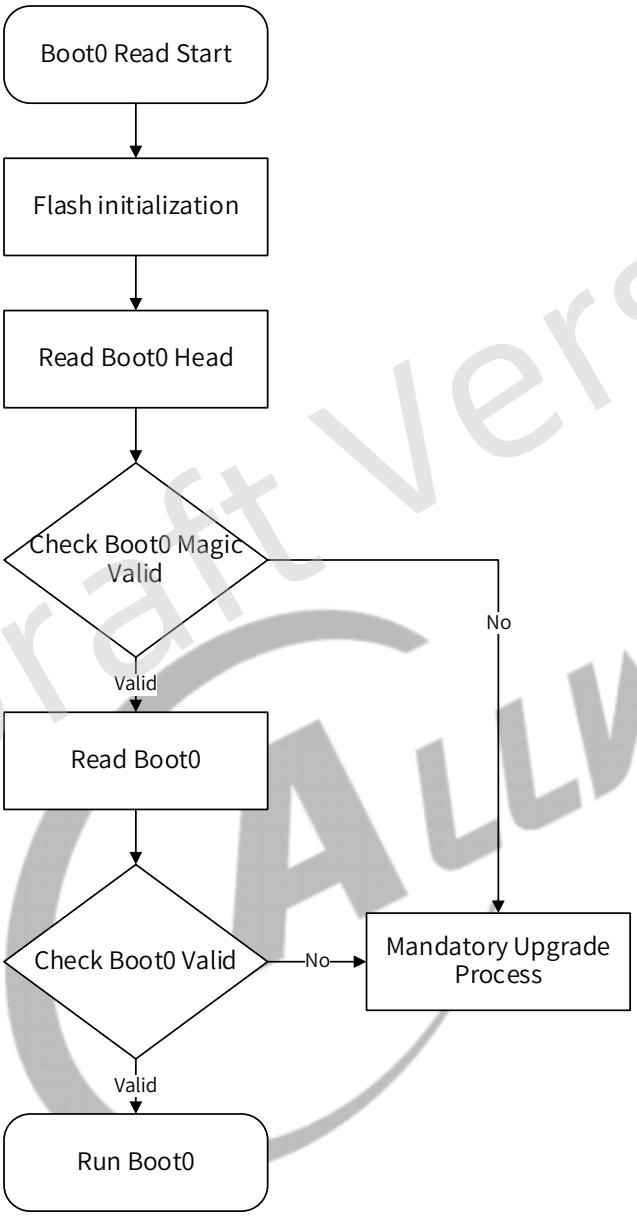
The GPIO boot select process and the efuse boot select process will read the boot0 , and check the validity of boot0.

The following figure shows the boot0 read process.

Figure 3-2 Boot0 Read Process



3.1.4.2 Secure BROM Algorithms

The Secure Boot has the following some requirements:

- Support cryptographic algorithms
  - ECDSA-256/384/521
  - RSA-2048/3072/4096
  - SHA-256

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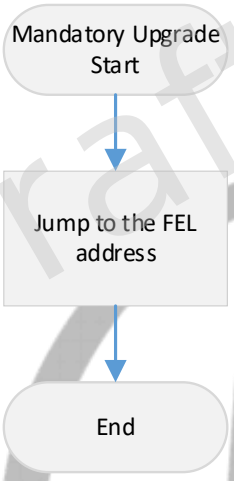
- Support OTP/eFuse
- Support lifecycle

Before runing Security Boot software, check the software whether it has been modified or replaced. The system will check and verify the integrity of the certificate as the certificate has used the RSA or EDCSA algorithm signature. The system also uses the Crypto Engine (CE) hardware module to accelerate the speed of encryption and decryption. Using standard cryptography ensures that the firmware images can be trusted, so the Secure Boot Process ensure the system security state is as expected.

3.1.4.3 Mandatory Upgrade Process

If the FEL pin is detected to pull low, the system will jump to the mandatory upgrade process. The following figure shows the mandatory upgrade process.

Figure 3-3 Mandatory Upgrade Process



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4 Power, Reset, and Clock Management

4.1 Clock Controller Unit (CCU)

4.1.1 Overview

The clock controller unit (CCU) controls the PLL configurations and most of the clock generation, division, distribution, synchronization, and gating in system. The input signals of the CCU include the external clock for the reference frequency (24 MHz). The outputs from the CCU are mostly clocks to other blocks in the system.

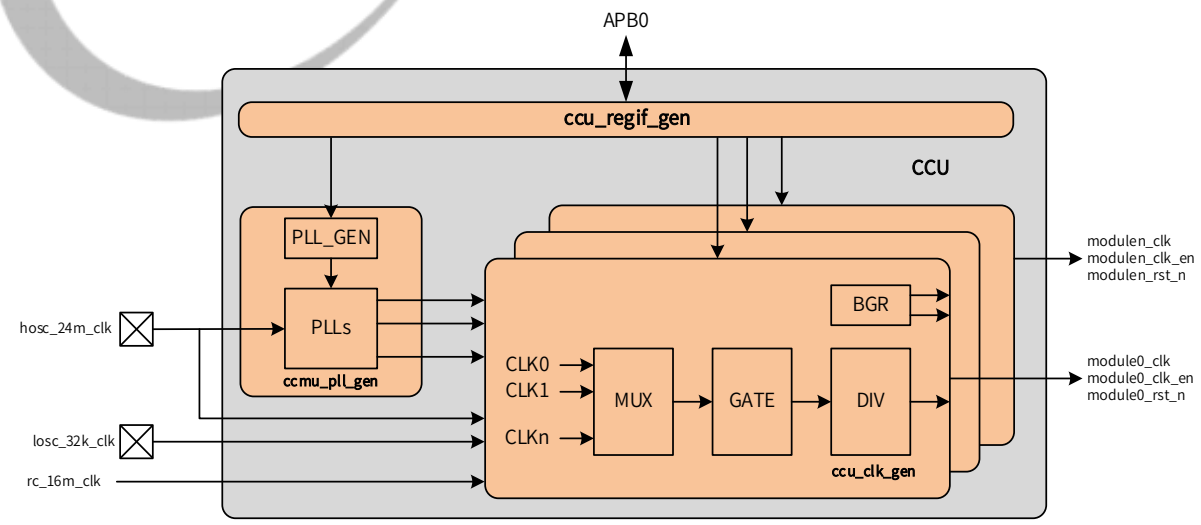
The CCU includes the following features:

- Clock sources: DCXO, external 32KHz, and RC 16MHz
- Bus source and divisions
- Clock output control
- Configurable modules clock
- Bus clock gating
- Bus software reset
- Phase-Locked Loop (PLL) management (5 PLLs: PLL_PERI0, PLL_PERI1, PLL_VIDEO0, PLL_AUDIO0, PLL_CPU)

4.1.2 Block Diagram

The following figure shows the functional block diagram of the CCU.

Figure 4-1 CCU Block Diagram



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4.1.3 External Signals

Table 4-1 CCU External Signals

Signal Name	Description	Type
CLK-FANOUT0	Internal Clock Fanout 0 Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, 50 MHz, and so on. For detailed information, see section 4.1.5.7 Clock Fanout.	O
CLK-FANOUT1	Internal Clock Fanout 1 Optional Frequency: 32 kHz, 12 MHz, 16 MHz, 24 MHz, 25 MHz, 27 MHz, 50 MHz, and so on. For detailed information, see section 4.1.5.7 Clock Fanout.	O

4.1.4 Functional Description

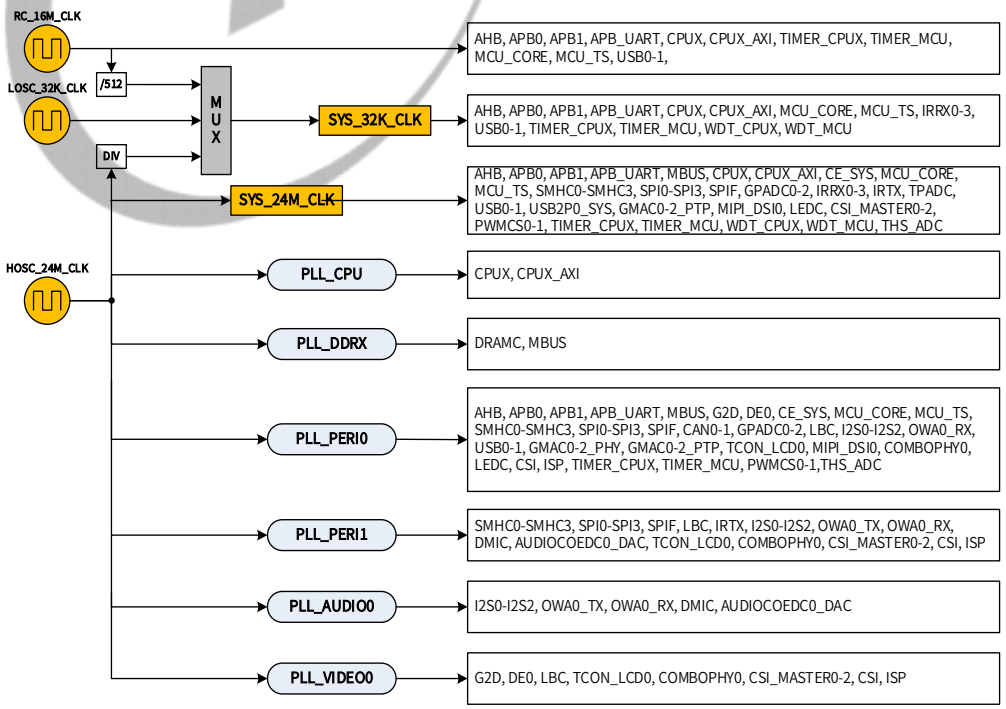
NOTE

This device has total 8 PLLs. 7 PLLs are described in CCU. See section 15.3 MIPI DSI for DISPLL description.

4.1.4.1 PLL Distribution

The following figure shows the PLL distribution for buses and modules.

Figure 4-2 PLL Distribution



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4.1.4.2 PLL Specifications

The specifications of each PLL are shown in the following table.

Table 4-2 PLL Specifications

PLL	Stable Operating Frequency	Spread Spectrum	Linear FM	Pk-Pk	Lock Time
PLL_CPU	480 MHz-2.016 GHz	Yes	Yes	50ps	1.5ms
PLL_DDR0	1 GHz-2.8 GHz	Yes	No	5% of the PLL_DDR output clock cycle	0.1ms
PLL_DDR1	1 GHz-1.8 GHz	Yes	No		0.1ms
PLL_PERI0	1.26 GHz-2.52 GHz	Yes	No	125ps	500us
PLL_PERI1	1.26 GHz-2.52 GHz	Yes	No	125ps	500us
PLL_VIDEO0	1.26 GHz-2.52 GHz	Yes	No	70ps@1 GHz	500us
PLL_AUDIO0	1.56 GHz-3.12 GHz	Yes	No	30ps at 3.12 GHz (50ppm)	500us

4.1.4.3 PLL VCO Frequency Calculation

The following is the VCO frequency calculation formula:

$F_{VCO}= (F_{DCXO}/M1) *N$

Where,

F_{VCO} is the VCO frequency;

F_{DCXO} is the DCXO frequency, F_{DCXO}=24 MHz;

M1 is the pre-frequency division factor of PLL;

N is the frequency doubling factor of PLL.

4.1.4.4 PLL Spread Spectrum/Fractional Frequency Division

PLL Spread Spectrum and fractional frequency division are configured by PLL pattern0 control registers and PLL pattern1 control registers respectively. The two kinds of registers are immediately effective, so the procedures of generic PLLs and linear FM PLLs are the same when configuring their Spread Spectrum and fractional frequency division. In the pattern registers, the main parameters are as follows.

- Static Parameters

The static parameters (such as: SPR_FREQ_MODE and DITHER) are used for adjusting the order, phase interpolation, and so on of fractional frequency division loop. To modify these parameters,

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PLL spread spectrum and fractional frequency division should be disabled until the modification finishes.

The static parameters are as follows:

Parameters	Registers	Description
SPR_FREQ_MODE	PLL pattern0 control registers	SDM Mode Select When selecting 00 mode, PLL_SDM_EN needs to be configured as 1 simultaneously. (Linear FM PLL doesn't have 00 mode.)
DITHER	PLL pattern1 control registers	Scrambling Control A pseudo-random sequence will be superimposed on the frequency modulation curve at the beginning.

- **Control Parameters**

The control parameters are used for controlling frequency curves and support dynamic switch.

The control parameters are as follows:

Parameters	Registers	Description
WAVE_STEP	PLL pattern0 control registers	PLL Spread Spectrum Step The equation is WAVE_STEP = 2^17*(x2-x1) *2/(Fclk/FREQ) where, x2 is the up limit of spread spectrum; x1 is the down limit of spread spectrum; Fclk is PLL reference clock frequency; FREQ is the frequency of spread spectrum; The frequency multiplication factor corresponding to PLL is from N.x1 to N.x2.
WAVE_BOT	PLL pattern0 control registers	PLL Spread Spectrum Initiation The equation is WAVE_BOT = x1*2^17 (up spread spectrum) WAVE_BOT = x2*2^17 (down spread spectrum, only linear FM PLL)
FREQ (only generic PLL)	PLL pattern0 control registers	Spread Spectrum Frequency When dynamic switching FREQ, WAVE_STEP should be switched simultaneously.
SDM_CYCLE (only linear FM PLL)	PLL pattern1 control registers	SDM Cycle It refers to the number of working clock cycles required for one FM period and is used to specify the turning point of the triangle wave. When configuring, you need to combine WAVE_STEP and WAVE_BOT for calculation.

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Parameters	Registers	Description
SDM_DIRECTION (only linear FM PLL)	PLL pattern1 control registers	SDM Direction Linear FM PLL supports up spread spectrum and down spread spectrum. When this parameter is configured as down spread spectrum, WAVE_BOT needs to be configured as the up limit of spread spectrum.
FRAC_EN	PLL pattern1 control registers	PLL Fractional Frequency Division Enable It supports dynamic switch.
FRAC_IN	PLL pattern1 control registers	Fractional Frequency Division Points It supports dynamic switch. The equation is FRAC_IN = x*2^17

- Function Master Enable Parameter

Parameter	Register	Description
SIG_DELT_PAT_EN	PLL pattern0 control registers	Sigma Delta Pattern Enable To use PLL spread spectrum and fractional frequency division, SIG_DELT_PAT_EN should be enabled.

- SDM Clock Source Select Parameter

Parameter	Register	Description
SDM_CLK_SEL	PLL pattern0 control registers	SDM Clock Source Select When PLL_INPUT_DIV2 is 1, SDM_CLK_SEL needs to be configured as 1.

The following table shows the configuration of PLL spread spectrum and fractional frequency division modes.

SIG_DELT_PAT_EN	FRAC_EN	WAVE_STEP	FRAN_IN	MODE
1	0	Not equal to 0	/	Spread spectrum
1	0	Equal to 0	/	Fractional frequency division (as effective as spread spectrum)
1	1	/	Arbitrary	Fractional frequency division

4.1.4.5 PLL Linear Frequency Modulation

The linear frequency modulation is to solve the problem of relock when switching the N factor in the integer frequency multiplication mode. When in linear frequency modulation, you can directly configure the N factor, and the PLL will smoothly change the N factor to the target frequency point with the configured frequency modulation slope.

The linear frequency modulation relies on fractional frequency division. When modulating frequency of linear FM PLLs, the static parameters in the decimal frequency division need to be

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configured manually, and the control parameters are automatically controlled by linear frequency modulation.

4.1.5 Programming Guidelines

4.1.5.1 Generic PLL Configuration

Enable the PLL

Step 1 Write 1 to the LDO_EN of the PRCM_PLL_CTRL_REG1 register in Section 4.2.6.7.

Step 2 Write 1 to the PLL_OUTPUT_GATE of PLL_XXX_CTRL_REG.

Step 3 Configure the N, M, and, P of PLL_XXX__CTRL_REG.

Step 4 Write 1 to the PLL_EN of PLL PLL_XXX_CTRL_REG.

Step 5 Write 1 to the LOCK_ENABLE of PLL_XXX__CTRL_REG.

Step 6 Read LOCK_STATUS every 3 us for total three times. The PLL is locked if the value of every read is 1.

Step 7 Delay 20us.

Disable the PLL

Step 1 Ensure that the clock using the PLL to be disabled is switched to other clock sources, or the clock is disabled.

Step 2 Write 0 to the PLL_OUTPUT_GATE of PLL_XXX_CTRL_REG.

Step 3 Write 0 to the LOCK_ENABLE of PLL_XXX_CTRL_REG.

Step 4 Write 0 to the PLL_EN of PLL_XXX_CTRL_REG.

Step 5 Write 0 to the LDO_EN of the PRCM_PLL_CTRL_REG1 register in Section 4.2.6.7.

NOTE

- Frequent PLL enable/disable is not recommended, as the PLL enable/disable will arise the mutual interference of PLLs and system instability. To disable PLL, it is recommended to write 0 to the PLL_OUTPUT_GATE.
- The LDO_EN of PRCM_PLL_CTRL_REG1 register controls the enable of all system PLL's LDO. The LDO_EN needs to be configured as 1, unless system is power-off or all PLLs are disabled.

Implement Spread Spectrum

The spread spectrum technology is to convert a narrowband signal into a wideband signal. It helps to reduce the effect of electromagnetic interference (EMI) associated with the fundamental frequency of the signal.

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For the general PLL frequency, the calculation formula is as follows:

$$f = \frac{N+1+X}{P\cdot(M0+1)\cdot(M1+1)}\cdot 24MHz, 0<X<1$$

Where,

P is the frequency division factor of module or PLL;

M0 is the post-frequency division factor of PLL;

M1 is the pre-frequency division factor of PLL;

N is the frequency doubling factor of PLL;

X is the amplitude coefficient of the spread spectrum.

The parameters N, P, M1, and M0 are for the frequency division.

When M1 = 0, M0 = 0, and P = 1 (no frequency division), the calculation formula of PLL frequency can be simplified as follows:

$$f = (N+1+X)\cdot 24MHz,\ 0<X<1$$

$$[f_1, f_2] = (N+1+[X_1, X_2])\cdot 24MHz$$

$$WAVE_BOT = 2^{17}\cdot X_1$$

$$\mathrm{WAVE_STEP} = 2^{17}\cdot(X_2 - X_1)/(24\ \mathrm{MHz}/FREQ)\cdot 2$$

Where, WAVE_BOT and WAVE_STEP are bits of the PLL pattern control register, and FREQ is the frequency of the spread spectrum.

NOTE

Different PLLs have different calculate formulas, refer to the CTRL register of the corresponding PLL in Section 4.1.7 Register Description.

Follow the steps below to implement the spread spectrum:

Step 1 Configure the control register of the corresponding PLL

a) Calculate the factor N and decimal value X according to the PLL frequency and PLL frequency formula. Refer to the control register of the corresponding PLL (named PLL_xxx_CTRL_REG, where xxx is the module name) in Section 4.1.7 Register Description for the corresponding PLL frequency formula.

b) Write M0, M1, N, and PLL frequency to the PLL_XXX_CTRL_REG.

c) Configure the PLL_SDM_EN bit (bit [24]) of the PLL_XXX_CTRL_REG to 1 to enable the spread spectrum function.

Step 2 Configure the pattern control register of the corresponding PLL

a) Calculate the WAVE_BOT and WAVE_STEP of the pattern control register according to decimal value X and spread spectrum frequency.

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b) Write WAVE_BOT and WAVE_STEP to the PLL_xxx_PAT0_CTRL_REG register.

c) Configure the FREQ of PLL_xxx_PAT0_CTRL_REG register.

d) Configure the spread spectrum mode (SPR_FREQ_MODE) to 2 or 3.

e) If the PLL_INPUT_DIV2 of the PLL_XXX_CTRL_REG is 1, configure the spread spectrum clock source select bit (SDM_CLK_SEL) of the PLL pattern control register to 1. Otherwise, configure SDM_CLK_SEL to the default value 0.

f) Configure the SIG_DELT_PAT_EN bit (bit [31]) of this register to 1.

Step 3 Delay 20 us.

Configure the PLL Frequency

Step 1 Make sure the PLL is enabled. If not, refer to Section Enable the PLL to enable the PLL.

Step 2 Write 0 to the PLL_OUTPUT_GATE of PLL_XXX_CTRL_REG.

Step 3 Modify the values of N and M factors. (It is not recommended to configure M1 factor.)

Step 4 Write 0 and then write 1 to the LOCK_ENABLE of PLL_XXX_CTRL_REG.

Step 5 Read LOCK_STATUS every 3 us for total three times. The PLL is locked if the value of all three reads is 1.

Step 6 Delay 20us.

Step 7 Write 1 to the PLL_OUTPUT_GATE of PLL_XXX_CTRL_REG.

4.1.5.2 Linear FM PLL Configuration

There are two types of configuration parameters for linear FM PLL:

Immediate effect type: This kind of registers are immediately effective after parameters are configured and have the same read value as configuration value.

Manual update type: After register configuration, the PLL_UPDATE bit of PLL_CPU_CTRL_REG needs to be set to 1 to update the value of configured parameters. This kind of parameters include PLL_N, PLL_SSC, and PLL_SSC_MODE. All other parameters for linear FM PLL are immediate effect type.

The parameters of manual update type depend on PLL output clock. The UPDATE operation is invalid when PLL_EN is disabled.

It is not suggested to enable or disable the PLLs during usage. When the clock is not required, it is recommended to configure the PLL_OUTPUT_EN bit of PLL control register as 0.

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Enable the PLL

NOTE

The first enable of PLL_CPU should start with default configuration.

Step 1 Write 1 to the LDO_EN of the PRCM_PLL_CTRL_REG1 register in Section 4.2.6.7.

Step 2 Write 1 to the PLL_OUTPUT_GATE of the PLL_CPU_CTRL_REG register.

Step 3 Write 1 to the PLL_PLL_EN of the PLL_CPU_CTRL_REG register.

Step 4 Write 1 to the LOCK_ENABLE of the PLL_CPU_CTRL_REG register.

Step 5 Read LOCK_STATUS every 3 us for total three times. The PLL is locked if the value of all three reads is 1.

Step 6 Delay 20 us

Step 7 Configure the N, M0, M1, and P factors of the PLL_CPU_CTRL_REG register.

Step 8 Write 1 to the PLL_UPDATE of the PLL_CPU_CTRL_REG register.

Step 9 Wait for the PLL_UPDATE of the PLL_CPU_CTRL_REG register to change to 0.

Step 10 Wait for the status of the Lock to change to 1.

Step 11 Delay 20 us

Disable the PLL

Follow the steps below to disable the PLL:

Step 1 Ensure that the clock using the PLL to be disabled is switched to other clock sources, or the clock is disabled.

Step 2 Write 0 to the PLL_OUTPUT_GATE of the PLL_CPU_CTRL_REG register.

Step 3 Write 0 to the LOCK_ENABLE of the PLL_CPU_CTRL_REG register.

Step 4 Write 0 to the PLL_PLL_EN and PLL_LDO_EN of the PLL_CPU_CTRL_REG register.

Step 5 Write 0 to the LDO_EN of the PRCM_PLL_CTRL_REG1 register in Section 4.2.6.7.

NOTE

- Frequent PLL enable/disable is not recommended, as the PLL enable/disable will arise the mutual interference of PLLs and system instability. To disable PLL, it is recommended to write 0 to the PLL_OUTPUT_GATE.
- The LDO_EN of PRCM_PLL_CTRL_REG1 register controls the enable of all system PLL's LDO. The LDO_EN needs to be configured as 1, unless system is power-off or all PLLs are disabled.

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Configure Generic Frequency Modulation

Step 1 Make sure the PLL is enabled. If not, refer to Section Enable the PLL to enable the PLL. It is not suggested to enable or disable the PLL_CPU during usage.

Step 2 Configure the N, P, and M factors of the PLL_CPU_CTRL_REG register.

Step 3 Write 1 to the PLL_UPDATE of the PLL_CPU_CTRL_REG register.

Step 4 Write 0 and then write 1 to the LOCK_ENABLE of the PLL_CPU_CTRL_REG register.

Step 5 Read LOCK_STATUS every 3 us for total three times. The PLL is locked if the value of all three reads is 1.

Step 6 Delay 20 us and then PLL is available.

Configure Linear Frequency Modulation

Step 1 Make sure the PLL is enabled. If not, refer to Section Enable the PLL to enable the PLL. It is not suggested to enable or disable the PLL_CPU during usage.

Step 2 Configure the PLL_SSC_STEP bit of the PLL_CPU_SSC_REG register to select required frequency modulation slope. The configuration value of 0111 and 1000 are recommended.

Step 3 Configure the PLL_SSC bit of the PLL_CPU_SSC_REG register to set the SSC amplitude.

Step 4 Write 1 to the PLL_SSC_MODE bit of the PLL_CPU_SSC_REG register to enable linear frequency modulation.

Step 5 Configure the N factor of the PLL_CPU_CTRL_REG register.

Step 6 Write 1 to the PLL_UPDATE bit of the PLL_CPU_CTRL_REG register to update PLL configuration parameters.

Step 7 Wait for the value of the PLL_UPDATE bit to change to 0.

Step 8 Write 0 to the PLL_SSC_MODE bit of the PLL_CPU_SSC_REG register to disable linear frequency modulation.

Step 9 Write 1 to the PLL_UPDATE bit of the PLL_CPU_CTRL_REG register to update PLL configuration parameters.

Step 10 Wait for the value of the PLL_UPDATE bit to change to 0.

NOTE

- After PLL_SSC_MODE is set to 1, ensure that the initial value of PLL_UPDATE is 0 before writing 1 to it.

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- Before enabling linear frequency modulation, ensure the value of SPR_FREQ_MODE in PLL_CPU_PAT0_CTRL_REG register is 3.
- During the first enable of PLL_SSC_MODE, N factor should be set to the default value. The value of N factor could be changed after the PLL is enabled and locked.
- When the PLL is in spread spectrum/decimal division mode, linear FM mode can also be enabled. When Linear FM mode is enabled, the spread spectrum/decimal division mode will be stopped automatically. When Linear FM mode is disabled, the PLL will automatically return to spread spectrum/decimal division mode.

4.1.5.3 PLL_AUDIO0 Configuration

The frequency configuration formula of PLL_AUDIO0:

PLL_AUDIO0 = 24MHz/M1*N/P/M0

PLL_AUDIO does not support dynamic adjustment because changing any parameter of N, M0, M1, and P will affect the normal work of PLL, and the PLL will need to be relocked.

Generally, PLL_AUDIO0 needs two frequency points:

- 24.576 MHz and its clocks with different multiples
- 22.5792 MHz and its clocks with different multiples

You can use PLL_AUDIO0 to generate any one of the two frequency points mentioned above, and you should use PLL_AUDIO0 to generate 22.5792 MHz point and use PERI1_600M to generate 24.576 MHz point, when both of the two points are needed at a time.

22.5792 MHz Point Generated by PLL_AUDIO0

To implement 22.5792 MHz point, you need to enable the decimal frequency-division function. The configuration steps are the same as that of generic PLLs.

The following is a configuration example.

- M1=1
- N=67
- FRAC_IN=0x179A7
- P=18
- M0=1

According to above configuration, you can get 22.5792*4 MHz clock. To get 22.5792 MHz point, divide 22.5792*4 MHz through frequency divider on the clock path.

24.576 MHz Point Generated by PLL_AUDIO0

The following is a configuration example.

- M1=1
- N=128

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- P=5
- M0=1

According to above configuration, you can get 24.576*25MHz clock. To get 24.576 MHz point, divide 24.576*25MHz through frequency divider on the clock path.

24.576 MHz Point Generated by PERI1_600M

To implement 24.576 MHz point, you need to enable the decimal frequency-division function. The configuration steps are the same as that of generic PLLs.

The following is a configuration example.

- M1=1
- N=102
- P0=2
- P1=3
- P2=5
- FRAC_IN=0xCCCD

According to above configuration, you can get 24.576*25MHz clock. To get 24.576 MHz point, divide 24.576*25MHz through frequency divider on the clock path.

PLL_PERI1 can output the following frequency points:

PERI1PLL2X = 24MHz*N/M1/P0=1228.8 MHz

PERI1_800M = 24MHz*N/M1/P1=819.2 MHz

PERI1_480M = 24MHz*N/M1/P2=491.52 MHz

PERI1_600M = PERI1PLL2X/2=614.4 MHz

PERI1_400M = PERI1PLL2X/3=409.6 MHz

PERI1_300M = PERI1_600M/2=307.2 MHz

PERI1_200M = PERI1_400M/2=204.8 MHz

PERI1_160M = PERI1_480M/3=163.84 MHz

PERI1_150M = PERI1_300M/2=153.6 MHz

NOTE

When the P factor of PLL_AUDIO0 is an odd number, the clock output is an unequal-duty-cycle signal.

4.1.5.4 Glitch-free Switch Configuration

To use glitch-free switch, follow the rules below:

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- Both the original clock source and the target clock source are enabled and output stably.
- In the frequency modulation process, ensure that the output frequency does not exceed the target frequency (during frequency increasing) or original frequency (during frequency reducing).
- The switching interval needs to be longer than two clock periods of the slowest clock source.
- To switch from a lower frequency to a higher frequency: set the frequency division factor first, and then configure the clock source.
- To switch from a higher frequency to a lower frequency: configure the clock source first, and then set the frequency division factor.

4.1.5.5 Bus Clock Configuration

The Bus clock and CPU clock must use glitch-free switch. See section 0 for configuration rules. Besides, ensure that there will be no frequency exceeding the highest frequency clock of the bus when switching bus clock. The following clocks support glitch-free switch:

- AHB_CLK, APB0_CLK, APB1_CLK, APB_UART_CLK
- MBUS_CLK
- MCU_CORE_CLK

The bus frequency for each bus is as follows:

- CPU_AXI: default configuration is recommended (1/2 of CPU clock frequency)
- AHB: 200 MHz at max.
- APB0: 100MHz at max.
- APB1: 100MHz at max.
- APB_UART: 160MHz at max.
- MBUS: 480 MHz at max.

The bus clock adopts a three-level gating design. When enabling module clock, all bus clock gating in the path needs to be enabled.

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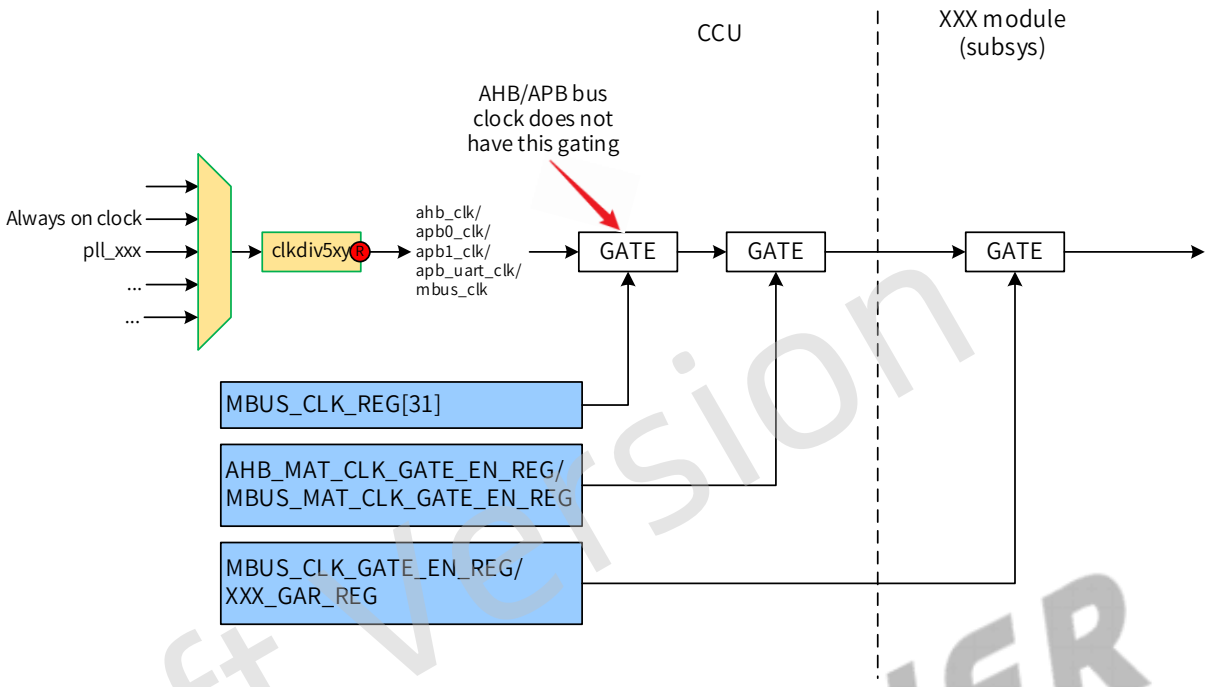
Figure 4-3 Bus Clock Gating Diagram



4.1.5.6 Module Clock Configuration

For the bus gating reset register of a module, the reset bit is de-asserted first, and then the gating clock is passed to avoid potential problems caused by the asynchronous release of the reset signal.

For all module clocks except the DRAM clocks, configure the clock source and frequency division factor first, and then release the clock gating (that is, set to 1). DRAM clocks use glitch-free switch, see Section 0 for configuration rules.

4.1.5.7 Clock Fanout

The following table shows the fanout clock frequencies, fanout signals (can be fanout through CLK-FANOUT pin), and their clock sources.

Output Frequency	Fanout Signal	Clock Sources
50 MHZ	CLK50M	PERI0_150M/3
PCLK	PCLK	APB0_CLK/M/N (M and N are configured in CLK_FAN_REG)
27 MHZ	CLK27M	VIDEO0PLL4X/M/N (M and N are configured in CLK27M_FAN_REG)
25 MHZ	CLK25M	PERI0_150M/6
24 MHZ	CLK24M	DCXO
16 MHZ	CLK16M	PERI0_160M/10 CLK16M_RC (The clock source of CLK16M is configured in CCU_FAN_REG.)

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Output Frequency	Fanout Signal	Clock Sources
12 MHZ	CLK12M	DCXO/2
32 kHz	FANOUT_32K_CLK	RTC

To implement clock fanout, follow the steps below:

Step 1 Configure CCU_FAN_REG register to select target clock frequency.

Step 2 If the clock source is PCLK or, configure CLK_FAN_REG register; if it is CLK27M, configure CLK27M_FAN_REG register; otherwise, configure CCU_FAN_GATE_REG register to enable target output clock.

Step 3 Configure CCU_FAN_GATE_REG to enable clock output.

Step 4 Configure the bit [23:21] of CCU_FAN_REG register to enable clock fanout.

4.1.6 Register List

Module Name	Base Address
CCU	0x02002000

Register Name	Offset	Description
PLL_PERI0_CTRL_REG	0x00A0	PLL_PERI0 Control Register
PLL_PERI0_PAT0_CTRL_REG	0x00A8	PLL_PERI0 Pattern0 Control Register
PLL_PERI0_PAT1_CTRL_REG	0x00AC	PLL_PERI0 Pattern1 Control Register
PLL_PERI0_BIAS_REG	0x00B0	PLL_PERI0 Bias Register
PLL_PERI1_CTRL_REG	0x00C0	PLL_PERI1 Control Register
PLL_PERI1_PAT0_CTRL_REG	0x00C8	PLL_PERI1 Pattern0 Control Register
PLL_PERI1_PAT1_CTRL_REG	0x00CC	PLL_PERI1 Pattern1 Control Register
PLL_PERI1_BIAS_REG	0x00D0	PLL_PERI1 Bias Register
PLL_VIDEO0_CTRL_REG	0x0120	PLL_VIDEO0 Control Register
PLL_VIDEO0_PAT0_CTRL_REG	0x0128	PLL_VIDEO0 Pattern0 Control Register
PLL_VIDEO0_PAT1_CTRL_REG	0x012C	PLL_VIDEO0 Pattern1 Control Register
PLL_VIDEO0_BIAS_REG	0x0130	PLL_VIDEO0 Bias Register
PLL_AUDIO0_CTRL_REG	0x0260	PLL_AUDIO0 Control Register
PLL_AUDIO0_PAT0_CTRL_REG	0x0268	PLL_AUDIO0 Pattern0 Control Register
PLL_AUDIO0_PAT1_CTRL_REG	0x026C	PLL_AUDIO0 Pattern1 Control Register
PLL_AUDIO0_BIAS_REG	0x0270	PLL_AUDIO0 Bias Register
PLL_CPU_CTRL_REG	0x0340	PLL_CPU Control Register
PLL_CPU_PAT0_CTRL_REG	0x0344	PLL_CPU Pattern0 Control Register
PLL_CPU_PAT1_CTRL_REG	0x0348	PLL_CPU Pattern1 Control Register
PLL_CPU_BIAS_REG	0x034C	PLL_CPU Bias Register
PLL_CPU_TUN1_REG	0x0350	PLL_CPU Tuning1 Control Register
PLL_CPU_SSC_REG	0x0354	PLL_CPU SSC Register
PLL_CPU_ECHO_REG	0x0358	PLL_CPU Echo Register

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Register Name	Offset	Description
AHB_CLK_REG	0x0500	AHB Clock Register
APB0_CLK_REG	0x0510	APB0 Clock Register
APB1_CLK_REG	0x0518	APB1 Clock Register
APB_UART_CLK_REG	0x0538	APB_UART Clock Register
MBUS_CLK_REG	0x0588	MBUS Clock Register
AHB_MAT_CLK_GATE_EN_REG	0x05C0	AHB Master Clock Gate Enable Register
PERI_MAT_CLK_GATE_EN_REG	0x05D0	PERI Master Clock Gate Enable Register
MBUS_CLK_GATE_EN_REG	0x05E0	MBUS Clock Gate Enable Register
MBUS_MAT_CLK_GATE_EN_REG	0x05E4	MBUS Master Clock Gate Enable Register
AHB_MAT_CLK_AUTO_GATE_EN_REG	0x05F0	AHB Master Clock Auto Gate Enable Register
MBUS_MAT_CLK_AUTO_GATE_EN_REG	0x05F4	MBUS Master Clock Auto Gate Enable Register
AHB_MAT_CLK_GATE_STAT_REG	0x05F8	AHB Master Clock Gate Status Register
MBUS_MAT_CLK_GATE_STAT_REG	0x05FC	MBUS Master Clock Gate Status Register
DMA0_GAR_REG	0x0704	DMA0 Gating And Reset Register
DMA1_GAR_REG	0x070C	DMA1 Gating And Reset Register
SPINLOCK_GAR_REG	0x0724	SPINLOCK Gating And Reset Register
MSGBOX_CPUX_GAR_REG	0x0744	MSGBOX_CPUX Gating And Reset Register
MSGBOX_CORE0_GAR_REG	0x074C	MSGBOX_CORE0 Gating And Reset Register
MSGBOX_CORE1_GAR_REG	0x0754	MSGBOX_CORE1 Gating And Reset Register
MSGBOX_CORE2_GAR_REG	0x075C	MSGBOX_CORE2 Gating And Reset Register
MSGBOX_CORE3_GAR_REG	0x0764	MSGBOX_CORE3 Gating And Reset Register
MSGBOX_MCU_GAR_REG	0x076C	MSGBOX_MCU Gating And Reset Register
PWM2_GAR_REG	0x0794	PWM2 Gating And Reset Register
DCU_GAR_REG	0x07A4	DCU Gating And Reset Register
DAP_GAR_REG	0x07AC	DAP Gating And Reset Register
PWMCS0_CLK_REG	0x07C0	PWMCS0 Clock Register
PWMCS0_GAR_REG	0x07C4	PWMCS0 Gating And Reset Register
PWMCS1_CLK_REG	0x07C8	PWMCS1 Clock Register
PWMCS1_GAR_REG	0x07CC	PWMCS1 Gating And Reset Register
TIMER0_CPUX_0_CLK_REG	0x0800	TIMER0_CPUX_0 Clock Register
TIMER0_CPUX_1_CLK_REG	0x0804	TIMER0_CPUX_1 Clock Register
TIMER0_CPUX_2_CLK_REG	0x0808	TIMER0_CPUX_2 Clock Register
TIMER0_CPUX_3_CLK_REG	0x080C	TIMER0_CPUX_3 Clock Register
TIMER0_CPUX_4_CLK_REG	0x0810	TIMER0_CPUX_4 Clock Register
TIMER0_CPUX_5_CLK_REG	0x0814	TIMER0_CPUX_5 Clock Register
TIMER0_CPUX_6_CLK_REG	0x0818	TIMER0_CPUX_6 Clock Register
TIMER0_CPUX_7_CLK_REG	0x081C	TIMER0_CPUX_7 Clock Register
TIMER0_CPUX_GAR_REG	0x0850	TIMER0_CPUX Gating And Reset Register

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Register Name	Offset	Description
TIEMR0_MCU_0_CLK_REG	0x0860	TIEMR0_MCU_0 Clock Register
TIEMR0_MCU_1_CLK_REG	0x0864	TIEMR0_MCU_1 Clock Register
TIEMR0_MCU_2_CLK_REG	0x0868	TIEMR0_MCU_2 Clock Register
TIEMR0_MCU_3_CLK_REG	0x086C	TIEMR0_MCU_3 Clock Register
TIMER0_MCU_GAR_REG	0x0870	TIMER0_MCU Gating And Reset Register
DE0_CLK_REG	0x0A00	DE0 Clock Register
DE0_GAR_REG	0x0A04	DE0 Gating And Reset Register
G2D_CLK_REG	0x0A40	G2D Clock Register
G2D_GAR_REG	0x0A44	G2D Gating And Reset Register
CE_SYS_CLK_REG	0x0AC0	CE_SYS Clock Register
CE_SYS_GAR_REG	0x0AC4	CE_SYS Gating And Reset Register
MCU_CORE_CLK_REG	0x0B80	MCU_CORE Clock Register
MCU_TS_CLK_REG	0x0B88	MCU_TS Clock Register
MCU_SYS_GAR_REG	0x0B94	MCU_SYS Gating And Reset Register
MCU_CFG_GAR_REG	0x0B9C	MCU_CFG Gating And Reset Register
DRAMC_GAR_REG	0x0C0C	DRAMC Gating And Reset Register
SMHC0_CLK_REG	0x0D00	SMHC0 Clock Register
SMHC0_GAR_REG	0x0D0C	SMHC0 Gating And Reset Register
SMHC1_CLK_REG	0x0D10	SMHC1 Clock Register
SMHC1_GAR_REG	0x0D1C	SMHC1 Gating And Reset Register
SMHC2_CLK_REG	0x0D20	SMHC2 Clock Register
SMHC2_GAR_REG	0x0D2C	SMHC2 Gating And Reset Register
SMHC3_CLK_REG	0x0D30	SMHC3 Clock Register
SMHC3_GAR_REG	0x0D3C	SMHC3 Gating And Reset Register
UART0_GAR_REG	0x0E00	UART0 Gating And Reset Register
UART1_GAR_REG	0x0E04	UART1 Gating And Reset Register
UART2_GAR_REG	0x0E08	UART2 Gating And Reset Register
UART3_GAR_REG	0x0E0C	UART3 Gating And Reset Register
UART4_GAR_REG	0x0E10	UART4 Gating And Reset Register
UART5_GAR_REG	0x0E14	UART5 Gating And Reset Register
UART6_GAR_REG	0x0E18	UART6 Gating And Reset Register
UART7_GAR_REG	0x0E20	UART7 Gating And Reset Register
UART8_GAR_REG	0x0E24	UART8 Gating And Reset Register
UART9_GAR_REG	0x0E28	UART9 Gating And Reset Register
TWI0_GAR_REG	0x0E80	TWI0 Gating And Reset Register
TWI1_GAR_REG	0x0E84	TWI1 Gating And Reset Register
TWI2_GAR_REG	0x0E88	TWI2 Gating And Reset Register
TWI3_GAR_REG	0x0E8C	TWI3 Gating And Reset Register
TWI4_GAR_REG	0x0E90	TWI4 Gating And Reset Register
TWI5_GAR_REG	0x0E94	TWI5 Gating And Reset Register
SPI0_CLK_REG	0x0F00	SPI0 Clock Register
SPI0_GAR_REG	0x0F04	SPI0 Gating And Reset Register

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Register Name	Offset	Description
SPI1_CLK_REG	0x0F08	SPI1 Clock Register
SPI1_GAR_REG	0x0F0C	SPI1 Gating And Reset Register
SPI2_CLK_REG	0x0F10	SPI2 Clock Register
SPI2_GAR_REG	0x0F14	SPI2 Gating And Reset Register
SPIF_CLK_REG	0x0F18	SPIF Clock Register
SPIF_GAR_REG	0x0F1C	SPIF Gating And Reset Register
SPI3_CLK_REG	0x0F20	SPI3 Clock Register
SPI3_GAR_REG	0x0F24	SPI3 Gating And Reset Register
CAN0_CLK_REG	0x0F80	CAN0 Clock Register
CAN0_GAR_REG	0x0F84	CAN0 Gating And Reset Register
CAN1_CLK_REG	0x0F88	CAN1 Clock Register
CAN1_GAR_REG	0x0F8C	CAN1 Gating And Reset Register
CAN_SYS_GAR_REG	0x0FBC	CAN_SYS Gating And Reset Register
GPADC0_CLK_REG	0x0FC0	GPADC0 Clock Register
GPADC0_GAR_REG	0x0FC4	GPADC0 Gating And Reset Register
GPADC1_CLK_REG	0x0FC8	GPADC1 Clock Register
GPADC1_GAR_REG	0x0FCC	GPADC1 Gating And Reset Register
GPADC2_CLK_REG	0x0FD0	GPADC2 Clock Register
GPADC2_GAR_REG	0x0FD4	GPADC2 Gating And Reset Register
THS_ADC_CLK_REG	0x0FD8	THS_ADC Clock Register
THS_ADC_GAR_REG	0x0FDC	THS_ADC Gating And Reset Register
THS_GAR_REG	0x0FE4	THS Gating And Reset Register
IR_RX0_CLK_REG	0x1000	IR_RX0 Clock Register
IR_RX0_GAR_REG	0x1004	IR_RX0 Gating And Reset Register
IR_TX_CLK_REG	0x1008	IR_TX Clock Register
IR_TX_GAR_REG	0x100C	IR_TX Gating And Reset Register
TPADC_CLK_REG	0x1030	TPADC Clock Register
TPADC_GAR_REG	0x1034	TPADC Gating And Reset Register
LBC_CLK_REG	0x1040	LBC Clock Register
LBC_GAR_REG	0x104C	LBC Gating And Reset Register
IR_RX1_CLK_REG	0x1100	IR_RX1 Clock Register
IR_RX1_GAR_REG	0x1104	IR_RX1 Gating And Reset Register
IR_RX2_CLK_REG	0x1108	IR_RX2 Clock Register
IR_RX2_GAR_REG	0x110C	IR_RX2 Gating And Reset Register
IR_RX3_CLK_REG	0x1110	IR_RX3 Clock Register
IR_RX3_GAR_REG	0x1114	IR_RX3 Gating And Reset Register
I2S0_CLK_REG	0x1200	I2S0 Clock Register
I2S0_GAR_REG	0x120C	I2S0 Gating And Reset Register
I2S1_CLK_REG	0x1210	I2S1 Clock Register
I2S1_GAR_REG	0x121C	I2S1 Gating And Reset Register
I2S2_CLK_REG	0x1220	I2S2 Clock Register
I2S2_GAR_REG	0x122C	I2S2 Gating And Reset Register

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Register Name	Offset	Description
OWA0_TX_CLK_REG	0x1280	OWA0 TX Clock Register
OWA0_RX_CLK_REG	0x1284	OWA0 RX Clock Register
OWA0_GAR_REG	0x128C	OWA0 Gating And Reset Register
DMIC_CLK_REG	0x12C0	DMIC Clock Register
DMIC_GAR_REG	0x12CC	DMIC Gating And Reset Register
AUDIOCODEC0_DAC_CLK_REG	0x12E0	AUDIOCODEC0 DAC Clock Register
AUDIOCODEC0_GAR_REG	0x12EC	AUDIOCODEC0 Gating And Reset Register
USB0_CLK_REG	0x1300	USB0 Clock Register
USB0_GAR_REG	0x1304	USB0 Gating And Reset Register
USB1_CLK_REG	0x1308	USB1 Clock Register
USB1_GAR_REG	0x130C	USB1 Gating And Reset Register
USB2P0_SYS_PHY_REF_CLK_REG	0x1340	USB2P0_SYS PHY Reference Clock Register
USB2P0_SYS_GAR_REG	0x1344	USB2P0_SYS Gating And Reset Register
GMAC0_PHY_CLK_REG	0x1400	GMAC0 PHY Clock Register
GMAC0_PTP_REF_CLK_REG	0x1404	GMAC0 PTP Reference Clock Register
GMAC0_GAR_REG	0x140C	GMAC0 Gating And Reset Register
GMAC1_PHY_CLK_REG	0x1410	GMAC1 PHY Clock Register
GMAC1_PTP_REF_CLK_REG	0x1414	GMAC1 PTP Reference Clock Register
GMAC1_GAR_REG	0x141C	GMAC1 Gating And Reset Register
GMAC2_PHY_CLK_REG	0x1420	GMAC2 PHY Clock Register
GMAC2_PTP_REF_CLK_REG	0x1424	GMAC2 PTP Reference Clock Register
GMAC2_GAR_REG	0x142C	GMAC2 Gating And Reset Register
TCON_LCD0_CLK_REG	0x1500	TCON_LCD0 Clock Register
TCON_LCD0_GAR_REG	0x1504	TCON_LCD0 Gating And Reset Register
LVDS0_GAR_REG	0x1544	LVDS0 Gating And Reset Register
MIPI_DSI0_CLK_REG	0x1580	MIPI_DSI0 Clock Register
MIPI_DSI0_GAR_REG	0x1584	MIPI_DSI0 Gating And Reset Register
COMBOPHY0_CLK_REG	0x15C0	COMBOPHY0 Clock Register
VO0_REG_GAR_REG	0x16C4	VO0_REG Gating And Reset Register
VIDEO_OUT0_GAR_REG	0x16E4	VIDEO_OUT0 Gating And Reset Register
LEDC_CLK_REG	0x1700	LEDC Clock Register
LEDC_GAR_REG	0x1704	LEDC Gating And Reset Register
CSI_MASTER0_CLK_REG	0x1800	CSI Master0 Clock Register
CSI_MASTER1_CLK_REG	0x1804	CSI Master1 Clock Register
CSI_MASTER2_CLK_REG	0x1808	CSI Master2 Clock Register
CSI_CLK_REG	0x1840	CSI Clock Register
ISP_CLK_REG	0x1860	ISP Clock Register
VIDEO_IN_GAR_REG	0x1884	VIDEO_IN Gating And Reset Register
PERI0PLL_GATE_EN_REG	0x1908	PERI0PLL Gate Enable Register
PERI1PLL_GATE_EN_REG	0x190C	PERI1PLL Gate Enable Register
VIDEOPLL_GATE_EN_REG	0x1910	VIDEOPLL Gate Enable Register

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Register Name	Offset	Description
AUDIOPLL_GATE_EN_REG	0x191C	AUDIOPLL Gate Enable Register
PERI0PLL_GATE_STAT_REG	0x1988	PERI0PLL Gate Status Register
PERI1PLL_GATE_STAT_REG	0x198C	PERI1PLL Gate Status Register
VIDEOPLL_GATE_STAT_REG	0x1990	VIDEOPLL Gate Status Register
AUDIOPLL_GATE_STAT_REG	0x199C	AUDIOPLL Gate Status Register
PLL_OPG_BYPASS_REG	0x1A20	PLL Output Gate Bypass Register
AXI_MON_GAR_REG	0x1C00	AXI MON Gating And Reset Register
AHB_MON_GAR_REG	0x1C04	AHB MON Gating And Reset Register
CCU_SEC_SWITCH_REG	0x1F00	CCU Security Switch Register
DAP_REQ_CTRL_REG	0x1F10	DAP REQ Control Register
PLL_CFG0_REG	0x1F20	PLL Configuration0 Register
PLL_CFG1_REG	0x1F24	PLL Configuration1 Register
PLL_CFG2_REG	0x1F28	PLL Configuration2 Register
CCU_FAN_GATE_REG	0x1F30	CCU Fanout Clock Gate Register
CLK27M_FAN_REG	0x1F34	CLK27M Fanout Register
CLK_FAN_REG	0x1F38	CLK Fanout Register
CCU_FAN_REG	0x1F3C	CCU Fanout Register
CLK_DBG_REG	0x1F50	Clock Debug Register
FRE_DET_CTRL_REG	0x1F60	Frequency Detect Control Register
FRE_UP_LIM_REG	0x1F64	Frequency Up Limit Register
FRE_DOWN_LIM_REG	0x1F68	Frequency Down Limit Register

4.1.7 Register Description

4.1.7.1 0x00A0 PLL_PERI0 Control Register (Default Value: 0x0021_6310)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00A0			Register Name: PLL_PERI0_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x00A0			Register Name: PLL_PERI0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_EN PLL Enable 0: Disable 1: Enable PERI0PLL2X = 24MHz*N/M1/P0 PERI0_800M = 24MHz*N/M1/P1 PERI0_480M/PERI0_480M_BUS = 24MHz*N/M1/P2 PERI0_600M/PERI0_600M_BUS = PERI0PLL2X/2 PERI0_400M = PERI0PLL2X/3 PERI0_300M = PERI0_600M/2 PERI0_200M = PERI0_400M/2 PERI0_160M = PERI0_480M/3 PERI0_150M = PERI0_300M/2 When the SYS_24M_CLK is 24 MHz, the default frequency of PERI0PLL2X is 1.2 GHz, the default frequency of PERI0_800M is 800 MHz, and the default frequency of PERI0_480M is 480 MHz. The output clock of PERI0PLL2X is fixed to 1.2 GHz and not suggested to change the parameter.
30	/	/	/
29	R/W	0x0	LOCK_ENABLE Lock Enable 0: Disable 1: Enable
28	R	0x0	LOCK PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.)
27	R/W	0x0	PLL_OUTPUT_GATE PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26:25	/	/	/
24	R/W	0x0	PLL_SDM_EN PLL SDM Enable 0: Disable 1: Enable Enable spread spectrum and decimal division.
23	/	/	/

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Offset: 0x00A0			Register Name: PLL_PERI0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
22:20	R/W	0x2	PLL_P1 PLL Output Div P1 P1=PLL_P1 + 1 PLL_P1 is from 0 to 7.
19	/	/	/
18:16	R/W	0x1	PLL_P0 PLL Output Div P0 P0=PLL_P0 + 1 PLL_P0 is from 0 to 7.
15:8	R/W	0x63	PLL_N PLL N N= PLL_N +1 PLL_N is from 0 to 254. In application, PLL_N shall be more than or equal to 11.
7:6	R/W	0x0	PLL_UNLOCK_MDSEL PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles
5	R/W	0x0	PLL_LOCK_MDSEL PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4:2	R/W	0x4	PLL_P2 PLL Output Div P2 P2=PLL_P2 + 1 PLL_P2 is from 0 to 7.
1	R/W	0x0	PLL_INPUT_DIV2 PLL Input Div M1 M1=PLL_INPUT_DIV2 + 1 PLL_INPUT_DIV2 is from 0 to 1.
0	/	/	/

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4.1.7.2 0x00A8 PLL_PERI0 Pattern0 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00A8			Register Name: PLL_PERI0_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable.
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode 00: Pulse swallow 01: Triangular(1bit) 10: Triangular(2bit) 11: Triangular(3bit)
28:20	R/W	0x0	WAVE_STEP Wave Step
19	R/W	0x0	SDM_CLK_SEL SDM Clock Select 0: 24MHz 1: 12MHz When the PLL_INPUT_DIV2 is 1, the bit needs to be set to 1.
18:17	R/W	0x0	FREQ Frequency 00: 31.5KHz 01: 32KHz 10: 32.5KHz 11: 33KHz
16:0	R/W	0x0	WAVE_BOT Wave Bottom

4.1.7.3 0x00AC PLL_PERI0 Pattern1 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00AC	Register Name: PLL_PERI0_PAT1_CTRL_REG

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Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	DITHER_EN Dither Enable
23:21	/	/	/
20	R/W	0x0	FRAC_EN Fraction Enable
19:17	/	/	/
16:0	R/W	0x0	FRAC_IN Fraction In

4.1.7.4 0x00B0 PLL_PERI0 Bias Register (Default Value: 0x0004_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00B0			Register Name: PLL_PERI0_BIAS_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x4	PLL_CP PLL bias control
15:0	/	/	/

4.1.7.5 0x00C0 PLL_PERI1 Control Register (Default Value: 0x0021_6310)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00C0			Register Name: PLL_PERI1_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x00C0			Register Name: PLL_PERI1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_EN PLL Enable 0: Disable 1: Enable PERI1PLL2X = 24MHz*N/M1/P0 PERI1_800M = 24MHz*N/M1/P1 PERI1_480M = 24MHz*N/M1/P2 PERI1_600M = PERI1PLL2X/2 PERI1_400M = PERI1PLL2X/3 PERI1_300M = PERI1_600M/2 PERI1_200M = PERI1_400M/2 PERI1_160M = PERI1_480M/3 PERI1_150M = PERI1_300M/2 When the SYS_24M_CLK is 24 MHz, the default frequency of PERI1PLL2X is 1.2 GHz, the default frequency of PERI1_800M is 800 MHz, and the default frequency of PERI1_480M is 480 MHz. The output clock of PERI1PLL2X is fixed to 1.2 GHz and not suggested to change the parameter.
30	/	/	/
29	R/W	0x0	LOCK_ENABLE Lock Enable 0: Disable 1: Enable
28	R	0x0	LOCK PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.)
27	R/W	0x0	PLL_OUTPUT_GATE PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26:25	/	/	/
24	R/W	0x0	PLL_SDM_EN PLL SDM Enable 0: Disable 1: Enable Enable spread spectrum and decimal division.
23	/	/	/

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Offset: 0x00C0			Register Name: PLL_PERI1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
22:20	R/W	0x2	PLL_P1 PLL Output Div P1 P1=PLL_P1 + 1 PLL_P1 is from 0 to 7.
19	/	/	/
18:16	R/W	0x1	PLL_P0 PLL Output Div P0 P0=PLL_P0 + 1 PLL_P0 is from 0 to 7.
15:8	R/W	0x63	PLL_N PLL N N= PLL_N +1 PLL_N is from 0 to 254. In application, PLL_N shall be more than or equal to 11.
7:6	R/W	0x0	PLL_UNLOCK_MDSEL PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles
5	R/W	0x0	PLL_LOCK_MDSEL PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4:2	R/W	0x4	PLL_P2 PLL Output Div P2 P2=PLL_P2 + 1 PLL_P2 is from 0 to 7.
1	R/W	0x0	PLL_INPUT_DIV2 PLL Input Div M1 M1=PLL_INPUT_DIV2 + 1 PLL_INPUT_DIV2 is from 0 to 1.
0	/	/	/

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4.1.7.6 0x00C8 PLL_PERI1 Pattern0 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00C8			Register Name: PLL_PERI1_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode 00: Pulse swallow 01: Triangular(1bit) 10: Triangular(2bit) 11: Triangular(3bit)
28:20	R/W	0x0	WAVE_STEP Wave Step
19	R/W	0x0	SDM_CLK_SEL SDM Clock Select 0: 24MHz 1: 12MHz When the PLL_INPUT_DIV2 is 1, the bit needs to be set to 1.
18:17	R/W	0x0	FREQ Frequency 00: 31.5KHz 01: 32KHz 10: 32.5KHz 11: 33KHz
16:0	R/W	0x0	WAVE_BOT Wave Bottom

4.1.7.7 0x00CC PLL_PERI1 Pattern1 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00CC	Register Name: PLL_PERI1_PAT1_CTRL_REG

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Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	DITHER_EN Dither Enable
23:21	/	/	/
20	R/W	0x0	FRAC_EN Fraction Enable
19:17	/	/	/
16:0	R/W	0x0	FRAC_IN Fraction In

4.1.7.8 0x00D0 PLL_PERI1 Bias Register (Default Value: 0x0004_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x00D0			Register Name: PLL_PERI1_BIAS_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x4	PLL_CP PLL bias control
15:0	/	/	/

4.1.7.9 0x0120 PLL_VIDEO0 Control Register (Default Value: 0x0012_6200)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0120			Register Name: PLL_VIDEO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0120			Register Name: PLL_VIDEO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_EN PLL Enable 0: Disable 1: Enable For application, VIDEO0PLL4X = InputFreq *N/M1/P0. VIDEO0PLL3X = InputFreq *N/M1/P1. 当 SYS_24M_CLK 为 24M 时，VIDEO0PLL4X default is 1188MHz=24*99/2; VIDEO0PLL3X default is 792MHz=24*99/3. When the SYS_24M_CLK is 24 MHz, The default value of VIDEO0PLL4X is 1188MHz=24*99/2 and the default value of VIDEO0PLL3X is 792MHz=24*99/3.
30	/	/	/
29	R/W	0x0	LOCK_ENABLE Lock Enable 0: Disable 1: Enable
28	R	0x0	LOCK PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.)
27	R/W	0x0	PLL_OUTPUT_GATE PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26:25	/	/	/
24	R/W	0x0	PLL_SDM_EN PLL SDM Enable 0: Disable 1: Enable
23	/	/	/
22:20	R/W	0x1	PLL_P0 PLL Output Div P0 P0=PLL_P0+1 PLL_P0 is from 0 to 7.
19	/	/	/

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Offset: 0x0120			Register Name: PLL_VIDEO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
18:16	R/W	0x2	PLL_P1 PLL Output Div P1 P1=PLL_P1+1 PLL_P1 is from 0 to 7.
15:8	R/W	0x62	PLL_N PLL N N= PLL_N +1 PLL_N is from 0 to 254. In application, PLL_N shall be more than or equal to 11.
7:6	R/W	0x0	PLL_UNLOCK_MDSEL PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles
5	R/W	0x0	PLL_LOCK_MDSEL PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4:2	/	/	/
1	R/W	0x0	PLL_INPUT_DIV2 PLL Input Div M1 M1=PLL_INPUT_DIV2 + 1 PLL_INPUT_DIV2 is from 0 to 1.
0	/	/	/

4.1.7.10 0x0128 PLL_VIDEO0 Pattern0 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0128			Register Name: PLL_VIDEO0_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable

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Offset: 0x0128			Register Name: PLL_VIDEO0_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode 00: Pulse swallow 01: Triangular(1bit) 10: Triangular(2bit) 11: Triangular(3bit)
28:20	R/W	0x0	WAVE_STEP Wave Step
19	R/W	0x0	SDM_CLK_SEL SDM Clock Select 0: 24MHz 1: 12MHz When the PLL_INPUT_DIV2 is 1, the bit needs to be set to 1.
18:17	R/W	0x0	FREQ Frequency 00: 31.5KHz 01: 32KHz 10: 32.5KHz 11: 33KHz
16:0	R/W	0x0	WAVE_BOT Wave Bottom

4.1.7.11 0x012C PLL_VIDEO0 Pattern1 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x012C			Register Name: PLL_VIDEO0_PAT1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	DITHER_EN Dither Enable
23:21	/	/	/
20	R/W	0x0	FRAC_EN Fraction Enable
19:17	/	/	/

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Offset: 0x012C			Register Name: PLL_VIDEO0_PAT1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
16:0	R/W	0x0	FRAC_IN Fraction In

4.1.7.12 0x0130 PLL_VIDEO0 Bias Register (Default Value: 0x0004_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0130			Register Name: PLL_VIDEO0_BIAS_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x4	PLL_CP PLL Bias Control
15:0	/	/	/

4.1.7.13 0x0260 PLL_AUDIO0 Control Register (Default Value: 0x0004_7F00)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0260			Register Name: PLL_AUDIO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_EN PLL Enable 0: Disable 1: Enable AUDIO0PLL = 24MHz*N/M1/M0/P When SYS_24M_CLK is 24MHz, AUDIO0PLL default is 614.4MHz (24.576MHz*25).
30	/	/	/
29	R/W	0x0	LOCK_ENABLE Lock Enable 0: Disable 1: Enable

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Offset: 0x0260			Register Name: PLL_AUDIO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
28	R	0x0	LOCK PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.) The bit is only valid when the bit29 is set to 1.
27	R/W	0x0	PLL_OUTPUT_GATE PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26:25	/	/	/
24	R/W	0x0	PLL_SDM_EN PLL SDM Enable 0: Disable 1: Enable Enable spread spectrum and decimal division.
23:22	/	/	/
21:16	R/W	0x4	PLL_P PLL Post div P P= PLL_P +1 PLL_P is from 0 to 63.
15:8	R/W	0x7F	PLL_N PLL N N= PLL_N +1 PLL_N is from 0 to 254. In application, PLL_N shall be more than or equal to 11.
7:6	R/W	0x0	PLL_UNLOCK_MDSEL PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles
5	R/W	0x0	PLL_LOCK_MDSEL PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4:2	/	/	/

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Offset: 0x0260			Register Name: PLL_AUDIO0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	PLL_INPUT_DIV2 PLL Input Div M1 M1=PLL_INPUT_DIV2 + 1 PLL_INPUT_DIV2 is from 0 to 1.
0	R/W	0x0	PLL_OUTPUT_DIV2 PLL Output Div M0 M0=PLL_OUTPUT_DIV2 + 1 PLL_OUTPUT_DIV2 is from 0 to 1.

4.1.7.14 0x0268 PLL_AUDIO0 Pattern0 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0268			Register Name: PLL_AUDIO0_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode 00: Pulse swallow 01: Triangular(1bit) 10: Triangular(2bit) 11: Triangular(3bit)
28:20	R/W	0x0	WAVE_STEP Wave Step
19	R/W	0x0	SDM_CLK_SEL SDM Clock Select 0: 24MHz 1: 12MHz When the PLL_INPUT_DIV2 is 1, the bit needs to be set to 1.
18:17	R/W	0x0	FREQ Frequency 00: 31.5KHz 01: 32KHz 10: 32.5KHz 11: 33KHz

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Offset: 0x0268			Register Name: PLL_AUDIO0_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
16:0	R/W	0x0	WAVE_BOT Wave Bottom

4.1.7.15 0x026C PLL_AUDIO0 Pattern1 Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x026C			Register Name: PLL_AUDIO0_PAT1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	DITHER_EN Dither Enable
23:21	/	/	/
20	R/W	0x0	FRAC_EN Fraction Enable
19:17	/	/	/
16:0	R/W	0x0	FRAC_IN Fraction In

4.1.7.16 0x0270 PLL_AUDIO0 Bias Register (Default Value: 0x0005_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of CCU_SEC_SWITCH_REG.

Offset: 0x0270			Register Name: PLL_AUDIO0_BIAS_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x5	PLL_CP PLL bias control
15:0	/	/	/

4.1.7.17 0x0340 PLL_CPU Control Register (Default Value: 0x0000_1400)

Offset: 0x0340	Register Name: PLL_CPU_CTRL_REG

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Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_PLL_EN PLL Enable 0: Disable 1: Enable The PLL_CPU= InputFreq*N/P/(M0*M1) The PLL_CPU frequency range is from 480 MHz to 2.016 GHz. When the SYS_24M_CLK is 24 MHz, the default frequency of PLL_CPU is 480 MHz.
30	/	/	/
29	R/W	0x0	LOCK_ENABLE Lock Enable 0: Disable 1: Enable
28	R	0x0	LOCK PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.)
27	R/W	0x0	PLL_OUTPUT_GATE PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26	R/WAC	0x0	PLL_UPDATE Write q update the cfg to PLL, auto clear.
25:22	/	/	/
21:20	R/W	0x0	PLL_M0 PLL_M0 M0 = PLL_M0 + 1 PLL_FACTOR_M0 is from 0 to 3.
19	/	/	/
18:16	R/W	0x0	PLL_P PLL PREDIV P P = PLL_P + 1.
15:8	R/W	0x14	PLL_N PLL N N= PLL_N
7:6	R/W	0x0	PLL_UNLOCK_MDSEL PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles

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5	R/W	0x0	PLL_LOCK_MDSEL PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4	/	/	/
3:0	R/W	0x0	PLL_M1 PLL_M1 M1 = PLL_M1 + 1 PLL_M1 is from 0 to 15.

4.1.7.18 0x0344 PLL_CPU Pattern0 Control Register (Default Value: 0x0000_0000)

Offset: 0x0344			Register Name: PLL_CPU_PAT0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode 00: SDM 01: MASH-1(8 bit) 10: MASH-2(8 bit) 11: MASH-3(8 bit)
28:17	R/W	0x0	WAVE_STEP Wave Step
16:0	R/W	0x0	WAVE_BOT Wave Bottom

4.1.7.19 0x0348 PLL_CPU Pattern1 Control Register (Default Value: 0x0000_0000)

Offset: 0x0348			Register Name: PLL_CPU_PAT1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:22	R/W	0x0	SDM_CYCLE SDM Cycle sdm_cycle=sdm_fre/clk_sdm/2
21	/	/	/
20	R/W	0x0	SDM_DIRECTION SDM direction 0: UP 1: DOWM
19	/	/	/
18	R/W	0x0	DITHER_EN Dither Enable

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Offset: 0x0348			Register Name: PLL_CPU_PAT1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	FRAC_EN Fraction Enable
16:0	R/W	0x0	FRAC_IN Fraction In

4.1.7.20 0x034C PLL_CPU Bias Register (Default Value: 0x0002_0000)

Offset: 0x034C			Register Name: PLL_CPU_BIAS_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x2	PLL_CP PLL current bias control
15:0	/	/	/

4.1.7.21 0x0350 PLL_CPU Tuning1 Control Register (Default Value: 0x0000_0000)

Offset: 0x0350			Register Name: PLL_CPU_TUN1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_SDM_EN PLL_SDM_EN
30:0	/	/	/

4.1.7.22 0x0354 PLL_CPU SSC Register (Default Value: 0x4CCC_A000)

Offset: 0x0354			Register Name: PLL_CPU_SSC_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_SSC_MODE PLL_SSC_MODE 0: Normal Mode 1: Continuously Frequency Scale
30	R/W	0x1	PLL_SSC_RSTN SSC RSTN bake up
29	R/W	0x0	PLL_SSC_CLK_SEL SSC CLK Select 0: ref_clk 1: pll_clk_sdm

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Offset: 0x0354			Register Name: PLL_CPU_SSC_REG
Bit	Read/Write	Default/Hex	Description
28:12	R/W	0xCCCA	PLL_SSC PLL_SSC SSC amplitude must be an integer multiple of 2^step. spread spectrum amplitude = (SSC amplitude + 2^step) *24/(2^17), unit: MHz.
11:7	/	/	/
6:4	R/W	0x0	PLL_PHASE_COMPENSATE PLL_PHASE_COMPENSATE The value of bit[6:4] is based on 24M clock, then the default PLL phase compensate is (3/24000000)s
3:0	R/W	0x0	PLL_SSC_STEP PLL_SSC_STEP 0000:0.00439MHz/us (576/2^17) 0001:0.00879MHz/us (576/2^16) 0010:0.01758MHz/us (576/2^15) 0011:0.03516MHz/us (576/2^14) 0100:0.07031MHz/us (576/2^13) 0101:0.14062MHz/us (576/2^12) 0110:0.28125MHz/us (576/2^11) 0111:0.56250MHz/us (576/2^10) 1000:1.12500MHz/us (576/2^9) 1001:2.25000MHz/us (576/2^8) 1010:4.50000MHz/us (576/2^7) 1011:9.00000MHz/us (576/2^6) Others:0.00439MHz/us(576/2^17)

4.1.7.23 0x0358 PLL_CPU Echo Register (Default Value: 0x0000_1400)

Offset: 0x0358			Register Name: PLL_CPU_ECHO_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	PLL_CPU_EN_ECHO PLL Enable 0: Disable 1: Enable The PLL_CPU= InputFreq*N/P/(M0*M1). The PLL_CPU frequency range is from 480 MHz to 2.016 GHz. When the crystal oscillator is 24 MHz, the default frequency of PLL_CPU is 480 MHz.
30	/	/	/

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Offset: 0x0358			Register Name: PLL_CPU_ECHO_REG
Bit	Read/Write	Default/Hex	Description
29	R	0x0	PLL_CPU_LOCK_ENABLE_ECHO Lock Enable 0: Disable 1: Enable
28	R	0x0	PLL_CPU_LOCK_ECHO PLL Lock Info 0: Unlocked 1: Locked (It indicates that the PLL has been stable.)
27	R	0x0	PLL_CPU_OUTPUT_GATE_ECHO PLL Output Gating Enable 0: Disable 1: Enable The bit is used to control the output enable of the PLL.
26	R	0x0	PLL_CPU_UPDATE_ECHO Write q update the cfg to PLL, auto clear
25:22	/	/	/
21:20	R	0x0	PLL_CPU_M0_ECHO PLL_M0 M0 = PLL_M0 + 1 PLL_FACTOR_M0 is from 0 to 3.
19	/	/	/
18:16	R	0x0	PLL_CPU_P_ECHO PLL PREDIV P P = PLL_P + 1.
15:8	R	0x14	PLL_CPU_N_SDM_PLLCFG_ECHO PLL N N= PLL_N
7:6	R	0x0	PLL_CPU_UNLOCK_MDSEL_ECHO PLL Unlock Level 00: 21-29 Clock Cycles 01: 22-28 Clock Cycles 1X: 20-30 Clock Cycles
5	R	0x0	PLL_CPU_LOCK_MDSEL_ECHO PLL Lock Level 0: 24-26 Clock Cycles 1: 23-27 Clock Cycles
4	/	/	/

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Offset: 0x0358			Register Name: PLL_CPU_ECHO_REG
Bit	Read/Write	Default/Hex	Description
3:0	R	0x0	PLL_CPU_M1_ECHO PLL_M1 M1 = PLL_M1 + 1 PLL_M1 is from 0 to 15.

4.1.7.24 0x0500 AHB Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x0500			Register Name: AHB_CLK_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 00: SYS_24M_CLK 01: SYS_32K_CLK 10: RC_16M_CLK 11: PERI0_600M_BUS AHB_CLK = Clock Source/M. The clock MUX uses glitch-free switch and supports dynamic configuration.
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31. The clock DIV supports dynamic configuration.

4.1.7.25 0x0510 APB0 Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x0510	Register Name: APB0_CLK_REG

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Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 00: SYS_24M_CLK 01: SYS_32K_CLK 10: RC_16M_CLK 11: PERI0_600M_BUS The clock MUX uses glitch-free switch and supports dynamic configuration. APB0_CLK = Clock Source/M.
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31. The clock DIV supports dynamic configuration.

4.1.7.26 0x0518 APB1 Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x0518			Register Name: APB1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 00: SYS_24M_CLK 01: SYS_32K_CLK 10: RC_16M_CLK 11: PERI0_600M_BUS The clock MUX uses glitch-free switch and supports dynamic configuration. APB1_CLK = Clock Source/M.
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31. The clock DIV supports dynamic configuration.

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4.1.7.27 0x0538 APB_UART Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x0538			Register Name: APB_UART_CLK_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: SYS_32K_CLK 010: RC_16M_CLK 011: PERI0_600M_BUS 100: PERI0_480M_BUS The clock MUX uses glitch-free switch and supports dynamic configuration. APB_UART_CLK = Clock Source/M.
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31. The clock DIV supports dynamic configuration.

4.1.7.28 0x0588 MBUS Clock Register (Default Value: 0x8000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x0588			Register Name: MBUS_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	MBUS_CLK_GATING Gating Special Clock 0: Clock is OFF 1: Clock is ON MBUS_CLK = Clock Source/M
30:29	/	/	/

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Offset: 0x0588			Register Name: MBUS_CLK_REG
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	MBUS_DFS_EN MBUS CLK Dynamic Frequency Scaling Enable 0: Disable 1: Enable
27	R/WAC	0x0	MBUS_UPD MBUS Clock Configuration 0 update 0: Invalid 1: Valid Note: Set this bit will validate Configuration 0. It will be auto cleared after the Configuration 0 is valid.
26:24	R/W	0x0	MBUS_CLK_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_400M 011: PERI0_300M 100: HDR_CLK The clock MUX uses glitch-free switch and supports dynamic configuration. HDR_CLK = (ddrphypll0 and ddrphypll1 mux)/4
23:5	/	/	/
4:0	R/W	0x0	MBUS_DIV1 Factor M M= MBUS_DIV1 + 1 FACTOR M is from 0 to 31. The clock DIV supports dynamic configuration.

4.1.7.29 0x05C0 AHB Master Clock Gate Enable Register (Default Value: 0x1041_F1EC)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x05C0			Register Name: AHB_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	AHB_MONITOR_EN AHB bus auto clock gating function enable 1: Enable auto clock gate 0: Disable auto clock gate
30	/	/	/

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Offset: 0x05C0			Register Name: AHB_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
29	R/W	0x0	SD_MONITOR_EN SD bus auto clock gating function enable 1: Enable auto clock gate 0: Disable auto clock gate
28	R/W	0x1	PRCM_PERI0PLL_CLK_GATE_SW_CFG PRCM PERI0PLL Clock Gate Enable 0: Disable 1: Enable
27:23	/	/	/
22	R/W	0x1	SPIF_AHB_GATE_SW_CFG SPIF_AHB Clock Gate Enable 0: Disable 1: Enable
21:17	/	/	/
16	R/W	0x1	USB2P0_SYS_AHB_GATE_SW_CFG USB2P0_SYS_ AHB Clock Gate Enable 0: Disable 1: Enable
15	R/W	0x1	GMAC2_AHB_GATE_SW_CFG GMAC2 AHB Clock Gate Enable 0: Disable 1: Enable
14	R/W	0x1	GMAC1_AHB_GATE_SW_CFG GMAC1 AHB Clock Gate Enable 0: Disable 1: Enable
13	R/W	0x1	GMAC0_AHB_GATE_SW_CFG GMAC0 AHB Clock Gate Enable 0: Disable 1: Enable
12	R/W	0x1	MCU_SYS_AHB_GATE_SW_CFG MCU_SYS AHB Clock Gate Enable 0: Disable 1: Enable
11:9	/	/	/
8	R/W	0x1	SMHC3_AHB_GATE_SW_CFG SMHC3 AHB Clock Gate Enable 0: Disable 1: Enable

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Offset: 0x05C0			Register Name: AHB_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x1	SMHC2_AHB_GATE_SW_CFG SMHC2 AHB Clock Gate Enable 0: Disable 1: Enable
6	R/W	0x1	SMHC1_AHB_GATE_SW_CFG SMHC1 AHB Clock Gate Enable 0: Disable 1: Enable
5	R/W	0x1	SMHC0_AHB_GATE_SW_CFG SMHC0 AHB Clock Gate Enable 0: Disable 1: Enable
4	/	/	/
3	R/W	0x1	VIDEO_OUT0_AHB_GATE_SW_CFG Video Out0 AHB Clock Gate Enable 0: Disable 1: Enable
2	R/W	0x1	VIDEO_IN_AHB_GATE_SW_CFG Video in AHB Clock Gate Enable 0: Disable 1: Enable
1:0	/	/	/

4.1.7.30 0x05D0 PERI Master Clock Gate Enable Register (Default Value: 0x0001_0001)

Offset: 0x05D0			Register Name: PERI_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x1	PERI_APB0_GATE_SW_CFG PERI_APB0 Clock Gate Enable 0: Disable 1: Enable
15:1	/	/	/
0	R/W	0x1	PERI_AHB_GATE_SW_CFG PERI_AHB Clock Gate Enable 0: Disable 1: Enable

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4.1.7.31 0x05E0 MBUS Clock Gate Enable Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x05E0			Register Name: MBUS_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	CAN_MBUS_CLK_EN Gating MBUS Clock for CAN 0: Mask 1: Pass
16:14	/	/	/
13	R/W	0x0	GMAC2_MBUS_CLK_EN Gating MBUS Clock for GMAC2 0: Mask 1: Pass
12	R/W	0x0	GMAC1_MBUS_CLK_EN Gating MBUS Clock for GMAC1 0: Mask 1: Pass
11	R/W	0x0	GMAC0_MBUS_CLK_EN GMAC0 MBUS Clock Gate Enable 0: Mask 1: Pass
10	/	/	/
9	R/W	0x0	ISP_MBUS_CLK_EN ISP MBUS Clock Gate Enable 0: Mask 1: Pass
8	R/W	0x0	CSI_MBUS_CLK_EN CSI MBUS Clock Gate Enable 0: Mask 1: Pass
7:4	/	/	/
3	R/W	0x0	DMA1_MBUS_CLK_EN DMA1 MBUS Clock Gate Enable 0: Mask 1: Pass

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Offset: 0x05E0			Register Name: MBUS_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	CE_SYS_MBUS_CLK_EN CE_SYS MBUS Clock Gate Enable 0: Mask 1: Pass
1	/	/	/
0	R/W	0x0	DMA0_MBUS_CLK_EN DMA0 MBUS Clock Gate Enable 0: Mask 1: Pass

4.1.7.32 0x05E4 MBUS Master Clock Gate Enable Register (Default Value: 0x38BC_7800)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x05E4			Register Name: MBUS_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x1	DMA1_MBUS_GATE_SW_CFG DMA1 MBUS Clock Gate Enable 0: Disable 1: Enable
28	R/W	0x1	DMA0_MBUS_GATE_SW_CFG DMA0 MBUS Clock Gate Enable 0: Disable 1: Enable
27	R/W	0x1	LBC_MBUS_GATE_SW_CFG LBC MBUS Clock Gate Enable 0: Disable 1: Enable
26:24	/	/	/
23	R/W	0x1	VIDEO_OUT0_MBUS_GATE_SW_CFG VIDEO_OUT0 MBUS Clock Gate Enable 0: Disable 1: Enable
22	/	/	/

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Offset: 0x05E4			Register Name: MBUS_MAT_CLK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
21	R/W	0x1	MCU_SYS_MBUS_GATE_SW_CFG MCU_SYS MBUS Clock Gate Enable 0: Disable 1: Enable
20	R/W	0x1	VIDEO_IN_MBUS_GATE_SW_CFG VIDEO_IN MBUS Clock Gate Enable 0: Disable 1: Enable
19	R/W	0x1	CAN_MBUS_GATE_SW_CFG CAN MBUS Clock Gate Enable 0: Disable 1: Enable
18	R/W	0x1	CE_SYS_MBUS_GATE_SW_CFG CE_SYS MBUS Clock Gate Enable 0: Disable 1: Enable
17:15	/	/	/
14	R/W	0x1	GMAC2_MBUS_GATE_SW_CFG GMAC2 MBUS Clock Gate Enable 0: Disable 1: Enable
13	R/W	0x1	GMAC1_MBUS_GATE_SW_CFG GMAC1 MBUS Clock Gate Enable 0: Disable 1: Enable
12	R/W	0x1	GMAC0_MBUS_GATE_SW_CFG GMAC0 MBUS Clock Gate Enable 0: Disable 1: Enable
11	R/W	0x1	GMAC_MBUS_GATE_SW_CFG GMAC MBUS Clock Gate Enable 0: Disable 1: Enable
10:0	/	/	/

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4.1.7.33 0x05F0 AHB Master Clock Auto Gate Enable Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x05F0			Register Name: AHB_MAT_CLK_AUTO_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22	R/W	0x0	SPIF_AHB_AUTO_GATE_EN SPIF AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
21:17	/	/	/
16	R/W	0x0	USB2P0_SYS_AHB_AUTO_GATE_EN USB2P0_SYS AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
15	R/W	0x0	GMAC2_AHB_AUTO_GATE_EN GMAC2 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
14	R/W	0x0	GMAC1_AHB_AUTO_GATE_EN GMAC1 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
13	R/W	0x0	GMAC0_AHB_AUTO_GATE_EN GMAC0 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
12	R/W	0x0	MCU_SYS_AHB_AUTO_GATE_EN MCU_SYS AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
11:9	/	/	/
8	R/W	0x0	SMHC3_AHB_AUTO_GATE_EN SMHC3 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto

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Offset: 0x05F0			Register Name: AHB_MAT_CLK_AUTO_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	SMHC2_AHB_AUTO_GATE_EN SMHC2 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
6	R/W	0x0	SMHC1_AHB_AUTO_GATE_EN SMHC1 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
5	R/W	0x0	SMHC0_AHB_AUTO_GATE_EN SMHC0 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
4	/	/	/
3	R/W	0x0	VIDEO_OUT0_AHB_AUTO_GATE_EN VIDEO_OUT0 AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
2	R/W	0x0	VIDEO_IN_AHB_AUTO_GATE_EN VIDEO_IN AHB Clock Auto Gate Enable 0: No-Auto 1: Auto
1:0	/	/	/

4.1.7.34 0x05F4 MBUS Master Clock Auto Gate Enable Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x05F4			Register Name: MBUS_MAT_CLK_AUTO_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	DMA1_MBUS_AUTO_GATE_EN DMA1 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto

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Offset: 0x05F4			Register Name: MBUS_MAT_CLK_AUTO_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	DMA0_MBUS_AUTO_GATE_EN DMA0 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
27	R/W	0x0	LBC_MBUS_AUTO_GATE_EN LBC MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
26:24	/	/	/
23	R/W	0x0	VIDEO_OUT0_MBUS_AUTO_GATE_EN VIDEO_OUT0 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
22	/	/	/
21	R/W	0x0	MCU_SYS_MBUS_AUTO_GATE_EN MCU_SYS MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
20	R/W	0x0	VIDEO_IN_MBUS_AUTO_GATE_EN VIDEO_IN MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
19	R/W	0x0	CAN_MBUS_AUTO_GATE_EN CAN MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
18	R/W	0x0	CE_SYS_MBUS_AUTO_GATE_EN CE_SYS MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
17:15	/	/	/
14	R/W	0x0	GMAC2_MBUS_AUTO_GATE_EN GMAC2 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
13	R/W	0x0	GMAC1_MBUS_AUTO_GATE_EN GMAC1 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto

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Offset: 0x05F4			Register Name: MBUS_MAT_CLK_AUTO_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	GMAC0_MBUS_AUTO_GATE_EN GMAC0 MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
11	R/W	0x0	GMAC_MBUS_AUTO_GATE_EN GMAC MBUS Clock Auto Gate Enable 0: No-Auto 1: Auto
10:0	/	/	/

4.1.7.35 0x05F8 AHB Master Clock Gate Status Register (Default Value: 0x0041_F1EC)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x05F8			Register Name: AHB_MAT_CLK_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22	R	0x1	SPIF_AHB_GATE_STAT SPIF AHB Clock Gate Status 0: Disable 1: Enable
21:17	/	/	/
16	R	0x1	USB2P0_SYS_AHB_GATE_STAT USB2P0_SYS AHB Clock Gate Status 0: Disable 1: Enable
15	R	0x1	GMAC2_AHB_GATE_STAT GMAC2 AHB Clock Gate Status 0: Disable 1: Enable
14	R	0x1	GMAC1_AHB_GATE_STAT GMAC1 AHB Clock Gate Status 0: Disable 1: Enable

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Offset: 0x05F8			Register Name: AHB_MAT_CLK_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
13	R	0x1	GMAC0_AHB_GATE_STAT GMAC0 AHB Clock Gate Status 0: Disable 1: Enable
12	R	0x1	MCU_SYS_AHB_GATE_STAT MCU_SYS AHB Clock Gate Status 0: Disable 1: Enable
11:9	/	/	/
8	R	0x1	SMHC3_AHB_GATE_STAT SMHC3 AHB Clock Gate Status 0: Disable 1: Enable
7	R	0x1	SMHC2_AHB_GATE_STAT SMHC2 AHB Clock Gate Status 0: Disable 1: Enable
6	R	0x1	SMHC1_AHB_GATE_STAT SMHC1 AHB Clock Gate Status 0: Disable 1: Enable
5	R	0x1	SMHC0_AHB_GATE_STAT SMHC0 AHB Clock Gate Status 0: Disable 1: Enable
4	/	/	/
3	R	0x1	VIDEO_OUT0_AHB_GATE_STAT VIDEO_OUT0 AHB Clock Gate Status 0: Disable 1: Enable
2	R	0x1	VIDEO_IN_AHB_GATE_STAT VIDEO_IN AHB Clock Gate Status 0: Disable 1: Enable
1:0	/	/	/

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4.1.7.36 0x05FC MBUS Master Clock Gate Status Register (Default Value: 0x38BC_7800)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x05FC			Register Name: MBUS_MAT_CLK_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R	0x1	DMA1_MBUS_GATE_STAT DMA1 MBUS Clock Gate Status 0: Disable 1: Enable
28	R	0x1	DMA0_MBUS_GATE_STAT DMA0 MBUS Clock Gate Status 0: Disable 1: Enable
27	R	0x1	LBC_MBUS_GATE_STAT LBC MBUS Clock Gate Status 0: Disable 1: Enable
26:24	/	/	/
23	R	0x1	VIDEO_OUT0_MBUS_GATE_STAT VIDEO_OUT0 MBUS Clock Gate Status 0: Disable 1: Enable
22	/	/	/
21	R	0x1	MCU_SYS_MBUS_GATE_STAT MCU_SYS MBUS Clock Gate Status 0: Disable 1: Enable
20	R	0x1	VIDEO_IN_MBUS_GATE_STAT VIDEO_IN MBUS Clock Gate Status 0: Disable 1: Enable
19	R	0x1	CAN_MBUS_GATE_STAT CAN MBUS Clock Gate Status 0: Disable 1: Enable

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Offset: 0x05FC			Register Name: MBUS_MAT_CLK_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
18	R	0x1	CE_SYS_MBUS_GATE_STAT CAN MBUS Clock Gate Status 0: Disable 1: Enable
17:15	/	/	/
14	R	0x1	GMAC2_MBUS_GATE_STAT GMAC2 MBUS Clock Gate Status 0: Disable 1: Enable
13	R	0x1	GMAC1_MBUS_GATE_STAT GMAC1 MBUS Clock Gate Status 0: Disable 1: Enable
12	R	0x1	GMAC0_MBUS_GATE_STAT GMAC0 MBUS Clock Gate Status 0: Disable 1: Enable
11	R	0x1	GMAC_MBUS_GATE_STAT GMAC MBUS Clock Gate Status 0: Disable 1: Enable
10:0	/	/	/

4.1.7.37 0x0704 DMA0 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

DMA working clock is derived from AHB_CLK.

Offset: 0x0704			Register Name: DMA0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DMA0_RST_N DMA0 Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x0704			Register Name: DMA0_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	DMA0_AHB_CLK_EN DMA0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.38 0x070C DMA1 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

DMA working clock is derived from AHB_CLK.

Offset: 0x070C			Register Name: DMA1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DMA1_RST_N DMA1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	DMA1_AHB_CLK_EN DMA1_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.39 0x0724 SPINLOCK Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0724			Register Name: SPINLOCK_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPINLOCK_RST_N SPINLOCK Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPINLOCK_AHB_CLK_EN SPINLOCK_AHB_CLK Gating Control 0: Mask 1: Pass

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4.1.7.40 0x0744 MSGBOX_CPUX Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0744			Register Name: MSGBOX_CPUX_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MSGBOX_CPUX_RST_N CPU MSGBOX_CPUX Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	MSGBOX_CPUX_AHB_CLK_EN MSGBOX_CPUX_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.41 0x074C MSGBOX_CORE0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x074C			Register Name: MSGBOX_CORE0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MSGBOX_CORE0_RST_N MSGBOX_CORE0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	MSGBOX_CORE0_AHB_CLK_EN MSGBOX_CORE0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.42 0x0754 MSGBOX_CORE1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0754			Register Name: MSGBOX_CORE1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MSGBOX_CORE1_RST_N MSGBOX_CORE1 Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x0754			Register Name: MSGBOX_CORE1_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	MSGBOX_CORE1_AHB_CLK_EN MSGBOX_CORE1_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.43 0x075C MSGBOX_CORE2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x075C			Register Name: MSGBOX_CORE2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MSGBOX_CORE2_RST_N MSGBOX_CORE2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	MSGBOX_CORE2_AHB_CLK_EN MSGBOX_CORE2_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.44 0x0764 MSGBOX_CORE3 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0764			Register Name: MSGBOX_CORE3_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MSGBOX_CORE3_RST_N MSGBOX_CORE3 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	MSGBOX_CORE3_AHB_CLK_EN MSGBOX_CORE3_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.45 0x076C MSGBOX_MCU Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x076C			Register Name: MSGBOX_MCU_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x076C			Register Name: MSGBOX_MCU_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	MSGBOX_MCU_RST_N MSGBOX_MCU Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	MSGBOX_MCU_AHB_CLK_EN MSGBOX_MCU_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.46 0x0794 PWM2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0794			Register Name: PWM2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	PWM2_RST_N PWM2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	PWM2_APB_CLK_EN PWM2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.47 0x07A4 DCU Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

DBGSYS working clock is derived from SYS_24M_CLK.

Offset: 0x07A4			Register Name: DCU_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DCU_RST_N DCU Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x07A4			Register Name: DCU_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	DCU_CLK_EN DCU_CLK Gating Control 0: Mask 1: Pass

4.1.7.48 0x07AC DAP Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of CCU_SEC_SWITCH_REG.

Offset: 0x07AC			Register Name: DAP_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DAP_RST_N DAP Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	DAP_AHB_CLK_EN DAP_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.49 0x07C0 PWMCS0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x07C0			Register Name: PWMCS0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PWMCS0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON PWMCS0_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M
23:5	/	/	/

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Offset: 0x07C0			Register Name: PWMCS0_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.50 0x07C4 PWMCS0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x07C4			Register Name: PWMCS0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	PWMCS0_RST_N PWMCS0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	PWMCS0_APB_CLK_EN PWMCS0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.51 0x07C8 PWMCS1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x07C8			Register Name: PWMCS1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PWMCS1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON PWMCS1_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.52 0x07CC PWMCS1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x07CC			Register Name: PWMCS1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	PWMCS1_RST_N PWMCS1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	PWMCS1_APB_CLK_EN PWMCS1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.53 0x0800 TIMER0_CPUX_0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0800			Register Name: TIMER0_CPUX_0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_0_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_0_CLK=Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_0 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0800			Register Name: TIMER0_CPUX_0_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.54 0x0804 TIMER0_CPUX_1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0804			Register Name: TIMER0_CPUX_1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_1_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_1_CLK=Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_1 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0804			Register Name: TIMER0_CPUX_1_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.55 0x0808 TIMER0_CPUX_2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0808			Register Name: TIMER0_CPUX_2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_2_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_2_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 00: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_2 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0808			Register Name: TIMER0_CPUX_2_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.56 0x080C TIMER0_CPUX_3 Clock Register (Default Value: 0x0000_0000)

Offset: 0x080C			Register Name: TIMER0_CPUX_3_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_3_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_3_CLK=Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_3 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x080C			Register Name: TIMER0_CPUX_3_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.57 0x0810 TIMER0_CPUX_4 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0810			Register Name: TIMER0_CPUX_4_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_4_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_4_CLK=Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_4 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0810			Register Name: TIMER0_CPUX_4_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.58 0x0814 TIMER0_CPUX_5 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0814			Register Name: TIMER0_CPUX_5_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_5_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_5_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_5 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0814			Register Name: TIMER0_CPUX_5_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.59 0x0818 TIMER0_CPUX_6 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0818			Register Name: TIMER0_CPUX_6_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_6_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_6_CLK=Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_6 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0818			Register Name: TIMER0_CPUX_6_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.60 0x081C TIMER0_CPUX_7 Clock Register (Default Value: 0x0000_0000)

Offset: 0x081C			Register Name: TIMER0_CPUX_7_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIMER0_CPUX_7_CLK_GATING Gating Clock 0: Disable 1: Enable TIMER0_CPUX_7_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_CPUX_7 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x081C			Register Name: TIMER0_CPUX_7_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.61 0x0850 TIMER0_CPUX Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0850			Register Name: TIMER0_CPUX_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TIMER0_CPUX_RST_N TIMER0_CPUX Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TIMER0_CPUX_AHB_CLK_EN TIMER0_CPUX_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.62 0x0860 TIEMR0_MCU_0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0860			Register Name: TIEMR0_MCU_0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIEMR0_MCU_0_CLK_GATING Gating Clock 0: Disable 1: Enable TIEMR0_MCU_0_CLK=Clock Source/M
30:27	/	/	/

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Offset: 0x0860			Register Name: TIEMR0_MCU_0_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIMER0_MCU_0 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.63 0x0864 TIEMR0_MCU_1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0864			Register Name: TIEMR0_MCU_1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIEMR0_MCU_1_CLK_GATING Gating Clock 0: Disable 1: Enable TIEMR0_MCU_1_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIEMR0_MCU_1 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0864			Register Name: TIEMR0_MCU_1_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source

4.1.7.64 0x0868 TIEMR0_MCU_2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0868			Register Name: TIEMR0_MCU_2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIEMR0_MCU_2_CLK_GATING Gating Clock 0: Disable 1: Enable TIEMR0_MCU_2_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIEMR0_MCU_2 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x0868			Register Name: TIEMR0_MCU_2_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.65 0x086C TIEMR0_MCU_3 Clock Register (Default Value: 0x0000_0000)

Offset: 0x086C			Register Name: TIEMR0_MCU_3_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TIEMR0_MCU_3_CLK_GATING Gating Clock 0: Disable 1: Enable TIEMR0_MCU_3_CLK=Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_200M TIEMR0_MCU_3 clock input, the clock source is 24MHz, RC 16 MHz, 32 kHz, or 200 MHz.
23:3	/	/	/

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Offset: 0x086C			Register Name: TIEMR0_MCU_3_CLK_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	FACTOR_M Factor M 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128 Select the pre-scale of clock source.

4.1.7.66 0x0870 TIMER0_MCU Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0870			Register Name: TIMER0_MCU_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TIMER0_MCU_RST_N TIMER0_MCU Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TIMER0_MCU_AHB_CLK_EN TIMER0_MCU_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.67 0x0A00 DE0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0A00			Register Name: DE0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DE0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON DE_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x0A00			Register Name: DE0_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x00	CLK_SRC_SEL Clock Source Select 000: PERI0_600M 001: PERI0_480M 010: PERI0_400M 011: VIDEO0PLL4X 100: VIDEO0PLL3X
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.68 0x0A04 DE0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0A04			Register Name: DE0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DE0_RST_N DE0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	DE0_AHB_CLK_EN DE0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.69 0x0A40 G2D Clock Register (Default Value: 0x0000_0000)

Offset: 0x0A40			Register Name: G2D_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	G2D_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON G2D_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x0A40			Register Name: G2D_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: PERI0_600M 001: PERI0_480M 010: PERI0_400M 011: VIDEO0PLL4X 100: VIDEO0PLL3X
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.70 0x0A44 G2D Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0A44			Register Name: G2D_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	G2D_RST_N G2D Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	G2D_AHB_CLK_EN G2D_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.71 0x0AC0 CE_SYS Clock Register (Default Value: 0x0000_0000)

Offset: 0x0AC0			Register Name: CE_SYS_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CE_SYS_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CE_SYS_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x0AC0			Register Name: CE_SYS_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M 010: PERI0_480M 011: PERI0_600M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 15.

4.1.7.72 0x0AC4 CE_SYS Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0AC4			Register Name: CE_SYS_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	CE_SYS_RST_N CE_SYS Reset 0: Assert 1: De-assert
16:2	/	/	/
1	R/W	0x0	CE_SYS_CLK_EN CE_SYS_CLK Gating Control 0: Mask 1: Pass
0	R/W	0x0	CE_SYS_IP_AHB_CLK_EN CE_SYS_IP_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.73 0x0B80 MCU_CORE Clock Register (Default Value: 0x0000_0000)

Offset: 0x0B80			Register Name: MCU_CORE_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MCU_CORE_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON MCU_CORE_CLK = Clock Source/M E907_AXI_CLK=MCU_CORE_CLK/N

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Offset: 0x0B80			Register Name: MCU_CORE_CLK_REG
Bit	Read/Write	Default/Hex	Description
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK 011: PERI0_600M 100: PERI0_480M 101: PERI0_400M The clock MUX uses glitch-free switch and supports dynamic configuration.
23:10	/	/	/
9:8	R/W	0x0	E907_AXI_DIV_CFG Factor N N = FACTOR_N +1 FACTOR_N is from 1 to 3. The clock DIV supports dynamic configuration.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31. The clock DIV supports dynamic configuration.

4.1.7.74 0x0B88 MCU_TS Clock Register (Default Value: 0x0000_0000)

Offset: 0x0B88			Register Name: MCU_TS_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MCU_TS_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON MCU_TS_CLK = Clock Source
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: RC_16M_CLK 010: SYS_32K_CLK
23:0	/	/	/

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4.1.7.75 0x0B94 MCU_SYS Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0B94			Register Name: MCU_SYS_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	MCU_SYS_RST_N MCU Debug Bus Reset 0: Assert 1: De-assert
16	R/W	0x0	MCU_CORE_RST_N E907 Reset 0: Assert 1: De-assert
15:0	/	/	/

4.1.7.76 0x0B9C MCU_CFG Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0B9C			Register Name: MCU_CFG_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MCU_CFG_RST_N MCU Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	MCU_CFG_CLK_EN MCU_CFG_CLK Gating Control 0: Mask 1: Pass

4.1.7.77 0x0C0C DRAMC Gating and Reset Register (Default Value: 0x0000_0001)

Offset: 0x0C0C			Register Name: DRAMC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DRAMC_RST_N DRAMC Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x0C0C			Register Name: DRAMC_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	DRAMC_AHB_CLK_EN DRAMC_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.78 0x0D00 SMHC0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0D00			Register Name: SMHC0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SMHC0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SMHC0_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M 010: PERI0_300M 011: PERI1_400M 100: PERI1_300M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.79 0x0D0C SMHC0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0D0C			Register Name: SMHC0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0D0C			Register Name: SMHC0_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	SMHC0_RST_N SMHC0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SMHC0_AHB_CLK_EN SMHC0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.80 0x0D10 SMHC1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0D10			Register Name: SMHC1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SMHC1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SMHC1_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M 010: PERI0_300M 011: PERI1_400M 100: PERI1_300M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.81 0x0D1C SMHC1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0D1C	Register Name: SMHC1_GAR_REG

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Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SMHC1_RST_N SMHC1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SMHC1_AHB_CLK_EN SMHC1_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.82 0x0D20 SMHC2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0D20			Register Name: SMHC2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SMHC2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SMHC2_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_800M 010: PERI0_600M 011: PERI1_800M 100: PERI1_600M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.83 0x0D2C SMHC2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0D2C	Register Name: SMHC2_GAR_REG

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Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SMHC2_RST_N SMHC2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SMHC2_AHB_CLK_EN SMHC2_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.84 0x0D30 SMHC3 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0D30			Register Name: SMHC3_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SMHC3_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SMHC3_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_400M 010: PERI0_300M 011: PERI1_400M 100: PERI1_300M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.85 0x0D3C SMHC3 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0D3C	Register Name: SMHC3_GAR_REG

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Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SMHC3_RST_N SMHC3 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SMHC3_AHB_CLK_EN SMHC3_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.86 0x0E00 UART0 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E00			Register Name: UART0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	UART0_RST_N UART0 Reset, For UART0_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART0_APB_CLK_EN UART0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.87 0x0E04 UART1 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E04			Register Name: UART1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E04			Register Name: UART1_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	UART1_RST_N UART1 Reset, For UART1_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART1_APB_CLK_EN UART1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.88 0x0E08 UART2 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E08			Register Name: UART2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	UART2_RST_N UART2 Reset, For UART2_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART2_APB_CLK_EN UART2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.89 0x0E0C UART3 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E0C			Register Name: UART3_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E0C			Register Name: UART3_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	UART3_RST_N UART3 Reset, For UART3_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART3_APB_CLK_EN UART3_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.90 0x0E10 UART4 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E10			Register Name: UART4_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	UART4_RST_N UART4 Reset, For UART4_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART4_APB_CLK_EN UART4_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.91 0x0E14 UART5 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E14			Register Name: UART5_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E14			Register Name: UART5_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	UART5_RST_N UART5 Reset, For UART5_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART5_APB_CLK_EN UART5_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.92 0x0E18 UART6 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E18			Register Name: UART6_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	UART6_RST_N UART6 Reset, For UART6_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART6_APB_CLK_EN UART6_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.93 0x0E20 UART7 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E20			Register Name: UART7_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E20			Register Name: UART7_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	UART7_RST_N UART7 Reset, For UART7_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART7_APB_CLK_EN UART7_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.94 0x0E24 UART8 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E24			Register Name: UART8_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	UART8_RST_N UART8 Reset, For UART8_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART8_APB_CLK_EN UART8_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.95 0x0E28 UART9 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

UART working clock is derived from APB_UART.

Offset: 0x0E28			Register Name: UART9_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E28			Register Name: UART9_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	UART9_RST_N UART9 Reset, For UART9_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	UART9_APB_CLK_EN UART9_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.96 0x0E80 TWI0 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E80			Register Name: TWI0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TWI0_RST_N TWI0 Reset, For TWI0_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI0_APB_CLK_EN TWI0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.97 0x0E84 TWI1 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E84			Register Name: TWI1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E84			Register Name: TWI1_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	TWI1_RST_N TWI1 Reset, For TWI1_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI1_APB_CLK_EN TWI1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.98 0x0E88 TWI2 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E88			Register Name: TWI2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TWI2_RST_N TWI2 Reset, For TWI2_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI2_APB_CLK_EN TWI2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.99 0x0E8C TWI3 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E8C			Register Name: TWI3_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E8C			Register Name: TWI3_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	TWI3_RST_N TWI3 Reset, For TWI3_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI3_APB_CLK_EN TWI3_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.100 0x0E90 TWI4 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E90			Register Name: TWI4_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TWI4_RST_N TWI4 Reset, For TWI4_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI4_APB_CLK_EN TWI4_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.101 0x0E94 TWI5 Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

TWI working clock is derived from APB1.

Offset: 0x0E94			Register Name: TWI5_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0E94			Register Name: TWI5_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	TWI5_RST_N TWI5 Reset, For TWI5_APB_CLK Domain 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TWI5_APB_CLK_EN TWI5_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.102 0x0F00 SPI0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F00			Register Name: SPI0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SPI0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SPI0_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_300M 011: PERI0_200M 100: PERI1_480M 101: PERI1_300M 110: PERI1_200M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.103 0x0F04 SPI0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F04			Register Name: SPI0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPI0_RST_N SPI0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPI0_AHB_CLK_EN SPI0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.104 0x0F08 SPI1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F08			Register Name: SPI1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SPI1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SPI1_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_300M 011: PERI0_200M 100: PERI1_480M 101: PERI1_300M 110: PERI1_200M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/

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Offset: 0x0F08			Register Name: SPI1_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.105 0x0F0C SPI1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F0C			Register Name: SPI1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPI1_RST_N SPI1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPI1_AHB_CLK_EN SPI1_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.106 0x0F10 SPI2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F10			Register Name: SPI2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SPI2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SPI2_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_300M 011: PERI0_200M 100: PERI1_480M 101: PERI1_300M 110: PERI1_200M
23:13	/	/	/

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Offset: 0x0F10			Register Name: SPI2_CLK_REG
Bit	Read/Write	Default/Hex	Description
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.107 0x0F14 SPI2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F14			Register Name: SPI2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPI2_RST_N SPI2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPI2_AHB_CLK_EN SPI2_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.108 0x0F18 SPIF Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F18			Register Name: SPIF_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SPIF_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SPIF_CLK = Clock Source/M/N
30:27	/	/	/

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Offset: 0x0F18			Register Name: SPIF_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_400M 011: PERI0_300M 100: PERI1_480M 101: PERI1_400M 110: PERI1_300M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.109 0x0F1C SPIF Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F1C			Register Name: SPIF_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPIF_RST_N SPIF Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPIF_AHB_CLK_EN SPIF_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.110 0x0F20 SPI3 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F20			Register Name: SPI3_CLK_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0F20			Register Name: SPI3_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SPI3_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON SPI3_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_480M 010: PERI0_300M 011: PERI0_200M 100: PERI1_480M 101: PERI1_300M 110: PERI1_200M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.111 0x0F24 SPI3 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F24			Register Name: SPI3_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SPI3_RST_N SPI3 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	SPI3_AHB_CLK_EN SPI3_AHB_CLK Gating Control 0: Mask 1: Pass

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4.1.7.112 0x0F80 CAN0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F80			Register Name: CAN0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CAN0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CAN0_CLK = PERI0_400M/M
30:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.113 0x0F84 CAN0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F84			Register Name: CAN0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	CAN0_RST_N CAN0 Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	CAN0_AHB_CLK_EN CAN0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.114 0x0F88 CAN1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0F88			Register Name: CAN1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CAN1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CAN1_CLK = PERI0_400M/M
30:5	/	/	/

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Offset: 0x0F88			Register Name: CAN1_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.115 0x0F8C CAN1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0F8C			Register Name: CAN1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	CAN1_RST_N CAN1 Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	CAN1_AHB_CLK_EN CAN1_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.116 0x0FBC CAN_SYS Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FBC			Register Name: CAN_SYS_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	CAN_SYS_RST_N CAN_SYS Reset 0: Assert 1: DE-assert
15:0	/	/	/

4.1.7.117 0x0FC0 GPADC0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0FC0			Register Name: GPADC0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GPADC0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GPADC0_CLK = Clock Source/M.

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Offset: 0x0FC0			Register Name: GPADC0_CLK_REG
Bit	Read/Write	Default/Hex	Description
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: CLK48M 010: PERI0_480M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.118 0x0FC4 GPADC0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FC4			Register Name: GPADC0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	GPADC0_RST_N GPADC0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GPADC0_APB_CLK_EN GPADC0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.119 0x0FC8 GPADC1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0FC8			Register Name: GPADC1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GPADC1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GPADC1_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x0FC8			Register Name: GPADC1_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: CLK48M 010: PERI0_480M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.120 0x0FCC GPADC1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FCC			Register Name: GPADC1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	GPADC1_RST_N GPADC1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GPADC1_APB_CLK_EN GPADC1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.121 0x0FD0 GPADC2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x0FD0			Register Name: GPADC2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GPADC2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GPADC2_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: CLK48M 010: PERI0_480M

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Offset: 0x0FD0			Register Name: GPADC2_CLK_REG
Bit	Read/Write	Default/Hex	Description
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.122 0x0FD4 GPADC2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FD4			Register Name: GPADC2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	GPADC2_RST_N GPADC2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GPADC2_APB_CLK_EN GPADC2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.123 0x0FD8 THS_ADC Clock Register (Default Value: 0x0000_0000)

Offset: 0x0FD8			Register Name: THS_ADC_CLK_REG/
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	THS_ADC_CLK_GATING Gating Clock. 0: Clock is OFF 1: Clock is ON THS_ADC_CLK = Clock Source/M.
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: CLK48M 010: PERI0_480M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1. FACTOR_M is from 0 to 31.

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4.1.7.124 0x0FDC THS_ADC Gating And Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FDC			Register Name: THS_ADC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	THS_ADC_RST_N THS_ADC Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	THS_ADC_APB_CLK_EN THS_ADC_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.125 0x0FE4 THS Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x0FE4			Register Name: THS_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	THS_RST_N THS Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	THS_APB_CLK_EN THS_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.126 0x1000 IR_RX0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1000			Register Name: IR_RX0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	IR_RX0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON IR_RX0_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x1000			Register Name: IR_RX0_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_32K_CLK 001: SYS_24M_CLK
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.127 0x1004 IR_RX0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1004			Register Name: IR_RX0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	IR_RX0_RST_N IR_RX0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	IR_RX0_APB_CLK_EN IR_RX0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.128 0x1008 IR_TX Clock Register (Default Value: 0x0000_0000)

Offset: 0x1008			Register Name: IR_TX_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	IR_TX_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON IR_TX_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 0: SYS_24M_CLK 1: PERI1_600M
23:5	/	/	/

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Offset: 0x1008			Register Name: IR_TX_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.129 0x100C IR_TX Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x100C			Register Name: IR_TX_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	IR_TX_RST_N IR_TX Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	IR_TX_APB_CLK_EN IR_TX_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.130 0x1030 TPADC Clock Register (Default Value: 0x0000_0000)

Offset: 0x1030			Register Name: TPADC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TPADC_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON TPADC_CLK = SYS_24M_CLK/M
30:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.131 0x1034 TPADC Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1034			Register Name: TPADC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x1034			Register Name: TPADC_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	TPADC_RST_N TPADC Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	TPADC_APB_CLK_EN TPADC_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.132 0x1040 LBC Clock Register (Default Value: 0x0000_0000)

Offset: 0x1040			Register Name: LBC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LBC_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON LBC_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: PERI0_480M 001: PERI0_400M 010: PERI0_300M 011: PERI1_480M 100: PERI1_400M 101: PERI1_300M 110: VIDEO0PLL4X 111: VIDEO0PLL3X
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.133 0x104C LBC Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x104C			Register Name: LBC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	LBC_RST_N LBC Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	LBC_AHB_CLK_EN LBC_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.134 0x1100 IR_RX1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1100			Register Name: IR_RX1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	IR_RX1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON IR_RX1_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_32K_CLK 001: SYS_24M_CLK
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.135 0x1104 IR_RX1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1104			Register Name: IR_RX1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x1104			Register Name: IR_RX1_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	IR_RX1_RST_N IR_RX1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	IR_RX1_APB_CLK_EN IR_RX1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.136 0x1108 IR_RX2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1108			Register Name: IR_RX2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	IR_RX2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON IR_RX2_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_32K_CLK 001: SYS_24M_CLK
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.137 0x110C IR_RX2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x110C			Register Name: IR_RX2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	IR_RX2_RST_N IR_RX2 Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x110C			Register Name: IR_RX2_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	IR_RX2_APB_CLK_EN IR_RX2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.138 0x1110 IR_RX3 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1110			Register Name: IR_RX3_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	IR_RX3_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON IR_RX3_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_32K_CLK 001: SYS_24M_CLK
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.139 0x1114 IR_RX3 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1114			Register Name: IR_RX3_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	IR_RX3_RST_N IR_RX3 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	IR_RX3_APB_CLK_EN IR_RX3_APB_CLK Gating Control 0: Mask 1: Pass

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4.1.7.140 0x1200 I2S0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1200			Register Name: I2S0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	I2S0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON I2S0_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M 010: /
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.141 0x120C I2S0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x120C			Register Name: I2S0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	I2S0_RST_N I2S0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	I2S0_APB_CLK_EN I2S0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.142 0x1210 I2S1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1210			Register Name: I2S1_CLK_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x1210			Register Name: I2S1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	I2S1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON I2S1_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M 010: /
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.143 0x121C I2S1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x121C			Register Name: I2S1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	I2S1_RST_N I2SPCM1 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	I2S1_APB_CLK_EN I2S1_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.144 0x1220 I2S2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1220			Register Name: I2S2_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	I2S2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON I2S2_CLK = Clock Source/M

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Offset: 0x1220			Register Name: I2S2_CLK_REG
Bit	Read/Write	Default/Hex	Description
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M 010: /
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.145 0x122C I2S2 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x122C			Register Name: I2S2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	I2S2_RST_N I2S2 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	I2S2_APB_CLK_EN I2S2_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.146 0x1280 OWA0 TX Clock Register (Default Value: 0x0000_0000)

Offset: 0x1280			Register Name: OWA0_TX_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	OWA0_TX_CLK_GATING Gating Special Clock 0: Clock is OFF 1: Clock is ON OWA0_TX_CLK =Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M

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Offset: 0x1280			Register Name: OWA0_TX_CLK_REG
Bit	Read/Write	Default/Hex	Description
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.147 0x1284 OWA0 RX Clock Register (Default Value: 0x0000_0000)

Offset: 0x1284			Register Name: OWA0_RX_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	OWA0_RX_CLK_GATING Gating Special Clock 0: Clock is OFF 1: Clock is ON OWA0_RX_CLK =Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: PERI0_400M 001: PERI0_300M 010: / 011: /
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.148 0x128C OWA0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x128C			Register Name: OWA0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	OWA0_RST_N OWA0 Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x128C			Register Name: OWA0_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	OWA0_APB_CLK_EN OWA0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.149 0x12C0 DMIC Clock Register (Default Value: 0x0000_0000)

Offset: 0x12C0			Register Name: DMIC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DMIC_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON DMIC_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31

4.1.7.150 0x12CC DMIC Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x12CC			Register Name: DMIC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DMIC_RST_N DMIC Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	DMIC_APB_CLK_EN DMIC_APB_CLK Gating Control 0: Mask 1: Pass

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4.1.7.151 0x12E0 AUDIOCODEC0 DAC Clock Register (Default Value: 0x0000_0000)

Offset: 0x12E0			Register Name: AUDIOCODEC0_DAC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	AUDIOCODEC0_DAC_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON AUDIOCODEC0_DAC_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: AUDIO0PLL 001: PERI1_600M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31

4.1.7.152 0x12EC AUDIOCODEC0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x12EC			Register Name: AUDIOCODEC0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	AUDIOCODEC0_RST_N AUDIOCODEC0 Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	AUDIOCODEC0_APB_CLK_EN AUDIOCODEC0_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.153 0x1300 USB0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1300			Register Name: USB0_CLK_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x1300			Register Name: USB0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	USB0_CLKEN Gating Clock for OHCI0 0: Clock is OFF 1: Clock is ON
30:26	/	/	/
25:24	R/W	0x0	USB0_CLK12M_SEL OHCI0 12M Source Select 00: 12M divided from CLK48M 01: 12M divided from SYS_24M_CLK 10: SYS_32K_CLK 11: RC_16M_CLK
23:0	/	/	/

4.1.7.154 0x1304 USB0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1304			Register Name: USB0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	USB0_DEV_RST_N USB0 Device Reset 0: Assert 1: De-assert
23:21	/	/	/
20	R/W	0x0	USB0_EHCI_RST_N USB0 HOST EHCI Reset 0: Assert 1: De-assert
19:17	/	/	/
16	R/W	0x0	USB0_OHCI_RST_N USB0 HOST OHCI Reset 0: Assert 1: De-assert
15:9	/	/	/
8	R/W	0x0	USB0_DEV_AHB_CLK_EN USB0_DEV_AHB_CLK Gating Control 0: Mask 1: Pass
7:5	/	/	/

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Offset: 0x1304			Register Name: USB0_GAR_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	USB0_EHCI_AHB_CLK_EN USB0_EHCI_AHB_CLK Gating Control 0: Mask 1: Pass
3:1	/	/	/
0	R/W	0x0	USB0_OHCI_AHB_CLK_EN USB0_OHCI_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.155 0x1308 USB1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1308			Register Name: USB1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	USB1_CLKEN Gating Clock for OHCI1 0: Clock is OFF 1: Clock is ON
30:26	/	/	/
25:24	R/W	0x0	USB1_CLK12M_SEL OHCI1 12M Source Select 00: 12M divided from CLK48M 01: 12M divided from SYS_24M_CLK 10: SYS_32K_CLK 11: RC_16M_CLK
23:0	/	/	/

4.1.7.156 0x130C USB1 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x130C			Register Name: USB1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20	R/W	0x0	USB1_EHCI_RST_N USB1 HOST EHCI Reset 0: Assert 1: De-assert
19:17	/	/	/
16	R/W	0x0	USB1_OHCI_RST_N USB1 HOST OHCI Reset 0: Assert 1: De-assert

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Offset: 0x130C			Register Name: USB1_GAR_REG
Bit	Read/Write	Default/Hex	Description
15:5	/	/	/
4	R/W	0x0	USB1_EHCI_AHB_CLK_EN USB1_EHCI_AHB_CLK Gating Control 0: Mask 1: Pass
3:1	/	/	/
0	R/W	0x0	USB1_OHCI_AHB_CLK_EN USB1_OHCI_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.157 0x1340 USB2P0_SYS PHY Reference Clock Register (Default Value: 0x8000_0000)

Offset: 0x1340			Register Name: USB2P0_SYS_PHY_REF_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	USB2P0_SYS_PHY_REF_CLK_GATING Gating Reference Clock for USB0/USB1 0: Clock is OFF 1: Clock is ON USB2P0_SYS_PHY_REF_CLK = SYS_24M_CLK
30:0	/	/	/

4.1.7.158 0x1344 USB2P0_SYS Gating and Reset Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of CCU_SEC_SWITCH_REG.

Offset: 0x1344			Register Name: USB2P0_SYS_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	USB2P0_SYS_RST_N USB2.0 SYS(USB0/USB1) Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x1344			Register Name: USB2P0_SYS_GAR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	USB2P0_SYS_AHB_CLK_EN USB2P0_SYS AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.159 0x1400 GMAC0 PHY Clock Register (Default Value: 0x0000_0000)

Offset: 0x1400			Register Name: GMAC0_PHY_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC0_PHY_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC0_PHY_CLK =PERI0_150M/M
30:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.160 0x1404 GMAC0 PTP Reference Clock Register (Default Value: 0x0000_0000)

Offset: 0x1404			Register Name: GMAC0_PTP_REF_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC0_PTP_REF_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC0_PTP_REF_CLK =Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_200M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.161 0x140C GMAC0 Gating and Reset Register (Default Value: 0x0004_0000)

Offset: 0x140C			Register Name: GMAC0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/W	0x1	GMAC0_TOP_AHB_RST_N GMAC0_TOP AHB Reset 0: Assert 1: De-assert
17	R/W	0x0	GMAC0_AXI_RST_N GMAC0 AXI Reset 0: Assert 1: De-assert
16	R/W	0x0	GMAC0_AHB_RST_N GMAC0 AHB Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GMAC0_AHB_CLK_EN GMAC0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.162 0x1410 GMAC1 PHY Clock Register (Default Value: 0x0000_0000)

Offset: 0x1410			Register Name: GMAC1_PHY_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC1_PHY_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC1_PHY_CLK =PERI0_150M/M
30:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.163 0x1414 GMAC1 PTP Reference Clock Register (Default Value: 0x0000_0000)

Offset: 0x1414			Register Name: GMAC1_PTP_REF_CLK_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x1414			Register Name: GMAC1_PTP_REF_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC1_PTP_REF_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC1_PTP_REF_CLK =Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_200M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.164 0x141C GMAC1 Gating and Reset Register (Default Value: 0x0004_0000)

Offset: 0x141C			Register Name: GMAC1_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/W	0x1	GMAC1_TOP_AHB_RST_N GMAC1_TOP AHB Reset 0: Assert 1: De-assert
17	R/W	0x0	GMAC1_AXI_RST_N GMAC1 AXI Reset 0: Assert 1: De-assert
16	R/W	0x0	GMAC1_AHB_RST_N GMAC1 AHB Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GMAC1_AHB_CLK_EN GMAC1_AHB_CLK Gating Control 0: Mask 1: Pass

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4.1.7.165 0x1420 GMAC2 PHY Clock Register (Default Value: 0x0000_0000)

Offset: 0x1420			Register Name: GMAC2_PHY_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC2_PHY_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC2_PHY_CLK =PERI0_150M/M
30:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.166 0x1424 GMAC2 PTP Reference Clock Register (Default Value: 0x0000_0000)

Offset: 0x1424			Register Name: GMAC2_PTP_REF_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	GMAC2_PTP_REF_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON GMAC2_PTP_REF_CLK =Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_200M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.167 0x142C GMAC2 Gating and Reset Register (Default Value: 0x0004_0000)

Offset: 0x142C			Register Name: GMAC2_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/

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Offset: 0x142C			Register Name: GMAC2_GAR_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x1	GMAC2_TOP_AHB_RST_N GMAC2_TOP AHB Reset 0: Assert 1: De-assert
17	R/W	0x0	GMAC2_AXI_RST_N GMAC2 AXI Reset 0: Assert 1: De-assert
16	R/W	0x0	GMAC2_AHB_RST_N GMAC2 AHB Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	GMAC2_AHB_CLK_EN GMAC2_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.168 0x1500 TCON_LCD0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1500			Register Name: TCON_LCD0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TCON_LCD0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON TCON_LCD0_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: VIDEO0PLL4X 001: PERI0PLL2X 010: PERI1PLL2X 011: VIDEO0PLL3X
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.169 0x1504 TCON_LCD0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1504			Register Name: TCON_LCD0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	TCON_LCD0_RST_N VIDEOOUT0 TCON LCD0 Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	TCON_LCD0_AHB_CLK_EN TCON_LCD0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.170 0x1544 LVDS0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1544			Register Name: LVDS0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	LVDS0_RST_N LVDS0 Reset 0: Assert 1: De-assert
15:0	/	/	/

4.1.7.171 0x1580 MIPI_DSI0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1580			Register Name: MIPI_DSI0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_DSI0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON MIPI_DSI0_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: PERI0_200M 010: PERI0_150M
23:5	/	/	/

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Offset: 0x1580			Register Name: MIPI_DSI0_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.172 0x1584 MIPI_DSI0 Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1584			Register Name: MIPI_DSI0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	MIPI_DSI0_RST_N MIPI_DSI0 Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	MIPI_DSI0_AHB_CLK_EN MIPI_DSI0_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.173 0x15C0 COMBOPHY0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x15C0			Register Name: COMBOPHY0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	COMBOPHY0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON COMBOPHY0_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: VIDEO0PLL4X 001: PERI0PLL2X 010: PERI1PLL2X 011: VIDEO0PLL3X
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.174 0x16C4 VO0_REG Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x16C4			Register Name: VO0_REG_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	VO0_REG_RST_N VO0_REG Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x0	VO0_REG_AHB_CLK_EN VO0_REG_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.175 0x16E4 VIDEO_OUT0 Gating and Reset Register (Default Value: 0x0001_0000)

Offset: 0x16E4			Register Name: VIDEO_OUT0_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x1	VIDEO_OUT0_RST_N VIDEO_OUT0 Reset 0: Assert 1: DE-assert
15:0	/	/	/

4.1.7.176 0x1700 LEDC Clock Register (Default Value: 0x0000_0000)

Offset: 0x1700			Register Name: LEDC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LEDC_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON LEDC_CLK = Clock Source/M
30:25	/	/	/
24	R/W	0x0	CLK_SRC_SEL Clock Source Select 0: SYS_24M_CLK 1: PERI0_600M
23:5	/	/	/

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Offset: 0x1700			Register Name: LEDC_CLK_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.177 0x1704 LEDC Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1704			Register Name: LEDC_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	LEDC_RST_N LEDC Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	LEDC_APB_CLK_EN LEDC_APB_CLK Gating Control 0: Mask 1: Pass

4.1.7.178 0x1800 CSI Master0 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1800			Register Name: CSI_MASTER0_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CSI_MASTER0_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CSI_MASTER0_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: VIDEO0PLL4X 010: VIDEO0PLL3X 011: PERI1PLL2X 100: PERI1_480M
23:13	/	/	/

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Offset: 0x1800			Register Name: CSI_MASTER0_CLK_REG
Bit	Read/Write	Default/Hex	Description
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.179 0x1804 CSI Master1 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1804			Register Name: CSI_MASTER1_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CSI_MASTER1_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CSI_MASTER1_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: VIDEO0PLL4X 010: VIDEO0PLL3X 011: PERI1PLL2X 100: PERI1_480M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.180 0x1808 CSI Master2 Clock Register (Default Value: 0x0000_0000)

Offset: 0x1808	Register Name: CSI_MASTER2_CLK_REG

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Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CSI_MASTER2_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CSI_MASTER2_CLK = Clock Source/M/N
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: SYS_24M_CLK 001: VIDEO0PLL4X 010: VIDEO0PLL3X 011: PERI1PLL2X 100: PERI1_480M
23:13	/	/	/
12:8	R/W	0x0	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
7:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.181 0x1840 CSI Clock Register (Default Value: 0x0000_0000)

Offset: 0x1840			Register Name: CSI_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CSI_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON CSI_CLK = Clock Source/M
30:27	/	/	/

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Offset: 0x1840			Register Name: CSI_CLK_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: PERI0_300M 001: PERI0_400M 010: PERI0_480M 011: VIDEO0PLL4X 100: VIDEO0PLL3X 101: PERI1PLL2X 110: PERI1_480M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.182 0x1860 ISP Clock Register (Default Value: 0x0000_0000)

Offset: 0x1860			Register Name: ISP_CLK_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	ISP_CLK_GATING Gating Clock 0: Clock is OFF 1: Clock is ON ISP_CLK = Clock Source/M
30:27	/	/	/
26:24	R/W	0x0	CLK_SRC_SEL Clock Source Select 000: PERI0_300M 001: PERI0_400M 010: PERI0_480M 011: VIDEO0PLL4X 100: VIDEO0PLL3X 101: PERI1PLL2X 110: PERI1_480M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

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4.1.7.183 0x1884 VIDEO_IN Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1884			Register Name: VIDEO_IN_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	VIDEO_IN_RST_N VIDEO_IN Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	VIDEO_IN_AHB_CLK_EN VIDEO_IN_AHB_CLK Gating Control 0: Mask 1: Pass

4.1.7.184 0x1908 PERI0PLL Gate Enable Register (Default Value: 0x0FFF_0FFF)

Offset: 0x1908			Register Name: PERI0PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W	0x1	PERI0PLL2X_GATE_SW_CFG PERI0PLL2X Clock Gate Enable 0: Disable 1: Enable
26	R/W	0x1	PERI0_800M_GATE_SW_CFG PERI0 800M Clock Gate Enable 0: Disable 1: Enable
25	R/W	0x1	PERI0_600M_GATE_SW_CFG PERI0 600M Clock Gate Enable 0: Disable 1: Enable
24	R/W	0x1	PERI0_480M_GATE_ALL_CFG PERI0 480M Clock Gate Enable 0: Disable 1: Enable
23	R/W	0x1	PERI0_480M_GATE_SW_CFG PERI0 480M Clock Gate Enable 0: Disable 1: Enable

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Offset: 0x1908			Register Name: PERI0PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
22	R/W	0x1	PERI0_160M_GATE_SW_CFG PERI0 160M Clock Gate Enable 0: Disable 1: Enable
21	R/W	0x1	PERI0_300M_GATE_ALL_CFG PERI0 300M Clock Gate Enable 0: Disable 1: Enable
20	R/W	0x1	PERI0_300M_GATE_SW_CFG PERI0 300M Clock Gate Enable 0: Disable 1: Enable
19	R/W	0x1	PERI0_150M_GATE_SW_CFG PERI0 150M Clock Gate Enable 0: Disable 1: Enable
18	R/W	0x1	PERI0_400M_GATE_ALL_CFG PERI0 400M Clock Gate Enable 0: Disable 1: Enable
17	R/W	0x1	PERI0_400M_GATE_SW_CFG PERI0 400M Clock Gate Enable 0: Disable 1: Enable
16	R/W	0x1	PERI0_200M_GATE_SW_CFG PERI0 200M Clock Gate Enable 0: Disable 1: Enable
15:12	/	/	/
11	R/W	0x1	PERI0PLL2X_AUTO_GATE_EN PERI0PLL2X Clock Auto Gate Enable 0: Auto 1: No-Auto
10	R/W	0x1	PERI0_800M_AUTO_GATE_EN PERI0 800M Clock Auto Gate Enable 0: Auto 1: No-Auto
9	R/W	0x1	PERI0_600M_AUTO_GATE_EN PERI0 600M Clock Auto Gate Enable 0: Auto 1: No-Auto

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Offset: 0x1908			Register Name: PERI0PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x1	PERI0_480M_AUTO_GATE_EN_ALL PERI0 480M Clock Auto Gate Enable 0: Auto 1: No-Auto
7	R/W	0x1	PERI0_480M_AUTO_GATE_EN PERI0 480M Clock Auto Gate Enable 0: Auto 1: No-Auto
6	R/W	0x1	PERI0_160M_AUTO_GATE_EN PERI0 160M Clock Auto Gate Enable 0: Auto 1: No-Auto
5	R/W	0x1	PERI0_300M_AUTO_GATE_EN_ALL PERI0 300M Clock Auto Gate Enable 0: Auto 1: No-Auto
4	R/W	0x1	PERI0_300M_AUTO_GATE_EN PERI0 300M Clock Auto Gate Enable 0: Auto 1: No-Auto
3	R/W	0x1	PERI0_150M_AUTO_GATE_EN PERI0 150M Clock Auto Gate Enable 0: Auto 1: No-Auto
2	R/W	0x1	PERI0_400M_AUTO_GATE_EN_ALL PERI0 400M Clock Auto Gate Enable All 0: Auto 1: No-Auto
1	R/W	0x1	PERI0_400M_AUTO_GATE_EN PERI0 400M Clock Auto Gate Enable 0: Auto 1: No-Auto
0	R/W	0x1	PERI0_200M_AUTO_GATE_EN PERI0 200M Clock Auto Gate Enable 0: Auto 1: No-Auto

4.1.7.185 0x190C PERI1PLL Gate Enable Register (Default Value: 0x1FFF_1FFF)

Offset: 0x190C			Register Name: PERI1PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x190C			Register Name: PERI1PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x1	PERI1PLL2X_GATE_SW_CFG PERI1PLL2X Clock Gate Enable 0: Disable 1: Enable
27	R/W	0x1	PERI1_800M_GATE_SW_CFG PERI1 800M Clock Gate Enable 0: Disable 1: Enable
26	R/W	0x1	PERI1_600M_GATE_ALL_CFG PERI1 600M Clock Gate Enable 0: Disable 1: Enable
25	R/W	0x1	PERI1_600M_GATE_SW_CFG PERI1 600M Clock Gate Enable 0: Disable 1: Enable
24	R/W	0x1	PERI1_480M_GATE_ALL_CFG PERI1 480M Clock Gate Enable 0: Disable 1: Enable
23	R/W	0x1	PERI1_480M_GATE_SW_CFG PERI1 480M Clock Gate Enable 0: Disable 1: Enable
22	R/W	0x1	PERI1_160M_GATE_SW_CFG PERI1 160M Clock Gate Enable 0: Disable 1: Enable
21	R/W	0x1	PERI1_300M_GATE_ALL_CFG PERI1 300M Clock Gate Enable 0: Disable 1: Enable
20	R/W	0x1	PERI1_300M_GATE_SW_CFG PERI1 300M Clock Gate Enable 0: Disable 1: Enable
19	R/W	0x1	PERI1_150M_GATE_SW_CFG PERI1 150M Clock Gate Enable 0: Disable 1: Enable

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Offset: 0x190C			Register Name: PERI1PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x1	PERI1_400M_GATE_ALL_CFG PERI1 400M Clock Gate Enable 0: Disable 1: Enable
17	R/W	0x1	PERI1_400M_GATE_SW_CFG PERI1 400M Clock Gate Enable 0: Disable 1: Enable
16	R/W	0x1	PERI1_200M_GATE_SW_CFG PERI1 200M Clock Gate Enable 0: Disable 1: Enable
15:13	/	/	/
12	R/W	0x1	PERI1PLL2X_AUTO_GATE_EN PERI1PLL2X Clock Auto Gate Enable 0: Auto 1: No-Auto
11	R/W	0x1	PERI1_800M_AUTO_GATE_EN PERI1 800M Clock Auto Gate Enable 0: Auto 1: No-Auto
10	R/W	0x1	PERI1_600M_AUTO_GATE_EN_ALL PERI1 600M Clock Auto Gate Enable 0: Auto 1: No-Auto
9	R/W	0x1	PERI1_600M_AUTO_GATE_EN PERI1 600M Clock Auto Gate Enable 0: Auto 1: No-Auto
8	R/W	0x1	PERI1_480M_AUTO_GATE_EN_ALL PERI1 480M Clock Auto Gate Enable 0: Auto 1: No-Auto
7	R/W	0x1	PERI1_480M_AUTO_GATE_EN PERI1 480M Clock Auto Gate Enable 0: Auto 1: No-Auto
6	R/W	0x1	PERI1_160M_AUTO_GATE_EN PERI1 160M Clock Auto Gate Enable 0: Auto 1: No-Auto

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Offset: 0x190C			Register Name: PERI1PLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
5	R/W	0x1	PERI1_300M_AUTO_GATE_EN_ALL PERI1 300M Clock Auto Gate Enable 0: Auto 1: No-Auto
4	R/W	0x1	PERI1_300M_AUTO_GATE_EN PERI1 300M Clock Auto Gate Enable 0: Auto 1: No-Auto
3	R/W	0x1	PERI1_150M_AUTO_GATE_EN PERI1 150M Clock Auto Gate Enable 0: Auto 1: No-Auto
2	R/W	0x1	PERI1_400M_AUTO_GATE_EN_ALL PERI1 400M Clock Auto Gate Enable All 0: Auto 1: No-Auto
1	R/W	0x1	PERI1_400M_AUTO_GATE_EN PERI1 400M Clock Auto Gate Enable 0: Auto 1: No-Auto
0	R/W	0x1	PERI1_200M_AUTO_GATE_EN PERI1 200M Clock Auto Gate Enable 0: Auto 1: No-Auto

4.1.7.186 0x1910 VIDEOPLL Gate Enable Register (Default Value: 0x0011_0011)

Offset: 0x1910			Register Name: VIDEOPLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20	R/W	0x1	VIDEO0PLL3X_GATE_SW_CFG VIDEO0PLL3X Clock Gate Enable 0: Disable 1: Enable
19:17	/	/	/
16	R/W	0x1	VIDEO0PLL4X_GATE_SW_CFG VIDEO0PLL4X Clock Gate Enable 0: Disable 1: Enable
15:5	/	/	/

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Offset: 0x1910			Register Name: VIDEOPLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x1	VIDEO0PLL3X_AUTO_GATE_EN VIDEO0PLL3X Clock Auto Gate Enable 0: Auto 1: No-Auto
3:1	/	/	/
0	R/W	0x1	VIDEO0PLL4X_AUTO_GATE_EN VIDEO0PLL4X Clock Auto Gate Enable 0: Auto 1: No-Auto

4.1.7.187 0x191C AUDIOPLL Gate Enable Register (Default Value: 0x0001_0001)

Offset: 0x191C			Register Name: AUDIOPLL_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x1	AUDIO0PLL_GATE_SW_CFG AUDIO0PLL Gate Enable 0: Disable 1: Enable
15:1	/	/	/
0	R/W	0x1	AUDIO0PLL_AUTO_GATE_EN AUDIO0PLL Clock Auto Gate Enable 0: Auto 1: No-Auto

4.1.7.188 0x1988 PERI0PLL Gate Status Register (Default Value: 0x0FFF_0000)

Offset: 0x1988			Register Name: PERI0PLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R	0x1	PERI0PLL2X_GATE_STAT PERI0PLL2X Clock Gate Status 0: Disable 1: Enable
26	R	0x1	PERI0_800M_GATE_STAT PERI0 800M Clock Gate Status 0: Disable 1: Enable

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Offset: 0x1988			Register Name: PERI0PLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
25	R	0x1	PERI0_600M_GATE_STAT PERI0 600M Clock Gate Status 0: Disable 1: Enable
24	R	0x1	PERI0_480M_GATE_ALL_STAT PERI0 480M Clock Gate Status 0: Disable 1: Enable
23	R	0x1	PERI0_480M_GATE_STAT PERI0 480M Clock Gate Status 0: Disable 1: Enable
22	R	0x1	PERI0_160M_GATE_STAT PERI0 160M Clock Gate Status 0: Disable 1: Enable
21	R	0x1	PERI0_300M_GATE_ALL_STAT PERI0 300M Clock Gate Status 0: Disable 1: Enable
20	R	0x1	PERI0_300M_GATE_STAT PERI0 300M Clock Gate Status 0: Disable 1: Enable
19	R	0x1	PERI0_150M_GATE_STAT PERI0 150M Clock Gate Status 0: Disable 1: Enable
18	R	0x1	PERI0_400M_GATE_ALL_STAT PERI0 400M Clock Gate Status 0: Disable 1: Enable
17	R	0x1	PERI0_400M_GATE_STAT PERI0 400M Clock Gate Status 0: Disable 1: Enable
16	R	0x1	PERI0_200M_GATE_STAT PERI0 200M Clock Gate Status 0: Disable 1: Enable
15:0	/	/	/

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4.1.7.189 0x198C PERI1PLL Gate Status Register (Default Value: 0x1FFF_0000)

Offset: 0x198C			Register Name: PERI1PLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R	0x1	PERI1PLL2X_GATE_STAT PERI1PLL2X Clock Gate Status 0: Disable 1: Enable
27	R	0x1	PERI1_800M_GATE_STAT PERI1 800M Clock Gate Status 0: Disable 1: Enable
26	R	0x1	PERI1_600M_GATE_ALL_STAT PERI1 600M Clock Gate Status 0: Disable 1: Enable
25	R	0x1	PERI1_600M_GATE_STAT PERI1 600M Clock Gate Status 0: Disable 1: Enable
24	R	0x1	PERI1_480M_GATE_ALL_STAT PERI1 480M Clock Gate Status 0: Disable 1: Enable
23	R	0x1	PERI1_480M_GATE_STAT PERI1 480M Clock Gate Status 0: Disable 1: Enable
22	R	0x1	PERI1_160M_GATE_STAT PERI1 160M Clock Gate Status 0: Disable 1: Enable
21	R	0x1	PERI1_300M_GATE_ALL_STAT PERI1 300M Clock Gate Status 0: Disable 1: Enable
20	R	0x1	PERI1_300M_GATE_STAT PERI1 300M Clock Gate Status 0: Disable 1: Enable

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Offset: 0x198C			Register Name: PERI1PLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
19	R	0x1	PERI1_150M_GATE_STAT PERI1 150M Clock Gate Status 0: Disable 1: Enable
18	R	0x1	PERI1_400M_GATE_ALL_STAT PERI1 400M Clock Gate Status 0: Disable 1: Enable
17	R	0x1	PERI1_400M_GATE_STAT PERI1 400M Clock Gate Status 0: Disable 1: Enable
16	R	0x1	PERI1_200M_GATE_STAT PERI1 200M Clock Gate Status 0: Disable 1: Enable
15:0	/	/	/

4.1.7.190 0x1990 VIDEOPLL Gate Status Register (Default Value: 0x0011_0000)

Offset: 0x1990			Register Name: VIDEOPLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20	R	0x1	VIDEO0PLL3X_GATE_STAT VIDEO0PLL3X Clock Gate Status 0: Disable 1: Enable
19:17	/	/	/
16	R	0x1	VIDEO0PLL4X_GATE_STAT VIDEO0PLL4X Clock Gate Status 0: Disable 1: Enable
15:0	/	/	/

4.1.7.191 0x199C AUDIOPLL Gate Status Register (Default Value: 0x0001_0000)

Offset: 0x199C			Register Name: AUDIOPLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x199C			Register Name: AUDIOPLL_GATE_STAT_REG
Bit	Read/Write	Default/Hex	Description
16	R	0x1	AUDIO0PLL_GATE_STAT AUDIO0PLL Clock Gate Status 0: Disable 1: Enable
15:0	/	/	/

4.1.7.192 0x1A20 PLL Output Gate Bypass Register (Default Value: 0x0000_0000)

Offset: 0x1A20			Register Name: PLL_OPG_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	PLL_OUTPUT_GATE_BYPASS Control pll_output_gate module bypass 0: Disable 1: Enable

4.1.7.193 0x1C00 AXI MON Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1C00			Register Name: AXI_MON_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	RV_AXIMON_RST_N RISCV AXIMON Reset 0: Assert 1: DE-assert
15:1	/	/	/
0	R/W	0x0	RV_AXIMON_CLK_EN RISCV AXIMON All Clock Gating Control 0: Mask 1: Pass

4.1.7.194 0x1C04 AHB MON Gating and Reset Register (Default Value: 0x0000_0000)

Offset: 0x1C04			Register Name: AHB_MON_GAR_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	DCU_AHBMON_RST_N DCU AHBMON Reset 0: Assert 1: DE-assert

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Offset: 0x1C04			Register Name: AHB_MON_GAR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	CPU_SYS_AHBMON_RST_N CPU_SYS AHBMON Reset 0: Assert 1: DE-assert
15:2	/	/	/
1	R/W	0x0	DCU_AHBMON_CLK_EN DCU AHBMON All Clock Gating Control 0: Mask 1: Pass
0	R/W	0x0	CPU_SYS_AHBMON_CLK_EN CPU_SYS AHBMON All Clock Gating Control 0: Mask 1: Pass

4.1.7.195 0x1F00 CCU Security Switch Register (Default Value: 0x0000_0000)

Offset: 0x1F00			Register Name: CCU_SEC_SWITCH_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	MBUS_SEC MBUS clock register security 0: Secure 1: Non-secure
1	R/W	0x0	BUS_SEC Bus relevant registers security 0: Secure 1: Non-secure
0	R/W	0x0	PLL_SEC PLL relevant registers security 0: Secure 1: Non-secure

4.1.7.196 0x1F10 DAP REQ Control Register (Default Value: 0x0000_0001)

Offset: 0x1F10			Register Name: DAP_REQ_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/

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Offset: 0x1F10			Register Name: DAP_REQ_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	DAP_REQ_ENABLE 0: Disable 1: Enable Note: Enable sysdap set req to control CE and DAPSYS clock and reset

4.1.7.197 0x1F20 PLL Configuration0 Register (Default Value: 0x0000_0000)

Offset: 0x1F20			Register Name: PLL_CFG0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PLL_CONFIG0 PLL Control Reserved Register

4.1.7.198 0x1F24 PLL Configuration1 Register (Default Value: 0x0000_0000)

Offset: 0x1F24			Register Name: PLL_CFG1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PLL_CONFIG1 PLL Control Reserved Register

4.1.7.199 0x1F28 PLL Configuration2 Register (Default Value: 0x0000_0000)

Offset: 0x1F28			Register Name: PLL_CFG2_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PLL_CONFIG2 PLL Control Reserved Register

4.1.7.200 0x1F30 CCU FANOUT CLOCK GATE Register (Default Value:0x0000_0000)

Offset: 0x1F30			Register Name: CCU_FAN_GATE_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4	R/W	0x0	CLK50M_EN Gating for CLK50M 0: Clock is OFF 1: Clock is ON

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Offset: 0x1F30			Register Name: CCU_FAN_GATE_REG
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	CLK25M_EN Gating for CLK25M 0: Clock is OFF 1: Clock is ON
2	R/W	0x0	CLK16M_EN Gating for CLK16M 0: Clock is OFF 1: Clock is ON
1	R/W	0x0	CLK12M_EN Gating for CLK12M 0: Clock is OFF 1: Clock is ON
0	R/W	0x0	CLK24M_EN Gating for CLK24M 0: Clock is OFF 1: Clock is ON

4.1.7.201 0x1F34 CLK27M FANOUT Register (Default Value:0x0000_0000)

Offset: 0x1F34			Register Name: CLK27M_FAN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CLK27M_EN Gating for CLK27M 0: Clock is OFF 1: Clock is ON SCLK=Clock Source/M/N
30:26	/	/	/
25:24	R/W	0x0	CLK27M_SCR_SEL Clock Source Select 000: VIDEO0PLL4X 001: / 010: /
23:13	/	/	/
12:8	R/W	0x0	CLK27M_DIV1 Factor N N= FACTOR_N +1 FACTOR_M is from 0 to 31.
7:5	/	/	/

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Offset: 0x1F34			Register Name: CLK27M_FAN_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	CLK27M_DIV0 Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.202 0x1F38 CLK FANOUT Register (Default Value:0x0000_0000)

Offset: 0x1F38			Register Name: CLK_FAN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PCLK_DIV_EN Gating for PCLK 0: Clock is OFF 1: Clock is ON PCLK = APB0_CLK/M/N
30:10	/	/	/
9:5	R/W	0x0	PCLK_DIV1 Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
4:0	R/W	0x0	PCLK_DIV Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.

4.1.7.203 0x1F3C CCU FANOUT Register (Default Value:0x0000_0007)

Offset: 0x1F3C			Register Name: CCU_FAN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	FANOUT_RC_16M_CLK_SEL RC_16M_CLK Source Select 0: From PERI0_160M/10 1: From RC_16M
30:24	/	/	/
23	R/W	0x0	FANOUT_CLK2_EN Gating for FANOUT_CLK2 0: Clock is OFF 1: Clock is ON
22	R/W	0x0	FANOUT_CLK1_EN Gating for FANOUT_CLK1 0: Clock is OFF 1: Clock is ON

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Offset: 0x1F3C			Register Name: CCU_FAN_REG
Bit	Read/Write	Default/Hex	Description
21	R/W	0x0	FANOUT_CLK0_EN Gating for FANOUT_CLK0 0: Clock is OFF 1: Clock is ON
20:9	/	/	/
8:6	R/W	0x0	FANOUT_CLK2_SEL Clock Fanout2 Select 000: FANOUT_32K_CLK (From SYSRTC) 001: CLK12M (From DCXO/2) 010: CLK16M (From PERI0_160M/10) 011: CLK24M (From DCXO) 100: CLK25M (From PERI0_150M/6) 101: CLK27M 110: PCLK 111: CLK50M (From PERI0_150M/3) FANOUT_CLK2 can output from the above eight sources.
5:3	R/W	0x0	FANOUT_CLK1_SEL Clock Fanout1 Select 000: SYS_32K_CLK_FANOUT (From SYSRTC) 001: CLK12M (From DCXO/2) 010: CLK16M (From PERI0_160M/10) 011: CLK24M (From DCXO) 100: CLK25M (From PERI0_150M/6) 101: CLK27M 110: PCLK 111: CLK50M (From PERI0_150M/3) FANOUT_CLK1 can output from the above eight sources.
2:0	R/W	0x7	FANOUT_CLK0_SEL Clock Fanout0 Select 000: SYS_32K_CLK_FANOUT (From SYSRTC) 001: CLK12M (From DCXO/2) 010: CLK16M (From PERI0_160M/10) 011: CLK24M (From DCXO) 100: CLK25M (From PERI0_150M/6) 101: CLK27M 110: PCLK 111: CLK50M (From PERI0_150M/3) FANOUT_CLK0 can output from the above eight sources.

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4.1.7.204 0x1F50 Clock Debug Register (Default Value: 0x0000_0000)

Offset: 0x1F50			Register Name: CLK_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x0	MDL_CLK_DBG_DIV Module Clock Debug divide 00: DIV1 01: DIV2 10: DIV4 11: DIV8
23:19	/	/	/
18:16	R/W	0x0	MDL_CLK_DBG_SEL Module Clock Debug select 000: PRCM_BACKDOOR_CLK 001:/ 010: CPU_DEBUG_CLK
15:9	/	/	/
8:4	R/W	0x0	PSR_CLK_DBG_SEL Psensor Clock Debug select Select psensor test clock 0 to 19 0_0000: cpu_clu0_psensor_clk [0] 0_0001: cpu_clu0_psensor_clk [1] 0_0010: cpu_clu0_psensor_clk [2] 0_0011: cpu_clu0_psensor_clk [3] 0_0100:/ 0_0101:/ 0_0110:/ 0_0111:/ 0_1000:/ 0_1001:/ 0_1010:/ 0_1011:/ 0_1100:/ 0_1101:/ 0_1110:/ 0_1111:/ 1_0000:/ 1_0001:/ 1_0010:/ else:/
3	/	/	/

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Offset: 0x1F50			Register Name: CLK_DBG_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	BUS_CLK_DBG_SEL Bus Clock Debug select 000: AHB_CLK 001: APB0_CLK 010: APB1_CLK 011: APB_UART_CLK 100: MBUS_CLK 101: CE_SYS_CLK 110: /

4.1.7.205 0x1F60 Frequency Detect Control Register (Default Value: 0x0000_0020)

Offset: 0x1F60			Register Name: FRE_DET_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W0C	0x0	ERROR_FLAG Error Flag 0: Write 0 to clear 1: Error
30:9	/	/	/
8:4	R/W	0x2	DET_TIME Detect Time Time=1/32k*(2^RegValue) Note: RegValue is form 0 to 16.
3:2	/	/	/
1	R/W	0x0	FRE_DET_IRQ_EN Frequency Detect IRQ Enable 0: Disable 1: Enable
0	R/W	0x0	FRE_DET_FUN_EN Frequency Detect Function Enable 0: Disable 1: Enable

4.1.7.206 0x1F64 Frequency Up Limit Register (Default Value: 0x0000_0000)

Offset: 0x1F64			Register Name: FRE_UP_LIM_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	FRE_UP_LIM Frequency Up Limit The value must be an integer multiple of 32. Unit: kHz

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4.1.7.207 0x1F68 Frequency Down Limit Register (Default Value: 0x0000_0000)

Offset: 0x1F68			Register Name: FRE_DOWN_LIM_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	FRE_DOWN_LIM Frequency Down Limit The value must be an integer multiple of 32. Unit: kHz

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4.2 Power Reset Clock Management (PRCM)

4.2.1 Overview

The Power Reset Clock Management (PRCM) module provides a versatile supporting multiple power-management techniques and manages the gating and enabling of the clocks to the device modules in system.

The system-level reset management provides correct reset routing and sequencing when one or more devices are stacked together in the same package. The device-level reset management provides reset routing to relevant devices.

The PRCM has the following features:

- Module Clock Configuration
- APBS Clock Configuration
- Module BUS Gating and Reset
- System NMI interrupt configuration

4.2.2 External Signals

Table 4-3 PRCM External Signals

Signal Name	Description	Type
PLLTEST	PLL Test Signal	AO, OD

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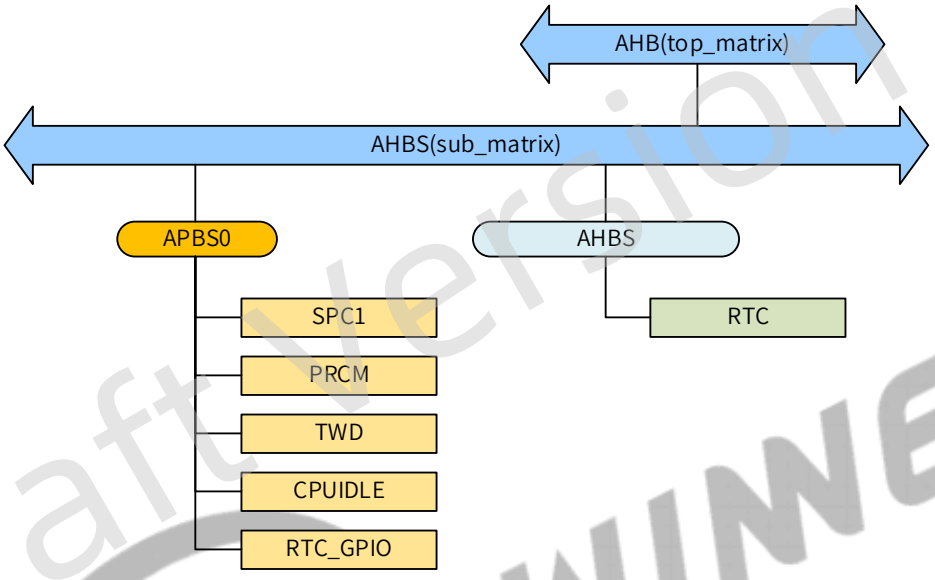
4.2.3 Functional Description

4.2.3.1 System Bus Tree

The following figures show the diagram of the System Bus Tree.

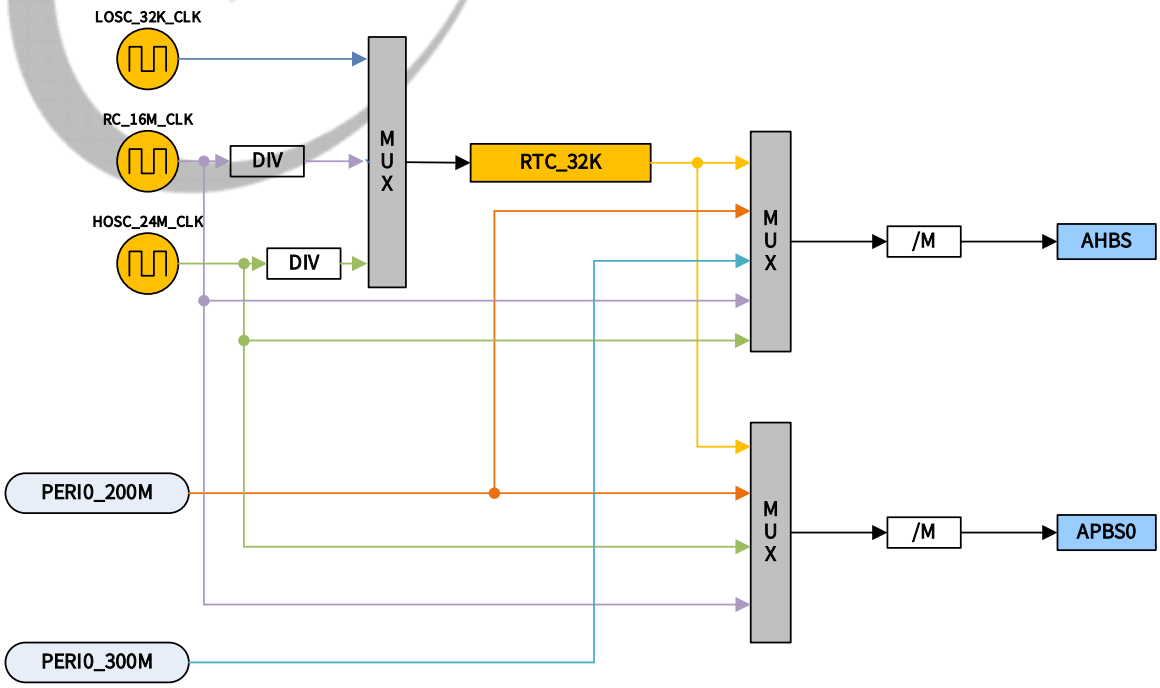
Figure 4-4 System Bus Tree



4.2.3.2 Bus Clock Tree

The following figures show a block diagram of the clock tree Diagram.

Figure 4-5 Bus Clock Tree



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4.2.4 Programming Guidelines

4.2.4.1 Configure Bus Clock

The bus clock supports dynamic switching, but the process of switching needs to follow the following two rules.

- From a higher frequency to a lower frequency: switch the clock source first, and then set the frequency division factor;
- From a lower frequency to a higher frequency: configure the frequency division factor first, and then switch the clock source.

The typical bus frequency for each bus is as follows:

- AHBS: 200 MHz
- APBS0: 100 MHz

4.2.4.2 Configure Module Clock

For the Bus Gating Reset register of a module, the reset bit is de-asserted first, and then the clock gating bit is enabled to avoid potential problems caused by the asynchronous release of the reset signal.

For all module clocks, configure the clock source and frequency division factor first, and then release the clock gating (that is, set to 1). For the configuration order of the clock source and frequency division factor, follow the rules below:

- With the increasing of the clock source frequency, configure the frequency division factor before the clock source.
- With the decreasing of the clock source frequency, configure the clock source before the frequency division factor.

4.2.4.3 System NMI Configuration

System NMI are detected in the PRCM by level changes on the NMI pin. You can select the interrupt type by configuring the NMI_INT_CTRL_REG register, open the interrupt detection function by configuring the NMI_INT_EN_REG register, read the interrupt status through the NMI_INT_PEND_REG register, and clear the interrupt status by writing 1 to NMI_INT_PEND_REG.

4.2.5 Register List

Module Name	Base Address
PRCM	0x07010000

Register Name	Offset	Description
AHBS_CLK_REG	0x0000	AHBS Clock Register

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Register Name	Offset	Description
APBS0_CLK_REG	0x000C	APBS0 Clock Register
TWD_BGR_REG	0x012C	TWD Bus Gating Reset Register
RTC_BGR_REG	0x020C	RTC Bus Gating Reset Register
CPUIDLE_BGR_REG	0x022C	CPUIDLE Bus Gating Reset Register
PRCM_PLL_CTRL_REG0	0x0240	PRCM PLL Control Register 0
PRCM_PLL_CTRL_REG1	0x0244	PRCM PLL Control Register 1
ANA_PWR_RST_REG	0x0254	ANALOG Power Off Gating Register
VDD_SYS_PWR_RST_REG	0x0260	VDD_SYS Power Domain Reset Register
RAMP2P_CFG_REG	0x0270	RAMP2P Configuration Register
RAM2P_CFG_REG	0x0274	RAM2P Configuration Register
RAMSP_CFG_REG	0x0278	RAMSP Configuration Register
ROM_CFG_REG	0x027C	ROM Configuration Register
PRCM_SEC_SWITCH_REG	0x0290	PRCM Security Switch Register
NMI_INT_CTRL_REG	0x0320	NMI Interrupt Control Register
NMI_INT_EN_REG	0x0324	NMI Interrupt Enable Register
NMI_INT_PEND_REG	0x0328	NMI Interrupt Pending Register
BUS_ACG_REG	0x033C	Bus Auto Clock Gating Register
AHBS_RDY_TOUT_CTRL_REG	0x0378	AHBS Ready Timeout Control Register

4.2.6 Register Description

4.2.6.1 0x0000 AHBS Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of PRCM_SEC_SWITCH_REG.

Offset: 0x0000			Register Name: AHBS_CLK_REG
Bit	R/W	Default/Hex	Description
31:27	/	/	/
26:24	R/W	0x0	AHBS_CLK_SRC_SEL AHBS Clock Source Select 000: DCXO 001: RTC_32K 010: RC16M 011: PERIPLL_DIV(200MHz) 100: PERI0_200M Others: / AHBS_HCLK = Clock Source/M
23:5	/	/	/

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Offset: 0x0000			Register Name: AHBS_CLK_REG
Bit	R/W	Default/Hex	Description
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31

4.2.6.2 0x000C APBS0 Clock Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of PRCM_SEC_SWITCH_REG.

Offset: 0x000C			Register Name: APBS0_CLK_REG
Bit	R/W	Default/Hex	Description
31:27	/	/	/
26:24	R/W	0x0	APBS0_CLK_SRC_SEL APBS0 Clock Source Select 000: DCXO 001: RTC_32K 010: RC16M 011: PERIPLL_DIV(200MHz) Others: / APBS0_CLK = Clock Source/M
23:5	/	/	/
4:0	R/W	0x0	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31

4.2.6.3 0x012C TWD Bus Gating Reset Register (Default Value: 0x0000_0001)

NOTE

This register is always secure.

Offset: 0x012C			Register Name: TWD_BGR_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/

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Offset: 0x012C			Register Name: TWD_BGR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	TWD_GATING Gating Clock for TWD 0: Mask 1: Pass

4.2.6.4 0x020C RTC Bus Gating Reset Register (Default Value: 0x0001_0001)

NOTE

This register is secure by default.

Offset: 0x020C			Register Name: RTC_BGR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x1	RTC_RST RTC Reset 0: Assert 1: De-assert
15:1	/	/	/
0	R/W	0x1	RTC_GATING Gating Clock for RTC 0: Mask 1: Pass

4.2.6.5 0x022C CPUIDLE Bus Gating Reset Register (Default Value: 0x0001_0001)

NOTE

This register is secure by default.

Offset: 0x022C			Register Name: CPUIDLE_BGR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x1	CPUIDLE_RST CPUIDLE Reset 0: Assert 1: De-assert
15:1	/	/	/

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Offset: 0x022C			Register Name: CPUIDLE_BGR_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	CPUIDLE_GATING Gating Clock for CPUIDLE 0: Mask 1: Pass

4.2.6.6 0x0240 PRCM PLL Control Register 0 (Default Value: 0x0000_0006)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of PRCM_SEC_SWITCH_REG.

Offset: 0x0240			Register Name: PRCM_PLL_CTRL_REG0
Bit	R/W	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	TEST_CLK_SEL PLL Reference Clock Select 0: DCXO 1: External Test Clock
23:3	/	/	/
2	R/W	0x1	GM1 XTAL Gain Control Bit1
1	R/W	0x1	GM0 XTAL Gain Control Bit0
0	/	/	/

4.2.6.7 0x0244 PRCM PLL Control Register 1 (Default Value: 0x0003_0005)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit1 of PRCM_SEC_SWITCH_REG.

Offset: 0x0244			Register Name: PRCM_PLL_CTRL_REG1
Bit	R/W	Default/Hex	Description
31:24	R/W	0x0	KEY_FIELD Key Field for LDO Enable bit If the key field value is 0xA7, the bit[23:0] can be modified.

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Offset: 0x0244			Register Name: PRCM_PLL_CTRL_REG1
Bit	R/W	Default/Hex	Description
23:19	/	/	/
18:16	R/W	0x3	PLLVDD_LDO_OUT_CTRL PLLVDD LDO Output Control 3'b000:0.9V 3'b001: 0.94V 3'b010: 0.98V 3'b011: 1.02V 3'b100: 1.06V 3'b101: 1.10V 3'b110: 1.14V 3'b111: 1.18V
15:5	/	/	/
4	R/W	0x0	MBIAS_EN Chip Master Bias Enable 0: From Internal Bias 1: From Audio Codec Bias
3	R/W	0x0	PLLTEST_EN PLLTEST Pin Output Enable For Verify (Back door clock PLLTEST enable) 0: Output clock is gated off. 1: Clock Output. The clock is the clock output to the PLL and the clock after frequency division through pins.
2	R/W	0x1	Reserved
1	/	/	/
0	R/W	0x1	LDO_EN LDO Enable 0: Disable 1: Enable Note: PLL Power enable (power source from AVCC).

4.2.6.8 0x0254 ANALOG Power Off Gating Register (Default Value: 0x0000_0001)

NOTE

This register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of PRCM_SEC_SWITCH_REG.

Offset: 0x0254			Register Name: ANA_PWR_RST_REG
Bit	R/W	Default/Hex	Description

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Offset: 0x0254			Register Name: ANA_PWR_RST_REG
Bit	R/W	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	PWROFF_ANA_EN_CPU Gating The CPU PLL ANA Module Input When CPUX_SYS Is Power Off. 0: Invalid, not gating 1: Valid, gating Note: When system is cold started, the value of this bit is 1 by default. Before using the related ANA module, this bit must be set to 0. This bit should be set to 1 during system standby. Related ANA module: CPUPLL
2	R/W	0x0	PWROFF_ANA_EN_DDR Gating The DDRPLL ANA Module Input When DRAMC Is Power Off. 0: Invalid, not gating 1: Valid, gating Note: When system is cold started, the value of this bit is 1 by default. Before using the related ANA module, this bit must be set to 0. This bit should be set to 1 during system standby. Related ANA module: DDRPLL
1	R/W	0x0	PWROFF_ANA_EN Gating The PLL ANA Module Input When VDD Is Power Off. 0: Invalid, not gating 1: Valid, gating Note: When system is cold started, the value of this bit is 0 by default; This bit should be set to 1 during system standby; After system exits standby mode, this bit must be set to 0 before using analog module. Related ANA module: PLL

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Offset: 0x0254			Register Name: ANA_PWR_RST_REG
Bit	R/W	Default/Hex	Description
0	R/W	0x1	SYS_ANA_VDDON_CTRL Gating The System Domain ANA Module Input When VDD Is Power Off. 0: Invalid, not gating 1: Valid, gating Note: When system is cold started, the value of this bit is 1 by default. Before using the related ANA module, this bit must be set to 0. This bit should be set to 1 during system standby. Relate ANA module: RES_CAL_DCAP/RES_TOP/USB0_USB1/CSI/LVDS/DSI/GPADC/ TPADC/AUDIO

4.2.6.9 0x0260 VDD_SYS Power Domain Reset Register (Default Value: 0x0000_0001)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit2 of PRCM_SEC_SWITCH_REG.

Offset: 0x0260			Register Name: VDD_SYS_PWR_RST_REG
Bit	R/W	Default/Hex	Description
31:1	/	/	/
0	R/W	0x1	MODULE_RST SYS Power Domain Modules should be reset before SYS power on. 0: Assert 1: De-assert

4.2.6.10 0x0270 RAMP2P Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0270			Register Name: RAMP2P_CFG_REG
Bit	R/W	Default/Hex	Description
31:0	R/W	0x0	RFP2P_MEM Rfp2p Configuration Information

4.2.6.11 0x0274 RAM2P Configuration Register (Default Value: 0x0000_2100)

Offset: 0x0274	Register Name: RAM2P_CFG_REG

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Bit	R/W	Default/Hex	Description
31:0	R/W	0x0000_2100	RF2P_MEM Rf2p Configuration Information

4.2.6.12 0x0278 RAMSP Configuration Register (Default Value: 0x0183_2C00)

Offset: 0x0278			Register Name: RAMSP_CFG_REG
Bit	R/W	Default/Hex	Description
31:0	R/W	0x0183_2C00	SPRAM_MEM SPRAM Configuration Information

4.2.6.13 0x027C ROM Configuration Register (Default Value: 0x0000_0002)

Offset: 0x027C			Register Name: ROM_CFG_REG
Bit	R/W	Default/Hex	Description
31:0	R/W	0x02	ROM_MEM ROM Configuration Information

4.2.6.14 0x0290 PRCM Security Switch Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is always secure and can only be accessed by secure masters. The security control is invalid in non-secure mode.

Offset: 0x0290			Register Name: PRCM_SEC_SWITCH_REG
Bit	R/W	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	POWER_SEC Security control for power relevant registers If the system is in non-secure mode, the related registers are always non-secure. 0: Secure 1: Non-secure Including: VDD_SYS power domain reset registers system domain power control registers

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Offset: 0x0290			Register Name: PRCM_SEC_SWITCH_REG
Bit	R/W	Default/Hex	Description
1	R/W	0x0	PLL_SEC Security control for PLL control relevant registers If the system is in non-secure mode, the related registers are always non-secure. 0: Secure 1: Non-secure Including: PLL control register 0/1.
0	R/W	0x0	SUB_SYS_BUS_CLK_SEC Security control for sub_sys bus clock relevant registers If the system is in non-secure mode, the related registers are always non-secure. 0: Secure 1: Non-secure Including: AHBS configuration register, APBS0 configuration register,

4.2.6.15 0x0320 NMI Interrupt Control Register (Default Value: 0x0000_0000)

Offset:0x0320			Register Name: NMI_INT_CTRL_REG
Bit	R/W	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	NMI_SRC_TYPE External NMI Interrupt Source Type External NMI pin can only output alarm signal if system is power down. (NMI_SRC_TYPE needs to be configured before NMI IRQ is enabled) 00: Low level sensitive 01: Negative edge trigged 10: High level sensitive 11: Positive edge sensitive

4.2.6.16 0x0324 NMI Interrupt Enable Register (Default Value: 0x8000_0000)

Offset:0x0324			Register Name: NMI_INT_EN_REG
Bit	R/W	Default/Hex	Description

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Offset:0x0324			Register Name: NMI_INT_EN_REG
Bit	R/W	Default/Hex	Description
31	R/W	0x1	NMI_INPUT_DET_EN NMI Input Detect Enable 0: Disable 1: Enable
30:1	/	/	/
0	R/W	0x0	NMIIRQ_SYS_EN Interrupt NMI Enable Bits 0: Interrupt is disabled. 1: Interrupt is enabled. This NMI interrupt is sent to CPUX

4.2.6.17 0x0328 NMI Interrupt Pending Register (Default Value: 0x0000_0000)

Offset:0x0328			Register Name: NMI_INT_PEND_REG
Bit	R/W	Default/Hex	Description
31:1	/	/	/
0	R/W1C	0x0	NMI_IRQ_PEND NMI IRQ Pending Interrupt NMI Pending/Clear Bit. Set 1 to the bit will clear it. (after first enabling NMI IRQ function, must set this bit to 1 to clear the unexpected pending once) 0: Corresponding interrupt is not pending. 1: Corresponding interrupt is pending

4.2.6.18 0x033C Bus Auto Clock Gating Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is secure by default and can only be accessed by secure masters. Security is configured in bit0 of PRCM_SEC_SWITCH_REG.

Offset: 0x033C			Register Name: BUS_ACG_REG
Bit	R/W	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	AHBS_AUTO_CLK_GATE_EN AHBS bus auto clock gating function enable 1: Enable auto clock gate 0: Disable auto clock gate, bus clock always on
23:1	/	/	/

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Offset: 0x033C			Register Name: BUS_ACG_REG
Bit	R/W	Default/Hex	Description
0	R/W	0x0	BM_ACG_MODE_EN BM Auto Clock Gating Mode Enable AHBS bus matrix module auto clock gating mode enable, if this bit is 1 and AHBS auto clock gating function is enable, AHBS bus matrix module hclk will close when AHBS bus clock close.

4.2.6.19 0x0378 AHBS Ready Timeout Control Register (Default Value: 0x0000_0000)

NOTE

In secure mode, this register is always secure by default and can only be accessed by secure masters.

Offset:0x0378			Register Name: AHBS_RDY_TOUT_CTRL_REG
Bit	R/W	Default/Hex	Description
31:25	/	/	/
24	R/W1C	0x0	AHBS_RDY_TIMEOUT_IRQ_PEND AHBS Hready Time Out IRQ Pending Write 1 Clear
23:9	/	/	/
8	R/W	0x0	AHBS_RDY_TIMEOUT_IRQ_EN AHBS Hready Time Out IRQ Enable 1: IRQ Enable 0: IRQ Disable
7	R/W	0x0	AHBS_TIMEOUT_INT_VAL_SEL AHBS Time Out Interval Value Select 0: interal_value* 2^10 + 2^10 -1 1: interal_value* 2^16 + 2^16 -1
6:1	R/W	0x0	AHBS_TIMEOUT_CNT_INT_VAL AHBS Time Out Counter Interval Value
0	R/W	0x0	AHBS_RDY_TIMEOUT_EN AHBS Hready Time Out Enable

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5 Processors

5.1 CPUX System

5.1.1 Overview

The CPU architecture of this device is based on ARMv7 and uses Cortex®-A7 as processor cores. A7 cores are reconstituted through the clusters of themselves and integrated in CPUX system in the end. The CPUX system consists of the following components:

- cpu_sys: system top, used to integrate GIC (see section 11.1 GIC), CoreSight, and other peripherals.
- ac327_subsystem: AC327 subsystem, containing cpu_pll registers, 4*A7, and other peripheral registers.

The configuration of each CPU and the cache size are showed below:

- Cluster is synchronous with core.
- Quad-core Cortex®-A7
  - CORE0:L1_DCACHE: 32KB, L1_ICACHE: 32KB
  - CORE1:L1_DCACHE: 32KB, L1_ICACHE: 32KB
  - CORE2:L1_DCACHE: 32KB, L1_ICACHE: 32KB
  - CORE3:L1_DCACHE: 32KB, L1_ICACHE: 32KB
  - L2_CACHE: 512KB

The CPUX SYS includes the following features:

- Full implementation of the ARMv7-A architecture instruction set.
- In-order pipeline with direct and indirect branch prediction.
- Harvard Level 1 (L1) memory system with a Memory Management Unit (MMU).
- Level 2 (L2) memory system.
- Supports Cortex®-A7 NEON MPE, it extends the Cortex®-A7 functionality to provide support for the ARMv7 Advanced SIMDv2 and Vector Floating-Pointv4 (VFPv4) instruction sets.
- Supports monitoring of CPUX read/write bandwidth and latency

The Cache includes the following features:

- L1 Instruction Cache
  - Instruction side cache line length of 32-bytes

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- Virtually indexed and physically tagged instruction cache.
- Pseudo random cache replacement policy
- 2-way set-associative instruction cache
- Export of memory attributes for external memory systems
- Support for Security Extensions
- Can be disabled independently, using the system control coprocessor.
- On a cache miss, critical word first filling of the cache is performed

- L1 Data Cache
  - Data side cache line length of 64-bytes.
  - Physically indexed and physically tagged data cache.
  - Pseudo random cache replacement policy.
  - 4-way set-associative data cache.
  - Two 32-byte linefill buffers and one 64-byte eviction buffer.
  - A 4-entry, 64-bit merging store buffer.
  - Can be disabled independently, using the system control coprocessor.
  - On a cache miss, critical word first filling of the cache is performed.
- L2 Cache
  - Fixed line length of 64 bytes
  - Physically indexed and tagged cache
  - 8-way set-associative cache structure
  - Pseudo-random cache replacement policy

The CPU_CFG includes the following features:

- CPU Reset system: core reset, debug circuit reset, and so on.
- Other control related to CPU: interface control, CP15 control, power-up/down control, and so on.
- CPU state inquiry: idle state, SMP state, interrupt state, and so on.
- CPU debug control and state register.

The TIMESTAMP **includes the following features:**

- The Timestamp uses a timing reference with a fixed frequency clock DCXO.
- The Timestamp is just as the clock source of the CPU local timer.

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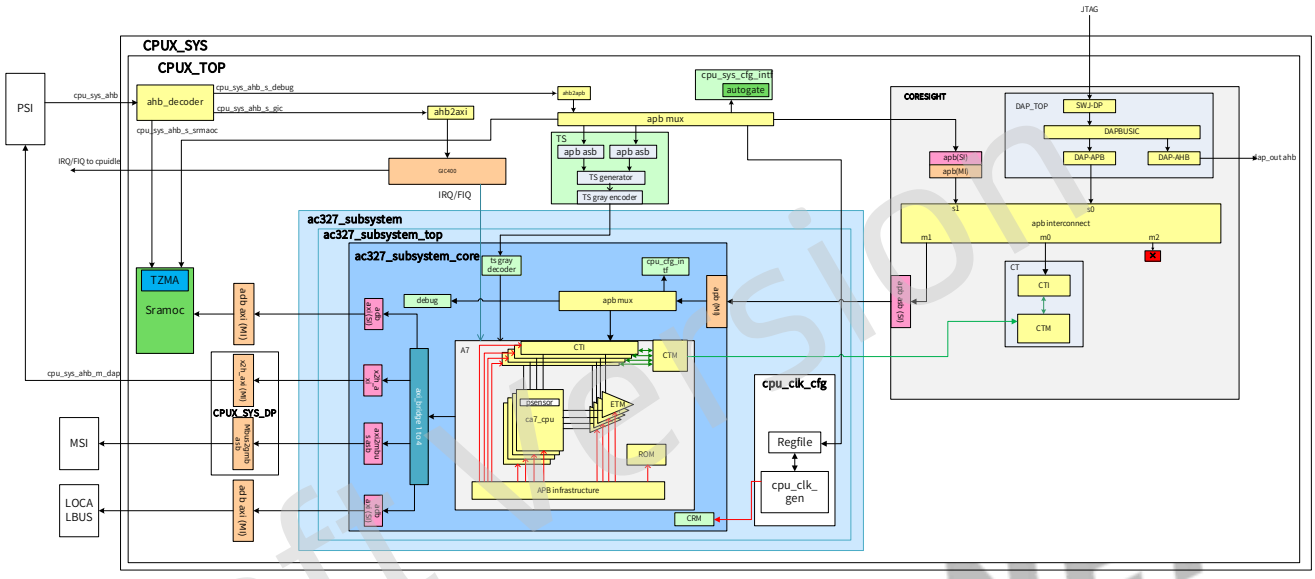
5.1.2 Block Diagram

The following figure shows the block diagram of CPUX system.

Figure 5-1 CPUX System Block Diagram



The following table describes the components of CPUX_SYS.

Table 5-1 CPUX_SYS Components

Sub-blocks	Description
ac327_subsystem	• Clusters composed of 4*A7 cores, L2 cache logic, components for debug, and so on. • APB bus for register configuration. • Asynchronous bridges for data transmission across power/clock domains.
cpux_sys_dp	Contains AHB/MSI asynchronous bridges for data transmission across power/clock domains.
CoreSight	• DAP for JTAG access. On-chip registers can be accessed and core operating status can be observed via off-chip debug tools. • APBIC, responsible for receiving access requests from DAP and AHB masters, and mapping them to the corresponding slaves based on address ranges. • Trigger-related logic including CTM and CTI, used to capture trigger information generated during core operating.
cpux_sys	• GIC module processing interrupt information. • Configuration registers for TZMA and SRAMOC.

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Sub-blocks	Description
SRAMOC	• Integrated 160 KB SRAMC supports simultaneous access by AXI and AHB bus, following the polling arbitration principle. • Integrated TZMA module supports security authentication for AXI and AHB access and supports isolation configuration based on master ID and address range.

5.1.3 Functional Descriptions

5.1.3.1 Signal Description

For details of A7 CPUX signals, refer to the technical reference manual of *DDI0464F_cortex_A7_mpcore_r0p5_trm.pdf*.

5.1.3.2 L2 Idle Mode

Enter the L2 idle mode by configuring the L2 cache of Cluster to enter the wait for interrupt (WFI) mode.

Follow the steps below:

Step 1 Make sure CPU [3:0] of Cluster has entered the WFI mode. You can check it through bit [19:16] of Cluster0 CPU Status Register.

Step 2 Pull high the ACINACTM of Cluster by writing 1 to the bit0 of Cluster 0 Control Register1.

Step 3 Check whether L2 enters idle status by checking whether the STANDBYWFIL2 is high.

Step 4 To exit the L2 idle mode, set the ACINACTM to low.

5.1.3.3 CPU Reset System

There are three levels of CPUX reset, with the scope size from small to large: core reset, power-on Reset, and H_Reset.

The following table describes all the reset signals in the CPUX reset system.

Table 5-2 Reset Signal Description

Reset Signal	Description
CORE_RST	This is the primary reset signal which resets the corresponding core logic, including NEON, VFP, Debug, ETM, breakpoint, and watchpoint logic. It maps to a warm reset that covers the reset of the processor logic.

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Reset Signal	Description
PWRON_RST	This power-on reset signal resets all the processor logic, including the Debug, ETM trace unit, breakpoint, watchpoint logic, and performance monitors logic. It maps to a cold reset that covers the reset of the processor logic and the integrated debug functionality. It does not reset debug logic in the debug power domain. PWRON_RST includes CORE_RST, ETM_RST, and DBG_RST.
H_RST	H_RST includes PWRON_RST, L2_RST, MBIST_RST, SOC_DBG_RST, and C0_CPUX_CFG.
AXI2MBUS_RST	This reset signal resets the AXI2MBUS interface logic circuit.
L2_RST	This single, cluster-wide signal resets the L2 memory system and the logic in the SCU.
ETM_RST	This reset signal resets ETM debug logic circuit.
DBG_RST	In the processor power domain, this reset signal resets only the debug, breakpoint, and watchpoint logic. In the debug power domain, it resets the debug logic for each processor.
SOC_DBG_RST	This reset signal resets all the debug logic. It includes DBG_RST.
MBIST_RST	This reset signal resets all resettable registers in the cluster, for entry into and exit from MBIST mode.
CPU_SUBSYS_RST	CPU_SUBSYS_RST includes C0_H_RST, GIC-400, and CPU_SUBSYS_CTRL.

5.1.3.4 TIMESTAMP

Timestamp generate and distribute a consistent time value to multiple processors and other IP in a SOC.

5.1.4 Register list

Module Name	Base Address	Description
CPUX_CS	0x08000000	CPUX CoreSight (4KB)
CPUX_SYS_CFG	0x08100000	CPUX System Control (4KB)
CPUX_TIMESTAMP_STA	0x08110000	Timestamp Status Register (4KB)
CPUX_TIMESTAMP_CTRL	0x08120000	Timestamp Control Registers (4KB)
CPUX_IDC_BASE	0x08130000	CPUX IDC Configure Register (4KB)
CPUX_PLL_CFG	0x08140000	CPUX PLL Configure Register(4KB)
CPUX_TZMA_CFG	0x08150000	CPUX TZMA Configure Register (4KB)
CPUX_CLU_CFG	0x09010000	Cluster0 CPUX configure (4KB)
CPUX_CLU_DEBUG	0x09400000	CPUX Cluster Debug Component Base Address (4MB)

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5.1.4.1 CPUX_SYS_CFG Register List

Module Name	Base Address	Description
CPUX_SYS_CFG	0x08100000	CPUX System Control (4KB)

Register Name	Offset	Description
GENER_CTRL_REG0	0x0000	General Control Register0
GIC_JTAG_RST_CTRL	0x000C	GIC and Jtag reset control Register
C0_INT_EN	0x0010	CLUSTER_0 Interrupt Enable Register
IRQ_FIQ_STATUS	0x0014	GIC IRQ/FIQ Status Register
GENER_CTRL_REG2	0x0018	General Control Register2
DBG_STATE	0x001C	Debug State Register
AUTOGATING_CTRL_REG	0x0020	Autogating Ctrl Register

5.1.4.2 CPUX_TIMESTAMP_STA Register List

Module Name	Base Address	Description
CPUX_TIMESTAMP_STA	0x08110000	Timestamp Status Register (4KB)

Register Name	Offset	Description
CNT_LOW_REG	0x0000	Counter Low Register
CNT_HI_REG	0x0004	Counter High Register

5.1.4.3 CPUX_TIMESTAMP_CTRL Register List

Module Name	Base Address	Description
CPUX_TIMESTAMP_CTRL	0x08120000	Timestamp Control Registers (4KB)

Register Name	Offset	Description
TSTAMP_CTRL_REG	0x0000	Timestamp Control Register
CNT_CTRL_LOW_REG	0x0008	Counter Low Register
CNT_CTRL_HI_REG	0x000C	Counter High Register
CNT_FREQID_REG	0x0020	Counter Base Frequency ID Register

5.1.4.4 CPUX_PLL_CFG Register List

Module Name	Base Address	Description
CPUX_PLL_CFG	0x08140000	CPUX PLL Configure Register(4KB)

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Register Name	Offset	Description
CLU_CLK_REG	0x0018	Cluster Clock Register
CLU_GATING_REG	0x001C	Cluster Gating Configuration Register
CLU_CLK_DIV_CFG_REG	0x0020	Cluster Clock Divide Configuration Register

5.1.4.5 CPUX_CLU_CFG Register List

Module Name	Base Address	Description
CPUX_CLU_CFG	0x09010000	Cluster0 CPUX configure (4KB)

Register Name	Offset	Description
C0_RST_CTRL	0x0000	Cluster 0 Reset Control Register
C0_CTRL_REG0	0x0010	Cluster 0 Control Register0
C0_CTRL_REG1	0x0014	Cluster 0 Control Register1
C0_CTRL_REG2	0x0018	Cluster 0 Control Register2
CLU0_ARM_SRAM_RAMCFG_REG	0x0020	Cluster 0 Arm Sram Ramcfg Register
CLU0_ARM_RF1P_RAMCFG_REG	0x0024	Cluster 0 Arm Rf1p Ramcfg Register
CLU0_FRD_SRAM_RAMCFG_REG	0x0028	Cluster 0 Frd Sram Ramcfg Register
CLU0_FRD_RF1P_RAMCFG_REG	0x002C	Cluster 0 Frd Rf1p ramcfg Register
CLU_CORE0_CTRL_REG	0x0060	Cluster Core0 Control Register
CLU_CORE1_CTRL_REG	0x0064	Cluster Core1 Control Register
CLU_CORE2_CTRL_REG	0x0068	Cluster Core2 Control Register
CLU_CORE3_CTRL_REG	0x006C	Cluster Core3 Control Register
C0_CPU_STATUS	0x0080	Cluster 0 CPU Status Register
L2_STATUS_REG	0x0084	L2 Status Register
DBG_REG0	0x00C0	Cluster 0 Debug Control Register0
AXI_MNT_CTRL_REG	0x00D0	CPU AXI MONITOR Control Register
AXI_MNT_PRD_REG	0x00D4	CPU AXI MONITOR Period Register
AXI_MNT_RLTCY_REG	0x00D8	CPU AXI MONITOR Read Total Latency Register
AXI_MNT_WLTCY_REG	0x00DC	CPU AXI MONITOR Write Total Latency Register
AXI_MNT_RREQ_REG	0x00E0	CPU AXI MONITOR Read Request times Register
AXI_MNT_WREQ_REG	0x00E4	CPU AXI MONITOR Write Request times Register
AXI_MNT_RBD_REG	0x00E8	CPU AXI MONITOR Read Bandwidth Register
AXI_MNT_WBD_REG	0x00EC	CPU AXI MONITOR Write Bandwidth Register

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5.1.5 CPUX_SYS_CFG Register Description

5.1.5.1 0x0000 General Control Register0 (Default Value:0x0000_0000)

Offset:0x0000			Register Name:GENER_CTRL_REG0
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	GIC_CFGSDISABLE. Disables write access to some secure GIC registers.

5.1.5.2 0x000C GIC and Jtag reset control Register (Default Value:0x0000_0F07)

Offset:0x000C			Register Name:GIC_JTAG_RST_CTRL
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	EXM_CLR Clear the status of interface. for debug. EXM_CLR[3:0]
15:12	/	/	/
11	R/W	0x1	CS_RST CoreSight Reset. 0: assert 1: de-assert.
10	R/W	0x1	DAP_RST DAP Reset. 0: assert 1: de-assert.
9	R/W	0x1	PORTRST Jtag Portrst. 0: assert 1: de-assert.
8	R/W	0x1	TRST. Jtag Trst. 0: assert 1: de-assert.
7:3	/	/	/
2	R/W	0x1	Reserved
1	R/W	0x1	IDC_RST Interrupt delay controller reset. 0: assert 1: de-assert.

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Offset:0x000C			Register Name:GIC_JTAG_RST_CTRL
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	GIC_RST. Gic_reset_cpu_reg 0: assert 1: de-assert.

5.1.5.3 0x0010 CLUSTER_0 Interrupt Enable Register (Default Value:0x0000_FFFF)

Offset:0x0010			Register Name:C0_INT_EN
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFFFF	C0_GIC_EN Interrupt Enable Control Register Mask irq_out/firq_out to system domain. Bit [3:0]: enable core0-core3 irq_out Bit [7:4]: enable core0-core3 firq_out Bit[15:8]: Reserved

5.1.5.4 0x0014 GIC IRQ/FIQ Status Register (Default Value:0x0000_FFFF)

Offset:0x0014			Register Name:IRQ_FIQ_STATUS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R	0xFF	FIQ_OUT FIQ Output signal
7:0	R	0xFF	IRQ_OUT IRQ Output signal

5.1.5.5 0x0018 General Control Register2 (Default Value:0x0000_0000)

Offset:0x0018			Register Name:GENER_CTRL_REG2
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	CDBGRSTACK Debug Reset ACK.
15:2	/	/	/
0	R/W	0x0	C0_TSCLKCHANGE Cluster 0 Time Stamp change bit.

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5.1.5.6 0x001C Debug State Register (Default Value:0x0000_0000)

Offset:0x001C			Register Name: DBG_STATE
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	CLU_PWRSW_STA Cluster 0 power switch status
23:1	/	/	/
0	R	0x0	C0_DBG_STATE Cluster 0 is in debug mode or normal mode

5.1.5.7 0x0020 Autogating Ctrl Register (Default Value:0x0000_0001)

Offset:0x0020			Register Name:AUTOGATING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:8	R/W	0x0	CPUX_SYS_CFG_LPCFG_PCYCLE Auto gating of CPUX_SYS_CFG_INTF delay shut down cycle If 0x0, 1 Cycle default.
1	R/W	0x0	CPUX_SYS_CFG_LPCFTG_EN Auto gating enable signal 0: Disable Autogating mode. 1: Enable Autogating mode.
0	R/W	0x1	CPUX_SYS_CFG_LPCFG_SW Force Bus Gating Software control signal for low power 0: Close clock gate. 1: Open clock gate.

5.1.6 CPUX_TIMESTAMP_STA Register Description

5.1.6.1 0x0000 Counter Low Register (Default Value:0x0000_0000)

Offset:0x0000			Register Name:CNT_LOW_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CNT_LOW_REG. Current value of counter[31:0]

5.1.6.2 0x0004 Counter High Register (Default Value:0x0000_0000)

Offset:0x0004			Register Name:CNT_HI_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x0004			Register Name:CNT_HI_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CNT_HI_REG. Current value of counter[64:32]

5.1.7 CPUX_TIMESTAMP_CTRL Register Description

5.1.7.1 0x0000 Timestamp Control Register (Default Value:0x0000_0001)

Offset:0x0000			Register Name: TSTAMP_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x1	EN. Timestamp Counter Enable. 0: Disable 1: Enable

5.1.7.2 0x0008 Counter Low Register (Default Value:0x0000_0000)

Offset:0x0008			Register Name:CNT_CTRL_LOW_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CNT_CTRL_LOW_REG. Current value of counter[31:0]

5.1.7.3 0x000C Counter High Register (Default Value:0x0000_0000)

Offset:0x000C			Register Name:CNT_CTRL_HI_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CNT_CTRL_HI_REG. Current value of counter[64:32]

5.1.7.4 0x0020 Counter Base Frequency ID Register (Default Value:0x0000_0000)

Offset:0x0020			Register Name:CNT_FREQID_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	FREQ. This register must be programmed to match the clock frequency of the timestamp generator, in ticks per second. For example, in our SoC for a 24 MHz clock, must program 0x16E3600.

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5.1.8 CPUX_PLL_CFG Register Description

5.1.8.1 0x0018 Cluster Clock Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: CLU_CLK_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:24	R/W	0x0	CPU_CLK_SEL. Clock Source Select. 000: HOSC 001: CLK32K 010: CLK16M_RC 011: CLU0_PLL/P 100: PERI0PLL2X 101: PERI0_600M CLU_CLK = Clock Source. The clock MUX uses glitch-free switch and supports dynamic configuration.
23:0	/	/	/

5.1.8.2 0x001C Cluster Gating Configuration Register (Default Value: 0x0000_0001)

Offset: 0x001C			Register Name: CLU_GATING_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	CPU_GATING_FIELD. CPU Gating Field. If CPU_GATING_FIELD == 16'h16AA, the bit [0] can be configured.
15:1	/	/	/
0	R/W	0x1	CLU_CLK_GATING. Gating Clock. 0: Clock is OFF 1: Clock is ON

5.1.8.3 0x0020 Cluster Clock Divide Configuration Register (Default Value: 0x0000_000D)

Offset: 0x0020			Register Name: CLU_CLK_DIV_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/

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Offset: 0x0020			Register Name: CLU_CLK_DIV_CFG_REG
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	CLU_PLL_OUT_EXT_DIVP. Factor P. 00:1 01:2 10:4 11:\ When the output clock is less than 480 MHz, it can divide P to get the required clock frequency. The clock DIV supports dynamic configuration.
15:4	/	/	/
3:2	R/W	0x3	CPU_APB_DIV_CFG. Factor N. (N = FACTOR_N +1) FACTOR_N is 3. apb_clk = CLU_CLK/N The clock DIV supports dynamic configuration.
1:0	R/W	0x1	CPU_AXI_DIV_CFG. Factor M:(M= FACTOR_M +1) FACTOR_M is from 1 to 3 aclk = CLU_CLK/M The clock DIV supports dynamic configuration.

5.1.9 CPUX_CLU_CFG Register Description

5.1.9.1 0x0000 Cluster 0 Reset Control Register (Default Value:0x1300_0100)

Offset:0x0000			Register Name:C0_RST_CTRL
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x1	DDR_RST AXI2MBUS logic circuit Reset. 0: assert 1: de-assert
27:26	/	/	/
25	R/W	0x1	MBIST_RST CPUBIST Reset. The reset signal for test. 0: assert 1: de-assert

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Offset:0x0000			Register Name:C0_RST_CTRL
Bit	Read/Write	Default/Hex	Description
24	R/W	0x1	SOC_DBG_RST. Cluster SOC Debug Reset 0: assert 1: de-assert.
23:9	/	/	/
8	R/W	0x1	L2_RST. Cluster L2 Cache Reset 0: assert 1: de-assert.
7:0	/	/	/

5.1.9.2 0x0010 Cluster 0 Control Register0 (Default Value:0x8000_0000)

Offset:0x0010			Register Name:C0_CTRL_REG0
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	SYSBAR_DISABLE. Disable broadcasting of barriers onto system bus: 0: Barriers are broadcast onto system bus; this requires an AMBA4 interconnect. 1: Barriers are not broadcast onto the system bus. This is compatible with an AXI3 interconnect.
30	R/W	0x0	BROADCAST_INNER. Enable broadcasting of Inner Shareable transactions: 0: Inner shareable transactions are not broadcasted externally. 1: Inner shareable transactions are broadcasted externally.
29	R/W	0x0	BROADCAST_OUTER. Enable broadcasting of outer shareable transactions: 0: Outer Shareable transactions are not broadcasted externally. 0: Outer Shareable transactions are broadcasted externally.

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Offset:0x0010			Register Name:C0_CTRL_REG0
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	BROADCAST_CACHE_MAINT Enable broadcasting of cache maintenance operations to downstream caches: 0: Cache maintenance operations are not broadcasted to downstream caches. 1: Cache maintenance operations are broadcasted to downstream caches.
27:5	/	/	/
4	R/W	0x0	L2_RST_DISABLE. Disable automatic L2 cache invalidate at reset: 0: L2 cache is reset by hardware. 1: L2 cache is not reset by hardware.
3:0	/	/	/

5.1.9.3 0x0014 Cluster 0 Control Register1 (Default Value:0x0000_0000)

Offset:0x0014			Register Name:C0_CTRL_REG1
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	ACINACTM. Snoop interface is inactive and no longer accepting requests. 0: Snoop interface is active 1: Snoop interface is inactive

5.1.9.4 0x0018 Cluster 0 Control Register2 (Default Value:0x0000_0000)

Offset:0x0018			Register Name:C0_CTRL_REG2
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	EVENTI Event input for processor wake up from WFE state. This bit must remain HIGH for at least one clock cycle to be visible by the cores.
23:0	/	/	/

5.1.9.5 0x0020 Cluster 0 Arm Sram Ramcfg Register (Default Value:0x0183_2C00)

Offset: 0x0020			Register Name: CLU0_ARM_SRAM_RAMCFG_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0020			Register Name: CLU0_ARM_SRAM_RAMCFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:0	R/W	0x1832c00	ARM_SRAM_RAMCFG Arm SRAM ramcfg configure.

5.1.9.6 0x0024 Cluster 0 Arm Rf1p Ramcfg Register (Default Value: 0x0183_2C00)

Offset: 0x0024			Register Name: CLU0_ARM_RF1P_RAMCFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:0	R/W	0x1832c00	ARM_RF1P_RAMCFG Arm rf1p ramcfg configure.

5.1.9.7 0x0028 Cluster 0 Frd Sram Ramcfg Register (Default Value:0x0000_0000)

Offset: 0x0028			Register Name: CLU0_FRD_SRAM_RAMCFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:0	R/W	0x0	FRD_SRAM_RAMCFG Frd SRAM ramcfg configure.

5.1.9.8 0x002C Cluster 0 Frd Rf1p ramcfg Register (Default Value:0x0000_0000)

Offset: 0x002C			Register Name: CLU0_FRD_RF1P_RAMCFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:0	R/W	0x0	FRD_RF1P_RAMCFG Frd rf1p ramcfg configure.

5.1.9.9 0x0060 Cluster Core0 Control Register (Default Value:0x0000_0107)

Offset:0x0060			Register Name:CLU_CORE0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R	0x0	CORE0_SMP_AMP CORE0 is in Symmetric Multiprocessing mode or Asymmetric Multiprocessing mode. 0: AMP mode 1: SMP mode

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Offset:0x0060			Register Name:CLU_CORE0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
17	R	0x0	CORE0_STANDBYWFI Indicates if Cluster CORE0 is in WFI standby mode: 0: Processor not in WFI standby mode. 1: Processor in WFI standby mode
16	R	0x0	CORE0_STANDBYWFE Indicates if Cluster CORE0 is in the WFE standby mode: 0: Processor not in WFE standby mode 1: Processor in WFE standby mode
15:14	/	/	/
13	R	0x0	CORE0_DBGNOPWRDWN No power-down request. Debugger has requested that processor is not powered down. Debug no power down.
12	R	0x0	CORE0_DBGPWRUPREQ Power up request. Debug power up request 0: Do not request that the core is powered up 1: Request that the core is powered up
11	/	/	/
10	R	0x0	CORE0_DBGRESTARTED Handshake for DBGRESTART.
9	R/W	0x0	CORE0_DBGRESTART External restart requests.
8	R/W	0x1	CORE0_DBGPWRDUP. Cluster Powered-up 0: Core is powered down 1: Core is powered up
7	/	/	/
6	R/W	0x0	CORE0_EXM_CLR Clear the status of interface.
5	R/W	0x0	CORE0_CP15S_DISABLE. Disable write access to some secure CP15 register.
4	R/W	0x0	CORE0_L1_RST_DISABLE. Disable automatic Cluster CORE0 L1 cache invalidate at reset: 0: L1 cache is reset by hardware. 1: L1 cache is not reset by hardware.
3	/	/	/

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Offset:0x0060			Register Name:CLU_CORE0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x1	CORE0_ETM_RST Cluster ETM Reset Assert. 0: assert 1: de-assert
1	R/W	0x1	CORE0_DBG_RST Cluster CORE0 Debug Reset Assert. 0: assert 1: de-assert
0	R/W	0x1	CORE0_CORE_RESET. Cluster CORE0 Reset Assert. 0: assert 1: de-assert.

5.1.9.10 0x0064 Cluster Core1 Control Register (Default Value:0x0002_0006)

Offset:0x0064			Register Name:CLU_CORE1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R	0x0	CORE1_SMP_AMP CORE1 is in Symmetric Multiprocessing mode or Asymmetric Multiprocessing mode. 0: AMP mode 1: SMP mode
17	R	0x1	CORE1_STANDBYWFI Indicates if Cluster CORE1 is in WFI standby mode: 0: Processor not in WFI standby mode. 1: Processor in WFI standby mode
16	R	0x0	CORE1_STANDBYWFE Indicates if Cluster CORE1 is in the WFE standby mode: 0: Processor not in WFE standby mode 1: Processor in WFE standby mode
15:14	/	/	/
13	R	0x0	CORE1_DBGNOPWRDWN. No power-down request. Debugger has requested that processor is not powered down. Debug no power down.

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Offset:0x0064			Register Name:CLU_CORE1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
12	R	0x0	CORE1_DBGPWRUPREQ. Power up request. Debug power up request 0: Do not request that the core is powered up 1: Request that the core is powered up
11	/	/	/
10	R	0x0	CORE1_DBGRESTARTED Handshake for DBGRESTART.
9	R/W	0x0	CORE1_DBGRESTART External restart requests.
8	R/W	0x0	CORE1_DBGPWRDUP. Cluster Powered-up 0: Core is powered down 1: Core is powered up
7	/	/	/
6	R/W	0x0	CORE1_EXM_CLR Clear the status of interface.
5	R/W	0x0	CORE1_CP15S_DISABLE. Disable write access to some secure CP15 register.
4	R/W	0x0	CORE1_L1_RST_DISABLE. Disable automatic Cluster CORE1 L1 cache invalidate at reset: 0: L1 cache is reset by hardware. 1: L1 cache is not reset by hardware.
3	/	/	/
2	R/W	0x1	CORE1_ETM_RST Cluster ETM Reset Assert. 0: assert 1: de-assert
1	R/W	0x1	CORE1_DBG_RST Cluster CORE1 Debug Reset Assert. 0: assert 1: de-assert
0	R/W	0x0	CORE1_CORE_RESET. Cluster CORE1 Reset Assert. 0: assert 1: de-assert.

5.1.9.11 0x0068 Cluster Core2 Control Register (Default Value:0x0002_0006)

Offset:0x0068	Register Name:CLU_CORE2_CTRL_REG

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Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R	0x0	CORE2_SMP_AMP CORE2 is in Symmetric Multiprocessing mode or Asymmetric Multiprocessing mode. 0: AMP mode 1: SMP mode
17	R	0x1	CORE2_STANDBYWFI Indicates if Cluster CORE2 is in WFI standby mode: 0: Processor not in WFI standby mode. 1: Processor in WFI standby mode
16	R	0x0	CORE2_STANDBYWFE Indicates if Cluster CORE2 is in the WFE standby mode: 0: Processor not in WFE standby mode 1: Processor in WFE standby mode
15:14	/	/	/
13	R	0x0	CORE2_DBGNOPWRDWN. No power-down request. Debugger has requested that processor is not powered down. Debug no power down.
12	R	0x0	CORE2_DBGPWRUPREQ. Power up request. Debug power up request 0: Do not request that the core is powered up 1: Request that the core is powered up
11	/	/	/
10	R	0x0	CORE2_DBGRESTARTED Handshake for DBGRESTART.
9	R/W	0x0	CORE2_DBGRESTART External restart requests.
8	R/W	0x0	CORE2_DBGPWRDUP. Cluster Powered-up 0: Core is powered down 1: Core is powered up
7	/	/	/
6	R/W	0x0	CORE2_EXM_CLR Clear the status of interface.
5	R/W	0x0	CORE2_CP15S_DISABLE. Disable write access to some secure CP15 register.

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4	R/W	0x0	CORE2_L1_RST_DISABLE. Disable automatic Cluster CORE2 L1 cache invalidate at reset: 0: L1 cache is reset by hardware. 1: L1 cache is not reset by hardware.
3	/	/	/
2	R/W	0x1	CORE2_ETM_RST Cluster ETM Reset Assert. 0: assert 1: de-assert
1	R/W	0x1	CORE2_DBG_RST Cluster CORE2 Debug Reset Assert. 0: assert 1: de-assert
0	R/W	0x0	CORE2_CORE_RESET. Cluster CORE2 Reset Assert. 0: assert 1: de-assert.

5.1.9.12 0x006C Cluster Core3 Control Register (Default Value:0x0002_0006)

Offset:0x006C			Register Name:CLU_CORE3_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R	0x0	CORE3_SMP_AMP CORE3 is in Symmetric Multiprocessing mode or Asymmetric Multiprocessing mode. 0: AMP mode 1: SMP mode
17	R	0x1	CORE3_STANDBYWFI Indicates if Cluster CORE3 is in WFI standby mode: 0: Processor not in WFI standby mode. 1: Processor in WFI standby mode
16	R	0x0	CORE3_STANDBYWFE Indicates if Cluster CORE3 is in the WFE standby mode: 0: Processor not in WFE standby mode 1: Processor in WFE standby mode
15:14	/	/	/
13	R	0x0	CORE3_DBGNOPWRDWN. No power-down request. Debugger has requested that processor is not powered down. Debug no power down.

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Offset:0x006C			Register Name:CLU_CORE3_CTRL_REG
Bit	Read/Write	Default/Hex	Description
12	R	0x0	CORE3_DBGPWRUPREQ. Power up request. Debug power up request 0: Do not request that the core is powered up 1: Request that the core is powered up
11	/	/	/
10	R	0x0	CORE3_DBGRESTARTED Handshake for DBGRESTART.
9	R/W	0x0	CORE3_DBGRESTART External restart requests.
8	R/W	0x0	CORE3_DBGPWRDUP. Cluster Powered-up 0: Core is powered down 1: Core is powered up
7	/	/	/
6	R/W	0x0	CORE3_EXM_CLR Clear the status of interface.
5	R/W	0x0	CORE3_CP15S_DISABLE. Disable write access to some secure CP15 register.
4	R/W	0x0	CORE3_L1_RST_DISABLE. Disable automatic Cluster CORE3 L1 cache invalidate at reset: 0: L1 cache is reset by hardware. 1: L1 cache is not reset by hardware.
3	/	/	/
2	R/W	0x1	CORE3_ETM_RST Cluster ETM Reset Assert. 0: assert 1: de-assert
1	R/W	0x1	CORE3_DBG_RST Cluster CORE3 Debug Reset Assert. 0: assert 1: de-assert
0	R/W	0x0	CORE3_CORE_RESET. Cluster CORE3 Reset Assert. 0: assert 1: de-assert.

5.1.9.13 0x0080 Cluster 0 CPU Status Register (Default Value:0x0000_0000)

Offset:0x0080	Register Name:C0_CPU_STATUS

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R	0x0	STANDBYWFIL2. L2 WFI standby mode. Indicates if the Cluster L2 memory system is in WFI standby mode. 0: Cluster L2 not in WFI standby mode 1: Cluster L2 in WFI standby mode

5.1.9.14 0x0084 L2 Status Register (Default Value:0x0000_0000)

Offset:0x0084			Register Name:L2_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
10	R	0x0	L2FLUSHDONE L2 Cache Flush has done.
9	R	0x0	EVENTO Event output. This bit is asserted HIGH for 3 clock cycles when any core in the cluster executes an SEV instruction.
8	R	0x0	CLREXMONACK Exclusive Monitor clear done ack.
7:0	/	/	/

5.1.9.15 0x00C0 Cluster 0 Debug Control Register0 (Default Value:0x0000_0000)

Offset:0x00C0			Register Name:DBG_REG0
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
16	R/W	0x0	DBGL1RSTDISABLE Do not invalidate L1 cache on reset.
15:0	/	/	/

5.1.9.16 0x00D0 CPU AXI MONITOR Control Register (Default Value:0x0000_0000)

Offset:0x00D0			Register Name:AXI_MNT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	WC	0x0	AXI_MONITOR_CLR AXI MONITOR function clear. 0: not affect 1: clear

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Offset:0x00D0			Register Name:AXI_MNT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	AXI_MONITOR_EN AXI MONITOR function enable. 0: not enable 1: enable

5.1.9.17 0x00D4 CPU AXI MONITOR Period Register (Default Value:0x0000_0000)

Offset:0x00D4			Register Name:AXI_MNT_PRD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PERIOD AXI MONITOR function monitoring period Unit is cycle.

5.1.9.18 0x00D8 CPU AXI MONITOR Read Total Latency Register (Default Value:0x0000_0000)

Offset:0x00D8			Register Name:AXI_MNT_RLTCY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LATENCY AXI MONITOR function READ channel total latency Unit is cycle.

5.1.9.19 0x00DC CPU AXI MONITOR Write Total Latency Register (Default Value:0x0000_0000)

Offset:0x00DC			Register Name:AXI_MNT_WLTCY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LATENCY AXI MONITOR function Write channel total latency Unit is cycle.

5.1.9.20 0x00E0 CPU AXI MONITOR Read Request Times Register (Default Value:0x0000_0000)

Offset:0x00E0			Register Name:AXI_MNT_RREQ_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TIMES AXI MONITOR function READ channel request total times.

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5.1.9.21 0x00E4 CPU AXI MONITOR Write Request Times Register (Default Value:0x0000_0000)

Offset:0x00E4			Register Name:AXI_MNT_WREQ_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TIMES AXI MONITOR function Write channel request total times.

5.1.9.22 0x00E8 CPU AXI MONITOR Read Bandwidth Register (Default Value:0x0000_0000)

Offset:0x00E8			Register Name:AXI_MNT_RBD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	BANDWIDTH AXI MONITOR function READ channel Bandwidth. Unit is KB.

5.1.9.23 0x00EC CPU AXI MONITOR Write Bandwidth Register (Default Value:0x0000_0000)

Offset:0x00EC			Register Name:AXI_MNT_WBD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	BANDWIDTH AXI_MONITOR function Write Channel Bandwidth. Unit is KB.

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5.2 MCU System

5.2.1 Overview

The MCU system includes RISC-V IP core and related peripheral devices (MCU_CFG, timestamp, watchdog, message box, timer, and so on), which are interconnected by RISCV BUS Matrix.

The MCU system has the following features:

- Integrated RISC-V E907 core
- The E907 start address can be configured via software
- Private TIMERSTAMP supports timing immediately after reset is released
- Private watchdog supports to reset the SoC system when the SoC system malfunctions
- Private message box supports communicating with other modules
- Supports private PMU check module
- Dynamic clock gating auto switch
- Bus hang detection and resume

NOTE

For CLIC description, see section 11.2 CLIC.

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5.2.2 Block Diagram

The following figure shows the block diagram of MCU system.

Figure 5-2 MCU System Block Diagram

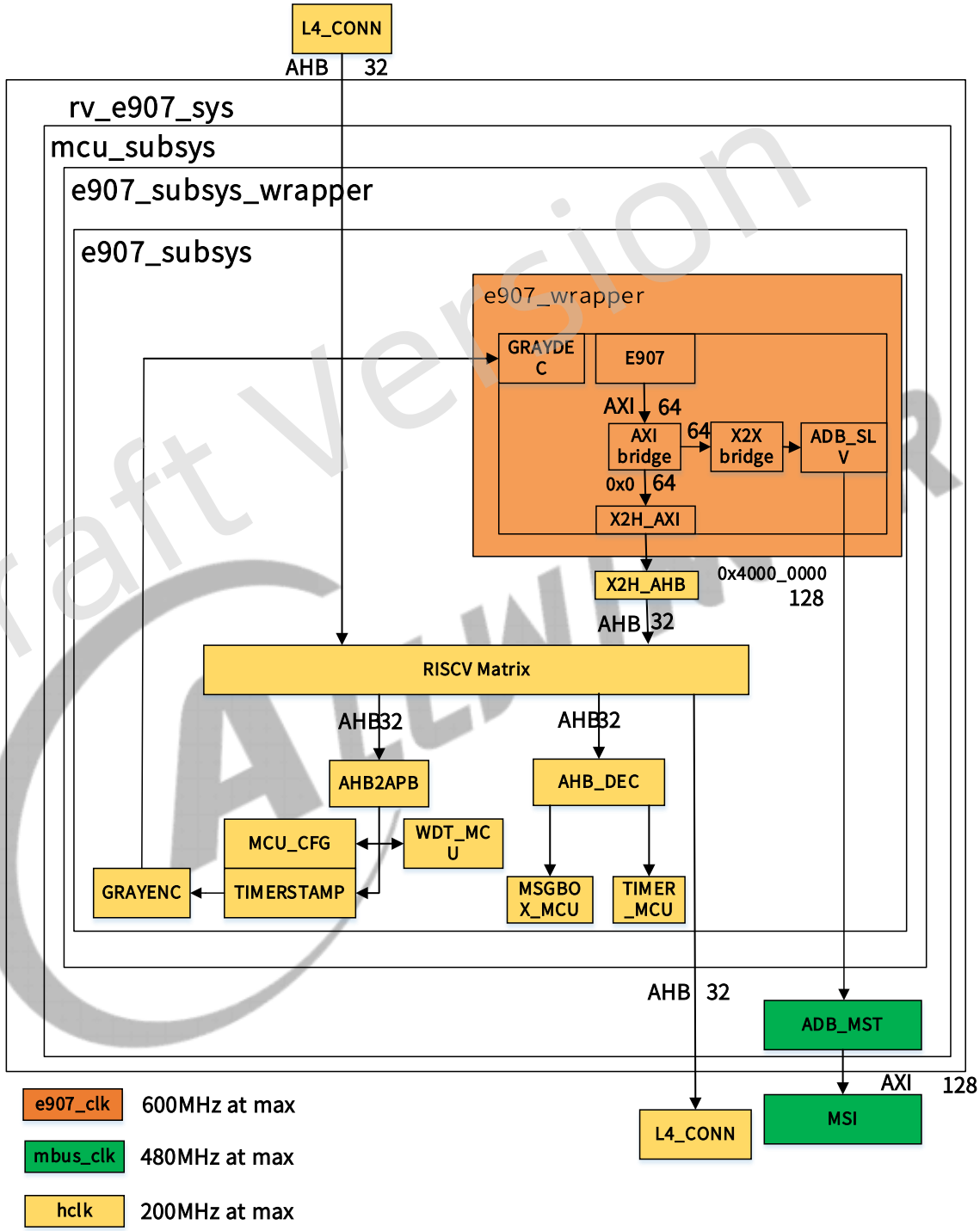

Table 5-3 MCU System Component

Sub-blocks	Description
rv_e907_sys	MCU system top

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Sub-blocks	Description
mcu_subsys	• **An AHB slave interface**: Used for system to access inner resources of MCU system. • **An AHB master interface**: Used for E907 to access external IO resources. • **An AXI master interface**: Connected to MSI bus. E907 can access DRAM space after bus protocol conversion.
e907_warapper	• **Single RISC-V E907 core** • **X2X Bridge**: Convert AXI64 into AXI128 and convert read WRAP instruction into INCR instruction. • **AXI Bridge**: Assign bus transactions to the appropriate address space. One route to IO space passes through X2H bridges, which convert AXI signals into AHB signals and connect them to system bus. The other leads to DRAM space. During an access, the signal passes through ADB400 asynchronous bridges and is then connected to MSI bus. The address of IO and DRAM space are as follows: IO space: 0x0000_0000-0x3FFF_FFFF: DRAM space: 0x4000_0000-0x23FFF_FFFF
RISCV Matrix	Connect the submodules in MCU system.
AHB2APB	Convert AHB signals into APB signals.
AHB_DEC	Decode AHB devices
MSGBOX_MCU	For communication between CPUX and MCU. See section 6.1 Message Box (MSGBOX)for details.
MCU_CFG	MCU system information configuration and read
TIMESTAMP	System 64-bit reference clock
TIMER_MCU	The timer for MCU system. See section 8.1 Timer for details.
WDT_MCU	The watchdog for MCU system. See section 8.2 Watchdog Timer (WDT) for details.
ADB_SLV ADB_MST	ADB400 asynchronous bridges, which are used for AXI signal synchronization.

5.2.3 Register list

Module Name	Base Address
MCU_CFG	0x01A00000

Register Name	Offset	Description
E907_DDR_REMAP_REG	0x0010	E907 DDR Remap Register
TS_TMODE_SEL_REG	0x0040	Timestamp Test Mode Select Register
E907_STA_ADD_REG	0x0204	E907 Start Address Register
E907_WAKEUP_AUTO_REG	0x0220	E907 Wakeup Auto Clock Gating Enable Register
E907_WAKEUP_IRQ_EN0_REG	0x0224	E907 Wakeup Interrupt Enable0 Register
E907_WAKEUP_IRQ_EN1_REG	0x0228	E907 Wakeup Interrupt Enable1 Register

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Register Name	Offset	Description
E907_WAKEUP_IRQ_EN2_REG	0x022C	E907 Wakeup Interrupt Enable2 Register
E907_WAKEUP_IRQ_EN3_REG	0x0230	E907 Wakeup Interrupt Enable3 Register
E907_WAKEUP_IRQ_EN4_REG	0x0234	E907 Wakeup Interrupt Enable4 Register
E907_WAKEUP_IRQ_EN5_REG	0x0238	E907 Wakeup Interrupt Enable5 Register
E907_WAKEUP_IRQ_EN6_REG	0x023C	E907 Wakeup Interrupt Enable6 Register
E907_WAKEUP_IRQ_EN7_REG	0x0240	E907 Wakeup Interrupt Enable7 Register
E907_WORK_MODE_REG	0x0248	E907 Work Mode Register
E907_AXI_PMU_CTRL	0x0304	E907 AXI PMU Control Register
E907_AXI_PMU_PRD	0x0308	E907 AXI PMU Period Register
E907_AXI_PMU_LAT_RD	0x030C	E907 AXI PMU Read Latency Register
E907_AXI_PMU_LAT_WR	0x0310	E907 AXI PMU Write Latency Register
E907_AXI_PMU_REQ_RD	0x0314	E907 AXI PMU Read Request Register
E907_AXI_PMU_REQ_WR	0x0318	E907 AXI PMU Write Request Register
E907_AXI_PMU_BW_RD	0x031C	E907 AXI PMU Read Bandwidth Register
E907_AXI_PMU_BW_WR	0x0320	E907 AXI PMU Write Bandwidth Register
RV_BUS_ACG_REG	0x0330	RV Bus Auto Clock Gating Register
E907_PAD_CPU_NMI_REG	0x0500	E907 PAD CPU NMI Control Register
CCU_CLOCK_GATE_EN_REG	0x0530	CCU Clock Gate Enable Register

5.2.4 Register Description

5.2.4.1 0x0010 E907 DDR Remap Register (Default Value: 0x0000_0000)

Offset:0x0010			Register Name: E907_DDR_REMAP_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2:0	R/W	0x0	E907_DDR_REMAP E907 Address Remap Control 000: DRAM Space 2G - 3G 001: DRAM Space 3G - 4G 010: DRAM Space 4G - 5G 011: DRAM Space 5G - 6G 100: DRAM Space 6G - 7G 101: DRAM Space 7G - 8G Others： DRAM Space 2G - 3G

5.2.4.2 0x0040 Timestamp Test Mode Select Register (Default Value: 0x0000_0000)

Offset:0x0040			Register Name: TS_TMODE_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/

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Offset:0x0040			Register Name: TS_TMODE_SEL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	E907_TS_TEST_MODE_EN E907 Timestamp Test Mode Enable 0: Normal Mode 1: Test Mode In Test Mode, this Counter Low/Hi registers will count simultaneously.
0	/	/	/

5.2.4.3 0x0204 E907 Start Address Register (Default Value: 0x0000_0000)

This register is the running PC address after RISCV releases reset. Before releasing reset, this register should be configured. This register does not support dynamic configuration.

Offset:0x0204			Register Name: E907_STA_ADD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	STA_ADD Start Address The bit 0 is fixed as 0 and could not be written.

5.2.4.4 0x0220 E907 Wakeup Auto Clock Gating Enable Register (Default Value: 0x0000_0000)

Offset:0x0220			Register Name: E907_WAKEUP_AUTO_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	WP_ATUO_CLK_GATE_EN E907 CORE Auto Clock Gating Enable 1: Enable auto clock gate 0: Disable auto clock gate When the clock auto-gating function is enabled, the E907 automatically disable the E907 CORE working clock after entering low power consumption mode. Wake-up interrupt can enable the E907 CORE working clock.

5.2.4.5 0x0224 E907 Wakeup Interrupt Enable0 Register (Default Value: 0x0000_0000)

The 0x0224 to 0x0240 registers corresponds to the wakeup enable bits of 240 interrupts. These wakeup enable bits are disabled by default. When some interrupt needs to be waken-up, the corresponding bit needs to be set to 1. For detailed interrupt sources, please refer to Section 11.2 CLIC.

Offset:0x0224	Register Name: E907_WAKEUP_IRQ_EN0_REG

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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN0 Wakeup Interrupt Enable0

5.2.4.6 0x0228 E907 Wakeup Interrupt Enable1 Register (Default Value: 0x0000_0000)

Offset:0x0228			Register Name: E907_WAKEUP_IRQ_EN1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN1 Wakeup Interrupt Enable1

5.2.4.7 0x022C E907 Wakeup Interrupt Enable2 Register (Default Value: 0x0000_0000)

Offset:0x022C			Register Name: E907_WAKEUP_IRQ_EN2_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN2 Wakeup Interrupt Enable2

5.2.4.8 0x0230 E907 Wakeup Interrupt Enable3 Register (Default Value: 0x0000_0000)

Offset:0x0230			Register Name: E907_WAKEUP_IRQ_EN3_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN3 Wakeup Interrupt Enable3

5.2.4.9 0x0234 E907 Wakeup Interrupt Enable4 Register (Default Value: 0x0000_0000)

Offset:0x0234			Register Name: E907_WAKEUP_IRQ_EN4_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN4 Wakeup Interrupt Enable4

5.2.4.10 0x0238 E907 Wakeup Interrupt Enable5 Register (Default Value: 0x0000_0000)

Offset:0x0238			Register Name: E907_WAKEUP_IRQ_EN5_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN5 Wakeup Interrupt Enable5

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5.2.4.11 0x023C E907 Wakeup Interrupt Enable6 Register (Default Value: 0x0000_0000)

Offset:0x023C			Register Name: E907_WAKEUP_IRQ_EN6_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00000000	WP_IRQ_EN6 Wakeup Interrupt Enable6

5.2.4.12 0x0240 E907 Wakeup Interrupt Enable7 Register (Default Value: 0x0000_0000)

Offset:0x0240			Register Name: E907_WAKEUP_IRQ_EN7_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0000	WP_IRQ_EN7 Wakeup Interrupt Enable7

5.2.4.13 0x0248 E907 Work Mode Register (Default Value: 0x0000_0003)

Offset: 0x0248			Register Name: E907_WORK_MODE_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R	0x0	LOCKUP_STA Lockup Status 0: Not Lockup 1: Lockup If CPU has a lockup, CPU will stop accessing data and fetching, and clear the pipeline. Reset processor unit and debug unit to unlock CPU.
2	R	0x0	DM_STA Debug Mode Status 0: Normal Mode 1: Debug Mode
1:0	R	0x3	LP_STA Low Power Status 00: Deep Sleep Low Power Mode 01: Light Sleep Low Power Mode 10: Reserved 11: Normal Mode

5.2.4.14 0x0304 E907 AXI PMU Control Register (Default Value: 0x0000_0000)

Offset:0x0304			Register Name: E907_AXI_PMU_CTRL
Bit	Read/Write	Default/Hex	Description

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Offset:0x0304			Register Name: E907_AXI_PMU_CTRL
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	WC	0x0	PMU_CLR PMU Clear 0: No operation 1: PMU cleared
0	R/W	0x0	PMU_EN PMU Enable 0: PMU disabled 1: PMU Enabled

5.2.4.15 0x0308 E907 AXI PMU Period Register (Default Value: 0x0000_0000)

Offset:0x0308			Register Name: E907_AXI_PMU_PRD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PRD Period. AXI cycle base, unit:1 us or 1 ms suggested.

5.2.4.16 0x030C E907 AXI PMU Read Latency Register (Default Value: 0x0000_0000)

Offset:0x030C			Register Name: E907_AXI_PMU_LAT_RD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RD_LAT Read Latency

5.2.4.17 0x0310 E907 AXI PMU Write Latency Register (Default Value: 0x0000_0000)

Offset:0x0310			Register Name: E907_AXI_PMU_LAT_WR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	WR_LAT Write Latency

5.2.4.18 0x0314 E907 AXI PMU Read Request Register (Default Value: 0x0000_0000)

Offset:0x0314			Register Name: E907_AXI_PMU_REQ_RD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RD_REQ Read Request

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5.2.4.19 0x0318 E907 AXI PMU Write Request Register (Default Value: 0x0000_0000)

Offset:0x0318			Register Name: E907_AXI_PMU_REQ_WR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	WR_REQ Write Request

5.2.4.20 0x031C E907 AXI PMU Read Bandwidth Register (Default Value: 0x0000_0000)

Offset:0x031C			Register Name: E907_AXI_PMU_BW_RD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RD_BW Read Bandwidth Unit: KB

5.2.4.21 0x0320 E907 AXI PMU Write Bandwidth Register (Default Value: 0x0000_0000)

Offset:0x0320			Register Name: E907_AXI_PMU_BW_WR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	WR_BW Write Bandwidth Unit: KB

5.2.4.22 0x0330 RV Bus Auto Clock Gating Register (Default Value: 0x0000_0000)

Offset:0x0330			Register Name: RV_BUS_ACG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MCU_AHB_AUTO_CLK_GATE_EN MCU AHB bus auto clock gating function enable 1: Enable auto clock gate 0: Disable auto clock gate
30:1	/	/	/
0	R/W	0x0	BM_ACG_MODE_EN BM Auto Clock Gating Mode Enable RISCV bus matrix module auto gating mode enable, if this bit is 1 and RISCV AHB auto clock gating function is enable, RISCV bus matrix module hclk will close when RISCV bus clock close.

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5.2.4.23 0x0500 E907 PAD CPU NMI Control Register (Default Value: 0x0000_0000)

Offset:0x0500			Register Name: E907_PAD_CPU_NMI_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	E907_PAD_CPU_NMI Non-Maskable Interrupt Enable Rising edge is valid.

5.2.4.24 0x0530 CCU Clock Gate Enable Register (Default Value: 0x0000_0000)

Offset:0x0530			Register Name: CCU_CLOCK_GATE_EN_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CCU_CLOCK_GATE_EN. CCU gating clock when E907 is in WFX mode.

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6 Interprocessor Communication

6.1 Message Box (MSGBOX)

6.1.1 Overview

The Message Box (MSGBOX) provides interrupt communication mechanism for microprocessors.

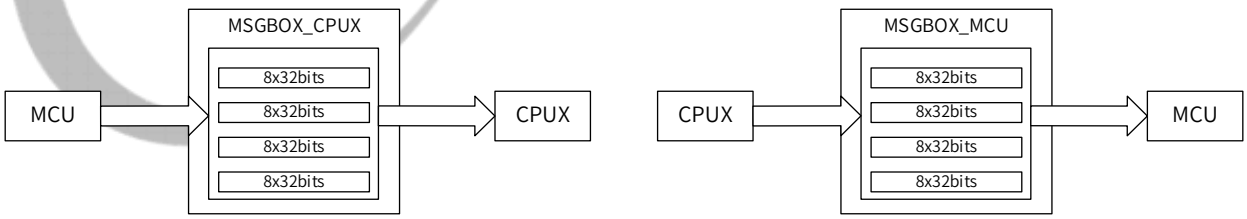
The MSGBOX has the following features:

- Up to six message boxes
  - Two message boxes for communication between CPUX system and MCU system through one way channels.
  - Four separate message boxes for priority adjustment in AMP mode among CPUX cores through one way channels
  - Each MSGBOX has one receiver and one or three transmitters
- 4 channels between every 2 users (the receiver or transmitter of one message box)
- The FIFO depth of a channel is 8
- The FIFO width of a channel is 32 bits
- Supports interrupts

6.1.2 Block Diagram

The following figure shows the block diagram of the message box.

Figure 6-1 Message Box Block Diagram



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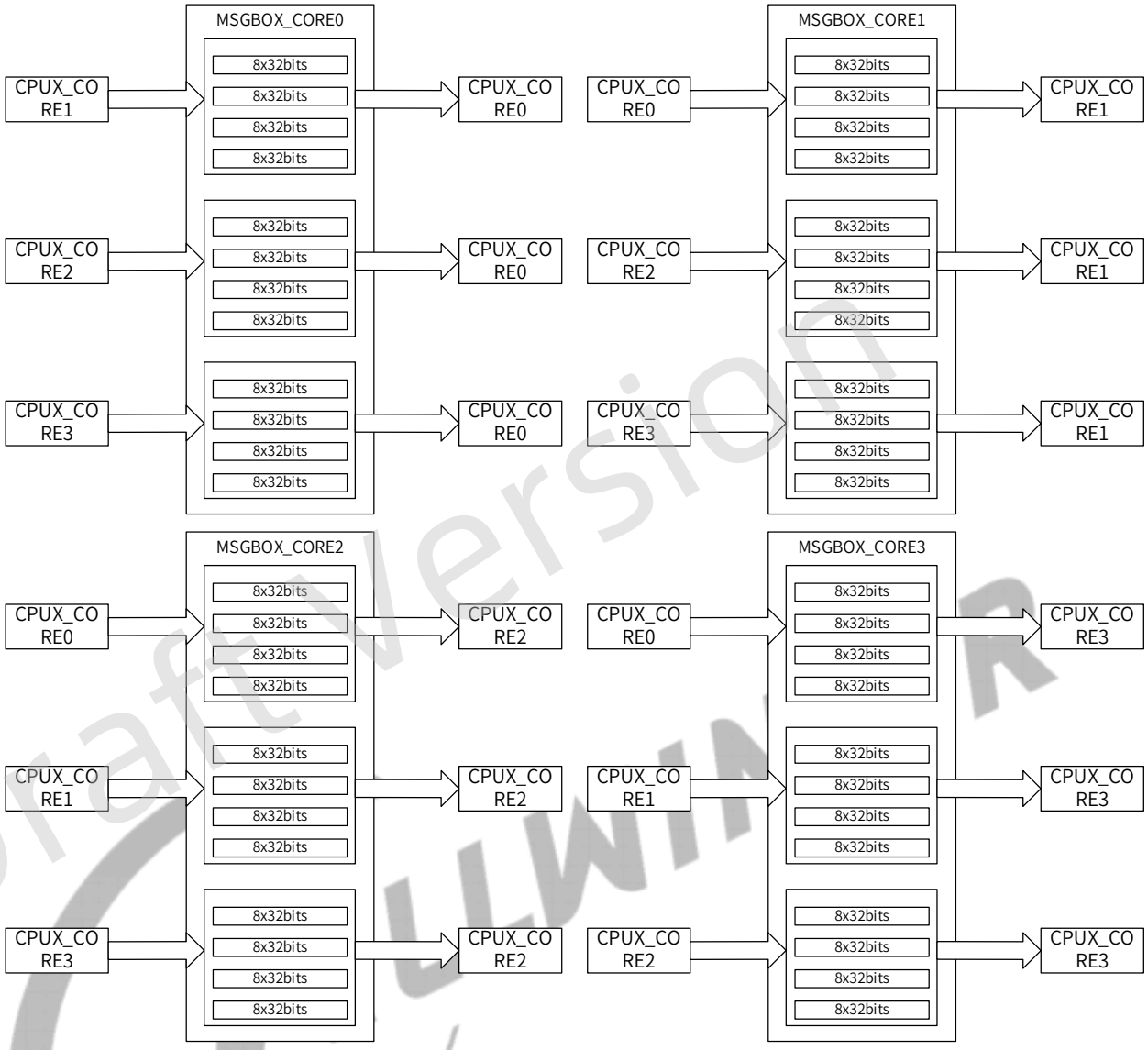

Each transmitter-receiver pair is assigned 4 channels. Two channels can be configured to be secure by software, the other two channels can be configured to be non-secure by software. The two secure channels or two non-secure channels can be configured as one synchronous box (Sending a message requires a response) or one asynchronous box (Sending a message doesn't require a response).

6.1.3 Functional Description

6.1.3.1 Clock and Reset

The MSGBOX is mounted on AHB. Before accessing the MSGBOX registers, you need to de-assert the MSGBOX reset signal on AHB bus and then open the MSGBOX gating signal on AHB bus.

6.1.3.2 Transmitter/Receiver Mode

For each MSGBOX, user0 is receiver by default and is fixed as receiver, other users are fixed as transmitter.

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6.1.3.3 Interrupt

Each channel can independently configure the interrupt enable bit. A read interrupt will be generated when the channel is not empty, while a write interrupt will be generated when the channel is not full.

For each message box, only one read interrupt will be generated, which is the OR operation result of the four read interrupts generated by the four channels. That is, only if one channel of a message box is not empty, a read interrupt will be generated by this message box. The channel that generates interrupts can be queried by the interrupt status register.

For MSGBOX_CPUX and MSGBOX_MCU, each message box can generate one write interrupt. For MSGBOX_COREx (x= 0-3), each message box can generate maximum 3 write interrupts, which are sent to corresponding transmitters separately. For example, 3 write interrupts are asserted by MSGBOX_CORE0 and sent to CPUX_CORE1-3. If CPUX_CORE1-3 accept the write interrupts, they can write data into the FIFO of MSGBOX_CORE0.

6.1.3.4 FIFO Status

When channel FIFO is non-full, the FIFO _NOT_AVA_FLAG is 0. At the moment, the FIFO can be written.

When channel FIFO is full, the FIFO_NOT_AVA_FLAG is 1. At the moment, if FIFO is written again, the previous data will be covered.

For message status, see MSGBOX_MSG_STATUS_REG.

6.1.4 Programming Guidelines

6.1.4.1 Check the Transfer Status via the Interrupt

Follow the steps below to check the transfer status:

Step 1 Enable the interrupts: Configure the interrupt enable bits of transmitter/receiver through MSGBOX_WR_IRQ_EN_REG/MSGBOX_RD_IRQ_EN_REG.

Step 2 Check the IRQ status of the corresponding queue through MSGBOX_WR_IRQ_STATUS_REG/MSGBOX_RD_IRQ_STATUS_REG.

- If the FIFO is not full, the channel will generate a transmission interrupt to remind the transmitter to transmit data. Write data to the FIFO in the interrupt handler, then clear the pending bit of the transmitter in MSGBOX_WR_IRQ_STATUS_REG and the enable bit of the transmitter in MSGBOX_WR_IRQ_EN_REG.
- If the FIFO is filled with new data, the channel generates a reception interrupt to remind the receiver to receive data. Read data from the FIFO in interrupt handler, then clear the pending bit of the receiver in MSGBOX_RD_IRQ_STATUS_REG and the enable bit of the receiver in MSGBOX_RD_IRQ_EN_REG.

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6.1.4.2 Check the Transfer Status via the FIFO

Follow the steps below to check the FIFO status of the corresponding queue:

- If the FIFO is not full, the transmitter fills the FIFO with 8*32 bits.
- If the FIFO is full, the receiver reads the FIFO data, and reads MSGBOX_FIFO_STATUS_REG to acquire the current FIFO data amount and the FIFO data amount before reading to ensure that no data is dropped.

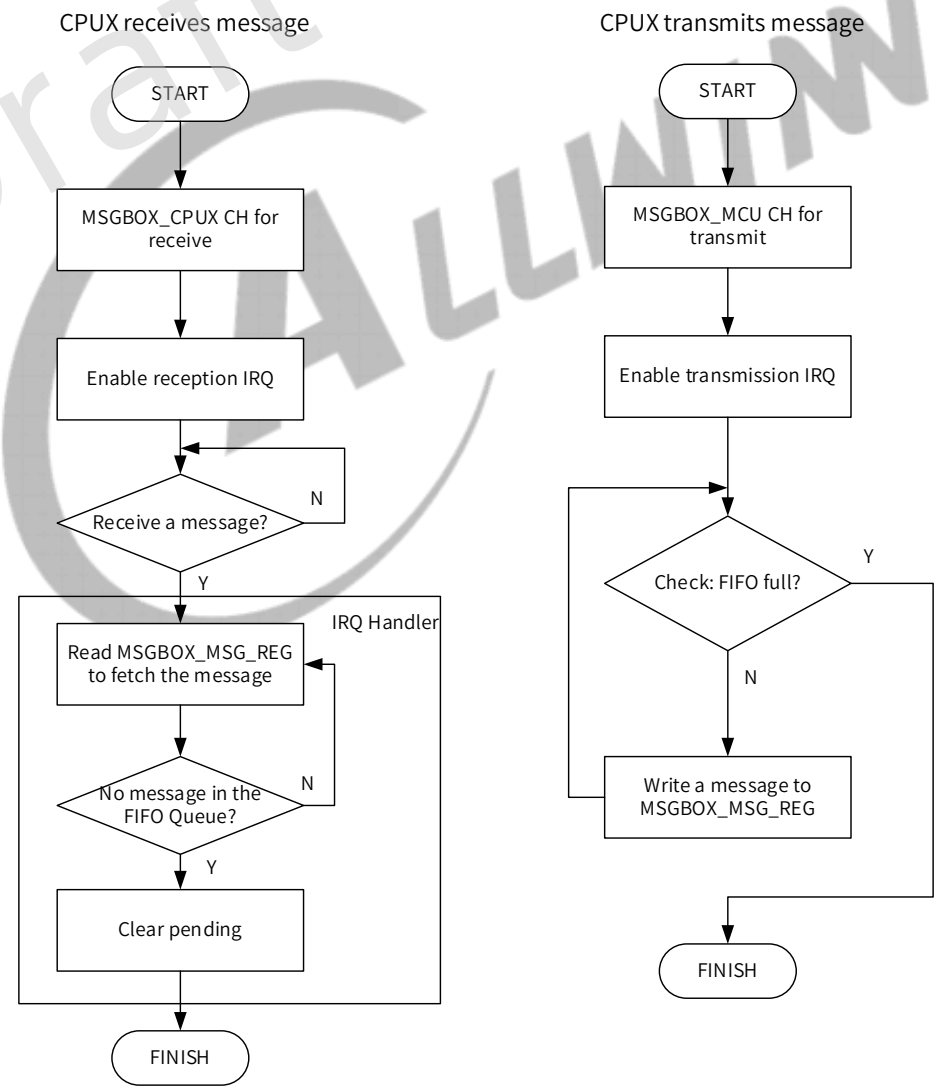
6.1.4.3 Transmit/Receive Message

The following figure takes the example of CPUX to show the message transmitting and receiving of CPUs.

CPUX for MSGBOX_CPUX: Receiving message

CPUX for MSGBOX_MCU: Transmitting message

Figure 6-2 The Communication Process



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6.1.5 Register List

Module Name	Base Address
MSGBOX_CPUX	0x03003000
MSGBOX_MCU	0x01A06000
MSGBOX_CORE0	0x03600000
MSGBOX_CORE1	0x03601000
MSGBOX_CORE2	0x03602000
MSGBOX_CORE3	0x03603000

Parameter	Description	Value
N(CPUX/MCU)	The number of other CPU communicating with the CPU	0
N(CORE0/CORE1/CORE2/CORE3)	The number of other cores communicating with the core	0-2
P	The channel number of two users	0-3

MSGBOX	USER	N Value
MSGBOX_CPUX	MCU->CPUX	N=0
MSGBOX_MCU	CPUX->MCU	N=0
MSGBOX_CORE0	CORE1->CORE0	N=0
MSGBOX_CORE0	CORE2->CORE0	N=1
MSGBOX_CORE0	CORE3->CORE0	N=2
MSGBOX_CORE1	CORE0->CORE1	N=0
MSGBOX_CORE1	CORE2->CORE1	N=1
MSGBOX_CORE1	CORE3->CORE1	N=2
MSGBOX_CORE2	CORE0->CORE2	N=0
MSGBOX_CORE2	CORE1->CORE2	N=1
MSGBOX_CORE2	CORE3->CORE2	N=2
MSGBOX_CORE3	CORE0->CORE3	N=0
MSGBOX_CORE3	CORE1->CORE3	N=1
MSGBOX_CORE3	CORE2->CORE3	N=2

Register Name	Offset	Description
MSGBOX_LPCFG_REG	0x0014+0x0100*N	MSGBOX Low Power Configuration Register
MSGBOX_RD_IRQ_EN_REG	0x0020+N*0x0100	MSGBOX Read IRQ Enable Register
MSGBOX_RD_IRQ_STATUS_REG	0x0024+N*0x0100	MSGBOX Read IRQ Status Register
MSGBOX_WR_IRQ_EN_REG	0x0030+N*0x0100	MSGBOX Write IRQ Enable Register
MSGBOX_WR_IRQ_STATUS_REG	0x0034+N*0x0100	MSGBOX Write IRQ Status Register
MSGBOX_DEBUG_REG	0x0040+N*0x0100	MSGBOX Debug Register

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Register Name	Offset	Description
MSGBOX_FIFO_STATUS_REG	0x0050+N*0x0100+P*0x0004(P=0-3)	MSGBOX FIFO Status Register
MSGBOX_MSG_STATUS_REG	0x0060+N*0x0100+P*0x0004(P=0-3)	MSGBOX Message Status Register
MSGBOX_MSG_REG	0x0070+N*0x0100+P*0x0004(P=0-3)	MSGBOX Message Queue Register
MSGBOX_WR_INT_THRESHOLD_REG	0x0080+N*0x0100+P*0x0004(P=0-3)	MSGBOX Write IRQ Threshold Register

6.1.6 Register Description

6.1.6.1 0x0014+0x0100*N MSGBOX Low Power Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0014+0x0100*N			Register Name: MSGBOX_LPCFG_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:8	R/W	0x0	MSGBOX_LPCFG_HCYCLE MSGBOX low power configuration AHB clk cycle, MSGBOX register configure AHB clk delay release cycle. 000: No delay, default 2 cycle 001: delay 1 cycle 010: delay 2 cycle 011: delay 3 cycle 100: delay 4 cycle others: increase in sequence
7:1	/	/	/
0	R/W	0x0	MSGBOX_LPCFG_EN MSGBOX Low Power Configuration Enable 0:MSGBOX low power mode disable 1:MSGBOX low power mode enable

6.1.6.2 0x0020+N*0x0100 MSGBOX Read IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x0020+N*0x0100			Register Name: MSGBOX_RD_IRQ_EN_REG
Bit	R/W	Default/Hex	Description
31:7	/	/	/

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Offset:0x0020+N*0x0100			Register Name: MSGBOX_RD_IRQ_EN_REG
Bit	R/W	Default/Hex	Description
6	R/W	0x0	RECEPTION_MQ3_IRQ_EN Reception Channel3 Interrupt Enable 0: Disable 1: Enable (It will notify receiver by interrupt when Message Queue 3 has received a new message.)
5	/	/	/
4	R/W	0x0	RECEPTION_MQ2_IRQ_EN Reception Channel2 Interrupt Enable 0: Disable 1: Enable (It will notify receiver by interrupt when Message Queue 2 has received a new message.)
3	/	/	/
2	R/W	0x0	RECEPTION_MQ1_IRQ_EN Reception Channel1 Interrupt Enable 0: Disable 1: Enable (It will notify receiver by interrupt when Message Queue 1 has received a new message.)
1	/	/	/
0	R/W	0x0	RECEPTION_MQ0_IRQ_EN Reception Channel0 Interrupt Enable 0: Disable 1: Enable (It will notify receiver by interrupt when Message Queue 0 has received a new message.)

6.1.6.3 0x0024+N*0x0100 MSGBOX Read IRQ Status Register (Default Value: 0x0000_0000)

Offset:0x0024+N*0x0100			Register Name: MSGBOX_RD_IRQ_STATUS_REG
Bit	R/W	Default/Hex	Description
31:3	/	/	/
6	R/W	0x0	RECEPTION_MQ3_IRQ_PEND Reception Channel3 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for receiver when Message Queue 3 has received a new message. Set one to this bit will clear it.
5	/	/	/

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Offset:0x0024+N*0x0100			Register Name: MSGBOX_RD_IRQ_STATUS_REG
Bit	R/W	Default/Hex	Description
4	R/W	0x0	RECEPTION_MQ2_IRQ_PEND Reception Channel2 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for receiver when Message Queue 2 has received a new message. Set one to this bit will clear it.
3	/	/	/
2	R/W	0x0	RECEPTION_MQ1_IRQ_PEND Reception Channel1 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for receiver when Message Queue 1 has received a new message. Set one to this bit will clear it.
1	/	/	/
0	R/W	0x0	RECEPTION_MQ0_IRQ_PEND Reception Channel0 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for receiver when Message Queue 0 has received a new message. Set one to this bit will clear it.

6.1.6.4 0x0030+N*0x0100 MSGBOX Write IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x0030+N*0x0100			Register Name: MSGBOX_WR_IRQ_EN_REG
Bit	R/W	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	TRANSMIT_MQ3_IRQ_EN Transmit Channel3 Interrupt Enable 0: Disable 1: Enable (It will notify transmitter by interrupt when Message Queue 3 empty level reach the configured threshold.)
6	/	/	/
5	R/W	0x0	TRANSMIT_MQ2_IRQ_EN Transmit Channel2 Interrupt Enable 0: Disable 1: Enable (It will notify transmitter by interrupt when Message Queue 2 empty level reach the configured threshold.)
4	/	/	/

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Offset:0x0030+N*0x0100			Register Name: MSGBOX_WR_IRQ_EN_REG
Bit	R/W	Default/Hex	Description
3	R/W	0x0	TRANSMIT_MQ1_IRQ_EN Transmit Channel1 Interrupt Enable 0: Disable 1: Enable (It will notify transmitter by interrupt when Message Queue 1 empty level reach the configured threshold.)
2	/	/	/
1	R/W	0x0	TRANSMIT_MQ0_IRQ_EN Transmit Channel0 Interrupt Enable 0: Disable 1: Enable (It will notify transmitter by interrupt when Message Queue 0 empty level reach the configured threshold.)
0	/	/	/

6.1.6.5 0x0034+N*0x0100 MSGBOX Write IRQ Status Register (Default Value: 0x0000_0000)

Offset:0x0034+N*0x0100			Register Name: MSGBOX_WR_IRQ_STATUS_REG
Bit	R/W	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	TRANSMIT_MQ3_IRQ_PEND Transmit Channel3 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for transmitter when Message Queue 3 empty level reach the configured threshold. Set one to this bit will clear it.
6	/	/	/
5	R/W	0x0	TRANSMIT_MQ2_IRQ_PEND Transmit Channel2 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for transmitter when Message Queue 2 empty level reach the configured threshold. Set one to this bit will clear it.
4	/	/	/
3	R/W	0x0	TRANSMIT_MQ1_IRQ_PEND Transmit Channel1 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for transmitter when Message Queue 1 empty level reach the configured threshold. Set one to this bit will clear it.
2	/	/	/

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Offset:0x0034+N*0x0100			Register Name: MSGBOX_WR_IRQ_STATUS_REG
Bit	R/W	Default/Hex	Description
1	R/W	0x0	TRANSMIT_MQ0_IRQ_PEND Transmit Channel0 Interrupt Pending 0: No effect, 1: Pending. This bit will be pending for transmitter when Message Queue 0 empty level reach the configured threshold. Set one to this bit will clear it.
0	/	/	/

6.1.6.6 0x0040+N*0x0100 MSGBOX Debug Register (Default Value: 0x0000_0000)

Offset:0x0040+N*0x0100			Register Name: MSGBOX_DEBUG_REG
Bit	R/W	Default/Hex	Description
31:12	/	/	/
11:8	R/W	0x0	FIFO_CTRL FIFO Control MQ [7:0] Control. In the debug mode, the corresponding FIFO channel will disable and only one register space valid for a message exchange. 0: Normal Mode 1: Disable the corresponding FIFO (Clear FIFO)
7:1	/	/	/
0	R/W	0x0	DEBUG_MODE Debug Mode In the Debug Mode, each user can transmit messages to itself through each Message Queue. 0: Normal Mode 1: Debug Mode

6.1.6.7 0x0050+N*0x0100+P*0x0004 (P=0-3) MSGBOX FIFO Status Register (Default Value: 0x0000_0000)

Offset:0x0050+N*0x0100+P*0x0004(P=0-3)			Register Name: MSGBOX_FIFO_STATUS_REG
Bit	R/W	Default/Hex	Description
31: 1	/	/	/

Draft Version

Offset:0x0050+N*0x0100+P*0x0004(P=0-3)			Register Name: MSGBOX_FIFO_STATUS_REG
Bit	R/W	Default/Hex	Description
0	R	0x0	FIFO_NOT_AVA_FLAG FIFO is not available flag 0: The Message FIFO queue empty level is not reached the configured threshold 1: The Message FIFO queue empty level reached the configured threshold This FIFO status register has the status related to the message queue.

6.1.6.8 0x0060+N*0x0100+P*0x0004 (P=0-3) MSGBOX Message Status Register (Default Value: 0x0000_0000)

Offset:0x0060+N*0x0100+P*0x0004(P=0-3)			Register Name: MSGBOX_MSG_STATUS_REG
Bit	R/W	Default/Hex	Description
31:4	/	/	/
3:0	R	0x0	MSG_NUM Message Number Number of unread messages in the message queue. Here, limited to eight messages per message queue. 0000: There is no message in the message FIFO queue. 0001: There is 1 message in the message FIFO queue. 0010: There are 2 messages in the message FIFO queue. 0011: There are 3 messages in the message FIFO queue. 0100: There are 4 messages in the message FIFO queue. 0101: There are 5 messages in the message FIFO queue. 0110: There are 6 messages in the message FIFO queue. 0111: There are 7 messages in the message FIFO queue. 1000: There are 8 messages in the message FIFO queue. 1001-1111:/

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6.1.6.9 0x0070+N*0x0100+P*0x0004 (P=0-3) MSGBOX Message Queue Register (Default Value: 0x0000_0000)

Offset:0x0070+N*0x0100+P*0x0004(P=0-3)			Register Name: MSGBOX_MSG_REG
Bit	R/W	Default/Hex	Description
31:0	R/W	0x0	MSG_QUE The message register stores the next to be read message of the message FIFO queue. Reads remove the message from the FIFO queue.

6.1.6.10 0x0080+N*0x0100+P*0x0004 (P=0-3) MSGBOX Write IRQ Threshold Register (Default Value: 0x0000_0000)

Offset:0x0080+N*0x0100+P*0x0004(P=0-3)			Register Name: MSGBOX_WR_INT_THRESHOLD_REG
Bit	R/W	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	MSG_WR_INT_THRESHOLD_CFG Config the FIFO empty level to trigger the write interrupt for transmitter. 00: 1 01: 2 10: 4 11: 8

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6.2 Spinlock

6.2.1 Overview

The spinlock provides hardware synchronization mechanism in multicore systems. With the lock operation, the spinlock prevents multiple processors from handling the sharing data simultaneously and thus ensure the coherence of data.

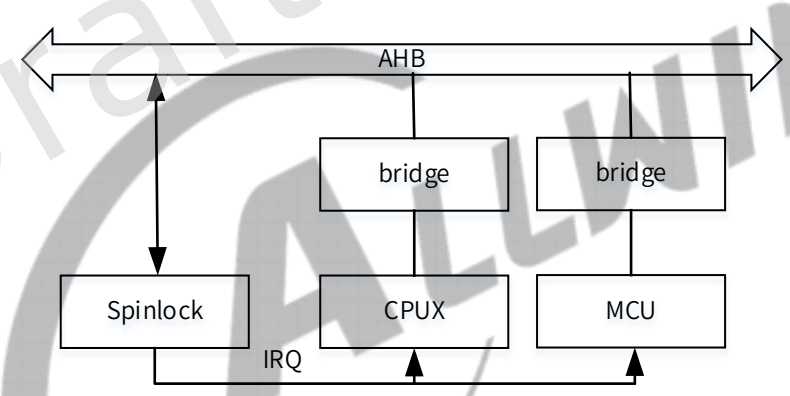
The spinlock has the following features:

- Supports 32 lock units
- Two kinds of lock status: locked and unlocked
- Lock time of the processor is predictable

6.2.2 Block Diagram

The following figure shows the block diagram of the spinlock.

Figure 6-3 Spinlock Block Diagram



6.2.3 Functional Description

6.2.3.1 Clock and reset

The spinlock is mounted on AHB. Before accessing the spinlock registers, you need to de-assert the reset signal on AHB bus and then open the corresponding gating signal on AHB bus.

6.2.3.2 Typical Application

The following figure shows how to take and release spinlock. A processor locks spinlock0 before executing specific codes, and then unlocks the codes. After the lock is free, other processors can read or write the data.

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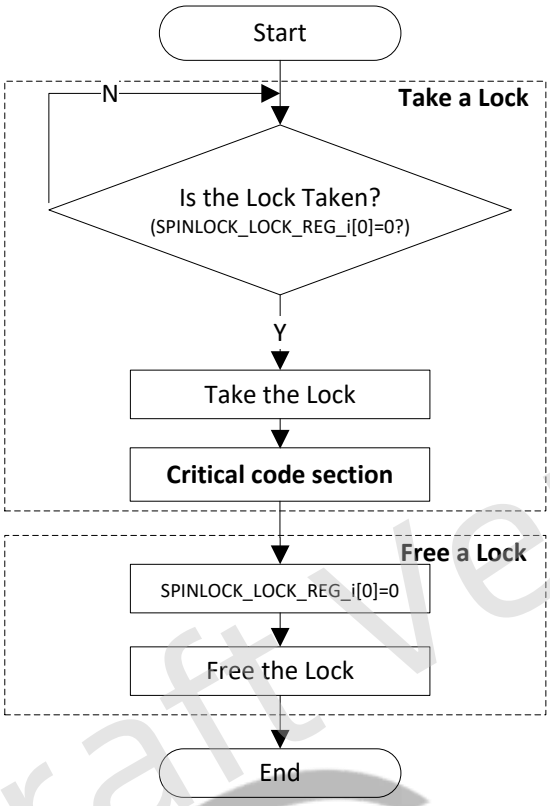
Figure 6-4 Take and Free Spinlock



6.2.3.3 Spinlock State Machine

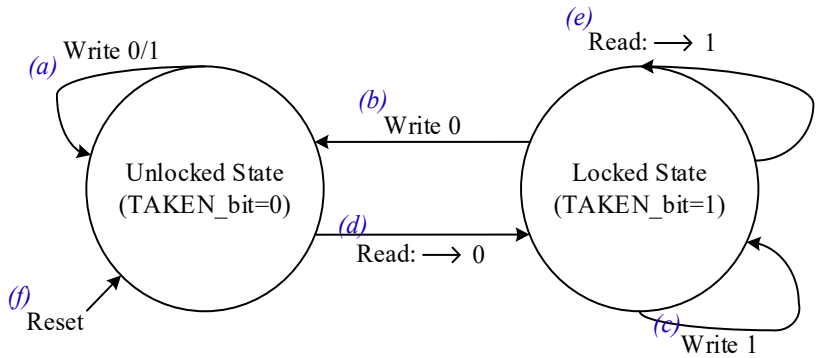
When a processor uses spinlock, it needs to acquire the spinlock status through SPINLOCK_STATUS_REG.

Reading operation: when the return value is 0, it indicates that the spinlock enters the locked status; reading this status bit again can return 1, it indicates that the spinlock is the locked status.

Writing operation: when the spinlock is in the locked status, writing 0 can convert the spinlock to the unlocked status, the writing operation for other status is invalid.

The following figure shows the spinlock state machine.

Figure 6-5 Spinlock State Machine



- When the spinlock is in the unlocked state, writing 0/1 has no effect;

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- When the spinlock is in the locked state, writing 0 can convert the corresponding spinlock to the unlocked state;
- When the spinlock is in the locked state, writing 1 has no effect;
- When the spinlock is in the unlocked state, reading the bit can return 0 (it indicates spinlock enters into the locked state);
- When the spinlock is in the locked state, reading the bit can return 1 (it indicates spinlock is in the locked state);
- After reset, the spinlock is in the unlock state by default.

6.2.4 Programming Guidelines

6.2.4.1 Spinlock Status Switch

Follow the steps below to switch the lock status of a spinlock between the locked and unlocked statuses.

Step 1 When the read value from SPINLOCKN_LOCK_REG (N=0–31) is 0, the spinlock comes into the locked status.

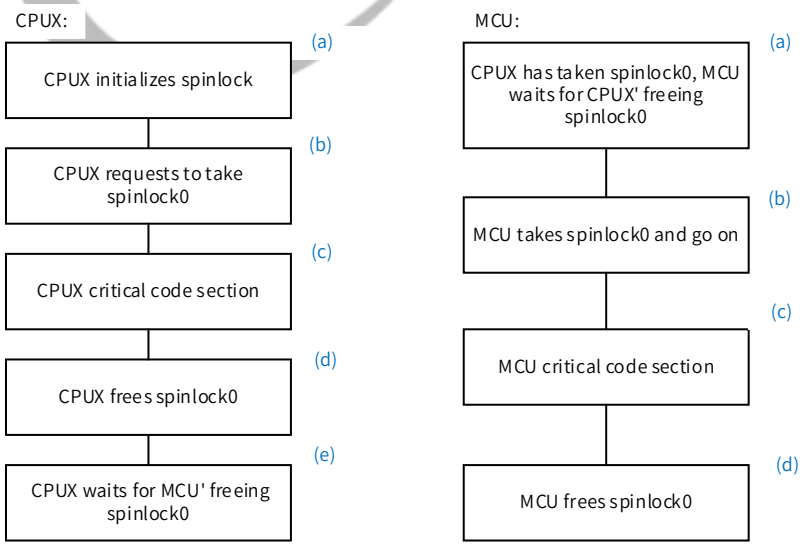
Step 2 Execute the application codes, and the status of SPINLOCK_STATUS_REG is 1.

Step 3 Write 0 to SPINLOCKN_LOCK_REG (N=0–31), the spinlock converts into the unlocked status, and the corresponding spinlock is released.

6.2.4.2 Spinlock Taking & Freeing

Take the synchronization between CPUX and MCU with Spinlock0 as an example, the CPUX and MCU perform the following steps.

Figure 6-6 CPUX and MCU Taking/Freeing Spinlock0 Process



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CPUX:

a) The CPUX initializes Spinlock.

b) Check lock register0 (SPINLOCK_STATUS_REG) status. If it is taken, check until CPUX frees spinlock0 and then request to take spinlock0. Otherwise, retry until the lock register0 is taken.

c) Execute CPUX critical code.

d) After executing CPUX critical code, the CPUX frees spinlock0.

e) The CPUX waits for MCU to free spinlock0.

MCU:

a) If the CPUX has taken spinlock0, the MCU waits for CPUX to free spinlock0.

b) The MCU requests to take spinlock0. If it fails, retry until the lock register0 is taken.

c) Execute MCU critical code.

d) After executing MCU critical code, the MCU frees spinlock0.

The following codes are for reference.

----------CPUX----------

Step 1 CPUX initializes Spinlock

```
put_wvalue(SPINLOCK_BGR_REG,0x00010000);
put_wvalue(SPINLOCK_BGR_REG,0x00010001);
```

Step 2 CPUX requests to take spinlock0

```
rdata=readl(SPINLOCK_STATUS_REG);              //Check lock register0 status
if (rdata != 0) writel (0, SPINLOCK_LOCK_REG);  //If it is taken, check till CPUX frees spinlock0
rdata=readl(SPINLOCK_LOCK_REG);                 //Request to take spinlock0
if (rdata != 0) rdata=readl(SPINLOCK_LOCK_REG);  //If it fails, retry till lock register0 is taken
```

---------- CPUX critical code section ----------

Step 3 CPUX frees spinlock0

```
writel (0, SPINLOCK_LOCK_REG);                  //CPUX frees spinlock0
```

Step 4 CPUX waits for MCU freeing spinlock0

```
writel(readl(SPINLOCK_STATUS_REG0) == 1);       //CPUX waits for MCU freeing spinlock0
```

----------MCU----------

Step 1 CPUX has taken spinlock0, MCU waits for CPUX freeing spinlock0

```
while(readl(SPINLOCK_STATUS_REG) == 1);         //MCU waits for CPUX freeing spinlock0
```

Step 2 MCU takes spinlock0 and go on

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```
rdata=readl(SPINLOCK_LOCK_REG);                    //Request to take spinlock0
if (rdata != 0) rdata=readl(SPINLOCK_LOCK_REG);    //If it fails, retry till lock register0 is taken
---------- MCU critical code section ----------
```

Step 3 MCU frees spinlock0

```
writel (0, SPINLOCK_LOCK_REG);                     //MCU frees spinlock0
```

6.2.4.3 Interrupt Processing

The spinlock generates an interrupt when a lock is freed (the lock status converts from the locked status to the unlocked status).

Follow the steps below to process the interrupt:

Step 1 Configure the interrupt enable bit of the corresponding spinlock in SPINLOCK_IRQ_EN_REG to enable the interrupt.

Step 2 The spinlock generates an interrupt when its status converts from the locked status to the unlocked status, and the corresponding bit of the SPINLOCK_IRQ_STA_REG turns to 1.

Step 3 Execute the interrupt handle function and clear the pending bit.

6.2.5 Register List

Module Name	Base Address
SPINLOCK	0x03005000

Register Name	Offset	Description
SPINLOCK_SYSTATUS_REG	0x0000	SpinLock System Status Register
SPINLOCK_STATUS_REG	0x0010	SpinLock Status Register
SPINLOCK_IRQ_EN_REG	0x0020	SpinLock Interrupt Enable Register
SPINLOCK_IRQ_STA_REG	0x0040	SpinLock Interrupt Status Register
SPINLOCK_LOCKID0_REG	0x0080	SpinLock Lockid0 Register
SPINLOCK_LOCKID1_REG	0x0084	SpinLock Lockid1 Register
SPINLOCK_LOCKID2_REG	0x0088	SpinLock Lockid2 Register
SPINLOCK_LOCKID3_REG	0x008C	SpinLock Lockid3 Register
SPINLOCK_LOCKID4_REG	0x0090	SpinLock Lockid4 Register
SPINLOCKN_LOCK_REG	0x0100+n*0x0004 (n=0-31)	SpinLock Register n

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6.2.6 Register Description

6.2.6.1 0x0000 SpinLock System Status Register (Default Value: 0x1000_0000)

Offset: 0x0000			Register Name: SPINLOCK_SYSTATUS_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R	0x1	LOCKS_NUM Number of lock registers implemented 00: This instance has 256 lock registers 01: This instance has 32 lock registers 10: This instance has 64 lock registers 11: This instance has 128 lock registers
27:9	/	/	/
8	R	0x0	IU0 Use Flag0 In-Use flag0, covering lock register0-31. 0: All lock register 0-31 are in the Not Taken state. 1: At least one of the lock register 0-31 is in the Taken state.
7:0	/	/	/

6.2.6.2 0x0010 SpinLock Status Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: SPINLOCK_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LOCK_REG_STATUS SpinLock Reg Status 0: The Spinlock is free 1: The Spinlock is taken Spinlock 0-31 bit.

6.2.6.3 0x0020 SpinLock Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: SPINLOCK_IRQ_EN_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LOCK_IRQ_EN SpinLock interrupt enable 0: Disable 1: Enable

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6.2.6.4 0x0040 SpinLock Interrupt Status Register (Default Value:0x0000_0000)

Offset: 0x0040			Register Name: SPINLOCK_IRQ_STA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W1C	0x0	LOCK_IRQ _STATUS SpinLock interrupt status 0: No effect 1: Pending Write 1 will clear this bit.

6.2.6.5 0x0080 SpinLock Lockid0 Register (Default Value: 0x7777_7777)

Offset: 0x0080			Register Name: SPINLOCK_LOCKID0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x77777777	LOCKID0

6.2.6.6 0x0084 SpinLock Lockid1 Register (Default Value: 0x7777_7777)

Offset: 0x0084			Register Name: SPINLOCK_LOCKID1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x77777777	LOCKID1

6.2.6.7 0x0088 SpinLock Lockid2 Register (Default Value: 0x7777_7777)

Offset: 0x0088			Register Name: SPINLOCK_LOCKID2_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x77777777	LOCKID2

6.2.6.8 0x008C SpinLock Lockid3 Register (Default Value: 0x7777_7777)

Offset: 0x008C			Register Name: SPINLOCK_LOCKID3_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x77777777	LOCKID3

6.2.6.9 0x0090 SpinLock Lockid4 Register (Default Value: 0x7777_7777)

Offset: 0x0090			Register Name: SPINLOCK_LOCKID4_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x77777777	LOCKID4

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6.2.6.10 0x0100+n*0x0004 (n=0-31) SpinLock Register n (Default Value:0x0000_0000)

Offset: 0x0100+n*0x0004 (n=0-31)			Register Name: SPINLOCKN_LOCK_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	TAKEN. Lock State. Read 0x0: The lock was previously Not Taken (free). The requester is granted the lock. Write 0x0: Set the lock to Not Taken (free). Read 0x1: The lock was previously Taken. The requester is not granted the lock and must retry. Write 0x1: No update to the lock value.

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7 Thermal Sensor

7.1 Thermal Sensor Controller (THS)

7.1.1 Overview

The thermal sensors are common elements in wide range of modern system on chips (SoCs) platform. The thermal sensors are used to constantly monitor the temperature on the chip.

The SoC embeds two thermal sensors in possible hot spots on the die, sensor0 is located in the DDR and sensor1 located in the CPU. These thermal sensors generate interrupt to lower temperature via DVFS, on reaching a certain thermal threshold.

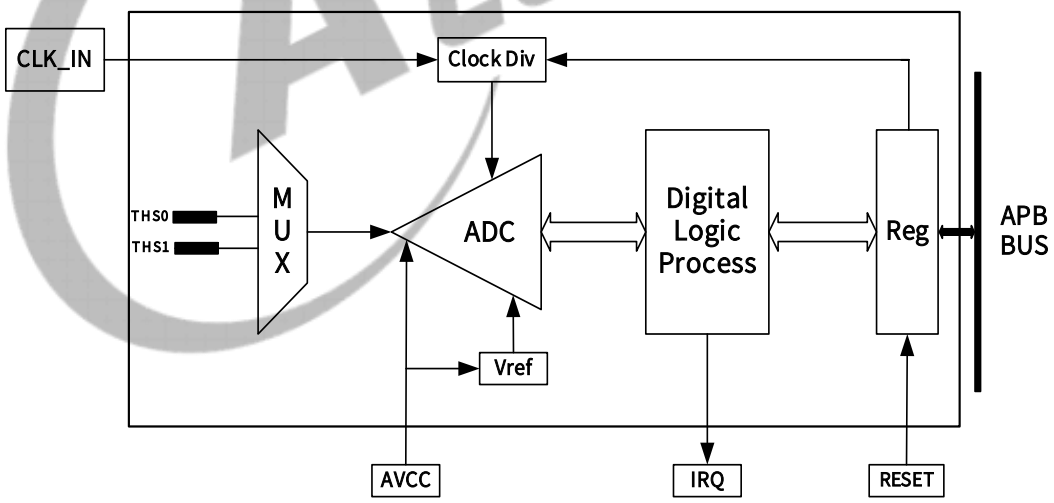
The THS has the following features:

- Averaging filter for thermal sensor reading
- Support over-temperature protection interrupt and over-temperature alarm interrupt
- Support ±3℃ error limit from 60℃ to 125℃,and ±5℃ from -40℃ to 60℃

7.1.2 Block Diagram

The following figures show the block diagrams of the THS.

Figure 7-1 THS Block Diagram



7.1.3 Functional Description

7.1.3.1 Clock Source

The THS controller has two clock sources: THS_ADC_CLK and PCLK. For details about clock configurations, refer to Section 4.1 Clock Controller Unit (CCU) .

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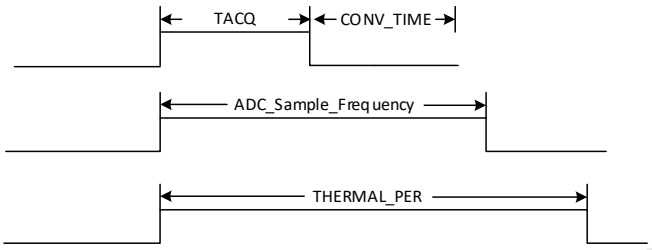
7.1.3.2 Timing Requirements

The following figure shows the timing requirements for the THS module.

Figure 7-2 Thermal Sensor Timing Requirement



CLK_IN = 24 MHz

CONV_TIME (Conversion Time) = 1/24 MHz x 14 Cycles = 0.583 us

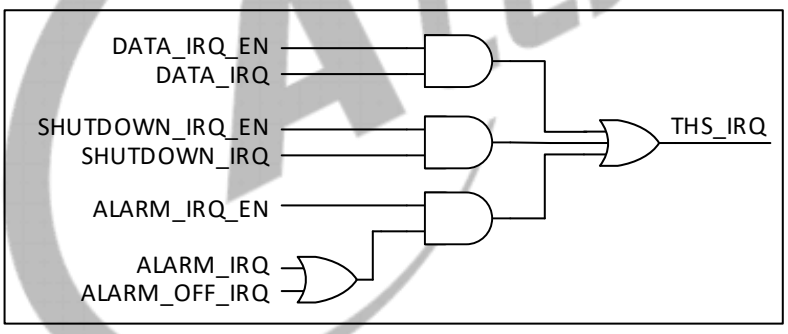
TACQ > 1/24 MHz x 24 Cycles

THERMAL_PER > ADC_Sample_Frequency > TACQ + CONV_TIME

7.1.3.3 Interrupts

The THS module has four interrupt sources: DATA_IRQ, SHUTDOWN_IRQ, ALARM_IRQ, and ALARM_OFF_IRQ. The following figure shows thermal sensor interrupt sources.

Figure 7-3 Thermal Sensor Controller Interrupt Source



DATA_IRQ

The interrupt is generated when the measured sensor_data is updated.

SHUTDOWN_IRQ

The interrupt is generated when the temperature is higher than the shutdown threshold.

ALARM_IRQ

The interrupt is generated when the temperature is higher than the Alarm_Threshold.

ALARM_OFF_IRQ

The interrupt is generated when the temperature is lower than the Alarm_Off_Thershold. It is triggered at the falling edge.

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7.1.3.4 THS Temperature Conversion Formula

T_{ft_cal} = (sensor_data$_{ft}$ - 2800)/(-15.73)

C = (T_{ft_cal} - T_{ft}-5)*15.73

-40°C -65°C T = (sensor_data+C-2800)/(-15.73)

65°C -125°C T= (sensor_data+C-2980)/(-18.3)

Unit of T_{ft_cal}, T_{ft}, and T : Celsius degree (°C).

Calculation parameters are explained below.

Table 7-1 Calculation Parameters for THS Temperature Conversion Formula

Parameters	Description
sensor_data	THS temperature measurement data, and it can be read from THSx (x=0 or 1)_DATA register.
sensor_data$_{ft}$	The calibration value of THS1 in Final Test (FT) stage. For register read address, see Table 7-2 eFuse Address of THS Calibration Value in FT Stage.
T_{ft_cal}	Calibration temperature value during FT stage.
T_{ft}	Room temperature during FT stage. For register read address, see Table 7-2 eFuse Address of THS Calibration Value in FT Stage.
C	Temperature compensation value during FT stage calibration.

Table 7-2 eFuse Address of THS Calibration Value in FT Stage

Base address for eFuse is 0x03006200.

Name	Offset Address	Description
T-sensor Calibration	0x002C-0x002D	The calibration value of THS1
	0x002D-0x002E	Room temperature
	0x002F-0x0030	The calibration value of THS0

7.1.4 Register List

Module Name	Base Address
THS	0x02086000

Register Name	Offset	Description
THS_CTRL	0x0000	THS Control Register
THS_EN	0x0004	THS Enable Register
THS_PER	0x0008	THS Period Control Register
THS_DATA_INTC	0x0010	THS Data Interrupt Control Register
THS_SHUT_INTC	0x0014	THS Shut Interrupt Control Register
THS_ALARM_INTC	0x0018	THS Alarm Interrupt Control Register
THS_DATA_INTS	0x0020	THS Data Interrupt Status Register
THS_SHUT_INTS	0x0024	THS Shut Interrupt Status Register

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Register Name	Offset	Description
THS_ALARMO_INTS	0x0028	THS Alarm off Interrupt Status Register
THS_ALARM_INTS	0x002C	THS Alarm Interrupt Status Register
THS_FILTER	0x0030	THS Median Filter Control Register
THS0_ALARM_CTRL	0x0040	THS0 Alarm Threshold Control Register
THS1_ALARM_CTRL	0x0044	THS1 Alarm Threshold Control Register
THS0_THS1_SHUTDOWN_CTRL	0x0080	THS0_THS1 Shutdown Threshold Control Register
THS0_DATA	0x00C0	THS0 Data Register
THS1_DATA	0x00C4	THS1 Data Register

7.1.5 Register Description

7.1.5.1 0x0000 THS Control Register (Default Value: 0x01DF_002F)

Offset: 0x0000			Register Name: THS_CTRL
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x1DF	TACQ ADC acquire time, default (20uS) CLK_IN/(n+1)
15:0	R/W	0x2F	Reserved

7.1.5.2 0x0004 THS Enable Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: THS_EN
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	THS1_EN Enable Temperature Measurement Sensor1 0: Disable 1: Enable
0	R/W	0x0	THS0_EN Enable Temperature Measurement Sensor0 0: Disable 1: Enable

7.1.5.3 0x0008 THS Period Control Register (Default Value: 0x0003_A000)

Offset: 0x0008			Register Name: THS_PER
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0008			Register Name: THS_PER
Bit	Read/Write	Default/Hex	Description
31:12	R/W	0x3A	THERMAL_PER Temperature Measurement Period, Default (10ms) 4096*(n+1)/CLK_IN
11:0	/	/	/

7.1.5.4 0x0010 THS Data Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: THS_DATA_INTC
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	THS1_DATA_IRQ_EN Selects Temperature Measurement Data of Sensor1 0: Disable 1: Enable
0	R/W	0x0	THS0_DATA_IRQ_EN Selects Temperature Measurement Data of Sensor0 0: Disable 1: Enable

7.1.5.5 0x0014 THS Shut Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: THS_SHUT_INTC
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	SHUT_INT1_EN Selects Shutdown Interrupt for Sensor1 0: Disable 1: Enable
0	R/W	0x0	SHUT_INT0_EN Selects Shutdown Interrupt for Sensor0 0: Disable 1: Enable

7.1.5.6 0x0018 THS Alarm Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: THS_ALARM_INTC
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/

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Offset: 0x0018			Register Name: THS_ALARM_INTC
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	ALARM_INT1_EN Selects Alarm Interrupt for Sensor1 0: Disable 1: Enable
0	R/W	0x0	ALARM_INT0_EN Selects Alarm Interrupt for Sensor0 0: Disable 1: Enable

7.1.5.7 0x0020 THS Data Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: THS_DATA_INTS
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W1C	0x0	THS1_DATA_IRQ_STS Data Interrupt Status for Sensor1 Write '1' to clear this interrupt.
0	R/W1C	0x0	THS0_DATA_IRQ_STS Data Interrupt Status for Sensor0 Write '1' to clear this interrupt.

7.1.5.8 0x0024 THS Shut Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: THS_SHUT_INTS
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W1C	0x0	SHUT_INT1_STS Shutdown Interrupt Status for Sensor1 Write '1' to clear this interrupt.
0	R/W1C	0x0	SHUT_INT0_STS Shutdown Interrupt Status for Sensor0 Write '1' to clear this interrupt.

7.1.5.9 0x0028 THS Alarm off Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: THS_ALARMO_INTS
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/

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Offset: 0x0028			Register Name: THS_ALARMO_INTS
Bit	Read/Write	Default/Hex	Description
1	R/W1C	0x0	ALARM_OFF1_STS Alarm Interrupt Off Pending for Sensor1 Write '1' to clear this interrupt.
0	R/W1C	0x0	ALARM_OFF0_STS Alarm Interrupt Off Pending for Sensor0 Write '1' to clear this interrupt.

7.1.5.10 0x002C THS Alarm Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: THS_ALARM_INTS
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W1C	0x0	ALARM_INT1_STS Alarm Interrupt Pending for Sensor1 Write '1' to clear this interrupt.
0	R/W1C	0x0	ALARM_INT0_STS Alarm Interrupt Pending for Sensor0 Write '1' to clear this interrupt.

7.1.5.11 0x0030 THS Median Filter Control Register (Default Value: 0x0000_0001)

Offset: 0x0030			Register Name: THS_FILTER
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	FILTER_EN Filter Enable 0: Disable 1: Enable
1:0	R/W	0x1	FILTER_TYPE Average Filter Type 00: 2 01: 4 10: 8 11: 16

7.1.5.12 0x0040 THS0 Alarm Threshold Control Register (Default Value: 0x05A0_0684)

Offset: 0x0040			Register Name: THS0_ALARM_CTRL
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/

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Offset: 0x0040			Register Name: THS0_ALARM_CTRL
Bit	Read/Write	Default/Hex	Description
27:16	R/W	0x5A0	ALARM0_T_HOT Thermal Sensor0 Alarm Threshold for Hot Temperature
15:12	/	/	/
11:0	R/W	0x684	ALARM0_T_HYST Thermal Sensor0 Alarm Threshold for Hysteresis Temperature

7.1.5.13 0x0044 THS1 Alarm Threshold Control Register (Default Value: 0x05A0_0684)

Offset: 0x0044			Register Name: THS1_ALARM_CTRL
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x5A0	ALARM1_T_HOT Thermal Sensor1 Alarm Threshold for Hot Temperature
15:12	/	/	/
11:0	R/W	0x684	ALARM1_T_HYST Thermal Sensor1 Alarm Threshold for Hysteresis Temperature

7.1.5.14 0x0080 THS0_THS1 Shutdown Threshold Control Register (Default Value: 0x04E9_04E9)

Offset: 0x0080			Register Name: THS0_THS1_SHUTDOWN_CTRL
Bit	Read/Write	Default/Hex	Description
27:16	R/W	0x4E9	SHUT1_T_HOT Thermal sensor1 Shutdown Threshold for Hot Temperature
15:12	/	/	/
11:0	R/W	0x4E9	SHUT0_T_HOT Thermal sensor0 Shutdown Threshold for Hot Temperature

7.1.5.15 0x00C0 THS0 Data Register (Default Value: 0x0000_0000)

Offset: 0x00C0			Register Name: THS0_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x0	THS0_DATA Temperature Measurement Data of Sensor0

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7.1.5.16 0x00C4 THS1 Data Register (Default Value: 0x0000_0000)

Offset: 0x00C4			Register Name: THS1_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x0	THS1_DATA Temperature Measurement Data of Sensor1

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8 Timers

8.1 Timer

8.1.1 Overview

The Timer module supports the timing and counting functions, which includes TIMER_CPUX and TIMER_MCU. There are 8 timers in TIMER_CPUX and 4 timers in TIMER_MCU.

The Timer module has the following features:

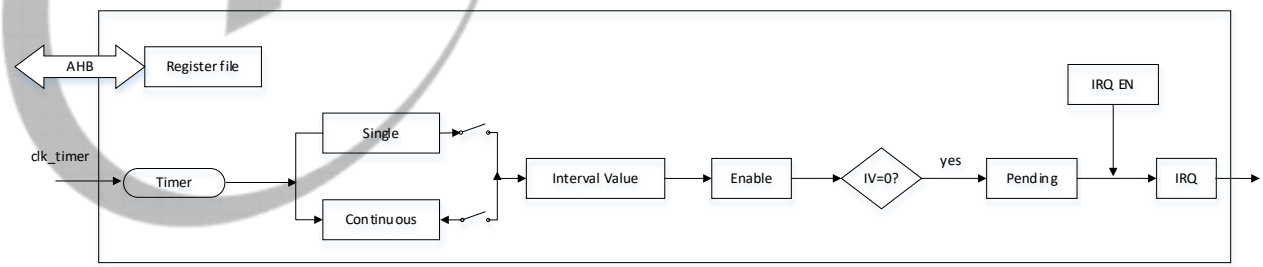
- AHB interface: configure timer registers
- Configurable count clock: the timer clock source can be switched to 32 kHz, 24 MHz, 16 MHz, and 200 MHz.
- Programmable 56-bit down timer
- Supports two timing modes: continuous mode and single mode
- Generates an interrupt after counting ends

8.1.2 Block Diagram

The timer is a 56-bit down counter. The counter value is decremented by 1 on each rising edge of the timer clock.

The following figure shows the block diagram for the timer.

Figure 8-1 Block Diagram for the Timer



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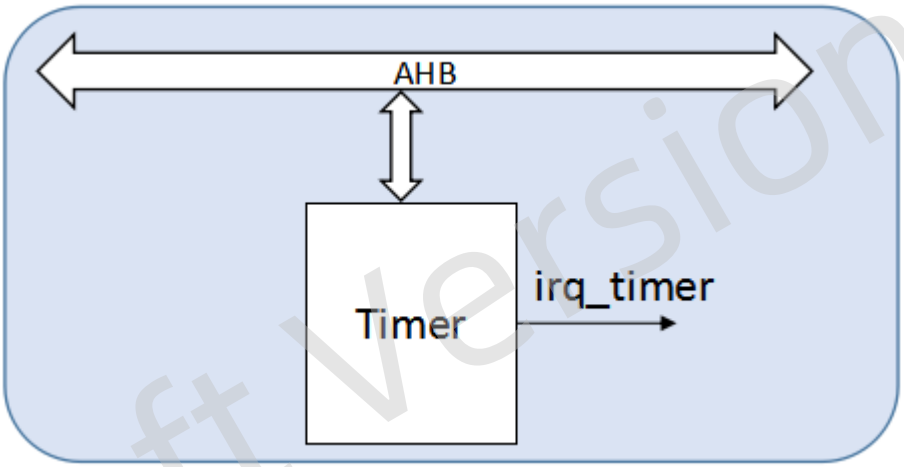
8.1.3 Functional Descriptions

8.1.3.1 Typical Application

The following figure shows the typical application of the Timer module.

Figure 8-2 Timer Typical Application



The Timer is mounted at the AHB bus. The system configures and controls the Timer via the AHB bus.

8.1.3.2 Time Calculation

$$T = \frac{\{TIMER_IVH, TIMER_IVL\} - \{TIMER_CVH, TIMER_CVL\}}{f_{clk_timer}}$$

Where,

TIMER_IVH is the higher 24 bits of the interval value, which could be configured by the lower 24 bits of the TIMER_IVH_REG register;

TIMER_IVL is the lower 32 bits of the interval value, which could be configured by the TIMER_IVL_REG register;

TIMER_CVH is the higher 24 bits of the current value, which could be configured by the lower 24 bits of the TIMER_CVH_REG register;

TIMER_CVL is the lower 32 bits of the current value, which could be configured by the TIMER_CVL_REG register;

fclk_timer is the frequency of the timer clock source;

{TIMER_IVH, TIMER_IVL} is the 56-bit interval value of the timer;

{TIMER_CVH, TIMER_CVL} is the 56-bit current value of the timer.

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8.1.3.3 Count Modes

Each timer is 56-bit down counter. The counter value is decreased by 1 on each rising edge of the count clock. The timer has two count modes: the continuous mode and the single mode. You can configure the count modes via the bit[7] of TIMER_CTRL_REG. The value 0 is for the period mode and value 1 is for the single counting mode.

- Continuous Mode

In the continuous mode, the timer will reload data from the Timer Interval Value and generate an interrupt when the counter is decreased to 0. Then, it will restart another round of counting.

- Single Mode

In the single mode, the timer will stop counting and generate an interrupt after the counter decreases to 0, and then stops counting. It starts to count again only when the interval value is reloaded.

8.1.4 Programming Guidelines

8.1.4.1 Initialization

Refer to the following steps to initialize the timer:

Step 1 Configure the timer parameters including the clock sources and timing modes by writing to TIMER_CTRL_REG. There is no sequence requirement of configuring these parameters.

Step 2 Write the interval value.

a) Write to TIMER_IVL bit of TIMER_IVL_REG register and TIMER_IVH bit of TIMER_IVH_REG register to configure the interval value for the timer.

b) Write to bit [1] of TIMER_CTRL_REG to load the interval value to the timer. The value of the bit will be cleared automatically after the interval value is loaded.

Step 3 Write to bit [0] of TIMER_CTRL_REG to start the timer. Read TIMER_CVL bit of TIMER_CVL_REG register and TIMER_CVH bit of TIMER_CVH_REG register to get the current value of the timer.

NOTE

When performing read or write operations on the current register, operate TIMER_CVL_REG register before TIMER_CVH_REG register.

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8.1.4.2 Interrupt Processing

Refer to the following steps to process the interrupt:

Step 1 Enable interrupts for the timer: write to the enable bit of the corresponding interrupt in TIMER_IRQ_REG for the timer. The timer will generate an interrupt once the count value reaches 0.

Step 2 After the software program enters the interrupt process, write to the pending bit of the corresponding interrupts in TIMER_STA_REG to clear the interrupt pending.

Step 3 Resume the interrupt and continue to execute the interrupted process.

8.1.5 Register List

Module Name	Base Address	The Value of N
TIMER0_CPUX	0x03008000	N=0-7
TIMER0_MCU	0x01A08000	N=0-3

Register Name	Offset	Description
TIMER_IRQ_REG	0x0000	Timer IRQ Enable Register
TIMER_STA_REG	0x0004	Timer Status Register
TIMER_SEC_REG	0x0008	Timer Secure Register
TIMER_CTRL_REG	0x0020+0x0020*N	Timer Control Register
TIMER_IVL_REG	0x0024+0x0020*N	Timer Low Interval Value Register
TIMER_CVL_REG	0x0028+0x0020*N	Timer Low Current Value Register
TIMER_IVH_REG	0x002C+0x0020*N	Timer High Interval Value Register
TIMER_CVH_REG	0x0030+0x0020*N	Timer High Current Value Register
TIMER_CV_READ_SEL_REG	0x0034+0x0020*N	Timer Current Value Read Select Register

8.1.6 Register Description

8.1.6.1 0x0000 Timer IRQ Enable Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: TIMER_IRQ_REG
Bit	Read/Write	Default/Hex	Description
31:NUM	/	/	/
NUM-1:0	R/W	0x0	TIMER_IRQ_EN Timer (0 to NUM-1) Interrupt Enable 1: Enable 0: Disable Note: The specific number is decided by the number of timers.

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8.1.6.2 0x0004 Timer Status Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: TIMER_STA_REG
Bit	Read/Write	Default/Hex	Description
31:NUM	/	/	/
NUM-1:0	R/WC	0x0	TIMER_IRQ_STAT Timer (0 to NUM-1) Status 1: Pending 0: No effect Note: The specific number is decided by the number of timers.

8.1.6.3 0x0008 Timer Secure Register (Default Value: 0x0000_00FF)

Offset: 0x0008			Register Name: TIMER_SEC_REG
Bit	Read/Write	Default/Hex	Description
31:NUM	/	/	/
NUM-1:0	R/W	0xFF	TIMER_SECURE Timer (0 to NUM-1) Secure Control 1: Secure 0: Non-secure Note: The specific number is decided by the number of timers.

8.1.6.4 0x0020+0x0020*N Timer Control Register (Default Value: 0x0000_0000)

Offset: 0x0020+0x0020*N			Register Name: TIMER_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	TIMER_MODE Timer Mode 0: Continuous mode. When interval value is reached, the timer will restart another round of counting automatically. 1: Single mode. When interval value is reached, the timer will stop counting.
6:2	/	/	/

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Offset: 0x0020+0x0020*N			Register Name: TIMER_CTRL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	TIMER_RELOAD. Timer 0 Reload. 0: No effect 1: Reload timer 0 Interval value After the bit is set, it cannot be written again before it's cleared automatically.
0	R/W	0x0	TIMER_EN. Timer 0 Enable. 0: Stop/Pause 1: Start If the timer is started, it will reload the interval value to internal register, and the current counter will count from interval value to 0. If the current counter does not reach the zero, the timer enable bit is set to "0", the current value counter will pause. Wait for at least 2 cycles, and then the start bit can be set to 1. In the pause state, the interval value register can be modified. If the timer is started again, and the Software hope the current value register to down-count from the new interval value. The reload bit and the enable bit should be set to 1 at the same time.

8.1.6.5 0x0024+0x0020*N Timer Low Interval Value Register (Default Value: 0x0000_0000)

Offset: 0x0024+0x0020*N			Register Name: TIMER_IVL_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TIMER_IVL Timer Interval Value bit[31:0] Register

8.1.6.6 0x0028+0x0020*N Timer Low Current Value Register (Default Value: 0x0000_0000)

Offset: 0x0028+0x0020*N			Register Name: TIMER_CVL_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TIMER_CVL Timer Current Value bit[31:0] Register

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8.1.6.7 0x002C+0x0020*N Timer High Interval Value Register (Default Value: 0x0000_0000)

Offset: 0x002C+0x0020*N			Register Name: TIMER_IVH_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	TIMER_IVH Timer Interval Value bit[55:32] Register

8.1.6.8 0x0030+0x0020*N Timer High Current Value Register (Default Value: 0x0000_0000)

Offset: 0x0030+0x0020*N			Register Name: TIMER_CVH_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	TIMER_CVH Timer Current Value bit[55:32] Register

NOTE

The current value register is a 56-bit register. When read or write the current value, the TIMER_CVL should be read or write first.

8.1.6.9 0x0034+0x0020*N Timer Current Value Read Select Register (Default Value: 0x0000_0000)

Offset: 0x0034+0x0020*N			Register Name: TIMER_CV_READ_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	TIMER_CV_READ_SEL Timer Read Current Value. 0: The Frequency of ahb_clk faster than 2 times of timer_clk. 1: The Relation of ahb_clk and timer_clk Frequency are not required.

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8.2 Watchdog Timer (WDT)

8.2.1 Overview

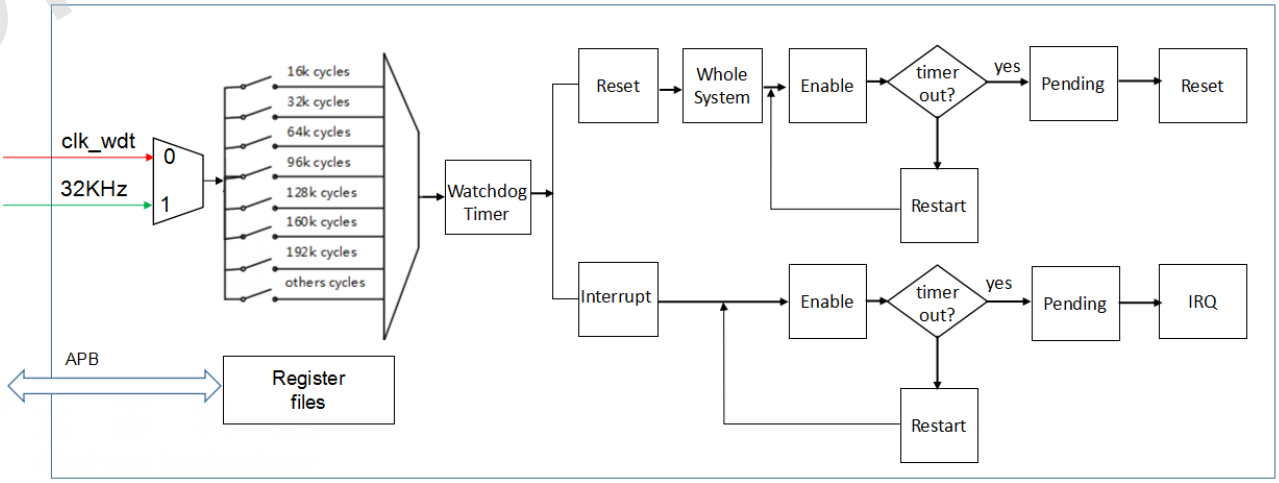
Watchdog Timer is used to transmit a reset signal to reset the entire system after an exception occurs, which includes two watchdog timers: WDT_CPUX and WDT_MCU. It has the following features:

- Two clock sources: SYS_32K_CLK or SYS_24M_CLK/750
- Over 1-minute timeout
- Generates timeout interrupts
- Generates timeout reset signals
- Supports system reset signals asserted via software
- Supports restarting the timing

8.2.2 Block Diagram

The following figure shows the functional block diagram of the watchdog module.

Figure 8-3 Watchdog Block Diagram



8.2.3 Functional Descriptions

8.2.3.1 Clock Sources

The clock source of the watchdog is either SYS_32K_CLK or SYS_24M_CLK/750 (24 MHz/750). Configure the bit [8] of the WDT_CFG_REG to select a clock source.

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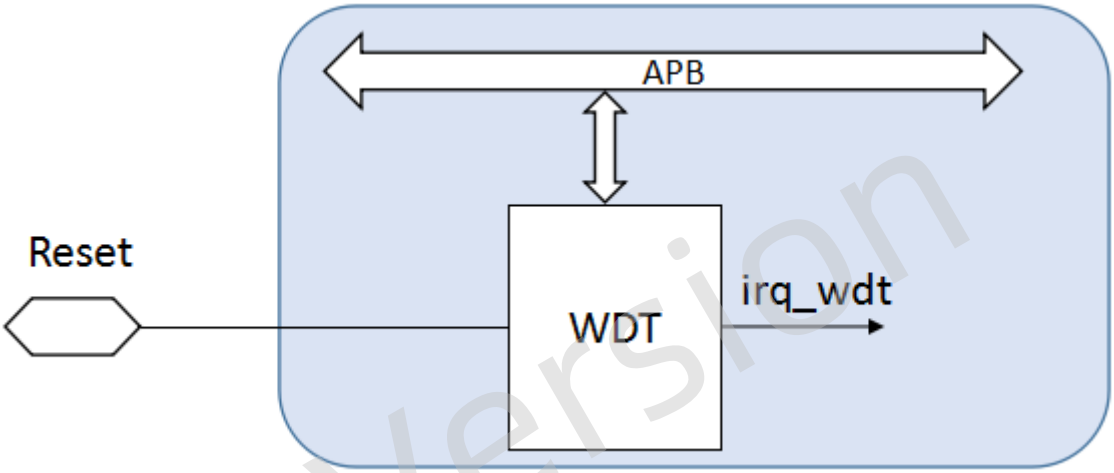
8.2.3.2 Typical Application

Figure 8-4 Watchdog Application Diagram



Watchdog configures register by APB bus.

The system configures the time of watchdog, if the system has no timing for restart watchdog (such as bus hang dead), then watchdog sends out watchdog reset external signal to reset system; meanwhile watchdog outputs signal to RESET pad to reset PMIC.

8.2.3.3 Operating Modes

The watchdog contains a 32-bit up counter and a 12-bit down counter. The counter value is increased or decreased by 1 on each rising edge of the count clock. The watchdog has two operating modes. The down counter only works in reset mode.

- Interrupt mode

The bit [1:0] of the WDT_CFG_REG is set to 0x2, when the counter value reaches the value of WDT_INTERVAL_VALUE of the WDT_MODE_REG register and the bit [0] of the WDT_IRQ_EN_REG is written to 1, the watchdog generates an interrupt, the watchdog enters into interrupt mode.

- Reset mode

The bit [1:0] of the WDT_CFG_REG is set to 0x1, when the counter value reaches the value of WDT_INTERVAL_VALUE of the WDT_MODE_REG register , the watchdog generates a reset signal to reset the entire system. The duration that the reset is held low is controlled by the WDT_OUT_CONFIG of the WDT_OCFG_REG.

You can make the watchdog to count from the initial value at any time by configuring the WDT_CTRL_REG: Write 0xA57 to bit[12:1], then write 1 to bit[0].

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8.2.4 Programming Guidelines

8.2.4.1 Initialization

Follow the steps below to initialize the watchdog:

Step 1 Configure the bit [1:0] of the WDT_CFG_REG to configure the generation of the interrupts or the output of reset signal.

Step 2 Configure the bit [7:4] of the WDT_MODE_REG to configure the initial count value.

Step 3 Write the bit [0] of the WDT_MODE_REG to 1 to enable the watchdog.

8.2.4.2 Interrupt Processing

Follow the steps below to process the interrupt:

Step 1 Write the bit [0] of the WDT_IRQ_EN_REG to 1 to enable the interrupt.

Step 2 After entering the interrupt process, write 1 to the bit [0] of the WDT_IRQ_STAT to clear the interrupt pending, and then process the interrupt.

Step 3 Resume the interrupt and continue to execute the interrupted process.

8.2.4.3 Watchdog Reset

Follow the steps below to reset the watchdog.

Step 1 Set the clock source as DCXO/750.

Step 2 Set the Interval Value as 1s.

Step 3 Configure Watchdog to reset the whole system.

```
write (0x16AA_0001, WDT_CFG_REG);
write (0x16AA_0010, WDT_MODE_REG);
write(read(WDT_MODE_REG) | (1<<0), WDT_MODE_REG);
```

8.2.4.4 Watchdog Restart

In the following instance making configurations for WDT:

Step 1 Set watchdog reset pulse width as 2ms.

Step 2 Set the interval value as 1s.

Step 3 Configure watchdog to reset the whole system.

In the normal dog feeding process, if the running time of other codes is longer than or equal to 1s, the watchdog will reset the whole system; if the running time of other codes is within 1s and the

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configuration of restart watchdog is implemented, the watchdog will be restarted. Note that the watchdog should also be restarted after changing the interval time.

The following are the procedures for the normal dog feeding and the interval value changing.

- Feed Dog Normally

```
writel(readl(WDT_CFG_REG) & (~ (0x1 << 0)) | (0x1 << 0), WDT_CFG_REG);
writel(readl(WDT_MODE_REG) & (~(0xF << 4)) | (0x1 << 4), WDT_MODE_REG);
writel(readl(WDT_MODE_REG) & (~(0x1 << 0)) | (0x1 << 0), WDOG_MODE_REG);
----other codes---
writel(readl(WDT_CTRL_REG) | (0xA57 << 1) & (~(0x1 << 0)) | (0x1 << 0), WDT_CTRL_REG);
```

- Change the Interval Value

```
writel(readl(WDT_CFG_REG) & (~(0x1 << 0)) | (0x1 << 0), WDT_CFG_REG);
writel(readl(WDT_MODE_REG) & (~(0xF << 4)) | (0x1 << 4), WDT_MODE_REG);
writel(readl(WDT_MODE_REG) & (~(0x1 << 0)) | (0x1 << 0), WDOG_MODE_REG);
----other codes---
writel(readl(WDT_MODE_REG) & (~(0xF << 4)), WDT_MODE_REG);
writel(readl(WDT_CTRL_REG) | (0xA57 << 1) & (~(0x1 << 0)) | (0x1 << 0), WDT_CTRL_REG);
```

8.2.5 Register List

Module Name	Base Address
WDT_CPUX	0x02050000
WDT_MCU	0x01A02000

Register Name	Offset	Description
WDT_IRQ_EN_REG	0x0000	WDT IRQ Enable Register
WDT_IRQ_STAT_REG	0x0004	WDT Status Register
WDT_SRST_REG	0x0008	WDT Software Reset Register
WDT_CTRL_REG	0x000C	WDT Control Register
WDT_CFG_REG	0x0010	WDT Configuration Register
WDT_MODE_REG	0x0014	WDT Mode Register
WDT_OCFG_REG	0x0018	WDT Output Configuration Register

8.2.6 Register Description

8.2.6.1 0x0000 WDT IRQ Enable Register (Default Value: 0x0000_0000)

Offset: 0x0000	Register Name: WDT_IRQ_EN_REG

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	WDT_IRQ_EN WDT Interrupt Enable 1: Enable 0: Disable

8.2.6.2 0x0004 WDT Status Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: WDT_IRQ_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/WAC	0x0	WDT_STAT WDT IRQ Pending. Setting 1 to the bit will clear it. 1: Pending. WDT interval value is reached. 0: No effect

8.2.6.3 0x0008 WDT Software Reset Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: WDT_SRST_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	KEY Field This field should be filled with 0x16AA, and then the bit 0 can be written with the new value.
15:1	/	/	/
0	R/W	0x0	SOFT_RST_EN Soft Reset Enable 0: De-assert 1: Reset System If this bit is used for system reset, the watchdog should be disabled.

8.2.6.4 0x000C WDT Control Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: WDT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:1	W	0x0	KEY_FIELD Watchdog Key Field This bit should be written at value 0xA57. Writing any other value in this field aborts the write operation.

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Offset: 0x000C			Register Name: WDT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W1S	0x0	WDT_RESTART Watchdog Restart 0: No effect 1: Restart the Watchdog 0

8.2.6.5 0x0010 WDT Configuration Register (Default Value: 0x0000_0001)

Offset: 0x0010			Register Name: WDT_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x16AA, and then the bit 15:0 can be written with the new value.
15:9	/	/	/
8	R/W	0x0	WDT_CLK_SEL WDT Clock select 0: SYS_24M_CLK/750 1: SYS_32K_CLK
7:2	/	/	/
1:0	R/W	0x1	WDT_CONFIG WDT Mode Configuration 00: To CPU core (only valid for WDT_MCU) 01: To whole system 10: Only interrupt 11: /

8.2.6.6 0x0014 WDT Mode Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: WDT_MODE_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x16AA, and then the bit 15:0 can be written with the new value.
15:10	/	/	/

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Offset: 0x0014			Register Name: WDT_MODE_REG
Bit	Read/Write	Default/Hex	Description
9:4	R/W	0x0	WDT_INTERVAL_VALUE WDT Interval Value Watchdog clock source is DCXO/750. If the clock source is turned off, WDT will not work. 000000: 16k cycles (0.5s) 000001: 32k cycles (1s) 000010: 64k cycles (2s) 000011: 96k cycles (3s) 000100: 128k cycles (4s) 000101: 160k cycles (5s) 000110: 192k cycles (6s) 000111: 256k cycles (8s) 001000: 320k cycles (10s) 001001: 384k cycles (12s) 001010: 448k cycles (14s) 001011: 512k cycles (16s) 001100: 576k cycles (18s) 001101: 640k cycles (20s) 001110: 704k cycles (22s) 001111: 768k cycles (24s) 010000: 832k cycles (26s) 010001: 896k cycles (28s) 010010: 960k cycles (30s) 010011: 1024k cycles (32s) 010000: 1088k cycles (34s) 010101: 1152k cycles (36s) 010110: 1216k cycles (38s) 010111: 1280k cycles (40s) 011000: 1344k cycles (42s) 011001: 1408k cycles (44s) 011010: 1472k cycles (46s) 011011: 1536k cycles (48s) 011100: 1600k cycles (50s) 011101: 1664k cycles (52s) 011110: 1728k cycles (54s) 011111: 1792k cycles (56s) 100000: 1856k cycles (58s) 100001: 1920k cycles (60s) 100010: 1984k cycles (62s) 100011: 2048k cycles (64s)
3:1	/	/	

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Offset: 0x0014			Register Name: WDT_MODE_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	WDT_EN Watchdog Enable 0: Disable 1: Enable

8.2.6.7 0x0018 WDT Output Configuration Register (Default Value: 0x0000_003F)

Offset: 0x0018			Register Name: WDT_OCFG_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x16AA, and then the bit 11:0 can be written with the new value.
15:12	/	/	/
11:0	R/W	0x3F	WDT_OUT_CONFIG WDT Reset Valid Time Configuration T=1/32ms*(N) Default 2ms

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9 RTC

9.1 RTC

9.1.1 Overview

The Real Time Clock (RTC) is used to implement the time counter and the timing wakeup functions. The RTC can display the year, month, day, week, hour, minute, second in real time. With the independent power, the RTC can still work when system powers off.

The RTC has the following features:

- Embedded with a 16-bit day counter, a 5-bit hour counter, a 6-bit minute counter, and a 6-bit second counter
- Uses a 32.867 kHz low-frequency oscillator as the count clock
- 1 kHz counting frequency
- Configurable initial value by software anytime
- Supports generating interrupt and waking up the external devices with the timing alarms function
- Supports fanout function of internal 32K clock
- 8 general purpose registers for storing the power-off information in RTC domain
- Multiple special registers for recording the BROM information
- Supports recording reboot events

NOTE

The register configuration of RTC is AHB bus. It only can support word operation, not byte operation and half-word operation.

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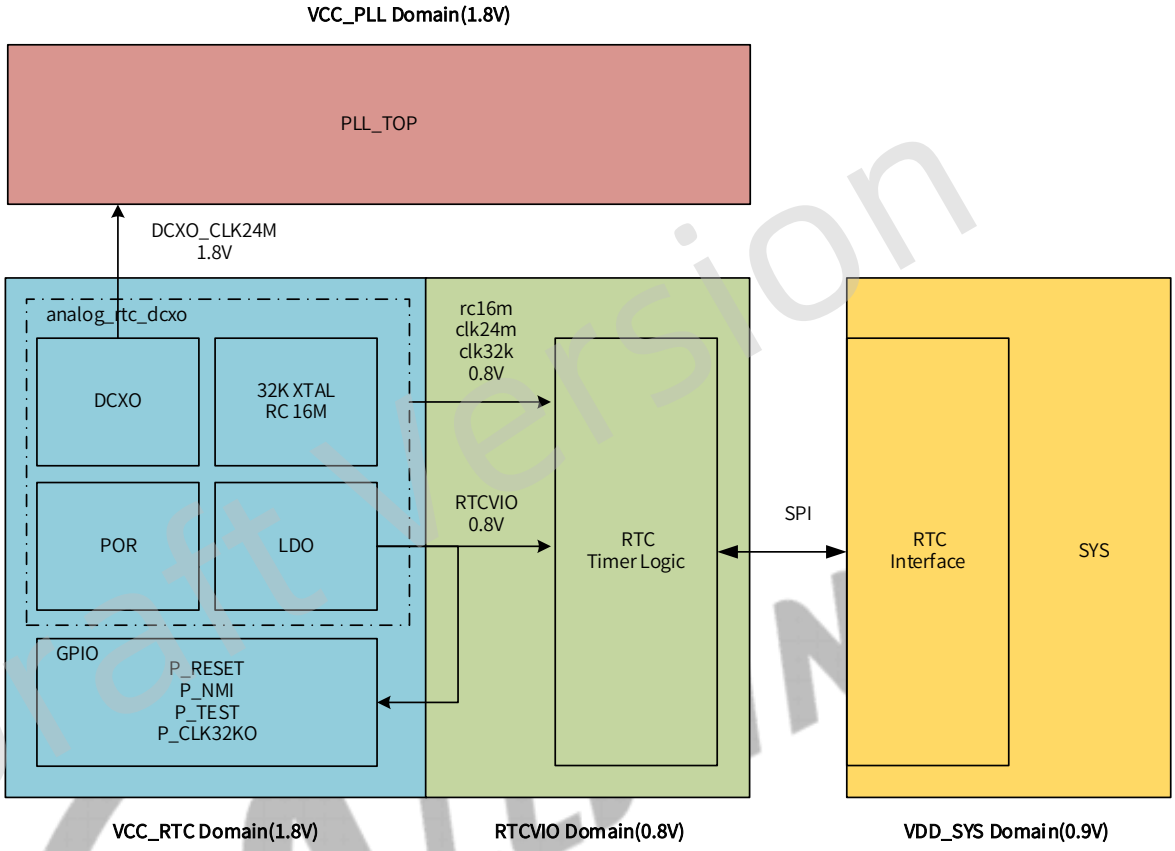
9.1.2 Block Diagram

The following figure shows the block diagram of the RTC.

Figure 9-1 RTC Block Diagram



9.1.3 External Signals

Table 9-1 RTC External Signals

Signal Name	Description	Type
X32KFOUT	32.768 kHz clock Fanout Provides low frequency clock for external devices	AO,OD
X32KIN	Clock Input of 32.768 kHz Crystal	AI
X32KOUT	Clock Output of 32.768 kHz Crystal	AO
DXIN	Digital Compensated Crystal Oscillator Input	AI
DXOUT	Digital Compensated Crystal Oscillator Output	AO
REFCLK-OUT	Digital Compensated Crystal Oscillator Clock Fanout	AO
RESET	Reset Signal (Low Active)	I/O, OD
VCC-RTC	RTC Power	P
TEST	Test Signal	I

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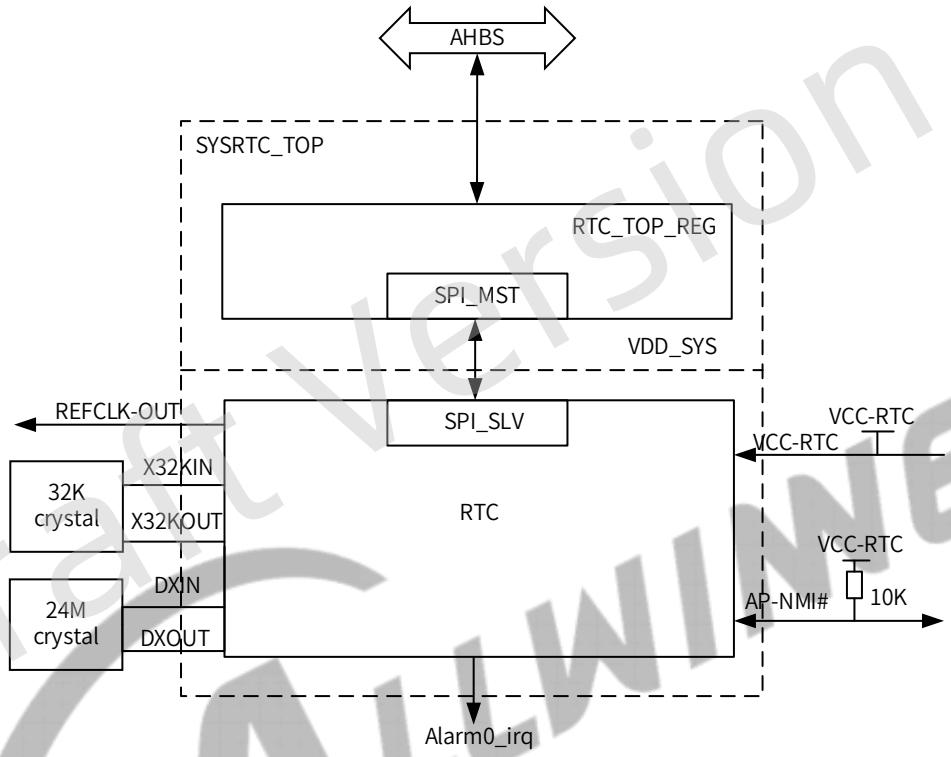
9.1.4 Functional Descriptions

9.1.4.1 Typical Application

Figure 9-2 RTC Application Diagram



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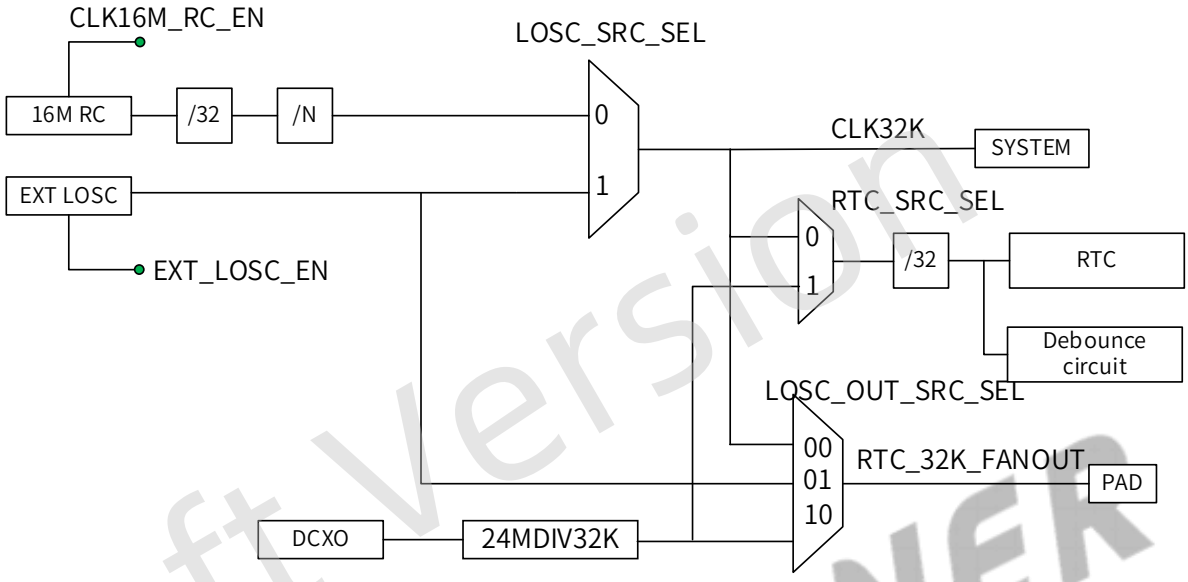
9.1.4.2 RTC Clock Tree

The following figure shows the clock tree of the RTC.

Figure 9-3 RTC Clock Tree



RTC

The clock sources of RTC can be selected by related switches, including 32K divided from internal 16 MHz RC, external 32.768 kHz crystal, and 32K divided from external DCXO.

System 32K

The clock sources of system 32K are from external 32.768 kHz crystal and 32K divided from the internal 16 MHz RC.

RTC_32K_FANOUT

The clock source of RTC_32K_FANOUT can select CLK32K, external 32.768 kHz crystal, or 32K divided from external DCXO.

9.1.4.3 RTC Clock Sources

There are three clock sources for RTC: Internal RC, external low-frequency oscillator, and DCXO divided clock.

The internal RC will be selected by default. After the system starts up, select the external low-frequency oscillator by software to provide more accurate clock.

When using internal RC, the clock can be changed by changing division ratio. When using external clock, the clock cannot be changed.

The RTC clock accuracy is related with external low-frequency oscillator accuracy (32.768K usually).

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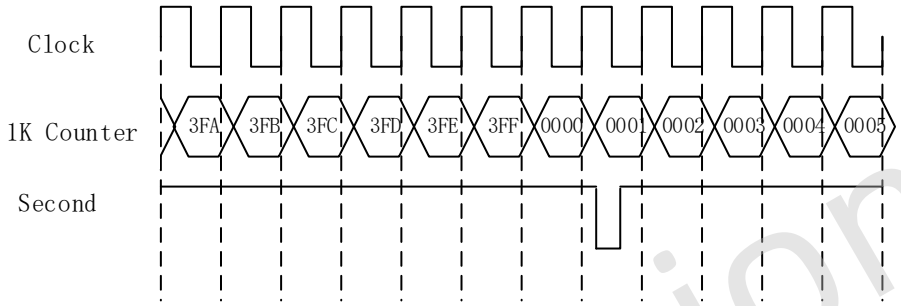
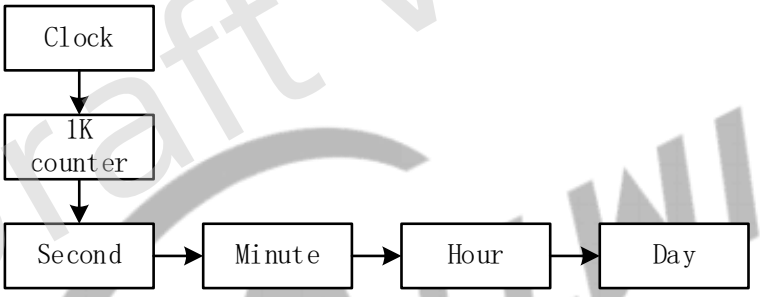
9.1.4.4 Real Time Clock

Figure 9-4 RTC Counter



The 1 kHz counter adds 1 on each rising edge of the clock. When the sampling clock number reaches 0x3FF, 1 kHz counter restarts to count from 0, and the second counter adds 1. The step structure of 1 kHz counter is as follows.

Figure 9-5 RTC 1 kHz Counter Step Structure



According to above implementation, the changing range of each counter is as follows.

Table 9-2 RTC Counter Changing Range

Counter	Range
Second	0 to 59
Minute	0 to 59
Hour	0 to 23
Day	0 to 65535 (The year, month, day need be transformed by software according to day counter)

CAUTION

As there is no error correction mechanism in the hardware, each counter configuration should not exceed a reasonable counting range.

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9.1.4.5 Alarm 0

The principle of alarm0 is a comparator. When RTC timer reaches scheduled time, the RTC generates the interrupt, or outputs low level signal by NMI pin to wakeup power management chip.

The RTC only generates one interrupt when RTC timer reached the scheduled day, hour, minute and second counter, then the RTC need set a new scheduled time, the next interrupt can be generated.

9.1.4.6 VDD-RTC

The RTC module has a LDO, the input source of the LDO is VCC-RTC, the output of the LDO is RTC_VIO. the RTC_VIO is mainly used for internal digital logic.

9.1.4.7 Reboot Event Record

RTC supports recoding reboot events caused by hardware reset, watchdog reset, abnormal power-off of system digital power and system IO power. The following table describes specific functions and their configuration registers.

Functions	Registers
Enable reboot record function Configure record function clock Mask reboot sources	RBT_RCD_CTRL_REG
Read the recorded number of reboot events	RBT_RCD_STAT_REG0/1
Read the recorded time of the latest reboot event (no records before cold boot are saved.)	RBT_RCD_LST_DATE_REG RBT_RCD_LST_TIM_REG
Clear the number and time of reboots	RBT_RCD_STAT_CLR_REG

9.1.4.8 NMI Pin Interrupt Generation

The NMI interrupt generation module collects ALARM interrupts, generates a cycle-configurable square-wave signal, and output it to the PMU through NMI pin in order to wake up the master control. This function can be masked, after masking the ALARM interrupt will be sent to the PMU through the NMI pin straightly.

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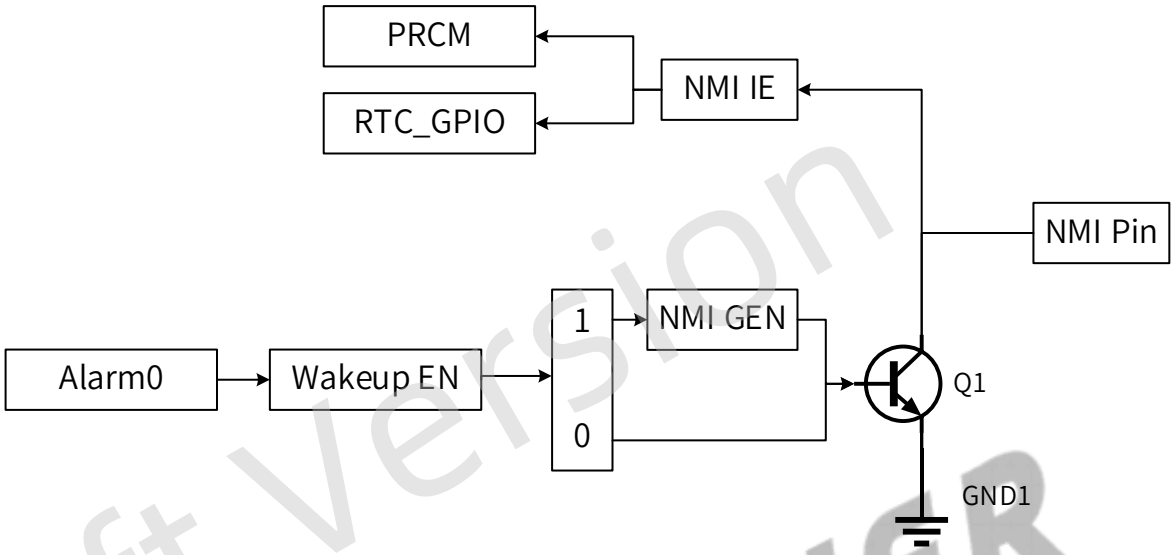
Figure 9-6 NMI Block Diagram



Parameters	Configuration Register
Wakeup EN	ALARM0_CONFIG_REG [0]
NMI GEN	PAD_CTRL_REG [8]
NMI IE	PWR_EN_CFG_REG [30]

9.1.5 Programming Guidelines

9.1.5.1 Clock Source Selection

Select 24MDIV32K

Set RTC_SRC_SEL of the LOSC_CTRL_REG register to 1.

Select 32K Divided from RC16M/EXT LOSC

- **Auto Switch to 32K Divided from RC16M/EXT LOSC**

Step 1 Write 1 to the LOSC_AUTO_SWT_32K_SEL_EN of the LOSC_CTRL_REG register.

Step 2 Write 1 to the EXT_LOSC_EN of the LOSC_CTRL_REG register.

Step 3 Write 0 to the LOSC_AUTO_SWT_FUNCTION of the LOSC_CTRL_REG register.

Step 4 Read the EXT_LOSC_STA of the LOSC_AUTO_SWT_STA_REG register for four times. Before every reading, you should delay at least 4 ms (such as 6 ms). If the value of every read is 0, it indicates that the external 32K has been enabled. Otherwise, repeat reading the EXT_LOSC_STA.

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Step 5 Write 1 to the LOSC_SRC_SEL of the LOSC_CTRL_REG register.

Step 6 Read the LOSC_SRC_SEL_STA of the LOSC_AUTO_SWT_STA_REG register. If the value of LOSC_SRC_SEL_STA equals to 1, it means the current low frequency clock source is external 32K. Writing 0 to LOSC_SRC_SEL can switch the clock source to the 32K divided from RC16M.

NOTE

If LOSC does not exist or is disabled, the value of LOSC_SRC_SEL_STA (the bit [0] of the LOSC_AUTO_SWT_STA_REG register) is 0, which indicates that the 32K divided from RC16M is selected as the clock source. In this case, writing 1 to the LOSC_SRC_SEL of the LOSC_CTRL_REG register will not change the clock source.

- **Manual Switch to 32K Divided from RC16M/EXT LOSC**

Step 1 Write 1 to the LOSC_AUTO_SWT_FUNCTION of the LOSC_CTRL_REG register to disable LOSC detection.

Step 2 Write 0 to the LOSC_AUTO_SWT_32K_SEL_EN of the LOSC_CTRL_REG register to disable auto-switch function.

NOTE

After Step 1 is performed, whether the EXT LOSC is lost will not be recorded. It is uncertain whether the LOSC exists. Therefore, ensure that the value of the following bits is 1 before performing Step 3:

EXT_LOSC_EN, the bit [4] of the LOSC_CTRL_REG register; the value is 1 by default.

CLK16M_RC_EN, the bit [0] of the DCXO_CTRL_REG register; the value is 1 by default.

Step 3 To select external 32K, write 1 to the LOSC_SRC_SEL; to select the 32K divided from RC16M, write 0 to the LOSC_SRC_SEL.

9.1.5.2 RTC Calendar

Step 1 Write time initial value: Write the current time to RTC_DAY_REG and RTC_HH_MM_SS_SET_REG.

Step 2 After updated time, the RTC restarts to count again. The software can read the current time anytime.

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NOTE

- The RTC can only provide day counter, so the current day counter need be converted to year, month, day and week by software.
- Ensure the bit [8:7] of LOSC_CTRL_REG is 0 before the next time configuration is performed.

Here are the basic code samples.

For example: set time to 21st, 07:08:09 and read it.

RTC_DAY_REG = 0x00000015;

RTC_HH_MM_SS_REG = 0x00070809; //0000 0000 000|0 0000(Hour) 00|00 0000(Minute) 00|00 0000(Second)

Read (RTC_DAY_REG);

Read (RTC_HH_MM_SS_REG);

9.1.5.3 Alarm0 Setting

Step 1 Enable alram0 interrupt by writing 1 to ALARM0_IRQ_EN.

Step 2 Write the count-down day, hour, minute, second number to ALARM0_DAY_SET_REG and ALARM0_HH_MM_SS_SET_REG.

Step 3 Enable alarm0 function by writing 1 to ALARM0_ENABLE_REG, then the software can query alarm count value in real time by RTC_DAY_REG and RTC_HH_MM_SS_SET_REG. When the setting time is reached, ALARM0_IRQ_STA_REG is set to 1 to generate an interrupt.

Step 4 After entering the interrupt process, write 0 to the ALARM0_IRQ_STA_REG to clear the interrupt pending, and execute the interrupt process.

Step 5 Resume the interrupt and continue to execute the interrupted process.

Step 6 The power-off wakeup is generated via SoC hardware and PMU, the software only needs to set the pending condition of alarm0, and set ALARM0_CONFIG_REG to 1.

Here are the basic code samples.

irq_request(GIC_SRC_R_Alarm0, Alm0_handler);

irq_enable(GIC_SRC_R_Alarm0);

writel(1, ALARM0_DAY_SET_REG);

writel(1, ALARM0_HH_MM_SS_SET_REG); //set 1 second corresponding to normal mode;

writel(1, ALM0_EN);

writel(1, ALM_CONFIG); //NMI output

while(!readl(ALM0_IRQ_STA));

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```
writel(1, ALM0_IRQ_EN);
while(readl(ALM0_IRQ_STA));
```

9.1.5.4 Clock Fanout

CLK32K Fanout

Set the LOSC_OUT_GATING bit (bit [0]) of CLK32K_FOUT_CTRL_REG register to 1, and ensure external pull-up resistor, voltage, and clock source are normal, then 32.768kHz square wave can be output.

CLK24M Fanout

To fanout CLK24M clock though REFCLK-OUT pin, configure the CLK_REQ_ENB bit (bit [31]) of DCXO_CTRL_REG register.

9.1.6 Register List

Module Name	Base Address
RTC	0x07090000

Register Name	Offset	Description
VDD-RTC Domain		
LOSC_CTRL_REG	0x0000	LOSC Control Register
LOSC_AUTO_SWT_STA_REG	0x0004	LOSC Auto Switch Status Register
INTOSC_CLK_PRESCAL_REG	0x0008	Internal OSC Clock Prescalar Register
RTC_DAY_REG	0x0010	RTC Year-Month-Day Register
RTC_HH_MM_SS_SET_REG	0x0014	RTC Hour-Minute-Second Register
ALARM0_DAY_SET_REG	0x0020	Alarm 0 Day Set Register
ALARM0_HH_MM_SS_SET_REG	0x0024	Alarm 0 Hour-Minute-Second Set Register
ALARM0_ENABLE_REG	0x0028	Alarm 0 Enable Register
ALARM0_IRQ_EN	0x002C	Alarm 0 IRQ Enable Register
ALARM0_IRQ_STA_REG	0x0030	Alarm 0 IRQ Status Register
TIMER_SOFT_RST_REG	0x0040	Timer Software Reset Register
ALARM0_CONFIG_REG	0x0050	Alarm 0 Config Register
CLK32K_FOUT_CTRL_REG	0x0060	CLK32K Fanout Control Register
GP_DATA_REGn	0x0100+N*0x0004 (N=0-7)	General Purpose Register
DCXO_CTRL_WP_REG	0x015C	DCXO Control Write Protect Register
DCXO_CTRL_REG	0x0160	DCXO Control Register
RTC_DBG_REG	0x0170	RTC Debug Register
VDD_RTC_REG	0x0190	VDD RTC Regulation Register
IC_CHARA_REG	0x01F0	IC Characteristic Register
VDD_OFF_GATING_CTRL_REG	0x01F4	VDD Off Gating Control Register

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Register Name	Offset	Description
EFUSE_HV_PWRSWT_CTRL_REG	0x0204	Efuse High Voltage Power Switch Control Register
PAD_CTRL_REG	0x0208	PAD Control Register
PWR_EN_CFG_REG	0x0230	Power en pin Control Register
PMC_BYP_ST_REG	0x0238	PMC Bypass Status Register
RBT_RCD_CTRL_REG	0x0240	Reboot Event Recording Control Register
RBT_RCD_STAT_REG0	0x0244	Reboot Event Recording State Register0
RBT_RCD_STAT_REG1	0x0248	Reboot Event Recording State Register1
RBT_RCD_LST_DATE_REG	0x0250	Last Reboot Event Recording Date Register
RBT_RCD_LST_TIM_REG	0x0254	Last Reboot Event Recording Time Register
RBT_RCD_STAT_CLR_REG	0x0258	Reboot Event Recording State Clear Register
VDD-SYS Domain		
RTC_SPI_CLK_CTRL_REG	0x0310	RTC SPI Clock Control Register

9.1.7 Register Description

9.1.7.1 0x0000 LOSC Control Register (Default Value: 0x0000_4010)

Offset: 0x0000			Register Name: LOSC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD KEY Field This field should be filled with 0x16AA, and then the bit 0 and bit 1 can be written with the new value.
15	R/W	0x0	LOSC_AUTO_SWT_FUNCTION LOSC auto switch function disable. 0: Enable 1: Disable
14	R/W	0x1	LOSC_AUTO_SWT_32K_SEL_EN LOSC auto switch 32K clk source sel enable. 0: Disable, when losc lost, 32k clk source will not change to RC 1: Enable, when losc lost, 32k clk source will change to RC (LOSC_SRC_SEL will be changed from 1 to 0)
13:9	/	/	/

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Offset: 0x0000			Register Name: LOSC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
8	R	0x0	RTC_HHMMSS_ACCE RTC Hour Minute Second access. After writing the RTC HH-MM-SS register, this bit is set and it will be cleared until the real writing operation is finished. After writing the RTC HH-MM-SS register, the HH-MM-SS register will be refreshed for at most one second. Note: Make sure that the bit is 0 for time configuration.
7	R	0x0	RTC_DAY_ACCE RTC DAY access. After writing the RTC DAY register, this bit is set and it will be cleared until the real writing operation is finished. After writing the RTC DAY register, the DAY register will be refreshed for at most one second. Note: Make sure that the bit is 0 for time configuration.
6:5	/	/	/
4	R/W	0x1	EXT_LOSC_EN External 32768Hz Crystal Enable. 0: Disable 1: Enable
3:2	R/W	0x0	EXT_LOSC_GSM External 32768Hz Crystal GSM. 00: Low 01: Reserved 10: Reserved 11: High When GSM is changed, the 32K oscillation circuit will arise transient instability. If the autoswitch function (bit 15) is enabled, 32K changes to RC16M with certain probability. The GSM can influence the time of 32K starting oscillation, the more the GSM, the shorter the time of starting oscillation. So modifying GSM is not recommended. If you need to modify the GSM, firstly disable the auto switch function (bit 15), with a delay of 50 us, then change the GSM, the 32K clock source is changed to external clock.

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Offset: 0x0000			Register Name: LOSC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	RTC_SRC_SEL RTC_TIMER Clock Source Select. 0: LOSC_SRC 1: 24MDIV32K Before switching the bit, make sure that the 24MDIV32K function is enabled, that is, the bit16 of the CLK32K_FOUT_CTRL_REG is 1.
0	R/W	0x0	LOSC_SRC_SEL LOSC Clock Source Select. 'N' is the value of Internal OSC Clock Prescalar register. 0: Low Frequency Clock from 16M RC 1: External 32.768KHz OSC.

9.1.7.2 0x0004 LOSC Auto Switch Status Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: LOSC_AUTO_SWT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R	0x0	EXT_LOSC_STA Work only when AUTO SWITCH function is enable. 0: External 32.768k OSC work normally. 1: External 32.768k OSC work abnormally.
1	R/W1C	0x0	LOSC_AUTO_SWT_PEND LOSC auto switch pending. 0: No effect 1: Auto switches pending, means LOSC_SRC_SEL is changed from 1 to 0.
0	R	0x0	LOSC_SRC_SEL_STA Checking LOSC Clock Source Status. 'N' is the value of Internal OSC Clock Prescalar register. 0: Low Frequency Clock from 16M RC 1: External 32.768KHz OSC

9.1.7.3 0x0008 Internal OSC Clock Prescalar Register (Default Value: 0x0000_000F)

Offset: 0x0008			Register Name: INTOSC_CLK_PRESCAL_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/

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Offset: 0x0008			Register Name: INTOSC_CLK_PRESCAL_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0xF	INTOSC_32K_CLK_PRESCAL Internal OSC 32K Clock Prescalar value N. The clock output = Internal RC/32 / N. 00000: 1 00001: 2 00002: 3 ..... 11111: 32

9.1.7.4 0x0010 RTC Year-Month-DAY Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: RTC_DAY_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	UDF	DAY Set DAY. Range from 0-65535.

9.1.7.5 0x0014 RTC Hour-Minute-Second Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: RTC_HH_MM_SS_SET_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	UDF	HOUR Set Hour. Range from 0-23
15:14	/	/	/
13:8	R/W	UDF	MINUTE Set Minute. Range from 0-59
7:6	/	/	/
5:0	R/W	UDF	SECOND Set second. Range from 0-59

9.1.7.6 0x0020 Alarm 0 Day Set Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: ALARM0_DAY_SET_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0020			Register Name: ALARM0_DAY_SET_REG
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	ALARM0_COUNTER Alarm 0 Counter is Based on Day.

9.1.7.7 0x0024 Alarm0 Hour-Minute-Second Set Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: ALARM0_HH_MM_SS_SET_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x0	HOUR Current hour. Range from 0-23
15:14	/	/	/
13:8	R/W	0x0	MINUTE Current minute. Range from 0-59
7:6	/	/	/
5:0	R/W	0x0	SECOND Current second. Range from 0-59

9.1.7.8 0x0028 Alarm 0 Enable Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: ALARM0_ENABLE_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	ALM_0_EN Alarm 0 Enable. 0: Disable 1: Enable

9.1.7.9 0x002C Alarm 0 IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x002C			Register Name: ALARM0_IRQ_EN
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	ALARM0_IRQ_EN Alarm 0 IRQ Enable. 0: Disable 1: Enable

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9.1.7.10 0x0030 Alarm 0 IRQ Status Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: ALARM0_IRQ_STA_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W1C	0x0	ALARM0_IRQ_PEND Alarm 0 IRQ Pending bit. 0: No effect 1: Pending, alarm 0 counter value is reached If alarm 0 irq enable is set to 1, the pending bit will be sent to the interrupt controller.

9.1.7.11 0x0040 Timer Software Reset Register (Default Value:0x0000_0000)

Offset: 0x0040			Register Name: TIMER_SOFT_RST_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	KEY FILED. Write Protect. To configure Timer soft reset write protection on software, first configure bits [31:16] to 0x16AA and write 0xAA16_XXX before writing bits [15:0].
15:1	/	/	/
0	R/WAC	0x0	TIMER_SOFT_RST Timer RTC logic software reset. 0: No-effect 1: Reset Note: Make sure that the bit is 0 for time configuration.

9.1.7.12 0x0050 Alarm 0 Config Register (Default Value: 0x0000_0000)

Offset: 0x0050			Register Name: ALARM0_CONFIG_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	ALARM0_WAKEUP Configuration of alarm wake up output. 0: Disable alarm wake up output 1: Enable alarm wake up output

9.1.7.13 0x0060 CLK32K Fanout Control Register (Default Value: 0x0000_0000)

Offset:0x0060	Register Name: CLK32K_FOUT_CTRL_REG

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Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	HOSC_TO_32K_DIVIDER_ENABLE Hosc to 32k divider enable. 1: Enable the HOSC 24M to 32k divider circuit 0: Disable the HOSC 24M to 32k divider circuit
15:3	/	/	/
2:1	R/W	0x0	LOSC_OUT_SRC_SEL Losc out source select. 00: RTC_32K (select by LOSC_AUTO_SWT_32K_SEL_EN & LOSC_SRC_SEL) 01: LOSC 10: HOSC divided 32K
0	R/W	0x0	LOSC_OUT_GATING Losc out gating enable. Configuration of LOSC output, and no LOSC output by default. 0: Mask LOSC output gating 1: Enable LOSC output gating

9.1.7.14 0x0100+N*0x0004 (N=0-7) General Purpose Register (Default Value: 0x0000_0000)

Offset: 0x0100+N*0x0004 (N=0-7)			Register Name: GP_DATA_REGn
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	GP_DATA Data.

NOTE

General purpose register 0 to 7 value can be stored if the RTC-VIO is larger than 0.7 V.

9.1.7.15 0x015C DCXO Control Write Protect Register (Default Value:0x0000_0000)

Offset: 0x015C			Register Name: DCXO_CTRL_WP_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	W	0x0	KEY FILED. Write protect for DCXO_CTRL_REG(0x0160). Write this field as 0x16AA before configuring DCXO_CTRL_REG(0x0160) and this field will be automatically cleared to 0 after DCXO_CTRL_REG is configured.

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9.1.7.16 0x0160 DCXO Control Register (Default Value: 0x883F_10F3)

Offset: 0x0160			Register Name: DCXO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	CLK_REQ_ENB Clk req enable. 0: Enable DCXO wake up function 1: Disable DCXO wake up function
30:28	/	/	/
27:24	R/W	0x8	DCXO_ICTRL DCXO current control value.
23	/	/	/
22:16	R/W	0x3F	DCXO_TRIM DCXO cap array value. cap cell is 55fF
15:13	/	/	/
12:8	R/W	0x10	DCXO_BG DCXO bandgap output voltage.
7	R/W	0x1	DCXO_LDO_INRUSHB DCXO LDO driving capacity signal active high.
6	R/W	0x1	XTAL_MODE Xtal mode enable signal active high. 0x0 for external clk input mode. 0x1 for normal mode.
5:4	R/W	0x3	DCXO_RFCLK_ENHANCE DCXO rfclk enhance. Enhance driving capacity of output OUT_RF_REFCLK, 0x0 for 5pF, 0x1 for 10pF, 0x2 for 15pF, 0x3 for 20pF
3:2	/	/	/
1	R/W	0x1	DCXO_EN DCXO enable. 1: Enable 0: Disable

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Offset: 0x0160			Register Name: DCXO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	CLK16M_RC_EN Clock RC16M enable. 1: Enable 0: Disable The related register configuration is necessary to ensure the reset debounce circuit has a stable clock source. The first time SoC starts up, by default, the reset debounce circuit of SoC uses 32K divided by RC16M. In power-off, software reads the related bit to ensure whether EXT32K is working normally, if it is normal, first switch the clock source of debounce circuit to EXT32K, then close RC16M. Without EXT32K scenario or external RTC scenario, software confirms firstly whether EXT32K is working normally before switching, or software does not close RC16M.

9.1.7.17 0x0170 RTC Debug Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: RTC_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4	R/W	0x0	DCXO_AUTO_EN. DCXO auto close function enable. 1: Enable 0: Disable If this bit set 1, DCXO will be closed when reset pull down.
3:2	/	/	/
1	R/W	0x0	RTC_TEST_MODE_CTRL RTC TEST Mode Control bit.
0	R/W	0x0	RTC_DEBUG RTC Simulation Control bit. 0: No effect 1: Simulation mode

9.1.7.18 0x0190 VDD RTC Regulation Register (Default Value: 0x0000_0003)

Offset: 0x0190			Register Name: VDD_RTC_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0190			Register Name: VDD_RTC_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4	R/W	0x0	V_SEL VDD select. 0: Resistance divider 1: Band bias
3	/	/	/
2:0	R/W	0x3	VDD_RTC_REGU RTC VIO voltage select. The RTC-VIO is provided power for RTC digital part. These bits are useful for regulating the RTC_VIO from 0.65V to 1.3V. 000: 1.0 V 001: 0.65 V 010: 0.7 V 011: 0.8 V 100: 0.9 V 101: 1.1 V 110: 1.2 V 111: 1.3 V

9.1.7.19 0x01F0 IC Characteristic Register (Default Value: 0x0000_0000)

Offset: 0x01F0			Register Name: IC_CHARA_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	KEY_FIELD Write protect. Should be written at value 0x16AA. Writing any other value in this field aborts the write operation.
15:0	R/W	0x0	ID_DATA IC_CHARA. Return 0x16AA only if the KEY_FIELD is set as 0x16AA when read those bits, otherwise return 0x0.

9.1.7.20 0x01F4 VDD Off Gating Control Register (Default Value: 0x0000_4021)

Offset: 0x01F4			Register Name: VDD_OFF_GATING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Write protect. This field should be filled with 0x16AA, and then the bit 15 can be configured.

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Offset: 0x01F4			Register Name: VDD_OFF_GATING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
15	WAC	0x0	PWROFF_GAT_RTC_CFG. Power off gating control signal. When use VDD_SYS to RTC isolation software control, write this bit 1, it will only be clear by resetb release. (For Debug Use Only)
14	R/W	0x1	POR_RST_OUT_EN POR reset output enable. 0: Disable 1: Enable
13:12	/	/	/
11:4	R/W	0x2	VCCIO_DET_SPARE VCCIO detect spare. **Bit [7:6]: Reserved** **Bit [5]: Test select output, default=0** 0: DCXO 24M CLK 1: RTC_VIO LDO **Bit [4]: Bypass debounce circuit, default=0** 0: Bypass 1: No bypass **Bit [3]: Enable control, default=0** 0: Disable VCC-IO detection 1: Force the detection output **Bit [2:0]: Gear adjustment** 000: Detection threshold is 2.5 V 001: Detection threshold is 2.6 V 010: Detection threshold is 2.7 V 011: Detection threshold is 2.8 V 100: Detection threshold is 2.9 V 101: Detection threshold is 3 V 110: N/A 111: N/A
3	R/W	0x0	DRAM_CKE_IO_SEL Select DRAM CKE PAD IO Status in auto-flush mode. 0: Tie0 1: HighZ

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Offset: 0x01F4			Register Name: VDD_OFF_GATING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	DRAM_VDDON_POR DRAM VDDON POR 0: Assert 1: Deassert Note: This bit should be set to 0 before VDD_SYS power off while it should be set to 1 after the VDD_SYS power on.
1	R/W	0x0	VDDSYS_DET_BYPASS_EN VDDSYS detect bypass enable 0: Not bypass 1: Bypass
0	R/W	0x1	VCCIO_DET_BYPASS_EN VCCIO detect bypass enable. 0: Not bypass 1: Bypass

9.1.7.21 0x0204 Efuse High Voltage Power Switch Control Register (Default Value: 0x0000_0000)

Offset: 0x0204			Register Name: EFUSE_HV_PWRSWT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	EFUSE_1P8V_POWER_SWITCH_CONTROL EFUSE power switch control. 1: Open power switch 0: Close power switch

9.1.7.22 0x0208 PAD Control Register (Default Value: 0x0000_00C0)

Offset:0x0208			Register Name: PAD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/WAC	0x0	NMI_PENDING IRQ Pending bit. 0: No effect 1: Pending, Set 1 to this bit will clear it.
15:9	/	/	/
8	R/W	0x0	NMI_GEN_EN 0: Disable 1: Enable

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Offset:0x0208			Register Name: PAD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
7:6	R/W	0x3	NMI_DLY_SEL 00: 8ms 01: 16ms 10: 32ms 11: 64ms
5	R/W	0x0	Reserved
4:0	/	/	/

9.1.7.23 0x0230 Power en pin Control Register(Default Value:0x00AA_0000)

Offset: 0x0230			Register Name:PWR_EN_CFG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	RESETB_IE Resetb input enable 0: Enable 1: Disable
30	R/W	0x0	NMI_IE NMI input enable 0: Enable 1: Disable
29:24	/	/	/
23:22	R/W	0x2	Reserved
21:20	R/W	0x2	Reserved
19:18	R/W	0x2	Reserved
17:16	R/W	0x2	Reserved
15:12	/	/	/
11	R/W	0x0	Reserved
10	R/W	0x0	Reserved
9	R/W	0x0	Reserved
8:0	/	/	/

9.1.7.24 0x0238 PMC Bypass Status Register (Default Value:0x0000_0002)

Offset: 0x0238			Register Name: PMC_BYP_ST_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	KEY FIELD. To configure PMC status write protection on software, first configure bits [31:16] to 0x16AA and write 0xAA16_XXX before writing bits [15:0].
15:2	/	/	/

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Offset: 0x0238			Register Name: PMC_BYP_ST_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x1	PMC_BYP_PULL PMC Bypass pull register. 0: NO PULL-UP 1: PULL-UP Set this bit to 0, otherwise the power consumption will increase during power-off process.
0	/	/	/

9.1.7.25 0x0240 Reboot Event Recording Control Register (Default Value: 0x0000_0000)

Offset: 0x0240			Register Name: RBT_RCD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R/W	0x0	RBT_SRC_MCUWDT_MASK Reboot Source MCU WDT Mask Bit. 0: Unmask 1: Mask
20	R/W	0x0	RBT_SRC_SYSWDT_MASK Reboot Source System WDT Mask Bit. 0: Unmask 1: Mask
19	R/W	0x0	RBT_SRC_HRDRST_MASK Reboot Source Hard Reset Mask Bit. 0: Unmask 1: Mask
18	R/W	0x0	RBT_SRC_VCCIODET_MASK Reboot Source VCCIO Detect Mask Bit. 0: Unmask 1: Mask
17	R/W	0x0	RBT_SRC_VDDSYSDET_MASK Reboot Source VDDSYS Detect Mask Bit. 0: Unmask 1: Mask
16	R/W	0x0	RBT_SRC_TWD_MASK Reboot Source TWD Mask Bit. 0: Unmask 1: Mask
15:11	/	/	/

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Offset: 0x0240			Register Name: RBT_RCD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	RBT_RCD_CLK_SRC_SEL Reboot Event Record Clock Source Select. 000: RC16M 111: DCXO Others: /
7:1	/	/	/
0	R/W	0x0	RBT_RCD_EN Reboot Event Record Enable. 0: Disable 1: Enable

9.1.7.26 0x0244 Reboot Event Recording State Register0 (Default Value: 0x0000_0000)

Offset: 0x0244			Register Name: RBT_RCD_STAT_REG0
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R	0x0	RBT_SRC_HRDRST_TIG_TIM Reboot Source Hard Reset Trigger Time. Support Record up to 32 times Hard Reset Event.
23:21	/	/	/
20:16	R	0x0	RBT_SRC_VCCIODET_TIG_TIM Reboot Source VCCIO Detect Trigger Time. Support Record up to 32 times VCCIO Detect Reset Event.
15:13	/	/	/
12:8	R	0x0	RBT_SRC_VDDSYSDET_TIG_TIM Reboot Source VDDSYS Detect Trigger Time. Support Record up to 32 times VDDSYS Detect Reset Event.
7:5	/	/	/
4:0	R	0x0	RBT_SRC_TWD_TIG_TIM Reboot Source TWD Trigger Time. Support Record up to 32 times TWD Reset Event.

9.1.7.27 0x0248 Reboot Event Recording State Register1 (Default Value: 0x0000_0000)

Offset: 0x0248			Register Name: RBT_RCD_STAT_REG1
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/

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Offset: 0x0248			Register Name: RBT_RCD_STAT_REG1
Bit	Read/Write	Default/Hex	Description
12:8	R	0x0	RBT_SRC_MCUWDT_TIG_TIM Reboot Source WDT_MCU Trigger Time. Support Record up to 32 times WDT Reset Event.
7:5	/	/	/
4:0	R	0x0	RBT_SRC_SYSWDT_TIG_TIM Reboot Source WDT_CPUX Trigger Time. Support Record up to 32 times WDT Reset Event.

9.1.7.28 0x0250 Last Reboot Event Recording Date Register (Default Value: 0x0000_0000)

Offset: 0x0250			Register Name: RBT_RCD_LST_DATE_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
15:0	R	0x0	RBT_RCD_LST_DATE Last Reboot Event Recording Date.

9.1.7.29 0x0254 Last Reboot Event Recording Time Register (Default Value: 0x0000_0000)

Offset: 0x0254			Register Name: RBT_RCD_LST_TIM_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R	0x0	RBT_RCD_LST_H Set Hour. Range from 0-23
15:14	/	/	/
13:8	R	0x0	RBT_RCD_LST_M Set Minute. Range from 0-59
7:6	/	/	/
5:0	R	0x0	RBT_RCD_LST_S Last Reboot Recording. Range from 0-59

9.1.7.30 0x0258 Reboot Event Recording State Clear Register (Default Value: 0x0000_0000)

Offset: 0x0258			Register Name: RBT_RCD_STAT_CLR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0258			Register Name: RBT_RCD_STAT_CLR_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	RBT_RCD_LST_TIM_CLR Last Reboot Record Time Clear. 1: Clear RBT_RCD_LST_TIM_REG 0: Invalid
15:11	/	/	/
10	R/W	0x0	RBT_SRC_MCUWDT_RCD_STAT_CLR Reboot Source WDT_MCU Recording State Clear. 1: Clear RBT_SRC_MCUWDT_TIG_TIM 0: Invalid
9	/	/	/
8	R/W	0x0	RBT_SRC_SYSWDT_RCD_STAT_CLR Reboot Source WDT_CPUX Recording State Clear. 1: Clear RBT_SRC_SYSWDT_TIG_TIM 0: Invalid
7			
6	R/W	0x0	RBT_SRC_HRDRST_RCD_STAT_CLR Reboot Source Hard Reset Recording State Clear. 1: Clear RBT_SRC_HRDRST_TIG_TIM 0: Invalid
5	/	/	/
4	R/W	0x0	RBT_SRC_VCCIODET_RCD_STAT_CLR Reboot Source VCCIO Detect Recording State Clear. 1: Clear RBT_SRC_VCCIODET_TIG_TIM 0: Invalid
3	/	/	/
2	R/W	0x0	RBT_SRC_VDDSYSDET_RCD_STAT_CLR Reboot Source VDDSYS Detect Recording State Clear. 1: Clear RBT_SRC_VDDSYSDET_TIG_TIM 0: Invalid
1	/	/	/
0	R/W	0x0	RBT_SRC_TWD_RCD_STAT_CLR Reboot Source WDT Recording State Clear. 1: Clear RBT_SRC_TWD_TIG_TIM 0: Invalid

9.1.7.31 0x0310 RTC SPI Clock Control Register (Default Value: 0x0000_0009)

Offset: 0x0310			Register Name: RTC_SPI_CLK_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0310			Register Name: RTC_SPI_CLK_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	RTC_SPI_GATE_EN RTC Reg CFG SPI Clock Gating. 0: Gating 1: Not Gating Before configuring RTC register, the clock divider of SPI needs be configured firstly, then clock gating needs be enabled. Note: Frequency division and clock gating cannot be set at the same time.
30:5	/	/	/
4:0	R/W	0x9	RTC_SPI_DIV RTC Reg CFG SPI Clock Divider M. Actual SPI Clock = AHBS1/(M+1), (0-15) AHBS default Freq is 200MHz, SPI Clock default Freq is 20MHz Note: The SPI clock cannot exceed 50 MHz, or else the RTC register may be abnormal.

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10 Memory Controllers

10.1 SDRAM Controller (DRAMC)

The DRAMC has the following features:

- 16-bit DDR3/DDR3L/DDR4 interface
- Supports 2 ranks
- Support up to 2 chip select
- Memory capacity up to 2G Bytes
- Support up to 17-bit address lines
- Support DDR3/DDR3L 3-bit Bank address
- Support DDR4 2-bit bank address and 2-bit bank group
- Clock frequency up to 1066 MHz for DDR3 and DDR3L
- Clock frequency up to 1200 MHz for DDR4
- Automatic initialization and sequence refreshing
- Configurable runtime parameters for application flexibility
- Random read/write operation
- Dynamic Frequency Setting(DFS) via hardware

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10.2 SD/MMC Host Controller (SMHC)

10.2.1 Overview

The SMHC (SD-MMC Host controller) can be configured to support Secure Digital memory (SD Memory), UHS-I Card, Secure Digital I/O (SDIO), Multimedia Cards (MMC), and eMMC.

The SMHC has the following features:

- Four SD/MMC Host Controller (SMHC) interfaces
  - SMHC0, compliant with the protocol Secure Digital Memory (up to SD3.0)
  - SMHC1 and SMHC3, compliant with the protocol Secure Digital I/O (up to SDIO3.0)
  - SMHC2, compliant with the protocol Multimedia Card (up to eMMC5.1)
- The SMHC0, SMHC1, and SMHC3 support the following:
  - 1-bit or 4-bit data width
  - Supports SDIO interrupts in 1-bit and 4-bit modes
  - Internal 4K Bytes FIFO for data transfer
  - Maximum performance:
    - SDR mode 150 MHz@1.8 V IO pad
    - DDR mode 50 MHz@1.8 V IO pad
    - SDR mode 50 MHz@3.3 V IO pad
- The SMHC2 supports the following:
  - 1-bit, 4-bit, or 8-bit data width
  - Supports eMMC boot operation and alternative boot operation
  - Supports eMMC Command Queue
  - Internal 1K Bytes FIFO for data transfer
  - Supports HS400 mode and HS200 mode
  - Maximum performance:
    - SDR mode 150 MHz@1.8V IO pad
    - DDR mode 150 MHz@1.8V IO pad
    - SDR mode 50 MHz@3.3V IO pad
    - DDR mode 50 MHz@3.3V IO pad
- Supports block size of 1 to 65535 bytes
- Supports hardware CRC generation and error detection
- Supports Command Completion signals and interrupts to host processor, and Command Completion signal disable feature

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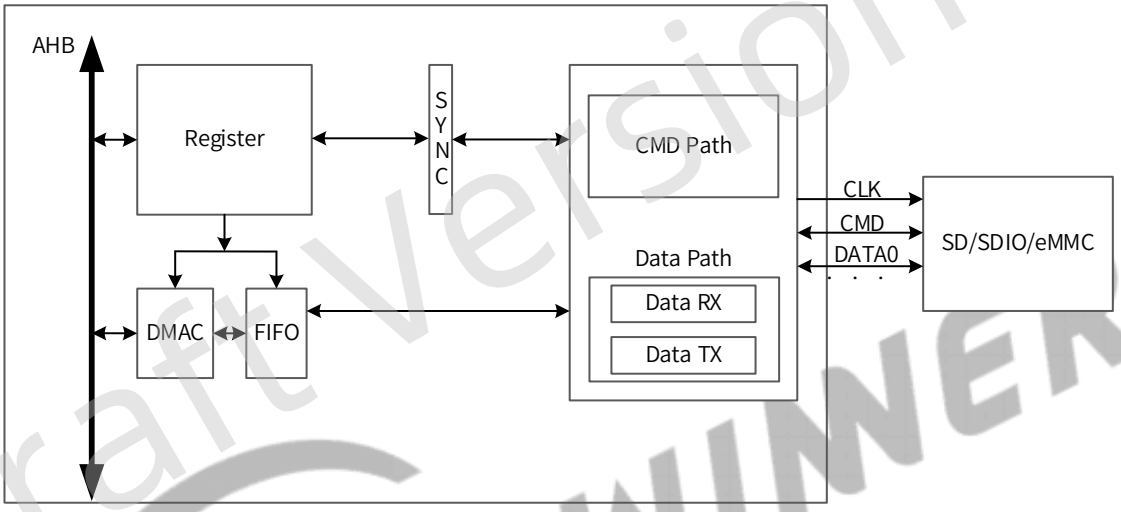
- Programmable baud rate
- Descriptor-based internal DMA controller
- Supports host pull-up control

10.2.2 Block Diagram

The following figure shows a block diagram of the SMHC.

Figure 10-1 SMHC Block Diagram



SMHC contains the following sub-blocks:

Table 10-1 SMHC Sub-blocks

Sub-block	Description
Register	It is used to configure the control signal for reading or writing the SD/SDIO/eMMC.
DMAC	The DMA controller that controls the data transfer between the memory and SMHC.
FIFO	A buffer for the data stream between the memory and the SMHC asynchronous clock domain.
SYNC	It synchronizes the signals from the AHB clock domain to the SMHC clock domain.
CMD Path	It sends commands to or receives commands from the SD/SDIO/eMMC.
Data Path	It consists of Data TX and Data RX sub-modules. The Data TX sends data blocks and the CRC codes to the SD/SDIO/eMMC. The Data RX receives data blocks and the CRC codes from the SD/SDIO/eMMC.

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10.2.3 External Signals

The following table describes the external signals of SMHC.

Table 10-2 SMHC External Signals

Signal Name	Description	Type
SMHC0		
SDC0-CMD	SD Card Command Output/Response Input	I/O, OD
SDC0-CLK	SD Card Clock Output	O
SDC0-D0	SD Card Data Input/ Output 0	I/O
SDC0-D1	SD Card Data Input/ Output 1	I/O
SDC0-D2	SD Card Data Input/ Output 2	I/O
SDC0-D3	SD Card Data Input/ Output 3	I/O
SMHC1		
SDC1-CMD	SDIO WIFI Command Output/Response Input	I/O, OD
SDC1-CLK	SDIO WIFI Clock Output	O
SDC1-D0	SDIO WIFI Data Input/ Output 0	I/O
SDC1-D1	SDIO WIFI Data Input/ Output 1	I/O
SDC1-D2	SDIO WIFI Data Input/ Output 2	I/O
SDC1-D3	SDIO WIFI Data Input/ Output 3	I/O
SMHC2		
SDC2-CMD	eMMC Command Output/Response Input	I/O, OD
SDC2-CLK	eMMC Clock Output	O
SDC2-D0	eMMC Data Input/ Output 0	I/O
SDC2-D1	eMMC Data Input/ Output 1	I/O
SDC2-D2	eMMC Data Input/ Output 2	I/O
SDC2-D3	eMMC Data Input/ Output 3	I/O
SDC2-D4	eMMC Data Input/ Output 4	I/O
SDC2-D5	eMMC Data Input/ Output 5	I/O
SDC2-D6	eMMC Data Input/ Output 6	I/O
SDC2-D7	eMMC Data Input/ Output 7	I/O
SDC2-RST	eMMC Reset	O
SDC2-DS	eMMC Data Strobe	I
SMHC3		
SDC3-CMD	SDIO WIFI Command Output/Response Input	I/O, OD
SDC3-CLK	SDIO WIFI Clock Output	O
SDC3-D0	SDIO WIFI Data Input/ Output 0	I/O
SDC3-D1	SDIO WIFI Data Input/ Output 1	I/O
SDC3-D2	SDIO WIFI Data Input/ Output 2	I/O
SDC3-D3	SDIO WIFI Data Input/ Output 3	I/O

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10.2.4 Functional Description

10.2.4.1 Data Path

The SMHC and SD/SDIO/eMMC contains the following interface buses: CLK, CMD, and DATA 1/4/8 (DATA8 is only for eMMC). During one clock cycle, the SMHC can transmit 1-bit command with one or two bits' data in 1-ch DATA mode, or four or eight bits' data in 4-ch DATA mode, and eight or sixteen bits' data in 8-ch DATA mode. The CMD is a bidirectional channel for initializing the SD/SDIO/eMMC and transmitting commands, and it can work in the open-drain mode and push-pull mode. The DATA is also a bidirectional channel, and it works in the push-pull mode.

- Reading Data from the SD/SDIO/eMMC

The register configures the signals for the read operation, and synchronizes the signals to the SMHC clock domain. Then the Data RX reads data from the SD/SDIO/eMMC via the CLK/CMD/DATA interface buses and writes the data in the FIFO. After that, the DMAC transfers the data from the FIFO to the memory.

- Writing Data to the SD/SDIO/eMMC

The register configures the signals for the write operation, and synchronize the signals to the SMHC clock domain. Then the DMAC reads data from the memory and writes the data to the FIFO. After that, the Data TX reads the data from the FIFO and writes the data to the SD/SDIO/eMMC via the CLK/CMD/DATA interface buses.

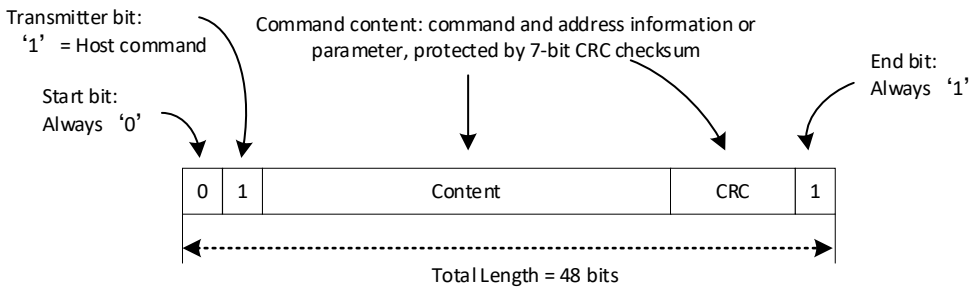
10.2.4.2 Package Format

Data transfer over the SD/SDIO/eMMC bus is based on command and data bitstreams that are initiated by a start bit and terminated by a stop bit. There are three types of SD/SDIO/eMMC packets: command token, response token, and data packet.

Command Tokens

The command token starts an operation. A command is sent from the host to a device. It is transferred serially on the CMD line. Command tokens have the following coding scheme:

Figure 10-2 Command Token Format



Each command token has 48 bits, preceded by a start bit ('0') and succeeded by an end bit ('1'). To detect transmission errors, each token is protected by CRC bits.

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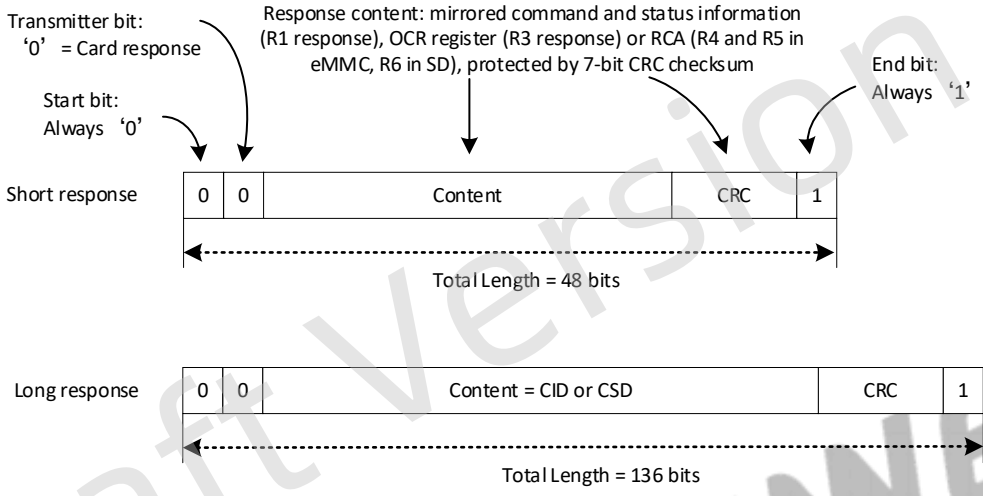
Response Tokens

After receiving a command, the card returns a 48-bit or 136-bit response based on the command type.

A response token is sent from the device to the host as an answer to a previously received command. It is transferred serially on the CMD line.

Figure 10-3 Response Token Format



Data Packet

Data can be transferred from the device to the host or vice versa. Data are transferred via the data lines.

Figure 10-4 Data Packet Format for SDR

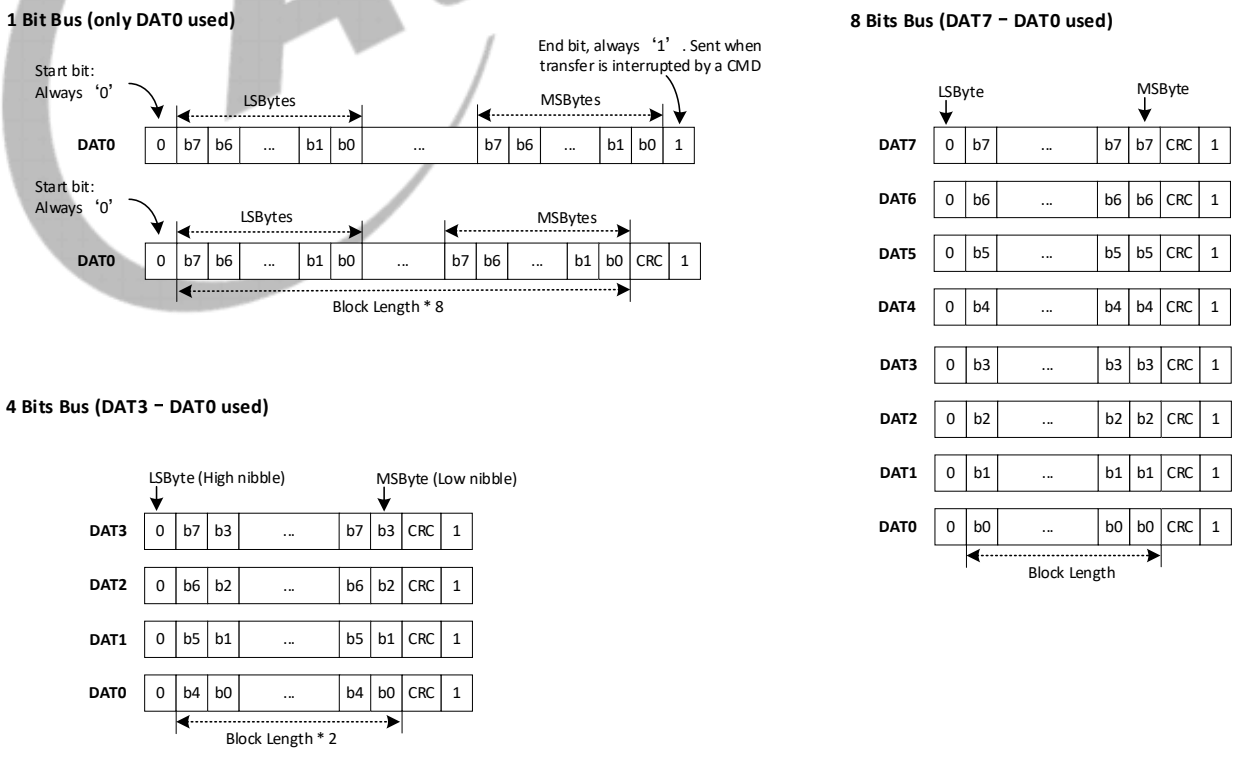


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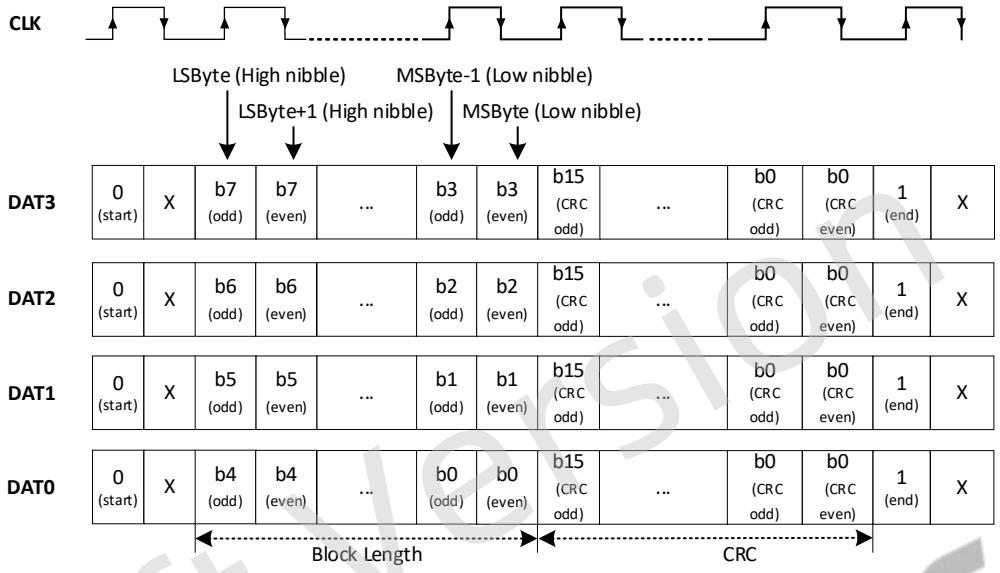
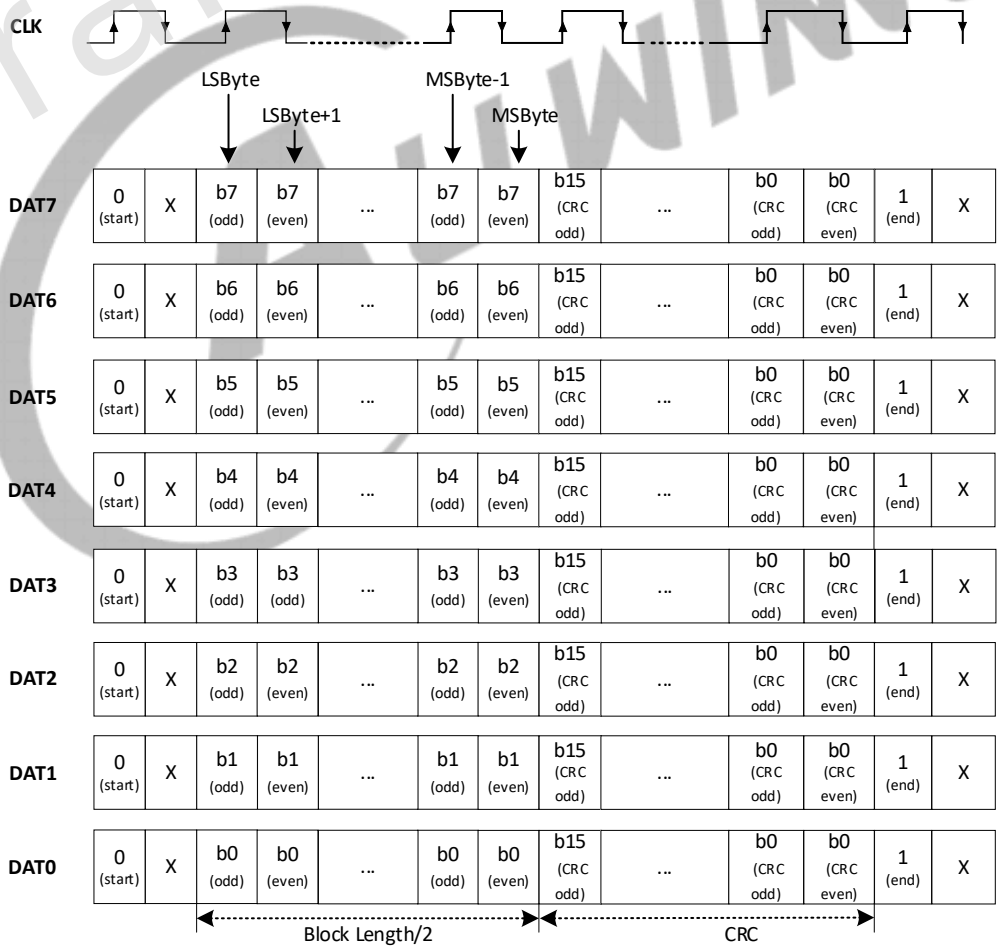
Figure 10-5 Data Packet Format for DDR

4 Bits Bus DDR (DAT3 – DAT0 used)



8 Bits Bus DDR (DAT7 – DAT0 used)



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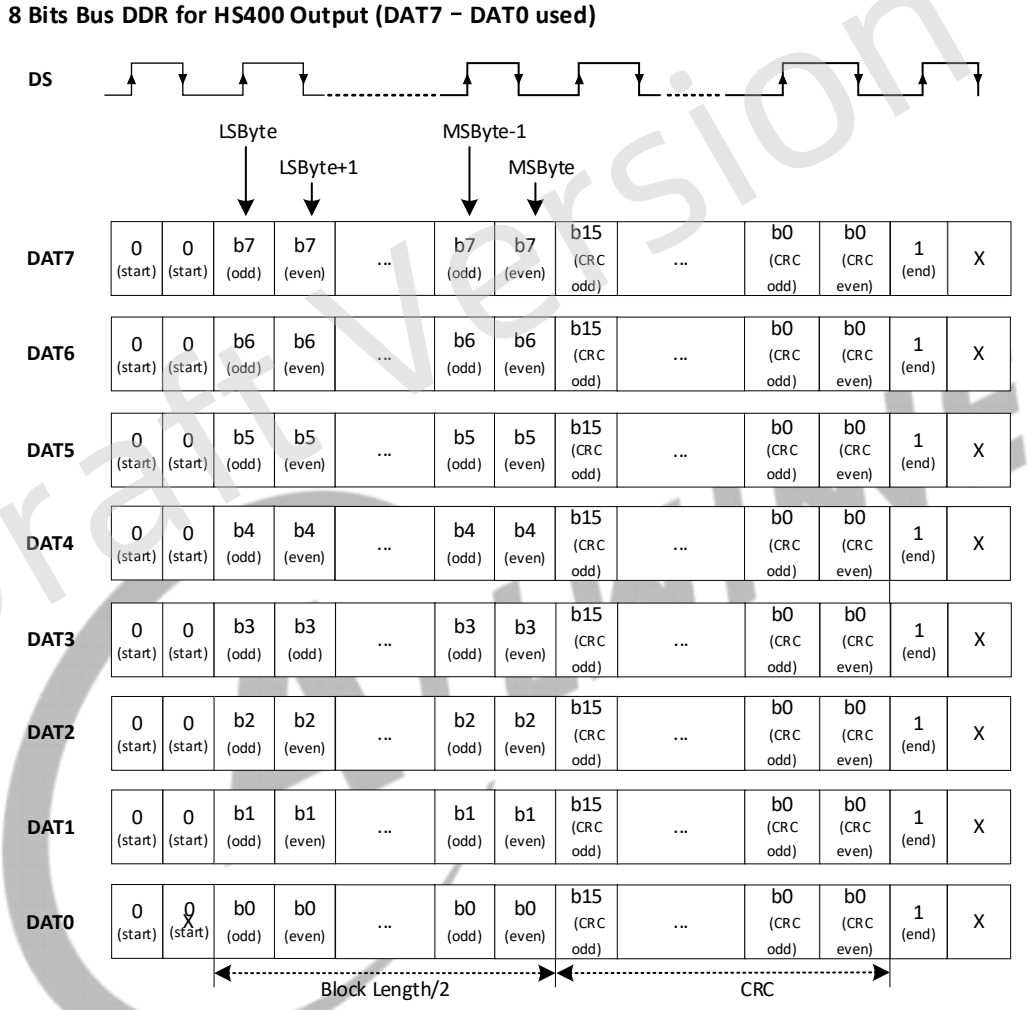
NOTE

- Bytes data are not interleaved but CRCs are interleaved.
- Start and end bits are only valid on the rising edge (“X” indicates “undefined”).

Figure 10-6 Data Packet Format for DDR in HS400 Mode



NOTE

- Bytes data are not interleaved while CRCs are interleaved.
- Start bits are valid when Data Strobe is High and Low.
- End bits are only valid when Data Strobe is High (“X” indicates “undefined”).

10.2.4.3 Data Transfer

Data transfers to or from the SD/eMMC card are done in blocks. Single and multiple block operations are widely used during data transfer.

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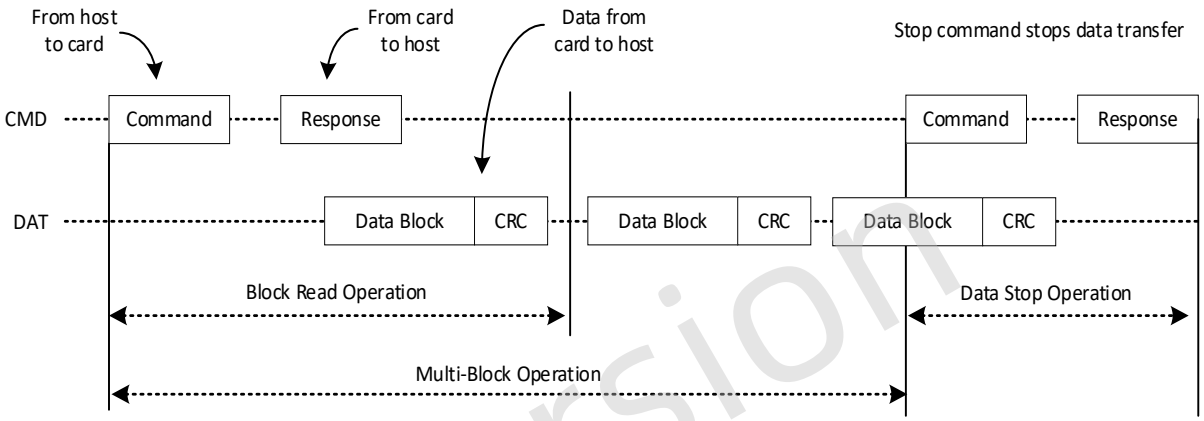
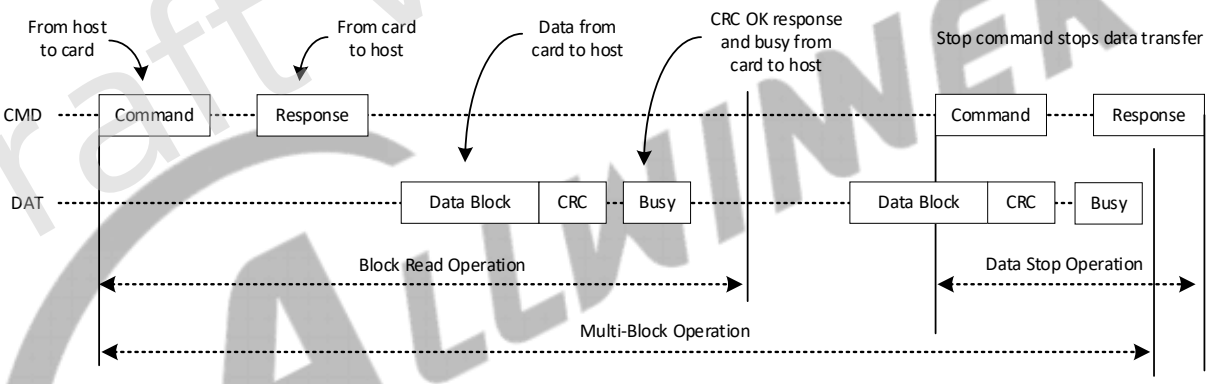
The following figure shows the single-block and multi-block read operation.

Figure 10-7 Single-Block and Multi-Block Read Operation



The following figure shows the single-block and multi-block write operation.

Figure 10-8 Single-Block and Multi-Block Read Operation



10.2.4.4 Internal DMA Controller

The SMHC has an internal DMA controller (IDMAC) to transfer data between the host memory and SMHC port. With a descriptor, the IDMAC can efficiently move data from the source to destination by automatically loading the next DMA transfer arguments, which needs less CPU intervention. Before transferring data in the IDMAC, the host driver should construct a descriptor list, configure arguments of every DMA transfer, and then launch the descriptor and start the DMA.

The IDMAC has an interrupt controller. When enabled, it generates an interrupt to the HOST CPU in situations such as data transmission is completed or some error happens.

IDMAC Descriptor Structure

The IDMAC uses a descriptor with a chain structure, and each descriptor points to a unique buffer and the next descriptor.

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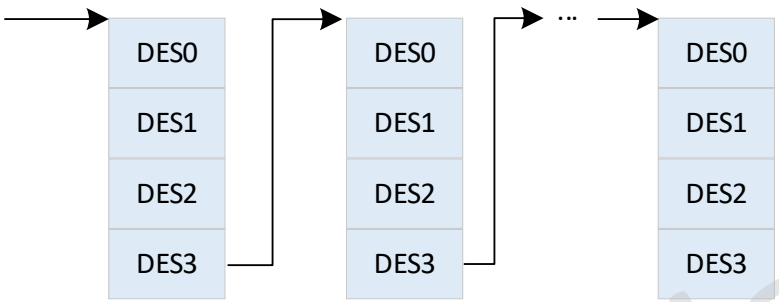
The following figure shows the internal formats of a descriptor.

Figure 10-9 IDMAC Descriptor Structure Diagram



This figure illustrates the internal formats of a descriptor. The descriptor address must be aligned to the bus width used for 32-bit buses. Each descriptor contains 16 bytes of control and status information.

DES0 corresponds to the [31:0] bits, DES1 corresponds to the [63:32] bits, DES2 corresponds to the [95:64] bits, and DES3 corresponds the [127:96] bits in a descriptor.

The following table shows the bit definition of DES0.

Table 10-3 DES0 Definition

Bits	Name	Description
31	HOLD	DES_OWN_FLAG When set, this bit indicates that the descriptor is owned by the IDMAC. When this bit is reset, it indicates that the descriptor is owned by the host. This bit is cleared when the transfer is over. The software should wait until it is cleared before using it.
30	ERROR	ERR_FLAG When some errors happen in transfer, this bit will be set to 1.
29:5	/	/
4	Chain Flag	CHAIN_MOD When set to 1, this bit indicates that the second address in the descriptor is the next descriptor address. It must be set to 1.
3	First DES Flag	FIRST_FLAG When set to 1, this bit indicates that this descriptor contains the first buffer of data. It must be set to 1 in the first DES.
2	Last DES Flag	LAST_FLAG When set to 1, this bit indicates that the buffers this descriptor points to are the last data buffer.
1	Disable Interrupt on completion	CUR_TXRX_OVER_INT_DIS When set to 1, this bit will prevent the setting of the TX/RX interrupt bit of the IDMAC status register for data that ends in the buffer the descriptor points to.
0	/	/

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The following table shows the bit definition of DES1.

Table 10-4 DES1 Definition

Bits	Name	Description
31:13	/	/
12:0	Buffer size	BUFF_SIZE The bits indicate the data buffer byte size, which must be a multiple of 4 bytes. If this field is 0, the DMA ignores this buffer and proceeds to the next descriptor.

The following table shows the bit definition of DES2.

Table 10-5 DES2 Definition

Bits	Name	Descriptor
31:0	Buffer address pointer	BUFF_ADDR The bits indicate the physical address of the data buffer. It is a word (4 bytes) address.

The following table shows the bit definition of DES3.

Table 10-6 DES3 Definition

Bits	Name	Descriptor
31:0	Next descriptor address	NEXT_DESP_ADDR The bits indicate the pointer to the physical memory where the next descriptor is present. It is a word (4 bytes) address.

10.2.4.5 Timing Specifications

Refer to the following relative specifications:

- Physical Layer Specification Ver3.00 Final, 2009.04.16
- SDIO Specification Ver2.00
- Multimedia Cards (MMC) version 4.2)
- JEDEC Standard (JESD84-44, Embedded Multimedia Card (eMMC) Card Product Standard
- JEDEC Standard (JESD84-B45, Embedded Multimedia Card (eMMC) Electrical Standard (4.5 Device)
- JEDEC Standard (JESD84-B50, Embedded Multimedia Card (eMMC) Electrical Standard (5.0)

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10.2.4.6 Serial CMDQ (CMDP) Mode

NOTE

SMHC1/3 doesn't supports CMDP mode.

To enable the CMDP mode, write 1 to the CQE_EN bit (bit [0]) and configure the field value of CMDQ_MODE bit (bit [5:4]) as 2'b00 in CQCFG register.

The CMDP mode only supports block data transfers for the following transfer modes and commands:

- Data transfer- single data block reading: CMD17
- Data transfer- single data block writing: CMD24
- Data transfer- pre-defined multiple data blocks reading: CMD23 --> CMD18
- Data transfer- pre-defined multiple data blocks writing: CMD23 --> CMD25
- Data transfer - open-ended multiple data blocks reading: CMD18 --> CMD12
- Data transfer - open-ended multiple data blocks writing: CMD25 --> CMD12

In CMDP mode, the transfer descriptor and DCMD descriptor are the same as these in CMDQ mode. The differences of task descriptor between CMDP mode and CMDQ mode are as follows.

Table 10-7 Differences of Task Descriptor Between CMDP Mode and CMDQ Mode

Field Name	Bit Loc.	Description	Encoded in CMDQ	Encoded in CMDP
Valid	0	1: The descriptor is effective and should be processed by hardware 0: The descriptor line should not be used	Valid	Valid
End	1	Should always be set to 1. Every Task Descriptor is standalone.	End	End
Int	2	Indicates the interrupt generation policy required for this task. 1: Hardware shall generate an interrupt upon the task s completion 0: Hardware shall count task s completion for interrupt coalescing	Int	Int
Act	5:3	Shall be set to b101, to indicate this is a Task Descriptor	Act	Act

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Field Name	Bit Loc.	Description	Encoded in CMDQ	Encoded in CMDP
Forced Programming	6	If 1, FP is enabled. Data shall be forcefully programmed to nonvolatile storage instead of volatile cache while cache is turned ON	CMD44	CMD23[24]
Context ID	10:7	/	CMD44	CMD23[28:25]
Tag Request	11	/	CMD44	CMD23[29]
Data Direction	12	Indicates the direction of data transfer 1: Device to Host (Read) 0: Host to Device (Write)	CMD44	1: Read 0: Write
Priority	13	1: High 0: Simple	CMD44	Multiple Block 1: Multiple block transfers 0: Single block transfer
Queue Barrier (QBR)	14	/	/	Transfer Mode 1: Pre-defined transfer 0: Open-ending transfer
Reliable Write	15	To avoid data in an unknown state	CMD44	CMD23[31]
Block Count	31:16	Number of blocks to be read/written	CMD44	CMD23[15:0]
Block Address	63:32	Data block address	CMD45	CMD17/18/24/25

10.2.4.7 Phase Offset of the Command and Data

To obtain the command phase or data phase in output timing, follow the steps blow:

Step 7 Configure the MODE_SEL (bit [31]) and HS400_NEW_SAM_EN (bit [0]) of SMHC_NTSR (offset: 0x005C) to choose the timing mode.

Step 8 Configure the DAT_DRV_PH_SEL (bit [17]) to select data drive phase offset and configure CMD_DRV_PH_SEL (bit [16]) to select command drive phase offset in SMHC_DRV_DL (offset: 0x0140).

Table 10-8 Phase Offset of Command and Data

Timing Mode	Drive Delay	Command	Data

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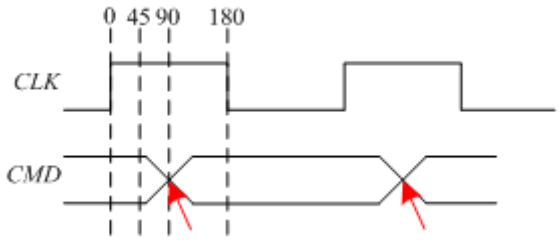
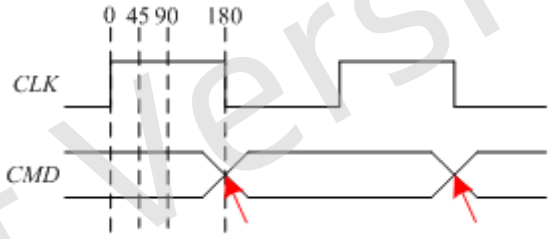
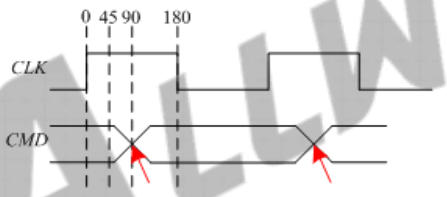
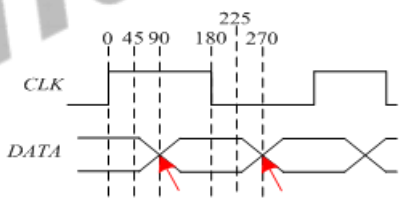
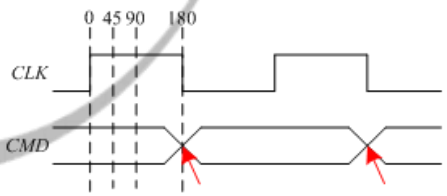
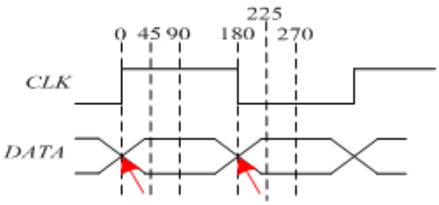
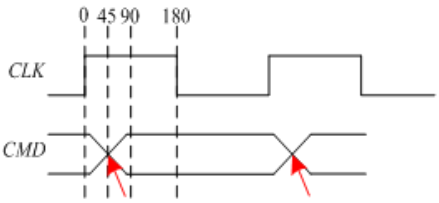
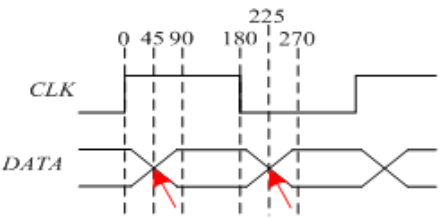
Timing Mode	Drive Delay	Command	Data
SDR	0		
		The command is updated in 90° clock position.	The data is updated in 90° clock position.
	1		
		The command is updated in 180° clock position.	The data is updated in 180° clock position.
DDR4 (1x mode: 0x5c[31]=0)	0		
		The command is updated in 90° clock position.	The data is updated in 90° clock position.
	1		
		The command is updated in 180° clock position.	The data is updated in 0° or 180° clock position.
DDR4 (2x mode: 0x5c[31]=1)	0		
		The command is updated in 45° clock position.	The data is updated in 45° clock position.

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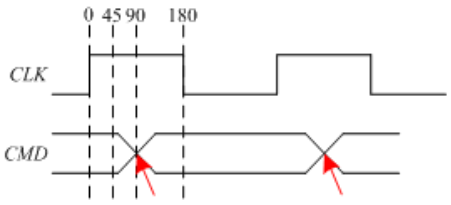
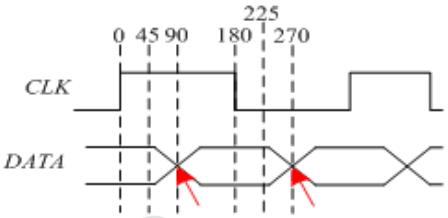
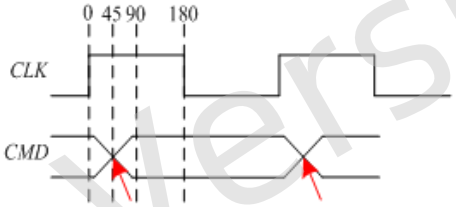
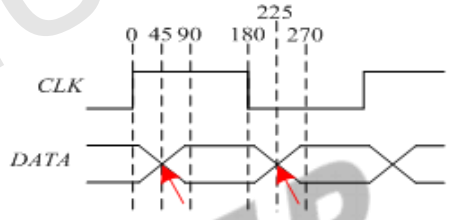
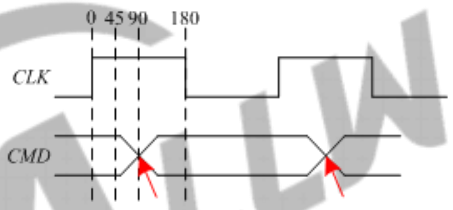
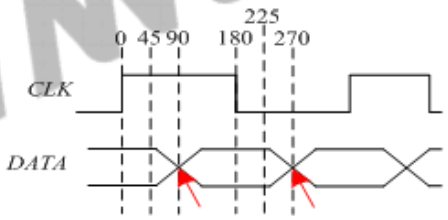
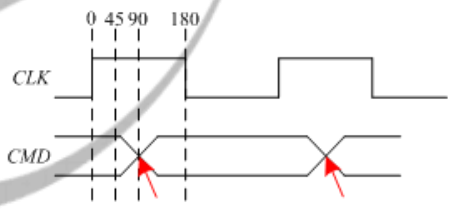
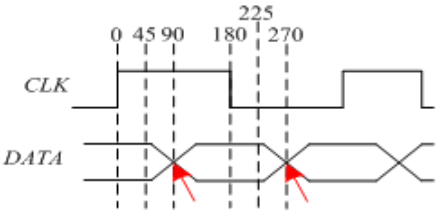
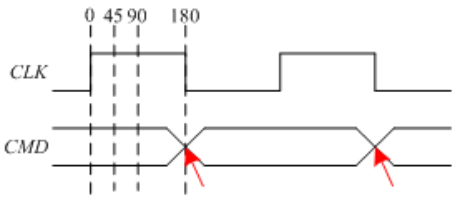
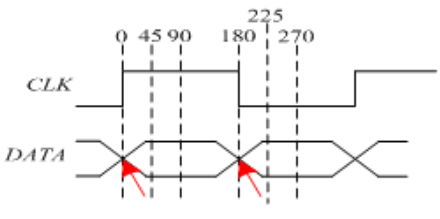
Timing Mode	Drive Delay	Command	Data
	1		
		The command is updated in 90° clock position.	The data is updated in 90° clock position.
DDR8	0		
		The command is updated in 45° clock position.	The data is updated in 45° clock position.
	1		
		The command is updated in 90° clock position.	The data is updated in 90° clock position.
HS400 (1x mode: 0x5c[0] =0)	0		
		The command is updated in 90° clock position.	The data is updated in 90° clock position.
	1		
		The command is updated in 180° clock position.	The data is updated in 0° or 180° clock position.

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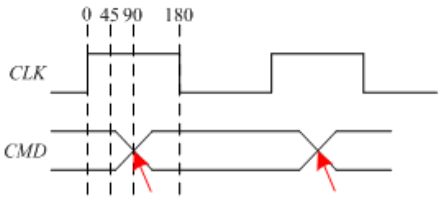
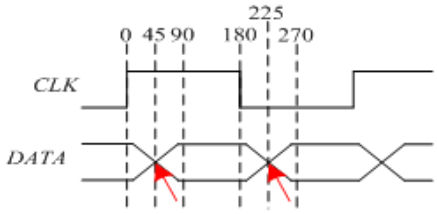
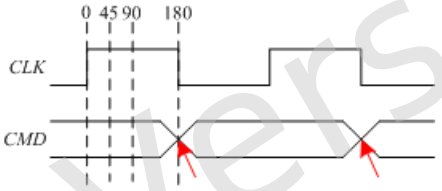
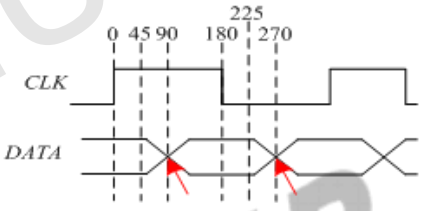
Timing Mode	Drive Delay	Command	Data
HS400 (2x mode: 0x5c[0] =1)	0		
		The command is updated in 90° clock position.	The data is updated in 45° clock position.
	1		
		The command is updated in 180° clock position.	The data is updated in 90° clock position.

10.2.5 Programming Guidelines

10.2.5.1 Initialization

Before data and commands are exchanged between a card and the SMHC, the SMHC needs to be initialized. Follow the steps below to initialize the SMHC:

Step 1 Configure the corresponding GPIO register as an SMHC in the section 17.9 GPIO.Reset clock by writing 1 to SMHCx_GAR_REG [SMHCx_RST_N](x=0-3), and open clock gating by writing 1 to SMHCx_GAR_REG [SMHCx_AHB_CLK_EN] (x=0-3); select clock sources and set the division factor by configuring the SMHCx_CLK_REG (x = 0-3) register.

Step 2 Configure SMHC_CTRL to reset the FIFO and controller, and enable the global interrupt; configure SMHC_INTMASK to 0xFFCE to enable the common interrupts and the interrupts caused by error/anomaly, and then register the interrupt function.

Step 3 Configure SMHC_CLKDIV to open clock for devices. Configure SMHC_CMD as the change clock command (for example 0x80202000). Send the update clock command to deliver clock to devices.

Step 4 Configure SMHC_CMD as a normal command. Configure SMHC_CMDARG to set command parameters. Configure SMHC_CMD to set parameters like the command sequence, the response receive, the response type, the response length, the response types to send the commands. According to the initialization process in the protocol, you can finish SMHC initialization by sending the corresponding command one by one.

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10.2.5.2 Single-block Write (CMD24)

To write a single block of data, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To write one block data to sector1, configure SMHC_BYTCNT[BYTE_CNT] to 0x200 and configure the descriptor according to the data size; set the data sector address of CMD24 (Single Data Block Write) to 0x1, write 0x80002758 to SMHC_CMD, and send CMD24 command to write data to the device.

Step 4 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successfully; otherwise, continue to wait until timeout, and then exit the process.

Step 5 Check whether SMHC_IDST[TX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

Step 6 Check whether SMHC_RINTSTS[DTC] is 1. If yes, the data transfer and CMD24 writing operations are completed. Otherwise, anomaly occurs. Read SMHC_RINTSTS and SMHC_STATUS to query the existing anomaly.

Step 7 Send CMD13 command to query whether the device writing operation is completed and returns to the idle status. For example, device RCA is 0x1234, first set SMHC_CMDARG to 0x12340000, write 0x8000014D to SMHC_CMD, go to step4 to ensure command transfer is completed, and then check whether the highest bit of SMHC_RESP0 (CMD13 response) is 1. If yes, the device is in the idle status, and the next command can be sent. Otherwise, the device is in the busy status. Continue to send CMD13 to wait for the device to enter the idle status until timeout.

10.2.5.3 Single-block Read (CMD17)

To read a single block of data, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

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Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To read one block data from sector1, configure SMHC_BYTCNT[BYTE_CNT] to 0x200 and configure the descriptor according to the data size; set the data sector address of CMD17 command (Single Data Block Read) to 0x1, write 0x80002351 to SMHC_CMD, and send CMD17 command to read data from the device to DRAM/SRAM.

Step 4 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successfully; otherwise, continue to wait until timeout, and then exit the process.

Step 5 Check whether SMHC_IDST[RX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

Step 6 Check whether SMHC_RINTSTS[DTC] is 1. If yes, data transfer and CMD17 reading operation are completed. Otherwise, anomaly occurs. Read SMHC_RINTSTS and SMHC_STATUS to query the existing abnormity.

10.2.5.4 Open-ended Multiple-Block Write (CMD25+Auto CMD12)

To write open-ended multiple data blocks, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To write three blocks of data to sectors begin with sector0, configure SMHC_BYTCNT[BYTE_CNT] to 0x600 and configure the descriptor according to the data size; set the data sector address of CMD25 command (Multiple Data Blocks Write) to 0x0, write 0x80003759 to SMHC_CMD, and send CMD25 command to read data from the device to DRAM/SRAM.

Step 4 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successfully; otherwise, continue to wait until timeout, and then exit the process.

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Step 5 Check whether SMHC_IDST[TX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

Step 6 Check whether SMHC_RINTSTS[ACD] and SMHC_RINTSTS[DTC] are both 1. If yes, the data transfer, CMD12 transfer, and CMD25 writing operations are completed. Otherwise, abnormity exists. Read SMHC_RINTSTS and SMHC_STATUS to query the existing abnormity.

Step 7 Send CMD13 command to query whether the device writing operation is completed and returns to the idle status. For example, device RCA is 0x1234, first set SMHC_CMDARG to 0x12340000, write 0x8000014D to SMHC_CMD, go to step4 to ensure command transfer completed, and then check whether the highest bit of SMHC_RESP0 (CMD13 response) is 1. If yes, the device is in the idle status, and the next command can be sent. Otherwise, the device is in the busy status. Continue to send CMD13 to wait for the device to enter the idle status until timeout.

10.2.5.5 Open-ended Multiple-block Read (CMD18+Auto CMD12)

To read open-ended multiple data blocks, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To read three blocks of data from sectors begin with sector0, configure SMHC_BYTCNT[BYTE_CNT] to 0x600 and configure the descriptor according to the data size; set the data sector address of CMD18 command (Multiple Data Blocks Read) to 0x0, write 0x80003352 to SMHC_CMD, and send CMD18 command to read data to the device. When the data transfer is completed, CMD12 will be sent automatically.

Step 4 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successfully; otherwise, continue to wait until timeout, and then exit the process.

Step 5 Check whether SMHC_IDST[RX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

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Step 6 Check whether SMHC_RINTSTS[ACD] and SMHC_RINTSTS[DTC] are both 1. If yes, data transfer, CMD12 transfer, and CMD18 reading operation are completed. Otherwise, abnormity exists. Read SMHC_RINTSTS and SMHC_STATUS to query the existing abnormity.

10.2.5.6 Pre-Defined Multiple-Block Write (CMD23+CMD25)

To write pre-defined multiple data blocks, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To write three blocks of data, configure SMHC_CMDARG to 0x3 to specify the number of data blocks as three. Then write 0x80000157 to SMHC_CMD to send the CMD23 command. Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successful; otherwise, continue to wait until timeout, and then exit the process.

Step 4 Configure SMHC_BYTCNT[BYTE_CNT] to 0x600 and configure the descriptor according to the data size; set the data sector address of CMD25 command (Multiple Data Blocks Write) to 0x0, write 0x80002759 to SMHC_CMD, and send CMD25 command to write data to the device.

Step 5 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successful; otherwise, continue to wait until timeout, and then exit the process.

Step 6 Check whether SMHC_IDST[TX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

Step 7 Check whether SMHC_RINTSTS[DTC] is 1. If yes, the data transfer and CMD25 writing operations are completed. Otherwise, abnormity exists. Read SMHC_RINTSTS and SMHC_STATUS to query the existing abnormity.

Step 8 Send CMD13 command to query whether the device writing operation is completed and returns to the idle status. For example, device RCA is 0x1234, first set SMHC_CMDARG to 0x12340000, write 0x8000014D to SMHC_CMD, go to step5 to ensure command transfer completed, and then check whether the highest bit of SMHC_RESP0 (CMD13 response) is 1. If yes, the device is in the idle status, and the next command can be sent. Otherwise,

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the device is in the busy status. Continue to send CMD13 to wait for the device to enter the idle status until timeout

10.2.5.7 Pre-Defined Multiple-Block Read (CMD23+CMD18)

To read pre-defined multiple data blocks, follow the steps below:

Step 1 Write 0x1 to SMHC_CTRL[DMA_RST] to reset the internal DMA controller; write 0x82 to SMHC_IDMAC to enable the IDMAC interrupt and configure AHB master burst transfers; configure SMHC_IDIE to enable the transfer interrupt, receive interrupt, and abnormal interrupt.

Step 2 Configure SMHC_FIFOTH to determine the burst size and TX/RX trigger level. For example, if SMHC_FIFOTH is configured as 0x300F00F0, it indicates the burst size is 16, TX_TL is 15, and RX_TL is 240. Configure SMHC_DLBA to determine the start address of the DMA descriptor.

Step 3 To read three blocks of data, configure SMHC_CMDARG to 0x3 to specify the number of data blocks as three. Then write 0x80000157 to SMHC_CMD to send the CMD23 command. Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successful; otherwise, continue to wait until timeout, and then exit the process.

Step 4 Configure SMHC_BYTCNT[BYTE_CNT] to 0x600 and configure the descriptor according to the data size; set the data sector address of CMD18 (Multiple Data Blocks Read) to 0x0, write 0x80002352 to SMHC_CMD, and send CMD18 command to read data from device to DRAM/SRAM.

Step 5 Check whether SMHC_RINTSTS[CC] is 1. If yes, the command is sent successful; otherwise, continue to wait until timeout, and then exit the process.

Step 6 Check whether SMHC_IDST[RX_INT] is 1. If yes, the data transfer for writing DMA is completed. Write 0x337 to SMHC_IDST to clear the interrupt flag. Otherwise, continue to wait until timeout, and then exit the process.

Step 7 Check whether SMHC_RINTSTS[DTC] is 1. If yes, the data transfer and CMD18 writing operations are completed. Otherwise, abnormity exists. Read SMHC_RINTSTS and SMHC_STATUS to query the existing abnormity.

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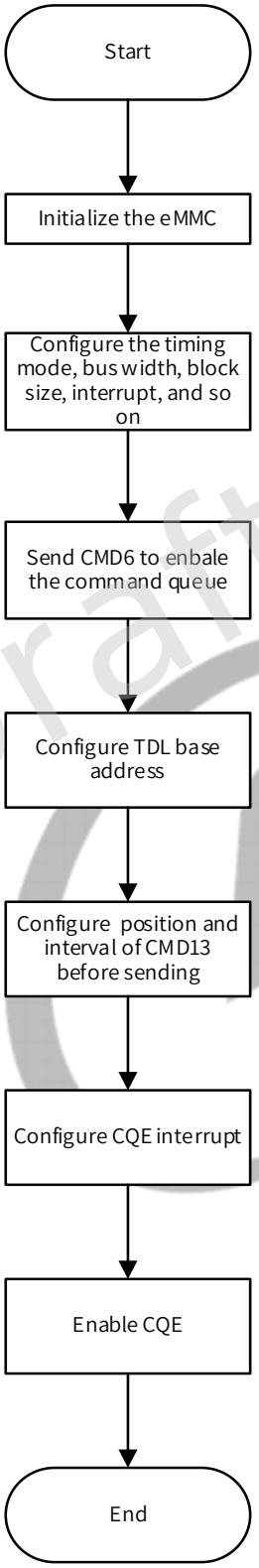
10.2.5.8 eMMC Command Queue

The following figure describes the initialization process of command queue.

Figure 10-10 Initialization Process of Command Queue



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Follow the steps below to initialize the command queue:

Step 1 Initialize the eMMC. For detailed initialization steps, refer to section 10.2.5.1 Initialization.

Step 2 Send CMD6 to configure the timing mode and bus width. Send CMD16 to configure the block size as 512 B.

NOTE

The block size must be set to 512 B. After the command queue is enabled, the block size cannot be modified.

Step 3 Send CMD6 to enable the command queue of the eMMC device.

Step 4 Configure CQTDLBA register and CQTDLBAU register to set the base address of the task descriptor list.

Step 5 Configure CQSSC1 register to set how to query the status of the device's task queue.

Step 6 Configure CQIC register to enable/disable interrupt, set interrupt count and timer protection.

Step 7 Configure CQRMEM register to set which errors may trigger a RED interrupt.

Step 8 Configure CQCFG register to enable CQE activity.

eMMC CMDQ supports multiple operation modes, which could be selected by configuring the CMDQ_MODE bit (bit [5:4]) of CQCFG register.

10.2.5.9 Voltage Switch

Step 1 Write 1 to SMHC_CMD[28] and write 0x0B to SMHC_CMD[5:0].

Step 2 Write 1 to SMHC_CMD[31].

Step 3 Write 0 to SMHC_CLKDIV[17] to disable the card clock.

Step 4 Write 1 to SMHC_CMD[28] and SMHC_CMD[13]. Then, write 0 to SMHC_CMD[5:0] and SMHC_CMDARG.

Step 5 Write 1 to SMHC_CMD[31].

Step 6 Inquiry SMHC_MINTSTS[10] to check if voltage switch has been finished.

Step 7 Write 1 to SMHC_CLKDIV[17] to enable the card clock.

10.2.6 Register List

Module Name	Base Address
SMHC0	0x04020000

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SMHC1	0x04021000
SMHC2	0x04022000
SMHC3	0x04023000

Register Name	Offset	Description
SMHC_CTRL	0x0000	SMHC Global Control Register
SMHC_CLKDIV	0x0004	SMHC Clock Control Register
SMHC_TMOUT	0x0008	SMHC Timeout Register
SMHC_CTYPE	0x000C	SMHC Bus Width Register
SMHC_BLKSIZ	0x0010	SMHC Block Size Register
SMHC_BYTCNT	0x0014	SMHC Byte Count Register
SMHC_CMD	0x0018	SMHC Command Register
SMHC_CMDARG	0x001C	SMHC Command Argument Register
SMHC_RESP0	0x0020	SMHC Response 0 Register
SMHC_RESP1	0x0024	SMHC Response 1 Register
SMHC_RESP2	0x0028	SMHC Response 2 Register
SMHC_RESP3	0x002C	SMHC Response 3 Register
SMHC_INTMASK	0x0030	SMHC Interrupt Mask Register
SMHC_MINTSTS	0x0034	SMHC Masked Interrupt Status Register
SMHC_RINTSTS	0x0038	SMHC Raw Interrupt Status Register
SMHC_STATUS	0x003C	SMHC Status Register
SMHC_FIFOTH	0x0040	SMHC FIFO Water Level Register
SMHC_FUNS	0x0044	SMHC FIFO Function Select Register
SMHC_TBC0	0x0048	SMHC Transferred Byte Count Register 0
SMHC_TBC1	0x004C	SMHC Transferred Byte Count Register 1
SMHC_CSDC	0x0054	SMHC CRC Status Detect Control Register
SMHC_A12A	0x0058	SMHC Auto Command 12 Argument Register
SMHC_NTSR	0x005C	SMHC New Timing Set Register
SMHC_HWRST	0x0078	SMHC Hardware Reset Register
SMHC_IDMAC	0x0080	SMHC IDMAC Control Register
SMHC_DLBA	0x0084	SMHC Descriptor List Base Address Register
SMHC_IDST	0x0088	SMHC IDMAC Status Register
SMHC_IDIE	0x008C	SMHC IDMAC Interrupt Enable Register
SMHC_THLD	0x0100	SMHC Card Threshold Control Register
SMHC_SFC	0x0104	SMHC Sample FIFO Control Register
SMHC_A23A	0x0108	SMHC Auto Command 23 Argument Register
EMMC_DDR_SBIT_DET	0x010C	SMHC eMMC4.5 DDR Start Bit Detection Control Register
SMHC_RESP_CRC	0x0110	SMHC Response CRC Register
SMHC_EXT_CMD	0x0138	SMHC Extended Command Register
SMHC_EXT_RESP	0x013C	SMHC Extended Response Register
SMHC_DRV_DL	0x0140	SMHC Drive Delay Control Register
SMHC_SAMP_DL	0x0144	SMHC Sample Delay Control Register

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Register Name	Offset	Description
SMHC_DS_DL	0x0148	SMHC Data Strobe Delay Control Register
SMHC_CIU_CNT	0x0154	SMHC Card Interface Counter Register
SMHC_WR_BUSY_TMOUT	0x015C	SMHC Write Busy Timeout Register
SMHC_DESKEW_DATA0_DL	0x0160	SMHC Deskew Data0 Delay Control Register
SMHC_DESKEW_DATA1_DL	0x0164	SMHC Deskew Data1 Delay Control Register
SMHC_DESKEW_DATA2_DL	0x0168	SMHC Deskew Data2 Delay Control Register
SMHC_DESKEW_DATA3_DL	0x016C	SMHC Deskew Data3 Delay Control Register
SMHC_DESKEW_DATA4_DL	0x0170	SMHC Deskew Data4 Delay Control Register
SMHC_DESKEW_DATA5_DL	0x0174	SMHC Deskew Data5 Delay Control Register
SMHC_DESKEW_DATA6_DL	0x0178	SMHC Deskew Data6 Delay Control Register
SMHC_DESKEW_DATA7_DL	0x017C	SMHC Deskew Data7 Delay Control Register
SMHC_DESKEW_DS_DL	0x0180	SMHC Deskew DS Delay Control Register
SMHC_DESKEW_CTRL	0x0184	SMHC Deskew Control Control Register
SMHC_CDATA_IN_DATA0_1_DL	0x0190	SMHC CDATA_IN Data0/1 Delay Control Register
SMHC_CDATA_IN_DATA2_3_DL	0x0194	SMHC CDATA_IN Data2/3 Delay Control Register
SMHC_CDATA_IN_DATA4_5_DL	0x0198	SMHC CDATA_IN Data4/5 Delay Control Register
SMHC_CDATA_IN_DATA6_7_DL	0x019C	SMHC CDATA_IN Data6/7 Delay Control Register
SMHC_CQE_CTRL	0x01A0	SMHC CQE Control Register
SMHC_FIFO	0x0200	SMHC FIFO Register
CQVER	0x0400	Command Queuing Version Register
CQCFG	0x0408	Command Queuing Configuration Register
CQCTL	0x040C	Command Queuing Control Register
CQIS	0x0410	Command Queuing Interrupt Status Register
CQISTE	0x0414	Command Queuing Interrupt Status Enable Register
CQISGE	0x0418	Command Queuing Interrupt Signal Enable Register
CQIC	0x041C	Command Queuing Interrupt Coalescing Register
CQTDLBA	0x0420	Command Queuing Task Descriptor List Base Address Register
CQTDLBAU	0x0424	Command Queuing Task Descriptor List Base Address Upper 32 bits
CQTDBR	0x0428	Command Queuing Task Doorbell Register
CQTCN	0x042C	Command Queuing Task Completion Notification Register
CQDQS	0x0430	Command Queuing Device Queue Status Register
CQDPT	0x0434	Command Queuing Device Pending Tasks Register
CQTCLR	0x0438	Command Queuing Task Clear Register

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Register Name	Offset	Description
CQSSC1	0x0440	Command Queuing Send Status Configuration 1 Register
CQSSC2	0x0444	Command Queuing Send Status Configuration 2 Register
CQCRDCT	0x0448	Command Queuing Command Response for Direct-Command Task Register
CQRMEM	0x0450	Command Queuing Response Mode Error Mask Register
CQTERRI	0x0454	Command Queuing Task Error Information Register
CQCRI	0x0458	Command Queuing Command Response Index Register
CQCRA	0x045C	Command Queuing Command Response Argument Register
CQFUNCCTL	0x0460	Command Queuing Function Control Register
CQDTD	0x0480	Command Queuing Descriptor Transfer Done Register
CQTCT	0x0484	Command Queuing Task Complete Total Register
CQSTATUS	0x0488	CQE Controller Status Register

10.2.7 Register Description

10.2.7.1 0x0000 SMHC Global Control Register (Default Value: 0x0000_0100)

Offset: 0x0000			Register Name: SMHC_CTRL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	FIFO_AC_MOD FIFO Access Mode 1: AHB bus 0: DMA bus
30	R/W	0x0	WAIT_HWEND Data transfer complete wait hwend 0: enable 1: disable
29	R	0x1	HWEND_INFO 0: AHB write 1: AHB write end or idle
28:18	/	/	/

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Offset: 0x0000			Register Name: SMHC_CTRL
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	SRAM_BIST_SEL SRAM BIST Select 00: for SMHC 4K SRAM(1024x32) BIST 01: for CQE SRAM(32x56) BIST 10: for EMCE SRAM BIST 11: Reserved
15:13	/	/	/
12	R/W	0x0	TIME_UNIT_CMD Time unit for command line Time unit used to calculate command line time out value defined in RTO_LMT 0: 1 card clock period 1: 256 card clock period
11	R/W	0x0	TIME_UNIT_DAT Time unit for data line Time unit used to calculate data line time out value defined in DTO_LMT. 0: 1 card clock period 1: 256 card clock period
10	R/W	0x0	DDR_MOD_SEL DDR Mode Select Although the HS400 speed mode of eMMC is 8-bit DDR, this field should be cleared when HS400_MD_EN is set. 0: SDR mode 1: DDR mode
9	/	/	/
8	R/W	0x1	CD_DBC_ENB Card Detect (Data [3] status) De-bounce Enable 0: Disable de-bounce 1: Enable de-bounce
7:6	/	/	/
5	R/W	0x0	DMA_ENB DMA Global Enable 0: Disable DMA to transfer data, using AHB bus 1: Enable DMA to transfer data
4	R/W	0x0	INT_ENB Global Interrupt Enable 0: Disable interrupts 1: Enable interrupts
3	/	/	/

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Offset: 0x0000			Register Name: SMHC_CTRL
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	DMA_RST DMA Reset
1	R/W	0x0	FIFO_RST FIFO Reset 0: No change 1: Reset FIFO This bit is auto-cleared after completion of reset operation.
0	R/W	0x0	SOFT_RST Software Reset 0: No change 1: Reset SD/MMC controller This bit is auto-cleared after completion of reset operation.

10.2.7.2 0x0004 SMHC Clock Control Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: SMHC_CLKDIV
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MASK_DAT0 Mask Data0 0: Do not mask data0 when updating clock 1: Mask data0 when updating clock
30:18	/	/	/
17	R/W	0x0	CCLK_CTRL Card Clock Output Control 0: Card clock always on 1 : Turn off card clock when FSM is in IDLE state
16	R/W	0x0	CCLK_ENB Card Clock Enable 0: Card Clock off 1: Card Clock on
15:8	/	/	/
7:0	R/W	0x0	CCLK_DIV Card clock divider n: Source clock is divided by 2*n.(n=0-255) When HS400_MD_EN is set, this field must be cleared.

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10.2.7.3 0x0008 SMHC Timeout Register (Default Value: 0xFFFF_FF40)

Offset: 0x0008			Register Name: SMHC_TMOUT
Bit	Read/Write	Default/Hex	Description
31:8	R/W	0xffffff	DTO_LMT Data Timeout Limit This field can set time of the Host wait for the data from the Device. To communicate with the Device, this field must be set to maximum that greater than the time N_{AC}. About the N_{AC}, the explanation is as follows: When the Host read data, data transmission from the Device starts after the access time delay N_{AC} beginning from the end bit of the read command (ACMD51, CMD8, CMD17, and CMD18). When the Host read multiple block(CMD18), the next block s data transmission from the Device starts after the access time delay N_{AC} beginning from the end bit of the previous block. When the Host write data, this value is ineffective. Note: Under the following 3 conditions, the timeout limit is half of the set time. 1> DDR8 mode 2> DDR4 and SMHC_NTSR[MODE_SELEC] (0x5C [31]) high 3> HS400 and SMHC_NTSR[HS400_NEW_SAMPLE_EN](0x5C[0]) high
7:0	R/W	0x40	RTO_LMT Response Timeout Limit

10.2.7.4 0x000C SMHC Bus Width Register (Default Value:0x0000_0000)

Offset: 0x000C			Register Name: SMHC_CTYPE
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	CARD_WID Card width 00: 1-bit width 01: 4-bit width 1x: 8-bit width

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10.2.7.5 0x0010 SMHC Block Size Register (Default Value:0x0000_0200)

Offset: 0x0010			Register Name: SMHC_BLKSIZ
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x200	BLK_SZ Block size

10.2.7.6 0x0014 SMHC Byte Count Register (Default Value:0x0000_0200)

Offset: 0x0014			Register Name: SMHC_BYTCNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x200	BYTE_CNT Byte counter Number of bytes to be transferred. It must be the integer multiple of Block Size (BLK_SZ) for block transfers.

10.2.7.7 0x0018 SMHC Command Register (Default Value:0x0000_0000)

Offset: 0x0018			Register Name: SMHC_CMD
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CMD_LOAD Start Command This bit is auto cleared when current command is sent. If there is no any response error happened, a command complete interrupt bit will be set in SMHC_RINTSTS register. You should not write any other command before this bit is cleared.
30:29	/	/	/
28	R/W	0x0	VOL_SW Voltage Switch 0: Normal command 1: Voltage switch command, set for CMD11 only
27	R/W	0x0	BOOT_ABT Boot Abort Setting this bit will terminate the boot operation.
26	R/W	0x0	EXP_BOOT_ACK Expect Boot Acknowledge When Software sets this bit along in mandatory boot operation, controller expects a boot acknowledge start pattern of 0-1-0 from the selected card.

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Offset: 0x0018			Register Name: SMHC_CMD
Bit	Read/Write	Default/Hex	Description
25:24	R/W	0x0	BOOT_MOD Boot Mode 00: Normal Command 01: Mandatory Boot operation 10: Alternate Boot operation 11: Reserved
23:22	/	/	/
21	R/W	0x0	PRG_CLK Change Clock 0: Normal Command 1: Change Card Clock When this bit is set, controller will change clock domain and clock output. No command will be sent.
20:18	/	/	/
17	R/W	0x0	R1B_BUSY_HW_DETECT R1b busy hardware detect enable 0: Disable hardware detect r1b busy 1: Enable hardware detect r1b busy
16	R/W	0x0	R1B_RESP R1b response 0: The response type isn't r1b for current command 1: The response type is r1b for current command Note: If STOP_CMD_FLAG is set, R1B_RESP will be set by controller when sending CMD12, and will be cleared by controller when sending next command.
15	R/W	0x0	SEND_INIT_SEQ Send Initialization 0: Normal command sending 1: Send initialization sequence before sending this command.
14	R/W	0x0	STOP_ABT_CMD Stop Abort Command 0: Normal command sending 1: Send Stop or abort command to stop current data transfer in progress.(CMD12, CMD52 for writing I/O Abort in SDIO CCCR)
13	R/W	0x0	WAIT_PRE_OVER Wait Data Transfer Over 0: Send command at once, do not care of data transferring 1: Wait for data transfer completion before sending current command

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Offset: 0x0018			Register Name: SMHC_CMD
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	STOP_CMD_FLAG Send Stop CMD Automatically (CMD12) 0: Do not send stop command at the end of data transfer 1: Send stop command automatically at the end of data transfer If set, the SMHC_RESP1 will record the response of auto CMD12.
11	R/W	0x0	TRANS_MODE Transfer Mode 0: Block data transfer command 1: Stream data transfer command
10	R/W	0x0	TRANS_DIR Transfer Direction 0: Read operation 1: Write operation
9	R/W	0x0	DATA_TRANS Data Transfer 0: Without data transfer 1: With data transfer
8	R/W	0x0	CHK_RESP_CRC Check Response CRC 0: Do not check response CRC 1: Check response CRC
7	R/W	0x0	LONG_RESP Response Type 0: Short Response (48 bits) 1: Long Response (136 bits)
6	R/W	0x0	RESP_RCV Response Receive 0: Command without Response 1: Command with Response
5:0	R/W	0x0	CMD_IDX CMD Index Command index value

10.2.7.8 0x001C SMHC Command Argument Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: SMHC_CMDARG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x001C			Register Name: SMHC_CMDARG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CMD_ARG Command argument

10.2.7.9 0x0020 SMHC Response 0 Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: SMHC_RESP0
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CMD_RESP0 Response 0 Bit[31:0] of response

10.2.7.10 0x0024 SMHC Response 1 Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: SMHC_RESP1
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CMD_RESP1 Response 1 Bit[63:31] of response

10.2.7.11 0x0028 SMHC Response 2 Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: SMHC_RESP2
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CMD_RESP2 Response 2 Bit[95:64] of response

10.2.7.12 0x002C SMHC Response 3 Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: SMHC_RESP3
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CMD_RESP3 Response 3 Bit[127:96] of response

10.2.7.13 0x0030 SMHC Interrupt Mask Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: SMHC_INTMASK
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0030			Register Name: SMHC_INTMASK
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CARD_REMOVAL_INT_EN Card Removed Interrupt Enable
30	R/W	0x0	CARD_INSERT_INT_EN Card Inserted Interrupt Enable
29:19	/	/	/
18	R/W	0x0	DATA_WR_BUSY_TO_INT_EN Data write busy timeout interrupt enable
17	R/W	0x0	R1B_BUSY_CLR_INT_EN R1B busy clear interrupt enable
16	R/W	0x0	SDIO_INT_EN SDIO Interrupt Enable
15	R/W	0x0	DEE_INT_EN Data End-bit Error Interrupt Enable
14	R/W	0x0	ACD_INT_EN Auto Command Done Interrupt Enable
13	R/W	0x0	DSE_BC_INT_EN Data Start Error Interrupt Enable
12	R/W	0x0	CB_IW_INT_EN Command Busy and Illegal Write Interrupt Enable
11	R/W	0x0	FU_FO_INT_EN FIFO Underrun/Overflow Interrupt Enable
10	R/W	0x0	DSTO_VSD_INT_EN Data Starvation Timeout/V1.8 Switch Done Interrupt Enable
9	R/W	0x0	DTO_BDS_INT_EN Data Timeout/Boot Data Start Interrupt Enable
8	R/W	0x0	RTO_BACK_INT_EN Response Timeout/Boot ACK Received Interrupt Enable
7	R/W	0x0	DCE_INT_EN Data CRC Error Interrupt Enable
6	R/W	0x0	RCE_INT_EN Response CRC Error Interrupt Enable
5	R/W	0x0	DRR_INT_EN Data Receive Request Interrupt Enable
4	R/W	0x0	DTR_INT_EN Data Transmit Request Interrupt Enable
3	R/W	0x0	DTC_INT_EN Data Transfer Complete Interrupt Enable
2	R/W	0x0	CC_INT_EN Command Complete Interrupt Enable

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Offset: 0x0030			Register Name: SMHC_INTMASK
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	RE_INT_EN Response Error Interrupt Enable
0	/	/	/

10.2.7.14 0x0034 SMHC Masked Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: SMHC_MINTSTS
Bit	Read/Write	Default/Hex	Description
31	R	0x0	M_CARD_REMOVAL_INT Card Removed
30	R	0x0	M_CARD_INSERT Card Inserted
29:19	/	/	/
18	R	0x0	DATA_WR_BUSY_TO_INT Data write busy timeout interrupt
17	R	0x0	R1B_BUSY_CLR_INT R1B busy clear interrupt
16	R	0x0	M_SDIO_INT SDIO Interrupt
15	R	0x0	M_DEE_INT Data End-bit Error When set during receiving data, it means that the host controller does not receive valid data end bit. When set during transmitting data, it means that the host controller does not receive CRC status token.
14	R	0x0	M_ACD_INT Auto Command Done When set, it means auto stop command(CMD12) completed.
13	R	0x0	M_DSE_BC_INT Data Start Error/busy clear When set during receiving data, it means that host controller found an error start bit. When set during transmitting data, it means that busy signal is cleared after the last block.
12	R	0x0	M_CB_IW_INT Command Busy and Illegal Write
11	R	0x0	M_FU_FO_INT FIFO Underrun/Overflow
10	R	0x0	M_DSTO_VSD_INT Data Starvation Timeout/V1.8 Switch Done

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Offset: 0x0034			Register Name: SMHC_MINTSTS
Bit	Read/Write	Default/Hex	Description
9	R	0x0	M_DTO_BDS_INT Data Timeout/Boot Data Start
8	R	0x0	M_RTO_BACK_INT Response Timeout/Boot ACK Received
7	R	0x0	M_DCE_INT Data CRC Error When set during receiving data, it means that the received data have data CRC error. When set during transmitting data, it means that the received CRC status token is negative.
6	R	0x0	M_RCE_INT Response CRC Error
5	R	0x0	M_DRR_INT Data Receive Request When set, it means that there are enough data in FIFO during receiving data.
4	R	0x0	M_DTR_INT Data Transmit Request When set, it means that there are enough space in FIFO during transmitting data.
3	R	0x0	M_DTC_INT Data Transfer Complete
2	R	0x0	M_CC_INT Command Complete
1	R	0x0	M_RE_INT Response Error When set, Transmit Bit error or End Bit error or CMD Index error may occurs.
0	/	/	/

10.2.7.15 0x0038 SMHC Raw Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: SMHC_RINTSTS
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	CARD_REMOVAL Card Removed This is write-1-to-clear bits.
30	R/W1C	0x0	CARD_INSERT Card Inserted This is write-1-to-clear bits.
29:19	/	/	/

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Offset: 0x0038			Register Name: SMHC_RINTSTS
Bit	Read/Write	Default/Hex	Description
18	R/W1C	0x0	DATA_WR_BUSY_TO_INT Data write busy timeout interrupt This is write-1-to-clear bits
17	R/W1C	0x0	R1B_BUSY_CLR R1b busy clear When set, it means that the r1b busy has finished or not set.
16	R/W1C	0x0	SDIOI_INT SDIO Interrupt This is write-1-to-clear bits.
15	R/W1C	0x0	DEE Data End-bit Error When set during receiving data, it means that host controller does not receive valid data end bit. When set during transmitting data, it means that host controller does not receive CRC status token. This is write-1-to-clear bits.
14	R/W1C	0x0	ACD Auto Command Done When set, it means auto stop command(CMD12) completed. This is write-1-to-clear bits.
13	R/W1C	0x0	DSE_BC Data Start Error/busy clear When set during receiving data, it means that host controller found an error start bit. It is valid at 4-bit or 8-bit bus mode, when it set, host found start bit at data0, but not find start bit at some or all of the other data lines. When set during transmitting data, it means that busy signal is cleared after the last block. This is write-1-to-clear bits.
12	R/W1C	0x0	CB_IW Command Busy and Illegal Write This is write-1-to-clear bits.
11	R/W1C	0x0	FU_FO FIFO Underrun/Overflow This is write-1-to-clear bits.
10	R/W1C	0x0	DSTO_VSD Data Starvation Timeout/V1.8 Switch Done This is write-1-to-clear bits.

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Offset: 0x0038			Register Name: SMHC_RINTSTS
Bit	Read/Write	Default/Hex	Description
9	R/W1C	0x0	DTO_BDS Data Timeout/Boot Data Start When set during receiving data, it means host did not find start bit on data0(1-bit bus) or data0-data3(4-bit bus) or data0-data7(8-bit bus). This is write-1-to-clear bits.
8	R/W1C	0x0	RTO_BACK Response Timeout/Boot ACK Received This is write-1-to-clear bits.
7	R/W1C	0x0	DCE Data CRC Error When set during receiving data, it means that the received data have data CRC error. When set during transmitting data, it means that the received CRC status token is negative. This is write-1-to-clear bits.
6	R/W1C	0x0	RCE Response CRC Error This is write-1-to-clear bits.
5	R/W1C	0x0	DRR Data Receive Request When set, it means that there are enough data in FIFO during receiving data. This is write-1-to-clear bits.
4	R/W1C	0x0	DTR Data Transmit Request When set, it means that there is enough space in FIFO during transmitting data. This is write-1-to-clear bits.
3	R/W1C	0x0	DTC Data Transfer Complete This is write-1-to-clear bits.
2	R/W1C	0x0	CC Command Complete When set, it means that current command completes even through error occurs. This is write-1-to-clear bits.
1	R/W1C	0x0	RE Response Error When set, Transmit Bit error or End Bit error or CMD Index error may occur. This is write-1-to-clear bits.

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Offset: 0x0038			Register Name: SMHC_RINTSTS
Bit	Read/Write	Default/Hex	Description
0	/	/	/

10.2.7.16 0x003C SMHC Status Register (Default Value: 0x0000_0006)

Offset: 0x003C			Register Name: SMHC_STATUS
Bit	Read/Write	Default/Hex	Description
31	R	0x0	DMA_REQ DMA Request DMA request signal state
30:28	/	/	/
27:17	R	0x0	FIFO_LEVEL FIFO Level Number of filled locations in FIFO
16:11	R	0x0	RESP_IDX Response Index Index of previous response, including any auto-stop sent by controller
10	R	0x0	FSM_BUSY Data FSM Busy Data transmit or receive state-machine is busy 0: Card FSM is idle 1: Card FSM is busy
9	R	0x0	CARD_BUSY Card data busy Inverted version of DATA [0] 0: Card data is idle 1: Card data is busy
8	R	0x0	CARD_PRESENT Data [3] status level of DATA [3]; checks whether card is present 0: Card is not present 1: Card is present

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Offset: 0x003C			Register Name: SMHC_STATUS
Bit	Read/Write	Default/Hex	Description
7:4	R	0x0	FSM_STA Command FSM states: 0000: Idle 0001: Send INIT sequence 0010: TX CMD start bit 0011: TX CMD TX bit 0100: TX CMD index + argument 0101: TX CMD CRC7 0110: TX CMD end bit 0111: RX response start bit 1000: RX response IRQ response 1001: RX response TX bit 1010: RX response CMD index 1011: RX response data 1100: RX response CRC7 1101: RX response end bit 1110: CMD path wait NCC 1111: Wait; CMD-to-response turnaround
3	R	0x0	FIFO_FULL FIFO full 1: FIFO full 0: FIFO not full
2	R	0x1	FIFO_EMPTY FIFO Empty 1: FIFO Empty 0: FIFO not Empty
1	R	0x1	FIFO_TX_LEVEL FIFO TX Water Level flag 0: FIFO doesn t reach transmit trigger level 1: FIFO reaches transmit trigger level
0	R	0x0	FIFO_RX_LEVEL FIFO TX Water Level flag 0: FIFO doesn t reach receive trigger level 1: FIFO reaches receive trigger level

10.2.7.17 0x0040 SMHC FIFO Water Level Register (Default Value: 0x000F_0000)

Offset: 0x0040			Register Name: SMHC_FIFOTH
Bit	Read/Write	Default/Hex	Description
31	/	/	/

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Offset: 0x0040			Register Name: SMHC_FIFOTH
Bit	Read/Write	Default/Hex	Description
30:28	R/W	0x0	BSIZE_OF_TRANS Burst size of multiple transaction 000: 1 transfers 001: 4 010: 8 011: 16 Others: Reserved Should be programmed same as DMA controller multiple transaction size. The units for transfers are the DWORD(4Byte). A single transfer would be signaled based on this value. Value should be sub-multiple of (RX_TL + 1) and (FIFO_DEPTH - TX_TL) Recommended: BSIZE_OF_TRANS = 3, MSize = 16, TX_TL = 240, RX_TL = 15 FIFO_DEPTH = 256 FIFO_SIZE = 256 * 32 = 1K
27	/	/	/
26:16	R/W	0xF	RX_TL RX Trigger Level **When BLOCK2K_EXIST is 1:** 0x0-0x3FE: RX Trigger Level is 0-1022(Unit is equal to the FIFO_WIDTH) 0x3FF reserved **When BLOCK2K_EXIST is 0:** 0x0-0xFE: RX Trigger Level is 0-254 (Unit is equal to the FIFO_WIDTH) 0xFF reserved FIFO threshold when FIFO request host to receive data from FIFO. When FIFO data level is greater than this value, DMA is request is raised if DMA enabled, or RX interrupt bit is set if interrupt enabled. At the end of packet, if the last transfer is less than this level, the value is ignored and relative request will be raised as usual. 15 (means greater than 15)
15:11	/	/	/

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Offset: 0x0040			Register Name: SMHC_FIFOTH
Bit	Read/Write	Default/Hex	Description
10:0	R/W	0x0	TX_TL TX Trigger Level **When BLOCK2K_EXIST is 1**: 0x1-0x3FF: TX Trigger Level is 1-1023 (Unit is equal to the FIFO_WIDTH) 0x0: No trigger **When BLOCK2K_EXIST is 0**: 0x1-0xFF: TX Trigger Level is 1-255 (Unit is equal to the FIFO_WIDTH) 0x0: No trigger FIFO threshold when FIFO requests host to transmit data to FIFO. When FIFO data level is less than or equal to this value, DMA TX request is raised if DMA enabled, or TX request interrupt bit is set if interrupt enabled. At the end of packet, if the last transfer is less than this level, the value is ignored and relative request will be raised as usual. Recommended: 240 (means less than or equal to 240)

10.2.7.18 0x0044 SMHC FIFO Function Select Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: SMHC_FUNS
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	ABT_RDATA Abort Read Data 0: Ignored 1: After suspend command is issued during read-transfer, software polls card to find when suspend happened. Once suspend occurs, software sets bit to reset data state-machine, which is waiting for next block of data. Used in SDIO card suspends sequence. This bit is auto-cleared once controller reset to idle state.
1	R/W	0x0	READ_WAIT Read Wait 0: Clear SDIO read wait 1: Assert SDIO read wait

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Offset: 0x0044			Register Name: SMHC_FUNS
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	HOST_SEND_MMC_IRQRESQ Host Send MMC IRQ Response 0: Ignored 1: Send auto IRQ response When the host is waiting MMC card interrupt response, setting this bit will make controller cancel wait state and return to idle state, at which time, controller will receive IRQ response sent by itself. This bit is auto-cleared after response is sent.

10.2.7.19 0x0048 SMHC Transferred Byte Count Register 0 (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: SMHC_TBC0
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TBC0 Transferred Count 0 Number of bytes transferred between card and internal FIFO. The register should be accessed in full to avoid read-coherency problems and read only after data transfer completes.

10.2.7.20 0x004C SMHC Transferred Byte Count Register 1 (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: SMHC_TBC1
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TBC1 Transferred Count 1 Number of bytes transferred between Host/DMA memory and internal FIFO. The register should be accessed in full to avoid read-coherency problems and read only after data transfer completes.

10.2.7.21 0x0054 SMHC CRC Status Detect Control Register (Default Value: 0x0000_0003)

Offset: 0x0054			Register Name: SMHC_CSDC
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x0054			Register Name: SMHC_CSDC
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x3	CRC_DET_PARA CRC Detect Para 110: HS400 speed mode 011: Other speed mode Others: Reserved

10.2.7.22 0x0058 SMHC Auto Command 12 Argument Register (Default Value: 0x0000_FFFF)

Offset: 0x0058			Register Name: SMHC_A12A
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0xFFFF	A12A Auto CMD12 Argument The argument of command 12 automatically send by controller with this field.

10.2.7.23 0x005C SMHC New Timing Set Register (Default Value: 0x8171_0000)

Offset: 0x005C			Register Name: SMHC_NTSR
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	MODE_SEL Mode Select 0: Old mode of Sample/Output Timing 1: New mode of Sample/Output Timing Default value : 1
30:26	/	/	/
25	R/W	0x0	BOOT_DAT_RX_PHASE_CLR After boot ACK, before receive boot data, clear data lines' input phase. 0: Disable 1: Enable
24	R/W	0x1	CMD_DAT_RX_PHASE_CLR Clear the input phase of command lines and data lines during the update clock operation. 0: Disable 1: Enable
23	/	/	/
22	R/W	0x1	DAT_CRC_STATUS_RX_PHASE_CLR Before receiving CRC status, clear data lines input phase. 0: Disable 1: Enable

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Offset: 0x005C			Register Name: SMHC_NTSR
Bit	Read/Write	Default/Hex	Description
21	R/W	0x1	DAT_TRANS_RX_PHASE_CLR Before transferring data, clear data lines input phase. 0: Disable 1: Enable
20	R/W	0x1	DAT_RECV_RX_PHASE_CLR Before receive data, clear data lines input phase clear 0: Disable 1: Enable
19:17	/	/	/
16	R/W	0x1	CMD_SEND_RX_PHASE_CLR Clear command RX phase before sending the command. 0: Disable 1: Enable
15:10	/	/	/
9:8	R/W	0x0	DAT_SAM_TIM_PHS Data Sample Timing Phase 00: Sample timing phase offset 90^0 01: Sample timing phase offset 180^0 10: Sample timing phase offset 270^0 11: Sample timing phase offset 0^0 only for SD2 hs400 mode Default value: 00
7:6	/	/	/
5:4	R/W	0x0	CMD_SAM_TIM_PHS Command Sample Timing Phase 00: Sample timing phase offset 90^0 01: Sample timing phase offset 180^0 10: Sample timing phase offset 270^0 11: Ignore Default value: 00
3:1	/	/	/
0	R/W	0x0	HS400_NEW_SAM_EN HS400 New Sample Enable 1: Enable hs400 new sample method 0: Disable hs400 new sample method

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10.2.7.24 0x0078 SMHC Hardware Reset Register (Default Value: 0x0000_0001)

Offset: 0x0078			Register Name: SMHC_HWRST
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x1	HW_RST Hardware Reset 1: Active mode 0: Reset These bits cause the cards to enter pre-idle state, which requires them to be re-initialized.

10.2.7.25 0x0080 SMHC IDMAC Control Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: SMHC_IDMAC
Bit	Read/Write	Default/Hex	Description
31	W	0x0	DES_LOAD_CTRL When IDMAC fetches a descriptor, if the valid bit of a descriptor is not set, IDMAC FSM will go to the suspend state. Setting this bit will make IDMAC re-fetch descriptor again and do the transfer normally.
30:8	/	/	/
7	R/W	0x0	IDMAC_ENB IDMAC Enable When set, the IDMAC is enabled. DE is read/write.
6:2	/	/	/
1	R/W	0x0	FIX_BUST_CTRL Fixed Burst Controls whether the AHB Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 during start of normal burst transfers. When reset, the AHB will use SINGLE and INCR burst transfer operations.
0	R/W	0x0	IDMAC_RST DMA Reset When set, the DMA Controller resets all its internal registers. SWR is read/write. It is automatically cleared after 1 clock cycle.

10.2.7.26 0x0084 SMHC Descriptor List Base Address Register (Default Value: 0x0000_0000)

Offset: 0x0084	Register Name: SMHC_DLBA

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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DES_BASE_ADDR Start of Descriptor List Contains the base address of the First Descriptor It is a word (4byte) address.

10.2.7.27 0x0088 SMHC IDMAC Status Register (Default Value: 0x0000_0000)

Offset: 0x0088			Register Name: SMHC_IDST
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:10	R	0x0	IDMAC_ERR_STA Error Bits Indicates the type of error that caused a Bus Error. Valid only with Fatal Bus Error bit (SMHC_ IDST [2]) set. This field does not generate an interrupt. 001: Host Abort received during transmission 010: Host Abort received during reception Others: Reserved This bit is read-only.
9	R/W1C	0x0	AIS Abnormal Interrupt Summary Logical OR of the following: SMHC_IDST [2]: Fatal Bus Interrupt SMHC_IDST [4]: Descriptor unavailable bit Interrupt SMHC_IDST [5]: Card Error Summary Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared. Writing a 1 clears this bit.
8	R/W1C	0x0	NIS Normal Interrupt Summary Logical OR of the following: SMHC_IDST [0]: Transmit Interrupt SMHC_IDST [1]: Receive Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes NIS to be set is cleared. Writing a 1 clears this bit.
7:6	/	/	/

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Offset: 0x0088			Register Name: SMHC_IDST
Bit	Read/Write	Default/Hex	Description
5	R/W1C	0x0	ERR_FLAG_SUM Card Error Summary Indicates the status of the transaction to/from the card; also present in RINTSTS. Indicates the logical OR of the following bits: EBE: End Bit Error RTO: Response Timeout RCRC: Response CRC SBE: Start Bit Error DRTO: Data Read Timeout DCRC: Data CRC for Receive RE: Response Error Writing a 1 clears this bit.
4	R/W1C	0x0	DES_UNAVL_INT Descriptor Unavailable Interrupt This bit is set when the descriptor is unavailable due to OWN bit = 0 (DES0[31] =0). Writing a 1 clears this bit.
3	/	/	/
2	R/W1C	0x0	FATAL_BERR_INT Fatal Bus Error Interrupt Indicates that a Bus Error occurred (SMHC_IDST [12:10]). When this bit is set, the DMA disables all its bus accesses. Writing a 1 clears this bit.
1	R/W1C	0x0	RX_INT Receive Interrupt Indicates the completion of data reception for a descriptor. Writing a 1 clears this bit.
0	R/W1C	0x0	TX_INT Transmit Interrupt Indicates that data transmission is finished for a descriptor. Writing a 1 clears this bit.

10.2.7.28 0x008C SMHC IDMAC Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x008C			Register Name: SMHC_IDIE
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/

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Offset: 0x008C			Register Name: SMHC_IDIE
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	ERR_SUM_INT_ENB Card Error Summary Interrupt Enable When set, it enables the Card Interrupt summary.
5	R/W	0x0	ERR_SUM_INT_ENB Card Error Summary Interrupt Enable When set, it enables the Card Interrupt summary.
4	R/W	0x0	DES_UNAVL_INT_ENB Descriptor Unavailable Interrupt When set along with Abnormal Interrupt Summary Enable, the Descriptor Unavailable interrupt is enabled.
3	/	/	/
2	R/W	0x0	FERR_INT_ENB Fatal Bus Error Enable When set with Abnormal Interrupt Summary Enable, the Fatal Bus Error Interrupt is enabled. When reset, Fatal Bus Error Enable Interrupt is disabled.
1	R/W	0x0	RX_INT_ENB Receive Interrupt Enable When set, Normal Interrupt Summary Enable, Receive Interrupt is enabled. When reset, Receive Interrupt is disabled.
0	R/W	0x0	TX_INT_ENB Transmit Interrupt Enable When set, Normal Interrupt Summary Enable, Transmit Interrupt is enabled. When reset, Transmit Interrupt is disabled.

10.2.7.29 0x0100 SMHC Card Threshold Control Register (Default Value: 0x0000_0005)

Offset: 0x0100			Register Name: SMHC_THLD
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	CARD_WR_THLD Card Read/write Threshold Size (Unit: byte)
15:3	/	/	/

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Offset: 0x0100			Register Name: SMHC_THLD
Bit	Read/Write	Default/Hex	Description
2	R/W	0x1	CARD_WR_THLD_ENB Card Write Threshold Enable 0: Card write threshold disable 1: Card write threshold enabled Host controller initiates write transfer only if card threshold amount of data is available in transmit FIFO
1	R/W	0x0	BCIG Busy Clear Interrupt Generation 0: Busy Clear Interrupt disabled 1: Busy Clear Interrupt Enabled The application can disable this feature if it does not want to wait for a Busy Clear Interrupt.
0	R/W	0x1	CARD_RD_THLD_ENB Card Read Threshold Enable 0: Card Read Threshold Disable 1: Card Read Threshold Enable Host controller initiates Read Transfer only if CARD_RD_THLD amount of space is available in receive FIFO

10.2.7.30 0x0104 SMHC Sample FIFO Control Register (Default Value: 0x0000_0006)

Offset: 0x0104			Register Name: SMHC_SFC
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4:1	R/W	0x3	STOP_CLK_CTRL Stop Clock Control When receiving data, if CARD_RD_THLD_ENB is set and CARD_RD_THLD is set same with BLK_SZ, the device clock may stop at block gap during data receiving. This field is used to control the position of stopping clock. The value can be change between 0x0 and 0xF, but actually the available value and the position of stopping clock must be decided by the actual situation. The value increase one in this field is linked to one cycle(two cycle in DDR mode) that the position of stopping clock moved up.

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Offset: 0x0104			Register Name: SMHC_SFC
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	BYPASS_EN Bypass enable When set, sample FIFO will be bypassed.

10.2.7.31 0x0108 SMHC Auto Command 23 Argument Register (Default Value: 0x0000_0000)

Offset: 0x0108			Register Name: SMHC_A23A
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	A23A Auto CMD23 Argument The argument of command 23 is automatically sent by controller with this field.

10.2.7.32 0x010C SMHC eMMC4.5 DDR Start Bit Detection Control Register (Default Value: 0x0000_0000)

Offset: 0x010C			Register Name: EMMC_DDR_SBIT_DET
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HS400_MD_EN HS400 Mode Enable 0: Disable 1: Enable It is required to set this bit to '1' before initiating any data transfer CMD in HS400 mode.
30:1	/	/	/
0	R/W	0x0	HALF_START_BIT Control for start bit detection mechanism inside host controller based on duration of start bit. For eMMC 4.5, start bit can be: 0: Full cycle 1: Less than one full cycle Set HALF_START_BIT=1 for eMMC 4.5 and above; set to 0 for SD applications.

10.2.7.33 0x0110 SMHC Response CRC Register (Default Value:0x0000_0000)

Offset: 0x0110			Register Name: SMHC_RESP_CRC
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/

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Offset: 0x0110			Register Name: SMHC_RESP_CRC
Bit	Read/Write	Default/Hex	Description
6:0	R	0x0	RESP_CRC Response CRC Response CRC from device

10.2.7.34 0x0138 SMHC Extended Command Register (Default Value: 0x0000_0000)

Offset: 0x0138			Register Name: SMHC_EXT_CMD
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	AUTO_CMD23_EN Send CMD23 Automatically When set this bit, send CMD23 automatically before send command specified in SMHC_CMD register. When SOFT_RST set, this field will be cleared.

10.2.7.35 0x013C SMHC Extended Response Register (Default Value: 0x0000_0000)

Offset: 0x013C			Register Name: SMHC_EXT_RESP
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SMHC_EXT_RESP When AUTO_CMD23_EN is set, this register stores the response of CMD23.

10.2.7.36 0x0140 SMHC Drive Delay Control Register (Default Value: 0x0001_0000)

Offset: 0x0140			Register Name: SMHC_DRV_DL
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/

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Offset: 0x0140			Register Name: SMHC_DRV_DL
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	DAT_DRV_PH_SEL Data Drive Phase Select 0: Data drive phase offset is: 90^0 at SDR mode, 45^0 at DDR8 mode, 90^0 at DDR4 mode when 0x5C [31] is low, 45^0 at DDR4 mode when 0x5C [31] is high, 90^0 at HS400 mode when 0x5C [0] is low, 45^0 at HS400 mode when 0x5C [0] is high. 1: Data drive phase offset is: 180^0 at SDR mode, 90^0 at DDR8 mode, 180^0 at DDR4 mode when 0x5C [31] is low, 90^0 at DDR4 mode when 0x5C [31] is high, 180^0 at HS400 mode when 0x5C [0] is low, 90^0 at HS400 mode when 0x5C[0] is high.
16	R/W	0x1	CMD_DRV_PH_SEL Command Drive Phase Select 0: Command drive phase offset is: 90^0 at SDR mode, 45^0 at DDR8 mode, 90^0 at DDR4 mode when 0x5C [31] is low, 45^0 at DDR4 mode when 0x5C [31] is high, 90^0 at HS400 mode 1: Command drive phase offset is: 180^0 at SDR mode, 90^0 at DDR8 mode, 180^0 at DDR4 mode when 0x5C [31] is low, 90^0 at DDR4 mode when 0x5C [31] is high, 180^0 at HS400 mode.
15:0	/	/	/

10.2.7.37 0x0144 SMHC Sample Delay Control Register (Default Value: 0x0000_2000)

Offset: 0x0144			Register Name: SMHC_SAMP_DL
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:8	R	0x20	Reserved

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Offset: 0x0144			Register Name: SMHC_SAMP_DL
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	SAMP_DL_SW_EN Sample Delay Software Enable When set, enable sample delay specified at SAMP_DL_SW
6	/	/	/
5:0	R/W	0x0	SAMP_DL_SW Sample Delay Software The relative delay between clock line and command line, data lines. It can be determined according to the value of SAMP_DL, the cycle of card clock and device s input timing requirement.

10.2.7.38 0x0148 SMHC Data Strobe Delay Control Register (Default Value: 0x0000_2000)

Offset: 0x0148			Register Name: SMHC_DS_DL
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:8	R	0x20	Reserved
7	R/W	0x0	DS_DL_SW_EN Sample Delay Software Enable
6	/	/	/
5:0	R/W	0x0	DS_DL_SW Data Strobe Delay Software

10.2.7.39 0x0154 SMHC Card Interface Counter Register (Default Value:0x0000_0000)

Offset: 0x0154			Register Name: SMHC_CIU_CNT
Bit	Read/Write	Default/Hex	Description
31:8	R	0x0	DATA_PATH_CNT Data Path Counter This field indicates the count value of counters for each stage when host read or write data. When Host read data, this field is one of the following values: The time of wait data start bit When Host write data, this field is one of the following values: The timer of wait busy clear

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Offset: 0x0154			Register Name: SMHC_CIU_CNT
Bit	Read/Write	Default/Hex	Description
7:0	/	/	/

10.2.7.40 0x015C SMHC Write Busy Timeout Register (Default Value:0x00FF_FFFF)

Offset: 0x015C			Register Name: SMHC_WR_BUSY_TMOUT
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:0	R/W	0xffffff	WR_BUSY_TO_LMT Data write busy timeout Limit

10.2.7.41 0x0160 SMHC Deskew Data0 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0160			Register Name: SMHC_DESKEW_DATA0_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA0_DL_CAL_START DESKEW_DATA0 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA0_DL_CAL_DONE DESKEW_DATA0 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA0_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA0_DL DESKEW_DATA0 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA0_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA0_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA0_DL_SW DESKEW_DATA0 Delay Software

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10.2.7.42 0x0164 SMHC Deskew Data1 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0164			Register Name: SMHC_DESKEW_DATA1_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA1_DL_CAL_START DESKEW_DATA1 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA1_DL_CAL_DONE DESKEW_DATA1 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA1_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA1_DL DESKEW_DATA1 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA1_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA1_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA1_DL_SW DESKEW_DATA1 Delay Software

10.2.7.43 0x0168 SMHC Deskew Data2 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0168			Register Name: SMHC_DESKEW_DATA2_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA2_DL_CAL_START DESKEW_DATA2 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA2_DL_CAL_DONE DESKEW_DATA2 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA2_DL.
13:12	/	/	/

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Offset: 0x0168			Register Name: SMHC_DESKEW_DATA2_DL
Bit	Read/Write	Default/Hex	Description
11:8	R	0x0	DESKEW_DATA2_DL DESKEW_DATA2 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA2_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA2_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA2_DL_SW DESKEW_DATA2 Delay Software

10.2.7.44 0x016C SMHC Deskew Data3 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x016C			Register Name: SMHC_DESKEW_DATA3_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA3_DL_CAL_START DESKEW_DATA3 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA3_DL_CAL_DONE DESKEW_DATA3 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA3_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA3_DL DESKEW_DATA3 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA3_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA3_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA3_DL_SW DESKEW_DATA3 Delay Software

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10.2.7.45 0x0170 SMHC Deskew Data4 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: SMHC_DESKEW_DATA4_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA4_DL_CAL_START DESKEW_DATA4 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA4_DL_CAL_DONE DESKEW_DATA4 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA4_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA4_DL DESKEW_DATA4 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA4_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA4_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA4_DL_SW DESKEW_DATA4 Delay Software

10.2.7.46 0x0174 SMHC Deskew Data5 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0174			Register Name: SMHC_DESKEW_DATA5_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA5_DL_CAL_START DESKEW_DATA5 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA5_DL_CAL_DONE DESKEW_DATA5 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA5_DL.
13:12	/	/	/

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Offset: 0x0174			Register Name: SMHC_DESKEW_DATA5_DL
Bit	Read/Write	Default/Hex	Description
11:8	R	0x0	DESKEW_DATA5_DL DESKEW_DATA5 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA5_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA5_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA5_DL_SW DESKEW_DATA5 Delay Software

10.2.7.47 0x0178 SMHC Deskew Data6 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0178			Register Name: SMHC_DESKEW_DATA6_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA6_DL_CAL_START DESKEW_DATA6 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA6_DL_CAL_DONE DESKEW_DATA6 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA6_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA6_DL DESKEW_DATA6 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when DESKEW_DATA6_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA6_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA6_DL_SW DESKEW_DATA6 Delay Software

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10.2.7.48 0x017C SMHC Deskew Data7 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x017C			Register Name: SMHC_DESKEW_DATA7_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DATA7_DL_CAL_START DESKEW_DATA7 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DATA7_DL_CAL_DONE DESKEW_DATA7 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DATA7_DL.
13:12	/	/	/
11:8	R	0x0	DESKEW_DATA7_DL DESKEW_DATA7 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DATA7_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DATA7_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DATA7_DL_SW DESKEW_DATA7 Delay Software

10.2.7.49 0x0180 SMHC Deskew DS Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0180			Register Name: SMHC_DESKEW_DS_DL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DESKEW_DS_DL_CAL_START DESKEW_DS Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	DESKEW_DS_DL_CAL_DONE DESKEW_DS Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in DESKEW_DS_DL.
13:12	/	/	/

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Offset: 0x0180			Register Name: SMHC_DESKEW_DS_DL
Bit	Read/Write	Default/Hex	Description
11:8	R	0x0	DESKEW_DS_DL DESKEW_DS Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when DESKEW_DS_DL_CAL_DONE is set.
7	R/W	0x0	DESKEW_DS_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	DESKEW_DS_DL_SW DESKEW_DS Delay Software

10.2.7.50 0x0184 SMHC Deskew Control Control Register (Default Value: 0x0000_0003)

Offset: 0x0184			Register Name: SMHC_DESKEW_CTRL
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	CDATA_IN_DL_EN CDATA input delay enable 0: Use original CDATA_IN before deskew 1: Use CDATA_IN delay chain before deskew
6:5	/	/	/
4	R/W	0x0	DESKEW_AUTO_MODE Deskew auto mode select 0: Deskew data only one time 1: Deskew data every RDATA BLK
3	R/W	0x0	DESKEW_DS_DL_SEL_SE_EN Deskew DS signal delay select 0: Deskew data strobe delay chain step use HW value 1: Deskew data strobe delay chain step use SW value(0x170[3:0])
2	R/W	0x0	DESKEW_DATA_DL_SEL_SE_EN Deskew DATA signal delay select 0: Deskew data delay chain step use HW value 1: Deskew data delay chain step use SW value (0x150-0x16C[3:0])

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Offset: 0x0184			Register Name: SMHC_DESKEW_CTRL
Bit	Read/Write	Default/Hex	Description
1	R/W	0x1	DESKEW_DS_DL_BYPASS Deskew DS select 0: Use deskew data strobe 1: Use original data strobe
0	R/W	0x1	DESKEW_DATA_DL_BYPASS Deskew DATA select 0: Use deskew CDATA_IN 1: Use original CDATA_IN

10.2.7.51 0x0190 SMHC CDATA_IN Data0/1 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0190			Register Name: SMHC_CDATA_IN_DATA0_1_DL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CDATA_IN_DATA1_DL_CAL_START CDATA_IN_DATA1 Delay Calibration Start When set, start sample delay chain calibration.
30	R	0x0	CDATA_IN_DATA1_DL_CAL_DONE CDATA_IN_DATA1 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA1_DL.
29:28	/	/	/
27:24	R	0x0	CDATA_IN_DATA1_DL CDATA_IN_DATA1 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when CDATA_IN_DATA1_DL_CAL_DONE is set.
23	R/W	0x0	CDATA_IN_DATA1_DL_SW_EN Sample Delay Software Enable
22:20	/	/	/
19:16	R/W	0x0	CDATA_IN_DATA1_DL_SW CDATA_IN_DATA1 Delay Software
15	R/W	0x0	CDATA_IN_DATA0_DL_CAL_START CDATA_IN_DATA0 Delay Calibration Start When set, start sample delay chain calibration.

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Offset: 0x0190			Register Name: SMHC_CDATA_IN_DATA0_1_DL
Bit	Read/Write	Default/Hex	Description
14	R	0x0	CDATA_IN_DATA0_DL_CAL_DONE CDATA_IN_DATA0 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA0_DL.
13:12	/	/	/
11:8	R	0x0	CDATA_IN_DATA0_DL CDATA_IN_DATA0 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC s clock nearly. This bit is valid only when CDATA_IN_DATA0_DL_CAL_DONE is set.
7	R/W	0x0	CDATA_IN_DATA0_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	CDATA_IN_DATA0_DL_SW CDATA_IN_DATA0 Delay Software

10.2.7.52 0x0194 SMHC CDATA_IN Data2/3 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0194			Register Name: SMHC_CDATA_IN_DATA2_3_DL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CDATA_IN_DATA3_DL_CAL_START CDATA_IN_DATA3 Delay Calibration Start When set, start sample delay chain calibration.
30	R	0x0	CDATA_IN_DATA3_DL_CAL_DONE CDATA_IN_DATA3 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA3_DL.
29:28	/	/	/
27:24	R	0x0	CDATA_IN_DATA3_DL CDATA_IN_DATA3 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA3_DL_CAL_DONE is set.

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Offset: 0x0194			Register Name: SMHC_CDATA_IN_DATA2_3_DL
Bit	Read/Write	Default/Hex	Description
23	R/W	0x0	CDATA_IN_DATA3_DL_SW_EN Sample Delay Software Enable
22:20	/	/	/
19:16	R/W	0x0	CDATA_IN_DATA3_DL_SW CDATA_IN_DATA3 Delay Software
15	R/W	0x0	CDATA_IN_DATA2_DL_CAL_START CDATA_IN_DATA2 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	CDATA_IN_DATA2_DL_CAL_DONE CDATA_IN_DATA2 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA2_DL.
13:12	/	/	/
11:8	R	0x0	CDATA_IN_DATA2_DL CDATA_IN_DATA2 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA2_DL_CAL_DONE is set.
7	R/W	0x0	CDATA_IN_DATA2_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	CDATA_IN_DATA2_DL_SW CDATA_IN_DATA2 Delay Software

10.2.7.53 0x0198 SMHC CDATA_IN Data4/5 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x0198			Register Name: SMHC_CDATA_IN_DATA4_5_DL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CDATA_IN_DATA5_DL_CAL_START CDATA_IN_DATA5 Delay Calibration Start When set, start sample delay chain calibration.
30	R	0x0	CDATA_IN_DATA5_DL_CAL_DONE CDATA_IN_DATA5 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA5_DL.
29:28	/	/	/

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Offset: 0x0198			Register Name: SMHC_CDATA_IN_DATA4_5_DL
Bit	Read/Write	Default/Hex	Description
27:24	R	0x0	CDATA_IN_DATA5_DL CDATA_IN_DATA5 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA5_DL_CAL_DONE is set.
23	R/W	0x0	CDATA_IN_DATA5_DL_SW_EN Sample Delay Software Enable
22:20	/	/	/
19:16	R/W	0x0	CDATA_IN_DATA5_DL_SW CDATA_IN_DATA5 Delay Software
15	R/W	0x0	CDATA_IN_DATA4_DL_CAL_START CDATA_IN_DATA4 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	CDATA_IN_DATA4_DL_CAL_DONE CDATA_IN_DATA4 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA4_DL.
13:12	/	/	/
11:8	R	0x0	CDATA_IN_DATA4_DL CDATA_IN_DATA4 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA4_DL_CAL_DONE is set.
7	R/W	0x0	CDATA_IN_DATA4_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/
3:0	R/W	0x0	CDATA_IN_DATA4_DL_SW CDATA_IN_DATA4 Delay Software

10.2.7.54 0x019C SMHC CDATA_IN Data6/7 Delay Control Register (Default Value: 0x0000_0000)

Offset: 0x019C			Register Name: SMHC_CDATA_IN_DATA6_7_DL
Bit	Read/Write	Default/Hex	Description

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Offset: 0x019C			Register Name: SMHC_CDATA_IN_DATA6_7_DL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CDATA_IN_DATA7_DL_CAL_START CDATA_IN_DATA7 Delay Calibration Start When set, start sample delay chain calibration.
30	R	0x0	CDATA_IN_DATA7_DL_CAL_DONE CDATA_IN_DATA7 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA7_DL.
29:28	/	/	/
27:24	R	0x0	CDATA_IN_DATA7_DL CDATA_IN_DATA7 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA7_DL_CAL_DONE is set.
23	R/W	0x0	CDATA_IN_DATA7_DL_SW_EN Sample Delay Software Enable
22:20	/	/	/
19:16	R/W	0x0	CDATA_IN_DATA7_DL_SW CDATA_IN_DATA7 Delay Software
15	R/W	0x0	CDATA_IN_DATA6_DL_CAL_START CDATA_IN_DATA6 Delay Calibration Start When set, start sample delay chain calibration.
14	R	0x0	CDATA_IN_DATA6_DL_CAL_DONE CDATA_IN_DATA6 Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in CDATA_IN_DATA6_DL.
13:12	/	/	/
11:8	R	0x0	CDATA_IN_DATA6_DL CDATA_IN_DATA6 Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of SMHC's clock nearly. This bit is valid only when CDATA_IN_DATA6_DL_CAL_DONE is set.
7	R/W	0x0	CDATA_IN_DATA6_DL_SW_EN Sample Delay Software Enable
6:4	/	/	/

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Offset: 0x019C			Register Name: SMHC_CDATA_IN_DATA6_7_DL
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	CDATA_IN_DATA6_DL_SW CDATA_IN_DATA6 Delay Software

10.2.7.55 0x01A0 SMHC CQE Control Register (Default Value: 0x0000_0003)

Offset: 0x01A0			Register Name: SMHC_CQE_CTRL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	DS_SAMP_RESP_EN datastrobe sample response enable It should be set when CQE work at HS400 enhance strobe mode
7:2	/	/	/
1	R/W	0x1	CQE_CLR_DTC_EN CQE clear data transfer complete(DTC) status select 0: CQE clear data transfer complete(DTC) status disable 1: CQE clear data transfer complete(DTC) status enable
0	R/W	0x1	CQE_CLR_CC_EN CQE clear command complete(CC) status select 0: CQE clear command complete(CC) status disable 1: CQE clear command complete(CC) status enable

10.2.7.56 0x0200 SMHC FIFO Register (Default Value: 0x0000_0000)

Offset: 0x0200			Register Name: SMHC_FIFO
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TX/RX_FIFO Data FIFO

10.2.7.57 0x0400 Command Queuing Version Register (Default Value: 0x0000_0510)

Offset: 0x0400			Register Name: CQVER
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/

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Offset: 0x0400			Register Name: CQVER
Bit	Read/Write	Default/Hex	Description
11:8	R	0x5	CQE_VER_MAJ eMMC Major Version Number (digit left of decimal point), in BCD format
07:4	R	0x1	CQE_VER_MIN e •MMC Minor Version Number (digit right of decimal point), in BCD format
03:0	R	0x0	CQE_VER_SUF e •MMC Version Suffix (2nd digit right of decimal point), in BCD format

10.2.7.58 0x0408 Command Queuing Configuration Register (Default Value: 0x0200_0000)

Offset: 0x0408			Register Name: CQCFG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x200	CQE_BLOCK_SIZE CQE Block Size Configuration It must be 0x200 when CMDQ_MODE is CMDQ or CMDQ_HALF or CMDQ_ALMOST.
15:13	/	/	/
12	R/W	0x0	CQE_DCMD_EN Direct Command (DCMD) Enable This bit indicates to the hardware whether the Task Descriptor in slot #31 of the TDL is a Data Transfer Task Descriptor, or a Direct Command Task Descriptor. CQE uses this bit when a task is issued in slot #31, to determine how to decode the Task Descriptor. Bit Value Description 1: Task descriptor in slot #31 is a DCMD Task Descriptor 0: Task descriptor in slot #31 is a Data Transfer Task Descriptor
11:9	/	/	/

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Offset: 0x0408			Register Name: CQCFG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	CQE_TASK_DESC_SIZE Task Descriptor Size This bit indicates whether the task descriptor size is 128 bits or 64 bits as detailed in Data Structures section. This bit can only be configured when Command Queuing Enable bit is 0 (command queuing is disabled) Bit Value Description 1: Task descriptor size is 128 bits 0: Task descriptor size is 64 bits
7:6	/	/	/
5:4	R/W	0x0	CMDQ_MODE Command Queue Mode select This field is used to select CMDQ mode, there are four kinds of CMDQ mode: ● CMDP mode (serial CMDQ mode) CMD and data can't be transmitted at the same time, and the current task can t be sent to the device before the previous task is completed. ● CMDQ_HALF mode CMD and data can't be transmitted at the same time, but the current task can be sent to the device before the previous task is completed. ● CMDQ_ALMOST mode CMD can be sent at the start of block, and the current task can be sent to the device before the previous task is completed. The Field Value Description 2'b00: CMDP mode 2'b01: CMDQ_HALF mode 2'b10: CMDQ_ALMOST mode 2'b11: Reserved Note: The CARD_WR_THLD_ENB must be set and the value of CARD_WR_THLD must be greater than or equal CQE_BLOCK_SIZE when CQE_MODE is CMDQ_ALMOST.
3:1	/	/	/

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Offset: 0x0408			Register Name: CQCFG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CQE_EN Command Queuing Enable Software shall write '1' this bit when in order to enable command queuing mode (i.e., enable CQE). When this bit is 0, CQE is disabled and software controls the eMMC bus using the legacy eMMC host controller. Before software writes '1' to this bit, software shall verify that the eMMC host controller is in idle state and there are no commands or data transfers ongoing. When software wants to exit command queuing mode, it shall clear all previous tasks if such exist before setting this bit to 0.

10.2.7.59 0x040C Command Queuing Control Register (Default Value: 0x0000_0000)

Offset: 0x040C			Register Name: CQCTL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/WAC	0x0	CQE_TCL_ALL CQE Task Cleared All Software shall write '1' this bit when it wants to clear all the tasks sent to the device. This bit can only be written when CQE is in halt state (i.e., Halt bit is 1). When software writes 1, the value of the register is updated to '1', and CQE shall reset CQTDBR register and all other context information for all unfinished tasks. Then CQE will clear this bit. Software should poll on this bit until it is set to back 0 and may then resume normal operation, by clearing the Halt bit. CQE does not communicate to the device that the tasks were cleared. It is software s responsibility to order the device to discard the tasks in its queue using CMDQ_TASK_MGMT command. Writing '0' to this register shall have no effect.
7:1	/	/	/

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Offset: 0x040C			Register Name: CQCTL
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CQE_HALT Host software shall write '1' to the bit when it wants to acquire software control over the eMMC bus and disable CQE from issuing commands on the bus. For example, issuing a Discard Task command (CMDQ_TASK_MGMT) When software writes '1', CQE shall complete the ongoing task if such a task is in progress. Once the task is completed and CQE is in idle state, CQE shall not issue new commands and shall indicate so to software by setting this bit to 1. Software may poll on this bit until it is set to 1, and may only then send commands on the eMMC bus. In order to exit halt state (i.e., resume CQE activity), software shall clear this bit (write '0'). Writing '0' when the value is already '0' shall have no effect.

10.2.7.60 0x0410 Command Queuing Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0410			Register Name: CQIS
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W1C	0x0	CQIS_TCL Task Cleared This status bit is asserted (if CQISTE.TCL=1) when a task clear operation is completed by CQE. The completed task clear operation is either an individual task clear (CQTCLR) or clearing of all tasks (CQCTL).
2	R/W1C	0x0	CQIS_RED Response Error Detected Interrupt This status bit is asserted (if CQISTE.RED=1) when a response is received with an error bit set in the device status field. Software uses CQRMEM register to configure which device status bit fields may trigger an interrupt, and which are masked.

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Offset: 0x0410			Register Name: CQIS
Bit	Read/Write	Default/Hex	Description
1	R/W1C	0x0	CQIS_TCC Task Complete Interrupt This status bit is asserted (if CQISTE.TCC=1) when at least one of the following two conditions are met: A task is completed and the INT bit is set in its Task Descriptor Interrupt caused by Interrupt Coalescing logic (see 0)
0	R/W1C	0x0	CQIS_HAC Halt Complete Interrupt This status bit is asserted (if CQISTE.HAC=1) when halt bit in CQCTL register transitions from 0 to 1 indicating that host controller has completed its current ongoing task and has entered halt state.

10.2.7.61 0x0414 Command Queuing Interrupt Status Enable Register (Default Value: 0x0000_0000)

Offset: 0x0414			Register Name: CQISTE
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	CQISTE_TCL Task Cleared Status Enable 1: CQIS.TCL will be set when its interrupt condition is active 0: CQIS.TCL is disabled
2	R/W	0x0	CQISTE_RED Response Error Detected Status Enable 1: CQIS.RED will be set when its interrupt condition is active 0: CQIS.RED is disabled
1	R/W	0x0	CQISTE_TCC Task Complete Status Enable 1: CQIS.TCC will be set when its interrupt condition is active 0: CQIS.TCC is disabled
0	R/W	0x0	CQISTE_HAC Halt Complete Status Enable 1: CQIS.HAC will be set when its interrupt condition is active 0: CQIS.HAC is disabled

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10.2.7.62 0x0418 Command Queuing Interrupt Signal Enable Register (Default Value: 0x0000_0000)

Offset: 0x0418			Register Name: CQISGE
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	CQISGE_TCL Task Cleared Signal Enable When set and CQIS.TCL is asserted, the CQE shall generate an interrupt
2	R/W	0x0	CQISGE_RED Response Error Detected Signal Enable When set and CQIS.RED is asserted, the CQE shall generate an interrupt
1	R/W	0x0	CQISGE_TCC Task Complete Signal Enable When set and CQIS.TCC is asserted, the CQE shall generate an interrupt
0	R/W	0x0	CQISGE_HAC Halt Complete Signal Enable When set and CQIS.HAC is asserted, the CQE shall generate an interrupt

10.2.7.63 0x041C Command Queuing Interrupt Coalescing Register (Default Value: 0x0000_0000)

Offset: 0x041C			Register Name: CQIC
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CQIC_EN Interrupt Coalescing Enable/Disable When set to '0' by software, command responses are neither counted nor timed. Interrupts are still triggered by completion of tasks with INT=1 in the Task Descriptor. When set to '1', the interrupt coalescing mechanism is enabled and coalesced interrupts are generated
30:21	/	/	/

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Offset: 0x041C			Register Name: CQIC
Bit	Read/Write	Default/Hex	Description
20	R	0x0	CQIC_SB Interrupt Coalescing Status Bit This bit indicates to software whether any tasks (with INT=0) have completed and counted towards interrupt coalescing (i.e., ICSB is set if and only if IC counter > 0). Bit Value Description 1: At least one task completion has been counted (IC counter >0) 0: No task completions have occurred since last counter reset (IC counter =0)
19:17	/	/	/
16	W	0x0	CQIC_CTR Counter and Timer Reset When host driver writes '1', the interrupt coalescing timer and counter are reset
15	W	0x0	CQIC_CTH_WEN Interrupt Coalescing Counter Threshold Write Enable When software writes '1', the value ICCTH is updated with the contents written at the same cycle. When software writes '0', the value in ICCTH is not updated. Note: Write operations to ICCTH are only allowed when the task queue is empty.
14:13	/	/	/
12:8	R/W	0x0	CQIC_CTH Interrupt Coalescing Counter Threshold Software uses this field to configure the number of task completions (only tasks with INT=0 in the Task Descriptor) which are required in order to generate an interrupt. Counter Operation: As data transfer tasks with INT=0 complete, they are counted by CQE. The counter is reset by software during the interrupt service routine. The counter stops counting when it reaches the value configured in ICCTH. The maximum allowed value is 31 Note: • When ICCTH is 0, task completions are not counted, and counting-based interrupts are not generated. • In order to write to this field, the ICCTHWEN bit must be set at the same write operation.

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Offset: 0x041C			Register Name: CQIC
Bit	Read/Write	Default/Hex	Description
7	W	0x0	CQIC_TOVAL_WEN Interrupt Coalescing Timeout Value Write Enable When software writes '1', the value ICTOVAL is updated with the contents written at the same cycle. When software writes '0', the value in ICTOVAL is not updated. Note: Write operations to ICTOVAL are only allowed when the task queue is empty.
6:0	R/W	0x0	CQIC_TOVAL Interrupt Coalescing Timeout Value Software uses this field to configure the maximum time allowed between the completion of a task on the bus and the generation of an interrupt. Timer Operation: The timer is reset by software during the interrupt service routine. It starts running when a data transfer task with INT=0 is completed, after the timer was reset. When the timer reaches the value configured in ICTOVAL field it generates an interrupt and stops. The timer's unit is equal to 1024 clock periods of the clock whose frequency is specified in the Internal Timer Clock Frequency field CQCAP register. The minimum value is 01h (1024 clock periods) and the maximum value is 7Fh (127*1024 clock periods). For example, a CQCAP field value of 0 indicates a 19.2 MHz clock frequency (period = 52.08 ns). If the setting in ICTOVAL is 10h, the calculated polling period is 16*1024*52.08 ns= 853.33 us Note: When ICTOVAL is 0, the timer is not running, and timer-based interrupts are not generated. In order to write to this field, the ICTOVALWEN bit must be set at the same write operation.

10.2.7.64 0x0420 Command Queuing Task Descriptor List Base Address Register (Default Value: 0x0000_0000)

Offset: 0x0420			Register Name: CQTDLBA
Bit	Read/Write	Default/Hex	Description

Draft Version

Offset: 0x0420			Register Name: CQTDLBA
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CQE_TDLBA Task Descriptor List Base Address This register stores the LSB bits (bits 31:0) of the byte address of the head of the Task Descriptor List in system memory. The size of the task descriptor list is 32 * (Task Descriptor size + Transfer Descriptor size) as configured by Host driver. This address shall be set on 1 KByte boundary: The lower 10 bits of this register shall be set to 0 by software and shall be ignored by CQE.

10.2.7.65 0x0424 Command Queuing Task Descriptor List Base Address Upper 32 Bits Register (Default Value: 0x0000_0000)

Offset: 0x0424			Register Name: CQTDLBAU
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CQE_TDLBAU Task Descriptor List Base Address Upper This register stores the MSB bits (bits 63:32) of the byte address of the head of the Task Descriptor List in system memory. The size of the task descriptor list is 32 * (Task Descriptor size + Transfer Descriptor size) as configured by Host driver. This register is reserved when using 32-bit addressing mode.

10.2.7.66 0x0428 Command Queuing Task Doorbell Register (Default Value: 0x0000_0000)

Offset: 0x0428			Register Name: CQTDBR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0428			Register Name: CQTDBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W1S	0x0	CQE_TDBR Command Queuing Task Doorbell Software shall configure TDLBA and TDLBAU, and enable CQE in CQCFG before using this register. Writing 1 to bit n of this register triggers CQE to start processing the task encoded in slot n of the TDL. CQE always processes tasks in-order according to the order submitted to the list by CQTDBR write transactions. CQE processes Data Transfer tasks by reading the Task Descriptor and sending QUEUED_TASK_PARAMS (CMD44) and QUEUED_TASK_ADDRESS (CMD45) commands to the device. CQE processes DCMD tasks (in slot #31, when enabled) by reading the Task Descriptor, and generating the command encoded by its index and argument. The corresponding bit is cleared to '0' by CQE in one of the following events: When a task execution is completed (with success or error) The task is cleared using CQTCLR register All tasks are cleared using CQCTL register CQE is disabled using CQCFG register Software may initiate multiple tasks at the same time (batch submission) by writing 1 to multiple bits of this register in the same transaction. In the case of batch submission: CQE shall process the tasks in order of the task index, starting with the lowest index. If one or more tasks in the batch are marked with QBR, the ordering of execution will be based on said processing order. Writing 0 by software shall have no impact on the hardware, and will not change the value of the register bit.

10.2.7.67 0x042C Command Queuing Task Completion Notification Register (Default Value: 0x0000_0000)

Offset: 0x042C	Register Name: CQTCN

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Bit	Read/Write	Default/Hex	Description
31:0	R/W1C	0x0	CQE_TCN Task Complete Notification CQE shall set bit n of this register (at the same time it clears bit n of CQTDBR) when a task execution is completed (with success or error). When receiving interrupt for task completion, software may read this register to know which tasks have finished. After reading this register, software may clear the relevant bit fields by writing 1 to the corresponding bits.

10.2.7.68 0x0430 Command Queuing Device Queue Status Register (Default Value: 0x0000_0000)

Offset: 0x0430			Register Name: CQDQS
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CQE_DQS Device Queue Status Every time the Host controller receives a queue status register (QSR) from the device, it updates this register with the response of status command, i.e., the device's queue status.

10.2.7.69 0x0434 Command Queuing Device Pending Tasks Register (Default Value: 0x0000_0000)

Offset: 0x0434			Register Name: CQDPT
Bit	Read/Write	Default/He	Description
31:0	R	0x0	CQE_DPT Device Pending Tasks Bit n of this register is set if and only if QUEUED_TASK_PARAMS (CMD44) and QUEUED_TASK_ADDRESS (CMD45) were sent for this specific task and if this task hasn't been executed yet. CQE shall set this bit after receiving a successful response for CMD45. CQE shall clear this bit after the task has completed execution. Software needs to read this register in the task-discard procedure, when the controller is halted, to determine if the task is queued in the device. If the task is queued, the driver sends a CMDQ_TASK_MGMT (CMD48) to the device ordering it to discard the task. Then software clears the task in the CQE. Only then the software orders CQE to resume its operation using CQCTL register.

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10.2.7.70 0x0438 Command Queuing Task Clear Register (Default Value: 0x0000_0000)

Offset: 0x0438			Register Name: CQTCLR
Bit	Read/Write	Default/He	Description
31:0	R/W	0x0	CQE_TASK_CLR Command Queuing Task Clear Writing 1 to bit *n* of this register orders CQE to clear a task which software has previously issued. This bit can only be written when CQE is in Halt state as indicated in CQCTL register Halt bit. When software writes '1' to a bit in this register, CQE updates the value to '1', and starts clearing the data structures related to the task. CQE clears the bit fields (sets a value of 0) in CQTCLR and in CQTDBR once clear operation is complete. Software should poll on the CQTCLR until it is cleared to verify clear operation was complete. Writing to this register only clears the task in the CQE and does not have impact on the device. In order to discard the task in the device, host software shall send CMDQ_TASK _MGMT while CQE is still in Halt state. Host driver is not allowed to use this register to clear multiple tasks at the same time. Clearing multiple tasks can be done using CQCTL register. Writing 0 to a register bit shall have no impact.

10.2.7.71 0x0440 Command Queuing Send Status Configuration 1 Register (Default Value: 0x0000_1000)

Offset: 0x0440			Register Name: CQSSC1
Bit	Read/Write	Default/He	Description
31:16	/	/	/

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Offset: 0x0440			Register Name: CQSSC1
Bit	Read/Write	Default/He	Description
15:0	R/W	0x1000	CQE_SSC_CIT Send Status Command Idle Timer This field indicates to CQE the polling period to use when using periodic SEND_QUEUE_STATUS (CMD13) polling. Periodic polling is used when tasks are pending in the device, but no data transfer is in progress. When a SEND_QUEUE_STATUS response indicating that no task is ready for execution, CQE counts the configured time until it issues the next SEND_QUEUE_STATUS. Timer units are clock periods of the clock whose frequency is specified in the Internal Timer Clock Frequency field CQCAP register. The minimum value is 0001h (1 clock period) and the maximum value is FFFFh (65535 clock periods). Default interval is: 4096 clock periods. For example, a CQCAP field value of 0 indicates a 19.2 MHz clock frequency (period = 52.08 ns). If the setting in CQSSC1.CIT is 1000h, the calculated polling period is 4096*52.08 ns= 213.33 us

10.2.7.72 0x0444 Command Queuing Send Status Configuration 2 Register (Default Value: 0x0000_0000)

Offset: 0x0444			Register Name: CQSSC2
Bit	Read/Write	Default/He	Description
31:16	/	/	/
15:0	R/W	0x0	CQE_SSC_RCA Send Queue Status RCA This field provides CQE with the contents of the 16-bit RCA field in SEND_QUEUE_ STATUS (CMD13) command argument. CQE shall copy this field to bits 31:16 of the argument when transmitting SEND_ QUEUE_STATUS (CMD13) command.

10.2.7.73 0x0448 Command Queuing Command Response for Direct-Command Task Register (Default Value: 0x0000_0000)

Offset: 0x0448			Register Name: CQCRDCT
Bit	Read/Write	Default/He	Description

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Offset: 0x0448			Register Name: CQCRDCT
Bit	Read/Write	Default/He	Description
31:0	R	0x0	CQE_DCMD_LAST_RESP Direct Command Last Response This register contains the response of the command generated by the last direct- command (DCMD) task which was sent. CQE shall update this register when it receives the response for a DCMD task. This register is considered valid only after bit 31 of CQTDBR register is cleared by CQE.

10.2.7.74 0x0450 Command Queuing Response Mode Error Mask Register (Default Value: 0xFDF9_A080)

Offset: 0x0450			Register Name: CQRMEM
Bit	Read/Write	Default/He	Description
31:0	R/W	0xFDF9A080	CQE_RMEM Response Mode Error Mask This bit is used as in interrupt mask on the device status filed which is received in R1/R1b responses. Bit Value Description (for any bit 'i'): 1: When a R1/R1b response is received, with bit 'I' in the device status set, a RED interrupt is generated 0: When a R1/R1b response is received, bit /in the device status is ignored The reset value of this register is set to trigger an interrupt on all "Error" type bits in the device status. Note: Responses to CMD13 (SQS) encode the QSR, so they are ignored by this logic.

10.2.7.75 0x0454 Command Queuing Task Error Information Register (Default Value: 0x0000_0000)

Offset: 0x0454			Register Name: CQTERRI
Bit	Read/Write	Default/He	Description
31	R	0x0	CQE_DTE_VLD Data Transfer Error Fields Valid This bit is updated when an error is detected by CQE, or indicated by eMMC controller. If a data transfer is in progress when the error is detected/indicated, the bit is set to 1. If a no data transfer is in progress when the error is detected/indicated, the bit is cleared to 0.

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Offset: 0x0454			Register Name: CQTERRI
Bit	Read/Write	Default/He	Description
30:29	/	/	/
28:24	R	0x0	CQE_DTE_TASK_ID Data Transfer Error Task ID This field indicates the ID of the task which was executed on the data lines when an error occurred. The field is updated if a data transfer is in progress when an error is detected by CQE, or indicated by eMMC controller.
23:22	/	/	/
21:16	R	0x0	CQE_DTE_CMD_IDX Data Transfer Error Command Index This field indicates the index of the command which was executed on the data lines when an error occurred. The index shall be set to EXECUTE_READ_TASK (CMD46) or EXECUTE_WRITE_TASK (CMD47) according to the data direction. The field is updated if a data transfer is in progress when an error is detected by CQE, or indicated by eMMC controller.
15	R	0x0	CQE_RME_VLD Response Mode Error Fields Valid This bit is updated when an error is detected by CQE, or indicated by eMMC controller. If a command transaction is in progress when the error is detected/indicated, the bit is set to 1. If a no command transaction is in progress when the error is detected/indicated, the bit is cleared to 0.
14:13	/	/	/
12:8	R	0x0	CQE_RME_TASK_ID Response Mode Error Task ID This field indicates the ID of the task which was executed on the command line when an error occurred. The field is updated if a command transaction is in progress when an error is detected by CQE, or indicated by eMMC controller.
7:6	/	/	/

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Offset: 0x0454			Register Name: CQTERRI
Bit	Read/Write	Default/He	Description
5:0	R	0x0	CQE_RME_CMD_IDX Response Mode Error Command Index This field indicates the index of the command which was executed on the command line when an error occurred. The field is updated if a command transaction is in progress when an error is detected by CQE, or indicated by eMMC controller.

10.2.7.76 0x0458 Command Queuing Command Response Index Register (Default Value: 0x0000_0000)

Offset: 0x0458			Register Name: CQCRI
Bit	Read/Write	Default/He	Description
31:6	/	/	/
5:0	R	0x0	CQE_LAST_RESP_IDX Last Command Response index This field stores the index of the last received command response. CQE shall update the value every time a command response is received.

10.2.7.77 0x045C Command Queuing Command Response Argument Register (Default Value: 0x0000_0000)

Offset: 0x045C			Register Name: CQCRA
Bit	Read/Write	Default/He	Description
31:0	R	0x0	CQE_LAST_RESP_ARG Last Command Response Argument This field stores the argument of the last received command. CQE shall update the value every time a command response is received.

10.2.7.78 0x0460 Command Queuing Function Control Register (Default Value: 0x0000_0000)

Offset: 0x0460			Register Name: CQFUNCCTL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0460			Register Name: CQFUNCCTL
Bit	Read/Write	Default/Hex	Description
15:8	R/W	0x0	CQE_SMHC_ERR_MASK This field is used to mask errors that CQE does not need to consider. bit7 for Data End-bit error mask bit6 for Data Start-bit error mask bit5 for Data Starvation error mask bit4 for Data timeout mask bit3 for Response timeout mask bit2 for Data CRC error mask bit1 for Response CRC error mask bit0 for Response error mask For any bit: 0: CQE will consider this SMHC error 1: CQE will not consider this SMHC error
7:1	/	/	/
0	R/W	0x0	CQE_CLR_REG_EN This field is used to control whether to clear register when disable CQE 0: Not clear register when disable CQE 1: Clear register to reset value when disable CQE

10.2.7.79 0x0480 Command Queuing Descriptor Transfer Done Register (Default Value: 0x0000_0000)

Offset: 0x0480			Register Name: CQDTD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CQE_DTD Descriptor Transfer Done Bit Value Description (for any bit i): 1: The descriptor of task-i has been transferred from the system memory to cqe. 0: The descriptor of task-i not transferred from the system memory to cqe.

10.2.7.80 0x0484 Command Queuing Task Complete Total Register (Default Value: 0x0000_0000)

Offset: 0x0484			Register Name: CQTCT
Bit	Read/Wri	Default/	Description

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Offset: 0x0484			Register Name: CQTCT
Bit	Read/Wri	Default/	Description
31:0	R	0x0	CQE_TCT Task Complete Total This register will be add 1 when a task execution is completed (with success or error) by CQE.

10.2.7.81 0x0488 CQE Controller Status Register (Default Value: 0x0000_0000)

Offset: 0x0488			Register Name: CQSTATUS
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R	0x0	CQE_FSM_STATUS CQE_CNL FSM Status The format is: {2'b0, fsm_task_exe, 3'b0, fsm_task_chk, 2'b0, fsm_task_send}

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10.3 SPI Flash Controller (SPIF)

10.3.1 Overview

The SPI Flash Controller (SPIF) is a synchronous, serial communication interface which allows rapid data communication with fewer software interrupts. Different from SPI, this IP is typically designed for higher speed Flash devices and it only works at Master mode.

The SPI_Flash has the following features:

- One SPIF controller
- Supports multiple SPI mode
  - Standard SPI
  - Dual-Output/Dual-Input SPI and Dual I/O SPI
  - Quad-Output/Quad-Input SPI, Quad I/O, and QPI
  - Octal-Input/Octal-Output SPI, Octal-I/O SPI, and OPI
  - Programmable serial data frame length: 1 bit to 32 bits
- Supports XIP mode of operation for executing code from FLASH memory
- Supports Single Transfer Rate (STR) mode and Dual Transfer Rate (DTR) mode, and DTR mode supports DQS signal
- High Speed Clock Frequency
  - 150MHz for STR Mode
  - 100MHz for DTR Mode
- Software Write Protection
  - Write protection for all/portion of memory via software
  - Top/Bottom Block protection
- Programmable delay between transactions
- Support Mode0, Mode1, Mode2 and Mode3
- Support control signal configuration
  - One chip select to support multiple peripherals
  - Polarity and phase of the Chip Select (SPIF_CS) and SPI Clock (SPIF_CLK) are configurable
- Supports data scramble/descramble
- Supports private DMA
  - Supports chain transmission
  - Supports descriptor in 8-word length

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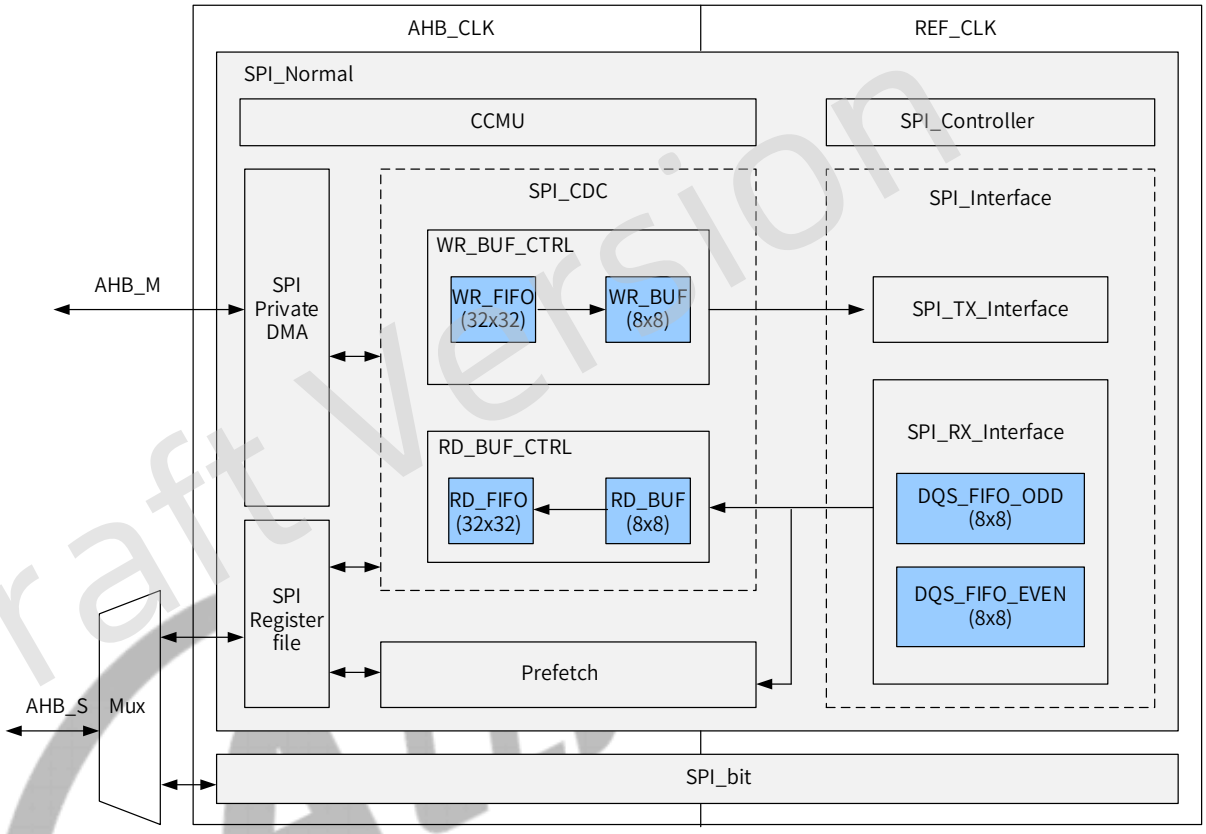
- Supports byte transfer

10.3.2 Block Diagram

The following figure shows a block diagram of the SPI_Flash.

Figure 10-11 SPI_Flash Block Diagram



SPI_Flash contains the following sub-blocks:

Table 10-9 SPI_Flash Sub-blocks

Sub-block	Description
SPI_Register_File	It is responsible for implementing registers configuration through AHB1 bus and CPU data prefetching.
SPI_Private_DMA	It is the private DMA for SPI, which contains AHB master and supports TXFIFO(WR_FIFO) and RXFIFO(RD_FIFO).
CMU	It is responsible for generating internal clock and communication clock. In addition, it Implements communication clock delay, clock sources selection, clock gating, and scan. It also supports synchronous release of asynchronous reset and scan.
SPI_CDC	SPI clock-domain-cross module, in which bit width transition finishes. (TX: 32bit->8bit; RX: 8bit->32bit) WR_BUF: Cache the write SPI data of AHB, and write them to WR_FIFO. RD_BUF: Cache the read SPI data of RD_FIFO to AHB. WR_FIFO: SPI write data FIFO (SRAM 32x32) RD_FIFO: SPI read data FIFO (SRAM 32x32) in the normal mode.

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Sub-block	Description
Prefetch	When prefetching instructions, the CPU can directly access SPI through this module.
SPI_Controller	SPI controlling center. It generates TX/RX controlling signal in the course of SPI communication.
SPI_Interface	Standard SPI interface. It is responsible for receiving and transmitting data with Devices.
SPI_bit	the processing module in SPI 3-wire mode.

10.3.3 External Signals

The following table describes the external signals of SPI_Flash. When using SPI_Flash, the corresponding PADs are selected as SPI_Flash function via GPIO.

Table 10-10 SPI_Flash External Signals

Signal Name	Description	Type
SPIF-CS0	SPI Peripheral Chip Select Signal, Low Active	O
SPIF-CLK	SPI Master Mode Clock Output	O
SPIF-MOSI	SPI Master Data Out, Slave Data In	I/O
SPIF-MISO	SPI Master Data In, Slave Data Out	I/O
SPIF-DQS	Data Strobe Signal	I
SPIF-D[7:4]	SPI Master Mode Data in Octal Mode.	I/O
SPIF-WP	SPI Write Protect, Low Active	I/O
SPIF-HOLD	SPI Hold Signal	I/O

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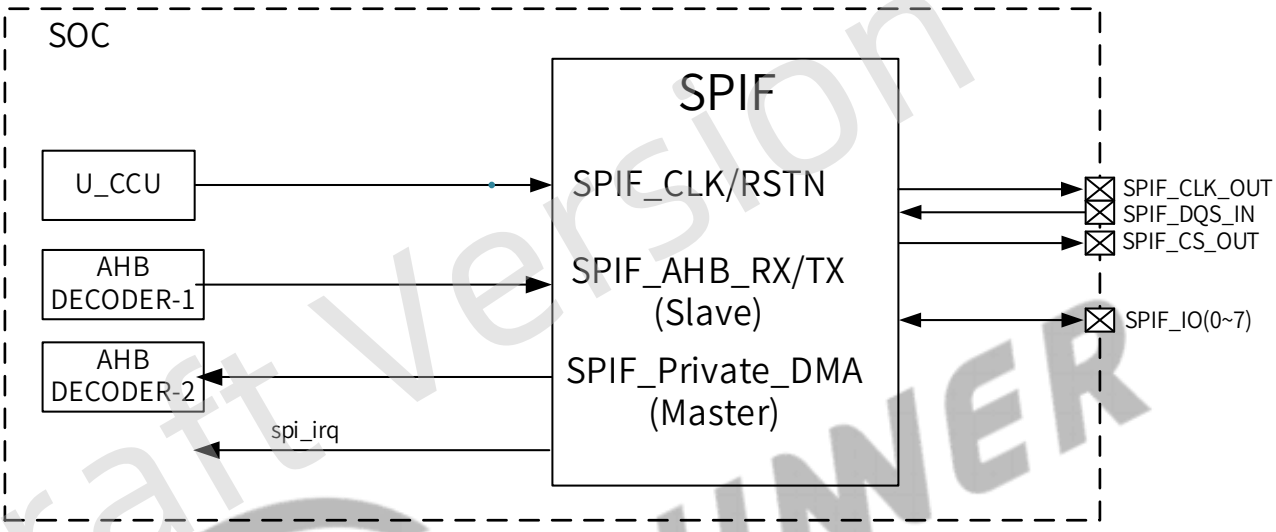
10.3.4 Functional Description

10.3.4.1 Typical Application

The following figure shows the application block diagram when the SPI master device is connected to a slave device.

Figure 10-12 Typical Application



The SPI_Flash is running in Master device. SPIF_CLK is generated and transmitted to external device. The data from the TX FIFO is routed to the MOSI pin. The data from slave is received on the MISO pin and sent to RX FIFO. Chip Select(SPIF_CS) signal is active in low level, which should be set to low before data are transmitted or received.

10.3.4.2 SPI_Flash Support List

Table 10-11 SPI_Flash Support List

SPI mode		Specification Items					
		STR	DTR	DTR-RX-DQS	SCK_MODE	Address Size	Remark
Bit Mode	3-wire	⋮	×	×	mode0	×	/
	4-wire						

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SPI mode		Specification Items					
		STR	DTR	DTR-RX-DQS	SCK_MODE	Address Size	Remark
Standard SPI	1-1-1-1	┆	┆	┆	• mode 0/1/2/3 (STR) • mode0 (DTR)	24bit/32 bit	• 1-1-1-1: cmd-addr-mode-data. • mode is optional and can be turned off.
Dual SPI	1-1-1-2	┆	┆	┆			/
	1-1-2-2						
	1-2-2-2						
	2-2-2-2						
Quad SPI	1-1-1-4	┆	┆	┆			/
	1-1-4-4						
	1-4-4-4						
	4-4-4-4						
Octal SPI	1-1-1-8	┆	┆	┆			8-wire DTR only supports the following: ADDR-24+MODE ADDR-32
	1-1-8-8						
	1-8-8-8						
	8-8-8-8						

10.3.4.3 SPI Clock

The SPI includes two clock domains: ahb_clk and spi_clk:

- ahb_clk supports parameter analysis (ahb_register) and DMA.
- spi_clk supports the communication between cross-domain clock, main control unit, and SPI-TX/RX and external devices.

The SPI_CMU divides the external SPI reference clock to get the SPI internal clock, which will be handled to get the sckt/sckr in accordance with the clock properties configured by the CPU. Then, the sckt/sckr will be used by the SPI_TX_INTERFACE/SPI_RX_INTERFACE.

The clock properties include:

- SPI Clock Mode
- DQS EN
- STR or DTR

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SPI Clock Mode

The SPI Flash supports 4 different modes for data transfer. Software can select one of the four modes in which the SPI works by setting the bit5(SPI_CPOL) and bit4(SPI_CPHA) of SPI Global Control Register[0x0004].

The SPI_CPOL defines the signal polarity when SPIF_CLK is in the idle state. The SPIF_CLK is high level when POL is '1 'and it is low level when POL is '0'. The SPI_CPHA decides whether the leading edge of SPIF_CLK is used for setup or sample data. The leading edge is used for setup data when PHA is '1' and for sample data when PHA is '0'. The four kind of modes are listed Table:

Table 10-12 SPI Flash Modes with Clock Polarity and Phase

SPI Mode	POL	PHA	Leading Edge	Trailing Edge
0	0	0	Rising, Sample	Falling, Setup
1	0	1	Rising, Setup	Falling, Sample
2	1	0	Falling, Sample	Rising, Setup
3	1	1	Failing, Setup	Rising, Sample

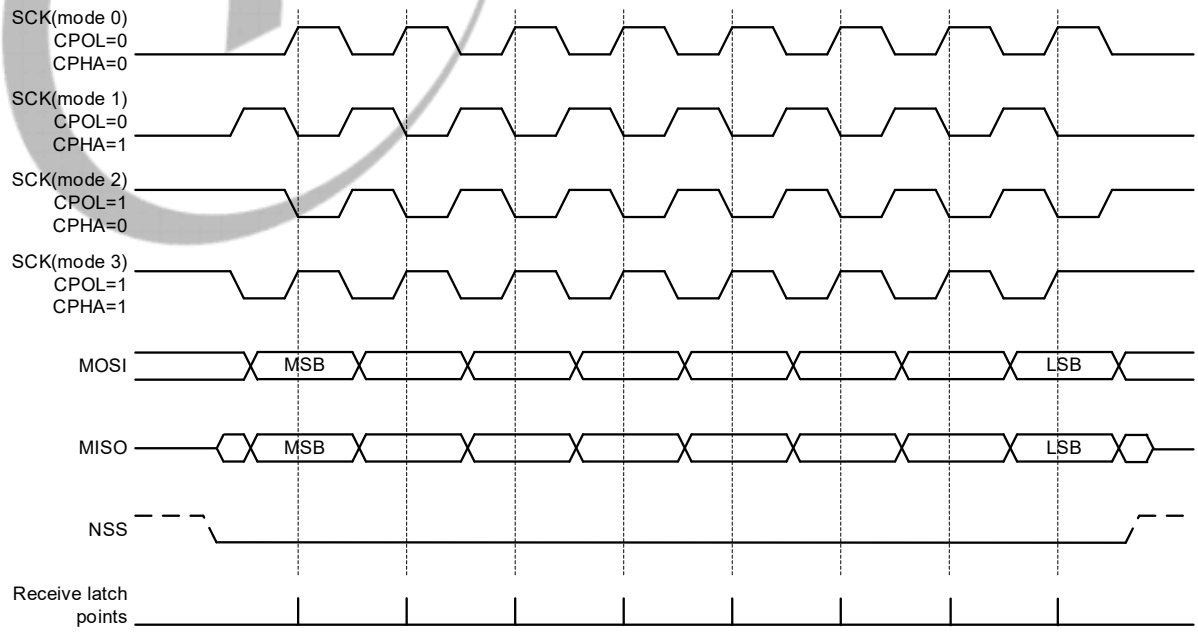
During Phase 0, Polarity 0 and Phase 1, Polarity 1 operations, output data changes on the falling edge and input data is shifted in on the rising edge.

During Phase 1, Polarity 0 and Phase 0, Polarity 1 operations, output data changes on the rising edges and is shifted in on falling edges.

The Bit Mode only supports Phase 0, Polarity 0 operation; the most significant bit (MSB) of data is transmitted first which is not configurable.

The following figure describe four waveforms for SPIF_CLK.

Figure 10-13 SPI Transfer Mode



For MSB first

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STR & DTR

- Single Transfer Rate (STR)

In the SPI mode 0 and mode 3, the input data of devices will be latched at the rising edge of SCK, and the output data can be used at the falling edge of SCK.

Figure 10-14 SPI STR Transfer

- Double Transfer Rate (DTR)

As with the STR command, the instruction bit is latched at the rising edge of the clock in the DTR command. However, after that, the address and input data are latched at the dual edge. After the instruction bit is latched at the falling edge of SCK, the first address bit will be latched at the next rising edge of SCK. The first output data bit will be sent at the falling edge of the last access latency period.

As with the STR command, the SCK period is obtained by measuring the cycle between two adjacent SCK falling edges. In the mode 0, the SCK is already at a low level when some command starts to be executed, thus the first SCK period during the command execution indicates the cycle from the falling edge of CS# to the first falling edge of SCK.

Figure 10-15 SPI DTR Transfer Example, 1-4-4

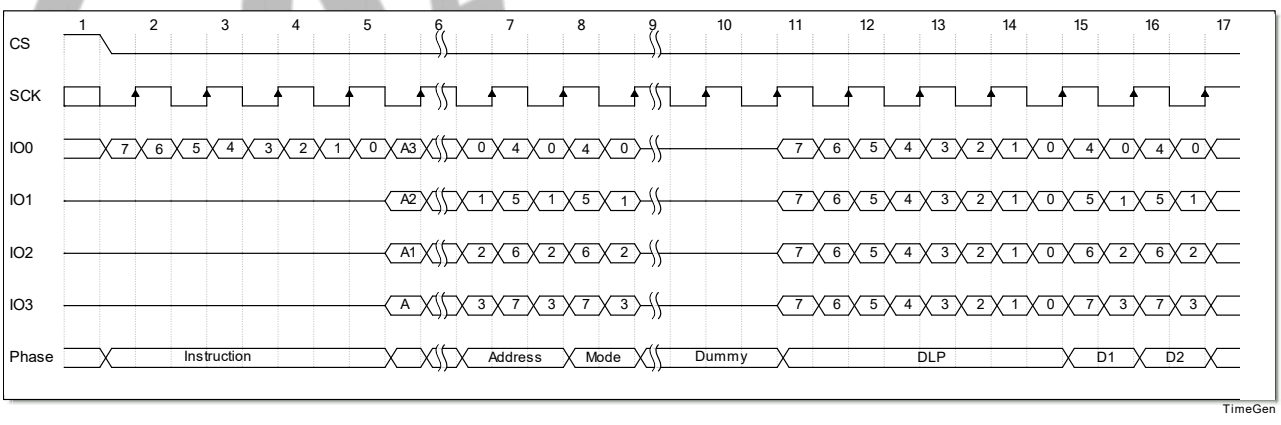

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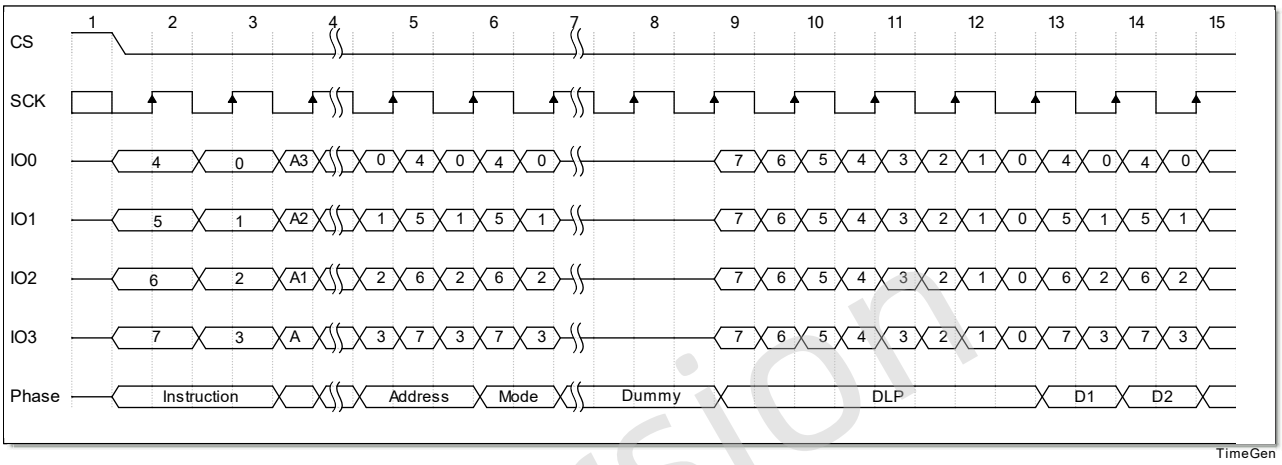
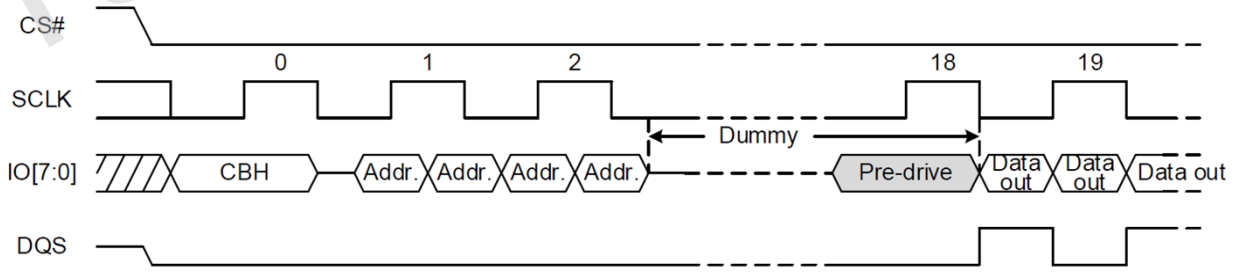
Figure 10-16 SPI DTR Transfer Example, 4-4-4



DQS

DQS signal indicates input/output data valid for DTR modes and is required to support high-speed data. When data strobe function is enabled, DQS signal will be driven to ground once CS# goes LOW till the device is driving output data, in which case DQS toggles to synchronize data output. When data strobe function is not enabled, DQS signal will not be driven.

Figure 10-17 SPI DTR Transfer with DQS Input Clock Signal



10.3.4.4 SPI Chip Select Signal

The SPI supports one chip select signal, which can be activated by the SPI_CS_SEL (bit[7:6]) of SPI_GLOBAL_CTRL. The polarities of CS signals can be configured by the SPI_CS_POL (bit[5]) of SPI_GLOBAL_CTRL.

The various intervals of CS signals can be configured by SPI_CS_DELAY. The specific configuration steps are as follows.

Step 8 Delay CSSOT and output the effective data after CS is enabled.

Step 9 Delay CSEOT at the rising edge of last valid data, and disable CS.

Step 10 After the CS is disabled, the interval between enabling CS and disabling CS is CSDA.

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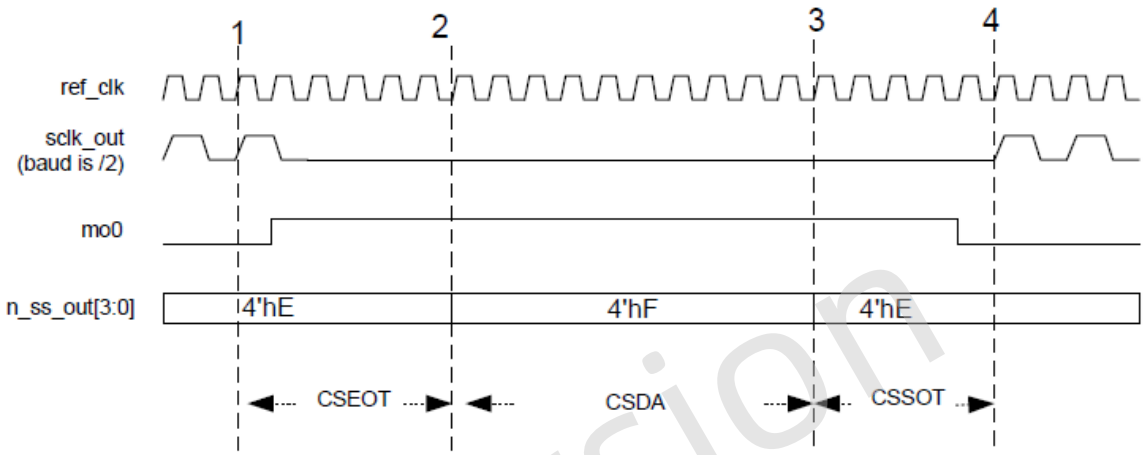
Figure 10-18 SPI CS Select Timing Relation



10.3.4.5 SPI Transfer Mode

SPI supports multiple transfer modes such as SPI Bit Mode, SPI Standard Mode, SPI Dual Mode, SPI Quad Mode, and SPI Octal Mode. Their main differences lie in the number of wires. Even in the same transfer mode, the subdivision modes will be further derived based on the number of wires used to data transfer. The number of data lines used by Command-Address-Data is specified on the subdivision pattern heading in the manner of X-X-X. For example, one Data line for Commend, two for Address, and two for data, are identified as (1-2-2).

SPI Bit Mode (3-Wire/4-Wire)

For some specific scenarios, the devices such as the sensor take the SPI interfaces as the communication protocol. Generally, their data volume is relatively small, and some devices support three wires, so the SPI Bit Mode is added, and 3-wire mode and 4-wire mode are subdivided, which includes SPIF_CS, SPIF_CLK, and 1/2 data line. The transmission length ranges from 1-32bit.

The 4-Wire Mode is selected when the Work Mode Select(bit[1:0]) is equal to 0x3 in the SPI Bit-Aligned Transfer Configure Register. In SPI 4-Wire Mode, the input data and output data use the independent two data lines. The MISO is used for inputting data, and the MOSI is used for outputting data.

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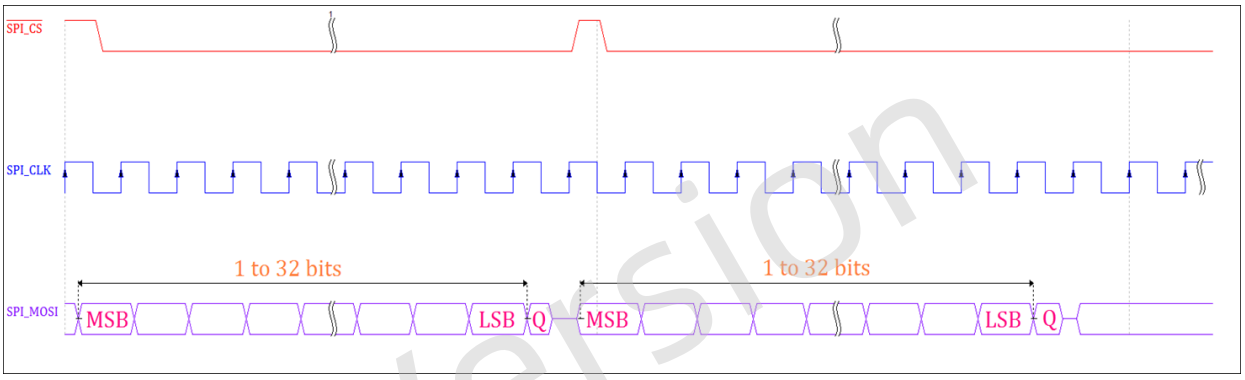
The SPI 3-Wire Mode is only valid when the SPI controller works as Master Device, and selected when the Work Mode Select(bit[1:0]) is equal to 0x2 in the SPI Bit-Aligned Transfer Configure Register. In the 3-Wire mode, the input data and the output data use the same single data line. The following figure describe this mode.

Figure 10-19 SPI 3-Wire Mode



SPI Standard Mode (4-Wire)

Signal Wire: CS#, SCK, IO0, and IO1.

Figure 10-20 SPI CMD with Single IO

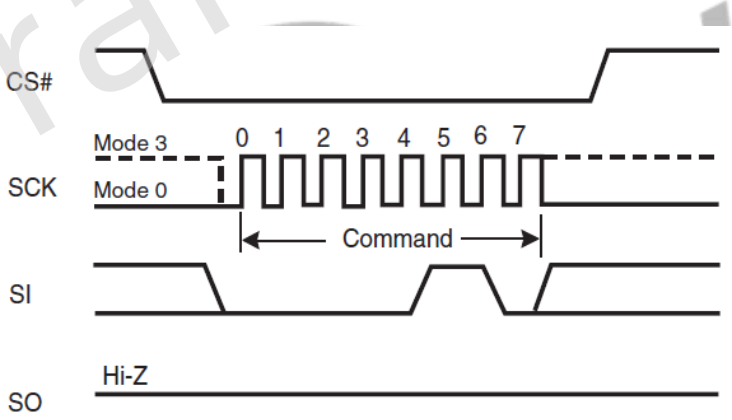


Figure 10-21 SPI Write CMD and DATA with One Wire

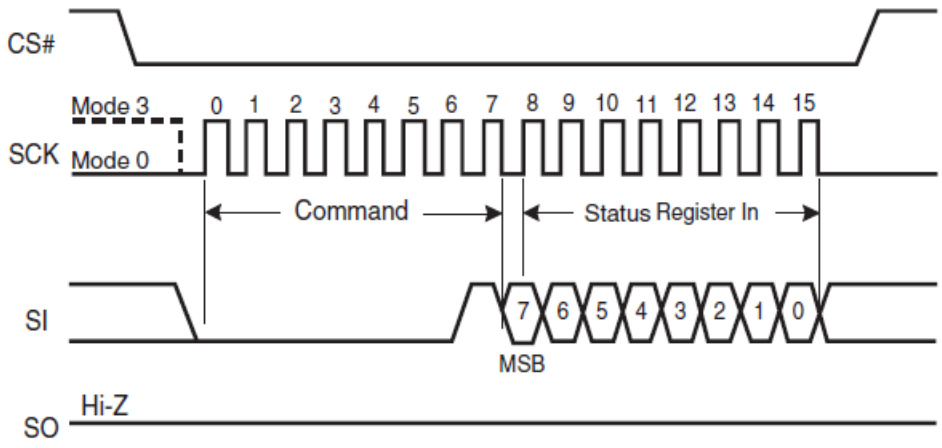


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Figure 10-22 SPI Write CMD and Read DATA with One Wire (No Dummy)

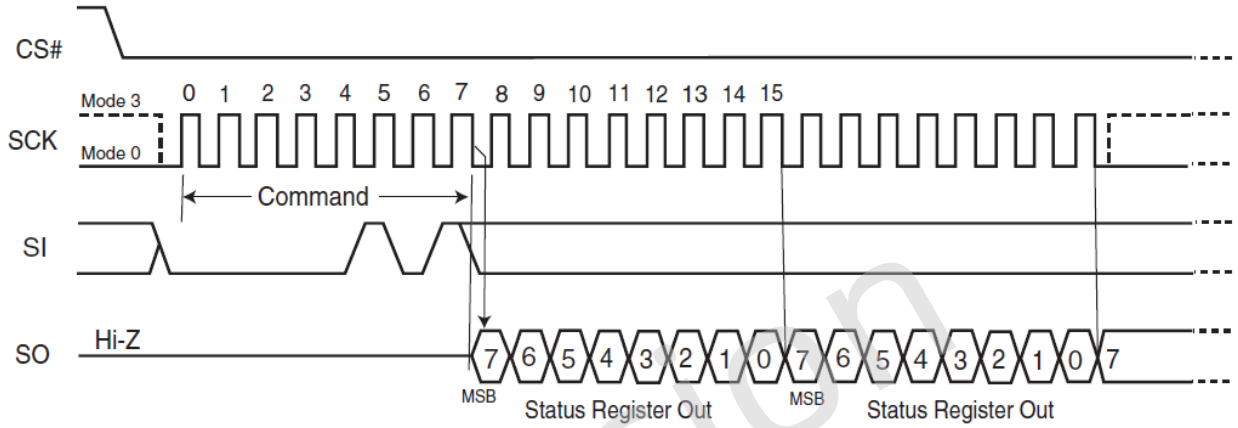
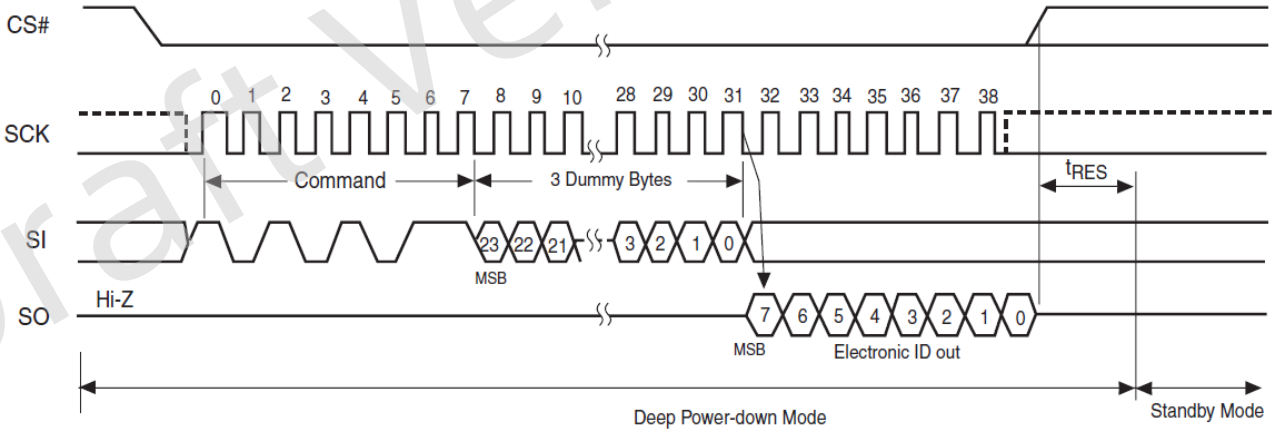


Figure 10-23 SPI Write CMD and Read DATA with One Wire (with Dummy)



SPI Dual Mode

Using the dual mode allows data to be transferred to or from the device at two times the rate of standard single mode SPI devices. Data can be read at a faster speed using two data bits (MOSI and MISO) at a time. The following describe the Dual Input/Dual Output SPI (1-1-2), the Dual IO SPI (1-2-2), and the (2-2-2) SPI Mode.

- SPI Dual Input/Dual Output Mode (1-1-2)

In the dual Input/dual Output SPI, the command, address, and the dummy bytes are output in the unit of a single bit in serial mode through SPIF_MOSI line. Only the data bytes are output (write) and input (read) in unit of dual bits through the SPIF_MOSI and SPIF_MISO.

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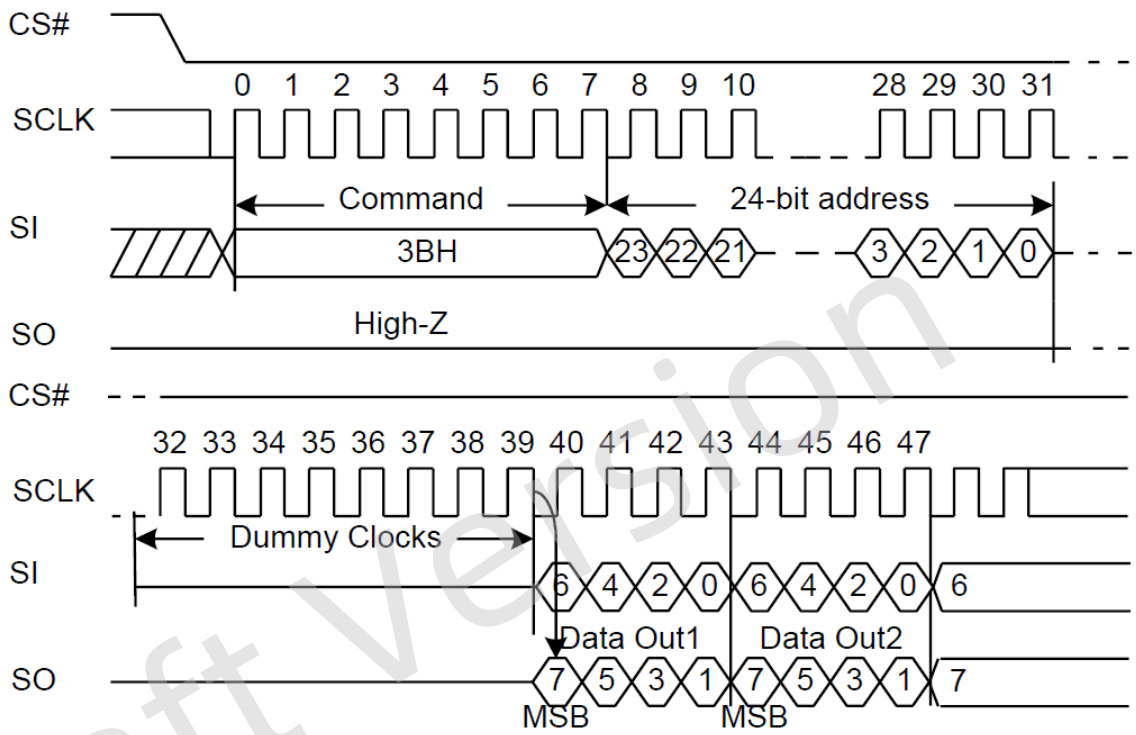
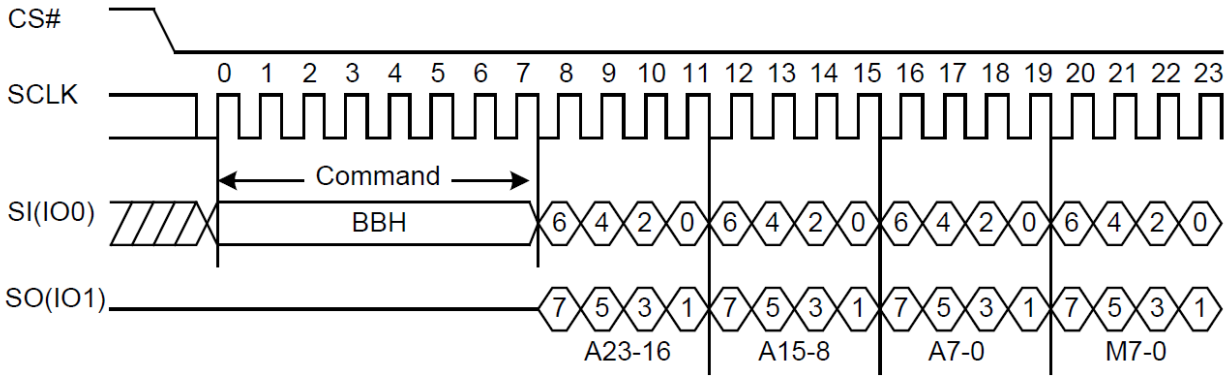
Figure 10-24 SPI Dual Input/Dual Output Mode (1-1-2



- SPI Dual IO Mode (1-2-2)

In the Dual IO SPI, only the command bytes are output in the unit of a single bit in serial mode through SPIF_MOSI line. The address bytes and the dummy bytes are output in the unit of dual bits through the SPIF_MOSI and SPIF_MISO. And the data bytes are output (write) and input (read) in the unit of dual bits through the SPIF_MOSI and SPIF_MISO.

Figure 10-25 SPI Dual Input/Dual Output Mode (1-2-2)



SPI Quad Mode

Using the quad mode allows data to be transferred to or from the device at 4 times the rate of standard single mode SPI devices, data can be read at fast speed using four data bits (MOSI, MISO, IO2(WP#) and IO3(HOLD#)) at the same time. The following describe the Quad Input/Quad Output SPI (1-1-4), 1-4-4 mode, and 4-4-4 mode.

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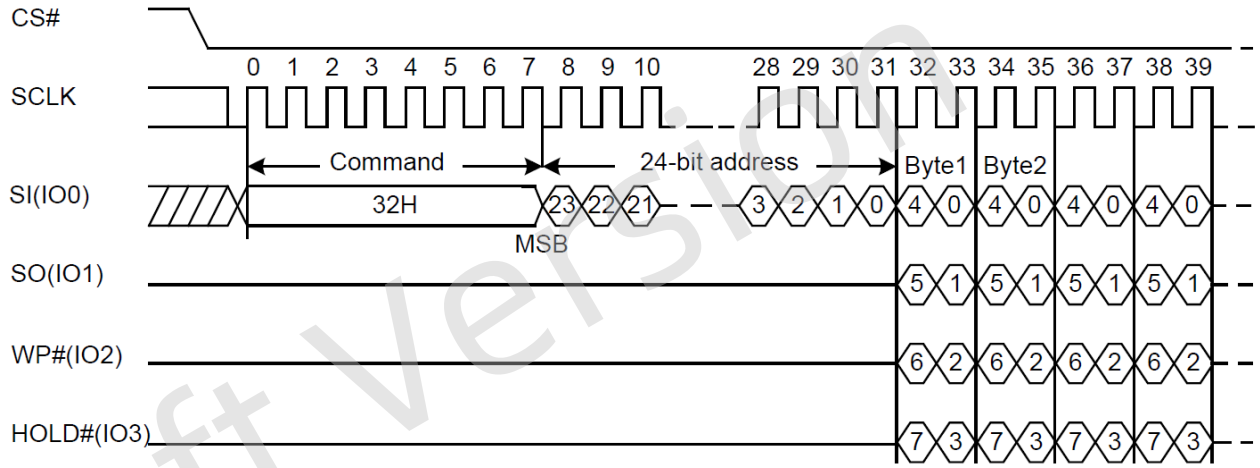
- Quad Input/Quad Output SPI (1-1-4)

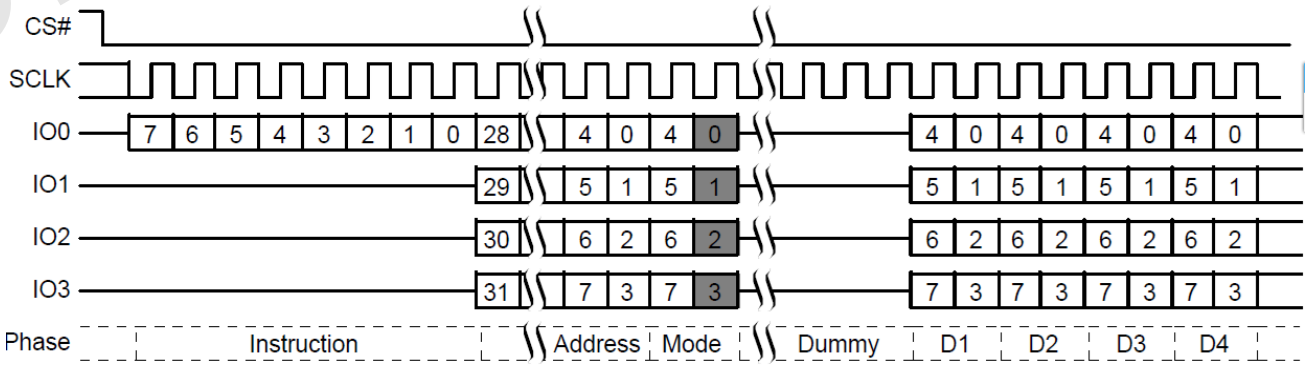
In the Quad Input/Quad Output SPI, the command, address, and the dummy bytes are output in unit of a single bit in serial mode through SPIF_MOSI line. Only the data bytes are output (write) and input(read) in unit of quad bits through the SPIF_MOSI, SPIF_MISO, SPIF_WP# and SPIF_HOLD#.

Figure 10-26 SPI Quad Input/Quad Output Mode (1-1-4)



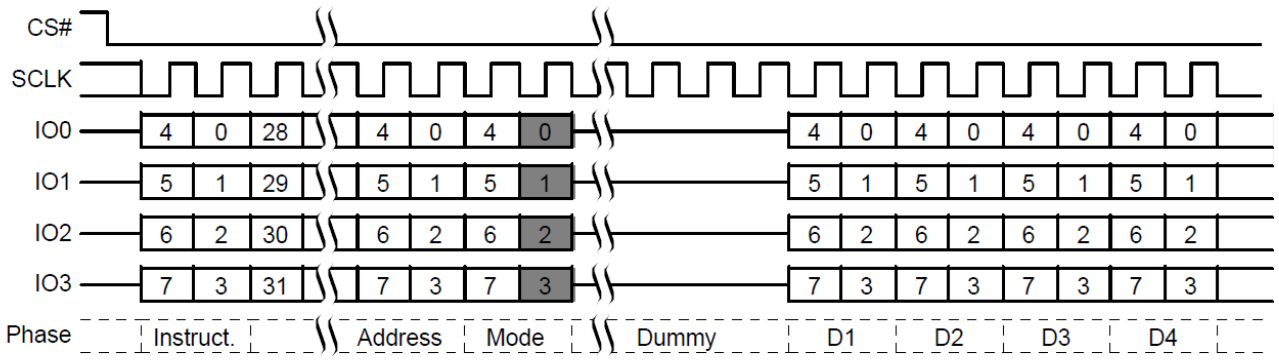
- 1-4-4 Mode

Figure 10-27 SPI Quad Input/ Quad Output Mode (1-4-4)



- 4-4-4 Mode

Figure 10-28 SPI Quad Input/ Quad Output Mode (4-4-4)



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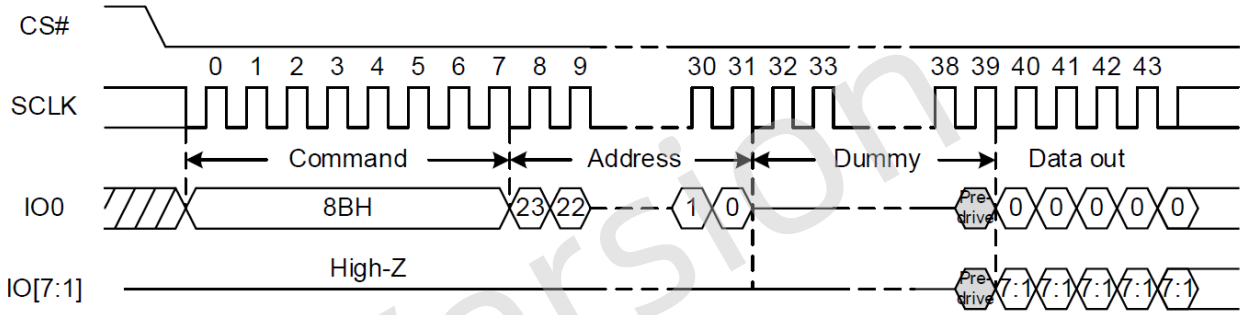
SPI Octal Mode

The Octal SPI Mode allow data to be transferred to or from the device at eight times the rate of the standard SPI. When using the Octal SPI, there are 8 data wires.

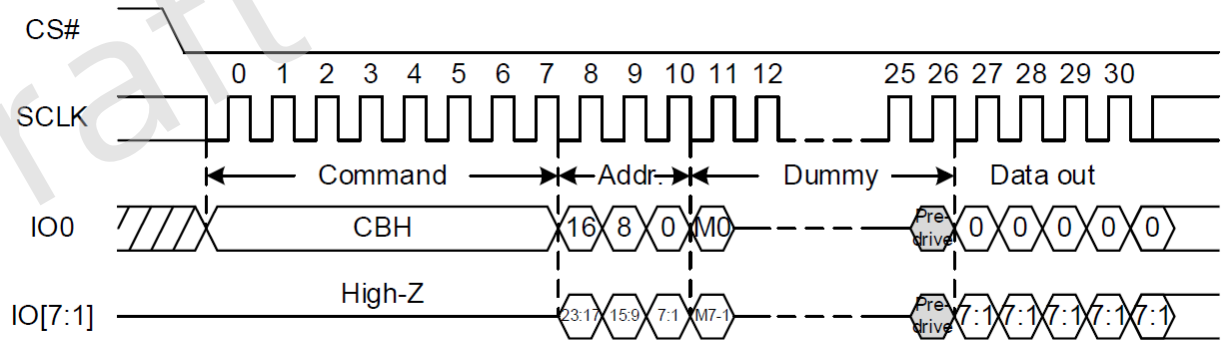
- SPI Octal 1-1-8

Figure 10-29 SPI Octal Input/ Octal Output Mode (1-1-8)



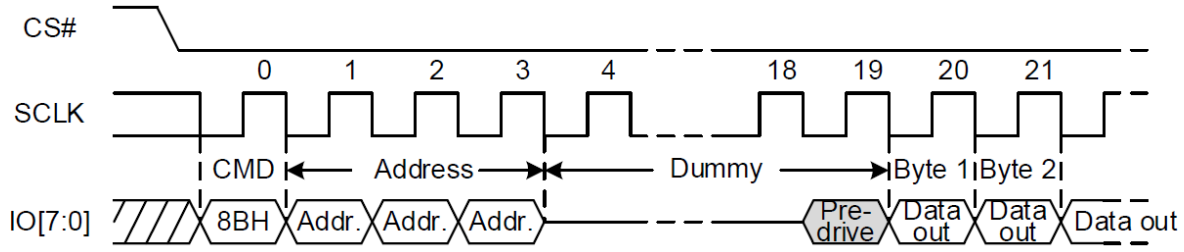
- SPI Octal 1-8-8

Figure 10-30 SPI Octal Input/ Octal Output Mode (1-8-8)



- SPI Octal 8-8-8

Figure 10-31 SPI Octal Input/ Octal Output Mode (8-8-8)



10.3.4.6 XIP Mode

If the CPU wants to directly access Flash, the prefetch mode is needed to improve efficiency. In the XIP mode, the CPU continues fetching instructions from flash and cache them (The maximum cache size is 128 Bytes.). Then HREADY_I should be controlled by handshake to send back the instructions to CPU through AHB.

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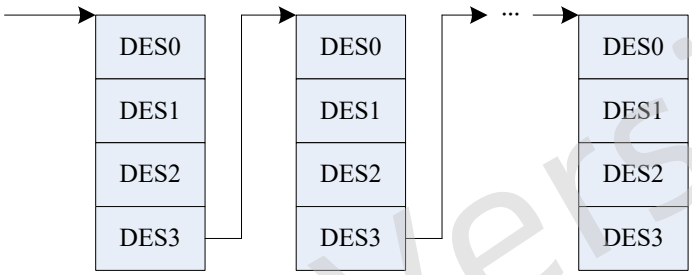
10.3.4.7 SPI Flash Private DMA

The private DMA adopts the chained descriptor structure. The address and size of the first descriptor are configured by registers. After getting the first descriptor, the hardware will obtain the data from the address recorded in the descriptors. When the current transfer finishes, the hardware will continue to obtain descriptors based on the address of the next descriptor given by the previous descriptor until the terminal character is found. Then, a DMA transfer ends.

Figure 10-32 Descriptor Structure Diagram



The private DMA also supports SPI parameter configuration. The SPI transfer can be configured by the parameters of the descriptors 4-7.

Descriptor0 Definition

Bits	Descriptor
31:7	Reserved
6:4	HBURST_TYPE indicate hburst len 000: SINGLE, hburst_len = 1 011: INCR4, hburst_len = 4 101: INCR8, hburst_len = 8 111: INCR16, hburst_len = 16
3:2	/
1	DMA_DIR DMA Write Process or Read Process 0: Read 1: Write
0	DMA_FINISH_FLAG DMA Finish Flag

Descriptor1 Definition

Bits	Descriptor

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Bits	Descriptor
31:24	DMA_BLK_LEN DMA Block Len Mode 0: 8Byte 1: 16Byte 2: 32Byte 3: 64Byte Recommended Configuration: The data volume of DMA_BLK_LEN is greater than or equal to that of HBURST_TYPE.
23:17	/
16:0	DMA_DATA_LEN Indicate the data byte number of current DMA operation.

Descriptor2 Definition

Bits	Descriptor
31:0	DMA_Buffer_ SADDR The real address is as below The word address is needed, namely, the byte address abandons the low 2 bits.

Descriptor3 Definition

Bits	Descriptor
31:0	NEXT_DESCRIPTOR_ADDR These bits indicate the pointer to the physical memory where the next descriptor is present, which are word (4byte) address (The lower two bits are deleted from the byte address).

Descriptor4 Definition

Bit	Description
31:30	/
29	CMD_DTR CMD DTR Control 1: CMD DTR 0: CMD not DTR CAUTION: When CMD DTR, DTR must be on and CMD2 is required.
28	COMMAND_TRANS_EN Set to '1' if command data need trans to device SPI Controller FSM Phase Enable 1: Enable 0: Disable
27:25	/

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Bit	Description
24	ADDRESS_TRANS_EN Set to '1' if address need trans to device SPI Controller FSM Phase Enable 1: Enable 0: Disable
23:21	/
20	MODE_BIT_TRANS_EN Set to '1' if Mode bit need trans after address SPI Controller FSM Phase Enable 1: Enable 0: Disable
19:17	/
16	DUMMY_BIT_TRANS_EN Dummy Bit State Enable SPI Controller FSM Phase Enable 1: Enable 0: Disable
15:13	/
12	TX_DATA_EN Set to '1' if Data Need Trans SPI Controller FSM Phase Enable 1: Enable 0: Disable
11:9	/
8	RX_DATA_EN Set to '1' if Data Need Receive SPI Controller FSM Phase Enable 1: Enable 0: Disable
7:0	/

Descriptor5 Definition

Bit	Description
31:0	ADDR_OPCODE Address Content Trans Through SPI

Descriptor6 Definition

This register should be setup while the controller is idle.

Bit	Description
31:24	CMD_OPCODE Command Content Trans Through SPI

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Bit	Description
23:16	MODE_OPCODE Mode Content Trans Through SPI
15:8	CMD_OPCODE2 Command2 Content Trans Through SPI
7:6	CMD_TRANS_TYPE Command Transfer Type 00: Command can be Shifted to the device on DQ0 01: Command can be Shifted to the device on DQ0 and DQ1 10: Command can be Shifted to the device on DQ0- DQ3 11 : Command can be Shifted to the device on DQ0-DQ7
5:4	ADDR_TRANS_TYPE Address Transfer Type 00: Address can be Shifted to the device on DQ0 01: Address can be Shifted to the device on DQ0 and DQ1 10: Address can be Shifted to the device on DQ0- DQ3 11: Address can be Shifted to the device on DQ0-DQ7
3:2	MODE_BIT_TRANS_TYPE Mode Bit Transfer Type 00: Mode Bit can be Shifted to the device on DQ0 01: Mode Bit can be Shifted to the device on DQ0 and DQ1 10: Mode Bit can be Shifted to the device on DQ0- DQ3 11: Mode Bit can be Shifted to the device on DQ0-DQ7
1:0	DATA_TRANS_TYPE Data Transfer Type 00: Opcode can be Shifted to the device on DQ0 only 01: Opcode can be Shifted to the device on DQ0 and DQ1 only 10: Opcode can be Shifted to the device on DQ0- DQ3 11: Opcode can be Shifted to the device on DQ0-DQ7

Descriptor7 Definition

Bit	Description
31	DATA_TRANS_NUM[16]
30:29	
28	SPI_NORMAL_EN If DMA configures SPI, this bit start SPI FSM.
27	/
26	CMD_LEN 0: 1byte 1: 2byte

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Bit	Description
25:24	ADDR_SIZE_MODE Address Size Mode 00: Address Size 8bit 01: Address Size 16bit 10: Address Size 24bit 11: Address Size 32bit
23:16	DUMMY_TRANS_NUM Number of Dummy Cycles A value of 0 = 1 Cycle A value of 1 = 1 Cycle ... A value of N = N Cycle
15:0	DATA_TRANS_NUM Num of Data Trans Through SPI(Byte) 0: Non-Write. 1: Write 1 Byte. 2: Write 2 Bytes. 3: Write 3 Bytes. ..... 65535: Write 65535 Bytes. Note: These Bits indicate the number of data bytes in a CHIP SELECT period. Notice the difference between DATA_TRANS_NUM here and DMA_DATA_LEN in Descriptor1.

10.3.5 Programming Guidelines

10.3.5.1 DMA Transfer

The software operation of the SPI DMA transfer is divided into 5 steps as follows.

a) System Setup

b) SPI Initialization

c) Channel Setup

d) DMA Setup

e) Enable SPI

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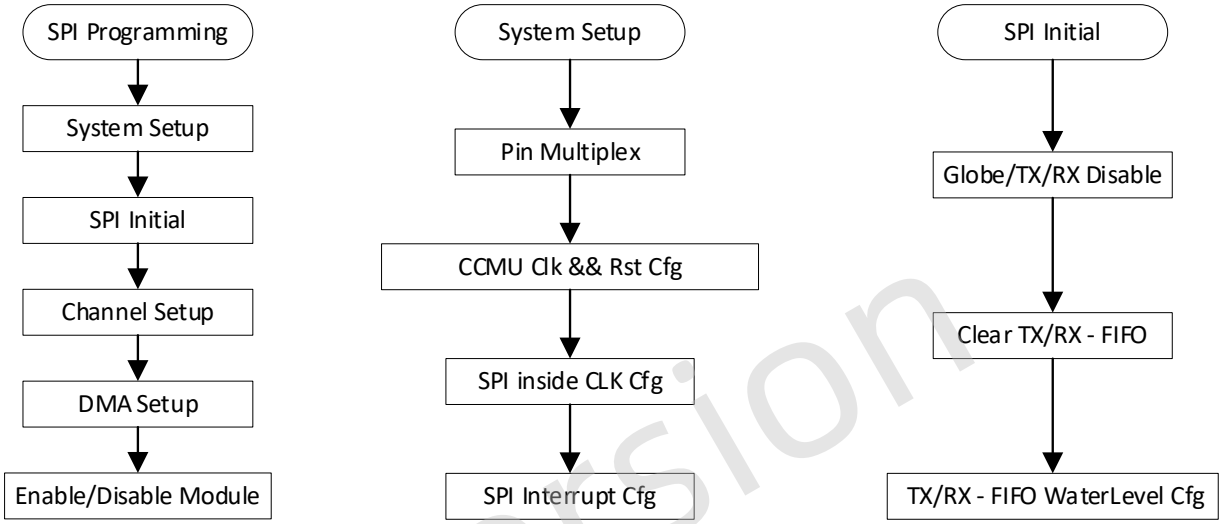
Figure 10-33 SPI Programming flow



System Setup

Step 1 Configure SPI Pin by programming the 17.9 GPIO.

Step 2 Configure SPI ref clock, AHB Clock, De-assert SPI ref Reset, and AHB reset in Section 4.1 Clock Controller Unit (CCU).

Step 3 Configure SPI Internal Clock

a) Configure clock source (SPI_TIMING_CFG)

- STR Mode

  Write 0 to the CLK_SPI_SRC_SEL bit (bit [24]), the CLK_SCK_SRC_SEL bit (bit [25]) bit, and the CLK_SCKOUT_SRC_SEL bit (bit [26]) bit.

- DTR Mode

  Write 0 to the CLK_SPI_SRC_SEL bit (bit [24]) and the CLK_SCK_SRC_SEL bit (bit [25]) bit.

  Write 1 to the CLK_SCKOUT_SRC_SEL bit (bit [26]) bit.

NOTE

When the value of the CLK_SCKOUT_SRC_SEL bit (bit [26]) bit is 1, the real output clock frequency of SPIF-CLK signal is the SPIFC clock frequency divided by 2.

b) Configure the sckr clock of SPI data receiving.

- Set the SCKR_DLY_MODE_SEL (bit[20]) in the SPI_TIMING_CFG.
- Set the DIGITAL_SCKR_DELAY_CFG bit[18:16] in the SPI_TIMING_CFG.
- Set the ANALOG_SAMP_DL_SW_VALUE bit[5:0] in the SPI_TIMING_CFG.

c) Generate SPI output clock

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- Select SPI working mode (SPI_MODE) in the SPI_CPOL (bit [5]) and SPI_CPHA (bit [4]) in the register SPI_GLOBAL_CTRL by referring to the section SPI Clock Mode.
- DTR_EN (bit [16]) in the SPI_GLOBAL_CTRL [0x0004] is closed by default. When it is closed, the frequency of output clock is equal to that of SCKR and SCKT. When it is open, the frequency of output clock is lowered to half that of SCKR and SCKT.

Step 4 Set SPI_INT_EN (0x0014[31:0]) as 0.

SPI Initialization

Step 1 Disable the SPI_GLOBAL_CTRL by writing 0 to SPI_DMA_MODE_EN (bit[2]), SPI_PMODE_EN (bit[3]), SPI_CPU_MODE_EN (bit[20]), and SPI_AUTO_READ_STATUS_MODE_EN (bit[21]).

Step 2 Reset TX/RX FIFO (SPI_GLOBAL_CTRL_ADD [0x0008])

Reset the CDC-BUF/FIFO of TX channel: Write 1 to CDC_WF_SRST to reset the WR_BUFF and WR_FIFO in SPI_WR_BUF_CTRL and the SWF in SPI_TX_INTERFACE.

Reset the CDC-BUF/FIFO of RX channel: Write 1 to CDC_RF_SRST.

Step 3 Configure the water level of FIFO in SPI_CDC_FIFO_TRIG_LEVEL.

Channel Setup

Step 1 Select the SPI channel parameter resource by setting the SPI_CFG_MODE(bit[0]) of SPI_GLOBAL_CTRL.

NOTE

If the parameters source from CPU, finish the configuration in reference to all the guidelines in the section Channel Setup. If the parameters source from DMA, skip the steps after CS Delay Configuration in this section to initiate the DMA. For the descriptions of DMA descriptors, refer to the section SPI Flash Private DMA.

Step 2 SPI Interface Configuration(c)Public Configuration

a) CS Delay Configuration: SPI_CS_DELAY[0x001C]

- CSDA (bit [23:16]): The intervals of two adjacent CS enable. The minimum interval is recommended to be 6sclk. The actual value is CSDA+1.
- CSEOT (bit [15:8]): After the SCLK_OUT is invalid, the CS signal will be de-asserted after CSEOT SCLK. The minimum interval is 1sclk. The actual value is CSEOT+1.
- CSSOT (bit [7:0]): After the CS is valid, the SCLK_OUT signal will be asserted after CSSOT SCLK. The minimum interval is 1sclk. The actual value is CSSOT+1.

b) Activate SPI Trans Phase (SPI_TRANS_PHA_CFG)

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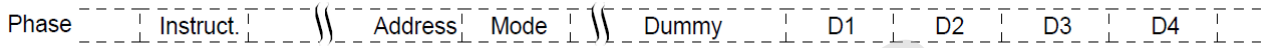
In a transmission of the SPI interface, the transferring content of I/O line is as follows. Enable the corresponding Trans Phase based on the content to be transferred by writing 1 to the COMMAND_TRANS_EN (bit [28]), ADDRESS_TRANS_EN (bit [24]), MODE_BIT_TRANS_EN (bit [20]), DUMMY_BIT_TRANS_EN (bit [16]), TX_DATA_EN (bit [12]), and RX_DATA_EN (bit [8]).

Figure 10-34 SPI Transfer Phase Flow Diagram



NOTE

RX_DATA_EN and TX_DATA_EN cannot be activated at the same time.

c) Configure the number of I/O used by TransPhase by setting the CMD_TRANS_TYPE (bit [13:12]), ADDR_TRANS_TYPE (bit [9:8]), MODE_BIT_TRANS_TYPE (bit [5:4]), and DATA_BIT_TRANS_TYPE (bit [1:0]) in the (SPI_TRANS_CFG2).

d) Configure the transferring volume of TransPhase in the SPI_TRANS_NUM[0x002C].

Step 3 SPI interface configuration—normal read and write

Configure the contents of all TransPhase by setting the following bits.

- ADDR: SPI_TRANS_CFG1 [0x0024], ADDR_OPCODE (bit [31:0])
- CMD: SPI_TRANS_CFG2 [0x0028], CMD_OPCODE (bit[31:24])
- CMD2 SPI_TRANS_CFG2 [0x0028], CMD_OPCODE2 (bit[15:8])
- MODE: SPI_TRANS_CFG2 [0x0028], CMD_OPCODE (bit[23:16])

Step 4 SPI Interface Configuration—XIP Mode

a) Configure whether to use address mapping mode by setting the XIP_ADDR_MAP_MODE (bit[1]) in the SPI_GLOBAL_CTRL.

b) Configure the contents of all TransPhase by setting the following bits.

- CMD: SPI_TRANS_CFG2[0x0028], CMD_OPCODE (bit[31:24])
- MODE: SPI_TRANS_CFG2[0x0028], CMD_OPCODE (bit[23:16])

c) Configure the effective range of XIP by setting the XIP_START_ADDR and the XIP_END_ADDR.

Step 5 SPI Interface Configuration(c)BIT Mode

Refer to the functions of SPI Bit Mode (3-Wire/4-Wire).

DMA Setup

Step 1 Configure the length and starting address of the first descriptor in the SPI_DMA_CTRL and SPI_DESCRIPTOR_SADDR to initiate DMA.

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Step 2 Configure the length and the starting address of storage location for the first descriptor. Then, finish the following operations to complete DMA Configuration.

- Descriptor0
  - AHB Master Burst Type Configuration

    HBURST_TYPE (bit [6:4]): Support SINGLE/INCR4/INCR8/INCR16 mode.
  - DMA Transfer Direction Configuration

    DMA_DIR (bit [1]): DMA read or DMA write.
  - DMA Finish Flag

    DMA_FINISH_FLAG (bit [0]): Currently the memory space allocated by the CPU to DMA may be a part of the total data volume to be transferred. For instance, the 256 Byte memory space is allocated to transfer 1 MB data. In this situation, a DMA descriptor merely points to the allocated 256 Byte data location, and it will also point to the next descriptor to fetch data. When the data of the last descriptor is fetched, the stop bit should be pulled up to terminate a DMA handling.
- Descriptor1
  - DMA_BLK_LEN ([31:16])

    DMA BLK length. It indicates the size of each DMA packet, with multiples of 8 Byte aligned to achieve the optimized rate when accessing storage.
  - DMA_DATA_LEN (bit [15:0])

    It indicates the data volume indicated by the current descriptor of DMA
- Descriptor2

  It indicates the data storage location of current descriptor.
- Descriptor3

  It indicates the storage location of the next descriptor.
- Descriptor4-7

  The SPI configuration parameters. In the course of setting parameters with DMA for SPI, the content of descriptors is configured to SPI.

Enable SPI

Only one of DMA mode CPU mode, AUTO_READ_STATUS_MODE mode and XIP mode can be enabled at a time. To enable DMA mode, write 1 to the SPI_NMODE_EN in the SPI_GLOBAL_CTRL to, then it will be auto pulled down.

10.3.5.2 CPU Transfer

The software operation of CPU transfer is almost the same as DMA transfer. However, there are still two main differences.

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One difference is that when the CPU transfers, channel parameter cannot come from DMA descriptors. SPI_GLOBAL_CTRL[0x0004], SPI_CFG_MODE(bit[0]) must be set to 0. All parameters should come from REGISTER FILE. SPI_GLOBAL_CTRL[0x0004], SPI_CPU_MODE_EN(bit[20]) is used to start CPU transfer (Different from descriptor and SPI_GLOBAL_CTRL[0x0004], SPI_NMODE_EN(bit[2])).

The other difference is that when reading data from Register [0x0210], it is recommended to read SPI_CDC_FIFO_STA[0x0050], RF_CNT([bit5-0]), especially when total read number is not integer multiple of trigger level. When writing data to Register [0x0220], it is always ready as long as trigger level is not reached.

NOTE

Register [0x0040] and [0x0044] are not used in CPU transfer.

10.3.5.3 Auto Status Reading

The software operation of STATUS READING is almost the same as CPU transfer. CMD_TRANS_EN and RX_DATA_EN must be set high because RX data path is used for status reading. MODE_BIT_TRANS_EN and DUMMY_BIT_TRANS_EN should be set accordingly. But STATUS OPCODE will not go through READ_FIFO and READ_BUFFER. Meanwhile, I/O Pins used should be set accordingly. SPI_STATUS_READ[0x0068] and SPI_STATUS_READ_2[0x006C] are used for STATUS READING. And to avoid endless reading and dead lock, the CPU should tell the maximum reading times. Then, to start STATUS READING, SPI_READ_STATUS_MODE_EN should be set. In this mode, DATA_TRANS_NUM (002C, [15:0]) is suggested to be 1 and DTR_EN [0x0004, bit 16] is suggested to be 0.

10.3.5.4 XIP

The software operation of XIP is almost the same as CPU transfer.

The System Setup part, SPI Initialization part and Channel Setup part is the same as DMA transfer.

After the basic initialization and setup, enable SPI_GLOBAL_CTRL [0x0004], SPI_PMODE_EN(bit 3) to enter XIP mode. Then directly read in XIP area (see Chapter 2 Memory Map) to get data.

10.3.6 Register List

Module Name	Base Address
SPIF	0x02800000

Register Name	Offset	Description
SPI_GLOBAL_CTRL	0x0004	SPI Global Control Register
SPI_GLOBAL_CTRL_ADD	0x0008	SPI Global Control Additional Register

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Register Name	Offset	Description
SPI_TIMING_CFG	0x000C	SPI Timing Configure Register
SPI_TIMING_DLY_STA	0x0010	SPI Timing Delay State Register
SPI_INT_EN	0x0014	SPI Interrupt Enable Register
SPI_INT_STA	0x0018	SPI Interrupt Status Register
SPI_CS_DELAY	0x001C	SPI Chipselect Delay Register
SPI_TRANS_PHA_CFG	0x0020	SPI Transfer Phase Configure Register
SPI_TRANS_CFG1	0x0024	SPI Trans Configure1 Register
SPI_TRANS_CFG2	0x0028	SPI Trans Configure2 Register
SPI_TRANS_NUM	0x002C	SPI Trans Number Register
SPI_XIP_STA	0x0030	SPI XIP State Register
SPI_XIP_START_ADDR	0x0034	SPI XIP Start Address Register
SPI_XIP_END_ADDR	0x0038	SPI XIP End Address Register
SPI_XIP_MAP_ADDR	0x003C	SPI XIP Map Address Register
SPI_DMA_CTRL	0x0040	SPI DMA Control Register
SPI_DESCRIPTOR_SADDR	0x0044	SPI DMA Descriptor Start Address Register
SPI_CDC_FIFO_TRIG_LEVEL	0x004C	SPI CDC FIFO Trigger Level Register
SPI_CDC_FIFO_STA	0x0050	SPI CDC FIFO Status Register
SPI_BATC	0x0054	SPI Bit-Aligned Transfer Configure Register
SPI_BA_CCR	0x0058	SPI Bit-Aligned CLOCK Configuration Register
SPI_TBR	0x005C	SPI TX Bit Register
SPI_RBR	0x0060	SPI RX Bit Register
SPI_STATUS	0x0064	SPI Status Register
SPI_STATUS_READ	0x0068	SPI Status Read Register
SPI_STATUS_READ2	0x006C	SPI Status Read Register_2
SPI_XIP_RX_DATA	0x0200	SPI XIP RX Data Register
SPI_RX_DATA	0x0210	SPI RX Data Register
SPI_TX_DATA	0x0220	SPI TX Data Register

10.3.7 Register Description

10.3.7.1 0x0004 SPI Global Control Register (Default Value: 0x0000_0100)

Offset: 0x0004			Register Name: SPI_GLOBAL_CTRL
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22	R/W	0x0	START_MODE 0: Old start mode 1: New start mode

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Offset: 0x0004			Register Name: SPI_GLOBAL_CTRL
Bit	Read/Write	Default/Hex	Description
21	R/WAC	0x0	SPI_READ_STATUS_MODE_EN SPI READ STATUS Mode Enable 0: Disable 1: Enable This bit is not used for data transfer. It is used for status confirm. And when this operation ends, the last returned byte value is sent to 0x0064[31:24].
20	R/WAC	0x0	SPI_CPU_MODE_EN SPI CPU Mode Enable 0: Disable 1: Enable CPU transfer mode.
19	R/W	0x0	SPI_CLK_OUTPUT 0: Disable 1: Enable When the SPI_CLK_OUTPUT is enabled, the SPI_FLASH will convey the sclk from CCU to the spi_sck_out pin. This bit cannot be enabled in the process of data transmission to avoid anomaly.
18	R/W	0x0	SPI_TX_CFG_FBS Transmit Interface First Transmit Bit Select. 0: MSB First 1: LSB First
17	R/W	0x0	SPI_RX_CFG_FBS Receive Interface First Transmit Bit Select. 0: MSB First 1: LSB First
16	R/W	0x0	DTR_EN Double Trans Rate Enable 0: Disable 1: Enable
15	R/W	0x0	SPI_WP_EN 0: Disable 1: Enable In Standard SPI or Dual SPI, SPI PIN Write Protect Function can be open In the open status, SPI_WP_IO2_OUT is enabled. The exact polarity is decided by the configuration of SPI_WP_POL. In the closed status, SPI_WP_IO2_OUT is controlled by hardware. The exact polarity is decided by the controller.

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Offset: 0x0004			Register Name: SPI_GLOBAL_CTRL
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	SPI_WP_POL SPI PIN Write Protect Polar 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle)
13	R/W	0x0	SPI_HOLD_EN 0: Disable 1: Enable In Standard SPI or Dual SPI, SPI PIN HOLD Function can be open. In the open status, SPI_HOLD_IO3_OUT is enabled. The exact polarity is decided by the configuration of SPI_HOLD_POL. In the closed status, SPI_HOLD_IO3_OUT is controlled by hardware. The exact polarity is decided by the controller.
12	R/W	0x0	SPI_HOLD_POL SPI PIN HOLD Polar 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle)
11	/	/	/
10	R/W	0x0	DQS_RX_EN Receive DQS Signal Enable 0: Disable 1: Enable
9	R/W	0x0	DUMMY_BIT_POL Dummy Bit Trans Value 0: Dummy bit Trans 0 1: Dummy bit Trans 1 In the dummy period, the output data of Hold and Write Protection will be controlled by the DUMMY_BIT_POPL.
8	R/W	0x1	SPI_CS_POL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = idle) 1: Active low polarity (1 = idle)
7:6	R/W	0x0	SPI_CS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPIF_CS0 will be asserted 01: SPIF_CS1 will be asserted (not uesed) 10: SPIF_CS2 will be asserted (not uesed) 11: SPIF_CS3 will be asserted (not uesed)

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Offset: 0x0004			Register Name: SPI_GLOBAL_CTRL
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	SPI_CPOL SPI Clock Polarity Control 0: Active High Polarity. 1: Active Low Polarity.
4	R/W	0x0	SPI_CPHA SPI Clock/Data Phase Control 0: Phase 0(Leading Edge for Sample Data) 1: Phase 1(Leading Edge For Setup Data)
3	R/W	0x0	SPI_PMODE_EN SPI XIP Mode Enable 0: Configuring AHB0-bus XIP 1: Configure AHB0-bus XIP Done When CPU needs to configure AHB0-bus XIP operation, Clear 0,then send configuration parameters. When CPU Configure AHB0-bus XIP Done , Set 1. When SPI Controller detect 'Enable XIP =1', parameters will be latched.
2	R/WAC	0x0	SPI_DMA_MODE_EN SPI DMA Mode Enable 0: Disable 1: Enable DMA transfer mode.
1	R/W	0x0	XIP_ADDR_MAP_MODE XIP Address Map Mode When the CPU accesses Flash through SPI, the address map mode can be selected by adding N to the address and then mapping the fetching address to the ADDR+N.
0	R/W	0x0	SPI_CFG_MODE SPI Parameter Config Source in normal(DMA) mode. Set to 0 When CPU Transfer. 0: From register 1: From DMA descriptor

10.3.7.2 0x0008 SPI Global Control Additional Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: SPI_GLOBAL_CTRL_ADD
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/

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Offset: 0x0008			Register Name: SPI_GLOBAL_CTRL_ADD
Bit	Read/Write	Default/Hex	Description
4	R/WAC	0x0	SPI_FLASH_DMA_END Terminate the DMA Chain after the present BLK is done. After the DMA chain terminates, set 1 to this bit to stop DMA transmission after the DMA module finishes the present BLK whether the current chain completes or the finish_flag is set to 1. DMA_END and SPI_SRST should be used together: Enable the DMA_END first, and then enable the SPI_FLASH_SRST after DMA_END is cleared.
3	R/WAC	0x0	SPI_FLASH_SRST Reset All the Logic in SPI Flash (DMA chain is not included) SPI is globally reset, and all the clock domain logic is reset
2	/	/	/
1	R/WAC	0x0	CDC_WF_SRST Transfer FIFO Software Reset, In SPI_WR_BUF_CTRL Module 0: Auto Clear to 0. 1: Reset Transfer FIFO. Writing 1 to this bit will reset the control portion of the transfer FIFO, and auto clear to 0 when completing reset operation. Writing 0 to this bit has no effect.
0	R/WAC	0x0	CDC_RF_SRST Receiver FIFO Software Reset, In SPI_RD_BUF_CTRL Module 0: Auto Clear to 0. 1: Reset Receiver FIFO. Writing 1 to this bit will reset the control portion of the receiver FIFO, and auto clear to 0 when completing reset operation, write 0 to this bit has no effect.

10.3.7.3 0x000C SPI Timing Configure Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: SPI_TIMING_CFG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/

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Offset: 0x000C			Register Name: SPI_TIMING_CFG
Bit	Read/Write	Default/Hex	Description
26	R/W	0x0	CLK_SCKOUT_SRC_SEL SPI Output Clk Select Select the rate of SPI output clock (SCK_OUT). 0: Original Clock 1: Two-frequency Clock
25	R/W	0x0	CLK_SCK_SRC_SEL SPI Interface Clock Select 0: Original Clock 1: Two-frequency Clock
24	R/W	0x0	CLK_SPI_SRC_SEL SPI Inter Process Clock Select 0: Original Clock 1: Two-frequency
23:22	/	/	/
21	R/W	0x0	SCKT_DELAY_MODE_SEL SCKT Delay Mode Select 0: Select the SCKT without delay 1: Select the SCKT after the analog delay chain adjustment
20	R/W	0x0	SCKR_DELAY_MODE_SEL SCKR Delay Mode Select 0: Digital delay 1: Digital delay + analog delay
19	/	/	/
18:16	R/W	0x0	DIGITAL_SCKR_DELAY_CFG Digital SCKR Delay Value Config, Step 0.5 spi_clk 000: Delay 0 clk 001: Delay 0.5 clk 010: Delay 1 clk 011: Delay 1.5 clk 100: Delay 2 clk 101: Delay 2.5 clk 110: Delay 3 clk 111: Delay 3.5 clk
15	R/W	0x0	ANALOG_SAMP_DL_CAL_START_TX Sample Delay Calibration Start When set, start sample delay chain calibration.
14	R/W	0x0	ANALOG_SAMP_DL_SW_EN_TX Sample Delay Software Enable When set, enable sample delay specified at SAMP_DL_SW

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Offset: 0x000C			Register Name: SPI_TIMING_CFG
Bit	Read/Write	Default/Hex	Description
13:8	R/W	0x0	ANALOG_SAMP_DL_SW_VALUE_TX Sample Delay Software Value The relative delay between clock line and command line, data lines. It can be determined according to the value of SAMP_DL, the cycle of card clock and device s input timing requirement.
7	R/W	0x0	ANALOG_SAMP_DL_CAL_START_RX Sample Delay Calibration Start When set, start sample delay chain calibration.
6	R/W	0x0	ANALOG_SAMP_DL_SW_EN_RX Sample Delay Software Enable When set, enable sample delay specified at SAMP_DL_SW
5:0	R/W	0x0	ANALOG_SAMP_DL_SW_VALUE_RX Sample Delay Software Value The relative delay between clock line and command line, data lines. It can be determined according to the value of SAMP_DL, the cycle of card clock and device s input timing requirement.

10.3.7.4 0x0010 SPI Timing Delay State Register (Default Value: 0x0000_2020)

Offset: 0x0010			Register Name: SPI_TIMING_DLY_STA
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R	0x0	ANALOG_SAMP_DL_CAL_DONE_TX Sample Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in SAMP_DL.
14	/	/	/

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Offset: 0x0010			Register Name: SPI_TIMING_DLY_STA
Bit	Read/Write	Default/Hex	Description
13:8	R	0x20	ANALOG_SAMP_DL_TX Sample Delay Value It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of card clock nearly. Generally, it is necessary to do drive delay calibration when card clock is changed. This bit is valid only when SAMP_DL_CAL_DONE is set.
7	R	0x0	ANALOG_SAMP_DL_CAL_DONE_RX Sample Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in SAMP_DL.
6	/	/	/
5:0	R	0x20	ANALOG_SAMP_DL_RX Sample Delay Value It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of card clock nearly. Generally, it is necessary to do drive delay calibration when card clock is changed. This bit is valid only when SAMP_DL_CAL_DONE is set.

10.3.7.5 0x0014 SPI Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: SPI_INT_EN
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x0	READ_STATUS_DONE_INT_EN 0: Disable. 1: Enable.
27	R/W	0x0	READ_STATUS_TIME_OUT_INT_EN 0: Disable. 1: Enable.

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Offset: 0x0014			Register Name: SPI_INT_EN
Bit	Read/Write	Default/Hex	Description
26	R/W	0x0	SPI_TRANS_DONE_INT_EN SPI Transmit Finish Interrupt Enable 0: Disable. 1: Enable.
25	R/W	0x0	XIP_READ_WORD_ACK_INT_EN XIP Mode Read back request word Interrupt Enable 0: Disable. 1: Enable.
24	R/W	0x0	DMA_TRANS_DONE_INT_EN DMA Chain Transmit Finish Interrupt Enable 0: Disable. 1: Enable.
23	R/W	0x0	DQS_RF_ODD_UDF_INT_EN DQS Receive FIFO Underflow Interrupt Enable 0: Disable. 1: Enable.
22	R/W	0x0	DQS_RF_ODD_OVF_INT_EN DQS Receive FIFO Overflow Interrupt Enable 0: Disable. 1: Enable.
21	R/W	0x0	DQS_RF_EVEN_UDF_INT_EN DQS Receive FIFO Underflow Interrupt Enable 0: Disable. 1: Enable.
20	R/W	0x0	DQS_RF_EVEN_OVF_INT_EN DQS Receive FIFO Overflow Interrupt Enable 0: Disable. 1: Enable.
19:12	/	/	/
11	R/W	0x0	WF_UDF_INT_EN Write FIFO Underflow Interrupt Enable 0: Disable. 1: Enable.
10	R/W	0x0	WF_OVF_INT_EN Write FIFO Overflow Interrupt Enable 0: Disable. 1: Enable.
9	R/W	0x0	RF_UDF_INT_EN Read FIFO Underflow Interrupt Enable 0: Disable. 1: Enable.

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Offset: 0x0014			Register Name: SPI_INT_EN
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	RF_OVF_INT_EN Read FIFO Overflow Interrupt Enable 0: Disable. 1: Enable.
7:5	/	/	/
4	R/W	0x0	WF_RDY_INT_EN Write FIFO Ready Request Interrupt Enable 0: Disable. 1: Enable.
3:1	/	/	/
0	R/W	0x0	RF_RDY_INT_EN Read FIFO Ready Request Interrupt Enable 0: Disable. 1: Enable.

10.3.7.6 0x0018 SPI Interrupt Status Register (Default Value: 0x0000_0010)

Offset: 0x0018			Register Name: SPI_INT_STA
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W1C	0x0	READ_STATUS_DONE_INT READ_STATUS MODE Done Interrupt. When set, this bit indicates that at least one wanted bit has got expected value.
27	R/W1C	0x0	READ_STATUS_TIME_OUT_INT READ_STATUS MODE TIME OUT Interrupt. When set, this bit indicates that READ_STATUS MODE ends and in the written period, no wanted bit gets expected value.
26	R/W1C	0x0	SPI_TRANS_DONE_INT SPI Transmit Finish Interrupt
25	R/W1C	0x0	XIP_READ_WORD_ACK_INT XIP Mode Read Back Request Word Interrupt When set, this bit indicates that Write FIFO has underflow. Writing “1” to this bit clears it.
24	R/W1C	0x0	DMA_TRANS_DONE_INT DMA Chain Transmit Finish Interrupt When set, this bit indicates that Write FIFO has underflow. Writing “1” to this bit clears it.

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Offset: 0x0018			Register Name: SPI_INT_STA
Bit	Read/Write	Default/Hex	Description
23	R/W1C	0x0	DQS_RF_ODD_UDF_INT DQS Receive FIFO underflow Interrupt Flag When set, this bit indicates that Write FIFO has underflow. Writing “1” to this bit clears it.
22	R/W1C	0x0	DQS_RF_ODD_OVF_INT DQS Receive FIFO Overflow Interrupt Flag When set, this bit indicates that Write FIFO has overflowed. Writing “1” to this bit clears it.
21	R/W1C	0x0	DQS_RF_EVEN_UDF_INT DQS Receive FIFO Underflow Interrupt Flag Note: When set, this bit indicates that Read FIFO has underflow. Writing “1” to this bit clears it.
20	R/W1C	0x0	DQS_RF_EVEN_OVF_INT DQS Receive FIFO Overflow Interrupt Flag When set, this bit indicates that Read FIFO has overflowed. Writing “1” to this bit clears it.
19:12	/	/	/
11	R/W1C	0x0	WF_UDF_INT Write FIFO underflow Interrupt Flag When set, this bit indicates that Write FIFO has underflow. Writing “1” to this bit clears it.
10	R/W1C	0x0	WF_OVF_INT Write FIFO Overflow Interrupt Flag When set, this bit indicates that Write FIFO has overflowed. Writing “1” to this bit clears it.
9	R/W1C	0x0	RF_UDF_INT Read FIFO Underflow Interrupt Flag When set, this bit indicates that Read FIFO has underflow. Writing “1” to this bit clears it.
8	R/W1C	0x0	RF_OVF_INT Read FIFO Overflow Interrupt Flag When set, this bit indicates that Read FIFO has overflowed. Writing “1” to this bit clears it.
7:5	/	/	/
4	R/W1C	0x1	WF_RDY_INT Write FIFO Ready Request Interrupt Flag 0: WF_WL > WF_EMPTY_TRIG_LEVEL. 1: WF_WL <= WF_EMPTY_TRIG_LEVEL. This bit is set any time if WF_WL <= WF_EMPTY_TRIG_LEVEL. Writing “1” to this bit clears it.
3:1	/	/	/

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Offset: 0x0018			Register Name: SPI_INT_STA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	RF_RDY_INT Read FIFO Ready Request Interrupt Flag 0: RF_WL < RX_EMPTY_TRIG_LEVEL and rest total RX data byte number is larger than RX_EMPTY_TRIG_LEVEL. 1: RF_WL >= RX_EMPTY_TRIG_LEVEL, or rest total RX data byte number is smaller than RX_EMPTY_TRIG_LEVEL meanwhile RF_WL >= 1. This bit is set any time if condition is reached. Writing "1" to this bit clears it.

10.3.7.7 0x001C SPI Chipselect Delay Register (Default Value: 0x0000_0606)

Offset: 0x001C			Register Name: SPI_CS_DELAY
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	DBG_SEL 0: The high 8 bits of debug signal are io*_oen, and the low 8 bits are 0. 1: The high 8 bits are WF_CNT, and the low 8 bits are RF_CNT. 2: The high 8 bits are WB_CNT, and the low 8 bits are RB_CNT. 3: The high 8 bits are DMA state machine, and the low 8 bits are SPI state machine. 4: Debug signal are read DRQ and write DRQ. 2'b0: dqs_rfifo_even water level; 4'b0: dqs_rfifo_odd water level.
23:16	R/W	0x0	CSDA Chip Select De-Assert The interval of two adjacent CS signal enable, namely, the interval of two SPI scheduling. The unit is the i_clk_spi_ref. The minimum interval A (CSDA=0) is 4 clk (used for ending SPI functions) The maximum A (CSDA=FB) is FF clk (used for ending SPI functions) When CSDA is X, the actual interval is X+A.

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Offset: 0x001C			Register Name: SPI_CS_DELAY
Bit	Read/Write	Default/Hex	Description
15:8	R/W	0x6	CSEOT Chip Select End of Transfer The interval between the rising edge of last output clock signal (SPI_CLK_OUT) and the CS signal disable. The unit is the reference clock (i_clk_spi_ref). The minimum interval B (CSEOT=0) is: DTR-TX: 3 clk DTR-RX: 5 clk STR-TX, CPOL=0: 3 clk STR-RX, CPOL=0: 4 clk STR-TX, CPOL=1: 2.5 clk STR-RX, CPOL=1: 3.5 clk When CSEOT is X, the actual interval is X+B. When using delay, the length of CSEOT should be larger than that of delay, or the data will be lost.
7:0	R/W	0x6	CSSOT Chip Select Start of Transfer The interval between the CS signal enable and the rising edge of the first output clock signal (SPI_CLK_OUT). The unit is the reference clock (i_clk_spi_ref). The minimum interval C (CSSOT=0) is: CPOL=0: 1 clk CPOL=1: 1.5 clk When CSSOT is X, the actual interval is X+C. The minimum interval is recommended to be 6, or the anomaly will occur in some special conditions. The minimum interval should be larger than 6 in NEW_START_MODE.

10.3.7.8 0x0020 SPI Transfer Phase Configure Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: SPI_TRANS_PHA_CFG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	CMD_DTR CMD DTR Control 1: CMD DTR 0: CMD not DTR CAUTION: When CMD DTR, DTR must be on and CMD_LEN must be 2 bytes.

Draft Version

Offset: 0x0020			Register Name: SPI_TRANS_PHA_CFG
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	COMMAND_TRANS_EN Set to 1 if command data need trans to device SPI Controller FSM Phase Enable 1: Enable 0: Disable
27:25	/	/	/
24	R/W	0x0	ADDRESS_TRANS_EN Set to 1 if address need trans to device SPI Controller FSM Phase Enable 1: Enable 0: Disable
23:21	/	/	/
20	R/W	0x0	MODE_BIT_TRANS_EN Set to 1 if Mode bit need trans after address SPI Controller FSM Phase Enable 1: Enable 0: Disable
19:17	/	/	/
16	R/W	0x0	DUMMY_BIT_TRANS_EN Dummy Bit State Enable SPI Controller FSM Phase Enable 1: Enable 0: Disable
15:13	/	/	/
12	R/W	0x0	TX_DATA_EN Set to 1 if Data Need Trans SPI Controller FSM Phase Enable 1: Enable 0: Disable
11:9	/	/	/
8	R/W	0x0	RX_DATA_EN Set to '1' if Data Need Receive SPI Controller FSM Phase Enable 1: Enable 0: Disable
7:0	/	/	/

10.3.7.9 0x0024 SPI Trans Configure1 Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: SPI_TRANS_CFG1
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0024			Register Name: SPI_TRANS_CFG1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADDR_OPCODE Address Content Trans Through SPI **When the length of ADDR is configured as 24bit.** a) If the MSB first is configured, the low 24 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [23:0]. The sending sequence is from bit23 to bit0. b) If the LSB first is configured, the high 24 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [31:8]. The sending sequence is from bit8 to bit31. **When the length of ADDR is configured as 16bit.** c) If the MSB first is configured, the low 16 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [15:0]. The sending sequence is from bit15 to bit0. d) If the LSB first is configured, the high 16 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [31:16]. The sending sequence is from bit16 to bit31. **When the length of ADDR is configured as 8bit.** e) If the MSB first is configured, the low 8 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [7:0]. The sending sequence is from bit7 to bit0. f) If the LSB first is configured, the high 8 bits of ADDR_OPCODE is used, namely, ADDR_OPCODE [31:24]. The sending sequence is from bit24 to bit31.

10.3.7.10 0x0028 SPI Trans Configure2 Register (Default Value: 0x0000_0000)

NOTE

This register should be setup while the controller is idle.

Offset: 0x0028			Register Name: SPI_TRANS_CFG2
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	CMD_OPCODE Command Content Trans Through SPI

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Offset: 0x0028			Register Name: SPI_TRANS_CFG2
Bit	Read/Write	Default/Hex	Description
23:16	R/W	0x0	MODE_OPCODE Mode Content Trans Through SPI
15:8	R/W	0x0	CMD_OPCODE2 Command2 Content Trans Through SPI
7:6	R/W	0x0	CMD_TRANS_TYPE Command Transfer Type 00: Command can be Shifted to the device on DQ0 01: Command can be Shifted to the device on DQ0 and DQ1 10: Command can be Shifted to the device on DQ0- DQ3 11: Command can be Shifted to the device on DQ0-DQ7
5:4	R/W	0x0	ADDR_TRANS_TYPE Address Transfer Type 00: Address can be Shifted to the device on DQ0 01: Address can be Shifted to the device on DQ0 and DQ1 10: Address can be Shifted to the device on DQ0- DQ3 11 : Address can be Shifted to the device on DQ0-DQ7
3:2	R/W	0x0	MODE_BIT_TRANS_TYPE Mode Bit Transfer Type 00: Mode Bit can be Shifted to the device on DQ0 01: Mode Bit can be Shifted to the device on DQ0 and DQ1 10: Mode Bit can be Shifted to the device on DQ0- DQ3 11 : Mode Bit can be Shifted to the device on DQ0-DQ7
1:0	R/W	0x0	DATA_TRANS_TYPE Data Transfer Type 00: Opcode can be Shifted to the device on DQ0 only 01: Opcode can be Shifted to the device on DQ0 and DQ1 only 10: Opcode can be Shifted to the device on DQ0- DQ3 11 : Opcode can be Shifted to the device on DQ0-DQ7

10.3.7.11 0x002C SPI Trans Number Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: SPI_TRANS_NUM
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DATA_TRANS_NUM_HIGH Bit [16] of DATA_TRANS_NUM

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Offset: 0x002C			Register Name: SPI_TRANS_NUM
Bit	Read/Write	Default/Hex	Description
30:27	/	/	/
26	R/W	0x0	CMD_LEN 0: 1byte 1: 2byte
25:24	R/W	0x0	ADDR_SIZE_MODE Address Size Mode 00: Address Size 8bit. 01: Address Size 16bit. 10: Address Size 24bit. 11: Address Size 32bit.
23:16	R/W	0x0	DUMMY_TRANS_NUM Number of Dummy Cycles A value of 0 = 1 Cycle A value of 1 = 1 Cycle ... A value of N = N Cycle
15:0	R/W	0x0	DATA_TRANS_NUM_LOW Bit [15:0] of DATA_TRANS_NUM. Num of Data Trans Through SPI(Byte) 0: Non-Write. 1: Write 1 Byte. 2: Write 2 Bytes. 3: Write 3 Bytes. ..... 65535: Write 65535 Bytes. Note: These Bits Indicate Current Operation is Write Data To Flash.

10.3.7.12 0x0030 SPI XIP State Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: SPI_XIP_STA
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:0	R	0x0	XIP_HIT_POSITION XIP hit position

10.3.7.13 0x0034 SPI XIP Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: SPI_XIP_START_ADDR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0034			Register Name: SPI_XIP_START_ADDR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XIP_START_ADDR XIP Work Start Address

10.3.7.14 0x0038 SPI XIP End Address Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: SPI_XIP_END_ADDR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XIP_END_ADDR XIP Work End Address

10.3.7.15 0x003C SPI XIP Map Address Register (Default Value: 0x0000_0000)

Offset: 0x003C			Register Name: SPI_XIP_MAP_ADDR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XIP_MAP_ADDR XIP Map Address, N When the address mapping mode (0x0004,bit[1]) is activated, the actual access address is ADDR+N.

10.3.7.16 0x0040 SPI DMA Control Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: SPI_DMA_CTRL
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:4	R/W	0x0	DMA_DESCRIPTOR_LEN DMA Descriptor Length In the initial design, there are eight DMA descriptors and the length of each descriptor is 4 Byte. In this bit, the length of descriptor is set to 32=8*4Byte.
3:1	/	/	/
0	R/WAC	0x0	CFG_DMA_START Config DMA Start Signal When all the DMA parameters are configured well, pull up a pulse of this signal to enable DMA chain.

10.3.7.17 0x0044 SPI DMA Descriptor Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: SPI_DESCRIPTOR_SADDR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0044			Register Name: SPI_DESCRIPTOR_SADDR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	FIRST_DESCRIPTOR_SADDR First Descriptor Start Address The WORD address is needed. Namely, the low 2 bit of byte address being removed.

10.3.7.18 0x004C SPI CDC FIFO Trigger Level Register (Default Value: 0x1001_1a10)

Offset: 0x004C			Name: SPI_CDC_FIFO_TRIG_LEVEL
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:24	R/W	0x10	WF_DMA_TRIG_LEVEL Write FIFO DMA Request Trigger Level (WORD) In the case of DMA transfer, this bit should be set to be less than or equal to 16 to avoid the efficiency reduction.
23:22	/	/	/
21:16	R/W	0x01	WF_CPU_TRIG_LEVEL Write FIFO CPU Request Trigger Level (WORD) In the case of DMA transfer, when WF_CNT is less than WF_CPU_TRIG_LEVEL, WF_RDY interrupts will be asserted.
15:14	/	/	/
13:8	R/W	0x1a	RF_FULL_TRIG_LEVEL Read FIFO Full Request Trigger Level (WORD) It is recommended to be less than or equal to 26 to avoid FIFO anomaly. It only determines the volume of RX_FIFO and has no effect on the RX_RDY_INT.
7:6	/	/	/
5:0	R/W	0x10	RF_RDY_TRIG_LEVEL Read FIFO RDY Request Trigger Level (WORD). In the case of CPU reading data, when the rest data volume is larger than RF_RDY_TRIG_LEVEL, and RF_CN is greater than RF_RDY_TRIG_LEVEL, or the rest data volume is less than RF_RDY_TRIG_LEVEL, and RF_CNT is larger than 1, the RF_RDY_INT will be sent out.

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10.3.7.19 0x0050 SPI CDC FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x0050			Name: SPI_CDC_FIFO_STA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:26	R	0x0	Flash Controller State Debug Reg.
25	R	0x0	WB_WR Write Buffer Could or Not Write.
24:21	R	0x0	WB_CNT Write Buffer Counter(Byte) These Bits Indicate Number of Bytes in Write Buffer.
20	R	0x0	RB_RD_RDY_STATE Read Buffer Could or Not Read
19:16	R	0x0	RB_CNT Read Buffer Counter(Byte) These Bits Indicate Number of Bytes in Read Buffer.
15:14	/	/	/
13:8	R	0x0	WF_CNT Write FIFO Counter (Level) These Bits Indicate Water Level in Write FIFO. 0: 0 level in Write FIFO. 1: 1 level in Write FIFO. (p) 32: 32 level in Write FIFO.
7:6	/	/	/
5:0	R	0x0	RF_CNT Read FIFO Counter (Level) These Bits Indicate Water Level in Read FIFO. 0: 0 level in Read FIFO. 1: 1 level in Read FIFO. (p) 32: 32 level in Read FIFO.

10.3.7.20 0x0054 SPI Bit-Aligned Transfer Configure Register (Default Value: 0x0000_00A0)

Offset: 0x0054			Register Name: SPI_BATC
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0054			Register Name: SPI_BATC
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	TCE Transfer Control Enable In master mode, it is used to start t1o transfer the serial bit's frame, it is only valid when Work Mode Select==0x10/0x11. 0: Idle 1: Initiates transfer. Writing 1 to this bit will start to transfer serial bits frame (the value comes from the SPI TX Bit Register or SPI RX Bit Register), and will auto clear after the bursts transfer completely. Write 0 to this bit has no effect.
30	R/W	0x0	MSMS Master Sample Standard 0: Standard Sample Mode 1: Delay Sample Mode In the Standard Sample Mode, SPI master samples the data at the standard rising edge of SCLK for each SPI mode. In the Delay Sample Mode, SPI master samples data at the edge that is half cycle delayed by the standard rising edge of SCLK defined in respective SPI mode.
29:26	/	/	
25	R/W1C	0x0	TBC Transfer Bits Completed When set, this bit indicates that the last bit of the serial data frame in SPI TX Bit Register (or SPI RX Bit Register) has been transferred completely. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed Note: It is only valid when Work Mode Select=0x10/0x11.
24	R/W	0x0	TBC_INT_EN Transfer Bits Completed Interrupt Enable 0: Disable 1: Enable Note: It is only valid when Work Mode Select=0x10/0x11.
23:22	/	/	/

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Offset: 0x0054			Register Name: SPI_BATC
Bit	Read/Write	Default/Hex	Description
21:16	R/W	0x00	RX_FEM_LEN Configure the length of serial data frame(burst) of RX 000000: 0bit 000001: 1bit (p) 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select==0x10/0x11, and can't be written when TCE=1.
15:14	/	/	/
13:8	R/W	0x00	TX_FEM_LEN Configure the length of serial data frame(burst) of TX 000000: 0bit 000001: 1bit (p) 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select==0x10/0x11, and can't be written when TCE=1.
7	R/W	0x1	SS_LEVEL When control CS signal manually, set this bit to 1 or 0 to control the level of CS signal. 0: Set CS to low 1: Set CS to high Note: It is only valid when Work Mode Select==0x10/0x11, and only work in Mode0, can't be written when TCE=1.
6	R/W	0x0	SS_OWNER CS Output Owner Select Usually, controller sends CS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of CS signal. 0: SPI controller 1: Software Note: It is only valid when Work Mode Select==0x10/0x11, and only work in Mode0, can't be written when TCE=1.

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Offset: 0x0054			Register Name: SPI_BATC
Bit	Read/Write	Default/Hex	Description
5	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: It is only valid when Work Mode Select==0x10/0x11, and only work in Mode0, can't be written when TCE=1.
4	/	/	/
3:2	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPIF_CS0 will be asserted 01: SPIF_CS1 will be asserted (not used) 10: SPIF_CS2 will be asserted (not used) 11: SPIF_CS3 will be asserted (not used) Note: It is only valid when Work Mode Select==0x10/0x11, and only work in Mode0, can't be written when TCE=1.
1:0	R/W	0x0	Work Mode Select 00: Data frame is byte aligned in Standard SPI, Dual-Output/Dual Input SPI, Dual IO SPI and Quad-Output/Quad-Input SPI. 01: Reserve 10: Data frame is bit aligned in 3-Wire SPI 11: Data frame is bit aligned in Standard SPI

10.3.7.21 0x0058 SPI Bit-Aligned CLOCK Configuration Register (Default Value: 0x0000_0000)

NOTE

This register is only valid when Work Mode Select=0x10/0x11.

Offset: 0x0058			Register Name: SPI_BA_CCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	CDR_N Clock Divide Rate (Master Mode Only) The SPI_SCLK is determined according to the following equation: SPI_CLK = Source_CLK / (2*(CDR_N + 1)).

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10.3.7.22 0x005C SPI TX Bit Register (Default Value: 0x0000_0000)

NOTE

This register is only valid when Work Mode Select=0x10/0x11.

Offset: 0x005C			Register Name: SPI_TBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VTB The Value of the Transmit Bits This register is used to store the value of the transmitted serial data frame. Note: In the process of transmission, the LSB is transmitted first.

10.3.7.23 0x0060 SPI RX Bit Register (Default Value: 0x0000_0000)

NOTE

This register is only valid when Work Mode Select=0x10/0x11.

Offset: 0x0060			Register Name: SPI_RBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VRB The Value of the Receive Bits This register is used to store the value of the received serial data frame. Note: In the process of transmission, the LSB is transmitted first.

10.3.7.24 0x0064 SPI Status Register (Default Value: 0x0000_0885)

Offset: 0x0064			Register Name: SPI_STATUS
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	READ_STATUS_RESULT
23:12	/	/	/
11:8	R	0x8	DQS_RF_ODD_WL Rest of space.
7: 4	R	0x8	DQS_RF_EVEN_WL Rest of space.

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Offset: 0x0064			Register Name: SPI_STATUS
Bit	Read/Write	Default/Hex	Description
3	R	0x0	WF_FULL Write FIFO Full Interrupt Flag 0: Write FIFO is Not Full. 1: Write FIFO is Full. Note: This bit is set when the Write FIFO is full.
2	R	0x1	WF_EMPTY Write FIFO Empty Request Interrupt Flag 0: Write FIFO always contains one or more bytes. 1: Write FIFO has been empty. Note: This bit is set if the Write FIFO is empty.
1	R	0x0	RF_FULL Read FIFO Full Interrupt Flag 0: Not Full. 1: Read FIFO has been Full. Note: This bit is set when Read FIFO is full.
0	R	0x1	RF_EMPTY Read FIFO Empty Interrupt Flag 0: Not Empty. 1: Read FIFO has been Empty. Note: This bit is set if the Read FIFO is empty.

10.3.7.25 0x0068 SPI Status Read Register (Default Value: 0x0000_0000)

Offset: 0x0068			Register Name: SPI_STATUS_READ
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	SPI_READ_STATUS_DETECTED_BIT_POSITION These 8 bits are used to distinguish the wanted bit in a STATUS BYTE. For every bit that CPU cares, put a high level in the corresponding bit position.
15:8	/	/	/
7:0	R/W	0x0	SPI_READ_STATUS_DETECTED_BIT_VALUE These 8 bits are used to distinguish the wanted bit in a STATUS BYTE. For every bit that CPU cares, put the level that CPU wants in the corresponding bit position. And when one bit is detected the same value, READ_STATUS_MODE shall stop and return a DONE interrupt. Then CPU shall check the exact result in 0x0064[31:24].

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10.3.7.26 0x006C SPI STATUS READ Register_2 (Default Value: 0x0000_0000)

Offset: 0x006C			Register Name: SPI_STATUS_READ2
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	MAXIMUM_STATUS_READ_TIME These 14 bits are used to tell the module that when reading FLASH status, how many times it should read before it stops reading and instead return a TIME_OUT INTERRUPT to avoid dead lock. Note: The suggested minimum value is 1.
15:14	/	/	/
13:0	R/W	0x0	READ_STATUS_INTERVAL The INTERVAL is counted with 32KHz clock. These 14 bits are used to tell that when READING STATUS, after how many rising edge of the 32K clock should the module start a READ STATUS visit to the connected FLASH. Note: The suggested minimum value is 1.

10.3.7.27 0x0200 SPI XIP RX Data Register (Default Value: 0x0000_0000)

Offset: 0x0200			Register Name: SPI_XIP_RX_DATA
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SPI_XIP_RX_DATA SPI XIP data.

10.3.7.28 0x0210 SPI RX Data Register (Default Value: 0x0000_0000)

Offset: 0x0210			Register Name: SPI_RX_DATA
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SPI_RX_DATA SPI RX Data. It cannot be read when DMA transfer.

10.3.7.29 0x0220 SPI TX Data Register (Default Value: 0x0000_0000)

Offset: 0x0220			Register Name: SPI_TX_DATA
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SPI_TX_DATA SPI TX Data. It cannot be written when DMA transfer.

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11 Interrupts

11.1 GIC

11.1.1 Overview

The GIC-400 is a high-performance, area-optimized interrupt controller with an Advanced Microcontroller Bus Architecture (AMBA) Advanced eXtensible Interface (AXI) interface. It detects, manages, and distributes interrupts in System on Chip (SoC) configurations. You can configure the GIC-400 to provide the optimum features, performance, and gate count required for your intended application. With the following software-configurable settings of the GIC-400, interrupts can be:

- Enabled or disabled.
- Assigned to one of two groups, Group 0 or Group 1.
- Prioritized.
- Signaled to different processors in multiprocessor implementations.
- Either level-sensitive or edge-triggered.

The GIC-400 implements:

- The GIC Security Extensions, that support:
  - Using Group 0 interrupts as Secure interrupts, and Group 1 interrupts as Non-secure interrupts.
  - Optionally, using the FIQ interrupt request to signal Secure interrupts to a connected processor. The GIC-400 always signals Group 1 interrupts using the IRQ interrupt request.
- The GIC Virtualization Extensions, that provide hardware support for managing virtualized interrupts.

The GIC-400 implements the interrupt types:

- 16 Software Generated Interrupts (SGIs).
- 6 external Private Peripheral Interrupts (PPIs) for each processor.
- 1 internal PPI for each processor.
- A configurable number of Shared Peripheral Interrupt (SPIs).

The GIC has the following features:

Implements the partitioning that the ARM Generic Interrupt Controller Architecture Specification describes.

- Enable and mask interrupts

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- Set interrupt trigger conditions
- Interrupt group management
- Implements the optional GIC Virtualization Extensions enabling it to manage virtual interrupts

11.1.2 Functional Description

11.1.2.1 CPUX Interrupt Sources

the following table describes the details of interrupt sources:

Table 11-1 Interrupt Sources in CPU System

Interrupt Number	Interrupt Source	Interrupt Vector	Description
0	SGI0	0x0000	SGI 0 interrupt
1	SGI1	0x0004	SGI 1 interrupt
2	SGI2	0x0008	SGI 2 interrupt
3	SGI3	0x000C	SGI 3 interrupt
4	SGI4	0x0010	SGI 4 interrupt
5	SGI5	0x0014	SGI 5 interrupt
6	SGI6	0x0018	SGI 6 interrupt
7	SGI7	0x001C	SGI 7 interrupt
8	SGI8	0x0020	SGI 8 interrupt
9	SGI9	0x0024	SGI 9 interrupt
10	SGI10	0x0028	SGI 10 interrupt
11	SGI11	0x002C	SGI 11 interrupt
12	SGI12	0x0030	SGI 12 interrupt
13	SGI13	0x0034	SGI 13 interrupt
14	SGI14	0x0038	SGI 14 interrupt
15	SGI15	0x003C	SGI 15 interrupt
32	CPUX_MBOX_R	0x0080	MSGBOX_CPUX READ IRQ FOR CPUX
33	DCXO_REF_ERR	0x0084	
34	UART0	0x0088	
35	UART1	0x008C	
36	UART2	0x0090	
37	UART3	0x0094	
38	UART4	0x0098	
39	UART5	0x009C	
40	UART6	0x00A0	
41	UART7	0x00A4	
42	TWI0	0x00A8	
43	TWI1	0x00AC	
44	TWI2	0x00B0	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
45	TWI3	0x00B4	
46	TWI4	0x00B8	
47	TWI5	0x00BC	
48	SPI0	0x00C0	
49	SPI1	0x00C4	
50	SPI2	0x00C8	
51	PWM0	0x00CC	PWMCS0 interrupt
52	SPI_FLASH	0x00D0	
53	SPI3	0x00D4	
54	/	0x00D8	
55	OWA	0x00DC	
56	DMIC	0x00E0	
57	ADDA	0x00E4	Audio codec interrupt
58	IRTX	0x00E8	
59	IRRX0	0x00EC	
60	LEDC	0x00F0	
61	USB0_DEVICE	0x00F4	
62	USB0_EHCI	0x00F8	
63	USB0_OHCI	0x00FC	
64	USB1_EHCI	0x0100	
65	USB1_OHCI	0x0104	
66	/	0x0108	
67	CAN0_TOP	0x010C	
68	CAN0_0	0x0110	
69	CAN0_1	0x0114	
70	/	0x0118	
71	THS0	0x011C	
72	SMHC0	0x0120	
73	SMHC1	0x0124	
74	SMHC2	0x0128	
75	MSI	0x012C	
76	SMC	0x0130	
77	PWM2	0x0134	
78	GMAC0_TOP	0x0138	
79	GMAC1_TOP	0x013C	
80	LBC	0x0140	
81	AHB_HREADY_TIME_OUT	0x0144	
82	DMAC0_CPUX_NS	0x0148	DMA0 channel 0-15non-secure interrupt
83	DMAC0_CPUX_S	0x014C	DMA0 channel 0-15 secure interrupt

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
84	CE_NS	0x0150	
85	CE_S	0x0154	
86	SPINLOCK	0x0158	
87	TIMER_0	0x015C	TIMER0_CPUX0 interrupt
88	TIMER_1	0x0160	TIMER0_CPUX1 interrupt
89	TIMER_2	0x0164	TIMER0_CPUX2 interrupt
90	TIMER_3	0x0168	TIMER0_CPUX3 interrupt
91	TIMER_4	0x016C	TIMER0_CPUX4 interrupt
92	TIMER_5	0x0170	TIMER0_CPUX5 interrupt
93	GPADC0	0x0174	
94	GPADC2	0x0178	
95	WDT0	0x017C	WDT_CPUX interrupt
96	GPADC1	0x0180	
97	GPIOA_OS1_NS	0x0184	
98	GPIOA_OS1_S	0x0188	
99	GPIOB_OS1_NS	0x018C	
100	GPIOB_OS1_S	0x0190	
101	GPIOC_OS1_NS	0x0194	
102	GPIOC_OS1_S	0x0198	
103	GPIOD_OS1_NS	0x019C	
104	GPIOD_OS1_S	0x01A0	
105	GPIOE_OS1_NS	0x01A4	
106	GPIOE_OS1_S	0x01A8	
107	GPIOF_OS1_NS	0x01AC	
108	GPIOF_OS1_S	0x01B0	
109	GPIOG_OS1_NS	0x01B4	
110	GPIOG_OS1_S	0x01B8	
111		0x01BC	
112		0x01C0	
113		0x01C4	
114		0x01C8	
115	GPIOJ_OS1_NS	0x01CC	
116	GPIOJ_OS1_S	0x01D0	
117		0x01D4	
118		0x01D8	
119	DE0	0x01DC	
120		0x01E0	
121		0x01E4	
122	TCON0_LCD0	0x01E8	
123	DSI0	0x01EC	
124	GMAC0_PWR_CLK_CTRL	0x01F0	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
125	GMAC0_PERCH_TX	0x01F4	
126	GMAC0_PERCH_RX	0x01F8	
127	GMAC1_PWR_CLK_CTRL	0x01FC	
128	GMAC1_PERCH_TX	0x0200	
129	GMAC1_PERCH_RX	0x0204	
130	GMAC2_PWR_CLK_CTRL	0x0208	
131	GMAC2_PERCH_TX	0x020C	
132	GMAC2_PERCH_RX	0x0210	
133	GMAC2_TOP	0x0214	
134	CPU_SYS_AHBMON	0x0218	
135	DCU_AHBMON	0x021C	
136	MCU_SYS_AXIMON	0x0220	
137	SID_SEC	0x0224	
138		0x0228	
139	SMHC3	0x022C	
140		0x0230	
141		0x0234	
142		0x0238	
143		0x023C	
144		0x0240	
145		0x0244	
146		0x0248	
147		0x024C	
148		0x0250	
149	IRRX1	0x0254	
150	IRRX2	0x0258	
151	IRRX3	0x025C	
152		0x0260	
153	MEMC_DFS	0x0264	DRAMC_DFS interrupt
154	CSI_DMA0	0x0268	
155	CSI_DMA1	0x026C	
156	CSI_DMA2	0x0270	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
157		0x0274	
158	CSI_VIPP0	0x0278	
159	CSI_VIPP1	0x027C	
160		0x0280	
161		0x0284	
162	CSI_PARSER0	0x0288	
163	CSI_PARSER1	0x028C	
164	CSI_PARSER2	0x0290	
165	CSI_ISP0	0x0294	
166	CSI_ISP1	0x0298	
167		0x029C	
168		0x02A0	
169	CSI_CMB	0x02A4	
170	CSI_TDM	0x02A8	
171	CSI_TOP_PKT	0x02AC	
172	GPIOK_OS1_NS	0x02B0	
173	GPIOK_OS1_S	0x02B4	
174	PWM1	0x02B8	PWMCS1 interrupt
175	G2D[0]	0x02BC	
176	CAN1_0	0x02C0	
177	CAN1_1	0x02C4	
178	CAN1_TOP	0x02C8	
179		0x02CC	
180	UART8	0x02D0	
181	UART9	0x02D4	
182	G2D[1]	0x02D8	
183	G2D[2]	0x02DC	
184	G2D[3]	0x02E0	
185		0x02E4	
186		0x02E8	
187		0x02EC	
188	TIMER_6	0x02F0	TIMER0_CPUX6 interrupt
189	TIMER_7	0x02F4	TIMER0_CPUX7 interrupt
190		0x02F8	
191	TPADC	0x02FC	
192	I2S0	0x0300	
193	I2S1	0x0304	
194	I2S2	0x0308	
195		0x030C	
196	DMAC1_CPUX_NS	0x0310	DMA1 channel 0~15 non-secure interrupt

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
197	DMAC1_CPUX_S	0x0314	DMA1 channel 0~15 secure interrupt
198	CORE0_MBOX_R	0x0318	MSGBOX_CORE0 READ IRQ
199	CORE1_MBOX_R	0x031C	MSGBOX_CORE1 READ IRQ
200	CORE2_MBOX_R	0x0320	MSGBOX_CORE2 READ IRQ
201	CORE3_MBOX_R	0x0324	MSGBOX_CORE3 READ IRQ
202	CORE0_MBOX_CORE1_W	0x0328	MSGBOX_CORE0 Write IRQ to core1
203	CORE0_MBOX_CORE2_W	0x032C	MSGBOX_CORE0 Write IRQ to core2
204	CORE0_MBOX_CORE3_W	0x0330	MSGBOX_CORE0 Write IRQ to core3
205	CORE1_MBOX_CORE0_W	0x0334	MSGBOX_CORE1 Write IRQ to core0
206	CORE1_MBOX_CORE2_W	0x0338	MSGBOX_CORE1 Write IRQ to core2
207	CORE1_MBOX_CORE3_W	0x033C	MSGBOX_CORE1 Write IRQ to core3
208	CORE2_MBOX_CORE0_W	0x0340	MSGBOX_CORE2 Write IRQ to core0
209	CORE2_MBOX_CORE1_W	0x0344	MSGBOX_CORE2 Write IRQ to core1
210	CORE2_MBOX_CORE3_W	0x0348	MSGBOX_CORE2 Write IRQ to core3
211	CORE3_MBOX_CORE0_W	0x034C	MSGBOX_CORE3 Write IRQ to core0
212	CORE3_MBOX_CORE1_W	0x0350	MSGBOX_CORE3 Write IRQ to core1
213	CORE3_MBOX_CORE2_W	0x0354	MSGBOX_CORE3 Write IRQ to core2
214		0x0358	
215	RV_MBOX_CPUX_W	0x035C	MSGBOX_MCU Write IRQ to CPUX
216		0x0360	
217	RV_WDG	0x0364	WDT_MCU interrupt
218	RV_TIMER_0	0x0368	TIMER0_MCU0 interrupt
219	RV_TIMER_1	0x036C	TIMER0_MCU1 interrupt
220	RV_TIMER_2	0x0370	TIMER0_MCU2 interrupt
221	RV_TIMER_3	0x0374	TIMER0_MCU3 interrupt
222		0x0378	
223		0x037C	
224	NMI	0x0380	
225		0x0384	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
226	R_TWD	0x0388	TWD interrupt
227		0x038C	
228		0x0390	
229		0x0394	
230		0x0398	
231		0x039C	
232	R_ALARM0	0x03A0	RTC alarm interrupt
233		0x03A4	
234		0x03A8	
235		0x03AC	
236		0x03B0	
237		0x03B4	
238		0x03B8	
239		0x03BC	
240		0x03C0	
241		0x03C4	
242		0x03C8	
243		0x03CC	
244		0x03D0	
245	R_AHB_HREADY_TIME_OUT	0x03D4	
246		0x03D8	
247	PCK600_CPU	0x03DC	
248	GPIO_NMI_S	0x03E0	RTC_GPIO Secure IRQ
249	GPIO_NMI_NS	0x03E4	RTC_GPIO Non-Secure IRQ
250		0x03E8	
251		0x03EC	
252		0x03F0	
253		0x03F4	
254		0x03F8	
255		0x03FC	
256	GPIOA_OS0_NS	0x0400	
257	GPIOA_OS0_S	0x0404	
258	GPIOB_OS0_NS	0x0408	
259	GPIOB_OS0_S	0x040C	
260	GPIOC_OS0_NS	0x0410	
261	GPIOC_OS0_S	0x0414	
262	GPIOD_OS0_NS	0x0418	
263	GPIOD_OS0_S	0x041C	
264	GPIOE_OS0_NS	0x0420	
265	GPIOE_OS0_S	0x0424	
266	GPIOF_OS0_NS	0x0428	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
267	GPIOF_OS0_S	0x042C	
268	GPIOG_OS0_NS	0x0430	
269	GPIOG_OS0_S	0x0434	
270	GPIOJ_OS0_NS	0x0438	
271	GPIOJ_OS0_S	0x043C	
272	GPIOK_OS0_NS	0x0440	
273	GPIOK_OS0_S	0x0444	
274	GPIOA_OS2_NS	0x0448	
275	GPIOA_OS2_S	0x044C	
276	GPIOB_OS2_NS	0x0450	
277	GPIOB_OS2_S	0x0454	
278	GPIOC_OS2_NS	0x0458	
279	GPIOC_OS2_S	0x045C	
280	GPIOD_OS2_NS	0x0460	
281	GPIOD_OS2_S	0x0464	
282	GPIOE_OS2_NS	0x0468	
283	GPIOE_OS2_S	0x046C	
284	GPIOF_OS2_NS	0x0470	
285	GPIOF_OS2_S	0x0474	
286	GPIOG_OS2_NS	0x0478	
287	GPIOG_OS2_S	0x047C	
288	GPIOJ_OS2_NS	0x0480	
289	GPIOJ_OS2_S	0x0484	
290	GPIOK_OS2_NS	0x0488	
291	GPIOK_OS2_S	0x048C	
292	GPIOA_OS3_NS	0x0490	
293	GPIOA_OS3_S	0x0494	
294	GPIOB_OS3_NS	0x0498	
295	GPIOB_OS3_S	0x049C	
296	GPIOC_OS3_NS	0x04A0	
297	GPIOC_OS3_S	0x04A4	
298	GPIOD_OS3_NS	0x04A8	
299	GPIOD_OS3_S	0x04AC	
300	GPIOE_OS3_NS	0x04B0	
301	GPIOE_OS3_S	0x04B4	
302	GPIOF_OS3_NS	0x04B8	
303	GPIOF_OS3_S	0x04BC	
304	GPIOG_OS3_NS	0x04C0	
305	GPIOG_OS3_S	0x04C4	
306	GPIOJ_OS3_NS	0x04C8	
307	GPIOJ_OS3_S	0x04CC	
308	GPIOK_OS3_NS	0x04D0	

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Interrupt Number	Interrupt Source	Interrupt Vector	Description
309	GPIOK_OS3_S	0x04D4	
CPUX Related			
320	CORE0_CTIIRQ	0x0500	
321	CORE1_CTIIRQ	0x0504	
322	CORE2_CTIIRQ	0x0508	
323	CORE3_CTIIRQ	0x050C	
324	CORE0_COMMTX	0x0510	
325	CORE1_COMMTX	0x0514	
326	CORE2_COMMTX	0x0518	
327	CORE3_COMMTX	0x051C	
328	CORE0_COMMRX	0x0520	
329	CORE1_COMMRX	0x0524	
330	CORE2_COMMRX	0x0528	
331	CORE3_COMMRX	0x052C	
332	CORE0_PMUIRQ	0x0530	
333	CORE1_PMUIRQ	0x0534	
334	CORE2_PMUIRQ	0x0538	
335	CORE3_PMUIRQ	0x053C	
336	CLU_nEXTERRIRQ	0x0540	
337	/	0x0544	
338	CLU_AXI_WR_IRQ	0x0548	
339	CLU_AXI_RD_IRQ	0x054C	
340	CORE0_DBGRSTREQ	0x0550	
341	CORE1_DBGRSTREQ	0x0554	
342	CORE2_DBGRSTREQ	0x0558	
343	CORE3_DBGRSTREQ	0x055C	
344		0x0560	
345		0x0564	
346		0x0568	
347		0x056C	
348		0x0570	
349		0x0574	
350	CONFLICT_IRQ	0x0578	
351	CPU_TZMA_ERR_IRQ	0x057C	

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11.1.2.2 PPIs Interrupt Sources

Interrupt Index	Interrupt Source	Description
PPI-20	/	
PPI-21	/	
PPI-22	nCOMMIRQ[3:0]	Interrupt-driven debug communications operation
PPI-23	nPMUIRQ[3:0]	PMU interrupt request
PPI-24	CTIIRQ[3:0]	CTI interrupt
PPI-25	nVCPUMNTIRQ[3:0]	Virtual CPU interface maintenance interrupt
PPI-26	nCNTHPIRQ[3:0]	Hypervisor physical timer event
PPI-27	nCNTVIRQ[3:0]	Virtual timer event
PPI-28	nCNTHVIRQ[3:0]	Hypervisor virtual timer event
PPI-29	nCNTPSIRQ[3:0]	Secure physical timer event
PPI-30	nCNTPNSIRQ[3:0]	Nonsecure physical timer event
PPI-31	/	

11.1.3 Register List

Module Name	Base Address
GIC	0x03400000

11.1.4 Register Description

For detailed register description, please refer to *DDI0471B_gic400_r0p1_trm.pdf*.

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11.2 CLIC

11.2.1 Overview

The Core-Local Interrupt Controller (CLIC) is only used for sampling, priority arbitration and distribution for external interrupt sources.

- Support RISC-V Core-Local Interrupt Controller Version 0.8 specification
- Up to 240 external interrupt souces and 16 internal interrupts compatible with CLINT
- Support level interrupts and pulse interrupts
- 32 levels of interrupt priority
- 4 memory-mapped control registers for each interrupt
- Each attribute of this interrupt source can be configured by writing the control register of the corresponding interrupt source.

11.2.2 Functional Description

The following table describes the detail of interrupt sources.

Table 11-2 Interrupt Sources in E907 System

Interrupt Number	Interrupt Sources	Interrupt Vector	Description
E907 Internal Interrupt			
0	Reserved	0x0000	Not used
1	Reserved	0x0004	Not used
2	Reserved	0x0008	Not used
3	CLINT_CPU_MS_INT	0x000C	Machine mode software interrupt
4	Reserved	0x0010	Not used
5	Reserved	0x0014	Not used
6	Reserved	0x0018	Not used
7	CLINT_CPU_MT_INT	0x001C	Machine mode timer interrupt
8	Reserved	0x0020	Not used
9	Reserved	0x0024	Not used
10	Reserved	0x0028	Not used
11		0x002C	
12	Reserved	0x0030	Not used
13	Reserved	0x0034	Not used
14	Reserved	0x0038	Not used
15	Reserved	0x003C	Not used
E907 External Interrupt			
16	/	0x0040	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
17	DCXO_REF_ERR	0x0044	
18	UART0	0x0048	
19	UART1	0x004C	
20	UART2	0x0050	
21	UART3	0x0054	
22	UART4	0x0058	
23	UART5	0x005C	
24	UART6	0x0060	
25	UART7	0x0064	
26	TWI0	0x0068	
27	TWI1	0x006C	
28	TWI2	0x0070	
29	TWI3	0x0074	
30	TWI4	0x0078	
31	TWI5	0x007C	
32	SPI0	0x0080	
33	SPI1	0x0084	
34	SPI2	0x0088	
35	PWM0	0x008C	PWMCS0 interrupt
36	SPI_FLASH	0x0090	
37	SPI3	0x0094	
38		0x0098	
39	OWA	0x009C	
40	DMIC	0x00A0	
41	AUDIO CODEC	0x00A4	
42	IRTX	0x00A8	
43	IRRX0	0x00AC	
44	LEDC	0x00B0	
45	USB0_DEVICE	0x00B4	
46	USB0_EHCI	0x00B8	
47	USB0_OHCI	0x00BC	
48	USB1_EHCI	0x00C0	
49	USB1_OHCI	0x00C4	
50		0x00C8	
51	CAN0_TOP	0x00CC	
52	CAN0_0	0x00D0	
53	CAN0_1	0x00D4	
54		0x00D8	
55	THS0	0x00DC	
56	SMHC0	0x00E0	
57	SMHC1	0x00E4	
58	SMHC2	0x00E8	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
59	MSI	0x00EC	
60	SMC	0x00F0	
61	PWM2	0x00F4	
62	GMAC0_TOP	0x00F8	
63	GMAC1_TOP	0x00FC	
64	LBC	0x0100	
65	AHB_HREADY_TIME_OUT	0x0104	
66		0x0108	
67		0x010C	
68	CE_NS	0x0110	
69	CE_S	0x0114	
70	SPINLOCK	0x0118	
71	TIMER CPUX0	0x011C	TIMER0_CPUX0 interrupt
72	TIMER CPUX1	0x0120	TIMER0_CPUX1 interrupt
73	TIMER CPUX2	0x0124	TIMER0_CPUX2 interrupt
74	TIMER CPUX3	0x0128	TIMER0_CPUX3 interrupt
75	TIMER CPUX4	0x012C	TIMER0_CPUX4 interrupt
76	TIMER CPUX5	0x0130	TIMER0_CPUX5 interrupt
77	GPADC0	0x0134	
78	GPADC2	0x0138	
79	WDT_CPUX	0x013C	
80	GPADC1	0x0140	
81	GPIOA_OS1_NS	0x0144	
82	GPIOA_OS1_S	0x0148	
83	GPIOB_OS1_NS	0x014C	
84	GPIOB_OS1_S	0x0150	
85	GPIOC_OS1_NS	0x0154	
86	GPIOC_OS1_S	0x0158	
87	GPIOD_OS1_NS	0x015C	
88	GPIOD_OS1_S	0x0160	
89	GPIOE_OS1_NS	0x0164	
90	GPIOE_OS1_S	0x0168	
91	GPIOF_OS1_NS	0x016C	
92	GPIOF_OS1_S	0x0170	
93	GPIOG_OS1_NS	0x0174	
94	GPIOG_OS1_S	0x0178	
95		0x017C	
96		0x0180	
97		0x0184	
98		0x0188	
99	GPIOJ_OS1_NS	0x018C	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
100	GPIOJ_OS1_S	0x0190	
101		0x0194	
102		0x0198	
103	DE0	0x019C	
104		0x01A0	
105		0x01A4	
106	TCON_LCD0	0x01A8	
107	DSI0	0x01AC	
108	GMAC0_PWR_CLK_CTRL	0x01B0	
109	GMAC0_PERCH_TX	0x01B4	
110	GMAC0_PERCH_RX	0x01B8	
111	GMAC1_PWR_CLK_CTRL	0x01BC	
112	GMAC1_PERCH_TX	0x01C0	
113	GMAC1_PERCH_RX	0x01C4	
114	GMAC2_PWR_CLK_CTRL	0x01C8	
115	GMAC2_PERCH_TX	0x01CC	
116	GMAC2_PERCH_RX	0x01D0	
117	GMAC2_TOP	0x01D4	
118	CPU_SYS_AHBMON	0x01D8	
119	DCU_AHBMON	0x01DC	
120	MCU_SYS_AXIMON	0x01E0	
121	SID_SEC	0x01E4	
122		0x01E8	
123	SMHC3	0x01EC	
124		0x01F0	
125		0x01F4	
126		0x01F8	
127		0x01FC	
128		0x0200	
129		0x0204	
130		0x0208	
131		0x020C	
132		0x0210	
133	IRRX1	0x0214	
134	IRRX2	0x0218	
135	IRRX3	0x021C	
136		0x0220	
137	DRAMC_DFS	0x0224	
138	CSI_DMA0	0x0228	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
139	CSI_DMA1	0x022C	
140	CSI_DMA2	0x0230	
141		0x0234	
142	CSI_VIPP0	0x0238	
143	CSI_VIPP1	0x023C	
144		0x0240	
145		0x0244	
146	CSI_PARSER0	0x0248	
147	CSI_PARSER1	0x024C	
148	CSI_PARSER2	0x0250	
149	CSI_ISP0	0x0254	
150	CSI_ISP1	0x0258	
151		0x025C	
152		0x0260	
153	CSI_CMB	0x0264	
154	CSI_TDM	0x0268	
155	CSI_TOP_PKT	0x026C	
156	GPIOK_OS1_NS	0x0270	
157	GPIOK_OS1_S	0x0274	
158	PWM1	0x0278	PWMCS1 interrupt
159	G2D	0x027C	
160	CAN1_0	0x0280	
161	CAN1_1	0x0284	
162	CAN1_TOP	0x0288	
163		0x028C	
164	UART8	0x0290	
165	UART9	0x0294	
166		0x0298	
167		0x029C	
168		0x02A0	
169		0x02A4	
170		0x02A8	
171		0x02AC	
172	TIMER CPUX6	0x02B0	
173	TIMER CPUX7	0x02B4	
174		0x02B8	
175	TPADC	0x02BC	
176	I2S0	0x02C0	
177	I2S1	0x02C4	
178	I2S2	0x02C8	
179		0x02CC	
180		0x02D0	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
181		0x02D4	
182	MCU_MSGBOX_R	0x02D8	
183	WDT_MCU	0x02DC	
184	TIMER_MCU0	0x02E0	
185	TIMER_MCU1	0x02E4	
186	TIMER_MCU2	0x02E8	
187	TIMER_MCU3	0x02EC	
188		0x02F0	
189		0x02F4	
190	NMI	0x02F8	
191		0x02FC	
192	R_TWD	0x0300	TWD interrupt
193		0x0304	
194		0x0308	
195		0x030C	
196		0x0310	
197		0x0314	
198	R_ALARM0	0x0318	RTC alarm interrupt
199		0x031C	
200		0x0320	
201		0x0324	
202		0x0328	
203		0x032C	
204		0x0330	
205		0x0334	
206		0x0338	
207		0x033C	
208		0x0340	
209		0x0344	
210		0x0348	
211	R_AHB_HREADY_TIME_OUT	0x034C	
212		0x0350	
213	PCK600_CPU	0x0354	
214	GPIO_NMI_S	0x0358	RTC_GPIOSecure IRQ
215	GPIO_NMI_NS	0x035C	RTC_GPIO Non-Secure IRQ
216		0x0360	
217		0x0364	
218		0x0368	
219		0x036C	
220		0x0370	
221		0x0374	

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Interrupt Number	Interrupt Sources	Interrupt Vector	Description
222	CPUX_MSGBOX_MCU_W	0x0378	
223	DMAC_IRQ1_NS	0x037C	
224	DMAC_IRQ1_S	0x0380	
225	DMAC1_IRQ1_NS	0x0384	
226	DMAC1_IRQ1_S	0x0388	
227		0x038C	
228		0x0390	
229		0x0394	
230		0x0398	
231		0x039C	
232		0x03A0	
233		0x03A4	
234		0x03A8	
235		0x03AC	
236		0x03B0	
237		0x03B4	
238		0x03B8	
239		0x03BC	
240		0x03C0	
241		0x03C4	
242		0x03C8	
243		0x03CC	
244		0x03D0	
245		0x03D4	
246		0x03D8	
247		0x03DC	
248		0x03E0	
249		0x03E4	
250		0x03E8	
251		0x03EC	
252		0x03F0	
253		0x03F4	
254		0x03F8	
255		0x03FC	

11.2.3 Register List

Module Name	Base Address
MCU_CLIC	0x30800000

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Register Name	Offset	Description
CLIC_CFG_REG	0x0000	CLIC Configuration Register
CLIC_INFO_REG	0x0004	CLIC Information Register
CLIC_MINTTHRESH_REG	0x0008	CLIC MINTTHRESH Register
CLIC_INT_REGn	0x1000+n*4	CLIC Interrupt Register n

11.2.4 Register description

For detailed register description, please refer to the section 8.2 in *XuanTie E907RIS3 User Manual*.

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12 DMA

12.1 DMA Controller (DMAC)

12.1.1 Overview

The direct memory access (DMA) is a method of transferring data between peripherals and memories (including the SRAM and DRAM) without using the CPU. The data can be transferred between a peripheral and a memory, between peripherals, or between memories through DMA. It is an efficient way to offload data transfer duties from the CPU.

The DMAC has the following features:

- Two DMA controllers:
  - DMAC0: 16 channels
  - DMAC1: 8 channels
- Independent Interrupt generating of each channel
- Supports channel and interrupt partition
  - DMA channels can be partitioned into maximum 2 groups
  - The number of channels in each group can be configured arbitrarily
  - The interrupt can be generated independently by each channel.
- Provides 64 peripheral DMA requests for data reading/writing
- Supports the following transfer mode:
  - Transfer data with linked list
  - IO type Transfer
  - Memory to memory, memory to peripheral, peripheral to memory, peripheral to peripheral
- Flexible data width: 8 bits, 16 bits, and 32 bits
- Programmable DMA burst length
- DRQ response includes waiting mode and handshake mode
- Supports non-aligned transform for memory devices
- DMA channels that support the following:
  - Pausing DMA
  - Byte mode and I/O speed mode
  - DMA timeout

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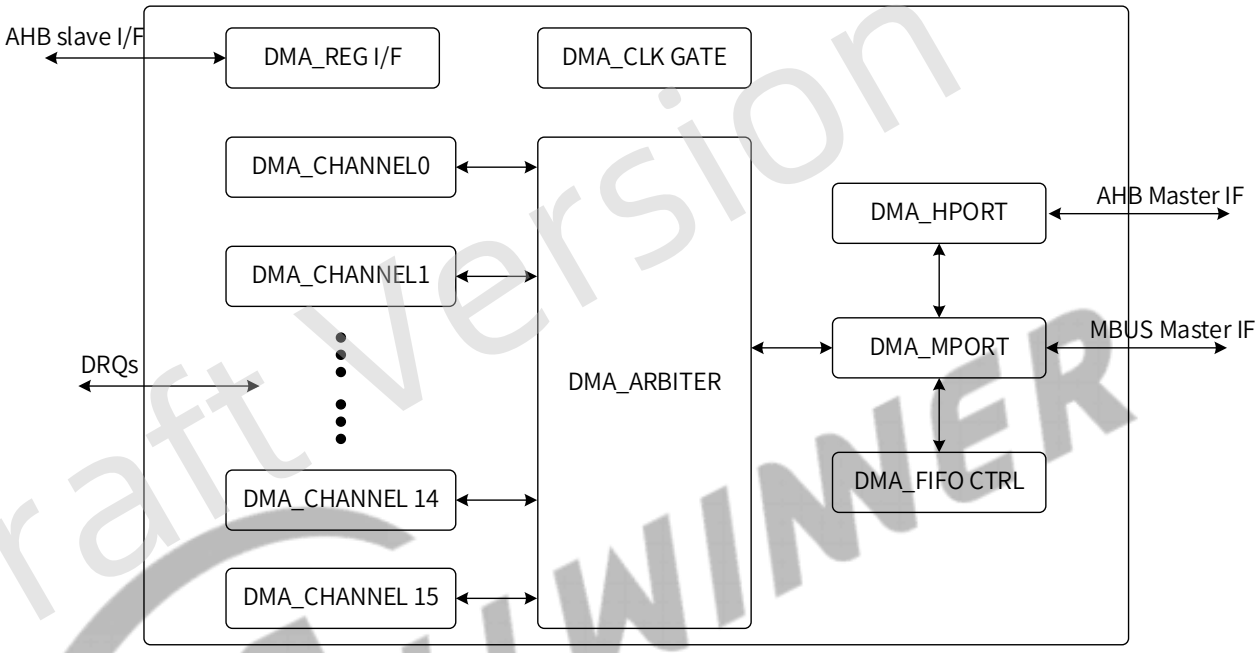
- Supports descriptor wait mode
- Supports unknown length mode

12.1.2 Block Diagram

The following figure shows a block diagram of DMAC.

Figure 12-1 DMAC Block Diagram



DMAC contains the following sub-blocks:

Table 12-1 DMAC Sub-blocks

Sub-block	Description
DMA_ARBITER	The port arbitrates the DMA read/write requests from all channels, and converts the requests to the read/write requests of ports.
DMA_CHANNELs	DMA transfer engine. Each channel is independent. The system uses the polling mechanism to decide the priorities of DMA channels. When the DMA requests from multiple peripherals are valid simultaneously, the channel with the higher priority starts data transfer first. When DMA_ARBITER is idle, channel 0 has the highest priority, whereas channel 15 has the lowest priority. When DMA_ARBITER is busy processing the request from channel n, channel (n+1) has the highest priority. For n = 15, the channel (n + 1) should be channel 0.
DRQs	DMA requests. Peripherals use the DMA request signals to request a data transfer.

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Sub-block	Description
DMA_MPORT	The port receives the read/write requests from DMA_ARBITER, and converts them to the corresponding MBUS access requests. It is mainly used for accessing the DRAM.
DMA_HPORT	The port for accessing the AHB Master. It is mainly used for accessing the SRAM and IO devices.
DMA_FIFO CTRL	Internal FIFO cell control module.
DMA_REG Interface	DMA_REG is the common register module that is mainly used to resolve AHB1 demands.
DMA_CLKGATE	The control module for hardware auto clock gating.

The DMAC integrates 16 independent DMA channels and each channel has an independent FIFO controller. When the DMA channel starts, the DMAC gets a DMA descriptor from the DMA_DESC_ADDR_REG and uses it as the configuration information for the data transfer of the current DMA package. Then the DMAC can transfer data between the specified devices. After transferring a DMA package, the DMAC judges whether the current channel transfer is finished via the linked address in the descriptor. If the linked address shows all the packages are transferred, the DMAC will end the chain transmission and close the channel.

12.1.3 Functional Description

12.1.3.1 DRQ Port Types

The following table shows the source DRQ types and destination DRQ types of different ports.

Table 12-2 DMA DRQ Types

Source DRQ Type		Destination DRQ Type	
port0	SRAM	port0	SRAM
port1	DRAM	port1	DRAM
port2	OWA0-RX	port2	OWA0-TX
port3	I2S0_RX	port3	I2S0_TX
port4	I2S1_RX	port4	I2S1_TX
port5	I2S2_RX	port5	I2S2_TX
port6		port6	
port7		port7	ADDA0
port8	DMIC0	port8	
port9		port9	
port10		port10	
port11		port11	
port12	GPADC0	Port12	
port13	GPADC1	port13	IR_TX
port14	UART0-RX	port14	UART0-TX
port15	UART1-RX	port15	UART1-TX
port16	UART2-RX	port16	UART2-TX

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Source DRQ Type		Destination DRQ Type	
port17	UART3-RX	port17	UART3-TX
port18	UART4-RX	port18	UART4-TX
port19	UART5-RX	port19	UART5-TX
port20	UART6-RX	Port20	UART6-TX
port21	UART7-RX	port21	UART7-TX
port22	SPI0-RX	port22	SPI0-TX
port23	SPI1-RX	port23	SPI1-TX
port24	SPI2-RX	port24	SPI2-TX
port25	SPI3-RX	port25	SPI3-TX
port26		port26	
port27		port27	
port28		port28	
port29		port29	
Port30	USB0_DEVICE_EP1	Port30	USB0_DEVICE_EP1
Port31	USB0_DEVICE_EP2	Port31	USB0_DEVICE_EP2
Port32	USB0_DEVICE_EP3	Port32	USB0_DEVICE_EP3
Port33	USB0_DEVICE_EP4	Port33	USB0_DEVICE_EP4
Port34	USB0_DEVICE_EP5	Port34	USB0_DEVICE_EP5
Port35	USB0_DEVICE_EP6	Port35	USB0_DEVICE_EP6
Port36	USB0_DEVICE_EP7	Port36	USB0_DEVICE_EP7
Port37	USB0_DEVICE_EP8	Port37	USB0_DEVICE_EP8
Port38		Port38	
Port39		Port39	
Port40	GPADC2	Port40	
Port41		Port41	
Port42	TPADC	Port42	LEDC
Port43	TWI0	Port43	TWI0
Port44	TWI1	Port44	TWI1
Port45	TWI2	Port45	TWI2
Port46	TWI3	Port46	TWI3
Port47	TWI4	Port47	TWI4
Port48	TWI5	Port48	TWI5
Port49		Port49	
Port50		Port50	
Port51		Port51	
Port52		Port52	
Port53		Port53	
Port54		Port54	
Port55	UART8-RX	Port55	UART8-TX
Port56	URAT9-RX	Port56	UART9-TX
Port57		Port57	PWMCS0
Port58		Port58	PWMCS1

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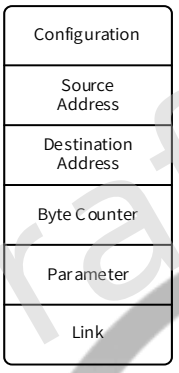
Source DRQ Type		Destination DRQ Type	
Port59		Port59	
Port60		Port60	
Port61		Port61	
Port62		Port62	
Port63	LBC_RX	Port63	LBC_TX

12.1.3.2 DMAC Descriptor

The DMAC descriptor is the configuration information of DMA transfer and decides the DMA working mode. Each descriptor includes 6 words: Configuration, Source Address, Destination Address, Byte Counter, Parameter, and Link.

Figure 12-2 DMA Descriptor



- **Configuration:** It is mainly used for configuring the following information by DMAC_CFG_REG.
  - **DRQ type**: DRQ type of the source and destination devices.
  - **Address counting mode**: For both the source and destination devices, there are two address counting modes: the IO mode and linear mode. The IO mode is for IO devices whose address is fixed during the data transfer and the linear mode is for the memory whose address is increasing during the data transfer.
  - **Transferred block length**: The amount of data that non-memory peripherals can transfer in a valid DRQ. The block length supports 1 beat, 4 beats, 8 beats, and 16 beats.
  - **Transferred data width**: The data width of operating the non-memory peripherals. The data width supports 8 bits, 16 bits, 32 bits, and 64 bits.
  - **BMODE mode:** It is used in the following scenario: the source is an IO device, and the destination is a memory device. In this mode, the data amount of DMA block transmission can be limited to that when the DRQ threshold of source IO device is 1.
- **Source Address**: It configures the address of the source device.
- **Destination Address**: It configures the address of the destination device. DMA reads data from the source address and then writes data to the destination address.

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- **Byte counter**: It configures the data amount of a package. The maximum value is (2^25-1) bytes. If the data amount of the package reaches the maximum value, even if DRQ is valid, the DMA will stop the current transfer.
- **Parameter**: It configures the interval of each data block. When detecting that the DRQ holds high, the DMA will start block transmission regardless of the DRQ change. When transmission ends, DMA will wait the set period and continue to the next DRQ detection.

  In addition, the Parameter is responsible for enabling and configuring TIMEOUT. In the case that the source device is an I/O device and the destination device is a memory device, the waiting time of a DRQ signal triggered by the source device can be set when TIMEOUT is enabled. When time is out, an interrupt signal of TIMEOUT will be generated by DMA. There are three sub-functions of TIMEOUT to be enabled (TIMEOUT will only generate interrupts if they are disabled): suspend the transmission of the current channel after an interrupt is generated; end the transmission of the current channel after an interrupt is generated; skip to the next descriptor for transmission after an interrupt is generated.

NOTE

- TIMEOUT only can be enabled in BMODE.
- Before using the ending and skipping function of TIMEOUT, enable the bit[4] of the DMAC_MODE_REG register first.

  The Parameter also configures whether the IOspeed is enabled or not. If IOspeed is enabled, DMA will transmit the data of the I/O device from the source device or the destination device, or both of them at a faster speed. The larger block indicates a faster speed. However, when the block is 1, the speed won't change a lot even if the IO speed is enabled.

- **Link**: If the value of the address bit in link and parameter is 0xFFFFF800 ({parameter [9], link} =0xFFFFF800), the current package is at the end of the linked list. The DMAC will stop the data transfer after transferring the package; otherwise, the value of the address bit in link and parameter is considered as the descriptor address of the next package.

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Figure 12-3 DMA Linked List Transfer

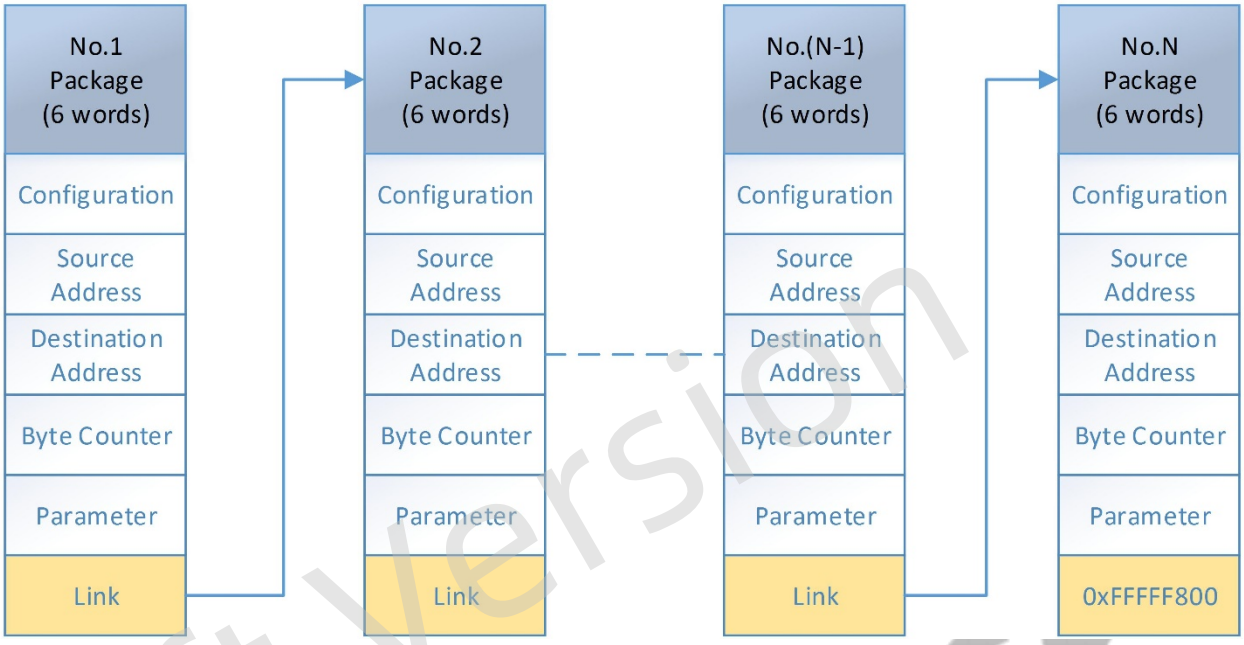

Table 12-3 DMA Address Descriptor

Descriptor Group	Bit	Description
Source Address	31:0	DMA transfers the lower 32 bits of the 35-bit source address
Destination Address	31:0	DMA transfers the lower 32 bits of the 35-bit destination address
Parameter	31	TIMEOUT Enable When using this function, BMODE or unknown length mode should be enabled, and IOspeed should be disabled.
	30:29	TIMEOUT Configuration 00: Not use sub-functions 01: Generate an interrupt and suspend the transmission after timeout. 10: Generate an interrupt and end the transmission after timeout. 11: Generate an interrupt and skip to the next descriptor for transmission after timeout.
	28:20	TIMEOUT Configuration Timer time of channels.
	19:18	Reserved
	17:15	DMA transfers the higher 3 bits of the 35-bit destination address.
	14	Reserved
	13:11	DMA transfers the high 3 bits of the 35-bit source address
	10	Reserved
	9	The highest bit of the word address in the next group of

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Descriptor Group	Bit	Description
		descriptor (word address bit [33])
	8	When DMA transfer the data of IO devices, enable this bit to obtain a higher speed. **Note: IOspeed and BMODE cannot be enabled at the same time.**
	7:0	Wait Clock Cycles Set the waiting time in DRQ mode
Link	31:2	The lower 30 bits of the word address in the next group of descriptor
	1:0	The higher 2 bits of the word address in the next group of descriptor

From the above table:

Real DMA source address (in byte mode) = {Parameter [13:11], Source Address [31:0]};

Real DMA destination address (in byte mode) = {Parameter [17:15], Destination Address [31:0]};

Real link address (in byte mode) = {parameter [9], Link [1:0], Link [31:2], 2'b00}.

12.1.3.3 Clock Gating

The DMA_CLK_GATE module can auto control clock gating by hardware, which is mainly used for generating clocks for DMAC sub-modules and some of their circuits.

The DMA_CLK_GATE module consists of two parts: the channel clock gate and the common clock gate.

Channel Clock Gate: When the system accesses the registers of the enabled DMA channels, the channel clock gate will be automatically open. When the system finishes accessing or the DMA data transfer is completed, the channel clock gate will be automatically closed after a 16-HCLK-cycle delay; also, the related circuit clocks, such as channel control and FIFO control, will be closed.

Common Clock Gate: When all the DMA channels are closed, the clock gates of common circuits will be automatically closed. The common circuits include the common circuits of FIFO control modules, MPORT modules, and MBUS.

Whether DMA clock gating can support all the functions stated above or not is decided by software.

12.1.3.4 Interrupts

There are four kinds of DMA interrupts: the half package interrupt, package end interrupt, queue end interrupt, and timeout interrupt.

- Half package interrupt

When enabled, the DMAC sends out a half package interrupt after transferring half of a package.

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- Package end interrupt

When enabled, the DMAC sends out a package end interrupt after transferring a complete package.

- Queue end interrupt

When enabled, the DMAC sends out a queue end interrupt after transferring a complete queue.

- Timeout interrupt

If TIMEOUT is enabled, DMA will generate a timeout interrupt after timeout. If the unknown length mode is enabled, a stop interrupt will be generated if a timeout interrupt isn't dealt before the package end interrupt of next descriptor is generated.

Notice that when CPU does not respond to the interrupts timely, or two DMA interrupts are generated very closely, the later interrupt may override the former one. As for CPU, the DMAC has only a system interrupt source.

12.1.3.5 Security

The DMAC supports system TrustZone and the security mode configuration of the DMA channel.

Each DMA channel is secure by default. When the TrustZone is enabled, the DMAC is secure, only the secure devices can access the DMAC.

When a DMA channel is configured to the non-secure mode, this channel can only access the non-secure area of the memory, that is, the DMAC cannot write data to the secure area of the memory, and the read-back data from the secure area of the memory is 0.

12.1.3.6 Transfer Mode

The peripherals initiate data transfer by transmitting DMA request signals to the DMAC. After receiving the request signal, the DMAC converts it to the internal DRQ signal and controls the DMA data transfer.

The DMAC supports two data transfer modes: the waiting mode and handshake mode.

Waiting Mode

a) When the DMAC detects a valid external request signal, the DMAC starts to operate the peripheral device. The internal DRQ always holds high before the transferred data amount reaches the transferred block length.

b) When the transferred data amount reaches the transferred block length, the internal DRQ will pull low automatically.

c) The internal DRQ holds low for certain clock cycles (WAIT_CYC), and then the DMAC restarts to detect the external requests. If the external request signal is valid, then the next transfer starts.

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Handshake Mode

a) When the DMAC detects a valid external request signal, the DMAC starts to operate the peripheral device. The internal DRQ always holds high before the transferred data amount reaches the transferred block length.

b) When the transferred data amount reaches the transferred block length, the internal DRQ will be pulled down automatically. For the last data transfer of the block, the DMAC sends a DMA Last signal with the DMA commands to the peripheral device. The DMA Last signal will be packed as part of the DMA commands and transmitted on the bus. It is used to inform the peripheral device that it is the end of the data transfer for the current DRQ.

c) When the peripheral device receives the DMA Last signal, it can judge that the data transfer for the current DRQ is finished. To continue the data transfer, it sends a DMA Active signal to the DMAC.

NOTE

A DMA Active signal will be converted to one DRQ signal in the DMA module. To generate multiple DRQs, the peripheral devices need to send out multiple DMA Active signals via the bus protocol.

d) When the DMAC receives the DMA Active signal, it will send back a DMA ACK signal to the peripheral device.

e) When the peripheral device receives the DMA ACK signal, it will wait for all the operations on the local device to be completed, and both the FIFO and DRQ status are refreshed. Then it invalidates the DMA Active signal.

f) When the DMAC detects the falling edge of the DMA Active signal, it invalidates the corresponding DMA ACK signal, and restarts to detect the external request signals. If a valid request signal is detected, the next data transfer starts.

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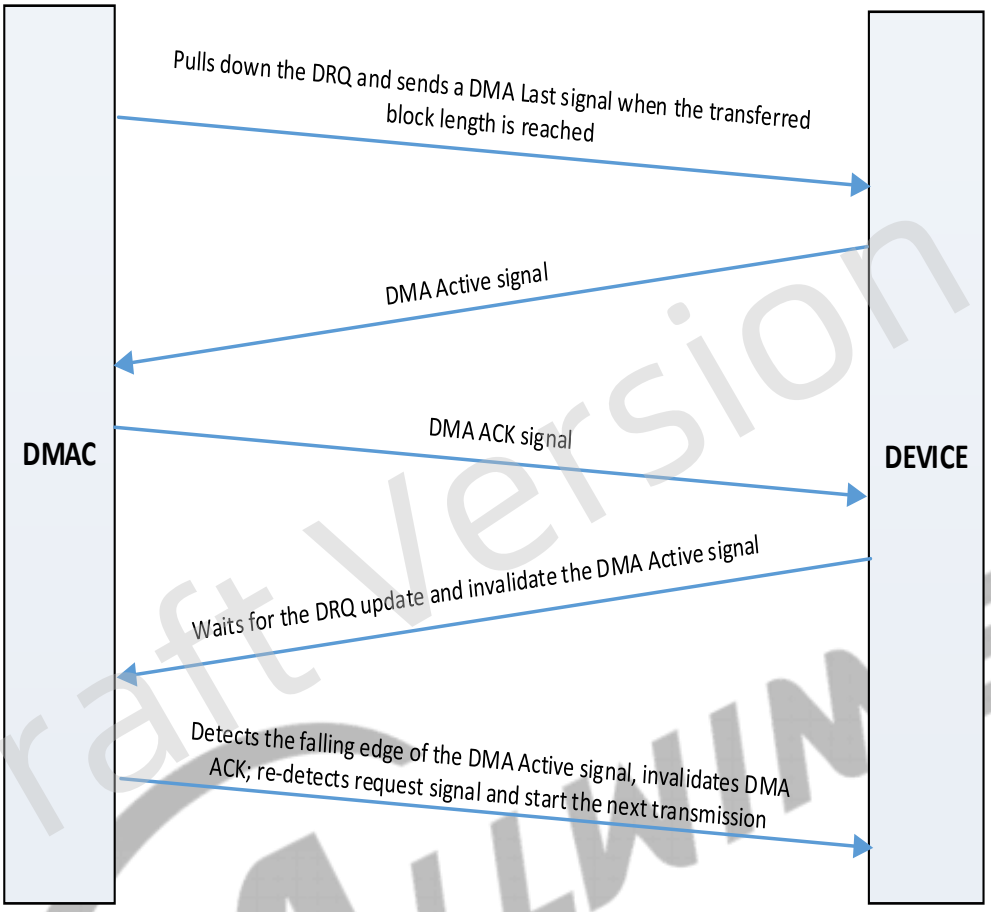
The following figure shows the workflow of the handshake mode.

Figure 12-4 Workflow of the DMAC Handshake Mode



12.1.3.7 Address Auto-Alignment

For the non-IO devices whose start address is not 32-byte-aligned, the DMAC will align the address to the 32-byte through the burst transfer within 32 bytes, which will improve the DRAM access efficiency.

The following example shows how the DMAC adjusts the address: when the peripheral device of a DMA channel is a non-IO device whose start address is 0x86 (not 32-byte-aligned), the DMAC firstly uses a 26-byte burst transfer to align the address to 0xA0 (32-byte-aligned), and then transfers data by 64-byte burst (the maximum transfer amount that MBUS allows).

The IO devices do not support address alignment, so the bit width of IO devices must match the address offset; otherwise, the DMAC will ignore the inconsistency and directly transmit data of the corresponding bit width to the address.

The address of the DMA descriptor does not support the address auto-alignment. Make sure the address is word-aligned; otherwise the DMAC cannot identify the descriptor.

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12.1.3.8 Unknown Length Mode

When the source device is an IO device and the destination device is a non-IO device, DMAC supports enabling unknown length mode if the data length of source device is unknown.

In the unknown length mode, DMAC continues transferring based on the descriptor configurations and performs TIMEOUT detection on the DRQ of IO devices. If TIMEOUT occurs as IO devices don't trigger DRQ for a long period, it will be considered as descriptor transmission completion. At this time, DMAC will write the data cached in the DMAC FIFO to non-IO devices, and then end the current descriptor transmission and enable TIMEOUT Interrupt and Package end interrupt. In this scene, you can end the chained transmission by closing channels or continue to transfer through next descriptor.

Before next descriptor transmission completes, CPU needs to access DMAC_UNKNOWN_LENGTH_STAGE_REG to clear TIMEOUT_REMAIN_FLAG. Then, you can obtain the data left in the DMAC through MA_REMAIN_LEFT. Or, the STOP interrupt will be triggered and the channel transmission will be disabled when next descriptor transmission, whether normal transmission or TIMEOUT transmission, completes.

12.1.3.9 Channel and Interrupt Partition

NOTE

For more function description of multi operating system, see section 17.9.3.8 GPIO Partition.

- Channel Partition

In a multi-master system, there may be scenarios where multiple masters share a single DMA, in which case DMA channels need to be respectively assigned to different OS. One master with channel ownership can configure the corresponding channel to initiate a DMA transfer. If the initiator is a master without ownership, the channel will not respond and will not transmit.

The following is an example. Channel0 to channel7 are assigned to OS0 and channel8 to channel15 are assigned to OS1 (DMA channels can be arbitrarily assigned to OS0 or OS1 during actual configuration). In this case, OS0 only responds transmission configuration requests from channel0 to channel7. After transfer is finished, interrupts will only return to OS0.

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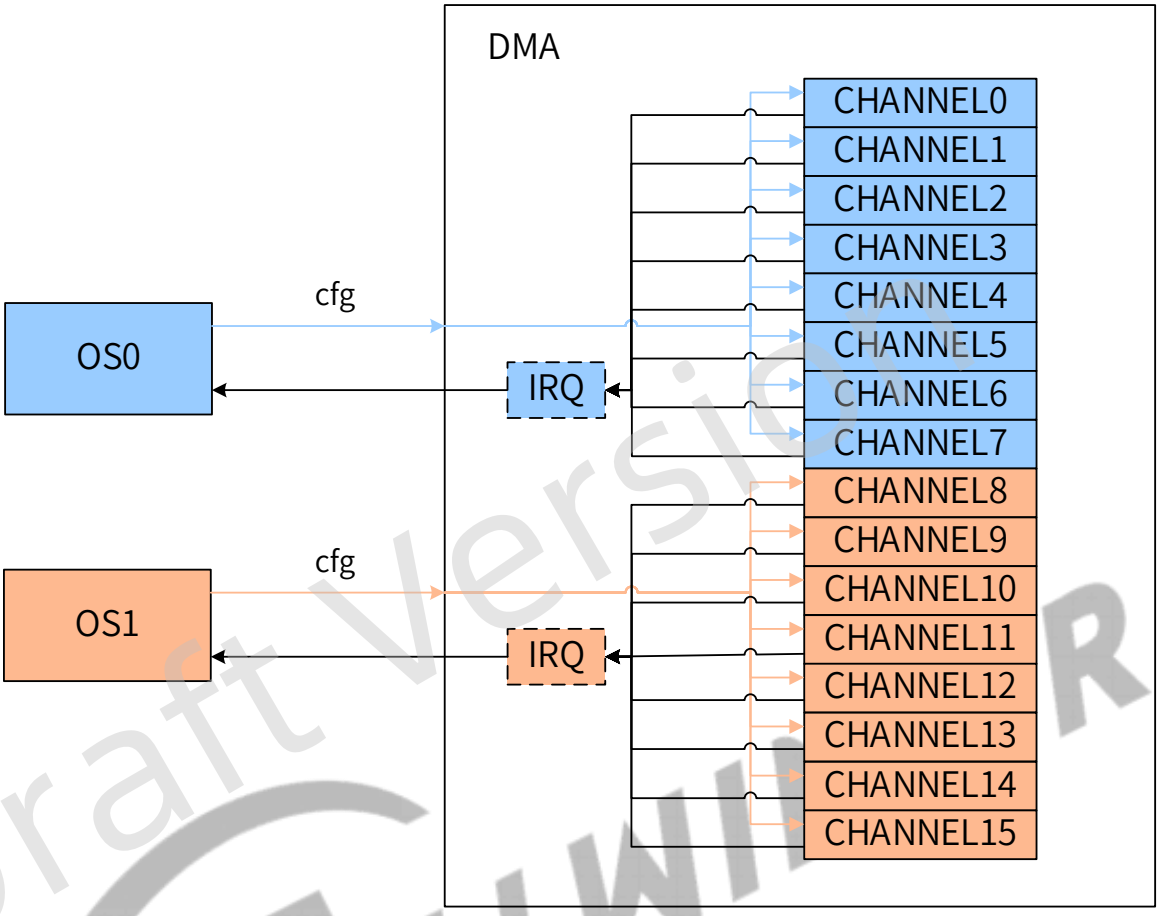
Figure 12-5 DMA Channel Partition



- **Interrupt Partition**

In a multi-master system, when a DMA transfer is completed, the interrupt is returned only to the master with channel ownership (because only the master with channel ownership can initiate the transfer and generate the interrupt).

For all four cores of the CPUX, the DMA can mask the master ID through the MASK mechanism, so that DMA will eventually see only one master ID.

When channel and interrupt partition function is disabled, all masters can access any channel, and interrupts generated during DMA transfers will be returned to all masters at the same time, requiring the DMA peripheral interrupt control module to mask the interrupts.

12.1.4 Programming Guidelines

12.1.4.1 DMAC Transfer Procedure

The DMAC transfer procedure is as follows.

Step 1 Assign DRQ signals to DMAC0/1

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Step 2 Configure DMAC_CHANNEL_MASTER_CFG_REG, DMA_CHAMMEL_MASK_CFG*_REG, DMA_MASTER_ID_REG to set channel and interrupt partition, and select the ownership of DMA channels. See section 12.1.4.6 Configure Channel and Interrupt Partition.

Step 3 Request DMA channel, and check if the DMA channel is idle by checking if it is enabled. A disabled channel indicates it is idle, while an enabled channel indicates it is busy.

Step 4 Write the descriptor with 6 words into the memory. The descriptor must be word-aligned. For more details, refer to section 12.1.3.2 DMAC Descriptor.

Step 5 Write the start address of the descriptor to DMAC_DESC_ADDR_REG and DMAC_DESC_HIG_ADDR_REG.

Step 6 Enable the DMA channel, and write the corresponding channel to DMAC_EN_REG.

Step 7 The DMA obtains the descriptor information.

Step 8 Start to transmit a package. When half of the package is completed, the DMA sends a Half Package Transfer Interrupt; when a total package is completed, the DMA sends a Package End Transfer Interrupt. This interrupt status can be read by DMAC_IRQ_PEND_REG.

Step 9 Set DMAC_PAU_REG to pause or resume the data transfer.

Step 10 After completing a total package transfer, the DMA decides to start the next package transfer or end the transfer by the link of the descriptor. If the link is 0xFFFFF800, the transfer ends; otherwise, the next package starts to transmit. When the transfer ends, the DMA sends a Queue End Transfer Interrupt.

Step 11 Disable the DMA channel.

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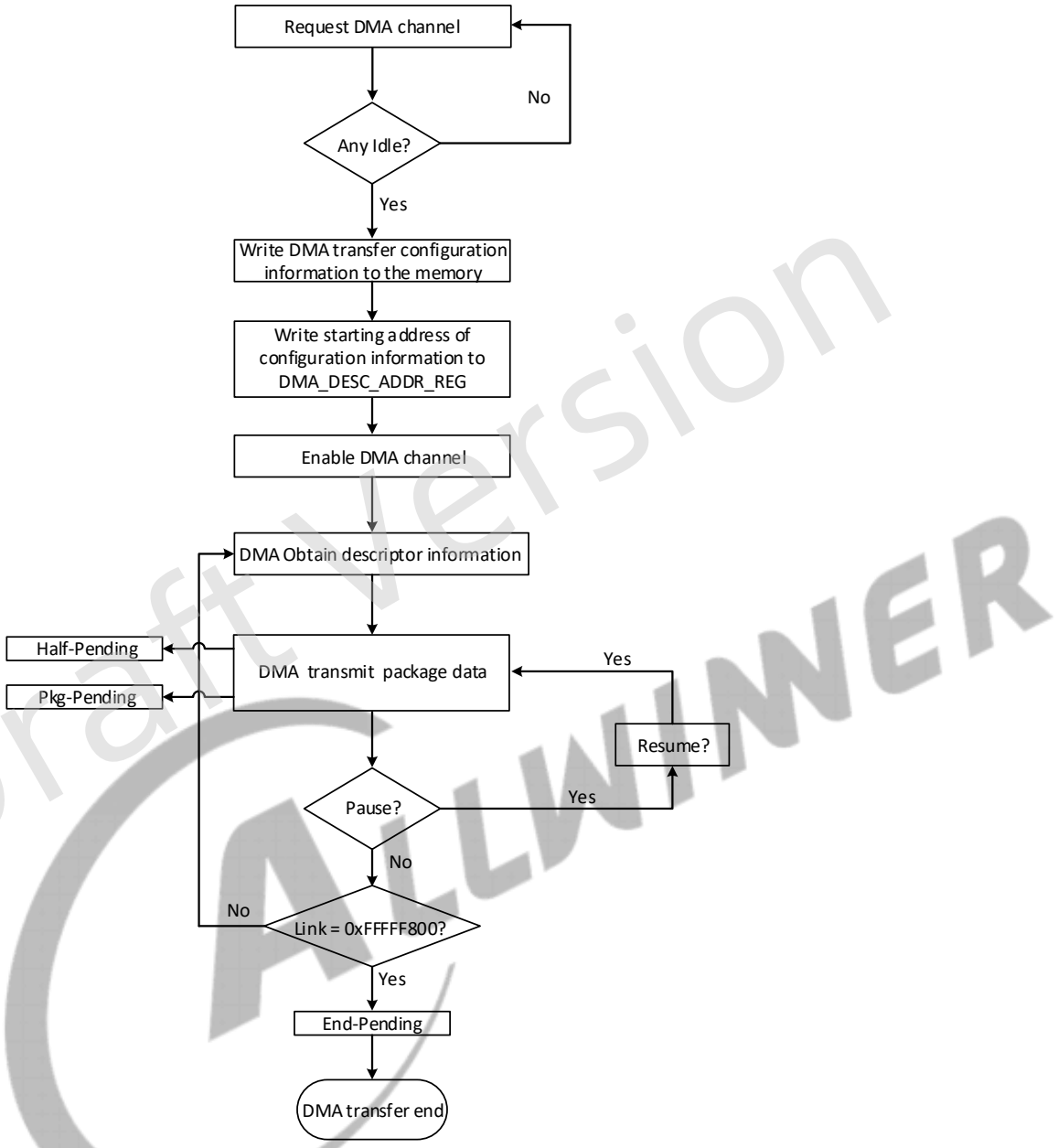
Figure 12-6 DMAC Transfer Process



12.1.4.2 DMAC Clock Control

If the DMAC clock is synchronous with the AHB clock, enable the DMAC gating bit of AHB clock before accessing the DMAC register.

If the reset input signal of the DMAC is asynchronous with AHB0 and is low valid by default, de-assert the reset signal of the DMAC before accessing the DMA register.

To avoid the indefinite state within registers, de-assert the reset signal first, and then enable the gating bit of AHB0.

As the DMAC supports Clock Auto Gating function, it will automatically disable the DMAC clock in the idle state to reduce power consumption. The Clock Auto Gating is enabled by default.

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12.1.4.3 Interrupt Processing

Follow the steps below to process the DMAC interrupt:

Step 1 Enable interrupt: write the corresponding interrupt enable bit of DMAC_IRQ_EN_REG. The system generates an interrupt when the corresponding condition is satisfied.

Step 2 After entering the interrupt process, write to clear the interrupt pending and execute the process of waiting for the interrupt.

Step 3 Resume the interrupt and continue to execute the interrupted process.

12.1.4.4 Configure Descriptor Wait Mode

Step 1 Configure DESC_WAIT_MODE as 1 before transmission.

Step 2 After the DMA starts working, if the write-back bit of the descriptor read by the DMA in a certain transfer is 1 (that is, descriptor [31] =1), it means that the descriptor read has not been updated by the software. At this time, DMA will enter descriptor wait mode. In this mode, the DMA will suspend work and generate a descriptor wait interrupt, and the status bit DESC_WAIT_STA=1.

Step 3 After DMA enter descriptor wait mode, The software reads the DMAC_FDESC_ADDR_REG register to obtain which descriptor has not been updated.

Step 4 When the data to be transmitted is ready, the software updates the descriptor and configures descriptor[31] to 0, and then writes the address storing the descriptor location into the DMAC_DESC_ADDR_REG register and the DMAC_DESC_HIG_ADDR_REG register.

Step 5 The software writes 1 to the DESC_READY bit, which indicates that the DMA descriptor has been updated.

Step 6 DMA reads the descriptor again through the address stored in DMAC_DESC_ADDR_REG and DMAC_DESC_HIG_ADDR_REG, and starts a new transfer.

12.1.4.5 Configure Unknown Length Mode

Step 1 Write 1 to the bit [6] of DMA_MODE_REG to enable the unknown length mode.

Step 2 Enable TIMEOUT function and disable BMODE function in descriptor.

Step 3 DMA transfer

- If TIMEOUT is triggered during DMA transfer, access DMA_UNKNOWN_LENGTH_STAGE_REG before next PKG interrupt is pending and read the value of DMA_REMAIN_LEFT, which is the remain data of the last descriptor. Then, read DMA_TIMEOUT_REMAIN_FLAG

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- If ARM CPU or RISC-V CPU does not respond TIMEOUT interrupt, STOP interrupt will be pulled high while the second PKG interrupt is triggered, and DMA channel will be disabled. In this case, the value of DMA_REMAIN_LEFT is the remain data of the last descriptor, and the value of DMA_BCNT_LEFT in DMAC_BCNT_LEFT_REG is the remain data of current descriptor.

12.1.4.6 Configure Channel and Interrupt Partition

Channel and interrupt partition function is disabled by default.

- **To enable channel and interrupt partition function**

Step 1 Configure DMAC_MASTER_ID_REG. The master ID is 0x00 (00000 in binary) for ARM CPU and is 0x09 (01001 in binary) for RISC-V CPU.

Step 2 Configure DMAC_CHANNEL_MASTER_CFG_REG according to the usage scenarios to assign DMA channels to masters.

Step 3 Configure DMA_CHANNEL_MASK_CONFIG0/1/2/3_REG.

When the master is ARM CPU, write 0x1C (11100 in binary) to registers above.

When the master is RISC-V CPU, write 0x1F (11111 in binary) to registers above.

These registers above need to be configured before the DMA channel configuration registers are configured, otherwise by default all DMA channels can be accessed by any master and initiate transfers.

- **To disable channel and interrupt partition function**

Write 0 to DMAC_CHANNEL_MASTER_CFG_RE, DMAC_CHANNEL_MASK_CONFIG0/1/2/3_REG, and DMAC_MASTER_ID_REG.

12.1.5 Register List

Module Name	Base Address
DMAC0	0x03001000
DMAC1	0x04000000

Register Name	Offset	Description
DMAC_SEC_REG	0x0020	DMAC Security Register
DMAC_AUTO_GATE_REG	0x0028	DMAC Auto Gating Register
DMAC_STA_REG	0x0030	DMAC Status Register
DMAC_CHANNEL_MASTER_CFG_REG	0x0070	DMAC Channel Master Configure Register
DMAC_CHANNEL_MASK_CFG0_REG	0x0074	DMAC Channel Mask Configure0 Register

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Register Name	Offset	Description
DMAC_CHANNEL_MASK_CFG1_REG	0x0078	DMAC Channel Mask Configure1 Register
DMAC_CHANNEL_MASK_CFG2_REG	0x007C	DMAC Channel Mask Configure2 Register
DMAC_CHANNEL_MASK_CFG3_REG	0x0080	DMAC Channel Mask Configure3 Register
DMAC_MASTER_ID_REG	0x0084	DMAC Master ID Register
DMAC_EN_REG	0x0100+0x0000+N*0x0040 (N=0-15)	DMAC Channel Enable Register
DMAC_PAU_REG	0x0100+0x0004+N*0x0040 (N=0-15)	DMAC Channel Pause Register
DMAC_DESC_ADDR_REG	0x0100+0x0008+N*0x0040 (N=0-15)	DMAC Channel Descriptor Address Register
DMAC_CFG_REG	0x0100+0x000C+N*0x0040 (N=0-15)	DMAC Channel Configuration Register
DMAC_CUR_SRC_REG	0x0100+0x0010+N*0x0040 (N=0-15)	DMAC Channel Current Source Address Register
DMAC_CUR_DEST_REG	0x0100+0x0014+N*0x0040 (N=0-15)	DMAC Channel Current Destination Address Register
DMAC_BCNT_LEFT_REG	0x0100+0x0018+N*0x0040 (N=0-15)	DMAC Channel Byte Counter Left Register
DMAC_PARA_REG	0x0100+0x001C+N*0x0040 (N=0-15)	DMAC Channel Parameter Register
DMAC_UNKNOWN_LENGTH_STAGE_REG	0x0100+0x0024+N*0x0040 (N=0-15)	DMAC Unknown Length Stage Register
DMAC_MODE_REG	0x0100+0x0028+N*0x0040 (N=0-15)	DMAC Mode Register
DMAC_FDESC_ADDR_REG	0x0100+0x002C+N*0x0040 (N=0-15)	DMAC Former Descriptor Address Register
DMAC_PKG_NUM_REG	0x0100+0x0030+N*0x0040 (N=0-15)	DMAC Package Number Register
DMAC_IRQ_EN_REG	0x0100+0x0034+N*0x0040 (N=0-15)	DMAC IRQ Enable Register
DMAC_IRQ_PEND_REG	0x0100+0x0038+N*0x0040 (N=0-15)	DMAC IRQ Pending Status Register
DMAC_DESC_HIG_ADDR_REG	0x0100+0x003C+N*0x0040 (N=0-15)	DMAC Channel Descriptor Highest Address Register

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12.1.6 Register Description

12.1.6.1 0x0020 DMAC Security Register (Default Value: 0x0000_FFFF)

Offset: 0x0020			Register Name: DMAC_SEC_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x1	DMA15_SEC DMA Channel 15 Security 0: Secure 1: Non-secure
14	R/W	0x1	DMA14_SEC DMA Channel 14 Security 0: Secure 1: Non-secure
13	R/W	0x1	DMA13_SEC DMA Channel 13 Security 0: Secure 1: Non-secure
12	R/W	0x1	DMA12_SEC DMA Channel 12 Security 0: Secure 1: Non-secure
11	R/W	0x1	DMA11_SEC DMA Channel 11 Security 0: Secure 1: Non-secure
10	R/W	0x1	DMA10_SEC DMA Channel 10 Security 0: Secure 1: Non-secure
9	R/W	0x1	DMA9_SEC DMA Channel 9 Security 0: Secure 1: Non-secure
8	R/W	0x1	DMA8_SEC DMA Channel 8 Security 0: Secure 1: Non-secure
7	R/W	0x1	DMA7_SEC DMA Channel 7 Security 0: Secure 1: Non-secure

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Offset: 0x0020			Register Name: DMAC_SEC_REG
Bit	Read/Write	Default/Hex	Description
6	R/W	0x1	DMA6_SEC DMA Channel 6 Security 0: Secure 1: Non-secure
5	R/W	0x1	DMA5_SEC DMA Channel 5 Security 0: Secure 1: Non-secure
4	R/W	0x1	DMA4_SEC DMA Channel 4 Security 0: Secure 1: Non-secure
3	R/W	0x1	DMA3_SEC DMA Channel 3 Security 0: Secure 1: Non-secure
2	R/W	0x1	DMA2_SEC DMA Channel 2 Security 0: Secure 1: Non-secure
1	R/W	0x1	DMA1_SEC DMA Channel 1 Security 0: Secure 1: Non-secure
0	R/W	0x1	DMA0_SEC DMA Channel 0 Security 0: Secure 1: Non-secure

12.1.6.2 0x0028 DMAC Auto Gating Register (Default Value: 0x00000000)

NOTE

When initializing DMA Controller, bit-2 should be set up to enable auto gating.

Offset: 0x0028			Register Name: DMAC_AUTO_GATE_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/

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Offset: 0x0028			Register Name: DMAC_AUTO_GATE_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	DMA_MCLK_CIRCUIT DMA MCLK interface circuit auto gating bit 0: Auto gating enable 1: Auto gating disable
1	R/W	0x0	DMA_COMMON_CIRCUIT DMA common circuit auto gating bit 0: Auto gating enable 1: Auto gating disable
0	R/W	0x0	DMA_CHAN_CIRCUIT DMA channel circuit auto gating bit 0: Auto gating enable 1: Auto gating disable

12.1.6.3 0x0030 DMAC Status Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: DMAC_STA_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	MBUS_FIFO_STATUS MBUS FIFO Status 0: Empty 1: Not Empty
30:16	/	/	/
15	R	0x0	DMA15_STATUS DMA Channel 15 Status 0: Idle 1: Busy
14	R	0x0	DMA14_STATUS DMA Channel 14 Status 0: Idle 1: Busy
13	R	0x0	DMA13_STATUS DMA Channel 13 Status 0: Idle 1: Busy
12	R	0x0	DMA12_STATUS DMA Channel 12 Status 0: Idle 1: Busy

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Offset: 0x0030			Register Name: DMAC_STA_REG
Bit	Read/Write	Default/Hex	Description
11	R	0x0	DMA11_STATUS DMA Channel 11 Status 0: Idle 1: Busy
10	R	0x0	DMA10_STATUS DMA Channel 10 Status 0: Idle 1: Busy
9	R	0x0	DMA9_STATUS DMA Channel 9 Status 0: Idle 1: Busy
8	R	0x0	DMA8_STATUS DMA Channel 8 Status 0: Idle 1: Busy
7	R	0x0	DMA7_STATUS DMA Channel 7 Status 0: Idle 1: Busy
6	R	0x0	DMA6_STATUS DMA Channel 6 Status 0: Idle 1: Busy
5	R	0x0	DMA5_STATUS DMA Channel 5 Status 0: Idle 1: Busy
4	R	0x0	DMA4_STATUS DMA Channel 4 Status 0: Idle 1: Busy
3	R	0x0	DMA3_STATUS DMA Channel 3 Status 0: Idle 1: Busy
2	R	0x0	DMA2_STATUS DMA Channel 2 Status 0: Idle 1: Busy

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Offset: 0x0030			Register Name: DMAC_STA_REG
Bit	Read/Write	Default/Hex	Description
1	R	0x0	DMA1_STATUS DMA Channel 1 Status 0: Idle 1: Busy
0	R	0x0	DMA0_STATUS DMA Channel 0 Status 0: Idle 1: Busy

12.1.6.4 0x0070 DMAC Channel Master Configure Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x0070			Register Name: DMAC_CHANNEL_MASTER_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	DMA_CHANNEL15_MASTER DMA Channel15 Master Register 00: Master0 01: Master1
29:28	R/W	0x0	DMA_CHANNEL14_MASTER DMA Channel14 Master Register 00: Master0 01: Master1
27:26	R/W	0x0	DMA_CHANNEL13_MASTER DMA Channel13 Master Register 00: Master0 01: Master1
25:24	R/W	0x0	DMA_CHANNEL12_MASTER DMA Channel12 Master Register 00: Master0 01: Master1
23:22	R/W	0x0	DMA_CHANNEL11_MASTER DMA Channel11 Master Register 00: Master0 01: Master1

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Offset:0x0070			Register Name: DMAC_CHANNEL_MASTER_CFG_REG
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x0	DMA_CHANNEL10_MASTER DMA Channel10 Master Register 00: Master0 01: Master1
19:18	R/W	0x0	DMA_CHANNEL9_MASTER DMA Channel9 Master Register 00: Master0 01: Master1
17:16	R/W	0x0	DMA_CHANNEL8_MASTER DMA Channel8 Master Register 00: Master0 01: Master1
15:14	R/W	0x0	DMA_CHANNEL7_MASTER DMA Channel7 Master Register 00: Master0 01: Master1
13:12	R/W	0x0	DMA_CHANNEL6_MASTER DMA Channel6 Master Register 00: Master0 01: Master1
11:10	R/W	0x0	DMA_CHANNEL5_MASTER DMA Channel5 Master Register 00: Master0 01: Master1
9:8	R/W	0x0	DMA_CHANNEL4_MASTER DMA Channel4 Master Register 00: Master0 01: Master1
7:6	R/W	0x0	DMA_CHANNEL3_MASTER DMA Channel3 Master Register 00: Master0 01: Master1
5:4	R/W	0x0	DMA_CHANNEL2_MASTER DMA Channel2 Master Register 00: Master0 01: Master1
3:2	R/W	0x0	DMA_CHANNEL1_MASTER DMA Channel1 Master Register 00: Master0 01: Master1

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Offset:0x0070			Register Name: DMAC_CHANNEL_MASTER_CFG_REG
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	DMA_CHANNEL0_MASTER DMA Channel0 Master Register 00: Master0 01: Master1

12.1.6.5 0x0074 DMAC Channel Mask Configure0 Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x0074			Register Name: DMAC_CHANNEL_MASK_CFG0_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	DMA_CHANNEL3_MASK DMA Channel3 Mask Register
23:21	/	/	/
20:16	R/W	0x0	DMA_CHANNEL2_MASK DMA Channel2 Mask Register
15:13	/	/	/
12:8	R/W	0x0	DMA_CHANNEL1_MASK DMA Channel1 Mask Register
7:5	/	/	/
4:0	R/W	0x0	DMA_CHANNEL0_MASK DMA Channel0 Mask Register

12.1.6.6 0x0078 DMAC Channel Mask Configure1 Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x0078			Register Name: DMAC_CHANNEL_MASK_CFG1_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/

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Offset:0x0078			Register Name: DMAC_CHANNEL_MASK_CFG1_REG
Bit	Read/Write	Default/Hex	Description
28:24	R/W	0x0	DMA_CHANNEL7_MASK DMA Channel7 Mask Register
23:21	/	/	/
20:16	R/W	0x0	DMA_CHANNEL6_MASK DMA Channel6 Mask Register
15:13	/	/	/
12:8	R/W	0x0	DMA_CHANNEL5_MASK DMA Channel5 Mask Register
7:5	/	/	/
4:0	R/W	0x0	DMA_CHANNEL4_MASK DMA Channel4 Mask Register

12.1.6.7 0x007C DMAC Channel Mask Configure2 Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x007C			Register Name: DMAC_CHANNEL_MASK_CFG2_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	DMA_CHANNEL11_MASK DMA Channel11 Mask Register
23:21	/	/	/
20:16	R/W	0x0	DMA_CHANNEL10_MASK DMA Channel10 Mask Register
15:13	/	/	/
12:8	R/W	0x0	DMA_CHANNEL9_MASK DMA Channel9 Mask Register
7:5	/	/	/
4:0	R/W	0x0	DMA_CHANNEL8_MASK DMA Channel8 Mask Register

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12.1.6.8 0x0080 DMAC Channel Mask Configure3 Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x0080			Register Name: DMAC_CHANNEL_MASK_CFG3_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	DMA_CHANNEL15_MASK DMA Channel15 Mask Register
23:21	/	/	/
20:16	R/W	0x0	DMA_CHANNEL14_MASK DMA Channel14 Mask Register
15:13	/	/	/
12:8	R/W	0x0	DMA_CHANNEL13_MASK DMA Channel135 Mask Register
7:5	/	/	/
4:0	R/W	0x0	DMA_CHANNEL12_MASK DMA Channel12 Mask Register

12.1.6.9 0x0084 DMAC Master ID Register (Default Value: 0x0000_0000)

NOTE

This register is only configured by secure master.

Offset:0x0084			Register Name: DMAC_MASTER_ID_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:8	R/W	0x00000	DMA _MASTERID_1 DMA Master ID 1 Register (RISC-V CPU)
7:5	/	/	/
4:0	R/W	0x00000	DMA_MASTERID_0 DMA Master ID 0 Register (ARM CPU)

12.1.6.10 0x0100+0x0000+N*0x0040 (N=0-15) DMAC Channel Enable Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0000+N*0x0040 (N=0-15)	Register Name: DMAC_EN_REG

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	DMA_EN DMA Channel Enable 0: Disable 1: Enable

12.1.6.11 0x0100+0x0004+N*0x0040 (N=0-15) DMAC Channel Pause Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0004+N*0x0040 (N=0-15)			Register Name: DMAC_PAU_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	DMA_PAUSE Pausing DMA Channel Transfer Data 0: Resume Transferring 1: Pause Transferring

12.1.6.12 0x0100+0x0008+N*0x0040 (N=0-15) DMAC Channel Descriptor Address Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0008+N*0x0040 (N=0-15)			Register Name: DMAC_DESC_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:2	R/W	0x0	DMA_DESC_LOW_ADDR DMA Channel Descriptor Word Address, Low 30bits. The Descriptor Address must be word-aligned.
1:0	R/W	0x0	DMA_DESC_HIGH_ADDR DMA Channel Descriptor High Address, High 2bits The highest bit is the bit [0] of the DMA_DESC_HIG_ADDR_REG register. The DMA descriptor address should be configured as {DMA_DESC_HIG_ADDR_REG[0], bit[1:0], bit[31:2], 2'b00}.

12.1.6.13 0x0100+0x000C+N*0x0040 (N=0-15) DMAC Channel Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x000C+N*0x0040 (N=0-15)			Register Name: DMAC_CFG_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0100+0x000C+N*0x0040 (N=0-15)			Register Name: DMAC_CFG_REG
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30	R	0x0	BMODE_SEL Mode select 0: Normal Mode 1: BMODE
29:27	/	/	/
26:25	R	0x0	DMA_DEST_DATA_WIDTH DMA Destination Data Width 00: 8-bit 01: 16-bit 10: 32-bit 11: 64-bit
24	R	0x0	DMA_ADDR_MODE DMA Destination Address Mode 0: Linear Mode 1: IO Mode
23:22	R	0x0	DMA_DEST_BLOCK_SIZE DMA Destination Block Size 00: 1 01: 4 10: 8 11: 16
21:16	R	0x0	DMA_DEST_DRQ_TYPE DMA Destination DRQ Type The details in DRQ Type and Port Corresponding Relation
15:11	/	/	/
10:9	R	0x0	DMA_SRC_DATA_WIDTH DMA Source Data Width 00: 8-bit 01: 16-bit 10: 32-bit 11: 64-bit
8	R	0x0	DMA_SRC_ADDR_MODE DMA Source Address Mode 0: Linear Mode 1: IO Mode

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Offset: 0x0100+0x000C+N*0x0040 (N=0-15)			Register Name: DMAC_CFG_REG
Bit	Read/Write	Default/Hex	Description
7:6	R	0x0	DMA_SRC_BLOCK_SIZE DMA Source Block Size 00: 1 01: 4 10: 8 11: 16
5:0	R	0x0	DMA_SRC_DRQ_TYPE DMA Source DRQ Type The details in DRQ Type and Port Corresponding Relation.

12.1.6.14 0x0100+0x0010+N*0x0040 (N=0-15) DMAC Channel Current Source Address Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0010+N*0x0040 (N=0-15)			Register Name: DMAC_CUR_SRC_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	DMA_CUR_SRC DMA Channel Current Source Address, read only.

12.1.6.15 0x0100+0x0014+N*0x0040 (N=0-15) DMAC Channel Current Destination Address Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0014+N*0x0040 (N=0-15)			Register Name: DMAC_CUR_DEST_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	DMA_CUR_DEST DMA Channel Current Destination Address, read only.

12.1.6.16 0x0100+0x0018+N*0x0040 (N=0-15) DMAC Channel Byte Counter Left Register (Default Value: 0x0000_0000)

NOTE

The data are not real-time. It will arrive the destination address after delaying for some time.

Offset: 0x0100+0x0018+N*0x0040 (N=0-15)	Register Name: DMAC_BCNT_LEFT_REG

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Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:0	R	0x0	DMA_BCNT_LEFT DMA Channel Byte Counter Left, read only.

12.1.6.17 0x0100+0x001C+N*0x0040 (N=0-15) DMAC Channel Parameter Register (Default Value: 0x0000_0000)

NOTE

- In the bits 20-28, Time cycle=Timing duration*clock frequency (AHB clock frequency)/4096. (The time cycle should be integer)
- When the clock frequency is 200MHz, the timing step will be 20.48us (10.46ms at max).

Offset: 0x0100+0x001C+N*0x0040 (N=0-15)			Register Name: DMAC_PARA_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	DMA_TO_EN Timeout function Enable Bit 1: Enable 0: Disable
30:29	R	0x0	DMA_TO_FUN Timeout Function Configuration 01: Enable pause channel transfer function after timeout 10: Enable stop channel transfer function after timeout 11: Enable jump next opcode function after timeout
28:20	R	0x0	DMA_TO_WAIT_CYC The time cycle of waiting the start device to restart transferring data
19:18	/	/	/
17:15	R	0x0	DMA_CUR_SRC_H DMA Channel Current Destination Address [34:32] , read only
14	/	/	/
13:11	R	0x0	DMA_CUR_DEST_H DMA Channel Current Source Address [34:32], read only
10	/	/	/
9	R	0x0	DMA_NXT_DESC_H DMA Next Descriptor Address[33](word address)

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Offset: 0x0100+0x001C+N*0x0040 (N=0-15)			Register Name: DMAC_PARA_REG
Bit	Read/Write	Default/Hex	Description
8	R	0x0	DMA_IOSPEED_EN Accelerate DMA speed when transferring the data of IO device 1: Enable 0: Disable
7:0	R	0x0	WAIT_CYC Wait Clock Cycles

12.1.6.18 0x0100+0x0024+N*0x0040 (N=0-15) DMAC Unknown Length Stage Register (Default Value: 0x0000_0001)

Offset:0x0100+0x0024+N*0x0040 (N=0-15)			Register Name: DMAC_UNKNOWN_LENGTH_STAGE_REG
Bit	Read/Write	Default/Hex	Description
31	RC	0x0	DMA_TIMEOUT_REMAIN_FLAG DMA Timeout Remain Flag If this register is 1, it means there is some data remain in DMA.CPU should read this register before next PKG interrupt is pending. This register will be clear when it is read.
30	/	/	/
29:5	R	0x0	DMA_REMAIN_LEFT. DMA Remain Byte Counter, read only. This value should be read before next PKG interrupt is pending.
4:1	/	/	/
0	R/W1C	0x1	DMA_CMD_DONE DMA Command Done If there is no transfer, it will be 1.

12.1.6.19 0x0100+0x0028+N*0x0040 (N=0-15) DMAC Mode Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0028+N*0x0040 (N=0-15)			Register Name: DMAC_MODE_REG
Bit	R/W	Default/Hex	Description
31	R/W	0x0	DESC_WAIT_MODE 0: Disable 1: Enable In descriptor wait mode, if this bit is enabled, DMA will be paused when the write back bit of descriptor that DMA read from DDR/SRAM is 1.

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Offset: 0x0100+0x0028+N*0x0040 (N=0-15)			Register Name: DMAC_MODE_REG
Bit	R/W	Default/Hex	Description
30	R	0x0	DESC_WAIT_STA 0: Current channel is not in descriptor wait state 1: Current channel is in descriptor wait state When DESC_WAIT_MODE=1 and the write back bit of descriptor is 1, this bit means the current channel is in descriptor wait state. Note: This bit can not be taken as a sign that the current channel is running.
29	R/WAC	0x0	DESC_READY When the DMA channel is in descriptor wait mode, seting this bit as 1 means the descriptor is updated and DMA channel will read the descriptor again from address that is in the DMAC_DESC_ADDR_REG register and the DMAC_DESC_HIG_ADDR_REG register. Note: This bit should not be set as 1 unless DESC_WAIT_STA=1
28:7	/	/	/
6	R/W	0x0	DMA_UNKNOWN_LENGTH_MODE DMA Unknown Length Mode Enable.(ULEN_mode and BMODE cannot be enabled at the same time) 0: Disable 1: Enable
5	R/W	0x0	LAG_TIME_MODE In wait mode, this bit can be enabled. DMA will not count the wait cycle until DMA receives or transmits data successfully. It's recommended to enable this bit before using DMA when the DMA has enable more than one channels which access IO and memory equipment. 0: Disable 1: Enable
4	R/W	0x0	DMA_BLOCK_ZERO_EN Enable pre-read source DRQ signals when source block is configured 0. It needs to be disabled when timeout function (end mode and jump mode) is enabled and source block is configured 0. 0: Enable 1: Disable

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Offset: 0x0100+0x0028+N*0x0040 (N=0-15)			Register Name: DMAC_MODE_REG
Bit	R/W	Default/Hex	Description
3	R/W	0x0	DMA_DST_MODE Destination communication Mode Select 0: Wait mode 1: Handshake mode
2	R/W	0x0	DMA_SRC_MODE Source communication Mode Select 0: Wait mode 1: Handshake mode
1:0	/	/	/

12.1.6.20 0x0100+0x002C+N*0x0040 (N=0-15) DMAC Former Descriptor Address Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x002C+N*0x0040 (N=0-15)			Register Name: DMAC_FDESC_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	DMA_FDESC_ADDR This register is used to storing the former value of DMA Channel Descriptor Address Register

12.1.6.21 0x0100+0x0030+N*0x0040 (N=0-15) DMAC Package Number Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0030+N*0x0040 (N=0-15)			Register Name: DMAC_PKG_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	DMA_PKG_NUM This register is used for recording the number of packages which has been completed in one transmission.

12.1.6.22 0x0100+0x0034+N*0x0040 (N=0-15) DMAC IRQ Enable Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0034+N*0x0040 (N=0-15)			Register Name: DMAC_IRQ_EN_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R/W	0x0	DMA_OS_SEL This bit is used to show that which master occupies current DMA channel. 00: OS 0 (operating system) 01: OS 1 (operating system)

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Offset: 0x0100+0x0034+N*0x0040 (N=0-15)			Register Name: DMAC_IRQ_EN_REG
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	DMA_STOP_IRQ_EN DMA Stop Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	DMA_DESC_WAIT_IRQ_EN DMA Descriptor Wait Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	DMA_TIMEOUT_IRQ_EN DMA Timeout Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	DMA_QUEUE_IRQ_EN DMA Queue End Transfer Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	DMA_PKG_IRQ_EN DMA Package End Transfer Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	DMA_HLAF_IRQ_EN DMA Half Package Transfer Interrupt Enable 0: Disable 1: Enable

12.1.6.23 0x0100+0x0038+N*0x0040 (N=0-15) DMAC IRQ Pending Status Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0038+N*0x0040 (N=0-15)			Register Name: DMAC_IRQ_PEND_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W1C	0x0	DMA_STOP_IRQ_PEDN DMA Stop Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending
4	R/W1C	0x0	DMA_DESC_WAIT_IRQ_PEND DMA Descriptor Wait Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending

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Offset: 0x0100+0x0038+N*0x0040 (N=0-15)			Register Name: DMAC_IRQ_PEND_REG
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	DMA_TIMEOUT_IRQ_PEND DMA Timeout Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending
2	R/W1C	0x0	DMA_QUEUE_IRQ_PEND DMA Queue End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending
1	R/W1C	0x0	DMA_PKG_IRQ_PEND DMA Package End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending
0	R/W1C	0x0	DMA_HLAF_IRQ_PEND DMA Half Package Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect 1: Pending

12.1.6.24 0x0100+0x003C+N*0x0040 (N=0-15) DMAC Channel Descriptor Highest Address Register (Default Value: 0x0000_0000)

Offset:0x0100+0x003C+N*0x0040 (N=0-15)			Register Name: DMAC_DESC_HIG_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R	0x0	DMA_FDESC_ADDR_HIGHEST. This register is used to storing the former value of DMA Channel Descriptor Highest Address[34].
15:1	/	/	/
0	R/W	0x0	DMA Channel Descriptor Highest Address (descriptor[34])

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13 Video & Graphics Engine

13.1 Display Engine (DE)

The Display Engine (DE) is a hardware composer to transfer image layers from a local bus or a video buffer to the LCD interface. The DE supports many overlay windows to blend, and supports video post-processing in the video channel.

The DE has the following features:

- One display output, size up to 1080P
- Two alpha blending channels optional
- Four overlay layers in UI channel,one overlay layers in video channel.
- Video channel and UI channel share one scaler function.
- Supports potter-duff compatible blending operation
- Supports dither output to TCON
- Supports 3DFIFO fot Dual Link LVDs to display
- Input format
  - Semi-planar of YUV422/YUV420/YUV411
  - Planar of YUV422/YUV420/ YUV411
  - ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555/RGB565
- Support Frame Packing/Top-and-Bottom/Side-by-Side Full/Side-by-Side Half 3D format data.
- SmartColor2.0 for excellent display experience
  - Flesh tone protection
  - Hue gain, saturation gain, and value gain controller
  - Fully programmable color matrix
  - Dynamic gamma
  - 17x17x17 programmeable 3D LUT
- Supports write back for verification

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13.2 Graphic 2D (G2D)

The Graphic 2D (G2D) engine is hardware accelerator for 2D graphic.

The G2D has the following features:

- Supports mixer layer size up to 2048x2048 pixels, rotate size up to 4096x4096 pixels
- Supports gradient color fill
- Supports pre-multiply alpha image data
- Supports color key
- Supports two pipes Porter-Duff alpha blending
- Supports multiple video formats 4:2:0, 4:2:2, 4:1:1 and multiple pixel formats (8/16/24/32 bits graphics layer)
- Supports memory scan order option
- Supports any format convert function above
- Supports 1/16x to 32x resize ratio
- Supports 32-phase 8-tap horizontal anti-alias filter, 32-phase 4-tap vertical anti-alias filter.
- Supports window clip
- Supports FillRectangle, FillGradient, BitBlt, StretchBlt and MaskBlt
- Supports horizontal and vertical flip, clockwise 0/90/180/270 degree rotate
- Supports 4 Thread RCQ function for mixer and rotate
- Supports dither

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14 Video Input Interface

14.1 CSIC

14.1.1 Overview

The CMOS Sensor Interface Controller (CSIC) is an image or video data receiver, which can receive image or video data via camera interface. The controller can transfer valid data to embedded ISP or store the data in memory directly.

The CSIC includes the following features:

- Supports crop function
- Supports frame rate down
- Supports senor frame synchronization
- Supports distinguishing odd field and even field based on frame number in interlaced input

MIPI CSI Interface

- Supports 4-lane, 2+2-lane MIPI Interface
  - MIPI CSI2 V1.1
  - MIPI DPHY V1.0
  - 1.0 Gbit/s per lane
- Pixel formats
  - RAW 8/10/12/14/16 bit
  - YUV422 8 bit
  - RGB888 and RGB565
- Three types of virtual channel timings: MIPI VC, MIPI USER Data Type, and MIPI DOL
  - Up to 4-ch 720P@30fps. Two of the four channles' data need to be in RAW format
  - Up to 2-ch 1080P@30fps

Parallel CSI Interface

- 8/10/12/16-bit width
- Digital Camera (DC) protocol, BT.656, BT.601, and BT.1120 interface
- Supports BT.656 2-channel time-multiplexed format
- DDR sample mode with pixel clock up to 148.5MHz
- ITU-R BT.656 up to 2-ch 720P@30fps
- TU-R BT.1120 up to 2-ch 1080P@30fps

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- RAW up to 4-ch 720P@30fps

Parser

- Parser0
  - Supports 2-lane MIPI PHYA
  - Supports 4-lane PHYA+PHYB
- Parser1 supports 2-lane MIPI PHYB
- Parser2 supports DVP

BK

- 3 BKs (BK0/1/2)
- BK0/1/2 supports up to 2M@30fps output
- Supports independent frame-done interrupts
- BK0/1 supports 2-lane time-multiplexing
- YUV422/420 8-bit planner/ combine format output
- RAW8/10/12/14/16 format output
- RGB888/565 output
- Supports Only Y output
- Horizontal and vertical flip for YUV422/YUV420/RAW format input

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14.1.2 Block Diagram

The following figure shows block diagram of the CSIC.

Figure 14-1 CSIC Block Diagram

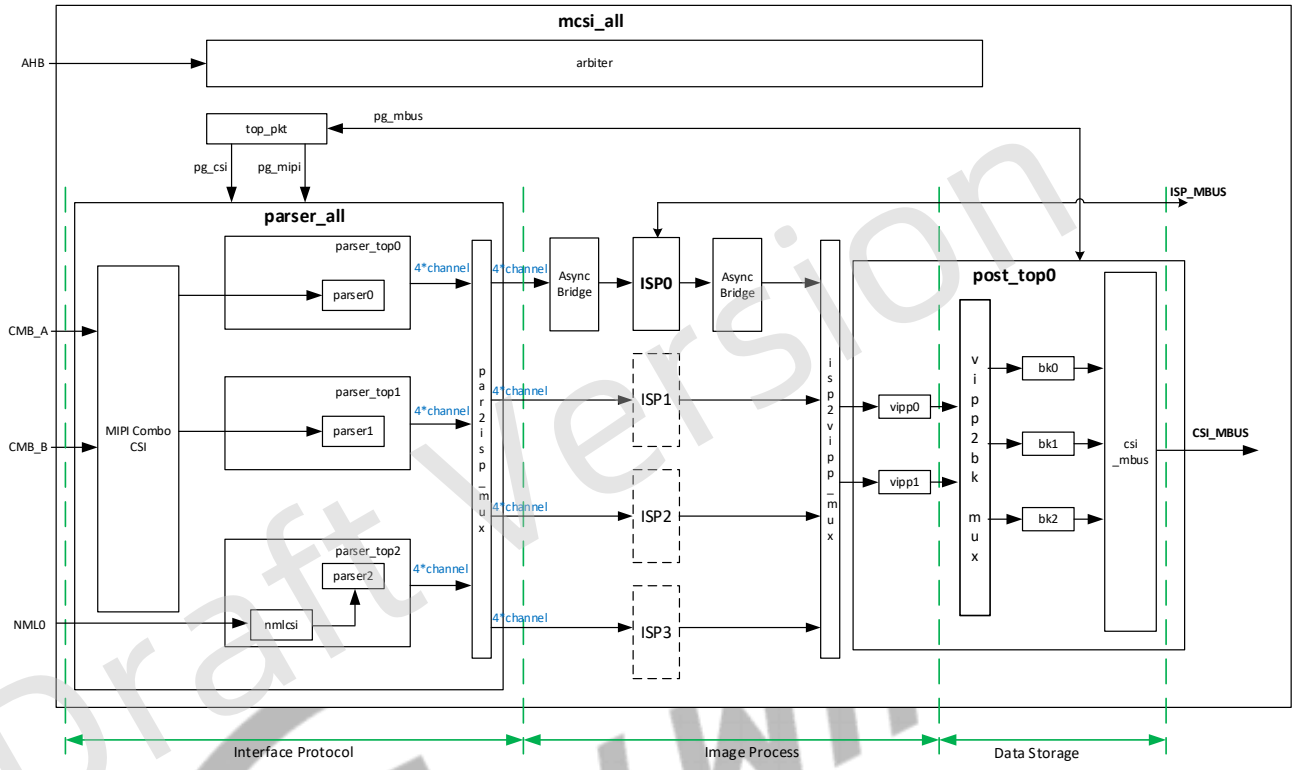

Table 14-1 CSIC Sub-blocks

Sub-blocks	Description
MIPI Combo	Combo is the CMOS input interface module. It supports MIPI CSI RX interface and multiple lane combination modes.
Parser	Parser is used to covert the MIPI data format into internal common format.
nmlcsi	Normal CSI. It is used to receive the digital image input from a digital parallel port, and convert it into the output of AIDB BUS in the mcsi_clk clock domain and parser.
Async Bridge	Asynchronous bridge is used to deal with the problem of clock domain crossing between parser and ISP, ISP and Post.
ISP	Image Signal Processor. It is used to process the sensor raw data, including lens compensation, and the removal of bad spots, gain, gamma and so on. (ISP1/2/3 is merely used as the data strobe path. Only ISP0 process data)
VIPP	Image post-processing. It can scale down the image.
BK	Write Back. It is used to write the image data back to the controller module of SDRAM.

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14.1.3 External Signals

Table 14-2 CSIC External Signals

Signal Name	Description	Type
MIPI CSI		
MCSI0-MCLK	MIPI CSI A Master Clock	O
MCSIA-CKN	MIPI CSI A Differential Receive Clock Signal (Negative)	AI
MCSIA-CKP	MIPI CSI A Differential Receive Clock Signal (Positive)	AI
MCSIA-D0N	MIPI CSI A Data Line 0 Input (Negative)	AI
MCSIA-D0P	MIPI CSI A Data Line 0 Input (Positive)	AI
MCSIA-D1N	MIPI CSI A Data Line 1 Input (Negative)	AI
MCSIA-D1P	MIPI CSI A Data Line 1 Input (Positive)	AI
MCSI1-MCLK	MIPI CSI B Master Clock	O
MCSIB-CKN	MIPI CSI B Differential Receive Clock Signal (Negative)	AI
MCSIB-CKP	MIPI CSI B Differential Receive Clock Signal (Positive)	AI
MCSIB-D0N/MCSIA-D3N	MIPI CSI B Data Line 0 Input (Negative)/MIPI CSI A Data Line 3 Input (Negative)	AI
MCSIB-D0P/MCSIA-D3P	MIPI CSI B Data Line 0 Input (Positive)/MIPI CSI A Data Line 3 Input (Positive)	AI
MCSIB-D1N/ MCSIA-D2N	MIPI CSI B Data Line 1 Input (Negative)/MIPI CSI A Data Line 2 Input (Negative)	AI
MCSIB-D1P/ MCSIA-D2P	MIPI CSI B Data Line 1 Input (Positive)/MIPI CSI A Data Line 2 Input (Positive)	AI
CSI0-XHS	CSI Horizontal SYNC 0	O
CSI1-XHS	CSI Horizontal SYNC 1	O
CSI0-XVS-FSYNC	CSI Vertical SYNC/Frame SYNC 0	I/O
CSI1-XVS-FSYNC	CSI Vertical SYNC/Frame SYNC 1	I/O
Parallel CSI		
NCSI-PCLK	Parallel CSI Pixel Clock	I
NCSI-MCLK	Parallel CSI Master Clock	O
NCSI-HSYNC	Parallel CSI Horizontal Synchronous	I
NCSI-VSYNC	Parallel CSI Vertical Synchronous	I
NCSI-D[15:0]	Parallel CSI Data Bit	I

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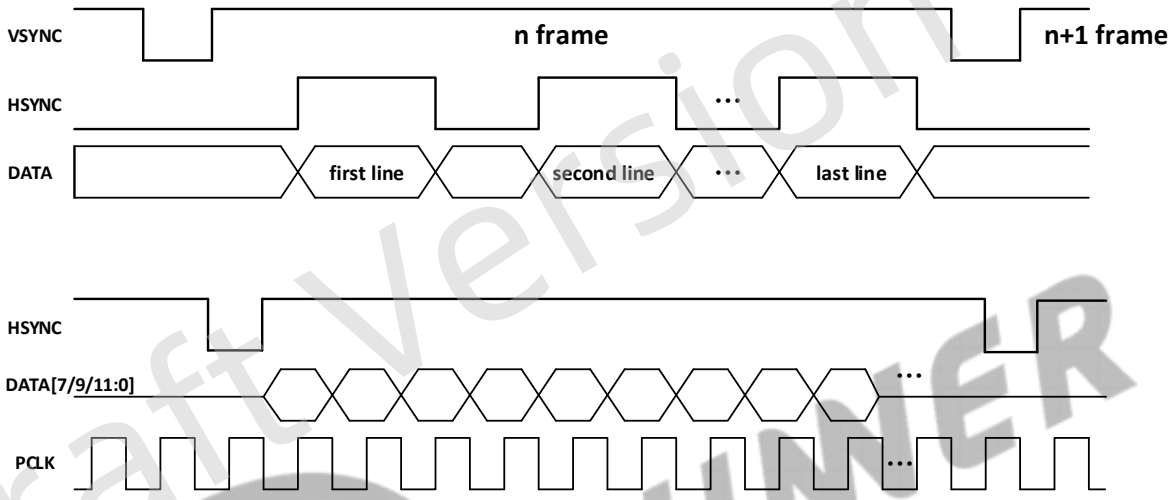
14.1.4 Functional Description

14.1.4.1 CSIC Input Timing

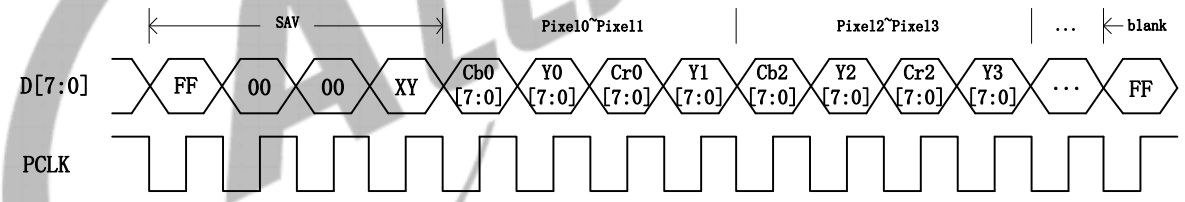
The following figure shows the timing of 8bit CMOS Sensor Interface, in this figure clock active at the rising edge, VSYNC valid at positive, HSYNC valid at positive.

Figure 14-2 8-bit DC Sensor Interface Timing



The following figure shows the timing of 8-bit YCbCr4:2:2 with embedded syncs (BT.656).

Figure 14-3 8-bit YCbCr4:2:2 with Embedded Syncs (BT.656)



The following table shows the header code of BT.656.

Table 14-3 BT.656 Header Code

Data Bit	First Word(0xFF)	Second Word(0x00)	Third Word(0x00)	Fourth Word
CS D[7] (MSB)	1	0	0	1
CS D[6]	1	0	0	F
CS D[5]	1	0	0	V
CS D[4]	1	0	0	H
CS D[3]	1	0	0	P3
CS D[2]	1	0	0	P2
CS D[1]	1	0	0	P1
CS D[0]	1	0	0	P0

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The following table shows the Header Data Bit Definition of BT.656.

Table 14-4 BT.656 Header Data Bit Definition

Decode	F	V	H	P3	P2	P1	P0
Field 1 start of active video (SAV)	0	0	0	0	0	0	0
Field 1 end of active video (EAV)	0	0	1	1	1	0	1
Field 1 SAV (digital blanking)	0	1	0	1	0	1	1
Field 1 EAV (digital blanking)	0	1	1	0	1	1	0
Field 2 SAV	1	0	0	0	1	1	1
Field 2 EAV	1	0	1	1	0	1	0
Field 2 SAV (digital blanking)	1	1	0	1	1	0	0
Field 2 EAV (digital blanking)	1	1	1	0	0	0	1

NOTE

P0-P3 are used for verification in 1-channel transmission and used for channel ID marking in 2/4-channel transmission.

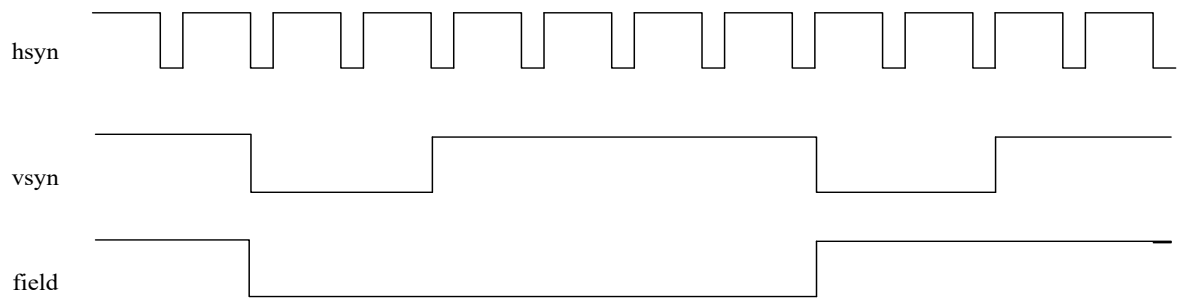
The BT.1120, except having a data bus width of 16 bits, is the same as BT.656 protocol (including EAV and SAV encoding). In the 16-bit data bus of BT.1120, the lower 8 bits are used to transmit the UV component and the higher 8 bits are used to transmit the Y component (or vice versa). The EAV and SAV encoding are used for the lower 8 bits. In addition, BT.1120 supports time multiplexing for 2-channel or 4-channel image data transmission.

The BT.601 supports 8-bit signal lines to transmit interlaced YUV422 data, and also supports separate signal lines to transmit synchronization information, such as field signal, vsyn signal, and hsyn signal. Both BT.601 and standard CSI support vf_mode. The difference from standard CSI is that the BT.601 also supports v2f_mode and f2v_mode.

In the vf_mode, the field signal and vsyn signal exist independently of each other.

In the v2f_mode, the field signal is derived from the phase difference between the vsyn signal and the hsyn signal. If the vsyn signal falls near the falling edge of the hsyn signal, the next field is field0; otherwise, it is field1.

Figure 14-4 BT.601 v2f_mode Timing



In the f2v_mode, the vsyn signal is derived from the field signal. The vsyn signal is invalid within a few cycles after the field signal jumps, and is valid the rest of the time. Currently, there are no practical applications of this mode.

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14.1.4.2 CSIC FIFO Distribution

Table 14-5 CSIC FIFO Distribution

Input format	YUV422/YUV420		Raw
Output format	Planar	UV combined	Raw
FIFO0	Y	Y	All pixels data
FIFO1	Cb (U)	CbCr (UV)	-
FIFO2	Cr (V)	-	-

14.1.4.3 Pixel Format Arrangement

The following figures show the Pixel Format Arrangement.

Figure 14-5 RAW-8 Format

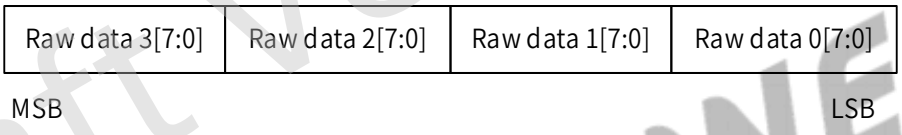
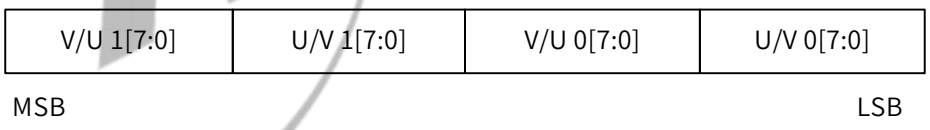


Figure 14-6 Y Format

Figure 14-7 UV-Combined Format



14.1.4.4 Offset Definition

Offset in horizontal and vertical can be added when receiving image. Unit is pixel.

- For YUV422 format, pixel unit is a YU/YV combination.
- For YUV420 format, pixel unit is a YU/YV combination in YC line, and only a Y in Y line.
- For Bayer and RAW format, pixel unit is a R/G/B single component.

14.1.4.5 Flip Definition

Both horizontal and vertical flip are supported at the same time. This function is implemented in the process of each FIFO writing data to memory, only flipping the data of separate FIFO, not changing component to FIFO distribution.

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If horizontal flip is enabled, one or more pixels will be took as a unit:

For YUV format, a unit of Y0U0Y1V1 will parser and flip the Y component in one channel, and UV will be treated as a whole. In planar output mode, U and V will be flipped separately. In UV combined output mode, UV will be flipped as a whole. So, a sequence of Y1U0Y0V1 will be.

For Bayer_raw format, situation is much like. A GR/BG sequence will be changed to BG/RG. A unit of square has four pixels.

14.1.4.6 Field Mode and Frame Mode

w below refers to the width of each line, h below refers to the number of lines per frame

- **Field mode for Interlaced input**

Number of frame packets per second: frame rate * 2

Size of each frame packet: w*(h/2)

Parity fields are packetized separately, and are distinguished by frame number. frame number information is included in the header of MIPI short packet for each field.

- **Frame mode for Interlaced input**

The number of frame packets per second is the same value as the frame rate.

Each frame packet w*h contains two fields, the upper (h/2) row is the odd field and the lower (h/2) is the even field.

14.1.5 Programming Guidelines

14.1.5.1 Async Bridge Configuration

- If ISP uses work clock with lower frequency than that of parser and post.

Step 1 Configure the field ISP_BRIDGE_EN (bit [3]) of CSIC_TOP_EN_REG as 1'b1 to enable async bridge

Step 2 Configure the field MCSI_PARSER_CLK_MODE (bit [0]) of CCU_CLK_MODE_REG as 1'b1 to make CSI parser work in CSI clock.

Step 3 Configure the field MCSI_POST_CLK_MODE (bit [1]) of CCU_CLK_MODE_REG as 1'b1 to make CSI post work in CSI clock.

- If ISP uses work clock with the same frequency as parser and post

Step 1 Configure the field ISP_BRIDGE_EN (bit [3]) of CSIC_TOP_EN_REG as 1'b0 to disable and bypass async bridge.

Step 2 Configure the field MCSI_PARSER_CLK_MODE (bit [0]) of CCU_CLK_MODE_REG as 1'b0 to make CSI parser work in ISP core clock.

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Step 3 Configure the field MCSI_POST_CLK_MODE (bit [1]) of CCU_CLK_MODE_REG as 1'b0 to make CSI post work in ISP core clock.

14.1.5.2 CSI Clock Frequency Configuration

The minimum clock configuration requirements for CSI_CLK are as follows:

One-pixel mode: CSI_CLK = lanespeed/8*lanenumber

Two-pixel mode: CSI_CLK = lanespeed/8*lanenumber/2

Unit: lanespeed – Gbit/s or Mbit/s; CSI_CLK — GHz or MHz

NOTE

Only 4-lane CSI supports two-pixel mode.

14.1.6 Register List

Module Name	Base Address	Description
CSIC_CCU	0x05800000	Do not support byte operations
CSIC_TOP	0x05800800	
CSIC_COMBO_PHY		
CSI_COMBO_PHY_COMMON	0x05810000	
CSI_COMBO_PHYA_DIGITAL_LAYER	0x05810100	
CSI_COMBO_PHYB_DIGITAL_LAYER	0x05810200	
CSI_COMBO_PHY_PORT0_PAYLOAD_LAYER	0x05811000	
CSI_COMBO_PHY_PORT1_PAYLOAD_LAYER	0x05811400	
CSIC_PARSER		
CSIC_PARSER0	0x05820000	
CSIC_PARSER1	0x05821000	
CSIC_PARSER2	0x05822000	
CSIC_DMA		
CSIC_DMA0	0x05830000	
CSIC_DMA1	0x05831000	
CSIC_DMA2	0x05832000	

14.1.6.1 CSIC_CCU Register List

Module Name	Base Address
CSIC_CCU	0x05800000

Register Name	Offset	Description

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Register Name	Offset	Description
CSIC_CCU_CLK_MODE_REG	0x0000	CSIC CCU Clock Mode Register
CSIC_CCU_PARSER_CLK_EN_REG	0x0004	CSIC CCU Parser Clock Enable Register
CCU_ISP_CLK_EN_REG	0x0008	CSIC CCU ISP Clock Enable Register
CSIC_CCU_POST0_CLK_EN_REG	0x000C	CSIC CCU Post0 Clock Enable Register
CSIC_CCU_CHFREQ_CLK_CTRL_REG	0x0014	CSIC CCU Chfreq Clock Control Register
CSIC_CCU_BK_INTPOOL_CLK_CTRL_REG	0x0018	CSIC CCU BK Intpool Clock Control Register

14.1.6.2 CSIC_TOP Register List

Module Name	Base Address
CSIC_TOP	0x05800800

Register Name	Offset	Description
CSIC_TOP_EN_REG	0x0000	CSIC TOP Enable Register
CSIC_PTN_GEN_EN_REG	0x0004	CSIC Pattern Generation Enable Register
CSIC_PTN_CTRL_REG	0x0008	CSIC Pattern Control Register
CSIC_PTN_VBLANK_CYCLE	0x0010	CSIC Pattern Vblank Cycle Register
CSIC_PTN_LEN_REG	0x0020	CSIC Pattern Generation Length Register
CSIC_PTN_ADDR_REG	0x0024	CSIC Pattern Generation Address Register
CSIC_PTN_ISP_SIZE_REG	0x0028	CSIC Pattern ISP Size Register
CSIC_ISP0_INPUT0_SEL_REG	0x0030	CSIC ISP0 Input0 Select Register
CSIC_ISP0_INPUT1_SEL_REG	0x0034	CSIC ISP0 Input1 Select Register
CSIC_ISP0_INPUT2_SEL_REG	0x0038	CSIC ISP0 Input2 Select Register
CSIC_ISP0_INPUT3_SEL_REG	0x003C	CSIC ISP0 Input3 Select Register
CSIC_ISP1_INPUT0_SEL_REG	0x0040	CSIC ISP1 Input0 Select Register
CSIC_ISP1_INPUT1_SEL_REG	0x0044	CSIC ISP1 Input1 Select Register
CSIC_ISP1_INPUT2_SEL_REG	0x0048	CSIC ISP1 Input2 Select Register
CSIC_ISP1_INPUT3_SEL_REG	0x004C	CSIC ISP1 Input3 Select Register
CSIC_ISP2_INPUT0_SEL_REG	0x0050	CSIC ISP2 Input0 Select Register
CSIC_ISP2_INPUT1_SEL_REG	0x0054	CSIC ISP2 Input1 Select Register
CSIC_ISP2_INPUT2_SEL_REG	0x0058	CSIC ISP2 Input2 Select Register
CSIC_ISP2_INPUT3_SEL_REG	0x005C	CSIC ISP2 Input3 Select Register
CSIC_ISP3_INPUT0_SEL_REG	0x0060	CSIC ISP3 Input0 Select Register
CSIC_ISP3_INPUT1_SEL_REG	0x0064	CSIC ISP3 Input1 Select Register
CSIC_ISP3_INPUT2_SEL_REG	0x0068	CSIC ISP3 Input2 Select Register
CSIC_ISP3_INPUT3_SEL_REG	0x006C	CSIC ISP3 Input3 Select Register
CSIC_ISP_BRG_BUF_MAXUSE_CNT_CLR_REG	0x0070	CSIC ISP Bridge Buffer Maxuse Counter Clear Register
CSIC_ISP0_BRG01_BUF_MAXUSE_CNT_REG	0x0074	CSIC ISP0 Bridge01 Buffer Maxuse Counter Register

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Register Name	Offset	Description
CSIC_ISP0_BRG23_BUF_MAXUSE_CNT_REG	0x0078	CSIC ISP0 Bridge23 Buffer Maxuse Counter Register
CSIC_ISP0_BRG_INT_EN_REG	0x0084	CSIC ISP0 Bridge Interrupt Enable Register
CSIC_ISP0_BRG_INT_PD_REG	0x008C	CSIC ISP0 Bridge Interrupt Pending Register
CSIC_VIPP0_INPUT_SEL_REG	0x00A0	CSIC VIPP0 Input Select Register
CSIC_VIPP1_INPUT_SEL_REG	0x00A4	CSIC VIPP1 Input Select Register
CSIC_VIPP2_INPUT_SEL_REG	0x00A8	CSIC VIPP2 Input Select Register
CSIC_DMA0_INPUT_SEL_REG	0x00D0	CSIC DMA0 Input Select Register
CSIC_DMA1_INPUT_SEL_REG	0x00D4	CSIC DMA1 Input Select Register
CSIC_DMA2_INPUT_SEL_REG	0x00D8	CSIC DMA2 Input Select Register
CSIC_MULF_MOD_REG	0x0100	CSIC Multi-Frame Mode Register
CSIC_MULF_INT_REG	0x0104	CSIC Multi-Frame Interrupt Register
CSIC_CHFREQ_CFG0_REG	0x0108	CSIC Change Frequency Configuration 0 Register
CSIC_CHFREQ_CFG1_REG	0x010C	CSIC Change Frequency Configuration 1 Register
CSIC_CHFREQ_OBS_REG	0x0110	CSIC Change Frequency Observation Register
CSIC_CHFREQ_INT_REG	0x0114	CSIC Change Frequency Interrupt Register
CSIC_CHFREQ_DBG_OBS_REG	0x0118	CSIC Change Frequency Debug Observation Register
CSIC_ISP_BYP_CFG_REG	0x0120	CSIC ISP Bypass Software Configuration Register
CSIC_MBUS_REQ_MAX_REG	0x01E0	CSIC MBUS REQ MAX Register
CSIC_MBUS_WAIT_MAX_REG	0x01E8	CSIC MBUS WAIT MAX Register
CSIC_MBUS_DEBUG_REG	0x01EC	CSIC MBUS DEBUG Register

14.1.6.3 CSIC_COMBO_PHY Register List

PHY COMMON

Module Name	Base Address
CSI_COMBO_PHY_COMMON	0x05810000

Register Name	Offset	Description
PHY_TOP_CTL_REG	0x0000	PHY Top Control Register
PHY_TRESCAL_REG	0x0004	PHY Terminal Resistor Calibration Register

PHY DIGITAL LAYER

Module Name	Base Address
CSI_COMBO_PHYA_DIGITAL_LAYER	0x05810100
CSI_COMBO_PHYB_DIGITAL_LAYER	0x05810200

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Register Name	Offset	Description
PHY_CTL_REG	0x0000	PHY Control Register
PHY_EQ_REG	0x0004	PHY Equalization Register
PHY_OFSCAL_REG0	0x0008	PHY Offset Calibration Register0
PHY_OFSCAL_REG1	0x000C	PHY Offset Calibration Register1
PHY_OFSCAL_REG2	0x0010	PHY Offset Calibration Register2
PHY_DESKEW_REG0	0x0014	PHY Deskew Register0
PHY_DESKEW_REG1	0x0018	PHY Deskew Register1
PHY_DESKEW_REG2	0x001C	PHY Deskew Register2
PHY_TERM_CTL_REG	0x0020	PHY Terminal Control Register
PHY_HS_CTL_REG	0x0024	PHY High Speed Control Register
PHY_S2P_CTL_REG	0x0028	PHY Serial to Parallel Control Register
PHY_MIPIRX_CTL_REG	0x002C	PHY MIPIRX Control Register
PHY_MIPIRX_SYNC_TIMEOUT_REG	0x0030	PHY MIPIRX Synchronization Timeout Register
PHY_FREQ_CNT _REG	0x0034	PHY Frequency Counter Register
PHY_MIPI_LP_TIMEOUT_REG	0x0038	PHY MIPI LP Timeout Register
PHY_MIPI_ULPSEXIT_REG0	0x0040	PHY MIPI ULPSEXIT Register0
PHY_INT_EN_REG	0x0080	PHY Interrupt Enable Register
PHY_INT_PD_REG	0x0084	PHY Interrupt Pending Register
PHY_DESKEW_REG3	0x00E0	PHY Deskew Register3

PORT PAYLOAD LAYER

Module Name	Base Address
CSI_COMBO_PHY_PORT0_PAYLOAD_LAYER	0x05811000
CSI_COMBO_PHY_PORT1_PAYLOAD_LAYER	0x05811400

Register Name	Offset	Description
PORT_CTL_REG	0x0000	PORT Control Register
PORT_LANE_MAP_REG0	0x0004	PORT Lane Mapping Register0
PORT_WDR_MODE_REG	0x000C	PORT WDR Mode Register
PORT_FID_SEL_REG	0x0010	PORT Frame ID Select Register
PORT_MIPI_CFG_REG	0x0100	PORT MIPI Configuration Register
PORT_MIPI_NO_UNPAK_NUM_REG	0x0104	PORT MIPI No Unpacket Number Register
PORT_MIPI_DI_REG	0x0108	PORT MIPI Data Identity Register
PORT_MIPI_USER_DT_REG	0x010C	PORT MIPI User Data Type Register
PORT_MIPI_CH0_DT_TRIG_EN_REG	0x0110	PORT MIPI Channel0 Data Type Trigger Register
PORT_MIPI_CH0_INT_EN_REG	0x0114	PORT MIPI Channel0 Interrupt Enable Register

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Register Name	Offset	Description
PORT_MIPI_CH0_INT_PD_REG	0x0118	PORT MIPI Channel0 Interrupt Pending Register
PORT_MIPI_CH0_PH_REG	0x011C	PORT MIPI Channel0 Packet Header Register
PORT_MIPI_CH0_ECC_REG	0x0120	PORT MIPI Channel0 ECC Register
PORT_MIPI_CH0_CKSUM_REG	0x0124	PORT MIPI Channel0 Checksum Register
PORT_MIPI_CH0_FRM_NUM_REG	0x0128	PORT MIPI Channel0 Frame Number Register
PORT_MIPI_CH0_LINE_NUM_REG	0x012C	PORT MIPI Channel0 Line Number Register
PORT_MIPI_CH1_DT_TRIG_EN_REG	0x0130	PORT MIPI Channel1 Data Type Trigger Register
PORT_MIPI_CH1_INT_EN_REG	0x0134	PORT MIPI Channel1 Interrupt Enable Register
PORT_MIPI_CH1_INT_PD_REG	0x0138	PORT MIPI Channel1 Interrupt Pending Register
PORT_MIPI_CH1_PH_REG	0x013C	PORT MIPI Channel1 Packet Header Register
PORT_MIPI_CH1_ECC_REG	0x0140	PORT MIPI Channel1 ECC Register
PORT_MIPI_CH1_CKSUM_REG	0x0144	PORT MIPI Channel1 Checksum Register
PORT_MIPI_CH1_FRM_NUM_REG	0x0148	PORT MIPI Channel1 Frame Number Register
PORT_MIPI_CH1_LINE_NUM_REG	0x014C	PORT MIPI Channel1 Line Number Register
PORT_MIPI_CH2_DT_TRIG_EN_REG	0x0150	PORT MIPI Channel2 Data Type Trigger Register
PORT_MIPI_CH2_INT_EN_REG	0x0154	PORT MIPI Channel2 Interrupt Enable Register
PORT_MIPI_CH2_INT_PD_REG	0x0158	PORT MIPI Channel2 Interrupt Pending Register
PORT_MIPI_CH2_PH_REG	0x015C	PORT MIPI Channel2 Packet Header Register
PORT_MIPI_CH2_ECC_REG	0x0160	PORT MIPI Channel2 ECC Register
PORT_MIPI_CH2_CKSUM_REG	0x0164	PORT MIPI Channel2 Checksum Register
PORT_MIPI_CH2_FRM_NUM_REG	0x0168	PORT MIPI Channel2 Frame Number Register
PORT_MIPI_CH2_LINE_NUM_REG	0x016C	PORT MIPI Channel2 Line Number Register
PORT_MIPI_CH3_DT_TRIG_EN_REG	0x0170	PORT MIPI Channel3 Data Type Trigger Register
PORT_MIPI_CH3_INT_EN_REG	0x0174	PORT MIPI Channel3 Interrupt Enable Register
PORT_MIPI_CH3_INT_PD_REG	0x0178	PORT MIPI Channel3 Interrupt Pending Register
PORT_MIPI_CH3_PH_REG	0x017C	PORT MIPI Channel3 Packet Header Register
PORT_MIPI_CH3_ECC_REG	0x0180	PORT MIPI Channel3 ECC Register
PORT_MIPI_CH3_CKSUM_REG	0x0184	PORT MIPI Channel3 Checksum Register
PORT_MIPI_CH3_FRM_NUM_REG	0x0188	PORT MIPI Channel3 Frame Number Register
PORT_MIPI_CH3_LINE_NUM_REG	0x018C	PORT MIPI Channel3 Line Number Register
PORT_MIPI_LANE_ERR_INT_EN_REG	0x01F0	PORT MIPI Lane Error Interrupt Register
PORT_MIPI_LANE_ERR_INT_PD_REG	0x01F4	PORT MIPI Lane Error Interrupt Pending Register

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14.1.6.4 CSIC_PARSER Register List

Module Name	Base Address
CSIC_PARSER0	0x05820000
CSIC_PARSER1	0x05821000
CSIC_PARSER2	0x05822000

Register Name	Offset	Description
CSIC_PRS_EN_REG	0x0000	CSIC Parser Enable Register
CSIC_PRS_NCSIC_IF_CFG_REG	0x0004	CSIC Parser NCSIC Interface Configuration Register
CSIC_PRS_MCSI_IF_CFG_REG	0x0008	CSIC Parser MCSIC Interface Configuration Register
CSIC_PRS_CAP_REG	0x000C	CSIC Parser Capture Register
CSIC_PRS_SIGNAL_STA_REG	0x0010	CSIC Parser Signal Status Register
CSIC_PRS_CAP_FRM_MSK_0_REG	0x0018	CSIC Parser Capture Frame Mask 0 Register
CSIC_PRS_CAP_FRM_MSK_1_REG	0x001C	CSIC Parser Capture Frame Mask 1 Register
CSIC_PRS_CH0_IN_DMSK_PERIOD_REG	0x0020	CSIC Parser Channel_0 Input Dmask Setting Register
CSIC_PRS_CH0_INFMT_REG	0x0024	CSIC Parser Channel_0 Input Format Register
CSIC_PRS_CH0_OUTPUT_HSIZE_REG	0x0028	CSIC Parser Channel_0 Output Horizontal Size Register
CSIC_PRS_CH0_OUTPUT_VSIZE_REG	0x002C	CSIC Parser Channel_0 Output Vertical Size Register
CSIC_PRS_CH0_INPUT_PARA0_REG	0x0030	CSIC Parser Channel_0 Input Parameter0 Register
CSIC_PRS_CH0_INPUT_PARA1_REG	0x0034	CSIC Parser Channel_0 Input Parameter1 Register
CSIC_PRS_CH0_INPUT_PARA2_REG	0x0038	CSIC Parser Channel_0 Input Parameter2 Register
CSIC_PRS_CH0_INPUT_PARA3_REG	0x003C	CSIC Parser Channel_0 Input Parameter3 Register
CSIC_PRS_CH0_INT_EN_REG	0x0040	CSIC Parser Channel_0 Interrupt Enable Register
CSIC_PRS_CH0_INT_STA_REG	0x0044	CSIC Parser Channel_0 Interrupt Status Register
CSIC_PRS_CH0_LINE_TIME_REG	0x0048	CSIC Parser Channel_0 Line Time Register
CSIC_PRS_CH0_FRM_PRE_MSK_REG	0x004C	CSIC Parser Channel_0 Frame Pre Mask Setting Register
CSIC_PRS_CH1_IN_DMSK_PERIOD_REG	0x0120	CSIC Parser Channel_1 Input Dmask Setting Register
CSIC_PRS_CH1_INFMT_REG	0x0124	CSIC Parser Channel_1 Input Format Register

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Register Name	Offset	Description
CSIC_PRS_CH1_OUTPUT_HSIZE_REG	0x0128	CSIC Parser Channel_1 Output Horizontal Size Register
CSIC_PRS_CH1_OUTPUT_VSIZE_REG	0x012C	CSIC Parser Channel_1 Output Vertical Size Register
CSIC_PRS_CH1_INPUT_PARA0_REG	0x0130	CSIC Parser Channel_1 Input Parameter0 Register
CSIC_PRS_CH1_INPUT_PARA1_REG	0x0134	CSIC Parser Channel_1 Input Parameter1 Register
CSIC_PRS_CH1_INPUT_PARA2_REG	0x0138	CSIC Parser Channel_1 Input Parameter2 Register
CSIC_PRS_CH1_INPUT_PARA3_REG	0x013C	CSIC Parser Channel_1 Input Parameter3 Register
CSIC_PRS_CH1_INT_EN_REG	0x0140	CSIC Parser Channel_1 Interrupt Enable Register
CSIC_PRS_CH1_INT_STA_REG	0x0144	CSIC Parser Channel_1 Interrupt Status Register
CSIC_PRS_CH1_LINE_TIME_REG	0x0148	CSIC Parser Channel_1 Line Time Register
CSIC_PRS_CH1_FRM_PRE_MSK_REG	0x014C	CSIC Parser Channel_1 Frame Pre Mask Setting Register
CSIC_PRS_CH2_IN_DMSK_PERIOD_REG	0x0220	CSIC Parser Channel_2 Input Dmask Setting Register
CSIC_PRS_CH2_INFMT_REG	0x0224	CSIC Parser Channel_2 Input Format Register
CSIC_PRS_CH2_OUTPUT_HSIZE_REG	0x0228	CSIC Parser Channel_2 Output Horizontal Size Register
CSIC_PRS_CH2_OUTPUT_VSIZE_REG	0x022C	CSIC Parser Channel_2 Output Vertical Size Register
CSIC_PRS_CH2_INPUT_PARA0_REG	0x0230	CSIC Parser Channel_2 Input Parameter0 Register
CSIC_PRS_CH2_INPUT_PARA1_REG	0x0234	CSIC Parser Channel_2 Input Parameter1 Register
CSIC_PRS_CH2_INPUT_PARA2_REG	0x0238	CSIC Parser Channel_2 Input Parameter2 Register
CSIC_PRS_CH2_INPUT_PARA3_REG	0x023C	CSIC Parser Channel_2 Input Parameter3 Register
CSIC_PRS_CH2_INT_EN_REG	0x0240	CSIC Parser Channel_2 Interrupt Enable Register
CSIC_PRS_CH2_INT_STA_REG	0x0244	CSIC Parser Channel_2 Interrupt Status Register
CSIC_PRS_CH2_LINE_TIME_REG	0x0248	CSIC Parser Channel_2 Line Time Register
CSIC_PRS_CH2_FRM_PRE_MSK_REG	0x024C	CSIC Parser Channel_2 Frame Pre Mask Setting Register
CSIC_PRS_CH3_IN_DMSK_PERIOD_REG	0x0320	CSIC Parser Channel_3 Input Dmask Setting Register
CSIC_PRS_CH3_INFMT_REG	0x0324	CSIC Parser Channel_3 Input Format Register

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Register Name	Offset	Description
CSIC_PRS_CH3_OUTPUT_HSIZE_REG	0x0328	CSIC Parser Channel_3 Output Horizontal Size Register
CSIC_PRS_CH3_OUTPUT_VSIZE_REG	0x032C	CSIC Parser Channel_3 Output Vertical Size Register
CSIC_PRS_CH3_INPUT_PARA0_REG	0x0330	CSIC Parser Channel_3 Input Parameter0 Register
CSIC_PRS_CH3_INPUT_PARA1_REG	0x0334	CSIC Parser Channel_3 Input Parameter1 Register
CSIC_PRS_CH3_INPUT_PARA2_REG	0x0338	CSIC Parser Channel_3 Input Parameter2 Register
CSIC_PRS_CH3_INPUT_PARA3_REG	0x033C	CSIC Parser Channel_3 Input Parameter3 Register
CSIC_PRS_CH3_INT_EN_REG	0x0340	CSIC Parser Channel_3 Interrupt Enable Register
CSIC_PRS_CH3_INT_STA_REG	0x0344	CSIC Parser Channel_3 Interrupt Status Register
CSIC_PRS_CH3_LINE_TIME_REG	0x0348	CSIC Parser Channel_3 Line Time Register
CSIC_PRS_CH3_FRM_PRE_MSK_REG	0x034C	CSIC Parser Channel_3 Frame Pre Mask Setting Register
CSIC_PRS_NCSIC_RX_SIGNAL0_DLY_ADJ_REG	0x0500	CSIC Parser NCSIC RX Signal0 Delay Adjust Register
CSIC_PRS_NCSIC_RX_SIGNAL3_DLY_ADJ_REG	0x050C	CSIC Parser NCSIC RX Signal3 Delay Adjust Register
CSIC_PRS_NCSIC_RX_SIGNAL4_DLY_ADJ_REG	0x0510	CSIC Parser NCSIC RX Signal4 Delay Adjust Register
CSIC_PRS_NCSIC_RX_SIGNAL5_DLY_ADJ_REG	0x0514	CSIC Parser NCSIC RX Signal5 Delay Adjust Register
CSIC_PRS_NCSIC_RX_SIGNAL6_DLY_ADJ_REG	0x0518	CSIC Parser NCSIC RX Signal6 Delay Adjust Register
CSIC_PRS_SYNC_EN_REG	0x0520	CSIC Parser SYNC EN Register
CSIC_PRS_SYNC_CFG_REG	0x0524	CSIC Parser SYNC CFG Register
CSIC_PRS_VS_WAIT_N_REG	0x0528	CSIC Parser VS WAIT N Register
CSIC_PRS_VS_WAIT_M_REG	0x052C	CSIC Parser VS WAIT M Register
CSIC_PRS_XSYNC_ENABLE_REG	0x0540	CSIC Parser XSYNC ENABLE Register
CSIC_PRS_XVS_PERIOD_REG	0x0544	CSIC Parser XVS Period Register
CSIC_PRS_XHS_PERIOD_REG	0x0548	CSIC Parser XHS Period Register
CSIC_PRS_XVS_LENGTH_REG	0x054C	CSIC Parser XVS LENGTH Register
CSIC_PRS_XHS_LENGTH_REG	0x0550	CSIC Parser XHS LENGTH Register
CSIC_PRS_SYNC_DLY_REG	0x0554	CSIC Parser SYNC DELAY Register

14.1.6.5 CSIC_DMA Register List

Module Name	Base Address

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Module Name	Base Address
CSIC_DMA0	0x05830000
CSIC_DMA1	0x05831000
CSIC_DMA2	0x05832000

Register Name	Offset	Description
CSIC_DMA_TOP_REG	0x0000	CSIC DMA TOP Register
CSIC_DMA_MUL_CH_CFG_REG	0x0004	CSIC DMA Multi-Channel Configuration Register
CSIC_DMA_WRITE_CNT_REG	0x0008	CSIC DMA Write_Counter_Register
CSIC_DMA_FIFO_PT_REG	0x000C	CSIC DMA FIFO Pointer Register
CSIC_DMA_FRM_CLK_CNT_REG	0x0010	CSIC DMA Frame Rate Clock Counter Register
CSIC_DMA_ACC_ITNL_CLK_CNT_REG	0x0014	CSIC DMA Accumulated and Internal Clock Counter Register
CSIC_DMA_LINE_STAT_REG	0x0018	CSIC DMA Line Statistic Register
CSIC_DMA_PCLK_STAT_REG	0x001C	CSIC DMA PCLK Statistic Register
CSIC_DMA_FS_FRM_CNT_REG	0x0020	CSIC DMA Fsync Frame Counter Register
CSIC_DMA_VI_TO_TH0_REG	0x0040	CSIC DMA Video Input Timeout Threshold0 Register
CSIC_DMA_VI_TO_TH1_REG	0x0044	CSIC DMA Video Input Timeout Threshold1 Register
CSIC_DMA_VI_TO_CNT_VAL_REG	0x0048	CSIC DMA Video Input Timeout Counter Value Register
CSIC_DMA_VE_INFO_CTRL_REG	0x004C	CSIC DMA VE Information Control Register
CSIC_DMA_VE_FRM_CNT_REG	0x0050	CSIC DMA VE Frame Counter Value Register
CSIC_DMA_VE_LINE_CNT_REG	0x0054	CSIC DMA VE Line Counter Value Register
CSIC_DMA_VE_CUR_FRM_ADDR_REG	0x0058	CSIC DMA VE Current Frame Address Register
CSIC_DMA_VE_LAST_FRM_ADDR_REG	0x005C	CSIC DMA VE Last Frame Address Register
CSIC_DMA_VE_EMBED_ADDR_REG	0x0060	CSIC DMA VE Embed Address Register
CSIC_DMA_CAP_STA_REG	0x0064	CSIC DMA Capture Status Register
CSIC_DMA_TIME_STAMP0_REG	0x0068	CSIC DMA TDM Time Stamp0 Register
CSIC_DMA_TIME_STAMP1_REG	0x006C	CSIC DMA TDM Time Stamp1 Register
CSIC_DMA_TIME_STAMP2_REG	0x0070	CSIC DMA TDM Time Stamp2 Register
CSIC_DMA_FIFO_STAT_REG	0x0080	CSIC DMA FIFO Statistic Register
CSIC_DMA_FIFO_THRS_REG	0x0084	CSIC DMA FIFO Threshold Register
CSIC_DMA_TOP_INT_EN_REG	0x0100	CSIC DMA TOP Interrupt Enable Register
CSIC_DMA_TOP_INT_STA_REG	0x0104	CSIC DMA TOP Interrupt Status Register
CSIC_DMA_VE_ID_EN_REG	0x0110	CSIC DMA VE ID Enable Register
CSIC_DMA_CH0_EN_REG	0x0200	CSIC DMA Channel0 Enable Register
CSIC_DMA_CH0_CFG_REG	0x0204	CSIC DMA Channel0 Configuration Register

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Register Name	Offset	Description
CSIC_DMA_CH0_EMBD_ADDR_REG	0x0208	CSIC DMA Channel0 Embedded Address Register
CSIC_DMA_CH0_FRM_MSK_CFG_REG	0x020C	CSIC DMA Channel0 Frame Mask Configuration Register
CSIC_DMA_CH0_HSIZE_REG	0x0210	CSIC DMA Channel0 Horizontal Size Register
CSIC_DMA_CH0_VSIZE_REG	0x0214	CSIC DMA Channel0 Vertical Size Register
CSIC_DMA_CH0_CROP_CFG_REG	0x0218	CSIC DMA Channel0 Crop Mode Register
CSIC_DMA_CH0_LINE_CNT_REG	0x021C	CSIC DMA Channel0 Line Counter Register
CSIC_DMA_CH0_F0_BUFA_REG	0x0220	CSIC DMA Channel0 FIFO 0 Output Buffer-A Address Register
CSIC_DMA_CH0_F1_BUFA_REG	0x0224	CSIC DMA Channel0 FIFO 1 Output Buffer-A Address Register
CSIC_DMA_CH0_F2_BUFA_REG	0x0228	CSIC DMA Channel0 FIFO 2 Output Buffer-A Address Register
CSIC_DMA_CH0_BUF_LEN_REG	0x022C	CSIC DMA Channel0 Buffer Length Register
CSIC_DMA_CH0_FLIP_SIZE_REG	0x0230	CSIC DMA Channel0 Flip Size Register
CSIC_DMA_CH0_ABN_MSK_REG	0x0234	CSIC DMA Channel0 Abnormal Mask Register
CSIC_DMA_CH0_PIXELS_ADD_EN_REG	0x0238	CSIC DMA Channel0 Pixels Add Enable Register
CSIC_DMA_CH0_PIXELS_ADD_VAL_REG	0x023C	CSIC DMA Channel0 Pixels Add Value Register
CSIC_DMA_CH0_TOT_FRM_CNT_CFG_REG	0x0240	CSIC DMA Channel0 Total Frame Counter Config Register
CSIC_DMA_CH0_BUF_TH_REG	0x0244	CSIC DMA Channel0 BUF Threshold Register
CSIC_DMA_CH0_BUF_ADDR_FIFO0_ENTRY_REG	0x02E8	CSIC DMA Channel0 BUF Address FIFO0 Entry Register
CSIC_DMA_CH0_BUF_ADDR_FIFO1_ENTRY_REG	0x02EC	CSIC DMA Channel0 BUF Address FIFO1 Entry Register
CSIC_DMA_CH0_BUF_ADDR_FIFO2_ENTRY_REG	0x02F0	CSIC DMA Channel0 BUF Address FIFO2 Entry Register
CSIC_DMA_CH0_EMBED_ADDR_ENTRY_REG	0x02F4	CSIC DMA Channel0 Embed Address Entry Register
CSIC_DMA_CH0_INT_EN_REG	0x02F8	CSIC DMA Channel0 Interrupt Enable Register
CSIC_DMA_CH0_INT_STA_REG	0x02FC	CSIC DMA Channel0 Interrupt Status Register
CSIC_DMA_CH0_F0_BUFA_RESULT_REG	0x0300	CSIC DMA Channel0 FIFO 0 Output Buffer-A Address Result Register
CSIC_DMA_CH0_F1_BUFA_RESULT_REG	0x0304	CSIC DMA Channel0 FIFO 1 Output Buffer-A Address Result Register
CSIC_DMA_CH0_F2_BUFA_RESULT_REG	0x0308	CSIC DMA Channel0 FIFO 2 Output Buffer-A Address Result Register
CSIC_DMA_CH0_BUF_LEN_RESULT_REG	0x030C	CSIC DMA Channel0 Buffer Length Result Register

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Register Name	Offset	Description
CSIC_DMA_CH0_FLIP_SIZE_RESULT_REG	0x0310	CSIC DMA Channel0 Flip Size Result Register
CSIC_DMA_CH0_BUF_ADDR_FIFO_CON_REG	0x0314	CSIC DMA Channel0 BUF Address FIFO Content Register
CSIC_DMA_CH0_STORED_FRM_CNT_REG	0x0318	CSIC DMA Channel0 Stored Frame Counter Register
CSIC_DMA_CH0_TOT_FRM_CNT_REG	0x031C	CSIC DMA Channel0 Total Frame Counter Register
CSIC_DMA_CH0_FRM_LOST_CNT_REG	0x0324	CSIC DMA Channel0 Frame Lost Counter Register
CSIC_DMA_CH0_EMBED_ADDR_READ_REG	0x0328	CSIC DMA Channel0 Embed Address Read Register
CSIC_DMA_CH1_EN_REG	0x0400	CSIC DMA Channel1 Enable Register
CSIC_DMA_CH1_CFG_REG	0x0404	CSIC DMA Channel1 Configuration Register
CSIC_DMA_CH1_EMBD_ADDR_REG	0x0408	CSIC DMA Channel1 Embedded Address Register
CSIC_DMA_CH1_FRM_MSK_CFG_REG	0x040C	CSIC DMA Channel1 Frame Mask Configuration Register
CSIC_DMA_CH1_HSIZE_REG	0x0410	CSIC DMA Channel1 Horizontal Size Register
CSIC_DMA_CH1_VSIZE_REG	0x0414	CSIC DMA Channel1 Vertical Size Register
CSIC_DMA_CH1_CROP_CFG_REG	0x0418	CSIC DMA Channel1 Crop Mode Register
CSIC_DMA_CH1_LINE_CNT_REG	0x041C	CSIC DMA Channel1 Line Counter Register
CSIC_DMA_CH1_F0_BUFA_REG	0x0420	CSIC DMA Channel1 FIFO 0 Output Buffer-A Address Register
CSIC_DMA_CH1_F1_BUFA_REG	0x0424	CSIC DMA Channel1 FIFO 1 Output Buffer-A Address Register
CSIC_DMA_CH1_F2_BUFA_REG	0x0428	CSIC DMA Channel1 FIFO 2 Output Buffer-A Address Register
CSIC_DMA_CH1_BUF_LEN_REG	0x042C	CSIC DMA Channel1 Buffer Length Register
CSIC_DMA_CH1_FLIP_SIZE_REG	0x0430	CSIC DMA Channel1 Flip Size Register
CSIC_DMA_CH1_ABN_MSK_REG	0x0434	CSIC DMA Channel1 Abnormal Mask Register
CSIC_DMA_CH1_PIXELS_ADD_EN_REG	0x0438	CSIC DMA Channel1 Pixels Add Enable Register
CSIC_DMA_CH1_PIXELS_ADD_VAL_REG	0x043C	CSIC DMA Channel1 Pixels Add Value Register
CSIC_DMA_CH1_TOT_FRM_CNT_CFG_REG	0x0440	CSIC DMA Channel1 Total Frame Counter Config Register
CSIC_DMA_CH1_BUF_TH_REG	0x0444	CSIC DMA Channel1 BUF Threshold Register
CSIC_DMA_CH1_BUF_ADDR_FIFO0_ENTRY_REG	0x04E8	CSIC DMA Channel1 BUF Address FIFO0 Entry Register
CSIC_DMA_CH1_BUF_ADDR_FIFO1_ENTRY_REG	0x04EC	CSIC DMA Channel1 BUF Address FIFO1 Entry Register

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Register Name	Offset	Description
CSIC_DMA_CH1_BUF_ADDR_FIFO2_ENTRY_REG	0x04F0	CSIC DMA Channel1 BUF Address FIFO2 Entry Register
CSIC_DMA_CH1_EMBED_ADDR_ENTRY_REG	0x04F4	CSIC DMA Channel1 Embed Address Entry Register
CSIC_DMA_CH1_INT_EN_REG	0x04F8	CSIC DMA Channel1 Interrupt Enable Register
CSIC_DMA_CH1_INT_STA_REG	0x04FC	CSIC DMA Channel1 Interrupt Status Register
CSIC_DMA_CH1_F0_BUFA_RESULT_REG	0x0500	CSIC DMA Channel1 FIFO 0 Output Buffer-A Address Result Register
CSIC_DMA_CH1_F1_BUFA_RESULT_REG	0x0504	CSIC DMA Channel1 FIFO 1 Output Buffer-A Address Result Register
CSIC_DMA_CH1_F2_BUFA_RESULT_REG	0x0508	CSIC DMA Channel1 FIFO 2 Output Buffer-A Address Result Register
CSIC_DMA_CH1_BUF_LEN_RESULT_REG	0x050C	CSIC DMA Channel1 Buffer Length Result Register
CSIC_DMA_CH1_FLIP_SIZE_RESULT_REG	0x0510	CSIC DMA Channel1 Flip Size Result Register
CSIC_DMA_CH1_BUF_ADDR_FIFO_CON_REG	0x0514	CSIC DMA Channel1 BUF Address FIFO Content Register
CSIC_DMA_CH1_STORED_FRM_CNT_REG	0x0518	CSIC DMA Channel1 Stored Frame Counter Register
CSIC_DMA_CH1_TOT_FRM_CNT_REG	0x051C	CSIC DMA Channel1 Total Frame Counter Register
CSIC_DMA_CH1_FRM_LOST_CNT_REG	0x0524	CSIC DMA Channel1 Frame Lost Counter Register
CSIC_DMA_CH1_EMBED_ADDR_READ_REG	0x0528	CSIC DMA Channel1 Embed Address Read Register

14.1.7 CSIC CCU Register Description

14.1.7.1 0x0000 CSIC CCU Clock Mode Register (Default Value:0x0060_0003)

Offset: 0x0000			Register Name: CSIC_CCU_CLK_MODE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CCU_CLK_GATING_DISABLE CCU Clock Gating Disable 0: CCU Clock Gating Registers(0x0004-0x0010) effect 1: CCU Clock Gating Registers(0x0004-0x0010) not effect
30:28	/	/	/

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Offset: 0x0000			Register Name: CSIC_CCU_CLK_MODE_REG
Bit	Read/Write	Default/Hex	Description
27	R/W	0x0	MCSI_HCLK_AUTOGATE_EN MCSI Hclk Autogate Enable 0: Disable 1: Enable
26:24	/	/	/
23:20	R/W	0x6	MCSI_HCLK_AUTOGATE_CYCLE MCSI Hclk Autogate cycle If MCSI_HCLK_AUTOGATE_EN is enabled, HCLK is automatically turned off after MCSI_HCLK_AUTOGATE_CYCLE+2 clock cycles of software stopping access to the CSIC registers. Note: The value of this bit should be larger than or equal to 6. Otherwise, the register configuration error will occur when HCLK is auto-gated.
19:2	/	/	/
1	R/W	0x1	MCSI_POST_CLK_MODE MCSI Post Clock Mode 0: CSI Post works in ISP core clock 1: CSI Post works in CSI clock
0	R/W	0x1	MCSI_PARSER_CLK_MODE MCSI Parser Clock Mode 0: CSI Parser works in ISP core clock 1: CSI Parser works in CSI clock

14.1.7.2 0x0004 CSIC CCU Parser Clock Enable Register (Default Value:0x0000_0000)

Offset: 0x0004			Register Name: CSIC_CCU_PARSER_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	MCSI_COMBO0_CLK_ENABLE MCSI Combo0 Clock Enable 0: Combo0 clock disable 1: Combo0 clock enable
7:3	/	/	/
2	R/W	0x0	MCSI_PARSER2_CLK_ENABLE MCSI Parser2 Clock Enable 0: CSI Parser2 clock disable 1: CSI Parser2 clock enable

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Offset: 0x0004			Register Name: CSIC_CCU_PARSER_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	MCSI_PARSER1_CLK_ENABLE MCSI Parser1 Clock Enable 0: CSI Parser1 clock disable 1: CSI Parser1 clock enable
0	R/W	0x0	MCSI_PARSER0_CLK_ENABLE MCSI Parser0 Clock Gating 0: CSI Parser0 clock disable 1: CSI Parser0 clock enable

14.1.7.3 0x0008 CSIC CCU ISP Clock Enable Register (Default Value:0x0000_0000)

Offset: 0x0008			Register Name: CCU_ISP_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:26	R/W	0x0	MISP1_S2F1_BRIDGE_CH_CLK_EN Async Bridge Channel 0-3 from ISP1 to Post Clock Enable Bit [24]: Channel0 Bit [25]: Channel1 Bit [26]: Channel2 Bit [27]: Channel3 0: Disable 1: Enable
25:22	R/W	0x0	MISP1_F2S1_BRIDGE_CH_CLK_EN Async Bridge Channel 0-3 from Parser to ISP1 Clock Enable Bit [20]: Channel0 Bit [21]: Channel1 Bit [22]: Channel2 Bit [23]: Channel3 0: Disable 1: Enable
21	R/W	0x0	MISP1_BRIDGE_CH_CLK_GATING_DISABLE Async Bridge Channel Clock Gating Disable 0: Async Bridge channel 0-3 Registers (MISP1_F2S1_BRIDGE_CH_CLK_EN and MISP1_S2F1_BRIDGE_CH_CLK_EN) effect 1: Async Bridge channel 0-3 Registers (MISP1_F2S1_BRIDGE_CH_CLK_EN and MISP1_S2F1_BRIDGE_CH_CLK_EN) not effect

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Offset: 0x0008			Register Name: CCU_ISP_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
20	R/W	0x0	MISP0_BRIDGE_CH_CLK_GATING_DISABLE Async Bridge Channel Clock Gating Disable 0: Async Bridge channel 0-3 Registers (MISP0_F2S0_BRIDGE_CH_CLK_EN and MISP0_S2F0_BRIDGE_CH_CLK_EN) effect 1: Async Bridge channel 0-3 Registers (MISP0_F2S0_BRIDGE_CH_CLK_EN and MISP0_S2F0_BRIDGE_CH_CLK_EN) not effect
19:16	R/W	0x0	MISP0_S2F0_BRIDGE_CH_CLK_EN Async Bridge Channel 0-3 from ISP to Post Clock Enable Bit [16]: Channel0 Bit [17]: Channel1 Bit [18]: Channel2 Bit [19]: Channel3 0: Disable 1: Enable
15:12	R/W	0x0	MISP0_F2S0_BRIDGE_CH_CLK_EN Async Bridge Channel 0-3 from Parser to ISP Clock Enable Bit [12]: Channel0 Bit [13]: Channel1 Bit [14]: Channel2 Bit [15]: Channel3 0: Disable 1: Enable
11:10	/	/	/
9	R/W	0x0	MISP_MBUS_MEM_RST MISP MBUS MEM Reset 0: ISP0 MBUS MEM reset de-assert 1: ISP0 MBUS MEM reset assert
8	R/W	0x0	MISP_MBUS_VE_RST MISP MBUS VE Reset 0: ISP0 MBUS reset de-assert 1: ISP0 MBUS reset assert
7:6	/	/	/
5	R/W	0x0	MISP1_BRIDGE_CLK_ENABLE MISP Bridge Clock Enable 0: ISP1 bridge clock disable 1: ISP1 bridge clock enable

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Offset: 0x0008			Register Name: CCU_ISP_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	MISP0_BRIDGE_CLK_ENABLE MISP Bridge Clock Enable 0: ISP0 bridge clock disable 1: ISP0 bridge clock enable
3:2	/	/	/
1	R/W	0x0	MISP1_CLK_ENABLE MISP Clock Enable 0: ISP1 clock disable 1: ISP1 clock enable
0	R/W	0x0	MISP0_CLK_ENABLE MISP Clock Enable 0: ISP0 clock disable 1: ISP0 clock enable

14.1.7.4 0x000C CSIC CCU Post0 Clock Enable Register (Default Value:0x0000_0000)

Offset: 0x000C			Register Name: CSIC_CCU_POST0_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R/W	0x0	MCSI_POST0_MBUS_MEM_RST MCSI Post0 MBUS MEM Reset 0: POST0 MBUS MEM reset de-assert 1: POST0 MBUS MEM reset assert
20	R/W	0x0	MCSI_POST0_MBUS_VE_RST MCSI Post0 MBUS VE Reset 0: POST0 MBUS reset de-assert 1: POST0 MBUS reset assert
19:17	/	/	/
16	R/W	0x0	MCSI_POST0_CLK_ENABLE Post0 clock gating 0: POST0 clock disable 1: POST0 clock enable
15:12	/	/	/
11	R/W	0x0	MCSI_VIPP3_CLK_ENABLE VIPP3 Clock Gating 0: VIPP3 clock disable 1: VIPP3 clock enable, when MCSI_POST0_CLK_ENABLE is set to 1

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Offset: 0x000C			Register Name: CSIC_CCU_POST0_CLK_EN_REG
Bit	Read/Write	Default/Hex	Description
10	R/W	0x0	MCSI_VIPP2_CLK_ENABLE VIPP2 Clock Gating 0: VIPP2 clock disable 1: VIPP2 clock enable, when MCSI_POST0_CLK_ENABLE is set to 1
9	R/W	0x0	MCSI_VIPP1_CLK_ENABLE VIPP1 Clock Gating 0: VIPP1 clock disable 1: VIPP1 clock enable, when MCSI_POST0_CLK_ENABLE is 1
8	R/W	0x0	MCSI_VIPP0_CLK_ENABLE VIPP0 Clock Gating 0: VIPP0 clock disable 1: VIPP0 clock enable, when MCSI_POST0_CLK_ENABLE is set to 1
7:4	/	/	/
3	R/W	0x0	MCSI_BK3_CLK_ENABLE BK3 Clock Gating 0: BK3 clock disable 1: BK3 clock enable, when MCSI_POST0_CLK_ENABLE is 1
2	R/W	0x0	MCSI_BK2_CLK_ENABLE BK2 Clock Gating 0: BK2 clock disable 1: BK2 clock enable, when MCSI_POST0_CLK_ENABLE is 1
1	R/W	0x0	MCSI_BK1_CLK_ENABLE BK1 Clock Gating 0: BK1 clock disable 1: BK1 clock enable, when MCSI_POST0_CLK_ENABLE is 1
0	R/W	0x0	MCSI_BK0_CLK_ENABLE BK0 Clock Gating 0: BK0 clock disable 1: BK0 clock enable, when MCSI_POST0_CLK_ENABLE is 1

14.1.7.5 0x0014 CSIC CCU Chfreq Clock Control Register (Default Value:0x0000_1700)

Offset: 0x0014			Register Name: CSIC_CCU_CHFREQ_CLK_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0014			Register Name: CSIC_CCU_CHFREQ_CLK_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x00	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
15:13	/	/	/
12:8	R/W	0x17	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.
7:1	/	/	/
0	R/W	0x0	CHFREQ_CLK_GATING. Gating Clock 0: Clock is OFF 1: Clock is ON CHFREQ_CLK = Clock Source/M/N. Clock Source is CLK24M.

14.1.7.6 0x0018 CSIC CCU BK Intpool Clock Control Register (Default Value:0x0000_1700)

Offset: 0x0018			Register Name: CSIC_CCU_BK_INTPOOL_CLK_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x00	FACTOR_N Factor N N= FACTOR_N +1 FACTOR_N is from 0 to 31.
15:13	/	/	/
12:8	R/W	0x17	FACTOR_M Factor M M= FACTOR_M +1 FACTOR_M is from 0 to 31.
7:1	/	/	/
0	R/W	0x0	BK_INTPOOL_CLK_GATING. Gating Clock 0: Clock is OFF 1: Clock is ON BK_INTPOOL_CLK = Clock Source/M/N. Clock Source is CLK24M.

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14.1.8 CSIC TOP Register Description

14.1.8.1 0x0000 CSIC TOP Enable Register (Default Value:0x0000_0000)

Offset: 0x0000			Register Name: CSIC_TOP_EN_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:26	R/W	0x0	ISP_S2F1_BRIDGE_CH_EN Enable Async Bridge channel 0-3 from isp1 to post, when ISP uses different clock source from csi_top_clk. Bit [24]: Channel0 Bit [25]: Channel1 Bit [26]: Channel2 Bit [27]: Channel3 0: Disable 1: Enable
25:22	R/W	0x0	ISP_F2S1_BRIDGE_CH_EN Enable Async Bridge channel 0-3 from parser to isp1, when ISP uses different clock source from csi_top_clk. Bit [20]: Channel0 Bit [21]: Channel1 Bit [22]: Channel2 Bit [23]: Channel3 0: Disable 1: Enable
21	R/W	0x0	ISP1_BRIDGE_CH_EN_DISABLE ISP Async Bridge Channel Disable 0: Async Bridge channel 0-3 Registers (ISP_S2F1_BRIDGE_CH_EN and ISP_F2S1_BRIDGE_CH_EN) effect 1: Async Bridge channel 0-3 Registers (ISP_S2F1_BRIDGE_CH_EN and ISP_F2S1_BRIDGE_CH_EN) not effect
20	R/W	0x0	ISP_BRIDGE_CH_EN_DISABLE ISP Async Bridge Channel Disable 0: Async Bridge channel 0-3 Registers (ISP_S2F0_BRIDGE_CH_EN and ISP_F2S0_BRIDGE_CH_EN) effect 1: Async Bridge channel 0-3 Registers (ISP_S2F0_BRIDGE_CH_EN and ISP_F2S0_BRIDGE_CH_EN) not effect

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Offset: 0x0000			Register Name: CSIC_TOP_EN_REG
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	ISP_S2F0_BRIDGE_CH_EN Enable Async Bridge channel 0-3 from ISP to post, when ISP uses different clock source from csi_top_clk. Bit [16]: Channel0 Bit [17]: Channel1 Bit [18]: Channel2 Bit [19]: Channel3 0: Disable 1: Enable
15:12	R/W	0x0	ISP_F2S0_BRIDGE_CH_EN Enable Async Bridge channel 0-3 from parser to ISP, when ISP uses different clock source from csi_top_clk. Bit [12]: Channel0 Bit [13]: Channel1 Bit [14]: Channel2 Bit [15]: Channel3 0: Disable 1: Enable
11:4	/	/	/
3	R/W	0x0	ISP_BRIDGE_EN Enable Async Bridge from parser to ISP and ISP to post, when ISP uses different clock source from csi_top_clk 0: Disable 1: Enable
2:0	/	/	/

14.1.8.2 0x0004 CSIC Pattern Generation Enable Register (Default Value:0x0000_0000)

Offset: 0x0004			Register Name: CSIC_PTN_GEN_EN_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PTN_CYCLE Pattern generating cycle counter. The pattern in DRAM will be generated in cycles of PTN_CYCLE+1.
15:5	/	/	/

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Offset: 0x0004			Register Name: CSIC_PTN_GEN_EN_REG
Bit	Read/Write	Default/Hex	Description
4	R/WAC	0x0	PTN_START CSIC Pattern Generating Start 0: Finish other: Start Software writes this bit to “1” to start pattern generating from DRAM. When finished, the hardware will clear this bit to “0” automatically. Generating cycles depends on PTN_CYCLE.
3:1	/	/	/
0	R/W	0x0	PTN_GEN_EN Pattern Generation Enable

14.1.8.3 0x0008 CSIC Pattern Control Register (Default Value:0x0000_000F)

Offset: 0x0008			Register Name: CSIC_PTN_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x0	PTN_PORT_SEL_COMBO_ALL_ENABLE Pattern Generator Output Port Selection Combo All Enable 0: To disable PTN_PORT_SEL, write 4’h0110. To enable PTN_PORT_SEL, write any value in range from 4’h0111 to 4’h1001. 1: To disable PTN_PORT_SEL, write any value in range from 4’h0111 to 4’h1001. To enable PTN_PORT_SEL, write 4’h0110.

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Offset: 0x0008			Register Name: CSIC_PTN_CTRL_REG
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	PTN_PORT_SEL Pattern Generator Output Port Selection 0010: NCSIC in Parser0 0110: COMBO ALL (All PHY) 0111: COMBO PHYA 1000: COMBO PHYB Others: Reserved Write 1 to the PTN_PORT_SEL_COMBO_ALL_ENABLE bit before configuring this bit as 4'h0110 (COMBO ALL). Write 0 to the PTN_PORT_SEL_COMBO_ALL_ENABLE bit before configuring this bit as any value in range from 4'h0111 to 4'h1001 (COMBO PHYA/B).
23:22	/	/	/
21:20	R/W	0x0	PTN_GEN_DATA_WIDTH Pattern Generator output data width 00: 8bit 01: 10bit 10: 12bit 11: Reserved
19:16	R/W	0x0	PTN_MODE Pattern Mode Selection 0000: MIPI 1-lane 0001: MIPI 2-lane 0010: MIPI 3-lane 0011: MIPI 4-lane 0100: NCSIC with max 12-bit data ({field, vsyn, hsyn, 1'b0, data [11:0]}) 0101: NCSIC with max 16-bit data ({12'h0, field, vsyn, hsyn, 1'b0, data [15:0]}) 0110: NCSIC with max 24-bit data ({field, vsyn, hsyn, 5'h0, data [23:0]}) 0111: Reserved 1000: BT656 8 bits' width 1001: BT656 16 bits' width 1010: BT656 24 bits' width 1011: Reserved 1100: BAYER 12 bits for ISPFE 1101: UYVY422 12 bits for ISPFE 1110: UYVY420 12 bits for ISPFE 1111: Reserved

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Offset: 0x0008			Register Name: CSIC_PTN_CTRL_REG
Bit	Read/Write	Default/Hex	Description
15:10	/	/	/
9:8	R/W	0x0	PTN_GEN_CLK_DIV Packet Generator Clock Divider
7:0	R/W	0xF	PTN_GEN_DLY Clocks Delayed Before Pattern Generating Start.

14.1.8.4 0x0010 CSIC Pattern Vblank Cycle Register (Default Value:0x0000_0000)

Offset: 0x0010			Register Name: CSIC_PTN_VBLANK_CYCLE
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	PTN_VBLANK_CYCLE Set clock cycles between two pattern as vblank

14.1.8.5 0x0020 CSIC Pattern Generation Length Register (Default Value:0x0000_0000)

Offset: 0x0020			Register Name: CSIC_PTN_LEN_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PTN_LEN The pattern length in byte when generating pattern.

14.1.8.6 0x0024 CSIC Pattern Generation Address Register (Default Value:0x0000_0000)

Offset: 0x0024			Register Name: CSIC_PTN_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PTN_ADDR The pattern DRAM address when generating pattern.

14.1.8.7 0x0028 CSIC Pattern ISP Size Register (Default Value:0x0000_0000)

Offset: 0x0028			Register Name: CSIC_PTN_ISP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	Height Vertical size, only valid for ISP mode pattern generation.
15:13	/	/	/

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Offset: 0x0028			Register Name: CSIC_PTN_ISP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
12:0	R/W	0x0	Width Horizontal size, only valid for ISP mode pattern generation.

14.1.8.8 0x0030 CSIC ISP0 Input0 Select Register (Default Value:0x0000_0000)

Offset :0x0030			Register Name: CSIC_ISP0_INPUT0_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x0	ISP0_INPUT0_SEL ISP0 Input0 Select 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.9 0x0034 CSIC ISP0 Input1 Select Register (Default Value:0x0000_0001)

Offset :0x0034			Register Name: CSIC_ISP0_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset :0x0034			Register Name: CSIC_ISP0_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x1	ISP0_INPUT1_SEL ISP0 Input1 Select 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.10 0x0038 CSIC ISP0 Input2 Select Register (Default Value:0x0000_0002)

Offset :0x0038			Register Name: CSIC_ISP0_INPUT2_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x2	ISP0_INPUT2_SEL ISP0 Input2 Select 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.11 0x003C CSIC ISP0 Input3 Select Register (Default Value:0x0000_0003)

Offset :0x003C			Register Name: CSIC_ISP0_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description

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Offset :0x003C			Register Name: CSIC_ISP0_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x3	ISP0_INPUT3_SEL ISP0 Input3 Select 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.12 0x0040 CSIC ISP1 Input0 Select Register (Default Value:0x0000_0004)

Offset :0x0040			Register Name: CSIC_ISP1_INPUT0_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x4	ISP1_INPUT0_SEL ISP1 Input0 Select (note that no ISP1 exists, ISP1 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

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14.1.8.13 0x0044 CSIC ISP1 Input1 Select Register (Default Value:0x0000_0005)

Offset :0x0044			Register Name: CSIC_ISP1_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x5	ISP1_INPUT1_SEL ISP1 Input1 Select (note that no ISP1 exists, ISP1 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.14 0x0048 CSIC ISP1 Input2 Select Register (Default Value:0x0000_0006)

Offset :0x0048			Register Name: CSIC_ISP1_INPUT2_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset :0x0048			Register Name: CSIC_ISP1_INPUT2_SEL_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x6	ISP1_INPUT2_SEL ISP1 Input2 Select (note that no ISP1 exists, ISP1 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.15 0x004C CSIC ISP1 Input3 Select Register (Default Value:0x0000_0007)

Offset :0x004C			Register Name: CSIC_ISP1_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x7	ISP1_INPUT3_SEL ISP1 Input3 Select (note that no ISP1 exists, ISP1 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

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14.1.8.16 0x0050 CSIC ISP2 Input0 Select Register (Default Value:0x0000_0008)

Offset: 0x0050			Register Name: CSIC_ISP2_INPUT0_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x8	ISP2_INPUT0_SEL ISP2 Input0 Select (note that no ISP2 exists, ISP2 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.17 0x0054 CSIC ISP2 Input1 Select Register (Default Value:0x0000_0009)

Offset: 0x0054			Register Name: CSIC_ISP2_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x0054			Register Name: CSIC_ISP2_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x9	ISP2_INPUT1_SEL ISP2 Input1 Select (note that no ISP2 exists, ISP2 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.18 0x0058 CSIC ISP2 Input2 Select Register (Default Value:0x0000_000a)

Offset: 0x0058			Register Name: CSIC_ISP2_INPUT2_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0xA	ISP2_INPUT2_SEL ISP2 Input2 Select (note that no ISP2 exists, ISP2 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

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14.1.8.19 0x005C CSIC ISP2 Input3 Select Register (Default Value:0x0000_000b)

Offset: 0x005C			Register Name: CSIC_ISP2_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0xb	ISP2_INPUT3_SEL ISP2 Input3 Select (note that no ISP2 exists, ISP2 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.20 0x0060 CSIC ISP3 Input0 Select Register (Default Value:0x0000_000c)

Offset: 0x0060			Register Name: CSIC_ISP3_INPUT0_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x0060			Register Name: CSIC_ISP3_INPUT0_SEL_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xc	ISP3_INPUT0_SEL ISP3 Input0 Select (note that no ISP3 exists, ISP3 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.21 0x0064 CSIC ISP3 Input1 Select Register (Default Value:0x0000_000d)

Offset: 0x0064			Register Name: CSIC_ISP3_INPUT1_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0xd	ISP3_INPUT1_SEL ISP3 Input1 select (note that no ISP3 exists, ISP3 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

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14.1.8.22 0x0068 CSIC ISP3 Input2 Select Register (Default Value:0x0000_000e)

Offset: 0x0068			Register Name: CSIC_ISP3_INPUT2_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0xe	ISP3_INPUT2_SEL ISP3 Input2 Select (note that no ISP3 exists, ISP3 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.23 0x006C CSIC ISP3 Input3 Select Register (Default Value:0x0000_000f)

Offset: 0x006C			Register Name: CSIC_ISP3_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x006C			Register Name: CSIC_ISP3_INPUT3_SEL_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xf	ISP3_INPUT3_SEL ISP3 Input3 Select (note that no ISP3 exists, ISP3 is a virtual concept here and referring to a pass-through channel) 0000: Input from Parser0 CH0 0001: Input from Parser0 CH1 0010: Input from Parser0 CH2 0011: Input from Parser0 CH3 0100: Input from Parser1 CH0 0101: Input from Parser1 CH1 0110: Input from Parser1 CH2 0111: Input from Parser1 CH3 1000: Input from Parser2 CH0 1001: Input from Parser2 CH1 1010: Input from Parser2 CH2 1011: Input from Parser2 CH3 Others: Reserved

14.1.8.24 0x0070 CSIC ISP Bridge Buffer Maxuse Counter Clear Register (Default Value:0x0000_0000)

Offset :0x0070			Register Name: CSIC_ISP_BRG_BUF_MAXUSE_CNT_CLR_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	ISP0_BRG3_BUF_MAXUSE_CNT_CLR ISP0 Bridge3 Buffer Maxuse Count Clear 0: No effect 1: Clear
2	R/W	0x0	ISP0_BRG2_BUF_MAXUSE_CNT_CLR ISP0 Bridge2 Buffer Maxuse Count Clear 0: No effect 1: Clear
1	R/W	0x0	ISP0_BRG1_BUF_MAXUSE_CNT_CLR ISP0 Bridge1 Buffer Maxuse Count Clear 0: No effect 1: Clear
0	R/W	0x0	ISP0_BRG0_BUF_MAXUSE_CNT_CLR ISP0 Bridge0 Buffer Maxuse Count Clear 0: No effect 1: Clear

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14.1.8.25 0x0074 CSIC ISP0 Bridge01 Buffer Maxuse Counter Register (Default Value:0x0000_0000)

Offset :0x0074			Register Name: CSIC_ISP0_BRG01_BUF_MAXUSE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	ISP0_BRG1_BUF_MAXUSE_CNT ISP0 Bridge1 Buffer Maxuse Counter
15:0	R	0x0	ISP0_BRG0_BUF_MAXUSE_CNT ISP0 Bridge0 Buffer Maxuse Counter

14.1.8.26 0x0078 CSIC ISP0 Bridge23 Buffer Maxuse Counter Register (Default Value:0x0000_0000)

Offset :0x0078			Register Name: CSIC_ISP0_BRG23_BUF_MAXUSE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	ISP0_BRG3_BUF_MAXUSE_CNT ISP0 Bridge3 Buffer Maxuse Counter
15:0	R	0x0	ISP0_BRG2_BUF_MAXUSE_CNT ISP0 Bridge2 Buffer Maxuse Counter

14.1.8.27 0x0084 CSIC ISP0 Bridge Interrupt Enable Register (Default Value:0x0000_0000)

Offset :0x0084			Register Name: CSIC_ISP0_BRG_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W	0x0	ISP0_BRG3_S2F_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge3 Slow to Fast Side
26	R/W	0x0	ISP0_BRG2_S2F_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge2 Slow to Fast Side
25	R/W	0x0	ISP0_BRG1_S2F_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge1 Slow to Fast Side
24	R/W	0x0	ISP0_BRG0_S2F_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge0 Slow to Fast Side
23:20	/	/	/
19	R/W	0x0	ISP0_BRG3_F2S_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge3 Fast to Slow Side

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Offset :0x0084			Register Name: CSIC_ISP0_BRG_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	ISP0_BRG2_F2S_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge2 Fast to Slow Side
17	R/W	0x0	ISP0_BRG1_F2S_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge1 Fast to Slow Side
16	R/W	0x0	ISP0_BRG0_F2S_LW_MISMTCH_INT_EN Picture Line Width Mismatch detect in ISP0 Bridge0 Fast to Slow Side
15:12	/	/	/
11	R/W	0x0	ISP0_BRG3_BUF_OV_INT_EN ISP0 Bridge3 Buffer overflow interrupt enable
10	R/W	0x0	ISP0_BRG2_BUF_OV_INT_EN ISP0 Bridge2 Buffer overflow interrupt enable
9	R/W	0x0	ISP0_BRG1_BUF_OV_INT_EN ISP0 Bridge1 Buffer overflow interrupt enable
8	R/W	0x0	ISP0_BRG0_BUF_OV_INT_EN ISP0 Bridge0 Buffer overflow interrupt enable
7:4	/	/	/
3	R/W	0x0	ISP0_BRG3_RS_INT_EN ISP0 Bridge3 Read clock too slow interrupt enable
2	R/W	0x0	ISP0_BRG2_RS_INT_EN ISP0 Bridge2 Read clock too slow interrupt enable
1	R/W	0x0	ISP0_BRG1_RS_INT_EN ISP0 Bridge1 Read clock too slow interrupt enable
0	R/W	0x0	ISP0_BRG0_RS_INT_EN ISP0 Bridge0 Read clock too slow interrupt enable

14.1.8.28 0x008C CSIC ISP0 Bridge Interrupt Pending Register (Default Value:0x0000_0000)

Offset :0x008C			Register Name: CSIC_ISP0_BRG_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W1C	0x0	ISP0_BRG3_S2F_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge3 Slow to Fast Side
26	R/W1C	0x0	ISP0_BRG2_S2F_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge2 Slow to Fast Side

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Offset :0x008C			Register Name: CSIC_ISP0_BRG_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
25	R/W1C	0x0	ISP0_BRG1_S2F_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge1 Slow to Fast Side
24	R/W1C	0x0	ISP0_BRG0_S2F_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge0 Slow to Fast Side
23:20	/	/	/
19	R/W1C	0x0	ISP0_BRG3_F2S_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge3 Fast to Slow Side
18	R/W1C	0x0	ISP0_BRG2_F2S_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge2 Fast to Slow Side
17	R/W1C	0x0	ISP0_BRG1_F2S_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge1 Fast to Slow Side
16	R/W1C	0x0	ISP0_BRG0_F2S_LW_MISMTCH_INT_PD Picture Line Width Mismatch detect in ISP0 Bridge0 Fast to Slow Side
15:12	/	/	/
11	R/W1C	0x0	ISP0_BRG3_BUF_OV_INT_PD ISP0 Bridge3 Buffer overflow interrupt pending
10	R/W1C	0x0	ISP0_BRG2_BUF_OV_INT_PD ISP0 Bridge2 Buffer overflow interrupt pending
9	R/W1C	0x0	ISP0_BRG1_BUF_OV_INT_PD ISP0 Bridge1 Buffer overflow interrupt pending
8	R/W1C	0x0	ISP0_BRG0_BUF_OV_INT_PD ISP0 Bridge0 Buffer overflow interrupt pending
7:4	/	/	/
3	R/W1C	0x0	ISP0_BRG3_RS_INT_PD ISP0 Bridge3 Read clock too slow interrupt pending
2	R/W1C	0x0	ISP0_BRG2_RS_INT_PD ISP0 Bridge2 Read clock too slow interrupt pending
1	R/W1C	0x0	ISP0_BRG1_RS_INT_PD ISP0 Bridge1 Read clock too slow interrupt pending
0	R/W1C	0x0	ISP0_BRG0_RS_INT_PD ISP0 Bridge0 Read clock too slow interrupt pending

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14.1.8.29 0x00A0 CSIC VIPP0 Input Select Register (Default Value:0x0000_0000)

Offset :0x00A0			Register Name: CSIC_VIPP0_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4:0	R/W	0x0	VIPP0_INPUT_SEL VIPP0 Input Select 00000: Input from ISP0 CH0 00001: Input from ISP0 CH1 00010: Input from ISP0 CH2 00011: Input from ISP0 CH3 00100: Input from ISP1 CH0 00101: Input from ISP1 CH1 00110: Input from ISP1 CH2 00111: Input from ISP1 CH3 01000: Input from ISP2 CH0 01001: Input from ISP2 CH1 01010: Input from ISP2 CH2 01011: Input from ISP2 CH3 01100: Input from ISP3 CH0 01101: Input from ISP3 CH1 01110: Input from ISP3 CH2 01111: Input from ISP3 CH3 10000: Reserved 10001: Input from VIPP1 No ISP1/2/3 exists, ISP1/2/3 is a virtual concept here and referring to a pass-through channel. This register should be configured together with CSIC_ISP1/2/3_INPUT0/1/2/3_SEL_REG.

14.1.8.30 0x00A4 CSIC VIPP1 Input Select Register (Default Value:0x0000_0000)

Offset :0x00A4			Register Name: CSIC_VIPP1_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/

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Offset :0x00A4			Register Name: CSIC_VIPP1_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	VIPP1_INPUT_SEL VIPP1 Input Select 00000: Input from ISP0 CH0 00001: Input from ISP0 CH1 00010: Input from ISP0 CH2 00011: Input from ISP0 CH3 00100: Input from ISP1 CH0 00101: Input from ISP1 CH1 00110: Input from ISP1 CH2 00111: Input from ISP1 CH3 01000: Input from ISP2 CH0 01001: Input from ISP2 CH1 01010: Input from ISP2 CH2 01011: Input from ISP2 CH3 01100: Input from ISP3 CH0 01101: Input from ISP3 CH1 01110: Input from ISP3 CH2 01111: Input from ISP3 CH3 10000: Input from VIPP0 10001: Reserved No ISP1/2/3 exists, ISP1/2/3 is a virtual concept here and referring to a pass-through channel. This register should be configured together with CSIC_ISP1/2/3_INPUT0/1/2/3_SEL_REG.

14.1.8.31 0x00A8 CSIC VIPP2 Input Select Register (Default Value:0x0000_0000)

Offset :0x00A8			Register Name: CSIC_VIPP2_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/

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Offset :0x00A8			Register Name: CSIC_VIPP2_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	VIPP2_INPUT_SEL VIPP2 Input Select 00000: Input from ISP0 CH0 00001: Input from ISP0 CH1 00010: Input from ISP0 CH2 00011: Input from ISP0 CH3 00100: Input from ISP1 CH0 00101: Input from ISP1 CH1 00110: Input from ISP1 CH2 00111: Input from ISP1 CH3 01000: Input from ISP2 CH0 01001: Input from ISP2 CH1 01010: Input from ISP2 CH2 01011: Input from ISP2 CH3 01100: Input from ISP3 CH0 01101: Input from ISP3 CH1 01110: Input from ISP3 CH2 01111: Input from ISP3 CH3 Others: Reserved No ISP1/2/3 exists, ISP1/2/3 is a virtual concept here and referring to a pass-through channel. This register should be configured together with CSIC_ISP1/2/3_INPUT0/1/2/3_SEL_REG.

14.1.8.32 0x00D0 CSIC DMA0 Input Select Register (Default Value:0x0000_0000)

Offset: 0x00D0			Register Name: CSIC_DMA0_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	DMA0 INPUT SELECT 000: Input from VIPP0 001: Input from VIPP1 010: Input from VIPP2

14.1.8.33 0x00D4 CSIC DMA1 Input Select Register (Default Value:0x0000_0001)

Offset: 0x00D4			Register Name: CSIC_DMA1_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/

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Offset: 0x00D4			Register Name: CSIC_DMA1_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x1	DMA1 INPUT SELECT 000: Input from VIPP0 001: Input from VIPP1 010: Input from VIPP2

14.1.8.34 0x00D8 CSIC DMA2 Input Select Register (Default Value:0x0000_0002)

Offset: 0x00D8			Register Name: CSIC_DMA2_INPUT_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x2	DMA2 INPUT SELECT 000: Input from VIPP0 001: Input from VIPP1 010: Input from VIPP2

14.1.8.35 0x0100 CSIC Multi-Frame Mode Register (Default Value:0x0000_0000)

Offset: 0x0100			Register Name: CSIC_MULF_MOD_REG
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	MULF_STATUS Multi-Frame Mode Status indicates each DMA in frame done pending or not
23:16	/	/	/
15:8	R/W	0x0	MULF_CS Chip Selection for Multi-Frame Mode indicates which DMA is in selected
7:1	/	/	/
0	R/W	0x0	MULF_EN Multi-Frame Mode Enable

14.1.8.36 0x0104 CSIC Multi-Frame Interrupt Register (Default Value:0x0000_0000)

Offset: 0x0104			Register Name: CSIC_MULF_INT_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W1C	0x0	MULF_ERR_PD Multi-Frame Mode Frame Error Interrupt Pending
16	R/W1C	0x0	MULF_DONE_PD Multi-Frame Mode Frame Done Interrupt Pending
15:2	/	/	/

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Offset: 0x0104			Register Name: CSIC_MULF_INT_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	MULF_ERR_EN Multi-Frame Mode Frame Error Interrupt Enable
0	R/W	0x0	MULF_DONE_EN Multi-Frame Mode Frame Done Interrupt Enable

14.1.8.37 0x0108 CSIC Change Frequency Configuration 0 Register (Default Value:0x0001_0000)

Offset: 0x0108			Register Name: CSIC_CHFREQ_CFG0_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0001	CHFREQ_PAR2ISP_SEL Chfreq_par2isp select. Each bit corresponds to one channel. When the channel has data stream passing through, the corresponding bit must be set to 1. [19:16]->par2isp0 ch3, ch2, ch1, ch0 [23:20]->par2isp1 ch3, ch2, ch1, ch0 [27:24]->par2isp2 ch3, ch2, ch1, ch0 [31:28]->par2isp3 ch3,ch2,ch1,ch0
15:6	/	/	/
5	R/W	0x0	CHFREQ_BK_DONE_SRC Chfreq_bk done source select. 0: Select bk_frm_end as source 1: Select bk_frm_done as source
4	R/W	0x0	CHFREQ_PULSE_CNT_SCLR_EN Chfreq_pulse Cent Self-Clear Enable 0: Disable pulse cnt self clear 1: Enable pulse cnt self clear
3	R/W	0x0	CHFREQ_RDY_MASK 0: Unmask 1: Mask assert, keep chfreq_rdy output as 0
2	R/W	0x0	CHFREQ_SELECT Chfreq_rdy output source select 0: New implementation 1: Old implementation
1	R/W	0x0	CHFREQ_ALLOW Chfreq Allow Signal 0: Keep chfreq_rdy output as 0 1: Allow chfreq logic when CHFREQ_EN is asserted or all of bk*_en, mvipp*_en, and isp_en are set as 0, CHFREQ_ALLOW will not take effect.

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Offset: 0x0108			Register Name: CSIC_CHFREQ_CFG0_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CHFREQ_RST Soft reset for chfreq logic. 0: Assert 1: De-assert

14.1.8.38 0x010C CSIC Change Frequency Configuration 1 Register (Default Value:0x0000_0064)

Offset: 0x010C			Register Name: CSIC_CHFREQ_CFG1_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0000	CHFREQ_WAIT_DONE_TIME The time between hardware chfreq frm_done assert and MCSI chfrq_rdy assert. Take one chfreq_clk cycle as the unit, which is 1 us by default.
15:0	R/W	0x0064	CHFREQ_DDR_TIME The time for DDR to complete frequency change. Take one chfreq_clk cycle as a unit, which is 1 us by default.

14.1.8.39 0x0110 CSIC Change Frequency Observation Register (Default Value:0x0000_0000)

Offset: 0x0110			Register Name: CSIC_CHFREQ_OBS_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0000	CHFREQ_FRM_DONE_TIME The time for frm_done after frm_end, measured by hardware. Take one chfreq_clk cycle as the unit, which by default is 1 us.
15:0	R	0x0000	CHFREQ_VBLANK_LENGTH Vblank length measured by hardware. Take one chfreq_clk cycle as the unit, which by default is 1 us.

14.1.8.40 0x0114 CSIC Change Frequency Interrupt Register (Default Value:0x0000_0000)

Offset: 0x0114			Register Name: CSIC_CHFREQ_INT_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/

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Offset: 0x0114			Register Name: CSIC_CHFREQ_INT_REG
Bit	Read/Write	Default/Hex	Description
18	R/W1C	0x0	CHFREQ_FRM_DONE_NUM_ERR_PD Change Frequency Frame Done Number Error Interrupt Pending
17	R/W1C	0x0	CHFREQ_FRM_END_NUM_ERR_PD Change Frequency Frame End Number Error Interrupt Pending
16	R/W1C	0x0	CHFREQ_FRM_STR_NUM_ERR_PD Change Frequency Frame Start Number Error Interrupt Pending
15:3	/	/	/
2	R/W	0x0	CHFREQ_FRM_DONE_NUM_ERR_EN Change Frequency Frame Done Number Error Interrupt Enable
1	R/W	0x0	CHFREQ_FRM_END_NUM_ERR_EN Change Frequency Frame End Number Error Interrupt Enable
0	R/W	0x0	CHFREQ_FRM_STR_NUM_ERR_EN Change Frequency Frame Start Number Error Interrupt Enable

14.1.8.41 0x0118 CSIC Change Frequency Debug Observation Register (Default Value:0x0000_2101)

Offset: 0x0118			Register Name: CSIC_CHFREQ_DBG_OBS_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R	0x0	CHFREQ_RSTN_T Chfreq Resetn
26	R	0x0	ISP_RSTN_T ISP Resetn
25	R	0x0	PAR_RSTN_T Parser Resetn
24	R	0x0	POST_RSTN_T Post Resetn
23	R	0x0	ISP_EN_POST_S ISP Enable Post
22:18	R	0x0	FRM_DONE_SRC_NUM Frame Done Source Number
17:13	R	0x1	FRM_STR_SRC_NUM Frame Start Source Number
12:8	R	0x1	FRM_END_SRC_NUM Frame End Source Number

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Offset: 0x0118			Register Name: CSIC_CHFREQ_DBG_OBS_REG
Bit	Read/Write	Default/Hex	Description
7	R	0x0	POST_EN_S POST Enable After Sync
6	R	0x0	ISP_EN_S ISP Enable After Sync
5	R	0x0	CHFREQ_ALLOW_S Chfreq Allow After Sync
4	R	0x0	FIRST_FRM_END_RECEIVED First Frm End Received
3	R	0x0	VBLANK_LENTH_VALID Vblank Length Valid
2	R	0x0	IN_VBLANK In Vblank
1	R	0x0	FRM_DONE_TIME_VALID Frame Done Time Valid
0	R	0x1	CHFREQ_RDY Chfreq Ready

14.1.8.42 0x0120 CSIC ISP Bypass Software Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0120			Register Name: CSIC_ISP_BYP_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	MISP_BYP_EN ISP Bypass Enable 0: Disable 1: Enable
0	R/W	0x0	MISP_BYP_SEL Select the Way to Bypass ISP 0: Hardware 1: Software

14.1.8.43 0x01E0 CSIC MBUS REQ MAX Register (Default Value:0x0F03_0F0F)

Offset:0x01E0			Register Name: CSIC_MBUS_REQ_MAX_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0F	MISP_MEM_LNUM_REQ_MAX Maximum of request commands for the master granted in MISP_MEM_LNUM arbiter is N+1.
23:21	/	/	/

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Offset:0x01E0			Register Name: CSIC_MBUS_REQ_MAX_REG
Bit	Read/Write	Default/Hex	Description
20:16	R/W	0x03	MISP_MEM_REQ_MAX Maximum of request commands for the master granted in MISP_MEM arbiter is N+1.
15:13	/	/	/
12:8	R/W	0x0F	MCSI_MEM_1_REQ_MAX Maximum of request commands for the master granted in MCSI_MEM_1 arbiter is N+1.
7:5	/	/	/
4:0	R/W	0x0F	MCSI_MEM_REQ_MAX Maximum of request commands for the master granted in MCSI_MEM arbiter is N+1.

14.1.8.44 0x01E8 CSIC MBUS WAIT MAX Register (Default Value:0x0008_0000)

Offset:0x01E8			Register Name: CSIC_MBUS_WAIT_MAX_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x08	MISP_MEM_WAIT_MAX Maximum of wait cycles for the master granted in MISP_MEM arbiter is N when N is 0 it waits until a request accepted.
15:8	R/W	0x00	MCSI_MEM_1_WAIT_MAX Maximum of wait cycles for the master granted in MCSI_MEM_1 arbiter is N when N is 0 it waits until a request accepted.
7:0	R/W	0x00	MCSI_MEM_WAIT_MAX Maximum of wait cycles for the master granted in MCSI_MEM arbiter is N when N is 0 it waits until a request accepted.

14.1.8.45 0x01EC CSIC MBUS DEBUG Register (Default Value:0x0001_0101)

Offset:0x01EC			Register Name: CSIC_MBUS_DEBUG_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R	0x1	MISP_MBUS_IDLE
15:9	/	/	/
8	R	0x1	MCSI_POST1_MBUS_IDLE
7:1	/	/	/
0	R	0x1	MCSI_POST0_MBUS_IDLE

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14.1.9 PHY COMMON Register Description

14.1.9.1 0x0000 PHY Top Control Register (Default Value: 0x0036_0500)

Offset: 0x0000			Register Name: PHY_TOP_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30:29	R/W	0x0	PHY_LINK_MODE PHY Link Selection 2'b00: 2x2 -lane 2'b01: 1x4 -lane(PHYA+PHYB) Link Others: Reserved
28:27	/	/	/
26:24	R/W	0x0	PRBS_SEL PRBS Test Interface Selection 0x0: PHYA PRBS 0x1: PHYB PRBS Other: Reserved
23:16	R	0x36	Reserved
15:13	/	/	/
12	R/W	0x0	PHY_SEL_VREF_0P2 PHY 0.2V Terminal Resistor Reference Voltage Source Selection 0: The reference voltage comes from PHY. 1: The reference voltage comes from the external.
11:10	R/W	0x1	PHY_VREF_0P2 PHY 0.2V Terminal Resistor Reference Voltage Fine Trimming Every step is 25 mV.
9:8	R/W	0x1	PHY_VREF_0P9 PHY 0.9V LDO Reference Voltage Fine Trimming Every step is 50 mV.
7:4	/	/	/
3	R/W	0x0	PHY_LVLDO_EN PHY 0.9V LDO Enable (High Active)
2	R/W	0x0	PHY_VREF_EN PHY Reference Voltage Enable (High Active)
1	R/W	0x0	PHY_RSTN PHY Analog Layer Digital Circuit Reset (Low Active)
0	R/W	0x0	PHY_PWDNZ PHY Analog Layer Power Reset (Low Active)

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14.1.9.2 0x0004 PHY Terminal Resistor Calibration Register (Default Value: 0x0000_0002)

Offset:0x0004			Register Name: PHY_TRESCAL_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R	0x0	PHY_TRESCAL_RESULT PHY Terminal Resistor Calibration Result In software calibration mode, the result is the calibration value set via software.
15:4	/	/	/
3	R	0x0	PHY_TRESCAL_FLAG PHY Terminal Resistor Calibration Flag This bit is fixed as 1 in software calibration mode.
2	/	/	/
1	R/W	0x1	PHY_TRESCAL_SOFT PHY Terminal Resistor Software Calibration Enable (High Active)
0	/	/	/

14.1.10 PHY DIGITAL LAYER Register Description

14.1.10.1 0x0000 PHY Control Register (Default Value: 0x0004_0008)

Offset: 0x0000			Register Name: PHY_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	PHY0_WORK_MODE 0: Every lane works in MIPI mode. Others: Reserved
27	/	/	/
26	R/W	0x0	PHY0_IBIAS_EN PHY0 Reference Current Enable (High Active)
25:20	/	/	/
19	R/W	0x0	PHY0_VCM PHY0 Common Voltage Selection 0: 200mV (MIPI) 1: Reserved
18:16	R/W	0x4	PHY0_HS_REFI Obtain faster speed of every lane by adjusting PHY0 HSRX reference current. The larger the value of this bit, the faster the speed of every lane.
15:13	/	/	/

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Offset: 0x0000			Register Name: PHY_CTL_REG
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PHY0_LP_REFI Adjust the threshold voltage window size by adjusting LPRX reference current, which is able to filter LP noise signals effectively.
11:9	/	/	/
8	R/W	0x0	PHY_LANECK_EN PHY Clock Lane Enable
7:6	/	/	/
5:4	R/W	0x0	PHY_LANEDT_EN PHY Data Lane0-1 Enable (High Active) Bit4: Data Lane0 Enable Bit5: Data Lane1 Enable
3	R/W	0x1	PHY_TRND_EN PHY TREND Status Enable (High Active)
2:1	/	/	/
0	R/W	0x0	PHY_EN PHY Digital Layer Enable (High Active)

14.1.10.2 0x0004 PHY Equalization Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: PHY_EQ_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x0	PHY_EQ_LANECK0 PHY clock lane0 comparator equalization compensation value
23:20	/	/	/
19:18	R/W	0x0	PHY_EQ_LANED1 PHY data lane1 comparator equalization compensation value
17:16	R/W	0x0	PHY_EQ_LANED0 PHY data lane0 comparator equalization compensation value
15:5	/	/	/
4	R/W	0x0	PHY_EQ_CK_EN PHY clock lane comparator equalization enable (High Active)
3:2	/	/	/

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Offset: 0x0004			Register Name: PHY_EQ_REG
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	PHY_EQ_DT_EN PHY data lane0-1 comparator equalization enable (high active) Bit0: Data Lane0 Enable Bit1: Data Lane1 Enable

14.1.10.3 0x0008 PHY Offset Calibration Register0 (Default Value: 0x0010_0010)

Offset: 0x0008			Register Name: PHY_OFSCAL_REG0
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R	0x0	PHY0_OFSCAL_FLAG PHY0 Offset Calibration Flag In automatic calibration mode, when the value of this bit is 1, the automatic calibration is completed. This bit is fixed as 1 in software calibration mode.
27:21	/	/	/
20	R/W	0x1	PHY0_OFSCAL_RESETN PHY0 Offset Calibration Reset The comparator offset of every lane restarts to calibrate after reset. Note: This bit is only valid for automatic calibration mode.
19:13	/	/	/
12	R/W	0x0	PHY0_OFSCAL_SOFT PHY0 offset software calibration enable (high active)
7:5	/	/	/
4	R/W	0x1	PHY_OFSCAL_AUTO PHY0 offset automatic calibration enable (high active)
3:0	/	/	/

14.1.10.4 0x000C PHY Offset Calibration Register1 (Default Value: 0x0120_0000)

Offset: 0x000C			Register Name: PHY_OFSCAL_REG1
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/

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Offset: 0x000C			Register Name: PHY_OFSCAL_REG1
Bit	Read/Write	Default/Hex	Description
24:20	R/W	0x12	PHY0_OFSCAL_SET PHY0 offset software calibration set value
19:0	/	/	/

14.1.10.5 0x0010 PHY Offset Calibration Register2 (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: PHY_OFSCAL_REG2
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R	0x0	PHY0_OFSCAL_RESULT PHY0 offset calibration result In automatic calibration mode, the result is automatic calibration output value. In software calibration mode, the result is the value of PHY0_OFSCAL_SET bit.
19:0	/	/	/

14.1.10.6 0x0014 PHY Deskew Register0 (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: PHY_DESKEW_REG0
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	PHY0_DESKEW_PERIOD_STEP PHY0 data lane period deskew step precision 0: Every step is 1. 1: Every step is 2.
11:10	/	/	/
9:8	R/W	0x0	PHY0_DESKEW_STEP PHY0 data lane initial deskew step precision 00: Every step is 1. 01: Every step is 2. 10: Every step is 4. 11: Reserved
7:6	/	/	/
5:4	R/W	0x0	PHY_DESKEW_PERIOD_EN PHY data lane0-1 period deskew automatic delay enable (high active) Bit4: Data Lane0 Enable Bit5: Data Lane1 Enable
3:2	/	/	/

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Offset: 0x0014			Register Name: PHY_DESKEW_REG0
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	PHY_DESKEW_EN PHY data lane0-1 deskew automatic delay enable (high active) Bit0: Data Lane0 Enable Bit1: Data Lane1 Enable

14.1.10.7 0x0018 PHY Deskew Register1 (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: PHY_DESKEW_REG1
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R/W	0x0	PHY_DESKEW_LANECK0_SET PHY clock lane0 delay
19:10	/	/	/
9:5	R/W	0x0	PHY_DESKEW_LANED1_SET PHY data lane1 delay This bit is valid when PHY_DESKEW_EN[1] =0.
4:0	R/W	0x0	PHY_DESKEW_LANED0_SET PHY data lane0 delay This bit is valid when PHY_DESKEW_EN[0] =0.

14.1.10.8 0x001C PHY Deskew Register2 (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: PHY_DESKEW_REG2
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R	0x0	PHY_DESKEW_LANECK0_RESULT PHY clock lane0 deskew delay result The value of this bit is equal to the value of PHY_DESKEW_LANECK0_SET bit.
19:10	/	/	/
9:5	R	0x0	PHY_DESKEW_LANED1_RESULT PHY data lane1 deskew delay result The value of this bit is equal to the value of PHY_DESKEW_LANECK1_SET bit, when PHY_DESKEW_EN[1] =0.
4:0	R	0x0	PHY_DESKEW_LANED0_RESULT PHY data lane0 deskew delay result The value of this bit is equal to the value of PHY_DESKEW_LANECK0_SET bit, when PHY_DESKEW_EN[0] =0.

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14.1.10.9 0x0020 PHY Terminal Control Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: PHY_TERM_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PHY0_TERM_EN_DLY These bits are to set the delay of every clock lane and data lane from high-speed RX (LP11-LP01-LP00) request is received to the terminal resistor is enabled in hardware auto-enable mode. These bits are only valid for MIPI CSI. The unit is cmb_clk cycle.
15:5	/	/	/
4	R/W	0x0	PHY_TERMCK_EN PHY Clock Lane Terminal Resistor Software Enable 0: Hardware auto-enable 1: Software enable
3:2	/	/	/
1:0	R/W	0x0	PHY_TERMDT_EN PHY Data Lane 0-1 Terminal Resistor Software Enable Bit0: Data Lane0 Enable Bit1: Data Lane1 Enable 0: Hardware auto-enable 1: Software enable

14.1.10.10 0x0024 PHY High Speed Control Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: PHY_HS_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PHY0_HS_DLY These bits are to set the delay of every clock lane and data lane from high-speed RX (LP11-LP01-LP00) request is received to the high-speed comparator is enabled in hardware auto-enable mode. These bits are only valid for MIPI CSI. The unit is cmb_clk cycle.
15:13	/	/	/

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Offset: 0x0024			Register Name: PHY_HS_CTL_REG
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PHY_HSCK_POLAR PHY Clock Lane HSRX Signal Inversion 0: HSRX Signal is not inversed. 1: HSRX Signal is inversed.
11:10	/	/	/
9:8	R/W	0x0	PHY_HSDT_POLAR PHY Data Lane0-1 HSRX Signal Inversion Bit8: Data Lane0 Enable Bit9: Data Lane1 Enable 0: HSRX Signal is not inversed. 1: HSRX Signal is inversed.
7:5	/	/	/
4	R/W	0x0	PHY_HSCK_EN PHY Clock Lane High-Speed Mode Software Enable (High Active) 0: Hardware auto-enable 1: Software enable
3:2	/	/	/
1:0	R/W	0x0	PHY_HSDT_EN PHY Data Lane0-1 High-Speed Mode Software Enable (High Active) Bit0: Data Lane0 Enable Bit1: Data Lane1 Enable 0: Hardware auto-enable 1: Software enable

14.1.10.11 0x0028 PHY Serial to Parallel Control Register (Default Value: 0x0000_0A00)

Offset: 0x0028			Register Name: PHY_S2P_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PHY0_S2P_DLY These bits are to set the delay of every data lane from high-speed RX request is received to the S2P circuit is enabled in hardware auto-enable mode (HS_Settle). These bits are only valid for MIPI CSI. The unit is cmb_clk cycle.
15:12	/	/	/
11:10	R	0x2	Reserved

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Offset: 0x0028			Register Name: PHY_S2P_CTL_REG
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x2	PHY0_S2P_WIDTH PHY0 S2P Circuit Output Width 00: 2 bits 01: 4 bits 1X: 8 bits
7:2	/	/	/
1:0	R/W	0x0	PHY_S2P_EN PHY Data Lane0-1 S2P Circuit Software Enable Bit0: Data Lane0 Enable Bit1: Data Lane1 Enable 0: Hardware auto-enable 1: Software enable

14.1.10.12 0x002C PHY MIPIRX Control Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: PHY_MIPIRX_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	PHY0_MIPILP_DBC_EN PHY0 LPRX Debounce Circuit Enable (High Active)
15:13	/	/	/
12	R/W	0x0	PHY_MIPI_LPCK_EN PHY Clock Lane LPRX Enable 0: Disable LPRX 1: Enable LPRX
11:10	/	/	/
9:8	R/W	0x0	PHY_MIPI_LPDT_EN PHY Data Lane 0-1 LPRX Enable Bit8: Data Lane0 Enable Bit9: Data Lane1 Enable 0: Disable LPRX 1: Enable LPRX
7:5	/	/	/
4	R/W	0x0	PHY0_MIPIHS_8B9B 0: No 8B9B decoding when receiving bytes in PHY0 MIPI high-speed mode. 1: 8B9B decoding when receiving bytes in PHY0 MIPI high-speed mode.
3	/	/	/

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Offset: 0x002C			Register Name: PHY_MIPIRX_CTL_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PHY0_MIPIHS_SYNC_MODE 0: The Sync-Sequence must be 0xb8 in PHY0 MIPI high-speed mode. 1: The Sync-Sequence allows 1-bit difference from 0xb8 in PHY0 MIPI high-speed mode.
1	/	/	/
0	R/W	0x0	PHY0_MIPIHS_ENDIAN This bit is to set which one of the big-endian and little-endian to receive data in PHY0 MIPI high-speed mode. 0: LSB First 1: MSB First

14.1.10.13 0x0030 PHY MIPIRX Synchronization Timeout Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: PHY_MIPIRX_SYNC_TIMEOUT_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PHY0_MIPIHS_SYNC_TOSET The PHY_NOSYNC_ERR_D0_PD bit and PHY_NOSYNC_ERR_D1_PD bit will generate interrupt when the Sync-Sequence (0xb8) is unable to be detected after PHY0 data lane switching from LP mode to HS mode for a period of time configured by this bit. The unit is 16 byte_clk cycles.
15:1	/	/	/
0	R/W	0x0	PHY0_MIPIHS_SYNC_TOEN PHY0 data lane HS Sync-Sequence Detection Time Out Enable (High Active)

14.1.10.14 0x0034 PHY Frequency Counter Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: PHY_FREQ_CNT _REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PHY_FREQ_CNT The value of these bits is the number of cmb_clk cycle during 1000 PHY byte clk cycles, which is used for byte clk frequency conversion of PHY0 or PHY1. The unit is cmb_clk cycle.

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Offset: 0x0034			Register Name: PHY_FREQ_CNT _REG
Bit	Read/Write	Default/Hex	Description
15:1	/	/	/
0	R/W	0x0	PHY0_FREQ_EN PHY0 byte_clk frequency detection enable (high active)

14.1.10.15 0x0038 PHY MIPI LP Timeout Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: PHY_MIPI_LP_TIMEOUT_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	PHY0_MIPI_LP_TIMEOUT_SET By default, the LP value will be refreshed when PHY0 MIPI LP signal bounce occurs. After enabling PHY0_MIPI_LP_TIMEOUT_EN bit, even if the signal bounce does not occur, the LP value will be refreshed after the time set by this bit. The unit is cmb_clk cycle.
15:1	/	/	/
0	R/W	0x0	PHY0_MIPI_LP_TIMEOUT_EN By default, the LP value will be refreshed when PHY0 MIPI LP signal bounce occurs. After enabling this bit, even if the signal bounce does not occur, the LP value will be refreshed after the time set by the PHY0_MIPI_LP_TIMEOUT_SET bit.

14.1.10.16 0x0040 PHY MIPI ULPSEXIT Register0 (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: PHY_MIPI_ULPSEXIT_REG0
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PHY0_MIPI_ULPS_EN Enter ULPS status after the ULPS instruction is detected by PHY0 lane.
30:20	/	/	/
19:0	R/W	0x0	PHY0_MIPI_ULPSEXIT PHY0 lane needs to keep detecting Mark 1 signal for a certain period of time before exiting ULPS status. The unit is cmb_clk cycle.

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14.1.10.17 0x0080 PHY Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: PHY_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R/W	0x0	PHY_HSRX_AFIFO_WFULL_D1_EN When data lane1 HSRX AFIFO is full, a interrupt will be generated.
20	R/W	0x0	PHY_HSRX_AFIFO_WFULL_D0_EN When data lane0 HSRX AFIFO is full, a interrupt will be generated.
19:17	/	/	/
16	R/W	0x0	PHY_LP_CK0_CMD_EN PHY clock lane0 interrupt generation enable after receiving recognizable LP commands.
15:14	/	/	/
13	R/W	0x0	PHY_LP_D1_CMD_EN PHY data lane1 interrupt generation enable after receiving recognizable LP commands.
12	R/W	0x0	PHY_LP_D0_CMD_EN PHY data lane0 interrupt generation enable after receiving recognizable LP commands.
11:10	/	/	/
9	R/W	0x0	PHY_NOSYNC_ERR_D1_EN PHY Sync-Sequence Error Interrupt Generation Enable This bit is to enable the interrupt generation when the Sync-Sequence is unable to be detected after data lane1 switching from LP mode to HS mode for a period of time.
8	R/W	0x0	PHY_NOSYNC_ERR_D0_EN PHY Sync-Sequence Error Interrupt Generation Enable This bit is to enable the interrupt generation when the Sync-Sequence is unable to be detected after data lane0 switching from LP mode to HS mode for a period of time.
7:6	/	/	/
5	R/W	0x0	PHY_SYNC_ERR_D1_EN This bit is to enable the interrupt generation when 1-bit error is detected in the data lane1 Sync-Sequence.

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Offset: 0x0080			Register Name: PHY_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	PHY_SYNC_ERR_D0_EN This bit is to enable the interrupt generation when 1-bit error is detected in the data lane0 Sync-Sequence.
3:2	/	/	/
1	R/W	0x0	PHY_SOTDET_D1_EN Data lane1 generates an interrupt after detecting the Sync-Sequence.
0	R/W	0x0	PHY_SOTDET_D0_EN Data lane0 generates an interrupt after detecting the Sync-Sequence.

14.1.10.18 0x0084 PHY Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x0084			Register Name: PHY_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R/W1C	0x0	PHY_HSRX_AFIFO_WFULL_D1_PD PHY_HSRX_AFIFO_WFULL_D1 Interrupt Pending
20	R/W1C	0x0	PHY_HSRX_AFIFO_WFULL_D0_PD PHY_HSRX_AFIFO_WFULL_D0 Interrupt Pending
19:17	/	/	/
16	R/W1C	0x0	PHY_LP_CK0_CMD_PD PHY clock lane0 generates an interrupt after receiving recognizable LP commands.
15:14	/	/	/
13	R/W1C	0x0	PHY_LP_D1_CMD_PD PHY data lane1 generates an interrupt after receiving recognizable LP commands.
12	R/W1C	0x0	PHY_LP_D0_CMD_PD PHY data lane0 generates an interrupt after receiving recognizable LP commands.
11:10	/	/	/
9	R/W1C	0x0	PHY_NOSYNC_ERR_D1_PD Generate an interrupt when the Sync-Sequence is unable to be detected after data lane1 switching from LP mode to HS mode for a period of time.
8	R/W1C	0x0	PHY_NOSYNC_ERR_D0_PD Generate an interrupt when the Sync-Sequence is unable to be detected after data lane0 switching from LP mode to HS mode for a period of time.

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Offset: 0x0084			Register Name: PHY_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
7:6	/	/	/
5	R/W1C	0x0	PHY_SYNC_ERR_D1_PD Generate an interrupt when 1-bit error is detected in the data lane1 Sync-Sequence.
4	R/W1C	0x0	PHY_SYNC_ERR_D0_PD Generate an interrupt when 1-bit error is detected in the data lane0 Sync-Sequence.
3:2	/	/	/
1	R/W1C	0x0	PHY_SOTDET_D1_PD Data lane1 generates an interrupt after detecting the Sync-Sequence.
0	R/W1C	0x0	PHY_SOTDET_D0_PD Data lane0 generates an interrupt after detecting the Sync-Sequence.

14.1.10.19 0x00E0 PHY Deskew Register3 (Default Value: 0x0000_0000)

Offset:0x00E0			Register Name: PHY_DESKEW_REG3
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:8	R	0x0	PHY_LANE1_DLY_MATCH_LEN PHY Lane1 Delay Step Length
7:6	/	/	/
5:0	R	0x0	PHY_LANE0_DLY_MATCH_LEN PHY Lane0 Delay Step Length

14.1.11 PHY PORT PAYLOAD LAYER Register Description

14.1.11.1 0x0000 PORT Control Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: PORT_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PORT_OUT_NUM Port Out Number 0: Output 1 data every valid 1: Output 2 data every valid
30:18	/	/	/

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Offset: 0x0000			Register Name: PORT_CTL_REG
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	PORT_CHANNEL_NUM Port Channel Number 0: 1 channel 1: 2 channels 2: 3 channels 3: 4 channels
15:12	/	/	/
11:8	R/W	0x0	PORT_LANE_NUM Port Lane Number MIPI: 1-4 PORT0 supports maximum 4 lanes. PORT1 supports maximum 2 lanes.
7:6	/	/	/
5:4	R/W	0x0	PORT_WORK_MODE 00: PORT channels work in MIPI mode.
3:1	/	/	/
0	R/W	0x0	PORT_EN PORT Channel Enable (High Active)

14.1.11.2 0x0004 PORT Lane Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: PORT_LANE_MAP_REG0
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:12	R/W	0x0	PORT_LANE3_ID Port Lane3 ID 0x0: Set lane0 (PHYA lane0) as PORT LANE3 0x1: Set lane1 (PHYA lane1) as PORT LANE3 0x4: Set lane4 (PHYB lane0) as PORT LANE3 0x5: Set lane5 (PHYB lane1) as PORT LANE3 Others: Reserved
11:8	R/W	0x0	PORT_LANE2_ID Port Lane2 ID 0x0: Set lane0 (PHYA lane0) as PORT LANE2 0x1: Set lane1 (PHYA lane1) as PORT LANE2 0x4: Set lane4 (PHYB lane0) as PORT LANE2 0x5: Set lane5 (PHYB lane1) as PORT LANE2 Others: Reserved

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Offset: 0x0004			Register Name: PORT_LANE_MAP_REG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	PORT_LANE1_ID Port Lane1 ID 0x0: Set lane0 (PHYA lane0) as PORT LANE1 0x1: Set lane1 (PHYA lane1) as PORT LANE1 0x4: Set lane4 (PHYB lane0) as PORT LANE1 0x5: Set lane5 (PHYB lane1) as PORT LANE1 Others: Reserved
3:0	R/W	0x0	PORT_LANE0_ID Port Lane0 ID 0x0: Set lane0 (PHYA lane0) as PORT LANE0 0x1: Set lane1 (PHYA lane1) as PORT LANE0 0x4: Set lane4 (PHYB lane0) as PORT LANE0 0x5: Set lane5 (PHYB lane1) as PORT LANE0 Others: Reserved

14.1.11.3 0x000C PORT WDR Mode Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: PORT_WDR_MODE_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	PORT_WDR_MODE Port WDR Mode 0x0: Normal mode, No WDR, MIPI VC, or MIPIDT. 0x2: Pixel data mode, distinguish the multi-channel WDR/HR images through the first pixel data of each line. others: Reserved

14.1.11.4 0x0010 PORT Frame ID Select Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: PORT_FID_SEL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PORT_FID_MODE 0: Identify the Frame in bit cs mode. For example, identify Frame0 when FID [0] =1, identify Frame1 when FID [1] =1. 1: Identify the Frame in bit value mode. For example, identify Frame0 when FID [0] =0, identify Frame1 when FID [0] =1.
30:20	/	/	/

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Offset: 0x0010			Register Name: PORT_FID_SEL_REG
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	PORT_FID_MAP_EN FID Mapping Enable Bit19: FID3_MAP_EN Bit18: FID2_MAP_EN Bit17: FID1_MAP_EN Bit16: FID0_MAP_EN
15:12	R/W	0x0	PORT_FID3_MAP These bits indicate the bit position with a 16-bit high-order aligned address for FID3 in the first pixel data of every line.
11:8	R/W	0x0	PORT_FID2_MAP These bits indicate the bit position with a 16-bit high-order aligned address for FID2 in the first pixel data of every line.
7:4	R/W	0x0	PORT_FID1_MAP These bits indicate the bit position with a 16-bit high-order aligned address for FID1 in the first pixel data of every line.
3:0	R/W	0x0	PORT_FID0_MAP These bits indicate the bit position with a 16-bit high-order aligned address for FID0 in the first pixel data of every line.

14.1.11.5 0x0100 PORT MIPI Configuration Register (Default Value: 0x4920_0000)

Offset: 0x0100			Register Name: PORT_MIPI_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:26	R/W	0x12	MIPI_REAR_EMB_DTYPE MIPI Rear Embedded Data Type Value
25:20	R/W	0x12	MIPI_FRONT_EMB_DTYPE MIPI Front Embedded Data Type Value
19	R/W	0x0	MIPI_EMB_MAP The MIPI received embedded data is output in a certain format. 0: Output in the pixel format of the corresponding channel. 1: Output in RAW8 format.
18	/	/	/

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Offset: 0x0100			Register Name: PORT_MIPI_CFG_REG
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	MIPI_YUV_SEQ MIPI YUV Format Transmission Sequence 00: YUYV 01: YVYU 10: UYVY 11: VYUY
15:10	/	/	/
9	R/W	0x0	MIPI_FRAME_SYNC_ONLY_FS MIPI Frame Synchronization only for Frame Start 0: No support 1: Support
8	R/W	0x0	MIPI_LINE_SYNC_EN MIPI Line Synchronization Enable 0: Generate line sync signals based on long packets 1: Generate line sync signals based on short packets
7	R/W	0x0	MIPI_PL_BITORD MIPI RX Payload Bit Sequence 0: LSb First 1: MSb First
6	R/W	0x0	MIPI_PH_BITORD MIPI Packet Header Bit Sequence (for ECC checksum) 0: LSb First, for example, {WC [15:8], WC [7:0], DI [7:0]} 1: MSb First, for example,{WC[8:15],WC[0:7], DI[0:7]}
5:4	R/W	0x0	MIPI_PH_BYTEORD MIPI Packet Header Byte Sequence (for ECC check) 0: {WCH, WCL, DI} 1: {DI, WCH, WCL} 2: {WCL, WCH, DI} 3: {DI,WCL,WCH}
3	R/W	0x0	MIPI_USER_DEF_EN MIPI USER Define Data Type Enable
2	R/W	0x0	EMBED_EN embed data receiving enable and output in the pixel format of the corresponding channel.

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Offset: 0x0100			Register Name: PORT_MIPI_CFG_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	MIPI_NO_UNPACK_ALL **MIPI_UNPACK_EN=0**: Receive data directly without unpacking. **MIPI_NO_UNPACK_ALL=0**: Only receive effective high-speed data without unpacking and output them from PORT Channel0 in RAW8 format. **MIPI_NO_UNPACK_ALL =1**: Receive rxhs_active, rxhs_valid, and rxhs_data[7:0] simultaneously as RX data and output them from PORT Channel0 in {2 h0, rxhs_active, rxhs_valid, rxhs_data[7:0]} sequence in RAW12 format.
0	R/W	0x0	MIPI_UNPACK_EN MIPI CSI2 Protocol Unpack Enable

14.1.11.6 0x0104 PORT MIPI No Unpacket Number Register (Default Value: 0x0000_0000)

Offset: 0x0104			Register Name: PORT_MIPI_NO_UNPAK_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	MIPI_NO_UNPAK_NUM_REG This bit indicates the number of received raw data. The unit is pixel component.

14.1.11.7 0x0108 PORT MIPI Data Identity Register (Default Value: 0x0000_0000)

Offset: 0x0108			Register Name: PORT_MIPI_DI_REG
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	MIPI_CH3_VC MIPI Channel3 VC Number Set
29:24	R/W	0x0	MIPI_CH3_DT MIPI Channel3 Data Type Set
23:22	R/W	0x0	MIPI_CH2_VC MIPI channel2 VC Number Set
21:16	R/W	0x0	MIPI_CH2_DT MIPI Channel2 Data Type Set
15:14	R/W	0x0	MIPI_CH1_VC MIPI Channel1 VC Number Set
13:8	R/W	0x0	MIPI_CH1_DT MIPI Channel1 Data Type Set
7:6	R/W	0x0	MIPI_CH0_VC MIPI channel0 VC Number Set

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Offset: 0x0108			Register Name: PORT_MIPI_DI_REG
Bit	Read/Write	Default/Hex	Description
5:0	R/W	0x0	MIPI_CH0_DT MIPI Channel0 Data Type Set/Real Mapping of USER Define Data Type

14.1.11.8 0x010C PORT MIPI User Data Type Register (Default Value: 0x0000_0000)

Offset: 0x010C			Register Name: PORT_MIPI_USER_DT_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_USER3_DT_EN Channel3 MIPI USER Define Data Type Enable
30	/	/	/
29:24	R/W	0x0	MIPI_USER3_DT Channel3 MIPI USER Define Data Type This bit is to identify the data belonging to channel3, which is unpacked into the data type set by the MIPI_CH3_DT bit following CSI2 protocol.
23	R/W	0x0	MIPI_USER2_DT_EN Channel2 MIPI USER Define Data Type Enable
22	/	/	/
21:16	R/W	0x0	MIPI_USER2_DT Channel2 MIPI USER Define Data Type This bit is to identify the data belonging to channel2, which is unpacked into the data type set by the MIPI_CH2_DT bit following CSI2 protocol.
15	R/W	0x0	MIPI_USER1_DT_EN Channel1 MIPI USER Define Data Type Enable
14	/	/	/
13:8	R/W	0x0	MIPI_USER1_DT Channel1 MIPI USER Define Data Type This bit is to identify the data belonging to channel1, which is unpacked into the data type set by the MIPI_CH1_DT bit following CSI2 protocol.
7	R/W	0x0	MIPI_USER0_DT_EN Channel0 MIPI USER Define Data Type
6	/	/	/
5:0	R/W	0x0	MIPI_USER0_DT Channel0 MIPI USER Define Data Type This bit is to identify the data belonging to channel0, which is unpacked into the data type set by the MIPI_CH0_DT bit following CSI2 protocol.

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14.1.11.9 0x0110 PORT MIPI Channel0 Data Type Trigger Register (Default Value: 0x0000_0000)

Offset: 0x0110			Register Name: PORT_MIPI_CH0_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_FIELD_REV Field Polarity 0: Positive (frm num [0] =1 indicate odd, =0 indicate even) 1: Negative (frm num [0] =0 indicate odd, =1 indicate even) This bit is only valid when MIPI_SRC_IS_FIELD = 1.
30:29	R/W	0x0	MIPI_SRC_IS_FIELD 00: The MIPI input is progressive data. 10: The MIPI input is interlaced data, field mode. 11: The MIPI input is interlaced data, frame mode. When in frame mode, you should configure INTERLACE_LINE_NUM bit.
28	/	/	/
27	R/W	0x0	MIPI_RAW_EXTEND The data type of RAW24/20/16 is represented by the data type of RAW12/10/8. 0: Not support 1: Support
26:20	/	/	/
19	R/W	0x0	RAW RAW Long Packet Data Type 0x28 to 0x2F Trigger Enable: 0: Masked 1: Enabled
18	R/W	0x0	RGB RGB Long Packet Data Type 0x20 to 0x27 Trigger Enable: 0: Masked 1: Enabled
17	R/W	0x0	YUV YUV Long Packet Data Type 0x18 to 0x1F Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0110			Register Name: PORT_MIPI_CH0_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	GL Generic 8bit Long Packet Data Type 0x10 to 0x17 Trigger Enable: 0: Masked 1: Enabled
15	R/W	0x0	GS7 Generic 8bit Short Packet Data Type 0x0F Trigger Enable: 0: Masked 1: Enabled
14	R/W	0x0	GS6 Generic Short Packet Data Type 0x0E Trigger Enable: 0: Masked 1: Enabled
13	R/W	0x0	GS5 Generic Short Packet Data Type 0x0D Trigger Enable: 0: Masked 1: Enabled
12	R/W	0x0	GS4 Generic Short Packet Data Type 0x0C Trigger Enable: 0: Masked 1: Enabled
11	R/W	0x0	GS3 Generic Short Packet Data Type 0x0B Trigger Enable: 0: Masked 1: Enabled
10	R/W	0x0	GS2 Generic Short Packet Data Type 0x0A Trigger Enable: 0: Masked 1: Enabled
9	R/W	0x0	GS1 Generic Short Packet Data Type 0x09 Trigger Enable: 0: Masked 1: Enabled
8	R/W	0x0	GS0 Generic Short Packet Data Type 0x08 Trigger Enable: 0: Masked 1: Enabled
7:4	/	/	/

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Offset: 0x0110			Register Name: PORT_MIPI_CH0_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	LE Data Type 0x03 Trigger Enable: 0: Masked 1: Enabled
2	R/W	0x0	LS Data Type 0x02 Trigger Enable: 0: Masked 1: Enabled
1	R/W	0x0	FE Data Type 0x01 Trigger Enable: 0: Masked 1: Enabled
0	R/W	0x0	FS Data Type 0x00 Trigger Enable: 0: Masked 1: Enabled

14.1.11.10 0x0114 PORT MIPI Channel0 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0114			Register Name: PORT_MIPI_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EOT_ERR_INT_EN EOT error interrupt enable
11	R/W	0x0	CHKSUM_ERR_INT_EN Checksum error interrupt enable
10	R/W	0x0	ECC_ERR_INT_EN ECC error interrupt enable
9	R/W	0x0	ECC_WRN_INT_EN ECC warning interrupt enable
8	R/W	0x0	LINE_SYNC_ERR_INT_EN Line synchronization error interrupt enable
7	R/W	0x0	FRAME_SYNC_ERR_INT_EN Frame synchronization error interrupt enable
6	R/W	0x0	EMBED_SYNC_INT_EN Embedded data interrupt enable
5	R/W	0x0	LINE_END_SYNC_INT_EN LE synchronization interrupt enable
4	R/W	0x0	LINE_START_SYNC_INT_EN LS synchronization interrupt enable

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Offset: 0x0114			Register Name: PORT_MIPI_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	FRAME_END_SYNC_INT_EN FE synchronization interrupt enable
2	R/W	0x0	FRAME_START_SYNC_INT_EN FS synchronization interrupt enable 0: Disable 1: Enable
1	R/W	0x0	PF_SYNC_INT_EN Packet Footer detected interrupt enable 0: Disable 1: Enable
0	R/W	0x0	PH_SYNC_INT_EN Packet Header update interrupt enable 0: Disable 1: Enable

14.1.11.11 0x0118 PORT MIPI Channel0 Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x0118			Register Name: PORT_MIPI_CH0_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W1C	0x0	EOT_ERR_INT_PD EOT error interrupt
11	R/W1C	0x0	CHKSUM_ERR_INT_PD Checksum error interrupt
10	R/W1C	0x0	ECC_ERR_INT_PD ECC error interrupt
9	R/W1C	0x0	ECC_WRN_INT_PD ECC warning interrupt
8	R/W1C	0x0	LINE_SYNC_ERR_INT_PD Line synchronization error interrupt
7	R/W1C	0x0	FRAME_SYNC_ERR_INT_PD Frame synchronization error interrupt
6	R/W1C	0x0	EMBED_SYNC_INT_PD Embedded data interrupt
5	R/W1C	0x0	LINE_END_SYNC_INT_PD LE synchronization interrupt
4	R/W1C	0x0	LINE_START_SYNC_INT_PD LS synchronization interrupt
3	R/W1C	0x0	FRAME_END_SYNC_INT_PD FE synchronization interrupt

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Offset: 0x0118			Register Name: PORT_MIPI_CH0_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	FRAME_START_SYNC_INT_PD FS synchronization interrupt
1	R/W1C	0x0	PF_SYNC_INT_PD Packet Footer detected interrupt
0	R/W1C	0x0	PH_SYNC_INT_PD Packet Header update interrupt

14.1.11.12 0x011C PORT MIPI Channel0 Packet Header Register (Default Value: 0x0000_0000)

Offset: 0x011C			Register Name: PORT_MIPI_CH0_PH_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CUR_WC MIPI current Word Counter
15:8	/	/	/
7:6	R	0x0	MIPI_CUR_VC MIPI current Virtual Channel
5:0	R	0x0	PORT_MIPI_CUR_DT MIPI current Data Type

14.1.11.13 0x0120 PORT MIPI Channel0 ECC Register (Default Value: 0x0000_0000)

Offset: 0x0120			Register Name: PORT_MIPI_CH0_ECC_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	MIPI_CAL_ECC MIPI calculated ECC
15:8	/	/	/
7:0	R	0x0	MIPI_RXD_ECC MIPI received ECC

14.1.11.14 0x0124 PORT MIPI Channel0 Checksum Register (Default Value: 0x0000_0000)

Offset: 0x0124			Register Name: PORT_MIPI_CH0_CKSUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CAL_CKSUM MIPI calculated Checksum
15:0	R	0x0	MIPI_RXD_CKSUM MIPI received Checksum

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14.1.11.15 0x0128 PORT MIPI Channel0 Frame Number Register (Default Value: 0x0000_0000)

Offset: 0x0128			Register Name: PORT_MIPI_CH0_FRM_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	FRAME_NUM Update when FRAME_START_SYNC_INT_PD and FRAME_END_SYNC_INT_PD comes

14.1.11.16 0x012C PORT MIPI Channel0 Line Number Register (Default Value: 0x0000_0000)

Offset: 0x012C			Register Name: PORT_MIPI_CH0_LINE_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	INTERLACE_HALF_LINE_NUM The total number of lines per field in frame mode, which is half the total number of lines per frame This bit is valid when MIPI_SRC_IS_FIELD=2'b11.
15:0	R	0x0	LINE_NUM Update when LINE_START_SYNC_INT_PD and LINE_END_SYNC_INT_PD comes

14.1.11.17 0x0130 PORT MIPI Channel1 Data Type Trigger Register (Default Value: 0x0000_0000)

Offset: 0x0130			Register Name: PORT_MIPI_CH1_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_FIELD_REV Field Polarity (only valid when MIPI_SRC_IS_FIEL┤ 0) **Field mode**: 0: Positive (frm num [0] =1 indicate odd, =0 indicate even) 1: Negative (frm num [0] =0 indicate odd, =1 indicate even) **Frame mode**: 0: Positive (the first half field is even field/field 0, and the second half field is odd field/field 1) 1: Negative (the first half field is odd field/field 1, and the second half field is even field/field 0)

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Offset: 0x0130			Register Name: PORT_MIPI_CH1_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
30:29	R/W	0x0	MIPI_SRC_IS_FIELD 00: The MIPI input is progressive data. 10: The MIPI input is interlaced data, field mode. 11: The MIPI input is interlaced data, frame mode. When in frame mode, you should configure INTERLACE_LINE_NUM bit.
28	/	/	/
27	R/W	0x0	MIPI_RAW_EXTEND The data type of RAW24/20/16 is represented by the data type of RAW12/10/8. 0: Not support 1: Support
26:20	/	/	/
19	R/W	0x0	RAW RAW Long Packet Data Type 0x28 to 0x2F Trigger Enable: 0: Masked 1: Enabled
18	R/W	0x0	RGB RGB Long Packet Data Type 0x20 to 0x27 Trigger Enable: 0: Masked 1: Enabled
17	R/W	0x0	YUV YUV Long Packet Data Type 0x18 to 0x1F Trigger Enable: 0: Masked 1: Enabled
16	R/W	0x0	GL Generic 8bit Long Packet Data Type 0x10 to 0x17 Trigger Enable: 0: Masked 1: Enabled
15	R/W	0x0	GS7 Generic 8bit Short Packet Data Type 0x0F Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0130			Register Name: PORT_MIPI_CH1_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	GS6 Generic Short Packet Data Type 0x0E Trigger Enable: 0: Masked 1: Enabled
13	R/W	0x0	GS5 Generic Short Packet Data Type 0x0D Trigger Enable: 0: Masked 1: Enabled
12	R/W	0x0	GS4 Generic Short Packet Data Type 0x0C Trigger Enable: 0: Masked 1: Enabled
11	R/W	0x0	GS3 Generic Short Packet Data Type 0x0B Trigger Enable: 0: Masked 1: Enabled
10	R/W	0x0	GS2 Generic Short Packet Data Type 0x0A Trigger Enable: 0: Masked 1: Enabled
9	R/W	0x0	GS1 Generic Short Packet Data Type 0x09 Trigger Enable: 0: Masked 1: Enabled
8	R/W	0x0	GS0 Generic Short Packet Data Type 0x08 Trigger Enable: 0: Masked 1: Enabled
7:4	/	/	/
3	R/W	0x0	LE Data Type 0x03 Trigger Enable: 0: Masked 1: Enabled
2	R/W	0x0	LS Data Type 0x02 Trigger Enable: 0: Masked 1: Enabled
1	R/W	0x0	FE Data Type 0x01 Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0130			Register Name: PORT_MIPI_CH1_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	FS Data Type 0x00 Trigger Enable: 0: Masked 1: Enabled

14.1.11.18 0x0134 PORT MIPI Channel1 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0134			Register Name: PORT_MIPI_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EOT_ERR_INT_EN EOT error interrupt enable
11	R/W	0x0	CHKSUM_ERR_INT_EN Checksum error interrupt enable
10	R/W	0x0	ECC_ERR_INT_EN ECC error interrupt enable
9	R/W	0x0	ECC_WRN_INT_EN ECC warning interrupt enable
8	R/W	0x0	LINE_SYNC_ERR_INT_EN Line synchronization error interrupt enable
7	R/W	0x0	FRAME_SYNC_ERR_INT_EN Frame synchronization error interrupt enable
6	R/W	0x0	EMBED_SYNC_INT_EN Embedded data interrupt enable
5	R/W	0x0	LINE_END_SYNC_INT_EN LE synchronization interrupt enable
4	R/W	0x0	LINE_START_SYNC_INT_EN LS synchronization interrupt enable
3	R/W	0x0	FRAME_END_SYNC_INT_EN FE synchronization interrupt enable
2	R/W	0x0	FRAME_START_SYNC_INT_EN FS synchronization interrupt enable 0: Disable 1: Enable
1	R/W	0x0	PF_SYNC_INT_EN Packet Footer detected interrupt enable 0: Disable 1: Enable

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Offset: 0x0134			Register Name: PORT_MIPI_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PH_SYNC_INT_EN Packet Header update interrupt enable 0: Disable 1: Enable

14.1.11.19 0x0138 PORT MIPI Channel1 Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x0138			Register Name: PORT_MIPI_CH1_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W1C	0x0	EOT_ERR_INT_PD EOT error interrupt
11	R/W1C	0x0	CHKSUM_ERR_INT_PD Checksum error interrupt
10	R/W1C	0x0	ECC_ERR_INT_PD ECC error interrupt
9	R/W1C	0x0	ECC_WRN_INT_PD ECC warning interrupt
8	R/W1C	0x0	LINE_SYNC_ERR_INT_PD Line synchronization error interrupt
7	R/W1C	0x0	FRAME_SYNC_ERR_INT_PD Frame synchronization error interrupt
6	R/W1C	0x0	EMBED_SYNC_INT_PD Embedded data interrupt
5	R/W1C	0x0	LINE_END_SYNC_INT_PD LE synchronization interrupt
4	R/W1C	0x0	LINE_START_SYNC_INT_PD LS synchronization interrupt
3	R/W1C	0x0	FRAME_END_SYNC_INT_PD FE synchronization interrupt
2	R/W1C	0x0	FRAME_START_SYNC_INT_PD FS synchronization interrupt
1	R/W1C	0x0	PF_SYNC_INT_PD Packet Footer detected interrupt
0	R/W1C	0x0	PH_SYNC_INT_PD Packet Header update interrupt

14.1.11.20 0x013C PORT MIPI Channel1 Packet Header Register (Default Value: 0x0000_0000)

Offset: 0x013C	Register Name: PORT_MIPI_CH1_PH_REG

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Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CUR_WC MIPI current Word Counter
15:8	/	/	/
7:6	R	0x0	MIPI_CUR_VC MIPI current Virtual Channel
5:0	R	0x0	PORT_MIPI_CUR_DT MIPI current Data Type

14.1.11.21 0x0140 PORT MIPI Channel1 ECC Register (Default Value: 0x0000_0000)

Offset: 0x0140			Register Name: PORT_MIPI_CH1_ECC_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	MIPI_CAL_ECC MIPI calculated ECC
15:8	/	/	/
7:0	R	0x0	MIPI_RXD_ECC MIPI received ECC

14.1.11.22 0x0144 PORT MIPI Channel1 Checksum Register (Default Value: 0x0000_0000)

Offset: 0x0144			Register Name: PORT_MIPI_CH1_CKSUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CAL_CKSUM MIPI calculated Checksum
15:0	R	0x0	MIPI_RXD_CKSUM MIPI received Checksum

14.1.11.23 0x0148 PORT MIPI Channel1 Frame Number Register (Default Value: 0x0000_0000)

Offset: 0x0148			Register Name: PORT_MIPI_CH1_FRM_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	FRAME_NUM Update when FRAME_START_SYNC_INT_PD and FRAME_END_SYNC_INT_PD comes

14.1.11.24 0x014C PORT MIPI Channel1 Line Number Register (Default Value: 0x0000_0000)

Offset: 0x014C	Register Name: PORT_MIPI_CH1_LINE_NUM_REG

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Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	INTERLACE_HALF_LINE_NUM The total number of lines per field in frame mode, which is half the total number of lines per frame This bit is valid when MIPI_SRC_IS_FIELD=2'b11.
15:0	R	0x0	LINE_NUM Update when LINE_START_SYNC_INT_PD and LINE_END_SYNC_INT_PD comes

14.1.11.25 0x0150 PORT MIPI Channel2 Data Type Trigger Register (Default Value: 0x0000_0000)

Offset: 0x0150			Register Name: PORT_MIPI_CH2_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_FIELD_REV Field Polarity (only valid when MIPI_SRC_IS_FIEL┤ 0) **Field mode**: 0: Positive (frm num [0] =1 indicate odd, =0 indicate even) 1: Negative (frm num [0] =0 indicate odd, =1 indicate even) **Frame mode**: 0: Positive (the first half field is even field/field 0, and the second half field is odd field/field 1) 1: Negative (the first half field is odd field/field 1, and the second half field is even field/field 0)
30:29	R/W	0x0	MIPI_SRC_IS_FIELD 00: The MIPI input is progressive data. 10: The MIPI input is interlaced data, field mode. 11: The MIPI input is interlaced data, frame mode. When in frame mode, you should configure INTERLACE_LINE_NUM bit.
28	/	/	/
27	R/W	0x0	The data type of RAW24/20/16 is represented by the data type of RAW12/10/8. 0: Not support 1: Support
26:20	/	/	/
19	R/W	0x0	RAW RAW Long Packet Data Type 0x28 to 0x2F Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0150			Register Name: PORT_MIPI_CH2_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	RGB RGB Long Packet Data Type 0x20 to 0x27 Trigger Enable: 0: Masked 1: Enabled
17	R/W	0x0	YUV YUV Long Packet Data Type 0x18 to 0x1F Trigger Enable: 0: Masked 1: Enabled
16	R/W	0x0	GL Generic 8bit Long Packet Data Type 0x10 to 0x17 Trigger Enable: 0: Masked 1: Enabled
15	R/W	0x0	GS7 Generic 8bit Short Packet Data Type 0x0F Trigger Enable: 0: Masked 1: Enabled
14	R/W	0x0	GS6 Generic Short Packet Data Type 0x0E Trigger Enable: 0: Masked 1: Enabled
13	R/W	0x0	GS5 Generic Short Packet Data Type 0x0D Trigger Enable: 0: Masked 1: Enabled
12	R/W	0x0	GS4 Generic Short Packet Data Type 0x0C Trigger Enable: 0: Masked 1: Enabled
11	R/W	0x0	GS3 Generic Short Packet Data Type 0x0B Trigger Enable: 0: Masked 1: Enabled
10	R/W	0x0	GS2 Generic Short Packet Data Type 0x0A Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0150			Register Name: PORT_MIPI_CH2_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	GS1 Generic Short Packet Data Type 0x09 Trigger Enable: 0: Masked 1: Enabled
8	R/W	0x0	GS0 Generic Short Packet Data Type 0x08 Trigger Enable: 0: Masked 1: Enabled
7:4	/	/	/
3	R/W	0x0	LE Data Type 0x03 Trigger Enable: 0: Masked 1: Enabled
2	R/W	0x0	LS Data Type 0x02 Trigger Enable: 0: Masked 1: Enabled
1	R/W	0x0	FE Data Type 0x01 Trigger Enable: 0: Masked 1: Enabled
0	R/W	0x0	FS Data Type 0x00 Trigger Enable: 0: Masked 1: Enabled

14.1.11.26 0x0154 PORT MIPI Channel2 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0154			Register Name: PORT_MIPI_CH2_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EOT_ERR_INT_EN EOT error interrupt enable
11	R/W	0x0	CHKSUM_ERR_INT_EN Checksum error interrupt enable
10	R/W	0x0	ECC_ERR_INT_EN ECC error interrupt enable
9	R/W	0x0	ECC_WRN_INT_EN ECC warning interrupt enable

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Offset: 0x0154			Register Name: PORT_MIPI_CH2_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	LINE_SYNC_ERR_INT_EN Line synchronization error interrupt enable
7	R/W	0x0	FRAME_SYNC_ERR_INT_EN Frame synchronization error interrupt enable
6	R/W	0x0	EMBED_SYNC_INT_EN Embedded data interrupt enable
5	R/W	0x0	LINE_END_SYNC_INT_EN LE synchronization interrupt enable
4	R/W	0x0	LINE_START_SYNC_INT_EN LS synchronization interrupt enable
3	R/W	0x0	FRAME_END_SYNC_INT_EN FE synchronization interrupt enable
2	R/W	0x0	FRAME_START_SYNC_INT_EN FS synchronization interrupt enable 0: Disable 1: Enable
1	R/W	0x0	PF_SYNC_INT_EN Packet Footer detected interrupt enable 0: Disable 1: Enable
0	R/W	0x0	PH_SYNC_INT_EN Packet Header update interrupt enable 0: Disable 1: Enable

14.1.11.27 0x0158 PORT MIPI Channel2 Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x0158			Register Name: PORT_MIPI_CH2_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W1C	0x0	EOT_ERR_INT_PD EOT error interrupt
11	R/W1C	0x0	CHKSUM_ERR_INT_PD Checksum error interrupt
10	R/W1C	0x0	ECC_ERR_INT_PD ECC error interrupt
9	R/W1C	0x0	ECC_WRN_INT_PD ECC warning interrupt
8	R/W1C	0x0	LINE_SYNC_ERR_INT_PD Line synchronization error interrupt

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Offset: 0x0158			Register Name: PORT_MIPI_CH2_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
7	R/W1C	0x0	FRAME_SYNC_ERR_INT_PD Frame synchronization error interrupt
6	R/W1C	0x0	EMBED_SYNC_INT_PD Embedded data interrupt
5	R/W1C	0x0	LINE_END_SYNC_INT_PD LE synchronization interrupt
4	R/W1C	0x0	LINE_START_SYNC_INT_PD LS synchronization interrupt
3	R/W1C	0x0	FRAME_END_SYNC_INT_PD FE synchronization interrupt
2	R/W1C	0x0	FRAME_START_SYNC_INT_PD FS synchronization interrupt
1	R/W1C	0x0	PF_SYNC_INT_PD Packet Footer detected interrupt
0	R/W1C	0x0	PH_SYNC_INT_PD Packet Header update interrupt

14.1.11.28 0x015C PORT MIPI Channel2 Packet Header Register (Default Value: 0x0000_0000)

Offset: 0x015C			Register Name: PORT_MIPI_CH2_PH_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CUR_WC MIPI current Word Counter
15:8	/	/	/
7:6	R	0x0	MIPI_CUR_VC MIPI current Virtual Channel
5:0	R	0x0	PORT_MIPI_CUR_DT MIPI current Data Type

14.1.11.29 0x0160 PORT MIPI Channel2 ECC Register (Default Value: 0x0000_0000)

Offset: 0x0160			Register Name: PORT_MIPI_CH2_ECC_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	MIPI_CAL_ECC MIPI calculated ECC
15:8	/	/	/
7:0	R	0x0	MIPI_RXD_ECC MIPI received ECC

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14.1.11.30 0x0164 PORT MIPI Channel2 Checksum Register (Default Value: 0x0000_0000)

Offset: 0x0164			Register Name: PORT_MIPI_CH2_CKSUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CAL_CKSUM MIPI calculated Checksum
15:0	R	0x0	MIPI_RXD_CKSUM MIPI received Checksum

14.1.11.31 0x0168 PORT MIPI Channel2 Frame Number Register (Default Value: 0x0000_0000)

Offset: 0x0168			Register Name: PORT_MIPI_CH2_FRM_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	FRAME_NUM Update when FRAME_START_SYNC_INT_PD and FRAME_END_SYNC_INT_PD comes

14.1.11.32 0x016C PORT MIPI Channel2 Line Number Register (Default Value: 0x0000_0000)

Offset: 0x016C			Register Name: PORT_MIPI_CH2_LINE_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	INTERLACE_HALF_LINE_NUM The total number of lines per field in frame mode, which is half the total number of lines per frame This bit is valid when MIPI_SRC_IS_FIELD=2'b11.
15:0	R	0x0	LINE_NUM Update when LINE_START_SYNC_INT_PD and LINE_END_SYNC_INT_PD comes

14.1.11.33 0x0170 PORT MIPI Channel3 Data Type Trigger Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: PORT_MIPI_CH3_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0170			Register Name: PORT_MIPI_CH3_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MIPI_FIELD_REV Field Polarity (only valid when MIPI_SRC_IS_FIEL┤ 0) **Field mode**: 0: Positive (frm num [0] =1 indicate odd, =0 indicate even) 1: Negative (frm num [0] =0 indicate odd, =1 indicate even) **Frame mode**: 0: Positive (the first half field is even field/field 0, and the second half field is odd field/field 1) 1: Negative (the first half field is odd field/field 1, and the second half field is even field/field 0)
30:29	R/W	0x0	MIPI_SRC_IS_FIELD 00: The MIPI input is progressive data. 10: The MIPI input is interlaced data, field mode. 11: The MIPI input is interlaced data, frame mode. When in frame mode, you should configure INTERLACE_LINE_NUM bit.
28	/	/	/
27	R/W	0x0	MIPI_RAW_EXTEND The data type of RAW24/20/16 is represented by the data type of RAW12/10/8. 0: Not support 1: Support
26:20	/	/	/
19	R/W	0x0	RAW RAW Long Packet Data Type 0x28 to 0x2F Trigger Enable: 0: Masked 1: Enabled
18	R/W	0x0	RGB RGB Long Packet Data Type 0x20 to 0x27 Trigger Enable: 0: Masked 1: Enabled
17	R/W	0x0	YUV YUV Long Packet Data Type 0x18 to 0x1F Trigger Enable: 0: Masked 1: Enabled

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Offset: 0x0170			Register Name: PORT_MIPI_CH3_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	GL Generic 8bit Long Packet Data Type 0x10 to 0x17 Trigger Enable: 0: Masked 1: Enabled
15	R/W	0x0	GS7 Generic 8bit Short Packet Data Type 0x0F Trigger Enable: 0: Masked 1: Enabled
14	R/W	0x0	GS6 Generic Short Packet Data Type 0x0E Trigger Enable: 0: Masked 1: Enabled
13	R/W	0x0	GS5 Generic Short Packet Data Type 0x0D Trigger Enable: 0: Masked 1: Enabled
12	R/W	0x0	GS4 Generic Short Packet Data Type 0x0C Trigger Enable: 0: Masked 1: Enabled
11	R/W	0x0	GS3 Generic Short Packet Data Type 0x0B Trigger Enable: 0: Masked 1: Enabled
10	R/W	0x0	GS2 Generic Short Packet Data Type 0x0A Trigger Enable: 0: Masked 1: Enabled
9	R/W	0x0	GS1 Generic Short Packet Data Type 0x09 Trigger Enable: 0: Masked 1: Enabled
8	R/W	0x0	GS0 Generic Short Packet Data Type 0x08 Trigger Enable: 0: Masked 1: Enabled
7:4	/	/	/

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Offset: 0x0170			Register Name: PORT_MIPI_CH3_DT_TRIG_EN_REG
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	LE Data Type 0x03 Trigger Enable: 0: Masked 1: Enabled
2	R/W	0x0	LS Data Type 0x02 Trigger Enable: 0: Masked 1: Enabled
1	R/W	0x0	FE Data Type 0x01 Trigger Enable: 0: Masked 1: Enabled
0	R/W	0x0	FS Data Type 0x00 Trigger Enable: 0: Masked 1: Enabled

14.1.11.34 0x0174 PORT MIPI Channel3 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0174			Register Name: PORT_MIPI_CH3_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EOT_ERR_INT_EN EOT error interrupt enable
11	R/W	0x0	CHKSUM_ERR_INT_EN Checksum error interrupt enable
10	R/W	0x0	ECC_ERR_INT_EN ECC error interrupt enable
9	R/W	0x0	ECC_WRN_INT_EN ECC warning interrupt enable
8	R/W	0x0	LINE_SYNC_ERR_INT_EN Line synchronization error interrupt enable
7	R/W	0x0	FRAME_SYNC_ERR_INT_EN Frame synchronization error interrupt enable
6	R/W	0x0	EMBED_SYNC_INT_EN Embedded data interrupt enable
5	R/W	0x0	LINE_END_SYNC_INT_EN LE synchronization interrupt enable
4	R/W	0x0	LINE_START_SYNC_INT_EN LS synchronization interrupt enable

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Offset: 0x0174			Register Name: PORT_MIPI_CH3_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	FRAME_END_SYNC_INT_EN FE synchronization interrupt enable
2	R/W	0x0	FRAME_START_SYNC_INT_EN FS synchronization interrupt enable 0: Disable 1: Enable
1	R/W	0x0	PF_SYNC_INT_EN Packet Footer detected interrupt enable 0: Disable 1: Enable
0	R/W	0x0	PH_SYNC_INT_EN Packet Header update interrupt enable 0: Disable 1: Enable

14.1.11.35 0x0178 PORT MIPI Channel3 Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x0178			Register Name: PORT_MIPI_CH3_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W1C	0x0	EOT_ERR_INT_PD EOT error interrupt
11	R/W1C	0x0	CHKSUM_ERR_INT_PD Checksum error interrupt
10	R/W1C	0x0	ECC_ERR_INT_PD ECC error interrupt
9	R/W1C	0x0	ECC_WRN_INT_PD ECC warning interrupt
8	R/W1C	0x0	LINE_SYNC_ERR_INT_PD Line synchronization error interrupt
7	R/W1C	0x0	FRAME_SYNC_ERR_INT_PD Frame synchronization error interrupt
6	R/W1C	0x0	EMBED_SYNC_INT_PD Embedded data interrupt
5	R/W1C	0x0	LINE_END_SYNC_INT_PD LE synchronization interrupt
4	R/W1C	0x0	LINE_START_SYNC_INT_PD LS synchronization interrupt
3	R/W1C	0x0	FRAME_END_SYNC_INT_PD FE synchronization interrupt

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Offset: 0x0178			Register Name: PORT_MIPI_CH3_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	FRAME_START_SYNC_INT_PD FS synchronization interrupt
1	R/W1C	0x0	PF_SYNC_INT_PD Packet Footer detected interrupt
0	R/W1C	0x0	PH_SYNC_INT_PD Packet Header update interrupt

14.1.11.36 0x017C PORT MIPI Channel3 Packet Header Register (Default Value: 0x0000_0000)

Offset: 0x017C			Register Name: PORT_MIPI_CH3_PH_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CUR_WC MIPI current Word Counter
15:8	/	/	/
7:6	R	0x0	MIPI_CUR_VC MIPI current Virtual Channel
5:0	R	0x0	PORT_MIPI_CUR_DT MIPI current Data Type

14.1.11.37 0x0180 PORT MIPI Channel3 ECC Register (Default Value: 0x0000_0000)

Offset: 0x0180			Register Name: PORT_MIPI_CH3_ECC_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	MIPI_CAL_ECC MIPI calculated ECC
15:8	/	/	/
7:0	R	0x0	MIPI_RXD_ECC MIPI received ECC

14.1.11.38 0x0184 PORT MIPI Channel3 Checksum Register (Default Value: 0x0000_0000)

Offset: 0x0184			Register Name: PORT_MIPI_CH3_CKSUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	MIPI_CAL_CKSUM MIPI calculated Checksum
15:0	R	0x0	MIPI_RXD_CKSUM MIPI received Checksum

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14.1.11.39 0x0188 PORT MIPI Channel3 Frame Number Register (Default Value: 0x0000_0000)

Offset: 0x0188			Register Name: PORT_MIPI_CH3_FRM_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	FRAME_NUM Update when FRAME_START_SYNC_INT_PD and FRAME_END_SYNC_INT_PD comes

14.1.11.40 0x018C PORT MIPI Channel3 Line Number Register (Default Value: 0x0000_0000)

Offset: 0x018C			Register Name: PORT_MIPI_CH3_LINE_NUM_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	INTERLACE_HALF_LINE_NUM The total number of lines per field in frame mode, which is half the total number of lines per frame This bit is valid when MIPI_SRC_IS_FIELD=2'b11.
15:0	R	0x0	LINE_NUM Update when LINE_START_SYNC_INT_PD and LINE_END_SYNC_INT_PD comes

14.1.11.41 0x01F0 PORT MIPI Lane Error Interrupt Register (Default Value: 0x0000_0000)

Offset: 0x01F0			Register Name: PORT_MIPI_LANE_ERR_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W	0x0	MIPI_LANE_ACT_NO_VLD_INT_EN Data lane0-3 receive no valid data when act high interrupt enable
3:0	R/W	0x0	MIPI_LANE_FIFO_OVERFLOW_INT_EN Data lane0-3 FIFO overflow interrupt enable

14.1.11.42 0x01F4 PORT MIPI Lane Error Interrupt Pending Register (Default Value: 0x0000_0000)

Offset: 0x01F4			Register Name: PORT_MIPI_LANE_ERR_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W1C	0x0	MIPI_LANE_ACT_NO_VLD_INT_PD Data lane0-3 receive no valid data when act high interrupt

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Offset: 0x01F4			Register Name: PORT_MIPI_LANE_ERR_INT_PD_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W1C	0x0	MIPI_LANE_FIFO_OVERFLOW_INT_PD Data lane0-3 FIFO overflow interrupt

14.1.12 CSIC_PARSER Register Description

14.1.12.1 0x0000 CSIC Parser Enable Register (Default Value:0x0000_0000)

Offset: 0x0000			Register Name: CSIC_PRS_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	MCSIC_EN MCSIC Enable 0: Reset and disable the MCSIC module 1: Enable the MCSIC module
30:17	/	/	/
16	R/W	0x0	NCSIC_EN NCSI Controller Enable 0: Reset and disable the NCSIC module 1: Enable the NCSIC module
15	R/W	0x0	PCLK_EN NCSI Pixel Clock Gating 0: Gate PCLK input 1: Enable PCLK input
14:4	/	/	/
3:2	R/W	0x0	PRS_CH_MODE Parser channel mode 00: Parser output channel 0-3 corresponding from input channel 0-3 01: Parser output channel 0-3 all from input channel 0 10: Parser output channel0 and 2 from input channel 0, output channel1 and 3 from input channel 1 11: Reserved
1	R/W	0x0	PRS_MODE Parser Mode 0: NCSI 1: MCSI

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Offset: 0x0000			Register Name: CSIC_PRS_EN_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PRS_EN Parser Enable 0: Reset and disable the parser module 1: Enable the parser module

14.1.12.2 0x0004 CSIC Parser NCSIC Interface Configuration Register (Default Value:0x0105_0080)

Offset: 0x0004			Register Name: CSIC_PRS_NCSIC_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	YUV420_LINE_ORDER YUV420 Input Line Order 0: YUV420 input in Y-YC-Y-YC Line Order 1: YUV420 input in YC-Y-YC-Y Line Order
30	R/W	0x0	ONLY_ACT_CARE Only Care V Active 1: Only care V active period
29	R/W	0x0	YUV422_SEP When input format is YUV422 16bit or Bt1120, set this bit to transfer YU,YV(2 components for each clock cycle) to Y,U,Y,V(1 component for each clock cycle)
28	/	/	/
27:24	R/W	0x1	FIELD_DT_PCLK_SHIFT Only for vsync detected field mode, the odd field permitted pclk shift = 4* FIELD_DT_PCLK_SHIFT
23:20	R/W	0x0	SRC_TYPE Bit 20-23 corresponding to the Source types for channel0-3 0: Progressed 1: Interlaced
19	R/W	0x0	FIELD For YUV HV timing, Field polarity 0: Negative (field=0 indicate odd, field=1 indicate even) 1: Positive (field=1 indicate odd, field=0 indicate even) For BT656 timing, Field sequence 0: Normal sequence (field 0 first) 1: Inverse sequence (field 1 first)

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Offset: 0x0004			Register Name: CSIC_PRS_NCSIC_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x1	VREF_POL Vref polarity 0: Negative 1: Positive This register is not apply to CCIR656 interface.
17	R/W	0x0	HERF_POL Href polarity 0: Negative 1: Positive This register is not apply to CCIR656 interface.
16	R/W	0x1	CLK_POL Data clock type 0: Active in rising edge 1: Active in falling edge
15:14	R/W	0x0	FIELD_DT_MODE only valid when CSI_IF is YUV and source type is interlaced 00: By both field and vsync 01: By field 10: By vsync 11: Reserved
13	R/W	0x0	DDR_SAMPLE_MODE_EN Dual Data Rate Sample Mode Enable 0: Disable 1: Enable
12:11	R/W	0x0	SEQ_8PLUS2 When select IF_DATA_WIDTH to be 8+2bit, odd/even pixel byte at CSI-D [11:4] will be rearranged to D [11:2] +2'b0 at the actual CSI data bus according to these sequences: 00: 6'bx+D [9:8], D [7:0] 01: D [9:2], 6'bx+D [1:0] 10: D [7:0], D [9:8] +6'bx 11: D[7:0], 6'bx+D[9:8]

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Offset: 0x0004			Register Name: CSIC_PRS_NCSIC_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	IF_DATA_WIDTH Input Data Width 000: 8-bit data bus 001: 10-bit data bus 010: 12-bit data bus 011: 8+2bit data bus 100: 2x8/16bit data bus 101: 14-bit data bus 110: 20-bit data bus, only for packet generator 111: 24 bit data bus, only for packet generator
7:6	R/W	0x2	INPUT_SEQ Input data sequence, only valid for YUV422 and YUV420 input format. All data interleaved in one channel: 00: YUYV 01: YVYU 10: UYVY 11: VYUY Y and UV in separated channel: x0: UV x1: VU
5	/	/	/

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Offset: 0x0004			Register Name: CSIC_PRS_NCSIC_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
4:0	R/W	0x0	CSI_IF YUV (separate syncs): 00000: YUYV422/YUV420 or RAW (All data in one data bus) 00001: 2x8 bit YUYV422 00010: Reserved 00011: Reserved CCIR656(embedded syncs): 00100: BT656 1 channel 00101: 16bit BT656(BT1120 like) 1 channel 00110: Reserved 00111: Reserved 01100: BT656 2 channels (All data interleaved in one data bus) 01101: 16bit BT656(BT1120 like) 2 channels (All data interleaved in one data bus) 01110: BT656 4 channels (All data interleaved in one data bus) 01111: 16bit BT656(BT1120 like) 4 channels (All data interleaved in one data bus) Others: Reserved

14.1.12.3 0x0008 CSIC Parser MCSIC Interface Configuration Register (Default Value: 0x0000_0080)

Offset: 0x0008			Register Name: CSIC_PRS_MCSI_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	YUV420_LINE_ORDER YUV420 Line Order 0: YUV420 input in Y-YC-Y-YC Line Order 1: YUV420 input in YC-Y-YC-Y Line Order
30:8	/	/	/

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Offset: 0x0008			Register Name: CSIC_PRS_MCSI_IF_CFG_REG
Bit	Read/Write	Default/Hex	Description
7:6	R/W	0x2	INPUT_SEQ Input data sequence, only valid for YUV422 and YUV420 input format. All data interleaved in one channel: 00: YUYV 01: YVYU 10: UYVY 11: VYUY Y and UV in separated channel: x0: UV x1: VU
5:0	/	/	/

14.1.12.4 0x000C CSIC Parser Capture Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: CSIC_PRS_CAP_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:26	R/W	0x0	CH3_FPS_DS_PRD Fps down sample period 0: Period is 1 frames 1: Period is 2 frames 2: Period is 3 frames 3: Period is 4 frames 4: Period is 5 frames (p)(p) 15: Period is 16 frames Note: CH3_FPS_DS_PRD should be used together with CSIC_PRS_CAP_FRM_MSK_1_REG bit [31:16]. Show an example as following. When CH3_FPS_DS_PRD is configured as 4'd6, CSIC_PRS_CAP_FRM_MSK_1_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CAP_FRM_MSK_1_REG bit[22:16] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.

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Offset: 0x000C			Register Name: CSIC_PRS_CAP_REG
Bit	Read/Write	Default/Hex	Description
25	R/W	0x0	CH3_VCAP_ON Video capture control: Capture the video image data stream on channel 3. 0: Disable video capture If video capture is in progress, the CSI stops capturing image data at the end of the current frame, and all of the current frame data is wrote to output FIFO. 1: Enable video capture The CSI starts capturing image data at the start of the next frame.
24	R/WAC	0x0	CH3_SCAP_ON Still capture control: Capture a single still image frame on channel 3. 0: Disable still capture. 1: Enable still capture The CSI module starts capturing image data at the start of the next frame. The CSI module captures only one frame of image data. This bit is self-cleared.
23:22	/	/	/
21:18	R/W	0x0	CH2_FPS_DS_PRD Fps down sample period 0: Period is 1 frames 1: Period is 2 frames 2: Period is 3 frames 3: Period is 4 frames 4: Period is 5 frames (p)(p) 15: Period is 16 frames Note: CH2_FPS_DS_PRD should be used together with CSIC_PRS_CAP_FRM_MSK_1_REG bit [15:0]. Show an example as following. When CH2_FPS_DS_PRD is configured as 4'd6, CSIC_PRS_CAP_FRM_MSK_1_REG bit [6:0] will take effect and bit [15:7] will be ignored. If CSIC_PRS_CAP_FRM_MSK_1_REG bit[6:0] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.

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Offset: 0x000C			Register Name: CSIC_PRS_CAP_REG
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	CH2_VCAP_ON Video capture control: Capture the video image data stream on channel 2. 0: Disable video capture If video capture is in progress, the CSI stops capturing image data at the end of the current frame, and all of the current frame data is wrote to output FIFO. 1: Enable video capture The CSI starts capturing image data at the start of the next frame.
16	R/WAC	0x0	CH2_SCAP_ON Still capture control: Capture a single still image frame on channel 2. 0: Disable still capture. 1: Enable still capture The CSI module starts capturing image data at the start of the next frame. The CSI module captures only one frame of image data. This bit is self-cleared.
15:14	/	/	/
13:10	R/W	0x0	CH1_FPS_DS_PRD Fps down sample period 0: Period is 1 frames 1: Period is 2 frames 2: Period is 3 frames 3: Period is 4 frames 4: Period is 5 frames (p)(p) 15: Period is 16 frames Note: CH1_FPS_DS_PRD should be used together with CSIC_PRS_CAP_FRM_MSK_0_REG bit [31:16]. Show an example as following. When CH1_FPS_DS_PRD is configured as 4'd6, CSIC_PRS_CAP_FRM_MSK_0_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CAP_FRM_MSK_0_REG bit[22:16] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.

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Offset: 0x000C			Register Name: CSIC_PRS_CAP_REG
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	CH1_VCAP_ON Video capture control: Capture the video image data stream on channel 1. 0: Disable video capture If video capture is in progress, the CSI stops capturing image data at the end of the current frame, and all of the current frame data is wrote to output FIFO. 1: Enable video capture The CSI starts capturing image data at the start of the next frame.
8	R/WAC	0x0	CH1_SCAP_ON Still capture control: Capture a single still image frame on channel 1. 0: Disable still capture. 1: Enable still capture The CSI module starts capturing image data at the start of the next frame. The CSI module captures only one frame of image data. This bit is self-cleared.
7:6	/	/	/
5:2	R/W	0x0	CH0_FPS_DS_PRD Fps down sample period 0: Period is 1 frames 1: Period is 2 frames 2: Period is 3 frames 3: Period is 4 frames 4: Period is 5 frames (p)(p) 15: Period is 16 frames Note: CH0_FPS_DS_PRD should be used together with CSIC_PRS_CAP_FRM_MSK_0_REG bit [15:0]. Show an example as following. When CH0_FPS_DS_PRD is configured as 4'd6, CSIC_PRS_CAP_FRM_MSK_0_REG bit [6:0] will take effect and bit [15:7] will be ignored. If CSIC_PRS_CAP_FRM_MSK_0_REG bit[6:0] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.

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Offset: 0x000C			Register Name: CSIC_PRS_CAP_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	CH0_VCAP_ON Video capture control: Capture the video image data stream on channel 0. 0: Disable video capture If video capture is in progress, the CSI stops capturing image data at the end of the current frame, and all of the current frame data is wrote to output FIFO. 1: Enable video capture The CSI starts capturing image data at the start of the next frame.
0	R/WAC	0x0	CH0_SCAP_ON Still capture control: Capture a single still image frame on channel 0. 0: Disable still capture. 1: Enable still capture The CSI module starts capturing image data at the start of the next frame. The CSI module captures only one frame of image data. This bit is self-cleared.

14.1.12.5 0x0010 CSIC Parser Signal Status Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: CSIC_PRS_SIGNAL_STA_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	FIFO_FULL Indicates the NCSI IN Async FIFO FULL 0: Not Full 1: Full
30:28	R	0x0	PCLK_CNT Indicates the NCSI Pclk is toggle or not
27	R	0x0	VSYNC_STA Indicates the NCSI Vsync Signal status 0: Low 1: High
26	R	0x0	HSYNC_STA Indicates the NCSI Hsync Signal status 0: Low 1: High

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Offset: 0x0010			Register Name: CSIC_PRS_SIGNAL_STA_REG
Bit	Read/Write	Default/Hex	Description
25	R	0x0	FIELD_STA Indicates the NCSI Field Signal status 0: Low 1: High
24	R	0x0	DATA_VALID_STA Indicates the NCSI Data Valid Signal status(n=0-15), MSB for D15, LSB for D0 0: Low 1: High
23:0	R	0x0	DATA_STA Indicates the NCSI Data Signal status(n=0-15), MSB for D15, LSB for D0 0: Low 1: High

14.1.12.6 0x0018 CSIC Parser Capture Frame Mask 0 Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: CSIC_PRS_CAP_FRM_MSK_0_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	CH1_FRAME_DATA_MASK Only [CH1_FPS_DS_PRD+16:16] are valid
15:0	R/W	0x0	CH0_FRAME_DATA_MASK Only [CH0_FPS_DS_PRD:0] are valid

14.1.12.7 0x001C CSIC Parser Capture Frame Mask 1 Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: CSIC_PRS_CAP_FRM_MSK_1_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	CH3_FRAME_DATA_MASK Only [CH3_FPS_DS_PRD+16:16] are valid
15:0	R/W	0x0	CH2_FRAME_DATA_MASK Only [CH2_FPS_DS_PRD:0] are valid

14.1.12.8 0x0020 CSIC Parser Channel_0 Input Dmask Setting Register (Default Value:0x0000_0000)

Offset: 0x0020			Register Name: CSIC_PRS_CH0_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x0020			Register Name: CSIC_PRS_CH0_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	IN_DMSK_PERIOD Input Dmask Period 0000: 1 0001: 2 0010: 3 ... 1111: 16 CH0 IN_DMSK_PERIOD should be used together with CSIC_PRS_CH0_INFMT_REG bit [31:16]. For example, when IN_DMSK_PERIOD is configured as 4'd6, CSIC_PRS_CH0_INFMT_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CH0_INFMT_REG bit [22:16] is configured as 7'b1011011, dvld high cycle 0, 1, 3, 4, 6, 7, 8, 10, 11, 13...will be dropped. Depending on AIDB btype bit[8], one dvld high cycle may transfer one pixel or two pixels.

14.1.12.9 0x0024 CSIC Parser Channel_0 Input Format Register (Default Value:0x0000_0003)

Offset: 0x0024			Register Name: CSIC_PRS_CH0_INFMT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PIXEL_DATA_MASK Only [IN_DMSK_PERIOD+16:16] are valid pixel data masks for every (IN_DMSK_PERIOD+1) dvld high cycles of one line
15:4	/	/	/
3:0	R/W	0x3	INPUT_FMT Input data format 0000: RAW stream 0001: Reserved 0010: Reserved 0011: YUV422 0100: YUV420 Others: Reserved

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14.1.12.10 0x0028 CSIC Parser Channel_0 Output Horizontal Size Register (Default Value:0x0500_0000)

Offset: 0x0028			Register Name: CSIC_PRS_CH0_OUTPUT_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HOR_MASK_MODE Horizontal Mask Mode 0: The Area defined by HOR_START and HOR_LEN is not Masked 1: The Area defined by HOR_START and HOR_LEN is Masked
30	/	/	/
29:16	R/W	0x500	HOR_LEN Horizontal pixel unit length. Valid pixel of a line.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.12.11 0x002C CSIC Parser Channel_0 Output Vertical Size Register (Default Value: 0x02d0_0000)

Offset: 0x002C			Register Name: CSIC_PRS_CH0_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	VER_MASK_MODE Vertical Mask Mode 0: The Area defined by VER_START and VER_LEN is not Masked 1: The Area defined by VER_START and VER_LEN is Masked
30	/	/	/
29:16	R/W	0x2d0	VER_LEN Valid line number of a frame.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

14.1.12.12 0x0030 CSIC Parser Channel_0 Input Parameter0 Register (Default Value: 0x0000_0000)

Offset: 0x0030	Register Name: CSIC_PRS_CH0_INPUT_PARA0_REG

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R	0x0	INPUT_SRC_TYPE Actual Input Source Type of Channel 0 0: Progress 1: Interlace

14.1.12.13 0x0034 CSIC Parser Channel_0 Input Parameter1 Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: CSIC_PRS_CH0_INPUT_PARA1_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VT Input Frame Vertical Total INPUT_VT = INPUT_VB+INPUT_Y
15:14	/	/	/
13:0	R	0x0	INPUT_HT Input Frame Horizontal Total INPUT_HT = INPUT_HB+INPUT_X

14.1.12.14 0x0038 CSIC Parser Channel_0 Input Parameter2 Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: CSIC_PRS_CH0_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VB Input Frame Vertical Blanking
15:14	/	/	/
13:0	R	0x0	INPUT_HB Input Frame Horizontal Blanking

14.1.12.15 0x003C CSIC Parser Channel_0 Input Parameter3 Register (Default Value:0x0000_0000)

Offset: 0x003C			Register Name: CSIC_PRS_CH0_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_Y Input Frame Vertical Size
15:14	/	/	/

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Offset: 0x003C			Register Name: CSIC_PRS_CH0_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
13:0	R	0x0	INPUT_X Input Frame Horizontal Size

14.1.12.16 0x0040 CSIC Parser Channel_0 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: CSIC_PRS_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W	0x0	VSYN_END_INT_EN Vsync End Interrupt Enable Field end interrupt enable for Interlaced input (two fields per frame); Frame end interrupt enable for progressive input.
9	R/W	0x0	VSYN_START_INT_EN Vsync Start Interrupt Enable Field start interrupt enable for Interlaced input (two fields per frame); Frame start interrupt enable for progressive input.
8	R/W	0x0	FRM_VSYN_END_INT_EN Frame Vsync End Interrupt Enable Frame end interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W	0x0	FRM_VSYN_START_INT_EN Frame Vsync Start Interrupt Enable Frame start interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W	0x0	IN_VB_CHG_INT_EN In Vblanking Change Interrupt Enable
5	R/W	0x0	IN_Y_CHG_INT_EN In Y Change Interrupt Enable
4	R/W	0x0	IN_HB_CHG_INT_EN In Hblanking Change Interrupt Enable
3	R/W	0x0	IN_X_CHG_INT_EN In X Change Interrupt Enable
2	R/W	0x0	MUL_ERR_INT_EN Multi-channel writing error Indicates error has been detected for writing data to a wrong channel.

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Offset: 0x0040			Register Name: CSIC_PRS_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	INPUT_PARA1_INT_EN Input Parameter1 update interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	INPUT_PARA0_INT_EN Input Parameter0 update interrupt Enable 0: Disable 1: Enable

14.1.12.17 0x0044 CSIC Parser Channel_0 Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: CSIC_PRS_CH0_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W1C	0x0	VSYN_END_PD Vsync End Interrupt Field end interrupt for Interlaced input (two fields per frame); Frame end interrupt for progressive input.
9	R/W1C	0x0	VSYN_START_PD Vsync Start Interrupt Field start interrupt for Interlaced input (two fields per frame); Frame start interrupt for progressive input.
8	R/W1C	0x0	FRM_VSYN_END_PD Frame Vsync End Interrupt Frame end interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W1C	0x0	FRM_VSYN_START_PD Frame Vsync Start Interrupt Frame start interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W1C	0x0	IN_VB_CHG_PD In Vblanking Change Interrupt
5	R/W1C	0x0	IN_Y_CHG_PD In Y Change Interrupt
4	R/W1C	0x0	IN_HB_CHG_PD In Hblanking Change Interrupt

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Offset: 0x0044			Register Name: CSIC_PRS_CH0_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	IN_X_CHG_PD In X Change Interrupt
2	R/W1C	0x0	MUL_ERR_PD Multi-channel writing error
1	R/W1C	0x0	INPUT_SRC_PD1 When the parser input parameter0 register update, this flag set to 1. Write 1 to clear.
0	R/W1C	0x0	INPUT_SRC_PD0 When the parser input parameter1 register, parser input parameter2 register or parser input parameter3 register update, this flag set to 1. Write 1 to clear.

14.1.12.18 0x0048 CSIC Parser Channel_0 Line Time Register (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: CSIC_PRS_CH0_LINE_TIME_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PRS_CH0_HBLK_TIME Time of H Blanking when vsync valid, unit is csi_top_clk cycle
15:0	R	0x0	PRS_CH0_HSYN_TIME Time of H SYNC when vsync valid, unit is csi_top_clk cycle

14.1.12.19 0x004C CSIC Parser Channel_0 Frame Pre Mask Setting Register (Default Value:0x0000_0000)

Offset: 0x004C			Register Name: CSIC_PRS_CH0_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:4	R/W	0x0	FRM_PRE_MSK_NUM Frame Pre Mask Number 00000000: 0 00000001: 1 00000010: 2 ... 11111111: 255
3:1	/	/	/

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Offset: 0x004C			Register Name: CSIC_PRS_CH0_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	FRM_PRE_MSK_EN This bit needs to be set after configuring FRM_PRE_MSK_NUM bit. 0: Disable frame pre mask 1: Enable frame pre mask

14.1.12.20 0x0120 CSIC Parser Channel_1 Input Dmask Setting Register (Default Value: 0x0000_0000)

Offset: 0x0120			Register Name: CSIC_PRS_CH1_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x0	IN_DMSK_PERIOD Input Dmask Period 0000: 1 0001: 2 0010: 3 ... 1111: 16 CH1 IN_DMSK_PERIOD should be used together with CSIC_PRS_CH1_INFMT_REG bit [31:16]. For example, when IN_DMSK_PERIOD is configured as 4'd6, CSIC_PRS_CH1_INFMT_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CH1_INFMT_REG bit [22:16] is configured as 7'b1011011, dvld high cycle 0, 1, 3, 4, 6, 7, 8, 10, 11, 13...will be dropped. Depending on AIDB btype bit[8], one dvld high cycle may transfer one pixel or two pixels.

14.1.12.21 0x0124 CSIC Parser Channel_1 Input Format Register (Default Value: 0x0000_0003)

Offset: 0x0124			Register Name: CSIC_PRS_CH1_INFMT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PIXEL_DATA_MASK Only [IN_DMSK_PERIOD+16:16] are valid. Pixel data masks for every (IN_DMSK_PERIOD+1) dvld high cycles of one line
15:4	/	/	/

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Offset: 0x0124			Register Name: CSIC_PRS_CH1_INFMT_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x3	INPUT_FMT Input data format 0000: RAW stream 0001: Reserved 0010: Reserved 0011: YUV422 0100: YUV420 Others: Reserved

14.1.12.22 0x0128 CSIC Parser Channel_1 Output Horizontal Size Register (Default Value:0x0500_0000)

Offset: 0x0128			Register Name: CSIC_PRS_CH1_OUTPUT_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HOR_MASK_MODE Horizontal Mask Mode 0: The Area defined by HOR_START and HOR_LEN is not Masked 1: The Area defined by HOR_START and HOR_LEN is Masked
30	/	/	/
29:16	R/W	0x500	HOR_LEN Horizontal pixel unit length. Valid pixel of a line.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.12.23 0x012C CSIC Parser Channel_1 Output Vertical Size Register (Default Value: 0x02d0_0000)

Offset: 0x012C			Register Name: CSIC_PRS_CH1_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	VER_MASK_MODE Vertical Mask Mode 0: The Area defined by VER_START and VER_LEN is not Masked 1: The Area defined by VER_START and VER_LEN is Masked

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Offset: 0x012C			Register Name: CSIC_PRS_CH1_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
30	/	/	/
29:16	R/W	0x2d0	VER_LEN Valid line number of a frame.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

14.1.12.24 0x0130 CSIC Parser Channel_1 Input Parameter0 Register (Default Value:0x0000_0000)

Offset: 0x0130			Register Name: CSIC_PRS_CH1_INPUT_PARA0_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R	0x0	INPUT_SRC_TYPE Actual Input Source Type of Channel 1 0: Progress 1: Interlace

14.1.12.25 0x0134 CSIC Parser Channel_1 Input Parameter1 Register (Default Value: 0x0000_0000)

Offset: 0x0134			Register Name: CSIC_PRS_CH1_INPUT_PARA1_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VT Input Frame Vertical Total INPUT_VT = INPUT_VB+INPUT_Y
15:14	/	/	/
13:0	R	0x0	INPUT_HT Input Frame Horizontal Total INPUT_HT = INPUT_HB+INPUT_X

14.1.12.26 0x0138 CSIC Parser Channel_1 Input Parameter2 Register (Default Value: 0x0000_0000)

Offset: 0x0138			Register Name: CSIC_PRS_CH1_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0138			Register Name: CSIC_PRS_CH1_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
29:16	R	0x0	INPUT_VB Input Frame Vertical Blanking
15:14	/	/	/
13:0	R	0x0	INPUT_HB Input Frame Horizontal Blanking

14.1.12.27 0x013C CSIC Parser Channel_1 Input Parameter3 Register (Default Value:0x0000_0000)

Offset: 0x013C			Register Name: CSIC_PRS_CH1_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_Y Input Frame Vertical Size
15:14	/	/	/
13:0	R	0x0	INPUT_X Input Frame Horizontal Size

14.1.12.28 0x0140 CSIC Parser Channel_1 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0140			Register Name: CSIC_PRS_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W	0x0	VSYN_END_INT_EN Vsync End Interrupt Enable Field end interrupt enable for Interlaced input (two fields per frame); Frame end interrupt enable for progressive input.
9	R/W	0x0	VSYN_START_INT_EN Vsync Start Interrupt Enable Field start interrupt enable for Interlaced input (two fields per frame); Frame start interrupt enable for progressive input.
8	R/W	0x0	FRM_VSYN_END_INT_EN Frame Vsync End Interrupt Enable Frame end interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.

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Offset: 0x0140			Register Name: CSIC_PRS_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	FRM_VSYN_START_INT_EN Frame Vsync Start Interrupt Enable Frame start interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W	0x0	IN_VB_CHG_INT_EN In Vblanking Change Interrupt Enable
5	R/W	0x0	IN_Y_CHG_INT_EN In Y Change Interrupt Enable
4	R/W	0x0	IN_HB_CHG_INT_EN In Hblanking Change Interrupt Enable
3	R/W	0x0	IN_X_CHG_INT_EN In X Change Interrupt Enable
2	R/W	0x0	MUL_ERR_INT_EN Multi-channel writing error Indicates error has been detected for writing data to a wrong channel.
1	R/W	0x0	INPUT_PARA1_INT_EN Input Parameter1 update interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	INPUT_PARA0_INT_EN Input Parameter0 update interrupt Enable 0: Disable 1: Enable

14.1.12.29 0x0144 CSIC Parser Channel_1 Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0144			Register Name: CSIC_PRS_CH1_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W1C	0x0	VSYN_END_PD Vsync End Interrupt Field end interrupt for Interlaced input (two fields per frame); Frame end interrupt for progressive input.
9	R/W1C	0x0	VSYN_START_PD Vsync Start Interrupt Field start interrupt for Interlaced input (two fields per frame); Frame start interrupt for progressive input.

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Offset: 0x0144			Register Name: CSIC_PRS_CH1_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	FRM_VSYN_END_PD Frame Vsync End Interrupt Frame end interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W1C	0x0	FRM_VSYN_START_PD Frame Vsync Start Interrupt Frame start interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W1C	0x0	IN_VB_CHG_PD In Vblanking Change Interrupt
5	R/W1C	0x0	IN_Y_CHG_PD In Y Change Interrupt
4	R/W1C	0x0	IN_HB_CHG_PD In Hblanking Change Interrupt
3	R/W1C	0x0	IN_X_CHG_PD In X Change Interrupt
2	R/W1C	0x0	MUL_ERR_PD Multi-channel writing error
1	R/W1C	0x0	INPUT_SRC_PD1 When the parser input parameter0 register update, this flag set to 1. Write 1 to clear.
0	R/W1C	0x0	INPUT_SRC_PD0 When the parser input parameter1 register, parser input parameter2 register or parser input parameter3 register update, this flag set to 1. Write 1 to clear.

14.1.12.30 0x0148 CSIC Parser Channel_1 Line Time Register (Default Value: 0x0000_0000)

Offset: 0x0148			Register Name: CSIC_PRS_CH1_LINE_TIME_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PRS_CH1_HBLK_TIME Time of H Blanking when vsync valid, unit is csi_top_clk cycle.
15:0	R	0x0	PRS_CH1_HSYN_TIME Time of H SYNC when vsync valid, unit is csi_top_clk cycle.

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14.1.12.31 0x014C CSIC Parser Channel_1 Frame Pre Mask Setting Register (Default Value:0x0000_0000)

Offset: 0x014C			Register Name: CSIC_PRS_CH1_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:4	R/W	0x0	FRM_PRE_MSK_NUM Frame Pre Mask Number 00000000: 0 00000001: 1 00000010: 2 ... 11111111: 255
3:1	/	/	/
0	R/W	0x0	FRM_PRE_MSK_EN This bit needs to be set after configuring FRM_PRE_MSK_NUM bit. 0: Disable frame pre mask 1: Enable frame pre mask

14.1.12.32 0x0220 CSIC Parser Channel_2 Input Dmask Setting Register (Default Value: 0x0000_0000)

Offset: 0x0220			Register Name: CSIC_PRS_CH2_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/

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Offset: 0x0220			Register Name: CSIC_PRS_CH2_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	IN_DMSK_PERIOD Input Dmask Period 0000: 1 0001: 2 0010: 3 ... 1111: 16 CH2 IN_DMSK_PERIOD should be used together with CSIC_PRS_CH2_INFMT_REG bit [31:16]. For example, when IN_DMSK_PERIOD is configured as 4'd6, CSIC_PRS_CH2_INFMT_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CH2_INFMT_REG bit [22:16] is configured as 7'b1011011, dvld high cycle 0, 1, 3, 4, 6, 7, 8, 10, 11, 13...will be dropped. Depending on AIDB btype bit[8], one dvld high cycle may transfer one pixel or two pixels.

14.1.12.33 0x0224 CSIC Parser Channel_2 Input Format Register (Default Value: 0x0000_0003)

Offset: 0x0224			Register Name: CSIC_PRS_CH2_INFMT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PIXEL_DATA_MASK Only [IN_DMSK_PERIOD+16:16] are valid pixel data masks for every (IN_DMSK_PERIOD+1) dvld high cycles of one line
15:4	/	/	/
3:0	R/W	0x3	INPUT_FMT Input data format 0000: RAW stream 0001: Reserved 0010: Reserved 0011: YUV422 0100: YUV420 Others: Reserved

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14.1.12.34 0x0228 CSIC Parser Channel_2 Output Horizontal Size Register (Default Value:0x0500_0000)

Offset: 0x0228			Register Name: CSIC_PRS_CH2_OUTPUT_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HOR_MASK_MODE Horizontal Mask Mode 0: The Area defined by HOR_START and HOR_LEN is not Masked 1: The Area defined by HOR_START and HOR_LEN is Masked
30	/	/	/
29:16	R/W	0x500	HOR_LEN Horizontal pixel unit length. Valid pixel of a line.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.12.35 0x022C CSIC Parser Channel_2 Output Vertical Size Register (Default Value: 0x02d0_0000)

Offset: 0x022C			Register Name: CSIC_PRS_CH2_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	VER_MASK_MODE Vertical Mask Mode 0: The Area defined by VER_START and VER_LEN is not Masked 1: The Area defined by VER_START and VER_LEN is Masked
30	/	/	/
29:16	R/W	0x2d0	VER_LEN Valid line number of a frame.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

14.1.12.36 0x0230 CSIC Parser Channel_2 Input Parameter0 Register (Default Value:0x0000_0000)

Offset: 0x0230	Register Name: CSIC_PRS_CH2_INPUT_PARA0_REG

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R	0x0	INPUT_SRC_TYPE Actual Input Source Type of Channel 2 0: Progress 1: Interlace

14.1.12.37 0x0234 CSIC Parser Channel_2 Input Parameter1 Register (Default Value: 0x0000_0000)

Offset: 0x0234			Register Name: CSIC_PRS_CH2_INPUT_PARA1_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VT Input Frame Vertical Total INPUT_VT = INPUT_VB+INPUT_Y
15:14	/	/	/
13:0	R	0x0	INPUT_HT Input Frame Horizontal Total INPUT_HT = INPUT_HB+INPUT_X

14.1.12.38 0x0238 CSIC Parser Channel_2 Input Parameter2 Register (Default Value: 0x0000_0000)

Offset: 0x0238			Register Name: CSIC_PRS_CH2_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VB Input Frame Vertical Blanking
15:14	/	/	/
13:0	R	0x0	INPUT_HB Input Frame Horizontal Blanking

14.1.12.39 0x023C CSIC Parser Channel_2 Input Parameter3 Register (Default Value: 0x0000_0000)

Offset: 0x023C			Register Name: CSIC_PRS_CH2_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_Y Input Frame Vertical Size
15:14	/	/	/

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Offset: 0x023C			Register Name: CSIC_PRS_CH2_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
13:0	R	0x0	INPUT_X Input Frame Horizontal Size

14.1.12.40 0x0240 CSIC Parser Channel_2 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0240			Register Name: CSIC_PRS_CH2_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W	0x0	VSYN_END_INT_EN Vsync End Interrupt Enable Field end interrupt enable for Interlaced input (two fields per frame); Frame end interrupt enable for progressive input.
9	R/W	0x0	VSYN_START_INT_EN Vsync Start Interrupt Enable Field start interrupt enable for Interlaced input (two fields per frame); Frame start interrupt enable for progressive input.
8	R/W	0x0	FRM_VSYN_END_INT_EN Frame Vsync End Interrupt Enable Frame end interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W	0x0	FRM_VSYN_START_INT_EN Frame Vsync Start Interrupt Enable Frame start interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W	0x0	IN_VB_CHG_INT_EN In Vblanking Change Interrupt Enable
5	R/W	0x0	IN_Y_CHG_INT_EN In Y Change Interrupt Enable
4	R/W	0x0	IN_HB_CHG_INT_EN In Hblanking Change Interrupt Enable
3	R/W	0x0	IN_X_CHG_INT_EN In X Change Interrupt Enable
2	R/W	0x0	MUL_ERR_INT_EN Multi-channel writing error Indicates error has been detected for writing data to a wrong channel.

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Offset: 0x0240			Register Name: CSIC_PRS_CH2_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	INPUT_PARA1_INT_EN Input Parameter1 update interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	INPUT_PARA0_INT_EN Input Parameter0 update interrupt Enable 0: Disable 1: Enable

14.1.12.41 0x0244 CSIC Parser Channel_2 Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0244			Register Name: CSIC_PRS_CH2_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W1C	0x0	VSYN_END_PD Vsync End Interrupt Field end interrupt for Interlaced input (two fields per frame); Frame end interrupt for progressive input.
9	R/W1C	0x0	VSYN_START_PD Vsync Start Interrupt Field start interrupt for Interlaced input (two fields per frame); Frame start interrupt for progressive input.
8	R/W1C	0x0	FRM_VSYN_END_PD Frame Vsync End Interrupt Frame end interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W1C	0x0	FRM_VSYN_START_PD Frame Vsync Start Interrupt Frame start interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W1C	0x0	IN_VB_CHG_PD In Vblanking Change Interrupt
5	R/W1C	0x0	IN_Y_CHG_PD In Y Change Interrupt
4	R/W1C	0x0	IN_HB_CHG_PD In Hblanking Change Interrupt

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Offset: 0x0244			Register Name: CSIC_PRS_CH2_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	IN_X_CHG_PD In X Change Interrupt
2	R/W1C	0x0	MUL_ERR_PD Multi-channel writing error
1	R/W1C	0x0	INPUT_SRC_PD1 When the parser input parameter0 register update, this flag set to 1. Write 1 to clear.
0	R/W1C	0x0	INPUT_SRC_PD0 When the parser input parameter1 register, parser input parameter2 register or parser input parameter3 register update, this flag set to 1. Write 1 to clear.

14.1.12.42 0x0248 CSIC Parser Channel_2 Line Time Register (Default Value: 0x0000_0000)

Offset: 0x0248			Register Name: CSIC_PRS_CH2_LINE_TIME_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PRS_CH2_HBLK_TIME Time of H Blanking when vsync valid, unit is csi_top_clk cycle.
15:0	R	0x0	PRS_CH2_HSYN_TIME Time of H SYNC when vsync valid, unit is csi_top_clk cycle.

14.1.12.43 0x024C CSIC Parser Channel_2 Frame Pre Mask Setting Register (Default Value: 0x0000_0000)

Offset: 0x024C			Register Name: CSIC_PRS_CH2_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:4	R/W	0x0	FRM_PRE_MSK_NUM Frame Pre Mask Number 00000000: 0 00000001: 1 00000010: 2 ... 11111111: 255
3:1	/	/	/

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Offset: 0x024C			Register Name: CSIC_PRS_CH2_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	FRM_PRE_MSK_EN Should be set after configuring FRM_PRE_MSK_NUM 0: Disable frame pre mask 1: Enable frame pre mask

14.1.12.44 0x0320 CSIC Parser Channel_3 Input Dmask Setting Register (Default Value: 0x0000_0000)

Offset: 0x0320			Register Name: CSIC_PRS_CH3_IN_DMSK_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x0	IN_DMSK_PERIOD Input Dmask Period 0000: 1 0001: 2 0010: 3 ... 1111: 16 CH3 IN_DMSK_PERIOD should be used together with CSIC_PRS_CH3_INFMT_REG bit [31:16]. For example, when IN_DMSK_PERIOD is configured as 4'd6, CSIC_PRS_CH3_INFMT_REG bit [22:16] will take effect and bit [31:23] will be ignored. If CSIC_PRS_CH3_INFMT_REG bit [22:16] is configured as 7'b1011011, dvld high cycle 0, 1, 3, 4, 6, 7, 8, 10, 11, 13...will be dropped. Depending on AIDB btype bit[8], one dvld high cycle may transfer one pixel or two pixels.

14.1.12.45 0x0324 CSIC Parser Channel_3 Input Format Register (Default Value: 0x0000_0003)

Offset: 0x0324			Register Name: CSIC_PRS_CH3_INFMT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PIXEL_DATA_MASK Only [IN_DMSK_PERIOD+16:16] are valid pixel data masks for every (IN_DMSK_PERIOD+1) dvld high cycles of one line
15:4	/	/	/

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Offset: 0x0324			Register Name: CSIC_PRS_CH3_INFMT_REG
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x3	INPUT_FMT Input data format 0000: RAW stream 0001: Reserved 0010: Reserved 0011: YUV422 0100: YUV420 Others: Reserved

14.1.12.46 0x0328 CSIC Parser Channel_3 Output Horizontal Size Register (Default Value: 0x0500_0000)

Offset: 0x0328			Register Name: CSIC_PRS_CH3_OUTPUT_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HOR_MASK_MODE Horizontal Mask Mode 0: The Area defined by HOR_START and HOR_LEN is not Masked 1: The Area defined by HOR_START and HOR_LEN is Masked
30	/	/	/
29:16	R/W	0x500	HOR_LEN Horizontal pixel unit length. Valid pixel of a line.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.12.47 0x032C CSIC Parser Channel_3 Output Vertical Size Register (Default Value: 0x02d0_0000)

Offset: 0x032C			Register Name: CSIC_PRS_CH3_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	VER_MASK_MODE Vertical Mask Mode 0: The Area defined by VER_START and VER_LEN is not Masked 1: The Area defined by VER_START and VER_LEN is Masked

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Offset: 0x032C			Register Name: CSIC_PRS_CH3_OUTPUT_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
30	/	/	/
29:16	R/W	0x2d0	VER_LEN Valid line number of a frame.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

14.1.12.48 0x0330 CSIC Parser Channel_3 Input Parameter0 Register (Default Value: 0x0000_0000)

Offset: 0x0330			Register Name: CSIC_PRS_CH3_INPUT_PARA0_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R	0x0	INPUT_SRC_TYPE Actual Input Source Type of Channel 3 0: Progress 1: Interlace

14.1.12.49 0x0334 CSIC Parser Channel_3 Input Parameter1 Register (Default Value: 0x0000_0000)

Offset: 0x0334			Register Name: CSIC_PRS_CH3_INPUT_PARA1_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_VT Input Frame Vertical Total INPUT_VT = INPUT_VB+INPUT_Y
15:14	/	/	/
13:0	R	0x0	INPUT_HT Input Frame Horizontal Total INPUT_HT = INPUT_HB+INPUT_X

14.1.12.50 0x0338 CSIC Parser Channel_3 Input Parameter2 Register (Default Value: 0x0000_0000)

Offset: 0x0338			Register Name: CSIC_PRS_CH3_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0338			Register Name: CSIC_PRS_CH3_INPUT_PARA2_REG
Bit	Read/Write	Default/Hex	Description
29:16	R	0x0	INPUT_VB Input Frame Vertical Blanking
15:14	/	/	/
13:0	R	0x0	INPUT_HB Input Frame Horizontal Blanking

14.1.12.51 0x033C CSIC Parser Channel_3 Input Parameter3 Register (Default Value: 0x0000_0000)

Offset: 0x033C			Register Name: CSIC_PRS_CH3_INPUT_PARA3_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	INPUT_Y Input Frame Vertical Size
15:14	/	/	/
13:0	R	0x0	INPUT_X Input Frame Horizontal Size

14.1.12.52 0x0340 CSIC Parser Channel_3 Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0340			Register Name: CSIC_PRS_CH3_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W	0x0	VSYN_END_INT_EN Vsync End Interrupt Enable Field end interrupt enable for Interlaced input (two fields per frame); Frame end interrupt enable for progressive input.
9	R/W	0x0	VSYN_START_INT_EN Vsync Start Interrupt Enable Field start interrupt enable for Interlaced input (two fields per frame); Frame start interrupt enable for progressive input.
8	R/W	0x0	FRM_VSYN_END_INT_EN Frame Vsync End Interrupt Enable Frame end interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.

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Offset: 0x0340			Register Name: CSIC_PRS_CH3_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	FRM_VSYN_START_INT_EN Frame Vsync Start Interrupt Enable Frame start interrupt enable for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W	0x0	IN_VB_CHG_INT_EN In Vblanking Change Interrupt Enable
5	R/W	0x0	IN_Y_CHG_INT_EN In Y Change Interrupt Enable
4	R/W	0x0	IN_HB_CHG_INT_EN In Hblanking Change Interrupt Enable
3	R/W	0x0	IN_X_CHG_INT_EN In X Change Interrupt Enable
2	R/W	0x0	MUL_ERR_INT_EN Multi-channel writing error Indicates error has been detected for writing data to a wrong channel.
1	R/W	0x0	INPUT_PARA1_INT_EN Input Parameter1 update interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	INPUT_PARA0_INT_EN Input Parameter0 update interrupt Enable 0: Disable 1: Enable

14.1.12.53 0x0344 CSIC Parser Channel_3 Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0344			Register Name: CSIC_PRS_CH3_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10	R/W1C	0x0	VSYN_END_PD Vsync End Interrupt Field end interrupt for Interlaced input (two fields per frame); Frame end interrupt for progressive input.
9	R/W1C	0x0	VSYN_START_PD Vsync Start Interrupt Field start interrupt for Interlaced input (two fields per frame); Frame start interrupt for progressive input.

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Offset: 0x0344			Register Name: CSIC_PRS_CH3_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	FRM_VSYN_END_PD Frame Vsync End Interrupt Frame end interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
7	R/W1C	0x0	FRM_VSYN_START_PD Frame Vsync Start Interrupt Frame start interrupt for Interlaced input (two fields per frame); Invalid for progressive input.
6	R/W1C	0x0	IN_VB_CHG_PD In Vblanking Change Interrupt
5	R/W1C	0x0	IN_Y_CHG_PD In Y Change Interrupt
4	R/W1C	0x0	IN_HB_CHG_PD In Hblanking Change Interrupt
3	R/W1C	0x0	IN_X_CHG_PD In X Change Interrupt
2	R/W1C	0x0	MUL_ERR_PD Multi-channel writing error
1	R/W1C	0x0	INPUT_SRC_PD1 When the parser input parameter0 register update, this flag set to 1. Write 1 to clear.
0	R/W1C	0x0	INPUT_SRC_PD0 When the parser input parameter1 register, parser input parameter2 register or parser input parameter3 register update, this flag set to 1. Write 1 to clear.

14.1.12.54 0x0348 CSIC Parser Channel_3 Line Time Register (Default Value: 0x0000_0000)

Offset: 0x0348			Register Name: CSIC_PRS_CH3_LINE_TIME_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PRS_CH3_HBLK_TIME Time of H Blanking when vsync valid, unit is csi_top_clk cycle
15:0	R	0x0	PRS_CH3_HSYN_TIME Time of H SYNC when vsync valid, unit is csi_top_clk cycle

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14.1.12.55 0x034C CSIC Parser Channel_3 Frame Pre Mask Setting Register (Default Value: 0x0000_0000)

Offset: 0x034C			Register Name: CSIC_PRS_CH3_FRM_PRE_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:4	R/W	0x0	FRM_PRE_MSK_NUM Frame Pre Mask Number 00000000: 0 00000001: 1 00000010: 2 ... 11111111: 255
3:1	/	/	/
0	R/W	0x0	FRM_PRE_MSK_EN This bit needs to be set after the FRM_PRE_MSK_NUM bit is configured. 0: Disable frame pre mask 1: Enable frame pre mask

14.1.12.56 0x0500 CSIC Parser NCSIC RX Signal0 Delay Adjust Register (Default Value: 0x0000_0000)

Offset: 0x0500			Register Name: CSIC_PRS_NCSIC_RX_SIGNAL0_DLY_ADJ_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	FIELD_DLY 32 Step for adjust, 1 step = 0.2ns
23:21	/	/	/
20:16	R/W	0x0	VSYNC_DLY 32 Step for adjust, 1 step = 0.2ns
15:13	/	/	/
12:8	R/W	0x0	HSYNC_DLY 32 Step for adjust, 1 step = 0.2ns
7:5	/	/	/
4:0	R/W	0x0	PCLK_DLY 32 Step for adjust, 1 step = 0.2ns

14.1.12.57 0x050C CSIC Parser NCSIC RX Signal3 Delay Adjust Register (Default Value: 0x0000_0000)

Offset: 0x050C	Register Name: CSIC_PRS_NCSIC_RX_SIGNAL3_DLY_ADJ_REG

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Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	D15_DLY 32 Step for adjust, 1 step = 0.2ns
23:21	/	/	/
20:16	R/W	0x0	D14_DLY 32 Step for adjust, 1 step = 0.2ns
15:13	/	/	/
12:8	R/W	0x0	D13_DLY 32 Step for adjust, 1 step = 0.2ns
7:5	/	/	/
4:0	R/W	0x0	D12_DLY 32 Step for adjust, 1 step = 0.2ns

14.1.12.58 0x0510 CSIC Parser NCSIC RX Signal4 Delay Adjust Register (Default Value: 0x0000_0000)

Offset: 0x0510			Register Name: CSIC_PRS_NCSIC_RX_SIGNAL4_DLY_ADJ_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	D11_DLY 32 Step for adjust, 1 step = 0.2ns
23:21	/	/	/
20:16	R/W	0x0	D10_DLY 32 Step for adjust, 1 step = 0.2ns
15:13	/	/	/
12:8	R/W	0x0	D9_DLY 32 Step for adjust, 1 step = 0.2ns
7:5	/	/	/
4:0	R/W	0x0	D8_DLY 32 Step for adjust, 1 step = 0.2ns

14.1.12.59 0x0514 CSIC Parser NCSIC RX Signal5 Delay Adjust Register (Default Value: 0x0000_0000)

Offset: 0x0514			Register Name: CSIC_PRS_NCSIC_RX_SIGNAL5_DLY_ADJ_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	D7_DLY 32 Step for adjust, 1 step = 0.2ns
23:21	/	/	/
20:16	R/W	0x0	D6_DLY 32 Step for adjust, 1 step = 0.2ns

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Offset: 0x0514			Register Name: CSIC_PRS_NCSIC_RX_SIGNAL5_DLY_ADJ_REG
Bit	Read/Write	Default/Hex	Description
15:13	/	/	/
12:8	R/W	0x0	D5_DLY 32 Step for adjust, 1 step = 0.2ns
7:5	/	/	/
4:0	R/W	0x0	D4_DLY 32 Step for adjust, 1 step = 0.2ns

14.1.12.60 0x0518 CSIC Parser NCSIC RX Signal6 Delay Adjust Register (Default Value: 0x0000_0000)

Offset: 0x0518			Register Name: CSIC_PRS_NCSIC_RX_SIGNAL6_DLY_ADJ_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	D3_DLY 32 Step for adjust, 1 step = 0.2ns
23:21	/	/	/
20:16	R/W	0x0	D2_DLY 32 Step for adjust, 1 step = 0.2ns
15:13	/	/	/
12:8	R/W	0x0	D1_DLY 32 Step for adjust, 1 step = 0.2ns
7:5	/	/	/
4:0	R/W	0x0	D0_DLY 32 Step for adjust, 1 step = 0.2ns

14.1.12.61 0x0520 CSIC Parser SYNC EN Register (Default Value: 0x0000_0000)

Offset :0x0520			Register Name: CSIC_PRS_SYNC_EN_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
19:16	R/W	0x0	VSYNC_MODE Input Vsync Signal Source Select 0: Vsync signals all from 1 parser 1: Vsync signals from 2 parser others: Reserved
15:12	/	/	/

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Offset :0x0520			Register Name: CSIC_PRS_SYNC_EN_REG
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0x0	VSYNC_SEL Generate sync signal Benchmark select Bit8: USE VSYNC_Input0 Bit9: USE VSYNC_Input1 Bit10: USE VSYNC_Input2 Bit11: Reserved Set 1,Use input
7:4	R/W	0x0	VSYNC_USED Parser input vsync signal enable in sync mode Bit4: VSYNC_Input0 Bit5: VSYNC_Input1 Bit6: VSYNC_Input2 Bit7: Reserved Set 1,enable input
3	/	/	/
2	R/W	0x0	FSYNC_OUT_SEL Parser sent sync signal via by 0: FSYNC0 1: FSYNC1
1	R/W	0x0	FSYNC_MODE Parser sync signal source select 0: From outside 1: Generate by self
0	R/W	0x0	FSYNC_FUN_EN Enable Parser sent sync signal 0: Disable 1: Enable

14.1.12.62 0x0524 CSIC Parser SYNC CFG Register (Default Value: 0x0000_0000)

Offset : 0x0524			Register Name: CSIC_PRS_SYNC_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	FSYNC_PUL_WID Sync signal pulse width N*T_{24M} N*T_{24M} 4*Tpclk
15:0	R/W	0x0	FYSNC_DISTANCE The interval of two sync signal

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14.1.12.63 0x0528 CSIC Parser VS WAIT N Register (Default Value: 0x0000_0000)

Offset : 0x0528			Register Name: CSIC_PRS_VS_WAIT_N_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VS_WAIT_N When multi-channel vsync all come, the max wait time.

14.1.12.64 0x052C CSIC Parser VS WAIT M Register (Default Value: 0x0000_0000)

Offset : 0x052C			Register Name: CSIC_PRS_VS_WAIT_M_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VS_WAIT_M When in multi-channel mode, sync comes at the different time, these bits indicate the max wait time.

14.1.12.65 0x0540 CSIC Parser XSYNC ENABLE Register (Default Value: 0x0000_0000)

Offset: 0x0540			Register Name: CSIC_PRS_XSYNC_ENABLE_REG
Bit	Read/Write	Default/Hex	Description
31:8	R/W	0x0	XVS_TO_XHS_T The period of XHS delay to XVS, ${XVS_TO_XHS_T}+1 master clock cycles, no more than XHS_LEN
7:5	/	/	/
4	R/W	0x0	XVS_XHS_OUT_SEL When sensor works in slave mode, this bit select XVS and XHS to output. 0: XHS0, XVS0 1: XHS1,XVS1
3	R/W	0x0	XVS_POL When sensor works in slave mode, this bit set polarity of XVS. 0: Negative 1: Positive
2	R/W	0x0	XHS_POL When sensor works in slave mode, this bit set polarity of XHS. 0: Negative 1: Positive

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Offset: 0x0540			Register Name: CSIC_PRS_XSYNC_ENABLE_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	XVS_OUT_EN When sensor works in slave mode, this bit enables output XVS to sensor 0: Disable 1: Enable
0	R/W	0x0	XHS_OUT_EN When sensor works in slave mode, this bit enables output XHS to sensor 0: Disable 1: Enable

14.1.12.66 0x0544 CSIC Parser XVS Period Register (Default Value: 0x0000_0000)

Offset: 0x0544			Register Name: CSIC_PRS_XVS_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XVS_T The period of XVS signal, ${XVS_T}+2 master clock cycles

14.1.12.67 0x0548 CSIC Parser XHS Period Register (Default Value: 0x0000_0000)

Offset: 0x0548			Register Name: CSIC_PRS_XHS_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XHS_T The period of XHS signal, ${XHS_T}+2 master clock cycles

14.1.12.68 0x054C CSIC Parser XVS LENGTH Register (Default Value: 0x0000_0000)

Offset: 0x054C			Register Name: CSIC_PRS_XVS_LENGTH_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XVS_LEN The valid length of XVS signal, ${XVS_LEN}+1 master clock cycles

14.1.12.69 0x0550 CSIC Parser XHS LENGTH Register (Default Value: 0x0000_0000)

Offset: 0x0550			Register Name: CSIC_PRS_XHS_LENGTH_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0550			Register Name: CSIC_PRS_XHS_LENGTH_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	XHS_LEN The valid length of XHS signal, ${XHS_LEN}+1 master clock cycles

14.1.12.70 0x0554 CSIC Parser SYNC DELAY Register (Default Value: 0x0000_0000)

Offset: 0x0554			Register Name: CSIC_PRS_SYNC_DLY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SYNC_DLY The XHS/XVS will sent after, ${SYNC_DLY}+1 master clock cycles, no more than XVS_LEN

14.1.13 CSIC_DMA Register Description

14.1.13.1 0x0000 CSIC DMA TOP Register (Default Value:0x7000_0000)

Offset:0x0000			Register Name: CSIC_DMA_TOP_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	VER_EN CSIC DMA Version Register Read Enable: 0: Disable 1: Enable
30	R/W	0x1	VFLIP_BUF_ADDR_CFG_MODE Vflip buffer address set by software or calculated by hardware 0: Hardware 1: Software
29	R/W	0x1	BUF_LENGTH_CFG_MODE buffer length set by software or calculated by hardware 0: Hardware 1: Software
28	R/W	0x1	FLIP_SIZE_CFG_MODE FLIP SIZE set by software or calculated by hardware 0: Hardware 1: Software
27:18	/	/	/

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Offset:0x0000			Register Name: CSIC_DMA_TOP_REG
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	FLD_ORDER_SEL Indicate the field order in field mode 0: Field 0 in front and field 1 in back 1: Field 1 in front and field 0 in back field 0 means the even field, which contain frame line 0
16	R/W	0x0	FRM_END_SEL Frame End Select 0: Sel mcsi_bk_frm_end_in (External input AIDB frm_end) 1: Sel dma_frm_end (Vsyn negedge after bk pre-process)
15	/	/	/
14:13	R/W	0x0	VE_ONLINE_CH_SEL Select BK Channel for VE Online handshake
12	R/W	0x0	VE_ONLINE_HANDSHAKE_EN Set this bit to enable frame and line counter for VE online handshake
11:10	/	/	/
9:8	R/W	0x0	MIN_SDR_WR_SIZE Minimum size of SDRAM block write 0: 256 bytes (If hflip is enabled, 256 bytes will be always selected) 1: 512 bytes 2: 1k bytes 3: 2k bytes (not Support For DMA1\2)
7	WAC	0x0	BK CHANNEL REGS REFRESH Refresh BK channel registers from SRAM
6	R/W	0x0	VI_TO_CNT_EN Video Input Timeout Counter Enable Increased by 1 when there is no effective video input with a 12M clock, and cleared to 0 when effective video input is detected. 0: Disable 1: Enable
5	R/W	0x0	FS_FRM_CNT_EN When BK_TOP_EN is enabled, setting this bit to 1 indicates the frame counter starts to add. 0: Disable 1: Enable
4:3	/	/	/

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Offset:0x0000			Register Name: CSIC_DMA_TOP_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	FRM_RATE_CNT_SPL Sampling time for CLK counter per frame 0: Sampling clock counter every frame done 1: Sampling clock counter every vsync
1	R/W	0x0	FRM_RATE_CNT_EN Clk Count Per Frame Enable
0	R/W	0x0	BK_TOP_EN Module Enable 0: Disable 1: Enable

14.1.13.2 0x0004 CSIC DMA Multi-Channel Configuration Register (Default Value:0x0001_1000)

Offset: 0x0004			Register Name: CSIC_DMA_MUL_CH_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R	0x0	CSTATE Debug Info.BK write DDR block FSM state.
27:24	R	0x0	LSTATE Debug Info.BK write DDR line FSM state.
23:22	/	/	/
21:20	R	0x0	FSTATE Debug Info.BK write DDR frame FSM state.
19:16	R	0x1	TR_VLD_ST Debug Info.BK transport FSM state.
15:12	R	0x1	IN_ST Debug Info.BK capture FSM state.
11:10	/	/	/
9:8	R	0x0	CUR_OUT_CH When Multi-Channel enable, this field indicates the Current Output Channel ID
7:6	/	/	/
5:4	R	0x0	CUR_IN_CH When Multi-Channel enable, this field indicates the Current Input Channel ID
3:1	/	/	/
0	R/W	0x0	MUL_CH_EN DMA off-line multi-channel enable 0: Disable 1: Enable

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14.1.13.3 0x0008 CSIC DMA Write Counter Register (Default Value:0x0000_0000)

Offset: 0x0008			Register Name: CSIC_DMA_WRITE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:16	R	0x0	BCNT Debug Info.BK received brdy number.
15:11	/	/	/
10:0	R	0x0	FCNT Debug Info.BK send valid req number.

14.1.13.4 0x000C CSIC DMA FIFO Pointer Register (Default Value:0x0000_0000)

Offset: 0x000C			Register Name: CSIC_DMA_FIFO_PT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	DMA_FIFO_RPT BK data FIFO read pointer. When FIFO full int pull up, this pointer will lock the value of BK data FIFO read pointer until the FIFO full int W1C.
15:14	/	/	/
13:0	R	0x0	DMA_FIFO_WPT BK data FIFO writer pointer. When FIFO full int pull up, this pointer will lock the value of BK data FIFO writer pointer until the FIFO full int W1C.

14.1.13.5 0x0010 CSIC DMA Frame Rate Clock Counter Register (Default Value:0x0000_0000)

Offset: 0x0010			Register Name: CSIC_DMA_FRM_CLK_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R	0x0	FRM_RATE_CLK_CNT Counter value between each frame. For instant hardware frame rate statics. The internal counter is added by one every 12MHz clock cycle. When frame done or vsync comes, the internal counter value is sampled to FRM_RATE_CLK_CNT, and cleared to 0.

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14.1.13.6 0x0014 CSIC DMA Accumulated and Internal Clock Counter Register (Default Value:0x0000_0000)

Offset: 0x0014			Register Name: CSIC_DMA_ACC_ITNL_CLK_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/WC	0x0	ACC_CLK_CNT The accumulated value of FRM_CLK_CNT for software frame rate statics. Every interrupt of frame done, the software checks this accumulated value and clear it to 0. If the ACC_CLK_CNT is larger than 1, the software has frame lost. When frame done or vsync comes, ACC_CLK_CNT = ACC_CLK_CNT + 1, and will be cleared to 0 when this register is being written.
23:0	R	0x0	ITNL_CLK_CNT The instant value of internal frame clock counter. When frame done interrupt comes, the software can query this counter for judging whether it is the time for updating the double buffer address registers.

14.1.13.7 0x0018 CSIC DMA Line Statistic Register (Default Value:0x0000_0000)

Offset: 0x0018			Register Name: CSIC_DMA_LINE_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	LINE_INDEX Indicates the line index in current hsync.
15:0	/	/	/

14.1.13.8 0x001C CSIC DMA PCLK Statistic Register (Default Value:0x0000_7fff)

Offset: 0x001C			Register Name: CSIC_DMA_PCLK_STAT_REG
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30:16	R	0x0	PCLK_CNT_LINE_MAX Indicates maximum pixel clock counter value for each line. Update at every vsync or framedone.
15	/	/	/

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Offset: 0x001C			Register Name: CSIC_DMA_PCLK_STAT_REG
Bit	Read/Write	Default/Hex	Description
14:0	R	0x7fff	PCLK_CNT_LINE_MIN Indicates minimum pixel clock counter value for each line. Update at every vsync or framedone.

14.1.13.9 0x0020 CSIC DMA Fsync Frame Counter Register (Default Value:0x0000_0000)

Offset: 0x0020			Register Name: CSIC_DMA_FS_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	FS_FRM_CNT_CLR When the bit set to 1, Frame cnt clear to 0
30:16	R/W	0x0	FS_FRM_CNT_CLR_DISTANCE Frame cnt clear cycle $N*T_{SYNC}$
15	/	/	/
14:0	R	0x0	FS_FRM_CNT Counter value of frame. When frame done comes, the internal counter value will be increased by 1, and when the register is full, the counter value will be cleared to 0. When parser is sent a sync signal, the counter will be cleared to 0.

14.1.13.10 0x0040 CSIC DMA Video Input Timeout Threshold0 Register (Default Value:0x0000_0000)

Offset: 0x0040			Register Name: CSIC_DMA_VI_TO_TH0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	VI_TO_TH0 Video Input Timeout Threshold0 Set VIDEO_INPUT_TO_INT_PD when VI Counter reaches TH0 after VI_TO_CNT_EN is set , Time Unit is a 12M clock period.

14.1.13.11 0x0044 CSIC DMA Video Input Timeout Threshold1 Register (Default Value:0x0000_0000)

Offset: 0x0044			Register Name: CSIC_DMA_VI_TO_TH1_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0044			Register Name: CSIC_DMA_VI_TO_TH1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	VI_TO_TH1 Video Input Timeout Threshold1 Set VIDEO_INPUT_TO_INT_PD when VI Counter reaches TH1 after the first frame has been input, Time Unit is a 12M clock period.

14.1.13.12 0x0048 CSIC DMA Video Input Timeout Counter Value Register (Default Value:0x0000_0000)

Offset: 0x0048			Register Name: CSIC_DMA_VI_TO_CNT_VAL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R	0x0	VI_TCNT_VAL Video Input Timeout Counter Value Indicate the current value of Video Input Timeout Counter

14.1.13.13 0x004C CSIC DMA VE Information Control Register (Default Value:0x3210_0000)

Offset: 0x004C			Register Name: CSIC_DMA_VE_INFO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x3	DMA TO VE ID_CHN3 DMA to VE chn3 id, set by software
27:24	R/W	0x2	DMA TO VE ID_CHN2 DMA to VE chn2 id, set by software
23:20	R/W	0x1	DMA TO VE ID_CHN1 DMA to VE chn1 id, set by software
19:16	R/W	0x0	DMA TO VE ID_CHN0 DMA to VE chn0 id, set by software
15:7	/	/	/
6	R/W	0x0	VE_FRM_DONE_VLD_BK_DOOR 0: Bk2ve frame done vld will output after last line done vld 1: Old version
5	R/W	0x0	VE_FRM_DONE_MSK 0: Do not mask frame done vld to ve 1: Mask frame done vld to ve

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Offset: 0x004C			Register Name: CSIC_DMA_VE_INFO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	DMA TO VE ID MODE 0: Use original bk channel id send to ve 1: Use software configured id send to ve
3:0	R/W	0x0	VE_INFO_VALID_CYCLE Indicate how many cycle interval between valid signals to ve, only valid for BK0.

14.1.13.14 0x0050 CSIC DMA VE Frame Counter Value Register (Default Value:0x0000_0000)

Offset: 0x0050			Register Name: CSIC_DMA_VE_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	FRM_DONE_CNT Indicates the number of frames that have been stored in the DDR.
15:8	/	/	/
7:0	R	0x0	FRM_ST_CNT Indicates the Frame Number which is being stored.

14.1.13.15 0x0054 CSIC DMA VE Line Counter Value Register (Default Value:0x0000_0000)

Offset: 0x0054			Register Name: CSIC_DMA_VE_LINE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	LINE_DONE_CNT Indicates the number of lines that have been stored in the DDR.
15:14	/	/	/
13:0	R	0x0	LINE_ST_CNT Indicates the Line Number which is being stored.

14.1.13.16 0x0058 CSIC DMA VE Current Frame Address Register (Default Value:0x0000_0000)

Offset: 0x0058			Register Name: CSIC_DMA_VE_CUR_FRM_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CUR_FRM_ADDR Indicates the FIFO0 address of current frame being stored.

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14.1.13.17 0x005C CSIC DMA VE Last Frame Address Register (Default Value:0x0000_0000)

Offset: 0x005C			Register Name: CSIC_DMA_VE_LAST_FRM_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LAST_FRM_ADDR Indicates the FIFO0 Address of the last frame.

14.1.13.18 0x0060 CSIC DMA VE Embed Address Register (Default Value:0x0000_0000)

Offset: 0x0060			Register Name: CSIC_DMA_VE_EMBED_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	EMBED_ADDR_TO_VE Indicates the Embedded Address send to VE

14.1.13.19 0x0064 CSIC DMA Capture Status Register (Default Value:0x0000_0000)

Offset: 0x0064			Register Name: CSIC_DMA_CAP_STA_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R	0x0	FIELD_STA The status of the received field 0: Field 0 1: Field 1
1:0	/	/	/

14.1.13.20 0x0068 CSIC DMA TDM Time Stamp0 Register (Default Value:0x0000_0000)

Offset: 0x0068			Register Name: CSIC_DMA_TIME_STAMP0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TDM_BASE_TIME Indicates the TDM base time of the last frame

14.1.13.21 0x006C CSIC DMA TDM Time Stamp1 Register (Default Value:0x0000_0000)

Offset: 0x006C			Register Name: CSIC_DMA_TIME_STAMP1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TDM_DELTA_TIME Indicates the TDM delta time of the last frame

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14.1.13.22 0x0070 CSIC DMA TDM Time Stamp2 Register (Default Value:0x0000_0000)

Offset: 0x0070			Register Name: CSIC_DMA_TIME_STAMP2_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TDM_FRAME_COUNTER Indicates the TDM frame counter of the last frame

14.1.13.23 0x0080 CSIC DMA FIFO Statistic Register (Default Value:0x0000_0000)

Offset: 0x0080			Register Name: CSIC_DMA_FIFO_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R	0x0	FIFO_FRM_MAX Indicates the maximum depth of FIFO being occupied for whole frame. Update at every vsync or framedone. Unit is byte.

14.1.13.24 0x0084 CSIC DMA FIFO Threshold Register (Default Value:0x0000_0000)

Offset: 0x0084			Register Name: CSIC_DMA_FIFO_THRS_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R/W	0x0	FIFO_THRS When the occupied memory of FIFO exceeds the threshold, DRAM frequency cannot be changed. Unit is byte.

14.1.13.25 0x0100 CSIC DMA TOP Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x0100			Register Name: CSIC_DMA_TOP_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	DMA_TO_VE_VALID_LOST_EN Set an INT when DMA to VE valid signal is lost in top arbiter
2	R/W	0x0	VIDEO_INPUT_TO_INT_EN Set an INT when no video input exceeds the setting threshold time
1	R/W	0x0	CLR_FS_FRM_CNT_INT_EN Set a INT When Clear FS Frame cnt.

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Offset: 0x0100			Register Name: CSIC_DMA_TOP_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	FS_PUL_INT_EN Set an INT when a Fsync signal is received

14.1.13.26 0x0104 CSIC DMA TOP Interrupt Status Register (Default Value:0x0000_0000)

Offset: 0x0104			Register Name: CSIC_DMA_TOP_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W1C	0x0	DMA_TO_VE_VALID_LOST_PD Set an INT Pending when DMA to VE valid signal is lost in top arbiter
2	R/W1C	0x0	VIDEO_INPUT_TO_INT_PD Set an INT Pending when no video input exceeds the setting threshold time
1	R/W1C	0x0	CLR_FRAME_CNT_INT_PD Set a INT When Clear FS Frame cnt.
0	R/W1C	0x0	FS_PUL_INT_PD Set an INT when a Fsync signal is received

14.1.13.27 0x0110 CSIC DMA VE ID Enable Register (Default Value:0x0000_0000)

Offset: 0x0110			Register Name: CSIC_DMA_VE_ID_EN_REG
Bit	Read/Write	Default/Hex	Description
30:17	/	/	/
16	R/W	0x0	VE_ONLINE_ENC_MODE 0: Use new online encoding solution 1: Use old online encoding solution
15:10	/	/	/
9:8	R/W	0x0	BK_EMBEDDED_FRAME_SEL 0: Send the embedded address of current frame to VE. 1: Send the embedded address of last frame to VE. 2: Send the embedded address of 2 frame before current frame to VE. 3: Send the embedded address of 3 frame before current frame to VE.

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Offset: 0x0110			Register Name: CSIC_DMA_VE_ID_EN_REG
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x0	SENSOR_ID_EN From parser to bk, id0-7 data path is on(bit0-7 corresponding to id0-7).

14.1.13.28 0x0200 CSIC DMA Channel0 Enable Register (Default Value:0x0000_0000)

Offset:0x0200			Register Name: CSIC_DMA_CH0_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EMBEDDED_DATA_WR_EN 0: BK does not write any info of embedded data 1: BK writes all info of embedded data
11:9	/	/	/
8	R/W	0x0	FRM_DROP_EN Drop Frame when vsync comes but buffer address is not ready 0: Not Drop 1: Drop
7	R/W	0x0	BUF_ADDR_MODE Frame buffer Address configuration mode 0: Buffer Address Register Mode 1: Buffer Address FIFO Mode
6	WAC	0x0	BUFA_FIFO_INIT Write this bit 1 to init bufa_fifo, and it will self clear to 0
5:2	/	/	/
1	R/W	0x0	LBC_EN LBC Function Enable 0: Disable 1: Enable
0	R/W	0x0	CAP_EN Video in Capture Enable 0: Disable 1: Enable

14.1.13.29 0x0204 CSIC DMA Channel0 Configuration Register (Default Value:0x0000_0000)

Offset: 0x0204			Register Name: CSIC_DMA_CH0_CFG_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0204			Register Name: CSIC_DMA_CH0_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	PAD_VAL Padding value when OUTPUT_FMT is prgb888 0x00-0xff
23:22	/	/	/
21	R/W	0x0	YUV10TO8_EN YUV 10bit input cut to 8bit 0: Disable 1: Enable
20	R/W	0x0	YUV10_STO_FMT YUV 10bit store configure 0: YUV 10bit Stored in low 10bit of a 16bit-word 1: YUV 10bit Stored in high 10bit of a 16bit-word
19:16	R/W	0x0	OUTPUT_FMT Output data format When the input format is set RAW stream 0000: Raw-8 0001: Raw-10 0010: Raw-12 0011: Raw-14 0100: Raw-16 0101: Raw-20 0110: Raw-24 0111: Reserved 1000: Rgb565 1001: Rgb888 1010: Prgb888 others: Reserved When the input format is set YUV422 0000: Field planar YCbCr 422 0001: Field planar YCbCr 420 0010: Frame planar YCbCr 420 0011: Frame planar YCbCr 422 0100: Field planar YCbCr 422 UV combined (UV sequence) 0101: Field planar YCbCr 420 UV combined (UV sequence) 0110: Frame planar YCbCr 420 UV combined (UV sequence) 0111: Frame planar YCbCr 422 UV combined (UV sequence) 1000: Field planar YCbCr 422 UV combined (VU

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Offset: 0x0204			Register Name: CSIC_DMA_CH0_CFG_REG
Bit	Read/Write	Default/Hex	Description
			sequence) 1001: Field planar YCbCr 420 UV combined (VU sequence) 1010: Frame planar YCbCr 420 UV combined (VU sequence) 1011: Frame planar YCbCr 422 UV combined (VU sequence) 1100: Reserved 1101: Field YCbCr 400 1110: Reserved 1111: Frame YCbCr 400 When the input format is set YUV420 0000: LBC Mode Output 0001: Field planar YCbCr 420 0010: Frame planar YCbCr 420 0011: Reserved 0100: Reserved 0101: Field planar YCbCr 420 UV combined (UV sequence) 0110: Frame planar YCbCr 420 UV combined (UV sequence) 0111-1000: Reserved 1001: Field planar YCbCr 420 UV combined (VU sequence) 1010: Frame planar YCbCr 420 UV combined (VU sequence) 1011-1100: Reserved 1101: Field YCbCr 400 1110: Reserved 1111: Frame YCbCr 400
15:14	/	/	/
13	R/W	0x0	VFLIP_EN Vertical flip enable When enabled, the received data will be arranged in vertical flip. 0: Disable 1: Enable

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Offset: 0x0204			Register Name: CSIC_DMA_CH0_CFG_REG
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	HFLIP_EN Horizontal flip enable When enabled, the received data will be arranged in horizontal flip. 0: Disable 1: Enable
11:10	R/W	0x0	FIELD_SEL Field selection. 00: Capturing with field 0. 01: Capturing with field 1. 10: Capturing with either field. 11: Reserved
9:6	R/W	0x0	FPS_DS_PRD Fps down sample period 0: 1 frames 1: 2 frames 2: 3 frames 3: 4 frames 4: 5 frames ..... 15: 16 frames Note: CH0 FPS_DS_PRD should be used together with CSIC_DMA_CH0_FRM_MSK_CFG_REG bit [15:0]. Show an example as following. When CH0 FPS_DS_PRD is configured as 4'd6, CSIC_DMA_CH0_FRM_MSK_CFG_REG bit [6:0] will take effect and bit [15:7] will be ignored. If CSIC_DMA_CH0_FRM_MSK_CFG_REG bit[6:0] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.
5:2	R/W	0x0	CAP_FRONT_MASK_NUM Indicates the frame number masked after CAP_EN set.
1:0	/	/	/

14.1.13.30 0x0208 CSIC DMA Channel0 Embedded Address Register (Default Value:0x0000_0000)

Offset: 0x0208			Register Name: CSIC_DMA_CH0_EMBD_ADDR_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0208			Register Name: CSIC_DMA_CH0_EMBD_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	EMBEDDED DATA ADDRESS BK Received Embedded Data Write DDR Address. Unit: Word.

14.1.13.31 0x020C CSIC DMA Channel0 Frame Mask Configuration Register (Default Value:0x0000_0000)

Offset: 0x020C			Register Name: CSIC_DMA_CH0_FRM_MSK_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0	FRAME_DATA_MASK Only [FPS_DS_PRD:0] are valid

14.1.13.32 0x0210 CSIC DMA Channel0 Horizontal Size Register (Default Value:0x0000_0000)

Offset: 0x0210			Register Name: CSIC_DMA_CH0_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	HOR_LEN When BK_TOP_EN is enabled and LBC_EN is disabled, HOR_LEN will be enabled. These bits indicate the unit length of valid horizontal pixels of a line in DMA mode.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.13.33 0x0214 CSIC DMA Channel0 Vertical Size Register (Default Value:0x0000_0000)

Offset: 0x0214			Register Name: CSIC_DMA_CH0_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	VER_LEN When BK_TOP_EN enable and LBC_EN is disabled, VER_LEN will be enabled. These bits indicate valid line number of a frame in DMA mode.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

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14.1.13.34 0x0218 CSIC DMA Channel0 Crop Mode Register (Default Value:0x0000_0000)

Offset: 0x0218			Register Name: CSIC_DMA_CH0_CROP_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	FRM_LOST_CNT_EN When set 1, FRM_LOST_CNT is enabled; when set 0, FRM_LOST_CNT is cleared.
7:5	/	/	/
4	R/W	0x0	REDUNDANT_ERR_MSK_EN 0: Disable redundant error mask 1: Enable redundant error mask When input picture size exceed bk set width or height, set this bit to 0 to pull up redundant error interrupt and do crop. Set this bit to 1 to do crop and not pull up redundant error interrupt.
3:1	/	/	/
0	R/W	0x0	FLD_FRM_VCROP_EN Field Frame Vcrop Enable 0: Disable field frame vcrop 1: Enable field frame vcrop When input frame is of field type and V crop wants to be implemented, this bit should be configured as 1. (note: Only even lines crop is supported for field input V crop)

14.1.13.35 0x021C CSIC DMA Channel0 Line Counter Register (Default Value:0x0000_0000)

Offset: 0x021C			Register Name: CSIC_DMA_CH0_LINE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:0	R/W	0x0	LINE_CNT_NUM The LINE_CNT_NUM value is set by user, when internal line counter reach the set value, the LC_PD will set.

14.1.13.36 0x0220 CSIC DMA Channel0 FIFO 0 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0220			Register Name: CSIC_DMA_CH0_F0_BUFA_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0220			Register Name: CSIC_DMA_CH0_F0_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F0_BUFA When LBC_EN disable, these bits indicate FIFO 0 output buffer-A address in DMA mode. When LBC_EN enable, these bits indicate the output buffer address in LBC mode.

14.1.13.37 0x0224 CSIC DMA Channel0 FIFO 1 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0224			Register Name: CSIC_DMA_CH0_F1_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F1_BUFA FIFO 1 output buffer-A address.

14.1.13.38 0x0228 CSIC DMA Channel0 FIFO 2 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0228			Register Name: CSIC_DMA_CH0_F2_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F2_BUFA FIFO 2 output buffer-A address

14.1.13.39 0x022C CSIC DMA Channel0 Buffer Length Register (Default Value:0x0000_0000)

Offset: 0x022C			Register Name: CSIC_DMA_CH0_BUF_LEN_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	BUF_LEN_C DMA_MODE: Buffer length of chroma C in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y in ONLY Y line. Unit is byte.
15:0	R/W	0x0	BUF_LEN DMA_MODE: Buffer length of luminance Y in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y and chroma C in YC line. Unit is byte.

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14.1.13.40 0x0230 CSIC DMA Channel0 Flip Size Register (Default Value:0x0000_0000)

Offset: 0x0230			Register Name: CSIC_DMA_CH0_FLIP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	VER_LEN Vertical line number when in VFLIP mode. Unit is line. Only Readable when FLIP_SIZE_CFG_MODE set 0
15	/	/	/
14:0	R/W	0x0	VALID_LEN Valid components of a line when in HFLIP mode. Unit is pixel component. Only Readable when FLIP_SIZE_CFG_MODE set 0

14.1.13.41 0x0234 CSIC DMA Channel0 Abnormal Mask Register (Default Value:0x0000_0000)

Offset: 0x0234			Register Name: CSIC_DMA_CH0_ABN_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	FRM_DONE_INT_MSK_EN Decide whether the FD_INT of this frame will be masked or not once any abnormal condition happens during this frame. 0: Disable MSK 1: Enable MSK
4	R/W	0x0	LINE_REDUNDANT_MSK_EN Decide whether the remaining data in this frame will be dropped or not once line redundant error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
3	R/W	0x0	LINE_MISS_MSK_EN Decide whether the remaining data in this frame will be dropped or not once line miss error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.

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Offset: 0x0234			Register Name: CSIC_DMA_CH0_ABN_MSK_REG
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PIXEL_REDUNDANT_MSK_EN Decide whether the remaining data in this frame will be dropped or not once pixel redundant error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
1	R/W	0x0	PIXEL_MISS_MSK_EN Decide whether the remaining data in this frame will be dropped or not once pixel miss error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
0	R/W	0x0	VSYN_ABN_MSK_EN Decide whether the remaining data in this frame will be dropped or not once abnormal happens in this frame. 0: Disable MSK 1: Enable MSK Note: When VSYN_ABN_MSK_EN is set as 1 and one of the following conditions happens, the remaining data in this frame will be dropped. And the next frame will be received as normal. 1, when trying to fill dummy for pixel miss and LACK_HBLANK happens. 2, when PIXEL_REDUNDANT happens and PIXEL_REDUNDANT_MSK_EN is set as 1. 3, when PIXEL_MISS happens, LACK_HBLANK no happen and PIXEL_MISS_MSK_EN is set as 1. 4, when LINE_REDUNDANT happens and LINE_REDUNDANT_MSK_EN is set as 1. 5, when LINE_MISS happens and LINE_MISS_MSK_EN is set as 1.

14.1.13.42 0x0238 CSIC DMA Channel0 Pixels Add Enable Register (Default Value:0x0000_0000)

Offset: 0x0238	Register Name: CSIC_DMA_CH0_PIXELS_ADD_EN_REG

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Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	PIXELS_ADD_EN Decide whether dummy pixels will be added in the end of the line or not once pixels' miss error happens. 0: Disable ADD 1: Enable ADD

14.1.13.43 0x023C CSIC DMA Channel0 Pixels Add Value Register (Default Value:0x0000_0000)

Offset: 0x023C			Register Name: CSIC_DMA_CH0_PIXELS_ADD_VAL_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PIXELS_ADD_VALUE Decide the value of dummy pixels when PIXELS_ADD_EN is set as 1

14.1.13.44 0x0240 CSIC DMA Channel0 Total Frame Counter Config Register (Default Value:0x0000_0000)

Offset: 0x0240			Register Name: CSIC_DMA_CH0_TOT_FRM_CNT_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:2	R/W	0x0	TOTAL_FRM_CNT_SEL Choose which source to trigger frame counter 0: Frame done 1: Frame start (vsyn pos edge) 2: Frame_end(vsyn neg edge)
1	W1C	0x0	TOTAL_FRM_CNT_CLR When set this bit to 1, the total frame counter will clear to 0
0	R/W	0x0	TOTAL_FRM_CNT_EN Total frame counter enable, if you want hardware frame counter, you should set this bit to 1. 0: Disable Total Frame Counter 1: Enable Total Frame Counter

14.1.13.45 0x0244 CSIC DMA Channel0 BUF Threshold Register (Default Value:0x0000_0000)

Offset: 0x0244			Register Name: CSIC_DMA_CH0_BUF_TH_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0244			Register Name: CSIC_DMA_CH0_BUF_TH_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:16	R/W	0x0	CSIC_DMA_STORED_FRM_THRESHOLD when stored frame counter value reaches the threshold , counter is cleared to 0 , only used in Buffer Addr FIFO Mode
15:6	/	/	/
5:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO_THRESHOLD when content in Buffer Address FIFO no more than the threshold, an interrupt is set, only used in Buffer Addr FIFO Mode

14.1.13.46 0x02E8 CSIC DMA Channel0 BUF Address FIFO0 Entry Register (Default Value:0x0000_0000)

Offset: 0x02E8			Register Name: CSIC_DMA_CH0_BUF_ADDR_FIFO0_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO0_ENTRY FIFO Entry of Buffer Address FIFO0 for input frames to be stored, only used in Buffer Addr FIFO Mode

14.1.13.47 0x02EC CSIC DMA Channel0 BUF Address FIFO1 Entry Register (Default Value:0x0000_0000)

Offset: 0x02EC			Register Name: CSIC_DMA_CH0_BUF_ADDR_FIFO1_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO1_ENTRY FIFO Entry of Buffer Address FIFO1 for input frames to be stored, only used in Buffer Addr FIFO Mode

14.1.13.48 0x02F0 CSIC DMA Channel0 BUF Address FIFO2 Entry Register (Default Value:0x0000_0000)

Offset: 0x02F0			Register Name: CSIC_DMA_CH0_BUF_ADDR_FIFO2_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO2_ENTRY FIFO Entry of Buffer Address FIFO2 for input frames to be stored, only used in Buffer Addr FIFO Mode

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14.1.13.49 0x02F4 CSIC DMA Channel0 Embed Address Entry Register (Default Value:0x0000_0000)

Offset: 0x02F4			Register Name: CSIC_DMA_CH0_EMBED_ADDR_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_EMBED_ADDR_ENTRY FIFO Entry of Buffer Embed Address for input frames to be stored, only used in Buffer Addr FIFO Mode

14.1.13.50 0x02F8 CSIC DMA Channel0 Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x02F8			Register Name: CSIC_DMA_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	LACK_HBLANK_INT_EN Set an INT when trying filling dummy for missed pixels at the end of the line but hblank is too short.
22	R/W	0x0	LINE_REDUNDANT_INT_EN Set an INT when the amount of lines in one frame is more than expected.
21	R/W	0x0	LINE_MISS_INT_EN Set an INT when the amount of lines in one frame is less than expected.
20	R/W	0x0	PIXEL_REDUNDANT_INT_EN Set an INT when the amount of pixels in one line is more than expected.
19	R/W	0x0	PIXEL_MISS_INT_EN Set an INT when the amount of pixels in one line is less than expected.
18	R/W	0x0	BUF_ADDR_OVERFLOW_INT_EN Set an INT when buffer address is overwrite before used
17	R/W	0x0	BUF_ADDR_UNDERFLOW_INT_EN Set an INT when new frame comes but buffer address is not ready
16	R/W	0x0	LBC_HBLKMIN_INT_EN Set an INT when hblanking less than 48 bk_clk cycles, only used in LBC Mode
15	R/W	0x0	FRM_LOST_INT_EN Set an INT once frame is in when last frame processing

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Offset: 0x02F8			Register Name: CSIC_DMA_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	STORED_FRM_CNT_INT_EN Set an INT when the value of CSIC_DMA_STORED_FRM_CNT reaches CSIC_DMA_STORED_FRM_THRESHOLD , only used in BUF Address FIFO MODE
13	R/W	0x0	BUF_ADDR_FIFO_INT_EN Set an INT when content in BUF Address FIFO less than CSIC_DMA_BUFA_FIFO_THRESHOLD, only used in BUF Address FIFO MODE
12:9	/	/	/
8	R/W	0x0	PARA_LOAD_UNFINISH_INT_EN BK read regs from SRAM process is unfinished, only could happen in offline mode
7	R/W	0x0	VS_INT_EN vsync flag The bit is set when vsync come. And at this time load the buffer address for the coming frame. So after this IRQ come, change the buffer address could only effect next frame
6	R/W	0x0	LI_OF_INT_EN Line information FIFO(16 lines) overflow.
5	R/W	0x0	LC_INT_EN Line counter flag The bit is set when the specific line has been written to dram every frame. The line number is set in the line counter register.
4:3	/	/	/
2	R/W	0x0	FIFO_OF_INT_EN FIFO overflow The bit is set when the FIFO become overflow.
1	R/W	0x0	FD_INT_EN Frame done Indicates the CSI has finished capturing an image frame. Applies to video capture mode. The bit is set after each completed frame capturing data is wrote to buffer as long as video capture remains enabled.

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Offset: 0x02F8			Register Name: CSIC_DMA_CH0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CD_INT_EN Capture done Indicates the CSI has completed capturing the image data. For still capture, the bit is set when one frame data has been written to buffer. For video capture, the bit is set when the last frame has been written to buffer after video capture has been disabled. For CCIR656 interface, if the output format is frame planar YCbCr 420 mode, the frame end means the field2 end, the other frame end means field end.

14.1.13.51 0x02FC CSIC DMA Channel0 Interrupt Status Register (Default Value:0x0000_0000)

Offset: 0x02FC			Register Name: CSIC_DMA_CH0_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	LACK_HBLANK_INT_PD Set an INT when trying filling dummy for missed pixels at the end of the line but hblank is too short.
22	R/W1C	0x0	LINE_REDUNDANT_INT_PD Set an INT when the amount of lines in one frame is more than expected.
21	R/W1C	0x0	LINE_MISS_INT_PD Set an INT when the amount of lines in one frame is less than expected.
20	R/W1C	0x0	PIXEL_REDUNDANT_INT_PD Set an INT when the amount of pixels in one line is more than expected.
19	R/W1C	0x0	PIXEL_MISS_INT_PD Set an INT when the amount of pixels in one line is less than expected.
18	R/W1C	0x0	BUF_ADDR_OVERFLOW_INT_PD Set an INT when buffer address is overwrite before used
17	R/W1C	0x0	BUF_ADDR_UNDERFLOW_INT_PD Set an INT when new frame comes but buffer address is not ready

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Offset: 0x02FC			Register Name: CSIC_DMA_CH0_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
16	R/W1C	0x0	LBC_HBLKMIN_INT_PD Set an INT when hblanking less than 48 bk_clk cycles, only use in LBC Mode
15	R/W1C	0x0	FRM_LOST_INT_PD Set an INT once frame is in when last frame processing
14	R/W1C	0x0	STORED_FRM_CNT_INT_PD Set an INT when the value of CSIC_DMA_STORED_FRM_CNT reaches CSIC_DMA_STORED_FRM_THRESHOLD, only use in BUF Address FIFO MODE
13	R/W1C	0x0	BUF_ADDR_FIFO_INT_PD Set an INT when content in BUF Address FIFO no more than CSIC_DMA_BUFA_FIFO_THRESHOLD, only use in BUF Address FIFO MODE
12:9	/	/	/
8	R/W1C	0x0	PARA_LOAD_UNFINISH_PD BK read regs from sram process is unfinished, only could happen in offline mode
7	R/W1C	0x0	VS_PD Vsync flag
6	R/W1C	0x0	LI_OF_PD Line information FIFO(16 lines) overflow
5	R/W1C	0x0	LC_PD Line counter flag
4:3	/	/	/
2	R/W1C	0x0	FIFO_OF_PD FIFO 0 overflow
1	R/W1C	0x0	FD_PD Frame done
0	R/W1C	0x0	CD_PD Capture done

14.1.13.52 0x0300 CSIC DMA Channel0 FIFO 0 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0300			Register Name: CSIC_DMA_CH0_F0_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0300			Register Name: CSIC_DMA_CH0_F0_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F0_BUFA_RESULT Indicate the final F0_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

14.1.13.53 0x0304 CSIC DMA Channel0 FIFO 1 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0304			Register Name: CSIC_DMA_CH0_F1_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F1_BUFA_RESULT Indicate the final F1_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

14.1.13.54 0x0308 CSIC DMA Channel0 FIFO 2 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0308			Register Name: CSIC_DMA_CH0_F2_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F2_BUFA_RESULT Indicate the final F2_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

14.1.13.55 0x030C CSIC DMA Channel0 Buffer Length Result Register (Default Value:0x0000_0000)

Offset: 0x030C			Register Name: CSIC_DMA_CH0_BUF_LEN_RESULT_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x030C			Register Name: CSIC_DMA_CH0_BUF_LEN_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	BUF_LEN_C_RESULT DMA_MODE: Buffer length of chroma C in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y in ONLY Y line. Unit is byte.
15:0	R	0x0	BUF_LEN_RESULT DMA_MODE: Buffer length of luminance Y in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y and chroma C in YC line. Unit is byte.

14.1.13.56 0x0310 CSIC DMA Channel0 Flip Size Result Register (Default Value:0x02d0_0500)

Offset: 0x0310			Register Name: CSIC_DMA_CH0_FLIP_SIZE_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	VER_LEN_RESULT Vertical line number when in VFLIP mode. Unit is line. Only Readable when FLIP_SIZE_CFG_MODE set 0
15	/	/	/
14:0	R	0x0	VALID_LEN_RESULT Valid components of a line when in HFLIP mode. Unit is pixel component. Only Readable when FLIP_SIZE_CFG_MODE set 0

14.1.13.57 0x0314 CSIC DMA Channel0 BUF Address FIFO Content Register (Default Value:0x0000_0000)

Offset: 0x0314			Register Name: CSIC_DMA_CH0_BUF_ADDR_FIFO_CON_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:24	R	0x0	CSIC_DMA_BUF_ADDR_EMBED_CONTENT FIFO Content of address buffered in Buffer Address Embedded, only used in Buffer Addr FIFO Mode
23:22	/	/	/

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Offset: 0x0314			Register Name: CSIC_DMA_CH0_BUF_ADDR_FIFO_CON_REG
Bit	Read/Write	Default/Hex	Description
21:16	R	0x0	CSIC_DMA_BUF_ADDR_FIFO2_CONTENT FIFO Content of address buffered in Buffer Address FIFO2, only used in Buffer Addr FIFO Mode
15:14	/	/	/
13:8	R	0x0	CSIC_DMA_BUF_ADDR_FIFO1_CONTENT FIFO Content of address buffered in Buffer Address FIFO1, only used in Buffer Addr FIFO Mode
7:6	/	/	/
5:0	R	0x0	CSIC_DMA_BUF_ADDR_FIFO0_CONTENT FIFO Content of address buffered in Buffer Address FIFO0, only used in Buffer Addr FIFO Mode

14.1.13.58 0x0318 CSIC DMA Channel0 Stored Frame Counter Register (Default Value:0x0000_0000)

Offset: 0x0318			Register Name: CSIC_DMA_CH0_STORED_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	CSIC_DMA_STORED_FRM_CNT Indicates value of stored frames counter, when counter value reaches CSIC_DMA_STORED_FRM_THRESHOLD, counter is cleared to 0, only used in Buffer Addr FIFO Mode

14.1.13.59 0x031C CSIC DMA Channel0 Total Frame Counter Register (Default Value:0x0000_0000)

Offset: 0x031C			Register Name: CSIC_DMA_CH0_TOT_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TOTAL_FRM_CNT Hardware Frame Counter

14.1.13.60 0x0324 CSIC DMA Channel0 Frame Lost Counter Register (Default Value:0x0000_0000)

Offset: 0x0324			Register Name: CSIC_DMA_CH0_FRM_LOST_CNT_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0324			Register Name: CSIC_DMA_CH0_FRM_LOST_CNT_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	CAP_STA capture in progress Indicates the CSI is capturing image data. The bit is set at the start of the first frame after enabling frame capture.
30:20	/	/	/
19:16	R	0x0	AI ISP_FRM_NUM Indicates this AI ISP Frame number, lock TOTAL_FRM_CNT low 4bit when error occurred, release when frm_done int is clear
15:12	R	0x0	ISP_ERROR_FRM_NUM Indicates this ERROR Frame number, lock TOTAL_FRM_CNT low 4bit when error occurred, release when frm_done int is clear
11:10	/	/	/
9	R	0x0	AI_ISP_FLAG Indicates this Frame is AI ISP Frame, will lock until frm_done int is clear
8	R	0x0	ISP_ERROR_FLAG Indicates this Frame is ISP ERROR Frame, will lock until frm_done int is clear
7:0	R	0x0	FRM_LOST_CNT Once a new frame comes when the buffer address not ready and FRM_DROP_EN set, or when the last frame still processing, the new frame is dropped and the FRM_LOST_CNT increases when FRM_LOST_CNT_EN set.

14.1.13.61 0x0328 CSIC DMA Channel0 Embed Address Read Register (Default Value:0x0000_0000)

Offset: 0x0328			Register Name: CSIC_DMA_CH0_EMBED_ADDR_READ_REG
Bit	Read/Write	Default/Hex	Description
31:00	R	0x0	EMBEDDED_ADDR Embedded Address of Current Frame

14.1.13.62 0x0400 CSIC DMA Channel1 Enable Register (Default Value:0x0000_0000)

Offset:0x0400			Register Name: CSIC_DMA_CH1_EN_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x0400			Register Name: CSIC_DMA_CH1_EN_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	EMBEDDED_DATA_WR_EN 0: BK not write any info of embedded data 1: BK write all info of embedded data
11:9	/	/	/
8	R/W	0x0	FRM_DROP_EN Drop Frame when vsync comes but buffer address is not ready 0: Not Drop 1: Drop
7	R/W	0x0	BUF_ADDR_MODE Frame buffer Address configuration mode 0: Buffer Address Register Mode 1: Buffer Address FIFO Mode
6:2	/	/	/
1	R/W	0x0	LBC_EN LBC Function Enable 0: Disable 1: Enable
0	R/W	0x0	CAP_EN Video in Capture Enable 0: Disable 1: Enable

14.1.13.63 0x0404 CSIC DMA Channel1 Configuration Register (Default Value:0x0000_0000)

Offset: 0x0404			Register Name: CSIC_DMA_CH1_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	PAD_VAL Padding value when OUTPUT_FMT is prgb888 0x00-0xff
23:22	/	/	/
21	R/W	0x0	YUV10TO8_EN YUV 10bit input cut to 8bit 0: Disable 1: Enable
20	R/W	0x0	YUV10_STO_FMT YUV 10bit store configure 0: YUV 10bit Stored in low 10bit of a 16bit-word 1: YUV 10bit Stored in high 10bit of a 16bit-word
19:16	R/W	0x0	OUTPUT_FMT

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Offset: 0x0404			Register Name: CSIC_DMA_CH1_CFG_REG
Bit	Read/Write	Default/Hex	Description
			Output data format When the input format is set RAW stream 0000: Raw-8 0001: Raw-10 0010: Raw-12 0011: Raw-14 0100: Raw-16 0101: Raw-20 0110: Raw-24 0111: Reserved 1000: Rgb565 1001: Rgb888 1010: Prgb888 others: Reserved When the input format is set YUV422 0000: Field planar YCbCr 422 0001: Field planar YCbCr 420 0010: Frame planar YCbCr 420 0011: Frame planar YCbCr 422 0100: Field planar YCbCr 422 UV combined (UV sequence) 0101: Field planar YCbCr 420 UV combined (UV sequence) 0110: Frame planar YCbCr 420 UV combined (UV sequence) 0111: Frame planar YCbCr 422 UV combined (UV sequence) 1000: Field planar YCbCr 422 UV combined (VU sequence) 1001: Field planar YCbCr 420 UV combined (VU sequence) 1010: Frame planar YCbCr 420 UV combined (VU sequence) 1011: Frame planar YCbCr 422 UV combined (VU sequence) 1100: Reserved 1101: Field YCbCr 400 1110: Reserved 1111: Frame YCbCr 400 When the input format is set YUV420

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Offset: 0x0404			Register Name: CSIC_DMA_CH1_CFG_REG
Bit	Read/Write	Default/Hex	Description
			0000: LBC Mode Output 0001: Field planar YCbCr 420 0010: Frame planar YCbCr 420 0011: Reserved 0100: Reserved 0101: Field planar YCbCr 420 UV combined (UV sequence) 0110: Frame planar YCbCr 420 UV combined (UV sequence) 0111-1000: Reserved 1001: Field planar YCbCr 420 UV combined (VU sequence) 1010: Frame planar YCbCr 420 UV combined (VU sequence) 1011-1100: Reserved 1101: Field YCbCr 400 1110: Reserved 1111: Frame YCbCr 400
15:14	/	/	/
13	R/W	0x0	VFLIP_EN Vertical flip enable When enabled, the received data will be arranged in vertical flip. 0: Disable 1: Enable
12	R/W	0x0	HFLIP_EN Horizontal flip enable When enabled, the received data will be arranged in horizontal flip. 0: Disable 1: Enable
11:10	R/W	0x0	FIELD_SEL Field selection. 00: Capturing with field 0. 01: Capturing with field 1. 10: Capturing with either field. 11: Reserved

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Offset: 0x0404			Register Name: CSIC_DMA_CH1_CFG_REG
Bit	Read/Write	Default/Hex	Description
9:6	R/W	0x0	FPS_DS_PRD Fps down sample period 0: 1 frames 1: 2 frames 2: 3 frames 3: 4 frames 4: 5 frames ..... 15: 16 frames Note: CH1 FPS_DS_PRD should be used together with CSIC_DMA_CH1_FRM_MSK_CFG_REG bit [15:0]. Show an example as following. When CH1 FPS_DS_PRD is configured as 4'd6, CSIC_DMA_CH1_FRM_MSK_CFG_REG bit [6:0] will take effect and bit [15:7] will be ignored. If CSIC_DMA_CH1_FRM_MSK_CFG_REG bit[6:0] is configured as 7'b1011011, frame 0,frame 1, frame 3,frame 4, frame 6, frame 7,frame 8, frame 10,frame 11, frame 13...will be dropped.
5:2	R/W	0x0	CAP_FRONT_MASK_NUM Indicates the frame number masked after CAP_EN set.
1:0	/	/	/

14.1.13.64 0x0408 CSIC DMA Channel1 Embedded Address Register (Default Value:0x0000_0000)

Offset: 0x0408			Register Name: CSIC_DMA_CH1_EMBD_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	EMBEDDED DATA ADDRESS BK Received Embedded Data Write DDR Address. Unit: Word;

14.1.13.65 0x040C CSIC DMA Channel1 Frame Mask Configuration Register (Default Value:0x0000_0000)

Offset: 0x040C			Register Name: CSIC_DMA_CH1_FRM_MSK_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0	FRAME_DATA_MASK Only [FPS_DS_PRD:0] are valid

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14.1.13.66 0x0410 CSIC DMA Channel1 Horizontal Size Register (Default Value:0x0000_0000)

Offset: 0x0410			Register Name: CSIC_DMA_CH1_HSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	HOR_LEN When BK_TOP_EN enable, DMA_EN enable, these bits indicate Horizontal pixel unit length. Valid pixel of a line in DMA mode.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

14.1.13.67 0x0414 CSIC DMA Channel1 Vertical Size Register (Default Value:0x0000_0000)

Offset: 0x0414			Register Name: CSIC_DMA_CH1_VSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	VER_LEN When BK_TOP_EN enable, DMA_EN enable, these bits indicate Valid line number of a frame in DMA mode.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

14.1.13.68 0x0418 CSIC DMA Channel1 Crop Mode Register (Default Value:0x0000_0000)

Offset: 0x0418			Register Name: CSIC_DMA_CH1_CROP_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	FRM_LOST_CNT_EN When set 1, FRM_LOST_CNT is enable; when set 0, FRM_LOST_CNT is cleared
7:5	/	/	/
4	R/W	0x0	REDUNDANT_ERR_MSK_EN 0: Disable redundant error mask. 1: Enable redundant error mask. When input picture size exceeds bk set width or height, set this bit to 0 to pull up redundant error interrupt and do crop. Set this bit to 1 to do crop and not pull up redundant error interrupt.

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Offset: 0x0418			Register Name: CSIC_DMA_CH1_CROP_CFG_REG
Bit	Read/Write	Default/Hex	Description
3:1	/	/	/
0	R/W	0x0	FLD_FRM_VCROP_EN Field Frame Vcrop Enable 0: Disable field frame vcrop 1: Enable field frame vcrop When input frame is field type and V crop wants to be implemented, this bit should be configured as 1. (note: Only even lines crop is supported for field input V crop)

14.1.13.69 0x041C CSIC DMA Channel1 Line Counter Register (Default Value:0x0000_0000)

Offset: 0x041C			Register Name: CSIC_DMA_CH1_LINE_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:0	R/W	0x0	LINE_CNT_NUM The LINE_CNT_NUM value is set by user, when internal line counter reach the set value, the LC_PD will set.

14.1.13.70 0x0420 CSIC DMA Channel1 FIFO 0 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0420			Register Name: CSIC_DMA_CH1_F0_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F0_BUFA When LBC_EN is disabled, these bits indicate FIFO 0 output buffer-A address in DMA mode. When LBC_EN is enabled, these bits indicate the output buffer address in LBC mode.

14.1.13.71 0x0424 CSIC DMA Channel1 FIFO 1 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0424			Register Name: CSIC_DMA_CH1_F1_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F1_BUFA FIFO 1 output buffer-A address

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14.1.13.72 0x0428 CSIC DMA Channel1 FIFO 2 Output Buffer-A Address Register (Default Value:0x0000_0000)

Offset: 0x0428			Register Name: CSIC_DMA_CH1_F2_BUFA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	F2_BUFA FIFO 2 output buffer-A address

14.1.13.73 0x042C CSIC DMA Channel1 Buffer Length Register (Default Value:0x0000_0000)

Offset: 0x042C			Register Name: CSIC_DMA_CH1_BUF_LEN_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	BUF_LEN_C DMA_MODE: Buffer length of chroma C in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y in ONLY Y line. Unit is byte.
15:0	R/W	0x0	BUF_LEN DMA_MODE: Buffer length of luminance Y in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y and chroma C in YC line. Unit is byte.

14.1.13.74 0x0430 CSIC DMA Channel1 Flip Size Register (Default Value:0x0000_0000)

Offset: 0x0430			Register Name: CSIC_DMA_CH1_FLIP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	VER_LEN Vertical line number when in VFLIP mode. Unit is line. Only Readable when FLIP_SIZE_CFG_MODE set 0
15	/	/	/
14:0	R/W	0x0	VALID_LEN Valid components of a line when in HFLIP mode. Unit is pixel component. Only Readable when FLIP_SIZE_CFG_MODE set 0

14.1.13.75 0x0434 CSIC DMA Channel1 Abnormal Mask Register (Default Value:0x0000_0000)

Offset: 0x0434			Register Name: CSIC_DMA_CH1_ABN_MSK_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0434			Register Name: CSIC_DMA_CH1_ABN_MSK_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	FRM_DONE_INT_MSK_EN Decide whether the FD_INT of this frame will be masked or not once any abnormal condition happens during this frame. 0: Disable MSK 1: Enable MSK
4	R/W	0x0	LINE_REDUNDANT_MSK_EN Decide whether the remaining data in this frame will be dropped or not once line redundant error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
3	R/W	0x0	LINE_MISS_MSK_EN Decide whether the remaining data in this frame will be dropped or not once line miss error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
2	R/W	0x0	PIXEL_REDUNDANT_MSK_EN Decide whether the remaining data in this frame will be dropped or not once pixel redundant error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.
1	R/W	0x0	PIXEL_MISS_MSK_EN Decide whether the remaining data in this frame will be dropped or not once pixel miss error happens in this frame. 0: Disable MSK 1: Enable MSK Note: This field takes effect only when VSYN_ABN_MSK_EN is set as 1.

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Offset: 0x0434			Register Name: CSIC_DMA_CH1_ABN_MSK_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	VSYN_ABN_MSK_EN Decide whether the remaining data in this frame will be dropped or not once abnormal happens in this frame. 0: Disable MSK 1: Enable MSK Note: When VSYN_ABN_MSK_EN is set as 1 and one of the following conditions happens, the remaining data in this frame will be dropped. And the next frame will be received as normal. 1, when trying to fill dummy for pixel miss and LACK_HBLANK happens. 2, when PIXEL_REDUNDANT happens and PIXEL_REDUNDANT_MSK_EN is set as 1. 3, when PIXEL_MISS happens, LACK_HBLANK no happen and PIXEL_MISS_MSK_EN is set as 1. 4, when LINE_REDUNDANT happens and LINE_REDUNDANT_MSK_EN is set as 1. 5, when LINE_MISS happens and LINE_MISS_MSK_EN is set as 1.

14.1.13.76 0x0438 CSIC DMA Channel1 Pixels Add Enable Register (Default Value:0x0000_0000)

Offset: 0x0438			Register Name: CSIC_DMA_CH1_PIXELS_ADD_EN_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	PIXELS_ADD_EN Decide whether dummy pixels will be added in the end of the line or not once pixels miss error happens. 0: Disable ADD 1: Enable ADD

14.1.13.77 0x043C CSIC DMA Channel1 Pixels Add Value Register (Default Value:0x0000_0000)

Offset: 0x043C			Register Name: CSIC_DMA_CH1_PIXELS_ADD_VAL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x043C			Register Name: CSIC_DMA_CH1_PIXELS_ADD_VAL_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PIXELS_ADD_VALUE Decide the value of dummy pixels when PIXELS_ADD_EN is set as 1

14.1.13.78 0x0440 CSIC DMA Channel1 Total Frame Counter Config Register (Default Value:0x0000_0000)

Offset: 0x0440			Register Name: CSIC_DMA_CH1_TOT_FRM_CNT_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:2	R/W	0x0	TOTAL_FRM_CNT_SEL Choose which source to trigger frame counter 0: Frame done 1: Frame start (vsyn pos edge) 2: Frame_end(vsyn neg edge)
1	W1C	0x0	TOTAL_FRM_CNT_CLR When set this bit to 1, the total frame counter will clear to 0
0	R/W	0x0	TOTAL_FRM_CNT_EN Total frame counter enable, if you want hardware frame counter, you should set this bit to 1 0: Disable Total Frame Counter 1: Enable Total Frame Counter

14.1.13.79 0x0444 CSIC DMA Channel1 BUF Threshold Register (Default Value:0x0000_0000)

Offset: 0x0444			Register Name: CSIC_DMA_CH1_BUF_TH_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:16	R/W	0x0	CSIC_DMA_STORED_FRM_THRESHOLD when stored frame counter value reaches the threshold , counter is cleared to 0 , only used in Buffer Addr FIFO Mode.
15:6	/	/	/
5:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO_THRESHOLD when content in Buffer Address FIFO no more than the threshold, an interrupt is set, only used in Buffer Addr FIFO Mode

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14.1.13.80 0x04E8 CSIC DMA Channel1 BUF Address FIFO0 Entry Register (Default Value:0x0000_0000)

Offset: 0x04E8			Register Name: CSIC_DMA_CH1_BUF_ADDR_FIFO0_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO0_ENTRY FIFO Entry of Buffer Address FIFO0 for input frames to be stored, only used in Buffer Addr FIFO Mode.

14.1.13.81 0x04EC CSIC DMA Channel1 BUF Address FIFO1 Entry Register (Default Value:0x0000_0000)

Offset: 0x04EC			Register Name: CSIC_DMA_CH1_BUF_ADDR_FIFO1_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO1_ENTRY FIFO Entry of Buffer Address FIFO1 for input frames to be stored, only used in Buffer Addr FIFO Mode

14.1.13.82 0x04F0 CSIC DMA Channel1 BUF Address FIFO2 Entry Register (Default Value:0x0000_0000)

Offset: 0x04F0			Register Name: CSIC_DMA_CH1_BUF_ADDR_FIFO2_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_BUF_ADDR_FIFO2_ENTRY FIFO Entry of Buffer Address FIFO2 for input frames to be stored, only used in Buffer Addr FIFO Mode.

14.1.13.83 0x04F4 CSIC DMA Channel1 Embed Address Entry Register (Default Value:0x0000_0000)

Offset: 0x04F4			Register Name: CSIC_DMA_CH1_EMBED_ADDR_ENTRY_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CSIC_DMA_EMBED_ADDR_ENTRY FIFO Entry of Buffer Embed Address for input frames to be stored, only used in Buffer Addr FIFO Mode.

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14.1.13.84 0x04F8 CSIC DMA Channel1 Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x04F8			Register Name: CSIC_DMA_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	LACK_HBLANK_INT_EN Set an INT when trying filling dummy for missed pixels at the end of the line but hblank is too short.
22	R/W	0x0	LINE_REDUNDANT_INT_EN Set an INT when the amount of lines in one frame is more than expected.
21	R/W	0x0	LINE_MISS_INT_EN Set an INT when the amount of lines in one frame is less than expected.
20	R/W	0x0	PIXEL_REDUNDANT_INT_EN Set an INT when the amount of pixels in one line is more than expected.
19	R/W	0x0	PIXEL_MISS_INT_EN Set an INT when the amount of pixels in one line is less than expected.
18	R/W	0x0	BUF_ADDR_OVERFLOW_INT_EN Set an INT when buffer address is overwrite before used
17	R/W	0x0	BUF_ADDR_UNDERFLOW_INT_EN Set an INT when new frame comes but buffer address is not ready
16	R/W	0x0	LBC_HBLKMIN_INT_EN Set an INT when hblanking less than 48 bk_clk cycles, only use in LBC Mode
15	R/W	0x0	FRM_LOST_INT_EN Set an INT once frame is in when last frame processing
14	R/W	0x0	STORED_FRM_CNT_INT_EN Set an INT when the value of CSIC_DMA_STORED_FRM_CNT reaches CSIC_DMA_STORED_FRM_THRESHOLD , only use in BUF Address FIFO MODE
13	R/W	0x0	BUF_ADDR_FIFO_INT_EN Set an INT when content in BUF Address FIFO less than CSIC_DMA_BUFA_FIFO_THRESHOLD, only use in BUF Address FIFO MODE
12:9	/	/	/

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Offset: 0x04F8			Register Name: CSIC_DMA_CH1_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PARA_LOAD_UNFINISH_INT_EN BK read regs from sram process is unfinished, only could happen in offline mode
7	R/W	0x0	VS_INT_EN vsync flag The bit is set when vsync come. And at this time load the buffer address for the coming frame. So after this IRQ come, change the buffer address could only effect next frame
6	R/W	0x0	LI_OF_INT_EN Line information FIFO(16 lines) overflow.
5	R/W	0x0	LC_INT_EN Line counter flag The bit is set when the specific line has been written to dram every frame. The line number is set in the line counter register.
4:3	/	/	/
2	R/W	0x0	FIFO_OF_INT_EN FIFO overflow The bit is set when the FIFO become overflow.
1	R/W	0x0	FD_INT_EN Frame done Indicates the CSI has finished capturing an image frame. Applies to video capture mode. The bit is set after each completed frame capturing data is wrote to buffer as long as video capture remains enabled.
0	R/W	0x0	CD_INT_EN Capture done Indicates the CSI has completed capturing the image data. For still capture, the bit is set when one frame data has been written to buffer. For video capture, the bit is set when the last frame has been written to buffer after video capture has been disabled. For CCIR656 interface, if the output format is frame planar YCbCr 420 mode, the frame end means the field2 end, the other frame end means field end.

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14.1.13.85 0x04FC CSIC DMA Channel1 Interrupt Status Register (Default Value:0x0000_0000)

Offset: 0x04FC			Register Name: CSIC_DMA_CH1_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	LACK_HBLANK_INT_PD Set an INT when trying filling dummy for missed pixels at the end of the line but hblank is too short.
22	R/W1C	0x0	LINE_REDUNDANT_INT_PD Set an INT when the amount of lines in one frame is more than expected.
21	R/W1C	0x0	LINE_MISS_INT_PD Set an INT when the amount of lines in one frame is less than expected.
20	R/W1C	0x0	PIXEL_REDUNDANT_INT_PD Set an INT when the amount of pixels in one line is more than expected.
19	R/W1C	0x0	PIXEL_MISS_INT_PD Set an INT when the amount of pixels in one line is less than expected.
18	R/W1C	0x0	BUF_ADDR_OVERFLOW_INT_PD Set an INT when buffer address is overwrite before used
17	R/W1C	0x0	BUF_ADDR_UNDERFLOW_INT_PD Set an INT when new frame comes but buffer address is not ready
16	R/W1C	0x0	LBC_HBLKMIN_INT_PD Set an INT when hblanking less than 48 bk_clk cycles, only use in LBC Mode
15	R/W1C	0x0	FRM_LOST_INT_PD Set an INT once frame is in when last frame processing
14	R/W1C	0x0	STORED_FRM_CNT_INT_PD Set an INT when the value of CSIC_DMA_STORED_FRM_CNT reaches CSIC_DMA_STORED_FRM_THRESHOLD, only use in BUF Address FIFO MODE
13	R/W1C	0x0	BUF_ADDR_FIFO_INT_PD Set an INT when content in BUF Address FIFO no more than CSIC_DMA_BUFA_FIFO_THRESHOLD, only use in BUF Address FIFO MODE
12:9	/	/	/

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Offset: 0x04FC			Register Name: CSIC_DMA_CH1_INT_STA_REG
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	PARA_LOAD_UNFINISH_PD BK read regs from sram process is unfinished, only could happen in offline mode
7	R/W1C	0x0	VS_PD Vsync flag
6	R/W1C	0x0	LI_OF_PD Line information FIFO(16 lines) overflow
5	R/W1C	0x0	LC_PD Line counter flag
4:3	/	/	/
2	R/W1C	0x0	FIFO_OF_PD FIFO 0 overflow
1	R/W1C	0x0	FD_PD Frame done
0	R/W1C	0x0	CD_PD Capture done

14.1.13.86 0x0500 CSIC DMA Channel1 FIFO 0 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0500			Register Name: CSIC_DMA_CH1_F0_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F0_BUFA_RESULT Indicate the final F0_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

14.1.13.87 0x0504 CSIC DMA Channel1 FIFO 1 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0504			Register Name: CSIC_DMA_CH1_F1_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F1_BUFA_RESULT Indicate the final F1_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

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14.1.13.88 0x0508 CSIC DMA Channel1 FIFO 2 Output Buffer-A Address Result Register (Default Value:0x0000_0000)

Offset: 0x0508			Register Name: CSIC_DMA_CH1_F2_BUFA_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	F2_BUFA_RESULT Indicate the final F2_BUFA address used for DMA after software configuration or hardware calculation from Buffer-A address register or buffer address FIFO. Only used for debug.

14.1.13.89 0x050C CSIC DMA Channel1 Buffer Length Result Register (Default Value:0x0000_0000)

Offset: 0x050C			Register Name: CSIC_DMA_CH1_BUF_LEN_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	BUF_LEN_C_RESULT DMA_MODE: Buffer length of chroma C in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y in ONLY Y line. Unit is byte.
15:0	R	0x0	BUF_LEN_RESULT DMA_MODE: Buffer length of luminance Y in a line. Unit is byte. LBC_MODE: Buffer length Stride of luminance Y and chroma C in YC line. Unit is byte.

14.1.13.90 0x0510 CSIC DMA Channel1 Flip Size Result Register (Default Value:0x04d0_0500)

Offset: 0x0510			Register Name: CSIC_DMA_CH1_FLIP_SIZE_RESULT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	VER_LEN_RESULT Vertical line number when in VFLIP mode. Unit is line. Only Readable when FLIP_SIZE_CFG_MODE set 0.
15	/	/	/
14:0	R	0x0	VALID_LEN_RESULT Valid components of a line when in HFLIP mode. Unit is pixel component. Only Readable when FLIP_SIZE_CFG_MODE set 0.

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14.1.13.91 0x0514 CSIC DMA Channel1 BUF Address FIFO Content Register (Default Value:0x0000_0000)

Offset: 0x0514			Register Name: CSIC_DMA_CH1_BUF_ADDR_FIFO_CON_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:24	R	0x0	CSIC_DMA_BUF_ADDR_EMBED_CONTENT FIFO Content of address buffered in Buffer Address Embedded, only used in Buffer Addr FIFO Mode.
23:22	/	/	/
21:16	R	0x0	CSIC_DMA_BUF_ADDR_FIFO2_CONTENT FIFO Content of address buffered in Buffer Address FIFO2, only used in Buffer Addr FIFO Mode.
15:14	/	/	/
13:8	R	0x0	CSIC_DMA_BUF_ADDR_FIFO1_CONTENT FIFO Content of address buffered in Buffer Address FIFO1, only used in Buffer Addr FIFO Mode.
7:6	/	/	/
5:0	R	0x0	CSIC_DMA_BUF_ADDR_FIFO0_CONTENT FIFO Content of address buffered in Buffer Address FIFO0, only used in Buffer Addr FIFO Mode.

14.1.13.92 0x0518 CSIC DMA Channel1 Stored Frame Counter Register (Default Value:0x0000_0000)

Offset: 0x0518			Register Name: CSIC_DMA_CH1_STORED_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	CSIC_DMA_STORED_FRM_CNT Indicates value of stored frames counter, when counter value reaches CSIC_DMA_STORED_FRM_THRESHOLD, counter is cleared to 0, only used in Buffer Addr FIFO Mode.

14.1.13.93 0x051C CSIC DMA Channel1 Total Frame Counter Register (Default Value:0x0000_0000)

Offset: 0x051C			Register Name: CSIC_DMA_CH1_TOT_FRM_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TOTAL_FRM_CNT Hardware Frame Counter

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14.1.13.94 0x0524 CSIC DMA Channel1 Frame Lost Counter Register (Default Value:0x0000_0000)

Offset: 0x0524			Register Name: CSIC_DMA_CH1_FRM_LOST_CNT_REG
Bit	Read/Write	Default/Hex	Description
31	R	0x0	CAP_STA capture in progress Indicates the CSI is capturing image data. The bit is set at the start of the first frame after enabling frame capture.
30:20	/	/	/
19:16	R	0x0	AI ISP_FRM_NUM Indicates this AI ISP Frame number, lock TOTAL_FRM_CNT low 4bit when error occurred, release when frm_done int is clear.
15:12	R	0x0	ISP_ERROR_FRM_NUM Indicates this ERROR Frame number, lock TOTAL_FRM_CNT low 4bit when error occurred, release when frm_done int is clear.
11:10	/	/	/
9	R	0x0	AI_ISP_FLAG Indicates this Frame is AI ISP Frame, will lock when frm_done int is cleared
8	R	0x0	ISP_ERROR_FLAG Indicates this Frame is ISP ERROR Frame, will lock when frm_done int is cleared
7:0	R	0x0	FRM_LOST_CNT Once a new frame comes when the buffer address not ready and FRM_DROP_EN set, or when the last frame still processing, the new frame is dropped and the FRM_LOST_CNT increases when FRM_LOST_CNT_EN is set.

14.1.13.95 0x0528 CSIC DMA Channel1 Embed Address Read Register (Default Value:0x0000_0000)

Offset: 0x0528			Register Name: CSIC_DMA_CH1_EMBED_ADDR_READ_REG
Bit	Read/Write	Default/Hex	Description
31:00	R	0x0	EMBEDDED_ADDR Embedded Address of Current Frame

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15 Video Out Interfaces

15.1 TCON LCD

15.1.1 Overview

The Timing Controller_LCD (TCON_LCD) is a module that processes video signals received from system through a complicated arithmetic and then generates control signals and transmits them to the LCD panel driver IC.

The TCON_LCD includes the following features:

- One TCON_LCD controller, supporting LCD/DSI/LVDS display
- One panel interface
  - RGB interface with DE/SYNC RGB mode, up to1920x1080@60fps
  - Serial RGB interface, up to 800 x 480@60fps
  - i8080 interface, up to 800x480@60fps
  - Supports BT.656 interface
  - RGB888, RGB666 and RGB565 pixel formats
- Supports outputting pulses to achieve synchronization between sensor and TCON_LCD.

15.1.2 Block Diagram

Figure 15-1 TCON_LCD Block Diagram

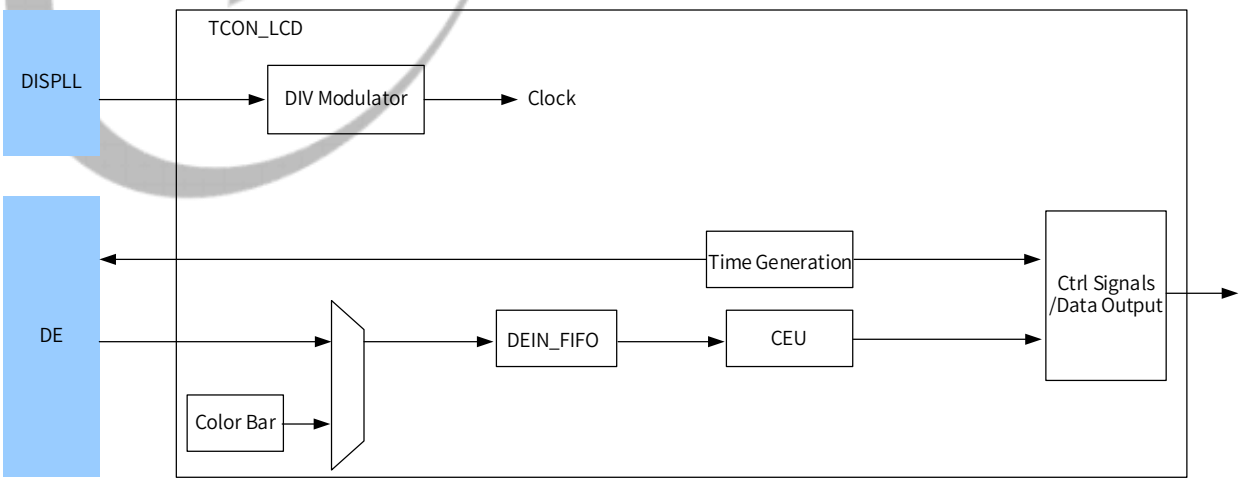

Table 15-1 TCON_LCD Sub-blocks

Sub-blocks	Description

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Sub-blocks	Description
ColorBar (Vdbist)	The circuit of embedded test data source in TCON. It provides colorbar pattern, grey pattern, lattice pattern, and grid pattern.
Div Modulator	It is responsible for clock division and phase adjustment.
DEFIFO	It supports temporary storage and backpressure on the DE data stream. It can implement data interaction between TCON_LCD and DE.
CEU	It supports strengthening or adjusting the RGB color, and convert from RGB to YUV.
Timing Generation	It generates the corresponding timing based on the configured parameters and the row state machine and column state machine in the module.

15.1.3 External Signals

The following table describes the external I/O signals of LCD.

Table 15-2 TCON_LCD External Signals

Signal Name	Description	Type
TCON-FSYNC0	Frame Synchronization Signal for TCON and Sensor	O
LCD-D[23:0]	LCD Data Input/Output	I/O
LCD-CLK	LCD Clock The pixel data are synchronized by this clock	O
LCD-VSYNC	LCD Vertical Sync It indicates one new frame	O
LCD-HSYNC	LCD Horizontal Sync It indicates one new scan line	O
LCD-DE	LCD Data Output Enable	O

For parallel RGB, the data of LCD is high-aligned. The correspondence is as follows.

Table 15-3 The Correspondence between LCD and RGB

LCD I/O	Parallel RGB I/O		
	RGB565	RGB666	RGB888
LCD-D23	R4	R5	R7
LCD-D22	R3	R4	R6
LCD-D21	R2	R3	R5
LCD-D20	R1	R2	R4
LCD-D19	R0	R1	R3
LCD-D18	-	R0	R2
LCD-D17	-	-	R1
LCD-D16	-	-	R0
LCD-D15	G5	G5	G7
LCD-D14	G4	G4	G6
LCD-D13	G3	G3	G5
LCD-D12	G2	G2	G4

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LCD I/O	Parallel RGB I/O		
	RGB565	RGB666	RGB888
LCD-D11	G1	G1	G3
LCD-D10	G0	G0	G2
LCD-D9	-	-	G1
LCD-D8	-	-	G0
LCD-D7	B4	B5	B7
LCD-D6	B3	B4	B6
LCD-D5	B2	B3	B5
LCD-D4	B1	B2	B4
LCD-D3	B0	B1	B3
LCD-D2	-	B0	B2
LCD-D1	-	-	B1
LCD-D0	-	-	B0

The multiplex relationship among LCD, LVDS, and DSI is shown as follows.

Table 15-4 The Correspondence among LCD, LVDS, and DSI.

LCD I/O	LVDS I/O	DSI I/O
LCD-D0	/	/
LCD-D1	/	/
LCD-D2	LVDS0-D0P	DSI-D0P
LCD-D3	LVDS0-D0N	DSI-D0N
LCD-D4	LVDS0-D1P	DSI-D1P
LCD-D5	LVDS0-D1N	DSI-D1N
LCD-D6	LVDS0-D2P	DSI-CKP
LCD-D7	LVDS0-D2N	DSI-CKN
LCD-D8	/	/
LCD-D9	/	/
LCD-D10	LVDS0-CKP	DSI-D2P
LCD-D11	LVDS0-CKN	DSI-D2N
LCD-D12	LVDS0-D3P	DSI-D3P
LCD-D13	LVDS0-D3N	DSI-D3N
LCD-D14	LVDS1-D0P	/
LCD-D15	LVDS1-D0N	/
LCD-D16	/	/
LCD-D17	/	/
LCD-D18	LVDS1-D1P	/
LCD-D19	LVDS1-D1N	/
LCD-D20	LVDS1-D2P	/
LCD-D21	LVDS1-D2N	/
LCD-D22	LVDS1-CKP	/
LCD-D23	LVDS1-CKN	/
LCD-CLK	LVDS1-D3P	/

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LCD I/O	LVDS I/O	DSI I/O
LCD-DE	LVDS1-D3N	/
LCD-HSYNC	/	/
LCD-VSYNC	/	/

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15.1.4 Functional Description

15.1.4.1 Control signal and data port mapping

NOTE

The meaning of 256K in CPU mode is 2^18/1024=256K.

In CPU TRIG mode, TCON_TRIG signal is multiplexed with VSYNC signal.

		SYNC RGB				YUV		DC	CPU Cmd	CPU 18-bit	CPU 18bit							CPU 16 bit	CPU 18bit			CPU 16bit		CPU 18bit		LVDS			DSI
External I/O	Internal Pin	Para RGB	Serial RGB			BT656	BT601	YUV422		256K	256K							64K	256K			64K		256K		Single Link		Dual Link	
			1st	2nd	3rd						1st	2nd	3rd	1st	2nd	1st	2nd		1st	2nd	3rd	1st	2nd	1st	2nd	LVDS 0	LVDS 1		
LCD_VSYNC	IO0	VSYNC					VSYNC	VSYNC	CS																				
LCD_HSYNC	IO1	HSYNC					HSYNC		RD																				
LCD_CLK	IO2	DCLK				DCLK	DCLK	DCLK	WR																		D3N	D3N	
LCD_DE	IO3	DE					DE	HSYNC	RS																		D3P	D3P	
LCD_D23	D23	R7							D23	R5	R5	B5	G5	R5		R5	B5	R4									CKN	CKN	
LCD_D22	D22	R6							D22	R4	R4	B4	G4	R4		R4	B4	R3									CKP	CKP	
LCD_D21	D21	R5							D21	R3	R3	B3	G3	R3		R3	B3	R2									D2N	D2N	
LCD_D20	D20	R4							D20	R2	R2	B2	G2	R2		R2	B2	R1									D2P	D2P	
LCD_D19	D19	R3							D19	R1	R1	B1	G1	R1		R1	B1	R0									D1N	D1N	
LCD_D18	D18	R2							D18	R0	R0	B0	G0	R0		R0	B0	G5									D1P	D1P	
LCD_D17	D17	R1							D17																				
LCD_D16	D16	R0							D16																				
LCD_D15	D15	G7							D15	G5								G4									D0N	D0N	
LCD_D14	D14	G6							D14	G4								G3									D0P	D0P	
LCD_D13	D13	G5							D13	G3																D3N		D3N	D3N
LCD_D12	D12	G4	D71	D72	D73	D7	D7	D7	D12	G2	G5	R5	B5	G5	B5	G5		G2	R5	G5	B5	R4	G2	R5	G2	D3P		D3P	D3P
LCD_D11	D11	G3	D61	D62	D63	D6	D6	D6	D11	G1	G4	R4	B4	G4	B4	G4		G1	R4	G4	B4	R3	G1	R4	G1	CKN		CKN	D2N
LCD_D10	D10	G2	D51	D52	D53	D5	D5	D5	D10	G0	G3	R3	B3	G3	B3	G3		G0	R3	G3	B3	R2	G0	R3	G0	CKP		CKP	D2P
LCD_D9	D9	G1							D9																				
LCD_D8	D8	G0							D8																				
LCD_D7	D7	B7	D41	D42	D43	D4	D4	D4	D7	B5	G2	R2	B2	G2	B2	G2		B4	R2	G2	B2	R1	B4	R2	B5	D2N		D2N	CKN
LCD_D6	D6	B6	D31	D32	D33	D3	D3	D3	D6	B4	G1	R1	B1	G1	B1	G1		B3	R1	G1	B1	R0	B3	R1	B4	D2P		D2P	CKP
LCD_D5	D5	B5	D21	D22	D23	D2	D2	D2	D5	B3	G0	R0	B0	G0	B0	G0		B2	R0	G0	B0	G5	B2	R0	B3	D1N		D1N	D1N
LCD_D4	D4	B4	D11	D12	D13	D1	D1	D1	D4	B2								B1				G4	B1	G5	B2	D1P		D1P	D1P
LCD_D3	D3	B3	D01	D02	D03	D0	D0	D0	D3	B1								B0				G3	B0	G4	B1	D0N		D0N	D0N
LCD_D2	D2	B2							D2	B0														G3	B0	D0P		D0P	D0P
LCD_D1	D1	B1							D1																				
LCD_D0	D0	B0							D0																				

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15.1.4.2 HV interface (Sync+DE mode)

HV I/F is also known as Sync + DE mode, which is widely used in TFT LCD module for PMP/MP4 applications.

Its signals are defined as:

Table 15-5 HV Panel Signals

Signal Name	Description	Type
Vsync	Vertical sync, indicates one new frame	O
Hsync	Horizontal sync, indicates one new scan line	O
DCLK	Dot clock, pixel data are sync by this clock	O
DE	LCD data enable	O
D[23..0]	24-bit RGB output from input FIFO for panel	O

The timing diagram of HV interface is as follows.

Figure 15-2 HV Interface Vertical Timing

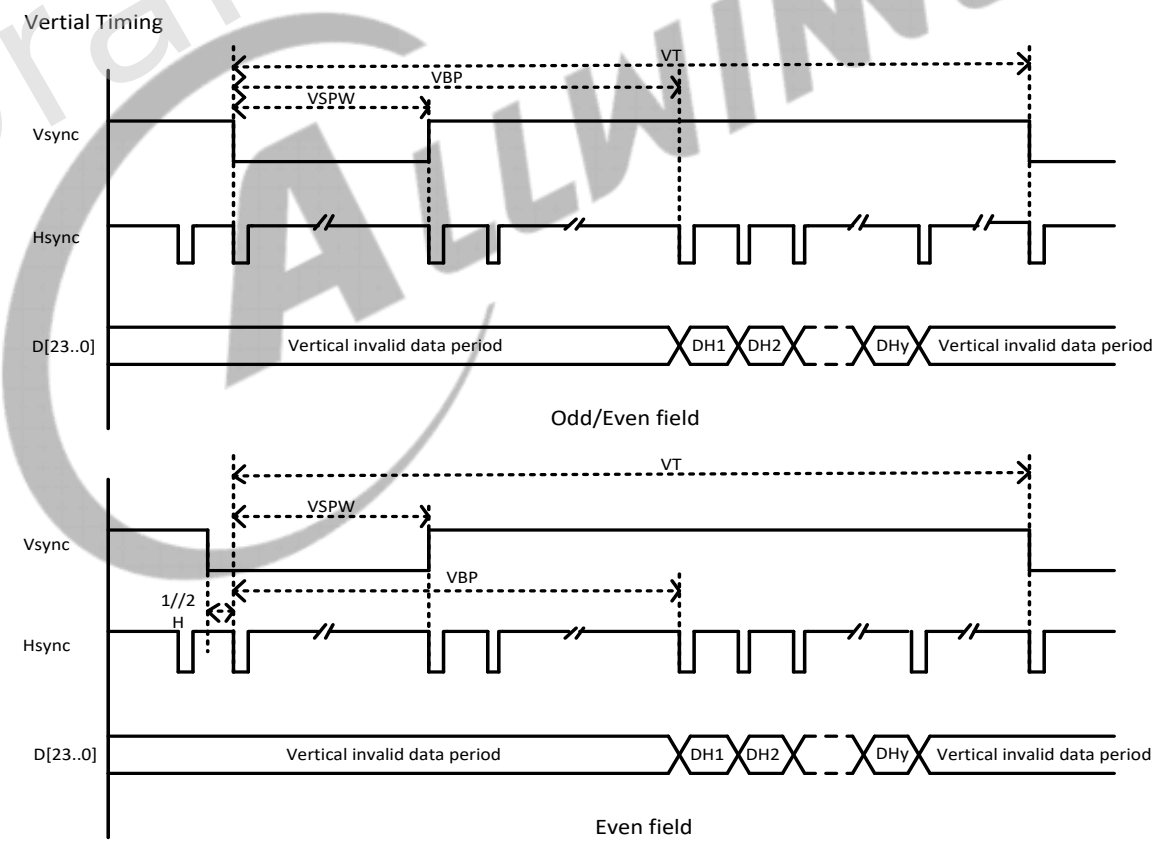

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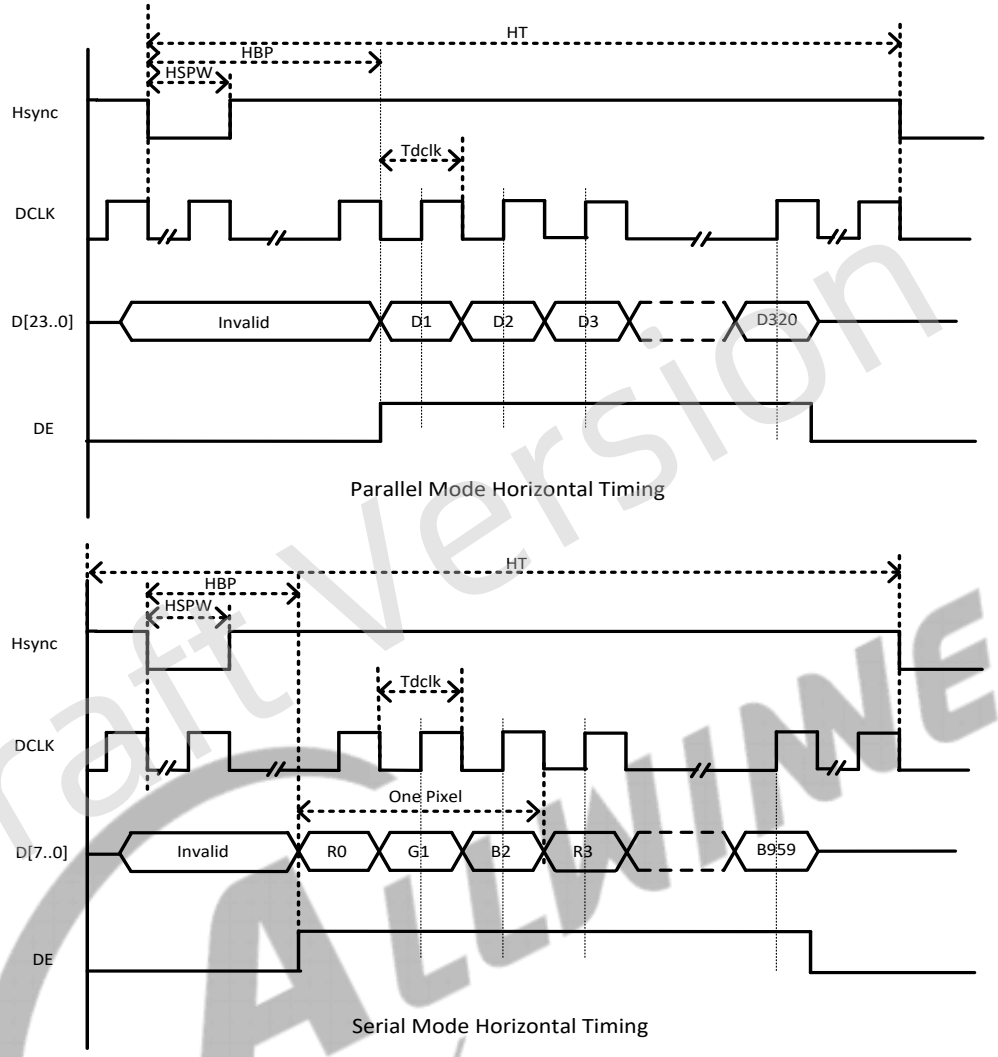
Figure 15-3 HV Interface Horizontal Timing



15.1.4.3 BT.656 Interface

The BT.656 transmission protocol is needed when outputting the data in YUV format in the HV mode. Compared with the standard CSI interface, BT.656 has no SYNC signals like Field, VSYN, and HSYN. Instead, it embeds the field synchronization information of image transmission in the data stream transmitted by Date Bus. The Data Bus bit width of BT.656 is 8bit. And the transmitting image data type is the interlaced YUV422. The encoding format for each row is as follows:

Row=End of Active Video (EAV) + horizontal blanking data (80H/10H) + Start of Active Video (SAV) + valid data (UYVY)

Both of the embedded EAV and SAV are four bytes in the format of FF 00 00 XY. The first three types are fixed to 8'hFF 8'h00 8'h00 in hexadecimal. The last bytes XY are used as the control word. The specific image row and field synchronization information is as follows.

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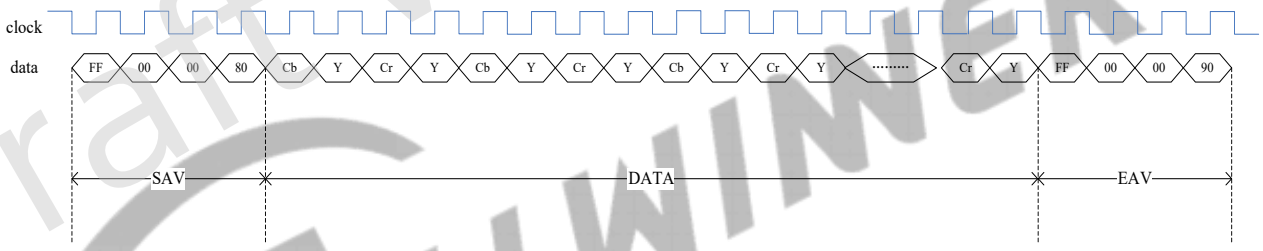
Table 15-6 Analysis of XY Signal

BIT	7	6	5	4	3	2	1	0
XY	1'b1	F	V	H	P3	P2	P1	P0
F	0: During field 1 1: During field 2							
V	1: During field blanking 0: Field active							
H	0: In the start of active video (SAV) 1: In the end of active video (EAV)							
P3-P0	Protection Bits. Single channel: P3=V^H, P2=F^H, P1=F^V, P0=F^V^H Multi-Channel: Channel ID							

During single-channel data transmission, P3-P0 is used for check. The transmission timing is shown in the following figure.

Figure 15-4 Single-channel Transmission Timing



In the programming, the final output format is successfully set in the RGB2YUV. Specifically, the 8bit data output by the RGB2YUV is shown in the above figure. The ctrl_data_out module performs pin mapping for the data.

15.1.4.4 i8080 Interface

The i8080 I/F LCD panel is most common interface for small size, low resolution LCD panels. The CPU control signals are active low.

Table 15-7 CPU Panel Signals

Signal Name	Description	Type
CS	Chip select, active low	O
WR	Write strobe, active low	O
RD	Read strobe, active low	O
A1	Address bit, controlled by "LCD_CPUI/F" BIT25, ctrl_out[3]	O
D[23..0]	Digital RGB output signal	I/O

The following figure shows the relationship between basic timing and CPU timing. WR is fixed to 1 when reading. And it is the reverse phase of dclk1 when writing, in which it is low active and only has 1 period of low level. CS is active when pixel data is valid. RD is always set to 1 when

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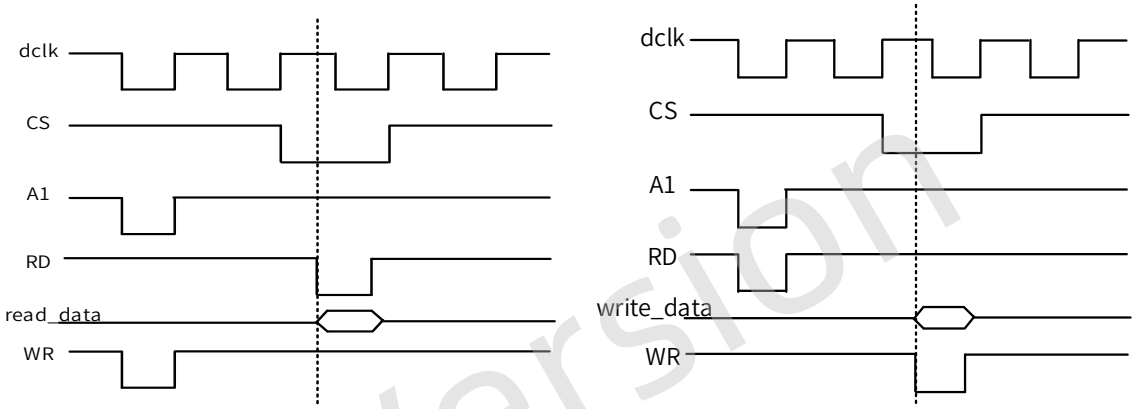
writing. And it is the reverse phase of dclk1 when writing, in which it is low active and only has 1 period of low level. 1 is set by LCD_CPUI/F . Set to 0 or 1 when reading, and set to 1 when reading.

Figure 15-5 CPU interface timing (left read, right write)



When CPU I/F is in the idle state, it can generate WR/RD timing by setting LCD_CPUI/F . The CS strobe is one DCLK width, and the WR/RD strobe is half DCLK width.

15.1.4.5 CEU Module

This module enhances color data from DE.

R = (((Rr*R + Rg*G + Rb*B + 8)/16) + Rc + 8)/16

G = (((Gr*R + Gg*G + Gb*B + 8)/16) + Gc + 8)/16

B = (((Br*R + Bg*G + Bb*B + 8)/16) + Bc + 8)/16

Rr, Rg, Rb, Gr, Gg, Gb, Br, Bg, Bb s13 (-16, 16)

Rc, Gc, Bc s19 (-16384, 16384)

R, G, B u8 [0-255]

R has the range of [Rmin, Rmax]

G has the range of [Rmin, Rmax]

B has the range of [Rmin, Rmax]

15.1.4.6 MIPI DSI

The requirements on MIPI DSI mode are as follows.

(1). When using MIPI DSI as display interface, the data clk of TCON needs be started firstly.

(2). When it is used with DSI video mode, the setting of block space needs to meet the following relationship.

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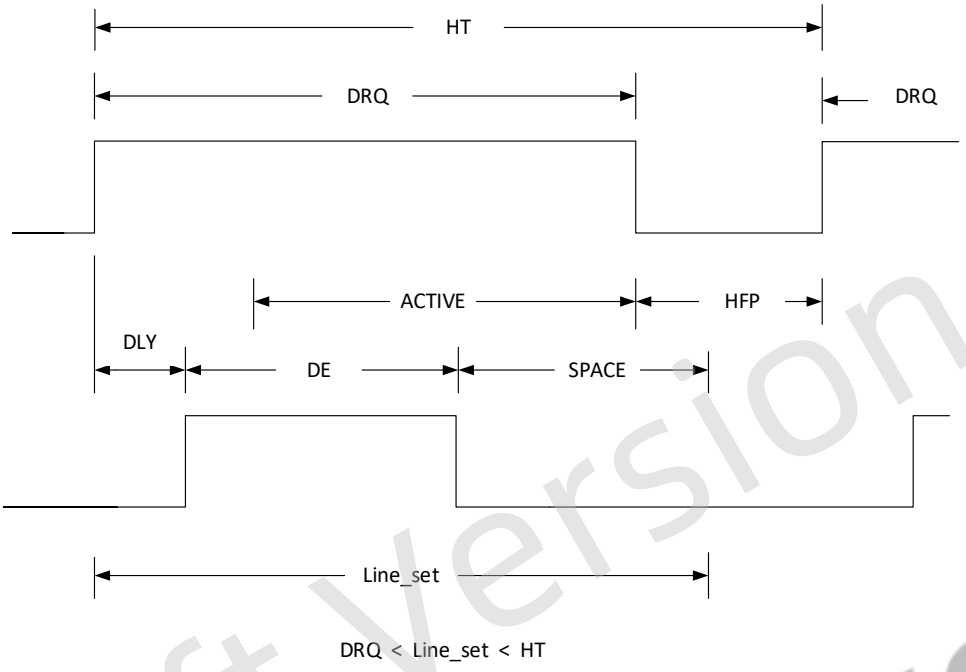
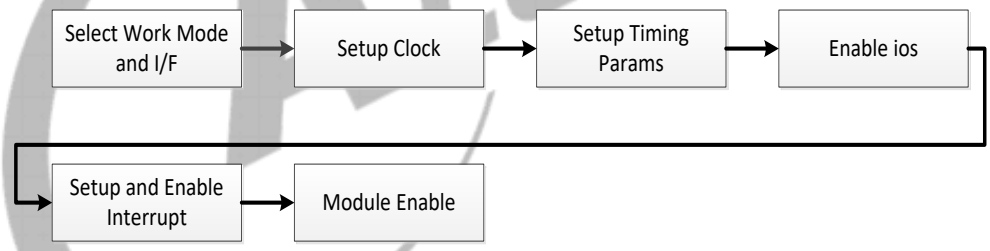
Figure 15-6 The Data Timing of MIPI DSI Video Mode



15.1.5 Programming Guidelines

15.1.5.1 HV Mode Configuration

Figure 15-7 HV Mode Initial Process



Step 1 Select HV interface type

Step 2 Set the clock, if you need to use the phase adjustment function, please set bit [31:28] of LCD_DCLK_REG to 0xf.

Step 3 Set the timing parameters x, ht, hbp, hspw, y, vt, vbp, vspw. Note that hbp contains hspw and vbp contains vspw. vt needs to be set to 2 times the actual value.

Step 4 Enable IO output.

Step 5 Set and enable the interrupt function. Note that when using line interrupt, you need to set LCD_LINE_INT_NUM in LCD_GINT1_REG first, and then enable LCD_LINE_INT_EN in LCD_GINT0_REG for line interrupt.

Step 6 Module enable

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Table 15-8 HV Mode Configuration Example

Step	Register	Typical Value of Simulation	Description
1	LCD_HV_IF_REG (0x0058)	32'h0000_0000	Select parallel RGB
	LCD_CTL_REG (0x0040)	32'h0000_0041	Select HV (Sync+DE) mode
	/	/	Enable some functions such as CEU enable.
2	LCD_DCLK_REG (0x0044)	32'hf000_0002	/
3	LCD_BASIC0_REG (0x0048)	32'h0063_000f	Set x and y.
	LCD_BASIC1_REG (0x004C)	32'h0095_0013	Set ht and hbp.
	LCD_BASIC2_REG (0x0050)	32'h0030_0001	Set vt and vbp.
	LCD_BASIC3_REG (0x0054)	32'h0009_0000	Set HSPW and VSPW to get the configured value. The actual value is equal to the configured value plus one.
4	LCD_IO_POL_REG (0x0088)	32'h0000_0000	/
	LCD_IO_TRI_REG (0x008C)	32'he000_0000	1: Disable 0: Enable
5	LCD_GINT1_REG (0x0008)	32'h0010_0000	/
	LCD_GINT0_REG (0x0004)	32'h2000_0000	/
6	LCD_CTL_REG (0x0040)	32'h8000_0041	Open module enable.
	LCD_GCTL_REG (0x0000)	32'h8000_0000	Enable vs and hs to count.

15.1.5.2 BT.656 Mode Configuration

The BT.656 mode configuration process is similar to the HV mode. The following table is an example of typical parameter configuration.

Table 15-9 BT.656 Mode Configuration Example

Step	Register	Typical Value of Simulation	Description
1	LCD_HV_IF_REG (0x0058)	32'h0000_0000	Select parallel RGB
	LCD_CTL_REG (0x0040)	32'h0000_0041	Select HV (Sync+DE) mode
	/	/	Enable some functions such as 3DFIFO and CEU enable.

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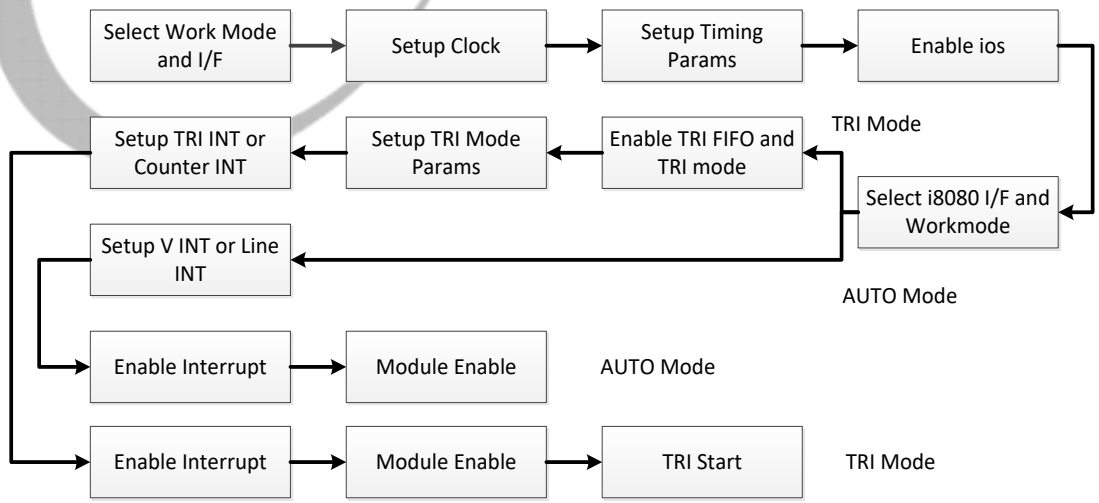
Step	Register	Typical Value of Simulation	Description
2	LCD_DCLK_REG (0x0044)	32'hf000_0002	/
3	LCD_BASIC0_REG (0x0048)	32'h0063_000f	Set x and y.
	LCD_BASIC1_REG (0x004C)	32'h0095_0013	Set ht and hbp.
	LCD_BASIC2_REG (0x0050)	32'h0030_0001	Set vt and vbp.
	LCD_BASIC3_REG (0x0054)	32'h0009_0000	Set HSPW and VSPW to get the configured value. The actual value is equal to the configured value plus one.
4	LCD_IO_POL_REG (0x0088)	32'h0000_0000	/
	LCD_IO_TRI_REG (0x008C)	32'he000_0000	1: Disable 0: Enable
5	LCD_GINT1_REG (0x0008)	32'h0010_0000	/
	LCD_GINT0_REG (0x0004)	32'h2000_0000	/
6	LCD_CTL_REG (0x0040)	32'h8000_0041	Open module enable.
	LCD_GCTL_REG (0x0000)	32'h8000_0000	Enable vs and hs to count.

15.1.5.3 i8080 Mode Configuration

Figure 15-8 i8080 Mode Initial Process



Step 1 Select i8080 interface type.

Step 2 The step is the same as HV mode, but pulse adjustment function is invalid.

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Step 3 The step is the same as HV mode. When using TRI mode, it is best to configure LCD timing parameters in HV mode, or a handful of functions such as CMAP will not be able to apply.

Step 4 The step is the same as HV mode.

Step 5 Select type and operating mode of i8080, the operating mode includes TRI mode and AUTO mode, and the two operating modes are different.

----------If for TRI mode----------

Step 6 Open TRI FIFO switch, and TRI mode function.

Step 7 Set parameters of TRI mode, including block size, block space and block number.

NOTE

- When output interface is parallel mode, then the setting value of block space parameter is not less than 20.
- When output interface is 2 cycle serial mode, then the setting value of block space parameter is not less than 40.
- When output interface is 3 cycle serial mode, then the setting value of block space parameter is not less than 60.
- When output interface is 4 cycle serial mode, then the setting value of block space parameter is not less than 80.

Step 8 Set the tri interrupt or counter interrupt. When using the two interrupts, mainly in the interrupt service function the tri start operation need be operated (the bit1 of LCD_CPU_IF_REG is set to"1"). If using TE trigger interrupt, you select the external input pin as a trigger signal, the 24-bit for offset 0x8C register is set to "1", to open up input of pad.

Step 9 Open the total switch of interrupt.

Step 10 Open the total enable of interrupt.

Step 11 Operate “tri start” operation (the bit1 of LCD_CPU_IF_REG is set to "1").

----------If for Auto mode----------

Step 6 Set and open V interrupt or Line interrupt, the step is the same as HV mode.

Step 7 Open module total enable.

The following table is an example of typical parameter configuration.

Table 15-10 i8080 Mode Configuration Example

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Step	Register	Typical Value of Simulation	Description
1	LCD_HV_IF_REG (0x0058)	32'h0000_0000	Select parallel RGB
	LCD_CTL_REG (0x0040)	32'h8100_0001	Select data source.
	/	/	Enable some functions such as 3DFIFO and CEU enable.
2	LCD_DCLK_REG (0x0044)	32'hf000_0002	/
3	LCD_BASIC0_REG (0x0048)	32'h0063_000f	Set x and y.
	LCD_BASIC1_REG (0x004C)	32'h0095_0013	Set ht and hbp.
	LCD_BASIC2_REG (0x0050)	32'h0030_0001	Set vt and vbp.
	LCD_BASIC3_REG (0x0054)	32'h0009_0000	Set HSPW and VSPW to get the configured value. The actual value is equal to the configured value plus one.
4	LCD_IO_POL_REG (0x0088)	32'h0000_0000	/
	LCD_IO_TRI_REG (0x008C)	32'he000_0000	1: Disable 0: Enable
5	LCD_CPU_IF_REG (0x0060)	32'h0401_0005	Select the type and working mode of the i8080 interface.
TRI mode			
6	LCD_CPU_IF_REG (0x0060)	32'h0401_0005	Enable the trigger FIFO and trigger mode.
7	LCD_CPU_TRI0_REG (0x0160)	32'h00ff_031f	Configure parameters in TRI mode.
	LCD_CPU_TRI1_REG (0x0164)	32'h0000_01df	
8	LCD_GINT0_REG (0x0004)	Write 1 to bit[26] or bit[27]	Select trig interrupt or counter interrupt.
9	LCD_CTL_REG (0x0040)	32'h8000_0000	Open module enable.
	LCD_GCTL_REG (0x0000)	32'h8000_0000	
10	LCD_CPU_IF_REG (0x0060)	32'h0401_0007	Tri start.
AUTO mode			
6	LCD_GINT1_REG (0x0008)	32'h0010_0000	
	LCD_GINT0_REG (0x0004)	32'h2000_0000	
7	LCD_CTL_REG (0x0040)	32'h8100_0040	Open module enable.

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Step	Register	Typical Value of Simulation	Description
	LCD_GCTL_REG (0x0000)	32'h8000_0000	Enable vs and hs to start counting

15.1.6 Register List

Module Name	Base Address
TCON_LCD0	0x05501000

Register Name	Offset	Description
LCD_GCTL_REG	0x0000	LCD Global Control Register
LCD_GINT0_REG	0x0004	LCD Global Interrupt Register0
LCD_GINT1_REG	0x0008	LCD Global Interrupt Register1
LCD_CTL_REG	0x0040	LCD Control Register
LCD_DCLK_REG	0x0044	LCD Data Clock Register
LCD_BASIC0_REG	0x0048	LCD Basic Timing Register0
LCD_BASIC1_REG	0x004C	LCD Basic Timing Register1
LCD_BASIC2_REG	0x0050	LCD Basic Timing Register2
LCD_BASIC3_REG	0x0054	LCD Basic Timing Register3
LCD_HV_IF_REG	0x0058	LCD HV Panel Interface Register
LCD_CPU_IF_REG	0x0060	LCD CPU Panel Interface Register
LCD_CPU_WR_REG	0x0064	LCD CPU Panel Write Data Register
LCD_CPU_RD0_REG	0x0068	LCD CPU Panel Read Data Register0
LCD_CPU_RD1_REG	0x006C	LCD CPU Panel Read Data Register1
LCD_IO_POL_REG	0x0088	LCD IO Polarity Register
LCD_IO_TRI_REG	0x008C	LCD IO Control Register
LCD_DEBUG_REG	0x00FC	LCD Debug Register
LCD_CEU_CTL_REG	0x0100	LCD CEU Control Register
LCD_CEU_COEF_MUL_REG0	0x0110+N*0x04(N=0-2)	LCD CEU Coefficient Register0
LCD_CEU_COEF_MUL_REG1	0x0120+N*0x04(N=0-2)	LCD CEU Coefficient Register1
LCD_CEU_COEF_MUL_REG2	0x0130+N*0x04(N=0-2)	LCD CEU Coefficient Register2
LCD_CEU_COEF_ADD_REG	0x011C+N*0x10(N=0-2)	LCD CEU Coefficient Register1
LCD_CEU_COEF_RANG_REG	0x0140+N*0x04(N=0-2)	LCD CEU Coefficient Register2
LCD_CPU_TRI0_REG	0x0160	LCD CPU Panel Trigger Register0
LCD_CPU_TRI1_REG	0x0164	LCD CPU Panel Trigger Register1
LCD_CPU_TRI2_REG	0x0168	LCD CPU Panel Trigger Register2
LCD_CPU_TRI3_REG	0x016C	LCD CPU Panel Trigger Register3
LCD_CPU_TRI4_REG	0x0170	LCD CPU Panel Trigger Register4
LCD_CPU_TRI5_REG	0x0174	LCD CPU Panel Trigger Register5
LCD_SAFE_PERIOD_REG	0x01F0	LCD Safe Period Register
FSYNC_GEN_CTRL_REG	0x0228	LCD FSYNC Generate Control Register
FSYNC_GEN_DLY_REG	0x022C	LCD FSYNC Generate Delay Register

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Register Name	Offset	Description
LCD_SYNC_CTL_REG	0x0230	LCD Sync Control Register

15.1.7 Register Description

15.1.7.1 0x0000 LCD Global Control Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: LCD_GCTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LCD_GLB_EN When it's disabled, the module will be reset to idle state (except configuration logic). 0: Disable 1: Enable
30:0	/	/	/

15.1.7.2 0x0004 LCD Global Interrupt Register0 (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: LCD_GINT0_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LCD_VB_INT_EN Enable the Vb interrupt. 0: Disable 1: Enable
30	/	/	/
29	R/W	0x0	LCD_LINE_INT_EN Enable the line interrupt. 0: Disable 1: Enable
28	/	/	/
27	R/W	0x0	LCD_TRI_FINISH_INT_EN Enable the trigger finish interrupt (TRI mode). 0: Disable 1: Enable
26	R/W	0x0	LCD_TRI_COUNTER_INT_EN Enable the trigger counter interrupt 0: Disable 1: Enable Before enabling, configure the TRI_INT_MODE bit of the LCD_CPU_TRI3_REG (offset: 0x016C) register first.
25:16	/	/	/

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Offset: 0x0004			Register Name: LCD_GINT0_REG
Bit	Read/Write	Default/Hex	Description
15	R/W0C	0x0	LCD_VB_INT_FLAG Asserted during vertical no-display period every frame. Write 0 to clear it.
14	/	/	/
13	R/W0C	0x0	LCD_LINE_INT_FLAG Trigger when SY0 match the current LCD scan line. Write 0 to clear it.
12	/	/	/
11	R/W0C	0x0	LCD_TRI_FINISH_INT_FLAG Trigger when CPU trigger mode finish. Write 0 to clear it.
10	R/W0C	0x0	LCD_TRI_COUNTER_INT_FLAG Trigger when tri counter reaches this value Write 0 to clear it. 00: Disable 01: Interrupt trigger flag in counter mode 10: Flag for Interrupt trigger in logic high level of GPIO TRIGGLE signal 11: Flag for Interrupt trigger in logic low level of GPIO TRIGGLE signal
9	R/W0C	0x0	LCD_TRI_UNDERFLOW_FLAG Only used in DSI video mode, tri when SYNC by DSI but not finish Write 0 to clear it.
8:3	/	/	/
2	R/W	0x0	FSYNC_INT_INV Enable the FSYNC interrupt set signal inverse polarity. When FSYNC is positive, this bit must be 1. And vice versa.
1	R/W	0x0	DE_INT_FLAG Asserted at the first valid line in every frame. Write 0 to clear it.
0	R/W	0x0	FSYNC_INT_FLAG Asserted at the FSYNC signal in every frame. Write 0 to clear it.

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15.1.7.3 0x0008 LCD Global Interrupt Register1 (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: LCD_GINT1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	LCD_LINE_INT_NUM Scan line for LCD line trigger (including inactive lines). Setting it for the specified line for trigger0. Configure this bit before writing 1 to the LCD_LINE_INT_EN in the LCD_GINT0_REG register Note: SY0 is writable only when LINE_TRG0 disable.
15:0	/	/	/

15.1.7.4 0x0040 LCD Control Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: LCD_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LCD_EN It executes at the beginning of the first blank line of LCD timing. 0: Disable 1: Enable
30:26	/	/	/
25:24	R/W	0x0	LCD_IF Set the interface type of LCD controller. 00: HV(Sync+DE) 01: 8080 I/F 1x: Reserved
23	R/W	0x0	LCD_RB_SWAP Enable the function to swap red data and blue data in FIFO1. 0: Default 1: Swap RED and BLUE data
22	/	/	/
21	R/WAC	0x0	LCD_FIFO1_RST LCD FIFO1 Reset Signal (High Active) Write 1 and then 0 at this bit will reset FIFO 1 Note: 1 holding time must more than 1 DCLK
20:9	/	/	/

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Offset: 0x0040			Register Name: LCD_CTL_REG
Bit	Read/Write	Default/Hex	Description
8:4	R/W	0x0	LCD_START_DLY The unit of delay is T_{line}. NOTE: Valid only when LCD_EN == 1 (This LCD_EN is for LCD timing enable)
3	/	/	/
2:0	R/W	0x0	LCD_SRC_SEL LCD Source Select 000: DE 001: Color Check 010: Grayscale Check 011: Black by White Check 100: Test Data all 0 101: Test Data all 1 110: Reversed 111: Gridding Check

15.1.7.5 0x0044 LCD Data Clock Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: LCD_DCLK_REG
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	LCD_DCLK_EN LCD clock enable. 0000: dclk_en = 0; dclk1_en = 0; dclk2_en = 0; 0001/0010: dclk_en = 1; dclk1_en = 0; dclk2_en = 0; 0011: dclk_en = 1; dclk1_en = 1; dclk2_en = 0; 0101: dclk_en = 1; dclk1_en = 0; dclk2_en = 1; 1111: dclk_en = 1; dclk1_en = 1; dclk2_en = 1; Others: Reversed dclk_en is used to configure dclk (pixel clock with phase 0). dclk1_en is used to configure dclk1 (pixel clock with phase 1/3). dclk2_en is used to configure dclk2 (pixel clock with phase 2/3).
27:7	/	/	/
6:0	R/W	0x0	LCD_DCLK_DIV LCD DCLK Diving Value, the range is 1-32 fdclk = flcd_clk/LCD_DCLK_DIV If dclk1&dclk2 used, LCD_DCLK_DIV >=6.

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15.1.7.6 0x0048 LCD Basic Timing Register0 (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: LCD_BASIC0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	WIDTH_X Panel width is X+1
15:12	/	/	/
11:0	R/W	0x0	HEIGHT_Y Panel height is Y+1

15.1.7.7 0x004C LCD Basic Timing Register1 (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: LCD_BASIC1_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HT Thcycle = (HT+1) * Tdclk Computation: 1) parallel: HT = X + BLANK Limitation: 1) parallel: HT >= (HBP +1) + (X+1) +2 2) serial 1: HT >= (HBP +1) + (X+1) *3+2 3) serial 2: HT >= (HBP +1) + (X+1) *3/2+2
15:12	/	/	/
11:0	R/W	0x0	HBP Horizontal back porch (in dclk) Thbp = (HBP +1) * Tdclk

15.1.7.8 0x0050 LCD Basic Timing Register2 (Default Value: 0x0000_0000)

Offset: 0x0050			Register Name: LCD_BASIC2_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/

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Offset: 0x0050			Register Name: LCD_BASIC2_REG
Bit	Read/Write	Default/Hex	Description
28:16	R/W	0x0	VT TVT = (VT)/2 * Thsync VT/2 >= (VBP+1) + (Y+1) +2 VT [0] is used to identify the used scanning mode. If VT [0] ==0, the current transmission uses progressive scanning. if VT[0]==1, the current transmission uses interlaced scanning.
15:12	/	/	/
11:0	R/W	0x0	VBP Tvbp = (VBP +1) * Thsync

15.1.7.9 0x0054 LCD Basic Timing Register3 (Default Value: 0x0000_0000)

Offset: 0x0054			Register Name: LCD_BASIC3_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:16	R/W	0x0	HSPW Thspw = (HSPW+1) * Tdclk Note: HT > (HSPW+1)
15:10	/	/	/
9:0	R/W	0x0	VSPW Tvspw = (VSPW+1) * Thsync Note: VT/2 > (VSPW+1)

15.1.7.10 0x0058 LCD HV Panel Interface Register (Default Value: 0x0000_0000)

Offset: 0x0058			Register Name: LCD_HV_IF_REG
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	HV_MODE Set the HV mode of LCD controller. 0000: 24bit/1cycle parallel mode 1000: 8bit/3cycle RGB serial mode (RGB888) 1010: 8bit/4cycle Dummy RGB (DRGB) 1011: 8bit/4cycle RGB Dummy (RGBD) 1100: 8bit/2cycle YUV serial mode (BT656)

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Offset: 0x0058			Register Name: LCD_HV_IF_REG
Bit	Read/Write	Default/Hex	Description
27:26	R/W	0x0	RGB888_ODD_ORDER Serial RGB888 mode Output sequence at odd lines of the panel (line 1, 3, 5, 7...). 00: R→G→B 01: B→R→G 10: G→B→R 11: R→G→B
25:24	R/W	0x0	RGB888_EVEN_ORDER Serial RGB888 mode Output sequence at even lines of the panel (line 2, 4, 6, 8...). 00: R→G→B 01: B→R→G 10: G→B→R 11: R→G→B
23:22	R/W	0x0	YUV_SM Serial YUV mode Output sequence 2-pixel-pair of every scan line. 00: UYVY 01: VYUY 10: YVYU 11: YUYV
21:20	R/W	0x0	YUV_EAV_SAV_F_LINE_DLY When the BT656_F_MASK bit of LCD_CEU_CTL_REG (offset: 0x0100) register is configured as 0, the F signal timing in BT.656 transmission is configured in this bit, and the value could be configured as maximum 3 lines. (The F signal is the odd and even field flag in the current transmission) 00: When a field counting ends, the F signal flips immediately. 01: When both the field counting value and the line counting value are 2, the F signal flips. 10: When both the field counting value and the line counting value are 3, the F signal flips. 11: Reserved.
19	R/W	0x0	CCIR_CSC_DIS Select '0' LCD convert source from RGB to YUV. 0: Enable 1: Disable Only valid when HV mode is "1100".
18:0	/	/	/

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15.1.7.11 0x0060 LCD CPU Panel Interface Register (Default Value: 0x0000_0000)

Offset: 0x0060			Register Name: LCD_CPU_IF_REG
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	CPU_MODE Set the CPU interface work mode. 0000: 18bit/256K mode (output 1 pixel in 1 cycle) 0010: 18bit/256K mode (1st, 2nd, 3rd) 0100: 18bit/256K mode (1st and 2nd use D[12:5]) 0110: 18bit/256K mode (1st and 2rd use D[23:18]) 1000: 16bit 64K mode (output 1 pixel in 1 cycle) 1010: 18bit 256K mode (output 1 pixel in 2 cycles) 1100: 18bit 256K mode (output 1 pixel in 3 cycles) 1110: 16bit 64K mode (output 1 pixel in 2 cycles) xxx1: 24bit for DSI
27	/	/	/
26	R/W	0x0	DA CPU Mode **AUTO Transmission** In the active region, the IO [3] port outputs the DA value. in the non-active region, if the WR/RD operation is not executed, the IO [3] port outputs the DA value. **FLUSH/TRI Transmission** When LCD_FIFO is not empty, the IO [3] port outputs the DA value. when LCD_FIFO is empty, if the WR/RD operation is not executed, the IO[3] outputs the DA value.
25	R/W	0x0	CA CPU Mode **AUTO Transmission** In the non-active region, if WR/RD operation is executed, the IO [3] port outputs the CA value; **FLUSH/TRI Transmission** When LCD_FIFO is empty and WR/RD operation is executed, the IO[3] port outputs the CA value.
24	/	/	/

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Offset: 0x0060			Register Name: LCD_CPU_IF_REG
Bit	Read/Write	Default/Hex	Description
23	R	0x0	WR_FLAG The status of write operation. 0: Write operation is finishing 1: Write operation is pending
22	R	0x0	RD_FLAG The status of read operation. 0: Read operation is finishing 1: Read operation is pending
21:18	/	/	/
17	R/W	0x0	AUTO Auto Transfer Mode: Auto transmission: If configured as 1, the current frame data is transmitted in the active region, and the transmission is controlled by HV timing (only valid in 8080 mode). This bit is sampled at the end of the active region.
16	R/W	0x0	FLUSH Direct transfer mode: Flush transmission: When AUTO is 0 and FLUSH is configured as 1, if LCD_FIFO is not empty, pixel data can be output to the subsequent module without being controlled by HV timing (only valid in 8080 mode). Data output rate control by DCLK.
15:3	/	/	/
2	R/W	0x0	TRI_FIFO_EN Enable the trigger FIFO. 0: Disable 1: Enable
1	R/W1S	0x0	TRI_START Software must make sure write '1' only when this flag is '0'. Write '1' to start a frame flush, write'0' has no effect. This flag indicated frame flush is running.
0	R/W	0x0	TRI_EN Enable trigger mode. 0: Trigger mode disable 1: Trigger mode enable

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15.1.7.12 0x0064 LCD CPU Panel Write Data Register (Default Value: 0x0000_0000)

Offset: 0x0064			Register Name: LCD_CPU_WR_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	W	0x0	DATA_WR Under Auto transmission, when in the VBlank region and WR is valid, the DATA_WR value is output through D [23:0]; Under Flush/TRIG transmission, when lcd_fifo is empty and WR is valid, the DATA_WR value is output through D[23:0].

15.1.7.13 0x0068 LCD CPU Panel Read Data Register0 (Default Value: 0x0000_0000)

Offset: 0x0068			Register Name: LCD_CPU_RD0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R	0x0	DATA_RD0 Data read on 8080 bus. Launch a new read operation on 8080 bus.

15.1.7.14 0x006C LCD CPU Panel Read Data Register1 (Default Value: 0x0000_0000)

Offset: 0x006C			Register Name: LCD_CPU_RD1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R	0x0	DATA_RD1 Data read on 8080 bus, without a new read operation on 8080 bus.

15.1.7.15 0x0088 LCD IO Polarity Register (Default Value: 0x0000_0000)

Offset: 0x0088			Register Name: LCD_IO_POL_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0088			Register Name: LCD_IO_POL_REG
Bit	Read/Write	Default/Hex	Description
29:28	R/W	0x0	DCLK_SEL Set the phase offset of clock and data in HV mode. 00: Use DCLK0 (normal phase offset) 01: Use DCLK1 (1/3 phase offset) 10: Use DCLK2 (2/3 phase offset) Others: Reserved
27	R/W	0x0	IO3_INV Enable invert function of IO3. 0: Not invert 1: Invert
26	R/W	0x0	IO2_INV Enable invert function of IO2. 0: Not invert 1: Invert
25	R/W	0x0	IO1_INV Enable invert function of IO1. 0: Not invert 1: Invert
24	R/W	0x0	IO0_INV Enable invert function of IO0. 0: Not invert 1: Invert
23:0	R/W	0x0	Data_INV LCD output port D [23:0] polarity control, with independent bit control. 0: Normal polarity 1: Invert the specify output

15.1.7.16 0x008C LCD IO Control Register (Default Value: 0x0FFF_FFFF)

Offset: 0x008C			Register Name: LCD_IO_TRI_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x0	RGB_ENDIAN Set the endian of data bits. 0: Normal 1: Bits_invert (The RGB byte order remains unchanged, and the bits within the bytes are inverted.)

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Offset: 0x008C			Register Name: LCD_IO_TRI_REG
Bit	Read/Write	Default/Hex	Description
27	R/W	0x1	IO3_OUTPUT_TRI_EN Enable the output of IO3. 1: Disable 0: Enable
26	R/W	0x1	IO2_OUTPUT_TRI_EN Enable the output of IO2. 1: Disable 0: Enable
25	R/W	0x1	IO1_OUTPUR_TRI_EN Enable the output of IO1. 1: Disable 0: Enable
24	R/W	0x1	IO0_OUTPUT_TRI_EN Enable the output of IO0. 1: Disable 0: Enable
23:0	R/W	0xFFFFFF	DATA_OUTPUT_TRI_EN LCD output port D [23:0] output enable, with independent bit control. 1: Disable 0: Enable

15.1.7.17 0x00FC LCD Debug Register (Default Value: 0x2000_0000)

Offset: 0x00FC			Register Name: LCD_DEBUG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W0C	0x0	LCD_FIFO_UNDERFLOW The flag shows whether the FIFOs in underflow status. 0: Not underflow 1: Underflow
30	/	/	/
29	R	0x1	LCD_FIELD_POL LCD field indication signal, initial value is 1. The flag indicates the current field polarity. 0: Second field 1: First field
28	/	/	/

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Offset: 0x00FC			Register Name: LCD_DEBUG_REG
Bit	Read/Write	Default/Hex	Description
27:16	R	0x0	LCD_CURRENT_LINE Scan lines of the current field (line count, starting from the vfp region with an initial value of 1).
15	R/W0C	0x0	WR Consistent with 0x060 WR_FLAG register, write screen register flag bit.
14	R/W0C	0x0	RD Consistent with 0x060 RD_FLAG register, read screen register flag bit.
13:0	/	/	/

15.1.7.18 0x0100 LCD CEU Control Register (Default Value: 0x0000_0000)

Offset: 0x0100			Register Name: LCD_CEU_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CEU_EN Enable CEU function. 0: Bypass 1: Enable
30	R/W	0x0	BT656_F_MASK The F signal generation mode selection enable during BT656 transmission. BT656 F Mask 0: Disable (F signal timing is configured in the YUV_EAV_SAV_F_LINE_DLY bit of LCD_HV_IF_REG register) 1: Enable (F signal timing is configured in the BT656_F_MASK_VALUE bit of LCD_CEU_CTL_REG register)
29	R/W	0x0	BT656_F_MASK_VALUE BT656 F Mask Value 0/1
28:0	/	/	/

15.1.7.19 0x0110+N*0x04(N=0-2) LCD CEU Coefficient Register0 (Default Value: 0x0000_0000)

Offset: 0x0110+N*0x04(N=0-2)			Register Name: LCD_CEU_COEF_MUL_REG0
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/

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Offset: 0x0110+N*0x04(N=0-2)			Register Name: LCD_CEU_COEF_MUL_REG0
Bit	Read/Write	Default/Hex	Description
12:0	R/W	0x0	CEU_COEF_MUL_VALUE Signed 13bit value, range of (-16,16). N=0: Rr N=1: Rg N=2: Rb

15.1.7.20 0x0120+N*0x04(N=0-2) LCD CEU Coefficient Register1 (Default Value: 0x0000_0000)

Offset: 0x0120+N*0x04(N=0-2)			Register Name: LCD_CEU_COEF_MUL_REG1
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	CEU_COEF_MUL_VALUE Signed 13bit value, range of (-16,16). N=0: Gr N=1: Gg N=2: Gb

15.1.7.21 0x0130+N*0x04(N=0-2) LCD CEU Coefficient Register2 (Default Value: 0x0000_0000)

Offset: 0x0130+N*0x04(N=0-2)			Register Name: LCD_CEU_COEF_MUL_REG2
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	CEU_COEF_MUL_VALUE Signed 13bit value, range of (-16,16). N=0: Br N=1: Bg N=2: Bb

15.1.7.22 0x011C+N*0x10(N=0-2) LCD CEU Coefficient Register1 (Default Value: 0x0000_0000)

Offset: 0x011C+N*0x10(N=0-2)			Register Name: LCD_CEU_COEF_ADD_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18:0	R/W	0x0	CEU_COEF_ADD_VALUE Signed 19bit value, range of (-16384, 16384). N=0: Rc N=1: Gc N=2: Bc

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15.1.7.23 0x0140+N*0x04(N=0-2) LCD CEU Coefficient Register2 (Default Value: 0x0000_0000)

Offset: 0x0140+N*0x04(N=0-2)			Register Name: LCD_CEU_COEF_RANG_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	CEU_COEF_RANGE_MIN Unsigned 8bit value, range of [0,255].
15:8	/	/	/
7:0	R/W	0x0	CEU_COEF_RANGE_MAX Unsigned 8bit value, range of [0,255].

15.1.7.24 0x0160 LCD CPU Panel Trigger Register0 (Default Value: 0x0000_0000)

Offset: 0x0160			Register Name: LCD_CPU_TRI0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	BLOCK_SPACE Trigger mode: The delay after reading a line from TRIGFIFO. Counting starts when a line of transmission ends. When the count value is equal to BLOCK_SPACE, TRIGFIFO can initiate the next line of reading. The spaces between data blocks. It should be set >20*pixel.
15:12	/	/	/
11:0	R/W	0x0	BLOCK_SIZE Triggle mode: The configuration value is the size of the effective display region X. The size of data block. It is usually set as X.

15.1.7.25 0x0164 LCD CPU Panel Trigger Register1 (Default Value: 0x0000_0000)

Offset: 0x0164			Register Name: LCD_CPU_TRI1_REG
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	BLOCK_CURRENT_NUM Shows the current data block transmitting to panel.
15:0	R/W	0x0	BLOCK_NUM The number of data blocks. It is usually set as Y.

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15.1.7.26 0x0168 LCD CPU Panel Trigger Register2 (Default Value: 0x0020_0000)

Offset: 0x0168			Register Name: LCD_CPU_TRI2_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x20	START_DLY T_{dly} = (Start_Delay +1) * be_clk*8.
15	R/W	0x0	TRANS_START_MODE Select the FIFOs used in CPU mode. 0: 3D_FIFO+TRI_FIFO 1: TRI_FIFO
14:13	R/W	0x0	SYNC_MODE Set the sync mode in CPU interface. 0x: Auto 10: 0 11: 1
12:0	R/W	0x0	TRANS_START_SET Transmission Start Trigger Value. When the amount of data in the buffer (selected in the TRANS_START_MODE bit) is not less than the configured value, the pixel data in TRIGFIFO can be transferred to LCD_FIFO. Usual set as the length of a line.

15.1.7.27 0x016C LCD CPU Panel Trigger Register3 (Default Value: 0x0000_0000)

Offset: 0x016C			Register Name: LCD_CPU_TRI3_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	TRI_INT_MODE Configure the CPU to trigger interrupt mode. When configured as 01, it is Counter mode. The counting range is configured by CONTER_N and COUNTER_M, and the interrupt period is (Count_N+1) *(Count_M+1) *4dclk. 00: Disable 01: Counter mode 10: GPIO TRIGGLE signal high level trigger mode 11: GPIO TRIGGLE signal low level trigger mode
27:24	/	/	/
23:8	R/W	0x0	COUNTER_N Count value in Counter mode, the counting period is COUNTER_N * (2*COUNTER_M*dclk).

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Offset: 0x016C			Register Name: LCD_CPU_TRI3_REG
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x0	COUNTER_M Count value in Counter mode, the counting period is COUNTER_M*dclk.

15.1.7.28 0x0170 LCD CPU Panel Trigger Register4 (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: LCD_CPU_TRI4_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	A1_First_Valid Valid in first Block (non-active region)
23:0	R/W	0x0	D23_TO_D0_First_Valid Valid in first Block

15.1.7.29 0x0174 LCD CPU Panel Trigger Register5 (Default Value: 0x0000_0000)

Offset: 0x0174			Register Name: LCD_CPU_TRI5_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24	R/W	0x0	A1_NON_First_Valid Valid in Block except first (active region)
23:0	R/W	0x0	D23_TO_D0_NON_First_Valid Valid in Block except first.

15.1.7.30 0x01F0 LCD Safe Period Register (Default Value: 0x0000_0000)

Offset: 0x01F0			Register Name: LCD_SAFE_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	SAFE_PERIOD_FIFO_NUM When the data length in line buffer is more than SAFE_PERIOD_FIFO_NUM, the LCD controller will allow DRAM controller to stop working to change frequency.
15:4	R/W	0x0	SAFE_PERIOD_LINE Set a fixed line and during the line time, the LCD controller allow DRAM controller to change frequency. The fixed line should be set in the blanking area.
3	/	/	/

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Offset: 0x01F0			Register Name: LCD_SAFE_PERIOD_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	SAFE_PERIOD_MODE Select the save mode 000: Unsafe 001: Safe 010: Safe at FIFO_CURR_NUM > SAFE_PERIOD_FIFO_NUM 011: Safe at 2 and safe at sync active 100: Safe at line

15.1.7.31 0x0228 LCD FSYNC Generate Control Register (Default Value: 0x0000_0000)

Offset: 0x0228			Register Name: FSYNC_GEN_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18:8	R/W	0x0	SENSOR_DIS_TIME Delay 0–2047 Hsync Period When HSYNC_POL_SEL is 0, the actual delay is sensor_dis_time-1. When HSYNC_POL_SEL is 1, the actual delay is sensor_dis_time.
7	/	/	/
6	R/W	0x0	SENSOR_ACT1_VALUE Sensor Active1 Value 0: Fsync active_1 period output 0 1: Fsync active_1 period output 1
5	R/W	0x0	SENSOR_ACT0_VALUE Sensor Active0 Value 0: Fsync active_0 period output 0 1: Fsync active_0 period output 1
4	R/W	0x0	SENSOR_DIS_VALUE Sensor Disable Value 0: Fsync disable period output 0 1: Fsync disable period output 1
3	/	/	/
2	R/W	0x0	HSYNC_POL_SEL Hsync Polarity Select 0: Normal 1: Opposite hsync to hsync counter

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Offset: 0x0228			Register Name: FSYNC_GEN_CTRL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	SEL_VSYNC_EN Select Vsync Enable 0: Select vsync falling edge to start state machine 1: Select vsync rising edge to start state machine
0	R/W	0x0	FSYNC_GEN_EN Fsync Generate Enable 0: Disable 1: Enable

15.1.7.32 0x022C LCD FSYNC Generate Delay Register (Default Value: 0x0000_0000)

Offset: 0x022C			Register Name: FSYNC_GEN_DLY_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	SENSOR_ACT0_TIME Delay 0-4095 Pixel clk Period The actual delay is sensor_act0_time+1.
15:12	/	/	/
11:0	R/W	0x0	SENSOR_ACT1_TIME Delay 0-4095 Pixel clk Period The actual delay is sensor_act1_time+1.

15.1.7.33 0x0230 LCD Sync Control Register (Default Value: 0x0000_0000)

Offset: 0x0230			Register Name: LCD_SYNC_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	LCD_CTRL_WORK_MODE LCD Controller Work mode 0: Single DSI mode 1: Dual DSI mode
7:0	/	/	/

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15.2 LVDS

15.2.1 Overview

LVDS implements the conversion of HV timing data from TCON_LCD into LVDS protocol format and output them to panels with LVDS interfaces.

The LVDS module has the following features:

LVDS Host

- Compatible with TIA/EIA 644-A standard
- Compatible with VESA/JEIDA format
- 8/6-bit format output
- Single LVDS driving one panel, up to 1366x768@60fps
- Dual LVDS
  - Drives one panel, up to 1920x1080@60fps
  - Drives two panels, up to 1366x768@60fps
- LVDS Lane remap

NOTE

For detailed description about single/dual-link mode for LVDS, see section 15.2.4.2.

LVDS PHY

- Up to 600 MHz's bandwidth for LVDS data lane transmission
- Polarity reversing for output signal
- LVDS PHY, DSI PHY, and RGB PHY combo

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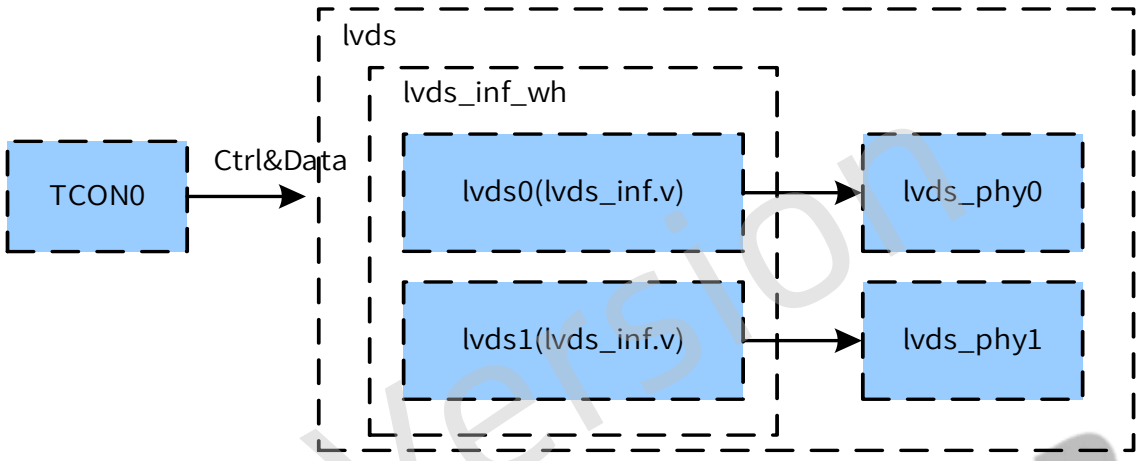
15.2.2 Block Diagram

The following figure shows a block diagram of the LVDS.

Figure 15-9 LVDS Block Diagram



15.2.3 External Signals

The following table describes the external signals of LVDS.

Table 15-7 LVDS External Signals

Signal Name	Description	Type
LVDS0-CKP	LVDS0 Differential Clock (Positive)	AO
LVDS0-CKN	LVDS0 Differential Clock (Negative)	AO
LVDS0-D0P	LVDS0 Differential Data 0 (Positive)	AO
LVDS0-D0N	LVDS0 Differential Data 0 (Negative)	AO
LVDS0-D1P	LVDS0 Differential Data 1 (Positive)	AO
LVDS0-D1N	LVDS0 Differential Data 1 (Negative)	AO
LVDS0-D2P	LVDS0 Differential Data 2 (Positive)	AO
LVDS0-D2N	LVDS0 Differential Data 2 (Negative)	AO
LVDS0-D3P	LVDS0 Differential Data 3 (Positive)	AO
LVDS0-D3N	LVDS0 Differential Data 3 (Negative)	AO
LVDS1-CKP	LVDS1 Differential Clock (Positive)	AO
LVDS1-CKN	LVDS1 Differential Clock (Negative)	AO
LVDS1-D0P	LVDS1 Differential Data 0 (Positive)	AO
LVDS1-D0N	LVDS1 Differential Data 0 (Negative)	AO
LVDS1-D1P	LVDS1 Differential Data 1 (Positive)	AO
LVDS1-D1N	LVDS1 Differential Data 1 (Negative)	AO
LVDS1-D2P	LVDS1 Differential Data 2 (Positive)	AO
LVDS1-D2N	LVDS1 Differential Data 2 (Negative)	AO
LVDS1-D3P	LVDS1 Differential Data 3 (Positive)	AO
LVDS1-D3N	LVDS1 Differential Data 3 (Negative)	AO
VCC18-LVDS	LVDS 1.8V Power Supply	P

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15.2.4 Functional Description

15.2.4.1 LVDS Interface Timing

8/6-bit Mode

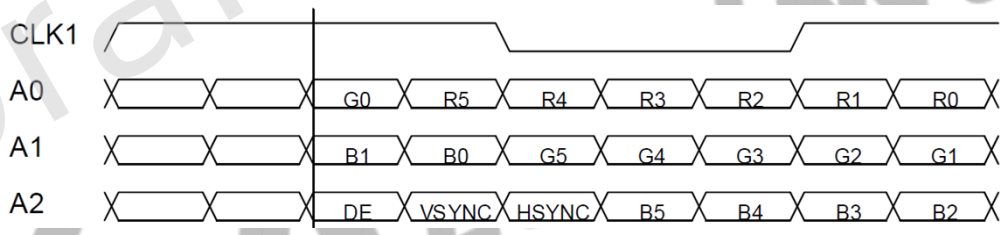
LVDS protocol requires that one set of RGB data transmission is implemented in 7 clock cycles. There is 7-bit video data transmitted in one clock cycle and the duty cycle of clock high/low logic level is 4:3. With more differential pairs, the volume of data that is able to be sent at a time can become larger.

Allwinner design fixes one clock lane and four data lanes for one LVDS interface, supporting 8-bit (RGB: 24-bit) mode and 6-bit (RGB: 18-bit) mode. Data lanes transmit both video data and video synchronization signals {DE, VSYNC, HSYNC} (see Figure 15-10).

In 8-bit mode, four data lanes are able to send 28-bit (4*7 bits) data totally, which contains 24-bit pixel information, 3-bit timing information, and 1-bit invalid bit.

In 6-bit mode, only three data lanes are available. In this case, RGB [7:2] in 8-bit mode are transmitted taken as RGB [5:0] in 6-bit mode.

Figure 15-10 VESA 6-Bit format



JEDIA/VSEA Format

LVDS protocol supports two video timing formats: JEDIA mode and VSEA mode. As Figure 15-11 and Figure 15-12 shown below, the two video data format modes have different data sending bits with a similar sending method. For VSEA mode transmission, RGB data and video synchronization signals should be remapped to perform serial output in 7 clock cycles. See Figure 15-11 for output sequence. For JEDIA mode transmission, you can get the same transmission sequence as that of VSEA mode by setting the lower 2 bits of video data to the higher 2 bits.

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Figure 15-11 Single LVDS VSEA Mode

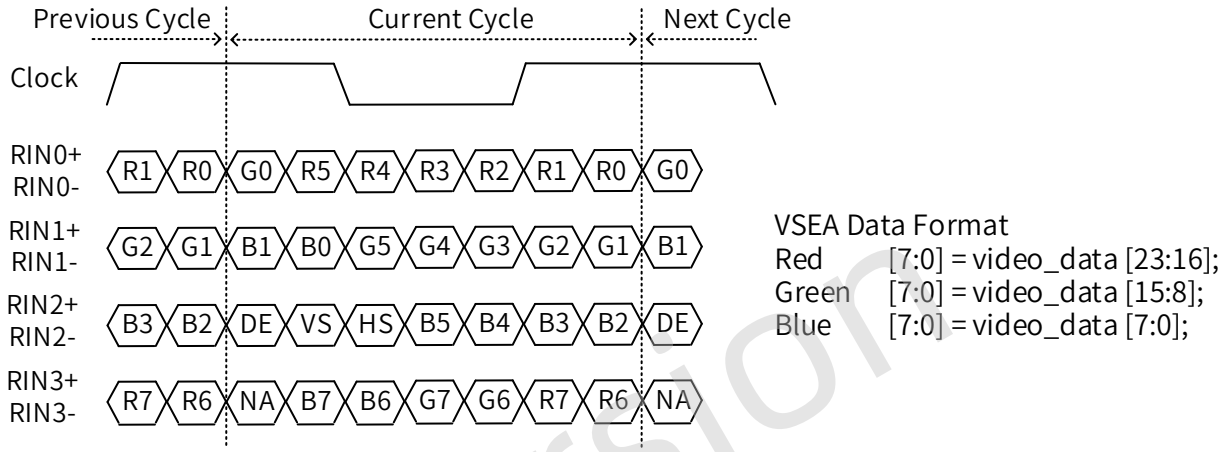
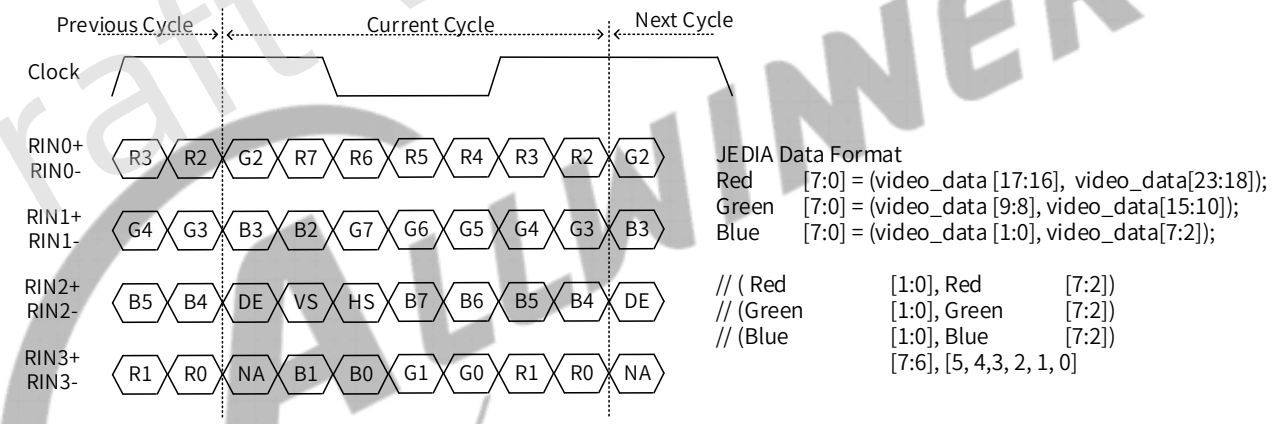


Figure 15-12 Single LVDS JEDIA Mode



15.2.4.2 Single-link/Dual-link Mode

Allwinner LVDS can work either in single link or dual-link mode. In the former mode, single LVDS outputs pixel data or dual LVDS outputs the same data. In the latter mode, the dual LVDS output even pixels and odd pixels respectively. Allwinner LVDS is able to achieve the same resolution with half of pixel clock frequency used in single-link LVDS display. Figure 15-13 shows that odd and even pixel data converted from TCON video data stream are respectively distributed to LVDS0 and LVDS1. They will be serialized before output.

The following table shows the display functions for the two modes mentioned above.

Work mode	Usage Scenarios
Single-link mode	• Single LVDS drives single panel. • Dual LVDS drives single panel. • Dual LVDS drives dual panel, which has identical display.

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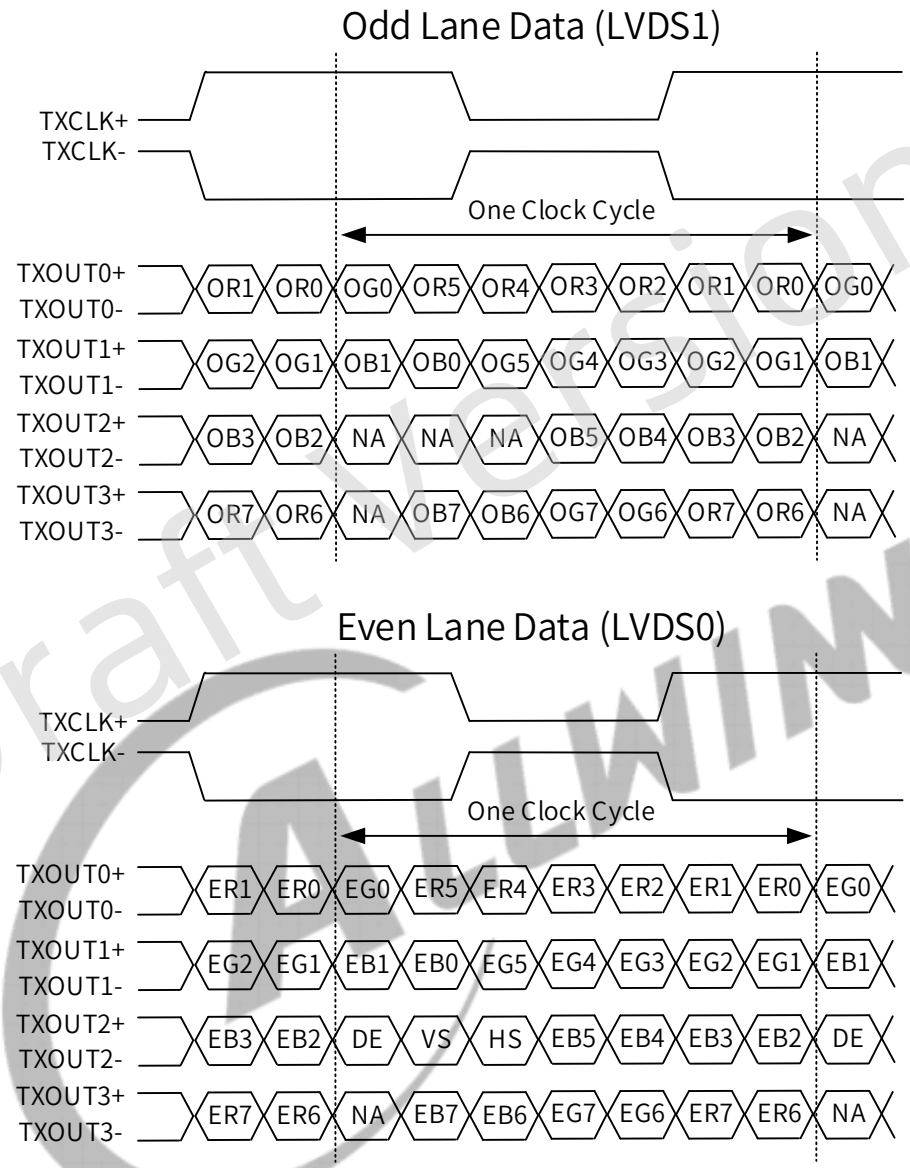
Work mode	Usage Scenarios
Dual-link mode	Dual LVDS drives dual panel, which has different display.

Figure 15-13 Dual-link LVDS VSEA Mode



15.2.4.3 Lane Remap Function

LVDS module supports 5-lane random mapping between digital input LVDS_IN [4:0] and analog PHY output, and supports mapping relationship changing through register configuration to achieve more flexibility of PAD output sequence. The following figure shows an example to explain lane mapping. The input sequence is {A, B, C, D, E} while the output sequence is remapped as {C, A, E, B, D}.

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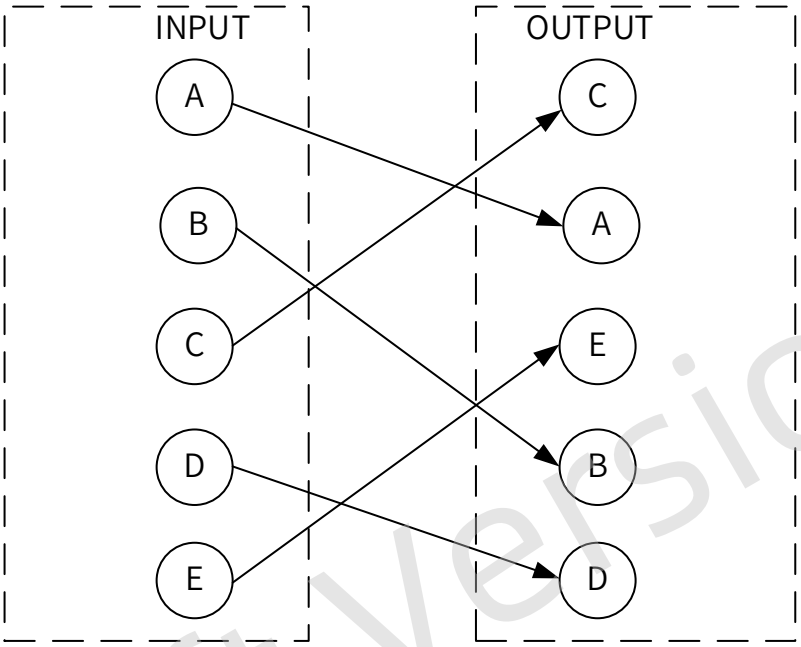
Figure 15-14 Interface Mapping



15.2.5 Programming Guidelines

NOTE

LVDS controller implements data conversion from RGB timing to LVDS protocol. TCON_LCD should be configured as HV timing and LVDS controller configuration should be finished before TCON_LCD is enabled.

Step 1 Configure LVDS controller.

a) Configure LCD_LVDS_IF_REG.LCD_LVDS_DATA_POL to set data/clock polarity of LVDS.

b) Configure LCD_LVDS_IF_REG.LCD_LVDS_BITWIDTH to set the bit width of pixel data.

c) Configure LCD_LVDS_IF_REG.LCD_LVDS_MODE to set the LVDS data format.

d) Configure LCD_LVDS_IF_REG.LCD_LVDS_DIR to set the LVDS output direction.

e) Configure LCD_LVDS_IF_REG.LCD_LVDS_LINK to select LVDS work mode as single link mode or dual link.

f) Configure LCD_LVDS_IF_REG.LCD_LVDS_EN t as 0x1 to enable LVDS controller.

Step 2 Take PHY0 as an example to configure analog LVDS PHY. The procedure for PHY1 (LCD_LVDS_ANA1_REG) is consistent with that of PHY0.

a) Configure LCD_LVDS_ANA0_REG.LVDS_REG_PLR to select whether the polarity of LVDS data channel [3:0] is reversed.

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b) Configure LCD_LVDS_ANA0_REG.LVDS_REG_PLRC to select whether the polarity of LVDS clock channel is reversed.

c) Configure LCD_LVDS_ANA0_REG.LVDS_REG_R to set the amplitude of the common signals.

d) Configure LCD_LVDS_ANA0_REG.LVDS_REG_C to set the amplitude of the differential signals.

e) Configure LCD_LVDS_ANA0_REG.LVDS_HPREN_DRV as 0xf to enable data channel [3:0] drive.

f) Configure LCD_LVDS_ANA0_REG.LVDS_HPREN_DRVC as 0x1 to enable clock channel drive.

g) Configure LCD_LVDS_ANA0_REG.LVDS_DUAL_CHANEL_SRC to set the data output mode of dual LVDS.

h) Configure LCD_LVDS_ANA0_REG.LVDS_EN_MB as 0x1 to enable analog circuit of LVDS PHY.

15.2.6 Register List

Module Name	Base Address	Comments
LVDS	0x05501000	Share the same address space with TCON_LCD.

Register Name	Offset	Description
LCD_LVDS_IF_REG	0x0084	LCD LVDS Interface Register
LCD_LVDS_ANA0_REG	0x0220	LCD LVDS Analog Register0
LCD_LVDS_ANA1_REG	0x0224	LCD LVDS Analog Register1
LCD_LVDS_LMAP0_REG	0x0248	LCD LVDS Lane Map Register0
LCD_LVDS_LMAP1_REG	0x024C	LCD LVDS Lane Map Register1

15.2.7 Register Description

15.2.7.1 0x0084 LCD LVDS Interface Register (Default Value: 0x0000_0000)

Offset: 0x0084			Register Name: LCD_LVDS_IF_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LCD_LVDS_EN Enable LVDS Interface. 0: Disable 1: Enable

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Offset: 0x0084			Register Name: LCD_LVDS_IF_REG
Bit	Read/Write	Default/Hex	Description
30	R/W	0x0	LCD_LVDS_LINK Select whether LVDS works in single link mode or dual link mode. 0: Single link 1: Dual link
29	R/W	0x0	LCD_LVDS_EVEN_ODD_DIR Set the order of even field and odd field. 0: Normal: lvds0-Odd, lvds1-Even 1: Reverse: lvds0-Even, lvds1-Odd Note: Valid when in Dual Mode.
28	R/W	0x0	LCD_LVDS_DIR Set the LVDS direction 0: Normal 1: Reverse
27	R/W	0x0	LCD_LVDS_MODE Set the LVDS data mode 0: NS mode(VESA) 1: JEIDA mode
26	R/W	0x0	LCD_LVDS_BITWIDTH Set the bit width of data 0: 24bit 1: 18bit
25	R/W	0x0	LCD_LVDS_DEBUG_EN Enable LVDS debug function 0: Disable 1: Enable
24	R/W	0x0	LCD_LVDS_DEBUG_MODE Set the output signal in debug mode. 0: Mode0 Random data (generated by PRBS10) 1: Mode1 Output CLK period=7/2 LVDS CLK period
23	R/W	0x0	LCD_LVDS_CORRECT_MODE Set the LVDS correct mode. 0: Mode0. Phase calibration can be performed in any region. 1: Mode1. Phase calibration can be performed in the blank region of the image.
22:21	/	/	/
20	R/W	0x0	LCD_LVDS_CLK_SEL Select the clock source of LVDS. 0: Revered 1: LCD CLK

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Offset: 0x0084			Register Name: LCD_LVDS_IF_REG
Bit	Read/Write	Default/Hex	Description
19:5	/	/	/
4:0	R/W	0x0	LCD_LVDS_DATA_POL Set the data/clk polarity of LVDS. Bit4: LVDS data [3] channel polarity Bit3: LVDS clock channel polarity Bit2: LVDS data [2] channel polarity Bit1: LVDS data [1] channel polarity Bit0: LVDS data [0] channel polarity 0: Reverse 1: Normal Note: When LCD_LVDS_DIR=1, the channel order corresponding to these bits will be reversed.

15.2.7.2 0x0220 LCD LVDS Analog Register0 (Default Value: 0x0000_0000)

Offset: 0x0220			Register Name: LCD_LVDS_ANA0_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LVDS_EN_MB Enable the bias circuit of the LVDS_Ana module. 0: Disable 1: Enable
30	R/W	0x0	LVDS_DUAL_CHANEL_SRC_SEL Dual LVDS Source Selection 0: LVDS0 outputs even pixels from TCON source. LVDS1 outputs odd pixels from TCON source. 1: LVDS0 and LVDS1 output the same pixels.
29	R/W	0x0	EN_LVDS Enable LVDS
28	R/W	0x0	EN_24M Enable the 24M clock
27:25	/	/	/
24	R/W	0x0	LVDS_HPREN_DRVC Enable clock channel drive. 0: Disable 1: Enable
23:20	R/W	0x0	LVDS_HPREN_DRV Enable data channel [3:0] drive. 0: Disable 1: Enable

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Offset: 0x0220			Register Name: LCD_LVDS_ANA0_REG
Bit	Read/Write	Default/Hex	Description
19:17	R/W	0x0	LVDS_REG_C Adjust current flowing through Rload of Rx to change the differential signals amplitude. 000: 216 mV 001: 252 mV 010: 276 mV 011: 312 mV 100: 336 mV 101: 372 mV 110: 395 mV 111: 432 mV
16	R/W	0x0	LVDS_REG_DENC Choose data output or PLL test clock output in LVDS_TX.
15:12	R/W	0x0	LVDS_REG_DEN Choose data output or PLL test clock output in LVDS_TX.
11	/	/	/
10:8	R/W	0x0	LVDS_REG_R Adjust current flowing through Rload of Rx to change the common signals amplitude. 000: 0.925 V 001: 0.950 V 010: 0.975 V 011: 1.000 V 100: 1.025 V 101: 1.050 V 110: 1.075 V 111: 1.100 V
7	R/W	0x0	LVDS_TEST_EN Test Enable 0: Disable 1: Enable. After enabled, LVDS output depends on the configuration of LVDS_TDATA_SEL. Note: LCD_LVDS_DIR should be set to 0x0.
6:5	R/W	0x0	LVDS_TDATA_SEL 00: LVDS data channel output is 0. 01: LVDS data channel output is 1. 10: LVDS data channel output is Vsync. 11: LVDS data channel output is Hsync.

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Offset: 0x0220			Register Name: LCD_LVDS_ANA0_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	LVDS_REG_PLRC LVDS clock channel direction. 0: Normal 1: Reverse
3:0	R/W	0x0	LVDS_REG_PLR LVDS data channel [3:0] direction. 0: Normal 1: Reverse

15.2.7.3 0x0224 LCD LVDS Analog Register1 (Default Value: 0x0000_0000)

Offset: 0x0224			Register Name: LCD_LVDS_ANA1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LVDS_EN_MB Enable the bias circuit of the LVDS_Ana module. 0: Disable 1: Enable
30	/	/	/
29	R/W	0x0	EN_LVDS Enable LVDS.
28	R/W	0x0	EN_24M Enable the 24M clock.
27:25	/	/	/
24	R/W	0x0	LVDS_HPREN_DRVC Enable clock channel drive. 0: Disable 1: Enable
23:20	R/W	0x0	LVDS_HPREN_DRV Enable data channel [3:0] drive. 0: Disable 1: Enable

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Offset: 0x0224			Register Name: LCD_LVDS_ANA1_REG
Bit	Read/Write	Default/Hex	Description
19:17	R/W	0x0	LVDS_REG_C Adjust current flowing through Rload of Rx to change the differential signals amplitude. 000: 216 mV 001: 252 mV 010: 276 mV 011: 312 mV 100: 336 mV 101: 372 mV 110: 395 mV 111: 432 mV
16	R/W	0x0	LVDS_REG_DENC Choose data output or PLL test clock output in LVDS_TX.
15:12	R/W	0x0	LVDS_REG_DEN Choose data output or PLL test clock output in LVDS_TX.
11	/	/	/
10:8	R/W	0x0	LVDS_REG_R Adjust current flowing through Rload of Rx to change the common signals amplitude. 000: 0.925 V 001: 0.950 V 010: 0.975 V 011: 1.000 V 100: 1.025 V 101: 1.050 V 110: 1.075 V 111: 1.100 V
7	R/W	0x0	LVDS_TEST_EN Test Enable 0: Disable 1: Enable. After enabled, LVDS output depends on the configuration of LVDS_TDATA_SEL. Note: LCD_LVDS_DIR should be set to 0x0.
6:5	R/W	0x0	LVDS_TDATA_SEL v00: LVDS data channel output is 0. 01: LVDS data channel output is 1. 10: LVDS data channel output is Vsync. 11: LVDS data channel output is Hsync.

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Offset: 0x0224			Register Name: LCD_LVDS_ANA1_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	LVDS_REG_PLRC LVDS clock channel direction. 0: Normal 1: Reverse
3:0	R/W	0x0	LVDS_REG_PLR LVDS data channel [3:0] direction. 0: Normal 1: Reverse

15.2.7.4 0x0248 LCD LVDS Lane Map Register0 (Default Value:0x0000_0000)

Offset: 0x0248			Register Name: LCD_LVDS_LMAP0_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LVDS_LMAP_EN 0: Disable, disable channel mapping logic. 1: Enable, enable channel mapping logic.
30:19	/	/	/
18:16	R/W	0x0	LANE4_MAP_SOURCE Physical channel lane4 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
15	/	/	/
14:12	R/W	0x0	LANE3_MAP_SOURCE Physical channel lane3 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
11	/	/	/

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Offset: 0x0248			Register Name: LCD_LVDS_LMAP0_REG
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	LANE2_MAP_SOURCE Physical channel lane2 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
7	/	/	/
6:4	R/W	0x0	LANE1_MAP_SOURCE Physical channel lane1 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
3	/	/	/
2:0	R/W	0x0	LANE0_MAP_SOURCE Physical channel lane0 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved

15.2.7.5 0x024C LCD LVDS Lane Map Register1 (Default Value:0x0000_0000)

Offset: 0x024C			Register Name: LCD_LVDS_LMAP1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LVDS_LMAP_EN 0: Disable, disable channel mapping logic. 1: Enable, enable channel mapping logic.
30:19	/	/	/

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Offset: 0x024C			Register Name: LCD_LVDS_LMAP1_REG
Bit	Read/Write	Default/Hex	Description
18:16	R/W	0x0	LANE4_MAP_SOURCE Physical channel lane4 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
15	/	/	/
14:12	R/W	0x0	LANE3_MAP_SOURCE Physical channel lane3 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
11	/	/	/
10:8	R/W	0x0	LANE2_MAP_SOURCE Physical channel lane2 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
7	/	/	/
6:4	R/W	0x0	LANE1_MAP_SOURCE Physical channel lane1 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved
3	/	/	/

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Offset: 0x024C			Register Name: LCD_LVDS_LMAP1_REG
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	LANE0_MAP_SOURCE Physical channel lane0 mapping data source. 0: Data comes from address channel 0. 1: Data comes from address channel 1. 2: Data comes from address channel 2. 3: Data comes from address channel 3. 4: Data comes from address channel 4. Others: Reserved

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15.3 MIPI DSI

15.3.1 Overview

The Display Serial Interface is a high-speed interface between a host processor and peripheral devices that adhere to MIPI Alliance specifications for mobile device interfaces. This DSI module is composed of a DSI controller which is compliance with MIPI DSI specification V1.02 and a D-PHY module which is compliance with MIPI DPHY specification V1.0.

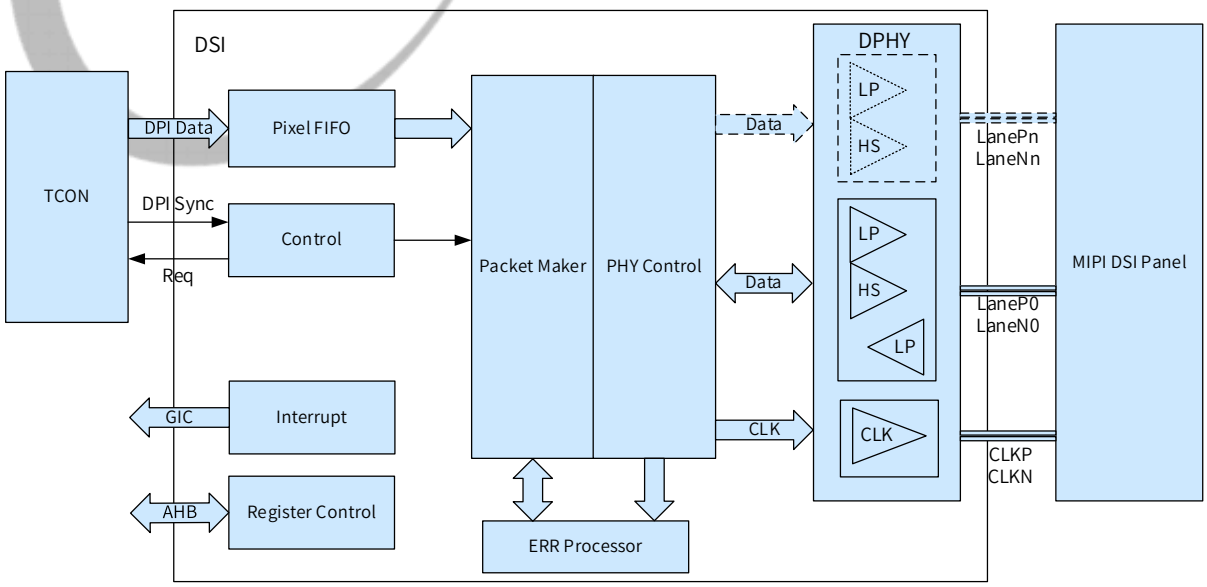
The MIPI DSI includes the following features:

- Compliance with MIPI DSI V1.02 and D-PHY v1.0
- Up to 1.0 Gbit/s for each lane
- Supports 4-lane MIPI DSI, up to 1920x1080@60fps
- Supports non-burst mode with sync pulse/sync event and burst mode
- Pixel format: RGB888, RGB666, RGB666 loosely packed and RGB565
- Continuous and non-continuous lane clock modes
- Generic commands support bidirectional communication in LP through data lane 0
- Supports low power data transmission
- Supports ULPS and escape modes
- Supports hardware checksum

15.3.2 Block Diagram

The following figure shows a block diagram of the MIPI DSI.

Figure 15-15 MIPI DSI Block Diagram



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15.3.3 External Signals

The following table describes the external signals of MIPI DSI.

Table 15-11 MIPI DSI External Signals

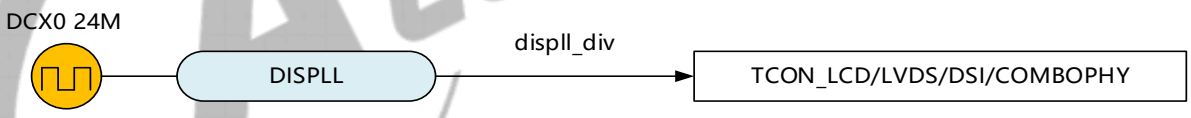
Signal Name	Description	Type
DSI-CKP	DSI Differential Clock (Positive)	AO
DSI-CKN	DSI Differential Clock (Negative)	AO
DSI-D0P	DSI Differential Data Line 0 (Positive)	A I/O
DSI-D0N	DSI Differential Data Line 0 (Negative)	A I/O
DSI-D1P	DSI Differential Data Line 1 (Positive)	AO
DSI-D1N	DSI Differential Data Line 1 (Negative)	AO
DSI-D2P	DSI Differential Data Line 2 (Positive)	AO
DSI-D2N	DSI Differential Data Line 2 (Negative)	AO
DSI-D3P	DSI Differential Data Line 3 (Positive)	AO
DSI-D3N	DSI Differential Data Line 3 (Negative)	AO

15.3.4 Functional Description

15.3.4.1 DISPLL Distribution

The following figure shows the DISPLL distribution for modules.

Figure 15-16 DISPLL Distribution



DISPLL is located inside the VIDEO_OUT_SYS system and mainly provides the function clock for the TCON_LCD/LVDS/DSI/COMBOPHY module.

DISPLL can output two clocks:

- clk_hs: 1 GHz at max. mainly used as DPHY serial clock (only used for output to the DSI interface).
- clk_ls: 1.2 GHz at max. It serves as the function clock for the TCON_LCD module (used to output RGB/LVDS/DSI interface timing). This clock is divided internally in TCON_LCD to obtain the DSI/LVDS module function clock.

15.3.4.2 DISPLL Specifications

The specifications of DISPLL are shown in the following table.

Table 15-12 DISPLL Specifications

PLL	Stable Operating Frequency	Actual Operating Frequency	Spread Spectrum	Linear FM	Pk-Pk	Lock Time

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PLL	Stable Operating Frequency	Actual Operating Frequency	Spread Spectrum	Linear FM	Pk-Pk	Lock Time
DISPLL	1.26 GHz-2.52 GHz	2.4 GHz	Yes	Yes	70ps@1GHz	0.1 ms

15.3.5 Programming Guidelines

Frequent PLL enable/disable is not recommended, as the PLL enable/disable will arise the mutual interference of PLLs and system instability.

15.3.5.1 Implement Spread Spectrum

The spread spectrum technology is to convert a narrowband signal into a wideband signal. It helps to reduce the effect of electromagnetic interference (EMI) associated with the fundamental frequency of the signal.

For the general PLL frequency, the calculation formula is as follows:

$$f = \frac{N+1+X}{P\cdot(M0+1)\cdot(M1+1)}\cdot 24MHz, 0<X<1$$

Where,

P is the frequency division factor of module or PLL;

M0 is the post-frequency division factor of PLL;

M1 is the pre-frequency division factor of PLL;

N is the frequency doubling factor of PLL;

X is the amplitude coefficient of the spread spectrum.

The parameters N, P, M1, and M0 are for the frequency division.

When M1 = 0, M0 = 0, and P = 1 (no frequency division), the calculation formula of PLL frequency can be simplified as follows:

$$f = (N+1+X)\cdot 24MHz,\ 0<X<1$$

$$[f_1, f_2] = (N+1+[X_1, X_2])\cdot 24MHz$$

$$WAVE_BOT = 2^{17}\cdot X_1$$

$$\mathrm{WAVE_STEP} = 2^{17}\cdot(X_2 - X_1)/(24\ \mathrm{MHz}/FREQ)\cdot 2$$

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Where, WAVE_BOT and WAVE_STEP are bits of the PLL pattern control register, and FREQ is the frequency of the spread spectrum.

Follow the steps below to implement the spread spectrum:

Step 1 Configure the control register of the corresponding PLL

a) Calculate the factor N and decimal value X according to the PLL frequency and PLL frequency formula. Refer to the control register of the corresponding PLL (named DPHY_PLL_REGx and DPHY_PLL_PATx_CTL_REG) for the PLL frequency formula.

b) Write M0, M1, N, and PLL frequency to the DPHY_PLL_REGx.

c) Configure the PLL_SDM_EN bit (bit [31]) of the DPHY_PLL_REG2 to 1 to enable the spread spectrum function.

Step 2 Configure the pattern control register of the corresponding PLL

a) Calculate the WAVE_BOT and WAVE_STEP of the pattern control register according to decimal value X and spread spectrum frequency.

b) Write the WAVE_BOT and WAVE_STEP to the DPHY_PLL_PAT0_CTL_REG register.

c) Configure the SDM_CYCLE of the DPHY_PLL_PAT1_CTL_REG register.

d) Configure the SPR_FREQ_MODE of the DPHY_PLL_PAT0_CTL_REG to 2 or 3.

e) Configure the SIG_DELT_PAT_EN bit (bit [31]) of DPHY_PLL_PAT0_CTL_REG register to 1.

Step 3 Delay 20 us.

15.3.5.2 Enable PLL

Step 1 Write 1 to the LDO_EN of the DPHY_PLL_REG0.

Step 2 Write 1 to PLL_CLK_LP_GATING of DPHY_PLL_REG1.

Step 3 Configure the N,M, and P factor in the DPHY_PLL_REG0.

Step 4 Write 1 to the PLL_EN of the DPHY_PLL_REG0.

Step 5 Write 1 to the LOCKDET_EN of the DPHY_PLL_REG1.

Step 6 Read LOCK_STATUS every 3 us for total three times. The PLL is locked if the value of all three reads is 1.

Step 7 Delay 20 us and the PLL is available.

15.3.5.3 Disable PLL

Step 1 Write 0 to PLL_CLK_LP_GATING of DPHY_PLL_REG1.

Step 2 Write 0 to the LOCKDET_EN of the DPHY_PLL_REG1.

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Step 3 Write 0 to the PLL_EN of the DPHY_PLL_REG0.

Step 4 Write 0 to the LDO_EN of the DPHY_PLL_REG0.

15.3.5.4 Configure clk_hs and clk_ls

clk_hs = fvco/(postdiv0*postdiv1)

clk_ls = fvco/(postdiv2*postdiv3)

Table 15-13 Parameter Configuration

Parameters	Configuration Register	
prediv = p<3:0>+1	p[3:0] is configured in bit [19:16].	DPHY_PLL_REG0 (Offset:0x1104)
postdiv0 = m0<1:0>+1	m0[1:0] is configured in bit [5:4].	
postdiv1 = m1<3:0>+1	m1[3:0] is configured in bit [3:0].	
postdiv2 = m2<1:0>+1	m2[1:0] is configured in bit [29:28].	
postdiv3 = m3<3:0>+1	m3[3:0] is configured in bit [27:24].	
fvco=fref*n<7:0>/prediv	n[7:0] is configured in bit [15:8].	

15.3.6 Register List

Module Name	Base Address
MIPI_DSI0	0x05506000

Register Name	Offset	Description
DPHY_CTL_REG	0x1000	DPHY Control Register
DPHY_TX_CTL_REG	0x1004	DPHY TX Control Register
DPHY_RX_CTL_REG	0x1008	DPHY RX Control Register
DPHY_TX_TIME0_REG	0x1010	DPHY TX Timing Parameter 0 Register
DPHY_TX_TIME1_REG	0x1014	DPHY TX Timing Parameter 1 Register
DPHY_TX_TIME2_REG	0x1018	DPHY TX Timing Parameter 2 Register
DPHY_TX_TIME3_REG	0x101C	DPHY TX Timing Parameter 3 Register
DPHY_RX_TIME0_REG	0x1030	DPHY RX Timing Parameter 0 Register
DPHY_RX_TIME1_REG	0x1034	DPHY RX Timing Parameter 1 Register
DPHY_RX_TIME3_REG	0x1040	DPHY RX Timing Parameter 3 Register
DPHY_ANA0_REG	0x104C	DPHY Analog 0 Register
DPHY_ANA1_REG	0x1050	DPHY Analog 1 Register
DPHY_ANA2_REG	0x1054	DPHY Analog 2 Register
DPHY_ANA3_REG	0x1058	DPHY Analog 3 Register
DPHY_ANA4_REG	0x105C	DPHY Analog 4 Register
DPHY_INT_EN0_REG	0x1060	DPHY Interrupt Enable 0 Register
DPHY_INT_EN1_REG	0x1064	DPHY Interrupt Enable 1 Register
DPHY_INT_EN2_REG	0x1068	DPHY Interrupt Enable 2 Register
DPHY_INT_PD0_REG	0x1070	DPHY Interrupt Pending 0 Register

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Register Name	Offset	Description
DPHY_INT_PD1_REG	0x1074	DPHY Interrupt Pending 1 Register
DPHY_INT_PD2_REG	0x1078	DPHY Interrupt Pending 2 Register
DPHY_DBG0_REG	0x10E0	DPHY Debug 0 Register
DPHY_DBG1_REG	0x10E4	DPHY Debug 1 Register
DPHY_DBG2_REG	0x10E8	DPHY Debug 2 Register
DPHY_DBG3_REG	0x10EC	DPHY Debug 3 Register
DPHY_DBG4_REG	0x10F0	DPHY Debug 4 Register
DPHY_DBG5_REG	0x10F4	DPHY Debug 5 Register
DPHY_TX_SKEW_REG0	0x10F8	DPHY TX Skew Register0
DPHY_TX_SKEW_REG1	0x10FC	DPHY TX Skew Register1
DPHY_TX_SKEW_REG2	0x1100	DPHY TX Skew Register2
DPHY_PLL_REG0	0x1104	DPHY PLL Register0
DPHY_PLL_REG1	0x1108	DPHY PLL Register1
DPHY_PLL_REG2	0x110C	DPHY PLL Register2
COMBO_PHY_REG0	0x1110	COMBO PHY Register0
COMBO_PHY_REG1	0x1114	COMBO PHY Register1
DPHY_PLL_REG3	0x1124	DPHY PLL Register3
DPHY_PLL_PAT0_CTL_REG	0x1130	DPHY PLL Patten0 Control Register
DPHY_PLL_PAT1_CTL_REG	0x1134	DPHY PLL Patten1 Control Register
DPHY_PLL_SSC_LIN_CFG_REG	0x1138	DPHY PLL SSC Linear Configuration Register

15.3.7 Register Description

15.3.7.1 0x1000 DPHY Control Register (Default Value: 0x0000_0000)

Offset:0x1000			Register Name: DPHY_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:4	R/W	0x0	LANE_NUM Lane Number 00: 1 data lane 01: 2 data lane 10: 3 data lane 11: 4 data lane Note: This bits determine lane number used for transmission.
3:1	/	/	/
0	R/W	0x0	MODULE_EN When it's disabled, the module will be reset to idle state. 0: Disable 1: Enable

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15.3.7.2 0x1004 DPHY TX Control Register (Default Value: 0x0000_0000)

Offset:0x1004			Register Name: DPHY_TX_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	ULPSTX_ENTER Ulpstx Enter 0: Direct (enter LP-00 after ULPS command) 1: Indirect (enter MARK-I after ULPS command then LP-00)
28	R/W	0x0	HSTX_CLK_CONT When this bit is enable, clk lane continue in HS mode. When this bit is disable, clk lane anto enter LP mode. 0: Disable 1: Enable
27	R/W	0x0	SKEWCAL_STP_SET When this bit is enable, data lane is forced to enter skewcal mode. 0: Disable 1: Enable
26	/	/	/
25	R/W	0x0	HSTX_CLK_EXIT HSTX Clock Exit 0: No effect 1: Force clock lane exit from HS
24	R/W	0x0	HSTX_DATA_EXIT HSTX Data Exit 0: No effect 1: Force all data lanes exit from HS
23:21	/	/	/
20	R/W	0x0	ULPSTX_CLK_EXIT Ulpstx Clk Exit 0: No effect 1: Force Clock lane exit from ULPS
19	R/W	0x0	ULPSTX_DATA3_EXIT Ulpstx Data3 Exit 0: No effect 1: Force data lane 3 exit from ULPS
18	R/W	0x0	ULPSTX_DATA2_EXIT Ulpstx Data2 Exit 0: No effect 1: Force data lane 2 exit from ULPS

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Offset:0x1004			Register Name: DPHY_TX_CTL_REG
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	ULPSTX_DATA1_EXIT Ulpstx Data1 Exit 0: No effect 1: Force data lane 1 exit from ULPS
16	R/W	0x0	ULPSTX_DATA0_EXIT Ulpstx Data0 Exit 0: No effect 1: Force data lane 0 exit from ULPS
15:13	/	/	/
12	R/W	0x0	FORCE_LP11 Force LP11 0: No effect 1: Force module work in LP11
11	R/W	0x0	HSTX_8B9B_EN HSTX 8B9B Enable 0: 8B9B Encoding disable 1: 8B9B Encoding enable
10	R/W	0x0	LPTX_8B9B_EN LPTX 8B9B Enable 0: 8B9B Encoding disable 1: 8B9B Encoding enable
9	R/W	0x0	HSTX_ENDIAN HSTX Endian 0: LSB first 1: MSB first
8	R/W	0x0	LPTX_ENDIAN LPTX Endian 0: LSB first 1: MSB first
7:5	/	/	/
4	R/W	0x0	TX_CLK_FORCE Write 1 force into TX default state(lp00), write 0 has no effect.
3	R/W	0x0	TX_D3_FORCE Write 1 force into TX default state(lp00), write 0 has no effect.
2	R/W	0x0	TX_D2_FORCE Write 1 force into TX default state(lp00), write 0 has no effect.
1	R/W	0x0	TX_D1_FORCE Write 1 force into TX default state(lp00), write 0 has no effect.

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Offset:0x1004			Register Name: DPHY_TX_CTL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	TX_D0_FORCE Write 1 force into TX default state(lp00), write 0 has no effect.

15.3.7.3 0x1008 DPHY RX Control Register (Default Value: 0x0000_0000)

Offset:0x1008			Register Name: DPHY_RX_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DBC_EN Debounce enable
30:25	/	/	/
24	R/W	0x0	RX_CLK_FORCE 0: No effect 1: Force RX
23	R/W	0x0	RX_D3_FORCE 0: No effect 1: Force RX
22	R/W	0x0	RX_D2_FORCE 0: No effect 1: Force RX
21	R/W	0x0	RX_D1_FORCE 0: No effect 1: Force RX
20	R/W	0x0	RX_D0_FORCE 0: No effect 1: Force RX
19:16	R/W	0x0	LPRX_TRND_MASK 0: Not mask 1: Mask, no ACK to turnaround in LPRX
15:11	/	/	/
10	R/W	0x0	LPRX_8B9B_EN 0: 8B9B Decoding disable 1: 8B9B Decoding enable
9	/	/	/
8	R/W	0x0	LPRX_ENDIAN 0: LSB first 1: MSB first
7:0	/	/	/

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15.3.7.4 0x1010 DPHY TX Timing Parameter 0 Register (Default Value: 0x0000_0000)

Offset:0x1010			Register Name: DPHY_TX_TIME0_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	HS_TRAIL_SET HS trail = (HS_TRAIL_SET + 2) * Tbyteclk + (2~3)*Tclk - (3~4)*Tclkd8.
23:16	R/W	0x0	HS_PRE_SET The setting of hs prepare time: Begin with LP00, end after (HS_PRE_SET + 4)*Tclk.
15:8	/	/	/
7:0	R/W	0x0	LPX_TM_SET LP pulse = (LPX_TM_SET + 1)*Tclk.

15.3.7.5 0x1014 DPHY TX Timing Parameter 1 Register (Default Value: 0x0000_0000)

Offset:0x1014			Register Name: DPHY_TX_TIME1_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	CK_POST_SET CK_Post_set cannot be 0,after the falling edge of entx,keep it for (CK_POST_SET + 2)* Tclk and then turn off the clk.
23:16	R/W	0x0	CK_PRE_SET Clk pre = ((CK_PRE_SET + 2)+ (0 ~ LPX_TM_SET))* Tclk
15:8	R/W	0x0	CK_ZERO_SET CK_Zero_set cannot be 0,when entxc is set 1,after (CK_ZERO_SET + 3)* Tclk, the enckq will be set as 1.
7:0	R/W	0x0	CK_PREP_SET Clk prepare = （CK_PREP_SET + 4)* clk

15.3.7.6 0x1018 DPHY TX Timing Parameter 2 Register (Default Value: 0x0000_0000)

Offset:0x1018			Register Name: DPHY_TX_TIME2_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/

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Offset:0x1018			Register Name: DPHY_TX_TIME2_REG
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	HS_DLY_MODE The setting of HS zero time depending on HS_DLYMODE: when HS_DLYMODE=0, HS_ZERO_TIME > 20*Tbyteclk When HS_DLYMODE=1,HS_ZERO_TIME = (HS_DLY_SET + 4)*Tclk - 2*(LPX_TM_SET + 1)*Tclk + (20~23)*Tbyteclk - (HS_PRE_SET + 4)*Tclk
27:24	/	/	/
23:8	R/W	0x0	HS_DLY_SET Use this value to prolong the time before LP switch to HScott down the length of HS-Zero.
7:0	R/W	0x0	CK_TRAIL_SET CK_TRAIL_SET clk tail =(CK_TRAIL_SET + 2)* Tclk

15.3.7.7 0x101C DPHY TX Timing Parameter 3 Register (Default Value: 0x0000_0000)

Offset:0x101C			Register Name: DPHY_TX_TIME3_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	LPTX_ULPS_EXIT_SET ULPS exit Cycles.

15.3.7.8 0x1030 DPHY RX Timing Parameter 0 Register (Default Value: 0x0000_0000)

Offset:0x1030			Register Name: DPHY_RX_TIME0_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	LPRX_TO The timeout counter for receiver in LP mode.
7:1	/	/	/
0	R/W	0x0	LPRX_TO_EN LP RX timeout enable 0: Disable 1: Enable

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15.3.7.9 0x1034 DPHY RX Timing Parameter 1 Register (Default Value: 0x0000_0000)

Offset:0x1034			Register Name: DPHY_RX_TIME1_REG
Bit	Read/Write	Default/Hex	Description
31:20	R/W	0x0	RX_DLY The time that DPHY process HS FIFO data and pull off data valid signal after DPHY has detected the lane state from HS to LP. It should be more than 8 byte clock cycle time.
19:0	R/W	0x0	LPRX_ULPS_WP When receivers detect MARK-I for ULPS Wakeup Cycles, the corresponding ULPS WP is pending. (Valid for clock lane and all data lane)

15.3.7.10 0x1040 DPHY RX Timing Parameter 3 Register (Default Value: 0x0000_0000)

Offset:0x1040			Register Name: DPHY_RX_TIME3_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	LPRST_DLY The added delay time that reset the HS-FIFO after the DPHY has detected the HS-to-LP.
15:0	/	/	/

15.3.7.11 0x104C DPHY Analog 0 Register (Default Value: 0x0000_0044)

Offset:0x104C			Register Name: DPHY_ANA0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	DSI_REG_PREEMPH0 Data Lane0 HSTX Preemphasis Tuning xx00: 0dB xx01: 1.42dB xx10: 3.24dB xx11: 4.8dB Note: Bit23/22 floating.
19:16	R/W	0x0	DSI_REG_PREEMPH1 Data Lane1 HSTX Preemphasis Tuning. xx00: 0dB xx01: 1.42dB xx10: 3.24dB xx11: 4.8dB Note: Bit19/18 floating.

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Offset:0x104C			Register Name: DPHY_ANA0_REG
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0x0	DSI_REG_PREEMPH2 Data Lane2 HSTX Preemphasis Tuning. xx00: 0dB xx01: 1.42dB xx10: 3.24dB xx11: 4.8dB Note: Bit15/14 floating.
11:8	R/W	0x0	DSI_REG_PREEMPH3 Data Lane3 HSTX Preemphasis Tuning. xx00: 0dB xx01: 1.42dB xx10: 3.24dB xx11: 4.8dB Note: Bit11/10 floating.
7	/	/	/
6:4	R/W	0x4	DSI_REG_LPTX_SETC LPTX Current Bias Tuning. 000: 7.14uA 001: 7.69uA 010: 8.33uA 011: 9.09uA 100: 10.0uA 101: 11.1uA 110: 12.5uA 111: 14.3uA
3	/	/	/
2:0	R/W	0x4	DSI_REG_LPTX_SETR LPTX Output Resistance Tuning 000: 118ohm 001: 105ohm 010: 94ohm 011: 85ohm 100: 79ohm 101: 73ohm 110: 68ohm 111: 63ohm

15.3.7.12 0x1050 DPHY Analog 1 Register (Default Value: 0x0000_0000)

Offset:0x1050			Register Name: DPHY_ANA1_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x1050			Register Name: DPHY_ANA1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	ENVTT_MODE VTT mode select.
30:27	/	/	/
26	R/W	0x0	DSI_EN_PHBACK_LOCK Back sync clk lock enable.
25	R/W	0x0	DSI_EN_PHBACK_SYNC Back sync clk enable.
24	R/W	0x0	DSI_EN_SOFT_PHSEL Resister control enable for sync clk.
23:22	R/W	0x0	DSI_REG_SOFT_PHSEL Resister tuning for sync clk. 00: 0° 01: 90° 10: 180° 11: 270°
21:20	R/W	0x0	DSI_REG_PHSYNC_SET digital clock phase set for parallel data. 00: 0° 01: 90° 10: 180° 11: 270°
19:16	R/W	0x0	DSI_EN_CLKPATD HSTX clk pattern enable for data lane0 to lane3.
15	/	/	/
14:11	R/W	0x0	DSI_EN_VTT_TEST HSTX VTT test enable for data lane0 to lane3.
10	R/W	0x0	DSI_EN_VTT_TESTC HSTX VTT test enable for clk lane.
9:6	R/W	0x0	DSI_REG_HSTX_PLR Data lane0 to lane3 HS TX polar reverse.
5	R/W	0x0	DSI_REG_HSTXCK_PLR Clk lane HS TX polar reverse.
4:1	R/W	0x0	DSI_EN_VLPTX_TEST Data lane0 to lane3 LPTX analog point test enable, only use bit0.
0	R/W	0x0	DSI_EN_VLPTX_TESTC Clk lane LPTX analog point test enable.

15.3.7.13 0x1054 DPHY Analog 2 Register (Default Value: 0x0000_0000)

Offset:0x1054	Register Name: DPHY_ANA2_REG

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Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:24	R/W	0x0	EN_P2S Enable signal for HS p2s module in Data Lane<3:0>
23:20	R/W	0x0	EN_RX Debug signal, HS RX module data lane enable Only EN_RX [0] could output to debug.
19:16	R/W	0x0	EN_TERM Debug signal, HS RX module terminal resistance enable in data lane. Only EN_TREM[0] could output to debug.
15	/	/	/
14	R/W	0x0	EN_RXC Debug signal, HS RX module clock lane enable
13	R/W	0x0	EN_TERMC Debug signal, HS RX module terminal resistance enable in clock lane.
12	/	/	/
11:8	R/W	0x0	EN_TXTERM Enable signal for HSTX termination in Data Lane<3:0>
7	/	/	/
6	R/W	0x0	EN_CLKQ Only when ANA_CPU_EN = 1, this bit will be valid. 1: Combophy output enable 0: Combophy output disable, that is the differential pair of the output clock is 0.
5	R/W	0x0	EN_TXTREMC Enable signal for HSTX termination in Clock Lane
4	R/W	0x0	EN_CLK clock module enable.
3:2	/	/	/
1	R/W	0x0	EN_IREF LVDS clk lane test clock select
0	R/W	0x0	ANA_CPU_EN Enable all register with CPU, when disable, those bits are controlled by circuit automatically.

15.3.7.14 0x1058 DPHY Analog 3 Register (Default Value: 0x0000_0000)

Offset:0x1058			Register Name: DPHY_ANA3_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x1058			Register Name: DPHY_ANA3_REG
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EN_VDRVLDO_ON Enable signal for HS TX LDO in Data Lane<3:0> Set as "1111" when HSTX
27	R/W	0x0	EN_VDRVLDO_ONC Enable signal for HS TX LDO in Clock Lane Set as "1" when HSTX
26	R/W	0x0	EN_DIV2 Enable signal for input clock divided by 2. Set as "1"
25	R/W	0x0	EN_LVCK Enable signal for clock power LDO is active DSI set as "1";CSI set as "0"
24	R/W	0x0	EN_LV Enable signal for data power LDO is active Set as "1"
23	R/W	0x0	EN_ULPSC CLK Lane LP contention detect threshold tuning.
22:19	R/W	0x0	EN_ULPS Data Lane LP contention detect threshold tuning.
18	R/W	0x0	EN_VREF Enable signal for reference power LDO is active Set as "1"
17:15	/	/	/
14	R/W	0x0	EN_LPCDC Enable signal for LP cd module is active in Clock Lane
13	R/W	0x0	EN_LPTXC Enable signal for LP TX module is active in Clock Lane
12	R/W	0x0	EN_LPRXC Enable signal for LP RX module is active in Clock Lane
11:8	R/W	0x0	EN_LPCD Enable signal for LP CD module is active in Data Lane<3:0>
7:4	R/W	0x0	EN_LPRX Enable signal for LP RX module is active in Data Lane<3:0>
3:0	R/W	0x0	EN_LPTX Enable signal for LP TX module is active in Data Lane<3:0>

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15.3.7.15 0x105C DPHY Analog 4 Register (Default Value: 0x04B3_3408)

Offset:0x105C			Register Name: DPHY_ANA4_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DSI_EN_MIPI MIPI DSI Enable
30	R/W	0x0	DSI_EN_COMTEST Common Test Enable
29:28	R/W	0x0	DSI_REG_COMTEST Common Test Configuration 00: vp 01: vpck 10: vres 11: ib5u_test
27	/	/	/
26:24	R/W	0x4	DSI_REG_IB Common Ibias Tuning 000: 2.500uA 001: 3.125uA 010: 3.750uA 011: 4.375uA 100: 5.000uA 101: 5.625uA 110: 6.250uA 111: 6.875uA
23	R/W	0x1	REG_VREF_SOURCE COMMON Vbias Tuning 0: 1.2v (from anabias) 1: 1.8v (from vph)
22:20	R/W	0x3	DSI_REG_VRES_SET Vref0p2 Tuning 000: 170mV 001: 180mV 010: 190mV 011: 200mV 100: 210mV 101: 220mV 110: 230mV 111: 240mV
19	/	/	/

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Offset:0x105C			Register Name: DPHY_ANA4_REG
Bit	Read/Write	Default/Hex	Description
18:16	R/W	0x3	DSI_REG_VTT_SET Vref0p4 Tuning 000: 380mV 001: 400mV 010: 430mV 011: 460mV 100: 500mV 101: 520mV 110: 550mV 111: 580mV
15	/	/	/
14:12	R/W	0x3	DSI_REG_VLPTX_SET LPtx Reference Voltage Tuning 000: 1.02V 001: 1.10V 010: 1.15V 011: 1.20V 100: 1.25V 101: 1.30V 110: 1.35V 111: 1.40V
11	/	/	/
10:8	R/W	0x4	DSI_REG_VLV_SET Vref0p9 Tuning 000: 860mV 001: 880mV 010: 900mV 011: 920mV 100: 940mV 101: 960mV 110: 980mV 111: 1V
7:4	/	/	/
3:0	R/W	0x8	Reserved

15.3.7.16 0x1060 DPHY Interrupt Enable 0 Register (Default Value: 0x0000_0000)

Offset:0x1060			Register Name: DPHY_INT_EN0_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CD_LP1_ERR_D3_INT Contention Detection detects LP1 error interrupt

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Offset:0x1060			Register Name: DPHY_INT_EN0_REG
Bit	Read/Write	Default/Hex	Description
30	R/W	0x0	CD_LP0_ERR_D3_INT Contention Detection detects LP0 error interrupt
29	R/W	0x0	CD_LP1_ERR_D2_INT Contention Detection detects LP1 error interrupt
28	R/W	0x0	CD_LP0_ERR_D2_INT Contention Detection detects LP0 error interrupt
27	R/W	0x0	CD_LP1_ERR_D1_INT Contention Detection detects LP1 error interrupt
26	R/W	0x0	CD_LP0_ERR_D1_INT Contention Detection detects LP0 error interrupt
25	R/W	0x0	CD_LP1_ERR_D0_INT Contention Detection detects LP1 error interrupt
24	R/W	0x0	CD_LP0_ERR_D0_INT Contention Detection detects LP0 error interrupt
23	R/W	0x0	CD_LP1_ERR_CLK_INT Contention Detection detects LP1 error interrupt
22	R/W	0x0	CD_LP0_ERR_CLK_INT Contention Detection detects LP0 error interrupt
21:16	/	/	/
15	R/W	0x0	RX_ALG_ERR_D3_INT The payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN
14	R/W	0x0	RX_ALG_ERR_D2_INT The payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN
13	R/W	0x0	RX_ALG_ERR_D1_INT The payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN
12	R/W	0x0	RX_ALG_ERR_D0_INT The payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN.
11:0	/	/	/

15.3.7.17 0x1064 DPHY Interrupt Enable 1 Register (Default Value: 0x0000_0000)

Offset:0x1064			Register Name: DPHY_INT_EN1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	FALSE_CTL_D3_INT A false control occurs at Data Lane 3 interrupt.
30	R/W	0x0	FALSE_CTL_D2_INT A false control occurs at Data Lane 2 interrupt.

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Offset:0x1064			Register Name: DPHY_INT_EN1_REG
Bit	Read/Write	Default/Hex	Description
29	R/W	0x0	FALSE_CTL_D1_INT A false control occurs at Data Lane 1 interrupt.
28	R/W	0x0	FALSE_CTL_D0_INT A false control occurs at Data Lane 0 interrupt.
27	R/W	0x0	ESC_CMD_ERR_D3_INT DPHY RX received cannot recognize the Escape Entry Command at Data Lane 3 interrupt.
26	R/W	0x0	ESC_CMD_ERR_D2_INT DPHY RX received cannot recognize the Escape Entry Command at Data Lane 2 interrupt.
25	R/W	0x0	ESC_CMD_ERR_D1_INT DPHY RX received cannot recognize the Escape Entry Command at Data Lane 1 interrupt.
24	R/W	0x0	ESC_CMD_ERR_D0_INT DPHY RX received cannot recognize the Escape Entry Command at Data Lane 0 interrupt.
23	R/W	0x0	RST_D3_INT DPHY RX detected Reset Trigger sequence at Data Lane 3 interrupt.
22	R/W	0x0	RST_D2_INT DPHY RX detected Reset Trigger sequence at Data Lane 2 interrupt.
21	R/W	0x0	RST_D1_INT DPHY RX detected Reset Trigger sequence at Data Lane 1 interrupt.
20	R/W	0x0	RST_D0_INT DPHY RX detected Reset Trigger sequence at Data Lane 0 interrupt.
19	R/W	0x0	UNDEF5_D0_INT DPHY RX received undefined 5 sequence at Data Lane 0 interrupt.
18	R/W	0x0	UNDEF4_D0_INT DPHY RX received undefined 4 sequence at Data Lane 0 interrupt.
17	R/W	0x0	UNDEF3_D0_INT DPHY RX received undefined 3 sequence at Data Lane 0 interrupt.
16	R/W	0x0	UNDEF2_D0_INT DPHY RX received undefined 2 sequence at Data Lane 0 interrupt.

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Offset:0x1064			Register Name: DPHY_INT_EN1_REG
Bit	Read/Write	Default/Hex	Description
15	R/W	0x0	UNDEF1_D0_INT DPHY RX received undefined 1 sequence at Data Lane 0 interrupt.
14	R/W	0x0	TX_TRND_ERR_D0_INT DPHY TX turn around error at Data Lane 0 interrupt.
13	R/W	0x0	RX_TRND_D0_INT DPHY RX received turn around sequence at Data Lane 0 interrupt.
12	R/W	0x0	LPDT_D0_INT DPHY RX complete LPDT transfer at Data Lane 0 interrupt.
11:10	/	/	/
9	R/W	0x0	ULPS_WP_CLK_INT DPHY RX detected ULPS wakeup at Clock Lane interrupt.
8	R/W	0x0	ULPS_CLK_INT DPHY RX detected ULPS sequence at Clock Lane interrupt.
7	R/W	0x0	ULPS_WP_D3_INT DPHY RX detected ULPS wakeup at Data Lane 3 interrupt.
6	R/W	0x0	ULPS_WP_D2_INT DPHY RX detected ULPS wakeup at Data Lane 2 interrupt.
5	R/W	0x0	ULPS_WP_D1_INT DPHY RX detected ULPS wakeup at Data Lane 1 interrupt.
4	R/W	0x0	ULPS_WP_D0_INT DPHY RX detected ULPS wakeup at Data Lane 0 interrupt.
3	R/W	0x0	ULPS_D3_INT DPHY RX detected ULPS sequence at Data Lane 3 interrupt.
2	R/W	0x0	ULPS_D2_INT DPHY RX detected ULPS sequence at Data Lane 2 interrupt.
1	R/W	0x0	ULPS_D1_INT DPHY RX detected ULPS sequence at Data Lane 1 interrupt.

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Offset:0x1064			Register Name: DPHY_INT_EN1_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	ULPS_D0_INT DPHY RX detected ULPS sequence at Data Lane 0 interrupt.

15.3.7.18 0x1068 DPHY Interrupt Enable 2 Register (Default Value: 0x0000_0000)

Offset:0x1068			Register Name: DPHY_INT_EN2_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	DISPLL_UNLOCK_INT 0: Disable 1: Enable DISPLL detected lock signal unstable interrupt.

15.3.7.19 0x1070 DPHY Interrupt Pending 0 Register (Default Value: 0x0000_0000)

Offset:0x1070			Register Name: DPHY_INT_PD0_REG
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	CD_LP1_ERR_D3_PD Contention Detection detects LP1 error interrupt pending.
30	R/W1C	0x0	CD_LP0_ERR_D3_PD Contention Detection detects LP0 error interrupt pending.
29	R/W1C	0x0	CD_LP1_ERR_D2_PD Contention Detection detects LP1 error interrupt pending.
28	R/W1C	0x0	CD_LP0_ERR_D2_PD Contention Detection detects LP0 error interrupt pending.
27	R/W1C	0x0	CD_LP1_ERR_D0_PD Contention Detection detects LP1 error interrupt pending.
26	R/W1C	0x0	CD_LP0_ERR_D0_PD Contention Detection detects LP0 error interrupt pending.
25	R/W1C	0x0	CD_LP1_ERR_D1_PD Contention Detection detects LP1 error interrupt pending.

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Offset:0x1070			Register Name: DPHY_INT_PD0_REG
Bit	Read/Write	Default/Hex	Description
24	R/W1C	0x0	CD_LP0_ERR_D1_PD Contention Detection detects LP0 error interrupt pending.
23	R/W1C	0x0	CD_LP1_ERR_CLK_PD Contention Detection detects LP1 error interrupt pending.
22	R/W1C	0x0	CD_LP0_ERR_CLK_PD Contention Detection detects LP0 error interrupt pending.
21:16	/	/	/
15	R/W1C	0x0	RX_ALG_ERR_D3_PD Asserted if the payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN.
14	R/W1C	0x0	RX_ALG_ERR_D2_PD Asserted if the payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN.
13	R/W1C	0x0	RX_ALG_ERR_D1_PD Asserted if the payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN.
12	R/W1C	0x0	RX_ALG_ERR_D0_PD Asserted if the payload that DPHY RX received is not aligned to 8bit or 9bit, which depends on HSRX_8B9B_EN.
11:0	/	/	/

15.3.7.20 0x1074 DPHY Interrupt Pending 1 Register (Default Value: 0x0000_0000)

Offset:0x1074			Register Name: DPHY_INT_PD1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	FALSE_CTL_D3_PD Asserted if a LP-Rqst (LP-10) is not followed by the remainder of a valid Escape sequence or a HS-Rqst (LP-01) is not correctly followed by a Bridge State (LP-00).

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Offset:0x1074			Register Name: DPHY_INT_PD1_REG
Bit	Read/Write	Default/Hex	Description
30	R/W1C	0x0	FALSE_CTL_D2_PD Asserted if a LP-Rqst (LP-10) is not followed by the remainder of a valid Escape sequence or a HS-Rqst (LP-01) is not correctly followed by a Bridge State (LP-00).
29	R/W1C	0x0	FALSE_CTL_D1_PD Asserted if a LP-Rqst (LP-10) is not followed by the remainder of a valid Escape sequence or a HS-Rqst (LP-01) is not correctly followed by a Bridge State (LP-00).
28	R/W1C	0x0	FALSE_CTL_D0_PD Asserted if a LP-Rqst (LP-10) is not followed by the remainder of a valid Escape sequence or a HS-Rqst (LP-01) is not correctly followed by a Bridge State (LP-00).
27	R/W1C	0x0	ESC_CMD_ERR_D3_PD Asserted if DPHY RX received cannot recognize the Escape Entry Command at Data Lane 3.
26	R/W1C	0x0	ESC_CMD_ERR_D2_PD Asserted if DPHY RX received cannot recognize the Escape Entry Command at Data Lane 2.
25	R/W1C	0x0	ESC_CMD_ERR_D1_PD Asserted if DPHY RX received cannot recognize the Escape Entry Command at Data Lane 1.
24	R/W1C	0x0	ESC_CMD_ERR_D0_PD Asserted if DPHY RX received cannot recognize the Escape Entry Command at Data Lane 0.
23	R/W1C	0x0	RST_D3_PD Asserted if DPHY RX detected Reset Trigger sequence at Data Lane3.
22	R/W1C	0x0	RST_D2_PD Asserted if DPHY RX detected Reset Trigger sequence at Data Lane2.
21	R/W1C	0x0	RST_D1_PD Asserted if DPHY RX detected Reset Trigger sequence at Data Lane1.
20	R/W1C	0x0	RST_D0_PD Asserted if DPHY RX detected Reset Trigger sequence at Data Lane0.

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Offset:0x1074			Register Name: DPHY_INT_PD1_REG
Bit	Read/Write	Default/Hex	Description
19	R/W1C	0x0	UNDEF5_D0_PD Asserted if DPHY RX received undefined 5 sequence at Data Lane0. "10100000"
18	R/W1C	0x0	UNDEF4_D0_PD Asserted if DPHY RX received undefined 4 sequence at Data Lane0. "00100001"
17	R/W1C	0x0	UNDEF3_D0_PD Asserted if DPHY RX received undefined 3 sequence at Data Lane0. "01011101"
16	R/W1C	0x0	UNDEF2_D0_PD Asserted if DPHY RX received undefined 2 sequence at Data Lane0. "11011110"
15	R/W1C	0x0	UNDEF1_D0_PD Asserted if DPHY RX received undefined 1 sequence at Data Lane0. "10011111"
14	R/W1C	0x0	TX_TRND_ERR_D0_PD Asserted if DPHY TX turn around error at Data Lane 0.
13	R/W1C	0x0	RX_TRND_D0_PD Asserted if DPHY RX received turn around sequence at Data Lane0.
12	R/W1C	0x0	LPDT_D0_PD Asserted if DPHY RX complete LPDT transfer at Data Lane 0. "11100001"
11:10	/	/	/
9	R/W1C	0x0	ULPS_WP_CLK_PD Asserted if DPHY RX detected ULPS wakeup at Clock Lane.
8	R/W1C	0x0	ULPS_CLK_PD Asserted if DPHY RX detected ULPS sequence at Clock Lane.
7	R/W1C	0x0	ULPS_WP_D3_PD Asserted if DPHY RX detected ULPS wakeup at Data Lane 3.

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Offset:0x1074			Register Name: DPHY_INT_PD1_REG
Bit	Read/Write	Default/Hex	Description
6	R/W1C	0x0	ULPS_WP_D2_PD Asserted if DPHY RX detected ULPS wakeup at Data Lane 2.
5	R/W1C	0x0	ULPS_WP_D1_PD Asserted if DPHY RX detected ULPS wakeup at Data Lane 1.
4	R/W1C	0x0	ULPS_WP_D0_PD Asserted if DPHY RX detected ULPS wakeup at Data Lane 0.
3	R/W1C	0x0	ULPS_D3_PD Asserted if DPHY RX detected ULPS sequence at Data Lane 3.
2	R/W1C	0x0	ULPS_D2_PD Asserted if DPHY RX detected ULPS sequence at Data Lane 2.
1	R/W1C	0x0	ULPS_D1_PD Asserted if DPHY RX detected ULPS sequence at Data Lane 1.
0	R/W1C	0x0	ULPS_D0_PD Asserted if DPHY RX detected ULPS sequence at Data Lane 0. "00011110"

15.3.7.21 0x1078 DPHY Interrupt Pending 2 Register (Default Value: 0x0000_0000)

Offset:0x1078			Register Name: DPHY_INT_PD2_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W1C	0x0	DISPLL_UNLOCK_PD DISPLL detects lock unstable interrupt pending.

15.3.7.22 0x10E0 DPHY Debug 0 Register (Default Value: 0x0000_0000)

Offset: 0x10E0			Register Name: DPHY_DBG0_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R	0x0	DIR Direction 0: No lane switch into HS transmission 1: At least one lane switch into HS transmission

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Offset: 0x10E0			Register Name: DPHY_DBG0_REG
Bit	Read/Write	Default/Hex	Description
27	/	/	/
26	R	0x0	LOCK 0: Unlocked 1: Locked(It indicates that the PLL has been stable)
25:19	/	/	/
18:16	R	0x0	LPTX_STA_CLK LPTX Status Clock 000: STP_ST 001: HSRQ_ST 010: ESRQ_ST 011: TNRD_ST 100: ULPS_ST 101: HS_ST 110: ESP_ST 111: RX_ST
15	/	/	/
14:12	R	0x0	LPTX_STA_D3 LPTX Status D3 000: STP_ST 001: HSRQ_ST 010: ESRQ_ST 011: TNRD_ST 100: ULPS_ST 101: HS_ST 110: ESP_ST 111: RX_ST
11	/	/	/
10:8	R	0x0	LPTX_STA_D2 LPTX Status D2 000: STP_ST 001: HSRQ_ST 010: ESRQ_ST 011: TNRD_ST 100: ULPS_ST 101: HS_ST 110: ESP_ST 111: RX_ST
7	/	/	/

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Offset: 0x10E0			Register Name: DPHY_DBG0_REG
Bit	Read/Write	Default/Hex	Description
6:4	R	0x0	LPTX_STA_D1 LPTX Status D1 000: STP_ST 001: HSRQ_ST 010: ESRQ_ST 011: TNRD_ST 100: ULPS_ST 101: HS_ST 110: ESP_ST 111: RX_ST
3	/	/	/
2:0	R	0x0	LPTX_STA_D0 LPTX Status D0 000: STP_ST 001: HSRQ_ST 010: ESRQ_ST 011: TNRD_ST 100: ULPS_ST 101: HS_ST 110: ESP_ST 111: RX_ST

15.3.7.23 0x10E4 DPHY Debug 1 Register (Default Value: 0x0000_0000)

Offset: 0x10E4			Register Name: DPHY_DBG1_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DIV_DBG_EN
30	/	/	/
29:28	R/W	0x0	LOCK_STEP
27:22	/	/	/
21	R/W	0x0	LPC_SET
20	R/W	0x0	TURN_DISMODEC
19	R/W	0x0	TURN_DISMODE3
18	R/W	0x0	TURN_DISMODE2
17	R/W	0x0	TURN_DISMODE1
16	R/W	0x0	TURN_DISMODE0
15:14	/	/	/

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Offset: 0x10E4			Register Name: DPHY_DBG1_REG
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x0	LPTX_SET_CK Clock Lane output state as follow: 00: LP-00 01: LP-01 10: LP-10 11: LP-11
11:10	R/W	0x0	LPTX_SET_D3 Data Lane 3 output state as follow: 00: LP-00 01: LP-01 10: LP-10 11: LP-11
9:8	R/W	0x0	LPTX_SET_D2 Data Lane 2 output state as follow: 00: LP-00 01: LP-01 10: LP-10 11: LP-11
7:6	R/W	0x0	LPTX_SET_D1 Data Lane 1 output state as follow: 00: LP-00 01: LP-01 10: LP-10 11: LP-11
5:4	R/W	0x0	LPTX_SET_D0 Data Lane 0 output state as follow: 00: LP-00 01: LP-01 10: LP-10 11: LP-11
3	/	/	/
2	R/W	0x0	DSI_PRBS_EN
1	R/W	0x0	HSTX_DBG_EN HSTX Debug Enable 0: Normal mode 1: Debug mode
0	R/W	0x0	LPTX_DBG_EN LPTX Debug Enable 0: Normal mode 1: Debug mode

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15.3.7.24 0x10E8 DPHY Debug 2 Register (Default Value: 0x0000_0000)

Offset: 0x10E8			Register Name: DPHY_DBG2_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	HSTX_Data Bit31-24: D3_DATA Bit23-16: D2_DATA Bit15-08: D1_DATA Bit07-00: D0_DATA

15.3.7.25 0x10EC DPHY Debug 3 Register (Default Value: 0x0000_0000)

Offset:0x10EC			Register Name: DPHY_DBG3_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:16	R	0x0	LPRX_STA_CLK LPRX Status Clock 0000: LP00S 0001: LP01S 0010: LP10S 0011: LP11S 0100: TRNDS 0101: HS_S 0110: LPDTS 0111: ESCPS 1000: ULPSS 1001: RSETS 1010: UDF1S 1011: UDF2S 1100: UDF3S 1101: UDF4S 1110: UDF5S Others: Reserved

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Offset:0x10EC			Register Name: DPHY_DBG3_REG
Bit	Read/Write	Default/Hex	Description
15:12	R	0x0	LPRX_STA_D3 LPRX Status D3 0000: LP00S 0001: LP01S 0010: LP10S 0011: LP11S 0100: TRNDS 0101: HS_S 0110: LPDTS 0111: ESCPS 1000: ULPSS 1001: RSETS 1010: UDF1S 1011: UDF2S 1100: UDF3S 1101: UDF4S 1110: UDF5S Others: Reserved
11:8	R	0x0	LPRX_STA_D2 LPRX Status D2 0000: LP00S 0001: LP01S 0010: LP10S 0011: LP11S 0100: TRNDS 0101: HS_S 0110: LPDTS 0111: ESCPS 1000: ULPSS 1001: RSETS 1010: UDF1S 1011: UDF2S 1100: UDF3S 1101: UDF4S 1110: UDF5S Others: Reserved

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Offset:0x10EC			Register Name: DPHY_DBG3_REG
Bit	Read/Write	Default/Hex	Description
7:4	R	0x0	LPRX_STA_D1 LPRX Status D1 0000: LP00S 0001: LP01S 0010: LP10S 0011: LP11S 0100: TRNDS 0101: HS_S 0110: LPDTS 0111: ESCPS 1000: ULPSS 1001: RSETS 1010: UDF1S 1011: UDF2S 1100: UDF3S 1101: UDF4S 1110: UDF5S Others: Reserved
3:0	R	0x0	LPRX_STA_D0 LPRX Status D0 0000: LP00S 0001: LP01S 0010: LP10S 0011: LP11S 0100: TRNDS 0101: HS_S 0110: LPDTS 0111: ESCPS 1000: ULPSS 1001: RSETS 1010: UDF1S 1011: UDF2S 1100: UDF3S 1101: UDF4S 1110: UDF5S Others: Reserved

15.3.7.26 0x10F0 DPHY Debug 4 Register (Default Value: 0x0000_0000)

Offset:0x10F0			Register Name: DPHY_DBG4_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x10F0			Register Name: DPHY_DBG4_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R	0x0	LPCD_PHY_CLK LPCD PHY Clock 00: LP-00 01: LP-01 10: LP-10 11: LP-11
23:22	R	0x0	LPCD_PHY_D3 LPCD PHY D3 00: LP-00 01: LP-01 10: LP-10 11: LP-11
21:20	R	0x0	LPCD_PHY_D2 LPCD PHY D2 00: LP-00 01: LP-01 10: LP-10 11: LP-11
19:18	R	0x0	LPCD_PHY_D1 LPCD PHY D1 00: LP-00 01: LP-01 10: LP-10 11: LP-11
17:16	R	0x0	LPCD_PHY_D0 LPCD PHY D0 00: LP-00 01: LP-01 10: LP-10 11: LP-11
15:10	/	/	/
9:8	R	0x0	LPRX_PHY_CLK LPRX PHY Clock 00: LP-00 01: LP-01 10: LP-10 11: LP-11

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Offset:0x10F0			Register Name: DPHY_DBG4_REG
Bit	Read/Write	Default/Hex	Description
7:6	R	0x0	LPRX_PHY_D3 LPRX PHY D3 00: LP-00 01: LP-01 10: LP-10 11: LP-11
5:4	R	0x0	LPRX_PHY_D2 LPRX PHY D2 00: LP-00 01: LP-01 10: LP-10 11: LP-11
3:2	R	0x0	LPRX_PHY_D1 LPRX PHY D1 00: LP-00 01: LP-01 10: LP-10 11: LP-11
1:0	R	0x0	LPRX_PHY_D0 LPRX PHY D0 00: LP-00 01: LP-01 10: LP-10 11: LP-11

15.3.7.27 0x10F4 DPHY Debug 5 Register (Default Value: 0x0000_0000)

Offset:0x10F4			Register Name: DPHY_DBG5_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HSRX_DATA Bit31-24: LANE_D3 Bit23-16: LANE_D2 Bit15-08: LANE_D1 Bit07-00: LANE_D0

15.3.7.28 0x10F8 DPHY TX Skew Register0 (Default Value: 0x0000_0000)

Offset:0x10F8			Register Name: DPHY_TX_SKEW_REG0
Bit	Read/Write	Default/Hex	Description

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Offset:0x10F8			Register Name: DPHY_TX_SKEW_REG0
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	SKEWCAL_ZERO_SET HS-Zero duration in Skew Calibration mode, unit: byteclk
23:16	R/W	0x0	SKEWCAL_TRAIL_SET HS-Trail duration in Skew Calibration mode, unit: byteclk
15:8	R/W	0x0	REG_SKEWCAL Clock Pattern Content : 8'b0101_0101 / 8'b1010_1010
7:0	R/W	0x0	REG_SKEWCAL_SYNC Skewcal sync Pattern Content : 8'b1111_1111

15.3.7.29 0x10FC DPHY TX Skew Register1 (Default Value: 0x0000_0000)

Offset:0x10FC			Register Name: DPHY_TX_SKEW_REG1
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	SKEWCAL_SYNC_SET SkewCal-Sync Duration, unit: byteclk.
23:16	R/W	0x0	SKEWCAL_PEDIC_SET Periodic Mode SkewCal Duration, unit: byteclk.
15:0	R/W	0x0	SKEWCAL_INIT_SET Initial mode SkewCal Duration, unit: byteclk.

15.3.7.30 0x1100 DPHY TX Skew Register2 (Default Value: 0x0000_0000)

Offset:0x1100			Register Name: DPHY_TX_SKEW_REG2
Bit	Read/Write	Default/Hex	Description
10	R/W	0x0	EN_SKEWCAL_INIT Enable Skewcal Init
9	R/W	0x0	EN_SKEWCAL_PERDIC Enable Skewcal Perdic
8	R/W	0x0	SKEWCAL_TRAIL_INV Skewcal Trail Inv
7:0	R/W	0x0	SKEWCAL_PREPARE_LP00 LP-00 duration, unit: byteclk.

15.3.7.31 0x1104 DPHY PLL Register0 (Default Value: 0x0F60_6330)

Offset:0x1104			Register Name: DPHY_PLL_REG0
Bit	Read/Write	Default/Hex	Description

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Offset:0x1104			Register Name: DPHY_PLL_REG0
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	REG_UPDATE write q update the cfg to pll,auto clear
30	R/W	0x0	Reserved
29:28	R/W	0x0	POST_DIV0_CLK_LS M2, post divider0= 1/2/3/4
27:24	R/W	0xf	POST_DIV1_CLK_LS m3, post divider1= m3+1
23	R/W	0x0	CP36_EN Enable Signal of CP for 3.3v Voltage
22	R/W	0x1	LDO_EN Enable Signal of 0.9V LDO for V &DV
21	R/W	0x1	EN_LVS Enable Gating For l2h Level_Shift
20	R/W	0x0	PLL_EN Enable Signal for PLL
19:16	R/W	0x0	PRE_DIV p, pre divider = p+1
15:8	R/W	0x63	FB_DIV n,feedback divider = n (4,5,8,9,10,12-255)
7	R/W	0x0	N_DET N Detection
6	R/W	0x0	T_DIV T Divider
5:4	R/W	0x3	POST_DIV0_CLK_HS m0, post divider0= 1/2/3/4
3:0	R/W	0x0	POST_DIV1_CLK_HS m1, post divider1= m1+1

15.3.7.32 0x1108 DPHY PLL Register1 (Default Value: 0x0003_0B46)

Offset:0x1108			Register Name: DPHY_PLL_REG1
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R/W	0x0	PLL_CLK_LS_GATING Clock gating. 0: Clock is OFF 1: Clock is ON
20:16	R/W	0x3	PLL_CP Charge Pump Current Adjustment

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Offset:0x1108			Register Name: DPHY_PLL_REG1
Bit	Read/Write	Default/Hex	Description
15:14	R/W	0x0	UNLOCK_MDSEL Unlock Mdsel 00: 21-29 clock cycles 01: 22-28 clock cycles
13	R/W	0x0	LOCKMDSEL Lockmdsel 0: 24-26 clock cycles 1: 23-27 clock cycles
12	R/W	0x0	LOCKDET_EN Lockdet Enable 0: Disable 1: Enable
11:9	R/W	0x5	VSETA LDO AV Voltage Control Signals 000: 0.82 V 001: 0.86 V 010: 0.90 V 011: 0.94 V 100: 0.98 V 101: 1.02 V 110: 1.06 V 111: 1.10 V
8:6	R/W	0x5	VSETD LDO DV Voltage Control Signals 000: 0.82 V 001: 0.86 V 010: 0.90 V 011: 0.94 V 100: 0.98 V 101: 1.02 V 110: 1.06 V 111: 1.10 V
5	R/W	0x0	LPF_SW Charge Pump Current Regulation
4:3	R/W	0x0	ICP_SEL Charge Pump Current Sel & Regulate: 1: Current Regulate 0: Ibias Current 1 for self-bias current

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Offset:0x1108			Register Name: DPHY_PLL_REG1
Bit	Read/Write	Default/Hex	Description
2:1	R/W	0x3	ATEST_SEL Atest Select 00: DV 01: V 10: ICP 11: VC_OD
0	R/W	0x0	TEST_EN Test Enable For Ck_Test & Ana_Test

15.3.7.33 0x110C DPHY PLL Register2 (Default Value: 0x0003_2800)

Offset:0x110C			Register Name: DPHY_PLL_REG2
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SDM_EN Enable For SDM
30	R/W	0x0	FF_EN Enable For Fractional
29	R/W	0x0	SS_EN Enable For SSC
28:20	R/W	0x0	SS_FRAC Fractional Part For SSC
19:12	R/W	0x32	SS_INT Integer Part For SSC
11:0	R/W	0x800	FRAC Registers For Fractional

15.3.7.34 0x1110 COMBO PHY Register0 (Default Value: 0x0000_1F00)

Offset:0x1110			Register Name: COMBO_PHY_REG0
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x1	Reserved
11:8	R/W	0xf	Reserved
7:4	/	/	/
3	R/W	0x0	EN_MIPI MIPI PHY Enable
2	R/W	0x0	EN_LVDS LVDS PHY Enable
1	/	/	/

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Offset:0x1110			Register Name: COMBO_PHY_REG0
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	EN_CP MIPI LDO test enable

15.3.7.35 0x1114 COMBO PHY Register1 (Default Value: 0x0000_0053)

Offset:0x1114			Register Name: COMBO_PHY_REG1
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x5	DSI_REG_C Adjust current flowing through Rload of Rx to change the differential signals amplitude. 000: 216 mV 001: 252 mV 010: 276 mV 011: 312 mV 100: 336 mV 101: 372 mV 110: 395 mV 111: 432 mV
3	/	/	/
2:0	R/W	0x3	DSI_REG_R Adjust current flowing through Rload of Rx to change the common signals amplitude. 000: 0.925 V 001: 0.950 V 010: 0.975 V 011: 1.000 V 100: 1.025 V 101: 1.050 V 110: 1.075 V 111: 1.100 V

15.3.7.36 0x1124 DPHY PLL Register3 (Default Value: 0x8000_0000)

Offset:0x1124			Register Name: DPHY_PLL_REG3
Bit	Read/Write	Default/Hex	Description
31	R/W	0x1	SSC RSTN Back up
30	R/W	0x0	SSC_CLK_MUX SSC CLK MUX Option 1: REF_CLK 0: PLL_CLK_SDM

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Offset:0x1124			Register Name: DPHY_PLL_REG3
Bit	Read/Write	Default/Hex	Description
29:0	/	/	/

15.3.7.37 0x1130 DPHY PLL Patten0 Control Register (Default Value:0x0000_0000)

Offset:0x1130			Register Name: DPHY_PLL_PAT0_CTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	SIG_DELT_PAT_EN Sigma Delta Pattern Enable.
30:29	R/W	0x0	SPR_FREQ_MODE Spread Frequency Mode. 00: DC=0 01: DC=1 10: Triangular (1 bit) 11: Triangular(n bit)
28:17	R/W	0x0	WAVE_STEP Wave STEP
16:0	R/W	0x0	WAVE_BOT Wave Bottom

15.3.7.38 0x1134 DPHY PLL Patten1 Control Register (Default Value:0x0000_0000)

Offset:0x1134			Register Name: DPHY_PLL_PAT1_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:22	R/W	0x0	SDM_CYCLE sdm_cycle=sdm_fre/clk_sdm/2
21	/	/	/
20	R/W	0x0	SDM_DIRECTION 0: Up 1: Down
19	/	/	/
18	R/W	0x0	DITH
17	R/W	0x0	FRC_EN
16:0	R/W	0x0	FRC_IN

15.3.7.39 0x1138 DPHY PLL SSC Linear Configuration Register (Default Value:0x0CCC_A000)

Offset:0x1138			Register Name: DPHY_PLL_SSC_LIN_CFG_REG
Bit	Read/Write	Default/Hex	Description

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Offset:0x1138			Register Name: DPHY_PLL_SSC_LIN_CFG_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	PLL_MODE 0: Normal Mode 1: Continuously Frequency Sacle
30:29	/	/	/
28:12	R/W	0xCCCA	PLL_SSC SSC amplitude must be an integer multiple of 2^step. spread spectrum amplitude = (SSC amplitude + 2^step) *24/(2^17), unit: MHz.
11:7	/	/	/
6:4	R/W	0x0	PLL_PHASE_COMPENSATE The value of bit[6:4] is based on 24M clock, then the default PLL phase compensate is (3/24000000)s
3:0	R/W	0x0	PLL_STEP. 0000:0.00439MHz/us (576/2^17) 0001:0.00879MHz/us (576/2^16) 0010:0.01758MHz/us (576/2^15) 0011:0.03516MHz/us (576/2^14) 0100:0.07031MHz/us (576/2^13) 0101:0.14062MHz/us (576/2^12) 0110:0.28125MHz/us (576/2^11) 0111:0.56250MHz/us (576/2^10) 1000:1.12500MHz/us (576/2^9) 1001:2.25000MHz/us (576/2^8) 1010:4.50000MHz/us (576/2^7) 1011:9.00000MHz/us (576/2^6) Others:0.00439MHz/us(576/2^17)

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16 Audio Interfaces

16.1 Audio Codec

16.1.1 Overview

The Audio Codec is high-performance audio decoder module which supports Dynamic Range Controller (DRC) and Digital Volume Controller (DVC) functions.

The Audio Codec has the following features:

- One audio differential lineout output: LINEOUTP/N (it can be configured as a single-ended output)
- One audio digital-to-analog converter (DAC) channel
  - 16-bit and 20-bit sample resolution
  - 8 kHz to 192 kHz DAC sample rate
  - 95 ± 3 dB SNR@A-weight, -85 ± 3 dB THD+N
- Supports Dynamic Range Controller adjusting the DAC playback
- TX_FIFO with 128 depths for DAC data transmit
- Programmable FIFO thresholds
- Supports interrupts and DMA

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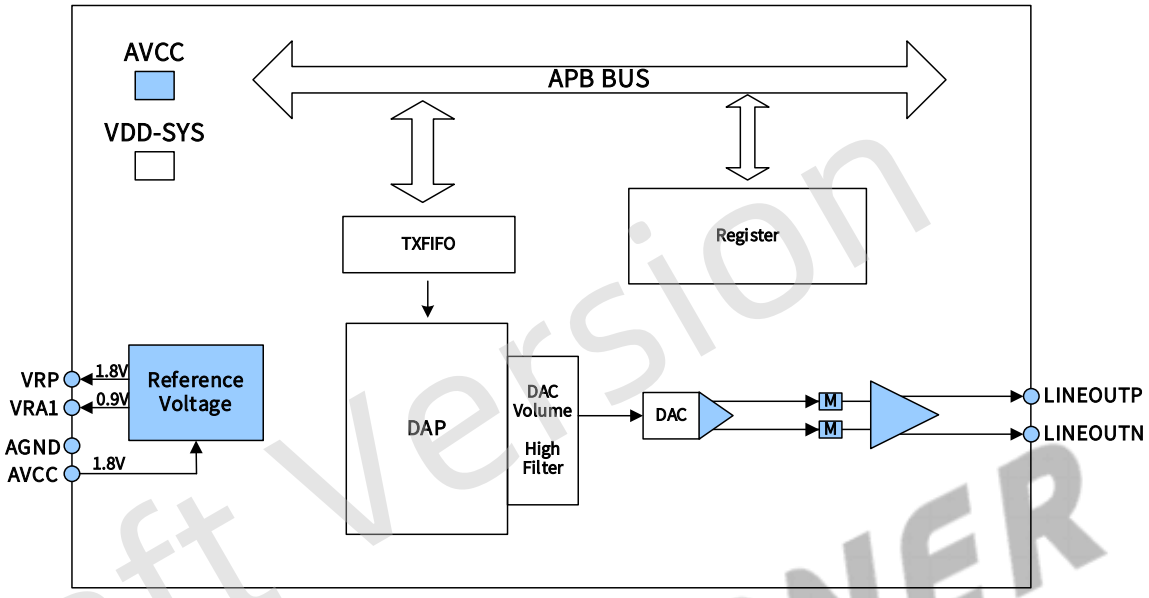
16.1.2 Block Diagram

The following figure shows the block diagram of Audio Codec.

Figure 16-1 Audio Codec Block Diagram



16.1.3 External Signals

Table 16-1 Audio Codec External Signals

Signal Name	Description	Type
LINEOUTN	Lineout Differential Output (Negative)	AO
LINEOUTP	Lineout Differential Output (Positive)	AO
AVCC	Power Supply for Analog Part	P
VRA1	Internal Reference Voltage	AO
AGND	Analog Ground	G

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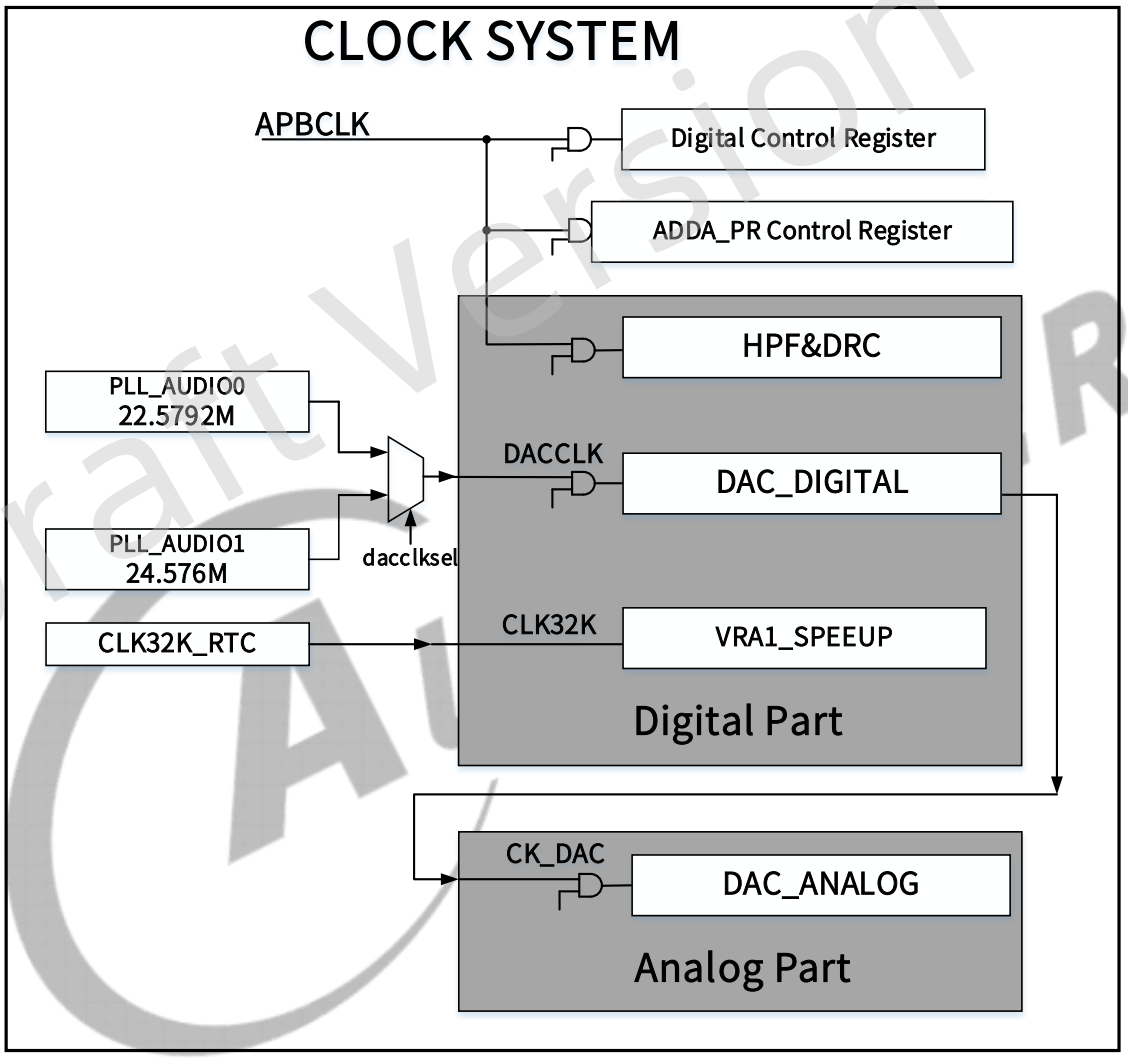
16.1.4 Functional Description

16.1.4.1 Clock Sources

The following figure describes the clock source of Audio Codec. For clock setting, configuration, and gating information, refer to Section 4.1 Clock Controller Unit (CCU).

Figure 16-2 Audio Codec Clock Diagram



- Digital Part

The clock sources for the digital DAC are the PLL_AUDIO0 and PLL_AUDIO1. The PK-PK jitter of PLL_AUDIO0 and PLL_AUDIO1 should be less than 200 ps. PLL_AUDIO0 and PLL_AUDIO1 need use decimal division.

The clock source for VRA1_SPEEDUP is CLK32K from RTC.

- Analog Part

The clock source for the analog part is the CK_DAC, which is divided from the digital part.

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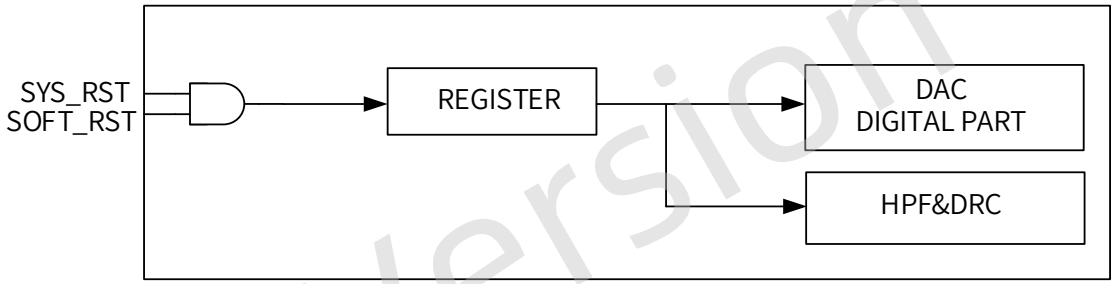
16.1.4.2 Reset System

Digital Part Reset System

The following figure shows the reset system of the audio codec digital part.

Figure 16-3 Audio Codec Digital Part Reset System

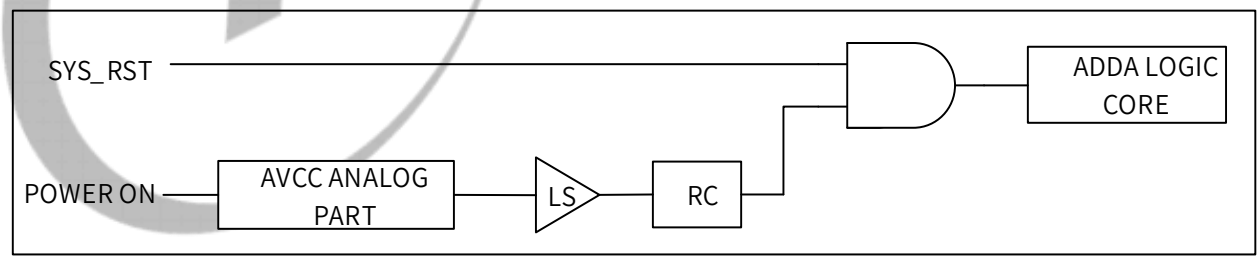


The SYS_RST reset comes from the VDD-SYS domain and is produced by the RTC domain which is controlled by S_PRCM. Each domain has the de-bounce to confirm the reset system is strong. For the codec register part, MIX can be reset by the SYS_RST when being powered on or the system soft is writing the reset control logic. The other parts can be reset by the soft configuration through writing the register.

Analog Part Reset System

The following figure shows the reset system of the audio codec analog part.

Figure 16-4 Audio Codec Analog Part Reset System



When AVCC is powered on, it sends the AVCC_POR signal. The AVCC_POR signal passes the level shift and RC filter part to the Audio Codec logic core.

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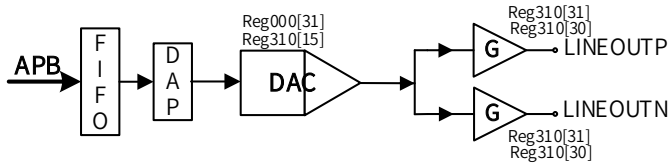
16.1.4.3 Data Path Diagram

The following figure shows a data path of the Audio Codec.

Figure 16-5 Audio Codec Data Path Diagram



16.1.4.4 Mono DAC

The mono DAC sample rate can be configured by setting the register. To save power, the analog DAC can be enabled or disabled by setting the bit [31] of the DAC_REG register. The digital DAC part can be enabled or disabled by the bit [31] of the AC_DAC_DPC register.

16.1.4.5 Analog Audio Output Path

The Audio Codec has one type of analog output ports:

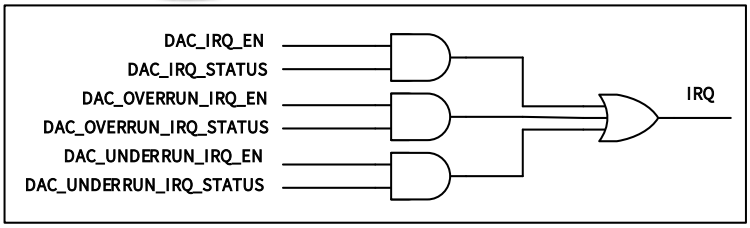
- LINEOUTP
- LINEOUTN

LINEOUT provides one differential output to drive line signals to external audio equipment. The volume control is logarithmic with a 42 dB rang in 6 dB step from -42 dB to 0 dB. The LINEOUT output buffer power up or down by bit [30] of DAC_REG.

16.1.4.6 Interrupt

The Audio Codec has two groups of interrupt. The following figure describes the Audio Codec interrupt system.

Figure 16-6 Audio Codec Interrupt System



16.1.4.7 Digital Audio Processor (DAP)

The DAP module is used to remove the DC offset and automatically adjusts the volume to a flatten volume level. It mainly consists of two HPF and one DRC.

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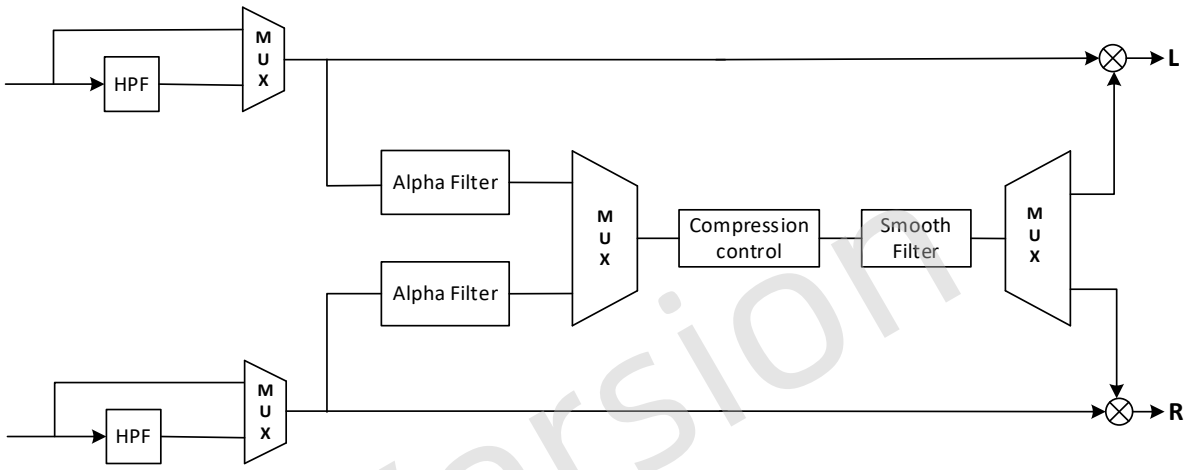
The following figure shows the DAP data flow.

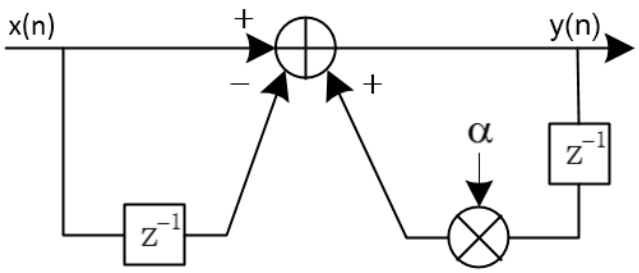
Figure 16-7 DAP Data Flow



HPF Function

The DAP has individual channel high pass filter (HPF, -3dB cutoff < 1Hz) that can be enabled and disabled. The filter cutoff frequency is less than 1 Hz that can be removed DC offset. The HPF can also be bypassed.

Figure 16-8 HPF Logic Structure



DRC Function

The DRC scheme has three thresholds, three offsets, and four slopes (all programmable). There is one ganged DRC for the left and right channels. The following figure shows the diagram of DRC input/output.

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Figure 16-9 DRC static Curve Parameters

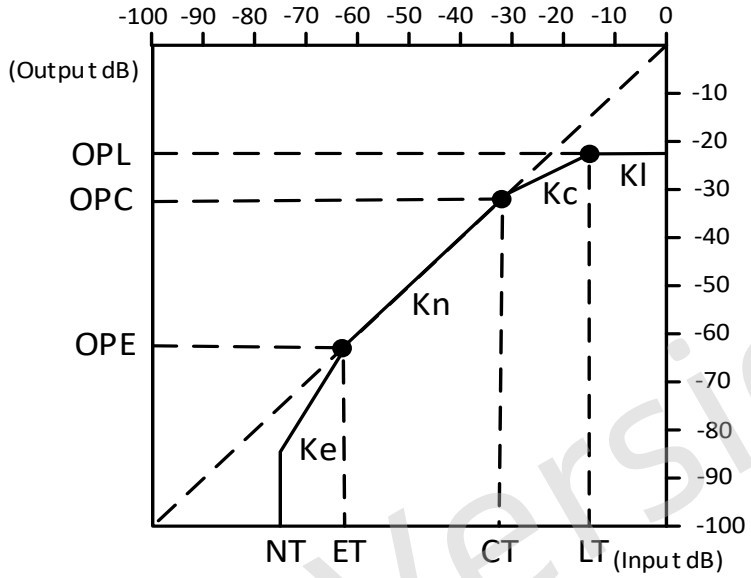
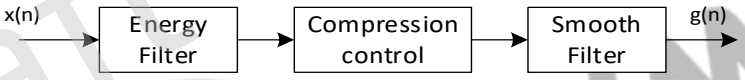


Figure 16-10 DRC Block Diagram



Professional-quality dynamic range compression automatically adjusts volume to flatten volume level.

One DRC for left/right and one DRC for subwoofer.

Each DRC has the adjustable threshold, offset, compression levels, programmable energy, attack, and decay time constants.

Transparent compression: Compressors can attack fast enough to avoid apparent clipping before engaging, and decay times can be set slow enough to avoid pumping.

Configure the DRC parameters according to the following guidelines:

- Number format

The Number format is N.M which means there are N bits to the left of the decimal point including the sign bit and M bits to the right of the decimal point. For example, Numbers formatted 9.23 means that there are 9 bits at the left of the decimal point and 23 bits at the right decimal point.

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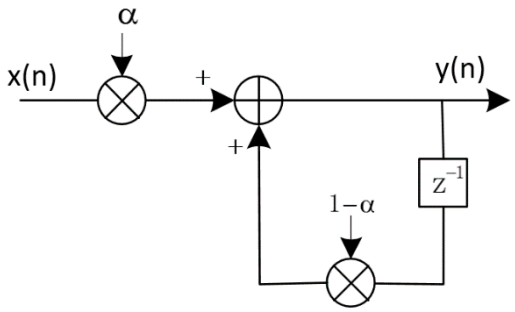
- Energy Filter

The following figure shows the structure of the energy filter.

Figure 16-11 Energy Filter Structure



The Energy Filter is to estimate of the RMS value of the audio data stream into DRC, and has two parameters, which determine the time window over which RMS to be made. The parameter is computed by $\alpha = 1 - e^{-2.2Ts/ta}$.

- Compression Control

This element has six parameters (ET, CT, LT, Ke, Kn, Kc, Kl, OPL, OPC, OPE), which are all programmable, and the computation will be explained as follows.

- Threshold Parameter Computation (T parameter)

  The threshold is the value that determines the signal to be compressed or not. When the signal's RMS is larger than the threshold, the signal will be compressed. The value of threshold input to the coefficient register is computed by

  $$Tin = -\frac{T_{dB}}{6.0206}$$

  Where, T_{dB} must less than zero, the positive value is illegal.

  For example, it is desired to set CT=-40dB, then the Tin require to set CT to -40dB is CTin = - (-40dB)/6.0206 = 6.644, CTin is entered as a 32-bit number in 8.24 format.

  Therefore, CTin = 6.644 = 0000 0110.1010 0100 1101 0011 1100 0000 = 0x06A4 D3C0 in 8.24 format.

- Slope Parameter Computation (K parameter)

  The K is the slope within compression region. For example, a n:1 compression means that an output increase 1dB is for n dB RMS input. The k input to the coefficient ram is computed by $K = \frac{1}{n}$.

  Where, n is from 1 to 50, and must be integer.

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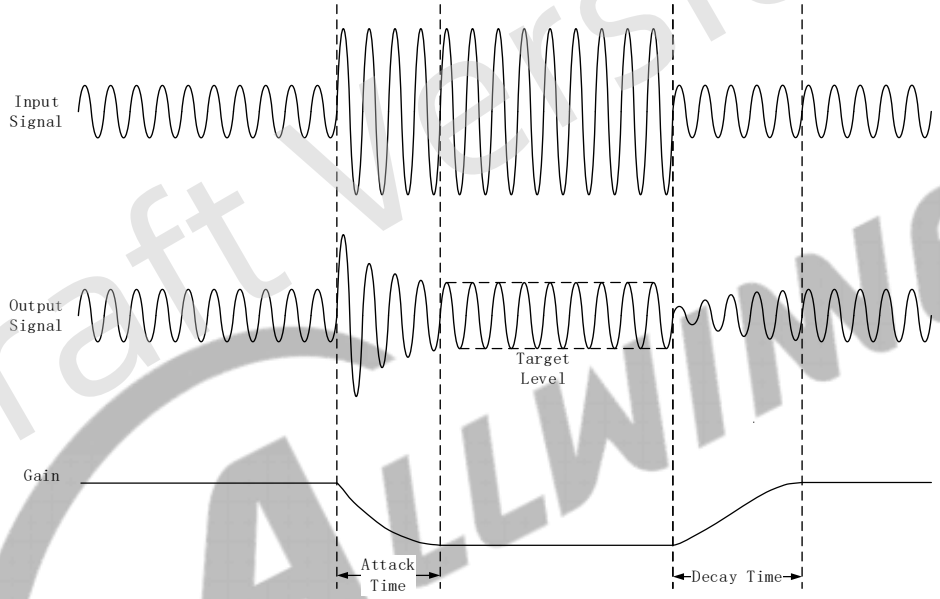
For example, it is desired to set 2:1, then the Kc require to set to 2:1 is Kc = 1/2 = 0.5, Kc is entered as a 32-bit number in 8.24 format.

Therefore, Kc = 0.5 = 0000 0000.1000 0000 0000 0000 0000 0000 = 0x0080 0000 in 8.24 format.

- **Gain Smooth Filter**

The Gain Smooth Filter is to smooth the gain and control the ratio of gain increase and decrease. The decay time and attack is shown in the following table. The structure of the Gain Smooth filter is also the Alpha filter, so the rise time computation is the same as the Energy filter which is $\alpha = 1 - e^{-2.2Ts/ta}$.

Figure 16-12 Gain Smooth Filter



16.1.5 Programming Guidelines

16.1.5.1 Playing

Step 1 Codec initialization: configure AUDIOCODEC0_GAR_REG and AUDIOCODEC0_DAC_CLK_REG open the audio codec bus clock gating, release the bus reset, and open the PLL_AUDIO DAC clock gating; set up PLL_AUDIO0_CTRL_REG to configure PLL_AUDIO0 frequency and enable PLL_AUDIO0. For details of PLL_AUDIO0, refer to Section 4.1 Clock Controller Unit (CCU).

Step 2 Set up the sample rate and data transfer format, then open the DAC.

Step 3 Configure the DMA and DMA request.

Step 4 Enable the DAC DRQ and DMA.

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16.1.6 Register List

Module Name	Base Address
AUDIO CODEC	0x02030000

Register Name	Offset	Description
AC_DAC_DPC	0x0000	DAC Digital Part Control Register
DAC_VOL_CTRL	0x0004	DAC Volume Control Register
AC_DAC_FIFOC	0x0010	DAC FIFO Control Register
AC_DAC_FIFOS	0x0014	DAC FIFO Status Register
AC_DAC_TXDATA	0x0020	DAC TX DATA Register
AC_DAC_CNT	0x0024	DAC TX FIFO Counter Register
AC_DAC_DG	0x0028	DAC Debug Register
VRA1SPEEDUP_DOWN_CTRL	0x0054	VRA1Speedup Down Control Register
ANA_DEBUG_REG1	0x0058	Analog Debug Register1
ANA_DEBUG_REG2	0x005C	Analog Debug Register2
AC_DAC_DAP_CTRL	0x00F0	DAC DAP Control Register
AC_DAC_DRC_HHPFC	0x0100	DAC DRC High HPF Coef Register
AC_DAC_DRC_LHPFC	0x0104	DAC DRC Low HPF Coef Register
AC_DAC_DRC_CTRL	0x0108	DAC DRC Control Register
AC_DAC_DRC_LPFHAT	0x010C	DAC DRC Left Peak Filter High Attack Time Coef Register
AC_DAC_DRC_LPFLAT	0x0110	DAC DRC Left Peak Filter Low Attack Time Coef Register
AC_DAC_DRC_RPFHAT	0x0114	DAC DRC Right Peak Filter High Attack Time Coef Register
AC_DAC_DRC_RPFLAT	0x0118	DAC DRC Left Peak Filter Low Attack Time Coef Register
AC_DAC_DRC_LPFHRT	0x011C	DAC DRC Left Peak Filter High Release Time Coef Register
AC_DAC_DRC_LPFLRT	0x0120	DAC DRC Left Peak Filter Low Release Time Coef Register
AC_DAC_DRC_RPFHRT	0x0124	DAC DRC Right Peak Filter High Release Time Coef Register
AC_DAC_DRC_RPFLRT	0x0128	DAC DRC Right Peak Filter Low Release Time Coef Register
AC_DAC_DRC_LRMSHAT	0x012C	DAC DRC Left RMS Filter High Coef Register
AC_DAC_DRC_LRMSLAT	0x0130	DAC DRC Left RMS Filter Low Coef Register
AC_DAC_DRC_RRMSHAT	0x0134	DAC DRC Right RMS Filter High Coef Register
AC_DAC_DRC_RRMSLAT	0x0138	DAC DRC Right RMS Filter Low Coef Register
AC_DAC_DRC_HCT	0x013C	DAC DRC Compressor Threshold High Setting Register
AC_DAC_DRC_LCT	0x0140	DAC DRC Compressor Slope High Setting Register

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Register Name	Offset	Description
AC_DAC_DRC_HKC	0x0144	DAC DRC Compressor Slope High Setting Register
AC_DAC_DRC_LKC	0x0148	DAC DRC Compressor Slope Low Setting Register
AC_DAC_DRC_HOPC	0x014C	DAC DRC Compressor High Output at Compressor Threshold Register
AC_DAC_DRC_LOPC	0x0150	DAC DRC Compressor Low Output at Compressor Threshold Register
AC_DAC_DRC_HLT	0x0154	DAC DRC Limiter Threshold High Setting Register
AC_DAC_DRC_LLT	0x0158	DAC DRC Limiter Threshold Low Setting Register
AC_DAC_DRC_HKl	0x015C	DAC DRC Limiter Slope High Setting Register
AC_DAC_DRC_LKl	0x0160	DAC DRC Limiter Slope Low Setting Register
AC_DAC_DRC_HOPL	0x0164	DAC DRC Limiter High Output at Limiter Threshold
AC_DAC_DRC_LOPL	0x0168	DAC DRC Limiter Low Output at Limiter Threshold
AC_DAC_DRC_HET	0x016C	DAC DRC Expander Threshold High Setting Register
AC_DAC_DRC_LET	0x0170	DAC DRC Expander Threshold Low Setting Register
AC_DAC_DRC_HKE	0x0174	DAC DRC Expander Slope High Setting Register
AC_DAC_DRC_LKE	0x0178	DAC DRC Expander Slope Low Setting Register
AC_DAC_DRC_HOPE	0x017C	DAC DRC Expander High Output at Expander Threshold
AC_DAC_DRC_LOPE	0x0180	DAC DRC Expander Low Output at Expander Threshold
AC_DAC_DRC_HKN	0x0184	DAC DRC Linear Slope High Setting Register
AC_DAC_DRC_LKN	0x0188	DAC DRC Linear Slope Low Setting Register
AC_DAC_DRC_SFHAT	0x018C	DAC DRC Smooth Filter Gain High Attack Time Coef Register
AC_DAC_DRC_SFLAT	0x0190	DAC DRC Smooth Filter Gain Low Attack Time Coef Register
AC_DAC_DRC_SFHRT	0x0194	DAC DRC Smooth Filter Gain High Release Time Coef Register
AC_DAC_DRC_SFLRT	0x0198	DAC DRC Smooth Filter Gain Low Release Time Coef Register
AC_DAC_DRC_MXGHS	0x019C	DAC DRC MAX Gain High Setting Register
AC_DAC_DRC_MXGLS	0x01A0	DAC DRC MAX Gain Low Setting Register
AC_DAC_DRC_MNGHS	0x01A4	DAC DRC MIN Gain High Setting Register
AC_DAC_DRC_MNGLS	0x01A8	DAC DRC MIN Gain Low Setting Register
AC_DAC_DRC_EPSHC	0x01AC	DAC DRC Expander Smooth Time High Coef Register
AC_DAC_DRC_EPSLC	0x01B0	DAC DRC Expander Smooth Time Low Coef Register
AC_DAC_DRC_HPFHGAIN	0x01B8	DAC DRC HPF Gain High Coef Register
AC_DAC_DRC_HPFLGAIN	0x01BC	DAC DRC HPF Gain Low Coef Register
DAC_REG	0x0310	DAC Analog Control Register

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Register Name	Offset	Description
RAMP_REG	0x031C	Ramp Control Register

16.1.7 Register Description

16.1.7.1 0x0000 DAC Digital Part Control Register (Default Value: 0x0000_0100)

Offset: 0x0000			Register Name: AC_DAC_DPC
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	EN_DA DAC Digital Part Enable 0: Disable 1: Enable
30:29	/	/	/
28:25	R/W	0x0	MODQU Internal DAC Quantization Levels Levels= [7*(21+MODQU [3:0])]/128 Default levels=7*21/128=1.15
24	R/W	0x0	DWA DWA Function Disable 0: Enable 1: Disable
23:19	/	/	/
18	R/W	0x0	HPF_EN High Pass Filter Enable 0: Disable 1: Enable
17:12	R/W	0x0	DVOL Digital volume control: DVC, ATT=DVC [5:0] *(-1.16dB) 64 steps, -1.16dB/step
11	/	/	/
10:8	R/W	0x1	DITHER_SGM Dither Sigma 000: Reserved 001: 1/2 010: $1/2^2$ 011: $1/2^3$ 100: $1/2^4$ 101: $1/2^5$ 110: $1/2^6$ 111: Reserved

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Offset: 0x0000			Register Name: AC_DAC_DPC
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	DITHER_SFT Dither Shift Can increase the output amplitude of dither. 0000: $x2^0$ 0001: $x2^1$ 0010: $x2^2$ 0011: $x2^3$ ... 1011: $x2^{11}$ 1100: $x2^{12}$ 1101—1111: Reserved
3:2	/	/	/
1	R/W	0x0	DITHER_EN DSM Dither Enable 0: Disable 1: Enable
0	R/W	0x0	HUB_EN Audio Hub Enable Only if EN_DA set 1, HUB_EN can take effect. SYS Domain： ADDA/ I2S0/ I2S1/ I2S2/ OWA TXFIFO Hub Enable. 0: Disable 1: Enable

16.1.7.2 0x0004 DAC Volume Control Register (Default Value: 0x0000_A0A0)

Offset: 0x0004			Register Name: DAC_VOL_CTRL
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	DAC_VOL_SEL DAC Volume Control Selection Enable 0: Disable 1: Enable

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Offset: 0x0004			Register Name: DAC_VOL_CTRL
Bit	Read/Write	Default/Hex	Description
15:8	R/W	0xA0	DAC_VOL_L DAC left channel volume (-119.25dB to 71.25dB, 0.75dB/Step) 0x00: Mute 0x01: -119.25dB …… 0x9F = -0.75dB 0xA0 = 0dB 0xA1 = 0.75dB …… 0xFF = 71.25dB
7:0	R/W	0xA0	DAC_VOL_R DAC Right Channel Volume (-119.25dB to 71.25dB, 0.75dB/Step) 0x00: Mute 0x01: -119.25dB …… 0x9F = -0.75dB 0xA0 = 0dB 0xA1 = 0.75dB …… 0xFF = 71.25dB

16.1.7.3 0x0010 DAC FIFO Control Register (Default Value: 0x0000_4000)

Offset: 0x0010			Register Name: AC_DAC_FIFOC
Bit	Read/Write	Default/Hex	Description
31:29	R/W	0x0	DAC_FS Sample Rate of DAC 000: 48kHz 010: 24kHz 100: 12kHz 110: 192kHz 001: 32kHz 011: 16kHz 101: 8kHz 111: 96kHz 44.1kHz/22.05kHz/11.025kHz can be supported by Audio PLL Configure Bit

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Offset: 0x0010			Register Name: AC_DAC_FIFOC
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	FIR_VER FIR Version 0: 64-Tap FIR 1: 32-Tap FIR
27	/	/	/
26	R/W	0x0	SEND_LASAT Audio sample select when TX FIFO under run 0: Sending zero 1: Sending last audio sample
25:24	R/W	0x0	FIFO_MODE For 20-bits transmitted audio sample: 00/10: FIFO_I [19:0] = {TXDATA [31:12]} 01/11: FIFO_I [19:0] = {TXDATA [19:0]} For 16-bits transmitted audio sample: 00/10: FIFO_I [19:0] = {TXDATA [31:16], 4'b0} 01/11: FIFO_I[19:0] = {TXDATA[15:0], 4'b0}
23	/	/	/
22:21	R/W	0x0	DAC_DRQ_CLR_CNT When TX FIFO available room less than or equal N, DRQ Request will be de-asserted. N is defined here: 00: IRQ/DRQ De-asserted when WLEVEL > TXTL 01: 4 10: 8 11: 16
20:15	/	/	/
14:8	R/W	0x40	TX_TRIG_LEVEL TX FIFO Empty Trigger Level (TXTL [12:0]) Interrupt and DMA request trigger level for TX FIFO normal condition. IRQ/DRQ Generated when WLEVEL ≤ TXTL Notes: WLEVEL represents the number of valid samples in the TX FIFO Only TXTL[6:0] valid when TXMODE = 0
7	/	/	/
6	R/W	0x0	DAC_MONO_EN DAC Mono Enable 0: Stereo, 64 levels FIFO 1: Mono, 128 levels FIFO When enabled, L & R channel send same data

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Offset: 0x0010			Register Name: AC_DAC_FIFOC
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	TX_SAMPLE_BITS Transmitting Audio Sample Resolution 0: 16 bits 1: 20 bits
4	R/W	0x0	DAC_DRQ_EN DAC FIFO Empty DRQ Enable 0: Disable 1: Enable
3	R/W	0x0	DAC_IRQ_EN DAC FIFO Empty IRQ Enable 0: Disable 1: Enable
2	R/W	0x0	FIFO_UNDERRUN_IRQ_EN DAC FIFO Under Run IRQ Enable 0: Disable 1: Enable
1	R/W	0x0	FIFO_OVERRUN_IRQ_EN DAC FIFO Over Run IRQ Enable 0: Disable 1: Enable
0	R/WC	0x0	FIFO_FLUSH DAC FIFO Flush Write '1' to flush TX FIFO, self clear to '0'

16.1.7.4 0x0014 DAC FIFO Status Register (Default Value: 0x0080_8008)

Offset: 0x0014			Register Name: AC_DAC_FIFOS
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R	0x1	TX_EMPTY TX FIFO Empty 0: No room for new sample in TX FIFO 1: More than one room for new sample in TX FIFO (>= 1 word)
22:8	R	0x80	TXE_CNT TX FIFO Empty Space Word Counter
7:4	/	/	/

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Offset: 0x0014			Register Name: AC_DAC_FIFOS
Bit	Read/Write	Default/Hex	Description
3	R/WC	0x1	TXE_INT TX FIFO Empty Pending Interrupt 0: No Pending IRQ 1: FIFO Empty Pending Interrupt Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
2	R/WC	0x0	TXU_INT TX FIFO Under run Pending Interrupt 0: No Pending Interrupt 1: FIFO Under run Pending Interrupt Write '1' to clear this interrupt
1	R/WC	0x0	TXO_INT TX FIFO Overrun Pending Interrupt 0: No Pending Interrupt 1: FIFO Overrun Pending Interrupt Write '1' to clear this interrupt
0	/	/	/

16.1.7.5 0x0020 DAC TX DATA Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: AC_DAC_TXDATA
Bit	Read/Write	Default/Hex	Description
31:0	W	0x0	TX_DATA Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample.

16.1.7.6 0x0024 DAC TX FIFO Counter Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: AC_DAC_CNT
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0024			Register Name: AC_DAC_CNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TX_CNT TX Sample Counter The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial valve at any time. After been updated by the initial value, the counter register should count on base of this initial value. Notes: It is used for Audio/ Video Synchronization

16.1.7.7 0x0028 DAC Debug Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: AC_DAC_DG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	DAC_MODU_SELECT DAC Modulator Debug 0: DAC Modulator Normal Mode 1: DAC Modulator Debug Mode
10:9	R/W	0x0	DAC_PATTERN_SELECT DAC Pattern Select 00: Normal (Audio Sample from TX FIFO) 01: -6 dB Sin wave 10: -60 dB Sin wave 11: Silent wave
8	R/W	0x0	CODEC_CLK_SELECT CODEC Clock Source Select: Cksel 0: CODEC Clock from PLL 1: CODEC Clock from OSC (for Debug)
7:0	/	/	/

16.1.7.8 0x0054 VRA1Speedup Down Control Register (Default Value: 0x0000_0000)

Offset: 0x0054			Register Name: VRA1SPEEDUP_DOWN_CTRL
Bit	Read/Write	Default/Hex	Description
31:05	/	/	/
4	R	0x0	VRA1SPEEDUP_DOWN_STATE 0: VAR1Speedup_Down not work 1: VAR1Speedup_Down work
3:2	/	/	/

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Offset: 0x0054			Register Name: VRA1SPEEDUP_DOWN_CTRL
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	VRA1SPEEDUP_DOWN_CTRL VAR1Speedup Down Manual Control Enable 0: Disable, VAR1Speedup Down convert 1 after bus rst release 32ms. 1: Enable,VAR1Speedup Down convert 1 immediately.
0	R/W	0x0	VRA1SPEEDUP_DOWN_RST_CTRL VAR1Speedup Down RST Manual Control Enable 0: Disable, VAR1Speedup Down convert 1 after bus rst release 32ms. 1: Enable,VAR1Speedup Down reset 0 immediately.

16.1.7.9 0x0058 Analog Debug Register1 (Default Value: 0x0000_0000)

Offset: 0x0058			Register Name: ANA_DEBUG_REG1
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/W	0x0	DACMUTE_CTL Dacmute Control Mode 0: Normal Mode (dacmute signal from analog) 1: Debug Mode(dacmute signal from reg config)
17	R/W	0x0	DACMUTE_DBG_EN Dacmute Debug Mode Enable 0: Disable 1: Enable
16:0	/	/	/

16.1.7.10 0x005C Analog Debug Register2 (Default Value: 0x0000_0000)

Offset: 0x005C			Register Name: ANA_DEBUG_REG2
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14:8	R	0x0	DAC_DATA Dac Data
7:1	R/W	0x0	DAC_DBG_DATA Dac Debug Data Note: Only valid when the value of DACDATA_CTRL is 1.

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Offset: 0x005C			Register Name: ANA_DEBUG_REG2
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	DACDATA_CTRL DAC Control Mode 0: Normal Mode 1: Debug Mode

16.1.7.11 0x00F0 DAC DAP Control Register (Default Value: 0x0000_0000)

Offset: 0x00F0			Register Name: AC_DAC_DAP_CTRL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DDAP_EN DAP for DRC Enable 0: Bypass 1: Enable
30	/	/	/
29	R/W	0x0	DDAP_DRC_EN DRC enable control 0: Disable 1: Enable
28	R/W	0x0	DDAP_HPF_EN HPF enable control 0: Disable 1: Enable
27:0	/	/	/

16.1.7.12 0x0100 DAC DRC High HPF Coef Register (Default Value: 0x0000_00FF)

Offset: 0x0100			Register Name: AC_DAC_DRC_HHPFC
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R/W	0xFF	DAC_DRC_HHPF_COE HPF coefficient setting and the data is 3.24 format.

16.1.7.13 0x0104 DAC DRC Low HPF Coef Register (Default Value: 0x0000_FAC1)

Offset: 0x0104			Register Name: AC_DAC_DRC_LHPFC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFAC1	DAC_DRC_LHPF_COE HPF coefficient setting and the data is 3.24 format.

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16.1.7.14 0x0108 DAC DRC Control Register (Default Value: 0x0000_0080)

Offset: 0x0108			Register Name: AC_DAC_DRC_CTRL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R	0x0	DAC_DRC_DELAY_BUF_STATE DRC delay buffer data output state when DRC delay function is enable and the DRC function disable. After disable DRC function and this bit go to 0, the user should write the DRC delay function bit to 0; 0: Not complete 1: Is complete
14:10	/	/	/
13:8	R/W	0x0	DAC_DRC_SIGNAL_DELAY_TIME_SET Signal delay time setting 6'h00: (8x1) fs 6'h01: (8x2) fs 6'h02: (8x3) fs ---------- 6'h2e: (8*47) fs 6'h2f: (8*48) fs 6'h30 -- 6'h3f: (8*48) fs Delay time = 8*(n+1) fs, n<6'h30; When the delay function is disable, the signal delay time is unused.
7	R/W	0x1	DAC_DRC_DELAY_BUF_EN The delay buffer use or not when the DRC disable and the DRC buffer data output completely 0: Don't use the buffer 1: Use the buffer
6	R/W	0x0	DAC_DRC_GAIN_MAX_LIMIT_EN DRC gain max limit enable 0: Disable 1: Enable
5	R/W	0x0	DAC_DRC_GAIN_MIN_LIMIT_EN DRC gain min limit enable. when this function enables, it will overwrite the noise detect function. 0: Disable 1: Enable
4	R/W	0x0	DAC_DRC_DETECT_NOISE_EN Control the DRC to detect noise when ET enable 0: Disable 1: Enable

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Offset: 0x0108			Register Name: AC_DAC_DRC_CTRL
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	DAC_DRC_SIGNAL_FUNC_SEL Signal function Select 0: RMS filter 1: Peak filter When Signal Function Select Peak filter, the RMS parameter is unused. (AC_DRC_LRMSHAT / AC_DRC_LRMSLAT / AC_DRC_LRMSHAT / AC_DRC_LRMSLAT) When Signal function Select RMS filter, the Peak filter parameter is unused.(AC_DRC_LPFHAT / AC_DRC_LPFLAT / AC_DRC_RPFHAT / AC_DRC_RPFLAT / AC_DRC_LPFHRT / AC_DRC_LPFLRT / AC_DRC_RPFHRT / AC_DRC_RPFLRT)
2	R/W	0x0	DAC_DRC_DELAY_FUNC_EN Delay function enable 0: Disable 1: Enable When the Delay function enable is disable, the Signal delay time is unused.
1	R/W	0x0	DAC_DRC_LT_EN DRC LT enable 0: Disable 1: Enable When the DRC LT is disable the LT, Kl and OPL parameter is unused.
0	R/W	0x0	DAC_DRC_ET_EN DRC ET enable 0: Disable 1: Enable When the DRC ET is disable the ET, Ke and OPE parameter is unused.

16.1.7.15 0x010C DAC DRC Left Peak Filter High Attack Time Coef Register (Default Value: 0x0000_000B)

Offset: 0x010C			Register Name: AC_DAC_DRC_LPFHAT
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/

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Offset: 0x010C			Register Name: AC_DAC_DRC_LPFHAT
Bit	Read/Write	Default/Hex	Description
10:0	R/W	0x000B	DAC_DRC_LPFHAT The left peak filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/ta). The format is 3.24. (The default value is 1ms)

16.1.7.16 0x0110 DAC DRC Left Peak Filter Low Attack Time Coef Register (Default Value: 0x0000_77BF)

Offset: 0x0110			Register Name: AC_DAC_DRC_LPFLAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x77BF	DAC_DRC_LPFLAT The left peak filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/ta). The format is 3.24. (The default value is 1ms)

16.1.7.17 0x0114 DAC DRC Right Peak Filter High Attack Time Coef Register (Default Value: 0x0000_000B)

Offset: 0x0114			Register Name: AC_DAC_DRC_RPFHAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x000B	DAC_DRC_RPFHAT The right peak filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/ta). The format is 3.24. (The default value is 1ms)

16.1.7.18 0x0118 DAC DRC Left Peak Filter Low Attack Time Coef Register (Default Value: 0x0000_77BF)

Offset: 0x0118			Register Name: AC_DAC_DRC_RPFLAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0118			Register Name: AC_DAC_DRC_RPFLAT
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x77BF	DAC_DRC_RPFLAT The left peak filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/ta). The format is 3.24. (The default value is 1ms)

16.1.7.19 0x011C DAC DRC Left Peak Filter High Release Time Coef Register (Default Value: 0x0000_00FF)

Offset: 0x011C			Register Name: AC_DAC_DRC_LPFHRT
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R/W	0x00FF	DAC_DRC_LPFHRT The left peak filter release time parameter setting, which determine by the equation that RT = exp(-2.2Ts/tr). The format is 3.24. (The default value is 100ms)

16.1.7.20 0x0120 DAC DRC Left Peak Filter Low Release Time Coef Register (Default Value: 0x0000_E1F8)

Offset: 0x0120			Register Name: AC_DAC_DRC_LPFLRT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xE1F8	DAC_DRC_LPFLRT The left peak filter release time parameter setting, which determine by the equation that RT = exp(-2.2Ts/tr). The format is 3.24. (The default value is 100ms)

16.1.7.21 0x0124 DAC DRC Right Peak Filter High Release Time Coef Register (Default Value: 0x0000_00FF)

Offset: 0x0124			Register Name: AC_DAC_DRC_RPFHRT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0124			Register Name: AC_DAC_DRC_RPFHRT
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x00FF	DAC_DRC_RPFHRT The right peak filter attack time parameter setting, which determine by the equation that RT = exp(-2.2Ts/tr). The format is 3.24. (The default value is 100ms)

16.1.7.22 0x0128 DAC DRC Right Peak Filter Low Release Time Coef Register (Default Value: 0x0000_E1F8)

Offset: 0x0128			Register Name: AC_DAC_DRC_RPFLRT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xE1F8	DAC_DRC_RPFLRT The right peak filter release time parameter setting, which determine by the equation that AT = exp(-2.2Ts/tr). The format is 3.24. (The default value is 100ms)

16.1.7.23 0x012C DAC DRC Left RMS Filter High Coef Register (Default Value: 0x0000_0001)

Offset: 0x012C			Register Name: AC_DAC_DRC_LRMSHAT
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R/W	0x0001	DAC_DRC_LRMSHAT The left RMS filter average time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tav). The format is 3.24. (The default value is 10ms)

16.1.7.24 0x0130 DAC DRC Left RMS Filter Low Coef Register (Default Value: 0x0000_2BAF)

Offset: 0x0130			Register Name: AC_DAC_DRC_LRMSLAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x2BAF	DAC_DRC_LRMSLAT The left RMS filter average time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tav). The format is 3.24. (The default value is 10ms)

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16.1.7.25 0x0134 DAC DRC Right RMS Filter High Coef Register (Default Value: 0x0000_0001)

Offset: 0x0134			Register Name: AC_DAC_DRC_RRMSHAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0001	DAC_DRC_RRMSHAT The right RMS filter average time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tav). The format is 3.24.(The default value is 10ms)

16.1.7.26 0x0138 DAC DRC Right RMS Filter Low Coef Register (Default Value: 0x0000_2BAF)

Offset: 0x0138			Register Name: AC_DAC_DRC_RRMSLAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x2BAF	DAC_DRC_RRMSLAT The right RMS filter average time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tav). The format is 3.24.(10ms)

16.1.7.27 0x013C DAC DRC Compressor Threshold High Setting Register (Default Value: 0x0000_06A4)

Offset: 0x013C			Register Name: AC_DAC_DRC_HCT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x06A4	DAC_DRC_HCT The compressor threshold setting, which set by the equation that CTin = -CT/6.0206. The format is 8.24. (The default value is -40dB)

16.1.7.28 0x0140 DAC DRC Compressor Slope High Setting Register (Default Value: 0x0000_D3C0)

Offset: 0x0140			Register Name: AC_DAC_DRC_LCT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xD3C0	DAC_DRC_LCT The compressor threshold setting, which set by the equation that CTin = -CT/6.0206. The format is 8.24. (The default value is -40dB)

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16.1.7.29 0x0144 DAC DRC Compressor Slope High Setting Register (Default Value: 0x0000_0080)

Offset: 0x0144			Register Name: AC_DAC_DRC_HKC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0080	DAC_DRC_HKC The slope of the compressor which determine by the equation that Kc = 1/R, there, R is the ratio of the compressor, which always is integer. The format is 8.24. (The default value is <2 : 1>)

16.1.7.30 0x0148 DAC DRC Compressor Slope Low Setting Register (Default Value: 0x0000_0000)

Offset: 0x0148			Register Name: AC_DAC_DRC_LKC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0000	DAC_DRC_LKC The slope of the compressor which determine by the equation that Kc = 1/R, there, R is the ratio of the compressor, which always is integer. The format is 8.24. (The default value is <2 : 1>)

16.1.7.31 0x014C DAC DRC Compressor High Output at Compressor Threshold Register (Default Value: 0x0000_F95B)

Offset: 0x014C			Register Name: AC_DAC_DRC_HOPC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xF95B	DAC_DRC_HOPC The output of the compressor which determine by the equation OPCin=OPC/6.0206. The format is 8.24. (The default value is -40dB)

16.1.7.32 0x0150 DAC DRC Compressor Low Output at Compressor Threshold Register (Default Value: 0x0000_2C3F)

Offset: 0x0150			Register Name: AC_DAC_DRC_LOPC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0150			Register Name: AC_DAC_DRC_LOPC
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x2C3F	DAC_DRC_LOPC The output of the compressor which determine by the equation OPCin=OPC/6.0206. The format is 8.24. (The default value is -40dB)

16.1.7.33 0x0154 DAC DRC Limiter Threshold High Setting Register (Default Value: 0x0000_01A9)

Offset: 0x0154			Register Name: AC_DAC_DRC_HLT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x01A9	DAC_DRC_HLT The limiter threshold setting, which set by the equation that LTin = -LT/6.0206. The format is 8.24. (The default value is -10dB)

16.1.7.34 0x0158 DAC DRC Limiter Threshold Low Setting Register (Default Value: 0x0000_34F0)

Offset: 0x0158			Register Name: AC_DAC_DRC_LLT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x34F0	DAC_DRC_LLT The limiter threshold setting, which set by the equation that LTin = -LT/6.0206. The format is 8.24. (The default value is -10dB)

16.1.7.35 0x015C DAC DRC Limiter Slope High Setting Register (Default Value: 0x0000_0005)

Offset: 0x015C			Register Name: AC_DAC_DRC_HKl
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0005	DAC_DRC_HKl The slope of the limiter which determine by the equation that Kl = 1/R, there, R is the ratio of the limiter, which always is integer. The format is 8.24. (The default value is <50:1>)

16.1.7.36 0x0160 DAC DRC Limiter Slope Low Setting Register (Default Value: 0x0000_1EB8)

Offset: 0x0160	Register Name: AC_DAC_DRC_LKl

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Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x1EB8	DAC_DRC_LKl The slope of the limiter which determine by the equation that Kl = 1/R, there, R is the ratio of the limiter, which always is integer. The format is 8.24. (The default value is <50 :1>)

16.1.7.37 0x0164 DAC DRC Limiter High Output at Limiter Threshold (Default Value: 0x0000_FBD8)

Offset: 0x0164			Register Name: AC_DAC_DRC_HOPL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFBD8	DAC_DRC_HOPL The output of the limiter which determine by equation OPTin=OPT/6.0206. The format is 8.24. (The default value is -25dB)

16.1.7.38 0x0168 DAC DRC Limiter Low Output at Limiter Threshold (Default Value: 0x0000_FBA7)

Offset: 0x0168			Register Name: AC_DAC_DRC_LOPL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFBA7	DAC_DRC_LOPL The output of the limiter which determine by equation OPTin=OPT/6.0206. The format is 8.24. (The default value is -25dB)

16.1.7.39 0x016C DAC DRC Expander Threshold High Setting Register (Default Value: 0x0000_0BA0)

Offset: 0x016C			Register Name: AC_DAC_DRC_HET
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0BA0	DAC_DRC_HET The expander threshold setting, which set by the equation that ETin = -ET/6.0206, The format is 8.24. (The default value is -70dB)

16.1.7.40 0x0170 DAC DRC Expander Threshold Low Setting Register (Default Value: 0x0000_7291)

Offset: 0x0170			Register Name: AC_DAC_DRC_LET
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0170			Register Name: AC_DAC_DRC_LET
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x7291	DAC_DRC_LET The expander threshold setting, which set by the equation that ETin = -ET/6.0206, The format is 8.24. (The default value is -70dB)

16.1.7.41 0x0174 DAC DRC Expander Slope High Setting Register (Default Value: 0x0000_0500)

Offset: 0x0174			Register Name: AC_DAC_DRC_HKE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0500	DAC_DRC_HKE The slope of the expander which determine by the equation that Ke = 1/R, there, R is the ratio of the expander, which always is integer and the ke must larger than 50. The format is 8.24. (The default value is <1:5>)

16.1.7.42 0x0178 DAC DRC Expander Slope Low Setting Register (Default Value: 0x0000_0000)

Offset: 0x0178			Register Name: AC_DAC_DRC_LKE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0000	DAC_DRC_LKE The slope of the expander which determine by the equation that Ke = 1/R, there, R is the ratio of the expander, which always is integer and the ke must larger than 50. The format is 8.24. (The default value is <1:5>)

16.1.7.43 0x017C DAC DRC Expander High Output at Expander Threshold (Default Value: 0x0000_F45F)

Offset: 0x017C			Register Name: AC_DAC_DRC_HOPE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xF45F	DAC_DRC_HOPE The output of the expander which determine by equation OPEin=OPE/6.0206. The format is 8.24. (The default value is -70dB)

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16.1.7.44 0x0180 DAC DRC Expander Low Output at Expander Threshold (Default Value: 0x0000_8D6E)

Offset: 0x0180			Register Name: AC_DAC_DRC_LOPE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x8D6E	DAC_DRC_LOPE The output of the expander which determine by equation OPEin=OPE/6.0206. The format is 8.24. (The default value is -70dB)

16.1.7.45 0x0184 DAC DRC Linear Slope High Setting Register (Default Value: 0x0000_0100)

Offset: 0x0184			Register Name: AC_DAC_DRC_HKN
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0100	DAC_DRC_HKN The slope of the linear which determine by the equation that Kn = 1/R, there, R is the ratio of the linear, which always is integer. The format is 8.24. (The default value is <1:1>)

16.1.7.46 0x0188 DAC DRC Linear Slope Low Setting Register (Default Value: 0x0000_0000)

Offset: 0x0188			Register Name: AC_DAC_DRC_LKN
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0000	DAC_DRC_LKN The slope of the linear which determine by the equation that Kn = 1/R, there, R is the ratio of the linear, which always is integer. The format is 8.24. (The default value is <1:1>)

16.1.7.47 0x018C DAC DRC Smooth Filter Gain High Attack Time Coef Register (Default Value: 0x0000_0002)

Offset: 0x018C			Register Name: AC_DAC_DRC_SFHAT
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/

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Offset: 0x018C			Register Name: AC_DAC_DRC_SFHAT
Bit	Read/Write	Default/Hex	Description
10:0	R/W	0x0002	DAC_DRC_SFHAT The smooth filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 5ms)

16.1.7.48 0x0190 DAC DRC Smooth Filter Gain Low Attack Time Coef Register (Default Value: 0x0000_5600)

Offset: 0x0190			Register Name: AC_DAC_DRC_SFLAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x5600	DAC_DRC_SFLAT The smooth filter attack time parameter setting, which determine by the equation that AT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 5ms)

16.1.7.49 0x0194 DAC DRC Smooth Filter Gain High Release Time Coef Register (Default Value: 0x0000_0000)

Offset: 0x0194			Register Name: AC_DAC_DRC_SFHRT
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R/W	0x0000	DAC_DRC_SFHRT The gain smooth filter release time parameter setting, which determine by the equation that RT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 200ms)

16.1.7.50 0x0198 DAC DRC Smooth Filter Gain Low Release Time Coef Register (Default Value: 0x0000_0F04)

Offset: 0x0198			Register Name: AC_DAC_DRC_SFLRT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0198			Register Name: AC_DAC_DRC_SFLRT
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0F04	DAC_DRC_SFLRT The gain smooth filter release time parameter setting, which determine by the equation that RT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 200ms)

16.1.7.51 0x019C DAC DRC MAX Gain High Setting Register (Default Value: 0x0000_FE56)

Offset: 0x019C			Register Name: AC_DAC_DRC_MXGHS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFE56	DAC_DRC_MXGHS The max gain setting which determine by equation MXGin=MXG/6.0206. The format is 8.24 and must -20dB <MXG < 30dB. (The default value is -10dB)

16.1.7.52 0x01A0 DAC DRC MAX Gain Low Setting Register (Default Value: 0x0000_CB0F)

Offset: 0x01A0			Register Name: AC_DAC_DRC_MXGLS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xCB0F	DAC_DRC_MXGLS The max gain setting which determine by equation MXGin=MXG/6.0206. The format is 8.24 and must -20dB <MXG < 30dB. (The default value is -10dB)

16.1.7.53 0x01A4 DAC DRC MIN Gain High Setting Register (Default Value: 0x0000_F95B)

Offset: 0x01A4			Register Name: AC_DAC_DRC_MNGHS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xF95B	DAC_DRC_MNGHS The min gain setting which determine by equation MNGin=MNG/6.0206. The format is 8.24 and must -60dB ≤MNG ≤ -40dB. (The default value is -40dB)

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16.1.7.54 0x01A8 DAC DRC MIN Gain Low Setting Register (Default Value: 0x0000_2C3F)

Offset: 0x01A8			Register Name: AC_DAC_DRC_MNGLS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x2C3F	DAC_DRC_MNGLS The min gain setting which determine by equation MNGin=MNG/6.0206. The format is 8.24 and must -60dB ≤MNG ≤ -40dB. (The default value is -40dB)

16.1.7.55 0x01AC DAC DRC Expander Smooth Time High Coef Register (Default Value: 0x0000_0000)

Offset: 0x01AC			Register Name: AC_DAC_DRC_EPSHC
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R/W	0x0000	DAC_DRC_EPSHC The gain smooth filter release and attack time parameter setting in expander region, which determine by the equation that RT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 30ms)

16.1.7.56 0x01B0 DAC DRC Expander Smooth Time Low Coef Register (Default Value: 0x0000_640C)

Offset: 0x01B0			Register Name: AC_DAC_DRC_EPSLC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x640C	DAC_DRC_EPSLC The gain smooth filter release and attack time parameter setting in expander region, which determine by the equation that RT = 1-exp(-2.2Ts/tr). The format is 3.24. (The default value is 30ms)

16.1.7.57 0x01B8 DAC DRC HPF Gain High Coef Register (Default Value: 0x0000_0100)

Offset: 0x01B8			Register Name: AC_DAC_DRC_HPFHGAIN
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/

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Offset: 0x01B8			Register Name: AC_DAC_DRC_HPFHGAIN
Bit	Read/Write	Default/Hex	Description
10:0	R/W	0x100	DAC_DRC_HPFHGAIN The gain of the HPF coefficient setting which format is 3.24.(gain = 1)

16.1.7.58 0x01BC DAC DRC HPF Gain Low Coef Register (Default Value: 0x0000_0000)

Offset: 0x01BC			Register Name: AC_DAC_DRC_HPFLGAIN
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0000	DAC_DRC_HPFLGAIN The gain of the HPF coefficient setting which format is 3.24.(gain = 1)

16.1.7.59 DAC Analog Control Register(Default Value: 0x0054_0000)

Offset: 0x0310			Register Name: DAC_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DAC_EN DAC Enable 0: Disable 1: Enable
30	R/W	0x0	LINEOUTEN LINEOUT Enable 0: Disable 1: Enable
29	R/W	0x0	MUTE DAC to LINEOUT Mute Control 0: Mute 1: Not mute
28	R/W	0x0	LINEOUT_DIFFEN LINEOUT Output Control 0: Single-End 1: Differential
27	/	/	/

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Offset: 0x0310			Register Name: DAC_REG
Bit	Read/Write	Default/Hex	Description
26:24	R/W	0x0	LINEOUT_VOL_CTRL LINEOUT Volume Control, Total 8 level from 0x0 to 0x7 with the volume -42dB to 0dB, -6dB/step 000:-42dB 001:-36dB 010:-30dB 011:-24dB 100:-18dB 101:-12dB 110:-6dB 111:0dB
23:22	R/W	0x1	IOPVRS VRA2 Buffer OP Bias Current Select 00: 6uA 01: 7uA 10: 8uA 11: 9uA
21:20	R/W	0x1	LINEOUTAMPS LINEOUT AMP Bias Current Select 00: 6uA 01: 7uA 10: 8uA 11: 9uA
19:18	R/W	0x1	IOPDACS OPDAC Bias Current Select 00: 6uA 01: 7uA 10: 8uA 11: 9uA
17:16	R/W	0x0	CKDAC_DELAY_SET clock delay time after CKDAC 0: Min 3: Max
15	R/W	0x0	DAC_CHOPPER_ENABLE DAC chopper enable 0: Chopper disable 1: Chopper enable
14	R/W	0x0	DAC_CHOPPER_NOL_ENABLE DAC chopper non-overlapping clock enable 0: Chopper non-overlapping clock disabled 1: Chopper non-overlapping clock enabled

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Offset: 0x0310			Register Name: DAC_REG
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x0	DAC_CHOPPER_CKSET DAC chopper clock frequency Fs:6.144MHz 0=96kHz 1=192kHz 2=384kHz 3=768kHz
11:10	R/W	0x0	DAC_CHOPPER_DELAY_SET Control DAC chopper clock delay time after CKDAC 0: Min 3: Max
9:8	R/W	0x0	DAC_CHOPPER_NOL_DELAY_SET Control DAC chopper clock non-overlapping time 0: Min when DAC_CHOPPER_NOL_ENABLE=1 3: Max when DAC_CHOPPER_NOL_ENABLE=1
7	R/W	0x0	LINEOUT_CHOPPER_ENABLE LINEOUT chopper enable 0: Chopper disable 1: Chopper enable
6	R/W	0x0	LINEOUT_CHOPPER_NOL_ENABLE LINEOUT chopper non-overlapping clock enable 0: Chopper non-overlapping clock disabled 1: Chopper non-overlapping clock enabled
5:4	R/W	0x0	LINEOUT_CHOPPER_CKSET LINEOUT chopper clock frequency Fs:6.144MHz 0=96kHz 1=192kHz 2=384kHz 3=768kHz
3:2	R/W	0x0	LINEOUT_CHOPPER_DELAY_SET Control LINEOUT chopper clock delay time after CKDAC 0: Min 3: Max
1:0	R/W	0x0	LINEOUT_CHOPPER_NOL_DELAY_SET Control LINEOUT chopper clock non-overlapping time 0: Min when LINEOUT_CHOPPER_NOL_ENABLE=1 3: Max when LINEOUT_CHOPPER_NOL_ENABLE=1

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16.1.7.60 0x031C Ramp Control Register (Default Value: 0x0018_0000)

Offset:0x031C			Register Name: RAMP_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	RAMP_RISE_INT_EN 0: Enable 1: Disable
30	R/W1C	0x0	RAMP_RISE_INT RK Increase upward Finish and Rampen Pull Down Instruction 0: No Pending IRQ 1: Ramp Rise Finish Pending Interrupt Write '1' to clear this interrupt
29	R/W	0x0	RAMP_FALL_INT_EN 0: Enable 1: Disable
28	R/W1C	0x0	RAMP_FALL_INT RK Downward Decrease Finish and Rampen Pull Down Instruction 0: No Pending IRQ 1: Ramp Fall Finish Pending Interrupt Write '1' to clear this interrupt
27:25	/	/	/
24	R/WAC	0x0	RAMP_SOFT_RESET 0: Disable 1: Enable
23:21	/	/	/
20:16	R/W	0x18	RAMP_CLK_DIV_M Analog Ramp Clk Div Freq Value: M (from 0 to 31, Default:24). Ana_Ramp_Clk= 24MHz/(M+1) Default Ramp Clk Freq: 24MHz/(24+1)=960KHz
15	R/W	0x0	HP_PULL_OUT_EN 0: Disable 1: Enable

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Offset:0x031C			Register Name: RAMP_REG
Bit	Read/Write	Default/Hex	Description
14:12	R/W	0x0	RAMP_HOLD_STEP 000: 9600 001: 19200 010: 38400 011: 76800 100: 96000 101: 115200 110: 153600 111: 192000 Ramp Hold Time = Ramp Hold Step/Ramp Clk Freq When Ramp Clk Freq equal to 960KHz, Corresponding Ramp Hold time of each gear: 000: 9600/960KHz=10ms 001: 19200/960KHz=20ms 010: 38400/960KHz=40ms 011: 76800/960KHz=80ms 100: 96000/960KHz=100ms 101: 115200/960KHz=120ms 110: 153600/960KHz=160ms 111: 192000/960KHz=200ms
11:10	/	/	/
9:8	R/W	0x0	GAP_STEP 00: Ramp step 01: Ramp step*2 10: Ramp step*3 11: Ramp step*4
7	/	/	/

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Offset:0x031C			Register Name: RAMP_REG
Bit	Read/Write	Default/Hex	Description
6:4	R/W	0x0	RAMP_STEP RK Frequency Gear, Control Ramp Rise/Fall Total Time 000: 20 001: 30 010: 40 011: 60 100: 80 101: 120 110: 160 111: 240 Ramp Rise/Fall Total Time = (Ramp Step/Ramp Clk Freq) *4096 When Default Ramp Clk Freq equal to 960KHz, Corresponding time of each gear: 000: (20/960KHz) *4096=85.3ms 001: (30/960KHz) *4096=128ms 010: (40/960KHz) *4096=170.6ms 011: (60/960KHz) *4096=256ms 100: (80/960KHz) *4096=341.3ms 101: (120/960KHz) *4096=512ms 110: (160/960KHz) *4096=682.6ms 111: (240/960KHz)*4096=1024ms
3	R/W	0x0	RMD_EN Ramp Manual Down Enable 0: Disable 1: Enable
2	R/W	0x0	RMU_EN Ramp Manual Up Enable 0: Disable 1: Enable
1	R/W	0x0	RMC_EN Ramp Manual Control Enable 0: Disable 1: Enable If this bit is set to “0”, DAC will not output signals.
0	R/W	0x0	RD_EN Ramp Digital Enable 0: Disable 1: Enable

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16.2 I2S/PCM

16.2.1 Overview

The I2S/PCM controller is designed to transfer streaming audio-data between the system memory and the codec chip. The controller supports standard I2S format, Left-justified mode format, Right-justified mode format, PCM mode format, and TDM mode format.

The I2S/PCM controller includes the following features:

- 3x I2S/PCM (I2S0-2)
  - I2S0 is used for voice input sampling
  - I2S1 is used for Bluetooth audio call.
  - I2S2 are used for digital power amplifier.
  - I2S0/2 support 4-lane appilication and I2S1 supports 2-lane application
- Compliant with standard Philips Inter-IC sound (I2S) bus specification
  - Left-justified, Right-justified, PCM mode, and Time Division Multiplexing (TDM) format
  - Programmable PCM frame width: 1 BCLK width (short frame) and 2 BCLKs width (long frame)
- FIFOs for transmitting and receiving data
  - Programmable FIFO thresholds
  - 128 depth x 32-bit width TXFIFO and 64 depth x 32-bit width RXFIFO
- Supports multiple function clocks
  - Clock up to 24.576 MHz Data Output of I2S/PCM in Master mode (Only if the IO PAD and Peripheral I2S/PCM satisfy Timing Parameters)
  - Clock up to 12.288 MHz Data Input of I2S/PCM in Master mode (Only if the IO PAD and Peripheral I2S/PCM satisfy Timing Parameters)
- Supports TX/RX DMA slave interface
- Supports multiple application scenarios
  - Up to 16 channels (fs = 48 kHz) which has adjustable width from 8-bit to 32-bit
  - Sample rate from 8 kHz to 384 kHz (sample rate*channels*slot width<=24.576MHz)
  - 8-bit u-law and 8-bit A-law companded sample
- Supports master/slave mode

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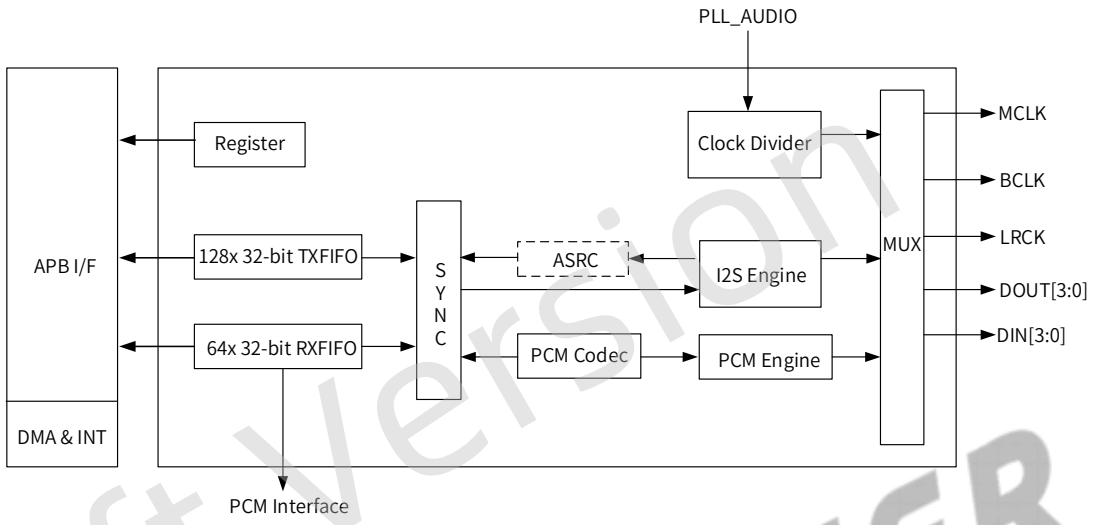
16.2.2 Block Diagram

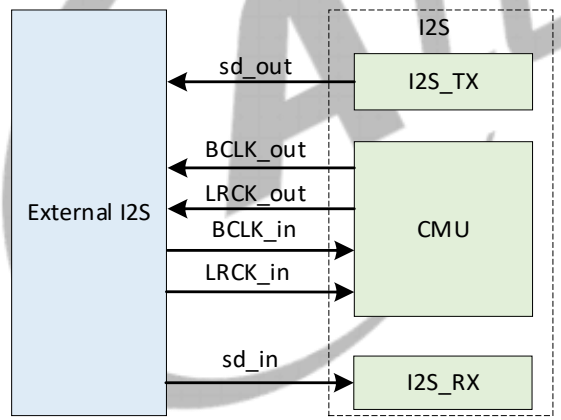
The following figure shows the functional block diagram of the I2S/PCM interface.

Figure 16-13 I2S/PCM Interface System Block Diagram



The following figure shows the typical application of the I2S/PCM interface.

Figure 16-14 Typical Application of I2S/PCM Interface



The I2S/PCM interface system integrates one I2S_TX and one I2S_RX.

- The I2S_TX is for playing music in I2S or PCM format.
- The I2S_RX is for receiving data in I2S or PCM format.
- When the I2S works in the master mode, the external I2S module provides BCLK_in and LRCK_in for the clock management unit (CMU), and the I2S_TX and I2S_RX work with the two external clocks.
- When the I2S works in the slave mode, the CMU provides clocks BCLK_out and LRCK_out for the external I2S module, and the I2S_TX and I2S_RX work with the internal clocks.

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16.2.3 External Signals

The following table describes the external signals of the I2S/PCM interface.

LRCK and BCLK are bidirectional I/O. When the I2S/PCM interface works in the Master mode, LRCK and BCLK are output pins. When the I2S/PCM interface works in the Slave mode, LRCK and BCLK are input pins.

MCLK is an output pin for external devices. DOUT is the serial data output pin, and DIN is the serial data input pin. For details about General Purpose I/O port, refer to the Section 17.9 GPIO.

Table 16-2 I2S/PCM External Signals

Signal Name	Description	Type
I2S0-DOUT[3:0]	I2S0/PCM0 Serial Data Output Channel [3:0]	O
I2S0-DIN[3:0]	I2S0/PCM0 Serial Data Input Channel [3:0]	I
I2S0-MCLK	I2S0 Master Clock	O
I2S0-LRCK	I2S0/PCM0 Sample Rate Clock/Sync	I/O
I2S0-BCLK	I2S0/PCM0 Bit Rate Clock	I/O
I2S1-DOUT[1:0]	I2S1/PCM1 Serial Data Output Channel [1:0]	O
I2S1-DIN[1:0]	I2S1/PCM1 Serial Data Input Channel [1:0]	I
I2S1-MCLK	I2S1 Master Clock	O
I2S1-LRCK	I2S1/PCM01Sample Rate Clock/Sync	I/O
I2S1-BCLK	I2S1/PCM1 Bit Rate Clock	I/O
I2S2-DOUT[3:0]	I2S2/PCM2 Serial Data Output Channel [3:0]	O
I2S2-DIN[3:0]	I2S2/PCM2 Serial Data Input Channel [3:0]	I
I2S2-MCLK	I2S2 Master Clock	O
I2S2-LRCK	I2S2/PCM2 Sample Rate Clock/Sync	I/O
I2S2-BCLK	I2S2/PCM2 Bit Rate Clock	I/O

16.2.4 Functional Description

16.2.4.1 I2S/PCM Transmit Modes

The I2S/PCM supports standard I2S mode, Left-justified I2S mode, Right-justified I2S mode, PCM mode, and TDM mode. The software can select the modes by setting I2S/PCM_CTL. The following figures describe the waveforms for SYNC, BCLK, DOUT, and DIN in different modes.

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Figure 16-15 I2S Standard Mode Timing

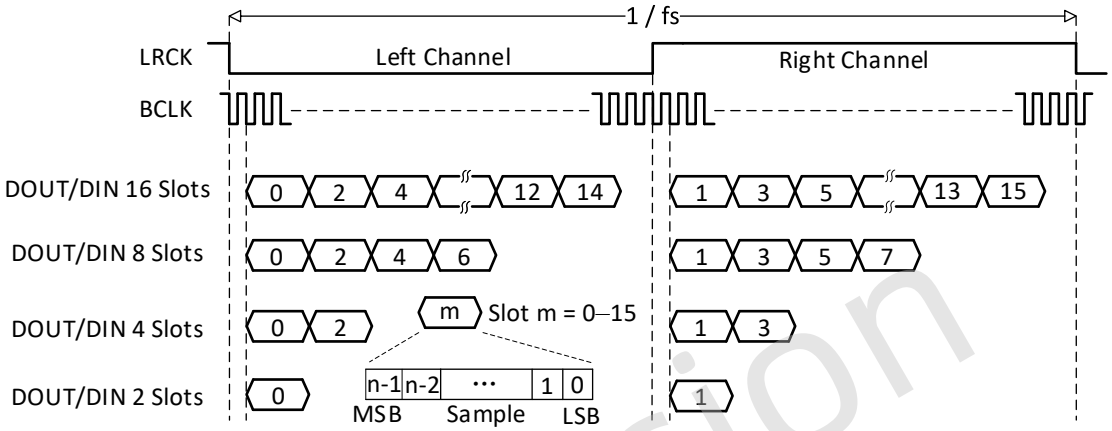

Figure 16-16 Left-Justified Mode Timing

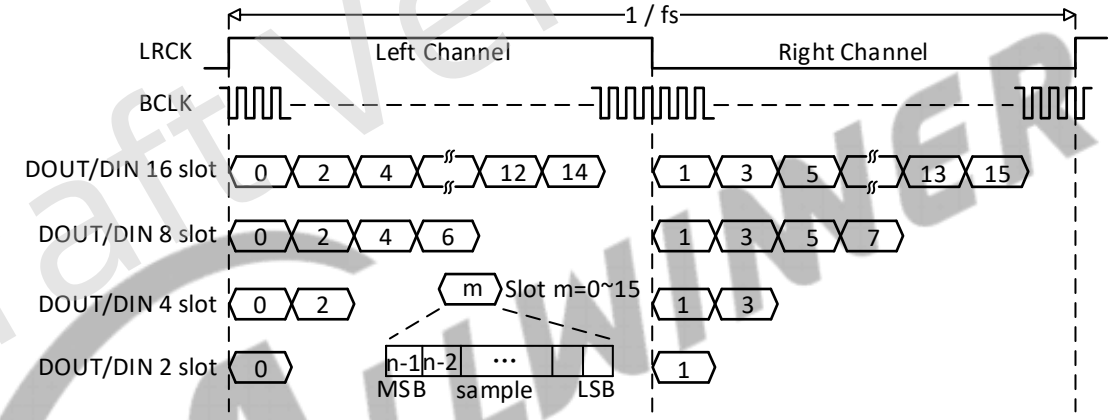

Figure 16-17 Right-Justified Mode Timing

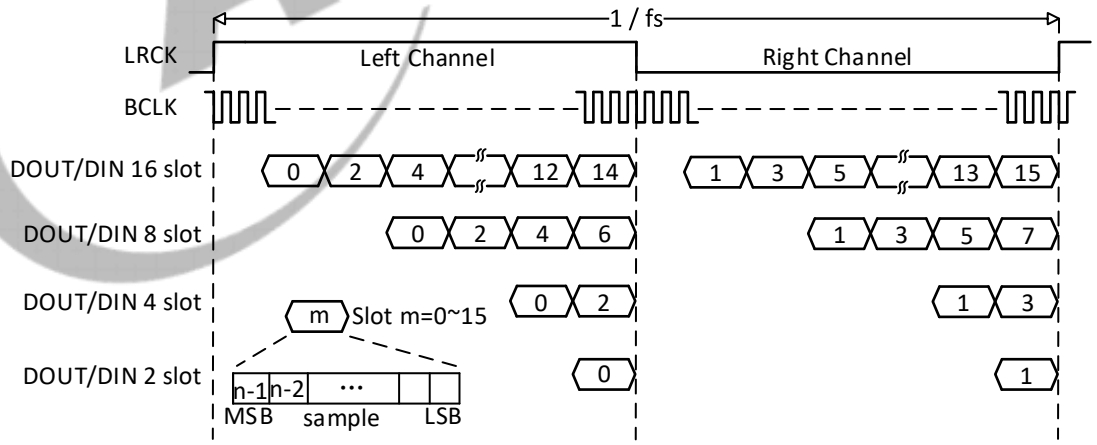

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Figure 16-18 PCM Long Frame Mode Timing

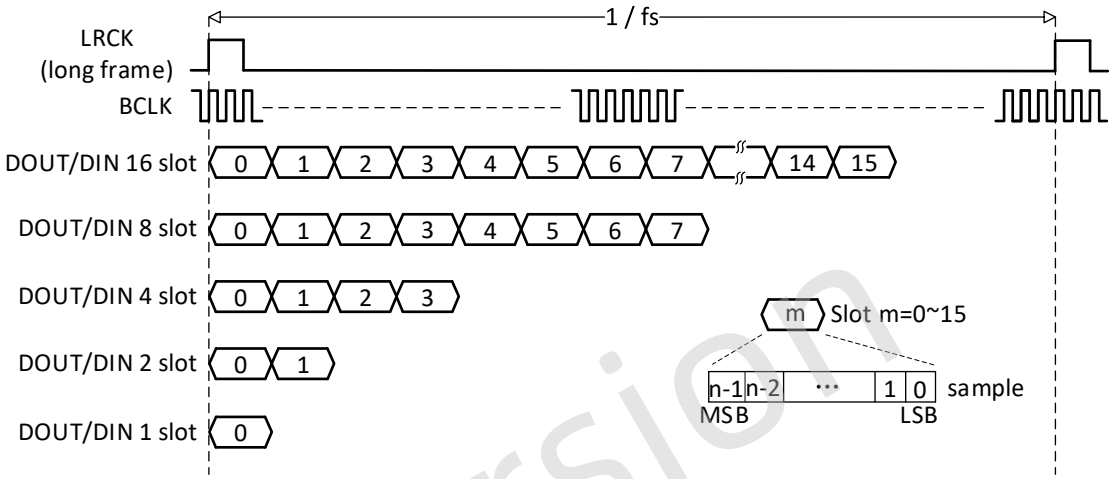
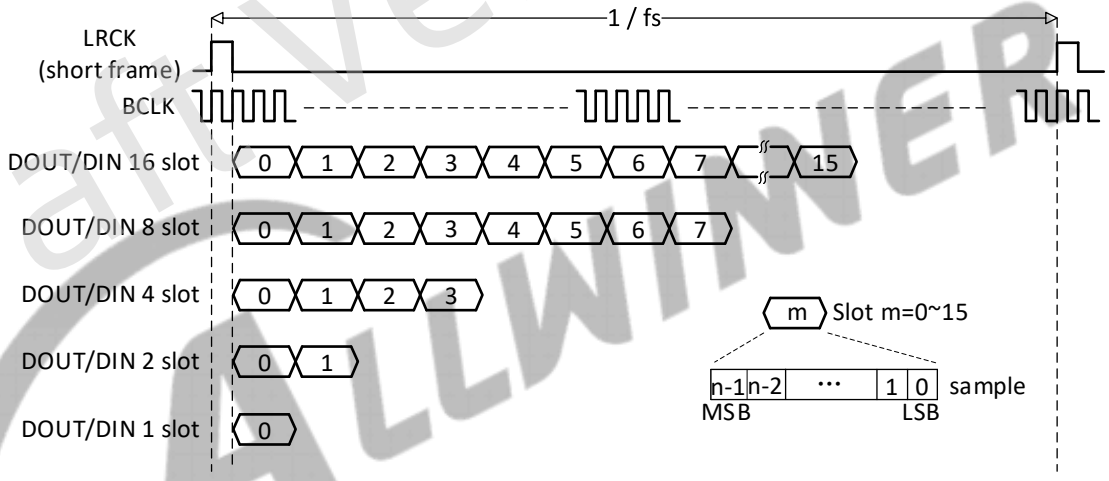


Figure 16-19 PCM Short Frame Mode Timing

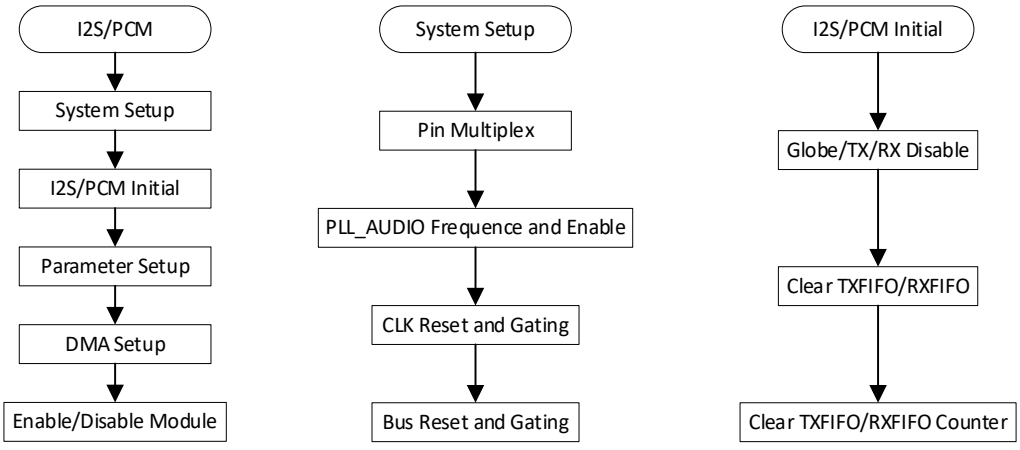


16.2.5 Programming Guidelines

The software operation of the I2S/PCM is divided into five steps: system setup, I2S/PCM initialization, the channel setup, DMA setup, and Enable/Disable module.

The following figure shows the whole operation flow of I2S/PCM.

Figure 16-20 I2S/PCM Operation Flow



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16.2.5.1 System Setup and I2S/PCM Initialization

The first step in the system setup is properly programming the GPIO. Because the I2S/PCM port is a multiplex pin. You can find the function in the pin multiplex specification.

The clock source for the I2S/PCM should be followed. Firstly, disable the PLL_AUDIO through PLL_AUDIOx Control Register[PLL_ENABLE] in the CCU. Secondly, set up the frequency of the PLL_AUDIO in the PLL_AUDIOx Control Register. After that, enable the I2S/PCM gating through the I2Sx_CLK_REG when you check out that the PLL_AUDIOx Control Register[LOCK] becomes to 1. At last, reset and enable the I2S/PCM bus gating by setting I2Sx_GAR_REG.

After the system setup, the register of I2S/PCM can be setup. Firstly, initialize the I2S/PCM. You should close the Globe Enable bit (I2S/PCM_CTL[0]), Transmitter Block Enable bit (I2S/PCM_CTL[2]), and Receiver Block Enable bit (I2S/PCM_CTL[1]) by writing 0. After that, clear the TX/RX FIFO by writing 0 to the bit[25:24] of I2S/PCM_FCTL. At last, you can clear the TX FIFO and RX FIFO counter by writing 0 to I2S/PCM_TXCNT and I2S/PCM_RXCNT.

16.2.5.2 Channel Setup and DMA Setup

First, you can set up the I2S/PCM of master and slave. The configuration can be referred to the protocol of I2S/PCM. Then, you can set up the translation mode, the sample resolution, the wide of the slot, the channel slot number, and the trigger level, and so on. The setup of the register can be found in the specification.

The I2S/PCM supports two methods to transfer the data. The most common way is DMA; the setup of DMA can be found in Section 12.1 DMA Controller (DMAC). In this module, you just enable the DRQ

16.2.5.3 Enabling and Disabling I2S/PCM

Step 1 Enable TX/RX by writing I2S/PCM_CTL[TXEN]/I2S/PCM_CTL[RXEN].

Step 2 Enable I2S/PCM by writing 1 to I2S/PCM_CTL[Globe Enable].

Step 3 Write 0 to the Globe Enable bit to disable I2S/PCM.

16.2.6 Register List

Module Name	Base Address
I2S0	0x02032000
I2S1	0x02033000
I2S2	0x02034000

Register Name	Offset	Description
I2S/PCM_CTL	0x0000	I2S/PCM Control Register
I2S/PCM_FMT0	0x0004	I2S/PCM Format Register 0

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Register Name	Offset	Description
I2S/PCM_FMT1	0x0008	I2S/PCM Format Register 1
I2S/PCM_ISTA	0x000C	I2S/PCM Interrupt Status Register
I2S/PCM_RXFIFO	0x0010	I2S/PCM RXFIFO Register
I2S/PCM_FCTL	0x0014	I2S/PCM FIFO Control Register
I2S/PCM_FSTA	0x0018	I2S/PCM FIFO Status Register
I2S/PCM_INT	0x001C	I2S/PCM DMA and Interrupt Control Register
I2S/PCM_TXFIFO	0x0020	I2S/PCM TXFIFO Register
I2S/PCM_CLKD	0x0024	I2S/PCM Clock Divide Register
I2S/PCM_TXCNT	0x0028	I2S/PCM TX Sample Counter Register
I2S/PCM_RXCNT	0x002C	I2S/PCM RX Sample Counter Register
I2S/PCM_CHCFG	0x0030	I2S/PCM Channel Configuration Register
I2S/PCM_TX0CHSEL	0x0034	I2S/PCM TX0 Channel Select Register
I2S/PCM_TX1CHSEL	0x0038	I2S/PCM TX1 Channel Select Register
I2S/PCM_TX2CHSEL	0x003C	I2S/PCM TX2 Channel Select Register
I2S/PCM_TX3CHSEL	0x0040	I2S/PCM TX3 Channel Select Register
I2S/PCM_TX0CHMAP0	0x0044	I2S/PCM TX0 Channel Mapping Register0
I2S/PCM_TX0CHMAP1	0x0048	I2S/PCM TX0 Channel Mapping Register1
I2S/PCM_TX1CHMAP0	0x004C	I2S/PCM TX1 Channel Mapping Register0
I2S/PCM_TX1CHMAP1	0x0050	I2S/PCM TX1 Channel Mapping Register1
I2S/PCM_TX2CHMAP0	0x0054	I2S/PCM TX2 Channel Mapping Register0
I2S/PCM_TX2CHMAP1	0x0058	I2S/PCM TX2 Channel Mapping Register1
I2S/PCM_TX3CHMAP0	0x005C	I2S/PCM TX3 Channel Mapping Register0
I2S/PCM_TX3CHMAP1	0x0060	I2S/PCM TX3 Channel Mapping Register1
I2S/PCM_RXCHSEL	0x0064	I2S/PCM RX Channel Select Register
I2S/PCM_RXCHMAP0	0x0068	I2S/PCM RX Channel Mapping Register0
I2S/PCM_RXCHMAP1	0x006C	I2S/PCM RX Channel Mapping Register1
I2S/PCM_RXCHMAP2	0x0070	I2S/PCM RX Channel Mapping Register2
I2S/PCM_RXCHMAP3	0x0074	I2S/PCM RX Channel Mapping Register3

16.2.7 Register Description

16.2.7.1 0x0000 I2S/PCM Control Register (Default Value: 0x0006_0000)

Offset: 0x0000			Register Name: I2S/PCM_CTL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/

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Offset: 0x0000			Register Name: I2S/PCM_CTL
Bit	Read/Write	Default/Hex	Description
21	R/W	0x0	RX_SYNC_EN_START Only if RX_SYNC_EN is set 1, RX_SYNC_EN_START can take effect. I2S0/I2S1/I2S2/DMIC/OWA RX Synchronize Enable Start. 0: Disable 1: Enable
20	R/W	0x0	RX_SYNC_EN I2S RX Synchronize Enable 0: Disable 1: Enable
19	/	/	/
18	R/W	0x1	BCLK_OUT Bit Clock Direction Select 0: Input 1: Output
17	R/W	0x1	LRCK_OUT LR Clock Direction Select 0: Input 1: Output
16:12	/	/	/
11	R/W	0x0	DOUT3_EN Data3 Output Enable 0: Disable, Hi-Z State 1: Enable
10	R/W	0x0	DOUT2_EN Data2 Output Enable 0: Disable, Hi-Z State 1: Enable
9	R/W	0x0	DOUT1_EN Data1 Output Enable 0: Disable, Hi-Z State 1: Enable
8	R/W	0x0	DOUT0_EN Data0 Output Enable 0: Disable, Hi-Z State 1: Enable
7	/	/	/
6	R/W	0x0	DOUT_MUTE_EN Data Output Mute Enable 0: Normal Transfer 1: Force DOUT to Output 0

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Offset: 0x0000			Register Name: I2S/PCM_CTL
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	MODE_SEL Mode Selection 00: PCM Mode offset 0: Long Frame offset 1: Short Frame 01: Left Mode offset 0: LJ Mode offset 1: I2S Mode 10: Right-Justified Mode 11: Reserved
3	R/W	0x0	LOOP Loop Back Test 0: Normal Mode 1: Loop Back Test When Set '1' , Connecting the DOUT with the DIN.
2	R/W	0x0	TXEN Transmitter Block Enable 0: Disable 1: Enable
1	R/W	0x0	RXEN Receiver Block Enable 0: Disable 1: Enable
0	R/W	0x0	GEN Globe Enable A disable on this bit overrides any other block or channel enables. 0: Disable 1: Enable

16.2.7.2 0x0004 I2S/PCM Format Register 0 (Default Value: 0x0000_0033)

Offset: 0x0004			Register Name: I2S/PCM_FMT0
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30	R/W	0x0	LRCK_WIDTH (Only Apply in PCM Mode) LRCK Width 0: LRCK = 1 BCLK Width (Short Frame) 1: LRCK = 2 BCLK Width (Long Frame)
29:20	/	/	/

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Offset: 0x0004			Register Name: I2S/PCM_FMT0
Bit	Read/Write	Default/Hex	Description
19	R/W	0x0	LRCK_POLARITY When Apply in I2S/Left-Justified/Right-Justified Mode: 0: Left Channel When LRCK is Low 1: Left Channel When LRCK is High When Apply in PCM Mode: 0: PCM LRCK Asserted at the Negative Edge 1: PCM LRCK Asserted at the Positive Edge
18	/	/	/
17:8	R/W	0x0	LRCK_PERIOD It is used to program the number of BCLKs per channel of sample frame. This value is interpreted as follow: PCM Mode: Number of BCLKs within (Left + Right) channel width. I2S/Left-Justified/Right-Justified Mode: Number of BCLKs within each individual channel width (Left or Right). N+1 For example: N = 7: 8 BCLKs width (p) N = 1023: 1024 BCLKs width
7	R/W	0x0	BCLK_POLARITY BCLK Polarity Select 0: Normal Mode. DOUT Drive Data at Negative Edge 1: Invert Mode. DOUT Drive Data at Positive Edge
6:4	R/W	0x3	SR Sample Resolution 000: Reserved 001: 8-bit 010: 12-bit 011: 16-bit 100: 20-bit 101: 24-bit 110: 28-bit 111: 32-bit

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Offset: 0x0004			Register Name: I2S/PCM_FMT0
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	EDGE_TRANSFER Edge Transfer 0: DOUT Drive Data and DIN Sample Data at the Different BCLK Edge 1: DOUT Drive Data and DIN Sample Data at the Same BCLK Edge BCLK_POLARITY = 0, EDGE_TRANSFER = 0, DIN Sample Data at Positive Edge; BCLK_POLARITY = 0, EDGE_TRANSFER = 1, DIN Sample Data at Negative Edge; BCLK_POLARITY = 1, EDGE_TRANSFER = 0, DIN Sample Data at Negative Edge; BCLK_POLARITY = 1, EDGE_TRANSFER = 1, DIN Sample Data at Positive Edge.
2:0	R/W	0x3	SW Slot Width Select 000: Reserved 001: 8-bit 010: 12-bit 011: 16-bit 100: 20-bit 101: 24-bit 110: 28-bit 111: 32-bit

16.2.7.3 0x0008 I2S/PCM Format Register 1 (Default Value: 0x0000_0030)

Offset: 0x0008			Register Name: I2S/PCM_FMT1
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	RX_MLS MSB / LSB First Select 0: MSB First 1: LSB First
6	R/W	0x0	TX_MLS MSB / LSB First Select 0: MSB First 1: LSB First

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Offset: 0x0008			Register Name: I2S/PCM_FMT1
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x3	SEXT Sign Extend in Slot [Sample Resolution < Slot Width] 00: Zeros or Audio Gain Padding at LSB Position 01: Sign Extension at MSB Position 10: Reserved 11: Transfer 0 after each Sample in each Slot
3:2	R/W	0x0	RX_PDM PCM Data Mode 00: Linear PCM 01: Reserved 10: 8-bits u-law 11: 8-bits A-law
1:0	R/W	0x0	TX_PDM PCM Data Mode 00: Linear PCM 01: Reserved 10: 8-bits u-law 11: 8-bits A-law

16.2.7.4 0x000C I2S/PCM Interrupt Status Register (Default Value: 0x0000_0010)

Offset: 0x000C			Register Name: I2S/PCM_ISTA
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6	R/W1C	0x0	TXU_INT TXFIFO Underrun Pending Interrupt 0: No Pending Interrupt 1: TXFIFO Underrun Pending Interrupt Write 1 to clear this interrupt.
5	R/W1C	0x0	TXO_INT TXFIFO Overrun Pending Interrupt 0: No Pending Interrupt 1: TXFIFO Overrun Pending Interrupt Write 1 to clear this interrupt.
4	R	0x1	TXE_INT TXFIFO Empty Pending Interrupt 0: No Pending IRQ 1: TXFIFO Empty Pending Interrupt When Data in TXFIFO are Less than TX Trigger Level
3	/	/	/

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Offset: 0x000C			Register Name: I2S/PCM_ISTA
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	RXU_INT RXFIFO Underrun Pending Interrupt 0: No Pending Interrupt 1: RXFIFO Underrun Pending Interrupt Write 1 to clear this interrupt.
1	R/W1C	0x0	RXO_INT RXFIFO Overrun Pending Interrupt 0: No Pending IRQ 1: RXFIFO Overrun Pending IRQ Write 1 to clear this interrupt.
0	R	0x0	RXA_INT RXFIFO Data Available Pending Interrupt 0: No Pending IRQ 1: Data Available Pending IRQ When Data in RXFIFO are More than RX Trigger Level

16.2.7.5 0x0010 I2S/PCM RXFIFO Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: I2S/PCM_RXFIFO
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RX_DATA RX Sample Host can get one sample by reading this register. The left channel sample data is first and then the right channel sample.

16.2.7.6 0x0014 I2S/PCM FIFO Control Register (Default Value: 0x0004_00F0)

Offset: 0x0014			Register Name: I2S/PCM_FCTL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HUB_EN Audio Hub Enable Only if TXEN set 1, HUB_EN can take effect. I2S0/I2S1/I2S2/Audio Codec/ OWA TXFIFO Hub Enable. 0: Disable 1: Enable
30:26	/	/	/
25	R/WAC	0x0	FTX Flush TX FIFO Write '1' to flush TXFIFO, self-clear to '0'.

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Offset: 0x0014			Register Name: I2S/PCM_FCTL
Bit	Read/Write	Default/Hex	Description
24	R/WAC	0x0	FRX Flush RX FIFO Write '1' to flush RXFIFO, self-clear to '0'.
23:19	/	/	/
18:12	R/W	0x40	TXTL TXFIFO Empty Trigger Level Interrupt and DMA request trigger level for TXFIFO normal condition. Trigger Level = TXTL
11:10	/	/	/
9:4	R/W	0xF	RXTL RXFIFO Trigger Level Interrupt and DMA request trigger level for RXFIFO normal condition. Trigger Level = RXTL + 1
3	/	/	/
2	R/W	0x0	TXIM TXFIFO Input Mode (Mode 0, 1) 0: Valid Data at the MSB of TXFIFO Register 1: Valid Data at the LSB of TXFIFO Register Example for 20-bits Transmitted Audio Sample: Mode 0: TXFIFO [31:0] = {APB_WDATA [31:12], 12'h0} Mode 1: TXFIFO[31:0] = {APB_WDATA[19:0], 12'h0}
1:0	R/W	0x0	RXOM RXFIFO Output Mode (Mode 0, 1, 2, 3) 00: Expanding '0' at LSB of RXFIFO Register 01: Expanding Received Sample Sign Bit at MSB of RXFIFO Register 10: Truncating Received Samples at High Half-word of RXFIFO Register and Low Half-word of RXFIFO Register is Filled by '0' 11: Truncating Received Samples at Low Half-word of RXFIFO Register and High Half-word of RXFIFO Register is Expanded by Its Sign Bit Example for 20-bits Received Audio Sample: Mode 0: APB_RDATA [31:0] = {RXFIFO [31:12], 12'h0} Mode 1: APB_RDATA [31:0] = {12{RXFIFO [31]}, RXFIFO [31:12]} Mode 2: APB_RDATA [31:0] = {RXFIFO [31:16], 16'h0} Mode 3: APB_RDATA[31:0] = {16{RXFIFO[31], RXFIFO[31:16]}

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16.2.7.7 0x0018 I2S/PCM FIFO Status Register (Default Value: 0x1080_0080)

Offset: 0x0018			Register Name: I2S/PCM_FSTA
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R	0x1	TXE TXFIFO Empty 0: No Room for New Sample in TXFIFO 1: More than One Room for New Sample in TXFIFO (>= 1 Word)
27:24	/	/	/
23:16	R	0x80	TXE_CNT TXFIFO Empty Space Word Counter
15:9	/	/	/
8	R	0x0	RXA RXFIFO Available 0: No Available Data in RXFIFO 1: More than One Sample in RXFIFO (>= 1 Word)
7	R	0x1	PLACE_HOLDER NO Meaning
6:0	R	0x0	RXA_CNT RXFIFO Available Sample Word Counter

16.2.7.8 0x001C I2S/PCM DMA and Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: I2S/PCM_INT
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	TX_DRQ TXFIFO Empty DRQ Enable 0: Disable 1: Enable
6	R/W	0x0	TXUI_EN TXFIFO Underrun Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	TXOI_EN TXFIFO Overrun Interrupt Enable 0: Disable 1: Enable When set to 1 , an interrupt happens when writing new audio data if TXFIFO is full.

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Offset: 0x001C			Register Name: I2S/PCM_INT
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	TXEI_EN TXFIFO Empty Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	RX_DRQ RXFIFO Data Available DRQ Enable 0: Disable 1: Enable When set to '1', RXFIFO DMA request line is asserted if data is available in RXFIFO.
2	R/W	0x0	RXUI_EN RXFIFO Underrun Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	RXOI_EN RXFIFO Overrun Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	RXAI_EN RXFIFO Data Available Interrupt Enable 0: Disable 1: Enable

16.2.7.9 0x0020 I2S/PCM TXFIFO Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: I2S/PCM_TXFIFO
Bit	Read/Write	Default/Hex	Description
31:0	W	0x0	TX_DATA TX Sample Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample.

16.2.7.10 0x0024 I2S/PCM Clock Divide Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: I2S/PCM_CLKD
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/

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Offset: 0x0024			Register Name: I2S/PCM_CLKD
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	MCLKO_EN MCLK Out Enable 0: Disable MCLK Output 1: Enable MCLK Output Notes: Whether in Slave or Master mode, when this bit is set to '1', MCLK should be output.
7:4	R/W	0x0	BCLKDIV BCLK Divide Ratio from PLL_AUDIO 0000: Reserved 0001: Divide by 1 0010: Divide by 2 0011: Divide by 4 0100: Divide by 6 0101: Divide by 8 0110: Divide by 12 0111: Divide by 16 1000: Divide by 24 1001: Divide by 32 1010: Divide by 48 1011: Divide by 64 1100: Divide by 96 1101: Divide by 128 1110: Divide by 176 1111: Divide by 192
3:0	R/W	0x0	MCLKDIV MCLK Divide Ratio from PLL_AUDIO 0000: Reserved 0001: Divide by 1 0010: Divide by 2 0011: Divide by 4 0100: Divide by 6 0101: Divide by 8 0110: Divide by 12 0111: Divide by 16 1000: Divide by 24 1001: Divide by 32 1010: Divide by 48 1011: Divide by 64 1100: Divide by 96 1101: Divide by 128 1110: Divide by 176 1111: Divide by 192

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16.2.7.11 0x0028 I2S/PCM TX Sample Counter Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: I2S/PCM_TXCNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TX_CNT TX Sample Counter The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial value at any time. After being updated by the initial value, the counter register should count on base of this initial value.

16.2.7.12 0x002C I2S/PCM RX Sample Counter Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: I2S/PCM_RXCNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RX_CNT RX Sample Counter The audio sample number of writing into RXFIFO. When one sample is written by Digital Audio Engine, the RX sample counter register increases by one. The RX sample counter register can be set to any initial value at any time. After been updated by the initial value, the counter register should count on base of this initial value.

16.2.7.13 0x0030 I2S/PCM Channel Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: I2S/PCM_CHCFG
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	TX_SLOT_HIZ TX Slot Value in Half Cycle of BCLK 0: Normal Mode for the Last Half Cycle of BCLK in the Slot 1: Turn to Hi-Z State for the Last Half Cycle of BCLK in the Slot

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Offset: 0x0030			Register Name: I2S/PCM_CHCFG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	TX_STATE The state of transmission line When there is No Transmission 0: Transfer Level 0 When not Transferring Slot 1: Turn to Hi-Z State (TDM) When not Transferring Slot
7:4	R/W	0x0	RX_SLOT_NUM RX Channel/Slot Number between CPU/DMA and RXFIFO 0000: 1 Channel or Slot ... 0111: 8 Channels or Slots 1000: 9 Channels or Slots ... 1111:16 Channels or Slots
3:0	R/W	0x0	TX_SLOT_NUM TX Channel/Slot Number Which between CPU/DMA and TXFIFO 0000: 1 Channel or Slot ... 0111: 8 Channels or Slots 1000: 9 Channels or Slots ... 1111:16 Channels or Slots

16.2.7.14 0x0034 I2S/PCM TX0 Channel Select Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: I2S/PCM_TX0CHSEL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	TX0_OFFSET TX0 offset Tune, TX0 Data offset to LRCK 0: No offset n: Data is offset by n BCLKs to LRCK

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Offset: 0x0034			Register Name: I2S/PCM_TX0CHSEL
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	TX0_CHSEL TX0 Channel (Slot) Number Select for each Output 0000:1 Channel / Slot (p) 7:8 Channels / Slots 8:9 Channels / Slots ... 15:16 Channels / Slots
15:0	R/W	0x0	TX0_CHEN TX0 Channel (Slot) Enable, bit [15:0] Refer to Slot [15:0]. When One or More Slot(s) is disabled, the Affected Slot(s) is set to disable State. 0: Disable 1: Enable

16.2.7.15 0x0038 I2S/PCM TX1 Channel Select Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: I2S/PCM_TX1CHSEL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	TX1_OFFSET TX1 offset Tune, TX1 Data offset to LRCK 0: No offset n: Data is offset by n BCLKs to LRCK
19:16	R/W	0x0	TX1_CHSEL TX1 Channel (Slot) Number Select for each Output 0000: 1 Channel/Slot (p) 0111: 8 Channels/Slots 1000: 9 Channels/Slots ... 1111: 16 Channels/Slots
15:0	R/W	0x0	TX1_CHEN TX1 Channel (Slot) Enable, bit [15:0] Refer to Slot [15:0]. When One or More Slot(s) is disabled, the Affected Slot(s) is set to Disable State. 0: Disable 1: Enable

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16.2.7.16 0x003C I2S/PCM TX2 Channel Select Register (Default Value: 0x0000_0000)

Offset: 0x003C			Register Name: I2S/PCM_TX2CHSEL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	TX2_OFFSET TX2 offset Tune, TX2 Data offset to LRCK 0: No offset n: Data is offset by n BCLKs to LRCK
19:16	R/W	0x0	TX2_CHSEL TX2 Channel (Slot) Number Select for each Output 0000: 1 Channel/Slot (p) 0111: 8 Channels/Slots 1000: 9 Channels/Slots ... 1111: 16 Channels/Slots
15:0	R/W	0x0	TX2_CHEN TX2 Channel (Slot) Enable, bit [15:0] Refer to Slot [15:0]. When One or More Slot(s) is(are) Disabled, the Affected Slot(s) is(are) Set to Disable State. 0: Disable 1: Enable

16.2.7.17 0x0040 I2S/PCM TX3 Channel Select Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: I2S/PCM_TX3CHSEL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	TX3_OFFSET TX3 offset Tune, TX Data offset to LRCK 0: No offset n: Data is offset by n BCLKs to LRCK
19:16	R/W	0x0	TX3_CHSEL TX3 Channel (Slot) Number Select for each Output 0000:1 Channel / Slot (p) 7:8 Channels / Slots 8:9 Channels / Slots ... 15:16 Channels / Slots

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Offset: 0x0040			Register Name: I2S/PCM_TX3CHSEL
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	TX3_CHEN TX3 Channel (Slot) Enable, bit [15:0] Refer to Slot [15:0]. When One or More Slot(s) is disabled, the Affected Slot(s) is set to Disable State. 0: Disable 1: Enable

16.2.7.18 0x0044 I2S/PCM TX0 Channel Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: I2S/PCM_TX0CHMAP0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX0_CH15_MAP TX0 Channel 15 Mapping 0: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX0_CH14_MAP TX0 Channel 14 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX0_CH13_MAP TX0 Channel 13 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0044			Register Name: I2S/PCM_TX0CHMAP0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	TX0_CH12_MAP TX0 Channel 12 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX0_CH11_MAP TX0 Channel 11 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX0_CH10_MAP TX0 Channel 10 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX0_CH9_MAP TX0 Channel 9 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX0_CH8_MAP TX0 Channel 8 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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16.2.7.19 0x0048 I2S/PCM TX0 Channel Mapping Register1 (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: I2S/PCM_TX0CHMAP1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX0_CH7_MAP TX0 Channel 7 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX0_CH6_MAP TX0 Channel 6 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX0_CH5_MAP TX0 Channel 5 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX0_CH4_MAP TX0 Channel 4 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX0_CH3_MAP TX0 Channel 3 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0048			Register Name: I2S/PCM_TX0CHMAP1
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0x0	TX0_CH2_MAP TX0 Channel 2 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX0_CH1_MAP TX0 Channel 1 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX0_CH0_MAP TX0 Channel 0 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.20 0x004C I2S/PCM TX1 Channel Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: I2S/PCM_TX1CHMAP0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX1_CH15_MAP TX1 Channel 15 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x004C			Register Name: I2S/PCM_TX1CHMAP0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	TX1_CH14_MAP TX1 Channel 14 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX1_CH13_MAP TX1 Channel 13 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX1_CH12_MAP TX1 Channel 12 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX1_CH11_MAP TX1 Channel 11 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX1_CH10_MAP TX1 Channel 10 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x004C			Register Name: I2S/PCM_TX1CHMAP0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	TX1_CH9_MAP TX1 Channel 9 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX1_CH8_MAP TX1 Channel 8 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.21 0x0050 I2S/PCM TX1 Channel Mapping Register1 (Default Value: 0x0000_0000)

Offset: 0x0050			Register Name: I2S/PCM_TX1CHMAP1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX1_CH7_MAP TX1 Channel 7 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX1_CH6_MAP TX1 Channel 6 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0050			Register Name: I2S/PCM_TX1CHMAP1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	TX1_CH5_MAP TX1 Channel 5 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX1_CH4_MAP TX1 Channel 4 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX1_CH3_MAP TX1 Channel 3 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX1_CH2_MAP TX1 Channel 2 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX1_CH1_MAP TX1 Channel 1 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0050			Register Name: I2S/PCM_TX1CHMAP1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	TX1_CH0_MAP TX1 Channel 0 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.22 0x0054 I2S/PCM TX2 Channel Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x0054			Register Name: I2S/PCM_TX2CHMAP0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX2_CH15_MAP TX2 Channel 15 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX2_CH14_MAP TX2 Channel 14 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX2_CH13_MAP TX2 Channel 13 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0054			Register Name: I2S/PCM_TX2CHMAP0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	TX2_CH12_MAP TX2 Channel 12 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX2_CH11_MAP TX2 Channel 11 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX2_CH10_MAP TX2 Channel 10 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX2_CH9_MAP TX2 Channel 9 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX2_CH8_MAP TX2 Channel 8 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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16.2.7.23 0x0058 I2S/PCM TX2 Channel Mapping Register1 (Default Value: 0x0000_0000)

Offset: 0x0058			Register Name: I2S/PCM_TX2CHMAP1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX2_CH7_MAP TX2 Channel 7 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX2_CH6_MAP TX2 Channel 6 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX2_CH5_MAP TX2 Channel 5 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX2_CH4_MAP TX2 Channel 4 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX2_CH3_MAP TX2 Channel 3 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0058			Register Name: I2S/PCM_TX2CHMAP1
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0x0	TX2_CH2_MAP TX2 Channel 2 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX2_CH1_MAP TX2 Channel 1 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX2_CH0_MAP TX2 Channel 0 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.24 0x005C I2S/PCM TX3 Channel Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x005C			Register Name: I2S/PCM_TX3CHMAP0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX3_CH15_MAP TX3 Channel 15 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x005C			Register Name: I2S/PCM_TX3CHMAP0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	TX3_CH14_MAP TX3 Channel 14 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:20	R/W	0x0	TX3_CH13_MAP TX3 Channel 13 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX3_CH12_MAP TX3 Channel 12 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX3_CH11_MAP TX3 Channel 11 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX3_CH10_MAP TX3 Channel 10 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x005C			Register Name: I2S/PCM_TX3CHMAP0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	TX3_CH9_MAP TX3 Channel 9 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
3:0	R/W	0x0	TX3_CH8_MAP TX3 Channel 8 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.25 0x0060 I2S/PCM TX3 Channel Mapping Register1 (Default Value: 0x0000_0000)

Offset: 0x0060			Register Name: I2S/PCM_TX3CHMAP1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	TX3_CH7_MAP TX3 Channel 7 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
27:24	R/W	0x0	TX3_CH6_MAP TX3 Channel 6 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0060			Register Name: I2S/PCM_TX3CHMAP1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	TX3_CH5_MAP TX3 Channel 5 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
19:16	R/W	0x0	TX3_CH4_MAP TX3 Channel 4 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:12	R/W	0x0	TX3_CH3_MAP TX3 Channel 3 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
11:8	R/W	0x0	TX3_CH2_MAP TX3 Channel 2 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:4	R/W	0x0	TX3_CH1_MAP TX3 Channel 1 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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Offset: 0x0060			Register Name: I2S/PCM_TX3CHMAP1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	TX3_CH0_MAP TX3 Channel 0 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.26 0x0064 I2S/PCM RX Channel Select Register (Default Value: 0x0000_0000)

Offset: 0x0064			Register Name: I2S/PCM_RXCHSEL
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	RX_OFFSET RX offset Tune, RX Data offset to LRCK 0000: No offset n: Data is offset by n BCLKs to LRCK
19:16	R/W	0x0	RX_CHSEL RX Channel (Slot) Number Select for Input 0000: 1 Channel / Slot (p) 0111: 8 Channels / Slots 1000: 9 Channels / Slots ... 1111: 16 Channels / Slots
15:0	/	/	/

16.2.7.27 0x0068 I2S/PCM RX Channel Mapping Register0 (Default Value: 0x0000_0000)

Offset: 0x0068			Register Name: I2S/PCM_RXCHMAP0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	RX_CH15_SELECT Rx channel 15 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3

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Offset: 0x0068			Register Name: I2S/PCM_RXCHMAP0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	RX_CH15_MAP RX Channel 15 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:22	/	/	/
21:20	R/W	0x0	RX_CH14_SELECT Rx channel 14 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
19:16	R/W	0x0	RX_CH14_MAP RX Channel 14 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:14	/	/	/
13:12	R/W	0x0	RX_CH13_SELECT Rx channel 13 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
11:8	R/W	0x0	RX_CH13_MAP RX Channel 13 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:6	/	/	/

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Offset: 0x0068			Register Name: I2S/PCM_RXCHMAP0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	RX_CH12_SELECT Rx channel 12 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
3:0	R/W	0x0	RX_CH12_MAP RX Channel 12 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.28 0x006C I2S/PCM RX Channel Mapping Register1 (Default Value: 0x0000_0000)

Offset: 0x006C			Register Name: I2S/PCM_RXCHMAP1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	RX_CH11_SELECT Rx channel 11 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
27:24	R/W	0x0	RX_CH11_MAP RX Channel 11 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:22	/	/	/
21:20	R/W	0x0	RX_CH10_SELECT Rx channel 10 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3

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Offset: 0x006C			Register Name: I2S/PCM_RXCHMAP1
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	RX_CH10_MAP RX Channel 10 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:14	/	/	/
13:12	R/W	0x0	RX_CH9_SELECT Rx channel 9 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
11:8	R/W	0x0	RX_CH9_MAP RX Channel 9 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:6	/	/	/
5:4	R/W	0x0	RX_CH8_SELECT Rx channel 8 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
3:0	R/W	0x0	RX_CH8_MAP RX Channel 8 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.29 0x0070 I2S/PCM RX Channel Mapping Register2 (Default Value: 0x0000_0000)

Offset: 0x0070	Register Name: I2S/PCM_RXCHMAP2

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Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	RX_CH7_SELECT Rx channel 7 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
27:24	R/W	0x0	RX_CH7_MAP RX Channel 7 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:22	/	/	/
21:20	R/W	0x0	RX_CH6_SELECT Rx channel 6 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
19:16	R/W	0x0	RX_CH6_MAP RX Channel 6 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:14	/	/	/
13:12	R/W	0x0	RX_CH5_SELECT Rx channel 5 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3

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11: 8	R/W	0x0	RX_CH5_MAP RX Channel 5 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:6	/	/	/
5:4	R/W	0x0	RX_CH4_SELECT Rx channel 4 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
3:0	R/W	0x0	RX_CH4_MAP RX Channel 4 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

16.2.7.30 0x0074 I2S/PCM RX Channel Mapping Register3 (Default Value: 0x0000_0000)

Offset: 0x0074			Register Name: I2S/PCM_RXCHMAP3
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	RX_CH3_SELECT Rx channel 4 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
27:24	R/W	0x0	RX_CH3_MAP RX Channel 3 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
23:22	/	/	/

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Offset: 0x0074			Register Name: I2S/PCM_RXCHMAP3
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x0	RX_CH2_SELECT Rx channel 2 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
19:16	R/W	0x0	RX_CH2_MAP RX Channel 2 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
15:14	/	/	/
13:12	R/W	0x0	RX_CH1_SELECT Rx channel 1 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3
11:8	R/W	0x0	RX_CH1_MAP RX Channel 1 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample
7:6	/	/	/
5:4	R/W	0x0	RX_CH0_SELECT Rx channel 0 select 00: SDI0 01: SDI1 10: SDI2 11: SDI3

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Offset: 0x0074			Register Name: I2S/PCM_RXCHMAP3
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	RX_CH0_MAP RX Channel 0 Mapping 0000: 1st Sample (p) 0111: 8th Sample 1000: 9th Sample ... 1111: 16th Sample

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16.3 DMIC

16.3.1 Overview

The Digital MIC controller supports one 8-channel digital microphone interface and can output 128 fs or 64 fs (fs = ADC sample rate).

The DMIC controller includes the following features:

- One DMIC interface
- Support PDM master receiver mode
- Receive Data FIFO
  - RX_FIFO with 128 depth for audio data receiving
  - Programmable FIFO thresholds
- Support DMA slave interface
- Support interrupts
- Supports multiple application scenarios
  - Up to 8 channels
  - Each channel can be enabled and disabled individually
  - Support sample rate: 8 kHz to 48 kHz
  - Support oversample rate 64*fs/128*fs
  - Support sample resolution 16bits/24bits
- Support programmable channels exchange and mapping
- Support High Pass Filter and Digital Volume Controller

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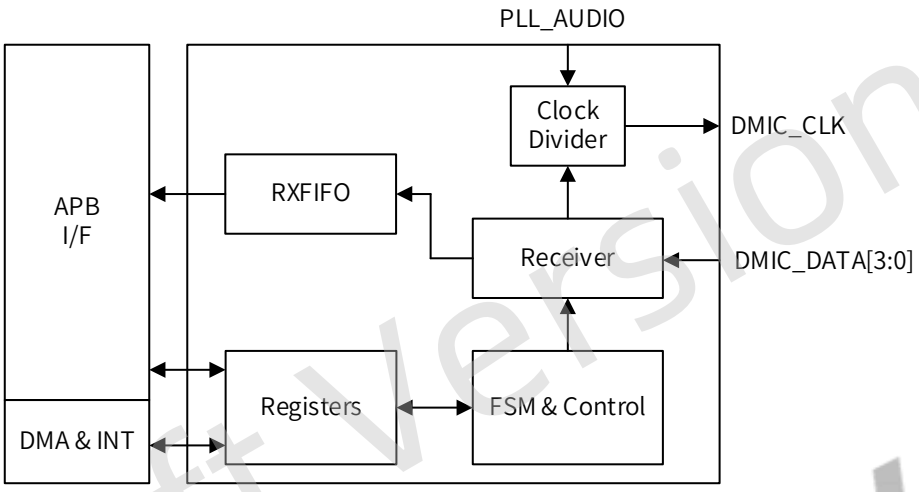
16.3.2 Block Diagram

The following figure shows a block diagram of the DMIC.

Figure 16-21 DMIC Block Diagram



16.3.3 External Signals

The following table describes the external signals of DMIC.

Table 16-3 DMIC External Signals

Signal Name	Description	Type
DMIC-DATA[3:0]	Digital Microphone Data Input	I
DMIC-CLK	Digital Microphone Clock Output	O

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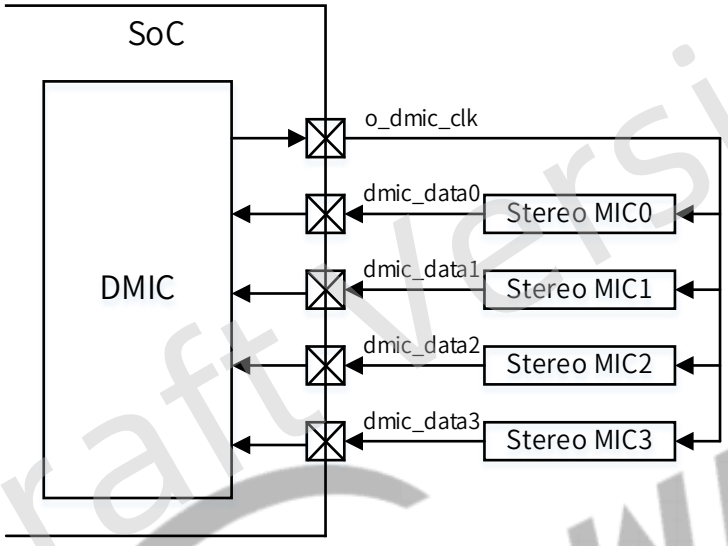
16.3.4 Functional Description

16.3.4.1 Typical Application

Stereo Microphone Application

The following figure shows stereo microphone application diagram of the DMIC.

Figure 16-22 DMIC Stereo Microphones Application Diagram



NOTE

DMIC supports simultaneous connections of maximum 4 stereo microphones. The number of connected microphones is configurable in different scenarios.

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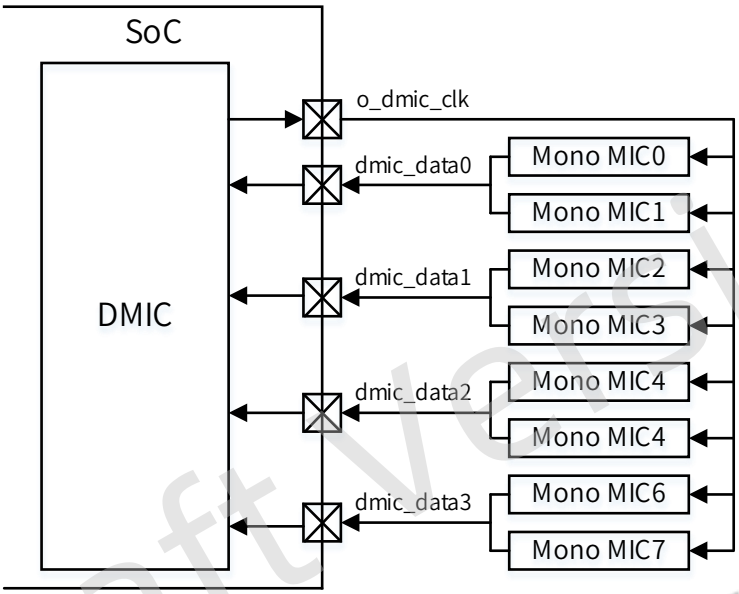
Mono Microphones Application

The following shows Mono Microphones Application Diagram of the DMIC.

Figure 16-23 DMIC Mono Microphones Application Diagram



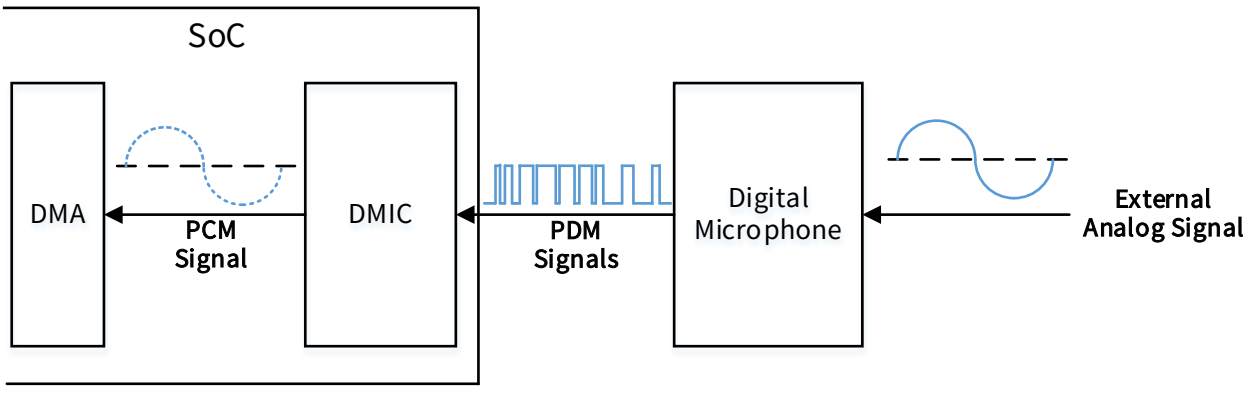
NOTE

DMIC supports simultaneous connections of maximum 8 mono microphones. The number of connected microphones is configurable in different scenarios.

16.3.4.2 Pulse Density Modulation (PDM) Format

PDM is a modulation method which uses binary number 0/1 to express analog signals. The amplitude of analog signals corresponds to output pulse density. The following figure shows the audio data format conversion in DMIC digital path.

Figure 16-24 PDM Format



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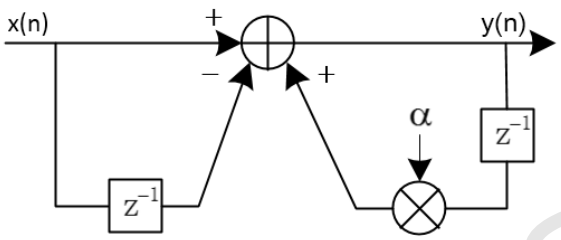
16.3.4.3 High Pass Filter (HPF) Function

HPF is mainly responsible for filtering the DC component of audio signals. This function can be enabled or disabled individually.

Figure 16-25 HPF Logic Structure



16.3.4.4 Digital Volume Control (DVC) Function

The gain adjusting range of DVC is from -119.25dB to 71.25dB (default 0 dB) with a step of 0.75dB. DVC is able to prevent Pop-Click noise and can be enabled or disabled individually.

Table 16-4 DVC Gain Configuration

Desired Gain(dB)	DVC.Reg(Dec)	DVC.Reg(Hex)
+71.25	255	FF
+70.50	254	FE
...	...	...
+0.75	161	A1
0 (default)	160	A0
-0.75	159	9F
...	...	...
-118.50	2	2
-119.25	1	1
Mute	0	0

16.3.5 Programming Guidelines

The software operation of the DMIC is divided into five steps: system setup, DMIC initialization, channel setup, DMA setup, and Enable/Disable module.

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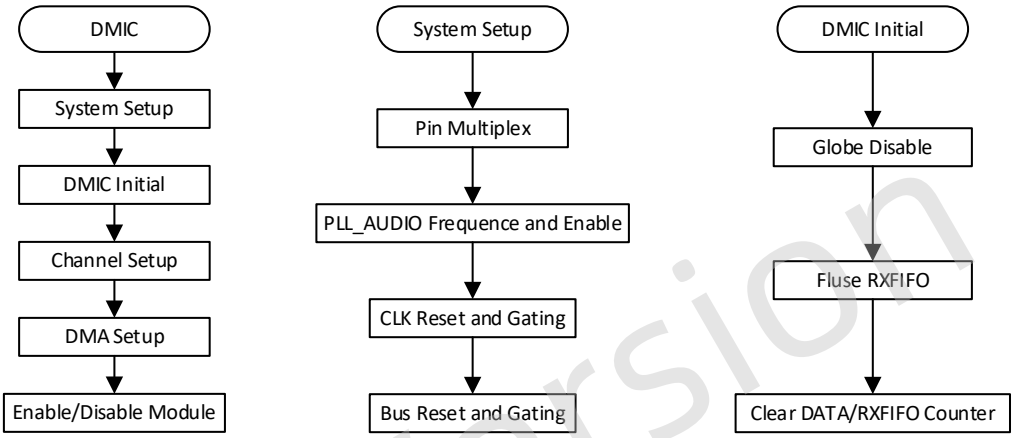
The following figure shows the flow chart of the whole operation, the system setup, and the DMIC initialization.

Figure 16-26 DMIC Operation Mode



16.3.5.1 System Setup and DMIC Initialization

The first step in the system setup is properly programming the GPIO because the DMIC port is a multiplex pin. For functions of the multiplex pins, refer to the Section 17.9 GPIO

Perform the following steps for the clock source. Firstly, disable the PLL_AUDIO through PLL_AUDIOx Control Register[PLL_ENABLE]. Secondly, set up the frequency of the PLL_AUDIO in PLL_AUDIOx Control Register. Then enable PLL_AUDIO. After that, enable the DMIC gating through DMIC_CLK_REG when you check out that the LOCK bit of PLL_AUDIOx Control Register becomes 1. At last, reset and enable the DMIC bus gating by DMIC_GAR_REG.

After the system setup, the register of DMIC can be setup. Firstly, initialize the DMIC. You should close the globe enable bit (DMIC_EN[8]), data channel enable bit (DMIC_EN[7:0]) by writing 0 to it. After that, flush the RXFIFO by writing 1 to DMIC_RXFIFO_CTR[31]. At last, you can clear the Data/RXFIFO counter by writing 1 to DMIC_RXFIFO_STA, DMIC_CNT.

16.3.5.2 Channel Setup and DMA Setup

You can set up the sample rate, the sample resolution, the over-sample rate, the channel number, the RXFIFO output mode, the RXFIFO trigger level, and so on. The setup of the register can be found in the specification.

The DMIC supports two methods to transfer the data. The most common way is DMA; the setup of DMA can be found in the Section 12.1 DMA Controller (DMAC). In this module, you just enable the DRQ.

16.3.5.3 Enabling and Disabling DMIC

Step 1 Enable the data channel enable bit (DMIC_EN[7:0]) by writing 1 to it.

Step 2 Enable DMIC by writing 1 to the Globe Enable bit (DMIC_EN[8]).

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Step 3 Write 0 to DMIC_EN[8] to disable DMIC.

16.3.6 Register List

Module Name	Base Address
DMIC	0x02031000

Register Name	Offset	Description
DMIC_EN	0x0000	DMIC Enable Control Register
DMIC_SR	0x0004	DMIC Sample Rate Register
DMIC_CTR	0x0008	DMIC Control Register
DMIC_DATA	0x0010	DMIC DATA Register
DMIC_INTC	0x0014	DMIC Interrupt Control Register
DMIC_INTS	0x0018	DMIC Interrupt Status Register
DMIC_RXFIFO_CTR	0x001C	DMIC RXFIFO Control Register
DMIC_RXFIFO_STA	0x0020	DMIC RXFIFO Status Register
DMIC_CH_NUM	0x0024	DMIC Channel Number Register
DMIC_CH_MAP	0x0028	DMIC Channel Mapping Register
DMIC_CNT	0x002C	DMIC Counter Register
DATA0_DATA1_VOL_CTR	0x0030	DATA0 and DATA1 Volume Control Register
DATA2_DATA3_VOL_CTR	0x0034	DATA2 and DATA3 Volume Control Register
HPF_EN_CTR	0x0038	High Pass Filter Enable Control Register
HPF_COEF_REG	0x003C	High Pass Filter Coef Register
HPF_GAIN_REG	0x0040	High Pass Filter Gain Register

16.3.7 Register Description

16.3.7.1 0x0000 DMIC Enable Control Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: DMIC_EN
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	RX_SYNC_EN_START Audio Subsys RX Synchronize Enable Start Includes I2S0/I2S1/I2S2/ DMIC/OWA. The bit will take effect only when RX_SYNC_EN is set to 1. 0: Disable 1: Enable

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Offset: 0x0000			Register Name: DMIC_EN
Bit	Read/Write	Default/Hex	Description
28	R/W	0x0	RX_SYNC_EN DMIC RX Synchronize Enable 0: Disable 1: Enable
27:9	/	/	/
8	R/W	0x0	GLOBE_EN DMIC Globe Enable 0: Disable 1: Enable
7	R/W	0x0	DATA3_CHR_EN DATA3 Right Channel Enable 0: Disable 1: Enable
6	R/W	0x0	DATA3_CHL_EN DATA3 Left Channel Enable 0: Disable 1: Enable
5	R/W	0x0	DATA2_CHR_EN DATA2 Right Channel Enable 0: Disable 1: Enable
4	R/W	0x0	DATA2_CHL_EN DATA2 Left Channel Enable 0: Disable 1: Enable
3	R/W	0x0	DATA1_CHR_EN DATA1 Right Channel Enable 0: Disable 1: Enable
2	R/W	0x0	DATA1_CHL_EN DATA1 Left Channel Enable 0: Disable 1: Enable
1	R/W	0x0	DATA0_CHR_EN DATA0 Right Channel Enable 0: Disable 1: Enable
0	R/W	0x0	DATA0_CHL_EN DATA0 Left Channel Enable 0: Disable 1: Enable

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16.3.7.2 0x0004 DMIC Sample Rate Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: DMIC_SR
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2:0	R/W	0x0	DMIC_SR Sample Rate of DMIC 000: 48 kHz 010: 24 kHz 100: 12 kHz 110: Reserved 001: 32 kHz 011: 16 kHz 101: 8 kHz 111: Reserved 44.1 kHz/22.05 kHz/11.025 kHz can be supported by Audio PLL Configure Bit.

16.3.7.3 0x0008 DMIC Control Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: DMIC_CTR
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:9	R/W	0x0	DMICFDT DMIC RXFIFO Delay Time for Writing Data after GLOBE_EN 00: 5 ms 01: 10 ms 10: 20 ms 11: 30 ms
8	R/W	0x0	DMICDFEN DMIC RXFIFO Delay Function for Writing Data after GLOBE_EN 0: Disable 1: Enable
7	R/W	0x0	DATA3_LR_SWAP_EN DATA3 Left Data and Right Data Swap Enable 0: Disable 1: Enable
6	R/W	0x0	DATA2_LR_SWAP_EN DATA2 Left Data and Right Data Swap Enable 0: Disable 1: Enable

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Offset: 0x0008			Register Name: DMIC_CTR
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	DATA1_LR_SWAP_EN DATA1 Left Data and Right Data Swap Enable 0: Disable 1: Enable
4	R/W	0x0	DATA0_LR_SWAP_EN DATA0 Left Data and Right Data Swap Enable 0: Disable 1: Enable
3:1	/	/	/
0	R/W	0x0	DMIC Oversample Rate 0: 128 (Supports 8 kHz to 24 kHz) 1: 64 (Supports 16 kHz to 48 kHz)

16.3.7.4 0x0010 DMIC DATA Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: DMIC_DATA
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	DMIC_DATA

16.3.7.5 0x0014 DMIC Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: DMIC_INTC
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	RXFIFO_DRQ_EN DMIC RXFIFO Data Available DRQ Enable 0: Disable 1: Enable
1	R/W	0x0	RXFIFO_OVERRUN_IRQ_EN DMIC RXFIFO Overrun IRQ Enable 0: Disable 1: Enable
0	R/W	0x0	DATA_IRQ_EN DMIC RXFIFO Data Available IRQ Enable 0: Disable 1: Enable

16.3.7.6 0x0018 DMIC Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0018	Register Name: DMIC_INTS

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Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W1C	0x0	RXFIFO_OVERRUN_IRQ_PENDING DMIC RXFIFO Overrun Pending Interrupt 0: No pending IRQ 1: RXFIFO overrun pending IRQ Writing '1' to clear this interrupt.
0	R/ W1C	0x0	RXFIFO_DATA_IRQ_PENDING DMIC RXFIFO Data Available Pending Interrupt 0: No pending IRQ 1: Data available pending IRQ Writing '1' to clear this interrupt or automatically clear if fifodeep >= rftlvl (RXFIFO Trigger Level).

16.3.7.7 0x001C DMIC RXFIFO Control Register (Default Value: 0x0000_0040)

Offset: 0x001C			Register Name: DMIC_RXFIFO_CTR
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	DMIC_RXFIFO_FLUSH DMIC RXFIFO Flush Writing '1' to flush RXFIFO, self-clear to '0'.
30:10	/	/	/
9	R/W	0x0	RXFIFO_MODE RXFIFO Output Mode (Mode 0, 1) 0: Expanding '0' at LSB of RXFIFO register 1: Expanding received sample sign bit at MSB of RXFIFO register For 24-bit received audio sample: Mode 0: APB_RDATA [31:0] = {RXFIFO_O [20:0], 11'h0} Mode 1: APB_RDATA [31:0] = {8{RXFIFO_O [20]}, RXFIFO_O [20:0], 3'h0} For 16-bit received audio sample: Mode 0: APB_RDATA [31:0] = {RXFIFO_O [20:5], 16'h0} Mode 1: APB_RDATA[31:0] = {16{RXFIFO_O[20]}, RXFIFO_O[20:5]}
8	R/W	0x0	SR Sample Resolution 0: 16-bit 1: 24- bit

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Offset: 0x001C			Register Name: DMIC_RXFIFO_CTR
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x40	RXFIFO_TRG_LEVEL RXFIFO Trigger Level (TRLV [7:0]) Interrupt and DMA request trigger level for DMIC RXFIFO normal condition IRQ/DRQ Generated when WLEVEL > TRLV [7:0]) Note: WLEVEL represents the number of valid samples in the DMIC RXFIFO

16.3.7.8 0x0020 DMIC RXFIFO Status Register (Default Value: 0x0000_0100)

Offset: 0x0020			Register Name: DMIC_RXFIFO_STA
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R	0x1	PLACEHOLDER Meaningless
7:0	R	0x0	DMIC_DATA_CNT DMIC RXFIFO Available Sample Word Counter

16.3.7.9 0x0024 DMIC Channel Number Register (Default Value: 0x0000_0001)

Offset: 0x0024			Register Name: DMIC_CH_NUM
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2:0	R/W	0x1	DMIC_CH_NUM DMIC enable channel numbers are (N + 1).

16.3.7.10 0x0028 DMIC Channel Mapping Register (Default Value: 0x7654_3210)

Offset: 0x0028			Register Name: DMIC_CH_MAP
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x7	DMIC_CH7_MAP DMIC Channel 7 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel

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Offset: 0x0028			Register Name: DMIC_CH_MAP
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x6	DMIC_CH6_MAP DMIC Channel 6 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel
23:20	R/W	0x5	DMIC_CH5_MAP DMIC Channel 5 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel
19:16	R/W	0x4	DMIC_CH4_MAP DMIC Channel 4 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel
15:12	R/W	0x3	DMIC_CH3_MAP DMIC Channel 3 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel

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Offset: 0x0028			Register Name: DMIC_CH_MAP
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0x2	DMIC_CH2_MAP DMIC Channel 2 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel
7:4	R/W	0x1	DMIC_CH1_MAP DMIC Channel 1 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel
3:0	R/W	0x0	DMIC_CH0_MAP DMIC Channel0 Mapping 0000: DATA0 Left Channel 0001: DATA0 Right Channel 0010: DATA1 Left Channel 0011: DATA1 Right Channel 0100: DATA2 Left Channel 0101: DATA2 Right Channel 0110: DATA3 Left Channel 0111: DATA3 Right Channel

16.3.7.11 0x002C DMIC Counter Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: DMIC_CNT
Bit	Read/Write	Default/Hex	Description

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Offset: 0x002C			Register Name: DMIC_CNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DMIC_CNT RX Sample Counter The audio sample number of writing into RXFIFO. When one sample is read by DMA or by host IO (example: MAD), the RX sample counter register increases by one. The RX sample counter register can be set to any initial valve at any time. After being updated by the initial value, the counter register should count on the base of this initial value. Note: It is used for Audio/Video Synchronization.

16.3.7.12 0x0030 DATA0 and DATA1 Volume Control Register (Default Value: 0xA0A0_A0A0)

Offset: 0x0030			Register Name: DATA0_DATA1_VOL_CTR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0xA0	DATA1L_VOL Data1 Left Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB
23:16	R/W	0xA0	DATA1R_VOL Data1 Right Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB

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Offset: 0x0030			Register Name: DATA0_DATA1_VOL_CTR
Bit	Read/Write	Default/Hex	Description
15:8	R/W	0xA0	DATA0L_VOL Data0 Left Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB
7:0	R/W	0xA0	DATA0R_VOL Data0 Right Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB

16.3.7.13 0x0034 DATA2 and DATA3 Volume Control Register (Default Value: 0xA0A0_A0A0)

Offset: 0x0034			Register Name: DATA2_DATA3_VOL_CTR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0xA0	DATA3L_VOL Data3 Left Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB

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Offset: 0x0034			Register Name: DATA2_DATA3_VOL_CTR
Bit	Read/Write	Default/Hex	Description
23:16	R/W	0xA0	DATA3R_VOL Data3 Right Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB
15:8	R/W	0xA0	DATA2L_VOL Data2 Left Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB
7:0	R/W	0xA0	DATA2R_VOL Data2 Right Channel Volume Control (-119.25 dB to 71.25 dB, 0.75 dB/Step) 0x00: Mute 0x01: -119.25 dB ... 0x9F: -0.75 dB 0xA0: 0 dB 0xA1: 0.75 dB ... 0xFF: 71.25 dB

16.3.7.14 0x0038 High Pass Filter Enable Control Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: HPF_EN_CTR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x0038			Register Name: HPF_EN_CTR
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	HPF_DATA3_CHR_EN High Pass Filter DATA3 Right Channel Enable 0: Disable 1: Enable
6	R/W	0x0	HPF_DATA3_CHL_EN High Pass Filter DATA3 Left Channel Enable 0: Disable 1: Enable
5	R/W	0x0	HPF_DATA2_CHR_EN High Pass Filter DATA2 Right Channel Enable 0: Disable 1: Enable
4	R/W	0x0	HPF_DATA2_CHL_EN High Pass Filter DATA2 Left Channel Enable 0: Disable 1: Enable
3	R/W	0x0	HPF_DATA1_CHR_EN High Pass Filter DATA1 Right Channel Enable 0: Disable 1: Enable
2	R/W	0x0	HPF_DATA1_CHL_EN High Pass Filter DATA1 Left Channel Enable 0: Disable 1: Enable
1	R/W	0x0	HPF_DATA0_CHR_EN High Pass Filter DATA0 Right Channel Enable 0: Disable 1: Enable
0	R/W	0x0	HPF_DATA0_CHL_EN High Pass Filter DATA0 Left Channel Enable 0: Disable 1: Enable

16.3.7.15 0x003C High Pass Filter Coef Register (Default Value: 0x00FF_AA45)

Offset: 0x003C			Register Name: HPF_COEF_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00FFAA45	HPF_COE High Pass Filter Coefficient

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16.3.7.16 0x0040 High Pass Filter Gain Register (Default Value: 0x00FF_D522)

Offset: 0x0040			Register Name: HPF_GAIN_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00FFD522	HPF_GAIN High Pass Filter Gain

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16.4 One Wire Audio (OWA)

16.4.1 Overview

The One Wire Audio (OWA) provides a serial bus interface for audio data. This interface is widely used for consumer audio connection.

The OWA includes the following features:

- One OWA TX and One OWA RX
- Compliance with S/PDIF interface
- Compatible with standard IEC-60958 and IEC-61937
  - IEC-60958 supports data formats: 16 bits, 20 bits, and 24 bits
- TXFIFO and RXFIFO
  - One 128×24bits TXFIFO and one 64×24bits RXFIFO for audio data transfer
  - Programmable FIFO thresholds
- Supports TX/RX DMA slave interface
- Multiple function clock
  - Separate clock for OWA TX and OWA RX
  - OWA TX function clock includes a series of 24.576MHz/49.152MHz and 22.579MHz frequency (TX function clock 49.152MHz support ample rate 384KHz)
  - OWA RX function clock includes a series of 400MHz frequency (RX function clock 400MHz support CDR of sample rate from 8KHz to 384KHz)
- Supports hardware parity on TX/RX
  - Hardware parity generation on the transmitter
  - Hardware parity checking on the receiver
- Supports channel status capture on the receiver
- Supports channel sample rate capture on the receiver
- Supports insertion detection for the receiver
- Supports channel status insertion for the transmitter

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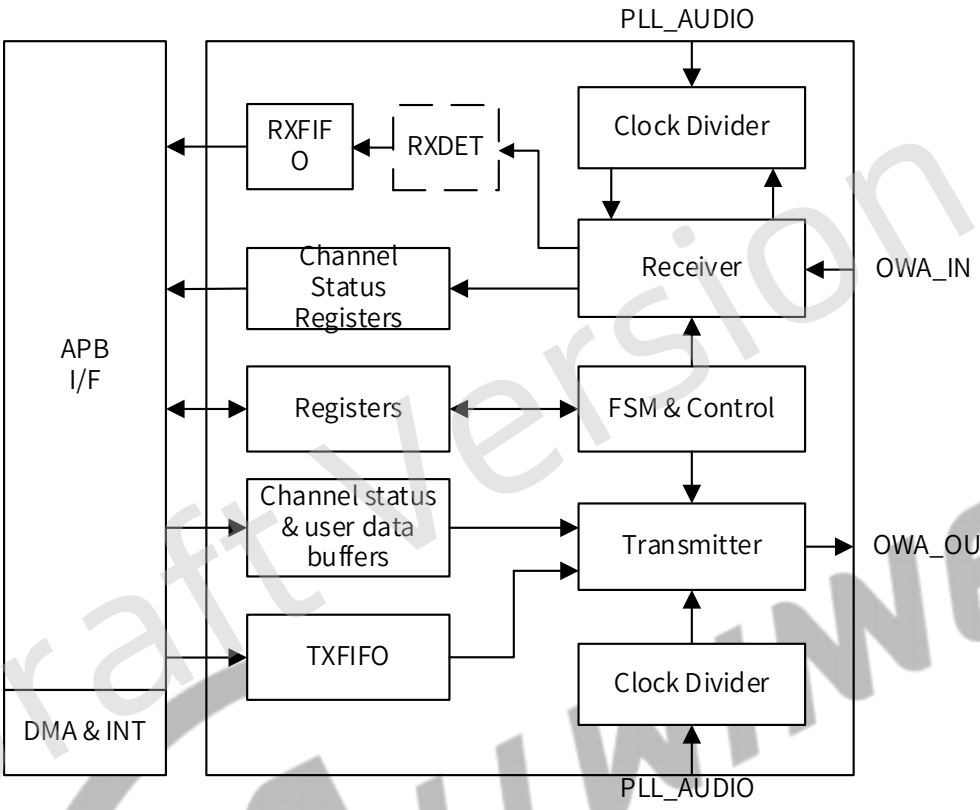
16.4.2 Block Diagram

The following figure shows the OWA block diagram.

Figure 16-27 OWA Block Diagram



16.4.3 External Signals

The OWA is a Biphasic-Mark Encoding Digital Audio Transfer protocol. In this protocol, the CLK signal and data signal are transferred in the same line. The following table describes the external signals of OWA. OWA-OUT is the output pin for the output CLK and DATA, and OWA-IN is the input pin for the input CLK and DATA.

Table 16-5 OWA External Signals

Signal Name	Description	Type
OWA-IN	One Wire Audio Input	I
OWA-OUT	One Wire Audio Output	O

16.4.4 Functional Description

16.4.4.1 Biphase-Mark Code (BMC)

In the OWA format, the digital signal is coded using the biphase-mark code (BMC). The clock, frame, and data are embedded in only one signal(c)the data pin. In the BMC system, each data bit is encoded into two logical states (00, 01, 10, or 11) at the pin. The following figure and table show how data is encoded to the BMC format.

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The frequency of the clock is twice the data bit rate, as shown in the following figure. Also, the clock is always programmed to 128 x fs, where fs is the sample rate. The device receiving in the OWA format can recover the clock and frame information from the BMC signal.

Figure 16-28 OWA Biphase-Mark Code

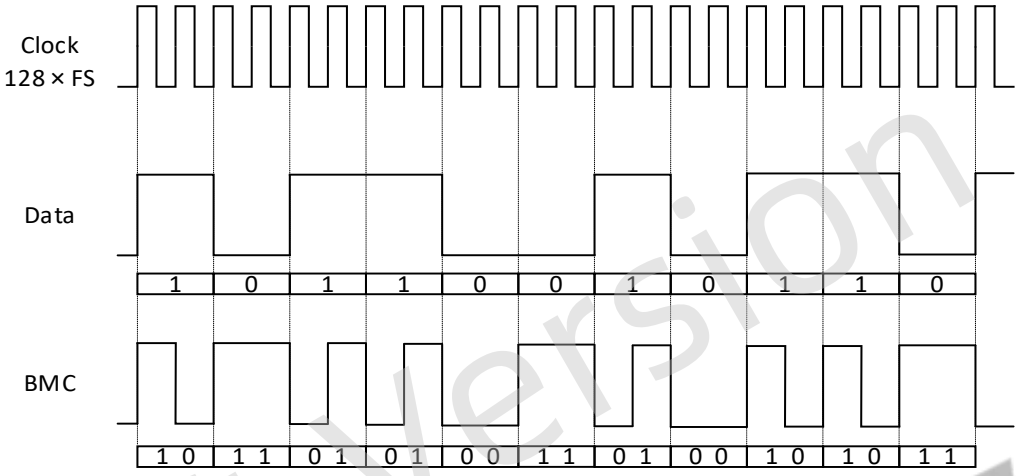


Table 16-6 Biphase-Mark Encoder

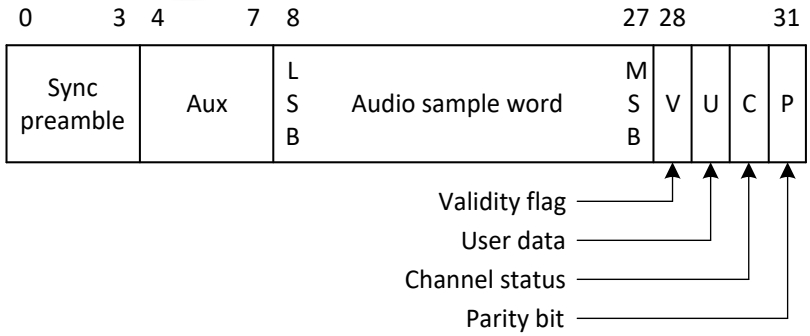
Data	Previous State	BMC
0	0	11
0	1	00
1	0	10
1	1	01

16.4.4.2 IEC60958 Transmit Format

The OWA supports digital audio data transfer and receive. It also supports full-duplex synchronous work mode. The software can set the work mode by the OWA Control Register.

Every audio sample transmitted in a sub-frame consists of 32-bit, numbered from 0 to 31. The following figure shows a sub-frame.

Figure 16-29 OWA Sub-Frame Format



Bits 0-3 carry one of the four permitted preambles to signify the type of audio sample in the current sub-frame. The preamble is not encoded in BMC format, and therefore the preamble code can contain more than two consecutive 0 or 1 logical states in a row.

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Bits 4-27 carry the audio sample word in linear 2s-complement representation. The most-significant bit (MSB) is carried by bit 27. When a 24-bit coding range is used, the least-significant bit (LSB) is in bit 4. When a 20-bit coding range is used, bit [8:27] carry the audio sample word with the LSB in bit 8. Bit [4:7] may be used for other applications and are designated auxiliary sample bits.

If the source provides fewer bits than the interface allows (either 20 or 24), the unused LSBs are set to logical 0. For a nonlinear PCM audio application or a data application, the main data field may carry any other information.

Bit 28 carries the validity bit (V) associated with the main data field in the sub-frame.

Bit 29 carries the user data channel (U) associated with the main data field in the sub-frame.

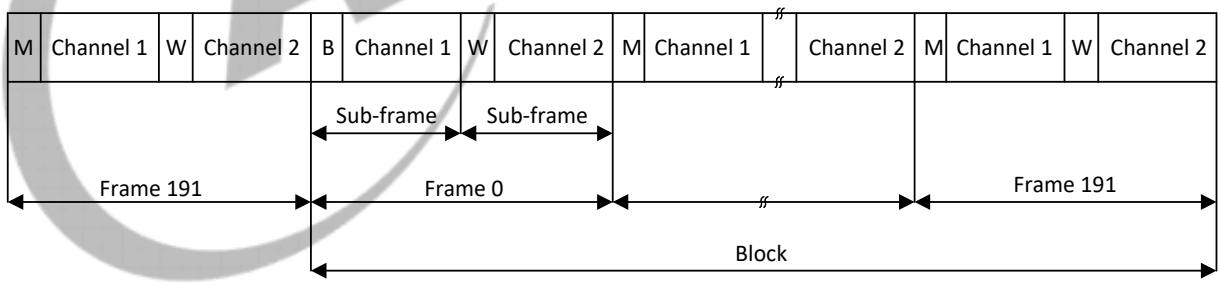
Bit 30 carries the channel status information (C) associated with the main data field in the sub-frame. The channel status indicates if the data in the sub-frame is a digital audio or some other type of data.

Bit 31 carries a parity bit (P) such that bit 4-31 carry an even number of 1s and an even number of 0s (even parity). As shown in the following table, the preambles (bit 0-3) are also defined with even parity.

Table 16-7 Preamble Codes

Preamble Code	Previous Logical State	Logical State	Description
B (or Z)	0	1110 1000	Start of a block and sub-frame 1
M (or X)	0	1110 0010	Sub-frame 1
W (or Y)	0	1110 0100	Sub-frame 2

Figure 16-30 OWA Frame/Block Format



16.4.4.3 IEC61937 Transmit Format

IEC 61937 applies to the digital audio interface by using the IEC 60958 series for the conveying of non-linear PCM encoded audio bitstreams. The non-linear PCM encoded audio bitstream is transferred by using the basic 16-bit data area of the IEC 60958 subframes, i.e. in time-slots 12 to 27. Because the non-linear PCM encoded audio bitstream to be transported is at a lower data rate than that supported by the IEC 60958 interface, the audio bitstream is broken into a sequence of discrete data-bursts, and stuffing between the data-bursts is necessary.

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IEC 60958 Data Burst

The method of placing the data into the IEC 60958 bitstream is to format the data to be transmitted into data-bursts and to send each data-burst in a continuous sequence of IEC 60958 frames.

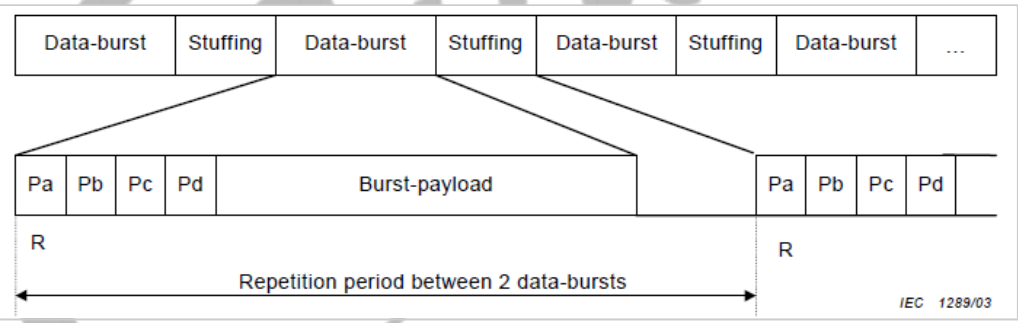
Table 16-8 Bit Allocation of Data-Burst in IEC 60958 Subframes

Subframe	Bit of subframes				
	MSB b27	b26	b25 ..... b14	b13	LSB b12
Frame 0; subframe B or M	0	1		14	15
Frame 0; subframe W	16	17		30	31
Frame 1; subframe B or M	32	33		46	47
Frame 1; subframe W	48	49		62	63
Frame 2; subframe B or M	64	65		78	79
----------			----------		
Last subframe B or M of data-burst	n-32	n-31		n-18	n-17
Last subframe W of data-burst	n-16	n-15		n-2	n-1

Data Burst Format

Each data-burst contains a burst-preamble consisting of four 16-bit words (Pa, Pb, Pc and Pd) followed by the burst-payload which contains data of an encoded audio frame.

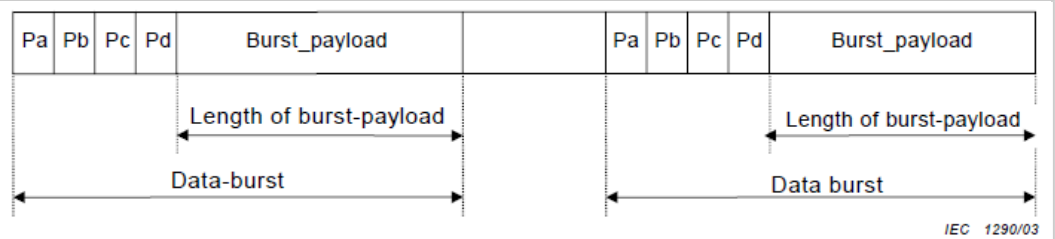
Figure 16-31 Data-Burst Format



- Burst-preamble

The burst-preamble consists of four mandatory fields. Pa and Pb represent a synchronization word. Pc gives information about the type of data, and some information/control for the receiver. Pd gives the length of the burst-payload, and is limited to 65535 bits in the case of Pd represent bits' length, or is limited to 65535 bytes in the case of Pd represent bytes' length.

Figure 16-32 Data-burst Preamble



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The four preamble words are contained in two sequential IEC 60958 frames. The frame beginning the data-burst contains preamble word Pa in subframe 1, and Pb in subframe 2. The next frame contains Pc in subframe 1 and Pd in subframe 2. When placed into an IEC 60958 subframe, the MSB of a 16-bit burst-preamble word is placed into time-slot 27 and the LSB is placed into time-slot 12.

Figure 16-33 Data-burst Preamble words

Preamble word	Length of field	Contents	Value MSB..LSB
Pa	16-bit	Sync word 1	F872h
Pb	16-bit	Sync word 2	4E1Fh
Pc	16-bit	Burst-info	Table 5
Pd	16-bit	Length-code	Number of bits or number of bytes according to data-type

- Burst-information

The 16-bit burst-information contains information about the data which will be found in the data-burst.

Figure 16-34 Fields of Burst-information

Bits of Pc	Value	Contents	Remark
0 – 6		Data-type	See IEC 61937-2
7	0	Error-flag indicating a valid burst-payload	
	1	Error-flag indicating that the burst-payload may contain errors	
8 – 12		Data-type-dependent info	
13 – 15	0	Bitstream-number	
NOTE The repetition period of pause data-bursts depends on the application in which IEC 60958 is used to convey encoded audio bitstreams.			

The 7-bit data-type is defined in bits 0-6 of the burst-preamble Pc, the bit 6 is the MSB. This data-type field indicates the format of the burst-payload, which will be conveyed in the data-burst. Typical properties of a data-type are the reference point and repetition period of the burst, which is the number of sampling periods of the audio between the reference point of the current data-burst and the reference point of the next data-burst. The reference point is inherently defined for each data-type.

The error-flag bit is available to indicate if the contents of the data-burst contain data errors. If a data-burst is thought to be error-free, or if the data source does not know if the data contains errors, then the value of this bit is set to a '0'. If the data source does know that a particular data-burst contains some errors this bit may be set to a '1'. The usage of this bit by receiver is optional.

The meaning of the 5-bit data-type-dependent info depends on the value of the data-type.

The 3-bit bitstream-number indicates to which bitstream the data-burst belongs. Eight codes (0-7) are available so that up to eight independent bitstreams may be multiplexed in one bitstream in a time multiplex. Each independent bitstream shall use a unique bit-streamnumber.

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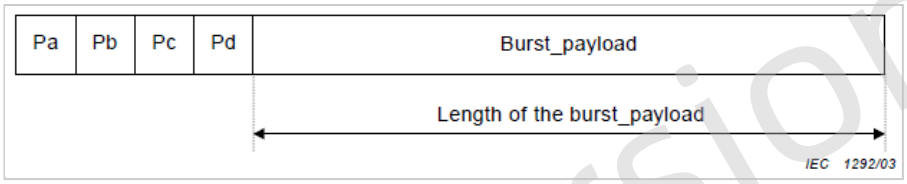
- Length-code

The length-code indicates the number of bits or bytes according to data-type within the databurst, from 0 to 65535. The size of the Pa, Pb, Pc and Pd is not counted in the value of the length-code. In other words, the length-code indicates the number of bits of the burst-payload in bits, plus the conditional length of Pe and Pf, or the number of bytes of the burst-payload in bytes, plus the conditional length of Pe and Pf if exists.

Figure 16-35 Length of the Burst-Payload Specified by Pd



16.4.4.4 Audio Sample Ratio Detection

The sampling rate is calculated according to the data pulse back-stepping method. In the first phase lock of the CDR, find the period of 1 Frame. Then count by using the high-speed sampling clock and read the counting value of the pulse width of Period Max, then the sampling rate can be calculated.

Table 16-9 The Relation between Different System Clock and Sample Ratio

	Sample Clock Cycles			
TX Sample Rate(kHz)	196.608 MHz-SysClk	200 MHz-SysClk	300 MHz-SysClk	400Mhz-SysClk
22.05	8916(±5)	9070(±5)	13605(±5)	18140(±5)
24	8192(±5)	8333(±5)	12500(±5)	16666(±5)
32	6144(±5)	6250(±5)	9375(±5)	12500(±5)
44.1	4458(±5)	4535(±5)	6802(±5)	9070(±5)
48	4096(±5)	4166(±5)	6250(±5)	8332(±5)
96	2048(±5)	2083(±5)	3125(±5)	4166(±5)
176.4	1114(±5)	1133(±5)	1700(±5)	2266(±5)
192	1024(±5)	1041(±5)	1562(±5)	2082(±5)

16.4.5 Programming Guidelines

The software operation of the OWA is divided into five steps: system setup, OWA initialization, channel setup, DMA setup and enable/disable module. The following sections describe these five steps.

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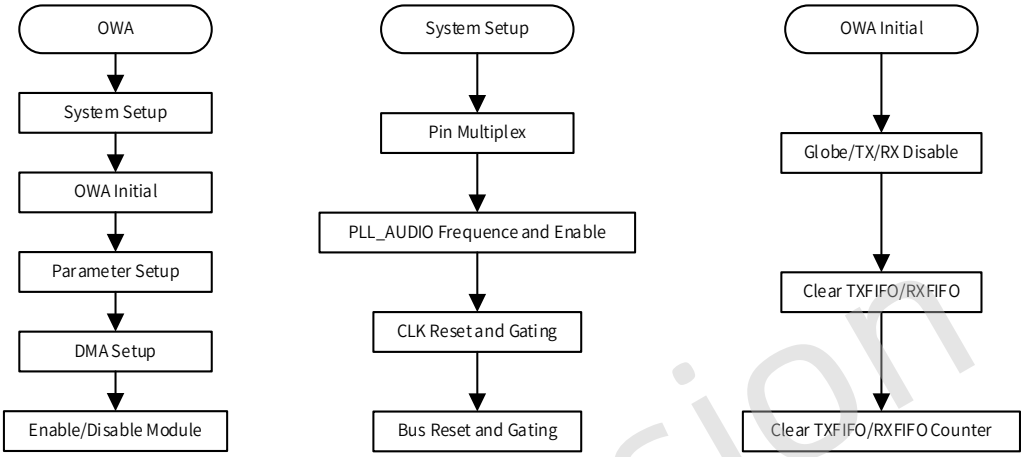
Figure 16-36 OWA Operation Flow



16.4.5.1 System Setup and OWA Initialization

The first step in the OWA initialization is properly programming the GPIO because the OWA port is a multiplex pin. You can find the function in Section 17.9 GPIO.

The clock source for the OWA should be followed. Firstly, reset the audio PLL in PLL_AUDIOx Control Register. Secondly, set up the frequency of the Audio PLL in the PLL_AUDIOx Control Register. After that, enable the OWA gating. Lastly, enable the OWA bus gating.

After the system setup, the register of OWA can be set up. Firstly, reset the OWA by writing 1 to OWA_GENCTL[0] and clear the TX/RX FIFO by writing 1 to OWA_FCTL[17:16]. After that, enable the globe enable bit by writing 1 to OWA_GENCTL[1] and clear the interrupt and TX/RX counter by setting OWA_ISTA and OWA_TX_CNT/OWA_RX_CNT.

16.4.5.2 Parameter Setup and DMA Setup

You can set up the audio type, clock divider ratio, the sample format, and the trigger level, and so on. The setup of the register can be found in the specification.

The OWA supports two methods to transfer the data. The most common way is DMA; the configuration of DMA can be found in Section 12.1 DMA Controller (DMAC). In this module, you just enable the DRQ in OWA_INT[7].

16.4.5.3 Enabling and Disabling OWA

Step 1 Enable TX/RX by writing OWA_TX_CFIG[31]/OWA_RX_CFIG[0].

Step 2 Enable OWA by writing 1 to OWA_GENCTL[1].

Step 3 Write 0 to OWA_GENCTL[1] to disable process.

16.4.6 Register List

Module Name	Base Address

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Module Name	Base Address
OWA0	0x02036000

Register Name	Offset	Description
OWA_GENCTL	0x0000	OWA General Control Register
OWA_TXCFIG	0x0004	OWA TX Configuration Register
OWA_RXCFIG	0x0008	OWA RX Configuration Register
OWA_ISTA	0x000C	OWA Interrupt Status Register
OWA_RXFIFO	0x0010	OWA RXFIFO Register
OWA_FCTL	0x0014	OWA FIFO Control Register
OWA_FSTA	0x0018	OWA FIFO Status Register
OWA_INT	0x001C	OWA Interrupt Control Register
OWA_TXFIFO	0x0020	OWA TX FIFO Register
OWA_TX_CNT	0x0024	OWA TX Counter Register
OWA_RX_CNT	0x0028	OWA RX Counter Register
OWA_TX_CHSTA0	0x002C	OWA TX Channel Status Register0
OWA_TX_CHSTA1	0x0030	OWA TX Channel Status Register1
OWA_RX_CHSTA0	0x0034	OWA RX Channel Status Register0
OWA_RX_CHSTA1	0x0038	OWA RX Channel Status Register1
OWA_INSERT_REDET_CTL	0x003C	OWA Insert Redetect Control Register
OWA_EXP_CTL	0x0040	OWA Expand Control Register
OWA_EXP_ISTA	0x0044	OWA Expand Interrupt Status Register
OWA_EXP_INFO_0	0x0048	OWA Expand Information Register 0
OWA_EXP_INFO_1	0x004C	OWA Expand Information Register 1
OWA_EXP_DBG_0	0x0050	OWA Expand Debug Register 0
OWA_EXP_DBG_1	0x0054	OWA Expand Debug Register 1
OWA_EXP_CDR_DBG	0x005C	OWA Expand CDR Debug Register

16.4.7 Register Description

16.4.7.1 0x0000 OWA General Control Register (Default Value: 0x0000_0080)

Offset: 0x0000			Register Name: OWA_GENCTL
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/WAC	0x0	RST_RX Reset RX 0: Normal 1: Reset Self-clear to '0'.
11:10	/	/	/

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Offset: 0x0000			Register Name: OWA_GENCTL
Bit	Read/Write	Default/Hex	Description
9:5	R/W	0x4	MCLKDIV MCLK Clock Divide Ratio MCLK Divide Ratio from PLL_AUDIO 00000: Divide by 128 00001: Divide by 2 00010: Divide by 4 00011: Divide by 6 00100: Divide by 8 00101: Divide by 10 00110: Divide by 12 00111: Divide by 14 01000: Divide by 16 01001: Divide by 18 01010: Divide by 20 01011: Divide by 22 01100: Divide by 24 (p)(p)(p)(p)(p)(p) 11111: Divide by 62
4	/	/	/
3	R/W	0x0	MCLKEN MCLK Enable 0: Disable 1: Enable
2	R/W	0x0	LOOP Loopback Test 0: Normal Mode 1: Loopback Test When the bit is set to '1', the DOUT and DIN need be connected.
1	R/W	0x0	GEN Global Enable A disable on this bit overrides any other block or channel enables and flushes all FIFOs. 0: Disable 1: Enable
0	R/WAC	0x0	RST_TX Reset TX 0: Normal 1: Reset Self-clear to 0.

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16.4.7.2 0x0004 OWA TX Configuration Register (Default Value: 0x0000_00F0)

Offset: 0x0004			Register Name: OWA_TXCFIG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	TX_SINGLE_MODE TX Single Channel Mode 0: Disable 1: Enable
30:18	/	/	/
17	R/W	0x0	ASS Audio Sample Select when TX FIFO Underrun 0: Sending 0 1: Sending the last audio Note: This bit is only valid in PCM mode.
16	R/W	0x0	TX_AUDIO TX Data Type 0: Linear PCM (Valid bit of both sub-frame set to 0) 1: Non-audio (Valid bit of both sub-frame set to 1)
15:9	/	/	/
8:4	R/W	0xF	TX_RATIO TX Clock Divide Ratio Clock divide ratio = TX_TATIO + 1 Fs = PLL_AUDIO/[(TX_TATIO + 1)*64*2]
3:2	R/W	0x0	TX_SF TX Sample Format 00: 16 bits 01: 20 bits 10: 24 bits 11: Reserved
1	R/W	0x0	TX_CHM CHSTMODE 0: Channel status A and B set to 0 1: Channel status A and B generated from TX_CHSTA
0	R/W	0x0	TXEN TX Enable 0: Disable 1: Enable

16.4.7.3 0x0008 OWA RX Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: OWA_RXCFIG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/

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Offset: 0x0008			Register Name: OWA_RXCFIG
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	RX_CDR_MANUAL RX CDR mode 0: Automatic (in this mode, RX_CDR_PREAMBLE does not need to be configured) 1: Manual (in this mode, RX_CDR_PREAMBLE needs to be configured)
4	R	0x0	RX_LOCK_FLAG RX Lock Flag 0: Unlock 1: Lock
3	R/W	0x0	RX_CHST_SRC RX Channel State Source Select 0: RX_CH_STA register holds status from Channel A 1: RX_CH_STA register holds status from Channel B
2	R/W	0x0	RX_CDR_PREAMBLE RX channel CDR Preamble Type 0: Support CDR from preamble X (11100010) 1: Support CDR from preamble X (00011101)
1	R/WAC	0x0	CHST_CP Channel Status Capture 0: Idle or Capture End 1: Capture Channel Status Start Note: a) The field must be set to 1 at each operation (such as recording). b) When set to '1', the channel status information is capturing, the bit will clear to '0' after captured.
0	R/W	0x0	RXEN RX Enable 0: Disabled 1: Enabled

16.4.7.4 0x000C OWA Interrupt Status Register (Default Value: 0x0000_0010)

Offset: 0x000C			Register Name: OWA_ISTA
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/

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Offset: 0x000C			Register Name: OWA_ISTA
Bit	Read/Write	Default/Hex	Description
18	R/W1C	0x0	RX_LOCK_INT RX Lock Interrupt 0: No Pending IRQ 1: RX Lock Pending Interrupt (RX_LOCK_FLAG turns from 0 to 1) Write '1' to clear this interrupt.
17	R/W1C	0x0	RX_UNLOCK_INT RX Unlock Pending Interrupt 0: No Pending IRQ 1: RX Unlock Pending Interrupt (RX_LOCK_FLAG turns from 1 to 0) Write 1 to clear this interrupt.
16	R/W1C	0x0	RX_PARERRI_INT RX Parity Error Pending Interrupt 0: No Pending IRQ 1: RX Parity Error Pending Interrupt Write "1" to clear this interrupt.
15:7	/	/	/
6	R/W1C	0x0	TXU_INT TX FIFO Underrun Pending Interrupt 0: No Pending IRQ 1: TXFIFO Underrun Pending Interrupt Writing "1" to clear this interrupt.
5	R/W1C	0x0	TXO_INT TX FIFO Overrun Pending Interrupt 0: No Pending IRQ 1: TXFIFO Overrun Pending Interrupt Writing "1" to clear this interrupt.
4	R/W1C	0x1	TXE_INT TX FIFO Empty Pending Interrupt 0: No Pending IRQ 1: FIFO Empty Pending Interrupt Write "1" to clear this interrupt or automatically clear if the interrupt condition fails.
3:2	/	/	/
1	R/W1C	0x0	RXO_INT RXFIFO Overrun Pending Interrupt 0: RXFIFO Overrun Pending Write '1' to clear this interrupt.

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Offset: 0x000C			Register Name: OWA_ISTA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	RXA_INT RXFIFO Available Pending Interrupt 0: No Pending IRQ 1: Data Available Pending IRQ Write '1' to clear this interrupt or automatically clear if the interrupt condition fails.

16.4.7.5 0x0010 OWA RXFIFO Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: OWA_RXFIFO
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RX_DATA The host can get one sample by reading this register. The first is A channel data and the latter is B channel data.

16.4.7.6 0x0014 OWA FIFO Control Register (Default Value: 0x0004_0200)

Offset: 0x0014			Register Name: OWA_FCTL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	HUB_EN Audio Hub Enable The bit will take effect only when the TXEN is set to 1. Audio Codec/I2S0/I2S1/I2S2/OWA TXFIFO Hub Enable. 0: Disable 1: Enable
30	R/W1C	0x0	FTX Write '1' to flush TXFIFO, self-clear to '0'.
29	R/W1C	0x0	FRX Write '1' to flush RXFIFO, self-clear to '0'.
28:22	/	/	/
21	R/W	0x0	RX_SYNC_EN_START The bit will take effect only when the RX_SYNC_EN is set to 1. I2S0/I2S1/I2S2/DMIC/OWA RX Synchronize Enable Start. 0: Disable 1: Enable

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Offset: 0x0014			Register Name: OWA_FCTL
Bit	Read/Write	Default/Hex	Description
20	R/W	0x0	RX_SYNC_EN OWA RX Synchronize Enable 0: Disable 1: Enable
19:12	R/W	0x40	TXTL TX FIFO Empty Trigger Level Interrupt and DMA request trigger level for TX FIFO normal condition. Trigger Level = TXTL
11	/	/	/
10:4	R/W	0x20	RXTL RX FIFO Trigger Level Interrupt and DMA request trigger level for RX FIFO normal condition. Trigger Level = RXTL + 1
3	/	/	/
2	R/W	0x0	TXIM TXFIFO Input Mode (Mode 0, 1) 0: Valid data at the MSB of TXFIFO Register 1: Valid data at the LSB of TXFIFO Register Example for 20-bit transmitted audio sample: Mode 0: TXFIFO [23:0] = {APB_WDATA [31:12], 4 h0} Mode 1: TXFIFO[23:0] = {APB_WDATA[19:0], 4 h0}
1:0	R/W	0x0	RXOM RXFIFO Output Mode (Mode 0, 1, 2, 3) 00: Expanding 0 at LSB of RXFIFO Register 01: Expanding received sample sign bit at MSB of RXFIFO Register 10: Truncating received samples at high half-word of RXFIFO Register and low half-word of RXFIFO Register is filled by 0 . 11: Truncating received samples at low half-word of RXFIFO Register and high half-word of RXFIFO. Register is expanded by its sigh bit Mode 0: APB_RDATA [31:0] = {RXFIFO [23:0], 8 h0} Mode 1: APB_RDATA [31:0] = {8 RXFIFO [23], RXFIFO [23:0]} Mode 2: APB_RDATA [31:0] = {RXFIFO [23:8], 16 h0} Mode 3: APB_RDATA[31:0] = {16 RXFIFO[23], RXFIFO[23:8]}

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16.4.7.7 0x0018 OWA FIFO Status Register (Default Value: 0x8080_0000)

Offset: 0x0018			Register Name: OWA_FSTA
Bit	Read/Write	Default/Hex	Description
31	R	0x1	TXE TXFIFO Empty (indicate the TXFIFO is not full) 0: No room for new sample in TXFIFO 1: More than one room for new sample in TXFIFO (>= 1 Word)
30:24	/	/	/
23:16	R	0x80	TXE_CNT TXFIFO Empty Space Word Counter
15	R	0x0	RXA RXFIFO Available 0: No available data in RXFIFO 1: More than one sample in RXFIFO (>= 1 Word)
14:7	/	/	/
6:0	R	0x0	RXA_CNT RXFIFO Available Sample Word Counter

16.4.7.8 0x001C OWA Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: OWA_INT
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/W	0x0	RX_LOCKI_EN RX LOCK Interrupt Enable 0: Disable 1: Enable
17	R/W	0x0	RX_UNLOCKI_EN RX UNLOCK Interrupt Enable 0: Disable 1: Enable
16	R/W	0x0	RX_PARERRI_EN RX PARITY ERORR Interrupt Enable 0: Disable 1: Enable
15:8	/	/	/
7	R/W	0x0	TX_DRQ TXFIFO Empty DRQ Enable 0: Disable 1: Enable

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Offset: 0x001C			Register Name: OWA_INT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	TXUI_EN TXFIFO Underrun Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	TXOI_EN TXFIFO Overrun Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	TXEI_EN TXFIFO Empty Interrupt Enable 0: Disable 1: Enable
3	/	/	/
2	R/W	0x0	RX_DRQ RXFIFO Data Available DRQ Enable When set to 1 , RXFIFO DMA Request is asserted if data is available in RXFIFO. 0: Disable 1: Enable
1	R/W	0x0	RXOI_EN RXFIFO Overrun Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	RXAI_EN RXFIFO Data Available Interrupt Enable 0: Disable 1: Enable

16.4.7.9 0x0020 OWA TX FIFO Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: OWA_TXFIFO
Bit	Read/Write	Default/Hex	Description
31:0	W	0x0	TX_DATA Transmitting A, B channel data should be written by this register one by one. The first is A channel data, and the latter is B channel data.

16.4.7.10 0x0024 OWA TX Counter Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: OWA_TX_CNT
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0024			Register Name: OWA_TX_CNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TX_CNT TX Sample Counter The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial value at any time. After being updated by the initial value, the counter register should count on the base of this initial value.

16.4.7.11 0x0028 OWA RX Counter Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: OWA_RX_CNT
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RX_CNT RX Sample Counter The audio sample number of writing into RXFIFO. When one sample is written by Codec, the RX sample counter register increases by one. The RX counter register can be set to any initial value at any time. After being updated by the initial value, the counter register should count on the base of this value.

16.4.7.12 0x002C OWA TX Channel Status Register0 (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: OWA_TX_CHSTA0
Bit	Read/Write	Default/Hex	Description
31: 30	/	/	/
29:28	R/W	0x0	CA Clock Accuracy 00: Level 2 01: Level 1 10: Level 3 11: Not matched

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Offset: 0x002C			Register Name: OWA_TX_CHSTA0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	FREQ Sampling Frequency 0000: 44.1 kHz 0001: Not indicated 0010: 48 kHz 0011: 32 kHz 0100: 22.05 kHz 0101: Reserved 0110: 24 kHz 0111: Reserved 1000: Reserved 1001: 768 kHz 1010: 96 kHz 1011: Reserved 1100: 176.4 kHz 1101: Reserved 1110: 192 kHz 1111: Reserved
23:20	R/W	0x0	CN Channel Number
19:16	R/W	0x0	SN Source Number
15:8	R/W	0x0	CC Category Code Indicates the kind of equipment that generates the digital audio interface signal.
7:6	R/W	0x0	MODE Mode 00: Default Mode 01 to 11: Reserved

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Offset: 0x002C			Register Name: OWA_TX_CHSTA0
Bit	Read/Write	Default/Hex	Description
5:3	R/W	0x0	EMP Emphasis Additional format information For bit 1 = “0”, Linear PCM audio mode: 000: 2 audio channels without pre-emphasis 001: 2 audio channels with 50 μs/15 μs pre-emphasis 010: Reserved (for 2 audio channels with pre-emphasis) 011: Reserved (for 2 audio channels with pre-emphasis) 100 to 111: Reserved For bit 1 = “1”, other than Linear PCM applications: 000: Default state 001 to 111: Reserved
2	R/W	0x0	CP Copyright 0: Copyright is asserted 1: No copyright is asserted
1	R/W	0x0	TYPE Audio Data Type 0: Linear PCM samples 1: Non-linear PCM audio
0	R/W	0x0	PRO Application Type 0: Consumer application 1: Professional application This bit must be fixed to “0”.

16.4.7.13 0x0030 OWA TX Channel Status Register1 (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: OWA_TX_CHSTA1
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9:8	R/W	0x0	CGMS_A 00: Copying is permitted without restriction 01: One generation of copies may be made 10: Condition not be used 11: No copying is permitted

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Offset: 0x0030			Register Name: OWA_TX_CHSTA1
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	ORIG_FREQ Original Sampling Frequency 0000: Not indicated 0001: 192 kHz 0010: 12 kHz 0011: 176.4 kHz 0100: Reserved 0101: 96 kHz 0110: 8 kHz 0111: 88.2 kHz 1000: 16 kHz 1001: 24 kHz 1010: 11.025 kHz 1011: 22.05 kHz 1100: 32 kHz 1101: 48 kHz 1110: Reserved 1111: 44.1 kHz
3:1	R/W	0x0	WL Sample Word Length For bit 0 = 0 : 000: Not indicated 001: 16 bits 010: 18 bits 100: 19 bits 101: 20 bits 110: 17 bits 111: Reserved For bit 0 = 1 : 000: Not indicated 001: 20 bits 010: 22 bits 100: 23 bits 101: 24 bits 110: 21 bits 111: Reserved
0	R/W	0x0	MWL Max Word Length 0: Maximum audio sample word length is 20 bits 1: Maximum audio sample word length is 24 bits

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16.4.7.14 0x0034 OWA RX Channel Status Register0 (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: OWA_RX_CHSTA0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R	0x0	CA Clock Accuracy 00: Level 2 01: Level 1 10: Level 3 11: Not Matched
27:24	R	0x0	FREQ Sampling Frequency 0000: 44.1 kHz 0001: Not Indicated 0010: 48 kHz 0011: 32 kHz 0100: 22.05 kHz 0101: Reserved 0110: 24 kHz 0111: Reserved 1000: Reserved 1001: 768 kHz 1010: 96 kHz 1011: Reserved 1100: 176.4 kHz 1101: Reserved 1110: 192 kHz 1111: Reserved
23:20	R	0x0	CN Channel Number
19:16	R	0x0	SN Source Number
15:8	R	0x0	CC Category Code Indicates the Kind of Equipment that Generates the digital audio interface Signal.
7:6	R	0x0	MODE Mode 00: Default mode 01 to 11: Reserved

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Offset: 0x0034			Register Name: OWA_RX_CHSTA0
Bit	Read/Write	Default/Hex	Description
5:3	R	0x0	EMP Emphasis Additional Format Information For bit 1 = '0', Linear PCM Audio mode: 000: 2 Audio channels without pre-emphasis 001: 2 Audio channels with 50 μs/15 μs pre-emphasis 010: Reserved (For 2 Audio channels with pre-emphasis) 011: Reserved (For 2 Audio channels with pre-emphasis) 100 to 111: Reserved For bit 1 = '1', Other than Linear PCM applications: 000: Default state 001 to 111: Reserved
2	R	0x0	CP Copyright 0: Copyright is asserted 1: No Copyright is asserted
1	R	0x0	TYPE Audio Data Type 0: Linear PCM samples 1: Non-linear PCM audio
0	R	0x0	PRO Application Type 0: Consumer application 1: Professional application

16.4.7.15 0x0038 OWA RX Channel Status Register1 (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: OWA_RX_CHSTA1
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9:8	R	0x0	CGMS_A 00: Copying is permitted without restriction 01: One generation of copies may be made 10: Condition is not used 11: No copying is permitted

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Offset: 0x0038			Register Name: OWA_RX_CHSTA1
Bit	Read/Write	Default/Hex	Description
7:4	R	0x0	ORIG_FREQ Original Sampling Frequency 0000: Not indicated 0001: 192 kHz 0010: 12 kHz 0011: 176.4 kHz 0100: Reserved 0101: 96 kHz 0110: 8 kHz 0111: 88.2 kHz 1000: 16 kHz 1001: 24 kHz 1010: 11.025 kHz 1011: 22.05 kHz 1100: 32 kHz 1101: 48 kHz 1110: Reserved 1111: 44.1 kHz
3:1	R	0x0	WL Sample Word Length For bit 0 = '0': 000: Not indicated 001: 16 bits 010: 18 bits 100: 19 bits 101: 20 bits 110: 17 bits 111: Reserved For bit 0 = '1': 000: Not indicated 001: 20 bits 010: 22 bits 100: 23 bits 101: 24 bits 110: 21 bits 111: Reserved
0	R	0x0	MWL Max Word Length 0: Maximum Audio sample word length is 20 bits 1: Maximum Audio sample word length is 24 bits

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16.4.7.16 0x003C OWA Insert Redetect Control Register (Default Value:0x0000_0180)

Offset: 0x003C			Register Name: OWA_INSERT_REDET_CTL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x1	INSERT_REDET_MONITOR_ENABLE Insert Re-detect Monitor Enable 0: Disable 1: Enable
7:0	R/W	0x80	INSERT_REDET_MONITOR_LENGTH Insert Re-detect Monitor Length The unit is one 32KHz cycle.

16.4.7.17 0x0040 OWA Expand Control Register (Default Value: 0x0000_000F)

Offset: 0x0040			Register Name: OWA_EXP_CTL
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30	R/W	0x0	BURST_DATAOUT_SELECT Burst data output select 0: Burst preamble and payload 1: Burst payload
29:16	R/W	0x0	REPEAT_PERIOD_OF_FR_NUM The number for the repetition period of the burst frame Configure this field according to the type of RX data. A mismatch between the configuration data and the received data will result in an error interrupt.
15	R/W	0x0	UNIT_SELECT Unit Select Configure this field according to the RX data type 0: In the units of 16-bit 1: In the units of 2-byte
14	R/W	0x0	OWA_RX_MODE_MAN OWA RX Protocol Select 0: IEC60958 1: IEC61937
13	R/W	0x0	OWA_RX_MODE OWA RX Mode Select 0: Manual Ctrl. It is configured by OWA RX_MODE_MAN. 1: Auto Ctrl. It is configured by the channel status values resolved by hardware.

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Offset: 0x0040			Register Name: OWA_EXP_CTL
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	AUDIO_DATA_BITORDER_EN Audio Data Bitorder Enable 0: The audio data received by RX is stored directly into FIFO 1: The high and low bits of the audio data received by RX are reversed and then stored into FIFO
11	R/W	0x0	DATA_LENGTH_BITORDER_EN Data Length Bitorder Enable 0: The received PD data is used as the length of the valid audio data. 1: The high and low bits of the received PD data are reversed and then used as the length of valid audio data.
10	R/W	0x0	DATA_TYPE_BITORDER_EN Data Type Bitorder Enable 0: The received PC data is used as the data length of the valid audio. 1: The high and low bits of the received PC data are reversed and then used as the length of the valid audio data.
9	R/W	0x0	SYNCW_BITORDER_EN 0: Pa/Pb is the sync code of audio data. 1: The Pa/Pb whose high and low bits are reversed is the sync code of audio data.
8	R/W	0x0	INSERT_DETECTION_ENABLE Insert Detection Enable 0: Disable 1: Enable
7:0	R/W	0x0F	INSERT_DETECTION_NUM Insert Detection Number Configure how many jumping edges are detected to generate an insertion interrupt

16.4.7.18 0x0044 OWA Expand Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: OWA_EXP_ISTA
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/

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Offset: 0x0044			Register Name: OWA_EXP_ISTA
Bit	Read/Write	Default/Hex	Description
24	R/W	0x0	PD_CHANGE_INT_EN PD_LENGTH_CHANGE Interrupt Enable 0: Disable 1: Enable
23	R/W	0x0	PC_PAUSE_STOP_INT_EN PC_PAUSE_BURSTS_STOP Interrupt Enable 0: Disable 1: Enable
22	R/W	0x0	PC_BITSTRM_CHANGE_INT_EN PC_BITSTREAM_CHANGE Interrupt Enable 0: Disable 1: Enable
21	R/W	0x0	PC_ERR_FLAG_INT_EN PC_ERROR_FLAG Interrupt Enable 0: Disable 1: Enable
20	R/W	0x0	PC_DTYPE_CHANGE_INT_EN PC_DATATYPE_CHANGE Interrupt Enable 0: Disable 1: Enable
19	R/W	0x0	RPDB_ERR_INT_EN RPDB_ERROR Interrupt Enable 0: Disable 1: Enable
18	R/W	0x0	PCPD_CAP_INT_EN PCPD_CAP Interrupt Enable 0: Disable 1: Enable
17	R/W	0x0	PAPB_CAP_INT_EN PAPB_CAP Interrupt Enable 0: Disable 1: Enable
16	R/W	0x0	INSERT_INT_EN INSERT Interrupt Enable 0: Disable 1: Enable
15:9	/	/	/

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Offset: 0x0044			Register Name: OWA_EXP_ISTA
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	PD_CHANGE_INT PD CHANGE INT 0: No Pending IRQ 1: PD Data length information is changed. (except Pause/Null data burst type) Write 1 to clear this interrupt.
7	R/W1C	0x0	PC_PAUSE_STOP_INT Audio bitstream is interrupted. When stopped, the interface becomes idle. 0: No Pending IRQ 1: PC Pause burst Stop, frame sequence discontinued. Transmitters may optionally use the STOP value to indicate that the transmission of the current encoded Write 1 to clear this interrupt.
6	R/W1C	0x0	PC_BITSTRM_CHANGE_INT PC BITSTRM CHANGE INT 0: No Pending IRQ 1: PC Bitstream Number is change. Bitstream Number indicates which bitstream the data burst belongs. (except Pause/Null data bursts type) Write 1 to clear this interrupt.
5	R/W1C	0x0	PC_ERR_FLAG_INT PC ERR FLAG INT 0: No Pending IRQ 1: PC Error-flag is available to indicate if the contents of the data-burst contain data errors (except Pause/Null data bursts type). The using of this bit by receivers is optional. Write 1 to clear this interrupt.
4	R/W1C	0x0	PC_DTYPE_CHANGE_INT PC DTYPE CHANGE INT 0: No Pending IRQ 1: PC Datatype (except Pause/Null data type) information is changed. Write 1 to clear this interrupt.

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Offset: 0x0044			Register Name: OWA_EXP_ISTA
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	RPDB_ERR_INT RPDB ERR INT 0: No Pending IRQ 1: Hardware counts the repetition period of the burst frame is different from the register configuration number. Write 1 to clear this interrupt.
2	R/W1C	0x0	PCPD_CAP_INT PCPD CAP INT 0: No Pending IRQ 1: IEC61937 mode captures PC and PD Write 1 to clear this interrupt.
1	R/W1C	0x0	PAPB_CAP_INT PAPB CAP INT 0: No Pending IRQ 1: IEC61937 mode captures PA and PB Write 1 to clear this interrupt.
0	R/W1C	0x0	INSERT_INT INSERT INT 0: No Pending IRQ 1: OWA RX detects device insertion Write 1 to clear this interrupt.

16.4.7.19 0x0048 OWA Expand Information Register 0 (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: OWA_EXP_INFO_0
Bit	Read/Write	Default/Hex	Description
31:16	R	0x0	PC_DATA PC Data information
15:0	R	0x0	PD_DATA PD Data information

16.4.7.20 0x004C OWA Expand Information Register 1 (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: OWA_EXP_INFO_1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x004C			Register Name: OWA_EXP_INFO_1
Bit	Read/Write	Default/Hex	Description
29:16	R	0x0	REPET_PERIOD_OF_FR_VALUE Repetition period of the burst frame value Check whether the repetition period of the burst frame calculated by hardware is consistent with the configured.
15:0	R	0x0	SR_VALUE Sample Rate Value Read this value after RX_LOCK.

16.4.7.21 0x0050 OWA Expand Debug Register 0 (Default Value: 0x0000_0000)

Offset: 0x0050			Register Name: OWA_EXP_DBG_0
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18:16	R	0x0	IEC61937_DATA_CAP_FSM IEC61937 Data Captures State Machine 000: IDLE 001: SYNC_PA 010: SYNC_PB 011: DTYPE_PC 100: DLEN_PD 101: RX_ACTIVE
15:0	R	0x0	DATA_CAP_NUM Remains Data Counter Value See the value of the sampled valid data in real time.

16.4.7.22 0x0054 OWA Expand Debug Register 1 (Default Value: 0x0000_0000)

Offset: 0x0054			Register Name: OWA_EXP_DBG_1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R	0x0	REPET_PERIOD_OF_FR_CNT Repetition period of the burst frame counter See the value of repetition period counter in real time.
15:0	R	0x0	SR_CNT Sample Rate Counter See the value of audio sample ratio in real time.

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16.4.7.23 0x005C OWA Expand CDR Debug Register (Default Value: 0x0000_07FF)

Offset: 0x005C			Register Name: OWA_EXP_CDR_DBG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R	0x0	CDR_STATE State of CDR 0: Idle 1: cnt_start
27	/	/	/
26:16	R	0x000	PERIOD_MAX Max period of RX recovery clock. Take one OWA_rx sys_clock cycle as unit.
15:11	/	/	/
10:0	R	0x7FF	PERIOD_MIN Min period of RX recovery clock. Take one OWA_rx sys_clock cycle as unit.

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17 Peripherals

17.1 GMAC

17.1.1 Overview

The Gigabit Medium Access Controller (GMAC) enables a host to transmit and receive data over Ethernet in compliance with the IEEE 802.3-2015 standard.

The GMAC has the following features:

- Three 10/100/1000 Mbit/s Ethernet ports with RGMII and RMII interfaces (GMAC0, GMAC1, and GMAC2)
- Compatible with the IEEE 802.3-2015 standard
- Supports both full-duplex and half-duplex operation
- Full-duplex flow control
- IEEE 1588-2008 for precision networked clock synchronization
- Supports Ethernet packet timestamping as described in IEEE 1588-2002 and IEEE 1588-2008
- Programmable frame length to support Standard or Jumbo Ethernet frames with sizes up to 16 KB
- Source Address field and VLAN insertion or replacement, Double VLAN
- Transmit TCP/IP Checksum Offload
- Supports a variety of flexible address filtering modes(including Hash filter function)
- Optimization for packet-oriented DMA transfers with frame delimiters
- Supports linked-list descriptor list structure
- Descriptor architecture, allowing large blocks of data transfer with minimum CPU intervention: each descriptor can transfer up to 32 KB of data
- Comprehensive status reporting for normal operation and transfers with errors
- Supports 4KB TXFIFO for transmission packets and 8KB RXFIFO for reception packets
- Supports 16Descriptors Descriptor Pre-fetch cache for TXDMA and RX DMA
- Programmable interrupt options for different operational conditions
- Supports MDIO (Clause22 and Clause45) Interface for PHY device configuration and management
- Configurable big-endian and little-endian mode for Transmit and Receive paths
- Support statistics on the received and transmitted packets
- Split Header Support

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- Supports Energy Efficient Ethernet(EEE)

17.1.2 Block Diagram

The following figure shows the block diagram of GMAC.

Figure 17-1 GMAC Block Diagram

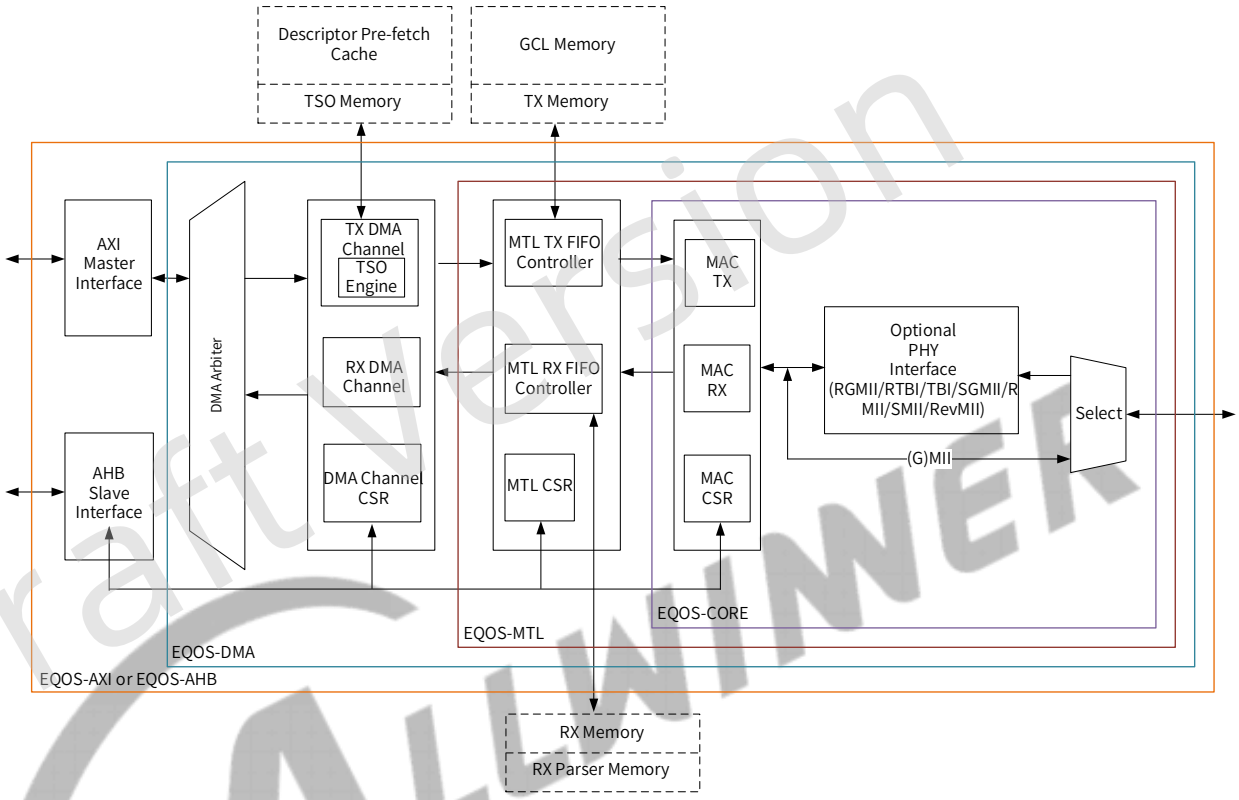

Table 17-1 GMAC Sub-blocks

Sub-blocks	Description
AHB Master Interface	The DMA controller interfaces with the application through the AMBA4 AXI interface. The AMBA4 AXI bus interface provides the characteristics to support highly-effective data traffic throughput.
AHB Slave Interface	The 32-bit AHB slave interface provides access to the DMA, MTL, and MAC CSR space.
PHY interface	The PHY interface can be configured to RGMII and RMII.
DMA	A DMA Channel has independent Transmit (TX) and Receive (RX) engines, and a CSR space. The TX engine transfers data from the system memory to the device port (MTL), whereas the RX engine transfers data from the device port to the system memory. The DMA engine uses descriptors to efficiently move data from source to destination with minimal application CPU intervention.

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Sub-blocks	Description
MTL	The MAC Transaction Layer (MTL) provides the FIFO memory Interface to buffer and regulate the packets between the application system memory and the MAC. It also enables the data to be transferred between the application clock and MAC clock domains. One MTL queue. The MTL layer has two data paths: Transmit path and Receive Path.
MAC	Media Access Controller. The MAC supports many interfaces towards the PHY chip.

17.1.3 External Signals

The following table describes the pin list of GMAC.

Table 17-2 GMAC External Signals

Signal Name	Description	Type
GMAC0		
RGMII0-RXCK/ RMII0-NULL	RGMII0 Receive Clock	I
RGMII0-TXCK/RMII0-TXCK	RGMII0/RMII0 Transmit Clock For RGMII, IO type is output; For RMII, IO type is input.	I/O
RGMII0-CLKIN/RMII0-RXER	RGMII0 Transmit Clock from External/RMII0 Receive Error	I
RGMII0-RXCTL/RMII0-CRS-DV	RGMII0 Receive Control/RMII0 Carrier Sense Receive Data Valid	I
RGMII0-TXCTL/RMII0-TXEN	RGMII0 Transmit Control/RMII0 Transmit Enable	O
RGMII0-MDC	RGMII0 Management Data Clock	O
RGMII0-MDIO	RGMII0 Management Data Input/ Output	I/O
RGMII0-EPHY-25/50M(1)	GMAC PHY 25 MHz or 50 MHz Clock Output	O
RGMII0-RXD0/RMII0-RXD0	RGMII0/RMII0 Receive Data 0	I
RGMII0-RXD1/RMII0-RXD1	RGMII0/RMII0 Receive Data 1	I
RGMII0-RXD2/RMII0-NULL	RGMII0 Receive Data 2	I
RGMII0-RXD3/RMII0-NULL	RGMII0 Receive Data 3	I
RGMII0-TXD0/RMII0-TXD0	RGMII0/RMII0 Transmit Data 0	O
RGMII0-TXD1/RMII0-TXD1	RGMII0/RMII0 Transmit Data 1	O
RGMII0-TXD2/RMII0-NULL	RGMII0 Transmit Data 2	O
RGMII0-TXD3/RMII0-NULL	RGMII0 Transmit Data 3	O
GMAC1		
RGMII1-RXCK/ RMII1-NULL	RGMII1 Receive Clock	I
RGMII1-TXCK/RMII1-TXCK	RGMII1/RMII1 Transmit Clock For RGMII, IO type is output; For RMII, IO type is input.	I/O
RGMII1-CLKIN/RMII1-RXER	RGMII1 Transmit Clock from External/RMII1 Receive Error	I

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Signal Name	Description	Type
RGMII1-RXCTL/RMII1-CRS-DV	RGMII1 Receive Control/RMII1 Carrier Sense Receive Data Valid	I
RGMII1-TXCTL/RMII1-TXEN	RGMII1 Transmit Control/RMII1 Transmit Enable	O
RGMII1-MDC	RGMII1 Management Data Clock	O
RGMII1-MDIO	RGMII1 Management Data Input/ Output	I/O
RGMII1-EPHY-25/50M(1)	GMAC PHY 25 MHz or 50 MHz Clock Output	O
RGMII1-RXD0/RMII1-RXD0	RGMII1/RMII1 Receive Data 0	I
RGMII1-RXD1/RMII1-RXD1	RGMII1/RMII1 Receive Data 1	I
RGMII1-RXD2/RMII1-NULL	RGMII1 Receive Data 2	I
RGMII1-RXD3/RMII1-NULL	RGMII1 Receive Data 3	I
RGMII1-TXD0/RMII1-TXD0	RGMII1/RMII1 Transmit Data 0	O
RGMII1-TXD1/RMII1-TXD1	RGMII1/RMII1 Transmit Data 1	O
RGMII1-TXD2/RMII1-NULL	RGMII1 Transmit Data 2	O
RGMII1-TXD3/RMII1-NULL	RGMII1 Transmit Data 3	O
GMAC2		
RGMII2-RXD0/RMII2-RXD0	RGMII2/RMII2 Receive Data 0	I
RGMII2-RXD1/RMII2-RXD1	RGMII2/RMII2 Receive Data 1	I
RGMII2-RXD2/RMII2-NULL	RGMII2 Receive Data 2	I
RGMII2-RXD3/RMII2-NULL	RGMII2 Receive Data 3	I
RGMII2-TXD0/RMII2-TXD0	RGMII2/RMII2 Transmit Data 0	O
RGMII2-TXD1/RMII2-TXD1	RGMII2/RMII2 Transmit Data 1	O
RGMII2-TXD2/RMII2-NULL	RGMII2 Transmit Data 2	O
RGMII2-TXD3/RMII2-NULL	RGMII2 Transmit Data 3	O
RGMII2-RXCK/RMII2-NULL	RGMII2 Receive Clock	I
RGMII2-TXCK/RMII2-TXCK	RGMII2/RMII2 Transmit Clock For RGMII, IO type is output; For RMII, IO type is input.	I/O
RGMII2-CLKIN/RMII2-RXER	RGMII2 Transmit Clock from External/RMII2 Receive Error	I
RGMII2-RXCTL/RMII2-CRS-DV	RGMII2 Receive Control/RMII2 Carrier Sense Receive Data Valid	I
RGMII2-TXCTL/RMII2-TXEN	RGMII2 Transmit Control/RMII2 Transmit Enable	O
RGMII2-MDC	RGMII2 Management Data Clock	O
RGMII2-MDIO	RGMII2 Management Data Input/ Output	I/O
RGMII2-EPHY-25/50M(1)	GMAC PHY 25 MHz or 50 MHz Clock Output	O

(1) RGMII0-EPHY-25/50M and RGMII1-EPHY-25/50M are external PHY reference clock signals. They are directly generated by CCU and output through GPIO. This function could be enabled by configuring the GMACx_PHY_CLK_REG (x=0 to 2) register in Section 4.1 Clock Controller Unit (CCU).

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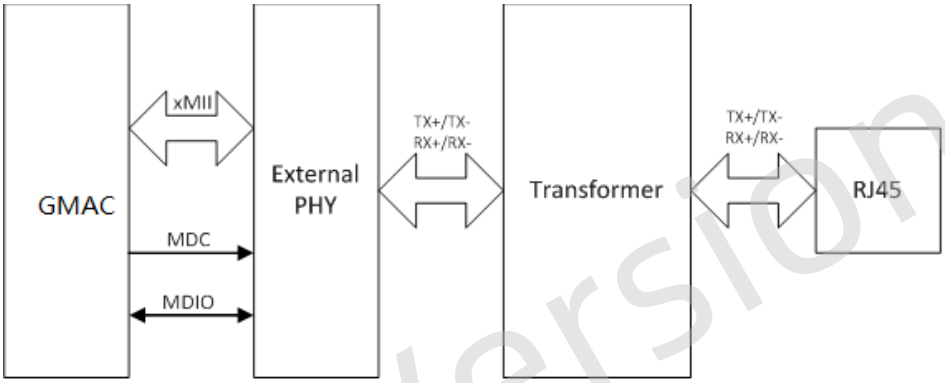
17.1.4 Functional Description

17.1.4.1 Typical Application

Figure 17-2 GMAC Typical Application



17.1.5 Register List

Module Name	Base Address
GMAC0_TOP	0x04508000——0x0450FFFF
GMAC1_TOP	0x04518000——0x0451FFFF
GMAC2_TOP	0x04528000——0x0452FFFF
GMAC0_IP	0x0450 0000——0x04507FFF
GMAC1_IP	0x0451 0000——0x04517FFF
GMAC2_IP	0x0452 0000——0x04527FFF

17.1.5.1 GMAC_TOP Register List

Register Name	Offset	Description
GMAC_CFG_REG	0x0000	GMAC Config Register
GMAC_PTP_TIMESTAMP_LOW_REG	0x0040	GMAC PTP Timestamp Low Register
GMAC_PTP_TIMESTAMP_HIGH_REG	0x0048	GMAC PTP Timestamp High Register
GMAC_STAT_INT_REG	0x004C	GMAC Low Power Interrupt Status Register
GMAC_CLK_GATE_CFG_REG	0x0080	GMAC Clock Gate Config Register

17.1.6 GMAC_TOP Register Description

17.1.6.1 0x0000 GMAC Config Register (Default Value: 0x0000_8000)

Offset:0x0000			Register Name: GMAC_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/

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Offset:0x0000			Register Name: GMAC_CFG_REG
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	ETXDC_H Configure GMAC Transmit Clock Delay Chain high 2bits.
15	R/W	0x1	PHY_SELECT. 0 : External PHY 1 : Internal PHY The default value is “1”. It should be configured as “0”.
14	R/W	0x0	Endianness_mode 0: Little 1: Big
13	R/W	0x0	RMII_EN RMII Enable 0 : Disable RMII Module 1 : Enable RMII Module When this bit is asserted, RGMII interface is disabled(This means bit13 is prior to bit2)、
12:10	R/W	0x0	ETXDC_L Configure GMAC Transmit Clock Delay Chain low 3bits.
9:5	R/W	0x0	ERXDC Configure GMAC Receive Clock Delay Chain.
4	R/W	0x0	ERXIE Enable GMAC Receive Clock Invertor. 0: Disable 1: Enable
3	R/W	0x0	ETXIE Enable GMAC Transmit Clock Invertor. 0: Disable 1: Enable
2	R/W	0x0	EPIT GMAC PHY Interface Type. 0: RMII 1: RGMII
1:0	R/W	0x0	ETCS GMAC Transmit Clock Source. 00: Transmit clock source for RMII 01: External transmit clock source for RGMII 10: Internal transmit clock source for RGMII 11: Reserved

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17.1.6.2 0x0040 GMAC PTP Timestamp Low Register (Default Value: 0x0000_0000)

Offset:0x0040			Register Name: GMAC_PTP_TIMESTAMP_LOW_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	PTP_TIMESTAMP_L ptp_timestamp_low

17.1.6.3 0x0048 GMAC PTP Timestamp High Register (Default Value: 0x0000_0000)

Offset:0x0048			Register Name: GMAC_PTP_TIMESTAMP_HIGH_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	PTP_TIMESTAMP_H ptp_timestamp_high

17.1.6.4 0x004C GMAC Low Power Interrupt Status Register (Default Value: 0x0000_0000)

Offset:0x004C			Register Name: GMAC_STAT_INT_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R	0x0	SBD_TX_CLK_GATING LPI Tx Clock Gating Control This signal is high after the MAC enters the Tx LPI mode. You can use this signal to control the Tx clock gating.
1	R	0x0	LPI_INT This signal is high when the MAC receiver exits the LPI state. You can use the lpi_intr_o signal to trigger the external clock-gating circuitry to restore the application clock to the MAC. If you do not want to gate-off the application clock during the Rx LPI state, you can leave the lpi_intr_o signal unconnected.
0	/	/	/

17.1.6.5 0x0080 GMAC Clock Gate Config Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: GMAC_CLK_GATE_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x0080			Register Name: GMAC_CLK_GATE_CFG_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	RX_CLK_GATING RX_CLK Gating Control 0: Clock gate disable(CLK is on) 1: Clock gate enable(CLK is off)
6	R/W	0x0	PTP_REFCLK_GATING PTP reference clock gate control 0: Clock gate disable(CLK is on) 1: Clock gate enable(CLK is off)
5	R/W	0x0	CSR_CLK_GATING CSR_CLK Gating Control 0: Clock gate disable(CLK is on) 1: Clock gate enable(CLK is off)
4	R/W	0x0	TX_CLK_GATING TX_CLK Gating Control 0: Clock gate disable(clk is on) 1: Clock gate enable(clk is off)
3	R/W	0x0	APP_CLK_GATING Application Clock Gating Control 0: Clock gate disable(CLK is on) 1: Clock gate enable(CLK is off)
2:0	/	/	/

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17.2 CAN-FD

17.2.1 Overview

The Controller Area Network (CAN) is an asynchronous serial communication interface. It is one of the most widely used field bus and is widely used in vehicle and industrial control area. It has fault tolerance and is working in real-time.

- Two CAN-FD ports (up to 64 data bytes)
- Conforms with ISO 11898-1:2015
- Supports CAN 2.0A and 2.0B protocol specification
- Supports AUTomotive Open System ARchitecture (AUTOSAR)
- Supports SAE J1939 standards
- CAN error logging
- Improved acceptance filtering
- Two configurable receive FIFOs
- Separate signaling on reception of high priority messages
- Up to 64 dedicated receive buffers
- Up to 32 dedicated transmit buffers
- Configurable transmit FIFO
- Configurable transmit queue
- Configurable transmit event FIFO
- Programmable loop-back test mode
- Maskable module interrupts
- Power-down support
- Debug on CAN support
- Direct message RAM access for host CPU
- Direct AHB slave interface to local SRAM
- Supports DMA Master
  - One channel for each CAN
  - MBUS Master Interface to common RAM
  - Memory Interface to local SRAM
  - Descriptor with finish flag, write back flag and next descriptor address
  - DMA TX initiated by descriptor and DMA RX initiated by REG from CMD FIFO

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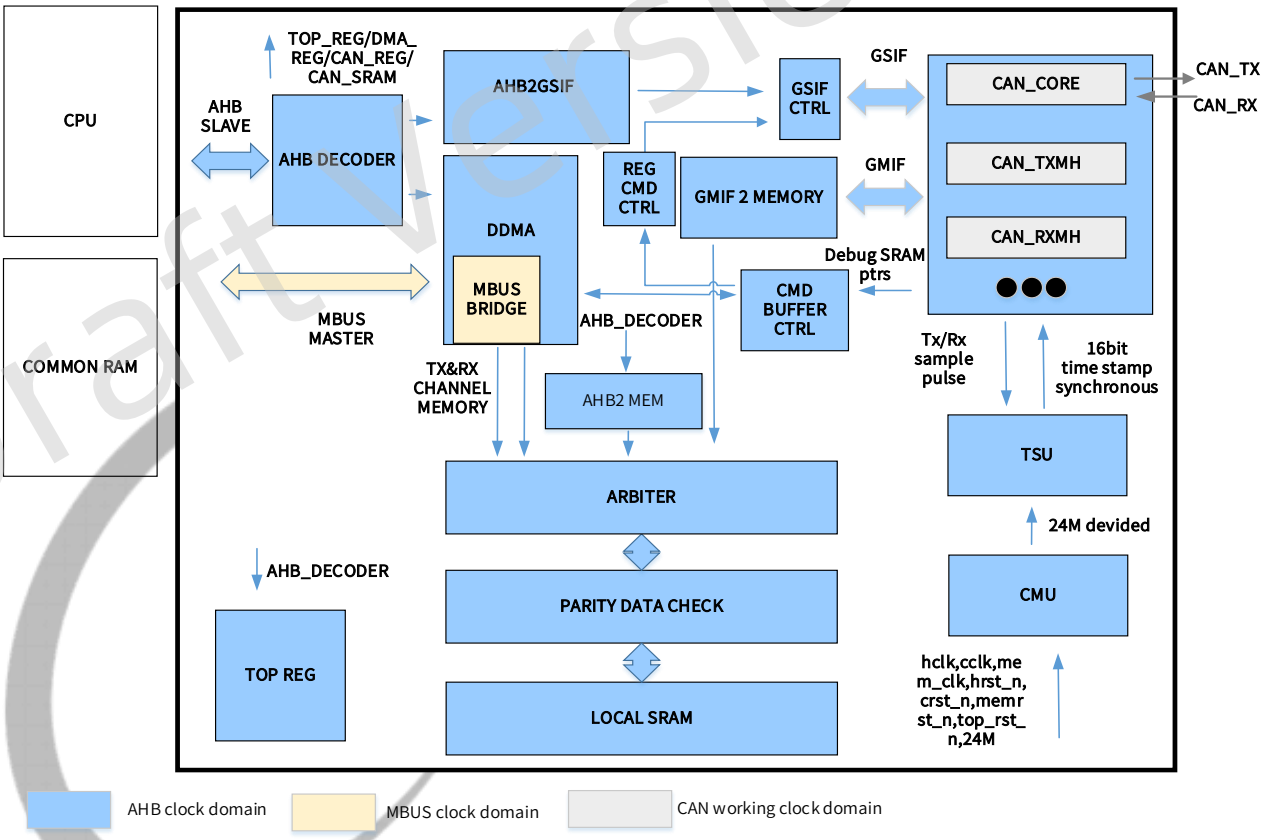
- CAN RX FIFO and TX FIFO mode (not supports RX dedicate buffer mode and TX queue mode)

- Parity data check for SRAM
- Supports arbiter between DMA master and GMIF to access local SRAM
- 16 bits' hardware timestamp

17.2.2 Block Diagram

The following figure shows the block diagram of the CAN-FD.

Figure 17-3 CAN-FD Block Diagram



17.2.3 External Signals

The following table describes the external signals of the CAN-FD.

Table 17-3 CAN-FD External Signals

Signal Name	Description	Type
CAN0-RX0	CAN0 Receive Data 0	I
CAN0-TX0	CAN0 Transmit Data 0	O
CAN1-RX0	CAN1 Receive Data 0	I
CAN1-TX0	CAN1 Transmit Data 0	O

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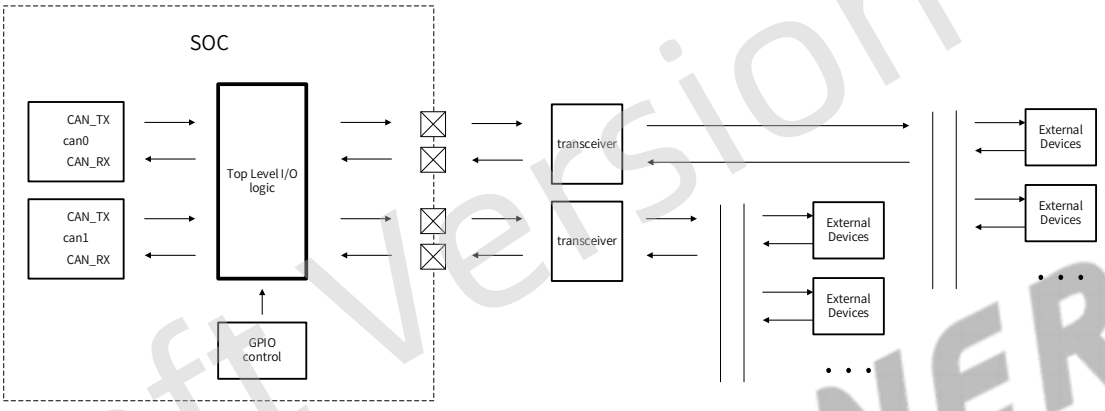
17.2.4 Functional Description

17.2.4.1 Typical Application

The following figure shows the IO signals of CAN controller. To connect external signals, you need to configure GPIO controller to enable CAN model.

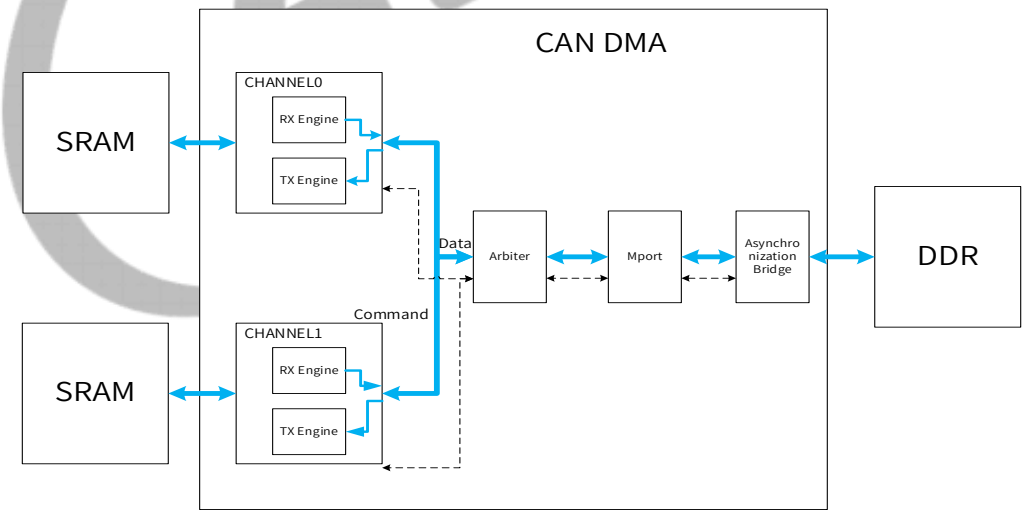
Figure 17-4 I/O Signals of CAN Controllers



17.2.4.2 CAN DMA

Overview

Figure 17-5 CAN DMA Block Diagram



This private DMA adopts a structure similar to the general DMA, assigning a private channel to each CAN, accessing the SRAM inside the CAN module through the memory bus, and accessing the RAM outside the module through MBUS.

The DMA module is divided into three modules: mport, arbiter and channel. Mport is mainly used to control the MBUS master and external access to DDR; arbiter is used to implement arbitration

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between channels; channel is used to implement the data buffering of the read and write buffers of each channel within the DMA, and initiate read and write access control to SRAM.

DMA TX (source address DDR, destination address SRAM) is initiated by the CPU and the configuration method is descriptor configuration. CPU first configures the descriptor chain information, and then configures the register to start the DMA TX transmission.

DMA RX (source address SRAM, destination address DDR) is initiated by the CAN node and the configuration method is similar to register configuration. The top layer of the CAN module analyzes the amount of data received by the CAN node, analyzes whether the SRAM is empty or full, and converts them into the DMA transmission configuration information.

CAN DMA Descriptor

The private DMA adopts a chain descriptor structure. The default descriptor length is 12 bytes. The address of the first descriptor is configured through registers. During transfer, after the hardware obtains the previous descriptor, it obtains data through the address recorded in the descriptor. When this transfer is completed, hardware continues to access the next descriptor address given by the previous descriptor. Hardware continually transfers data until the finish flag configuration is valid, then the DMA transfer ends.

Table 17-4 Descriptor0 Definition

Bits	Descriptor
31	This bit is used to operate write back behavior. This bit will be set to 1 by DMA when DMA has already completed current opcode.
30:29	/
28:26	Frame length 111: 72 bytes 110: 56 bytes 101: 40 bytes 100: 32 bytes 011: 28 bytes 010: 24 bytes 001: 20 bytes 000: 16 bytes Used to create a frame finish signal for CAN.
25	/
24:15	Data Length The data length is in word unit.
14:13	/
12:0	Destination Address

Table 17-5 Descriptor1 Definition

Bits	Descriptor

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Bits	Descriptor
31:0	NEXT_DESCRIPTOR_ADDR It needs to be a word address. When 32'hFFFFF800 is written, it means that current descriptor is the last descriptor. After execution, the DMA transfer will automatically stop.

Table 17-6 Descriptor2 Definition

Bits	Descriptor
31:0	SOURCE_ ADDR Source address in DDR It needs to be a word address.

17.2.5 Programming Guidelines

17.2.5.1 Global Configuration

Basic Configuration

The CPU configures the M_CAN by reading/writing the registers in the M_CAN. The basic things to be configured are:

- Bit timings
- Enabling interrupts and selecting the interrupt lines

Transmit with CPU mode

1. For the M_CAN to behave as a transmitting node, the CPU configures the fundamental Tx registers like TXBC, TXESC etc.
2. The messages that should be transmitted are first written into the Message RAM in the section allocated for the Tx Buffers.
3. In order to request the transmission of a message, the CPU has to write to the appropriate bits in the TXBAR register of the M_CAN.
4. When a request to transmit a message is received by the M_CAN, the M_CAN scans the Message RAM (Tx Buffers) for the highest priority message and transmits this message.

Receive with CPU mode

1. For the M_CAN to behave as a receiving node, the fundamental Rx registers like RXBC, RXFnC (n = 0,1), and RXESC etc. have to be configured by the CPU. These registers define among others where the messages are stored.
2. Acceptance filtering of the received messages can be done by configuring registers GFC, SIDFC and XIDFC.
3. If acceptance filtering is configured, the CPU has to write the standard/extended message ID filters into the Message RAM.

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4. When a message is received by the M_CAN, the M_CAN does the acceptance filtering according to the GFC, SIDFC and XIDFC configurations. It compares the ID of the received message with the ID's in the filter elements. If it finds a match, it stores/discards the received message according to the filter element configuration.
5. After filtering, if the message is to be stored, it will be stored either in an Rx FIFO or a Dedicated Rx buffer in the Message RAM, according to the filter element configuration.
6. The M_CAN can inform the CPU via interrupts (if configured) that a new message has arrived.
7. The CPU reads the Rx Buffer/FIFO status registers of the M_CAN to find out where the M_CAN had stored the message in the Message RAM.
8. The CPU reads the message from the Message RAM.
9. The CPU acknowledges the message by writing to an M_CAN register.

Transmit with DMA mode

1. For the M_CAN to behave as a transmitting node, the CPU configures the fundamental Tx registers like TXBC, TXESC etc.
2. Set TX descriptor in common RAM.
3. Configure CAN DMA TX CHANNEL registers, and enable DMA TX CHANNEL.
4. The messages that should be transmitted are first written by DMA into the Message RAM in the section allocated for the Tx Buffers.
5. The request of TX data transmission is automatically done.
6. When a request to transmit a message is received by the M_CAN, the M_CAN scans the Message RAM (Tx Buffers) for the highest priority message and transmits this message.

Receive with DMA mode

1. For the M_CAN to behave as a receiving node, the fundamental Rx registers like RXBC, RXFnC (n = 0,1), and RXESC etc. have to be configured by the CPU. These registers define among others where the messages are stored.
2. Acceptance filtering of the received messages can be done by configuring registers GFC, SIDFC and XIDFC.
3. If acceptance filtering is configured, the CPU has to write the standard/extended message ID filters into the Message RAM.
4. Configure CAN DMA RX CHANNEL registers, and enable DMA RX CHANNEL.
5. When a message is received by the M_CAN, the M_CAN does the acceptance filtering according to the GFC, SIDFC and XIDFC configurations. It compares the ID of the received message with the ID's in the filter elements. If it finds a match, it stores/discards the received message according to the filter element configuration.
6. After filtering, if the message is to be stored, it will be stored only in Rx FIFOs in the Message RAM, according to the filter element configuration.

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7. The M_CAN can inform the CPU via interrupts (if configured) that a new message has arrived or not.
8. DMA RX CHANNEL will send RX data in Rx FIFOs into common RAM, according to configured address area.
9. The CAN module will inform the CPU via interrupts that a number of frames of data is already in common RAM.
10. The CPU should check the DDR write pointer address (known as XXX_CUR_ADDRESS in registers) and DDR read pointer address (known as XXX_RPTR_ADDRESS).
11. The CPU should get frame data from DDR read pointer address to DDR write pointer address.
12. When sending frame data from local SRAM to DDR, instead of CPU, the CAN module acknowledges the message by writing to an M_CAN register.

17.2.5.2 M_CAN Configuration

Software Initialization

Software initialization is started by setting bit CCCR.INIT, either by software or by a hardware reset, when an uncorrected bit error was detected in the Message RAM, or by going Bus_Off. While CCCR.INIT is set, message transfer from and to the CAN bus is stopped, the status of the CAN bus output m_can_tx is recessive (HIGH). The counters of the Error Management Logic EML are unchanged. Setting CCCR.INIT does not change any configuration register. Resetting CCCR.INIT finishes the software initialization. Afterwards the Bit Stream Processor BSP synchronizes itself to the data transfer on the CAN bus by waiting for the occurrence of a sequence of 11 consecutive recessive bits (┤ Bus_Idle) before it can take part in bus activities and start the message transfer.

Access to the M_CAN configuration registers is only enabled when both bits CCCR.INIT and CCCR.CCE are set (protected write).

CCCR.CCE can only be set/reset while CCCR.INIT = '1'. CCCR.CCE is automatically reset when CCCR.INIT is reset.

The following registers are reset when CCCR.CCE is set

- HPMS - High Priority Message Status
- RXF0S - Rx FIFO 0 Status
- RXF1S - Rx FIFO 1 Status
- TXFQS - Tx FIFO/Queue Status
- TXBRP - Tx Buffer Request Pending
- TXBTO - Tx Buffer Transmission Occurred
- TXBCF - Tx Buffer Cancellation Finished
- TXEFS - Tx Event FIFO Status

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The Timeout Counter value TOCV.TOC is preset to the value configured by TOCC.TOP when CCCR.CCE is set.

In addition the state machines of the Tx Handler and Rx Handler are held in idle state while CCCR.CCE = '1'.

In addition the state machines of the Tx Handler and Rx Handler are held in idle state while CCCR.CCE = '1'.

The following registers are only writeable while CCCR.CCE = '0'

- TXBAR - Tx Buffer Add Request
- TXBCR - Tx Buffer Cancellation Request

CCCR.TEST and CCCR.MON can only be set by the Host while CCCR.INIT = '1' and CCCR.CCE = '1'. Both bits may be reset at any time. CCCR.DAR can only be set/reset while CCCR.INIT = '1' and CCCR.CCE = '1'.

NOTE

In case the Message RAM is equipped with parity or ECC functionality, it is recommended to initialize the Message RAM after hardware reset by writing e.g. 0x00000000 to each Message RAM word to create valid parity/ECC checksums. This avoids that reading from uninitialized Message RAM sections will activate interrupt IR.BEC (Bit Error Corrected) or IR.BEU (Bit Error Uncorrected).

Normal Operation

Once the M_CAN is initialized and CCCR.INIT is reset to zero, the M_CAN synchronizes itself to the CAN bus and is ready for communication.

After passing the acceptance filtering, received messages including Message ID and DLC are stored into a dedicated Rx Buffer or into Rx FIFO 0 or Rx FIFO 1.

For messages to be transmitted dedicated Tx Buffers and/or a Tx FIFO or a Tx Queue can be initialized or updated. Automated transmission on reception of remote frames is not implemented.

CAN FD Operation

There are two variants in the CAN FD frame transmission, first the CAN FD frame without bit rate switching. The second variant is the CAN FD frame where control field, data field, and CRC field are transmitted with a higher bit rate than the beginning and the end of the frame.

The previously reserved bit in CAN frames with 11-bit identifiers and the first previously reserved bit in CAN frames with 29-bit identifiers will now be decoded as FDF bit. FDF = recessive signifies a CAN FD frame, FDF = dominant signifies a Classic CAN frame. In a CAN FD frame, the two bits following FDF, res and BRS, decide whether the bit rate inside of this CAN FD frame is switched. A CAN FD bit rate switch is signified by res = dominant and BRS = recessive. The coding of res = recessive is reserved for future expansion of the protocol. In case the M_CAN receives a frame with FDF = recessive and res = recessive, it will signal a Protocol Exception Event by setting bit

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PSR.PXE. When Protocol Exception Handling is enabled (CCCR.PXHD = '0'), this causes the operation state to change from Receiver (PSR.ACT = "10") to Integrating (PSR.ACT = "00") at the next sample point. In case Protocol Exception Handling is disabled (CCCR.PXHD = '1'), the M_CAN will treat a recessive res bit as an form error and will respond with an error frame.

CAN FD operation is enabled by programming CCCR.FDOE. In case CCCR.FDOE = '1', transmission and reception of CAN FD frames is enabled. Transmission and reception of Classic CAN frames is always possible. Whether a CAN FD frame or a Classic CAN frame is transmitted can be configured via bit FDF in the respective Tx Buffer element. With CCCR.FDOE = '0', received frames are interpreted as Classic CAN frames, which leads to the transmission of an error frame when receiving a CAN FD frame. When CAN FD operation is disabled, no CAN FD frames are transmitted even if bit FDF of a Tx Buffer element is set. CCCR.FDOE and CCCR.BRSE can only be changed while CCCR.INIT and CCCR.CCE are both set.

With CCCR.FDOE = '0', the setting of bits FDF and BRS is ignored and frames are transmitted in Classic CAN format. With CCCR.FDOE = '1' and CCCR.BRSE = '0', only bit FDF of a Tx Buffer element is evaluated. With CCCR.FDOE = '1' and CCCR.BRSE = '1', transmission of CAN FD frames with bit rate switching is enabled. All Tx Buffer elements with bits FDF and BRS set are transmitted in CAN FD format with bit rate switching.

A mode change during CAN operation is only recommended under the following conditions:

- The failure rate in the CAN FD data phase is significant higher than in the CAN FD arbitration phase. In this case disable the CAN FD bit rate switching option for transmissions.
- During system start-up all nodes are transmitting Classic CAN messages until it is verified that they are able to communicate in CAN FD format. If this is true, all nodes switch to CAN FD operation.
- Wake-up messages in CAN Partial Networking have to be transmitted in Classic CAN format.
- End-of-line programming in case not all nodes are CAN FD capable. Non CAN FD nodes are held in Silent mode until programming has completed. Then all nodes switch back to Classic CAN communication.

In CAN FD frames, the bit timing will be switched inside the frame, after the BRS (Bit Rate Switch) bit, if this bit is recessive. Before the BRS bit, in the CAN FD arbitration phase, the nominal CAN bit timing is used as defined by the Nominal Bit Timing & Prescaler Register NBTP. In the following CAN FD data phase, the data phase bit timing is used as defined by the Data Bit Timing & Prescaler Register DBTP. The bit timing is switched back from the data phase timing at the CRC delimiter or when an error is detected, whichever occurs first.

The maximum configurable bit rate in the CAN FD data phase depends on the CAN clock frequency. Example: with a CAN clock frequency of 20MHz and the shortest configurable bit time of 4 tq, the bit rate in the data phase is 5 Mbit/s.

In both data frame formats, CAN FD and CAN FD with bit rate switching, the value of the bit ESI (Error Status Indicator) is determined by the transmitter's error state at the start of the transmission.

If the transmitter is error passive, ESI is transmitted recessive, else it is transmitted dominant.

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Restricted Operation Mode

In Restricted Operation Mode the node is able to receive data and remote frames and to give acknowledge to valid frames, but it does not send data frames, remote frames, active error frames, or overload frames. In case of an error condition or overload condition, it does not send dominant bits, instead it waits for the occurrence of bus idle condition to resynchronize itself to the CAN communication. The error counters (ECR.REC, ECR.TEC) are frozen while Error Logging (ECR.CEL) is active. The Host can set the M_CAN into Restricted Operation mode by setting bit CCCR.ASM. The bit can only be set by the Host when both CCCR.CCE and CCCR.INIT are set to '1'. The bit can be reset by the Host at any time.

Restricted Operation Mode is automatically entered when the Tx Handler was not able to read data from the Message RAM in time. To leave Restricted Operation Mode, the Host CPU has to reset CCCR.ASM.

The Restricted Operation Mode can be used in applications that adapt themselves to different CAN bit rates. In this case the application tests different bit rates and leaves the Restricted Operation Mode after it has received a valid frame.

NOTE

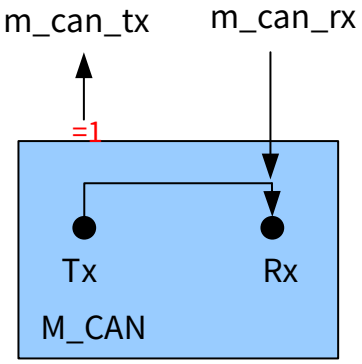
The Restricted Operation Mode must not be combined with the Loop Back Mode (internal or external).

Bus Monitor Mode

The M_CAN is set in Bus Monitoring Mode by programming CCCR.MON to one. In Bus Monitoring Mode (see *ISO 11898-1:2015, 10.14 Bus monitoring*), the M_CAN is able to receive valid data frames and valid remote frames, but cannot start a transmission. In this mode, it sends only recessive bits on the CAN bus. If the M_CAN is required to send a dominant bit (ACK bit, overload flag, active error flag), the bit is rerouted internally so that the M_CAN monitors this dominant bit, although the CAN bus may remain in recessive state. In Bus Monitoring Mode register TXBRP is held in reset state.

The Bus Monitoring Mode can be used to analyze the traffic on a CAN bus without affecting it by the transmission of dominant bits. The following figure shows the connection of signals m_can_tx and m_can_rx to the M_CAN in Bus Monitoring Mode.

Figure 17-6 Bus Monitor Mode



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Disabled Automatic Retransmission

According to the CAN Specification (see *ISO 11898-1:2015, 8.3.4 Recovery Management*), the M_CAN provides means for automatic retransmission of frames that have lost arbitration or that have been disturbed by errors during transmission. By default automatic retransmission is enabled.

To support time-triggered communication as described in *ISO 11898-1:2015, chapter 9.2*, the automatic retransmission may be disabled via CCCR.DAR.

Frame Transmission in DAR Mode

In DAR mode all transmissions are automatically cancelled after they started on the CAN bus. A Tx Buffer's Tx Request Pending bit TXBRP.TRPx is reset after successful transmission, when a transmission has not yet been started at the point of cancellation, has been aborted due to lost arbitration, or when an error occurred during frame transmission.

- Successful transmission:

  Corresponding Tx Buffer Transmission Occurred bit TXBTO.TOx set

  Corresponding Tx Buffer Cancellation Finished bit TXBCF.CFx not set
- Successful transmission in spite of cancellation:

  Corresponding Tx Buffer Transmission Occurred bit TXBTO.TOx set

  Corresponding Tx Buffer Cancellation Finished bit TXBCF.CFx set
- Arbitration lost or frame transmission disturbed:

  Corresponding Tx Buffer Transmission Occurred bit TXBTO.TOx not set

  Corresponding Tx Buffer Cancellation Finished bit TXBCF.CFx set

In case of a successful frame transmission, and if storage of Tx events is enabled, a Tx Event FIFO element is written with Event Type ET = "10" (transmission in spite of cancellation).

Power Down (Sleep Mode)

The M_CAN can be set into power down mode controlled by input signal m_can_clkstop_req or via CC Control Register CCCR.CSR. As long as the clock stop request signal m_can_clkstop_req is active, bit CCCR.CSR is read as one.

When all pending transmission requests have completed, the M_CAN waits until bus idle state is detected. Then the M_CAN sets then CCCR.INIT to one to prevent any further CAN transfers. Now the M_CAN acknowledges that it is ready for power down by setting output signal m_can_clkstop_ack to one and CCCR.CSA to one. Before the module clocks m_can_hclk and m_can_cclk are switched off, further register accesses can be made except for CCCR.INIT which is held at one.

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NOTE

In case of a heavily disturbed CAN bus, it may happen that idle state is never reached and CCCR.INIT is therefore not set by the M_CAN. This situation can be detected by polling PSR.ACT. In case the M_CAN does not enter idle state, the software can write CCCR.INIT = '1' which stops CAN communication of the M_CAN immediately, regardless whether there is a transmission/reception ongoing or not.

To leave power down mode, the application has to turn on the module clocks before resetting signal m_can_clkstop_req resp. CC Control Register flag CCCR.CSR. The M_CAN will acknowledge this by resetting output signal m_can_clkstop_ack and resetting CCCR.CSA. Afterwards, the application can restart CAN communication by resetting bit CCCR.INIT.

Test Modes

To enable write access to register TEST, bit CCCR.TEST has to be set to one.

This allows the configuration of the test modes and test functions.

Four output functions are available for the CAN transmit pin m_can_tx by programming TEST.TX.

Additionally to its default function - the serial data output - it can drive the CAN Sample Point signal to monitor the M_CAN's bit timing and it can drive constant dominant or recessive values. The actual value at pin m_can_rx can be read from TEST.RX. Both functions can be used to check the CAN bus' physical layer.

Due to the synchronization mechanism between CAN clock and Host clock domain, there may be a delay of several Host clock periods between writing to TEST.TX until the new configuration is visible at output pin m_can_tx. This applies also when reading input pin m_can_rx via TEST.RX.

NOTE

Test modes should be used for production tests or self test only. The software control for pin m_can_tx interferes with all CAN protocol functions. It is not recommended to use test modes for application.

- External Loop Back Mode

The M_CAN can be set in External Loop Back Mode by programming TEST.LBCK to one. In Loop Back Mode, the M_CAN treats its own transmitted messages as received messages and stores them (if they pass acceptance filtering) into an Rx Buffer or an Rx FIFO. Figure 3-2. shows the connection of signals m_can_tx and m_can_rx to the M_CAN in External Loop Back Mode.

This mode is provided for hardware self-test. To be independent from external stimulation, the M_CAN ignores acknowledge errors (recessive bit sampled in the acknowledge slot of a data/remote frame) in Loop Back Mode. In this mode the M_CAN performs an internal feedback from its Tx output to its Rx input. The actual value of the m_can_rx input pin is disregarded by the M_CAN. The transmitted messages can be monitored at the m_can_tx pin.

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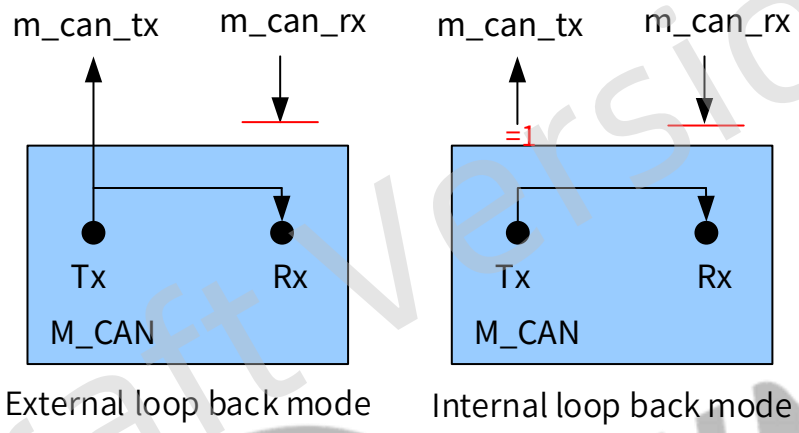
- Internal Loop Back Mode

Internal Loop Back Mode is entered by programming bits TEST.LBCK and CCCR.MON to one. This mode can be used for a “Hot Selftest”, meaning the M_CAN can be tested without affecting a running CAN system connected to the pins m_can_tx and m_can_rx. In this mode pin m_can_rx is disconnected from the M_CAN and pin m_can_tx is held recessive.

The following figure shows the connection of m_can_tx and m_can_rx to the M_CAN in case of Loop Back Mode.

Figure 17-7 Loop Back Mode



17.2.5.3 Local message RAM

Message RAM Configuration

The Message RAM has a width of 36 bits. 32bits for data, and 4 bits for data parity check. 4 parity bits are auto checked and is not visible to users. The total message ram size of one SRAM is 4352 levels with 36 bits width. The address of each small inner message ram is configurable through registers.

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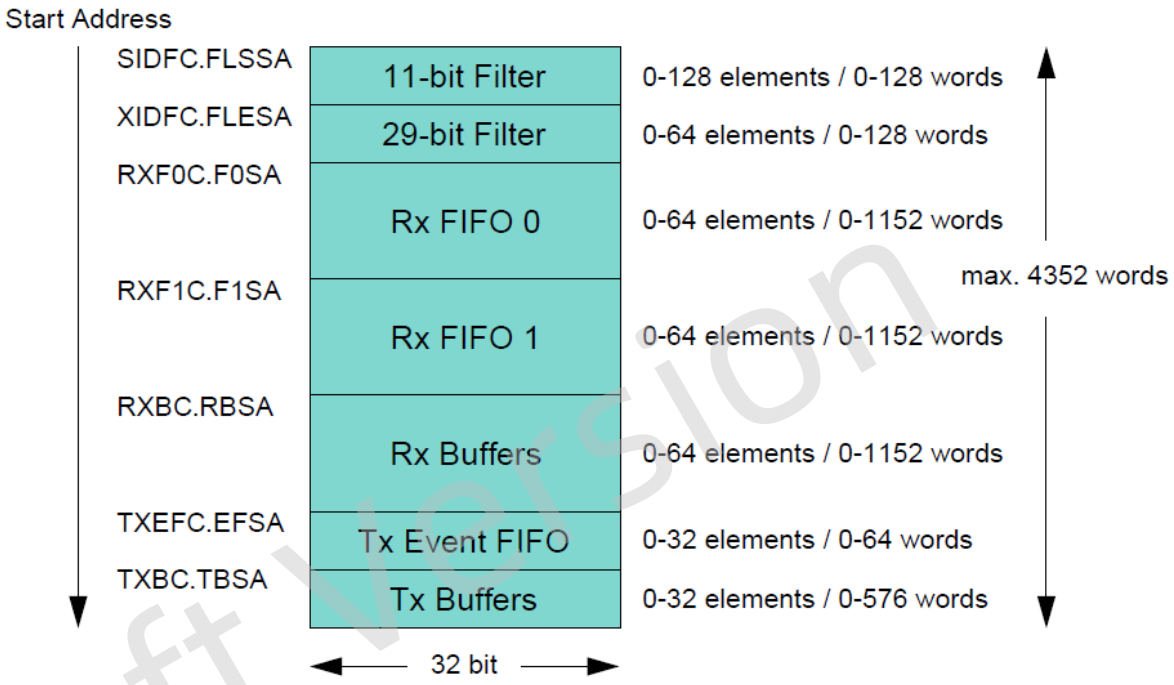
Figure 17-8 Message Ram Configuration Example



When operated in CAN FD mode the required Message RAM size strongly depends on the element size configured for Rx FIFO0, Rx FIFO1, Rx Buffers, and Tx Buffers via RXESC.F0DS, RXESC.F1DS, RXESC.RBDS, and TXESC.TBDS.

Rx Buffer and FIFO Element

Up to 64 Rx Buffers and two Rx FIFOs can be configured in the Message RAM. Each Rx FIFO section can be configured to store up to 64 received messages. The structure of a Rx Buffer / FIFO element is shown in Figure 17-9 below. The element size can be configured for storage of CAN FD messages with up to 64 bytes data field via register RXESC.

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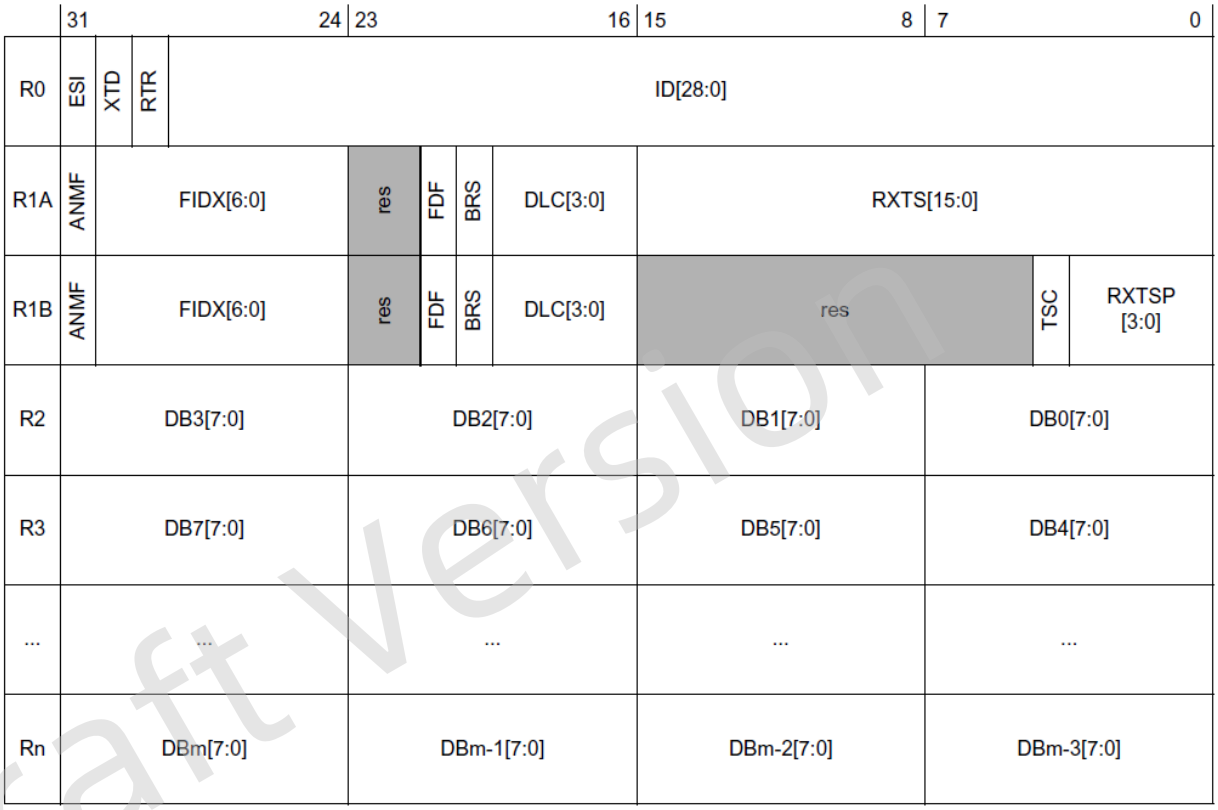
Figure 17-9 Rx Buffer/FIFO Element



- **R1A**: When no TSU is used (CCCR.UTSU = '0'), R1A.RXTS[15:0] holds the 16-bit timestamp generated by the M_CAN's internal timestamping logic.
- **R1B**: Not used in current design.
- **R0 Bit 31 ESI**: Error State Indicator

  0: Transmitting node is error active

  1: Transmitting node is error passive
- **R0 Bit 30 XTD**: Extended Identifier

  Signals to the Host whether the received frame has a standard or extended identifier.

  0: 11-bit standard identifier

  1: 29-bit extended identifier
- **R0 Bit 29 RTR**: Remote Transmission Request

  Signals to the Host whether the received frame is a data frame or a remote frame.

  0: Received frame is a data frame

  1: Received frame is a remote frame
- **R0 Bits 28:0 ID[28:0]**: Identifier

  Standard or extended identifier depending on bit XTD. A standard identifier is stored into ID[28:18].

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- **R1A/B Bit 31 ANMF**: Accepted Non-matching Frame

  Acceptance of non-matching frames may be enabled via GFC.ANFS and GFC.ANFE.

  0: Received frame matching filter index FIDX

  1: Received frame did not match any Rx filter element

- **R1A/B Bits 30:24 FIDX[6:0]**: Filter Index

  0-127:Index of matching Rx acceptance filter element (in valid if ANMF = '1').

  Range is 0 to SIDFC.LSS - 1 resp. XIDFC.LSE - 1.

- **R1A/B Bit 21 FDF**: FD Format

  0: Classical CAN frame format

  1: CAN FD frame format (new DLC-coding and CRC)

- **R1A/B Bit 20 BRS**: Bit Rate Switch

  0: Frame received without bit rate switching

  1: Frame received with bit rate switching

- **R1A/B Bits 19:16 DLC[3:0]**: Data Length Code

  0-8: CAN + CAN FD: received frame has 0-8 data bytes

  9-15: CAN: received frame has 8 data bytes

  9-15: CAN FD: received frame has 12/16/20/24/32/48/64 data bytes

- **R1A Bits 15:0 RXTS[15:0]**: Rx Timestamp

  Timestamp Counter value captured on start of frame reception. Resolution depending on configuration of the Timestamp Counter Prescaler TSCC.TCP.

- **R1B Bit 4 TSC**: Timestamp Captured

  0: No timestamp captured

  1: Timestamp captured and stored in TSU Timestamp register referenced by R1B.RXTSP

- **R1B Bits 3:0 RXTSP[3:0]**: Rx Timestamp Pointer

  Number of TSU Time Stamp register (TS0..15) where the related timestamp is stored.

- **R2 Bits 31:24 DB3[7:0]**: Data Byte 3
- **R2 Bits 23:16 DB2[7:0]**: Data Byte 2
- **R2 Bits 15:8 DB1[7:0]**: Data Byte 1
- **R2 Bits 7:0 DB0[7:0]**: Data Byte 0
- **R3 Bits 31:24 DB7[7:0]**: Data Byte 7
- **R3 Bits 23:16 DB6[7:0]**: Data Byte 6
- **R3 Bits 15:8 DB5[7:0]**: Data Byte 5
- **R3 Bits 7:0 DB4[7:0]**: Data Byte 4

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... ... : ...

- **Rn Bits 31:24 DBm[7:0]**: Data Byte m
- **Rn Bits 23:16 DBm-1[7:0]**: Data Byte m-1
- **Rn Bits 15:8 DBm-2[7:0]**: Data Byte m-2
- **Rn Bits 7:0 DBm-3[7:0]**: Data Byte m-3

NOTE

Depending on the configuration of the element size (RXESC), between two and sixteen 32-bit words (Rn = 2 ..17) are used for storage of a CAN message's data field.

Tx Buffer Element

The Tx Buffers section can be configured to hold dedicated Tx Buffers as well as a Tx FIFO / Tx Queue. In case that the Tx Buffers section is shared by dedicated Tx buffers and a Tx FIFO / Tx Queue, the dedicated Tx Buffers start at the beginning of the Tx Buffers section followed by the buffers assigned to the Tx FIFO or Tx Queue. The Tx Handler distinguishes between dedicated Tx Buffers and Tx FIFO / Tx Queue by evaluating the Tx Buffer configuration TXBC.TFQS and TXBC.NDTB. The element size can be configured for storage of CAN FD messages with up to 64 bytes data field via register TXESC.

Figure 17-10 TX Buffer element

	31 24	23 16	15 8	7 0
T0	ESI, XTD, RTR	ID[28:0]		
T1	MM[7:0]	EFC, TSCE, FDF, BRS, DLC[3:0]	MM[15:8]	res
T2	DB3[7:0]	DB2[7:0]	DB1[7:0]	DB0[7:0]
T3	DB7[7:0]	DB6[7:0]	DB5[7:0]	DB4[7:0]
...	...	...	...	...
Tn	DBm[7:0]	DBm-1[7:0]	DBm-2[7:0]	DBm-3[7:0]

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- **T0 Bit 31 ESI**: Error State Indicator

  0: ESI bit in CAN FD format depends only on error passive flag

  1: ESI bit in CAN FD format transmitted recessive

NOTE

The ESI bit of the transmit buffer is or'ed with the error passive flag to decide the value of the ESI bit in the transmitted FD frame. As required by the CAN FD protocol specification, an error active node may optionally transmit the ESI bit recessive, but an error passive node will always transmit the ESI bit recessive.

- **T0 Bit 30 XTD**: Extended Identifier

  0: 11-bit standard identifier

  1: 29-bit extended identifier

- **T0 Bit 29 RTR**: Remote Transmission Request

  0: Transmit data frame

  1: Transmit remote frame

NOTE

When RTR = 1, the M_CAN transmits a remote frame according to ISO 11898-1:2015, even if CCCR.FDOE enables the transmission in CAN FD format.

- **T0 Bits 28:0 ID[28:0]**: Identifier

  Standard or extended identifier depending on bit XTD. A standard identifier has to be written to ID[28:18].

- **T1 Bits 31:24 MM[7:0]**: Message Marker

  Written by CPU during Tx Buffer configuration. Copied into Tx Event FIFO element for identification of Tx message status.

- **T1 Bit 23 EFC**: Event FIFO Control

  0: Don't store Tx events

  1: Store Tx events

- **T1 Bit 22 TSCE**: Time Stamp Capture Enable for TSU

  Only available when CCCR.UTSU = '1'. When this bit is set and the message is transmitted, a pulse with the duration of one m_can_hclk period is generated at output m_can_tsrx to signal the transmission of a Sync message to the Timestamping Unit (TSU) connected to the M_CAN.

  0: Time Stamp Capture disabled

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1: Time Stamp Capture enabled(not available in this CHIP)

- **T1 Bit 21 FDF**: FD Format

  0: Frame transmitted in Classic CAN format

  1: Frame transmitted in CAN FD format

- **T1 Bit 20 BRS**: Bit Rate Switch

  0: CAN FD frames transmitted without bit rate switching

  1: CAN FD frames transmitted with bit rate switching

NOTE

Bits ESI, FDF, and BRS are only evaluated when CAN FD operation is enabled CCCR.FDOE = 1'. Bit BRS is only evaluated when in addition CCCR.BRSE = '1'.

- **T1 Bits 19:16 DLC[3:0]**: Data Length Code

  0-8: CAN + CAN FD: transmit frame has 0-8 data bytes

  9-15: CAN: transmit frame has 8 data bytes

  9-15:CAN FD: transmit frame has 12/16/20/24/32/48/64 data bytes

- **T1 Bits 15:8 MM[15:8]**: Message Marker

  High byte of Wide Message Marker, written by CPU during Tx Buffer configuration. Copied into Tx Event FIFO element for identification of Tx message status. Available only when CCCR.WMM = '1' or when CCCR.UTSU = '1'.

- **T2 Bits 31:24 DB3[7:0]**: Data Byte 3
- **T2 Bits 23:16 DB2[7:0]**: Data Byte 2
- **T2 Bits 15:8 DB1[7:0]**: Data Byte 1
- **T2 Bits 7:0 DB0[7:0]**: Data Byte 0
- **T3 Bits 31:24 DB7[7:0]**: Data Byte 7
- **T3 Bits 23:16 DB6[7:0]**: Data Byte 6
- **T3 Bits 15:8 DB5[7:0]**: Data Byte 5
- **T3 Bits 7:0 DB4[7:0]**: Data Byte 4

... ... : ...

- **Tn Bits 31:24 DBm[7:0]**: Data Byte m
- **Tn Bits 23:16 DBm-1[7:0]**: Data Byte m-1
- **Tn Bits 15:8 DBm-2[7:0]**: Data Byte m-2
- **Tn Bits 7:0 DBm-3[7:0]**: Data Byte m-3

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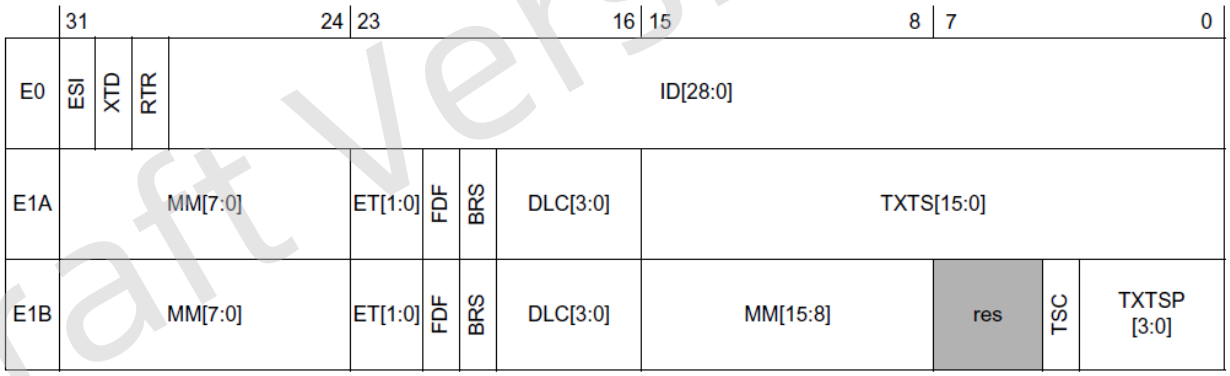
NOTE

Depending on the configuration of the element size (TXESC), between two and sixteen 32-bit words (Tn = 2 ..17) are used for storage of a CAN message's data field.

Tx Event FIFO Element

Each element stores information about transmitted messages. By reading the Tx Event FIFO the Host CPU gets this information in the order the messages were transmitted. Status information about the Tx Event FIFO can be obtained from register TXEFS.

Figure 17-11 TX Event FIFO Element



- **E1A**: When CCCR.WMM = '0' and no TSU is used (CCCR.UTSU = '0'), E1A.TXTS[15:0] holds the 16-bit timestamp generated by the M_CAN's internal timestamping logic.
- **E1B**: When 16-bit Message Markers are enabled (CCCR.WMM ='1'), E1B.MM[15:8] holds the upper 8 bit of the Wide Message Marker. In this design, E1B.TSC is always '0' and E1B.TXTSP[3:0] is not valid.
- **E0 Bit 31 ESI**: Error State Indicator

  0: Transmitting node is error active

  1: Transmitting node is error passive
- **E0 Bit 30 XTD**: Extended Identifier

  0: 11-bit standard identifier

  1: 29-bit extended identifier
- **E0 Bit 29 RTR**: Remote Transmission Request

  0: Data frame transmitted

  1: Remote frame transmitted
- **E0 Bits 28:0 ID[28:0]**: Identifier

  Standard or extended identifier depending on bit XTD. A standard identifier is stored into ID[28:18].

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- **E1A/B Bits 31:24 MM[7:0]**: Message Marker

  Copied from Tx Buffer into Tx Event FIFO element for identification of Tx message status.

- **E1A/B Bit 23:22 ET[1:0]**: Event Type

  00: Reserved

  01: Tx event

  10: Transmission in spite of cancellation (always set for transmissions in DAR mode)

  11: Reserved

- **E1A/B Bit 21 FDF**: FD Format

  0: Classical CAN frame format

  1: CAN FD frame format (new DLC-coding and CRC)

- **E1A/B Bit 20 BRS**: Bit Rate Switch

  0: Frame transmitted without bit rate switching

  1: Frame transmitted with bit rate switching

- **E1A/B Bits 19:16 DLC[3:0]**: Data Length Code

  0-8: CAN + CAN FD: frame with 0-8 data bytes transmitted

  9-15: CAN: frame with 8 data bytes transmitted

  9-15: CAN FD: frame with 12/16/20/24/32/48/64 data bytes transmitted

- **E1A Bits 15:0 TXTS[15:0]**: Tx Timestamp

  Timestamp Counter value captured on start of frame transmission. Resolution depending on configuration of the Timestamp Counter Prescaler TSCC.TCP.

- **E1B Bits 15:8 MM[15:8]**: Message Marker

  High byte of Wide Message Marker, written by CPU during Tx Buffer configuration. Copied into Tx Event FIFO element for identification of Tx message status.

- **E1B Bit 4 TSC**: Timestamp Captured

  0: No timestamp captured

  1: not available in this design

- **E1B Bits 3:0 TXTSP[3:0]**: not available in this design

Standard Message ID Filter Element

Up to 128 filter elements can be configured for 11-bit IDs. When accessing a Standard Message ID Filter element, its address is the Filter List Standard Start Address SIDFC.FLSSA plus the index of the filter element (0...127).

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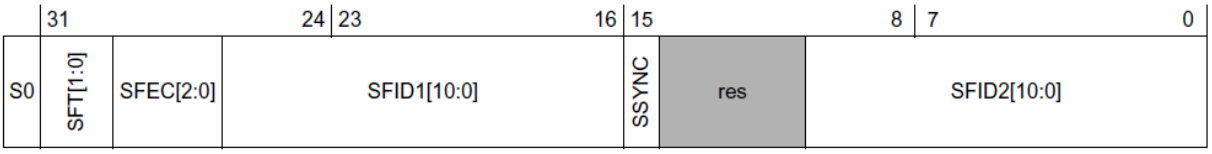
Figure 17-12 Standard Message ID Filter Element



- **Bits 31:30 SFT[1:0]**: Standard Filter Type

  00: Range filter from SFID1 to SFID2 (SFID2 SFID1)

  01: Dual ID filter for SFID1 or SFID2

  10: Classic filter: SFID1 = filter, SFID2 = mask

  11: Filter element disabled

NOTE

With SFT = "11" the filter element is disabled and the acceptance filtering continues (same behaviour as with SFEC = "000").

- **Bit 29:27 SFEC[2:0]**: Standard Filter Element Configuration

  All enabled filter elements are used for acceptance filtering of 11-bit ID frames. Acceptance filtering stops at the first matching enabled filter element or when the end of the filter list is reached. If SFEC = "100", "101", or "110" a match sets interrupt flag IR.HPM and, if enabled, an interrupt is generated.

  In this case register HPMS is updated with the status of the priority match.

  000: Disable filter element

  001: Store in Rx FIFO 0 if filter matches

  010: Store in Rx FIFO 1 if filter matches

  011: Reject ID if filter matches, not intended to be used with Sync messages

  100: Set priority if filter matches, not intended to be used with Sync messages, no storage

  101: Set priority and store in FIFO 0 if filter matches

  110: Set priority and store in FIFO 1 if filter matches

  111: Store into Rx Buffer or as debug message, configuration of SFT[1:0] ignored

  Bits 26:16 SFID1[10:0]: Standard Filter ID 1

  First ID of standard ID filter element. When filtering for Rx Buffers, Sync messages, or for debug messages this field defines the ID of the message to be stored. The received identifiers must match exactly, no masking mechanism is used.

- **Bit 15 SSYNC**: Standard Sync Message

  Only evaluated when CCCR.UTSU = '1'. When this bit is set and a matching message is received, a pulse with the duration of one m_can_hclk period is generated at output

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m_can_tsrx to signal the reception of a Sync message to the Timestamping Unit (TSU) connected to the M_CAN.

0: Timestamping for the matching Sync message disabled

1: Timestamping for the matching Sync message enabled

NOTE

The generation of a pulse at output m_can_tsrx is independent of whether the matching received message is stored or not. E.g. if a message is received for one of the two Rx FIFOs and the FIFO is full and operated in blocking mode, the pulse is generated but the message itself is not stored. In this case the timestamp captured by the TSU does not relate to a message in the M_CAN's Message RAM.

- **Bits 10:0 SFID2[10:0]**: Standard Filter ID 2

  This bit field has a different meaning depending on the configuration of SFEC:

  - SFEC = "001"..."110" Second ID of standard ID filter element
  - SFEC = "111" Filter for Rx Buffers or for debug messages

  SFID2[10:9] decides whether the received message is stored into an Rx Buffer or treated as message A, B, or C of the debug message sequence.

  00: Store message into an Rx Buffer

  01: Debug Message A

  10: Debug Message B

  11: Debug Message C

  SFID2[8:6] is used to control the filter event pins m_can_fe[2:0] at the Extension Interface. A one at the respective bit position enables generation of a pulse at the related filter event pin with the duration of one m_can_hclk period in case the filter matches.

  SFID2[5:0] defines the offset to the Rx Buffer Start Address RXBC.RBSA for storage of a matching message.

Extended Message ID Filter Element

Up to 64 filter elements can be configured for 29-bit IDs. When accessing an Extended Message ID Filter element, its address is the Filter List Extended Start Address XIDFC.FLESA plus two times the index of the filter element (0...63).

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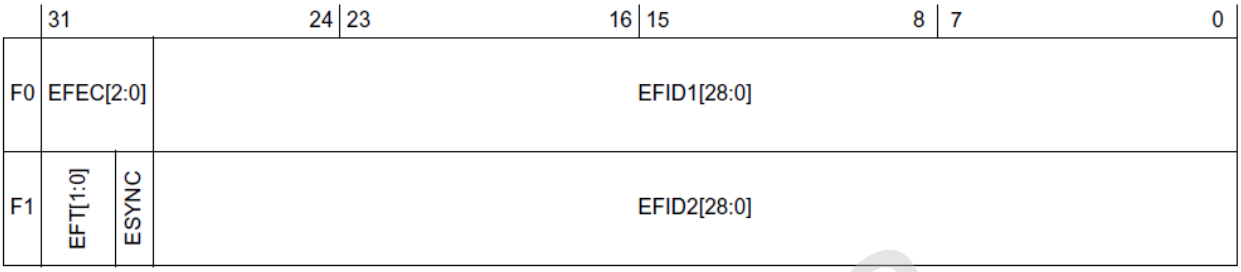
Figure 17-13 Extended Message ID Filter Element



- **F0 Bit 31:29 EFEC[2:0]**: Extended Filter Element Configuration

  All enabled filter elements are used for acceptance filtering of 29-bit ID frames. Acceptance filtering stops at the first matching enabled filter element or when the end of the filter list is reached. If EFEC = “100”, “101”, or “110” a match sets interrupt flag IR.HPM and, if enabled, an interrupt is generated.

  In this case register HPMS is updated with the status of the priority match.

  000: Disable filter element

  001: Store in Rx FIFO 0 if filter matches

  010: Store in Rx FIFO 1 if filter matches

  011: Reject ID if filter matches, not intended to be used with Sync messages

  100: Set priority if filter matches, not intended to be used with Sync messages, no storage

  101: Set priority and store in FIFO 0 if filter matches

  110: Set priority and store in FIFO 1 if filter matches

  111: Store into Rx Buffer or as debug message, configuration of EFT[1:0] ignored

- **F0 Bits 28:0 EFID1[28:0]**: Extended Filter ID 1

  First ID of extended ID filter element. When filtering for Rx Buffers, Sync messages, or for debug messages this field defines the ID of an extended message to be stored. The received identifiers must match exactly, only XIDAM masking mechanism is used.

- **F1 Bits 31:30 EFT[1:0]**: Extended Filter Type

  00: Range filter from EFID1 to EFID2 (EFID2 EFID1)

  01: Dual ID filter for EFID1 or EFID2

  10: Classic filter: EFID1 = filter, EFID2 = mask

  11: Range filter from EFID1 to EFID2 (EFID2 EFID1), XIDAM mask not applied

- **F1 Bit 29 ESYNC**: Extended Sync Message

  Always disabled in this design

- **F1 Bits 28:0 EFID2[28:0]**: Extended Filter ID 2

  This bit field has a different meaning depending on the configuration of EFEC:

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- EFEC = “001”...“110” Second ID of extended ID filter element
- EFEC = “111” Filter for Rx Buffers or for debug messages

EFID2[10:9] decides whether the received message is stored into an Rx Buffer or treated as message A, B, or C of the debug message sequence.

00: Store message into an Rx Buffer

01: Debug Message A

10: Debug Message B

11: Debug Message C

EFID2[8:6] is used to control the filter event pins m_can_fe[2:0] at the Extension Interface. A one at the respective bit position enables generation of a pulse at the related filter event pin with the duration of one m_can_hclk period in case the filter matches.

EFID2[5:0] defines the offset to the Rx Buffer Start Address RXBC.RBSA for storage of a matching message.

17.2.6 Register List

Module Name	Base Address
CAN_TOP0	0x04530000
CAN_TOP1	0x04531000
CAN_DMA0	0x04534000
CAN_DMA1	0x04535000
M_CAN0	0x0453C800
M_CAN1	0x04541800

17.2.6.1 CAN_TOP Register List

Module Name	Base Address
CAN_TOP0	0x04530000
CAN_TOP1	0x04531000

Register Name	Offset	Description
CAN0_GC_REG	0x0000	CAN Global Control Register
CANDMA_RF0_CHL0_FM_NUM	0x0004	CAN DMA Channel0 RF0 Frame Number Register
CANDMA_RF0_CHL0_DES_ADD	0x0008	CAN DMA Channel0 RF0 DES Address Register
CANDMA_RF0_CHL0_RPTR_ADD	0x000C	CAN DMA Channel0 RF0 RPTR Address Register
CANDMA_RF0_CHL0_WRAP_ADD	0x0010	CAN DMA Channel0 RF0 WRAP Address Register
CANDMA_RF0_CHL0_CUR_ADD	0x0014	CAN DMA Channel0 RF0 CURRENT Address Register
CANDMA_RF1_CHL0_FM_NUM	0x0018	CAN DMA Channel0 RF1 Frame Number Register
CANDMA_RF1_CHL0_DES_ADD	0x001C	CAN DMA Channel0 RF1 DES Address Register

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Register Name	Offset	Description
CANDMA_RF1_CHL0_RPTR_ADD	0x0020	CAN DMA Channel0 RF1 RPTR Address Register
CANDMA_RF1_CHL0_WRAP_ADD	0x0024	CAN DMA Channel0 RF1 WRAP Address Register
CANDMA_RF1_CHL0_CUR_ADD	0x0028	CAN DMA Channel0 RF1 CURRENT Address Register
CANDMA_TEF_CHL0_FM_NUM	0x002C	CAN DMA Channel0 TEF Frame Number Register
CANDMA_TEF_CHL0_DES_ADD	0x0030	CAN DMA Channel0 TEF DES Address Register
CANDMA_TEF_CHL0_RPTR_ADD	0x0034	CAN DMA Channel0 TEF RPTR Address Register
CANDMA_TEF_CHL0_WRAP_ADD	0x0038	CAN DMA Channel0 TEF WRAP Address Register
CANDMA_TEF_CHL0_CUR_ADD	0x003C	CAN DMA Channel0 TEF CURRENT Address Register
CANDMA_TB_CHL0_FM_NUM	0x0040	CAN DMA Channel0 TB Frame Number Register
CAN0_INT_EN_REG	0x0044	CAN0 Interrupt Request Enable Register
CAN0_INT_PEN_REG	0x0048	CAN0 Interrupt Request Pending Register
CANDMA_CHL0_RX_CF_WL	0x004C	CAN DMA Channel0 RX CMDFIFO WL Register
CANDMA_CHL0_DEBUG_STATUS	0x0050	CAN DMA Channel0 TSU Debug Status
CANDMA_CHL0_PAR_ERROR_ADDR	0x0054	CAN DMA Channel0 PAR ERROR ADDR Register
CANDMA_CHL0_ARB_PHASE_ADDR	0x0058	CAN DMA Channel0 Arbiter Phase Register
CANDMA_RF_TIMEOUT	0x005C	CAN DMA RF TIMEOUT Register

17.2.6.2 CAN_DMA Register List

Module Name	Base Address
CAN_DMA0	0x04534000
CAN_DMA1	0x04535000

Register Name	Offset	Description
CANDMA_TX_START	0x0000	CAN DMA TX Channel N Start Register
CANDMA_OP	0x0004	CAN DMA TX Channel N Opcode Register
CANDMA_TX_CMD_LEN	0x0008	CAN DMA TX Channel N Command Length Register
CANDMA_TX_LEN_LEFT	0x000C	CAN DMA TX Channel N Current Left Data Register
CANDMA_TX_OP_NUM	0x0010	CAN DMA TX Channel N Current Opcode executed Register
CANDMA_TX_OP_RD_1	0x001C	CAN DMA TX Channel N Read Current Opcode Register 1
CANDMA_TX_OP_RD_2	0x0020	CAN DMA TX Channel N Read Current Opcode Register 2
CANDMA_TX_OP_RD_3	0x0024	CAN DMA TX Channel N Read Current Opcode Register 3
CANDMA_TX_CUR_DESCR	0x0028	CAN DMA TX Channel N Current Descriptor Register

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Register Name	Offset	Description
CANDMA_TX_RESUME	0x002C	CAN DMA TX Channel N Resuming Register
CANDMA_RX_START	0x0080	CAN DMA RX Channel N Start Register

17.2.6.3 M_CAN Register List

Module Name	Base Address	Description
M_CAN0	0x0453C800	
M_CAN1	0x04541800	M_CAN1 registers are the same with M_CAN0.

Register Name	Offset	Description
CAN0_CREL	0x0000	CAN0 Core Release Register
CAN0_ENDN	0x0004	CAN0 Endian Register
CAN0_DBTP	0x000C	CAN0 Data Bit Timing & Prescaler Register
CAN0_TEST	0x0010	CAN0 Test Register
CAN0_RWD	0x0014	CAN0 RAM Watchdog
CAN0_CCCR	0x0018	CAN0 CC Control Register
CAN0_NBTP	0x001C	CAN0 Nominal Bit Timing & Prescaler Register
CAN0_TSCC	0x0020	CAN0 Timestamp Counter Configuration
CAN0_TSCV	0x0024	CAN0 Timestamp Counter Value
CAN0_TOCC	0x0028	CAN0 Timeout Counter Configuration
CAN0_TOCV	0x002C	CAN0 Timeout Counter Value
CAN0_ECR	0x0040	CAN0 Error Counter Register
CAN0_PSR	0x0044	CAN0 Protocol Status Register
CAN0_TDCR	0x0048	CAN0 Transmitter Delay Compensation Register
CAN0_IR	0x0050	CAN0 Interrupt Register
CAN0_IE	0x0054	CAN0 Interrupt Enable
CAN0_ILS	0x0058	CAN0 Interrupt Line Select
CAN0_ILE	0x005C	CAN0 Interrupt Line Enable
CAN0_GFC	0x0080	CAN0 Global Filter Configuration
CAN0_SIDFC	0x0084	CAN0 Standard ID Filter Configuration
CAN0_XIDFC	0x0088	CAN0 Extended ID Filter Configuration
CAN0_XIDAM	0x0090	CAN0 Extended ID AND Mask
CAN0_HPMS	0x0094	CAN0 High Priority Message Status
CAN0_New Data1	0x0098	CAN0 NDAT1
CAN0_New Data2	0x009C	CAN0 NDAT2
CAN0_RXF0C	0x00A0	CAN0 Rx FIFO 0 Configuration
CAN0_RXF0S	0x00A4	CAN0 Rx FIFO 0 Status
CAN0_RXF0A	0x00A8	CAN0 Rx FIFO 0 Acknowledge
CAN0_RXBC	0x00AC	CAN0 Rx Buffer Configuration
CAN0_RXF1C	0x00B0	CAN0 Rx FIFO 1 Configuration
CAN0_RXF1S	0x00B4	CAN0 Rx FIFO 1 Status

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Register Name	Offset	Description
CAN0_RXF1A	0x00B8	CAN0 Rx FIFO 1 Acknowledge
CAN0_RXESC	0x00BC	CAN0 Rx Buffer / FIFO Element Size Configuration
CAN0_TXBC	0x00C0	CAN0 Tx Buffer Configuration
CAN0_TXFQS	0x00C4	CAN0 Tx FIFO/Queue Status
CAN0_TXESC	0x00C8	CAN0 Tx Buffer Element Size Configuration
CAN0_TXBRP	0x00CC	CAN0 Tx Buffer Request Pending
CAN0_TXBAR	0x00D0	CAN0 Tx Buffer Add Request
CAN0_TXBCR	0x00D4	CAN0 Tx Buffer Cancellation Request
CAN0_TXBTO	0x00D8	CAN0 Tx Buffer Transmission Occurred
CAN0_TXBCF	0x00DC	CAN0 Tx Buffer Cancellation Finished
CAN0_TXBTIE	0x00E0	CAN0 Tx Buffer Transmission Interrupt Enable
CAN0_TXBCIE	0x00E4	CAN0 Tx Buffer Cancellation Finished Interrupt Enable
CAN0_TXEFC	0x00F0	CAN0 Tx Event FIFO Configuration
CAN0_TXEFS	0x00F4	CAN0 Tx Event FIFO Status
CAN0_TXEFA	0x00F8	CAN0 Tx Event FIFO Acknowledge

17.2.7 CAN_TOP Register Description

17.2.7.1 0x0000 CAN Global Control Register (Default Value:0x0000_0000)

Offset: 0x0000			Register Name: CAN0_GC_REG
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/WAC	0x0	RF0_INITIAL To set rf0 address and pointer status into initial state. 0: Not reset 1: Reset
17	R/WAC	0x0	RF1_INITIAL To set rf1 address and pointer status into initial state.
16	R/WAC	0x0	TEF_INITIAL To set tef address and pointer status into initial state.
15:11	/	/	/

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Offset: 0x0000			Register Name: CAN0_GC_REG
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	TSU_CLK_DIVIDE 0: 24M 1: 12M 2: 6M 3: 3M 4: 2M 5: 1M For CAN0 TSU CCU clock source and RF0/RF1 Timeout feature clock, they share the same TSU_CLK, and the same division factor.
7:2	/	/	/
1	R/W	0x0	TSU_WORK 0: IDLE 1: WORK
0	R/W	0x0	CAN_MODE 0: CPU_MODE 1: DMA_MODE

17.2.7.2 0x0004 CAN DMA Channel0 RF0 Frame Number Register (Default Value:0x0080_0040)

Offset: 0x0004			Register Name: CANDMA_RF0_CHL0_FM_NUM
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x80	RF0_TRIG_NUM The frame number in DDR that triggers a TRIG_INT. The actual trigger number is RF0_TRIG_NUM + 1. The number of actual trigger frames needs to be smaller than the applied DDR space size (RF0_WRAP_ADDRESS - RF0_DES_ADDRESS), otherwise efficiency and interrupt triggering may be affected.
15:9	/	/	/
8	R/W	0x0	RF0_TIMEOUT_EN 0: OFF 1: ON
7	/	/	/
6:0	R/W	0x40	RF0_Frame Number The maximum number of RX FIFO0 frames of CAN (in local SRAM).

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17.2.7.3 0x0008 CAN DMA Channel0 RF0 DES Address Register (Default Value:0x0000_0000)

Offset: 0x0008			Register Name: CANDMA_RF0_CHL0_DES_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF0_DES_ADDRESS The destination WORD address of CAN DMA RX FIFO0 channel0 (in common ram)

17.2.7.4 0x000C CAN DMA Channel0 RF0 RPTR Address Register (Default Value:0x0000_0000)

Offset: 0x000C			Register Name: CANDMA_RF0_CHL0_RPTR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF0_RPTR_ADDRESS The read pointer WORD address of CAN DMA RX FIFO0 channel0 (in common ram). When CAN DMA reaches the trigger address in destination DDR, it should stop writing into DDR until CPU updates the rptr address. Before the transfer starts, CPU should configure this the same value as WRAP Address Register, [0x0010]. Caution: The RPTR address needs to be an address aligned with the frame header.

17.2.7.5 0x0010 CAN DMA Channel0 RF0 WRAP Address Register (Default Value:0x0000_0000)

Offset: 0x0010			Register Name: CANDMA_RF0_CHL0_WRAP_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF0_WRAP_ADDRESS The wrap WORD address of CAN DMA RX FIFO0 channel0 (in common ram). Caution: The WRAP address needs to be the last address of the aligned frame header that can be written in the applied DDR space.

17.2.7.6 0x0014 CAN DMA Channel0 RF0 CUR Address Register (Default Value:0x0000_0000)

Offset: 0x0014			Register Name: CANDMA_RF0_CHL0_CUR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RF0_CUR_ADDRESS Current WORD address in common RAM that CAN DMA is writing to.

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17.2.7.7 0x0018 CAN DMA Channel0 RF1 Frame Number Register (Default Value:0x0080_0040)

Offset: 0x0018			Register Name: CANDMA_RF1_CHL0_FM_NUM
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x80	RF1_TRIG_NUM The frame number in DDR that triggers a TRIG_INT. The actual trigger number is RF1_TRIG_NUM + 1. The number of actual trigger frames needs to be smaller than the applied DDR space size (RF1_WRAP_ADDRESS - RF1_DES_ADDRESS), otherwise efficiency and interrupt triggering may be affected.
15:9	/	/	/
8	R/W	0x0	RF1_TIMEOUT_EN 0: OFF 1: ON
7	/	/	/
6:0	R/W	0x40	RF1_Frame Number The maximum number of RX FIFO1 frames of CAN (in local SRAM).

17.2.7.8 0x001C CAN DMA Channel0 RF1 DES Address Register (Default Value:0x0000_0000)

Offset: 0x001C			Register Name: CANDMA_RF1_CHL0_DES_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF1_DES_ADDRESS The destination WORD address of CAN DMA RX FIFO1 channel0 (in common ram)

17.2.7.9 0x0020 CAN DMA Channel0 RF1 RPTR Address Register (Default Value:0x0000_0000)

Offset: 0x0020			Register Name: CANDMA_RF1_CHL0_RPTR_ADD
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0020			Register Name: CANDMA_RF1_CHL0_RPTR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF1_RPTR_ADDRESS The read pointer WORD address of CAN DMA RX FIFO1 channel0 (in common ram). When CAN DMA reaches the trigger address in destination DDR, it should stop writing into DDR until CPU updates the rptr address. Before the transfer starts, CPU should configure this the same value as WRAP Address Register, [0x0024]. Caution: The RPTR address needs to be an address aligned with the frame header.

17.2.7.10 0x0024 CAN DMA Channel0 RF1 WRAP Address Register (Default Value:0x0000_0000)

Offset: 0x0024			Register Name: CANDMA_RF1_CHL0_WRAP_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	RF1_WRAP_ADDRESS The wrap WORD address of CAN DMA RX FIFO1 channel0 (in common ram). Caution: The WRAP address needs to be the last address of the aligned frame header that can be written in the applied DDR space.

17.2.7.11 0x0028 CAN DMA Channel0 RF1 CUR Address Register (Default Value:0x0000_0000)

Offset: 0x0028			Register Name: CANDMA_RF1_CHL0_CUR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RF1_CUR_ADDRESS Current WORD address in common RAM that CAN DMA is writing to.

17.2.7.12 0x002C CAN DMA Channel0 TEF Frame Number Register (Default Value:0x0080_0020)

Offset: 0x002C			Register Name: CANDMA_TEF_CHL0_FM_NUM
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x002C			Register Name: CANDMA_TEF_CHL0_FM_NUM
Bit	Read/Write	Default/Hex	Description
23:16	R/W	0x80	TEF_TRIG_NUM The frame number in DDR that triggers a TRIG_INT. The actual trigger number is TEF_TRIG_NUM + 1. The number of actual trigger frames needs to be smaller than the applied DDR space size (TEF_WRAP_ADDRESS - TEF_DES_ADDRESS), otherwise efficiency and interrupt triggering may be affected.
15:9	/	/	/
8	R/W	0x0	TEF ON 0: TEF OFF 1: TEF ON
7:6	/	/	/
5:0	R/W	0x20	TEF_Frame Number The maximum number of TX EVENT FIFO frames of CAN (in local SRAM).

17.2.7.13 0x0030 CAN DMA Channel0 TEF DES Address Register (Default Value:0x0000_0000)

Offset: 0x0030			Register Name: CANDMA_TEF_CHL0_DES_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TEF_DES_ADDRESS The destination WORD address of CAN DMA TX EVENT FIFO channel0 (in common ram)

17.2.7.14 0x0034 CAN DMA Channel0 TEF RPTR Address Register (Default Value:0x0000_0000)

Offset: 0x0034			Register Name: CANDMA_TEF_CHL0_RPTR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TEF_RPTR_ADDRESS The read pointer WORD address of CAN DMA TX EVENT FIFO channel0 (in common ram). When CAN DMA reaches the trigger address in destination DDR, it should stop writing into DDR until CPU updates the rptr address. Before the transfer starts, CPU should configure this the same value as WRAP Address Register, [0x0038]. Caution: The RPTR address needs to be an address aligned with the frame header.

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17.2.7.15 0x0038 CAN DMA Channel0 TEF WRAP Address Register (Default Value:0x0000_0000)

Offset: 0x0038			Register Name: CANDMA_TEF_CHL0_WRAP_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TEF_WRAP_ADDRESS The wrap WORD address of CAN DMA TX EVENT FIFO channel0 (in common ram). Caution: The WRAP address needs to be the last address of the aligned frame header that can be written in the applied DDR space.

17.2.7.16 0x003C CAN DMA Channel0 TEF CUR Address Register (Default Value:0x0000_0000)

Offset: 0x003C			Register Name: CANDMA_TEF_CHL0_CUR_ADD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TEF_CUR_ADDRESS Current WORD address in common RAM that CAN DMA is writing to.

17.2.7.17 0x0040 CAN DMA Channel0 TB Frame Number Register (Default Value:0x0000_0020)

Offset: 0x0040			Register Name: CANDMA_TB_CHL0_FM_NUM
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0x20	Frame Number The maximum number of TB frames of CAN (in local SRAM) When using DMA mode, Frame Number must be at least 2.

17.2.7.18 0x0044 CAN0 Interrupt Request Enable Register (Default Value:0x0000_0000)

Offset: 0x0044			Register Name: CAN0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W	0x0	DMA TX IRQ WAIT Interrupt Enable 0: Disable 1: Enable
26	R/W	0x0	DMA TX ONE DESCRIPTOR DONE Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0044			Register Name: CAN0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
25	R/W	0x0	DMA TX FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
24	R/W	0x0	DMA TX FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable
23	R/W	0x0	DMA RX CMD FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
22	R/W	0x0	DMA RX CMD FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable
21	R/W	0x0	DMA RX DATA FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
20	R/W	0x0	DMA RX DATA FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable
19	R/W	0x0	DMA CMD FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
18	R/W	0x0	DMA CMD FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable
17	R/W	0x0	DMA CMD STORE FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
16	R/W	0x0	DMA CMD STORE FIFO UNDERUN Interrupt Enable 0: Disable 1: Enable
15	R/W	0x0	REG RD FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
14	R/W	0x0	REG RD FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable
13	R/W	0x0	REG WR FIFO OVERRUN Interrupt Enable 0: Disable 1: Enable
12	R/W	0x0	REG WR FIFO UNDERRUN Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0044			Register Name: CAN0_INT_EN_REG
Bit	Read/Write	Default/Hex	Description
11	R/W	0x0	RF0 TIMEOUT Interrupt Enable 0: Disable 1: Enable
10	R/W	0x0	RF1 TIMEOUT Interrupt Enable 0: Disable 1: Enable
9	R/W	0x0	TX CAN DONE Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	PAR_ERROR Interrupt Enable 0: Disable 1: Enable
7	R/W	0x0	TSU WRAP Interrupt Enable 0: Disable 1: Enable
6	R/W	0x0	RF0 DDR TRIG Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	RF0 DDR WRAP Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	RF1 DDR TRIG Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	RF1 DDR WRAP Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	TEF DDR TRIG Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	TEF DDR WRAP Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	TX DMA DONE Interrupt Enable 0: Disable 1: Enable

17.2.7.19 0x0048 CAN0 Interrupt Request Pending Register (Default Value:0x0000_0000)

Offset: 0x0048			Register Name: CAN0_INT_PEN_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0048			Register Name: CAN0_INT_PEN_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W1C	0x0	DMA TX IRQ WAIT Interrupt PENDING 0: Not Interrupt 1: Interrupt
26	R/W1C	0x0	DMA TX ONE DESCRIPTOR DONE Interrupt PENDING 0: Not Interrupt 1: Interrupt
25	R/W1C	0x0	DMA TX FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
24	R/W1C	0x0	DMA TX FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
23	R/W1C	0x0	DMA RX CMD FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
22	R/W1C	0x0	DMA RX CMD FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
21	R/W1C	0x0	DMA RX DATA FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
20	R/W1C	0x0	DMA RX DATA FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
19	R/W1C	0x0	DMA CMD FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
18	R/W1C	0x0	DMA CMD FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
17	R/W1C	0x0	DMA CMD STORE FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
16	R/W1C	0x0	DMA CMD STORE FIFO UNDERUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
15	R/W1C	0x0	REG RD FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt

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Offset: 0x0048			Register Name: CAN0_INT_PEN_REG
Bit	Read/Write	Default/Hex	Description
14	R/W1C	0x0	REG RD FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
13	R/W1C	0x0	REG WR FIFO OVERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
12	R/W1C	0x0	REG WR FIFO UNDERRUN Interrupt PENDING 0: Not Interrupt 1: Interrupt
11	R/W1C	0x0	RF0 TIMEOUT Interrupt PENDING 0: Not Interrupt 1: Interrupt When RF0_TIMEOUT_EN is enabled, if RF0 RPTR ADDR and RF0 CUR ADDR do not overlap, timeout counting will start. When the count reaches the upper limit described by RF0_TIMEOUT_CNT_NUM, an interrupt will be generated.
10	R/W1C	0x0	RF1 TIMEOUT Interrupt PENDING 0: Not Interrupt 1: Interrupt When RF1_TIMEOUT_EN is enabled, if RF1 RPTR ADDR and RF1 CUR ADDR do not overlap, timeout counting will start. When the count reaches the upper limit described by RF1_TIMEOUT_CNT_NUM, an interrupt will be generated.
9	R/W1C	0x0	TX CAN DONE Interrupt PENDING 0: Not Interrupt 1: Interrupt This bit is used in TX DMA transfer. When can_mode is ON and both TX DMA and CAN have transferred all data of one descriptor chain, this bit is set. When CPU transfer, it is suggested to use interrupt signal CAN0_INT.
8	R/W1C	0x0	PAR ERROR Interrupt PENDING 0: Not Interrupt 1: Interrupt
7	R/W1C	0x0	TSU WRAP Interrupt PENDING 0: Not Interrupt 1: Interrupt

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Offset: 0x0048			Register Name: CAN0_INT_PEN_REG
Bit	Read/Write	Default/Hex	Description
6	R/W1C	0x0	RF0 DDR TRIG Interrupt PENDING 0: Not Interrupt 1: Interrupt Compared with RF0_RPTR_ADDRESS, whenever the number of frames currently written in DDR (corresponding to RF0_CUR_ADDRESS) is greater than or equal to the number of RF0 frames (RF0_TRIG_NUM), this interrupt is automatically pending once.
5	R/W1C	0x0	RF0 DDR WRAP Interrupt PENDING 0: Not Interrupt 1: Interrupt
4	R/W1C	0x0	RF1 DDR TRIG Interrupt PENDING 0: Not Interrupt 1: Interrupt Compared with RF1_RPTR_ADDRESS, whenever the number of frames currently written in DDR (corresponding to RF1_CUR_ADDRESS) is greater than or equal to the number of RF1 frames (RF1_TRIG_NUM), this interrupt is automatically pending once.
3	R/W1C	0x0	RF1 DDR WRAP Interrupt PENDING 0: Not Interrupt 1: Interrupt
2	R/W1C	0x0	TEF DDR TRIG Interrupt PENDING 0: Not Interrupt 1: Interrupt Compared with TEF_RPTR_ADDRESS, whenever the number of frames currently written in DDR (corresponding to TEF_CUR_ADDRESS) is greater than or equal to the number of TEF frames (TEF_TRIG_NUM), this interrupt is automatically pending once.
1	R/W1C	0x0	TEF DDR WRAP Interrupt PENDING 0: Not Interrupt 1: Interrupt
0	R/W1C	0x0	TX DMA DONE Interrupt PENDING 0: Not Interrupt 1: Interrupt This bit describes TX DMA behavior, it does not mean CAN has transmitted all TX data.

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17.2.7.20 0x004C CAN DMA Channel0 RX CMDFIFO WL Register (Default Value:0x0000_0000)

Offset: 0x004C			Register Name: CANDMA_CHL0_RX_CF_WL
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R	0x0	REG RD CMD FIFO WATER LEVEL Used to store the RD CMD that updates put index of SRAM.
23:21	/	/	/
20:16	R	0x0	REG WR CMD FIFO WATER LEVEL Used to store the WR CMD that updates put index of SRAM.
15:13	/	/	/
12:8	R	0x0	DMA CMD STORE FIFO WATER LEVEL Used to store the CMD that has been sent to CAN DMA.
7:5	/	/	/
4:0	R	0x0	DMA CMD FIFO WATER LEVEL Used to store the CMD that to be sent to CAN DMA.

17.2.7.21 0x0050 CAN DMA Channel0 Debug Status Register (Default Value:0x0000_0000)

Offset: 0x0050			Register Name: CANDMA_CHL0_DEBUG_STATUS
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17:16	R	0x0	LAST_RD_TYPE 00: / 01: RX fifo0 RD 10: RX fifo1 RD 11: TX buffer RD
15:13	/	/	/
12:0	R	0x0	LAST_RD_POINTER LAST_RD pointer in SRAM

17.2.7.22 0x0054 CAN DMA Channel0 PAR ERROR ADDR Register (Default Value:0x0000_0000)

Offset: 0x0054			Register Name: CANDMA_CHL0_PAR_ERROR_ADDR
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R	0x0	PARITY ERROR ADDR in SRAM

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17.2.7.23 0x0058 CAN DMA Channel0 Arbiter Phase Register (Default Value:0x1010_1000)

Offset: 0x0058			Register Name: CANDMA_CHL0_ARB_PHASE_ADDR
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x10	Arbiter phase for CPU For how many valid data phase, arbiter shall end current channel and arbitrate again.
23:21	/	/	/
20:16	R/W	0x10	Arbiter phase for DMA RX For how many valid data phase, arbiter shall end current channel and arbitrate again.
15:13	/	/	/
12:8	R/W	0x10	Arbiter phase for DMA TX For how many valid data phase, arbiter shall end current channel and arbitrate again.
7:0	/	/	/

17.2.7.24 0x005C CAN DMA RF TIMEOUT Register (Default Value:0x8000_8000)

Offset: 0x005C			Register Name: CANDMA_RF_TIMEOUT
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x8000	RF0_TIMEOUT_CNT_NUM The RF0 timeout counter will count in 24M clock (or 24M divided). And when it reaches the number, it will generate an interrupt and clear itself.
15:0	R/W	0x8000	RF1_TIMEOUT_CNT_NUM The RF1 timeout counter will count in 24M clock(or 24M divided). And when it reaches the number, it will generate an interrupt and clear itself.

17.2.8 CAN_DMA Register Description

17.2.8.1 0x00000 CAN DMA TX Channel N Start Register (Default Value:0x0000_0000)

Offset: 0x00000			Register Name: CANDMA_TX_START
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x00	TX_CHL_EN Enable CAN DMA TX channel

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17.2.8.2 0x0004 CAN DMA TX Channel N Opcode Register (Default Value:0x0000_0000)

Offset: 0x0004			Register Name: CANDMA_OP
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00	OPCODE_START_ADDR The address of opcode. Statement: When the next opcode address is 32'hFFFFF800, the CAN DMA TX channel will stop when the current opcode is finished. The next opcode address should write in opcode.

17.2.8.3 0x0008 CAN DMA TX Channel N Command Length Register (Default Value:0x0000_0000)

Offset: 0x0008			Register Name: CANDMA_TX_CMD_LEN
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x00	WAIT_EN Statement: WAIT_EN=1: It means the DMA will pause to execute descriptor if the write back bit of current descriptor is 1. WAIT_EN=0: It means the DMA will execute descriptor no matter the write back bit of current descriptor is 1 or 0.
0	R/W	0x00	TX_BLK_LENGTH The TX data transfer length. 1: 128 bytes 0: 64 bytes

17.2.8.4 0x000C CAN DMA TX Channel N Current Left Data Register (Default Value:0x0000_0000)

Offset: 0x000C			Register Name: CANDMA_TX_LEN_LEFT
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x00	OPCODE_LEFT_LENGTH The left data length of Current opcode

17.2.8.5 0x0010 CAN DMA TX Channel N Current Opcode Executed Register (Default Value:0x0000_0000)

Offset: 0x0010			Register Name: CANDMA_TX_OP_NUM
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0010			Register Name: CANDMA_TX_OP_NUM
Bit	Read/Write	Default/Hex	Description
31:0	R	0x00	OPCODE_DONE The opcodes has been executed.

17.2.8.6 0x001C CAN DMA TX Channel N Read Current Opcode Register1 (Default Value:0x0000_0000)

Offset: 0x001C			Register Name: CANDMA_TX_OP_RD_1
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	OP1 TX current opcode content, opcode[31:0]

17.2.8.7 0x0020 CAN DMA TX Channel N Read Current Opcode Register2 (Default Value:0x0000_0000)

Offset: 0x0020			Register Name: CANDMA_TX_OP_RD_2
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	OP2 TX current opcode content, opcode[63:32]

17.2.8.8 0x0024 CAN DMA TX Channel N Read Current Opcode Register3 (Default Value:0x0000_0000)

Offset: 0x0024			Register Name: CANDMA_TX_OP_RD_3
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	OP3 TX current opcode content, opcode[95:64]

17.2.8.9 0x0028 CAN DMA TX Channel N Current Descriptor Register (Default Value:0x0000_0000)

Offset: 0x0028			Register Name: CANDMA_TX_CUR_DESCR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x00	DESCR_CURRENT_ADDR The address of current descriptor. Statement: This register should no write until WAIT_EN=1. If WAIT_EN=1, writing this register before set DESC_RDY=1 can change the address of descriptor that DMA reads when resuming from wait state.

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17.2.8.10 0x002C CAN DMA TX Channel N Resuming Register (Default Value:0x0000_0000)

Offset: 0x002C			Register Name: CANDMA_TX_RESUME
Bit	Read/Write	Default/Hex	Description
31	R	0x00	WAIT_STATE The DMA will be in WAIT STATE if WAIT_EN=1 and the write back bit of current descriptor is 1. In WAIT STATE, the DMA will suspend transfer until user set RESUME_EN=1 to telling DMA that the current descriptor is refresh and DMA can regain the descriptor to transfer.
30:1	/	/	/
0	R/WAC	0x00	RESUME_EN Statement: When WAIT_EN=1, writing RESUME_EN=1 means the descriptor is refresh and DMA will resume from the wait mode. This bit only can be write when WAIT_STATE=1.

17.2.8.11 0x0080 CAN DMA RX Channel N Start Register (Default Value:0x0000_0000)

Offset: 0x0080			Register Name: CANDMA_RX_START
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x00	RX_CHL_EN Enable CAN DMA RX channel

17.2.9 M_CAN Register Description

17.2.9.1 0x0000 CAN0 Core Release Register (Default Value:0x3302_0801)

Offset: 0x0000			Register Name: CAN0_CREL
Bit	Read/Write	Default/Hex	Description
31:28	R	0x3	REL Core Release. One digit, BCD-coded.
27:24	R	0x3	STEP Step of Core Release. One digit, BCD-coded.
23:20	R	0x0	SUBSTEP Sub-step of Core Release. One digit, BCD-coded.
19:16	R	0x2	YEAR Time Stamp Year. One digit, BCD-coded.

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Offset: 0x0000			Register Name: CAN0_CREL
Bit	Read/Write	Default/Hex	Description
15:8	R	0x8	MON Time Stamp Month. Two digits, BCD-coded.
7:0	R	0x1	DAY Time Stamp Day. Two digits, BCD-coded.

17.2.9.2 0x0004 CAN0 Endian Register (Default Value:0x8765_4321)

Offset: 0x0004			Register Name: CAN0_ENDN
Bit	Read/Write	Default/Hex	Description
31:0	R	0x8765_4321	ETV Endianness Test Value

17.2.9.3 0x000C CAN0 Data Bit Timing & Prescaler Register (Default Value:0x0000_0A33)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set. The CAN bit time may be programmed in the range of 4 to 49 time quanta. The CAN time quantum may be programmed in the range of 1 to 32 m_can_cclk periods. tq = (DBRP + 1) mtq.

DTSEG1 is the sum of Prop_Seg and Phase_Seg1. DTSEG2 is Phase_Seg2.

Therefore the length of the bit time is (prog rammed values) [DTSEG1 + DTSEG2 + 3] tq or (functional values) [Sync_Seg + Prop_Seg + Phase_Seg1 + Phase_Seg2] tq.

The Information Processing Time (IPT) is zero, meaning the data for the next bit is available at the first clock edge after the sample point.

Offset: 0x000C			Register Name: CAN0_DBTP
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	TDC Transmitter Delay Compensation 0: Transmitter Delay Compensation disabled 1: Transmitter Delay Compensation enabled
22:21	/	/	/
21:16	R/W	0x0	DBRP Data Bit Rate Prescaler The value by which the clock frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 31. When TDC = '1', the range is limited to 0,1. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.

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Offset: 0x000C			Register Name: CAN0_DBTP
Bit	Read/Write	Default/Hex	Description
15:13	/	/	/
12:8	R/W	0xA	DTSEG1 Data time segment before sample point Valid values are 0 to 31. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
7:4	R/W	0x3	DTSEG2 Data time segment after sample point Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
3:0	R/W	0x3	DSJW Data (Re)Synchronization Jump Width Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.

17.2.9.4 0x0010 CAN0 TEST Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Write access to the Test Register has to be enabled by setting bit CCCR.TEST to '1'. All Test

Register functions are set to their reset values when bit CCCR.TEST is reset.

Loop Back Mode and software control of pin m_can_tx are hardware test modes. Programming of TX ≠ '00' may disturb the message transfer on the CAN bus.

Offset: 0x0010			Register Name: CAN0_TEST
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R	0x0	SVAL Started Valid 0: Value of TXBNS not valid 1: Value of TXBNS valid
20:16	R	0x0	TXBNS TX Buffer Number Started TX Buffer number of message whose transmission was started last. Valid when SVAL is set. Valid values are 0 to 31.
15:14	/	/	/

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Offset: 0x0010			Register Name: CAN0_TEST
Bit	Read/Write	Default/Hex	Description
13	R	0x0	PVAL. Prepared Valid 0: Value of TXBNP not valid 1: Value of TXBNP valid
12:8	R	0x0	TXBNP TX Buffer Number Prepared TX Buffer number of message that is ready for transmission. Valid when PVAL is set. Valid values are 0 to 31.
7	R	0x0	RX. Receive Pin Monitors the actual value of pin m_can_rx. 0: The CAN bus is dominant (m_can_rx = '0') 1: The CAN bus is recessive (m_can_rx = '1')
6:5	R/W	0x0	TX Control of Transmit Pin 00: Reset value, m_can_tx controlled by the CAN Core, updated at the end of the CAN bit time 01: Sample Point can be monitored at pin m_can_tx 10: Dominant ('0') level at pin m_can_tx 11: Recessive ('1') at pin m_can_tx
4	R/W	0x0	LBCK Loop Back Mode 0: Reset value, Loop Back Mode is disabled 1: Loop Back Mode is enabled
3:0	/	/	/

17.2.9.5 0x0014 CAN0 RAM Watchdog Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

The RAM Watchdog monitors the READY output of the Message RAM (m_can_aeim_ready). A Message RAM access via the M_CAN's Generic Master Interface (m_can_aeim_sel active) starts the Message RAM Watchdog Counter with the value configured by RWD.WDC. The counter is reloaded with RWD.WDC when the Message RAM signals successful completion by activating its READY output. In case there is no response from the Message RAM until the counter has counted down to zero, the counter stops and interrupt flag IR.WDI is set. The RAM Watchdog Counter is clocked by the Host clock (m_can_hclk).

Offset: 0x0014			Register Name: CAN0_RWD
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0014			Register Name: CAN0_RWD
Bit	Read/Write	Default/Hex	Description
15:8	R	0x0	WDV Watchdog Value Actual Message RAM Watchdog Counter Value.
7:0	R/W	0x0	WDC Watchdog Configuration Start value of the Message RAM Watchdog Counter. With the reset value of “00” the counter is disabled.

17.2.9.6 0x0018 CAN0 CC Control Register (Default Value:0x0000_0001)

Other bits of this register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0018			Register Name: CAN0_CCCR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	NISO Non ISO Operation If this bit is set, the M_CAN uses the CAN FD frame format as specified by the Bosch CAN FD Specification V1.0. 0: CAN FD frame format according to ISO 11898-1:2015 1: CAN FD frame format according to Bosch CAN FD Specification V1.0
14	R/W	0x0	TXP Transmit Pause If this bit is set, the M_CAN pauses for two CAN bit times before starting the next transmission after itself has successfully transmitted a frame . 0: Transmit pause disabled 1: Transmit pause enabled
13	R/W	0x0	EFBI Edge Filtering during Bus Integration 0: Edge filtering disabled 1: Two consecutive dominant tq required to detect an edge f or hard synchronization

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Offset: 0x0018			Register Name: CAN0_CCCR
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PXHD Protocol Exception Handling Disable 0: Protocol exception handling enabled 1: Protocol exception handling disabled Note: When protocol exception handling is disabled, the M_CAN will transmit an error frame when it detects a protocol exception condition.
11	R/W	0x0	WMM Wide Message Marker Enables the use of 16-bit Wide Message Markers. When 16-bit Wide Message Markers are used (WMM = '1'), 16-bit internal timestamping is disabled for the TX Event FIFO. 0: 8-bit Message Marker used 1: 16-bit Message Marker used, replacing 16-bit timestamps in TX Event FIFO
10	/	/	/
9	R/W	0x0	BRSE Bit Rate Switch Enable 0: Bit rate switching for transmissions disabled 1: Bit rate switching for transmissions enabled Note: When CAN FD operation is disabled(FDOE = '0'), BRSE is not evaluated.
8	R/W	0x0	FDOE FD Operation Enable 0: FD operation disabled 1: FD operation enabled
7	R/W	0x0	TEST Test Mode Enable 0: Normal operation, register TEST holds reset values 1: Test Mode, write access to register TEST enabled
6	R/W	0x0	DAR Disable Automatic Retransmission 0: Automatic retransmission of messages not transmitted successfully enabled 1: Automatic retransmission disabled

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Offset: 0x0018			Register Name: CAN0_CCCR
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	MON Bus Monitoring Mode Bit MON can only be set by the Host when both CCE and INIT are set to '1'. The bit can be reset by the Host at any time. 0: Bus Monitoring Mode is disabled 1: Bus Monitoring Mode is enabled
4	R/W	0x0	CSR Clock Stop Request 0: No clock stop is requested 1: Clock stop requested. When clock stop is requested, first INIT and then CSA will be set after all pending transfer requests have been completed and the CAN bus reached idle.
3	R/W	0x0	CSA Clock Stop Acknowledge 0: No clock stop acknowledged 1: M_CAN may be set in power down by stopping m_can_hclk and m_can_cclk
2	R/W	0x0	ASM Restricted Operation Mode Bit ASM can only be set by the Host when both CCE and INIT are set to '1'. The bit can be reset by the Host at any time. 0: Normal CAN operation 1: Restricted Operation Mode active
1	R/W	0x0	CCE Configuration Change Enable 0: The CPU has no write access to the protected configuration registers 1: The CPU has write access to the protected configuration registers (while CCCR.INIT = '1')
0	R/W	0x1	INIT Initialization 0: Normal Operation 1: Initialization is started Note: Due to the synchronization mechanism between the two clock domains, there may be a delay until the value written to INIT can be read back. Therefore the programmer has to assure that the previous value written to INIT has been accepted by reading INIT before setting INIT to a new value.

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17.2.9.7 0x001C CAN0 Nominal Bit Timing & Prescaler Register (Default Value:0x0600_0A03)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set. The CAN bit time may be programmed in the range of 4 to 385 time quanta. The CAN time quantum may be programmed in the range of 1 to 512 m_can_cclk periods. tq = (NBRP + 1) mtq.

NTSEG1 is the sum of Prop_Seg and Phase_Seg1. NTSEG2 is Phase_Seg2.

Therefore the length of the bit time is (prog rammed values) [NTSEG1 + NTSEG2 + 3] tq or (functional values) [Sync_Seg + Prop_Seg + Phase_Seg1 + Phase_Seg2] tq.

The Information Processing Time (IPT) is zero, meaning the data for the next bit is available at the first clock edge after the sample point.

Offset: 0x001C			Register Name: CAN0_NBTP
Bit	Read/Write	Default/Hex	Description
31:25	R/W	0x3	NSJW Nominal (Re)Synchronization Jump Width Valid values are 0 to 127. The actual interpretation by the hardware of this value is such that one more than the value prog rammed here is used.
24:16	R/W	0x0	NBRP Nominal Bit Rate Prescaler The value by which the oscillator frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 511. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
15:8	R/W	0xA	NTSEG1 Nominal Time segment before sample point Valid values are 1 to 255. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
7	/	/	/
6:0	R/W	0x3	NTSEG2 Nominal Time segment after sample point Valid values are 1 to 127. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.

17.2.9.8 0x0020 CAN0 Timestamp Counter Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0020	Register Name: CAN0_TSCC

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Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:16	R/W	0x0	TCP Timestamp Counter Prescaler Configures the timestamp and timeout counters time unit in multiples of CAN bit times [1...16]. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
15:2	/	/	/
1:0	R/W	0x0	TSS Timestamp Select 00: Timestamp counter value always 0x0000 01: Timestamp counter value incremented according to TCP 10: External timestamp counter value used 11: Same as “00”

17.2.9.9 0x0024 CAN0 Timestamp Counter Value Register (Default Value:0x0000_0000)

Offset: 0x0024			Register Name: CAN0_TSCV
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0	TSC Timestamp Counter The internal/external Timestamp Counter value is captured on start of frame (both Rx and TX). When TSCC.TSS = “01”, the Timestamp Counter is incremented in multiples of CAN bit times [1...16] depending on the configuration of TSCC.TCP. A wrap around sets interrupt flag IR.TSW. When TSCC.TSS = “10”, TSC reflects the external Timestamp Counter value. A write access has no impact. Note: Writing one of the register bytes 3/2/1/0 will reset the Timestamp Counter. Note: A “wrap around” is a change of the Timestamp Counter value from non-zero to zero not caused by write access to TSCV.

17.2.9.10 0x0028 CAN0 Timeout Counter Configuration Register (Default Value:0xFFFF_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

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Offset: 0x0028			Register Name: CAN0_TOCC
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0xFFFF	TOP Timeout Period Start value of the Timeout Counter (down-counter). Configures the Timeout Period.
15:3	/	/	/
2:1	R/W	0x0	TOS Timeout Select When operating in Continuous mode, a write to TOCV presets the counter to the value configured by TOCC.TOP and continues down-counting. When the Timeout Counter is controlled by one of the FIFOs, an empty FIFO presets the counter to the value configured by TOCC.TOP. Down-counting is started when the first FIFO element is stored. 00: Continuous operation 01: Timeout controlled by TX Event FIFO 10: Timeout controlled by Rx FIFO 0 11: Timeout controlled by Rx FIFO 1
0	R/W	0x0	ETOC Enable Timeout Counter 0: Timeout Counter disabled 1: Timeout Counter enabled

17.2.9.11 0x002C CAN0 Timeout Counter Value Register (Default Value:0x0000_FFFF)

Offset: 0x002C			Register Name: CAN0_TOCV
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0xFFFF	TOC Timeout Counter The Timeout Counter is decremented in multiples of CAN bit times [1...16] depending on the configuration of TSCC.TCP. When decremented to zero, interrupt flag IR.TOO is set and the Timeout Counter is stopped. Start and reset/restart conditions are configured via TOCC.TOS. Note: Byte access: When TOCC.TOS = “00” , writing one of the register bytes 3/2/1/0 will preset the Timeout Counter.

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17.2.9.12 0x0040 CAN0 Error Counter Register (Default Value:0x0000_0000)

Offset: 0x0040			Register Name: CAN0_ECR
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	CEL CAN Error Logging The counter is incremented each time when a CAN protocol error causes the 8-bit Transmit Error Counter TEC or the 7-bit Receive Error Counter REC to be incremented. The counter is also incremented when the Bus_Off limit is reached. It is not incremented when only RP is set without changing REC. The increment of CEL follows after the increment of REC or TEC. The counter is reset by read access to CEL. The counter stops at 0xFF; the next increment of TEC or REC sets interrupt flag IR.ELO.
15	R	0x0	RP Receive Error Passive 0: The Receive Error Counter is below the error passive level of 128 1: The Receive Error Counter has reached the error passive level of 128
14:8	R	0x0	REC Receive Error Counter Actual state of the Receive Error Counter, values between 0 and 127
7:0	R	0x0	TEC Transmit Error Counter Actual state of the Transmit Error Counter, values between 0 and 255 Note: When CCCR.ASM is set, the CAN protocol controller does not increment TEC and REC when a CAN protocol error is detected, but CEL is still incremented.

17.2.9.13 0x0044 CAN0 Protocol Status Register (Default Value:0x0000_0707)

Offset: 0x0044			Register Name: CAN0_PSR
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/

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Offset: 0x0044			Register Name: CAN0_PSR
Bit	Read/Write	Default/Hex	Description
22:16	R	0x0	TDCV Transmitter Delay Compensation Value Position of the secondary sample point, defined by the sum of the measured delay from m_can_tx to m_can_rx and TDCR.TDCO. The SSP position is, in the data phase, the number of mtq between the start of the transmitted bit and the secondary sample point. Valid values are 0 to 127 mtq.
15	/	/	/
14	R	0x0	PXE Protocol Exception Event 0: No protocol exception event occurred since last read access 1: Protocol exception event occurred
13	R	0x0	RFDF: Received a CAN FD Message This bit is set independent of acceptance filtering. 0: Since this bit was reset by the CPU, no CAN FD message has been received 1: Message in CAN FD format with FDF flag set has been received Note: Byte access: Reading byte 0 will reset RFDF, reading bytes 3/2/1 has no impact.
12	R	0x0	RBRS BRS flag of last received CAN FD Message This bit is set together with RFDF, independent of acceptance filtering. 0: Last received CAN FD message did not have its BRS flag set 1: Last received CAN FD message had its BRS flag set Note: Byte access: Reading byte 0 will reset RBRS, reading bytes 3/2/1 has no impact.
11	R	0x0	RESI ESI flag of last received CAN FD Message This bit is set together with RFDF, independent of acceptance filtering. 0: Last received CAN FD message did not have its ESI flag set 1: Last received CAN FD message had its ESI flag set Note: Byte access: Reading byte 0 will reset RESI, reading bytes 3/2/1 has no impact.

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Offset: 0x0044			Register Name: CAN0_PSR
Bit	Read/Write	Default/Hex	Description
10:8	R	0x7	DLEC Data Phase Last Error Code Type of last error that occurred in the data phase of a CAN FD format frame with its BRS flag set. Coding is the same as for LEC. This field will be cleared to zero when a CAN FD format frame with its BRS flag set has been transferred (reception or transmission) without error. Note: Byte access: Reading byte 0 will set DLEC to “111”, reading bytes 3/2/1 has no impact.
7	R	0x0	BO Bus_Off Status 0: The M_CAN is not Bus_Off 1: The M_CAN is in Bus_Off state
6	R	0x0	EW Warning Status 0: Both error counters are below the Error_Warning limit of 96 1: At least one of error counter has reached the Error_Warning limit of 96
5	R	0x0	EP Error Passive 0: The M_CAN is in the Error_Active state. It normally takes part in bus communication and sends an active error flag when an error has been detected 1: The M_CAN is in the Error_Passive state
4:3	R	0x0	ACT Activity Monitors the module’s CAN communication state. 00: Synchronizing - node is synchronizing on CAN communication 01: Idle - node is neither receiver nor transmitter 10: Receiver - node is operating as receiver 11: Transmitter - node is operating as transmitter Note: ACT is set to “00” by a Protocol Exception Event.
2:0	R	0x7	LEC Last Error Code The LEC indicates the type of the last error to occur on the CAN bus. This field will be cleared to ‘0’ when a message has been transferred (reception or

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Offset: 0x0044			Register Name: CAN0_PSR
Bit	Read/Write	Default/Hex	Description
			transmission) without error. 0: No Error: No error occurred since LEC has been reset by successful reception or transmission. 1: Stuff Error: More than 5 equal bits in a sequence have occurred in a part of a received message where this is not allowed. 2: Form Error: A fixed format part of a received frame has the wrong format. 3: AckError: The message transmitted by the M_CAN was not acknowledged by another node. 4: Bit1Error: During the transmission of a message (with the exception of the arbitration field), the device wanted to send a recessive level (bit of logical value '1'), but the monitored bus value was dominant. 5: Bit0Error: During the transmission of a message (or acknowledge bit, or active error flag, or overload flag), the device wanted to send a dominant level (data or identifier bit logical value '0'), but the monitored bus value was recessive. During Bus_Off recovery this status is set each time a sequence of 11 recessive bits has been monitored. This enables the CPU to monitor the proceeding of the Bus_Off recovery sequence (indicating the bus is not stuck at dominant or continuously disturbed). 6: CRCError: The CRC check sum of a received message was incorrect. The CRC of an incoming message does not match with the CRC calculated from the received data. 7: NoChange: Any read access to the Protocol Status Register re-initializes the LEC to '7'. When the LEC shows the value '7', no CAN bus event was detected since the last CPU read access to the Protocol Status Register. Note: When a frame in CAN FD format has reached the data phase with BRS flag set, the next CAN event (error or valid frame) will be shown in DLEC instead of LEC. An error in a fixed stuff bit of a CAN FD CRC sequence will be shown as a Form Error, not Stuff Error. Note: The Bus_Off recovery sequence (see *ISO 11898-1:2015*) cannot be shortened by setting or

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Offset: 0x0044			Register Name: CAN0_PSR
Bit	Read/Write	Default/Hex	Description
			resetting CCCR.INIT. If the device goes Bus_Off, it will set CCCR.INIT of its own accord, stopping all bus activities. Once CCCR.INIT has been cleared by the CPU, the device will then wait for 129 occurrences of Bus Idle (129 * 11 consecutive recessive bits) before resuming normal operation. At the end of the Bus_Off recovery sequence, the Error Management Counters will be reset. During the waiting time after the resetting of CCCR.INIT, each time a sequence of 11 recessive bits has been monitored, a Bit0Error code is written to PSR.LEC, enabling the CPU to readily check up whether the CAN bus is stuck at dominant or continuously disturbed and to monitor the Bus_Off recovery sequence. ECR.REC is used to count these sequences. Note: Byte access: Reading byte 0 will set LEC to “111”, reading bytes 3/2/1 has no impact.

17.2.9.14 0x0048 CAN0 Transmitter Delay Compensation Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0048			Register Name: CAN0_TDCR
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14:8	R/W	0x0	TDCO Transmitter Delay Compensation SSP Offset Offset value defining the distance between the measured delay from m_can_tx to m_can_rx and the secondary sample point. Valid values are 0 to 127 mtq.
7	/	/	/

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Offset: 0x0048			Register Name: CAN0_TDCR
Bit	Read/Write	Default/Hex	Description
6:0	R/W	0x0	TDCF Transmitter Delay Compensation Filter Window Length Defines the minimum value for the SSP position, dominant edges on m_can_rx that would result in an earlier SSP position are ignored for transmitter delay measurement. The feature is enabled when TDCF is configured to a value greater than TDCO. Valid values are 0 to 127 mtq.

17.2.9.15 0x0050 CAN0 Interrupt Register (Default Value:0x0000_0000)

Offset: 0x0050			Register Name: CAN0_IR
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W1C	0x0	ARA: Access to Reserved Address 0: No access to reserved address occurred 1: Access to reserved address occurred
28	R/W1C	0x0	PED: Protocol Error in Data Phase (Data Bit Time is used) 0: No protocol error in data phase 1: Protocol error in data phase detected (PSR.DLEC ≠ 0,7)
27	R/W1C	0x0	PEA: Protocol Error in Arbitration Phase (Nominal Bit Time is used) 0: No protocol error in arbitration phase 1: Protocol error in arbitration phase detected (PSR.LEC ≠ 0,7)
26	R/W1C	0x0	WDI: Watchdog Interrupt 0: No Message RAM Watchdog event occurred 1: Message RAM Watchdog event due to missing READY
25	R/W1C	0x0	BO: Bus_Off Status 0: Bus_Off status unchanged 1: Bus_Off status changed
24	R/W1C	0x0	EW: Warning Status 0: Error_Warning status unchanged 1: Error_Warning status changed

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Offset: 0x0050			Register Name: CAN0_IR
Bit	Read/Write	Default/Hex	Description
23	R/W1C	0x0	EP: Error Passive 0: Error_Passive status unchanged 1: Error_Passive status changed
22	R/W1C	0x0	ELO: Error Logging Overflow 0: CAN Error Logging Counter did not overflow 1: Overflow of CAN Error Logging Counter occurred
21	R/W1C	0x0	BEU: Bit Error Uncorrected Message RAM bit error detected, uncorrected. Controlled by input signal m_can_aeim_berr[1] generated by an optional external parity / ECC logic attached to the Message RAM. An uncorrected Message RAM bit error sets CCCR.INIT to '1'. This is done to avoid transmission of corrupted data. 0: No bit error detected when reading from Message RAM 1: Bit error detected, uncorrected (e.g. parity logic)
20	R/W1C	0x0	BEC: Bit Error Corrected Message RAM bit error detected and corrected. Controlled by input signal m_can_aeim_berr[0] generated by an optional external parity / ECC logic attached to the Message RAM. 0: No bit error detected when reading from Message RAM 1: Bit error detected and corrected (e.g. ECC)
19	R/W1C	0x0	DRX: Message stored to Dedicated Rx Buffer The flag is set whenever a received message has been stored into a dedicated Rx Buffer. 0: No Rx Buffer updated 1: At least one received message stored into an Rx Buffer
18	R/W1C	0x0	TOO: Timeout Occurred 0: No timeout 1: Timeout reached

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Offset: 0x0050			Register Name: CAN0_IR
Bit	Read/Write	Default/Hex	Description
17	R/W1C	0x0	MRAF: Message RAM Access Failure The flag is set, when the Rx Handler • has not completed acceptance filtering or storage of an accepted message until the arbitration field of the following message has been received. In this case acceptance filtering or message storage is aborted and the Rx Handler starts processing of the following message. • was not able to write a message to the Message RAM. In this case message storage is aborted. In both cases the FIFO put index is not updated resp. the New Data flag for a dedicated Rx Buffer is not set, a partly stored message is overwritten when the next message is stored to this location. The flag is also set when the Tx Handler was not able to read a message from the Message RAM in time. In this case message transmission is aborted. In case of a Tx Handler access failure the M_CAN is switched into Restricted Operation Mode. To leave Restricted Operation Mode, the Host CPU has to reset CCCR.ASM. 0: No Message RAM access failure occurred 1: Message RAM access failure occurred
16	R/W1C	0x0	TSW: Timestamp Wraparound 0: No timestamp counter wrap-around 1: Timestamp counter wrapped around
15	R/W1C	0x0	TEFL: Tx Event FIFO Element Lost 0: No Tx Event FIFO element lost 1: Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero
14	R/W1C	0x0	TEFF: Tx Event FIFO Full 0: Tx Event FIFO not full 1: Tx Event FIFO full
13	R/W1C	0x0	TEFW: Tx Event FIFO Watermark Reached 0: Tx Event FIFO fill level below watermark 1: Tx Event FIFO fill level reached watermark
12	R/W1C	0x0	TEFN: Tx Event FIFO New Entry 0: Tx Event FIFO unchanged 1: Tx Handler wrote Tx Event FIFO element
11	R/W1C	0x0	TFE: Tx FIFO Empty 0: Tx FIFO non-empty 1: Tx FIFO empty

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Offset: 0x0050			Register Name: CAN0_IR
Bit	Read/Write	Default/Hex	Description
10	R/W1C	0x0	TCF: Transmission Cancellation Finished 0: No transmission cancellation finished 1: Transmission cancellation finished
9	R/W1C	0x0	TC: Transmission Completed 0: No transmission completed 1: Transmission completed
8	R/W1C	0x0	HPM: High Priority Message 0: No high priority message received 1: High priority message received
7	R/W1C	0x0	RF1L: Rx FIFO 1 Message Lost 0: No Rx FIFO 1 message lost 1: Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero
6	R/W1C	0x0	RF1F: Rx FIFO 1 Full 0: Rx FIFO 1 not full 1: Rx FIFO 1 full
5	R/W1C	0x0	RF1W: Rx FIFO 1 Watermark Reached 0: Rx FIFO 1 fill level below watermark 1: Rx FIFO 1 fill level reached watermark
4	R/W1C	0x0	RF1N: Rx FIFO 1 New Message 0: No new message written to Rx FIFO 1 1: New message written to Rx FIFO 1
3	R/W1C	0x0	RF0L: Rx FIFO 0 Message Lost 0: No Rx FIFO 0 message lost 1: Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero
2	R/W1C	0x0	RF0F: Rx FIFO 0 Full 0: Rx FIFO 0 not full 1: Rx FIFO 0 full
1	R/W1C	0x0	RF0W: Rx FIFO 0 Watermark Reached 0: Rx FIFO 0 fill level below watermark 1: Rx FIFO 0 fill level reached watermark
0	R/W1C	0x0	RF0N: Rx FIFO 0 New Message 0: No new message written to Rx FIFO 0 1: New message written to Rx FIFO 0

17.2.9.16 0x0054 CAN0 Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x0054			Register Name: CAN0_IE
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0054			Register Name: CAN0_IE
Bit	Read/Write	Default/Hex	Description
29	R/W	0x0	ARAE: Access to Reserved Address Enable
28	R/W	0x0	PEDE: Protocol Error in Data Phase (Data Bit Time is used) Enable
27	R/W	0x0	PEAE: Protocol Error in Arbitration Phase (Nominal Bit Time is used) Enable
26	R/W	0x0	WDIE: Watchdog Interrupt Enable
25	R/W	0x0	BOE: Bus_Off Status Enable
24	R/W	0x0	EWE: Warning Status Enable
23	R/W	0x0	EPE: Error Passive Enable
22	R/W	0x0	ELOE: Error Logging Overflow Enable
21	R/W	0x0	BEUE: Bit Error Uncorrected Enable
20	R/W	0x0	BECE: Bit Error Corrected Enable
19	R/W	0x0	DRXE: Message stored to Dedicated Rx Buffer Enable
18	R/W	0x0	TOOE: Timeout Occurred Enable
17	R/W	0x0	MRAFE: Message RAM Access Failure Enable
16	R/W	0x0	TSWE: Timestamp Wraparound Enable
15	R/W	0x0	TEFLE: Tx Event FIFO Element Lost Enable
14	R/W	0x0	TEFFE: Tx Event FIFO Full Enable
13	R/W	0x0	TEFWE: Tx Event FIFO Watermark Reached Enable
12	R/W	0x0	TEFNE: Tx Event FIFO New Entry Enable
11	R/W	0x0	TFEE: Tx FIFO Empty Enable
10	R/W	0x0	TCFE: Transmission Cancellation Finished Enable
9	R/W	0x0	TCE: Transmission Completed Enable
8	R/W	0x0	HPME: High Priority Message Enable
7	R/W	0x0	RF1LE: Rx FIFO 1 Message Lost Enable
6	R/W	0x0	RF1FE: Rx FIFO 1 Full Enable
5	R/W	0x0	RF1WE: Rx FIFO 1 Watermark Reached Enable
4	R/W	0x0	RF1NE: Rx FIFO 1 New Message Enable
3	R/W	0x0	RF0LE: Rx FIFO 0 Message Lost Enable
2	R/W	0x0	RF0FE: Rx FIFO 0 Full Enable
1	R/W	0x0	RF0WE: Rx FIFO 0 Watermark Reached Enable
0	R/W	0x0	RF0NE: Rx FIFO 0 New Message Enable

17.2.9.17 0x0058 CAN0 Interrupt Line Register (Default Value:0x0000_0000)

The Interrupt Line Select register assigns an interrupt generated by a specific interrupt flag from the Interrupt Register to one of the two module interrupt lines. For interrupt generation the respective interrupt line has to be enabled via ILE.EINT0 and ILE.EINT1.

0: Interrupt assigned to interrupt line m_can_int0

1: Interrupt assigned to interrupt line m_can_int1

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Offset: 0x0058			Register Name: CAN0_IL
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	ARAL: Access to Reserved Address Line
28	R/W	0x0	PEDL: Protocol Error in Data Phase (Data Bit Time is used) Line
27	R/W	0x0	PEAL: Protocol Error in Arbitration Phase (Nominal Bit Time is used) Line
26	R/W	0x0	WDIL: Watchdog Interrupt Line
25	R/W	0x0	BOL: Bus_Off Status Line
24	R/W	0x0	EWL: Warning Status Line
23	R/W	0x0	EPL: Error Passive Line
22	R/W	0x0	ELOL: Error Logging Overflow Line
21	R/W	0x0	BEUL: Bit Error Uncorrected Line
20	R/W	0x0	BECL: Bit Error Corrected Line
19	R/W	0x0	DRXL: Message stored to Dedicated Rx Buffer Line
18	R/W	0x0	TOOL: Timeout Occurred Line
17	R/W	0x0	MRAFL: Message RAM Access Failure Line
16	R/W	0x0	TSWL: Timestamp Wraparound Line
15	R/W	0x0	TEFLL: Tx Event FIFO Element Lost Line
14	R/W	0x0	TEFFL: Tx Event FIFO Full Line
13	R/W	0x0	TEFWL: Tx Event FIFO Watermark Reached Line
12	R/W	0x0	TEFNL: Tx Event FIFO New Entry Line
11	R/W	0x0	TFEL: Tx FIFO Empty Line
10	R/W	0x0	TCFL: Transmission Cancellation Finished Line
9	R/W	0x0	TCL: Transmission Completed Line
8	R/W	0x0	HPML: High Priority Message Line
7	R/W	0x0	RF1LL: Rx FIFO 1 Message Lost Line
6	R/W	0x0	RF1FL: Rx FIFO 1 Full Line
5	R/W	0x0	RF1WL: Rx FIFO 1 Watermark Reached Line
4	R/W	0x0	RF1NL: Rx FIFO 1 New Message Line
3	R/W	0x0	RF0LL: Rx FIFO 0 Message Lost Line
2	R/W	0x0	RF0FL: Rx FIFO 0 Full Line
1	R/W	0x0	RF0WL: Rx FIFO 0 Watermark Reached Line
0	R/W	0x0	RF0NL: Rx FIFO 0 New Message Line

17.2.9.18 0x005C CAN0 Interrupt Line Enable Register (Default Value:0x0000_0000)

Offset: 0x005C			Register Name: CAN0_ILE
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/

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Offset: 0x005C			Register Name: CAN0_ILE
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	EINT1: Enable Interrupt Line 1 0: Interrupt line m_can_int1 disabled 1: Interrupt line m_can_int1 enabled
0	R/W	0x0	EINT0: Enable Interrupt Line 0 0: Interrupt line m_can_int0 disabled 1: Interrupt line m_can_int0 enabled

17.2.9.19 0x0080 CAN0 Global Filter Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0080			Register Name: CAN0_GFC
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:4	R/W	0x0	ANFS Accept Non-matching Frames Standard Defines how received messages with 11-bit IDs that do not match any element of the filter list are treated. 00: Accept in Rx FIFO 0 01: Accept in Rx FIFO 1 10: Reject 11: Reject
3:2	R/W	0x0	ANFE Accept Non-matching Frames Extended Defines how received messages with 29-bit IDs that do not match any element of the filter list are treated. 00: Accept in Rx FIFO 0 01: Accept in Rx FIFO 1 10: Reject 11: Reject
1	R/W	0x0	RRFS Reject Remote Frames Standard 0: Filter remote frames with 11-bit standard IDs 1: Reject all remote frames with 11-bit standard IDs
0	R/W	0x0	RRFE Reject Remote Frames Extended 0: Filter remote frames with 29-bit extended IDs 1: Reject all remote frames with 29-bit extended IDs

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17.2.9.20 0x0084 CAN0 Standard ID Filter Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0084			Register Name: CAN0_SIDFC
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	LSS List Size Standard 0: No standard Message ID filter 1-128: Number of standard Message ID filter elements >128: Values greater than 128 are interpreted as 128
15:2	R/W	0x0	FLSSA Filter List Standard Start Address Start address of standard Message ID filter list
1:0	/	/	/

17.2.9.21 0x0088 CAN0 Extend ID Filter Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0088			Register Name: CAN0_XIDFC
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22:16	R/W	0x0	LSE List Size Extended 0: No extended Message ID filter 1-64: Number of extended Message ID filter elements >64: Values greater than 64 are interpreted as 64
15:2	R/W	0x0	FLESA Filter List Extended Start Address Start address of extended Message ID filter list
1:0	/	/	/

17.2.9.22 0x0090 CAN0 Extended ID AND Mask Register (Default Value:0x1FFF_FFFF)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x0090			Register Name: CAN0_XIDAM
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/

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Offset: 0x0090			Register Name: CAN0_XIDAM
Bit	Read/Write	Default/Hex	Description
28:0	R/W	0x1FFF_FFFF	EIDM Extended ID Mask For acceptance filtering of extended frames the Extended ID AND Mask is ANDed with the Message ID of a received frame. Intended for masking of 29-bit IDs in SAE J1939. With the reset value of all bits set to one the mask is not active .

17.2.9.23 0x0094 CAN0 High Priority Message Status Register (Default Value:0x0000_0000)

This register is updated every time a Message ID filter element configured to generate a priority event matches. This can be used to monitor the status of incoming high priority messages and to enable fast access to these messages.

Offset: 0x0094			Register Name: CAN0_HPMS
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R	0x0	FLST Filter List Indicates the filter list of the matching filter element. 0: Standard Filter List 1: Extended Filter List
14:8	R	0x0	FIDX Filter Index Index of matching filter element. Range is 0 to SIDFC.LSS - 1 resp. XIDFC.LSE - 1.
7:6	R	0x0	MSI Message Storage Indicator 00: No FIFO selected 01: FIFO message lost 10: Message stored in FIFO 0 11: Message stored in FIFO 1
5:0	R	0x0	BIDX Buffer Index Index of Rx FIFO element to which the message w as stored. Only valid when MSI[1] = ‘1’.

17.2.9.24 0x0098 CAN0 New Data 1 Register (Default Value:0x0000_0000)

Offset: 0x0098	Register Name: CAN0_NDAT1

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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ND[31:0] New Data[31:0] The register holds the New Data flags of Rx Buffers 0 to 31. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them. A flag is cleared by writing a '1' to the corresponding bit position. Writing a '0' has no effect. A hard reset will clear the register. 0: Rx Buffer not updated 1: Rx Buffer updated from new message

17.2.9.25 0x009C CAN0 New Data 2 Register (Default Value:0x0000_0000)

Offset: 0x009C			Register Name: CAN0_NDAT2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ND[63:32] New Data[63:32] The register holds the New Data flags of Rx Buffers 32 to 64. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them. A flag is cleared by writing a '1' to the corresponding bit position. Writing a '0' has no effect. A hard reset will clear the register. 0: Rx Buffer not updated 1: Rx Buffer updated from new message

17.2.9.26 0x00A0 CAN0 RX FIFO0 Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00A0			Register Name: CAN0_RXF0C
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	F0OM FIFO 0 Operation Mode FIFO 0 can be operated in blocking or in overwrite mode 0: FIFO 0 blocking mode 1: FIFO 0 overwrite mode

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Offset: 0x00A0			Register Name: CAN0_RXF0C
Bit	Read/Write	Default/Hex	Description
30:24	R/W	0x0	F0WM Rx FIFO 0 Watermark 0: Watermark interrupt disabled 1-64: Level for Rx FIFO 0 watermark interrupt (IR.RF0W) >64: Watermark interrupt disabled
23	/	/	/
22:16	R/W	0x0	F0S Rx FIFO 0 Size 0: No Rx FIFO 0 1-64: Number of Rx FIFO 0 elements >64: Values greater than 64 are interpreted as 64 The Rx FIFO 0 elements are indexed from 0 to F0S-1
15:2	R/W	0x0	F0SA Rx FIFO 0 Start Address Start address of Rx FIFO 0 in Message RAM
1:0	/	/	/

17.2.9.27 0x00A4 CAN0 RX FIFO0 Status Register (Default Value:0x0000_0000)

Offset: 0x00A4			Register Name: CAN0_RXF0S
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25	R	0x0	RF0L Rx FIFO 0 Message Lost This bit is a copy of interrupt flag IR.RF0L. When IR.RF0L is reset, this bit is also reset. 0: No Rx FIFO 0 message lost 1: Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero Note: Overwriting the oldest message when RXF0C.F0OM = '1' will not set this flag.
24	R	0x0	F0F Rx FIFO 0 Full 0: Rx FIFO 0 not full 1: Rx FIFO 0 full
23:22	/	/	/
21:16	R	0x0	F0PI Rx FIFO 0 Put Index Rx FIFO 0 write index pointer, range 0 to 63.
15:14	/	/	/

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Offset: 0x00A4			Register Name: CAN0_RXF0S
Bit	Read/Write	Default/Hex	Description
13:8	R	0x0	F0GI Rx FIFO 0 Get Index Rx FIFO 0 read index pointer, range 0 to 63.
7:6	/	/	/
5:0	R	0x0	F0FL Rx FIFO 0 Fill Level Number of elements stored in Rx FIFO 0, range 0 to 64.

17.2.9.28 0x00A8 CAN0 RX FIFO0 Acknowledge Register (Default Value:0x0000_0000)

Offset: 0x00A8			Register Name: CAN0_RXF0A
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0x0	F0AI Rx FIFO 0 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 0 it has to write the buffer index of the last element read from Rx FIFO 0 to F0AI. This will set the Rx FIFO 0 Get Index RXF0S.F0GI to F0AI + 1 and update the FIFO 0 Fill Level RXF0S.F0FL.

17.2.9.29 0x00AC CAN0 RX Buffer Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00AC			Register Name: CAN0_RXBC
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:2	R/W	0x0	RBSA Rx Buffer Start Address Configures the start address of the Rx Buffers section in the Message RAM (32-bit word address). Also used to reference debug messages A,B,C.
1:0	/	/	/

17.2.9.30 0x00B0 CAN0 RX FIFO1 Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00B0	Register Name: CAN0_RXF1C

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Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	F1OM FIFO 1 Operation Mode FIFO 1 can be operated in blocking or in overwrite mode 0: FIFO 1 blocking mode 1: FIFO 1 overwrite mode
30:24	R/W	0x0	F0WM Rx FIFO 1 Watermark 0: Watermark interrupt disabled 1-64: Level for Rx FIFO 1 watermark interrupt (IR.RF1W) >64: Watermark interrupt disabled
23	/	/	/
22:16	R/W	0x0	F1S Rx FIFO 1 Size 0: No Rx FIFO 1 1-64: Number of Rx FIFO 1 elements >64: Values greater than 64 are interpreted as 64 The Rx FIFO 1 elements are indexed from 0 to F1S-1
15:2	R/W	0x0	F1SA Rx FIFO 1 Start Address Start address of Rx FIFO 1 in Message RAM
1:0	/	/	/

17.2.9.31 0x00B4 CAN0 RX FIFO1 Status Register (Default Value:0x0000_0000)

Offset: 0x00B4			Register Name: CAN0_RXF1S
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25	R	0x0	RF1L Rx FIFO 1 Message Lost This bit is a copy of interrupt flag IR.RF1L. When IR.RF1L is reset, this bit is also reset. 0: No Rx FIFO 1 message lost 1: Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero Note: Overwriting the oldest message when RXF1C.F1OM = '1' will not set this flag.
24	R	0x0	F0F Rx FIFO 1 Full 0: Rx FIFO 1 not full 1: Rx FIFO 1 full

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Offset: 0x00B4			Register Name: CAN0_RXF1S
Bit	Read/Write	Default/Hex	Description
23:22	/	/	/
21:16	R	0x0	F1PI Rx FIFO 1 Put Index Rx FIFO 1 write index pointer, range 0 to 63.
15:14	/	/	/
13:8	R	0x0	F0GI Rx FIFO 1 Get Index Rx FIFO 1 read index pointer, range 0 to 63.
7:6	/	/	/
5:0	R	0x0	F0FL Rx FIFO 1 Fill Level Number of elements stored in Rx FIFO 1, range 0 to 64.

17.2.9.32 0x00B8 CAN0 RX FIFO1 Acknowledge Register (Default Value:0x0000_0000)

Offset: 0x00B8			Register Name: CAN0_RXF1A
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0x0	F1AI Rx FIFO 1 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 1 it has to write the buffer index of the last element read from Rx FIFO 1 to F1AI. This will set the Rx FIFO 1 Get Index RXF1S.F1GI to F1AI + 1 and update the FIFO 1 Fill Level RXF1S.F1FL.

17.2.9.33 0x00BC CAN0 RX Buffer/FIFO Element Size Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00BC			Register Name: CAN0_RXESC
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/

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Offset: 0x00BC			Register Name: CAN0_RXESC
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	RBDS Rx Buffer Data Field Size 000: 8 byte data field 001: 12 byte data field 010: 16 byte data field 011: 20 byte data field 100: 24 byte data field 101: 32 byte data field 110: 48 byte data field 111: 64 byte data field
7	/	/	/
6:4	R/W	0x0	F1DS Rx FIFO1 Data Field Size 000: 8 byte data field 001: 12 byte data field 010: 16 byte data field 011: 20 byte data field 100: 24 byte data field 101: 32 byte data field 110: 48 byte data field 111: 64 byte data field
3	/	/	/
2:0	R/W	0x0	F0DS Rx FIFO0 Data Field Size 000: 8 byte data field 001: 12 byte data field 010: 16 byte data field 011: 20 byte data field 100: 24 byte data field 101: 32 byte data field 110: 48 byte data field 111: 64 byte data field

17.2.9.34 0x00C0 CAN0 TX Buffer Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00C0			Register Name: CAN0_TXBC
Bit	Read/Write	Default/Hex	Description
31	/	/	/

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Offset: 0x00C0			Register Name: CAN0_TXBC
Bit	Read/Write	Default/Hex	Description
30	R/W	0x0	TFQM Tx FIFO/Queue Mode 0: Tx FIFO operation 1: Tx Queue operation
29:24	R/W	0x0	TFQS Transmit FIFO/Queue Size 0: No Tx FIFO/Queue 1-32: Number of Tx Buffers used for Tx FIFO/Queue >32: Values greater than 32 are interpreted as 32
23:22	/	/	/
21:16	R/W	0x0	NDTB Number of Dedicated Transmit Buffers 0: No Dedicated Tx Buffers 1-32: Number of Dedicated Tx Buffers >32: Values greater than 32 are interpreted as 32 Note: Be aware that the sum of TFQS and NDTB may be not greater than 32. There is no check for erroneous configurations. The Tx Buffers section in the Message RAM starts with the dedicated Tx Buffers.
15:2	R/W	0x0	TBSA Tx Buffers Start Address Start address of Tx Buffers section in Message RAM
1:0	/	/	/

17.2.9.35 0x00C4 CAN0 TX FIFO/Queue Status Register (Default Value:0x0000_0000)

Offset: 0x00C4			Register Name: CAN0_TXFQS
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21	R	0x0	TFQF Tx FIFO/Queue Full 0: Tx FIFO/Queue not full 1: Tx FIFO/Queue full
20:16	R	0x0	TFQPI Tx FIFO/Queue Put Index Tx FIFO/Queue write index pointer, range 0 to 31.
15:13	/	/	/

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Offset: 0x00C4			Register Name: CAN0_TXFQS
Bit	Read/Write	Default/Hex	Description
12:8	R	0x0	TFGI Tx FIFO Get Index Tx FIFO read index pointer, range 0 to 31. Read as zero when Tx Queue operation is configured(TXBC.TFQM = '1').
15:2	/	/	/
1:0	R	0x0	TFFL Tx FIFO Free Level Number of consecutive free Tx FIFO elements starting from TFGI, range 0 to 32. Read as zero when Tx Queue operation is configured (TXBC.TFQM = '1') Note: In case of mixed configurations where dedicated Tx Buffers are combined with a Tx FIFO or a Tx Queue, the Put and Get Indices indicate the number of the Tx Buffer starting with the first dedicated Tx Buffers. Example: For a configuration of 12 dedicated Tx Buffers and a Tx FIFO of 20 Buffers a Put Index of 15 points to the fourth buffer of the Tx FIFO.

17.2.9.36 0x00C8 CAN0 TX Buffer Element Size Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00C8			Register Name: CAN0_TXESC
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
2:0	R/W	0x0	TBDS Tx Buffer Data Field Size 000: 8 byte data field 001: 12 byte data field 010: 16 byte data field 011: 20 byte data field 100: 24 byte data field 101: 32 byte data field 110: 48 byte data field 111: 64 byte data field Note: In case the data length code DLC of a Tx Buffer element is configured to a value higher than the Tx Buffer data field size TXESC.TBDS, the bytes not defined by the Tx Buffer are transmitted as "0xCC" (padding bytes).

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17.2.9.37 0x00CC CAN0 TX Buffer Request Pending Register (Default Value:0x0000_0000)

Offset: 0x00CC			Register Name: CAN0_TXBRP
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TRP Transmission Request Pending Each Tx Buffer has its own Transmission Request Pending bit. The bits are set via register TXBAR. The bits are reset after a requested transmission has completed or has been canceled via register TXBCR. TXBRP bits are set only for those Tx Buffers configured via TXBC. After a TXBRP bit has been set, a Tx scan is started to check for the pending Tx request with the highest priority (Tx Buffer with lowest Message ID). A cancellation request resets the corresponding transmission request pending bit of register TXBRP. In case a transmission has already been started when a cancellation is requested, this is done at the end of the transmission, regardless whether the transmission was successful or not. The cancellation request bits are reset directly after the corresponding TXBRP bit has been reset. After a cancellation has been requested, a finished cancellation is signaled via TXBCF • after successful transmission together with the corresponding TXBTO bit • when the transmission has not yet been started at the point of cancellation • when the transmission has been aborted due to lost arbitration • when an error occurred during frame transmission In DAR mode all transmissions are automatically canceled if they are not successful. The corresponding TXBCF bit is set for all unsuccessful transmissions. 0: No transmission request pending 1: Transmission request pending Note: TXBRP bits which are set while a Tx scan is in progress are not considered during this particular Tx scan. In case a cancellation is requested for such a Tx Buffer, this Add Request is canceled immediately, the corresponding TXBRP bit is reset.

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17.2.9.38 0x00D0 CAN0 TX Buffer Add Request Register (Default Value:0x0000_0000)

Offset: 0x00D0			Register Name: CAN0_TXBAR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	AR Add Request Each Tx Buffer has its own Add Request bit. Writing a '1' will set the corresponding Add Request bit; writing a '0' has no impact. This enables the Host to set transmission requests for multiple Tx Buffers with one write to TXBAR. TXBAR bits are set only for those Tx Buffers configured via TXBC. When no Tx scan is running, the bits are reset immediately, else the bits remain set until the Tx scan process has completed. 0: No transmission request added 1: Transmission requested added Note: If an add request is applied for a Tx Buffer with pending transmission request (corresponding TXBRP bit already set), this add request is ignored.

17.2.9.39 0x00D4 CAN0 TX Buffer Cancellation Request Register (Default Value:0x0000_0000)

Offset: 0x00D4			Register Name: CAN0_TXBCR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CR Cancellation Request Each Tx Buffer has its own Cancellation Request bit. Writing a '1' will set the corresponding Cancellation Request bit; writing a '0' has no impact. This enables the Host to set cancellation requests for multiple Tx Buffers with one write to TXBCR. TXBCR bits are set only for those Tx Buffers configured via TXBC. The bits remain set until the corresponding bit of TXBRP is reset. 0: No cancellation pending 1: Cancellation pending

17.2.9.40 0x00D8 CAN0 TX Buffer Transmission Occurred Register (Default Value:0x0000_0000)

Offset: 0x00D8			Register Name: CAN0_TXBTO
Bit	Read/Write	Default/Hex	Description

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Offset: 0x00D8			Register Name: CAN0_TXBTO
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	TO Transmission Occurred Each Tx Buffer has its own Transmission Occurred bit. The bits are set when the corresponding TXBRP bit is cleared after a successful transmission. The bits are reset when a new transmission is requested by writing a '1' to the corresponding bit of register TXBAR. 0: No transmission occurred 1: Transmission occurred

17.2.9.41 0x00DC CAN0 TX Buffer Cancellation Finished Register (Default Value:0x0000_0000)

Offset: 0x00DC			Register Name: CAN0_TXBCF
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CF Cancellation Finished Each Tx Buffer has its own Cancellation Finished bit. The bits are set when the corresponding TXBRP bit is cleared after a cancellation was requested via TXBCR. In case the corresponding TXBRP bit was not set at the point of cancellation, CF is set immediately. The bits are reset when a new transmission is requested by writing a '1' to the corresponding bit of register TXBAR. 0: No transmit buffer cancellation 1: Transmit buffer cancellation finished

17.2.9.42 0x00E0 CAN0 TX Buffer Transmission Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x00E0			Register Name: CAN0_TXBTIE
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TIE Transmission Interrupt Enable Each Tx Buffer has its own Transmission Interrupt Enable bit. 0: Transmission interrupt disabled 1: Transmission interrupt enable

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17.2.9.43 0x00E4 CAN0 TX Buffer Cancellation Interrupt Enable Register (Default Value:0x0000_0000)

Offset: 0x00E4			Register Name: CAN0_TXBCIE
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CFIE Cancellation Finished Interrupt Enable Each Tx Buffer has its own Cancellation Finished Interrupt Enable bit. 0: Cancellation finished interrupt disabled 1: Cancellation finished interrupt enabled

17.2.9.44 0x00F0 CAN0 TX Event FIFO Configuration Register (Default Value:0x0000_0000)

This register is only writable if bits CCCR.CCE and CCCR.INIT are set.

Offset: 0x00F0			Register Name: CAN0_TXEFC
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:24	R/W	0x0	EFWM Event FIFO Watermark 0: Watermark interrupt disabled 1-32: Level for Tx Event FIFO watermark interrupt (IR.TEFW) >32: Watermark interrupt disabled
23:22	/	/	/
21:16	R/W	0x0	EFS Event FIFO Size 0: Tx Event FIFO disabled 1-32: Number of Tx Event FIFO elements >32: Values greater than 32 are interpreted as 32 The Tx Event FIFO elements are indexed from 0 to EFS - 1
15:2	R/W	0x0	EFSA Event FIFO Start Address Start address of Tx Event FIFO in Message RAM
1:0	/	/	/

17.2.9.45 0x00F4 CAN0 TX Event FIFO Status Register (Default Value:0x0000_0000)

Offset: 0x00F4			Register Name: CAN0_TXEFS
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/

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Offset: 0x00F4			Register Name: CAN0_TXEFS
Bit	Read/Write	Default/Hex	Description
25	R	0x0	TEFL Tx Event FIFO Element Lost This bit is a copy of interrupt flag IR.TEFL. When IR.TEFL is reset, this bit is also reset. 0: No Tx Event FIFO element lost 1: Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero.
24	R	0x0	EFF Event FIFO Full 0: Tx Event FIFO not full 1: Tx Event FIFO full
23:21	/	/	/
20:16	R	0x0	EFPI Event FIFO Put Index Tx Event FIFO write index pointer, range 0 to 31.
15:13	/	/	/
12:8	R	0x0	EFGI Event FIFO Get Index Tx Event FIFO read index pointer, range 0 to 31.
7:6	/	/	/
5:0	R	0x0	EFFL Event FIFO Fill Level Number of elements stored in Tx Event FIFO, range 0 to 32.

17.2.9.46 0x00F8 CAN0 Tx Event FIFO Acknowledge Register (Default Value:0x0000_0000)

Offset: 0x00F8			Register Name: CAN0_TXEFA
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4:0	R/W	0x0	EFAI Event FIFO Acknowledge Index After the Host has read an element or a sequence of elements from the Tx Event FIFO it has to write the index of the last element read from Tx Event FIFO to EFAI. This will set the Tx Event FIFO Get Index TXEFS.EFGI to EFAI + 1 and update the Event FIFO Fill Level TXEFS.EFFL.

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17.3 Local Bus Controller (LBC)

17.3.1 Overview

Based on the data transmission requirements of high bandwidth and low latency in industrial data transmission, Local Bus Controller (LBC) implements the private Localbus protocol for synchronous transmission, supports direct CPU access, indirect CPU access, and private DMA access.

The Local Bus Controller (LBC) has the following features:

- Supports one set of Localbus controller, using privatized synchronous Transport Protocol
  - Maximum rate is150 MHz@16bits, 100MHz@32bits
  - Supports address/data multiplexing, and 8/16/32 data bit width
  - Supports mode 0/ mode 1 two transmission timing
  - Supports big endian and little endian
  - Supports data parity check
  - FIX/INCR type address transmission with configurable burst length (maximum 128 beats in one transmission)
  - Supports slave data transmission through the ready signal reverse voltage master. You can choose no reverse voltage, data start reverse voltage, or data real-time reverse voltage.
- Supports a maximum of 4 chip selects for transmission:
  - The timing parameters of the four chip selects can be configured independently, and the polarity of the control signal can be configured
  - Support address binding, CPU binding, fixed access and other methods to select the corresponding chip select transmission
  - Supports interface clock off/always on. Only clock configured by CS0 is available when interface clock is always-on.
- Supports multiple access methods:
  - Supports direct access, indirect access, and private DMA access
  - Indirect access supports data transfer through system DMA
  - Private DMA supports external data DRQ requests
  - Private DMA supports 4-word descriptor, and supports descriptor invalidation and descriptor write-back
  - Private DMA supports data transfer with linked list

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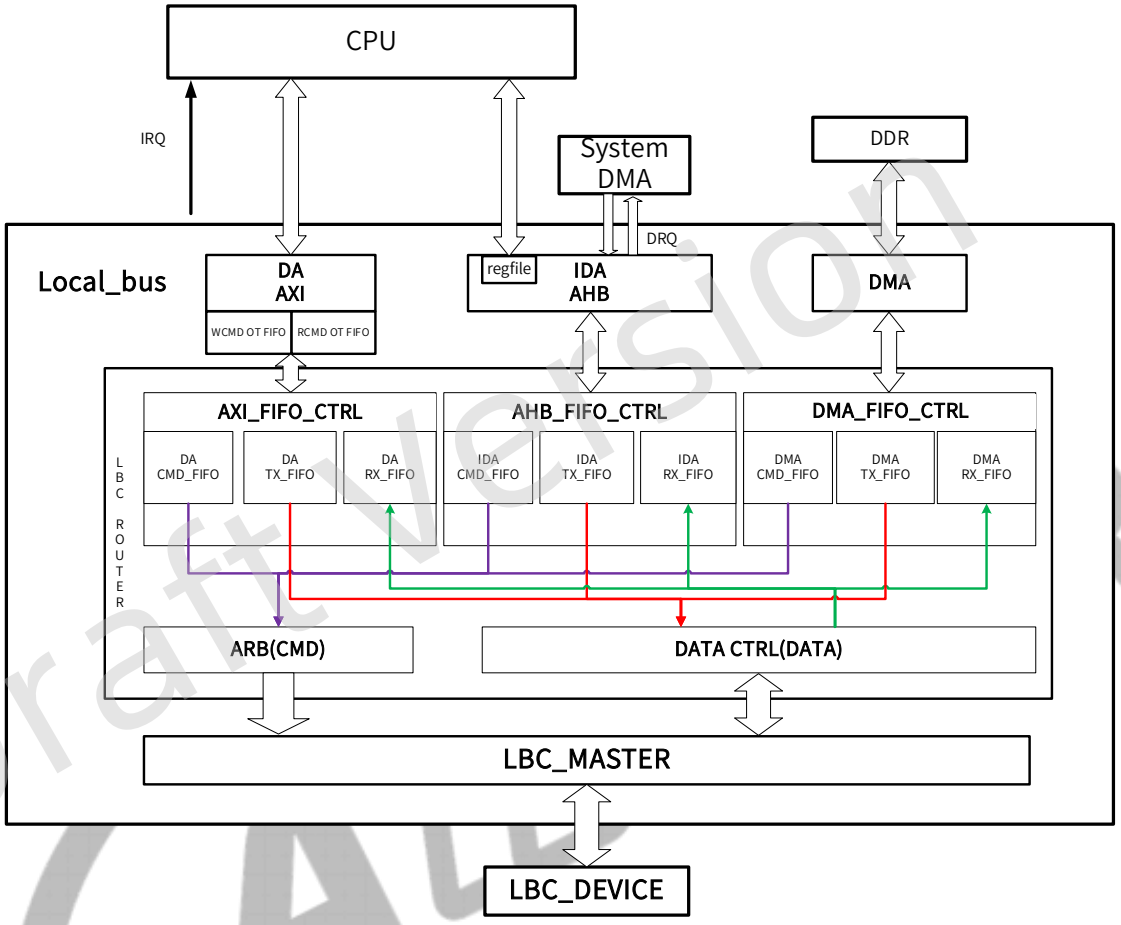
17.3.2 Block Diagram

The following figure shows the block diagram of LOCAL_BUS_CTRL module.

Figure 17-14 LOCAL_BUS_CTRL Block Diagram



17.3.3 External Signals

The following table describes the external signals of the LOCAL_BUS_CTRL.

Table 17-7 LOCAL_BUS_CTRL External Signals

Signal Name	PIN Priority	I/O	Localbus Sync Mode 0	Localbus Sync Mode1
LBUS-CLK	Required	O	CLK	CLK
LBUS-CS[3:0]	Required	O	CS[3:0]	CS[3:0]
LBUS-ALE	Required	O	ALE	ALE
LBUS-LBE[3:0]	Optional	O	LBE	LBE[3:0]
LBUS-WR	Required	O	WR/RD	WR/RD
LBUS-OE	Required	O	BURST[0]	CLE
LBUS-LD[31:15]	Required	I/O	LD[31:15]	LD[31:15]

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Signal Name	PIN Priority	I/O	Localbus Sync Mode 0	Localbus Sync Mode1
LBUS-LD[15:0]	Required	I/O	LD[15:0]	LD[15:0]
LBUS-DP[3:0]	Optional	I/O	DP[3:0]	DP[3:0]
LBUS-WAIT	Optional	O	BURST[1]	WAIT
LBUS-READY[3:0]	Optional	I	READY[3:0]	READY[3:0]
LBUS-DE	Optional	O	BURST[2]	DE
LBUS-DRQ[1:0]	Optional	I	DRQ[1:0]	DRQ[1:0]

17.3.4 Functional Description

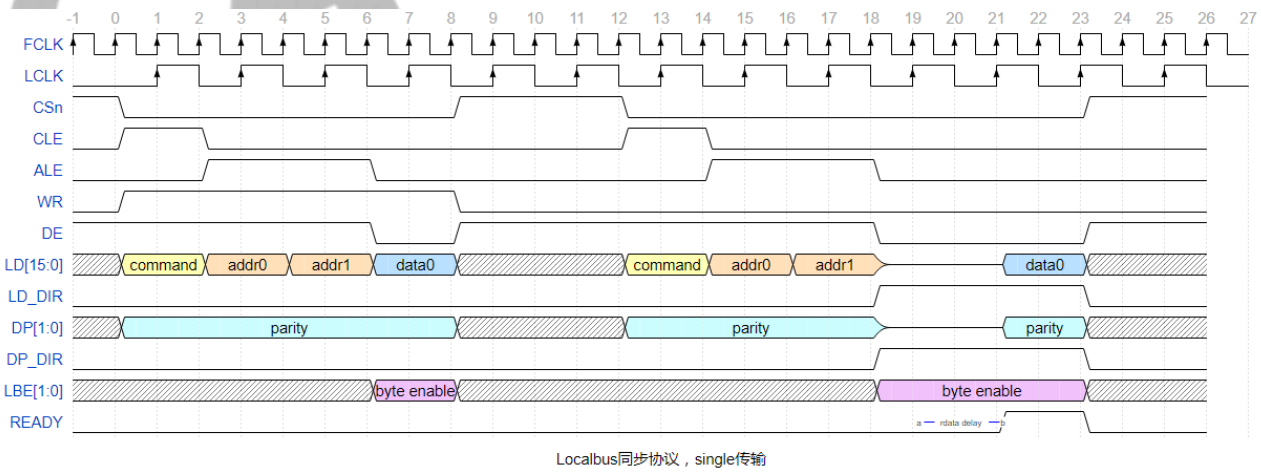
17.3.4.1 Typical application

Private LocalBus Protocol

Timing diagrams

Local_bus supports a private LocalBus synchronous transfer protocol with configurable data bus widths (8-bit, 16-bit, 32-bit) and multiplexed address/data buses. Below are typical 16-bit bus timing diagrams.

Figure 17-15 LocalBus Protocol Timing (Single Transfer, Mode 1)



Localbus同步协议，single传输

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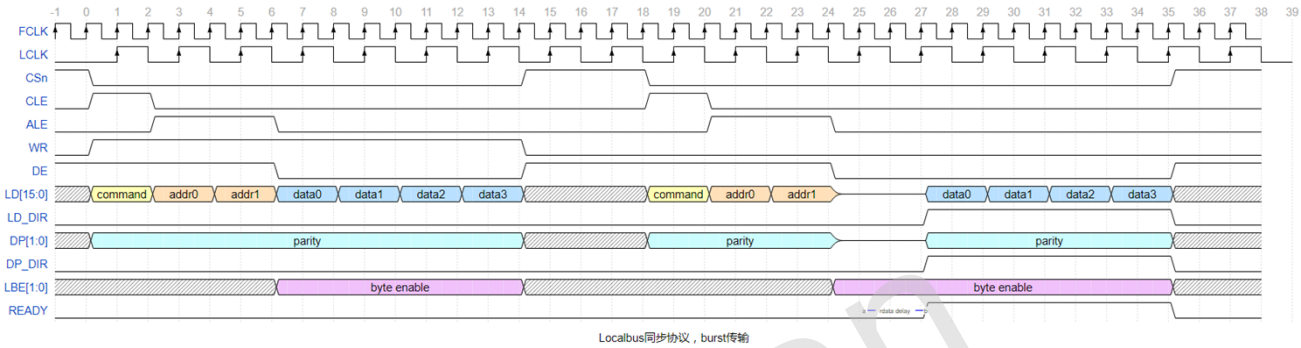
Figure 17-16 LocalBus Protocol Timing (Burst Transfer, Mode 1)



Signal Descriptions

- FCLK: Clock input from CCU or external oscillator.
- LCLK: Clock derived from FCLK via frequency division (max 150 MHz).
- CSn: Chip select (active level configurable). Remains active throughout the transaction (command, address, data phases). Supports up to 4 chip selects.
- CLE: Command latch enable (active high). Valid during command phase.
- ALE: Address latch enable (active high). Valid during address phase. Supports up to 2 address phases.
- WR: Transfer direction (high = write, low = read). Remains stable during the transaction.
- LD_OUT/LD_IN: Bidirectional data/address bus (32-bit). Multiplexed for address and data.
- DP_OUT/DP_IN: Data parity bus for byte-wise or full-data parity. Aligned with data bus.
- LBE: Data byte enable (active high). Indicates valid bytes during writes; fully valid during reads.
- DE: Data enable (active low). Marks start of data phase. Optional for adding delay between address/data phases.
- READY: Data valid input (active high). Used by devices to indicate data validity.

Bus Behavior

- Command Phase:

  The command phase is used for the host to inform the slave of the transmission type. When CLE is high, the instruction phase is valid. The format is as follows:

Table 17-8 Command Phase Format (8-bit)

Bit	Parameter	Description
[7]	BURST_TYPE	0: FIX (same address) 1: INCR (incremental address)
[6:0]	BURST_LENGTH	Burst length. The unit is cycle. The value ranges from 0 to 127. The actual value ranges for 1 to 128.

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Since it is a 16-bit bus, and the instruction is only 8 bits, the high position is 0 when the bus is transmitted, which means that the format of the transmitted instruction is as follows:

Table 17-9 Command Phase Format (16-bit)

Bit	Parameter	Description
[15:8]	0	Zero-padded (participates in parity)
[7]	BURST_TYPE	
[6:0]	BURST_LENGTH	

During the command phase, the read and write state should be determined according to the level of WR. When WR is high, it means Master writes; and when WR is low, it means WR reads.

- Address Phase:

  The address phase is used for the Master to inform the slave of the written/read address. When ALE is high, the address phase is effective. When the 16-bit bus is used, the 1/2 cycle address can be sent; when one cycle address is sent, the effective LBC address is 16 bits; when two cycle addresses are sent, the effective LBC address is 32 bits. If two cycle addresses are sent, MSB is sent first, and LSB is sent later, corresponding to the relationship in the timing diagram as follows:

  ADDR0: LBC_ADDR[31:16]

  ADDR1: LBC_ADDR[15:0]

  For one address cycle, the corresponding relationship is as follows:

  ADDR0: LBC_ADDR[15:0]

- Data Phase:

  The data phase is divided into the write phase and the reading phase. When writing, the Master outputs the data, while the slave outputs the data when reading. When writing, there is no interval between the address phase and the data phase. When reading, the data cannot be given immediately after receiving the address from the machine, and it needs to wait for the data to be prepared and then sent to the bus. After the Master receives all the data sent by the Slave, it will pull down the CS and end the transmission.

  The data phase can be divided into single and burst, the specific number of data, determined by the length instruction in the CMD instruction. The slave needs to receive or send enough data required by the instruction before the end of the transmission.

  When the Master writes data, it is accompanied by the Byte Enable signal. The BE signal indicates the validity of 2 byte data on the 16-bit bus. If BE is equal to 0x3, all 2 byte data are valid; if BE is equal to 0x2, the lower 8bit data is invalid. It should BE noted that during the Master read, the data sent by the Master will be valid and will not be read out of alignment.

- Parity Check:

  When LBC is transmitted, DP check signal is attached, which is used to provide parity check information on the bus. DP is divided into two modes, byte by byte check or whole check.

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Byte by byte check is to calculate the parity check code of 2byte on the 16-bit bus respectively, and output 2bit check information to the DP signal. The overall check is 16bit together to calculate the parity check code, and only 1bit of DP signal is output

It should be noted that the CMD of the 16-bit bus is actually only valid for the low 8 bits, and the high 8 bits are zeros, but when DP is checked, it is checked according to the 16 bits of valid data, which means that the zeros of the high 8 bits in CMD are also involved in the check; DP is used to represent the error behavior in the transmission, the CMD phase, according to the 16-bit data transmission.

The DP check method is to recalculate the received bus data and compare it with the transmitted DP on the bus. If the data is consistent, the transmission parity check passes. If the data is inconsistent, the transmission error occurs. In the write phase, data DP is provided by the Master, and in the read phase, data DP is provided by the Slave.

- Endianness:

  LBC supports communication in big-end mode and little-end mode. The Master side can be configured, and the Slave side also needs to be configured. The mode is converted by Byte, and the conversion relationship is as follows:

Table 17-10 Big-Endian Data Mapping

Signal	Field	Mapping
LD	[15:8]	DATA[7:0]
	[7:0]	DATA[15:8]
LBE	[1]	DATA_BE[0]
	[0]	DATA_BE[1]
DP	[1]	DATA_DP[0]
	[0]	DATA_DP[1]

In big-endian mode, the BE is also big-endian mapped, and the DP signal is also calculated according to the mapped DATA.

In addition, CMD is arranged in big-end mode into: {burst_type,burst_length[6:0], 8 'h0}, and will also carry out big-end mapping, so CMD can be understood as 16-bit CMD data participating in bus transmission, only low 8bit effective.

17.3.4.2 Functions

Sampling Phase Adjustment

The LBC supports the Delay chain to adjust the phase of input clock domain data to ensure correct data sampling. There are 64 levels of Delay cells in the Delay chain. You can obtain the Delay of each level of Delay cell by calibrating the Delay chain.

The procedure for starting Delay chain calibration is as follows:

Step 1 Start the Local_bus module, start the localbus clock in the section 4.1 Clock Controller Unit (CCU), and release the lbc_rst_n and lbc_ahb_clk_en.

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Step 2 Start the lbc_clk clock in the section 4.1 Clock Controller Unit (CCU). This clock is the reference clock of the delay chain.

Step 3 Set the initial delay time. Write 0xA0 to SMP_DELAY_CTRL_REG, set the initial delay value 0x20, and write 0x00 to SMP_DELAY_CTRL_REG to clear the value.

Step 4 Write 0x8000 to SMP_DELAY_CTRL_REG to enable delay chain calibration.

Step 5 Wait until bit14 of SMP_DELAY_CTRL_REG is set to 1, indicating that the calibration is complete. At this time, bit8-14 can be used to read the number of delay cells.

Step 6 Calculate the delay cell time based on the clock cycle of the lbc_clk and the calibration result.

Transfer Instruction Mapping

The Local_bus transfer is based on internal instructions. After the three channels are transmitted, they are converted into independent LBC commands and arbitrated. After the arbitration, the authorized LBC commands can be issued and executed. The internal instructions are transformed based on AXI protocol information under direct access. In indirect access, the conversion is based on register configuration. Under DMA access, the conversion is based on the DMA descriptor.

Instructions are 64-bit and can be read back through registers during transmission. The standard instruction format is as follows:

Figure 17-17 LBC Command Format

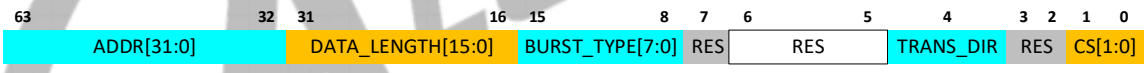

Table 17-11 LBC Command Format

CMD	Bits	Length	Description
ADDR	[63:32]	32	LBC Device address
DATA_LENGTH	[31:16]	16	Data Length. The value is a maximum of 512 bytes based on the actual number of data to be transmitted or the arbitration particle size.
BURST_TYPE	[15:8]	8	Burst type (FIX/INCR) with configurable length, ranges from 1 to 128.
TRANS_DIR	[4]	1	Direction (0 = read, 1 = write)
CS	[1:0]	2	Select LBC Device chip

- ADDR

ADDR indicates the address of the transmitted LBC Device. In the direct channel, ADDR is the address obtained from the AXI_ADDR of the AXI bus after the internal SADDR offset. In practice, due to the need for four CS mappings of AXI addresses, the address entry of the AXI bus is 1G according to chapter 2 Memory Map. Therefore, the entry address (AXI_ADDR) allocated to each CS is less than 1G (generally, one CS is allocated 256M entry address space). If the actual address

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(LBC_ADDR) space of the LBC Device is not in the assigned entry description, the SADDR of the corresponding CS can be configured. When AXI_ADDR + SADDR = LBC_ADDR is met, you can access any 1 GB address space.

For example, if the mapping address of CS1 is set to 256-511M and the actual location of CS1 is 768-1023M, the SADDR of CS1 can be set to 512M. In this case, CS1 can access 768-1023M. It should be noted that the bus bit width of the entry address is 32bit. Therefore, if the mapping address of the entry address of CS1 is 256-511M, and the actual location of the CS1 device is 0-255M, SADDR needs to be configured as 0xFFFF_FDAA based on the AXI entry address. Access to the 0-255M address space is achieved through 32bit address wrapping.

In the indirect path, ADDR is the LBC Device address configured through the register.

In the DMA access path, ADDR is the address indicated in the descriptor, located in the fourth word of the descriptor.

- DATA_LENGTH

Data transmission length: The minimum data transmission length supported by a single command is 1Byte and the maximum data transmission length is 512 bytes.

In the direct path, the data length transmitted by a single instruction is calculated according to the size and len of the AXI instruction, and the amount of data that the current AXI instruction wants to transmit is obtained. In this project, the maximum amount of data transmitted by a single AXI instruction supported is 64 bytes.

In the indirect path, the data length of a single command transmission is generated according to the configured indirect access data length and arbitration granularity of the indirect path. If the total length of indirect access data is less than the arbitration granularity, the data length of a single command transmission is the total length of indirect access data. If the total length of the indirect access data is greater than the arbitration granularity, the data length transmitted by a single command is the arbitration granularity length. Until the remaining data amount is less than the arbitration granularity length, the remaining data amount is transmitted.

In the DMA access path, similar to the indirect path, the granularity is determined by the length of the transmitted data configured by the descriptor and the DMA path.

- BURST TYPE

BURST_TYPE indicates the transmission type, which can be FIX_ADDR or INCR_ADDR. The BURST length can range from 1 to 128.

In indirect access, configure the source register; Under DMA access, configure from the DMA descriptor; Under direct access, it's different.

Under direct access, the BURST_TYPE signal can be generated in two ways. One is to automatically parse the amount of transmitted data through the bus protocol (AXI fix type converted to FIX_ADDR type; The types of AXI incr and wrap are converted to INCR_ADDR), and LBC BURST signals are generated according to the bit width of the target device (LBC transmission length first calculates the total number of bytes transmitted by AXI, and then truncates (divides) the total number of bytes according to the LBC transmission Data_width to obtain the LBC transmission length.) ; The other is to receive the configuration from the software. If the length of the data transmitted by the configured BURST_TYPE is greater than the length of

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the data represented by the AXI protocol, the BURST length indicated by the corresponding BURST_TYPE will be sent in full, and the LBE corresponding to the additional invalid data will be set to 0, representing the invalid data.

The configuration of BURST_TYPE needs to be flexibly configured according to the LBC Device. If the LBC_Device mode is LBC1, it can be automatically generated through the bus protocol. If the LBC Device is an LBC0, configure it according to the BURST_TYPE supported by the device and use the software configuration mode.

- TRANS DIR

In the direct access path, the transmission direction depends on whether the current AXI instruction is read or write. In the indirect access path, the transmission direction depends on the read and write direction configured in the register. In the DMA access path, the transmission direction depends on the read/write direction configured in the descriptor.

- CS

The CS supports flexible configuration based on the operating host, address interval, and data path.

The following three configurations are mutually exclusive.

  - Address mapping

LBC supports automatic CS mapping based on the transport address of the LBC Device. The upper and lower limits of each CS address mapping range can be configured. After the configuration, the transmission address is compared with the upper and lower limits of the address mapping range. (The address space allocated to the LBC Device is 1G. And the upper and lower limits of each CS address must not coincide, otherwise multiple CS will be hit), only the access between the upper and lower limits of the entry address can be initiated normally, otherwise it will be sent to CS0 by default;

  - USERID mapping

Determine whether the USERID configured for each channel is the same as the USERID incoming from the bus. According to the comparison result, if the user ID is the same, the corresponding channel number is output. In the USERID mapping mode, the DMA channel directly uses the CS configured by the descriptor. (The AHB channel can only be configured to access CS0, and the AXI and DMA channels can be configured freely);

  - Configure software

Each path supports the configuration of binding CS, and directly outputs the CS number configured by the software.

Concurrent Triple-Channel Transfers

LBC supports concurrent triple-channel transfers, and you can configure AXI direct access, CPU indirect access, and DMA access simultaneously. After the configuration, wait for transmission, but there is only one set of LBC bus exits, and only one path can be transmitted at a time, so it is necessary to arbitrate the triple-channel transfers. Supports three priority levels, corresponding

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to three channel. Note that if you want to modify the priority information, you can modify it only after all three channels are transmitted and LBC is idle.

Arbitration Mechanism

The arbitration mode of priority +RR supports RR with higher priorities and RR with the same priorities. For example, if the AXI and AHB channels are set to high priorities and the DMA channels are set to low priorities, the AXI and AHB channels use RR arbitration to decide the authorization status. The DMA can obtain the authorization only when neither AXI nor AHB requests it.

Arbitrated Particle Size

LBC transfer is based on the transfer instruction after mapping. The AXI channel is converted according to the AXI protocol. The maximum command length supported by the AXI channel is 64 bytes, and the maximum transmission data length configured by the AHB channel and the DMA channel is 65536 bytes. Therefore, in order to ensure reasonable arbitration waiting, the data length of the AHB and DMA channel is disassembled into multiple commands according to the arbitration granularity. Disassembly granularity supports 64/128/256/512bytes.

Dedicated DMA Transmission

Chain Transmission

Dedicated DMA uses a chain descriptor structure, through the register to configure the first descriptor address (descriptor can only be placed in DDR), hardware to obtain the descriptor. And then obtain data from the address recorded in the descriptor. When the handling is completed, complete the write back operation, and according to the first descriptor given by the next descriptor address, continue to obtain the descriptor. This is repeated until a terminator is found, ending a DMA move.

Figure 17-18 DMA Descriptor Format

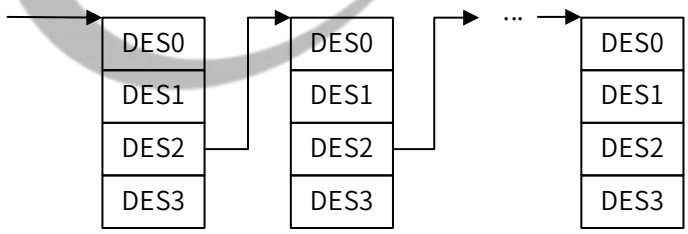

Table 17-12 DMA Descriptor0 Definition

Bits	Descriptor
31	DMA_DES_VLD 0: Current descriptor is not valid 1: Current descriptor is valid When not valid, DMA will return to IDLE and send an IRQ. Then it will stop until been restarted.

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Bits	Descriptor
30	DMA_WRITE_BACK_FLAG 0: not write back 1: write back When this bit is set high, DMA will write back and clean the WRITE_BACK_FLAG bit after this descriptor is done.
29	/
28	DMA_DIR DMA Write Process or Read Process 0: Read 1: Write This bit describes DMA direction, and the DMA direction here is opposite with LB direction in REGFILE.
27	DMA_FINISH_FLAG DMA Finish Flag
26:25	DMA Channel Select DMA CS
24	/
23:16	DMA_BURST_TYPE Burst type 00000000 - 01111111: Fix type,burst length from1 to 128Beat 10000000 - 11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.
15:0	IODL Input/Output Data Length CPU should tell the CTRL that in present address, how many bytes of data are going to write/read. 0-65535: Data = IODL + 1 IODL must be align with data width.

Table 17-13 DMA Descriptor1 Definition

Bits	Descriptor
31:0	DMA_Buffer_SADDR The starting address of data in DDR. Could be any 2word-align address.

Table 17-14 DMA Descriptor2 Definition

Bits	Descriptor
31:0	NEXT_DESCRIPTOR_ADDR These bits indicate the pointer to the physical memory where the next descriptor is present. Could be any 2word-align address.

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Table 17-15 DMA Descriptor3 Definition

Bits	Descriptor
31:0	TSA Target Starting Address Local bus starting address.

DMA trigger mode

DMA supports two trigger modes of transmission, one is register triggered read and write, the other is DRQ triggered read and write.

In regist-triggered read and write mode, after the software configuration of DMA transmission is enabled, the DMA host retrieves the descriptor and begins to send instructions, transport data, send instructions to participate in arbitration, and start data interaction with the LBC device.

In DRQ triggered read and write mode, after DMA transmission is enabled in the software, the DMA host retrieves the descriptor and sends commands to move data. However, the sent commands cannot participate in arbitration and will not be executed until the LBC Device initiates DRQ. One DRQ of an external device corresponds to one descriptor. If a new DRQ is pulled up before the execution of the current descriptor ends, an invalid DRQ will be tried.

Timeout detection

In the transmission process, if the downstream device does not return data or accept data, the timeout detection is performed. If timeout is triggered, the lbc_master terminates the transmission of the current command and returns the corresponding timeout interrupt according to the current command delivery path. The CPU can read the LBC_*_TO_CMD_REG of the corresponding path to obtain information about the timeout stuck instruction. If the current command is AXI Direct Access, reset the AXI direct access path or clr operation can be performed only after Direct Access completes the AXI protocol. Otherwise, the CPU will hang. If you access the AHB or DMA channel, you can reset the corresponding channel directly.

After a path triggers timeout, unless the current path is reset (or the *_TIMEOUT_CLR signal of the corresponding path is configured), all instructions of the current path cannot be sent, and the transmission of other paths is not affected.

In the process of protocol completion of the AXI channel, the AXI channel will neither receive any new instructions from the CPU, nor send any arbitration requests downstream. And the AXI channel will complete all the instructions that have been backlog in the LBC falsely (write instruction data is discarded, read instruction data is filled with 0). The CPU needs to determine that the AXI protocol is completed according to the waterline of the OT fifo on the AXI channel. In this case, you can restart the AXI channel in either of the following ways:

1. Reset the AXI channel.
2. Clear all FIFOs on the channel. After clearing the FIFO, set the da_timeout_clr signal to 1 and read back the signal until it becomes 1. (Since AXI's instruction and data FIFO are still residual at this time, neither AXI channel should be allowed to send an arbitration request to downstream until the FIFO is cleared.)

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System DMA Transmission

AHB indirect access supports transmission through the system DMA. To use the system DMA for transmission, you need to configure related signals in the Local bus first. After configuring the transfer information indirectly accessed by AHB, enable the system DMA mode for LBC, and then configure the FIFO trigger waterline for TX and RX to match the amount of data to be moved by the DRQ in the system DMA.

After configuring the LBC, configure the system DMA, configure the system DMA descriptor according to the transmission information configured in the LBC, and enable the system DMA transmission.

Note that whether the system DMA is used or the CPU is transferred through indirect access, the operation host of the next indirect access must be switched after the current AHB indirect access is complete.

Resetting each channel independently

The Local Bus supports independent resetting of each channel. Resetting any channel does not affect transmission of other channels. However, it should be noted that due to the need to ensure the protocol integrity of the AXI channel, the reset needs to ensure that the OT FIFO waterline of TX and RX on the AXI channel is 0 before the reset.

The reset structure is shown in the figure below:

Figure 17-19 LBC Reset Domain Diagram

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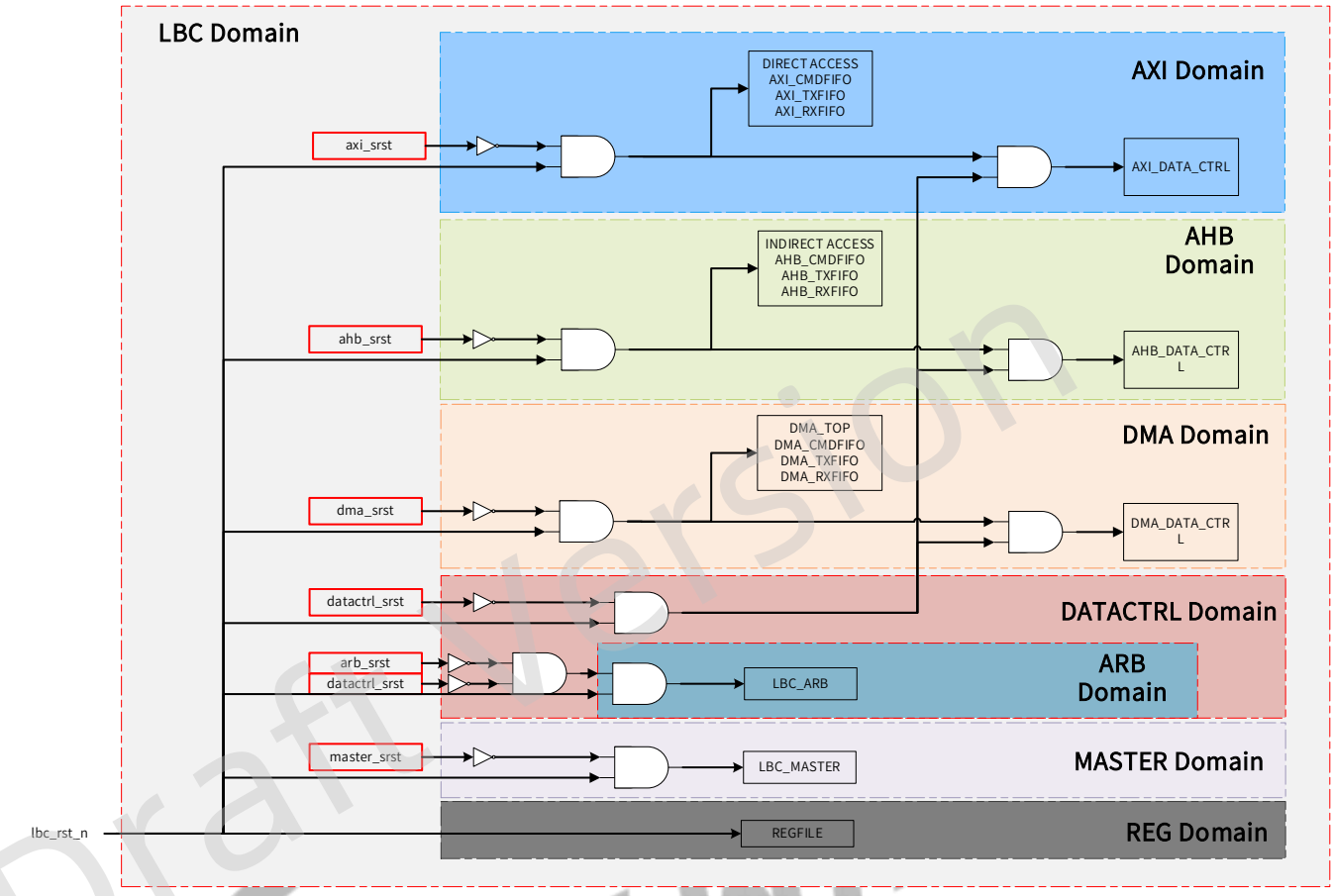
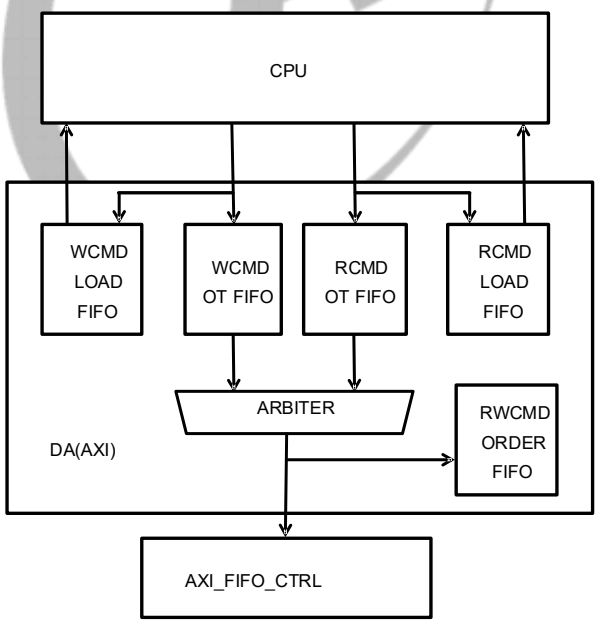


Direct Access Command FIFO

Figure 17-20 Direct Access FIFO Structure



There are multiple FIFOs within the direct access path. OT FIFO is used to cache AXI commands. Read and write commands are arbitrated and sent downstream. LOAD_FIFO is used to cache instruction information and return instruction completion information to the CPU according to the content of the LOAD FIFO when the instruction is completed. ORDER FIFO is used to store the

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order of the instructions to be sent. The FIFO is 1bit wide and contains the read/write direction of the instructions.

17.3.5 Programming Guidelines

17.3.5.1 Starting and Stopping the LBC

Starts LBC

To start the Local Bus, you need to configure the clock for starting the local bus and reset it.

Step 1 Ensure that the MBUS clock of the system is enabled.

Step 2 Configure the LBC_GAR_REG register in the CCU, release the LBC_RST_N, and enable the AHB clock function of the LBC module. (The AHB clock function here is also the MBUS clock function, because the AHB clock in the LBC module is the same clock as the MBUS clock.)

Step 3 Configure the LBC_CLK_REG register in the CCU and configure the LBC_CLK register.

Step 4 Configure the LBC_CLK_REG register in Local_bus, and configure clock gating (CCU clock or external clock) according to the usage scenario.

Closes LBC

Step 1 Configure the LBC_GAR_REG register in the CCU, disable the LBC_RST_N, and disable the AHB clock function of the LBC module.

Step 2 Configure LBC_CLK_REG in local_bus and disable clock gating.

Step 3 Configure LBC_CLK_REG in the CCU and disable LBC_CLK.

17.3.5.2 Initial Configuration

Before carry out data transfer through LBC, you need to initialize the parameters according to the target device firstly, and configure the transmission control register of the corresponding channel after the general parameters are configured.

Protocol Timing Configuration

Registers 0x000-0x070 are LBC timing related registers. After the configuration is complete, the next transmission will generate the corresponding timing according to the configuration. Note that there are four sets of registers in this set, corresponding to four CS, so four CS are supported to send through different protocols. The specific definition of each slice selection parameter is as follows:

Table 17-16 LBC_MODE_CTRL_REG Definition

LBC_MODE_CTRL_REG (0x0000+0x0080*N(N=0-3))

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Name	Width	Description
ADDR_MUX_TYPE	2	Address multiplexing mode. It includes non-multiplexed mode, AD multiplexed mode and AAD multiplexed mode. It is used to indicate the multiplexing of address and data bus. In non-multiplexing mode, no address cycle is sent. In AD multiplexing mode, one address cycle is sent through the data bus. In AAD multiplexing mode, two address cycles are sent through the data bus.
TIME_MODE	2	Bus mode. Supports LBC1 mode, and LBC0 mode.
BUS_ENDIAN	1	Data size end mode, the size end can be adjusted only between Byte and Byte, and the default little end arrangement within a single Byte;
BUS_DATA_WIDTH	2	Data bus bit width. Supports 1byte, 2byte, 4byte
BUS_ADDR_OFFSET	1	Reserved
RD_SYNC_TYPE	1	Asynchronous/synchronous read transmission. Supports LBC0/LBC1 synchronization.
WR_SYNC_TYPE	1	Asynchronous/synchronous write transmission. Supports LBC0/LBC1 synchronization.

Table 17-17 LBC_TIMESCALE_CTRL_REG Definition

LBC_TIMESCALE_CTRL_REG (0x0004+0x0080*N(N=0-3))		
Name	Width	Description
TIME_GRANULARITY	2	Reserved
CLK_DIVIDER	2	Clock frequency division. The frequency relationship between the high frequency clock lbc_clk and the final output lbc_clk_o. LBC1 modes support frequency division configuration, and bus information is generated based on internal high frequency clock. In LBC0 mode, frequency division is not supported, and the output is based on lbc_clk.
CLK_DELAY_TIME	3	Clock phase delay. Supports phase matching of the frequency-division clock and source clock, and supports a maximum of eight source clock cycles.

Table 17-18 LBC_READY_CTRL_REG Definition

LBC_READY_CTRL_REG (0x0008+0x0080*N(N=0-3))		
Name	Width	Description
READY_TIMEOUT_TIME	16	READY timeout cycle. The READY signal of the LBC_DEVICE can be timed out. If the READY signal is not returned for a long time, the timeout is interrupted. Timeout is based on internal high frequency clock lbc_clk.

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LBC_READY_CTRL_REG (0x0008+0x0080*N(N=0-3))		
Name	Width	Description
RD_READY_EN	1	The read READY function is enabled. If the read READY function is enabled, the read data is valid. If the READY function is valid, the read data is valid (see RD_READY_MODE for the READY rule).
WR_READY_EN	1	Write READY is enabled. After the write READY function is enabled, data can be written based on the READY function. If the READY function is valid, data can be written (see WR_READY_MODE for the READY rule).
READY_DELAY_TIME	2	READY delay time. If READY is returned before the data, set the READY delay time to align with the data.
RD_READY_MODE	1	Read the READY rule, which can be configured to determine only the first batch of READY or every batch of READY in a single transmission.
WR_READY_MODE	1	Write the READY rule, which can be configured to determine only the first batch of READY or every batch of READY in a single transmission.
READY_POLARITY	1	READY polarity configuration, which can be set to high or low level valid.

Table 17-19 LBC_DP_CTRL_REG Definition

LBC_DP_CTRL_REG (0x000C+0x0080*N(N=0-3))		
Name	Width	Description
DP_MODE	1	Select the data parity check mode, which can be Byte by byte or global
DP_PARITY	1	DP check type, supporting odd check or even check.
DP_EN	1	Enable data parity.

Table 17-20 LBC_BE_CTRL_REG Definition

LBC_BE_CTRL_REG (0x0010+0x0080*N(N=0-3))		
Name	Width	Description
BE_PARITY_EN	1	The BE parity check mode is enabled to enable parity check between the BE signal and the data indicated by the BE.
DE_POLARITY	1	DE polarity configuration, can be configured as high or low level effective.
BE_POLARITY	1	BE polarity configuration, can be configured as high or low level effective.

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Table 17-21 LBC_CS_TIMING_CTRL_REG Definition

LBC_CS_TIMING_CTRL_REG (0x0018+0x0080*N(N=0~3))		
Name	Width	Description
CS_RDOFF_TIME	8	Read transmission CS invalid time.
CS_WROFF_TIME	8	Write transmission CS invalid time..
CS_ON_TIME	8	Transmission CS effective time.
CS_POLARITY	1	CS polarity configuration, can be configured as high or low level effective.

Table 17-22 LBC_ALE_TIMING0_CTRL_REG Definition

LBC_ALE_TIMING0_CTRL_REG (0x001C+0x0080*N(N=0~3))		
Name	Width	Description
ALE_RDOFF_TIME	8	Invalid read transmission address cycle.
ALE_WROFF_TIME	8	Invalid write transmission address cycle.
ALE_ON_TIME	8	Valid time of the transmission address cycle
ALE_POLARITY	1	ALE polarity configuration, can be configured as high or low level effective;

Table 17-23 LBC_ALE_TIMING1_CTRL_REG Definition

LBC_ALE_TIMING1_CTRL_REG (0x0020+0x0080*N(N=0~3))		
Name	Width	Description
ALE_AAD_RDOFF_TIME	8	In the AAD multiplexing mode of LBC1 mode, the second address cycle reads the invalid time of transmission address cycle.
ALE_AAD_WROFF_TIME	8	In the AAD multiplexing mode of LBC1 mode, the second address cycle writes the invalid time of transmission address cycle.
ALE_AAD_ON_TIME	8	In the AAD multiplexing mode of LBC1 mode, the second address cycle transmits the valid time of the address cycle.

Table 17-24 LBC_WE_TIMING_CTRL_REG Definition

LBC_WE_TIMING_CTRL_REG (0x0024+0x0080*N(N=0~3))		
Name	Width	Description
WE_OFF_TIME	8	WE invalid time
WE_ON_TIME	8	WE valid time
WE_POLARITY	1	WE polarity configuration, can be configured as high or low level effective

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Table 17-25 LBC_OE_TIMING0_CTRL_REG Definition

LBC_OE_TIMING0_CTRL_REG (0x0028+0x0080*N(N=0~3))		
Name	Width	Description
OE_OFF_TIME	8	OE invalid time
OE_ON_TIME	8	OE valid time
OE_POLARITY	1	OE polarity configuration, can be configured as high or low level effective

Table 17-26 LBC_OE_TIMING1_CTRL_REG Definition

LBC_OE_TIMING1_CTRL_REG (0x002C+0x0080*N(N=0~3))		
Name	Width	Description
OE _AAD_OFF_TIME	8	In AAD multiplexing mode of LBC1 mode, the second OE invalid moment is used for read transmission under AAD multiplexing.
OE_AAD_ON_TIME	8	In the AAD multiplexing mode of LBC1 mode, the second OE valid moment is used for read transmission under AAD multiplexing.

Table 17-27 LBC_CYCLE_TIMING_CTRL_REG Definition

LBC_CYCLE_TIMING_CTRL_REG (0x0034+0x0080*N(N=0~3))		
Name	Width	Description
RD_CYCLE_TIME	8	Read transmission duration. The transmission cycle depends on this time. If the advance timing ends, the CS will be pulled down ahead of time
WR_CYCLE_TIME	8	Write transmission duration. The transmission cycle depends on this time. If the advance timing ends, the CS will be pulled down ahead of time

Table 17-28 LBC_ACCESS_TIMING_CTRL_REG Definition

LBC_ACCESS_TIMING_CTRL_REG (0x0038+0x0080*N(N=0~3))		
Name	Width	Description
RD_ACCESS_TIME	8	Read data sampling time. If the read data sampling time is not ready, this parameter needs to be configured because the slave cannot return data immediately.
WR_ACCESS_TIME	8	Write data sampling time

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Table 17-29 LBC_PAGE_TIMING_CTRL_REG Definition

LBC_PAGE_TIMING_CTRL_REG (0x003C+0x0080*N(N=0~3))		
Name	Width	Description
PAGE_ACCESS_TIME	8	The number of data cycle

Table 17-30 LBC_DATAMUX_TIMING_CTRL_REG Definition

LBC_DATAMUX_TIMING_CTRL_REG (0x0040+0x0080*N(N=0~3))		
Name	Width	Description
WR_DATA_ON_TIME	8	Address duration. The address can keep multiple cycles. Even if ALE has been invalid, the address is still valid. After this time, the multiplexed bus will become the data bus; In LBC mode, it can be configured as the address validity time.

Table 17-31 LBC_CLE_TIMING0_CTRL_REG Definition

LBC_CLE_TIMING0_CTRL_REG (0x0044+0x0080*N(N=0~3))		
Name	Width	Description
CLE_OFF_TIME	8	CLE invalid time
CLE_ON_TIME	8	CLE valid time
CLE_MODE	1	CLE mode. In single mode, it can be configured to send or not send CLE cycles.
CLE_POLARITY	1	CLE polarity configuration, can be configured as high or low level effective.

Table 17-32 LBC_TURNAROUND_CTRL_REG Definition

LBC_TURNAROUND _CTRL_REG (0x0060+0x0080*N(N=0~3))		
Name	Width	Description
TURNAROUND_TIME	8	Protocol recovery time. After the end of the transmission, expand the number of clock cycles sent to ensure that the Slave end of the data transmission.

The following are LBC1 16bit, Burst2, instruction cycle 2, address cycle 3, and data cycle 2 for configuration.

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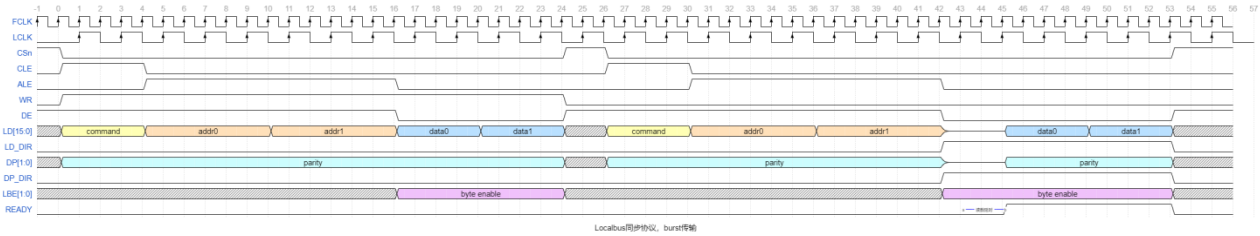
Figure 17-21 LBC1 Sequence Diagram



Step 1 Set ADDR_MUX_TYPE to 10 and send two IP addresses.

Step 2 Set TIME_MODE to 01, which indicates LBC mode 1.

Step 3 Set BUS_ENDIAN to 1 and the little-end mode.

Step 4 Set BUS_DATA_WIDTH to 01,2 bytes.

Step 5 Set RD_SYNC_TYPE and WR_SYNC_TYPE to 1, indicating synchronous transmission.

Step 6 Set CLK_DIVIDER to 01, indicating that the internal clock frequency division ratio is 2.

Step 7 Set CLK_DELAY_TIME to 0, indicating that no delay is required.

Step 8 Configure READY_TIMEOUT_TIME as required.

Step 9 If RD_READY_EN is set to 1, the ready signal needs to be determined.

Step 10 If WR_READY_EN is set to 0, the write transmission signal is not ready.

Step 11 Set READY_DELAY_TIME based on the SLAVE condition (READY and DATA delay or not).

Step 12 If RD_READY_MODE is set to 1, the READY signal is detected for each snapshot.

Step 13 If WR_READY_MODE is set to 1 (if WR_READY_EN is set to 0, the signal is invalid), the READY signal is detected for each snapshot.

Step 14 Set READY_POLARITY to 1: The polarity is high.

Step 15 Set DP_MODE to 1 for global verification.

Step 16 Configure DP_PARITY parity parity check as required.

Step 17 Set DP_EN to 1 and enable parity check.

Step 18 Configure BE_PARITY and enable the BE check.

Step 19 Set DE_POLARITY to 0 to enable low polarity.

Step 20 Set BE_POLARITY to 1 to enable high polarity.

Step 21 Set CS_RD_OFF_TIME to 23, which indicates that the CS fails at the 23th clock cycle during read transmission.

Step 22 Set CS_WR_OFF_TIME to 20, which indicates that the CS fails at the 20th clock cycle during write transmission.

Step 23 If CS_ON_TIME is set to 0, CS takes effect at cycle 0.

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Step 24 Set CS_POLARITY to 0, which means that a low level is active in ADD mode.

Step 25 Set ALE_ADD_ON_TIME to 4, which indicates that the 4th cycle of the first address is valid in ADD mode.

Step 26 Set ALE_ADD_RDOFF_TIME to 10, which indicates that the 10th cycle of the first address is invalid during read transmission in ADD mode.

Step 27 Set ALE_ADD_WDOFF_TIME to 10, which indicates that the 10th cycle of the first address is invalid during write transmission in ADD mode.

Step 28 Set ALE_RDOFF_TIME to 16, which indicates that ALE fails at the 16th cycle during read transmission in ADD mode.

Step 29 Set ALE_WROFF_TIME to 16, indicating that ALE becomes invalid at the 16th cycle during write transmission in ADD mode.

Step 30 Set ALE_ON_TIME to 10, indicating that ALE takes effect at the 10th cycle in ADD mode.

Step 31 Set ALE_POLARITY to 1: The polarity is active at a high level.

Step 32 Set WE_OFF_TIME to 20, which indicates that WE fails at the 20th cycle during write transmission. But in burst mode, it will add PAGE_ACCESS_TIME timing, so it will expand to the 24th cycle invalid.

Step 33 Set WE_ON_TIME to 0, which means that WE takes effect at the 0th cycle of write transmission.

Step 34 Set WE_POLARITY to 1: The polarity is active at a high level.

Step 35 If CLE_OFF_TIME is set to 4, it indicates that CLE fails at the fourth cycle.

Step 36 If CLE_ON_TIME is set to 0, it indicates that CLE takes effect at cycle 0.

Step 37 Set CLE_POLARITY to 1: A high level is active.

Step 38 Set PAGE_ACCESS_TIME to 4, which indicates that each data lasts 4 cycles.

Step 39 Set RD_CYCLE_TIME to 20, indicating that the read transmission cycle is 20 cycles. But in burst mode, it will add PAGE_ACCESS_TIME timing, and the timing will stop untill the slave return ready signal after 3 cycles, and so it will expand to the 27th cycle invalid.

Step 40 Set WR_CYCLE_TIME to 20, indicating that the write transmission cycle is 20 cycles. But in burst mode, it will add PAGE_ACCESS_TIME timing, so it will expand to the 24th cycle invalid.

Step 41 Set TURNAROUND_TIME based on the delay between Local_bus and LBC Device.

Other parameters not mentioned are registered as other mode parameters and do not need to be configured under LBC1.

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Chip Selection Configuration

Registers 0x200-0x254, which are CS-related registers, can be configured to flexibly generate and configure CS. CS can be generated by operating host, address interval, data path, and operating host and address interval.

The configuration process is as follows:

Operating a CPU to Generate CS:

Step 1 Set CH_MODE_CTRL_REG to 0x00010000.

Step 2 Configure USER_CS_REG and configure the operable host of the corresponding CS. AHB can access CS0 only, and AXI and DMA can access any of them. In this mode, you do not need to configure USER_CS_REG for DMA.

Generate CS through an Address Interval:

Step 1 Set CH_MODE_CTRL_REG to 0x00000001.

Step 2 Configure CS*_DA_HADDR_REG and CS*_DA_LADDR_REG. Set the address range of each CS in the direct path. LADDR is the low address and HADDR is the high address.

Step 3 Configure CS*_IDA_HADDR_REG and CS*_IDA_LADDR_REG. Configure the address interval of each CS in the indirect path (DMA shares the CS address interval of the indirect path). LADDR is the low address and HADDR is the high address.

Step 4 Configure DA_CS*_SADDR_REG. Configure the start address of the direct path.

Generate CS through a data Path:

Step 1 Set CH_MODE_CTRL_REG to 0x00000100.

Step 2 Configure FIX_CS_REG to set the CS for each path.

Configure the Arbitration Priority

Configure LBC_ARB_PRI_CFG_REG to configure the priorities of three channels. Each channel can be configured as any priority level. The channel with a higher priority is transmitted preferentially. If the priorities are the same, authorization is performed according to the RR polling arbitration mechanism.

Interrupt Configuration

Step 1 Configure GIC (configure interrupt mask) to ensure that LBC interrupts can be reported to the CPU.

Step 2 Configure LBC_INT*_EN_REG to enable the corresponding interrupt function. If the specified interrupt condition is met, an interrupt is generated.

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Step 3 After entering the interrupt thread, write LBC_INT*_PENDING_REG to clear the interrupt pending process and process the interrupt process.

17.3.5.3 Direct CPU Access

Step 1 Configure DA_BST_SEL_REG to select the BURST signal generation mode of different CS.

Step 2 Configure DA_BST_TYPE_REG and select the BURST type generated by the software.

Step 3 The CPU initiates the read or write transfer through AXI and checks the transfer status through the interrupt or DA_INTF_STA_REG or DA_FIFO_STA_REG registers.

17.3.5.4 Indirect CPU Access

Step 1 Configure IDA_FIFO_TRIGGER_LVL_REG. Configure the fifo fill/read threshold for AHB channel data. (After the configuration, if the TX channel data volume is lower than the threshold, interruption will be triggered to remind the CPU to fill data. If the RX channel data volume exceeds the threshold, an interrupt will be triggered to remind the CPU to read the data.

Step 2 Configure IDA_TS_ADDR_REG. Configure the start transmission address.

Step 3 Set IDA_TS_DATA_LEN_REG to set the length of the data to be transmitted.

Step 4 Configure IDA_TS_CTRL_REG to configure transmission control signals (After configuring other parameters, configure bit0 to enable transmission).

Step 5 Clear bit[31:30] in LBC_INT0_PENDING_REG to clear the interrupt flag.

Step 6 Open bit[31:30] in LBC_INT0_EN_REG to enable interrupt.

Step 7 Write data to IDA_WDATA_REG to send data to the device. Initiate a read operation on IDA_RDATA_REG to read data back from the device (use triglvl to ensure that data conditions in the data fifo are met before reading or writing).

Step 8 After the data length is sufficient, configure bit[31:30] in LBC_INT0_EN_REG to disable interrupt.

17.3.5.5 Dedicated DMA access

Step 1 Select a piece of address space configuration descriptor in DDR, and a single descriptor is 4word long;

Step 2 Configure DMA_DES_ADDR_REG to configure the start address of the descriptor.

Step 3 Configure DMA_MODE_CFG_REG to configure transmission control signals. (If external drq triggers are supported, set bit1 to 1. After configuring other parameters, configure bit0 to enable transmission).

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17.3.5.6 LBC Device Timeout

If timeout is triggered during transmission, first check the LBC_INT*_PENDING_REG register to determine which channel triggers timeout to pull the interrupt.

AXI Channel Timeout

Step 1 Read LBC_DA_TO_CMD*_REG back to obtain information about the Timeout command.

Step 2 Read back DA_INTF_STA_REG. If DA_RX_OT_FIFO_WL and DA_TX_OT_FIFO_WL are both 0, it indicates that the AXI path protocol is completed.

Step 3 Release the AXI channel in two ways:

a) Reset AXI channel, configure bit0 of LBC_SOFT_RST_REG, reset AXI channel, after reset, AXI channel can be used again;

b) Configure DA_*_CLR in LBC_FIFO_CLR_REG, and confirm to clear all fifo states of AXI channel; Then set bit0 of LBC_TIMEOUT_CLR_REG to 1 and read it back. When the bit becomes 0, the AXI path can be used again.

AHB Path Timeout

Step 1 Read LBC_IDA_TO_CMD*_REG back to obtain information about the Timeout command.

Step 2 Release the AHB channel in either of the following ways:

a) Reset the AHB path, configure bit1 of LBC_SOFT_RST_REG, reset the AHB path, and after the reset, the AHB path can be used again;

b) Configure IDA_*_REG in LBC_FIFO_CLR_REG and clear all fifo states in the AHB channel. Then set bit1 of LBC_TIMEOUT_CLR_REG to 1 and read it back. When the bit changes to 0, the AHB path can be used again.

DMA Path Timeout

Step 1 Read LBC_DMA_TO_CMD*_REG back to obtain information about the Timeout command.

Step 2 Release the DMA channel in two ways:

a) Reset the DMA channel, configure bit2 of LBC_SOFT_RST_REG, reset the DMA channel, and after the reset, the DMA channel can be used again;

b) Configure DMA_*_REG in LBC_FIFO_CLR_REG, and confirm to clear all fifo states in the DMA channel; Then set bit2 of LBC_TIMEOUT_CLR_REG to 1 and read it back. When the bit becomes 0, the DMA channel can be used again.

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17.3.5.7 System DMA Transmission

Step 1 The CPU indirect path supports system DMA for transport. Both LBC and system DMA need to be configured.

Step 2 Configure IDA_FIFO_TRIGGER_LVL_REG. Configure the FIFO fill/read threshold for AHB channel data. (After the configuration, if the TX channel data volume is lower than the threshold, a DRQ is initiated to remind the DMA fill data. When the RX channel data volume exceeds the threshold, an interrupt is triggered to remind DMA to read data. It should be noted that triglvl is properly configured to avoid the read and write operation that DMA drq will directly cause overflow to fifo.)

Step 3 Configure IDA_TS_ADDR_REG. Configure the start transmission address.

Step 4 Configure IDA_TS_DATA_LEN_REG to set the length of the data to be transmitted. The data length configured by the system DMA descriptor is the same as the data length.

Step 5 Configure IDA_TS_CTRL_REG. Configure transmission signals, enable IDA_DMA_ON, and then configure IDA_TS_START to enable transmission.

Step 6 Configure the system DMA. For details, see section 12.1 DMA Controller (DMAC).

17.3.6 Cautions and notes

1. In the address mapping, the upper and lower limits of CS addresses configured cannot coincide. Otherwise, multiple CS will be matched.
2. The AXI direct access path triggers a timeout. The AXI path RESET or CLR operation can be performed only after the AXI protocol is completed. Otherwise, the CPU will panic.
3. In User ID mapping mode, the AHB path can access CS0 only, while the AXI path and the DMA path can be flexibly accessed according to the configuration.
4. After the CPU indirect access mode transfer starts, you need to manually clear all the interrupt pending.
5. Because the AHB indirect access path has no back pressure, it is necessary to pay attention to fifo read and write overflow, which will report interruption after overflow.
6. The CPU indirectly accesses the TX direction without limitation, and the RX direction requires data length word alignment.
7. The DMA descriptor address needs to be aligned with 2Word, the DMA data address needs to be aligned with 2Word, and the data length needs to be aligned with lbc_data_width.
8. If the arbitration threshold of the DMA channel is set to 512Byte, the DMA data address must be 64 bytes aligned for transmission. When the arbitration threshold is 64/128/256 bytes, there is no alignment limit.

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17.3.7 Register List

Module Name	Base Address
LOCAL_BUS_CTRL	0x02810000

Register Name	Offset	Description
LBC_MODE_CTRL_REG	0x0000+0x0080*N(N=0-3)	LocalBus Mode Ctrl Register
LBC_TIMESCALE_CTRL_REG	0x0004+0x0080*N(N=0-3)	LocalBus Timescale Ctrl Register
LBC_READY_CTRL_REG	0x0008+0x0080*N(N=0-3)	LocalBus Ready Ctrl Register
LBC_DP_CTRL_REG	0x000C+0x0080*N(N=0-3)	LocalBus Data Parity Ctrl Register
LBC_BE_CTRL_REG	0x0010+0x0080*N(N=0-3)	LocalBus Byte Enable Ctrl Register
LBC_CS_TIMING_CTRL_REG	0x0018+0x0080*N(N=0-3)	LocalBus CS Timing Ctrl Register
LBC_ALE_TIMING0_CTRL_REG	0x001C+0x0080*N(N=0-3)	LocalBus ALE Timing0 Ctrl Register
LBC_ALE_TIMING1_CTRL_REG	0x0020+0x0080*N(N=0-3)	LocalBus ALE Timing1 Ctrl Register
LBC_WE_TIMING_CTRL_REG	0x0024+0x0080*N(N=0-3)	LocalBus WE Timing Ctrl Register
LBC_OE_TIMING0_CTRL_REG	0x0028+0x0080*N(N=0-3)	LocalBus OE Timing0 Ctrl Register
LBC_OE_TIMING1_CTRL_REG	0x002C+0x0080*N(N=0-3)	LocalBus OE Timing1 Ctrl Register
LBC_CYCLE_TIMING _CTRL_REG	0x0034+0x0080*N(N=0-3)	LocalBus Cycle Timing Ctrl Register
LBC_ACCESS _TIMING _CTRL_REG	0x0038+0x0080*N(N=0-3)	LocalBus Access Timing Ctrl Register
LBC_PAGE_TIMING _CTRL_REG	0x003C+0x0080*N(N=0-3)	LocalBus Page Timing Ctrl Register
LBC_DATAMUX_TIMING _CTRL_REG	0x0040+0x0080*N(N=0-3)	LocalBus Data Mux Timing Ctrl Register
LBC_CLE_TIMING_CTRL_REG	0x0044+0x0080*N(N=0-3)	LocalBus CLE Timing Ctrl Register
LBC_TURNAROUND_CTRL_REG	0x0060+0x0080*N(N=0-3)	LocalBus Turnaround Ctrl Register
CH_MODE_CTRL_REG	0x0200	Channel Mode Ctrl Register
CS0_DA_HADDR_REG	0x0210	Channel0 DA High Address Register

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Register Name	Offset	Description
CS0_DA_LADDR_REG	0x0214	Channel0 DA Low Address Register
CS0_IDA_HADDR_REG	0x0218	Channel0 IDA High Address Register
CS0_IDA_LADDR_REG	0x021C	Channel0 IDA Low Address Register
CS1_DA_HADDR_REG	0x0220	Channel1 DA High Address Register
CS1_DA_LADDR_REG	0x0224	Channel1 DA Low Address Register
CS1_IDA_HADDR_REG	0x0228	Channel1 IDA High Address Register
CS1_IDA_LADDR_REG	0x022C	Channel1 IDA Low Address Register
CS2_DA_HADDR_REG	0x0230	Channel2 DA High Address Register
CS2_DA_LADDR_REG	0x0234	Channel2 DA Low Address Register
CS2_IDA_HADDR_REG	0x0238	Channel2 IDA High Address Register
CS2_IDA_LADDR_REG	0x023C	Channel2 IDA Low Address Register
CS3_DA_HADDR_REG	0x0240	Channel3 DA High Address Register
CS3_DA_LADDR_REG	0x0244	Channel3 DA Low Address Register
CS3_IDA_HADDR_REG	0x0248	Channel3 IDA High Address Register
CS3_IDA_LADDR_REG	0x024C	Channel3 IDA Low Address Register
FIX_CS_REG	0x0250	Channel Fix Select Register
USER_CS_REG	0x0254	Channel User Select Register
LBC_CLK_REG	0x0280	LBC Clock Register
LBC_SOFT_RST_REG	0x0288	LBC Soft Reset Register
SMP_DELAY_CTRL_REG	0x0290	Sample Delay Control Register
IDA_TS_CTRL_REG	0x0300	Indirect Access Transfer Control Register
IDA_TS_ADDR_REG	0x0304	Indirect Access Transfer Address Register
IDA_TS_DATA_LEN_REG	0x0308	Indirect Access Transfer Length Register
IDA_WDATA_REG	0x0340	Indirect Access Write Data Register

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Register Name	Offset	Description
IDA_RDATA_REG	0x0350	Indirect Access Read Data Register
DA_BST_SEL_REG	0x0400	Direct Access Burst Select Register
DA_BST_TYPE_REG	0x0404	Direct Access Burst Type Register
DA_CS0_SADDR_REG	0x0410	Direct Access CS0 Start Address Register
DA_CS1_SADDR_REG	0x0414	Direct Access CS1 Start Address Register
DA_CS2_SADDR_REG	0x0418	Direct Access CS2 Start Address Register
DA_CS3_SADDR_REG	0x041C	Direct Access CS3 Start Address Register
DA_RX_TO_FIX_DONE_REG	0x0420	Direct Access RX Timeout Fix Done Register
LBC_TIMEOUT_CLR_REG	0x0424	LBC Timeout Clear Register
LBC_ARB_PRI_CFG_REG	0x0440	LocalBus Arbiter Priority Register
DMA_MODE_CFG_REG	0x0460	DMA Mode Config Register
DMA_DES_ADDR_REG	0x0464	DMA Descriptor Start Address Register
DMA_DES0_DBG_REG	0x04D0	DMA Descriptor0 Debug Register
DMA_DES1_DBG_REG	0x04D4	DMA Descriptor1 Debug Register
DMA_DES2_DBG_REG	0x04D8	DMA Descriptor2 Debug Register
DMA_DES3_DBG_REG	0x04DC	DMA Descriptor3 Debug Register
LBC_INT0_EN_REG	0x0500	LocalBus Interrupt0 Enable Register
LBC_INT1_EN_REG	0x0504	LocalBus Interrupt1 Enable Register
LBC_INT0_PENDING_REG	0x0508	LocalBus Interrupt0 Pending Register
LBC_INT1_PENDING_REG	0x050C	LocalBus Interrupt1 Pending Register
DA_INTF_STA_REG	0x0510	Direct Access Interface Status Register
DA_FIFO_STA_REG	0x0514	Direct Access FIFO Status Register
IDA_FIFO_STA_REG	0x0518	Indirect Access FIFO Status Register
DMA_FIFO_STA_REG	0x051C	DMA FIFO Status Register
DA_FIFO_TRIGGER_LVL_REG	0x0524	DA FIFO Trigger Level Register
IDA_FIFO_TRIGGER_LVL_REG	0x0528	IDA FIFO Trigger Level Register
DMA_FIFO_TRIGGER_LVL_REG	0x052C	DMA FIFO Trigger Level Register
LBC_FIFO_CLR_REG	0x0530	LBC FIFO Clear Register
LBC_IDA_MIRROR_REG	0x0600	LBC IDA MIRROR Register
LBC_DA_CMD0_REG	0x0604	LBC DA Command0 Register

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Register Name	Offset	Description
LBC_DA_CMD1_REG	0x0608	LBC DA Command1 Register
LBC_CMD_BUSY_REG	0x060C	LBC Command Busy Register
LBC_CMD0_REG	0x0610	LBC Command0 Register
LBC_CMD1_REG	0x0614	LBC Command1 Register
LBC_CMD2_REG	0x0618	LBC Command2 Register
LBC_DA_TO_CMD0_REG	0x0620	LBC DA TIMEOUT Command0 Register
LBC_DA_TO_CMD1_REG	0x0624	LBC DA TIMEOUT Command1 Register
LBC_IDA_TO_CMD0_REG	0x0628	LBC IDA TIMEOUT Command0 Register
LBC_IDA_TO_CMD1_REG	0x062C	LBC IDA TIMEOUT Command1 Register
LBC_DMA_TO_CMD0_REG	0x0630	LBC DMA TIMEOUT Command0 Register
LBC_DMA_TO_CMD1_REG	0x0634	LBC DMA TIMEOUT Command1 Register
LBC_MASTER_DBG0_REG	0x0640	LBC Master Debug0 Signal Register
LBC_MASTER_DBG1_REG	0x0644	LBC Master Debug1 Signal Register

17.3.8 Register Description

17.3.8.1 0x0000+0x0080*N(N=0-3) LocalBus Mode Ctrl Register (Default Value: 0x1000_0000)

Offset: 0x0000+0x0080*N(N=0-3)			Register Name: LBC_MODE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	ADDR_MUX_TYPE Address multiplexing mode selection. 8-bit bus supports a maximum of 16 bits of address, 16-bit bus supports a maximum of 32 bits of address. 00: Non Mux 01: AD Mux 10: AAD Mux
27:26	/	/	/

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Offset: 0x0000+0x0080*N(N=0-3)			Register Name: LBC_MODE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
25:24	R/W	0x0	TIME_MODE Timing mode selection. Four pieces can be configured to different modes 00: LBC Mode0 01: LBC Mode1
23:17	/	/	/
16	R/W	0x0	BUS_ENDIAN 0: Big-endian 1: Little-endian
15:14	/	/	/
13:12	R/W	0x0	BUS_DATA_WIDTH 00:1byte 01:2byte 10:4byte
11:5	/	/	/
4	R/W	0x0	RD_SYNC_TYPE Select the read synchronous or asynchronous mode. LBC does not support the asynchronous mode. 0: Async 1: Sync
3:1	/	/	/
0	R/W	0x0	WR_SYNC_TYPE Select the write synchronous or asynchronous mode. LBC does not support the asynchronous mode. 0: Async 1: Sync

17.3.8.2 0x0004+0x0080*N(N=0-3) LocalBus Timescale Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0004+0x0080*N(N=0-3)			Register Name: LBC_TIMESCALE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:10	/	/	/

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Offset: 0x0004+0x0080*N(N=0-3)			Register Name: LBC_TIMESCALE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x0	CLK_DIVIDER Select the clock frequency division to configure the interface clock and internal function clock frequency division ratio 00: LBC clk = FCLK/1 01: LBC clk = FCLK/2 10: LBC clk = FCLK/4 11: LBC clk = FCLK/8
7:3	/	/	/
2:0	R/W	0x0	CLK_DELAY_TIME Configure the phase relationship between interface clock and function clock. If 0 is set, FCLK and interface clock phase are aligned. If 1 is set, FCLK clock phase is 1 FCLK cycle later than FCLK clock phase. The phase delay value cannot exceed the frequency division ratio (for example, when the frequency division ratio is 4, the delay does not exceed 3 FCLK); 000:0 FCLK Cycle 001:1 FCLK Cycle 010:2 FCLK Cycle 011:3 FCLK Cycle 100:4 FCLK Cycle 101:5 FCLK Cycle 110:6 FCLK Cycle 111:7 FCLK Cycle

17.3.8.3 0x0008+0x0080*N(N=0-3) LocalBus Ready Ctrl Register (Default Value: 0x0400_0000)

Offset: 0x0008+0x0080*N(N=0-3)			Register Name: LBC_READY_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x400	READY_TIMEOUT_TIME The unit is the number of FCLK cycles. When the counting threshold is reached, the ready timeout action is triggered. The timeout cycle ranges from 0 to 65535.
15:13	/	/	/
12	R/W	0x0	RD_READY_EN Enable the read transmission READY detection 0: Disable 1: Enable

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Offset: 0x0008+0x0080*N(N=0-3)			Register Name: LBC_READY_CTRL_REG
Bit	Read/Write	Default/Hex	Description
11:9	/	/	/
8	R/W	0x0	WR_READY_EN Enable write transmission READY detection. If write transmission READY detection is enabled, the Slave end needs to adjust the READY sampling phase; otherwise, write READY detection errors may occur due to the difference in interface propagation latency. 0: Disable 1: Enable
7:6	R/W	0x0	READY_DELAY_TIME Configure READY signal delay. Delay detection ready time, used for sampling phase fine-tuning. The unit is FCLK cycles; 00: Delay 0 cycle 01: Delay 1 cycle 10: Delay 2 cycle 11: Delay 3 cycle
5	R/W	0x0	RD_READY_MODE Select read transmission READY detection mode. 0: Detect ready at first trans 1: Detect ready every trans
4	R/W	0x0	WR_READY_MODE Select write transmission READY detection mode. 0: Detect ready at first trans 1: Detect ready every trans
3:1	/	/	/
0	R/W	0x0	READY_POLARITY 0: Active low 1: Active high

17.3.8.4 0x000C+0x0080*N(N=0-3) LocalBus Data Parity Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x000C+0x0080*N(N=0-3)			Register Name: LBC_DP_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	DP_MODE 0: One DP per byte 1: One DP only
11:9	/	/	/

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Offset: 0x000C+0x0080*N(N=0-3)			Register Name: LBC_DP_CTRL_REG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	DP_PARITY 0: Odd data parity 1: Even data parity
7:1	/	/	/
0	R/W	0x0	DP_EN 0: Disable 1: Enable

17.3.8.5 0x0010+0x0080*N(N=0-3) LocalBus Byte Enable Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0010+0x0080*N(N=0-3)			Register Name: LBC_BE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	BE_PARITY_EN Enable BE signals to participate in parity check. When the parity is enabled, LBE signals participate in data check. 0: Disable 1: Enable
15:9	/	/	/
8	R/W	0x0	DE_POLARITY 0: Active low 1: Active high
7:1	/	/	/
0	R/W	0x0	BE_POLARITY 0: Active low 1: Active high

17.3.8.6 0x0018+0x0080*N(N=0-3) LocalBus CS Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0018+0x0080*N(N=0-3)			Register Name: LBC_CS_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	CS_RDOFF_TIME The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.

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Offset: 0x0018+0x0080*N(N=0-3)			Register Name: LBC_CS_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
23:16	R/W	0x0	CS_WROFF_TIME The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.
15:8	R/W	0x0	CS_ON_TIME It can cooperate with the phase of the interface clock, so that the CS can be pulled down first, and then the timing of the interface clock can be output. The value ranges from 0 to 255.
7:1	/	/	/
0	R/W	0x0	CS_POLARITY 0: Active low 1: Active high

17.3.8.7 0x001C+0x0080*N(N=0-3) LocalBus ALE Timing0 Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x001C+0x0080*N(N=0-3)			Register Name: LBC_ALE_TIMING0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	ALE_RDOFF_TIME The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.
23:16	R/W	0x0	ALE_WROFF_TIME The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.
15:8	R/W	0x0	ALE_ON_TIME The valid time of the ALE signal. The ALE timing is calculated according to the presence of CLE signal in LBC mode 1; In AAD mode, it is the valid time of the second address cycle. The value ranges from 0 to 255.
7:1	/	/	/
0	R/W	0x0	ALE_POLARITY 0: Active low 1: Active high

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17.3.8.8 0x0020+0x0080*N(N=0-3) LocalBus ALE Timing1 Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0020+0x0080*N(N=0-3)			Register Name: LBC_ALE_TIMING1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	ALE_AAD_RDOFF_TIME ALE signal read transmission invalid time. The first address phase in AAD multiplexing mode is invalid. The unit is FCLK. This parameter is invalid in AD mode. The value ranges from 0 to 255.
23:16	R/W	0x0	ALE_AAD_WROFF_TIME ALE signal write transmission invalid time.The first address phase in AAD multiplexing mode is invalid. The unit is FCLK. This parameter is invalid in AD mode. The value ranges from 0 to 255.
15:8	R/W	0x0	ALE_AAD_ON_TIME ALE signal write valid time. The first address phase in AAD multiplexing mode is invalid. The unit is FCLK. This parameter is invalid in AD mode. The value ranges from 0 to 255.
7:0	/	/	/

17.3.8.9 0x0024+0x0080*N(N=0-3) LocalBus WE Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0024+0x0080*N(N=0-3)			Register Name: LBC_WE_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	WE_OFF_TIME WE signal invalid time. The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.
15:8	R/W	0x0	WE_ON_TIME WE signal valid time. The unit is FCLK. The value ranges from 0 to 255.
7:1	/	/	/
0	R/W	0x0	WE_POLARITY 0: Active low 1: Active high

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17.3.8.10 0x0028+0x0080*N(N=0-3) LocalBus OE Timing0 Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0028+0x0080*N(N=0-3)			Register Name: LBC_OE_TIMING0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	OE_OFF_TIME OE signal invalid time. The unit is FCLK, and the single transmission timing parameter is configured, and the Burst timing is automatically superimposed during Burst transmission. The value ranges from 0 to 255.
15:8	R/W	0x0	OE_ON_TIME OE signal valid time. The unit is FCLK. The value ranges from 0 to 255.
7:1	/	/	/
0	R/W	0x0	OE_POLARITY 0: Active low 1: Active high

17.3.8.11 0x002C+0x0080*N(N=0-3) LocalBus OE Timing1 Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x002C+0x0080*N(N=0-3)			Register Name: LBC_OE_TIMING1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	OE_AAD_OFF_TIME OE invalid time, expressed in FCLK. In AAD mode, it is the first invalid time. In AD mode, this parameter is invalid. The value ranges from 0 to 255.
15:8	R/W	0x0	OE_AAD_ON_TIME OE valid time, expressed in FCLK. In AAD mode, it is the first valid time. In AD mode, this parameter is invalid. The value ranges from 0 to 255.
7:0	/	/	/

17.3.8.12 0x0034+0x0080*N(N=0-3) LocalBus Cycle Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0034+0x0080*N(N=0-3)			Register Name: LBC_CYCLE_TIMING _CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0034+0x0080*N(N=0-3)			Register Name: LBC_CYCLE_TIMING _CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	RD_CYCLE_TIME Configuration for Read Transmission Total Cycle Time (timing configuration parameter during single transmission) Unit: FCLK, range: 0-65535 Automatically overlaied with burst timing during burst transmission.
15:0	R/W	0x0	WR_CYCLE_TIME Configuration for Write Transmission Total Cycle Time (timing configuration parameter during single transmission) Unit: FCLK, range: 0-65535 Automatically overlaied with burst timing during burst transmission.

17.3.8.13 0x0038+0x0080*N(N=0-3) LocalBus Access Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0038+0x0080*N(N=0-3)			Register Name: LBC_ACCESS _TIMING _CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	RD_ACCESS_TIME The first data sampling time of read transmission, expressed in FCLK. If the READY detection is enabled, the READY test starts at the sampling time and completes the first data sampling when the READY is effective. After the sampling time and the waiting time when the READY is invalid, the interface timing is paused and is not included in the timing configuration cycle. The sampling time ranges from 0 to 65535.
15:0	R/W	0x0	WR_ACCESS_TIME The first data sampling time of write transmission, expressed in FCLK; If the READY detection is enabled, the READY test starts at the sampling time and completes the first data sampling when the READY is effective. After the sampling time and the waiting time when the READY is invalid, the interface timing is paused and is not included in the timing configuration cycle. The sampling time ranges from 0 to 65535.

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17.3.8.14 0x003C+0x0080*N(N=0-3) LocalBus Page Timing Ctrl Register (Default Value: 0x0000_0001)

Offset: 0x003C+0x0080*N(N=0-3)			Register Name: LBC_PAGE_TIMING _CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x1	PAGE_ACCESS_TIME The duration of each Burst of data during burst transmission, expressed in FCLK. The duration ranges from 0 to 255.

17.3.8.15 0x0040+0x0080*N(N=0-3) LocalBus Data Mux Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0040+0x0080*N(N=0-3)			Register Name: LBC_DATAMUX_TIMING _CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0	WR_DATA_ON_TIME In AD/AAD write mode, the time when the first data is placed on the bus, expressed in FCLK. The transmission time ranges from 0 to 65535.

17.3.8.16 0x0044+0x0080*N(N=0-3) LocalBus CLE Timing Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0044+0x0080*N(N=0-3)			Register Name: LBC_CLE_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	CLE_OFF_TIME Invalid time of CLE signal. It is valid only in LBC mode 1, and the unit is FCLK; This parameter is configured according to the default CLE enable. when CLE is not transmitted in single mode, the timing is automatically matched. The value of invalid time ranges from 0 to 255.
15:8	R/W	0x0	CLE_ON_TIME Valid time of CLE signal. It is valid only in LBC mode 1. The unit is FCLK. This parameter is configured according to the default CLE enable. when CLE is not transmitted in single mode, the timing is automatically matched. When valid, the value ranges from 0 to 255.
7:5	/	/	/

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Offset: 0x0044+0x0080*N(N=0-3)			Register Name: LBC_CLE_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	CLE_MODE CLE signal mode selection. Single transmission also sends CLE signals when enabled. When disabled, the single transmission does not send CLE to improve transmission efficiency, but only sends CLE during BURST transmission. 0: Disable in single transfer 1: Enable in single transfer
3:1	/	/	/
0	R/W	0x0	CLE_POLARITY 0: Active low 1: Active high

17.3.8.17 0x0060+0x0080*N(N=0-3) LocalBus Turnaround Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0060+0x0080*N(N=0-3)			Register Name: LBC_TURNAROUND_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	TURNAROUND_TIME Configure bus turnaround time in FCLK. It is used to configure the interval between two transfers. The time ranges from 0 to 255.

17.3.8.18 0x0200 Channel Mode Ctrl Register (Default Value: 0x0000_0000)

Offset: 0x0200			Register Name: CH_MODE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	USER_SEL_EN Enable host ID mapping mode 0: Disable 1: Enable
15:9	/	/	/
8	R/W	0x0	FIX_SEL_EN Enable path fixed mapping mode 0: Disable 1: Enable
7:1	/	/	/

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Offset: 0x0200			Register Name: CH_MODE_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	ADDR_SEL_EN Enable address mapping mode 0: Disable 1: Enable

17.3.8.19 0x0210 Channel0 DA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0210			Register Name: CS0_DA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS0_DA_HADDR CS0 Upper limit of direct access address range

17.3.8.20 0x0214 Channel0 DA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x0214			Register Name: CS0_DA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS0_DA_LADDR CS0 Lower limit of direct access address range

17.3.8.21 0x0218 Channel0 IDA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0218			Register Name: CS0_IDA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS0_IDA_HADDR CS0 Upper limit of IDA/DMA address range

17.3.8.22 0x021C Channel0 IDA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x021C			Register Name: CS0_IDA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS0_IDA_LADDR CS0 Lower limit of IDA/DMA address range

17.3.8.23 0x0220 Channel1 DA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0220			Register Name: CS1_DA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS1_DA_HADDR CS1 Upper limit of direct access address range

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17.3.8.24 0x0224 Channel1 DA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x0224			Register Name: CS1_DA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS1_DA_LADDR CS1 Lower limit of direct access address range

17.3.8.25 0x0228 Channel1 IDA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0228			Register Name: CS1_IDA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS1_IDA_HADDR CS1 Upper limit of indirect/DMA access address range

17.3.8.26 0x022C Channel1 IDA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x022C			Register Name: CS1_IDA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS1_IDA_LADDR CS1 Lower limit of IDA/DMA address range

17.3.8.27 0x0230 Channel2 DA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0230			Register Name: CS2_DA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS2_DA_HADDR CS2 Upper limit of direct access address range

17.3.8.28 0x0234 Channel2 DA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x0234			Register Name: CS2_DA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS2_DA_LADDR CS2 Lower limit of direct access address range

17.3.8.29 0x0238 Channel2 IDA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0238			Register Name: CS2_IDA_HADDR_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0238			Register Name: CS2_IDA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS2_IDA_HADDR CS2 Upper limit of IDA/DMA address range

17.3.8.30 0x023C Channel2 IDA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x023C			Register Name: CS2_IDA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS2_IDA_LADDR CS2 Lower limit of IDA/DMA address range

17.3.8.31 0x0240 Channel3 DA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0240			Register Name: CS3_DA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS3_DA_HADDR CS3 Upper limit of direct access address range

17.3.8.32 0x0244 Channel3 DA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x0244			Register Name: CS3_DA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS3_DA_LADDR CS3 Lower limit of direct access address range

17.3.8.33 0x0248 Channel3 IDA High Address Register (Default Value: 0x0000_0000)

Offset: 0x0248			Register Name: CS3_IDA_HADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS3_IDA_HADDR CS3 Upper limit of IDA/DMA address range

17.3.8.34 0x024C Channel3 IDA Low Address Register (Default Value: 0x0000_0000)

Offset: 0x024C			Register Name: CS3_IDA_LADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	CS3_IDA_LADDR CS3 Lower limit of IDA/DMA address range

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17.3.8.35 0x0250 Channel Fix Select Register (Default Value: 0x0000_0000)

Offset: 0x0250			Register Name: FIX_CS_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17:16	R/W	0x0	DMA_FIX_CS CS selection of DMA in route fixed mapping mode 00: Fix to Channel0 01: Fix to Channel1 10: Fix to Channel2 11: Fix to Channel3
15:10	/	/	/
9:8	R/W	0x0	IDA_FIX_CS CS selection of indirect access in route fixed mapping mode 00: Fix to Channel0 01: Fix to Channel1 10: Fix to Channel2 11: Fix to Channel3
7:2	/	/	/
1:0	R/W	0x0	DA_FIX_CS CS selection of direct access in route fixed mapping mode 00: Fix to Channel0 01: Fix to Channel1 10: Fix to Channel2 11: Fix to Channel3

17.3.8.36 0x0254 Channel User Select Register (Default Value: 0x0000_0000)

Offset: 0x0254			Register Name: USER_CS_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:24	R/W	0x0	CS3_USER_SEL Configure CS3 binding to a host in host ID mapping mode 000: CPU0 001: CPU1 010: CPU2 011: CPU3
23:19	/	/	/

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Offset: 0x0254			Register Name: USER_CS_REG
Bit	Read/Write	Default/Hex	Description
18:16	R/W	0x0	CS2_USER_SEL Configure CS2 binding to a host in host ID mapping mode 000: CPU0 001: CPU1 010: CPU2 011: CPU3
15:11	/	/	/
10:8	R/W	0x0	CS1_USER_SEL Configure CS1 binding to a host in host ID mapping mode 000: CPU0 001: CPU1 010: CPU2 011: CPU3
7:3	/	/	/
2:0	R/W	0x0	CS0_USER_SEL Configure CS0 binding to a host in host ID mapping mode 000: CPU0 001: CPU1 010: CPU2 011: CPU3

17.3.8.37 0x0280 LBC Clock Register (Default Value: 0x0000_0000)

Offset: 0x0280			Register Name: LBC_CLK_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/

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Offset: 0x0280			Register Name: LBC_CLK_REG
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	LBC_CLK_MODE When the LBC clock mode is set to 0, the clock is turned off after the end of each transfer. The four chips can be configured with different clocks. If the Slave is too late to complete the sampling, the BUS_TURNAROUND_TIME can be increased to send more clock cycles. If the clock is set to 1, the clock is always on, and all four chip must use the same clock. The clock division ratio only uses the configuration parameter of chip options 0, and the clock delay configuration of four chip options is still valid. 0: Close when idle 1: Always on
15:4	/	/	/
3	R/W	0x0	LBC_RX_NEGATIVE_SEL Whether the LBC input signal uses negative clock sampling 0: Not Negative 1: Negative
2	R/W	0x0	LBC_TX_NEGATIVE_SEL Whether the LBC output signal is output using a negative clock. 0: Not Negative 1: Negative
1	R/W	0x0	LBC_FCLK_SEL Select LBC_FCLK high frequency clock source 0: From CCU 1: From GPIO
0	R/W	0x0	LBC_FCLK_GATE Enable LBC_FCLK High Frequency clock gating 0: Disable 1: Enable

17.3.8.38 0x0288 LBC Soft Reset Register (Default Value: 0x0000_0000)

Offset: 0x0288			Register Name: LBC_SOFT_RST_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/

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Offset: 0x0288			Register Name: LBC_SOFT_RST_REG
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	LBC_MASTER_SRST Soft reset LBC_MASTER 0: Not reset 1: Reset
4	R/W	0x0	LBC_DATACTRL_SRST Soft reset LocalBus data controller 0: Not reset 1: Reset
3	R/W	0x0	LBC_ARB_SRST Soft reset LocalBus arbitration 0: Not reset 1: Reset
2	R/W	0x0	DMA_SRST Soft reset DMA 0: Not reset 1: Reset
1	R/W	0x0	IDA_SRST Soft resetIDA 0: Not reset 1: Reset
0	R/W	0x0	DA_SRST Soft reset DA 0: Not reset 1: Reset

17.3.8.39 0x0290 Sample Delay Control Register (Default Value: 0x0000_2000)

Offset: 0x0290			Register Name: SMP_DELAY_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	SAMP_DL_BYPASS Enable Sample Delay Chain Bypass. In bypass mode, the delaychain is not used and the input signal is directly sampled. 0: No bypass 1: Bypass
15	R/W	0x0	SAMP_DL_CAL_START Sample Delay Calibration Start When set, start sample delay chain calibration. 0: Disable sample delay chain calibration. 1: Enable sample delay chain calibration.

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Offset: 0x0290			Register Name: SMP_DELAY_CTRL_REG
Bit	Read/Write	Default/Hex	Description
14	R	0x0	SAMP_DL_CAL_DONE Sample Delay Calibration Done When set, it means that sample delay chain calibration is done and the result of calibration is shown in SAMP_DL. 0: Sample delay chain calibration not done yet. 1: Sample delay chain calibration done.
13:8	R	0x20	SAMP_DL_CAL_VALUE Sample Delay It indicates the number of delay cells corresponding to current card clock. The delay time generated by these delay cells is equal to the cycle of clock nearly. Generally, it is necessary to do drive delay calibration when card clock is changed. This bit is valid only when SAMP_DL_CAL_DONE is set.
7	R/W	0x0	SAMP_DL_SW_EN Sample Delay Software Enable When set, enable sample delay specified at SAMP_DL_SW. 0: Disable sample delay chain function. 1: Enable sample delay chain function.
6	/	/	/
5:0	R/W	0x0	SAMP_DL_SW_VALUE Sample Delay Software The relative delay between clock line and command line, data lines. It can be determined according to the value of SAMP_DL_CAL_VALUE, the cycle of card clock and device's input timing requirement.

17.3.8.40 0x0300 Indirect Access Transfer Control Register (Default Value: 0x0000_0000)

Offset: 0x0300			Register Name: IDA_TS_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/

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Offset: 0x0300			Register Name: IDA_TS_CTRL_REG
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	IDA_BURST_ARB_GRAIN IDA BURST Arbitration Grain 00: 64 Byte 01: 128 Byte 10: 256 Byte 11: 512 Byte
15:8	R/W	0x0	IDA_BURST_TYPE 00000000 - 01111111: Fix type,burst length from1 to 128Beat 10000000 - 11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.
7:3	/	/	/
2	R/W	0x0	IDA_DMA_ON Enable IDA DMA 0: Off 1: On
1	R/W	0x0	IDA_TS_DIR IDA transfer direction 0: Read 1: Write
0	R/WAC	0x0	IDA_TS_START IDA transfer start 0: None 1: Start

17.3.8.41 0x0304 Indirect Access Transfer Address Register (Default Value: 0x0000_0000)

Offset: 0x0304			Register Name: IDA_TS_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	IDA_TS_ADDR IDA transder address

17.3.8.42 0x0308 Indirect Access Transfer Length Register (Default Value: 0x0000_0000)

Offset: 0x0308			Register Name: IDA_TS_DATA_LEN_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0308			Register Name: IDA_TS_DATA_LEN_REG
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	IDA_TS_DATA_LEN IDA transfer data length 0~65535: Data = IDA_TS_DATA_LEN + 1 Note: IDA_TS_DATA_LEN must be align with data width.

17.3.8.43 0x0340 Indirect Access Write Data Register (Default Value: 0x0000_0000)

Offset: 0x0340			Register Name: IDA_WDATA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0000_0000	IDA_WDATA_REG IDA write data

17.3.8.44 0x0350 Indirect Access Read Data Register (Default Value: 0x0000_0000)

Offset: 0x0350			Register Name: IDA_RDATA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0000_0000	IDA_RDATA_REG IDA read data

17.3.8.45 0x0400 Direct Access Burst Select Register (Default Value: 0x0000_0000)

Offset: 0x0400			Register Name: DA_BST_SEL_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	DA_CS3_BURST_SEL Select DA CS3 Burst mapping mode. When 0 is selected, it will be automatically converted to LBC type according to AXI transmission type; when 1 is selected, LBC type shall be subject to. 0: Config by axi 1: Config by soft
2	R/W	0x0	DA_CS2_BURST_SEL Select DA CS2 Burst mapping mode. When 0 is selected, it will be automatically converted to LBC type according to AXI transmission type; when 1 is selected, LBC type shall be subject to. 0: Config by axi 1: Config by soft

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Offset: 0x0400			Register Name: DA_BST_SEL_REG
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	DA_CS1_BURST_SEL Select DA CS1 Burst mapping mode. When 0 is selected, it will be automatically converted to LBC type according to AXI transmission type; when 1 is selected, LBC type shall be subject to. 0: Config by axi 1: Config by soft
0	R/W	0x0	DA_CS0_BURST_SEL Select DA CS0 Burst mapping mode. When 0 is selected, it will be automatically converted to LBC type according to AXI transmission type; when 1 is selected, LBC type shall be subject to. 0: Config by axi 1: Config by soft

17.3.8.46 0x0404 Direct Access Burst Type Register (Default Value: 0x0000_0000)

Offset: 0x0404			Register Name: DA_BST_TYPE_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	DA_CS3_BURST_TYPE 00000000-01111111: Fix type,burst length from1 to 128Beat 10000000-11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.
23:16	R/W	0x0	DA_CS2_BURST_TYPE 00000000-01111111: Fix type,burst length from1 to 128Beat 10000000-11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.
15:8	R/W	0x0	DA_CS1_BURST_TYPE 00000000-01111111: Fix type,burst length from1 to 128Beat 10000000-11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.

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Offset: 0x0404			Register Name: DA_BST_TYPE_REG
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x0	DA_CS0_BURST_TYPE 00000000-01111111: Fix type,burst length from1 to 128Beat 10000000-11111111: Incr type,burst_len 1-128Beat Note: The highest bit config burst type,0-fix,1-incr;lower 7 bit config burst_length.

17.3.8.47 0x0410 Direct Access CS0 Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0410			Register Name: DA_CS0_SADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DA_CS0_SADDR_REG Direct access CS0 start address

17.3.8.48 0x0414 Direct Access CS1 Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0414			Register Name: DA_CS1_SADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DA_CS1_SADDR_REG Direct access CS1 start address

17.3.8.49 0x0418 Direct Access CS2 Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0418			Register Name: DA_CS2_SADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DA_CS2_SADDR_REG Direct access CS2 start address

17.3.8.50 0x041C Direct Access CS3 Start Address Register (Default Value: 0x0000_0000)

Offset: 0x041C			Register Name: DA_CS3_SADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DA_CS3_SADDR_REG Direct access CS3 start address

17.3.8.51 0x0420 Direct Access RX Timeout Fix Done Register (Default Value: 0x0000_0000)

Offset: 0x0420			Register Name: DA_RX_TO_FIX_DONE_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0420			Register Name: DA_RX_TO_FIX_DONE_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W1C	0x0	DA_RX_TO_FIX_DONE_REG Direct Access RX Timeout Fix Done signal. When the signal becomes 1, RX access timeout transmission is completed.

17.3.8.52 0x0424 LBC Timeout Clear Register (Default Value: 0x0000_0000)

Offset: 0x0424			Register Name: LBC_TIMEOUT_CLR_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
2	R/WAC	0x0	DMA_TIMEOUT_CLR_REG DMA timeout clear signal
1	R/WAC	0x0	IDA_TIMEOUT_CLR_REG IDA timeout clear signal
0	R/WAC	0x0	DA_TIMEOUT_CLR_REG DA timeout clear signal

17.3.8.53 0x0440 LocalBus Arbiter Priority Register (Default Value: 0x0000_0000)

Offset: 0x0440			Register Name: LBC_ARB_PRI_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9:8	R/W	0x0	DMA_PRIORITY 10: High 01: Middle 00: Low
7:6	/	/	/
5:4	R/W	0x0	IDA_PRIORITY 10: High 01: Middle 00: Low
3:2	/	/	/
1:0	R/W	0x0	DA_PRIORITY 10: High 01: Middle 00: Low

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17.3.8.54 0x0460 DMA Mode Config Register (Default Value: 0x0000_0000)

Offset: 0x0460			Register Name: DMA_MODE_CFG_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:4	R/W	0x0	DMA_BURST_ARB_GRAIN DMA BURST arbitration grain 00: 64 Byte 01: 128 Byte 10: 256 Byte 11: 512 Byte
3:2	/	/	/
1	R/W	0x0	DMA_DRQ_EN Enable DMA DRQ 0: Disable 1: Enable
0	R/WAC	0x0	DMA_MODE_ON Turn on DMA mode 0: OFF 1: ON

17.3.8.55 0x0464 DMA Descriptor Start Address Register (Default Value: 0x0000_0000)

Offset: 0x0464			Register Name: DMA_DES_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	DMA_DES_SADDR Could be any 2word align address

17.3.8.56 0x04D0 DMA Descriptor0 Debug Register (Default Value: 0x0000_0000)

Offset: 0x04D0			Register Name: DMA_DES0_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_DES0 DMA descriptor Word0 read back register

17.3.8.57 0x04D4 DMA Descriptor1 Debug Register (Default Value: 0x0000_0000)

Offset: 0x04D4			Register Name: DMA_DES1_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_DES1 DMA descriptor Word1 read back register

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17.3.8.58 0x04D8 DMA Descriptor2 Debug Register (Default Value: 0x0000_0000)

Offset: 0x04D8			Register Name: DMA_DES2_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_DES2 DMA descriptor Word2 read back register

17.3.8.59 0x04DC DMA Descriptor3 Debug Register (Default Value: 0x0000_0000)

Offset: 0x04DC			Register Name: DMA_DES3_DBG_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_DES3 DMA descriptor Word3 read back register

17.3.8.60 0x0500 LocalBus Interrupt0 Enable Register (Default Value: 0x0000_0000)

Offset: 0x0500			Register Name: LBC_INT0_EN_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	LBC_IDA_RX_TRIG_INT_EN Enable localbus indirect access RX request 0x0 : Disable. 0x1 : Enable. Note: LocalBus Indirect Access RX Fifo Waterline Higher than Trigger Level Interrupt Enable
30	R/W	0x0	LBC_IDA_TX_TRIG_INT_EN Enable localbus indirect access TX request 0x0 : Disable. 0x1 : Enable. Note: LocalBus Indirect Access TX Fifo Waterline Lower than Trigger Level Interrupt Enable
29	R/W	0x0	LBC_DA_RX_TRIG_INT_EN Enable localbus direct access RX request 0x0 : Disable. 0x1 : Enable. Note: LocalBus Direct Access RX Fifo Waterline Higher than Trigger Level Interrupt Enable
28	R/W	0x0	LBC_DA_TX_TRIG_INT_EN Enable localbus direct access TX request 0x0 : Disable. 0x1 : Enable. Note: LocalBus Direct Access TX Fifo Waterline Lower than Trigger Level Interrupt Enable
27:8	/	/	/

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Offset: 0x0500			Register Name: LBC_INT0_EN_REG
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	LBC_DP_INT_EN Enable byte by byte parity interrupt 0x0 : Disable. 0x1 : Enable.
6	R/W	0x0	LBC_ODP_INT_EN Enable global parity interrupt 0x0 : Disable. 0x1 : Enable.
5	R/W	0x0	LBC_DMA_DES_INVLD_EN Enable invalid DMA descriptor interrupt 0x0 : Disable. 0x1 : Enable.
4	R/W	0x0	LBC_DMA_TRSF_DONE_INT_EN Enable DMA channel transfer completion interrupt 0x0 : Disable. 0x1 : Enable.
3	R/W	0x0	LBC_IDA_TRSF_DONE_INT_EN Enable IDA channel transfer completion interrupt 0x0 : Disable. 0x1 : Enable.
2	R/W	0x0	LBC_DA_TRSF_DONE_INT_EN Enable DA channel transfer completion interrupt 0x0 : Disable. 0x1 : Enable.
1	R/W	0x0	LBC_READY_TIMEOUT_INT_EN Enable READY timeout interrupt 0x0 : Disable. 0x1 : Enable.
0	R/W	0x0	LBC_CMD_DONE_INT_EN Enable LBC command done interrupt 0x0 : Disable. 0x1 : Enable.

17.3.8.61 0x0504 LocalBus Interrupt1 Enable Register (Default Value: 0x0000_0000)

Offset: 0x0504			Register Name: LBC_INT1_EN_REG
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/

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Offset: 0x0504			Register Name: LBC_INT1_EN_REG
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	LBC_DMA_TIMEOUT_INT_EN Enable DMA timeout interrupt 0x0 : Disable. 0x1 : Enable.
5	R/W	0x0	LBC_IDA_TIMEOUT_INT_EN Enable IDA timeout interrupt 0x0 : Disable. 0x1 : Enable.
4	R/W	0x0	LBC_DA_TIMEOUT_INT_EN Enable DA timeout interrupt 0x0 : Disable. 0x1 : Enable.
3	R/W	0x0	LBC_DMA_DES_DONE_INT_EN Enable DMA descriptor done interrupt 0x0 : Disable. 0x1 : Enable.
2	R/W	0x0	LBC_DATA_REQ_TIMEOUT_INT_EN Enable data request timeout interrupt 0x0 : Disable. 0x1 : Enable.
1	R/W	0x0	LBC_DMA_RX_TRIG_INT_EN Enable DMA RX request interrupt 0x0 : Disable. 0x1 : Enable. Note: LocalBus DMA RX Fifo Waterline Higher than Trigger Level Interrupt Enable
0	R/W	0x0	LBC_DMA_TX_TRIG_INT_EN Enable DMA TX request interrupt 0x0 : Disable. 0x1 : Enable. Note: LocalBus DMA TX Fifo Waterline Lower than Trigger Level Interrupt Enable

17.3.8.62 0x0508 LocalBus Interrupt0 Pending Register (Default Value: 0x0000_0000)

Offset: 0x0508			Register Name: LBC_INT0_PENDING_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0508			Register Name: LBC_INT0_PENDING_REG
Bit	Read/Write	Default/Hex	Description
31	R/W1C	0x0	LBC_IDA_RX_TRIG_INT_PENDING IDA RX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus Indirect Access RX Fifo Waterline Higher than Trigger Level Interrupt Pending
30	R/W1C	0x0	LBC_IDA_TX_TRIG_INT_PENDING IDA TX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus Indirect Access TX Fifo Waterline Lower than Trigger Level Interrupt Pending
29	R/W1C	0x0	LBC_DA_RX_TRIG_INT_PENDING DA RX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus Direct Access RX Fifo Waterline Higher than Trigger Level Interrupt Pending
28	R/W1C	0x0	LBC_DA_TX_TRIG_INT_PENDING DA TX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus Direct Access TX Fifo Waterline Lower than Trigger Level Interrupt Pending
27:8	/	/	/
7	R/W1C	0x0	LBC_DP_INT_PENDING Byte by byte parity interrupt pending 0x0 : No pending. 0x1 : Pending.
6	R/W1C	0x0	LBC_ODP_INT_PENDING Golbal parity interrupt pending 0x0 : No pending. 0x1 : Pending.
5	R/W1C	0x0	LBC_DMA_DES_INVLD_PENDING DMA descriptor invalid pending 0x0 : No pending. 0x1 : Pending.
4	R/W1C	0x0	LBC_DMA_TRSF_DONE_INT_PENDING DMA transfer done interrupt pending 0x0 : No pending. 0x1 : Pending.

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Offset: 0x0508			Register Name: LBC_INT0_PENDING_REG
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	LBC_IDA_TRSF_DONE_INT_PENDING IDA transfer done interrupt pending 0x0 : No pending. 0x1 : Pending.
2	R/W1C	0x0	LBC_DA_TRSF_DONE_INT_PENDING DA transfer done interrupt pending 0x0 : No pending. 0x1 : Pending.
1	R/W1C	0x0	LBC_READY_TIMEOUT_INT_PENDING READY timout interrupt pending 0x0 : No pending. 0x1 : Pending.
0	R/W1C	0x0	LBC_CMD_DONE_INT_PENDING LBC command done interrupt pending 0x0 : No pending. 0x1 : Pending.

17.3.8.63 0x050C LocalBus Interrupt1 Pending Register (Default Value: 0x0000_0000)

Offset: 0x050C			Register Name: LBC_INT1_PENDING_REG
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
6	R/W1C	0x0	LBC_DMA_TIMEOUT_INT_PENDING DMA timeout interrupt pending 0x0 : No pending. 0x1 : Pending.
5	R/W1C	0x0	LBC_IDA_TIMEOUT_INT_PENDING IDA timeout interrupt pending 0x0 : No pending. 0x1 : Pending.
4	R/W1C	0x0	LBC_DA_TIMEOUT_INT_PENDING DA timeout interrupt pending 0x0 : No pending. 0x1 : Pending.
3	R/W1C	0x0	LBC_DMA_DES_DONE_INT_PENDING DMA descriptor done interrupt pending 0x0 : No pending. 0x1 : Pending.

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Offset: 0x050C			Register Name: LBC_INT1_PENDING_REG
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	LBC_DATA_REQ_TIMEOUT_INT_PENDING LBC data request timeout interrupt pending 0x0 : No pending. 0x1 : Pending.
1	R/W1C	0x0	LBC_DMA_RX_TRIG_INT_PENDING DMA RX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus DMA RX Fifo Waterline Higher than Trigger Level Interrupt Pending
0	R/W1C	0x0	LBC_DMA_TX_TRIG_INT_PENDING DMA TX request interrupt pending 0x0 : No pending. 0x1 : Pending. Note: LocalBus DMA TX Fifo Waterline Lower than Trigger Level Interrupt Pending

17.3.8.64 0x0510 Direct Access Interface Status Register (Default Value: 0x0000_0000)

Offset: 0x0510			Register Name: DA_INTF_STA_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R	0x0	DA_ORDER_FIFO_WL DA order FIFO waterline
19:17	/	/	/
16	R	0x0	DA_ORDER_FIFO_DATA DA order FIFO data 0: Read 1: Write
15	/	/	/
14:12	R	0x0	DA_RX_OT_FIFO_WL DA RX Outstanding FIFO Waterline
11	/	/	/
10:8	R	0x0	DA_TX_OT_FIFO_WL DA TX Outstanding FIFO Waterline
7:4	R	0x0	DA_RX_LOAD_FIFO_WL DA RX Load FIFO Waterline
3:0	R	0x0	DA_TX_LOAD_FIFO_WL DA TX Load FIFO Waterline

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17.3.8.65 0x0514 Direct Access FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x0514			Register Name: DA_FIFO_STA_REG
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22:16	R	0x0	DA_RX_FIFO_WL DA RX Data FIFO Waterline
15:13	/	/	/
12:8	R	0x0	DA_TX_FIFO_WL DA TX Data FIFO Waterline
7:3	/	/	/
2:0	R	0x0	DA_CMD_FIFO_WL DA command FIFO Waterline

17.3.8.66 0x0518 Indirect Access FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x0518			Register Name: IDA_FIFO_STA_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:16	R	0x0	IDA_RX_FIFO_WL IDA RX Data FIFO Waterline
15:13	/	0x0	/
12:4	R	0x0	IDA_TX_FIFO_WL IDA TX Data FIFO Waterline
3	/	/	/
2:0	R	0x0	IDA_CMD_FIFO_WL IDA command FIFO Waterline

17.3.8.67 0x051C DMA FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x051C			Register Name: DMA_FIFO_STA_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R	0x0	DMA_RX_FIFO_WL DMA RX Data FIFO Waterline
15:8	R	0x0	DMA_TX_FIFO_WL DMA TX Data FIFO Waterline
7:3	/	/	/
2:0	R	0x0	DMA_CMD_FIFO_WL DMA Command FIFO Waterline

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17.3.8.68 0x0524 DA FIFO Trigger Level Register (Default Value: 0x0000_0000)

Offset: 0x0524			Register Name: DA_FIFO_TRIGGER_LVL_REG
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:8	R/W	0x0	DA_RX_TRIGGER_LVL Direct access RX data FIFO trigger level. The value ranges from 0 to 63. Note: When waterline higher than rx trigger level, it means rxfifo has enough data from device. One level data number depends on axi burst_size
7:4	/	/	/
3:0	R/W	0x0	DA_TX_TRIGGER_LVL Direct Access TX data FIFO trigger level. The value ranges from 0 to 15. Note: When waterline lower than tx trigger level, it means txfifo has enough space to accept data from master. One level data number depends on axi burst_size

17.3.8.69 0x0528 IDA FIFO Trigger Level Register (Default Value: 0x0000_0000)

Offset: 0x0528			Register Name: IDA_FIFO_TRIGGER_LVL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	IDA_RX_TRIGGER_LVL Indirect access RX data FIFO trigger level. The value ranges from 0 to 255. Note: When waterline higher than rx trigger level, it means rxfifo has enough data from device. One level data number is 1Word
15:8	/	/	/
7:0	R/W	0x0	IDA_TX_TRIGGER_LVL Indirect access TX data FIFO trigger level. The value ranges from 0 to 255. Note: When waterline lower than tx trigger level, it means txfifo has enough space to accept data from master. One level data number is 1Word

17.3.8.70 0x052C DMA FIFO Trigger Level Register (Default Value: 0x0000_0000)

Offset: 0x052C	Register Name: DMA_FIFO_TRIGGER_LVL_REG

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Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22:16	R/W	0x0	DMA_RX_TRIGGER_LVL DMA access RX data FIFO trigger level. The value ranges from 0 to 127. Note: When waterline higher than rx trigger level, it means rxfifo has enough data from device. One level data number is 2Word(When start addr is not 2Word align,first fifo level only have 1Word data)
15:7	/	/	/
6:0	R/W	0x0	DMA_TX_TRIGGER_LVL DMA access TX data FIFO trigger level. The value ranges from 0 to 127. Note: When waterline lower than tx trigger level, it means txfifo has enough space to accept data from master. One level data number is 2Word(When start addr is not 2Word align,first fifo level only have 1Word data)

17.3.8.71 0x0530 LBC FIFO Clear Register (Default Value: 0x0000_0000)

Offset: 0x0530			Register Name: LBC_FIFO_CLR_REG
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22	R/W	0x0	DA_RWCMD_ODFIFO_CLR Direct access path read and write command order FIFO clear signals 0: Not reset 1: Reset
21	R/W	0x0	DA_RCMD_LDFIFO_CLR Direct access route Load FIFO read clear signal 0: Not reset 1: Reset
20	R/W	0x0	DA_RCMD_OTFIFO_CLR Direct access route Outstanding FIFO read clear signal 0: Not reset 1: Reset
19	R/W	0x0	DA_WCMD_LDFIFO_CLR Direct access route Load FIFO write clear signal 0: Not reset 1: Reset

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Offset: 0x0530			Register Name: LBC_FIFO_CLR_REG
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	DA_WCMD_OTFIFO_CLR Direct access route Outstanding FIFO write clear signal 0: Not reset 1: Reset
17	R/W	0x0	DMA_RX_R_FIFO_CLR DMA route RX data FIFO read clear signal 0: Not reset 1: Reset
16	R/W	0x0	DMA_RX_W_FIFO_CLR DMA route RX data FIFO write clear signal 0: Not reset 1: Reset
15	R/W	0x0	DMA_TX_R_FIFO_CLR DMA route TX data FIFO read clear signal 0: Not reset 1: Reset
14	R/W	0x0	DMA_TX_W_FIFO_CLR DMA route TX data FIFO write clear signal 0: Not reset 1: Reset
13	R/W	0x0	DMA_CMD_R_FIFO_CLR DMA route command FIFO read clear signal 0: Not reset 1: Reset
12	R/W	0x0	DMA_CMD_W_FIFO_CLR DMA route command FIFO write clear signal 0: Not reset 1: Reset
11	R/W	0x0	IDA_RX_R_FIFO_CLR Indirect access route RX data FIFO read clear signal 0: Not reset 1: Reset
10	R/W	0x0	IDA_RX_W_FIFO_CLR Indirect access route RX data FIFO write clear signal 0: Not reset 1: Reset
9	R/W	0x0	IDA_TX_R_FIFO_CLR Indirect access route TX data FIFO read clear signal 0: Not reset 1: Reset

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Offset: 0x0530			Register Name: LBC_FIFO_CLR_REG
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	IDA_TX_W_FIFO_CLR Indirect access route TX data FIFO write clear signal 0: Not reset 1: Reset
7	R/W	0x0	IDA_CMD_R_FIFO_CLR Indirect access route command FIFO read clear signal 0: Not reset 1: Reset
6	R/W	0x0	IDA_CMD_W_FIFO_CLR Indirect access route command FIFO write clear signal 0: Not reset 1: Reset
5	R/W	0x0	DA_RX_R_FIFO_CLR Direct access route RX data FIFO read clear signal 0: Not reset 1: Reset
4	R/W	0x0	DA_RX_W_FIFO_CLR Direct access route RX data FIFO write clear signal 0: Not reset 1: Reset
3	R/W	0x0	DA_TX_R_FIFO_CLR Direct access route TX data FIFO read clear signal 0: Not reset 1: Reset
2	R/W	0x0	DA_TX_W_FIFO_CLR Direct access route TX data FIFO write clear signal 0: Not reset 1: Reset
1	R/W	0x0	DA_CMD_R_FIFO_CLR Direct access route command FIFO read clear signal 0: Not reset 1: Reset
0	R/W	0x0	DA_CMD_W_FIFO_CLR Direct access route command FIFO write clear signal 0: Not reset 1: Reset

17.3.8.72 0x0600 LBC IDA MIRROR Register (Default Value: 0x0000_0000)

Offset: 0x0600	Register Name: LBC_IDA_MIRROR_REG

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Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_IDA_MIRROR IDA debug signal

17.3.8.73 0x0604 LBC DA Command0 Register (Default Value: 0x0000_0000)

Offset: 0x0604			Register Name: LBC_DA_CMD0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DA_CMD0 LBC DA Command0 Register. Corresponding Command bit[31:0]

17.3.8.74 0x0608 LBC DA Command1 Register (Default Value: 0x0000_0000)

Offset: 0x0608			Register Name: LBC_DA_CMD1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DA_CMD1 LBC DA Command1 Register. Corresponding Command bit[63:32]

17.3.8.75 0x060C LBC Command Busy Register (Default Value: 0x0000_0000)

Offset: 0x060C			Register Name: LBC_CMD_BUSY_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R	0x0	LBC_DMA_CMD_BUSY LBC DMA Command Busy Status 0: FREE 1: BUSY
0	R	0x0	LBC_AHB_CMD_BUSY LBC AHB Command Busy Status 0: FREE 1: BUSY

17.3.8.76 0x0610 LBC Command0 Register (Default Value: 0x0000_0000)

Offset: 0x0610			Register Name: LBC_CMD0_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0610			Register Name: LBC_CMD0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_CMD0_REG LBC Command0 Register Note: This Register refers bit[31:0] of the lbc command.

17.3.8.77 0x0614 LBC Command1 Register (Default Value: 0x0000_0000)

Offset: 0x0614			Register Name: LBC_CMD1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_CMD1_REG LBC Command1 Register Note: This Register refers bit[63:32] of the lbc command.

17.3.8.78 0x0618 LBC Command2 Register (Default Value: 0x0000_0000)

Offset: 0x0618			Register Name: LBC_CMD2_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R	0x0	LBC_ROUTE_REG LBC Command2 Register 00: Non 01: AXI 10: AHB 11: DMA

17.3.8.79 0x0620 LBC DA TIMEOUT Command0 Register (Default Value: 0x0000_0000)

Offset: 0x0620			Register Name: LBC_DA_TO_CMD0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DA_TO_CMD0_REG LBC DA TIMEOUT Command0 Register Note: This Register refers bit[31:0] of the lbc command.

17.3.8.80 0x0624 LBC DA TIMEOUT Command1 Register (Default Value: 0x0000_0000)

Offset: 0x0624			Register Name: LBC_DA_TO_CMD1_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0624			Register Name: LBC_DA_TO_CMD1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DA_TO_CMD1_REG LBC DA TIMEOUT Command1 Register Note: This Register refers bit[63:32] of the lbc command.

17.3.8.81 0x0628 LBC IDA TIMEOUT Command0 Register (Default Value: 0x0000_0000)

Offset: 0x0628			Register Name: LBC_IDA_TO_CMD0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_IDA_TO_CMD0_REG LBC IDA TIMEOUT Command0 Register Note: This Register refers bit[31:0] of the lbc command.

17.3.8.82 0x062C LBC IDA TIMEOUT Command1 Register (Default Value: 0x0000_0000)

Offset: 0x062C			Register Name: LBC_IDA_TO_CMD1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_IDA_TO_CMD1_REG LBC IDA TIMEOUT Command1 Register Note: This Register refers bit[63:32] of the lbc command.

17.3.8.83 0x0630 LBC DMA TIMEOUT Command0 Register (Default Value: 0x0000_0000)

Offset: 0x0630			Register Name: LBC_DMA_TO_CMD0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_TO_CMD0_REG LBC DMA TIMEOUT Command0 Register Note: This Register refers bit[31:0] of the lbc command.

17.3.8.84 0x0634 LBC DMA TIMEOUT Command1 Register (Default Value: 0x0000_0000)

Offset: 0x0634			Register Name: LBC_DMA_TO_CMD1_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0634			Register Name: LBC_DMA_TO_CMD1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_DMA_TO_CMD1_REG LBC DMA TIMEOUT Command1 Register Note: This Register refers bit[63:32] of the lbc command.

17.3.8.85 0x0640 LBC Master Debug0 Signal Register (Default Value: 0x0000_0000)

Offset: 0x0640			Register Name: LBC_MASTER_DBG0_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_MASTER_DEBUG0_REG

17.3.8.86 0x0644 LBC Master Debug1 Signal Register (Default Value: 0x0000_0000)

Offset: 0x0644			Register Name: LBC_MASTER_DBG1_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LBC_MASTER_DEBUG1_REG

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17.4 USB2.0 System (USB2P0_SYS)

17.4.1 Overview

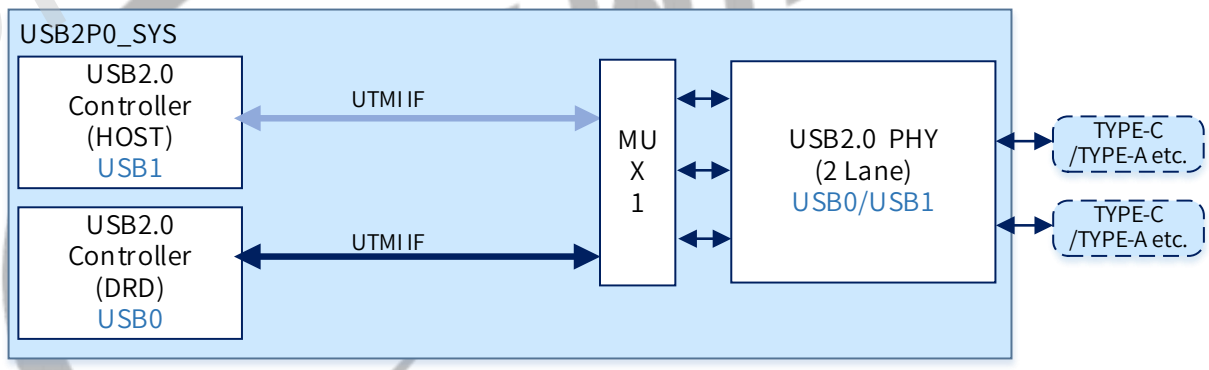
USB2P0_SYS contains one USB 2.0 DRD (USB0) controller, and one USB 2.0 Host (USB1) controller. This chapter mainly describes the registers for USB2.0 PHY. For features for USB2.0 DRD and USB2.0 Host, see section 17.5 USB2.0 DRD and section 17.6 USB2.0 HOST.

The USB2.0 PHY has the following features:

- Supports USB2.0, up to 480 Mbit/s
- Compatible with USB1.1, up to 12 Mbit/s
- Compatible with USB1.0, up to 1.5 Mbit/s
- Using external input clock as PHY reference clock
- Supports suspend, resume, and remote wakeup
- PHY COMMON is shared by USB0 PHY and USB1 PHY.

17.4.2 Block Diagram

Figure 17-22 USB2P0_SYS Block Diagram



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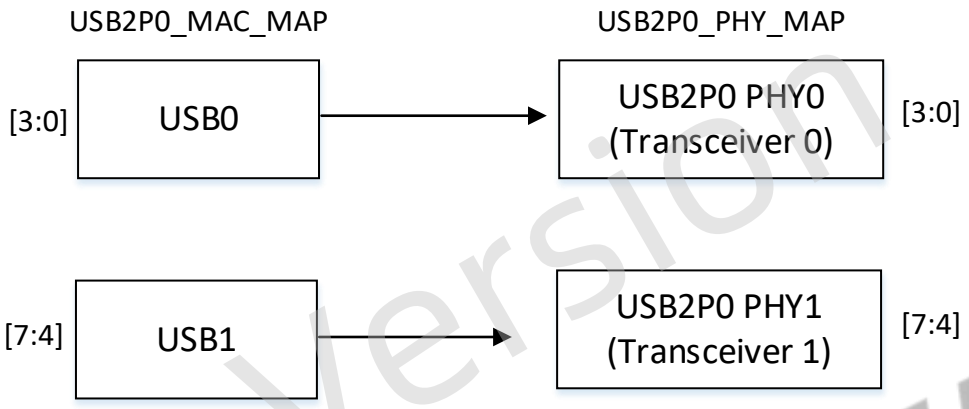
17.4.3 USB2.0 PHY Mapping

The following figure shows the default mapping between USB controller and USB2.0 PHY.

Figure 17-23 USB2.0 PHY Mapping



17.4.4 Register List

Module Name	Base Address	Description
USB2P0_PHY_APP	0x04100000	USB2P0_PHY application register
USB2P0_SYS_DIG_APP	0x04180000	USB2P0 subsystem digital application register

Register Name	Offset	Description
USB2P0_PHY_APP		
USB_PHY_CTRL	0x0010+N*0x0020 (N=0-2)	USB2P0 PHY Control Register
USB_PHY_STA	0x0024+N*0x0020 (N=0-2)	USB2P0 PHY Status Register
USB_RST_CTRL	0x0028+N*0x0020 (N=0-1)	USB2P0 PHY Reset Control Register

Register Name	Offset	Description
USB2P0_SYS_DIG_APP		
USB2P0_MAC_MAP	0x0000	USB2P0 MAC MAP Register
USB2P0_PHY_MAP	0x0004	USB2P0 PHY MAP Register
USB_DCTRL	0x0008	USB2P0 SYS Digtal Control Register

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17.4.5 USB2P0_PHY_APP Register Description

17.4.5.1 0x0010+N*0x0020 (N=0-2) USB2P0 PHY Control Register(Default Value: 0x0000_0008)

Offset: 0x0010+N*0x0020 (N=0-2)			Register Name: USB_PHY_CTRL
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	BIST_EN_A 0: bist_en disable 1: bist_en enable
15:8	R/W	0x0	VC_ADDR PHY VC BUS ADDR
7	R/W	0x0	VC_DI PHY VC BUS DATA IN
6:4	/	/	/
3	R/W	0x1	SIDDQ 0: Write 0 to enable PHY. 1: Write 1 to disable PHY. Setting the SIDDQ of either transceiver to 0 will enable COMMON block. The SIDDQ of COMMON does not need to be configured after enabling this bit.
2	/	/	/
1	R/W	0x0	VC_EN 0: Write 0 to disable PHY VC bus. 1: Write 1 to enable PHY VC bus. Note: only 1 VC bus of all the USB controllers could be enabled at a time.
0	R/W	0x0	VC_CLK VC Bus clock

17.4.5.2 0x0024+N*0x0020 (N=0-2) USB2P0 PHY Status Register(Default Value: 0x0000_0000)

Offset: 0x0024+N*0x0020 (N=0-2)			Register Name: USB_PHY_STA
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R	0x0	BIST_ERROR PHY Bist Done, Active High
16	R	0x0	BIST_DONE PHY Bist Error, Active High
15:1	/	/	/
0	R	0x0	VC_DO Vendor Control Interface Data Output

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17.4.5.3 0x0028+N*0x0020 (N=0-1) USB2P0 PHY Reset Control Register(Default Value: 0x0000_0000)

Offset: 0x0028+N*0x0020 (N=0-1)			Register Name: USB_RST_CTRL
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	PHY_RST Reset for USB2P0 PHY

17.4.6 USB2P0_SYS_DIG_APP Register Description

17.4.6.1 0x0000 USB2P0 MAC MAP Register(Default Value:0x0000_0210)

Offset: 0x0000			Register Name: USB2P0_MAC_MAP
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W	0x1	USB1_MAC_MAP_SEL 0: USB1 controller selects USB2P0 PHY0 (x_p0_transceiver) 1: USB1 controller selects USB2P0 PHY1 (x_p1_transceiver)
3:0	R/W	0x0	USB0_MAC_MAP_SEL 0: USB0 controller selects USB2P0 PHY0 (x_p0_transceiver) 1: USB0 controller selects USB2P0 PHY1 (x_p1_transceiver)

17.4.6.2 0x0004 USB2P0 PHY MAP Register(Default Value:0x0000_0210)

Offset: 0x0004			Register Name: USB2P0_PHY_MAP
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W	0x1	USB2P0_PHY1_MAP_SEL 0: USB2P0 PHY1 (x_p1_transceiver) selects USB0 controller 1: USB2P0 PHY1 (x_p1_transceiver) selects USB1 controller
3:0	R/W	0x0	USB2P0_PHY0_MAP_SEL 0: USB2P0 PHY0 (x_p0_transceiver) selects USB0 controller 1: USB2P0 PHY0 (x_p0_transceiver) selects USB1 controller

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17.4.6.3 0x0008 USB2P0 SYS Digtal Control Register(Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: USB_DCTRL
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	ULPI_RST_N ULPI Reset
0	R/W	0x0	U2_MAP_SEL 0: Reserved 1: Registers within USB2P0_SYS control the mapping between USB0/1 controller and USB2.0 PHY.

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17.5 USB2.0 DRD

17.5.1 Overview

The USB2.0 dual-role device (USB2.0 DRD) supports both device and host functions which can also be configured as a Host-only or Device-only controller.

The USB2.0 DRD has the following features:

- Compatible with USB2.0 Specification
- Support USB Host Function
  - Compatible with Enhanced Host Controller Interface (EHCI) Specification, Version 1.0
  - Compatible with Open Host Controller Interface (OHCI) Specification, Version 1.0a
  - Support High-Speed (HS, 480 Mbit/s)
  - Support Full-Speed (FS, 12 Mbit/s)
  - Support Low-Speed (LS, 1.5 Mbit/s)
  - Supports only 1 USB Root Port shared between EHCI and OHCI
- Support USB Device Function
  - Support High-Speed (HS, 480 Mbit/s)
  - Support Full-Speed (FS, 12 Mbit/s)
  - Support bi-directional Endpoint0 for Control transfer
  - Support 9 Endpoints IN (including control endpoint 0), 9 Endpoints OUT (including control endpoint 0)
  - Support up to (8KB+64Bytes) FIFO for EPs (Including EP0)
  - Support interface to an external Normal DMA controller for every EPs (Endpoint1, Endpoint2, Endpoint3, Endpoint4, Endpoint5, Endpoint6, Endpoint7, Endpoint8)
- Support an internal DMA Controller for data transfer with memory
- Support High-bandwidth Isochronous & Interrupt transfers
- Support Automated splitting/combining of packets for Bulk transfers
- Support point-to-point and point-to-multipoint transfer in both Host and Peripheral mode
- Support Automatic ping capabilities
- Support Soft connect/disconnect function
- Support all transaction scheduling in hardware
- Power Optimization and Power Management capabilities
- Device and Host controller share an 8K SRAM and a physical PHY

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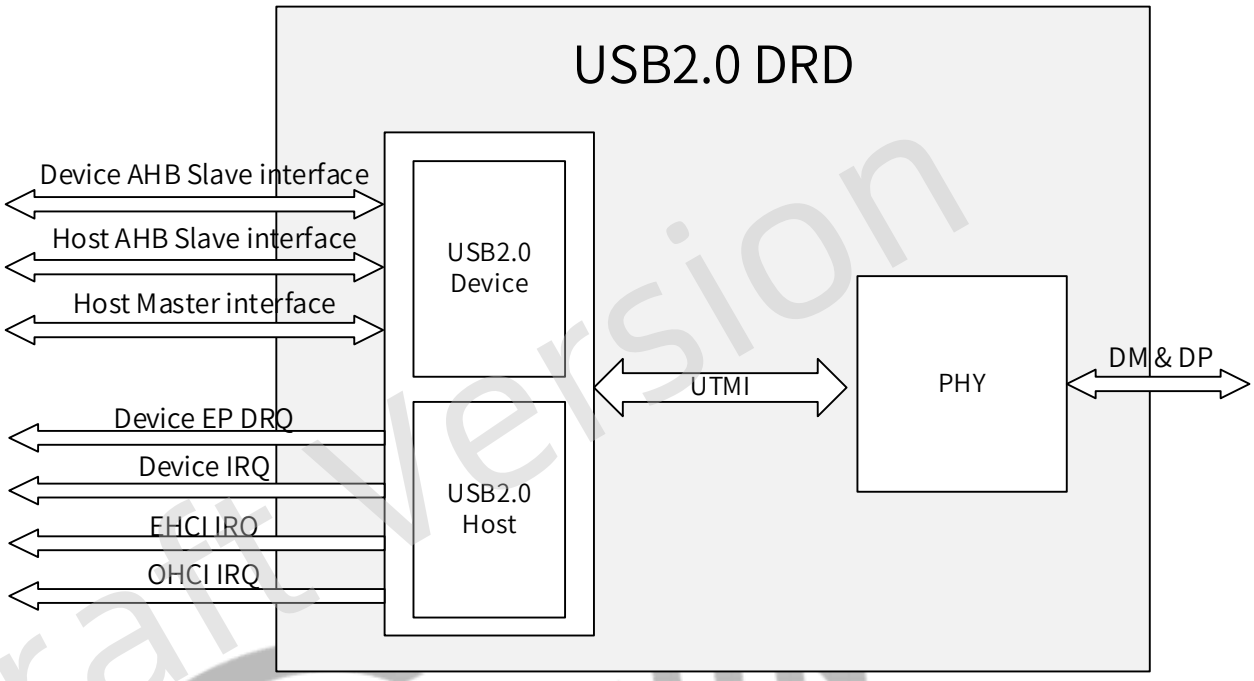
17.5.2 Block Diagram

The following figure shows the block diagram of USB2.0 DRD Controller.

Figure 17-24 USB2.0 DRD Controller Block Diagram



17.5.3 External signals

Table 17-33 USB2.0 DRD External Signals

Signal Name	Description	Type
USB0-DM	USB2.0 DRD Differential Data (Negative)	A I/O
USB0-DP	USB2.0 DRD Differential Data (Positive)	A I/O

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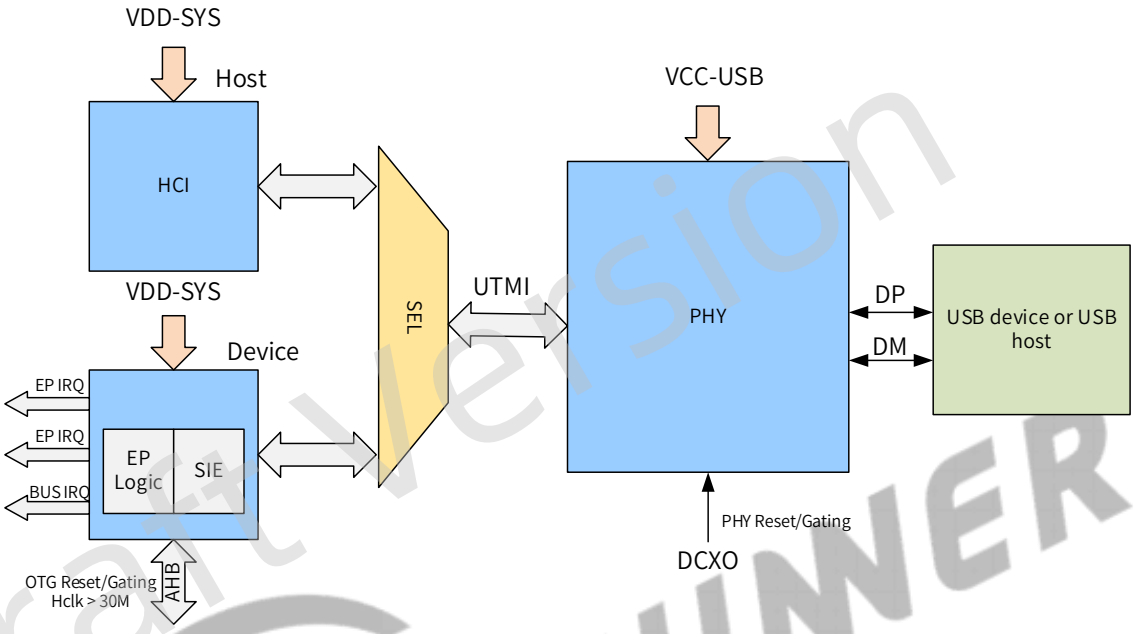
17.5.4 Functional Description

17.5.4.1 Controller and PHY Connection Diagram

Figure 17-25 USB2.0 DRD Controller and PHY Connection Diagram



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17.6 USB2.0 HOST

17.6.1 Overview

The USB Host Controller is fully compliant with USB 2.0 Specification, Enhanced Host Controller Interface (EHCI) Specification Revision 1.0 and Open Host Controller Interface (OHCI) Specification Release 1.0a. The controller supports high-speed, 480-Mbit/s transfers (40 times faster than USB 1.1 full-speed mode) using an EHCI Host Controller, as well as full and low speeds through one or more integrated OHCI Host Controllers.

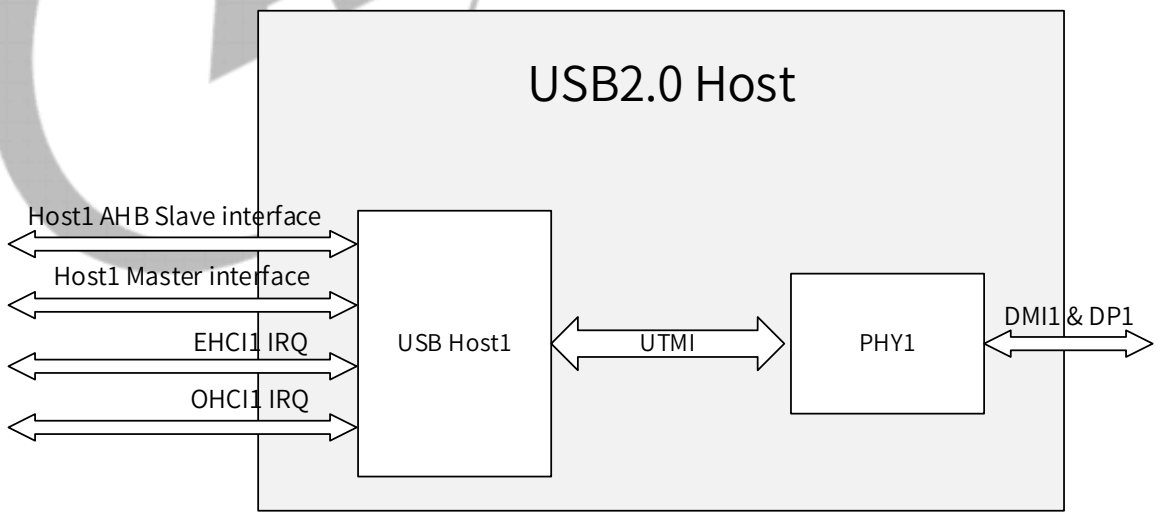
The USB2.0 host controller includes the following features:

- Supports industry-standard AMBA High-Performance Bus (AHB) and it is fully compliant with the AMBA Specification, Revision 2.0. Supports bus.
- Including an internal DMA Controller for data transfer with memory.
- Complies with Enhanced Host Controller Interface (EHCI) Specification, Version 1.0, and the Open Host Controller Interface (OHCI) Specification, Version 1.0a.
- Supports High-Speed (HS, 480-Mbit/s), Full-Speed (FS, 12-Mbit/s), and Low-Speed (LS, 1.5-Mbit/s) Device.
- Supports only 1 USB Root Port shared between EHCI and OHCI.

17.6.2 Block Diagram

The following figure shows the block diagram of USB2.0 Host Controller.

Figure 17-26 USB2.0 Host Controller Block Diagram



17.6.3 External signals

Table 17-34 USB2.0 Host External Signals

Signal Name	Description	Type
USB1-DM	USB2.0 Host Differential Data (Negative)	A I/O

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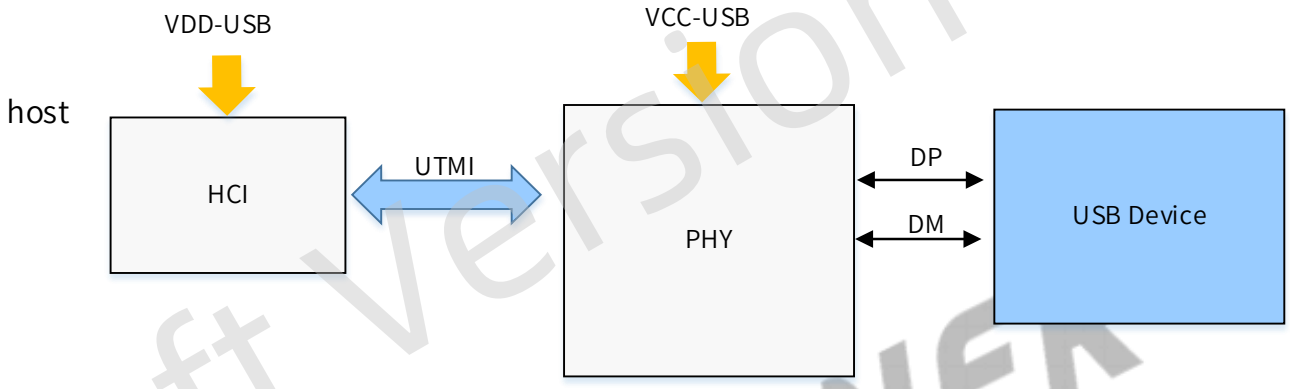
Signal Name	Description	Type
USB1-DP	USB2.0 Host Differential Data (Positive)	A I/O

17.6.4 Functional Description

17.6.4.1 Controller and PHY Connection Diagram

Figure 17-27 USB2.0 Host Controller and PHY Connection Diagram



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17.7 General Purpose ADC (GPADC)

17.7.1 Overview

The General Purpose ADC (GPADC) can convert the external signal into a certain proportion of digital value, to realize the measurement of analog signal, which can be applied to power detection and key detection.

The GPADC has the following features:

- Successive-Approximation Register ADC (SAR ADC) with 24 pads
  - GPADC0: 10 channels, multiplexed in PA4 pin to PA13 pin.
  - GPADC1: 10 channels, multiplexed in PA14 pin to PA23 pin.
  - GPADC2: 4 channels, multiplexed with TPADC signals in PA0 pin to PA3 pin. GPADC2 can be enabled by PWM.
- 64 FIFO depth of data register
- 12-bit sampling resolution
- Up to 1 MHz sampling frequency
- Input voltage: 0 V to 1.8 V.
- Supports data comparison and interrupts
- Three operation modes
  - Single conversion mode
  - Continuous conversion mode
  - Burst conversion mode
- Supports calibration averaging filter

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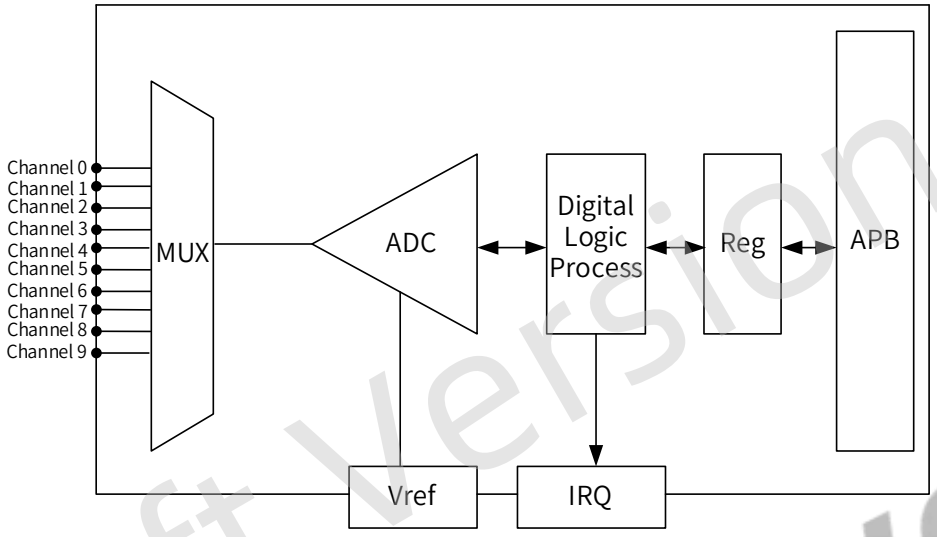
17.7.2 Block Diagram

The following table shows the block diagram of the GPADC, taking GPADC0 as an example.

Figure 17-28 GPADC Block Diagram



17.7.3 External Signals

The following table describes the external signals of the GPADC.

Table 17-35 GPADC External Signals

Signal Name	Description	Type
GPADC0-0	General Purpose ADC 0 Input Channel 0	AI
GPADC0-1	General Purpose ADC 0 Input Channel 1	AI
GPADC0-2	General Purpose ADC 0 Input Channel 2	AI
GPADC0-3	General Purpose ADC 0 Input Channel 3	AI
GPADC0-4	General Purpose ADC 0 Input Channel 4	AI
GPADC0-5	General Purpose ADC 0 Input Channel 5	AI
GPADC0-6	General Purpose ADC 0 Input Channel 6	AI
GPADC0-7	General Purpose ADC 0 Input Channel 7	AI
GPADC0-8	General Purpose ADC 0 Input Channel 8	AI
GPADC0-9	General Purpose ADC 0 Input Channel 9	AI
GPADC1-0	General Purpose ADC 1 Input Channel 0	AI
GPADC1-1	General Purpose ADC 1 Input Channel 1	AI
GPADC1-2	General Purpose ADC 1 Input Channel 2	AI
GPADC1-3	General Purpose ADC 1 Input Channel 3	AI
GPADC1-4	General Purpose ADC 1 Input Channel 4	AI
GPADC1-5	General Purpose ADC 1 Input Channel 5	AI
GPADC1-6	General Purpose ADC 1 Input Channel 6	AI
GPADC1-7	General Purpose ADC 1 Input Channel 7	AI
GPADC1-8	General Purpose ADC 1 Input Channel 8	AI
GPADC1-9	General Purpose ADC 1 Input Channel 9	AI

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Signal Name	Description	Type
GPADC2-0	General Purpose ADC 2 Input Channel 0	AI
GPADC2-1	General Purpose ADC 2 Input Channel 1	AI
GPADC2-2	General Purpose ADC 2 Input Channel 2	AI
GPADC2-3	General Purpose ADC 2 Input Channel 3	AI
AGND-ADC	Analog Ground	G

17.7.4 Functional Description

17.7.4.1 Operation Mode

- Single conversion mode

The GPADC samples and converts once in a specified channel, and then the converted data is updated at the data registers of the corresponding channels.

- External signal triggering mode

The GPADC captures the level signals of other modules (such as PWM) and uses the signal flips as starting information. After a period of counting (optional), GPADC will be enabled automatically and start voltage detection.

- Continuous conversion mode

The GPADC keeps sampling and converting in a specified channel until the ADC is disabled, and then the converted data is updated at the data registers of the corresponding channels.

- Burst conversion mode

The GPADC keeps sampling and converting in a specified channel until the ADC is disabled, and then the converted data is sequentially stored at the FIFO.

17.7.4.2 Clock and Timing Requirements

ADC Sample Frequency TACQ_TIME (ADC acquiring time) + CONV_TIME (Conversion Time) + WAIT_TIME (minimum spare time)

TACQ > 10RC (R is output resistor of ADC sample circuit, C = 6.4 pF)

Take 24 MHz clock as an example:

CONV_TIME = 13cycle times = 1/(24MHz/13Cycles) =0.542 (us).

WAIT_TIME= 3cycle times = 1/(24MHz/3Cycles) =0.125 (us).

If ADC sample frequency is 1 MHz, FS_DIV needs to be set to 21 (24Cycles) and TACQ should be less than or equal to 7 (8 cycles).

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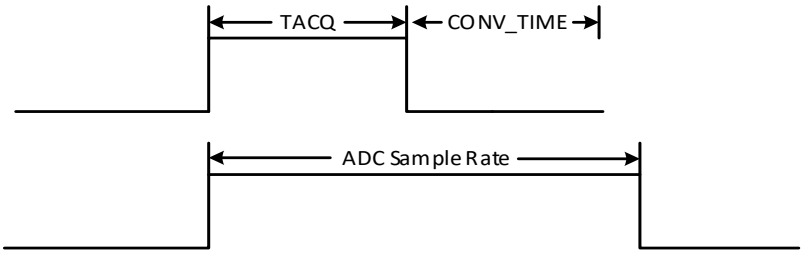
Figure 17-29 GPADC Timing Requirement



17.7.4.3 GPADC Calculation Formula

GPADC calculation formula: GPADC_DATA = V_{in}/V_{REF}*4095

Where:

V_{REF} = 1.8 V

17.7.5 Programming Guidelines

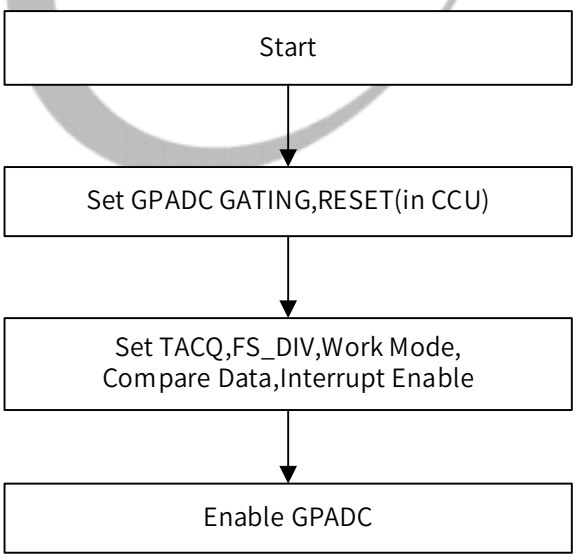
17.7.5.1 Initialization

NOTE

GPADC2 and TPADC are multiplexed in PA pins. Ensure that the TP_MODE_SELECT bit of TP_CTRL_REG1 register is set to 1 before enabling GPADC2.

The GPADC initial process is as follows.

Figure 17-30 GPADC Initial Process



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Query Mode

Take channel0 for example:

Step 1 Write 0x1 to the bit [16] of GPADCx_GAR_REG (x = 0-2) to deassert reset.

Step 2 Write 0x1 to the bit [0] of GPADCx_GAR_REG (x = 0-2) to enable the GPADC clock.

Step 3 Write 0x1 to the bit [31] of GPADCx_CLK_REG (x = 0-2) to enable GPADC function clock.

Step 4 Configure PA_CFGx to set pin multiplexing function as GPADC.

Step 5 Write 0x2F to the bit [15:0] of GP_SR_CON to set the acquiring time of ADC.

Step 6 Write 0x1DF to the bit [31:16] of GP_SR_CON to set the ADC sample frequency divider.

Step 7 Write 0x2 to the bit [19:18] of GP_CTRL to set the continuous conversion mode.

Step 8 Write 0x1 to the bit [0] of GP_CS_EN to enable the analog input channel.

Step 9 Write 0x1 to the bit [16] of GP_CTRL to enable the ADC function.

Step 10 Read the bit [0] of GP_DATA_INTS. If the bit is 1, then data conversion is complete.

Step 11 Read the bit [11:0] of GP_CH0_DATA, and calculate voltage value based on GPADC formula.

Interrupt Mode

Take channel0 for example:

Step 1 Write 0x1 to the bit [16] of GPADCx_GAR_REG (x = 0-2) to deassert reset.

Step 2 Write 0x1 to the bit [0] of GPADCx_GAR_REG (x = 0-2) to enable the GPADC clock.

Step 3 Write 0x1 to the bit [31] of GPADCx_CLK_REG (x = 0-2) to enable GPADC function clock.

Step 4 Configure PA_CFGx to set pin multiplexing function as GPADC.

Step 5 Write 0x2F to the bit [15:0] of GP_SR_CON to set the acquiring time of ADC.

Step 6 Write 0x1DF to the bit [31:16] of GP_SR_CON to set the ADC sample frequency divider.

Step 7 Write 0x2 to the bit [19:18] of GP_CTRL to set the continuous conversion mode.

Step 8 Write 0x1 to the bit [0] of GP_CS_EN to enable the analog input channel.

Step 9 Write 0x1 to the bit [0] of GP_DATA_INTC to enable the GPADC data interrupt.

Step 10 Set the GIC module based on the GPADC IRQ number.

Step 11 Put interrupt handler address into interrupt vector table based on the GPADC IRQ number.

Step 12 Write 0x1 to the bit16 of GP_CTRL to enable the ADC function.

Step 13 Read the bit [11:0] of GP_CH0_DATA from the interrupt handler, calculate voltage value based on GPADC formula.

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17.7.5.2 External Signal Triggering Mode

Assume that the ADC is triggered by PWM.

- **Enable**

Step 1 Configure ADC Voltage detection parameter.

Step 2 Configure GP_CNT_LVL bit in GP_CNT_LVL register to set the hardware counting value according to the delay and configured working clock. If the counting value is 0, GPADC will be auto enabled immediately after capturing PWM signal. The calculation formula is as follow:

$T_{delay}=(1000/F_{GPADC})*$ (GP_CNT_LVL+3)

T_{delay}: The unit is ns.

F_{GPADC}: GPADC output clock frequency in CCU. The unit is MHz.

GP_CNT_LVL: should be configured as 2^N (N = 0-14).

The counting clock frequency is the same as the GPADC operating clock frequency. After the count value reaches the value of GP_CNT_LVL, GPADC will start transmitting.

Step 3 Configure GP_CNT_LVL register to enable EXT_TRIG_EN bit.

Step 4 Configure the PCR register to enable ADC_WORK bit.

Step 5 After PWM generates the trigger signal, wait for hardware count to end.

Step 6 GPADC captures PWM enable signal and start detecting external voltage.

- **Disable**

Step 1 Configure GP_CNT_LVL register to disable EXT_TRIG_EN bit.

Step 2 Configure the PCR register to disable ADC_WORK bit

17.7.5.3 Sampling Frequency Configuration

The sampling frequency is only configurable in continue conversion mode.

When 24 MHz clock is selected, the sampling frequency for each channel can be configured as a multiple of 32 kHz and the total sampling frequency of all channels should be less than or equal to 1 MHz.

When 48 MHz clock is selected, the sampling frequency for each channel can be configured as a multiple of 64 kHz and the total sampling frequency of all channels should be less than or equal to 2 MHz.

The sampling frequency configuration steps are as follows.

Step 1 If GPADC function is enabled, configure the ADC_EN bit (bit [16]) of GP_CTRL (offset: 0x0004) register to disable ADC function.

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Step 2 Configure the FS_DIV bit (bit [31:16]) of GP_SR_CON (offset: 0x0000) register to set the initial sampling frequency of each channel.

Step 3 Configure GP_SMP_TMSx (x = 0-2) register to set GP_SMP_TMS value. The initial sampling frequency is multiplied by this value to get the actual sampling frequency.

NOTE

All GPADC channels should be configured simultaneously.

The following are the all available sampling frequencies for each channel and corresponding GP_SMP_TMS values, when GPADC operating clock is 24MHz.

Table 17-36 GP_SMP_TMS Value Corresponding to Each Sampling Frequency

Sampling frequency	GP_SMP_TMS Value	Sampling frequency	GP_SMP_TMS Value
32 kHz	1	544 kHz	17
64 kHz	2	576 kHz	18
96 kHz	3	608 kHz	19
128 kHz	4	640 kHz	20
160 kHz	5	672 kHz	21
192 kHz	6	704 kHz	22
224 kHz	7	736 kHz	23
256 kHz	8	768 kHz	24
288 kHz	9	800 kHz	25
320 kHz	10	832 kHz	26
352 kHz	11	864 kHz	27
384 kHz	12	896 kHz	28
416 kHz	13	928 kHz	29
448 kHz	14	960 kHz	30
480 kHz	15	992 kHz	31
512 kHz	16	/	/

Step 4 Configure the GP_CH_SMP_BYP bit (bit {0}) of GP_SMP_BYP (offset: 0x00D0) register as 0.

Step 5 Configure the ADC_EN bit (bit [16]) of GP_CTRL (offset: 0x0004) register to enable ADC function.

17.7.6 Register List

Module Name	Base Address
GPADC0	0x02087000
GPADC1	0x02088000
GPADC2	0x02089000

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Register Name	Offset	Description
GP_SR_CON	0x0000	GPADC Sample Rate Configure Register
GP_CTRL	0x0004	GPADC Control Register
GP_CS_EN	0x0008	GPADC Compare and Select Enable Register
GP_FIFO_INTC	0x000C	GPADC FIFO Interrupt Control Register
GP_FIFO_INTS	0x0010	GPADC FIFO Interrupt Status Register
GP_FIFO_DATA	0x0014	GPADC FIFO Data Register
GP_CDATA	0x001C	GPADC Calibration Data Register
GP_DATAL_INTC	0x0020	GPADC Data Low Interrupt Configure Register
GP_DATAH_INTC	0x0024	GPADC Data High Interrupt Configure Register
GP_DATA_INTC	0x0028	GPADC Data Interrupt Configure Register
GP_DATAL_INTS	0x0030	GPADC Data Low Interrupt Status Register
GP_DATAH_INTS	0x0034	GPADC Data High Interrupt Status Register
GP_DATA_INTS	0x0038	GPADC Data Interrupt Status Register
GP_CH0_CMP_DATA	0x0040	GPADC CH0 Compare Data Register
GP_CH1_CMP_DATA	0x0044	GPADC CH1 Compare Data Register
GP_CH2_CMP_DATA	0x0048	GPADC CH2 Compare Data Register
GP_CH3_CMP_DATA	0x004C	GPADC CH3 Compare Data Register
GP_CH4_CMP_DATA	0x0050	GPADC CH4 Compare Data Register
GP_CH5_CMP_DATA	0x0054	GPADC CH5 Compare Data Register
GP_CH6_CMP_DATA	0x0058	GPADC CH6 Compare Data Register
GP_CH7_CMP_DATA	0x005C	GPADC CH7 Compare Data Register
GP_CH8_CMP_DATA	0x0060	GPADC CH8 Compare Data Register
GP_CH9_CMP_DATA	0x0064	GPADC CH9 Compare Data Register
GP_CH0_DATA	0x0080	GPADC CH0 Data Register
GP_CH1_DATA	0x0084	GPADC CH1 Data Register
GP_CH2_DATA	0x0088	GPADC CH2 Data Register
GP_CH3_DATA	0x008C	GPADC CH3 Data Register
GP_CH4_DATA	0x0090	GPADC CH4 Data Register
GP_CH5_DATA	0x0094	GPADC CH5 Data Register
GP_CH6_DATA	0x0098	GPADC CH6 Data Register
GP_CH7_DATA	0x009C	GPADC CH7 Data Register
GP_CH8_DATA	0x00A0	GPADC CH8 Data Register
GP_CH9_DATA	0x00A4	GPADC CH9 Data Register
GP_SMP_TMS0	0x00C0	GPADC Sampling Times0 Register
GP_SMP_TMS1	0x00C4	GPADC Sampling Times1 Register
GP_SMP_TMS2	0x00C8	GPADC Sampling Times2 Register
GP_SMP_BYP	0x00D0	GPADC Channel Sample Rate Setting Bypass Register
GP_CNT_LVL	0x01F0	GPADC External Trigger Register

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17.7.7 Register Description

17.7.7.1 0x0000 GPADC Sample Rate Configure Register (Default Value: 0x01DF_002F)

Offset: 0x0000			Register Name: GP_SR_CON
Bit	Read/Write	Default/Hex	Description
31: 16	R/W	0x1DF	FS_DIV ADC Sample Frequency Divider CLK_IN/(n+3) Default value: 50K
15:0	R/W	0x2F	TACQ ADC Acquiring Time (n+1)/CLK_IN Default value: 2 us

17.7.7.2 0x0004 GPADC Control Register (Default Value: 0x0080_0000)

Offset: 0x0004			Register Name: GP_CTRL
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	ADC_FIRST_DLY ADC First Convert Delay setting ADC conversion of each channel is delayed by N samples.
23	R/W	0x1	ADC_AUTOCALI_EN ADC Auto Calibration 0: The ADC channel starts sampling data immediately after ADC_EN is enabled. 1: GPADC automatically performs an calibration operation before the ADC channel starts sampling data after ADC_EN is enabled.
22	/	/	/
21:20	R/W	0x0	ADC_OP_BIAS ADC OP Bias Adjust the bandwidth of the ADC amplifier. Bit20 is used for GPADC sampling timing revert logic. 0: Revert to old sampling timing 1: Default new sampling timing.

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Offset: 0x0004			Register Name: GP_CTRL
Bit	Read/Write	Default/Hex	Description
19:18	R/W	0x0	GP_MODE GPADC Work Mode 00: Single conversion mode 01: Reserved 10: Continuous conversion mode 11: Burst conversion mode
17	R/WAC	0x0	ADC_CALI_EN ADC Manual Calibration 1: Start manual calibration, it is cleared to 0 after calibration You should disable THSx_EN (x=0 or 1) and ADC_EN before writing 1 to this bit, which can be enabled after manual calibration.
16	R/W	0x0	ADC_EN ADC Function Enable Before the bit is enabled, configure ADC parameters including the work mode and channel number, etc. 0: Disable 1: Enable When selecting a single conversion mode, the bit can be cleared automatically after the conversion is completed.
15	/	/	/
14:13	R/W	0x0	ADC_CALI_TYPE ADC Calibration Filter Type Select 00: No filtering. Sample the ADC internal base value one time and use it as calibration value. 01: Sample the ADC internal base value 2 times for mean filtering. 10: Sample the ADC internal base value 4 times for mean filtering. 11: Sample the ADC internal base value 8 times for mean filtering.
12:0	/	/	/

17.7.7.3 0x0008 GPADC Compare and Select Enable Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: GP_CS_EN
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/

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Offset: 0x0008			Register Name: GP_CS_EN
Bit	Read/Write	Default/Hex	Description
25	R/W	0x0	ADC_CH9_CMP_EN Channel 9 Compare Enable 0: Disable 1: Enable
24	R/W	0x0	ADC_CH8_CMP_EN Channel 8 Compare Enable 0: Disable 1: Enable
23	R/W	0x0	ADC_CH7_CMP_EN Channel 7 Compare Enable 0: Disable 1: Enable
22	R/W	0x0	ADC_CH6_CMP_EN Channel 6 Compare Enable 0: Disable 1: Enable
21	R/W	0x0	ADC_CH5_CMP_EN Channel 5 Compare Enable 0: Disable 1: Enable
20	R/W	0x0	ADC_CH4_CMP_EN Channel 4 Compare Enable 0: Disable 1: Enable
19	R/W	0x0	ADC_CH3_CMP_EN Channel 3 Compare Enable 0: Disable 1: Enable
18	R/W	0x0	ADC_CH2_CMP_EN Channel 2 Compare Enable 0: Disable 1: Enable
17	R/W	0x0	ADC_CH1_CMP_EN Channel 1 Compare Enable 0: Disable 1: Enable
16	R/W	0x0	ADC_CH0_CMP_EN Channel 0 Compare Enable 0: Disable 1: Enable
15:10	/	/	/

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Offset: 0x0008			Register Name: GP_CS_EN
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	ADC_CH9_SELECT Analog input channel 9 Select 0: Disable 1: Enable
8	R/W	0x0	ADC_CH8_SELECT Analog input channel 8 Select 0: Disable 1: Enable
7	R/W	0x0	ADC_CH7_SELECT Analog input channel 7 Select 0: Disable 1: Enable
6	R/W	0x0	ADC_CH6_SELECT Analog input channel 6 Select 0: Disable 1: Enable
5	R/W	0x0	ADC_CH5_SELECT Analog input channel 5 Select 0: Disable 1: Enable
4	R/W	0x0	ADC_CH4_SELECT Analog input channel 4 Select 0: Disable 1: Enable
3	R/W	0x0	ADC_CH3_SELECT Analog input channel 3 Select 0: Disable 1: Enable
2	R/W	0x0	ADC_CH2_SELECT Analog input channel 2 Select 0: Disable 1: Enable
1	R/W	0x0	ADC_CH1_SELECT Analog input channel 1 Select 0: Disable 1: Enable
0	R/W	0x0	ADC_CH0_SELECT Analog input channel 0 Select 0: Disable 1: Enable

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17.7.7.4 0x000C GPADC FIFO Interrupt Control Register (Default Value: 0x0000_1F00)

Offset: 0x000C			Register Name: GP_FIFO_INTC
Bit	Read/Write	Default/Hex	Description
31:19	/	/	/
18	R/W	0x0	FIFO_DATA_DRQ_EN ADC FIFO Date DRQ Enable 0: Disable 1: Enable
17	R/W	0x0	FIFO_OVERRUN_IRQ_EN ADC FIFO Over Run IRQ Enable 0: Disable 1: Enable
16	R/W	0x0	FIFO_DATA_IRQ_EN ADC FIFO Data Available IRQ Enable 0: Disable 1: Enable
15:14	/	/	/
13:8	R/W	0x1F	FIFO_TRIG_LEVEL Interrupt trigger level for ADC Trigger Level = TXTL + 1
7:5	/	/	/
4	R/WAC	0x0	FIFO_FLUSH ADC FIFO Flush Write 1 to flush TX FIFO, self-clear to 0 .
3:0	/	/	/

17.7.7.5 0x0010 GPADC FIFO Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: GP_FIFO_INTS
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W1C	0x0	FIFO_OVERRUN_PENDING. ADC FIFO Over Run IRQ pending 0: No Pending IRQ 1: FIFO Overrun Pending IRQ Write '1' to clear this interrupt
16	R/W1C	0x0	FIFO_DATA_PENDING. ADC FIFO Data Available Pending Bit 0: NO Pending IRQ 1: FIFO Available Pending IRQ Write '1' to clear this interrupt
15	/	/	/

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Offset: 0x0010			Register Name: GP_FIFO_INTS
Bit	Read/Write	Default/Hex	Description
14:8	R	0x0	RXA_CNT. ADC FIFO available Sample Word Counter
7:0	/	/	/

17.7.7.6 0x0014 GPADC FIFO Data Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: GP_FIFO_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x0	GP_FIFO_DATA GPADC Data in FIFO

17.7.7.7 0x001C GPADC Calibration Data Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: GP_CDATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R/W	0x000	GP_CDATA GPADC Calibration Data

17.7.7.8 0x0020 GPADC Data Low Interrupt Configure Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: GP_DATAL_INTC
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	CH9_LOW_IRQ_EN Channel 9 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	CH8_LOW_IRQ_EN Channel 8 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
7	R/W	0x0	CH7_LOW_IRQ_EN Channel 7 Voltage Low Available Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0020			Register Name: GP_DATAL_INTC
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	CH6_LOW_IRQ_EN Channel 6 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	CH5_LOW_IRQ_EN Channel 5 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	CH4_LOW_IRQ_EN Channel 4 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	CH3_LOW_IRQ_EN Channel 3 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	CH2_LOW_IRQ_EN Channel 2 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	CH1_LOW_IRQ_EN Channel 1 Voltage Low Available Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	CH0_LOW_IRQ_EN Channel 0 Voltage Low Available Interrupt Enable 0: Disable 1: Enable

17.7.7.9 0x0024 GPADC Data High Interrupt Configure Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: GP_DATAH_INTC
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	CH9_HIG_IRQ_EN Channel 9 Voltage High Available Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0024			Register Name: GP_DATAH_INTC
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	CH8_HIG_IRQ_EN Channel 8 Voltage High Available Interrupt Enable 0: Disable 1: Enable
7	R/W	0x0	CH7_HIG_IRQ_EN Channel 7 Voltage High Available Interrupt Enable 0: Disable 1: Enable
6	R/W	0x0	CH6_HIG_IRQ_EN Channel 6 Voltage High Available Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	CH5_HIG_IRQ_EN Channel 5 Voltage High Available Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	CH4_HIG_IRQ_EN Channel 4 Voltage High Available Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	CH3_HIG_IRQ_EN Channel 3 Voltage High Available Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	CH2_HIG_IRQ_EN Channel 2 Voltage High Available Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	CH1_HIG_IRQ_EN Channel 1 Voltage High Available Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	CH0_HIG_IRQ_EN Channel 0 Voltage High Available Interrupt Enable 0: Disable 1: Enable

17.7.7.10 0x0028 GPADC Data Interrupt Configure Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: GP_DATA_INTC
Bit	Read/Write	Default/Hex	Description

Draft Version

Offset: 0x0028			Register Name: GP_DATA_INTC
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	CH9_DATA_IRQ_EN Channel 9 Data Available Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	CH8_DATA_IRQ_EN Channel 8 Data Available Interrupt Enable 0: Disable 1: Enable
7	R/W	0x0	CH7_DATA_IRQ_EN Channel 7 Data Available Interrupt Enable 0: Disable 1: Enable
6	R/W	0x0	CH6_DATA_IRQ_EN Channel 6 Data Available Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	CH5_DATA_IRQ_EN Channel 5 Data Available Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	CH4_DATA_IRQ_EN Channel 4 Data Available Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	CH3_DATA_IRQ_EN Channel 3 Data Available Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	CH2_DATA_IRQ_EN Channel 2 Data Available Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	CH1_DATA_IRQ_EN Channel 1 Data Available Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	CH0_DATA_IRQ_EN Channel 0 Data Available Interrupt Enable 0: Disable 1: Enable

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17.7.7.11 0x0030 GPADC Data Low Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: GP_DATAL_INTS
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W1C	0x0	CH9_LOW_PENGDING Channel 9 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 9 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
8	R/W1C	0x0	CH8_LOW_PENGDING Channel 8 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 8 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
7	R/W1C	0x0	CH7_LOW_PENGDING Channel 7 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 7 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
6	R/W1C	0x0	CH6_LOW_PENGDING Channel 6 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 6 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
5	R/W1C	0x0	CH5_LOW_PENGDING Channel 5 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 5 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
4	R/W1C	0x0	CH4_LOW_PENGDING Channel 4 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 4 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

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Offset: 0x0030			Register Name: GP_DATAL_INTS
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	CH3_LOW_PENGDING Channel 3 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 3 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
2	R/W1C	0x0	CH2_LOW_PENGDING Channel 2 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 2 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
1	R/W1C	0x0	CH1_LOW_PENGDING Channel 1 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 1 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
0	R/W1C	0x0	CH0_LOW_PENGDING Channel 0 Voltage Low Available Interrupt Status 0: NO Pending IRQ 1: Channel 0 Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

17.7.7.12 0x0034 GPADC Data High Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: GP_DATAH_INTS
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W1C	0x0	CH9_HIG_PENGDING Channel 9 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 9 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

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Offset: 0x0034			Register Name: GP_DATAH_INTS
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	CH8_HIG_PENGDING Channel 8 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 8 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
7	R/W1C	0x0	CH7_HIG_PENGDING Channel 7 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 7 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
6	R/W1C	0x0	CH6_HIG_PENGDING Channel 6 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 6 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
5	R/W1C	0x0	CH5_HIG_PENGDING Channel 5 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 5 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
4	R/W1C	0x0	CH4_HIG_PENGDING Channel 4 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 4 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
3	R/W1C	0x0	CH3_HIG_PENGDING Channel 3 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 3 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
2	R/W1C	0x0	CH2_HIG_PENGDING Channel 2 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 2 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

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Offset: 0x0034			Register Name: GP_DATAH_INTS
Bit	Read/Write	Default/Hex	Description
1	R/W1C	0x0	CH1_HIG_PENGDING Channel 1 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 1 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
0	R/W1C	0x0	CH0_HIG_PENGDING Channel 0 Voltage High Available Interrupt Status 0: No Pending IRQ 1: Channel 0 Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

17.7.7.13 0x0038 GPADC Data Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: GP_DATA_INTS
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W1C	0x0	CH9_DATA_PENGDING Channel 9 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 9 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
8	R/W1C	0x0	CH8_DATA_PENGDING Channel 8 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 8 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
7	R/W1C	0x0	CH7_DATA_PENGDING Channel 7 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 7 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

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Offset: 0x0038			Register Name: GP_DATA_INTS
Bit	Read/Write	Default/Hex	Description
6	R/W1C	0x0	CH6_DATA_PENGDING Channel 6 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 6 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
5	R/W1C	0x0	CH5_DATA_PENGDING Channel 5 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 5 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
4	R/W1C	0x0	CH4_DATA_PENGDING Channel 4 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 4 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
3	R/W1C	0x0	CH3_DATA_PENGDING Channel 3 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 3 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
2	R/W1C	0x0	CH2_DATA_PENGDING Channel 2 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 2 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
1	R/W1C	0x0	CH1_DATA_PENGDING Channel 1 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 1 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails
0	R/W1C	0x0	CH0_DATA_PENGDING Channel 0 Data Available Interrupt Status 0: No Pending IRQ 1: Channel 0 Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails

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17.7.7.14 0x0040 GPADC CH0 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0040			Register Name: GP_CH0_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH0_CMP_HIG_DATA Channel 0 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH0_CMP_LOW_DATA Channel 0 Voltage Low Value

17.7.7.15 0x0044 GPADC CH1 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0044			Register Name: GP_CH1_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH1_CMP_HIG_DATA Channel 1 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH1_CMP_LOW_DATA Channel 1 Voltage Low Value

17.7.7.16 0x0048 GPADC CH2 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0048			Register Name: GP_CH2_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH2_CMP_HIG_DATA Channel 2 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH2_CMP_LOW_DATA Channel 2 Voltage Low Value

17.7.7.17 0x004C GPADC CH3 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x004C			Register Name: GP_CH3_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH3_CMP_HIG_DATA Channel 3 Voltage High Value
15:12	/	/	/

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Offset: 0x004C			Register Name: GP_CH3_CMP_DATA
Bit	Read/Write	Default/Hex	Description
11:0	R/W	0x400	CH3_CMP_LOW_DATA Channel 3 Voltage Low Value

17.7.7.18 0x0050 GPADC CH4 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0050			Register Name: GP_CH4_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH4_CMP_HIG_DATA Channel 4 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH4_CMP_LOW_DATA Channel 4 Voltage Low Value

17.7.7.19 0x0054 GPADC CH5 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0054			Register Name: GP_CH5_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH5_CMP_HIG_DATA Channel 5 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH5_CMP_LOW_DATA Channel 5 Voltage Low Value

17.7.7.20 0x0058 GPADC CH6 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0058			Register Name: GP_CH6_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH6_CMP_HIG_DATA Channel 6 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH6_CMP_LOW_DATA Channel 6 Voltage Low Value

17.7.7.21 0x005C GPADC CH7 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x005C	Register Name: GP_CH7_CMP_DATA

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Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH7_CMP_HIG_DATA Channel 7 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH7_CMP_LOW_DATA Channel 7 Voltage Low Value

17.7.7.22 0x0060 GPADC CH8 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0060			Register Name: GP_CH8_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH8_CMP_HIG_DATA Channel 8 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH8_CMP_LOW_DATA Channel 8 Voltage Low Value

17.7.7.23 0x0064 GPADC CH9 Compare Data Register (Default Value: 0x0BFF_0400)

Offset: 0x0064			Register Name: GP_CH9_CMP_DATA
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0xBFF	CH9_CMP_HIG_DATA Channel 9 Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CH9_CMP_LOW_DATA Channel 9 Voltage Low Value

17.7.7.24 0x0080 GPADC CH0 Data Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: GP_CH0_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH0_DATA Channel 0 Data

17.7.7.25 0x0084 GPADC CH1 Data Register (Default Value: 0x0000_0000)

Offset: 0x0084	Register Name: GP_CH1_DATA

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Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH1_DATA Channel 1 Data

17.7.7.26 0x0088 GPADC CH2 Data Register (Default Value: 0x0000_0000)

Offset: 0x0088			Register Name: GP_CH2_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH2_DATA Channel 2 Data

17.7.7.27 0x008C GPADC CH3 Data Register (Default Value: 0x0000_0000)

Offset: 0x008C			Register Name: GP_CH3_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH3_DATA Channel 3 Data

17.7.7.28 0x0090 GPADC CH4 Data Register (Default Value: 0x0000_0000)

Offset: 0x0090			Register Name: GP_CH4_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH4_DATA Channel 4 Data

17.7.7.29 0x0094 GPADC CH5 Data Register (Default Value: 0x0000_0000)

Offset: 0x0094			Register Name: GP_CH5_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH5_DATA Channel 5 Data

17.7.7.30 0x0098 GPADC CH6 Data Register (Default Value: 0x0000_0000)

Offset: 0x0098			Register Name: GP_CH6_DATA
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0098			Register Name: GP_CH6_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH6_DATA Channel 6 Data

17.7.7.31 0x009C GPADC CH7 Data Register (Default Value: 0x0000_0000)

Offset: 0x009C			Register Name: GP_CH7_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH7_DATA Channel 7 Data

17.7.7.32 0x00A0 GPADC CH8 Data Register (Default Value: 0x0000_0000)

Offset: 0x00A0			Register Name: GP_CH8_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH8_DATA Channel 8 Data

17.7.7.33 0x00A4 GPADC CH9 Data Register (Default Value: 0x0000_0000)

Offset: 0x00A4			Register Name: GP_CH9_DATA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x000	GP_CH9_DATA Channel 9 Data

17.7.7.34 0x00C0 GPADC Sampling Times0 Register (Default Value 0x0000_0000)

Offset: 0x00C0			Register Name: GP_SMP_TMS0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x00C0			Register Name: GP_SMP_TMS0
Bit	Read/Write	Default/Hex	Description
29:24	R/W	0x0	GP_ CH3_SMP_TMS GPADC Channel 3 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
23: 22	/	/	/
21:16	R/W	0x0	GP_ CH2_SMP_TMS GPADC Channel 2 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
15: 14	/	/	/
13:8	R/W	0x0	GP_ CH1_SMP_TMS GPADC Channel 1 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
7:6	/	/	/
5:0	R/W	0x0	GP_ CH0_SMP_TMS GPADC Channel 0 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved

17.7.7.35 0x00C4 GPADC Sampling Times1 Register (Default Value 0x0000_0000)

Offset: 0x00C4			Register Name: GP_SMP_TMS1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x00C4			Register Name: GP_SMP_TMS1
Bit	Read/Write	Default/Hex	Description
29:24	R/W	0x0	GP_ CH7_SMP_TMS GPADC Channel 7 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
23: 22	/	/	/
21:16	R/W	0x0	GP_ CH6_SMP_TMS GPADC Channel 6 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
15: 14	/	/	/
13:8	R/W	0x0	GP_ CH5_SMP_TMS GPADC Channel 5 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
7:6	/	/	/
5:0	R/W	0x0	GP_ CH4_SMP_TMS GPADC Channel 4 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved

17.7.7.36 0x00C8 GPADC Sampling Times2 Register (Default Value 0x0000_0000)

Offset: 0x00C8			Register Name: GP_SMP_TMS2
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/

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Offset: 0x00C8			Register Name: GP_SMP_TMS2
Bit	Read/Write	Default/Hex	Description
13:8	R/W	0x0	GP_ CH9_SMP_TMS GPADC Channel 9 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved
7:6	/	/	/
5:0	R/W	0x0	GP_ CH8_SMP_TMS GPADC Channel 8 Sampling Times Register 0: 0 (channel disable) 1: 1 (p) 31: 31 Other: Reserved

17.7.7.37 0x00D0 GPADC Channel Sample Rate Setting Bypass Register (Default Value 0x0000_0001)

Offset: 0x00D0			Register Name: GP_SMP_BYP
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x1	GP_ CH_SMP_BYP GPADC Channel Sample Rate Setting Bypass Register 0: Enable sampling rate setting 1: Disable sampling rate setting

17.7.7.38 0x01F0 GPADC External Trigger Register (Default Value 0x0000_0000)

Offset: 0x01F0			Register Name: GP_CNT_LVL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	EXT_TRIG_EN External Signal Trigger Enable. If this bit is set to 0, GPADC does not respond to external trigger.
30:15	/	/	/

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Offset: 0x01F0			Register Name: GP_CNT_LVL
Bit	Read/Write	Default/Hex	Description
14:0	R/W	0x0	GP_CNT_LVL GPADC AUTO Count Level M. **M=0**: auto enable GPADC after catching PWM signal. **M≠ 0 and M=2^N (N=0-14)**: auto enable GPADC after catching PWM signal and counting to M.

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17.8 TPADC

17.8.1 Overview

The Touch Panel ADC (TPADC) is a 4-wire resistive touch screen controller, including a 12-bit SAR type A/D converter.

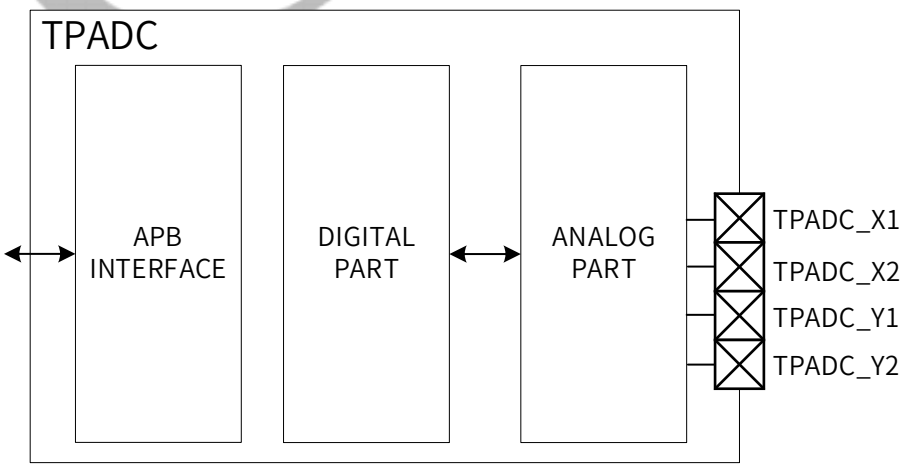
The TPADC has the following features:

- 12-bit SAR type A/D converter
- Configurable sample frequency up to 750 kHz
- A/D can buffer up to 32 sample results
- Supports DMA slave interface
- Supports 4-wire resistive touch panel input detection
  - Supports pen down detection with programmable sensitivity
  - Supports single touch coordinate measurement
  - Supports dual touch detection
  - Supports touch pressure measurement with programmable threshold
  - Supports median and averaging filter for noise reduction
  - Supports X and Y coordinate exchange function
- Supports Aux ADC with up to 4 channels
- TPADC has 4 pads, multiplexing GPADC2 and GPIOA(0-3)

17.8.2 Block Diagram

The following figure shows the block diagram of the TPADC.

Figure 17-31 TPADC Block Diagram



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17.8.3 External Signals

The following table describes the external signals of the TPADC. The TPADC pin is the analog input signal.

Table 17-37 TPADC External Signals

Signal Name	Description	Type
TP-X1	Touch Panel X1 Input	AI
TP-X2	Touch Panel X2 Input	AI
TP-Y1	Touch Panel Y1 Input	AI
TP-Y2	Touch Panel Y2 Input	AI

17.8.4 Functional Description

17.8.4.1 Single-ended Mode and Differential Mode

The controller is a typical type of successive approximation ADC (SAR ADC) which contains a sample/hold, analog-to-digital conversion, serial data output functions.

The analog inputs (X+, X-, Y+, Y-) enter the ADC through the control register, the ADC can work in single-ended or differential mode. Selecting Aux ADC should work in single-ended mode; for a touch screen application, it works in a differential mode, which can effectively eliminate the impact on conversion accuracy caused by the parasitic resistance of the driver switch and external interference.

The following figure shows TPADC Single-Ended Mode for the measurement of Aux, using the 1.8V reference source as the ADC reference voltage.

Figure 17-32 TPADC Single-Ended Mode for AUX ADC

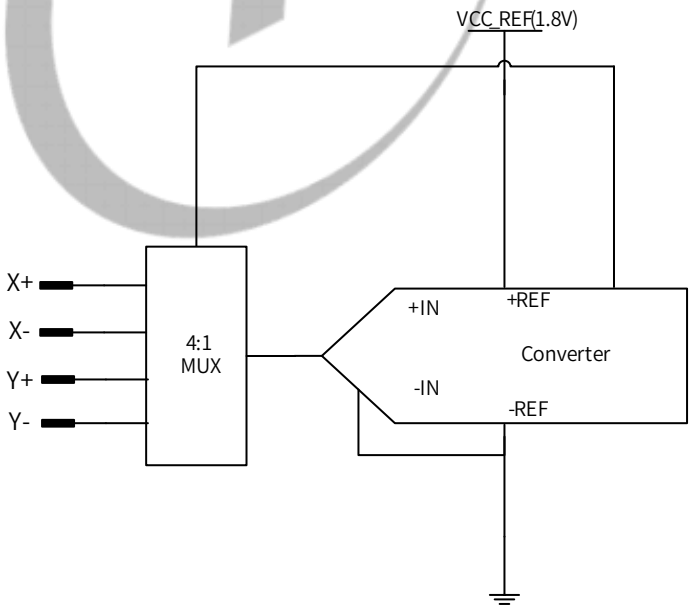

Figure 17-33 shows TPADC differential mode for the measurement of X/Y/Z coordinate of Touch Panel. The advantage of differential mode: +REF and –REF can directly input to the Y+ and Y-(or X+ and X-), which can eliminate the measurement error of X+/X-(or Y+/Y-) because of the switch

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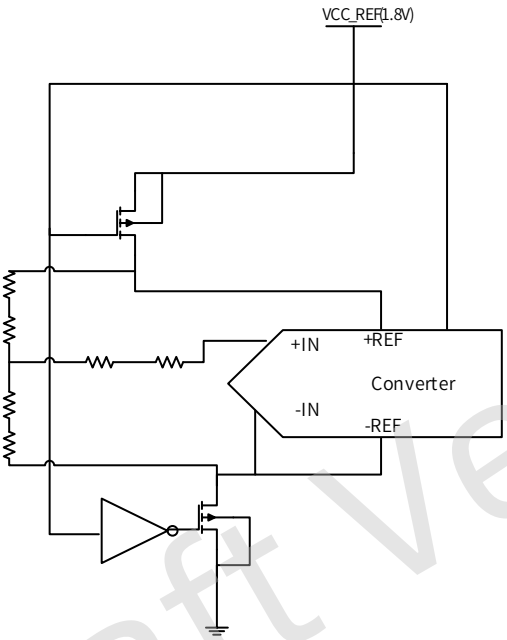
on resistance. The disadvantage is that: both the sample or conversion process, the driver will need to be enabled. Compared with single-ended mode, the power consumption increases.

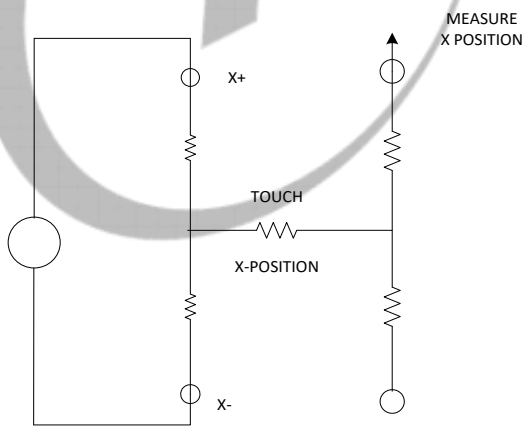
Figure 17-33 TPADC Differential Mode for Touch Panel



17.8.4.2 Single Touch Measurement

The following figure shows the operation principle of the single touch X-Coordinate measurement.

Figure 17-34 Single Touch X-Coordinate Measurement for Touch Panel



For an X coordinate measurement, the X+ pin is internally switched to VCC_REF and X- to GND. The X plate becomes a potential divider, and the voltage at the point of contact is proportional to its X co-ordinate. This voltage is measured on the Y+, which carries no current (hence there is no voltage drop in RY+ or RY-). Due to the ratio metric measurement method, the supply voltage does not affect measurement accuracy. The voltage references VREF+ and VREF- are taken from

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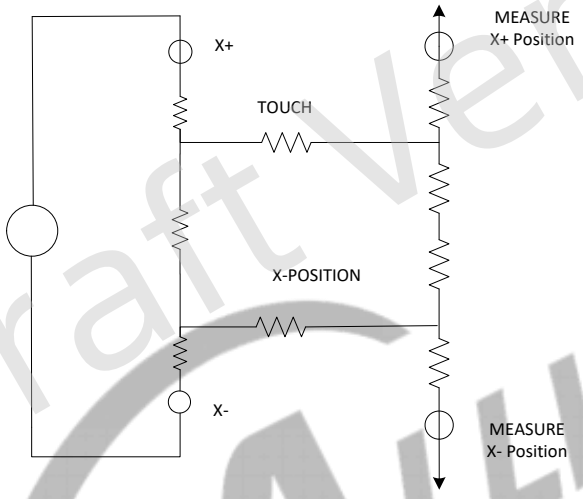
after the matrix switches, so that any voltage drop in these switches has no effect on the ADC measurement.

Y coordinate measurements are similar to X coordinate measurements, with the X and Y plates interchanged.

17.8.4.3 Dual Touch Measurement

In single touch mode, it only needs to test X+ and Y+ signals. But in dual touch mode, it needs to test X+, X-, Y+, and Y- signals. The following figure shows the operation principle of dual touch detection for touch panel.

Figure 17-35 Dual Touch Detection for Touch Panel



For X coordinates measurement, the X+ pin is internally switched to 3 V and X- to GND. The controller needs to test Y+ and Y-, Y coordinates measurement is similar. And record ぱX=|X+ - X-|, ぱY= | Y+ - Y-|. In practice, we can set a threshold. If ぱX or ぱY greater than the threshold, we consider it as a dual touch, otherwise as a single touch.

17.8.4.4 Touch Pressure Measurement

The pressure applied to the touch screen by a pen or finger to filter unavailable can also be measured by the controller using some simple calculations. The contact resistance between the X and Y plates is measured, which provides a good indication of the size of the depressed area and the applied pressure. The area of the touch spot t is proportional to the size of the object touching it. And the value of this resistance (Rtouch) can be calculated using two different methods.

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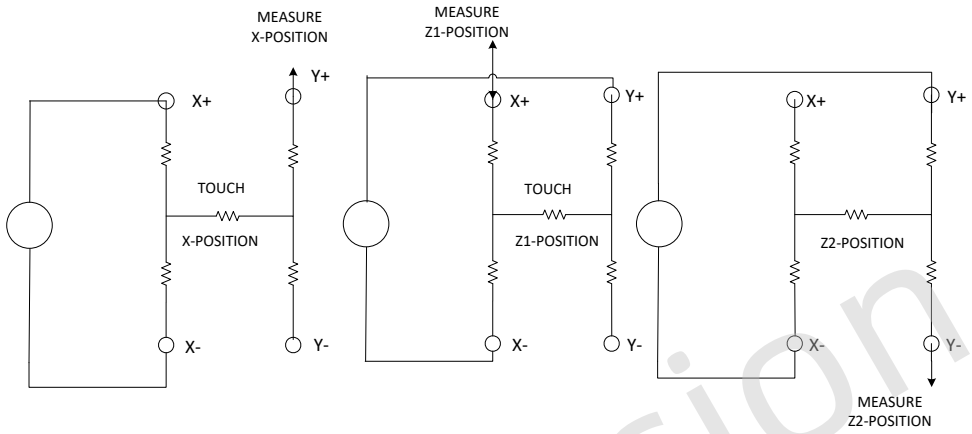
Figure 17-36 Touch Pressure Measurement for Touch Panel



- First Method

The first method requires the user to know the total resistance of the X plate tablet (R_{XPLATE}). Three touch screen conversions are required: measurement of the X position, XPOSITION(Y+ input); measurement of the X+ input with the excitation voltage applied to Y+ and X– (Z1 measurement); and measurement of the Y– input with the excitation voltage applied to Y+ and X– (Z2 measurement). These three measurements are illustrated in following Figure. The controller have two special ADC channel settings to configure the X and Y switches for the Z1 and Z2 measurements and store the results in the Z1 and Z2 result registers. The touch resistance (R_{TOUCH}) can then be calculated using the following equation.

$R_{TOUCH} = (R_{XPLATE}) \times (X_{POSITION}/4096) \times [(Z2/Z1) - 1]$

- Second Method

The second method requires the user to know the resistance of the X-plate and Y-plate tablets. Three touch screen conversions are required: a measurement of the X position ($X_{POSITION}$), the Y position ($Y_{POSITION}$), and the Z1 position. The following equation also calculates the touch resistance (R_{TOUCH}).

$R_{TOUCH} = R_{XPLATE} \times (X_{POSITION}/4096) \times [(4096/Z1) - 1] - RYPLATE \times [1 - (Y_{POSITION}/4096)]$

17.8.4.5 Pen Down Detection

Pen down detection is used as an interrupt to the host. RIRQ is an internal pull-up resistor with a programmable value from 6 kΩ to 96 kΩ (default 48 kΩ).

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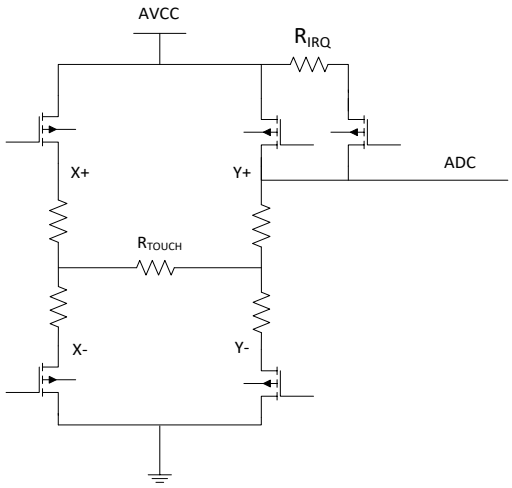
Figure 17-37 Pen Down Detection for Touch Panel



The pen down IRQ output is pulled high by an internal pull-up. In the pen down detection, the Y– driver is enabled and connected to GND, and the pen down IRQ output is connected to the X+ input. When the panel is touched, the X+ input is pulled to ground through the touch screen, and the pen down IRQ output goes low because of the current path through the panel to GND, initiating an interrupt to the processor.

During the measurement cycle for X-, Y-, and Z-position, the X+ input is disconnected from the pen down IRQ pull-down transistor to eliminate any pull-up resistor leakage current from flowing through the touch screen, thus causing no errors.

17.8.4.6 Median and Averaging Filter

Touch screens are composed of two resistive layers, normally placed over an LCD screen. Because these layers are in close proximity to the LCD screen, noise can be coupled from the screen onto these resistive layers, causing errors in the touch screen positional measurements.

The controller contain a filtering block to process the data and discard the spurious noise before sending the information to the host. The purpose of this block is not only the suppression of noise; the on-chip filtering also greatly reduces the host processing loading.

The processing function consists of two filters that are applied to the converted results: the median filter and the averaging filter. The median filter suppresses the isolated out-of-range noise and sets the number of measurements to be taken. These measurements are arranged in a temporary array, where the first value is the smallest measurement and the last value is the largest measurement. Then the averaging filter size determines the number of values to average. There are four choices which is configured by TP_CTRL3 register (bit 1 and bit 0) to filtrate the ADC sampling data.

Table 17-38 Median and Averaging Filter Size

Bit1	Bit0	Averaging Filter Size	Median Filter Size
0	0	2	4
0	1	3	5
1	0	4	8

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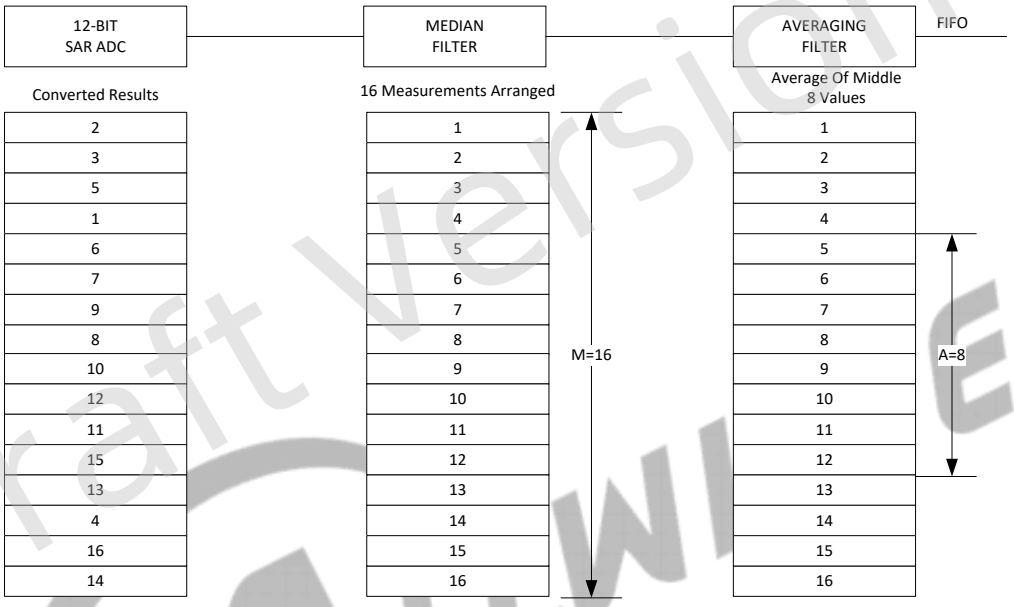
Bit1	Bit0	Averaging Filter Size	Median Filter Size
1	1	8	16

Example: In this example, the bit[1:0] of TP_CTRL_REG3 is configured as 2'b11. So the median filter has a window size of 16. This means that 16 measurements are taken and arranged in descending order in a temporary array. The averaging window size in this example is 8. The output is the average of the middle eight values of the 16 measurements taken with the median filter.

Table 17-39 Median and Averaging Filter Example



17.8.4.7 A/D Conversion Time

When the clock source is 24MHz and the prescalar value is 6, the conversion time calculation formula is as follows:

CLK_IN = 24MHz/6 = 4MHz

Conversion Time = 1 / (CLK_IN /13Cycles) = 3.25us.

FS_TIME (Frequency Scan Time) is calculated bases on TACQ and Touch Mode, they must meet the following in equation:

FS_TIME >= M*(TACQ + Conversion Time)

For example, if touch acquire time divider is 15, then TACQ = 4MHz /(16*(15+1)) = 64us.

When TP mode is in dual and pressure measurement mode, then M=6, and the FS_TIME must be great than or equal to 6*(64 + 3.25) us.

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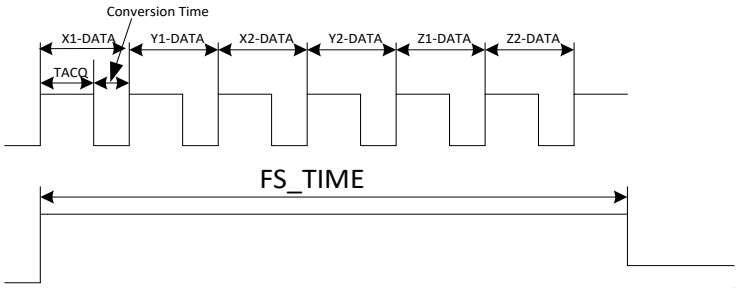
Figure 17-38 Dual Touch and Pressure Measurement



17.8.5 Register List

Module Name	Base Address
TPADC	0x0208C000

Register Name	Offset	Description
TP_CTRL_REG0	0x0000	TP Control Register 0
TP_CTRL_REG1	0x0004	TP Control Register 1
TP_CTRL_REG2	0x0008	TP Control Register 2
TP_CTRL_REG3	0x000C	TP Control Register 3
TP_INT_FIFO_CTRL_REG	0x0010	TP Interrupt FIFO Control Register
TP_INT_FIFO_STAT_REG	0x0014	TP Interrupt FIFO Status Register
TP_CALI_DATA_REG	0x001C	TP Calibration Data Register
TP_DATA_REG	0x0024	TP Data Register

17.8.6 Register Description

17.8.6.1 0x0000 TP Control Register 0 (Default Value:0x0F80_0000)

Offset: 0x0000			Register Name: TP_CTRL_REG0
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0xF	ADC_FIRST_DLY ADC First Convert Delay Time (T_FCDT) Setting Based on ADC first convert delay mode select (Bit 23) T_FCDT = ADC_FIRST_DLY * ADC_FIRST_DLY_MODE
23	R/W	0x1	ADC_FIRST_DLY_MODE ADC First Convert Delay Mode Select 0: CLK_IN/16 1: CLK_IN/16*256
22	/	/	/

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Offset: 0x0000			Register Name: TP_CTRL_REG0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x0	ADC_CLK_DIVIDER ADC Clock Divider (CLK_IN) 00: CLK/2 01: CLK/3 10: CLK/6 11: CLK/1
19:16	R/W	0x0	FS_DIV ADC Sample Frequency Divider 0000: CLK_IN/$2^{(20-0)}$ 0001: CLK_IN/$2^{(20-1)}$ 0010: CLK_IN/$2^{(20-2)}$ .... 1111: CLK_IN/$2^{(20-15)}$
15:0	R/W	0x0	TACQ Touch panel ADC acquire time CLK_IN/(16*(N+1))

17.8.6.2 0x0004 TP Control Register 1 (Default Value:0x0000_0101)

Offset: 0x0004			Register Name: TP_CTRL_REG1
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:12	R/W	0x0	STYLUS_UP_DEBOUNCE Stylus Up De-bounce Time Setting 0x00: 0 .... 0xFF: 2N*(CLK_IN/16*256)
11:10	/	/	/
9	R/W	0x0	STYLUS_UP_DEBOUCE_EN Stylus Up Debounce Function Select 0: Disable 1: Enable
8	R/W	0x1	CHOPPER_EN T-sensor Chopping Enable 0: Disable 1: Enable This field is not used when there is no T-sensor in TPADC
7	R/WAC	0x0	TOUCH_PAN_CALI_EN Touch Panel Calibration 1: Start calibration, it is cleared to 0 after calibration

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Offset: 0x0004			Register Name: TP_CTRL_REG1
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	TP_DUAL_EN Touch Panel Double Point Enable 0: Disable 1: Enable
5	R/W	0x0	TP_EN. TP Function Enable 0: Disable 1: Enable
4	R/W	0x0	TP_MODE_SELECT. Touch Panel Mode and Auxiliary ADC Mode Select 0: TP 1: Auxiliary ADC
3	R/W	0x0	ADC_CHAN3_SELECT Analog Input Channel 3 Select 0: Disable 1: Enable
2	R/W	0x0	ADC_CHAN2_SELECT Analog Input Channel 2 Select 0: Disable 1: Enable
1	R/W	0x0	ADC_CHAN1_SELECT Analog Input Channel 1 Select 0: Disable 1: Enable
0	R/W	0x1	ADC_CHAN0_SELECT Analog Input Channel 0 Select 0: Disable 1: Enable

CHAN0–3 can be selected at the same time. If N channel(s) (N=1 to 4) is selected, each channel has 1/N full speed of the ADC. If only one channel is selected, it has the full conversion rate. CHAN0–3 correspond to TP_YN, TP_YP, TP_XN, TP_XP.

17.8.6.3 0x0008 TP Control Register 2 (Default Value:0x8000_0FFF)

Offset: 0x0008			Register Name: TP_CTRL_REG2
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0008			Register Name: TP_CTRL_REG2
Bit	Read/Write	Default/Hex	Description
31:29	R/W	0x4	TP_SENSITIVE_ADJUST Internal Pull-up Resistor Control 0000: Least sensitive ..... 1111: Most sensitive This field is used to adjust sensitivity of pen down detection.
28	R/W	0x0	TP_FALLBACK_SELECT TP Sample Timing Fallback Enable 0: No fallback 1: Fallback to old sample timing
27:26	R/W	0x0	TP_FIFO_MODE_SELECT TP FIFO Access Data Mode Select 00: FIFO store X1,Y1 data for single touch no pressure mode 01: FIFO store X1,Y1, △X, △Y data for dual touch no pressure mode 10: FIFO store X1,Y1, X2,Y2 data for dual touch no pressure mode 11: FIFO store X1,Y1, X2,Y2,Z1,Z2 data for dual touch and pressure mode The ADC output data in single touch mode can store in FIFO with TP_FIFO_MODE_SELECT configured as 01,10,11. The data △X and △Y are theoretically equal to 0, and X2,Y2 are equal to X1,Y1. When PRE_MEA_EN is set and TP_FIFO_MODE_SELECT is not configured as 2'b11, X and Y data will not be stored unless x1*(z2-z1)/z1 < PRE_MEA_THRE_CNT. Z data will always be zero when TP_FIFO_MODE_SELECT is configured as 2'b11 but PRE_MEA_EN is not set.
25	/	/	/
24	R/W	0x0	PRE_MEA_EN TP Pressure Measurement Enable Control 0: Disable 1: Enable
23:0	R/W	0xFFF	PRE_MEA_THRE_CNT TP Pressure Measurement Threshold Control 0x000000: Least sensitive 0xFFFFFF: Most sensitive This field is used to adjust sensitivity of touch.

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17.8.6.4 0x000C TP Control Register 3 (Default Value:0x0000_0001)

Offset: 0x000C			Register Name: TP_CTRL_REG3
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	FILTER_EN Filter Enable 0: Disable 1: Enable
1:0	R/W	0x1	FILTER_TYPE Filter Type 00: 4/2 01: 5/3 10: 8/4 11: 16/8

17.8.6.5 0x0010 TP Interrupt& FIFO Control Register (Default Value:0x0000_0F00)

Offset: 0x0010			Register Name: TP_INT_FIFO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	TP_OVERRUN_IRQ_EN TP FIFO Overrun IRQ Enable 0: Disable 1: Enable
16	R/W	0x0	TP_DATA_IRQ_EN TP FIFO Data Available IRQ Enable 0: Disable 1: Enable
15:14	/	/	/
13	R/W	0x0	TP_DATA_XY_CHANGE TP FIFO X,Y Data Interchange Function Select 0: Disable 1: Enable
12:8	R/W	0xF	TP_FIFO_TRIG_LEVEL TP FIFO Data Available Trigger Level Interrupt and DMA request trigger level for TP or Auxiliary ADC Trigger Level = TXTL + 1
7	R/W	0x0	TP_DATA_DRQ_EN TP FIFO Data Available DRQ Enable 0: Disable 1: Enable

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Offset: 0x0010			Register Name: TP_INT_FIFO_CTRL_REG
Bit	Read/Write	Default/Hex	Description
6:5	/	/	/
4	R/WAC	0x0	TP_FIFO_FLUSH TP FIFO Flush Write '1' to flush TX FIFO, self clear to '0'
3:2	/	/	/
1	R/W	0x0	TP_UP_IRQ_EN Touch Panel Last Touch (Stylus Up) IRQ Enable 0: Disable 1: Enable
0	R/W	0x0	TP_DOWN_IRQ_EN Touch Panel First Touch (Stylus Down) IRQ Enable 0: Disable 1: Enable

17.8.6.6 0x0014 TP Interrupt& FIFO Status Register (Default Value:0x0000_0000)

Offset: 0x0014			Register Name: TP_INT_FIFO_STAT_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W1C	0x0	FIFO_OVERRUN_PENDING TP FIFO Overrun IRQ pending 0: No Pending IRQ 1: FIFO Overrun Pending IRQ Write '1' to clear this interrupt or automatically clear if interrupt condition fails.
16	R/W1C	0x0	FIFO_DATA_PENDING TP FIFO Data Available Pending Bit 0: NO Pending IRQ 1: FIFO Available Pending IRQ Write '1' to clear this interrupt or automatically clear if FIFO flushed.
15:14	/	/	/
13:8	R	0x0	RXA_CNT TP FIFO Available Sample Word Counter
7:3	/	/	/
2	R	0x0	TP_IDLE_FLG Touch Panel Idle Flag 0: Idle 1: Not idle

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Offset: 0x0014			Register Name: TP_INT_FIFO_STAT_REG
Bit	Read/Write	Default/Hex	Description
1	R/W1C	0x0	TP_UP_PENDING Touch Panel Last Touch (Stylus Up) IRQ Pending bit 0: No IRQ 1: IRQ Writing 1 to the bit clears it and its corresponding interrupt if the interrupt is enabled.
0	R/W1C	0x0	TP_DOWN_PENDING Touch Panel First Touch (Stylus Down) IRQ Pending bit 0: No IRQ 1: IRQ Writing 1 to the bit clears it and its corresponding interrupt if the interrupt is enabled.

17.8.6.7 0x001C TP Calibration Data Register (Default Value:0x0000_0800)

Offset: 0x001C			Register Name: TP_CALI_DATA_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R/W	0x800	TP_CDAT TP Common Data It is used to adjust the tolerance of the internal ADC.

17.8.6.8 0x0024 TP Data Register (Default Value:0x0000_0000)

In touch panel mode, the data stored in this register bases on TP_FIFO_MODE_SELECT. In Auxiliary ADC mode, the data stored in this register bases on ADC_CHAN_SELECT. If four channels are enabled, FIFO will access the input data in successive turn (ADC_CHAN0 -> ADC_CHAN1 -> ADC_CHAN2 -> ADC_CHAN3). If only two or three channels are selected, such as ADC_CHAN0 and ADC_CHAN3, firstly ADC_CHAN0 input data is accessed, then ADC_CHAN3 input data.

Offset: 0x0024			Register Name: TP_DATA_REG
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:0	R	0x0	TP_DATA Touch Panel X ,Y data or Auxiliary analogy input data converted by the internal ADC.

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17.9 GPIO

17.9.1 Overview

The General Purpose Input/Output (GPIO) is one of the blocks controlling the chip multiplexing pins. Each pin can be configured as input or output and these pins are used to generate input signals or output signals for special purposes.

The GPIO has the following features:

- 10 groups of ports (PA,PB,PC,PD,PE,PF,PG,PJ,PK, RTC_GPIO)
- Software control for each signal pin
- Each GPIO peripheral can produce an interrupt
- Pull-up/Pull-down/no-Pull register control
- Control the direction of every signal
- 4 drive strengths in each operating mode
- Up to 141 interrupts
- Configurable interrupt mode: Positive Edge/ Negative Edge/ High Level /Low Level /Double Edge(Positive/Negative)
- 32kHz/24MHz/60MHz interrupt sampling clock and 1/2/4/8/16/32/64/128 frequency division factor
- GPIO arbitrary grouping function (maximum 4 groups)
- Supports atomic operation and supports “0” or “1” output per individual I/O pin.
- Supports interrupt partition function

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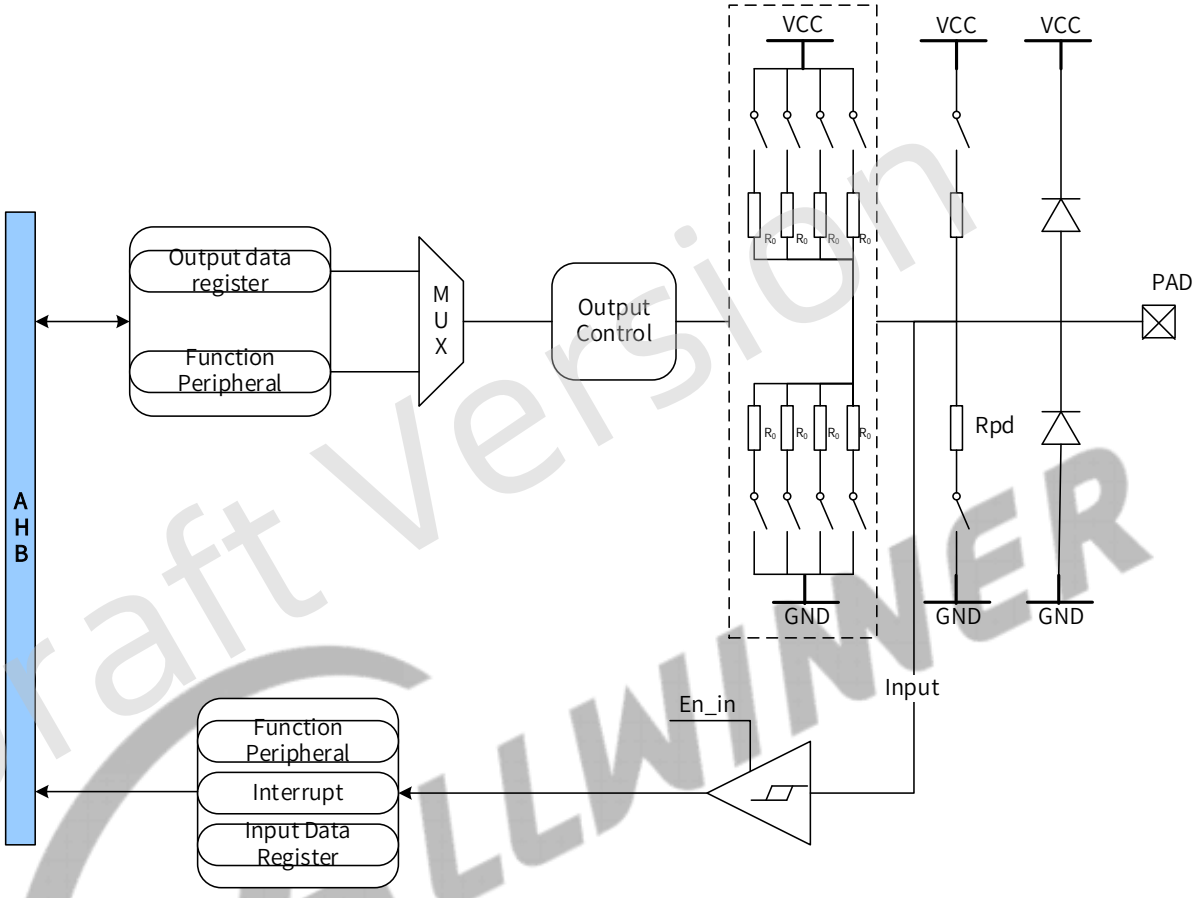
17.9.2 Block Diagram

The following figure shows the block diagram of the GPIO.

Figure 17-39 GPIO Block Diagram



The overall I/O ports consists of the digital part (GPIO, peripheral interfaces) and IO analog part (output buffer, dual pull down, pad). The output of digital part can be selected by MUX, and the analog part can be used to configure pull-up/pull-down registers, driving strengths and so on.

When in the read operation, read the current pin level to the bus of internal registers. When not in the read operation, the external pins aren't connected to the bus of internal registers, which is called the high-impedance status.

17.9.3 Functional Description

17.9.3.1 Port Function

Every GPIO can be configured as Input, Output, Function Peripheral, I/O disable or Interrupt function. The configuration instruction of every function is as follows.

Table 17-40 Port Function

Category	Function	Buffer Strength	Pull Up	Pull Down
Input	GPIO/Multiplexing Input	/	X	X

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Category	Function	Buffer Strength	Pull Up	Pull Down
Output	GPIO/Multiplexing Output	Y	X	X
Disable	Pull Up	/	Y	N
	Pull Down	/	N	Y
Interrupt	Trigger	/	X	X

NOTE

/: Configuration is unnecessary and invalid.

Y: Configuration is necessary. If no configuration, it will be configured to the default status.

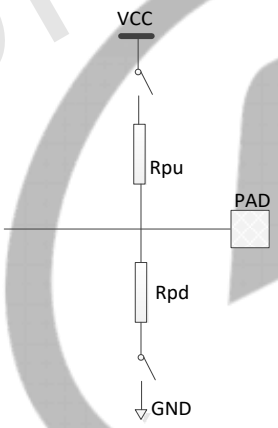
X: Configuration is selected according to the actual situation

N: Configuration is forbidden.

17.9.3.2 Pull-up, Pull-down, and High-impedance Input

Each IO can be configured as the internal pull-up, pull-down, or high-impedance input mode.

Figure 17-40 Pull-up, Pull-down, and High-impedance Input Mode



High-impedance: The output port is in the floating status. All the drives are turned off, and their levels are depended by the external levels.

Pull-up: The uncertain signals are pulled high by resistors, which performs the current-limiting function. In pull-up status, the IO is pulled up to VCC by Rpu.

Pull-down: The uncertain signals are pulled low by resistors. In pull-down status, the IO is pulled down to GND by Rpd.

NOTE

The pull-up/down resistor value of each IO is around 100K.

Whether pull-down, pull-up or high-impedance input is decided by the external circuit.

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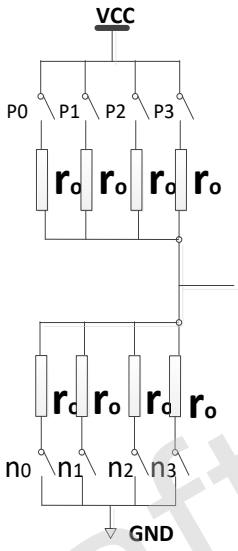
17.9.3.3 Output Driving Strength

Each IO can be set as different output driving strength. The IO driving strength diagram is as follows.

Figure 17-41 IO Buffer Strength Diagram



High Level Output

In the process of high level output, the n0, n1, n2, n3 of NMOS is off, and the p0, p1, p2, p3 of PMOS is on.

When the driving strength is set to 0, only the p0 is turned on, during which the output impedance is maximum, that is, the driving capability is the weakest. The output impedance is the turned-on r0.

When the driving strength is set to 1, only the p0 and p1 are turned on. The output impedance is r0/2, equivalent to the parallel-connected two r0.

When the driving strength is set to 2, three MOS of p0-p3 are turned on. The output impedance is r0/3, equivalent to the parallel-connected three p0.

When the driving strength is set to 3, p0-p3 are turned on. The output impedance is r0/4, equivalent to the parallel-connected four p0.

Low Level Output

In the process of low level output, the p0, p1, p2, p3 of PMOS is off, and the n0, n1, n2, n3 of NMOS is on.

When the driving strength is set to 0, only the n0 is turned on, during which the output impedance is maximum, that is, the driving capability is the weakest. The output impedance is the turned-on r0.

When the driving strength is set to 1, only the n0 and n1 are turned on. The output impedance is r0/2, equivalent to the parallel-connected two r0.

When the driving strength is set to 2, three MOS of n0-n3 are turned on. The output impedance is r0/3, equivalent to the parallel-connected three r0.

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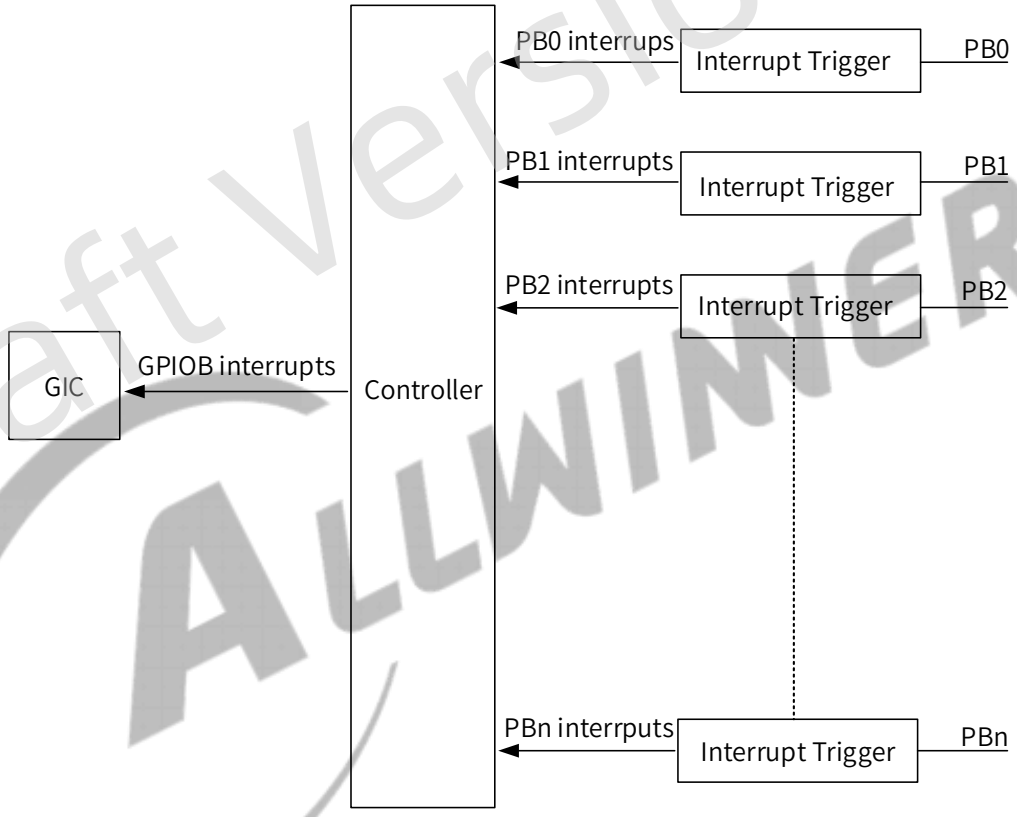
When the driving strength is set to 3, n0-n3 are turned on. The output impedance is r0/4, equivalent to the parallel-connected four r0.

17.9.3.4 Interrupts

There is an interrupt number for each group of IO, in which all IOs share one interrupt number. When an IO generates interrupts, the collector will send interrupt request to GIC. The user can query which IO generates interrupt in the external interrupt status registers. The following interrupt diagram takes PB as an example.

Figure 17-42 GPIOB Interrupt Diagram



The interrupt trigger of GPIO supports the following trigger types.

- **Positive Edge**: When a low level changes to a high level, the IO will generate interrupts.
- **Negative Edge**: When a high level changes to a low level, the IO will generate interrupts.
- **High Level**: The IO will generate interrupts in the high level.
- **Low Level**: The IO will generate interrupts in the low level.
- **Double Edge**: The IO will generate interrupts both in the positive edge and negative edge.

The interrupt trigger types can be configured in the external interrupt configure registers.

The GPIO interrupts support hardware debounce function configuration in the external interrupt debounce Registers. In this way, the trigger signal will be sampled with a lower-frequency

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sample clock. As the dither frequency of signals is higher than their sample frequency, the debounce effect is reached.

The sample clock source can be set in the GPIO_INT_CLK_SELECT, and the prescale factor can be set in DEB_CLK_PRE_SCALE.

17.9.3.5 Multi-function Port

This device includes 141 multi-functional input/output port pins. There are 10 ports as listed below.

Table 17-41 Multi-function Port

Port Name	Number of Pins	Input Driver	Output Driver	Multiplex Pins	Typical Power Supply
PA	24	Schmitt	CMOS	GPADC/ TPADC/ LBUS/ RGMII/ RMII/ NCSI/PWMCS/LCD/ UART/ TWI/ SPI	3.3V/1.8V
PB	15	Schmitt	CMOS	UART/ SPI/ JTAG/ PWM/ RJTAG/ RGMII/ RMII/ TWI/ I2S/ IR-RX/IR-TX/ OWA/ CAN/ CLK-FANOUT	3.3V
PC	12	Schmitt	CMOS	SDC/ SPI/ SPIF/ BOOT/ CAN/ UART	3.3V/1.8V
PD	24	Schmitt	CMOS	LCD/ LVDS/ DSI/ PWM/ UART/ LBUS/ PWMCS/ RGMII/ RMII/ CAN/ DMIC/ SPI/ DBI/ TWI/ CSI	3.3V/1.8V
PE	10	Schmitt	CMOS	MCSI/ NCSI/ CLK-FANOUT/ DMIC/ SDC/ TWI/ SPI/ PWM/ UART/ TCON	3.3V/1.8V
PF	7	Schmitt	CMOS	SDC/ JTAG/ UART/ RJTAG/ IR-RX/IR-TX / LBUS/ DMIC/ SPI	3.3V/1.8V
PG	16	Schmitt	CMOS	SDC/ UART/ RGMII/ RMII/ CAN/ TWI/ I2S/ DMIC/ LBUS	3.3V/1.8V
PJ	16	Schmitt	CMOS	PWM/ I2S/ TWI/ RGMII/ RMII/ DMIC/ UART/ LEDC/ IR-RX/IR-TX /CAN	3.3V/1.8V
PK	12	Schmitt	CMOS	MCSI/ UART/ TWI/ NCSI/ LBUS/ I2S/ LCD/ SPI/ RGMII/ RMII/ CLK/ PWM	3.3V/1.8V
RTC_GPIO	5	Schmitt	CMOS	NMI	1.8V

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The following tables show the multiplex function pins.

NOTE

For each GPIO, Function0 is input function; Function1 is output function; Function12 to Function13 are reserved.

Table 17-42 Port A Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PA0	I/O	GPADC2-0	TP-X1			LBUS-READY3	LBUS-LBE2					PA-EINT0
PA1	I/O	GPADC2-1	TP-X2			LBUS-READY2	LBUS-DP1					PA-EINT1
PA2	I/O	GPADC2-2	TP-Y1			LBUS-DP2						PA-EINT2
PA3	I/O	GPADC2-3	TP-Y2			LBUS-DP3						PA-EINT3
PA4	I/O	GPADC0-0		RGMII0-RXD1/RMII0-RXD1	NCSI-HSYNC	LBUS-CLK						PA-EINT4
PA5	I/O	GPADC0-1		RGMII0-RXD0/RMII0-RXD0	NCSI-VSYNC	LBUS-ALE						PA-EINT5
PA6	I/O	GPADC0-2		RGMII0-RXCTL/RMII0-CRS-DV	NCSI-MCLK	LBUS-CS1	PWMCS-I8		LCD-D0			PA-EINT6
PA7	I/O	GPADC0-3		RGMII0-CLKIN/RMII0-RXER	NCSI-PCLK	LBUS-CS0	PWMCS-I9		LCD-D1			PA-EINT7
PA8	I/O	GPADC0-4		RGMII0-TXD1/RMII0-TXD1	NCSI-D0	LBUS-DP0	PWMCS-IA		LCD-D8			PA-EINT8
PA9	I/O	GPADC0-5		RGMII0-TXD0/RMII0-TXD0	NCSI-D1	LBUS-DP1	PWMCS-IB	LBUS-LBE3	LCD-D9			PA-EINT9
PA10	I/O	GPADC0-6		RGMII0-TXCK/RMII0-TXCK	NCSI-D2	LBUS-LBE0	PWMCS-IC		LCD-D16			PA-EINT10
PA11	I/O	GPADC0-7		RGMII0-TXCTL/RMII0-TXEN	NCSI-D3	LBUS-LBE1	PWMCS-ID		LCD-D17			PA-EINT11
PA12	I/O	GPADC0-8		RGMII0-MDC	NCSI-D4	LBUS-LD8	PWMCS-IE					PA-EINT12
PA13	I/O	GPADC0-9		RGMII0-MDIO	NCSI-D5	LBUS-LD9	PWMCS-IF					PA-EINT13
PA14	I/O	GPADC1-0		RGMII0-EPHY-25/50M	NCSI-D6	LBUS-LD10	UART6-TX					PA-EINT14
PA15	I/O	GPADC1-1		RGMII0-RXD3/RMII0-NULL	NCSI-D7	LBUS-LD11	UART6-RX					PA-EINT15
PA16	I/O	GPADC1-2		RGMII0-RXD2/RMII0-NULL	NCSI-D8	LBUS-LD12	TWI3-SCK					PA-EINT16
PA17	I/O	GPADC1-3		RGMII0-RXCK/RMII0-NULL	NCSI-D9	LBUS-LD13	TWI3-SDA					PA-EINT17
PA18	I/O	GPADC1-4		RGMII0-TXD3/RMII0-NULL	NCSI-D10	LBUS-LD14						PA-EINT18
PA19	I/O	GPADC1-5		RGMII0-TXD2/RMII0-NULL	NCSI-D11	LBUS-LD15						PA-EINT19
PA20	I/O	GPADC1-6			NCSI-D12	LBUS-WAIT						PA-EINT20
PA21	I/O	GPADC1-7			NCSI-D13	LBUS-CS3	SPI2-CS1					PA-EINT21
PA22	I/O	GPADC1-8			NCSI-D14	LBUS-CS2						PA-EINT22
PA23	I/O	GPADC1-9			NCSI-D15	LBUS-DE						PA-EINT23

Table 17-43 Port B Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PB0	I/O	UART2-TX	SPI3-CS0	JTAG-MS	PWM0-6		RJTAG-MS		RGMII2-RXD1/RMII2-RXD1			PB-EINT0
PB1	I/O	UART2-RX	SPI3-CLK	JTAG-CK	PWM0-7		RJTAG-CK		RGMII2-RXD0/RMII2-RXD0			PB-EINT1
PB2	I/O	UART2-RTS	SPI3-MOSI	JTAG-DO	TWI2-SCK		RJTAG-DO		RGMII2-RXCTL/RMII2-CRS-DV			PB-EINT2
PB3	I/O	UART2-CTS	SPI3-MISO	JTAG-DI	TWI2-SDA		RJTAG-DI		RGMII2-CLKIN/RMII2-RXER			PB-EINT3
PB4	I/O	TWI1-SCK	I2S0-MCLK	SPI3-CS1	PWM1-0	IR0-RX	UART8-TX		RGMII2-TXD1/RMII2-TXD1			PB-EINT4
PB5	I/O	TWI1-SDA	I2S0-BCLK	SPI3-CS2	PWM1-1	IR1-RX	UART8-RX		RGMII2-TXD0/RMII2-TXD0			PB-EINT5
PB6	I/O		I2S0-LRCK		PWM1-2	IR-TX	UART8-RTS	TWI0-SCK	RGMII2-TXCK/RMII2-TXCK			PB-EINT6
PB7	I/O	OWA-IN	I2S0-DOUT0	I2S0-DIN1	PWM1-3	CAN1-TX0	UART8-CTS	TWI0-SDA	RGMII2-TXCTL/RMII2-TXEN			PB-EINT7

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Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PB8	I/O	OWA-OUT	I2S0-DIN0	I2S0-DOUT1	PWM0-0	CAN1-RX0	UART8-DCD					PB-EINT8
PB9	I/O	UART0-TX	I2S0-DIN2	I2S0-DOUT2		TWI0-SCK	UART8-DSR					PB-EINT9
PB10	I/O	UART0-RX	I2S0-DIN3	I2S0-DOUT3	PWM0-1	TWI0-SDA	UART8-DTR					PB-EINT10
PB11	I/O	TWI5-SCK	UART7-RTS	SPI3-CS0	PWM0-2		UART8-RI		RGMII2-MDC			PB-EINT11
PB12	I/O	TWI5-SDA	UART7-CTS	SPI3-CLK	PWM0-3				RGMII2-MDIO			PB-EINT12
PB13	I/O	TWI4-SCK	UART7-TX	SPI3-MOSI	PWM0-4	IR2-RX	CLK-FANOUT0		RGMII2-EPHY-25/50M			PB-EINT13
PB14	I/O	TWI4-SDA	UART7-RX	SPI3-MISO	PWM0-5	IR3-RX	CLK-FANOUT1					PB-EINT14

Table 17-44 Port C Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PC0	I/O	SDC2-CLK	SPI0-CLK	SPIF0-CLK								PC-EINT0
PC1	I/O	SDC2-CMD	SPI0-CS0	SPIF0-CS0								PC-EINT1
PC2	I/O	SDC2-D2	SPI0-MOSI	SPIF0-MOSI	BOOT-SEL0							PC-EINT2
PC3	I/O	SDC2-D1	SPI0-MISO	SPIF0-MISO	BOOT-SEL1							PC-EINT3
PC4	I/O	SDC2-D0	SPI0-WP	SPIF0-WP	CAN1-TX0							PC-EINT4
PC5	I/O	SDC2-D3	SPI0-HOLD	SPIF0-HOLD	CAN1-RX0							PC-EINT5
PC6	I/O	SDC2-D4	SPI0-CS1	SPIF0-D4	CAN0-TX0	UART6-TX						PC-EINT6
PC7	I/O	SDC2-D5		SPIF0-D5	CAN0-RX0	UART6-RX						PC-EINT7
PC8	I/O	SDC2-D6		SPIF0-D6	UART5-CTS							PC-EINT8
PC9	I/O	SDC2-D7		SPIF0-D7	UART5-TX							PC-EINT9
PC10	I/O	SDC2-DS		SPIF0-DQS	UART5-RX							PC-EINT10
PC11	I/O	SDC2-RST			UART5-RTS							PC-EINT11

Table 17-45 Port D Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PD0	I/O	LCD-D2	LVDS0-D0P	DSI-D0P	PWM0-0	UART2-CTS	UART9-TX	LBUS-LD20	PWMCS-O0		RGMII2-RXD1/RMII2-RXD1	PD-EINT0
PD1	I/O	LCD-D3	LVDS0-D0N	DSI-D0N	PWM0-1	UART2-RTS	UART9-RX	LBUS-LD21	PWMCS-O1		RGMII2-RXD0/RMII2-RXD0	PD-EINT1
PD2	I/O	LCD-D4	LVDS0-D1P	DSI-D1P	PWM0-2	UART2-TX	UART9-RTS	LBUS-LD22	PWMCS-O2		RGMII2-RXCTL/RMII2-CRS-DV	PD-EINT2
PD3	I/O	LCD-D5	LVDS0-D1N	DSI-D1N	PWM0-3	UART2-RX	UART9-CTS	LBUS-LD23	PWMCS-O3		RGMII2-CLKIN/RMII2-RXER	PD-EINT3
PD4	I/O	LCD-D6	LVDS0-D2P	DSI-CKP	PWM0-4	UART4-CTS	UART6-TX	LBUS-LD24	PWMCS-O4		RGMII2-TXD1/RMII2-TXD1	PD-EINT4
PD5	I/O	LCD-D7	LVDS0-D2N	DSI-CKN	PWM0-5	UART4-RTS	UART6-RX	LBUS-LD25	PWMCS-O5	LBUS-CS3	RGMII2-TXD0/RMII2-TXD0	PD-EINT5
PD6	I/O	LCD-D10	LVDS0-CKP	DSI-D2P	PWM0-6	UART4-TX	UART6-RTS	LBUS-LD26	PWMCS-O6	LBUS-CS2	RGMII2-TXCK/RMII2-TXCK	PD-EINT6
PD7	I/O	LCD-D11	LVDS0-CKN	DSI-D2N	PWM0-7	UART4-RX	UART6-CTS	LBUS-LD27	PWMCS-O7	LBUS-READY3	RGMII2-TXCTL/RMII2-TXEN	PD-EINT7
PD8	I/O	LCD-D12	LVDS0-D3P	DSI-D3P	PWM1-0		CAN1-TX0	LBUS-LD28	PWMCS-O8	LBUS-DE	RGMII2-MDC	PD-EINT8
PD9	I/O	LCD-D13	LVDS0-D3N	DSI-D3N	PWM1-1	DMIC-CLK	CAN1-RX0	LBUS-LD29	PWMCS-O9	LBUS-READY2	RGMII2-MDIO	PD-EINT9
PD10	I/O	LCD-D14	LVDS1-D0P	SPI1-CS0/DBI-CSX	PWM1-2	DMIC-DATA0	UART8-TX	LBUS-LD30	PWMCS-OA	LBUS-READY1		PD-EINT10
PD11	I/O	LCD-D15	LVDS1-D0N	SPI1-CLK/DBI-SCLK	PWM1-3	DMIC-DATA1	UART8-RX	LBUS-LD31	PWMCS-OB	LBUS-READY0		PD-EINT11
PD12	I/O	LCD-D18	LVDS1-D1P	SPI1-MOSI/DBI-SDO	PWM1-4	DMIC-DATA2	UART8-RTS	LBUS-LD0	PWMCS-OC			PD-EINT12
PD13	I/O	LCD-D19	LVDS1-D1N	SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX	PWM1-5	DMIC-DATA3	UART8-CTS	LBUS-LD1	PWMCS-OD			PD-EINT13
PD14	I/O	LCD-D20	LVDS1-D2P	SPI1-HOLD/DBI-DCX/DBI-WRX	PWM1-6	UART3-TX	UART8-DCD	LBUS-LD2	PWMCS-OE			PD-EINT14
PD15	I/O	LCD-D21	LVDS1-D2N	SPI1-WP/DBI-TE	PWM1-7	UART3-RX	UART8-DSR	LBUS-LD3	PWMCS-OF			PD-EINT15
PD16	I/O	LCD-D22	LVDS1-CKP		PWM2-0	UART3-RTS	UART8-DTR	LBUS-LD4	PWMCS-I0		RGMII2-EPHY-25/50M	PD-EINT16

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Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PD17	I/O	LCD-D23	LVDS1-CKN		PWM2-1	UART3-CTS	UART8-RI	LBUS-LD5	PWMCS-I1		RGMII2-RXD3/RMII2-NULL	PD-EINT17
PD18	I/O	LCD-CLK	LVDS1-D3P	TWI5-SCK	PWM2-2		CAN0-TX0	LBUS-LD6	PWMCS-I2		RGMII2-RXD2/RMII2-NULL	PD-EINT18
PD19	I/O	LCD-DE	LVDS1-D3N	TWI5-SDA	PWM2-3		CAN0-RX0	LBUS-LD7	PWMCS-I3		RGMII2-RXCK/RMII2-NULL	PD-EINT19
PD20	I/O	LCD-HSYNC	PWM2-4	UART3-TX	TWI5-SCK			LBUS-WR	PWMCS-I4		RGMII2-TXD3/RMII2-NULL	PD-EINT20
PD21	I/O	LCD-VSYNC	PWM2-5	UART3-RX	TWI5-SDA			LBUS-OE	PWMCS-I5		RGMII2-TXD2/RMII2-NULL	PD-EINT21
PD22	I/O		PWM2-6	UART3-RTS	TWI4-SCK	TWI0-SCK	CSI0-XHS	LBUS-DRQ0	PWMCS-I6			PD-EINT22
PD23	I/O		PWM2-7	UART3-CTS	TWI4-SDA	TWI0-SDA	CSI1-XHS	LBUS-DRQ1	PWMCS-I7			PD-EINT23

Table 17-46 Port E Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PE0	I/O	MCSI0-MCLK		NCSI-D8	CLK-FANOUT1		DMIC-CLK		SDC3-CLK			PE-EINT0
PE1	I/O	TWI1-SCK		NCSI-D9		SPI2-CS0	DMIC-DATA0					PE-EINT1
PE2	I/O	TWI1-SDA	PWM0-3	NCSI-D10		SPI2-CLK	DMIC-DATA1					PE-EINT2
PE3	I/O	TWI2-SCK	UART5-RTS	NCSI-D11	UART6-TX	SPI2-MOSI	DMIC-DATA2					PE-EINT3
PE4	I/O	TWI2-SDA	UART5-CTS	NCSI-D12	UART6-RX	SPI2-MISO	DMIC-DATA3					PE-EINT4
PE5	I/O	MCSI1-MCLK	UART5-TX	NCSI-D13	UART6-RTS	SPI2-WP			SDC3-CMD			PE-EINT5
PE6	I/O	TWI3-SCK	UART5-RX	NCSI-D14	UART6-CTS	SPI2-HOLD			SDC3-D0			PE-EINT6
PE7	I/O	TWI3-SDA	PWM0-2	NCSI-D15		SPI2-CS1			SDC3-D1			PE-EINT7
PE8	I/O		PWM0-4	CSI0-XVS-FSYNC	CLK-FANOUT0	SPI2-CS2			SDC3-D2			PE-EINT8
PE9	I/O	TCON-FSYNC0	PWM0-5	CSI1-XVS-FSYNC					SDC3-D3			PE-EINT9

Table 17-47 Port F Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PF0	I/O	SDC0-D1	JTAG-MS	UART1-TX			RJTAG-MS	IR0-RX	LBUS-DRQ0			PF-EINT0
PF1	I/O	SDC0-D0	JTAG-DI	UART1-RX			RJTAG-DI	IR1-RX	LBUS-DRQ1			PF-EINT1
PF2	I/O	SDC0-CLK	UART0-TX	DMIC-CLK		SPI3-CLK		IR2-RX				PF-EINT2
PF3	I/O	SDC0-CMD	JTAG-DO	DMIC-DATA0		SPI3-CS0	RJTAG-DO	IR3-RX				PF-EINT3
PF4	I/O	SDC0-D3	UART0-RX	DMIC-DATA1		SPI3-MOSI		IR-TX				PF-EINT4
PF5	I/O	SDC0-D2	JTAG-CK	DMIC-DATA2		SPI3-MISO	RJTAG-CK					PF-EINT5
PF6	I/O			DMIC-DATA3								PF-EINT6

Table 17-48 Port G Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PG0	I/O	SDC1-CLK		UART8-TX	RGMII1-RXD3/RMII1-NULL							PG-EINT0
PG1	I/O	SDC1-CMD		UART8-RX	RGMII1-RXD2/RMII1-NULL							PG-EINT1
PG2	I/O	SDC1-D0		UART8-RTS	RGMII1-RXCK/RMII1-NULL							PG-EINT2
PG3	I/O	SDC1-D1		UART8-CTS	RGMII1-TXD3/RMII1-NULL							PG-EINT3
PG4	I/O	SDC1-D2		CAN0-TX0	RGMII1-TXD2/RMII1-NULL							PG-EINT4
PG5	I/O	SDC1-D3		CAN0-RX0	RGMII1-EPHY-25/50M							PG-EINT5
PG6	I/O	UART1-TX	TWI4-SCK	UART9-CTS	RGMII1-RXD1/RMII1-RXD1	I2S2-MCLK						PG-EINT6
PG7	I/O	UART1-RX	TWI4-SDA	UART9-RTS	RGMII1-RXD0/RMII1-RXD0	I2S2-BCLK						PG-EINT7
PG8	I/O	UART1-RTS	TWI5-SCK	UART9-TX	RGMII1-RXCTL/RMII1-CRS-DV	I2S2-LRCK						PG-EINT8

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Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PG9	I/O	UART1-CTS	TWI5-SDA	UART9-RX	RGMII1-CLKIN/RMII1-RXER	I2S2-DOUT0	I2S2-DIN1					PG-EINT9
PG10	I/O	UART4-TX	I2S1-MCLK	TWI2-SCK	RGMII1-TXD1/RMII1-TXD1	I2S2-DIN0	I2S2-DOUT1					PG-EINT10
PG11	I/O	UART4-RX	I2S1-BCLK	TWI2-SDA	RGMII1-TXD0/RMII1-TXD0	DMIC-CLK						PG-EINT11
PG12	I/O	UART4-RTS	I2S1-LRCK		RGMII1-TXCK/RMII1-TXCK	DMIC-DATA0						PG-EINT12
PG13	I/O	UART4-CTS	I2S1-DOUT0	I2S1-DIN1	RGMII1-TXCTL/RMII1-TXEN	DMIC-DATA1						PG-EINT13
PG14	I/O	CAN1-TX0	I2S1-DIN0	I2S1-DOUT1	RGMII1-MDC	DMIC-DATA2		LBUS-DRQ0				PG-EINT14
PG15	I/O	CAN1-RX0			RGMII1-MDIO	DMIC-DATA3		LBUS-DRQ1				PG-EINT15

Table 17-49 Port J Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PJ0	I/O	PWM2-0	I2S2-MCLK	TWI4-SCK	RGMII0-RXD1/RMII0-RXD1	DMIC-CLK	UART5-TX					PJ-EINT0
PJ1	I/O	PWM2-1	I2S2-BCLK	TWI4-SDA	RGMII0-RXD0/RMII0-RXD0	DMIC-DATA0	UART5-RX					PJ-EINT1
PJ2	I/O	PWM2-2	I2S2-LRCK		RGMII0-RXCTL/RMII0-CRS-DV	DMIC-DATA1	UART5-RTS					PJ-EINT2
PJ3	I/O	PWM2-3	I2S2-DOUT0	I2S2-DIN1	RGMII0-CLKIN/RMII0-RXER	DMIC-DATA2	UART5-CTS					PJ-EINT3
PJ4	I/O	PWM2-4	I2S2-DIN0	I2S2-DOUT1	RGMII0-TXD1/RMII0-TXD1	DMIC-DATA3	UART7-TX					PJ-EINT4
PJ5	I/O	PWM2-5	I2S2-DIN2	I2S2-DOUT2	RGMII0-TXD0/RMII0-TXD0		UART7-RX					PJ-EINT5
PJ6	I/O	PWM2-6	I2S2-DIN3	I2S2-DOUT3	RGMII0-TXCK/RMII0-TXCK		UART7-RTS					PJ-EINT6
PJ7	I/O	PWM2-7	I2S1-MCLK		RGMII0-TXCTL/RMII0-TXEN		UART7-CTS					PJ-EINT7
PJ8	I/O	PWM2-8	I2S1-BCLK	LEDC	RGMII0-MDC		UART2-TX					PJ-EINT8
PJ9	I/O	PWM2-9	I2S1-LRCK	IR-TX	RGMII0-MDIO		UART2-RX					PJ-EINT9
PJ10	I/O	PWM2-10	I2S1-DOUT0	I2S1-DIN1	RGMII0-EPHY-25/50M		UART2-RTS					PJ-EINT10
PJ11	I/O	PWM2-11	I2S1-DIN0	I2S1-DOUT1	RGMII0-RXD3/RMII0-NULL		UART2-CTS					PJ-EINT11
PJ12	I/O	PWM2-12	CAN0-TX0	IR0-RX	RGMII0-RXD2/RMII0-NULL	TWI0-SCK	UART9-TX					PJ-EINT12
PJ13	I/O	PWM2-13	CAN0-RX0	IR1-RX	RGMII0-RXCK/RMII0-NULL	TWI0-SDA	UART9-RX					PJ-EINT13
PJ14	I/O		CAN1-TX0	IR2-RX	RGMII0-TXD3/RMII0-NULL	TWI5-SCK	UART9-RTS					PJ-EINT14
PJ15	I/O		CAN1-RX0	IR3-RX	RGMII0-TXD2/RMII0-NULL	TWI5-SDA	UART9-CTS					PJ-EINT15

Table 17-50 Port K Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
PK0	I/O		MCSIA-CKP		UART7-TX	TWI1-SCK	NCSI-D0	LBUS-READY1	I2S1-MCLK	LBUS-DRQ0	LCD-D0	PK-EINT0
PK1	I/O		MCSIA-CKN		UART7-RX	TWI1-SDA	NCSI-D1	LBUS-READY0	I2S1-BCLK	LBUS-DRQ1	LCD-D1	PK-EINT1
PK2	I/O		MCSIA-D0P		UART7-RTS	SPI1-CS0	NCSI-D2	LBUS-LD19	I2S1-LRCK	LBUS-LBE3	LCD-D8	PK-EINT2
PK3	I/O		MCSIA-D0N		UART7-CTS	SPI1-CLK	NCSI-D3	LBUS-LD18	I2S1-DOUT0	LBUS-DE	LCD-D9	PK-EINT3
PK4	I/O		MCSIA-D1P		TWI1-SCK	SPI1-MOSI	NCSI-D4	LBUS-LD17	I2S1-DIN0	LBUS-CS3	LCD-D16	PK-EINT4
PK5	I/O		MCSIA-D1N		TWI1-SDA	SPI1-MISO	NCSI-D5	LBUS-LD16		LBUS-CS2	LCD-D17	PK-EINT5
PK6	I/O		MCSIB-D1P/MCSIA-D2P		MCSI0-MCLK	SPI1-WP	NCSI-D6	LBUS-LD15	UART3-RTS		RGMII0-RXD3/RMII0-NULL	PK-EINT6
PK7	I/O		MCSIB-D1N/MCSIA-D2N		TWI2-SCK	SPI1-HOLD	NCSI-D7	LBUS-LD14			RGMII0-RXD2/RMII0-NULL	PK-EINT7
PK8	I/O		MCSIB-D0P/MCSIA-D3P		TWI2-SDA	SPI1-CS1	NCSI-PCLK	LBUS-LD13		CLK-FANOUT0	RGMII0-RXCK/RMII0-NULL	PK-EINT8
PK9	I/O		MCSIB-D0N/MCSIA-D3N		MCSI1-MCLK	SPI1-CS2	NCSI-MCLK	LBUS-LD12	UART3-TX	CLK-FANOUT1	RGMII0-TXD3/RMII0-NULL	PK-EINT9
PK10	I/O		MCSIB-CKP		TWI3-SCK	PWM0-6	NCSI-HSYNC	LBUS-WAIT	UART3-RX	LBUS-LBE2	RGMII0-TXD2/RMII0-NULL	PK-EINT10
PK11	I/O		MCSIB-CKN		TWI3-SDA	PWM0-7	NCSI-VSYNC	LBUS-CS2	UART3-CTS			PK-EINT11

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Table 17-51 RTC_GPIO Multiplex Function

Pin Name	IO Type	Function2	Function3	Function4	Function5	Function6	Function7	Function8	Function9	Function10	Function11	Function14
GPIO0	I/O, OD											IRQ-EINT0
GPIO1	I/O, OD											IRQ-EINT1
GPIO2	I/O, OD											IRQ-EINT2
GPIO4	I/O, OD											IRQ-EINT4
NMI	I/O, OD	NMI										NMI-EINTT0

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17.9.3.6 GPIO Group Withstand Voltage

The purpose of group withstand voltage configuration is to keep the internal withstand circuit voltage consistent with the group voltage. The withstand voltage of this device is auto detected by default without software operation.

- Debounce function

This function can realize GPIO voltage signal debounce, and the debounce time is configurable. If the level duration does not exceed the debounce time, the level will be considered as a glitch.

That is, after GPIO_POW_VAL voltage is obtained by digital design, the voltage will be assigned to GPIO_POW_MOD_STATE after debounce.

- Debounce bypass configuration

The debounce function of GPIO voltage signal supports bypass configuration: after the bypass is enabled, the input signal will be decoded and output directly.

That is, after the GPIO_POW_VAL voltage is obtained by the digital design, the voltage is directly assigned to GPIO_POW_MOD_STATE.

17.9.3.7 Secure

Register security for GPIOs is configured by register Px_SECURE, where 0 means non-secure and 1 means secure. The default is non-secure.

For registers that control each I/O port, PORTx_SEC_SEL in Px_SECURE controls the security attributes corresponding to each I/O port individually. This type of registers contains Px_CFG0-3, Px_DAT, PB_DAT_SET, PB_DAT_CLR, PB_DRV0-3, Px_PUL0-1, PB_INT_CFG0-3, PB_INT_CTL, and PB_INT_STA.

For registers that control a group of GPIOs, a register is secure as long as one of the Px_SECURE has PORTx_SEC_SEL configured as secure. This type of registers contains Px_INT_DEB, Px_INT_DEB_BYPASS, Px_PWR_MOD_STATE, and Px_PWR_VAL.

17.9.3.8 GPIO Partition

GPIO partition implements the following functions:

- GPIO pin partition: every GPIO pin can be assigned to a OS and can only be configured by its OS.
- GPIO interrupt partition: interrupts only return to the OS to which the GPIO pin belongs.

This device supports maximum four operating systems (OS0-OS3). If GPIO pins in one group are respectively configured by different OS, the operations or commands will not mutually conflict or be covered.

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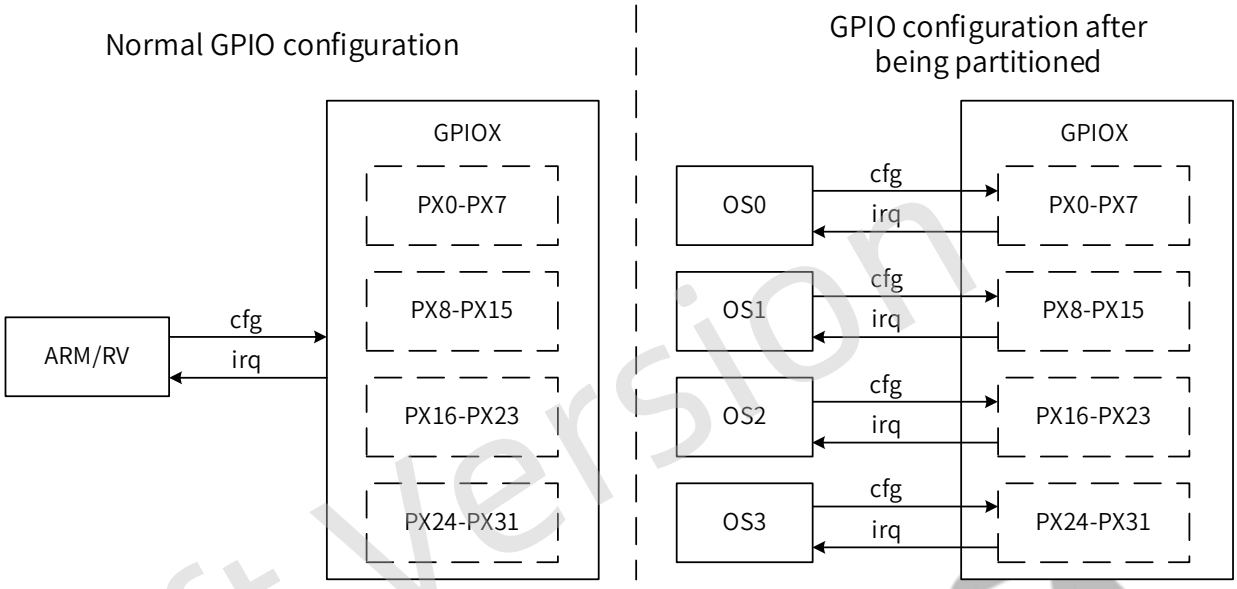
The following shows an GPIO partition example.

Figure 17-43 GPIO Partition



17.9.4 Programming Guidelines

17.9.4.1 Intra-Group Partition

If you do not use multi operating system, the registers for partition configuration (Px_OS_SEL0, Px_OS_SEL1, GPIO_OS_ID_CFG, GPIO_OS_ID_CFG_BYPASS) cannot be configured, and they all should keep the default configuration:

Step 1 All bits of GPIO_OS_ID_CFG should keep the default value 0.

Step 2 All bits of Px_OS_SEL0 and Px_OS_SEL1 should keep the default value 0;

Step 3 The GPIO_OS_ID_CFG_BYPASS of GPIO_OS_ID_CFG_BYPASS should keep the default value 1.

If you need to use multiple operating system for intra-group partition, configure as follows:

Step 1 Configure GPIO_OS_ID_CFG to assign OS0-3 to each master according to your requirements, and configure the bits of GPIO_OS_ID_x corresponding to their masters as 1. OS1 can be assigned to CPU COREx/RISC-V and other OS can be assigned to CPU COREx. The configuration example is as follows:

Master	OS	Bit	Configuration value
CPUX_CORE0	OS0	GPIO_OS_ID_0	5'b00011
CPUX_CORE1			
RISC-V E907	OS1	GPIO_OS_ID_1	5'b10000
CPUX_CORE2	OS2	GPIO_OS_ID_2	5'b00100
CPUX_CORE3	OS3	GPIO_OS_ID_3	5'b01000

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Step 2 Configure Px_OS_SEL0 and Px_OS_SEL1 registers to assign the OS to which the GPIO pins (including interrupts) belong;

Step 3 Configure GPIO_OS_ID_CFG_BYPASS as 0x23AA0000 to close the bypass path for intra-group partition.

When GPIO_OS_ID_CFG_BYPASS is 1, the generated interrupts are sent to ARM and RISC-V.

When GPIO_OS_ID_CFG_BYPASS is 0, the generated interrupts are only be sent to the OS to which the pin belongs, and will eventually be sent to only one CPU, not all CPUs.

NOTE

Before closing the bypass path for intra-group partition, you must set a valid ID master for GPIO_OS_ID_0 and cannot configure GPIO_OS_ID_0 as 5'b00000.

17.9.4.2 Disable I/O Pins

To disable I/O pins, follow the steps as below.

Step 1 Write 1111 to the Px_SELECT of the Px_CFG register to disable the I/O pins.

Step 2 If needed, set the pull-up/pull-down status of the I/O pins by setting the Px_PULL register. In the pull-up status, the pin level is 1 by default; in the pull-down status, the pin level is 0 by default.

17.9.4.3 Input Configuration

To configure I/O pins as the general input pins, follow the steps as below.

Step 1 Write 0000 to the Px_SELECT of the Px_CFG register to enable input function.

Step 2 If needed, set the pull-up/pull-down status of the I/O pins by setting the Px_PULL register. When the external input is driven, the external input value can be read; When the external input is not driven, the pin status (pullup or pulldown) can be read.

Step 3 Read the pin status in the Px_DAT register.

17.9.4.4 Output Configuration

To configure I/O pins as the general output pins, follow the steps as below.

Step 1 Write 0001 to the Px_SELECT of the Px_CFG register to enable output function.

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Step 2 If needed, configure the Px_DRV register to set the output driving strength. When it is set as 00, the driving strength is minimum; when it is set as 11, the driving strength is maximum. It is set as 01 by default.

Step 3 If needed, configure the Px_PULL register to set the pull-up and pull-down status. When the output function is enabled, the pin status is consistent with the output signal; when the output function is disabled, the pin status is pull-up or pull-down, which corresponds to 1 or 0.

Step 4 Output 1 or 0 to the I/O pins by writing the Px_DAT register or through atomic operations.

Step 5 During atomic operations, write 1 to the Px_DAT_SET register and then the output level will be set as 1; write 1 to the Px_DAT_CLR register and then the output level will be cleared as 0.

Take PA_DAT[0] bit as an example to explain the output configuration:

If PA0_SELECT of PA_CFG0 is configured as 0000 (input mode), reading PA_DAT[0]bit can get the pin status of PA0 pin;

If PA0_SELECT of PA_CFG0 is configured as 0001 (output mode), configuring PA_DAT[0]bit as 1 can output 1 to PA0 pin; configuring PA_DAT[0]bit as 0 can output 0 to PA0 pin.

17.9.4.5 Interrupt Configuration

To configure I/O pins as the interrupt pins, follow the steps as below.

Step 1 Write 1110 to the Px_SELECT of the Px_CFG register to enable the interrupt function.

Step 2 If needed, set the pull-up/pull-down status of the I/O pins by setting the Px_PULL register. In the pull-up status, the interrupt input level is 1 by default in case of no driving strength. In the pull-down status, the interrupt input level is 0 by default in case of no driving strength.

Step 3 Set the interrupt trigger types in the Px_INT_CFG register.

Step 4 Set the debounce clock in the Px_INT_CFG register.

Step 5 To use the pulse interrupt detection function, configure PX_INT_DEB_BYPASS to 0x23AA0001 to close the bypass path for interrupt debounce;

Interrupt sample time = DEB_CLK_PRE_SCALE / GPIO_INT_CLK_SELECT

Step 6 Clear the IRQ pending status in the Px_INT_STA register.

Step 7 Enable external interrupts in the Px_INT_CTL register.

Step 8 After receiving interrupts, CPU will handle them. After that, the pending status of interrupts can be cleared in the Px_INT_STA register.

Step 9 Disable external interrupts in the Px_INT_CTL register in case no interrupts are needed.

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NOTE

- It is not recommended to switch functions through Px_CFG in the interrupt mode. As the pins are in the input state, their input levels can be read in the Px_DAT.
- Disable the interrupts in the Px_INT_CTL before configuring new functions during the function switches. Before switching back to the interrupt functions, clear the pending states in the Px_INT_STA.
- If the interrupts aren't disabled in the course of function switches, an interrupt will be mistakenly triggered.
- The following describes the detection principle of pulse interrupt detection and high/low level interrupt detection:
  - When the input interrupt is sampled by the sampling clock to four consecutive 4-bit “0”, the interrupt is considered to be in a low level state and a low level interrupt can be triggered;
  - When the input interrupt is sampled by the sampling clock to four consecutive 4-bit “1”, the interrupt is considered to be in the high level state and the high level interrupt can be triggered;
  - When a switch from a high level state to a low level state is detected, it is considered to be a falling edge pulse interrupt;
  - When the switch from low level state to high level state is detected, it is considered to be a rising edge pulse interrupt.

17.9.4.6 Multi-Function Configuration

To configure I/O pins as the multiplexing function pins, follow the steps as below.

Step 1 Set the pin as the needed function through the Px_SELECT bit of the Px_CFG register.

Step 2 Set the driving strength in the Px_DRV register according to the selected function.

Step 3 Set the pull-up/down status in the Px_PULL register according to the selected function. When the pin is driven, the pin status is consistent with the driving signal; when the pin is not driven, the pin status is pull-up or pull-down, which corresponds to 1 or 0.

17.9.4.7 Power Configuration

Configuring Group Power Voltage

The SoC only supports setting group power voltage in PF ports, which is able to be set as 3.3 V or 1.8V by configuring GPIO_POW_VAL_SET_CTL register. The group power voltage in other ports is unconfigurable and depended on the peripheral circuit.

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Configuring Group Withstand Voltage

- Debounce configuration

Configure GPIO_DBC_CLK_SET register as 0x000F0040, indicating that debounce clock frequency is 24 MHz, debounce count value is 0x000F, the prescale fator for 24 MHz debounce clock is 16, the reference clock frequency is 1.5 MHz, the counting cycles are 15, and the debounce time is 10us.

GPIO_PWR_VAL readback value is the voltage value before debounce; GPIO_PWR_MOD_STATE readback value is the voltage value after debounce.

- Debounce bypass configuration

Configure GPIO_DBC_BYPASS register as 0x23AA0001. The upper 16bit is used as key field and this register can be configured only when the value of the upper 16bit is set to 0x23AA. To enable debounce bypass function, set bit 0 to1.

GPIO_PWR_VAL and GPIO_PWR_MOD_STATE readback values are voltage values without debounce.

17.9.4.8 NMI Configuration

NMI pin can act as a general-purpose I/O and generate interrupt signals by configuring RTC_GPIO_CFG register.

When NMI pin acts as non-maskable interrupt pin, it has two types of interrupt function:

- Interrupt generation and PMU wake up. For detailed description and register configuration, see section 9.1.4.8 NMI Pin Interrupt Generation
- System non-maskable interrupt enable. For detailed description and register configuration, see section 4.2.4.3 System NMI Configuration.

17.9.5 Register List

GPIO supports two group registers.

Module Name	Base Address
GPIO	0x03604000
RTC_GPIO	0x07096000

17.9.5.1 GPIO Register List

Module Name	Base Address
GPIO	0x03604000

Register Name	Offset	Description
GPIO_DBC_CLK_SET	0x0030	GPIO Debounce Clock Select Register
GPIO_DBC_BYPASS	0x0034	GPIO Debounce Bypass Register

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Register Name	Offset	Description
GPIO_POW_MOD_STATE	0x0040	GPIO Group Withstand Voltage Mode State Register
GPIO_POW_VAL	0x0048	GPIO Group Power Value Register
GPIO_POW_VAL_SET_CTL	0x0070	GPIO Group Power Voltage Select Control Register
PA_CFG0	0x0080	PA Configure Register 0
PA_CFG1	0x0084	PA Configure Register 1
PA_CFG2	0x0088	PA Configure Register 2
PA_DAT	0x0090	PA Data Register
PA_DAT_SET	0x0094	PA Data Set Register
PA_DAT_CLR	0x0098	PA Data Clear Register
PA_DRV0	0x00A0	PA Multi_Driving Register 0
PA_DRV1	0x00A4	PA Multi_Driving Register 1
PA_DRV2	0x00A8	PA Multi_Driving Register 2
PA_PUL0	0x00B0	PA Pull Register 0
PA_PUL1	0x00B4	PA Pull Register 1
PA_INT_CFG0	0x00C0	PA External Interrupt Configure Register 0
PA_INT_CFG1	0x00C4	PA External Interrupt Configure Register 1
PA_INT_CFG2	0x00C8	PA External Interrupt Configure Register 2
PA_INT_CTL	0x00D0	PA External Interrupt Control Register
PA_INT_STA	0x00D4	PA External Interrupt Status Register
PA_INT_DEB	0x00D8	PA External Debounce Configure Register
PA_INT_DEB_BYPASS	0x00DC	PA External Debounce Bypass Configure Register
PA_SECURE	0x00F0	PA Secure Configure Register
PB_CFG0	0x0100	PB Configure Register 0
PB_CFG1	0x0104	PB Configure Register 1
PB_DAT	0x0110	PB Data Register
PB_DAT_SET	0x0114	PB Data Set Register
PB_DAT_CLR	0x0118	PB Data Clear Register
PB_DRV0	0x0120	PB Multi_Driving Register 0
PB_DRV1	0x0124	PB Multi_Driving Register 1
PB_PUL0	0x0130	PB Pull Register 0
PB_INT_CFG0	0x0140	PB External Interrupt Configure Register 0
PB_INT_CFG1	0x0144	PB External Interrupt Configure Register 1
PB_INT_CTL	0x0150	PB External Interrupt Control Register
PB_INT_STA	0x0154	PB External Interrupt Status Register
PB_INT_DEB	0x0158	PB External Debounce Configure Register
PB_INT_DEB_BYPASS	0x015C	PB External Debounce Bypass Configure Register
PB_SECURE	0x0170	PB Secure Configure Register
PC_CFG0	0x0180	PC Configure Register 0
PC_CFG1	0x0184	PC Configure Register 1
PC_DAT	0x0190	PC Data Register
PC_DAT_SET	0x0194	PC Data Set Register
PC_DAT_CLR	0x0198	PC Data Clear Register

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Register Name	Offset	Description
PC_DRV0	0x01A0	PC Multi_Driving Register 0
PC_DRV1	0x01A4	PC Multi_Driving Register 1
PC_PUL0	0x01B0	PC Pull Register 0
PC_INT_CFG0	0x01C0	PC External Interrupt Configure Register 0
PC_INT_CFG1	0x01C4	PC External Interrupt Configure Register 1
PC_INT_CTL	0x01D0	PC External Interrupt Control Register
PC_INT_STA	0x01D4	PC External Interrupt Status Register
PC_INT_DEB	0x01D8	PC External Debounce Configure Register
PC_INT_DEB_BYPASS	0x01DC	PC External Debounce Bypass Configure Register
PC_SECURE	0x01F0	PC Secure Configure Register
PD_CFG0	0x0200	PD Configure Register 0
PD_CFG1	0x0204	PD Configure Register 1
PD_CFG2	0x0208	PD Configure Register 2
PD_DAT	0x0210	PD Data Register
PD_DAT_SET	0x0214	PD Data Set Register
PD_DAT_CLR	0x0218	PD Data Clear Register
PD_DRV0	0x0220	PD Multi_Driving Register 0
PD_DRV1	0x0224	PD Multi_Driving Register 1
PD_DRV2	0x0228	PD Multi_Driving Register 2
PD_PUL0	0x0230	PD Pull Register 0
PD_PUL1	0x0234	PD Pull Register 1
PD_INT_CFG0	0x0240	PD External Interrupt Configure Register 0
PD_INT_CFG1	0x0244	PD External Interrupt Configure Register 1
PD_INT_CFG2	0x0248	PD External Interrupt Configure Register 2
PD_INT_CTL	0x0250	PD External Interrupt Control Register
PD_INT_STA	0x0254	PD External Interrupt Status Register
PD_INT_DEB	0x0258	PD External Debounce Configure Register
PD_INT_DEB_BYPASS	0x025C	PD External Debounce Bypass Configure Register
PD_SECURE	0x0270	PD Secure Configure Register
PE_CFG0	0x0280	PE Configure Register 0
PE_CFG1	0x0284	PE Configure Register 1
PE_DAT	0x0290	PE Data Register
PE_DAT_SET	0x0294	PE Data Set Register
PE_DAT_CLR	0x0298	PE Data Clear Register
PE_DRV0	0x02A0	PE Multi_Driving Register 0
PE_DRV1	0x02A4	PE Multi_Driving Register 1
PE_PUL0	0x02B0	PE Pull Register 0
PE_INT_CFG0	0x02C0	PE External Interrupt Configure Register 0
PE_INT_CFG1	0x02C4	PE External Interrupt Configure Register 1
PE_INT_CTL	0x02D0	PE External Interrupt Control Register
PE_INT_STA	0x02D4	PE External Interrupt Status Register
PE_INT_DEB	0x02D8	PE External Debounce Configure Register

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Register Name	Offset	Description
PE_INT_DEB_BYPASS	0x02DC	PE External Debounce Bypass Configure Register
PE_SECURE	0x02F0	PE Secure Configure Register
PF_CFG0	0x0300	PF Configure Register 0
PF_DAT	0x0310	PF Data Register
PF_DAT_SET	0x0314	PF Data Set Register
PF_DAT_CLR	0x0318	PF Data Clear Register
PF_DRV0	0x0320	PF Multi_Driving Register 0
PF_PUL0	0x0330	PF Pull Register 0
PF_INT_CFG0	0x0340	PF External Interrupt Configure Register 0
PF_INT_CTL	0x0350	PF External Interrupt Control Register
PF_INT_STA	0x0354	PF External Interrupt Status Register
PF_INT_DEB	0x0358	PF External Debounce Configure Register
PF_INT_DEB_BYPASS	0x035C	PF External Debounce Bypass Configure Register
PF_SECURE	0x0370	PF Secure Configure Register
PG_CFG0	0x0380	PG Configure Register 0
PG_CFG1	0x0384	PG Configure Register 1
PG_DAT	0x0390	PG Data Register
PG_DAT_SET	0x0394	PG Data Set Register
PG_DAT_CLR	0x0398	PG Data Clear Register
PG_DRV0	0x03A0	PG Multi_Driving Register 0
PG_DRV1	0x03A4	PG Multi_Driving Register 1
PG_PUL0	0x03B0	PG Pull Register 0
PG_INT_CFG0	0x03C0	PG External Interrupt Configure Register 0
PG_INT_CFG1	0x03C4	PG External Interrupt Configure Register 1
PG_INT_CTL	0x03D0	PG External Interrupt Control Register
PG_INT_STA	0x03D4	PG External Interrupt Status Register
PG_INT_DEB	0x03D8	PG External Debounce Configure Register
PG_INT_DEB_BYPASS	0x03DC	PG External Debounce Bypass Configure Register
PG_SECURE	0x03F0	PG Secure Configure Register
PJ_CFG0	0x0500	PJ Configure Register 0
PJ_CFG1	0x0504	PJ Configure Register 1
PJ_DAT	0x0510	PJ Data Register
PJ_DAT_SET	0x0514	PJ Data Set Register
PJ_DAT_CLR	0x0518	PJ Data Clear Register
PJ_DRV0	0x0520	PJ Multi_Driving Register 0
PJ_DRV1	0x0524	PJ Multi_Driving Register 1
PJ_PUL0	0x0530	PJ Pull Register 0
PJ_INT_CFG0	0x0540	PJ External Interrupt Configure Register 0
PJ_INT_CFG1	0x0544	PJ External Interrupt Configure Register 1
PJ_INT_CTL	0x0550	PJ External Interrupt Control Register
PJ_INT_STA	0x0554	PJ External Interrupt Status Register
PJ_INT_DEB	0x0558	PJ External Debounce Configure Register

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Register Name	Offset	Description
PJ_INT_DEB_BYPASS	0x055C	PJ External Debounce Bypass Configure Register
PJ_SECURE	0x0570	PJ Secure Configure Register
PK_CFG0	0x0580	PK Configure Register 0
PK_CFG1	0x0584	PK Configure Register 1
PK_DAT	0x0590	PK Data Register
PK_DAT_SET	0x0594	PK Data Set Register
PK_DAT_CLR	0x0598	PK Data Clear Register
PK_DRV0	0x05A0	PK Multi_Driving Register 0
PK_DRV1	0x05A4	PK Multi_Driving Register 1
PK_PUL0	0x05B0	PK Pull Register 0
PK_INT_CFG0	0x05C0	PK External Interrupt Configure Register 0
PK_INT_CFG1	0x05C4	PK External Interrupt Configure Register 1
PK_INT_CTL	0x05D0	PK External Interrupt Control Register
PK_INT_STA	0x05D4	PK External Interrupt Status Register
PK_INT_DEB	0x05D8	PK External Debounce Configure Register
PK_INT_DEB_BYPASS	0x05DC	PK External Debounce Bypass Configure Register
PK_SECURE	0x05F0	PK Secure Configure Register
PA_OS_SEL0	0x0800	PA Operating System Select Register 0
PA_OS_SEL1	0x0804	PA Operating System Select Register 1
PB_OS_SEL0	0x0808	PB Operating System Select Register 0
PC_OS_SEL0	0x0810	PC Operating System Select Register 0
PD_OS_SEL0	0x0818	PD Operating System Select Register 0
PD_OS_SEL1	0x081C	PD Operating System Select Register 1
PE_OS_SEL0	0x0820	PE Operating System Select Register 0
PF_OS_SEL0	0x0828	PF Operating System Select Register 0
PG_OS_SEL0	0x0830	PG Operating System Select Register 0
PJ_OS_SEL0	0x0848	PJ Operating System Select Register 0
PK_OS_SEL0	0x0850	PK Operating System Select Register 0
GPIO_OS_ID_CFG	0x0860	GPIO Operating System ID Configure Register
GPIO_OS_ID_CFG_BYPASS	0x0864	GPIO Operating System ID Configure Bypass Register

17.9.5.2 RTC_GPIO Register List

Module Name	Base Address
RTC_GPIO	0x07096000

Register Name	Offset	Description
RTC_GPIO_CFG	0x0500	RTC_GPIO Configure Register
RTC_GPIO_DAT	0x0510	RTC_GPIO Data Register
RTC_GPIO_INT_CFG	0x0540	RTC_GPIO External Interrupt Configure Register

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Register Name	Offset	Description
RTC_GPIO_INT_CTL	0x0550	RTC_GPIO External Interrupt Control Register
RTC_GPIO_INT_STA	0x0554	RTC_GPIO External Interrupt Status Register
RTC_GPIO_INT_DEB	0x0558	RTC_GPIO External Debounce Configure Register
RTC_GPIO_SECURE	0x0570	RTC_GPIO Secure Configure Register

17.9.6 GPIO Register Description

17.9.6.1 0x0030 GPIO Debounce Clock Select Register (Default Value: 0x000F_0040)

Offset: 0x0030			Register Name: GPIO_DBC_CLK_SET
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:16	R/W	0x0F	DBC_CNT_VAL Debounce Count Value (If cnt value cannot satisfy the minimum time of debounce,cnt value will be locked within different clock freq: 6M:0x3C 3M:0x1E 1.5M:0x0F 750K:0x07 375K:0x03 187.5K:0x01)
15:7	/	/	/
6:4	R/W	0x4	DBC_CLK_PRE_SCALE Debounce 24M clock pre_scale n The selected 24M clock source is prescaled by 2^n. Note: n is from 0-7,if set n as 0or 1,the n will be tied in 4(the freq will be tied in 1.5M).
3:1	/	/	/
0	R/W	0x0	DBC_CLK_SEL Debounce Clock Select Control 0: Clk24M 1: Clk32K

17.9.6.2 0x0034 GPIO Debounce Bypass Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: GPIO_DBC_BYPASS
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0034			Register Name: GPIO_DBC_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x0	DBC_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.3 0x0040 GPIO Group Withstand Voltage Mode State Register (Default Value: 0x0114_1551)

Offset: 0x0040			Register Name: GPIO_POW_MOD_STATE
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R	0x1	VCCIO_PWR_MOD_STATE VCC_IO POWER MODE State 00: 1.8V 01: 3.3V
23:22	/	/	/
21:20	R	0x1	PK_PWR_MOD_STATE PK_POWER MODE State 00: 1.8V 01: 3.3V
19:18	R	0x1	PJ_PWR_MOD_STATE PJ_POWER MODE State 00: 1.8V 01: 3.3V
17:14	/	/	/
13:12	R	0x1	PG_PWR_MOD_STATE PG_POWER MODE State 00: 1.8V 01: 3.3V
11:10	R	0x1	PF_PWR_MOD_STATE PF_POWER MODE State 00: 1.8V 01: 3.3V
9:8	R	0x1	PE_PWR_MOD_STATE PE_POWER MODE State 00: 1.8V 01: 3.3V

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Offset: 0x0040			Register Name: GPIO_POW_MOD_STATE
Bit	Read/Write	Default/Hex	Description
7:6	R	0x1	PD_PWR_MOD_STATE PD_POWER MODE State 00: 1.8V 01: 3.3V
5:4	R	0x1	PC_PWR_MOD_STATE PC_POWER MODE State 00: 1.8V 01: 3.3V
3:2	/	/	/
1:0	R	0x1	PA_PWR_MOD_STATE PA_POWER MODE State 00: 1.8V 01: 3.3V

17.9.6.4 0x0048 GPIO Group Power Value Register (Default Value: 0x0114_1551)

Offset: 0x0048			Register Name: GPIO_POW_VAL
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R	0x1	VCCIO_PWR_VAL VCC_IO POWER VALUE 00: 1.8V 01: 3.3V
23:22	/	/	/
21:20	R	0x1	PK_PWR_VAL PK_PORT POWER VALUE 00: 1.8V 01: 3.3V
19:18	R	0x1	PJ_PWR_VAL PJ_PORT POWER VALUE 00: 1.8V 01: 3.3V
17:14	/	/	/
13:12	R	0x1	PG_PWR_VAL PG_PORT POWER VALUE 00: 1.8V 01: 3.3V
11:10	R	0x1	PF_PWR_VAL PF_PORT POWER VALUE 00: 1.8V 01: 3.3V

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Offset: 0x0048			Register Name: GPIO_POW_VAL
Bit	Read/Write	Default/Hex	Description
9:8	R	0x1	PE_PWR_VAL PE_PORT POWER VALUE 00: 1.8V 01: 3.3V
7:6	R	0x1	PD_PWR_VAL PD_PORT POWER VALUE 00: 1.8V 01: 3.3V
5:4	R	0x1	PC_PWR_VAL PC_PORT POWER VALUE 00: 1.8V 01: 3.3V
3:2	/	/	/
1:0	R	0x1	PA_PWR_VAL PA_PORT POWER VALUE 00: 1.8V 01: 3.3V

17.9.6.5 0x0070 GPIO Group Power Voltage Select Control Register (Default Value: 0x0000_0001)

Offset: 0x0070			Register Name: GPIO_POW_VAL_SET_CTL
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x1	VCC_PF_PWR_VOL_SEL VCC_PF Power Voltage Select Control 0: 1.8V 1: 3.3V

17.9.6.6 0x0080 PA Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0080			Register Name: PA_CFG0
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0080			Register Name: PA_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PA7_SELECT PA7 Select 0000: Input 0001: Output 0010: GPADC0-3 0011: Reserved 0100: RGMII0-CLKIN/RMII0-RXER 0101: NCSI-PCLK 0110: LBUS-CS0 0111: PWMCS-I9 1000: Reserved 1001: LCD-D1 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT7 1111: Disable
27:24	R/W	0xF	PA6_SELECT PA6 Select 0000: Input 0001: Output 0010: GPADC0-2 0011: Reserved 0100: RGMII0-RXCTL/RMII0-CRS-DV 0101: NCSI-MCLK 0110: LBUS-CS1 0111: PWMCS-I8 1000: Reserved 1001: LCD-D0 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT6 1111: Disable
23:20	R/W	0xF	PA5_SELECT PA5 Select 0000: Input 0001: Output 0010: GPADC0-1 0011: Reserved 0100: RGMII0-RXD0/RMII0-RXD0 0101: NCSI-VSYNC 0110: LBUS-ALE 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT5 1111: Disable

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Offset: 0x0080			Register Name: PA_CFG0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0xF	PA4_SELECT PA4 Select 0000: Input 0001: Output 0010: GPADC0-0 0011: Reserved 0100: RGMII0-RXD1/RMII0-RXD1 0101: NCSI-HSYNC 0110: LBUS-CLK 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT4 1111: Disable
15:12	R/W	0xF	PA3_SELECT PA3 Select 0000: Input 0001: Output 0010: GPADC2-3 0011: TP-Y2 0100: Reserved 0101: Reserved 0110: LBUS-DP3 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT3 1111: Disable
11:8	R/W	0xF	PA2_SELECT PA2 Select 0000: Input 0001: Output 0010: GPADC2-2 0011: TP-Y1 0100: Reserved 0101: Reserved 0110: LBUS-DP2 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT2 1111: Disable
7:4	R/W	0xF	PA1_SELECT PA1 Select 0000: Input 0001: Output 0010: GPADC2-1 0011: TP-X2 0100: Reserved 0101: Reserved 0110: LBUS-READY2 0111: LBUS-DP1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT1 1111: Disable

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Offset: 0x0080			Register Name: PA_CFG0
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PA0_SELECT PA0 Select 0000: Input 0001: Output 0010: GPADC2-0 0011: TP-X1 0100: Reserved 0101: Reserved 0110: LBUS-READY3 0111: LBUS-LBE2 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT0 1111: Disable

17.9.6.7 0x0084 PA Configure Register 1 (Default Value: 0xFFFF_FFFF)

Offset: 0x0084			Register Name: PA_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PA15_SELECT PA15 Select 0000: Input 0001: Output 0010: GPADC1-1 0011: Reserved 0100: RGMII0-RXD3/RMII0-NULL 0101: NCSI-D7 0110: LBUS-LD11 0111: UART6-RX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT15 1111: Disable
27:24	R/W	0xF	PA14_SELECT PA14 Select 0000: Input 0001: Output 0010: GPADC1-0 0011: Reserved 0100: RGMII0-EPHY-25/50M 0101: NCSI-D6 0110: LBUS-LD10 0111: UART6-TX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT14 1111: Disable

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Offset: 0x0084			Register Name: PA_CFG1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0xF	PA13_SELECT PA13 Select 0000: Input 0001: Output 0010: GPADC0-9 0011: Reserved 0100: RGMII0-MDIO 0101: NCSI-D5 0110: LBUS-LD9 0111: PWMCS-IF 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT13 1111: Disable
19:16	R/W	0xF	PA12_SELECT PA12 Select 0000: Input 0001: Output 0010: GPADC0-8 0011: Reserved 0100: RGMII0-MDC 0101: NCSI-D4 0110: LBUS-LD8 0111: PWMCS-IE 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT12 1111: Disable
15:12	R/W	0xF	PA11_SELECT PA11 Select 0000: Input 0001: Output 0010: GPADC0-7 0011: Reserved 0100: RGMII0-TXCTL/RMII0-TXEN 0101: NCSI-D3 0110: LBUS-LBE1 0111: PWMCS-ID 1000: Reserved 1001: LCD-D17 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT11 1111: Disable
11:8	R/W	0xF	PA10_SELECT PA10 Select 0000: Input 0001: Output 0010: GPADC0-6 0011: Reserved 0100: RGMII0-TXCK/RMII0-TXCK 0101: NCSI-D2 0110: LBUS-LBE0 0111: PWMCS-IC 1000: Reserved 1001: LCD-D16 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT10 1111: Disable

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Offset: 0x0084			Register Name: PA_CFG1
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0xF	PA9_SELECT PA9 Select 0000: Input 0001: Output 0010: GPADC0-5 0011: Reserved 0100: RGMII0-TXD0/RMII0-TXD0 0101: NCSI-D1 0110: LBUS-DP1 0111: PWMCS-IB 1000: LBUS-LBE3 1001: LCD-D9 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT9 1111: Disable
3:0	R/W	0xF	PA8_SELECT PA8 Select 0000: Input 0001: Output 0010: GPADC0-4 0011: Reserved 0100: RGMII0-TXD1/RMII0-TXD1 0101: NCSI-D0 0110: LBUS-DP0 0111: PWMCS-IA 1000: Reserved 1001: LCD-D8 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT8 1111: Disable

17.9.6.8 0x0088 PA Configure Register 2 (Default Value: 0xFFFF_FFFF)

Offset: 0x0088			Register Name: PA_CFG2
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PA23_SELECT PA23 Select 0000: Input 0001: Output 0010: GPADC1-9 0011: Reserved 0100: Reserved 0101: NCSI-D15 0110: LBUS-DE 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT23 1111: Disable

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Offset: 0x0088			Register Name: PA_CFG2
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PA22_SELECT PA22 Select 0000: Input 0001: Output 0010: GPADC1-8 0011: Reserved 0100: Reserved 0101: NCSI-D14 0110: LBUS-CS2 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT22 1111: Disable
23:20	R/W	0xF	PA21_SELECT PA21 Select 0000: Input 0001: Output 0010: GPADC1-7 0011: Reserved 0100: Reserved 0101: NCSI-D13 0110: LBUS-CS3 0111: SPI2-CS1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT21 1111: Disable
19:16	R/W	0xF	PA20_SELECT PA20 Select 0000: Input 0001: Output 0010: GPADC1-6 0011: Reserved 0100: Reserved 0101: NCSI-D12 0110: LBUS-WAIT 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT20 1111: Disable
15:12	R/W	0xF	PA19_SELECT PA19 Select 0000: Input 0001: Output 0010: GPADC1-5 0011: Reserved 0100: RGMII0-TXD2/RMII0-NULL 0101: NCSI-D11 0110: LBUS-LD15 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT19 1111: Disable

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Offset: 0x0088			Register Name: PA_CFG2
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PA18_SELECT PA18 Select 0000: Input 0001: Output 0010: GPADC1-4 0011: Reserved 0100: RGMII0-TXD3/RMII0-NULL 0101: NCSI-D10 0110: LBUS-LD14 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT18 1111: Disable
7:4	R/W	0xF	PA17_SELECT PA17 Select 0000: Input 0001: Output 0010: GPADC1-3 0011: Reserved 0100: RGMII0-RXCK/RMII0-NULL 0101: NCSI-D9 0110: LBUS-LD13 0111: TWI3-SDA 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT17 1111: Disable
3:0	R/W	0xF	PA16_SELECT PA16 Select 0000: Input 0001: Output 0010: GPADC1-2 0011: Reserved 0100: RGMII0-RXD2/RMII0-NULL 0101: NCSI-D8 0110: LBUS-LD12 0111: TWI3-SCK 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PA-EINT16 1111: Disable

17.9.6.9 0x0090 PA Data Register (Default Value: 0x0000_0000)

Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
23	R/W	0x0	PORT23_DAT PA23 Data If the function of PA23 port is configured as input, the PORT23_DAT bit indicates the sate of the PA23 pin; If the function of PA23 port is configured as output, the sate of the PA23 pin can be configured via software by modifying the value of the PORT23_DAT bit. In this case, the value obtained by reading PORT23_DAT bit is the value set by the software; If the PA23 port is configured as functional pin, this bit is invalid.
22	R/W	0x0	PORT22_DAT PA22 Data If the function of PA22 port is configured as input, the PORT22_DAT bit indicates the sate of the PA22 pin; If the function of PA22 port is configured as output, the sate of the PA22 pin can be configured via software by modifying the value of the PORT22_DAT bit. In this case, the value obtained by reading PORT22_DAT bit is the value set by the software; If the PA22 port is configured as functional pin, this bit is invalid.
21	R/W	0x0	PORT21_DAT PA21 Data If the function of PA21 port is configured as input, the PORT21_DAT bit indicates the sate of the PA21 pin; If the function of PA21 port is configured as output, the sate of the PA21 pin can be configured via software by modifying the value of the PORT21_DAT bit. In this case, the value obtained by reading PORT21_DAT bit is the value set by the software; If the PA21 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
20	R/W	0x0	PORT20_DAT PA20 Data If the function of PA20 port is configured as input, the PORT20_DAT bit indicates the sate of the PA20 pin; If the function of PA20 port is configured as output, the sate of the PA20 pin can be configured via software by modifying the value of the PORT20_DAT bit. In this case, the value obtained by reading PORT20_DAT bit is the value set by the software; If the PA20 port is configured as functional pin, this bit is invalid.
19	R/W	0x0	PORT19_DAT PA19 Data If the function of PA19 port is configured as input, the PORT19_DAT bit indicates the sate of the PA19 pin; If the function of PA19 port is configured as output, the sate of the PA19 pin can be configured via software by modifying the value of the PORT19_DAT bit. In this case, the value obtained by reading PORT19_DAT bit is the value set by the software; If the PA19 port is configured as functional pin, this bit is invalid.
18	R/W	0x0	PORT18_DAT PA18 Data If the function of PA18 port is configured as input, the PORT18_DAT bit indicates the sate of the PA18 pin; If the function of PA18 port is configured as output, the sate of the PA18 pin can be configured via software by modifying the value of the PORT18_DAT bit. In this case, the value obtained by reading PORT18_DAT bit is the value set by the software; If the PA18 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	PORT17_DAT PA17 Data If the function of PA17 port is configured as input, the PORT17_DAT bit indicates the sate of the PA17 pin; If the function of PA17 port is configured as output, the sate of the PA17 pin can be configured via software by modifying the value of the PORT17_DAT bit. In this case, the value obtained by reading PORT17_DAT bit is the value set by the software; If the PA17 port is configured as functional pin, this bit is invalid.
16	R/W	0x0	PORT16_DAT PA16 Data If the function of PA16 port is configured as input, the PORT16_DAT bit indicates the sate of the PA16 pin; If the function of PA16 port is configured as output, the sate of the PA16 pin can be configured via software by modifying the value of the PORT16_DAT bit. In this case, the value obtained by reading PORT16_DAT bit is the value set by the software; If the PA16 port is configured as functional pin, this bit is invalid.
15	R/W	0x0	PORT15_DAT PA15 Data If the function of PA15 port is configured as input, the PORT15_DAT bit indicates the sate of the PA15 pin; If the function of PA15 port is configured as output, the sate of the PA15 pin can be configured via software by modifying the value of the PORT15_DAT bit. In this case, the value obtained by reading PORT15_DAT bit is the value set by the software; If the PA15 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	PORT14_DAT PA14 Data If the function of PA14 port is configured as input, the PORT14_DAT bit indicates the sate of the PA14 pin; If the function of PA14 port is configured as output, the sate of the PA14 pin can be configured via software by modifying the value of the PORT14_DAT bit. In this case, the value obtained by reading PORT14_DAT bit is the value set by the software; If the PA14 port is configured as functional pin, this bit is invalid.
13	R/W	0x0	PORT13_DAT PA13 Data If the function of PA13 port is configured as input, the PORT13_DAT bit indicates the sate of the PA13 pin; If the function of PA13 port is configured as output, the sate of the PA13 pin can be configured via software by modifying the value of the PORT13_DAT bit. In this case, the value obtained by reading PORT13_DAT bit is the value set by the software; If the PA13 port is configured as functional pin, this bit is invalid.
12	R/W	0x0	PORT12_DAT PA12 Data If the function of PA12 port is configured as input, the PORT12_DAT bit indicates the sate of the PA12 pin; If the function of PA12 port is configured as output, the sate of the PA12 pin can be configured via software by modifying the value of the PORT12_DAT bit. In this case, the value obtained by reading PORT12_DAT bit is the value set by the software; If the PA12 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
11	R/W	0x0	PORT11_DAT PA11 Data If the function of PA11 port is configured as input, the PORT11_DAT bit indicates the sate of the PA11 pin; If the function of PA11 port is configured as output, the sate of the PA11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PA11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PA10 Data If the function of PA10 port is configured as input, the PORT10_DAT bit indicates the sate of the PA10 pin; If the function of PA10 port is configured as output, the sate of the PA10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PA10 port is configured as functional pin, this bit is invalid.
9	R/W	0x0	PORT9_DAT PA9 Data If the function of PA9 port is configured as input, the PORT9_DAT bit indicates the sate of the PA9 pin; If the function of PA9 port is configured as output, the sate of the PA9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PA9 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PORT8_DAT PA8 Data If the function of PA8 port is configured as input, the PORT8_DAT bit indicates the sate of the PA8 pin; If the function of PA8 port is configured as output, the sate of the PA8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PA8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PA7 Data If the function of PA7 port is configured as input, the PORT7_DAT bit indicates the sate of the PA7 pin; If the function of PA7 port is configured as output, the sate of the PA7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PA7 port is configured as functional pin, this bit is invalid.
6	R/W	0x0	PORT6_DAT PA6 Data If the function of PA6 port is configured as input, the PORT6_DAT bit indicates the sate of the PA6 pin; If the function of PA6 port is configured as output, the sate of the PA6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PA6 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_DAT PA5 Data If the function of PA5 port is configured as input, the PORT5_DAT bit indicates the sate of the PA5 pin; If the function of PA5 port is configured as output, the sate of the PA5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PA5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PA4 Data If the function of PA4 port is configured as input, the PORT4_DAT bit indicates the sate of the PA4 pin; If the function of PA4 port is configured as output, the sate of the PA4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PA4 port is configured as functional pin, this bit is invalid.
3	R/W	0x0	PORT3_DAT PA3 Data If the function of PA3 port is configured as input, the PORT3_DAT bit indicates the sate of the PA3 pin; If the function of PA3 port is configured as output, the sate of the PA3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PA3 port is configured as functional pin, this bit is invalid.

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Offset: 0x0090			Register Name: PA_DAT
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PORT2_DAT PA2 Data If the function of PA2 port is configured as input, the PORT2_DAT bit indicates the sate of the PA2 pin; If the function of PA2 port is configured as output, the sate of the PA2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PA2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PA1 Data If the function of PA1 port is configured as input, the PORT1_DAT bit indicates the sate of the PA1 pin; If the function of PA1 port is configured as output, the sate of the PA1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PA1 port is configured as functional pin, this bit is invalid.
0	R/W	0x0	PORT0_DAT PA0 Data If the function of PA0 port is configured as input, the PORT0_DAT bit indicates the sate of the PA0 pin; If the function of PA0 port is configured as output, the sate of the PA0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PA0 port is configured as functional pin, this bit is invalid.

17.9.6.10 0x0094 PA Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0094			Register Name: PA_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x0094			Register Name: PA_DAT_SET
Bit	Read/Write	Default/Hex	Description
23	R/W1S	0x0	PORT23_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
22	R/W1S	0x0	PORT22_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
21	R/W1S	0x0	PORT21_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
20	R/W1S	0x0	PORT20_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
19	R/W1S	0x0	PORT19_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
18	R/W1S	0x0	PORT18_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
17	R/W1S	0x0	PORT17_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
16	R/W1S	0x0	PORT16_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.

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Offset: 0x0094			Register Name: PA_DAT_SET
Bit	Read/Write	Default/Hex	Description
15	R/W1S	0x0	PORT15_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
14	R/W1S	0x0	PORT14_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
13	R/W1S	0x0	PORT13_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
12	R/W1S	0x0	PORT12_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.

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Offset: 0x0094			Register Name: PA_DAT_SET
Bit	Read/Write	Default/Hex	Description
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PA _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePA _DAT register.

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17.9.6.11 0x0098 PA Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0098			Register Name: PA_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	PORT23_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
22	R/W1C	0x0	PORT22_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
21	R/W1C	0x0	PORT21_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
20	R/W1C	0x0	PORT20_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
19	R/W1C	0x0	PORT19_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
18	R/W1C	0x0	PORT18_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
17	R/W1C	0x0	PORT17_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.

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Offset: 0x0098			Register Name: PA_DAT_CLR
Bit	Read/Write	Default/Hex	Description
16	R/W1C	0x0	PORT16_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
15	R/W1C	0x0	PORT15_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
14	R/W1C	0x0	PORT14_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
13	R/W1C	0x0	PORT13_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
12	R/W1C	0x0	PORT12_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.

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Offset: 0x0098			Register Name: PA_DAT_CLR
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.

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Offset: 0x0098			Register Name: PA_DAT_CLR
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PA_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PA_DAT register.

17.9.6.12 0x00A0 PA Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x00A0			Register Name: PA_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PA7_DRV PA7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PA6_DRV PA6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PA5_DRV PA5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PA4_DRV PA4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PA3_DRV PA3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PA2_DRV PA2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/

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Offset: 0x00A0			Register Name: PA_DRV0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x1	PA1_DRV PA1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PA0_DRV PA0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.13 0x00A4 PA Multi_Driving Register 1 (Default Value: 0x1111_1111)

Offset: 0x00A4			Register Name: PA_DRV1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PA15_DRV PA15 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PA14_DRV PA14 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PA13_DRV PA13 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PA12_DRV PA12 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PA11_DRV PA11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/

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Offset: 0x00A4			Register Name: PA_DRV1
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x1	PA10_DRV PA10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PA9_DRV PA9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PA8_DRV PA8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.14 0x00A8 PA Multi_Driving Register 2 (Default Value: 0x1111_1111)

Offset: 0x00A8			Register Name: PA_DRV2
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PA23_DRV PA23 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PA22_DRV PA22 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PA21_DRV PA21 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PA20_DRV PA20 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/

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Offset: 0x00A8			Register Name: PA_DRV2
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x1	PA19_DRV PA19 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PA18_DRV PA18 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PA17_DRV PA17 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PA16_DRV PA16 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.15 0x00B0 PA Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x00B0			Register Name: PA_PUL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PA15_PULL PA15 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
29:28	R/W	0x0	PA14_PULL PA14 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
27:26	R/W	0x0	PA13_PULL PA13 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
25:24	R/W	0x0	PA12_PULL PA12 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x00B0			Register Name: PA_PUL0
Bit	Read/Write	Default/Hex	Description
23:22	R/W	0x0	PA11_PULL PA11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PA10_PULL PA10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PA9_PULL PA9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PA8_PULL PA8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PA7_PULL PA7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PA6_PULL PA6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PA5_PULL PA5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PA4_PULL PA4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PA3_PULL PA3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PA2_PULL PA2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x00B0			Register Name: PA_PUL0
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	PA1_PULL PA1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PA0_PULL PA0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.16 0x00B4 PA Pull Register 1 (Default Value: 0x0000_0000)

Offset: 0x00B4			Register Name: PA_PUL1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:14	R/W	0x0	PA23_PULL PA23 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PA22_PULL PA22 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PA21_PULL PA21 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PA20_PULL PA20 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PA19_PULL PA19 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PA18_PULL PA18 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x00B4			Register Name: PA_PUL1
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	PA17_PULL PA17 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PA16_PULL PA16 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.17 0x00C0 PA External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x00C0			Register Name: PA_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x00C0			Register Name: PA_INT_CFG0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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17.9.6.18 0x00C4 PA External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x00C4			Register Name: PA_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT15_CFG External INT15 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT14_CFG External INT14 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT13_CFG External INT13 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT12_CFG External INT12 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x00C4			Register Name: PA_INT_CFG1
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.19 0x00C8 PA External Interrupt Configure Register 2 (Default Value: 0x0000_0000)

Offset: 0x00C8			Register Name: PA_INT_CFG2
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT23_CFG External INT23 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x00C8			Register Name: PA_INT_CFG2
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT22_CFG External INT22 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT21_CFG External INT21 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT20_CFG External INT20 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT19_CFG External INT19 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT18_CFG External INT18 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x00C8			Register Name: PA_INT_CFG2
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT17_CFG External INT17 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT16_CFG External INT16 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.20 0x00D0 PA External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x00D0			Register Name: PA_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	EINT23_CTL External INT23 Enable 0: Disable 1: Enable
22	R/W	0x0	EINT22_CTL External INT22 Enable 0: Disable 1: Enable
21	R/W	0x0	EINT21_CTL External INT21 Enable 0: Disable 1: Enable
20	R/W	0x0	EINT20_CTL External INT20 Enable 0: Disable 1: Enable

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Offset: 0x00D0			Register Name: PA_INT_CTL
Bit	Read/Write	Default/Hex	Description
19	R/W	0x0	EINT19_CTL External INT19 Enable 0: Disable 1: Enable
18	R/W	0x0	EINT18_CTL External INT18 Enable 0: Disable 1: Enable
17	R/W	0x0	EINT17_CTL External INT17 Enable 0: Disable 1: Enable
16	R/W	0x0	EINT16_CTL External INT16 Enable 0: Disable 1: Enable
15	R/W	0x0	EINT15_CTL External INT15 Enable 0: Disable 1: Enable
14	R/W	0x0	EINT14_CTL External INT14 Enable 0: Disable 1: Enable
13	R/W	0x0	EINT13_CTL External INT13 Enable 0: Disable 1: Enable
12	R/W	0x0	EINT12_CTL External INT12 Enable 0: Disable 1: Enable
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable

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Offset: 0x00D0			Register Name: PA_INT_CTL
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

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17.9.6.21 0x00D4 PA External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x00D4			Register Name: PA_INT_STA
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	EINT23_STATUS External INT23 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
22	R/W1C	0x0	EINT22_STATUS External INT22 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
21	R/W1C	0x0	EINT21_STATUS External INT21 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
20	R/W1C	0x0	EINT20_STATUS External INT20 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
19	R/W1C	0x0	EINT19_STATUS External INT19 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
18	R/W1C	0x0	EINT18_STATUS External INT18 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
17	R/W1C	0x0	EINT17_STATUS External INT17 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x00D4			Register Name: PA_INT_STA
Bit	Read/Write	Default/Hex	Description
16	R/W1C	0x0	EINT16_STATUS External INT16 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
15	R/W1C	0x0	EINT15_STATUS External INT15 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
14	R/W1C	0x0	EINT14_STATUS External INT14 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
13	R/W1C	0x0	EINT13_STATUS External INT13 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
12	R/W1C	0x0	EINT12_STATUS External INT12 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x00D4			Register Name: PA_INT_STA
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x00D4			Register Name: PA_INT_STA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.22 0x00D8 PA External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x00D8			Register Name: PA_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.23 0x00DC PA External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x00DC			Register Name: PA_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.24 0x00F0 PA Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x00F0			Register Name: PA_SECURE
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x00F0			Register Name: PA_SECURE
Bit	Read/Write	Default/Hex	Description
23	R/W	0x0	PORT23_SEC_SEL Port23 Secure Selection 0: Not secure 1: Secure
22	R/W	0x0	PORT22_SEC_SEL Port22 Secure Selection 0: Not secure 1: Secure
21	R/W	0x0	PORT21_SEC_SEL Port21 Secure Selection 0: Not secure 1: Secure
20	R/W	0x0	PORT20_SEC_SEL Port20 Secure Selection 0: Not secure 1: Secure
19	R/W	0x0	PORT19_SEC_SEL Port19 Secure Selection 0: Not secure 1: Secure
18	R/W	0x0	PORT18_SEC_SEL Port18 Secure Selection 0: Not secure 1: Secure
17	R/W	0x0	PORT17_SEC_SEL Port17 Secure Selection 0: Not secure 1: Secure
16	R/W	0x0	PORT16_SEC_SEL Port16 Secure Selection 0: Not secure 1: Secure
15	R/W	0x0	PORT15_SEC_SEL Port15 Secure Selection 0: Not secure 1: Secure
14	R/W	0x0	PORT14_SEC_SEL Port14 Secure Selection 0: Not secure 1: Secure

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Offset: 0x00F0			Register Name: PA_SECURE
Bit	Read/Write	Default/Hex	Description
13	R/W	0x0	PORT13_SEC_SEL Port13 Secure Selection 0: Not secure 1: Secure
12	R/W	0x0	PORT12_SEC_SEL Port12 Secure Selection 0: Not secure 1: Secure
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure

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Offset: 0x00F0			Register Name: PA_SECURE
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.25 0x0100 PB Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0100			Register Name: PB_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PB7_SELECT PB7 Select 0000: Input 0001: Output 0010: OWA-IN 0011: I2S0-DOUT0 0100: I2S0-DIN1 0101: PWM1-3 0110: CAN1-TX0 0111: UART8-CTS 1000: TWI0-SDA 1001: RGMII2-TXCTL/RMII2-TXEN 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT7 1111: Disable

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Offset: 0x0100			Register Name: PB_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PB6_SELECT PB6 Select 0000: Input 0001: Output 0010: Reserved 0011: I2S0-LRCK 0100: Reserved 0101: PWM1-2 0110: IR-TX 0111: UART8-RTS 1000: TWI0-SCK 1001: RGMII2-TXCK/RMII2-TXCK 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT6 1111: Disable
23:20	R/W	0xF	PB5_SELECT PB5 Select 0000: Input 0001: Output 0010: TWI1-SDA 0011: I2S0-BCLK 0100: SPI3-CS2 0101: PWM1-1 0110: IR1-RX 0111: UART8-RX 1000: Reserved 1001: RGMII2-TXD0/RMII2-TXD0 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT5 1111: Disable
19:16	R/W	0xF	PB4_SELECT PB4 Select 0000: Input 0001: Output 0010: TWI1-SCK 0011: I2S0-MCLK 0100: SPI3-CS1 0101: PWM1-0 0110: IR0-RX 0111: UART8-TX 1000: Reserved 1001: RGMII2-TXD1/RMII2-TXD1 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT4 1111: Disable

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Offset: 0x0100			Register Name: PB_CFG0
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PB3_SELECT PB3 Select 0000: Input 0001: Output 0010: UART2-CTS 0011: SPI3-MISO 0100: JTAG-DI 0101: TWI2-SDA 0110: Reserved 0111: RJTAG-DI 1000: Reserved 1001: RGMII2-CLKIN/RMII2-RXER 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT3 1111: Disable
11:8	R/W	0xF	PB2_SELECT PB2 Select 0000: Input 0001: Output 0010: UART2-RTS 0011: SPI3-MOSI 0100: JTAG-DO 0101: TWI2-SCK 0110: Reserved 0111: RJTAG-DO 1000: Reserved 1001: RGMII2-RXCTL/RMII2-CRS-DV 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT2 1111: Disable
7:4	R/W	0xF	PB1_SELECT PB1 Select 0000: Input 0001: Output 0010: UART2-RX 0011: SPI3-CLK 0100: JTAG-CK 0101: PWM0-7 0110: Reserved 0111: RJTAG-CK 1000: Reserved 1001: RGMII2-RXD0/RMII2-RXD0 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT1 1111: Disable

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Offset: 0x0100			Register Name: PB_CFG0
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PB0_SELECT PB0 Select 0000: Input 0001: Output 0010: UART2-TX 0011: SPI3-CS0 0100: JTAG-MS 0101: PWM0-6 0110: Reserved 0111: RJTAG-MS 1000: Reserved 1001: RGMII2-RXD1/RMII2-RXD1 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT0 1111: Disable

17.9.6.26 0x0104 PB Configure Register 1 (Default Value: 0x0FFF_FFFF)

Offset: 0x0104			Register Name: PB_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:24	R/W	0xF	PB14_SELECT PB14 Select 0000: Input 0001: Output 0010: TWI4-SDA 0011: UART7-RX 0100: SPI3-MISO 0101: PWM0-5 0110: IR3-RX 0111: CLK-FANOUT1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT14 1111: Disable
23:20	R/W	0xF	PB13_SELECT PB13 Select 0000: Input 0001: Output 0010: TWI4-SCK 0011: UART7-TX 0100: SPI3-MOSI 0101: PWM0-4 0110: IR2-RX 0111: CLK-FANOUT0 1000: Reserved 1001: RGMII2-EPHY-25/50M 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT13 1111: Disable

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Offset: 0x0104			Register Name: PB_CFG1	
Bit	Read/Write	Default/Hex	Description	
19:16	R/W	0xF	PB12_SELECT PB12 Select 0000: Input 0010: TWI5-SDA 0100: SPI3-CLK 0110: Reserved 1000: Reserved 1010: Reserved 1100: Reserved 1110: PB-EINT12	 0001: Output 0011: UART7-CTS 0101: PWM0-3 0111: Reserved 1001: RGMII2-MDIO 1011: Reserved 1101: Reserved 1111: Disable
15:12	R/W	0xF	PB11_SELECT PB11 Select 0000: Input 0010: TWI5-SCK 0100: SPI3-CS0 0110: Reserved 1000: Reserved 1010: Reserved 1100: Reserved 1110: PB-EINT11	 0001: Output 0011: UART7-RTS 0101: PWM0-2 0111: UART8-RI 1001: RGMII2-MDC 1011: Reserved 1101: Reserved 1111: Disable
11:8	R/W	0xF	PB10_SELECT PB10 Select 0000: Input 0010: UART0-RX 0100: I2S0-DOUT3 0110: TWI0-SDA 1000: Reserved 1010: Reserved 1100: Reserved 1110: PB-EINT10	 0001: Output 0011: I2S0-DIN3 0101: PWM0-1 0111: UART8-DTR 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable
7:4	R/W	0xF	PB9_SELECT PB9 Select 0000: Input 0010: UART0-TX 0100: I2S0-DOUT2 0110: TWI0-SCK 1000: Reserved 1010: Reserved 1100: Reserved 1110: PB-EINT9	 0001: Output 0011: I2S0-DIN2 0101: Reserved 0111: UART8-DSR 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable

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Offset: 0x0104			Register Name: PB_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PB8_SELECT PB8 Select 0000: Input 0001: Output 0010: OWA-OUT 0011: I2S0-DIN0 0100: I2S0-DOUT1 0101: PWM0-0 0110: CAN1-RX0 0111: UART8-DCD 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PB-EINT8 1111: Disable

17.9.6.27 0x0110 PB Data Register (Default Value: 0x0000_0000)

Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W	0x0	PORT14_DAT PB14 Data If the function of PB14 port is configured as input, the PORT14_DAT bit indicates the sate of the PB14 pin; If the function of PB14 port is configured as output, the sate of the PB14 pin can be configured via software by modifying the value of the PORT14_DAT bit. In this case, the value obtained by reading PORT14_DAT bit is the value set by the software; If the PB14 port is configured as functional pin, this bit is invalid.
13	R/W	0x0	PORT13_DAT PB13 Data If the function of PB13 port is configured as input, the PORT13_DAT bit indicates the sate of the PB13 pin; If the function of PB13 port is configured as output, the sate of the PB13 pin can be configured via software by modifying the value of the PORT13_DAT bit. In this case, the value obtained by reading PORT13_DAT bit is the value set by the software; If the PB13 port is configured as functional pin, this bit is invalid.

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Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PORT12_DAT PB12 Data If the function of PB12 port is configured as input, the PORT12_DAT bit indicates the sate of the PB12 pin; If the function of PB12 port is configured as output, the sate of the PB12 pin can be configured via software by modifying the value of the PORT12_DAT bit. In this case, the value obtained by reading PORT12_DAT bit is the value set by the software; If the PB12 port is configured as functional pin, this bit is invalid.
11	R/W	0x0	PORT11_DAT PB11 Data If the function of PB11 port is configured as input, the PORT11_DAT bit indicates the sate of the PB11 pin; If the function of PB11 port is configured as output, the sate of the PB11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PB11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PB10 Data If the function of PB10 port is configured as input, the PORT10_DAT bit indicates the sate of the PB10 pin; If the function of PB10 port is configured as output, the sate of the PB10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PB10 port is configured as functional pin, this bit is invalid.

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Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	PORT9_DAT PB9 Data If the function of PB9 port is configured as input, the PORT9_DAT bit indicates the sate of the PB9 pin; If the function of PB9 port is configured as output, the sate of the PB9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PB9 port is configured as functional pin, this bit is invalid.
8	R/W	0x0	PORT8_DAT PB8 Data If the function of PB8 port is configured as input, the PORT8_DAT bit indicates the sate of the PB8 pin; If the function of PB8 port is configured as output, the sate of the PB8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PB8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PB7 Data If the function of PB7 port is configured as input, the PORT7_DAT bit indicates the sate of the PB7 pin; If the function of PB7 port is configured as output, the sate of the PB7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PB7 port is configured as functional pin, this bit is invalid.

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Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PORT6_DAT PB6 Data If the function of PB6 port is configured as input, the PORT6_DAT bit indicates the sate of the PB6 pin; If the function of PB6 port is configured as output, the sate of the PB6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PB6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PB5 Data If the function of PB5 port is configured as input, the PORT5_DAT bit indicates the sate of the PB5 pin; If the function of PB5 port is configured as output, the sate of the PB5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PB5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PB4 Data If the function of PB4 port is configured as input, the PORT4_DAT bit indicates the sate of the PB4 pin; If the function of PB4 port is configured as output, the sate of the PB4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PB4 port is configured as functional pin, this bit is invalid.

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Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_DAT PB3 Data If the function of PB3 port is configured as input, the PORT3_DAT bit indicates the sate of the PB3 pin; If the function of PB3 port is configured as output, the sate of the PB3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PB3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PB2 Data If the function of PB2 port is configured as input, the PORT2_DAT bit indicates the sate of the PB2 pin; If the function of PB2 port is configured as output, the sate of the PB2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PB2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PB1 Data If the function of PB1 port is configured as input, the PORT1_DAT bit indicates the sate of the PB1 pin; If the function of PB1 port is configured as output, the sate of the PB1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PB1 port is configured as functional pin, this bit is invalid.

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Offset: 0x0110			Register Name: PB_DAT
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PORT0_DAT PB0 Data If the function of PB0 port is configured as input, the PORT0_DAT bit indicates the sate of the PB0 pin; If the function of PB0 port is configured as output, the sate of the PB0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PB0 port is configured as functional pin, this bit is invalid.

17.9.6.28 0x0114 PB Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0114			Register Name: PB_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1S	0x0	PORT14_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
13	R/W1S	0x0	PORT13_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
12	R/W1S	0x0	PORT12_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.

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Offset: 0x0114			Register Name: PB_DAT_SET
Bit	Read/Write	Default/Hex	Description
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.

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Offset: 0x0114			Register Name: PB_DAT_SET
Bit	Read/Write	Default/Hex	Description
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PB _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePB _DAT register.

17.9.6.29 0x0118 PB Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0118			Register Name: PB_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1C	0x0	PORT14_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
13	R/W1C	0x0	PORT13_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
12	R/W1C	0x0	PORT12_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.

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Offset: 0x0118			Register Name: PB_DAT_CLR
Bit	Read/Write	Default/Hex	Description
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.

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Offset: 0x0118			Register Name: PB_DAT_CLR
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PB_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PB_DAT register.

17.9.6.30 0x0120 PB Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x0120			Register Name: PB_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PB7_DRV PB7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PB6_DRV PB6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PB5_DRV PB5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PB4_DRV PB4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/

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Offset: 0x0120			Register Name: PB_DRV0
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x1	PB3_DRV PB3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PB2_DRV PB2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PB1_DRV PB1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PB0_DRV PB0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.31 0x0124 PB Multi_Driving Register 1 (Default Value: 0x0111_1111)

Offset: 0x0124			Register Name: PB_DRV1
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x1	PB14_DRV PB14 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PB13_DRV PB13 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PB12_DRV PB12 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/

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Offset: 0x0124			Register Name: PB_DRV1
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x1	PB11_DRV PB11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PB10_DRV PB10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PB9_DRV PB9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PB8_DRV PB8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.32 0x0130 PB Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x0130			Register Name: PB_PUL0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	PB14_PULL PB14 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
27:26	R/W	0x0	PB13_PULL PB13 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
25:24	R/W	0x0	PB12_PULL PB12 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
23:22	R/W	0x0	PB11_PULL PB11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0130			Register Name: PB_PUL0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x0	PB10_PULL PB10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PB9_PULL PB9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PB8_PULL PB8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PB7_PULL PB7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PB6_PULL PB6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PB5_PULL PB5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PB4_PULL PB4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PB3_PULL PB3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PB2_PULL PB2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PB1_PULL PB1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0130			Register Name: PB_PUL0
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	PB0_PULL PB0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.33 0x0140 PB External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x0140			Register Name: PB_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0140			Register Name: PB_INT_CFG0
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.34 0x0144 PB External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x0144			Register Name: PB_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/

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Offset: 0x0144			Register Name: PB_INT_CFG1
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT14_CFG External INT14 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT13_CFG External INT13 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT12_CFG External INT12 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0144			Register Name: PB_INT_CFG1
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.35 0x0150 PB External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0150			Register Name: PB_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W	0x0	EINT14_CTL External INT14 Enable 0: Disable 1: Enable
13	R/W	0x0	EINT13_CTL External INT13 Enable 0: Disable 1: Enable
12	R/W	0x0	EINT12_CTL External INT12 Enable 0: Disable 1: Enable
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable

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Offset: 0x0150			Register Name: PB_INT_CTL
Bit	Read/Write	Default/Hex	Description
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable

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Offset: 0x0150			Register Name: PB_INT_CTL
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.36 0x0154 PB External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0154			Register Name: PB_INT_STA
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1C	0x0	EINT14_STATUS External INT14 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
13	R/W1C	0x0	EINT13_STATUS External INT13 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
12	R/W1C	0x0	EINT12_STATUS External INT12 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0154			Register Name: PB_INT_STA
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0154			Register Name: PB_INT_STA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.37 0x0158 PB External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x0158			Register Name: PB_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.38 0x015C PB External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x015C			Register Name: PB_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.39 0x0170 PB Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: PB_SECURE
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/

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Offset: 0x0170			Register Name: PB_SECURE
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	PORT14_SEC_SEL Port14 Secure Selection 0: Not secure 1: Secure
13	R/W	0x0	PORT13_SEC_SEL Port13 Secure Selection 0: Not secure 1: Secure
12	R/W	0x0	PORT12_SEC_SEL Port12 Secure Selection 0: Not secure 1: Secure
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure

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Offset: 0x0170			Register Name: PB_SECURE
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.40 0x0180 PC Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0180			Register Name: PC_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PC7_SELECT PC7 Select 0000: Input 0001: Output 0010: SDC2-D5 0011: Reserved 0100: SPIF0-D5 0101: CAN0-RX0 0110: UART6-RX 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT7 1111: Disable

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Offset: 0x0180			Register Name: PC_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PC6_SELECT PC6 Select 0000: Input 0001: Output 0010: SDC2-D4 0011: SPI0-CS1 0100: SPIF0-D4 0101: CAN0-TX0 0110: UART6-TX 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT6 1111: Disable
23:20	R/W	0xF	PC5_SELECT PC5 Select 0000: Input 0001: Output 0010: SDC2-D3 0011: SPI0-HOLD 0100: SPIF0-HOLD 0101: CAN1-RX0 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT5 1111: Disable
19:16	R/W	0xF	PC4_SELECT PC4 Select 0000: Input 0001: Output 0010: SDC2-D0 0011: SPI0-WP 0100: SPIF0-WP 0101: CAN1-TX0 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT4 1111: Disable
15:12	R/W	0xF	PC3_SELECT PC3 Select 0000: Input 0001: Output 0010: SDC2-D1 0011: SPI0-MISO 0100: SPIF0-MISO 0101: BOOT-SEL1 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT3 1111: Disable

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Offset: 0x0180			Register Name: PC_CFG0
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PC2_SELECT PC2 Select 0000: Input 0001: Output 0010: SDC2-D2 0011: SPI0-MOSI 0100: SPIF0-MOSI 0101: BOOT-SEL0 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT2 1111: Disable
7:4	R/W	0xF	PC1_SELECT PC1 Select 0000: Input 0001: Output 0010: SDC2-CMD 0011: SPI0-CS0 0100: SPIF0-CS0 0101: Reserved 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT1 1111: Disable
3:0	R/W	0xF	PC0_SELECT PC0 Select 0000: Input 0001: Output 0010: SDC2-CLK 0011: SPI0-CLK 0100: SPIF0-CLK 0101: Reserved 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT0 1111: Disable

17.9.6.41 0x0184 PC Configure Register 1 (Default Value: 0x0000_FFFF)

Offset: 0x0184			Register Name: PC_CFG1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0184			Register Name: PC_CFG1
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PC11_SELECT PC11 Select 0000: Input 0001: Output 0010: SDC2-RST 0011: Reserved 0100: Reserved 0101: UART5-RTS 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT11 1111: Disable
11:8	R/W	0xF	PC10_SELECT PC10 Select 0000: Input 0001: Output 0010: SDC2-DS 0011: Reserved 0100: SPIF0-DQS 0101: UART5-RX 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT10 1111: Disable
7:4	R/W	0xF	PC9_SELECT PC9 Select 0000: Input 0001: Output 0010: SDC2-D7 0011: Reserved 0100: SPIF0-D7 0101: UART5-TX 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT9 1111: Disable
3:0	R/W	0xF	PC8_SELECT PC8 Select 0000: Input 0001: Output 0010: SDC2-D6 0011: Reserved 0100: SPIF0-D6 0101: UART5-CTS 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PC-EINT8 1111: Disable

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17.9.6.42 0x0190 PC Data Register (Default Value: 0x0000_0000)

Offset: 0x0190			Register Name: PC_DAT
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	PORT11_DAT PC11 Data If the function of PC11 port is configured as input, the PORT11_DAT bit indicates the sate of the PC11 pin; If the function of PC11 port is configured as output, the sate of the PC11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PC11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PC10 Data If the function of PC10 port is configured as input, the PORT10_DAT bit indicates the sate of the PC10 pin; If the function of PC10 port is configured as output, the sate of the PC10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PC10 port is configured as functional pin, this bit is invalid.
9	R/W	0x0	PORT9_DAT PC9 Data If the function of PC9 port is configured as input, the PORT9_DAT bit indicates the sate of the PC9 pin; If the function of PC9 port is configured as output, the sate of the PC9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PC9 port is configured as functional pin, this bit is invalid.

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Offset: 0x0190			Register Name: PC_DAT
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PORT8_DAT PC8 Data If the function of PC8 port is configured as input, the PORT8_DAT bit indicates the sate of the PC8 pin; If the function of PC8 port is configured as output, the sate of the PC8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PC8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PC7 Data If the function of PC7 port is configured as input, the PORT7_DAT bit indicates the sate of the PC7 pin; If the function of PC7 port is configured as output, the sate of the PC7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PC7 port is configured as functional pin, this bit is invalid.
6	R/W	0x0	PORT6_DAT PC6 Data If the function of PC6 port is configured as input, the PORT6_DAT bit indicates the sate of the PC6 pin; If the function of PC6 port is configured as output, the sate of the PC6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PC6 port is configured as functional pin, this bit is invalid.

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Offset: 0x0190			Register Name: PC_DAT
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_DAT PC5 Data If the function of PC5 port is configured as input, the PORT5_DAT bit indicates the sate of the PC5 pin; If the function of PC5 port is configured as output, the sate of the PC5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PC5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PC4 Data If the function of PC4 port is configured as input, the PORT4_DAT bit indicates the sate of the PC4 pin; If the function of PC4 port is configured as output, the sate of the PC4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PC4 port is configured as functional pin, this bit is invalid.
3	R/W	0x0	PORT3_DAT PC3 Data If the function of PC3 port is configured as input, the PORT3_DAT bit indicates the sate of the PC3 pin; If the function of PC3 port is configured as output, the sate of the PC3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PC3 port is configured as functional pin, this bit is invalid.

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Offset: 0x0190			Register Name: PC_DAT
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PORT2_DAT PC2 Data If the function of PC2 port is configured as input, the PORT2_DAT bit indicates the sate of the PC2 pin; If the function of PC2 port is configured as output, the sate of the PC2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PC2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PC1 Data If the function of PC1 port is configured as input, the PORT1_DAT bit indicates the sate of the PC1 pin; If the function of PC1 port is configured as output, the sate of the PC1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PC1 port is configured as functional pin, this bit is invalid.
0	R/W	0x0	PORT0_DAT PC0 Data If the function of PC0 port is configured as input, the PORT0_DAT bit indicates the sate of the PC0 pin; If the function of PC0 port is configured as output, the sate of the PC0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PC0 port is configured as functional pin, this bit is invalid.

17.9.6.43 0x0194 PC Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0194			Register Name: PC_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/

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Offset: 0x0194			Register Name: PC_DAT_SET
Bit	Read/Write	Default/Hex	Description
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.

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Offset: 0x0194			Register Name: PC_DAT_SET
Bit	Read/Write	Default/Hex	Description
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PC _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePC _DAT register.

17.9.6.44 0x0198 PC Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0198			Register Name: PC_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.

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Offset: 0x0198			Register Name: PC_DAT_CLR
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.

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Offset: 0x0198			Register Name: PC_DAT_CLR
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PC_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PC_DAT register.

17.9.6.45 0x01A0 PC Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x01A0			Register Name: PC_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PC7_DRV PC7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PC6_DRV PC6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PC5_DRV PC5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PC4_DRV PC4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PC3_DRV PC3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PC2_DRV PC2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/

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Offset: 0x01A0			Register Name: PC_DRV0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x1	PC1_DRV PC1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PC0_DRV PC0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.46 0x01A4 PC Multi_Driving Register 1 (Default Value: 0x0000_1111)

Offset: 0x01A4			Register Name: PC_DRV1
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:12	R/W	0x1	PC11_DRV PC11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PC10_DRV PC10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PC9_DRV PC9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PC8_DRV PC8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.47 0x01B0 PC Pull Register 0 (Default Value: 0x0000_0054)

Offset: 0x01B0			Register Name: PC_PUL0
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x01B0			Register Name: PC_PUL0
Bit	Read/Write	Default/Hex	Description
23:22	R/W	0x0	PC11_PULL PC11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PC10_PULL PC10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PC9_PULL PC9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PC8_PULL PC8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PC7_PULL PC7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PC6_PULL PC6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PC5_PULL PC5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PC4_PULL PC4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x1	PC3_PULL PC3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x1	PC2_PULL PC2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x01B0			Register Name: PC_PUL0
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x1	PC1_PULL PC1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PC0_PULL PC0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.48 0x01C0 PC External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x01C0			Register Name: PC_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x01C0			Register Name: PC_INT_CFG0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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17.9.6.49 0x01C4 PC External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x01C4			Register Name: PC_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.50 0x01D0 PC External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x01D0			Register Name: PC_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/

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Offset: 0x01D0			Register Name: PC_INT_CTL
Bit	Read/Write	Default/Hex	Description
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable

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Offset: 0x01D0			Register Name: PC_INT_CTL
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.51 0x01D4 PC External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x01D4			Register Name: PC_INT_STA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x01D4			Register Name: PC_INT_STA
Bit	Read/Write	Default/Hex	Description
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.52 0x01D8 PC External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x01D8			Register Name: PC_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/

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Offset: 0x01D8			Register Name: PC_INT_DEB
Bit	Read/Write	Default/Hex	Description
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.53 0x01DC PC External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x01DC			Register Name: PC_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.54 0x01F0 PC Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x01F0			Register Name: PC_SECURE
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure

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Offset: 0x01F0			Register Name: PC_SECURE
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.55 0x0200 PD Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0200			Register Name: PD_CFG0
Bit	Read/Write	Default/Hex	Description

Draft Version

Offset: 0x0200			Register Name: PD_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PD7_SELECT PD7 Select 0000: Input 0001: Output 0010: LCD-D11 0011: LVDS0-CKN 0100: DSI-D2N 0101: PWM0-7 0110: UART4-RX 0111: UART6-CTS 1000: LBUS-LD27 1001: PWMCS-O7 1010: LBUS-READY3 1011: RGMII2-TXCTL/RMII2-TXEN 1100: Reserved 1101: Reserved 1110: PD-EINT7 1111: Disable
27:24	R/W	0xF	PD6_SELECT PD6 Select 0000: Input 0001: Output 0010: LCD-D10 0011: LVDS0-CKP 0100: DSI-D2P 0101: PWM0-6 0110: UART4-TX 0111: UART6-RTS 1000: LBUS-LD26 1001: PWMCS-O6 1010: LBUS-CS2 1011: RGMII2-TXCK/RMII2-TXCK 1100: Reserved 1101: Reserved 1110: PD-EINT6 1111: Disable
23:20	R/W	0xF	PD5_SELECT PD5 Select 0000: Input 0001: Output 0010: LCD-D7 0011: LVDS0-D2N 0100: DSI-CKN 0101: PWM0-5 0110: UART4-RTS 0111: UART6-RX 1000: LBUS-LD25 1001: PWMCS-O5 1010: LBUS-CS3 1011: RGMII2-TXD0/RMII2-TXD0 1100: Reserved 1101: Reserved 1110: PD-EINT5 1111: Disable

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Offset: 0x0200			Register Name: PD_CFG0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0xF	PD4_SELECT PD4 Select 0000: Input 0001: Output 0010: LCD-D6 0011: LVDS0-D2P 0100: DSI-CKP 0101: PWM0-4 0110: UART4-CTS 0111: UART6-TX 1000: LBUS-LD24 1001: PWMCS-O4 1010: Reserved 1011: RGMII2-TXD1/RMII2-TXD1 1100: Reserved 1101: Reserved 1110: PD-EINT4 1111: Disable
15:12	R/W	0xF	PD3_SELECT PD3 Select 0000: Input 0001: Output 0010: LCD-D5 0011: LVDS0-D1N 0100: DSI-D1N 0101: PWM0-3 0110: UART2-RX 0111: UART9-CTS 1000: LBUS-LD23 1001: PWMCS-O3 1010: Reserved 1011: RGMII2-CLKIN/RMII2-RXER 1100: Reserved 1101: Reserved 1110: PD-EINT3 1111: Disable
11:8	R/W	0xF	PD2_SELECT PD2 Select 0000: Input 0001: Output 0010: LCD-D4 0011: LVDS0-D1P 0100: DSI-D1P 0101: PWM0-2 0110: UART2-TX 0111: UART9-RTS 1000: LBUS-LD22 1001: PWMCS-O2 1010: Reserved 1011: RGMII2-RXCTL/RMII2-CRS-DV 1100: Reserved 1101: Reserved 1110: PD-EINT2 1111: Disable

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Offset: 0x0200			Register Name: PD_CFG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0xF	PD1_SELECT PD1 Select 0000: Input 0001: Output 0010: LCD-D3 0011: LVDS0-D0N 0100: DSI-D0N 0101: PWM0-1 0110: UART2-RTS 0111: UART9-RX 1000: LBUS-LD21 1001: PWMCS-O1 1010: Reserved 1011: RGMII2-RXD0/RMII2-RXD0 1100: Reserved 1101: Reserved 1110: PD-EINT1 1111: Disable
3:0	R/W	0xF	PD0_SELECT PD0 Select 0000: Input 0001: Output 0010: LCD-D2 0011: LVDS0-D0P 0100: DSI-D0P 0101: PWM0-0 0110: UART2-CTS 0111: UART9-TX 1000: LBUS-LD20 1001: PWMCS-O0 1010: Reserved 1011: RGMII2-RXD1/RMII2-RXD1 1100: Reserved 1101: Reserved 1110: PD-EINT0 1111: Disable

17.9.6.56 0x0204 PD Configure Register 1 (Default Value: 0xFFFF_FFFF)

Offset: 0x0204			Register Name: PD_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PD15_SELECT PD15 Select 0000: Input 0001: Output 0010: LCD-D21 0011: LVDS1-D2N 0100: SPI1-WP/DBI-TE 0101: PWM1-7 0110: UART3-RX 0111: UART8-DSR 1000: LBUS-LD3 1001: PWMCS-OF 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT15 1111: Disable

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Offset: 0x0204			Register Name: PD_CFG1
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PD14_SELECT PD14 Select 0000: Input 0001: Output 0010: LCD-D20 0011: LVDS1-D2P 0100: SPI1-HOLD/DBI-DCX/DBI-WRX 0101: PWM1-6 0110: UART3-TX 0111: UART8-DCD 1000: LBUS-LD2 1001: PWMCS-OE 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT14 1111: Disable
23:20	R/W	0xF	PD13_SELECT PD13 Select 0000: Input 0001: Output 0010: LCD-D19 0011: LVDS1-D1N 0100: SPI1-MISO/DBI-SDI/DBI-TE/DBI-DCX 0101: PWM1-5 0110: DMIC-DATA3 0111: UART8-CTS 1000: LBUS-LD1 1001: PWMCS-OD 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT13 1111: Disable
19:16	R/W	0xF	PD12_SELECT PD12 Select 0000: Input 0001: Output 0010: LCD-D18 0011: LVDS1-D1P 0100: SPI1-MOSI/DBI-SDO 0101: PWM1-4 0110: DMIC-DATA2 0111: UART8-RTS 1000: LBUS-LD0 1001: PWMCS-OC 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT12 1111: Disable
15:12	R/W	0xF	PD11_SELECT PD11 Select 0000: Input 0001: Output 0010: LCD-D15 0011: LVDS1-D0N 0100: SPI1-CLK/DBI-SCLK 0101: PWM1-3 0110: DMIC-DATA1 0111: UART8-RX 1000: LBUS-LD31 1001: PWMCS-OB 1010: LBUS-READY0 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT11 1111: Disable

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Offset: 0x0204			Register Name: PD_CFG1
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PD10_SELECT PD10 Select 0000: Input 0001: Output 0010: LCD-D14 0011: LVDS1-D0P 0100: SPI1-CS0/DBI-CSX 0101: PWM1-2 0110: DMIC-DATA0 0111: UART8-TX 1000: LBUS-LD30 1001: PWMCS-OA 1010: LBUS-READY1 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT10 1111: Disable
7:4	R/W	0xF	PD9_SELECT PD9 Select 0000: Input 0001: Output 0010: LCD-D13 0011: LVDS0-D3N 0100: DSI-D3N 0101: PWM1-1 0110: DMIC-CLK 0111: CAN1-RX0 1000: LBUS-LD29 1001: PWMCS-O9 1010: LBUS-READY2 1011: RGMII2-MDIO 1100: Reserved 1101: Reserved 1110: PD-EINT9 1111: Disable
3:0	R/W	0xF	PD8_SELECT PD8 Select 0000: Input 0001: Output 0010: LCD-D12 0011: LVDS0-D3P 0100: DSI-D3P 0101: PWM1-0 0110: Reserved 0111: CAN1-TX0 1000: LBUS-LD28 1001: PWMCS-O8 1010: LBUS-DE 1011: RGMII2-MDC 1100: Reserved 1101: Reserved 1110: PD-EINT8 1111: Disable

17.9.6.57 0x0208 PD Configure Register 2 (Default Value: 0xFFFF_FFFF)

Offset: 0x0208			Register Name: PD_CFG2
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0208			Register Name: PD_CFG2
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PD23_SELECT PD23 Select 0000: Input 0001: Output 0010: Reserved 0011: PWM2-7 0100: UART3-CTS 0101: TWI4-SDA 0110: TWI0-SDA 0111: CSI1-XHS 1000: LBUS-DRQ1 1001: PWMCS-I7 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT23 1111: Disable
27:24	R/W	0xF	PD22_SELECT PD22 Select 0000: Input 0001: Output 0010: Reserved 0011: PWM2-6 0100: UART3-RTS 0101: TWI4-SCK 0110: TWI0-SCK 0111: CSI0-XHS 1000: LBUS-DRQ0 1001: PWMCS-I6 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PD-EINT22 1111: Disable
23:20	R/W	0xF	PD21_SELECT PD21 Select 0000: Input 0001: Output 0010: LCD-VSYNC 0011: PWM2-5 0100: UART3-RX 0101: TWI5-SDA 0110: Reserved 0111: Reserved 1000: LBUS-OE 1001: PWMCS-I5 1010: Reserved 1011: RGMII2-TXD2/RMII2-NULL 1100: Reserved 1101: Reserved 1110: PD-EINT21 1111: Disable
19:16	R/W	0xF	PD20_SELECT PD20 Select 0000: Input 0001: Output 0010: LCD-HSYNC 0011: PWM2-4 0100: UART3-TX 0101: TWI5-SCK 0110: Reserved 0111: Reserved 1000: LBUS-WR 1001: PWMCS-I4 1010: Reserved 1011: RGMII2-TXD3/RMII2-NULL 1100: Reserved 1101: Reserved 1110: PD-EINT20 1111: Disable

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Offset: 0x0208			Register Name: PD_CFG2
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PD19_SELECT PD19 Select 0000: Input 0001: Output 0010: LCD-DE 0011: LVDS1-D3N 0100: TWI5-SDA 0101: PWM2-3 0110: Reserved 0111: CAN0-RX0 1000: LBUS-LD7 1001: PWMCS-I3 1010: Reserved 1011: RGMII2-RXCK/RMII2-NULL 1100: Reserved 1101: Reserved 1110: PD-EINT19 1111: Disable
11:8	R/W	0xF	PD18_SELECT PD18 Select 0000: Input 0001: Output 0010: LCD-CLK 0011: LVDS1-D3P 0100: TWI5-SCK 0101: PWM2-2 0110: Reserved 0111: CAN0-TX0 1000: LBUS-LD6 1001: PWMCS-I2 1010: Reserved 1011: RGMII2-RXD2/RMII2-NULL 1100: Reserved 1101: Reserved 1110: PD-EINT18 1111: Disable
7:4	R/W	0xF	PD17_SELECT PD17 Select 0000: Input 0001: Output 0010: LCD-D23 0011: LVDS1-CKN 0100: Reserved 0101: PWM2-1 0110: UART3-CTS 0111: UART8-RI 1000: LBUS-LD5 1001: PWMCS-I1 1010: Reserved 1011: RGMII2-RXD3/RMII2-NULL 1100: Reserved 1101: Reserved 1110: PD-EINT17 1111: Disable

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Offset: 0x0208			Register Name: PD_CFG2
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PD16_SELECT PD16 Select 0000: Input 0001: Output 0010: LCD-D22 0011: LVDS1-CKP 0100: Reserved 0101: PWM2-0 0110: UART3-RTS 0111: UART8-DTR 1000: LBUS-LD4 1001: PWMCS-I0 1010: Reserved 1011: RGMII2-EPHY-25/50M 1100: Reserved 1101: Reserved 1110: PD-EINT16 1111: Disable

17.9.6.58 0x0210 PD Data Register (Default Value: 0x0000_0000)

Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	PORT23_DAT PD23 Data If the function of PD23 port is configured as input, the PORT23_DAT bit indicates the sate of the PD23 pin; If the function of PD23 port is configured as output, the sate of the PD23 pin can be configured via software by modifying the value of the PORT23_DAT bit. In this case, the value obtained by reading PORT23_DAT bit is the value set by the software; If the PD23 port is configured as functional pin, this bit is invalid.
22	R/W	0x0	PORT22_DAT PD22 Data If the function of PD22 port is configured as input, the PORT22_DAT bit indicates the sate of the PD22 pin; If the function of PD22 port is configured as output, the sate of the PD22 pin can be configured via software by modifying the value of the PORT22_DAT bit. In this case, the value obtained by reading PORT22_DAT bit is the value set by the software; If the PD22 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
21	R/W	0x0	PORT21_DAT PD21 Data If the function of PD21 port is configured as input, the PORT21_DAT bit indicates the sate of the PD21 pin; If the function of PD21 port is configured as output, the sate of the PD21 pin can be configured via software by modifying the value of the PORT21_DAT bit. In this case, the value obtained by reading PORT21_DAT bit is the value set by the software; If the PD21 port is configured as functional pin, this bit is invalid.
20	R/W	0x0	PORT20_DAT PD20 Data If the function of PD20 port is configured as input, the PORT20_DAT bit indicates the sate of the PD20 pin; If the function of PD20 port is configured as output, the sate of the PD20 pin can be configured via software by modifying the value of the PORT20_DAT bit. In this case, the value obtained by reading PORT20_DAT bit is the value set by the software; If the PD20 port is configured as functional pin, this bit is invalid.
19	R/W	0x0	PORT19_DAT PD19 Data If the function of PD19 port is configured as input, the PORT19_DAT bit indicates the sate of the PD19 pin; If the function of PD19 port is configured as output, the sate of the PD19 pin can be configured via software by modifying the value of the PORT19_DAT bit. In this case, the value obtained by reading PORT19_DAT bit is the value set by the software; If the PD19 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	PORT18_DAT PD18 Data If the function of PD18 port is configured as input, the PORT18_DAT bit indicates the sate of the PD18 pin; If the function of PD18 port is configured as output, the sate of the PD18 pin can be configured via software by modifying the value of the PORT18_DAT bit. In this case, the value obtained by reading PORT18_DAT bit is the value set by the software; If the PD18 port is configured as functional pin, this bit is invalid.
17	R/W	0x0	PORT17_DAT PD17 Data If the function of PD17 port is configured as input, the PORT17_DAT bit indicates the sate of the PD17 pin; If the function of PD17 port is configured as output, the sate of the PD17 pin can be configured via software by modifying the value of the PORT17_DAT bit. In this case, the value obtained by reading PORT17_DAT bit is the value set by the software; If the PD17 port is configured as functional pin, this bit is invalid.
16	R/W	0x0	PORT16_DAT PD16 Data If the function of PD16 port is configured as input, the PORT16_DAT bit indicates the sate of the PD16 pin; If the function of PD16 port is configured as output, the sate of the PD16 pin can be configured via software by modifying the value of the PORT16_DAT bit. In this case, the value obtained by reading PORT16_DAT bit is the value set by the software; If the PD16 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
15	R/W	0x0	PORT15_DAT PD15 Data If the function of PD15 port is configured as input, the PORT15_DAT bit indicates the sate of the PD15 pin; If the function of PD15 port is configured as output, the sate of the PD15 pin can be configured via software by modifying the value of the PORT15_DAT bit. In this case, the value obtained by reading PORT15_DAT bit is the value set by the software; If the PD15 port is configured as functional pin, this bit is invalid.
14	R/W	0x0	PORT14_DAT PD14 Data If the function of PD14 port is configured as input, the PORT14_DAT bit indicates the sate of the PD14 pin; If the function of PD14 port is configured as output, the sate of the PD14 pin can be configured via software by modifying the value of the PORT14_DAT bit. In this case, the value obtained by reading PORT14_DAT bit is the value set by the software; If the PD14 port is configured as functional pin, this bit is invalid.
13	R/W	0x0	PORT13_DAT PD13 Data If the function of PD13 port is configured as input, the PORT13_DAT bit indicates the sate of the PD13 pin; If the function of PD13 port is configured as output, the sate of the PD13 pin can be configured via software by modifying the value of the PORT13_DAT bit. In this case, the value obtained by reading PORT13_DAT bit is the value set by the software; If the PD13 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PORT12_DAT PD12 Data If the function of PD12 port is configured as input, the PORT12_DAT bit indicates the sate of the PD12 pin; If the function of PD12 port is configured as output, the sate of the PD12 pin can be configured via software by modifying the value of the PORT12_DAT bit. In this case, the value obtained by reading PORT12_DAT bit is the value set by the software; If the PD12 port is configured as functional pin, this bit is invalid.
11	R/W	0x0	PORT11_DAT PD11 Data If the function of PD11 port is configured as input, the PORT11_DAT bit indicates the sate of the PD11 pin; If the function of PD11 port is configured as output, the sate of the PD11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PD11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PD10 Data If the function of PD10 port is configured as input, the PORT10_DAT bit indicates the sate of the PD10 pin; If the function of PD10 port is configured as output, the sate of the PD10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PD10 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	PORT9_DAT PD9 Data If the function of PD9 port is configured as input, the PORT9_DAT bit indicates the sate of the PD9 pin; If the function of PD9 port is configured as output, the sate of the PD9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PD9 port is configured as functional pin, this bit is invalid.
8	R/W	0x0	PORT8_DAT PD8 Data If the function of PD8 port is configured as input, the PORT8_DAT bit indicates the sate of the PD8 pin; If the function of PD8 port is configured as output, the sate of the PD8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PD8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PD7 Data If the function of PD7 port is configured as input, the PORT7_DAT bit indicates the sate of the PD7 pin; If the function of PD7 port is configured as output, the sate of the PD7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PD7 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PORT6_DAT PD6 Data If the function of PD6 port is configured as input, the PORT6_DAT bit indicates the sate of the PD6 pin; If the function of PD6 port is configured as output, the sate of the PD6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PD6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PD5 Data If the function of PD5 port is configured as input, the PORT5_DAT bit indicates the sate of the PD5 pin; If the function of PD5 port is configured as output, the sate of the PD5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PD5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PD4 Data If the function of PD4 port is configured as input, the PORT4_DAT bit indicates the sate of the PD4 pin; If the function of PD4 port is configured as output, the sate of the PD4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PD4 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_DAT PD3 Data If the function of PD3 port is configured as input, the PORT3_DAT bit indicates the sate of the PD3 pin; If the function of PD3 port is configured as output, the sate of the PD3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PD3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PD2 Data If the function of PD2 port is configured as input, the PORT2_DAT bit indicates the sate of the PD2 pin; If the function of PD2 port is configured as output, the sate of the PD2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PD2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PD1 Data If the function of PD1 port is configured as input, the PORT1_DAT bit indicates the sate of the PD1 pin; If the function of PD1 port is configured as output, the sate of the PD1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PD1 port is configured as functional pin, this bit is invalid.

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Offset: 0x0210			Register Name: PD_DAT
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PORT0_DAT PD0 Data If the function of PD0 port is configured as input, the PORT0_DAT bit indicates the sate of the PD0 pin; If the function of PD0 port is configured as output, the sate of the PD0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PD0 port is configured as functional pin, this bit is invalid.

17.9.6.59 0x0214 PD Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0214			Register Name: PD_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1S	0x0	PORT23_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
22	R/W1S	0x0	PORT22_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
21	R/W1S	0x0	PORT21_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
20	R/W1S	0x0	PORT20_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.

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Offset: 0x0214			Register Name: PD_DAT_SET
Bit	Read/Write	Default/Hex	Description
19	R/W1S	0x0	PORT19_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
18	R/W1S	0x0	PORT18_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
17	R/W1S	0x0	PORT17_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
16	R/W1S	0x0	PORT16_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
15	R/W1S	0x0	PORT15_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
14	R/W1S	0x0	PORT14_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
13	R/W1S	0x0	PORT13_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
12	R/W1S	0x0	PORT12_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.

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Offset: 0x0214			Register Name: PD_DAT_SET
Bit	Read/Write	Default/Hex	Description
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.

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Offset: 0x0214			Register Name: PD_DAT_SET
Bit	Read/Write	Default/Hex	Description
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PD _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePD _DAT register.

17.9.6.60 0x0218 PD Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0218			Register Name: PD_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	PORT23_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
22	R/W1C	0x0	PORT22_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
21	R/W1C	0x0	PORT21_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.

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Offset: 0x0218			Register Name: PD_DAT_CLR
Bit	Read/Write	Default/Hex	Description
20	R/W1C	0x0	PORT20_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
19	R/W1C	0x0	PORT19_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
18	R/W1C	0x0	PORT18_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
17	R/W1C	0x0	PORT17_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
16	R/W1C	0x0	PORT16_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
15	R/W1C	0x0	PORT15_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
14	R/W1C	0x0	PORT14_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
13	R/W1C	0x0	PORT13_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.

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Offset: 0x0218			Register Name: PD_DAT_CLR
Bit	Read/Write	Default/Hex	Description
12	R/W1C	0x0	PORT12_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.

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Offset: 0x0218			Register Name: PD_DAT_CLR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PD_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PD_DAT register.

17.9.6.61 0x0220 PD Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x0220			Register Name: PD_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PD7_DRV PD7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PD6_DRV PD6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/

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Offset: 0x0220			Register Name: PD_DRV0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x1	PD5_DRV PD5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PD4_DRV PD4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PD3_DRV PD3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PD2_DRV PD2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PD1_DRV PD1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PD0_DRV PD0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.62 0x0224 PD Multi_Driving Register 1 (Default Value: 0x1111_1111)

Offset: 0x0224			Register Name: PD_DRV1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PD15_DRV PD15 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/

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Offset: 0x0224			Register Name: PD_DRV1
Bit	Read/Write	Default/Hex	Description
25:24	R/W	0x1	PD14_DRV PD14 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PD13_DRV PD13 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PD12_DRV PD12 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PD11_DRV PD11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PD10_DRV PD10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PD9_DRV PD9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PD8_DRV PD8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.63 0x0228 PD Multi_Driving Register 2 (Default Value: 0x1111_1111)

Offset: 0x0228			Register Name: PD_DRV2
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0228			Register Name: PD_DRV2
Bit	Read/Write	Default/Hex	Description
29:28	R/W	0x1	PD23_DRV PD23 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PD22_DRV PD22 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PD21_DRV PD21 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PD20_DRV PD20 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PD19_DRV PD19 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PD18_DRV PD18 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PD17_DRV PD17 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PD16_DRV PD16 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

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17.9.6.64 0x0230 PD Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x0230			Register Name: PD_PUL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PD15_PULL PD15 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
29:28	R/W	0x0	PD14_PULL PD14 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
27:26	R/W	0x0	PD13_PULL PD13 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
25:24	R/W	0x0	PD12_PULL PD12 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
23:22	R/W	0x0	PD11_PULL PD11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PD10_PULL PD10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PD9_PULL PD9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PD8_PULL PD8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PD7_PULL PD7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PD6_PULL PD6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0230			Register Name: PD_PUL0
Bit	Read/Write	Default/Hex	Description
11:10	R/W	0x0	PD5_PULL PD5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PD4_PULL PD4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PD3_PULL PD3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PD2_PULL PD2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PD1_PULL PD1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PD0_PULL PD0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.65 0x0234 PD Pull Register 1 (Default Value: 0x0000_0000)

Offset: 0x0234			Register Name: PD_PUL1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:14	R/W	0x0	PD23_PULL PD23 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PD22_PULL PD22 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0234			Register Name: PD_PUL1
Bit	Read/Write	Default/Hex	Description
11:10	R/W	0x0	PD21_PULL PD21 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PD20_PULL PD20 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PD19_PULL PD19 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PD18_PULL PD18 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PD17_PULL PD17 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PD16_PULL PD16 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.66 0x0240 PD External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x0240			Register Name: PD_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0240			Register Name: PD_INT_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0240			Register Name: PD_INT_CFG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.67 0x0244 PD External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x0244			Register Name: PD_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT15_CFG External INT15 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT14_CFG External INT14 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0244			Register Name: PD_INT_CFG1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	EINT13_CFG External INT13 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT12_CFG External INT12 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0244			Register Name: PD_INT_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.68 0x0248 PD External Interrupt Configure Register 2 (Default Value: 0x0000_0000)

Offset: 0x0248			Register Name: PD_INT_CFG2
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT23_CFG External INT23 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT22_CFG External INT22 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT21_CFG External INT21 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0248			Register Name: PD_INT_CFG2
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	EINT20_CFG External INT20 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT19_CFG External INT19 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT18_CFG External INT18 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT17_CFG External INT17 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT16_CFG External INT16 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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17.9.6.69 0x0250 PD External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0250			Register Name: PD_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	EINT23_CTL External INT23 Enable 0: Disable 1: Enable
22	R/W	0x0	EINT22_CTL External INT22 Enable 0: Disable 1: Enable
21	R/W	0x0	EINT21_CTL External INT21 Enable 0: Disable 1: Enable
20	R/W	0x0	EINT20_CTL External INT20 Enable 0: Disable 1: Enable
19	R/W	0x0	EINT19_CTL External INT19 Enable 0: Disable 1: Enable
18	R/W	0x0	EINT18_CTL External INT18 Enable 0: Disable 1: Enable
17	R/W	0x0	EINT17_CTL External INT17 Enable 0: Disable 1: Enable
16	R/W	0x0	EINT16_CTL External INT16 Enable 0: Disable 1: Enable
15	R/W	0x0	EINT15_CTL External INT15 Enable 0: Disable 1: Enable

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Offset: 0x0250			Register Name: PD_INT_CTL
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	EINT14_CTL External INT14 Enable 0: Disable 1: Enable
13	R/W	0x0	EINT13_CTL External INT13 Enable 0: Disable 1: Enable
12	R/W	0x0	EINT12_CTL External INT12 Enable 0: Disable 1: Enable
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable

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Offset: 0x0250			Register Name: PD_INT_CTL
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.70 0x0254 PD External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0254			Register Name: PD_INT_STA
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W1C	0x0	EINT23_STATUS External INT23 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
22	R/W1C	0x0	EINT22_STATUS External INT22 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
21	R/W1C	0x0	EINT21_STATUS External INT21 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0254			Register Name: PD_INT_STA
Bit	Read/Write	Default/Hex	Description
20	R/W1C	0x0	EINT20_STATUS External INT20 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
19	R/W1C	0x0	EINT19_STATUS External INT19 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
18	R/W1C	0x0	EINT18_STATUS External INT18 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
17	R/W1C	0x0	EINT17_STATUS External INT17 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
16	R/W1C	0x0	EINT16_STATUS External INT16 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
15	R/W1C	0x0	EINT15_STATUS External INT15 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
14	R/W1C	0x0	EINT14_STATUS External INT14 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
13	R/W1C	0x0	EINT13_STATUS External INT13 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0254			Register Name: PD_INT_STA
Bit	Read/Write	Default/Hex	Description
12	R/W1C	0x0	EINT12_STATUS External INT12 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0254			Register Name: PD_INT_STA
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.71 0x0258 PD External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x0258			Register Name: PD_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

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17.9.6.72 0x025C PD External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x025C			Register Name: PD_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.73 0x0270 PD Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x0270			Register Name: PD_SECURE
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	PORT23_SEC_SEL Port23 Secure Selection 0: Not secure 1: Secure
22	R/W	0x0	PORT22_SEC_SEL Port22 Secure Selection 0: Not secure 1: Secure
21	R/W	0x0	PORT21_SEC_SEL Port21 Secure Selection 0: Not secure 1: Secure
20	R/W	0x0	PORT20_SEC_SEL Port20 Secure Selection 0: Not secure 1: Secure
19	R/W	0x0	PORT19_SEC_SEL Port19 Secure Selection 0: Not secure 1: Secure
18	R/W	0x0	PORT18_SEC_SEL Port18 Secure Selection 0: Not secure 1: Secure

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Offset: 0x0270			Register Name: PD_SECURE
Bit	Read/Write	Default/Hex	Description
17	R/W	0x0	PORT17_SEC_SEL Port17 Secure Selection 0: Not secure 1: Secure
16	R/W	0x0	PORT16_SEC_SEL Port16 Secure Selection 0: Not secure 1: Secure
15	R/W	0x0	PORT15_SEC_SEL Port15 Secure Selection 0: Not secure 1: Secure
14	R/W	0x0	PORT14_SEC_SEL Port14 Secure Selection 0: Not secure 1: Secure
13	R/W	0x0	PORT13_SEC_SEL Port13 Secure Selection 0: Not secure 1: Secure
12	R/W	0x0	PORT12_SEC_SEL Port12 Secure Selection 0: Not secure 1: Secure
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure

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Offset: 0x0270			Register Name: PD_SECURE
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.74 0x0280 PE Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0280			Register Name: PE_CFG0
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0280			Register Name: PE_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PE7_SELECT PE7 Select 0000: Input 0001: Output 0010: TWI3-SDA 0011: PWM0-2 0100: NCSI-D15 0101: Reserved 0110: SPI2-CS1 0111: Reserved 1000: Reserved 1001: SDC3-D1 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT7 1111: Disable
27:24	R/W	0xF	PE6_SELECT PE6 Select 0000: Input 0001: Output 0010: TWI3-SCK 0011: UART5-RX 0100: NCSI-D14 0101: UART6-CTS 0110: SPI2-HOLD 0111: Reserved 1000: Reserved 1001: SDC3-D0 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT6 1111: Disable
23:20	R/W	0xF	PE5_SELECT PE5 Select 0000: Input 0001: Output 0010: MCSI1-MCLK 0011: UART5-TX 0100: NCSI-D13 0101: UART6-RTS 0110: SPI2-WP 0111: Reserved 1000: Reserved 1001: SDC3-CMD 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT5 1111: Disable
19:16	R/W	0xF	PE4_SELECT PE4 Select 0000: Input 0001: Output 0010: TWI2-SDA 0011: UART5-CTS 0100: NCSI-D12 0101: UART6-RX 0110: SPI2-MISO 0111: DMIC-DATA3 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT4 1111: Disable

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Offset: 0x0280			Register Name: PE_CFG0
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PE3_SELECT PE3 Select 0000: Input 0001: Output 0010: TWI2-SCK 0011: UART5-RTS 0100: NCSI-D11 0101: UART6-TX 0110: SPI2-MOSI 0111: DMIC-DATA2 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT3 1111: Disable
11:8	R/W	0xF	PE2_SELECT PE2 Select 0000: Input 0001: Output 0010: TWI1-SDA 0011: PWM0-3 0100: NCSI-D10 0101: Reserved 0110: SPI2-CLK 0111: DMIC-DATA1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT2 1111: Disable
7:4	R/W	0xF	PE1_SELECT PE1 Select 0000: Input 0001: Output 0010: TWI1-SCK 0011: Reserved 0100: NCSI-D9 0101: Reserved 0110: SPI2-CS0 0111: DMIC-DATA0 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT1 1111: Disable
3:0	R/W	0xF	PE0_SELECT PE0 Select 0000: Input 0001: Output 0010: MCSI0-MCLK 0011: Reserved 0100: NCSI-D8 0101: CLK-FANOUT1 0110: Reserved 0111: DMIC-CLK 1000: Reserved 1001: SDC3-CLK 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT0 1111: Disable

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17.9.6.75 0x0284 PE Configure Register 1 (Default Value: 0x0000_00FF)

Offset: 0x0284			Register Name: PE_CFG1
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W	0xF	PE9_SELECT PE9 Select 0000: Input 0001: Output 0010: TCON-FSYNC0 0011: PWM0-5 0100: CSI1-XVS-FSYNC 0101: Reserved 0110: Reserved 0111: Reserved 1000: Reserved 1001: SDC3-D3 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT9 1111: Disable
3:0	R/W	0xF	PE8_SELECT PE8 Select 0000: Input 0001: Output 0010: Reserved 0011: PWM0-4 0100: CSI0-XVS-FSYNC 0101: CLK-FANOUT0 0110: SPI2-CS2 0111: Reserved 1000: Reserved 1001: SDC3-D2 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PE-EINT8 1111: Disable

17.9.6.76 0x0290 PE Data Register (Default Value: 0x0000_0000)

Offset: 0x0290			Register Name: PE_DAT
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	PORT9_DAT PE9 Data If the function of PE9 port is configured as input, the PORT9_DAT bit indicates the sate of the PE9 pin; If the function of PE9 port is configured as output, the sate of the PE9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PE9 port is configured as functional pin, this bit is invalid.

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Offset: 0x0290			Register Name: PE_DAT
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PORT8_DAT PE8 Data If the function of PE8 port is configured as input, the PORT8_DAT bit indicates the sate of the PE8 pin; If the function of PE8 port is configured as output, the sate of the PE8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PE8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PE7 Data If the function of PE7 port is configured as input, the PORT7_DAT bit indicates the sate of the PE7 pin; If the function of PE7 port is configured as output, the sate of the PE7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PE7 port is configured as functional pin, this bit is invalid.
6	R/W	0x0	PORT6_DAT PE6 Data If the function of PE6 port is configured as input, the PORT6_DAT bit indicates the sate of the PE6 pin; If the function of PE6 port is configured as output, the sate of the PE6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PE6 port is configured as functional pin, this bit is invalid.

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Offset: 0x0290			Register Name: PE_DAT
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_DAT PE5 Data If the function of PE5 port is configured as input, the PORT5_DAT bit indicates the sate of the PE5 pin; If the function of PE5 port is configured as output, the sate of the PE5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PE5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PE4 Data If the function of PE4 port is configured as input, the PORT4_DAT bit indicates the sate of the PE4 pin; If the function of PE4 port is configured as output, the sate of the PE4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PE4 port is configured as functional pin, this bit is invalid.
3	R/W	0x0	PORT3_DAT PE3 Data If the function of PE3 port is configured as input, the PORT3_DAT bit indicates the sate of the PE3 pin; If the function of PE3 port is configured as output, the sate of the PE3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PE3 port is configured as functional pin, this bit is invalid.

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Offset: 0x0290			Register Name: PE_DAT
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PORT2_DAT PE2 Data If the function of PE2 port is configured as input, the PORT2_DAT bit indicates the sate of the PE2 pin; If the function of PE2 port is configured as output, the sate of the PE2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PE2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PE1 Data If the function of PE1 port is configured as input, the PORT1_DAT bit indicates the sate of the PE1 pin; If the function of PE1 port is configured as output, the sate of the PE1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PE1 port is configured as functional pin, this bit is invalid.
0	R/W	0x0	PORT0_DAT PE0 Data If the function of PE0 port is configured as input, the PORT0_DAT bit indicates the sate of the PE0 pin; If the function of PE0 port is configured as output, the sate of the PE0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PE0 port is configured as functional pin, this bit is invalid.

17.9.6.77 0x0294 PE Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0294			Register Name: PE_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/

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Offset: 0x0294			Register Name: PE_DAT_SET
Bit	Read/Write	Default/Hex	Description
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.

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Offset: 0x0294			Register Name: PE_DAT_SET
Bit	Read/Write	Default/Hex	Description
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PE _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePE _DAT register.

17.9.6.78 0x0298 PE Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0298			Register Name: PE_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.

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Offset: 0x0298			Register Name: PE_DAT_CLR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PE_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PE_DAT register.

17.9.6.79 0x02A0 PE Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x02A0			Register Name: PE_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PE7_DRV PE7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PE6_DRV PE6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/

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Offset: 0x02A0			Register Name: PE_DRV0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x1	PE5_DRV PE5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PE4_DRV PE4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PE3_DRV PE3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PE2_DRV PE2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PE1_DRV PE1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PE0_DRV PE0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.80 0x02A4 PE Multi_Driving Register 1 (Default Value: 0x0000_0011)

Offset: 0x02A4			Register Name: PE_DRV1
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:4	R/W	0x1	PE9_DRV PE9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/

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Offset: 0x02A4			Register Name: PE_DRV1
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x1	PE8_DRV PE8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.81 0x02B0 PE Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x02B0			Register Name: PE_PUL0
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:18	R/W	0x0	PE9_PULL PE9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PE8_PULL PE8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PE7_PULL PE7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PE6_PULL PE6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PE5_PULL PE5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PE4_PULL PE4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PE3_PULL PE3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x02B0			Register Name: PE_PUL0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	PE2_PULL PE2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PE1_PULL PE1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PE0_PULL PE0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.82 0x02C0 PE External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x02C0			Register Name: PE_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x02C0			Register Name: PE_INT_CFG0
Bit	Read/Write	Default/Hex	Description
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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17.9.6.83 0x02C4 PE External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x02C4			Register Name: PE_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.84 0x02D0 PE External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x02D0			Register Name: PE_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable

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Offset: 0x02D0			Register Name: PE_INT_CTL
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.85 0x02D4 PE External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x02D4			Register Name: PE_INT_STA
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x02D4			Register Name: PE_INT_STA
Bit	Read/Write	Default/Hex	Description
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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17.9.6.86 0x02D8 PE External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x02D8			Register Name: PE_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.87 0x02DC PE External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x02DC			Register Name: PE_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.88 0x02F0 PE Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x02F0			Register Name: PE_SECURE
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure

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Offset: 0x02F0			Register Name: PE_SECURE
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.89 0x0300 PF Configure Register 0 (Default Value: 0x0FFF_FFFF)

Offset: 0x0300			Register Name: PF_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/

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Offset: 0x0300			Register Name: PF_CFG0	
Bit	Read/Write	Default/Hex	Description	
27:24	R/W	0xF	PF6_SELECT PF6 Select 0000: Input 0010: Reserved 0100: DMIC-DATA3 0110: Reserved 1000: Reserved 1010: Reserved 1100: Reserved 1110: PF-EINT6	 0001: Output 0011: Reserved 0101: Reserved 0111: Reserved 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable
23:20	R/W	0xF	PF5_SELECT PF5 Select 0000: Input 0010: SDC0-D2 0100: DMIC-DATA2 0110: SPI3-MISO 1000: Reserved 1010: Reserved 1100: Reserved 1110: PF-EINT5	 0001: Output 0011: JTAG-CK 0101: Reserved 0111: RJTAG-CK 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable
19:16	R/W	0xF	PF4_SELECT PF4 Select 0000: Input 0010: SDC0-D3 0100: DMIC-DATA1 0110: SPI3-MOSI 1000: IR-TX 1010: Reserved 1100: Reserved 1110: PF-EINT4	 0001: Output 0011: UART0-RX 0101: Reserved 0111: Reserved 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable
15:12	R/W	0xF	PF3_SELECT PF3 Select 0000: Input 0010: SDC0-CMD 0100: DMIC-DATA0 0110: SPI3-CS0 1000: IR3-RX 1010: Reserved 1100: Reserved 1110: PF-EINT3	 0001: Output 0011: JTAG-DO 0101: Reserved 0111: RJTAG-DO 1001: Reserved 1011: Reserved 1101: Reserved 1111: Disable

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Offset: 0x0300			Register Name: PF_CFG0
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PF2_SELECT PF2 Select 0000: Input 0001: Output 0010: SDC0-CLK 0011: UART0-TX 0100: DMIC-CLK 0101: Reserved 0110: SPI3-CLK 0111: Reserved 1000: IR2-RX 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PF-EINT2 1111: Disable
7:4	R/W	0xF	PF1_SELECT PF1 Select 0000: Input 0001: Output 0010: SDC0-D0 0011: JTAG-DI 0100: UART1-RX 0101: Reserved 0110: Reserved 0111: RJTAG-DI 1000: IR1-RX 1001: LBUS-DRQ1 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PF-EINT1 1111: Disable
3:0	R/W	0xF	PF0_SELECT PF0 Select 0000: Input 0001: Output 0010: SDC0-D1 0011: JTAG-MS 0100: UART1-TX 0101: Reserved 0110: Reserved 0111: RJTAG-MS 1000: IR0-RX 1001: LBUS-DRQ0 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PF-EINT0 1111: Disable

17.9.6.90 0x0310 PF Data Register (Default Value: 0x0000_0000)

Offset: 0x0310			Register Name: PF_DAT
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/

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Offset: 0x0310			Register Name: PF_DAT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PORT6_DAT PF6 Data If the function of PF6 port is configured as input, the PORT6_DAT bit indicates the sate of the PF6 pin; If the function of PF6 port is configured as output, the sate of the PF6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PF6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PF5 Data If the function of PF5 port is configured as input, the PORT5_DAT bit indicates the sate of the PF5 pin; If the function of PF5 port is configured as output, the sate of the PF5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PF5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PF4 Data If the function of PF4 port is configured as input, the PORT4_DAT bit indicates the sate of the PF4 pin; If the function of PF4 port is configured as output, the sate of the PF4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PF4 port is configured as functional pin, this bit is invalid.

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Offset: 0x0310			Register Name: PF_DAT
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_DAT PF3 Data If the function of PF3 port is configured as input, the PORT3_DAT bit indicates the sate of the PF3 pin; If the function of PF3 port is configured as output, the sate of the PF3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PF3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PF2 Data If the function of PF2 port is configured as input, the PORT2_DAT bit indicates the sate of the PF2 pin; If the function of PF2 port is configured as output, the sate of the PF2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PF2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PF1 Data If the function of PF1 port is configured as input, the PORT1_DAT bit indicates the sate of the PF1 pin; If the function of PF1 port is configured as output, the sate of the PF1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PF1 port is configured as functional pin, this bit is invalid.

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Offset: 0x0310			Register Name: PF_DAT
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PORT0_DAT PF0 Data If the function of PF0 port is configured as input, the PORT0_DAT bit indicates the sate of the PF0 pin; If the function of PF0 port is configured as output, the sate of the PF0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PF0 port is configured as functional pin, this bit is invalid.

17.9.6.91 0x0314 PF Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0314			Register Name: PF_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.

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Offset: 0x0314			Register Name: PF_DAT_SET
Bit	Read/Write	Default/Hex	Description
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PF _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePF _DAT register.

17.9.6.92 0x0318 PF Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0318			Register Name: PF_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.

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Offset: 0x0318			Register Name: PF_DAT_CLR
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PF_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PF_DAT register.

17.9.6.93 0x0320 PF Multi_Driving Register 0 (Default Value: 0x0111_1111)

Offset: 0x0320			Register Name: PF_DRV0
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:24	R/W	0x1	PF6_DRV PF6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PF5_DRV PF5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PF4_DRV PF4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PF3_DRV PF3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/

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Offset: 0x0320			Register Name: PF_DRV0
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x1	PF2_DRV PF2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PF1_DRV PF1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PF0_DRV PF0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.94 0x0330 PF Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x0330			Register Name: PF_PUL0
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:12	R/W	0x0	PF6_PULL PF6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PF5_PULL PF5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PF4_PULL PF4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PF3_PULL PF3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PF2_PULL PF2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0330			Register Name: PF_PUL0
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	PF1_PULL PF1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PF0_PULL PF0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.95 0x0340 PF External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x0340			Register Name: PF_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0340			Register Name: PF_INT_CFG0
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.96 0x0350 PF External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0350			Register Name: PF_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/

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Offset: 0x0350			Register Name: PF_INT_CTL
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.97 0x0354 PF External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0354			Register Name: PF_INT_STA
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0354			Register Name: PF_INT_STA
Bit	Read/Write	Default/Hex	Description
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.98 0x0358 PF External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x0358			Register Name: PF_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/

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Offset: 0x0358			Register Name: PF_INT_DEB
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.99 0x035C PF External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x035C			Register Name: PF_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.100 0x0370 PF Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x0370			Register Name: PF_SECURE
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure

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Offset: 0x0370			Register Name: PF_SECURE
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.101 0x0380 PG Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0380			Register Name: PG_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PG7_SELECT PG7 Select 0000: Input 0001: Output 0010: UART1-RX 0011: TWI4-SDA 0100: UART9-RTS 0101: RGMII1-RXD0/RMII1-RXD0 0110: I2S2-BCLK 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT7 1111: Disable
27:24	R/W	0xF	PG6_SELECT PG6 Select 0000: Input 0001: Output 0010: UART1-TX 0011: TWI4-SCK 0100: UART9-CTS 0101: RGMII1-RXD1/RMII1-RXD1 0110: I2S2-MCLK 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT6 1111: Disable

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Offset: 0x0380			Register Name: PG_CFG0
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0xF	PG5_SELECT PG5 Select 0000: Input 0001: Output 0010: SDC1-D3 0011: Reserved 0100: CAN0-RX0 0101: RGMII1-EPHY-25/50M 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT5 1111: Disable
19:16	R/W	0xF	PG4_SELECT PG4 Select 0000: Input 0001: Output 0010: SDC1-D2 0011: Reserved 0100: CAN0-TX0 0101: RGMII1-TXD2/RMII1-NULL 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT4 1111: Disable
15:12	R/W	0xF	PG3_SELECT PG3 Select 0000: Input 0001: Output 0010: SDC1-D1 0011: Reserved 0100: UART8-CTS 0101: RGMII1-TXD3/RMII1-NULL 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT3 1111: Disable

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Offset: 0x0380			Register Name: PG_CFG0
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PG2_SELECT PG2 Select 0000: Input 0001: Output 0010: SDC1-D0 0011: Reserved 0100: UART8-RTS 0101: RGMII1-RXCK/RMII1-NULL 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT2 1111: Disable
7:4	R/W	0xF	PG1_SELECT PG1 Select 0000: Input 0001: Output 0010: SDC1-CMD 0011: Reserved 0100: UART8-RX 0101: RGMII1-RXD2/RMII1-NULL 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT1 1111: Disable
3:0	R/W	0xF	PG0_SELECT PG0 Select 0000: Input 0001: Output 0010: SDC1-CLK 0011: Reserved 0100: UART8-TX 0101: RGMII1-RXD3/RMII1-NULL 0110: Reserved 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT0 1111: Disable

17.9.6.102 0x0384 PG Configure Register 1 (Default Value: 0xFFFF_FFFF)

Offset: 0x0384			Register Name: PG_CFG1
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0384			Register Name: PG_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PG15_SELECT PG15 Select 0000: Input 0001: Output 0010: CAN1-RX0 0011: Reserved 0100: Reserved 0101: RGMII1-MDIO 0110: DMIC-DATA3 0111: Reserved 1000: LBUS-DRQ1 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT15 1111: Disable
27:24	R/W	0xF	PG14_SELECT PG14 Select 0000: Input 0001: Output 0010: CAN1-TX0 0011: I2S1-DIN0 0100: I2S1-DOUT1 0101: RGMII1-MDC 0110: DMIC-DATA2 0111: Reserved 1000: LBUS-DRQ0 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT14 1111: Disable
23:20	R/W	0xF	PG13_SELECT PG13 Select 0000: Input 0001: Output 0010: UART4-CTS 0011: I2S1-DOUT0 0100: I2S1-DIN1 0101: RGMII1-TXCTL/RMII1-TXEN 0110: DMIC-DATA1 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT13 1111: Disable
19:16	R/W	0xF	PG12_SELECT PG12 Select 0000: Input 0001: Output 0010: UART4-RTS 0011: I2S1-LRCK 0100: Reserved 0101: RGMII1-TXCK/RMII1-TXCK 0110: DMIC-DATA0 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT12 1111: Disable

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Offset: 0x0384			Register Name: PG_CFG1
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PG11_SELECT PG11 Select 0000: Input 0001: Output 0010: UART4-RX 0011: I2S1-BCLK 0100: TWI2-SDA 0101: RGMII1-TXD0/RMII1-TXD0 0110: DMIC-CLK 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT11 1111: Disable
11:8	R/W	0xF	PG10_SELECT PG10 Select 0000: Input 0001: Output 0010: UART4-TX 0011: I2S1-MCLK 0100: TWI2-SCK 0101: RGMII1-TXD1/RMII1-TXD1 0110: I2S2-DIN0 0111: I2S2-DOUT1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT10 1111: Disable
7:4	R/W	0xF	PG9_SELECT PG9 Select 0000: Input 0001: Output 0010: UART1-CTS 0011: TWI5-SDA 0100: UART9-RX 0101: RGMII1-CLKIN/RMII1-RXER 0110: I2S2-DOUT0 0111: I2S2-DIN1 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT9 1111: Disable

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Offset: 0x0384			Register Name: PG_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PG8_SELECT PG8 Select 0000: Input 0001: Output 0010: UART1-RTS 0011: TWI5-SCK 0100: UART9-TX 0101: RGMII1-RXCTL/RMII1-CRS-DV 0110: I2S2-LRCK 0111: Reserved 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PG-EINT8 1111: Disable

17.9.6.103 0x0390 PG Data Register (Default Value: 0x0000_0000)

Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	PORT15_DAT PG15 Data If the function of PG15 port is configured as input, the PORT15_DAT bit indicates the sate of the PG15 pin; If the function of PG15 port is configured as output, the sate of the PG15 pin can be configured via software by modifying the value of the PORT15_DAT bit. In this case, the value obtained by reading PORT15_DAT bit is the value set by the software; If the PG15 port is configured as functional pin, this bit is invalid.
14	R/W	0x0	PORT14_DAT PG14 Data If the function of PG14 port is configured as input, the PORT14_DAT bit indicates the sate of the PG14 pin; If the function of PG14 port is configured as output, the sate of the PG14 pin can be configured via software by modifying the value of the PORT14_DAT bit. In this case, the value obtained by reading PORT14_DAT bit is the value set by the software; If the PG14 port is configured as functional pin, this bit is invalid.

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Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
13	R/W	0x0	PORT13_DAT PG13 Data If the function of PG13 port is configured as input, the PORT13_DAT bit indicates the sate of the PG13 pin; If the function of PG13 port is configured as output, the sate of the PG13 pin can be configured via software by modifying the value of the PORT13_DAT bit. In this case, the value obtained by reading PORT13_DAT bit is the value set by the software; If the PG13 port is configured as functional pin, this bit is invalid.
12	R/W	0x0	PORT12_DAT PG12 Data If the function of PG12 port is configured as input, the PORT12_DAT bit indicates the sate of the PG12 pin; If the function of PG12 port is configured as output, the sate of the PG12 pin can be configured via software by modifying the value of the PORT12_DAT bit. In this case, the value obtained by reading PORT12_DAT bit is the value set by the software; If the PG12 port is configured as functional pin, this bit is invalid.
11	R/W	0x0	PORT11_DAT PG11 Data If the function of PG11 port is configured as input, the PORT11_DAT bit indicates the sate of the PG11 pin; If the function of PG11 port is configured as output, the sate of the PG11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PG11 port is configured as functional pin, this bit is invalid.

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Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
10	R/W	0x0	PORT10_DAT PG10 Data If the function of PG10 port is configured as input, the PORT10_DAT bit indicates the sate of the PG10 pin; If the function of PG10 port is configured as output, the sate of the PG10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PG10 port is configured as functional pin, this bit is invalid.
9	R/W	0x0	PORT9_DAT PG9 Data If the function of PG9 port is configured as input, the PORT9_DAT bit indicates the sate of the PG9 pin; If the function of PG9 port is configured as output, the sate of the PG9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PG9 port is configured as functional pin, this bit is invalid.
8	R/W	0x0	PORT8_DAT PG8 Data If the function of PG8 port is configured as input, the PORT8_DAT bit indicates the sate of the PG8 pin; If the function of PG8 port is configured as output, the sate of the PG8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PG8 port is configured as functional pin, this bit is invalid.

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Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	PORT7_DAT PG7 Data If the function of PG7 port is configured as input, the PORT7_DAT bit indicates the sate of the PG7 pin; If the function of PG7 port is configured as output, the sate of the PG7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PG7 port is configured as functional pin, this bit is invalid.
6	R/W	0x0	PORT6_DAT PG6 Data If the function of PG6 port is configured as input, the PORT6_DAT bit indicates the sate of the PG6 pin; If the function of PG6 port is configured as output, the sate of the PG6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PG6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PG5 Data If the function of PG5 port is configured as input, the PORT5_DAT bit indicates the sate of the PG5 pin; If the function of PG5 port is configured as output, the sate of the PG5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PG5 port is configured as functional pin, this bit is invalid.

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Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	PORT4_DAT PG4 Data If the function of PG4 port is configured as input, the PORT4_DAT bit indicates the sate of the PG4 pin; If the function of PG4 port is configured as output, the sate of the PG4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PG4 port is configured as functional pin, this bit is invalid.
3	R/W	0x0	PORT3_DAT PG3 Data If the function of PG3 port is configured as input, the PORT3_DAT bit indicates the sate of the PG3 pin; If the function of PG3 port is configured as output, the sate of the PG3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PG3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PG2 Data If the function of PG2 port is configured as input, the PORT2_DAT bit indicates the sate of the PG2 pin; If the function of PG2 port is configured as output, the sate of the PG2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PG2 port is configured as functional pin, this bit is invalid.

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Offset: 0x0390			Register Name: PG_DAT
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	PORT1_DAT PG1 Data If the function of PG1 port is configured as input, the PORT1_DAT bit indicates the sate of the PG1 pin; If the function of PG1 port is configured as output, the sate of the PG1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PG1 port is configured as functional pin, this bit is invalid.
0	R/W	0x0	PORT0_DAT PG0 Data If the function of PG0 port is configured as input, the PORT0_DAT bit indicates the sate of the PG0 pin; If the function of PG0 port is configured as output, the sate of the PG0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PG0 port is configured as functional pin, this bit is invalid.

17.9.6.104 0x0394 PG Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0394			Register Name: PG_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1S	0x0	PORT15_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
14	R/W1S	0x0	PORT14_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.

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Offset: 0x0394			Register Name: PG_DAT_SET
Bit	Read/Write	Default/Hex	Description
13	R/W1S	0x0	PORT13_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
12	R/W1S	0x0	PORT12_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.

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Offset: 0x0394			Register Name: PG_DAT_SET
Bit	Read/Write	Default/Hex	Description
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PG _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePG _DAT register.

17.9.6.105 0x0398 PG Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0398			Register Name: PG_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1C	0x0	PORT15_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.

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Offset: 0x0398			Register Name: PG_DAT_CLR
Bit	Read/Write	Default/Hex	Description
14	R/W1C	0x0	PORT14_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
13	R/W1C	0x0	PORT13_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
12	R/W1C	0x0	PORT12_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.

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Offset: 0x0398			Register Name: PG_DAT_CLR
Bit	Read/Write	Default/Hex	Description
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PG_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PG_DAT register.

17.9.6.106 0x03A0 PG Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x03A0			Register Name: PG_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x03A0			Register Name: PG_DRV0
Bit	Read/Write	Default/Hex	Description
29:28	R/W	0x1	PG7_DRV PG7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PG6_DRV PG6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PG5_DRV PG5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PG4_DRV PG4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PG3_DRV PG3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PG2_DRV PG2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PG1_DRV PG1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PG0_DRV PG0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

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17.9.6.107 0x03A4 PG Multi_Driving Register 1 (Default Value: 0x1111_1111)

Offset: 0x03A4			Register Name: PG_DRV1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PG15_DRV PG15 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PG14_DRV PG14 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PG13_DRV PG13 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PG12_DRV PG12 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PG11_DRV PG11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PG10_DRV PG10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PG9_DRV PG9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PG8_DRV PG8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

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17.9.6.108 0x03B0 PG Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x03B0			Register Name: PG_PUL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PG15_PULL PG15 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
29:28	R/W	0x0	PG14_PULL PG14 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
27:26	R/W	0x0	PG13_PULL PG13 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
25:24	R/W	0x0	PG12_PULL PG12 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
23:22	R/W	0x0	PG11_PULL PG11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PG10_PULL PG10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PG9_PULL PG9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PG8_PULL PG8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PG7_PULL PG7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PG6_PULL PG6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x03B0			Register Name: PG_PUL0
Bit	Read/Write	Default/Hex	Description
11:10	R/W	0x0	PG5_PULL PG5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PG4_PULL PG4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PG3_PULL PG3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PG2_PULL PG2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PG1_PULL PG1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PG0_PULL PG0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.109 0x03C0 PG External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x03C0			Register Name: PG_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x03C0			Register Name: PG_INT_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x03C0			Register Name: PG_INT_CFG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.110 0x03C4 PG External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x03C4			Register Name: PG_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT15_CFG External INT15 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT14_CFG External INT14 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x03C4			Register Name: PG_INT_CFG1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	EINT13_CFG External INT13 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT12_CFG External INT12 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x03C4			Register Name: PG_INT_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.111 0x03D0 PG External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x03D0			Register Name: PG_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	EINT15_CTL External INT15 Enable 0: Disable 1: Enable
14	R/W	0x0	EINT14_CTL External INT14 Enable 0: Disable 1: Enable
13	R/W	0x0	EINT13_CTL External INT13 Enable 0: Disable 1: Enable
12	R/W	0x0	EINT12_CTL External INT12 Enable 0: Disable 1: Enable
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable

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Offset: 0x03D0			Register Name: PG_INT_CTL
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

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17.9.6.112 0x03D4 PG External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x03D4			Register Name: PG_INT_STA
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1C	0x0	EINT15_STATUS External INT15 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
14	R/W1C	0x0	EINT14_STATUS External INT14 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
13	R/W1C	0x0	EINT13_STATUS External INT13 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
12	R/W1C	0x0	EINT12_STATUS External INT12 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x03D4			Register Name: PG_INT_STA
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x03D4			Register Name: PG_INT_STA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.113 0x03D8 PG External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x03D8			Register Name: PG_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.114 0x03DC PG External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x03DC			Register Name: PG_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.115 0x03F0 PG Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x03F0			Register Name: PG_SECURE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x03F0			Register Name: PG_SECURE
Bit	Read/Write	Default/Hex	Description
15	R/W	0x0	PORT15_SEC_SEL Port15 Secure Selection 0: Not secure 1: Secure
14	R/W	0x0	PORT14_SEC_SEL Port14 Secure Selection 0: Not secure 1: Secure
13	R/W	0x0	PORT13_SEC_SEL Port13 Secure Selection 0: Not secure 1: Secure
12	R/W	0x0	PORT12_SEC_SEL Port12 Secure Selection 0: Not secure 1: Secure
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure

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Offset: 0x03F0			Register Name: PG_SECURE
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.116 0x0500 PJ Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0500			Register Name: PJ_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PJ7_SELECT PJ7 Select 0000: Input 0001: Output 0010: PWM2-7 0011: I2S1-MCLK 0100: Reserved 0101: RGMII0-TXCTL/RMII0-TXEN 0110: Reserved 0111: UART7-CTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT7 1111: Disable

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Offset: 0x0500			Register Name: PJ_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PJ6_SELECT PJ6 Select 0000: Input 0001: Output 0010: PWM2-6 0011: I2S2-DIN3 0100: I2S2-DOUT3 0101: RGMII0-TXCK/RMII0-TXCK 0110: Reserved 0111: UART7-RTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT6 1111: Disable
23:20	R/W	0xF	PJ5_SELECT PJ5 Select 0000: Input 0001: Output 0010: PWM2-5 0011: I2S2-DIN2 0100: I2S2-DOUT2 0101: RGMII0-TXD0/RMII0-TXD0 0110: Reserved 0111: UART7-RX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT5 1111: Disable
19:16	R/W	0xF	PJ4_SELECT PJ4 Select 0000: Input 0001: Output 0010: PWM2-4 0011: I2S2-DIN0 0100: I2S2-DOUT1 0101: RGMII0-TXD1/RMII0-TXD1 0110: DMIC-DATA3 0111: UART7-TX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT4 1111: Disable

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Offset: 0x0500			Register Name: PJ_CFG0
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PJ3_SELECT PJ3 Select 0000: Input 0001: Output 0010: PWM2-3 0011: I2S2-DOUT0 0100: I2S2-DIN1 0101: RGMII0-CLKIN/RMII0-RXER 0110: DMIC-DATA2 0111: UART5-CTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT3 1111: Disable
11:8	R/W	0xF	PJ2_SELECT PJ2 Select 0000: Input 0001: Output 0010: PWM2-2 0011: I2S2-LRCK 0100: Reserved 0101: RGMII0-RXCTL/RMII0-CRS-DV 0110: DMIC-DATA1 0111: UART5-RTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT2 1111: Disable
7:4	R/W	0xF	PJ1_SELECT PJ1 Select 0000: Input 0001: Output 0010: PWM2-1 0011: I2S2-BCLK 0100: TWI4-SDA 0101: RGMII0-RXD0/RMII0-RXD0 0110: DMIC-DATA0 0111: UART5-RX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT1 1111: Disable

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Offset: 0x0500			Register Name: PJ_CFG0
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PJ0_SELECT PJ0 Select 0000: Input 0001: Output 0010: PWM2-0 0011: I2S2-MCLK 0100: TWI4-SCK 0101: RGMII0-RXD1/RMII0-RXD1 0110: DMIC-CLK 0111: UART5-TX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT0 1111: Disable

17.9.6.117 0x0504 PJ Configure Register 1 (Default Value: 0xFFFF_FFFF)

Offset: 0x0504			Register Name: PJ_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PJ15_SELECT PJ15 Select 0000: Input 0001: Output 0010: Reserved 0011: CAN1-RX0 0100: IR3-RX 0101: RGMII0-TXD2/RMII0-NULL 0110: TWI5-SDA 0111: UART9-CTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT15 1111: Disable
27:24	R/W	0xF	PJ14_SELECT PJ14 Select 0000: Input 0001: Output 0010: Reserved 0011: CAN1-TX0 0100: IR2-RX 0101: RGMII0-TXD3/RMII0-NULL 0110: TWI5-SCK 0111: UART9-RTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT14 1111: Disable

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Offset: 0x0504			Register Name: PJ_CFG1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0xF	PJ13_SELECT PJ13 Select 0000: Input 0001: Output 0010: PWM2-13 0011: CAN0-RX0 0100: IR1-RX 0101: RGMII0-RXCK/RMII0-NULL 0110: TWI0-SDA 0111: UART9-RX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT13 1111: Disable
19:16	R/W	0xF	PJ12_SELECT PJ12 Select 0000: Input 0001: Output 0010: PWM2-12 0011: CAN0-TX0 0100: IR0-RX 0101: RGMII0-RXD2/RMII0-NULL 0110: TWI0-SCK 0111: UART9-TX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT12 1111: Disable
15:12	R/W	0xF	PJ11_SELECT PJ11 Select 0000: Input 0001: Output 0010: PWM2-11 0011: I2S1-DIN0 0100: I2S1-DOUT1 0101: RGMII0-RXD3/RMII0-NULL 0110: Reserved 0111: UART2-CTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT11 1111: Disable

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Offset: 0x0504			Register Name: PJ_CFG1
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PJ10_SELECT PJ10 Select 0000: Input 0001: Output 0010: PWM2-10 0011: I2S1-DOUT0 0100: I2S1-DIN1 0101: RGMII0-EPHY-25/50M 0110: Reserved 0111: UART2-RTS 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT10 1111: Disable
7:4	R/W	0xF	PJ9_SELECT PJ9 Select 0000: Input 0001: Output 0010: PWM2-9 0011: I2S1-LRCK 0100: IR-TX 0101: RGMII0-MDIO 0110: Reserved 0111: UART2-RX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT9 1111: Disable
3:0	R/W	0xF	PJ8_SELECT PJ8 Select 0000: Input 0001: Output 0010: PWM2-8 0011: I2S1-BCLK 0100: LEDC 0101: RGMII0-MDC 0110: Reserved 0111: UART2-TX 1000: Reserved 1001: Reserved 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PJ-EINT8 1111: Disable

17.9.6.118 0x0510 PJ Data Register (Default Value: 0x0000_0000)

Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
15	R/W	0x0	PORT15_DAT PJ15 Data If the function of PJ15 port is configured as input, the PORT15_DAT bit indicates the sate of the PJ15 pin; If the function of PJ15 port is configured as output, the sate of the PJ15 pin can be configured via software by modifying the value of the PORT15_DAT bit. In this case, the value obtained by reading PORT15_DAT bit is the value set by the software; If the PJ15 port is configured as functional pin, this bit is invalid.
14	R/W	0x0	PORT14_DAT PJ14 Data If the function of PJ14 port is configured as input, the PORT14_DAT bit indicates the sate of the PJ14 pin; If the function of PJ14 port is configured as output, the sate of the PJ14 pin can be configured via software by modifying the value of the PORT14_DAT bit. In this case, the value obtained by reading PORT14_DAT bit is the value set by the software; If the PJ14 port is configured as functional pin, this bit is invalid.
13	R/W	0x0	PORT13_DAT PJ13 Data If the function of PJ13 port is configured as input, the PORT13_DAT bit indicates the sate of the PJ13 pin; If the function of PJ13 port is configured as output, the sate of the PJ13 pin can be configured via software by modifying the value of the PORT13_DAT bit. In this case, the value obtained by reading PORT13_DAT bit is the value set by the software; If the PJ13 port is configured as functional pin, this bit is invalid.

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	PORT12_DAT PJ12 Data If the function of PJ12 port is configured as input, the PORT12_DAT bit indicates the sate of the PJ12 pin; If the function of PJ12 port is configured as output, the sate of the PJ12 pin can be configured via software by modifying the value of the PORT12_DAT bit. In this case, the value obtained by reading PORT12_DAT bit is the value set by the software; If the PJ12 port is configured as functional pin, this bit is invalid.
11	R/W	0x0	PORT11_DAT PJ11 Data If the function of PJ11 port is configured as input, the PORT11_DAT bit indicates the sate of the PJ11 pin; If the function of PJ11 port is configured as output, the sate of the PJ11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PJ11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PJ10 Data If the function of PJ10 port is configured as input, the PORT10_DAT bit indicates the sate of the PJ10 pin; If the function of PJ10 port is configured as output, the sate of the PJ10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PJ10 port is configured as functional pin, this bit is invalid.

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	PORT9_DAT PJ9 Data If the function of PJ9 port is configured as input, the PORT9_DAT bit indicates the sate of the PJ9 pin; If the function of PJ9 port is configured as output, the sate of the PJ9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PJ9 port is configured as functional pin, this bit is invalid.
8	R/W	0x0	PORT8_DAT PJ8 Data If the function of PJ8 port is configured as input, the PORT8_DAT bit indicates the sate of the PJ8 pin; If the function of PJ8 port is configured as output, the sate of the PJ8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PJ8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PJ7 Data If the function of PJ7 port is configured as input, the PORT7_DAT bit indicates the sate of the PJ7 pin; If the function of PJ7 port is configured as output, the sate of the PJ7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PJ7 port is configured as functional pin, this bit is invalid.

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PORT6_DAT PJ6 Data If the function of PJ6 port is configured as input, the PORT6_DAT bit indicates the sate of the PJ6 pin; If the function of PJ6 port is configured as output, the sate of the PJ6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PJ6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PJ5 Data If the function of PJ5 port is configured as input, the PORT5_DAT bit indicates the sate of the PJ5 pin; If the function of PJ5 port is configured as output, the sate of the PJ5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PJ5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PJ4 Data If the function of PJ4 port is configured as input, the PORT4_DAT bit indicates the sate of the PJ4 pin; If the function of PJ4 port is configured as output, the sate of the PJ4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PJ4 port is configured as functional pin, this bit is invalid.

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_DAT PJ3 Data If the function of PJ3 port is configured as input, the PORT3_DAT bit indicates the sate of the PJ3 pin; If the function of PJ3 port is configured as output, the sate of the PJ3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PJ3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PJ2 Data If the function of PJ2 port is configured as input, the PORT2_DAT bit indicates the sate of the PJ2 pin; If the function of PJ2 port is configured as output, the sate of the PJ2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PJ2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PJ1 Data If the function of PJ1 port is configured as input, the PORT1_DAT bit indicates the sate of the PJ1 pin; If the function of PJ1 port is configured as output, the sate of the PJ1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PJ1 port is configured as functional pin, this bit is invalid.

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Offset: 0x0510			Register Name: PJ_DAT
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PORT0_DAT PJ0 Data If the function of PJ0 port is configured as input, the PORT0_DAT bit indicates the sate of the PJ0 pin; If the function of PJ0 port is configured as output, the sate of the PJ0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PJ0 port is configured as functional pin, this bit is invalid.

17.9.6.119 0x0514 PJ Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0514			Register Name: PJ_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1S	0x0	PORT15_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
14	R/W1S	0x0	PORT14_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
13	R/W1S	0x0	PORT13_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
12	R/W1S	0x0	PORT12_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.

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Offset: 0x0514			Register Name: PJ_DAT_SET
Bit	Read/Write	Default/Hex	Description
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.

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Offset: 0x0514			Register Name: PJ_DAT_SET
Bit	Read/Write	Default/Hex	Description
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PJ _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePJ _DAT register.

17.9.6.120 0x0518 PJ Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0518			Register Name: PJ_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1C	0x0	PORT15_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
14	R/W1C	0x0	PORT14_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
13	R/W1C	0x0	PORT13_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.

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Offset: 0x0518			Register Name: PJ_DAT_CLR
Bit	Read/Write	Default/Hex	Description
12	R/W1C	0x0	PORT12_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.

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Offset: 0x0518			Register Name: PJ_DAT_CLR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PJ_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PJ_DAT register.

17.9.6.121 0x0520 PJ Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x0520			Register Name: PJ_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PJ7_DRV PJ7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PJ6_DRV PJ6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/

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Offset: 0x0520			Register Name: PJ_DRV0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x1	PJ5_DRV PJ5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PJ4_DRV PJ4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PJ3_DRV PJ3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PJ2_DRV PJ2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PJ1_DRV PJ1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PJ0_DRV PJ0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.122 0x0524 PJ Multi_Driving Register 1 (Default Value: 0x1111_1111)

Offset: 0x0524			Register Name: PJ_DRV1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PJ15_DRV PJ15 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/

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Offset: 0x0524			Register Name: PJ_DRV1
Bit	Read/Write	Default/Hex	Description
25:24	R/W	0x1	PJ14_DRV PJ14 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/
21:20	R/W	0x1	PJ13_DRV PJ13 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PJ12_DRV PJ12 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PJ11_DRV PJ11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PJ10_DRV PJ10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PJ9_DRV PJ9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PJ8_DRV PJ8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.123 0x0530 PJ Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x0530			Register Name: PJ_PUL0
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0530			Register Name: PJ_PUL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PJ15_PULL PJ15 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
29:28	R/W	0x0	PJ14_PULL PJ14 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
27:26	R/W	0x0	PJ13_PULL PJ13 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
25:24	R/W	0x0	PJ12_PULL PJ12 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
23:22	R/W	0x0	PJ11_PULL PJ11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PJ10_PULL PJ10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PJ9_PULL PJ9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PJ8_PULL PJ8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PJ7_PULL PJ7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
13:12	R/W	0x0	PJ6_PULL PJ6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x0530			Register Name: PJ_PUL0
Bit	Read/Write	Default/Hex	Description
11:10	R/W	0x0	PJ5_PULL PJ5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PJ4_PULL PJ4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PJ3_PULL PJ3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PJ2_PULL PJ2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PJ1_PULL PJ1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PJ0_PULL PJ0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.124 0x0540 PJ External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x0540			Register Name: PJ_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0540			Register Name: PJ_INT_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0540			Register Name: PJ_INT_CFG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.125 0x0544 PJ External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x0544			Register Name: PJ_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT15_CFG External INT15 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
27:24	R/W	0x0	EINT14_CFG External INT14 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0544			Register Name: PJ_INT_CFG1
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	EINT13_CFG External INT13 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT12_CFG External INT12 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x0544			Register Name: PJ_INT_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.126 0x0550 PJ External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0550			Register Name: PJ_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	EINT15_CTL External INT15 Enable 0: Disable 1: Enable
14	R/W	0x0	EINT14_CTL External INT14 Enable 0: Disable 1: Enable
13	R/W	0x0	EINT13_CTL External INT13 Enable 0: Disable 1: Enable
12	R/W	0x0	EINT12_CTL External INT12 Enable 0: Disable 1: Enable
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable

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Offset: 0x0550			Register Name: PJ_INT_CTL
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

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17.9.6.127 0x0554 PJ External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0554			Register Name: PJ_INT_STA
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W1C	0x0	EINT15_STATUS External INT15 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
14	R/W1C	0x0	EINT14_STATUS External INT14 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
13	R/W1C	0x0	EINT13_STATUS External INT13 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
12	R/W1C	0x0	EINT12_STATUS External INT12 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0554			Register Name: PJ_INT_STA
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0554			Register Name: PJ_INT_STA
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.128 0x0558 PJ External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x0558			Register Name: PJ_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.129 0x055C PJ External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x055C			Register Name: PJ_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.130 0x0570 PJ Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x0570			Register Name: PJ_SECURE
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0570			Register Name: PJ_SECURE
Bit	Read/Write	Default/Hex	Description
15	R/W	0x0	PORT15_SEC_SEL Port15 Secure Selection 0: Not secure 1: Secure
14	R/W	0x0	PORT14_SEC_SEL Port14 Secure Selection 0: Not secure 1: Secure
13	R/W	0x0	PORT13_SEC_SEL Port13 Secure Selection 0: Not secure 1: Secure
12	R/W	0x0	PORT12_SEC_SEL Port12 Secure Selection 0: Not secure 1: Secure
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure

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Offset: 0x0570			Register Name: PJ_SECURE
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.131 0x0580 PK Configure Register 0 (Default Value: 0xFFFF_FFFF)

Offset: 0x0580			Register Name: PK_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0xF	PK7_SELECT PK7 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-D1N/MCSIA-D2N 0100: Reserved 0101: TWI2-SCK 0110: SPI1-HOLD 0111: NCSI-D7 1000: LBUS-LD14 1001: Reserved 1010: Reserved 1011: RGMII0-RXD2/RMII0-NULL 1100: Reserved 1101: Reserved 1110: PK-EINT7 1111: Disable

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Offset: 0x0580			Register Name: PK_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0xF	PK6_SELECT PK6 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-D1P/MCSIA-D2P 0100: Reserved 0101: MCSI0-MCLK 0110: SPI1-WP 0111: NCSI-D6 1000: LBUS-LD15 1001: UART3-RTS 1010: Reserved 1011: RGMII0-RXD3/RMII0-NULL 1100: Reserved 1101: Reserved 1110: PK-EINT6 1111: Disable
23:20	R/W	0xF	PK5_SELECT PK5 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-D1N 0100: Reserved 0101: TWI1-SDA 0110: SPI1-MISO 0111: NCSI-D5 1000: LBUS-LD16 1001: Reserved 1010: LBUS-CS2 1011: LCD-D17 1100: Reserved 1101: Reserved 1110: PK-EINT5 1111: Disable
19:16	R/W	0xF	PK4_SELECT PK4 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-D1P 0100: Reserved 0101: TWI1-SCK 0110: SPI1-MOSI 0111: NCSI-D4 1000: LBUS-LD17 1001: I2S1-DIN0 1010: LBUS-CS3 1011: LCD-D16 1100: Reserved 1101: Reserved 1110: PK-EINT4 1111: Disable
15:12	R/W	0xF	PK3_SELECT PK3 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-D0N 0100: Reserved 0101: UART7-CTS 0110: SPI1-CLK 0111: NCSI-D3 1000: LBUS-LD18 1001: I2S1-DOUT0 1010: LBUS-DE 1011: LCD-D9 1100: Reserved 1101: Reserved 1110: PK-EINT3 1111: Disable

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Offset: 0x0580			Register Name: PK_CFG0
Bit	Read/Write	Default/Hex	Description
11:8	R/W	0xF	PK2_SELECT PK2 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-D0P 0100: Reserved 0101: UART7-RTS 0110: SPI1-CS0 0111: NCSI-D2 1000: LBUS-LD19 1001: I2S1-LRCK 1010: LBUS-LBE3 1011: LCD-D8 1100: Reserved 1101: Reserved 1110: PK-EINT2 1111: Disable
7:4	R/W	0xF	PK1_SELECT PK1 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-CKN 0100: Reserved 0101: UART7-RX 0110: TWI1-SDA 0111: NCSI-D1 1000: LBUS-READY0 1001: I2S1-BCLK 1010: LBUS-DRQ1 1011: LCD-D1 1100: Reserved 1101: Reserved 1110: PK-EINT1 1111: Disable
3:0	R/W	0xF	PK0_SELECT PK0 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIA-CKP 0100: Reserved 0101: UART7-TX 0110: TWI1-SCK 0111: NCSI-D0 1000: LBUS-READY1 1001: I2S1-MCLK 1010: LBUS-DRQ0 1011: LCD-D0 1100: Reserved 1101: Reserved 1110: PK-EINT0 1111: Disable

17.9.6.132 0x0584 PK Configure Register 1 (Default Value: 0x0000_FFFF)

Offset: 0x0584			Register Name: PK_CFG1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x0584			Register Name: PK_CFG1
Bit	Read/Write	Default/Hex	Description
15:12	R/W	0xF	PK11_SELECT PK11 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-CKN 0100: Reserved 0101: TWI3-SDA 0110: PWM0-7 0111: NCSI-VSYNC 1000: LBUS-CS2 1001: UART3-CTS 1010: Reserved 1011: Reserved 1100: Reserved 1101: Reserved 1110: PK-EINT11 1111: Disable
11:8	R/W	0xF	PK10_SELECT PK10 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-CKP 0100: Reserved 0101: TWI3-SCK 0110: PWM0-6 0111: NCSI-HSYNC 1000: LBUS-WAIT 1001: UART3-RX 1010: LBUS-LBE2 1011: RGMII0-TXD2/RMII0-NULL 1100: Reserved 1101: Reserved 1110: PK-EINT10 1111: Disable
7:4	R/W	0xF	PK9_SELECT PK9 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-D0N/MCSIA-D3N 0100: Reserved 0101: MCSI1-MCLK 0110: SPI1-CS2 0111: NCSI-MCLK 1000: LBUS-LD12 1001: UART3-TX 1010: CLK-FANOUT1 1011: RGMII0-TXD3/RMII0-NULL 1100: Reserved 1101: Reserved 1110: PK-EINT9 1111: Disable

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Offset: 0x0584			Register Name: PK_CFG1
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0xF	PK8_SELECT PK8 Select 0000: Input 0001: Output 0010: Reserved 0011: MCSIB-D0P/MCSIA-D3P 0100: Reserved 0101: TWI2-SDA 0110: SPI1-CS1 0111: NCSI-PCLK 1000: LBUS-LD13 1001: Reserved 1010: CLK-FANOUT0 1011: RGMII0-RXCK/RMII0-NULL 1100: Reserved 1101: Reserved 1110: PK-EINT8 1111: Disable

17.9.6.133 0x0590 PK Data Register (Default Value: 0x0000_0000)

Offset: 0x0590			Register Name: PK_DAT
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	PORT11_DAT PK11 Data If the function of PK11 port is configured as input, the PORT11_DAT bit indicates the sate of the PK11 pin; If the function of PK11 port is configured as output, the sate of the PK11 pin can be configured via software by modifying the value of the PORT11_DAT bit. In this case, the value obtained by reading PORT11_DAT bit is the value set by the software; If the PK11 port is configured as functional pin, this bit is invalid.
10	R/W	0x0	PORT10_DAT PK10 Data If the function of PK10 port is configured as input, the PORT10_DAT bit indicates the sate of the PK10 pin; If the function of PK10 port is configured as output, the sate of the PK10 pin can be configured via software by modifying the value of the PORT10_DAT bit. In this case, the value obtained by reading PORT10_DAT bit is the value set by the software; If the PK10 port is configured as functional pin, this bit is invalid.

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Offset: 0x0590			Register Name: PK_DAT
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	PORT9_DAT PK9 Data If the function of PK9 port is configured as input, the PORT9_DAT bit indicates the sate of the PK9 pin; If the function of PK9 port is configured as output, the sate of the PK9 pin can be configured via software by modifying the value of the PORT9_DAT bit. In this case, the value obtained by reading PORT9_DAT bit is the value set by the software; If the PK9 port is configured as functional pin, this bit is invalid.
8	R/W	0x0	PORT8_DAT PK8 Data If the function of PK8 port is configured as input, the PORT8_DAT bit indicates the sate of the PK8 pin; If the function of PK8 port is configured as output, the sate of the PK8 pin can be configured via software by modifying the value of the PORT8_DAT bit. In this case, the value obtained by reading PORT8_DAT bit is the value set by the software; If the PK8 port is configured as functional pin, this bit is invalid.
7	R/W	0x0	PORT7_DAT PK7 Data If the function of PK7 port is configured as input, the PORT7_DAT bit indicates the sate of the PK7 pin; If the function of PK7 port is configured as output, the sate of the PK7 pin can be configured via software by modifying the value of the PORT7_DAT bit. In this case, the value obtained by reading PORT7_DAT bit is the value set by the software; If the PK7 port is configured as functional pin, this bit is invalid.

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Offset: 0x0590			Register Name: PK_DAT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PORT6_DAT PK6 Data If the function of PK6 port is configured as input, the PORT6_DAT bit indicates the sate of the PK6 pin; If the function of PK6 port is configured as output, the sate of the PK6 pin can be configured via software by modifying the value of the PORT6_DAT bit. In this case, the value obtained by reading PORT6_DAT bit is the value set by the software; If the PK6 port is configured as functional pin, this bit is invalid.
5	R/W	0x0	PORT5_DAT PK5 Data If the function of PK5 port is configured as input, the PORT5_DAT bit indicates the sate of the PK5 pin; If the function of PK5 port is configured as output, the sate of the PK5 pin can be configured via software by modifying the value of the PORT5_DAT bit. In this case, the value obtained by reading PORT5_DAT bit is the value set by the software; If the PK5 port is configured as functional pin, this bit is invalid.
4	R/W	0x0	PORT4_DAT PK4 Data If the function of PK4 port is configured as input, the PORT4_DAT bit indicates the sate of the PK4 pin; If the function of PK4 port is configured as output, the sate of the PK4 pin can be configured via software by modifying the value of the PORT4_DAT bit. In this case, the value obtained by reading PORT4_DAT bit is the value set by the software; If the PK4 port is configured as functional pin, this bit is invalid.

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Offset: 0x0590			Register Name: PK_DAT
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_DAT PK3 Data If the function of PK3 port is configured as input, the PORT3_DAT bit indicates the sate of the PK3 pin; If the function of PK3 port is configured as output, the sate of the PK3 pin can be configured via software by modifying the value of the PORT3_DAT bit. In this case, the value obtained by reading PORT3_DAT bit is the value set by the software; If the PK3 port is configured as functional pin, this bit is invalid.
2	R/W	0x0	PORT2_DAT PK2 Data If the function of PK2 port is configured as input, the PORT2_DAT bit indicates the sate of the PK2 pin; If the function of PK2 port is configured as output, the sate of the PK2 pin can be configured via software by modifying the value of the PORT2_DAT bit. In this case, the value obtained by reading PORT2_DAT bit is the value set by the software; If the PK2 port is configured as functional pin, this bit is invalid.
1	R/W	0x0	PORT1_DAT PK1 Data If the function of PK1 port is configured as input, the PORT1_DAT bit indicates the sate of the PK1 pin; If the function of PK1 port is configured as output, the sate of the PK1 pin can be configured via software by modifying the value of the PORT1_DAT bit. In this case, the value obtained by reading PORT1_DAT bit is the value set by the software; If the PK1 port is configured as functional pin, this bit is invalid.

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Offset: 0x0590			Register Name: PK_DAT
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PORT0_DAT PK0 Data If the function of PK0 port is configured as input, the PORT0_DAT bit indicates the sate of the PK0 pin; If the function of PK0 port is configured as output, the sate of the PK0 pin can be configured via software by modifying the value of the PORT0_DAT bit. In this case, the value obtained by reading PORT0_DAT bit is the value set by the software; If the PK0 port is configured as functional pin, this bit is invalid.

17.9.6.134 0x0594 PK Data Set Register (Default Value: 0x0000_0000)

Offset: 0x0594			Register Name: PK_DAT_SET
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W1S	0x0	PORT11_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
10	R/W1S	0x0	PORT10_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
9	R/W1S	0x0	PORT9_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
8	R/W1S	0x0	PORT8_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.

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Offset: 0x0594			Register Name: PK_DAT_SET
Bit	Read/Write	Default/Hex	Description
7	R/W1S	0x0	PORT7_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
6	R/W1S	0x0	PORT6_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
5	R/W1S	0x0	PORT5_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
4	R/W1S	0x0	PORT4_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
3	R/W1S	0x0	PORT3_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
2	R/W1S	0x0	PORT2_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
1	R/W1S	0x0	PORT1_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.
0	R/W1S	0x0	PORT0_DAT_SET Writing 1 to this bit will set the corresponding bit in the PK _DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in thePK _DAT register.

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17.9.6.135 0x0598 PK Data Clear Register (Default Value: 0x0000_0000)

Offset: 0x0598			Register Name: PK_DAT_CLR
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W1C	0x0	PORT11_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
10	R/W1C	0x0	PORT10_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
9	R/W1C	0x0	PORT9_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
8	R/W1C	0x0	PORT8_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
7	R/W1C	0x0	PORT7_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
6	R/W1C	0x0	PORT6_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
5	R/W1C	0x0	PORT5_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.

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Offset: 0x0598			Register Name: PK_DAT_CLR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	PORT4_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
3	R/W1C	0x0	PORT3_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
2	R/W1C	0x0	PORT2_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
1	R/W1C	0x0	PORT1_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.
0	R/W1C	0x0	PORT0_DAT_CLR Writing 1 to this bit will clear the corresponding bit in the PK_DAT register; writing 0 will have no effect. A read of this bit returns the value of the corresponding bit in the PK_DAT register.

17.9.6.136 0x05A0 PK Multi_Driving Register 0 (Default Value: 0x1111_1111)

Offset: 0x05A0			Register Name: PK_DRV0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x1	PK7_DRV PK7 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
27:26	/	/	/
25:24	R/W	0x1	PK6_DRV PK6 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
23:22	/	/	/

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Offset: 0x05A0			Register Name: PK_DRV0
Bit	Read/Write	Default/Hex	Description
21:20	R/W	0x1	PK5_DRV PK5 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
19:18	/	/	/
17:16	R/W	0x1	PK4_DRV PK4 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
15:14	/	/	/
13:12	R/W	0x1	PK3_DRV PK3 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/
9:8	R/W	0x1	PK2_DRV PK2 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PK1_DRV PK1 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PK0_DRV PK0 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.137 0x05A4 PK Multi_Driving Register 1 (Default Value: 0x0000_1111)

Offset: 0x05A4			Register Name: PK_DRV1
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:12	R/W	0x1	PK11_DRV PK11 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
11:10	/	/	/

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Offset: 0x05A4			Register Name: PK_DRV1
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x1	PK10_DRV PK10 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
7:6	/	/	/
5:4	R/W	0x1	PK9_DRV PK9 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3
3:2	/	/	/
1:0	R/W	0x1	PK8_DRV PK8 Multi-driving Select 00: Level0 01: Level1 10: Level2 11: Level3

17.9.6.138 0x05B0 PK Pull Register 0 (Default Value: 0x0000_0000)

Offset: 0x05B0			Register Name: PK_PUL0
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:22	R/W	0x0	PK11_PULL PK11 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
21:20	R/W	0x0	PK10_PULL PK10 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
19:18	R/W	0x0	PK9_PULL PK9 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
17:16	R/W	0x0	PK8_PULL PK8 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
15:14	R/W	0x0	PK7_PULL PK7 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

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Offset: 0x05B0			Register Name: PK_PUL0
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x0	PK6_PULL PK6 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
11:10	R/W	0x0	PK5_PULL PK5 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
9:8	R/W	0x0	PK4_PULL PK4 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
7:6	R/W	0x0	PK3_PULL PK3 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
5:4	R/W	0x0	PK2_PULL PK2 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
3:2	R/W	0x0	PK1_PULL PK1 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved
1:0	R/W	0x0	PK0_PULL PK0 Pull_up or down Select 00: Pull_up/down disable 01: Pull_up 10: Pull_down 11: Reserved

17.9.6.139 0x05C0 PK External Interrupt Configure Register 0 (Default Value: 0x0000_0000)

Offset: 0x05C0			Register Name: PK_INT_CFG0
Bit	Read/Write	Default/Hex	Description
31:28	R/W	0x0	EINT7_CFG External INT7 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x05C0			Register Name: PK_INT_CFG0
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	EINT6_CFG External INT6 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
23:20	R/W	0x0	EINT5_CFG External INT5 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	R/W	0x0	EINT4_CFG External INT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
15:12	R/W	0x0	EINT3_CFG External INT3 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG External INT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x05C0			Register Name: PK_INT_CFG0
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT1_CFG External INT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG External INT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.140 0x05C4 PK External Interrupt Configure Register 1 (Default Value: 0x0000_0000)

Offset: 0x05C4			Register Name: PK_INT_CFG1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:12	R/W	0x0	EINT11_CFG External INT11 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT10_CFG External INT10 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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Offset: 0x05C4			Register Name: PK_INT_CFG1
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x0	EINT9_CFG External INT9 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT8_CFG External INT8 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

17.9.6.141 0x05D0 PK External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x05D0			Register Name: PK_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	EINT11_CTL External INT11 Enable 0: Disable 1: Enable
10	R/W	0x0	EINT10_CTL External INT10 Enable 0: Disable 1: Enable
9	R/W	0x0	EINT9_CTL External INT9 Enable 0: Disable 1: Enable
8	R/W	0x0	EINT8_CTL External INT8 Enable 0: Disable 1: Enable

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Offset: 0x05D0			Register Name: PK_INT_CTL
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	EINT7_CTL External INT7 Enable 0: Disable 1: Enable
6	R/W	0x0	EINT6_CTL External INT6 Enable 0: Disable 1: Enable
5	R/W	0x0	EINT5_CTL External INT5 Enable 0: Disable 1: Enable
4	R/W	0x0	EINT4_CTL External INT4 Enable 0: Disable 1: Enable
3	R/W	0x0	EINT3_CTL External INT3 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL External INT2 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL External INT1 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL External INT0 Enable 0: Disable 1: Enable

17.9.6.142 0x05D4 PK External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x05D4			Register Name: PK_INT_STA
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/

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Offset: 0x05D4			Register Name: PK_INT_STA
Bit	Read/Write	Default/Hex	Description
11	R/W1C	0x0	EINT11_STATUS External INT11 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
10	R/W1C	0x0	EINT10_STATUS External INT10 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
9	R/W1C	0x0	EINT9_STATUS External INT9 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
8	R/W1C	0x0	EINT8_STATUS External INT8 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
7	R/W1C	0x0	EINT7_STATUS External INT7 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
6	R/W1C	0x0	EINT6_STATUS External INT6 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
5	R/W1C	0x0	EINT5_STATUS External INT5 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	R/W1C	0x0	EINT4_STATUS External INT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x05D4			Register Name: PK_INT_STA
Bit	Read/Write	Default/Hex	Description
3	R/W1C	0x0	EINT3_STATUS External INT3 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
2	R/W1C	0x0	EINT2_STATUS External INT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS External INT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS External INT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.6.143 0x05D8 PK External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x05D8			Register Name: PK_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n.
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 00: SYS_32K_CLK 01: SYS_24M_CLK 10: GPIO_60M_CLK

17.9.6.144 0x05DC PK External Debounce Bypass Configure Register (Default Value: 0x0000_0001)

Offset: 0x05DC			Register Name: PK_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description

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Offset: 0x05DC			Register Name: PK_INT_DEB_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/
0	R/W	0x1	DEB_BYPASS Debounce Bypass 0: No-bypass 1: Bypass

17.9.6.145 0x05F0 PK Secure Configure Register (Default Value: 0x0000_0000)

Offset: 0x05F0			Register Name: PK_SECURE
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11	R/W	0x0	PORT11_SEC_SEL Port11 Secure Selection 0: Not secure 1: Secure
10	R/W	0x0	PORT10_SEC_SEL Port10 Secure Selection 0: Not secure 1: Secure
9	R/W	0x0	PORT9_SEC_SEL Port9 Secure Selection 0: Not secure 1: Secure
8	R/W	0x0	PORT8_SEC_SEL Port8 Secure Selection 0: Not secure 1: Secure
7	R/W	0x0	PORT7_SEC_SEL Port7 Secure Selection 0: Not secure 1: Secure
6	R/W	0x0	PORT6_SEC_SEL Port6 Secure Selection 0: Not secure 1: Secure

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Offset: 0x05F0			Register Name: PK_SECURE
Bit	Read/Write	Default/Hex	Description
5	R/W	0x0	PORT5_SEC_SEL Port5 Secure Selection 0: Not secure 1: Secure
4	R/W	0x0	PORT4_SEC_SEL Port4 Secure Selection 0: Not secure 1: Secure
3	R/W	0x0	PORT3_SEC_SEL Port3 Secure Selection 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL Port2 Secure Selection 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL Port1 Secure Selection 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL Port0 Secure Selection 0: Not secure 1: Secure

17.9.6.146 0x0800 PA Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0800			Register Name: PA_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PA15_OS_SEL PA15 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
29:28	R/W	0x0	PA14_OS_SEL PA14 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0800			Register Name: PA_OS_SEL0
Bit	Read/Write	Default/Hex	Description
27:26	R/W	0x0	PA13_OS_SEL PA13 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
25:24	R/W	0x0	PA12_OS_SEL PA12 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
23:22	R/W	0x0	PA11_OS_SEL PA11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PA10_OS_SEL PA10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
19:18	R/W	0x0	PA9_OS_SEL PA9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PA8_OS_SEL PA8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PA7_OS_SEL PA7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0800			Register Name: PA_OS_SEL0
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x0	PA6_OS_SEL PA6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PA5_OS_SEL PA5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PA4_OS_SEL PA4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PA3_OS_SEL PA3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PA2_OS_SEL PA2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PA1_OS_SEL PA1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PA0_OS_SEL PA0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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17.9.6.147 0x0804 PA Operating System Select Register 1 (Default Value: 0x0000_0000)

Offset: 0x0804			Register Name: PA_OS_SEL1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:14	R/W	0x0	PA23_OS_SEL PA23 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PA22_OS_SEL PA22 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PA21_OS_SEL PA21 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PA20_OS_SEL PA20 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PA19_OS_SEL PA19 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PA18_OS_SEL PA18 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0804			Register Name: PA_OS_SEL1
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	PA17_OS_SEL PA17 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PA16_OS_SEL PA16 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.148 0x0808 PB Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0808			Register Name: PB_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:28	R/W	0x0	PB14_OS_SEL PB14 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
27:26	R/W	0x0	PB13_OS_SEL PB13 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
25:24	R/W	0x0	PB12_OS_SEL PB12 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0808			Register Name: PB_OS_SEL0
Bit	Read/Write	Default/Hex	Description
23:22	R/W	0x0	PB11_OS_SEL PB11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PB10_OS_SEL PB10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
19:18	R/W	0x0	PB9_OS_SEL PB9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PB8_OS_SEL PB8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PB7_OS_SEL PB7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PB6_OS_SEL PB6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PB5_OS_SEL PB5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0808			Register Name: PB_OS_SEL0
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x0	PB4_OS_SEL PB4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PB3_OS_SEL PB3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PB2_OS_SEL PB2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PB1_OS_SEL PB1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PB0_OS_SEL PB0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.149 0x0810 PC Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0810			Register Name: PC_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x0810			Register Name: PC_OS_SEL0
Bit	Read/Write	Default/Hex	Description
23:22	R/W	0x0	PC11_OS_SEL PC11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PC10_OS_SEL PC10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
19:18	R/W	0x0	PC9_OS_SEL PC9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PC8_OS_SEL PC8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PC7_OS_SEL PC7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PC6_OS_SEL PC6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PC5_OS_SEL PC5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0810			Register Name: PC_OS_SEL0
Bit	Read/Write	Default/Hex	Description
9:8	R/W	0x0	PC4_OS_SEL PC4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PC3_OS_SEL PC3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PC2_OS_SEL PC2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PC1_OS_SEL PC1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PC0_OS_SEL PC0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.150 0x0818 PD Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0818			Register Name: PD_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PD15_OS_SEL PD15 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0818			Register Name: PD_OS_SEL0
Bit	Read/Write	Default/Hex	Description
29:28	R/W	0x0	PD14_OS_SEL PD14 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
27:26	R/W	0x0	PD13_OS_SEL PD13 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
25:24	R/W	0x0	PD12_OS_SEL PD12 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
23:22	R/W	0x0	PD11_OS_SEL PD11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PD10_OS_SEL PD10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
19:18	R/W	0x0	PD9_OS_SEL PD9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PD8_OS_SEL PD8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0818			Register Name: PD_OS_SEL0
Bit	Read/Write	Default/Hex	Description
15:14	R/W	0x0	PD7_OS_SEL PD7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PD6_OS_SEL PD6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PD5_OS_SEL PD5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PD4_OS_SEL PD4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PD3_OS_SEL PD3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PD2_OS_SEL PD2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PD1_OS_SEL PD1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0818			Register Name: PD_OS_SEL0
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	PD0_OS_SEL PD0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.151 0x081C PD Operating System Select Register 1 (Default Value: 0x0000_0000)

Offset: 0x081C			Register Name: PD_OS_SEL1
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:14	R/W	0x0	PD23_OS_SEL PD23 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PD22_OS_SEL PD22 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PD21_OS_SEL PD21 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PD20_OS_SEL PD20 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x081C			Register Name: PD_OS_SEL1
Bit	Read/Write	Default/Hex	Description
7:6	R/W	0x0	PD19_OS_SEL PD19 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PD18_OS_SEL PD18 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PD17_OS_SEL PD17 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PD16_OS_SEL PD16 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.152 0x0820 PE Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0820			Register Name: PE_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:18	R/W	0x0	PE9_OS_SEL PE9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0820			Register Name: PE_OS_SEL0
Bit	Read/Write	Default/Hex	Description
17:16	R/W	0x0	PE8_OS_SEL PE8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PE7_OS_SEL PE7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PE6_OS_SEL PE6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PE5_OS_SEL PE5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PE4_OS_SEL PE4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PE3_OS_SEL PE3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PE2_OS_SEL PE2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0820			Register Name: PE_OS_SEL0
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	PE1_OS_SEL PE1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PE0_OS_SEL PE0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.153 0x0828 PF Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0828			Register Name: PF_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:12	R/W	0x0	PF6_OS_SEL PF6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PF5_OS_SEL PF5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PF4_OS_SEL PF4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0828			Register Name: PF_OS_SEL0
Bit	Read/Write	Default/Hex	Description
7:6	R/W	0x0	PF3_OS_SEL PF3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PF2_OS_SEL PF2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PF1_OS_SEL PF1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PF0_OS_SEL PF0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.154 0x0830 PG Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0830			Register Name: PG_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PG15_OS_SEL PG15 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
29:28	R/W	0x0	PG14_OS_SEL PG14 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0830			Register Name: PG_OS_SEL0
Bit	Read/Write	Default/Hex	Description
27:26	R/W	0x0	PG13_OS_SEL PG13 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
25:24	R/W	0x0	PG12_OS_SEL PG12 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
23:22	R/W	0x0	PG11_OS_SEL PG11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PG10_OS_SEL PG10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
19:18	R/W	0x0	PG9_OS_SEL PG9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PG8_OS_SEL PG8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PG7_OS_SEL PG7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0830			Register Name: PG_OS_SEL0
Bit	Read/Write	Default/Hex	Description
13:12	R/W	0x0	PG6_OS_SEL PG6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PG5_OS_SEL PG5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PG4_OS_SEL PG4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PG3_OS_SEL PG3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
5:4	R/W	0x0	PG2_OS_SEL PG2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PG1_OS_SEL PG1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PG0_OS_SEL PG0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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17.9.6.155 0x0848 PJ Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0848			Register Name: PJ_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:30	R/W	0x0	PJ15_OS_SEL PJ15 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
29:28	R/W	0x0	PJ14_OS_SEL PJ14 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
27:26	R/W	0x0	PJ13_OS_SEL PJ13 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
25:24	R/W	0x0	PJ12_OS_SEL PJ12 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
23:22	R/W	0x0	PJ11_OS_SEL PJ11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PJ10_OS_SEL PJ10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0848			Register Name: PJ_OS_SEL0
Bit	Read/Write	Default/Hex	Description
19:18	R/W	0x0	PJ9_OS_SEL PJ9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PJ8_OS_SEL PJ8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PJ7_OS_SEL PJ7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PJ6_OS_SEL PJ6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PJ5_OS_SEL PJ5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PJ4_OS_SEL PJ4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PJ3_OS_SEL PJ3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0848			Register Name: PJ_OS_SEL0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	PJ2_OS_SEL PJ2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PJ1_OS_SEL PJ1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PJ0_OS_SEL PJ0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.156 0x0850 PK Operating System Select Register 0 (Default Value: 0x0000_0000)

Offset: 0x0850			Register Name: PK_OS_SEL0
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:22	R/W	0x0	PK11_OS_SEL PK11 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
21:20	R/W	0x0	PK10_OS_SEL PK10 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0850			Register Name: PK_OS_SEL0
Bit	Read/Write	Default/Hex	Description
19:18	R/W	0x0	PK9_OS_SEL PK9 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
17:16	R/W	0x0	PK8_OS_SEL PK8 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
15:14	R/W	0x0	PK7_OS_SEL PK7 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
13:12	R/W	0x0	PK6_OS_SEL PK6 OS1 SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
11:10	R/W	0x0	PK5_OS_SEL PK5 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
9:8	R/W	0x0	PK4_OS_SEL PK4 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
7:6	R/W	0x0	PK3_OS_SEL PK3 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

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Offset: 0x0850			Register Name: PK_OS_SEL0
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	PK2_OS_SEL PK2 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
3:2	R/W	0x0	PK1_OS_SEL PK1 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3
1:0	R/W	0x0	PK0_OS_SEL PK0 OS SELECT 00: Use OS0 01: Use OS1 10: Use OS2 11: Use OS3

17.9.6.157 0x0860 GPIO Operating System ID Configure Register (Default Value: 0x0000_0000)

Offset: 0x0860			Register Name: GPIO_OS_ID_CFG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:24	R/W	0x0	GPIO_OS_ID_3 GPIO Operating System ID 3 Register bit[0]: CPU_CORE0 bit[1]: CPU_CORE1 bit[2]: CPU_CORE2 bit[3]: CPU_CORE3 bit[4]: RV-E907 Set one of bit[4:0] to 1 to assign the OS to the master corresponding to the bit.
23:21	/	/	/

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Offset: 0x0860			Register Name: GPIO_OS_ID_CFG
Bit	Read/Write	Default/Hex	Description
20:16	R/W	0x0	GPIO_OS_ID_2 GPIO Operating System ID 2 Register bit[0]: CPU_CORE0 bit[1]: CPU_CORE1 bit[2]: CPU_CORE2 bit[3]: CPU_CORE3 bit[4]: RV-E907 Set one of bit[4:0] to 1 to assign the OS to the master corresponding to the bit.
15:13	/	/	/
12:8	R/W	0x0	GPIO_OS_ID_1 GPIO Operating System ID 1 Register bit[0]: CPU_CORE0 bit[1]: CPU_CORE1 bit[2]: CPU_CORE2 bit[3]: CPU_CORE3 bit[4]: RV-E907 Set one of bit[4:0] to 1 to assign the OS to the master corresponding to the bit.
7:5	/	/	/
4:0	R/W	0x0	GPIO_OS_ID_0 GPIO Operating System ID 0 Register bit[0]: CPU_CORE0 bit[1]: CPU_CORE1 bit[2]: CPU_CORE2 bit[3]: CPU_CORE3 bit[4]: RV-E907 Set one of bit[4:0] to 1 to assign the OS to the master corresponding to the bit.

17.9.6.158 0x0864 GPIO Operating System ID Configure Bypass Register (Default Value: 0x0000_0001)

Offset: 0x0864			Register Name: GPIO_OS_ID_CFG_BYPASS
Bit	Read/Write	Default/Hex	Description
31:16	W	0x0	KEY_FIELD Key Field. This field should be filled with 0x23AA, and then the bit 15:0 can be written with the new value.
15:1	/	/	/

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Offset: 0x0864			Register Name: GPIO_OS_ID_CFG_BYPASS
Bit	Read/Write	Default/Hex	Description
0	R/W	0x1	GPIO_OS_ID_CFG_BYPASS GPIO Operating System ID Configure Register Bypass 0: No-bypass 1: Bypass

17.9.7 RTC_GPIO Register Description

17.9.7.1 0x0500 RTC_GPIO Configure Register (Default Value: 0x0022_2222)

Offset: 0x0500			Register Name: RTC_GPIO_CFG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x2	GPIO4_SELECT GPIO4 SELECT 0000: Input 0001: Output 0010: Reserved 1110: IRQ-EINT4 else: IO Disable
19:16	R/W	0x2	Reserved
15:12	R/W	0x2	GPIO2_SELECT GPIO2 SELECT 0000: INPUT 0001: Output 0010: Reserved 1110: IRQ-EINT2 else: IO Disable
11:8	R/W	0x2	GPIO1_SELECT GPIO1 SELECT 0000: INPUT 0001: Output 0010: Reserved 1110: IRQ-EINT1 else: IO Disable

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Offset: 0x0500			Register Name: RTC_GPIO_CFG
Bit	Read/Write	Default/Hex	Description
7:4	R/W	0x2	GPIO0_SELECT GPIO0 SELECT 0000: INPUT 0001: Output 0010: Reserved 1110: IRQ-EINT0 else: IO Disable
3:0	R/W	0x2	NMI_SELECT NMI SELECT 0000: INPUT 0001: Output 0010: NMI 1110: NMI-EINT0 else: IO Disable

17.9.7.2 0x0510 RTC_GPIO Data Register (Default Value: 0x0000_0000)

Offset: 0x0510			Register Name: RTC_GPIO_DAT
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0	RTC_GPIO_DAT RTC_GPIO Data If the port is configured as input, the corresponding bit is the pin state. If the port is configured as output, the pin state is the same as the corresponding bit. The read bit value is the value setup by software. If the port is configured as functional pin, the undefined value will be read. Bit4 is reserved.

17.9.7.3 0x0540 RTC_GPIO External Interrupt Configure Register (Default Value: 0x0000_0000)

Offset: 0x0540			Register Name: RTC_GPIO_INT_CFG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x0540			Register Name: RTC_GPIO_INT_CFG
Bit	Read/Write	Default/Hex	Description
23:20	R/W	0x0	EINT5_CFG IRQ-EINT4 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
19:16	/	/	/
15:12	R/W	0x0	EINT3_CFG IRQ-EINT2 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
11:8	R/W	0x0	EINT2_CFG IRQ-EINT1 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
7:4	R/W	0x0	EINT1_CFG IRQ-EINT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved
3:0	R/W	0x0	EINT0_CFG NMI-EINT0 Mode 0x0: Positive Edge 0x1: Negative Edge 0x2: High Level 0x3: Low Level 0x4: Double Edge(Positive/Negative) Others: Reserved

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17.9.7.4 0x0550 RTC_GPIO External Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0550			Register Name: RTC_GPIO_INT_CTL
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	EINT5_CTL IRQ-EINT4 Enable 0: Disable 1: Enable
4	/	/	/
3	R/W	0x0	EINT3_CTL IRQ-EINT2 Enable 0: Disable 1: Enable
2	R/W	0x0	EINT2_CTL IRQ-EINT1 Enable 0: Disable 1: Enable
1	R/W	0x0	EINT1_CTL IRQ-EINT0 Enable 0: Disable 1: Enable
0	R/W	0x0	EINT0_CTL NMI-EINT0 Enable 0: Disable 1: Enable

17.9.7.5 0x0554 RTC_GPIO External Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0554			Register Name: RTC_GPIO_INT_STA
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W1C	0x0	EINT5_STATUS IRQ-EINT4 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
4	/	/	/
3	R/W1C	0x0	EINT3_STATUS IRQ-EINT2 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

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Offset: 0x0554			Register Name: RTC_GPIO_INT_STA
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	EINT2_STATUS IRQ-EINT1 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
1	R/W1C	0x0	EINT1_STATUS IRQ-EINT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear
0	R/W1C	0x0	EINT0_STATUS NMI-EINT0 Pending Bit 0: No IRQ pending 1: IRQ pending Write '1' to clear

17.9.7.6 0x0558 RTC_GPIO External Debounce Configure Register (Default Value: 0x0000_0000)

Offset: 0x0558			Register Name: RTC_GPIO_INT_DEB
Bit	Read/Write	Default/Hex	Description
31:7	/	/	/
6:4	R/W	0x0	DEB_CLK_PRE_SCALE Debounce Clock Pre_scale n The selected clock source is prescaled by 2^n
3:1	/	/	/
0	R/W	0x0	GPIO_INT_CLK_SELECT GPIO Interrupt Clock Select 0: SYS_32K_CLK 1: SYS_24M_CLK

17.9.7.7 0x0570 RTC_GPIO SECURE Configure Register (Default Value: 0x0000_0000)

Offset: 0x0570			Register Name: RTC_GPIO_SECURE
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	PORT5_SEC_SEL GPIO4 secure select 0: Not secure 1: Secure
4	/	/	/

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Offset: 0x0570			Register Name: RTC_GPIO_SECURE
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	PORT3_SEC_SEL GPIO2 secure select 0: Not secure 1: Secure
2	R/W	0x0	PORT2_SEC_SEL GPIO1 secure select 0: Not secure 1: Secure
1	R/W	0x0	PORT1_SEC_SEL GPIO0 secure select 0: Not secure 1: Secure
0	R/W	0x0	PORT0_SEC_SEL NMI secure select 0: Not secure 1: Secure

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17.10 UART

17.10.1 Overview

The universal asynchronous receiver transmitter (UART) provides an asynchronous serial communication with external devices, modem (data carrier equipment, DCE). It performs serial-to-parallel conversion on the data received from peripherals and transmits the converted data to the internal bus. It also performs parallel-to-serial conversion on the data that is transmitted to peripherals.

The UART has the following features:

- Up to 10 UART controllers: UART0, UART1, UART2, UART3, UART4, UART5, UART6, UART7, UART8, UART9
- Compatible with industry-standard 16450/16550 UARTs
- Two separate FIFOs: one is RX FIFO, and the other is TX FIFO
  - Each of them is 64 bytes for UART0, UART6-9
  - Each of them is 128 bytes for UART1-5
- Supports baud rate generator from APB bus clock
  - Speed up to 10 Mbit/s with 160 MHz APB clock
  - Speed up to 5 Mbit/s with 80 MHz APB clock
  - Speed up to 3.75 Mbit/s with 60 MHz APB clock
  - Speed up to 1.5 Mbit/s with 24 MHz APB clock
- Supports 5-8 data bits and 1/1.5/2 stop bits
- Supports Even, Odd or No Parity
- Supports TX/RX DMA Slave interface
- Supports Software/ Hardware Flow Control
- Supports IrDA 1.0 SIR
- Supports RS-485/9-bit mode
- Supports Auto-Flow by using CTS & RTS (excluding UART0)
- Supports FIFIO test mode

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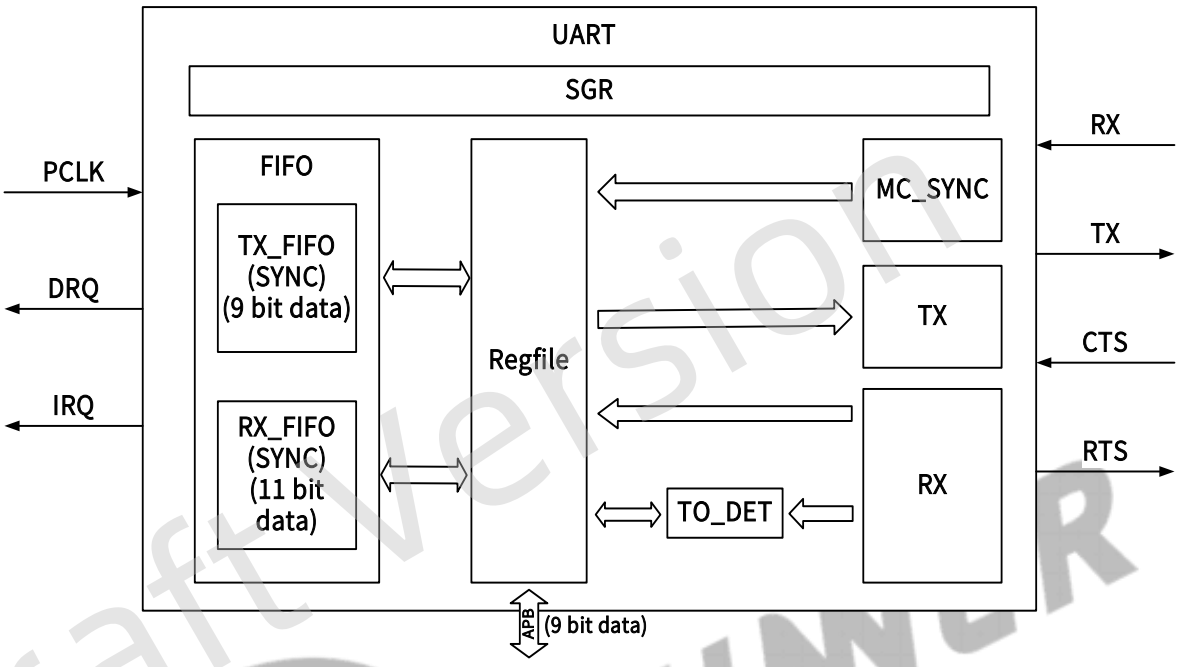
17.10.2 Block Diagram

The following figure shows a block diagram of the UART.

Figure 17-44 UART Block Diagram



17.10.3 External Signals

The following table describes the external signals of UART.

Table 17-52 UART External Signals

Signal Name	Description	Type
UART0		
UART0-RX	UART0 Data Receiver	I
UART0-TX	UART0 Data Transmitter	O
UART1		
UART1-CTS	UART1 Clear to Send	I
UART1-RTS	UART1 Request to Send	O
UART1-RX	UART1 Data Receiver	I
UART1-TX	UART1 Data Transmitter	O
UART2		
UART2-CTS	UART2 Clear to Send	I
UART2-RTS	UART2 Request to Send	O
UART2-RX	UART2 Data Receiver	I
UART2-TX	UART2 Data Transmitter	O
UART3		
UART3-CTS	UART3 Clear to Send	I
UART3-RTS	UART3 Request to Send	O
UART3-RX	UART3 Data Receiver	I
UART3-TX	UART3 Data Transmitter	O

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Signal Name	Description	Type
UART4		
UART4-CTS	UART4 Clear to Send	I
UART4-RTS	UART4 Request to Send	O
UART4-RX	UART4 Data Receiver	I
UART4-TX	UART4 Data Transmitter	O
UART5		
UART5-CTS	UART5 Clear to Send	I
UART5-RTS	UART5 Request to Send	O
UART5-RX	UART5 Data Receiver	I
UART5-TX	UART5 Data Transmitter	O
UART6		
UART6-CTS	UART6 Clear to Send	I
UART6-RTS	UART6 Request to Send	O
UART6-RX	UART6 Data Receiver	I
UART6-TX	UART6 Data Transmitter	O
UART7		
UART7-CTS	UART7 Clear to Send	I
UART7-RTS	UART7 Request to Send	O
UART7-RX	UART7 Data Receiver	I
UART7-TX	UART7 Data Transmitter	O
UART8		
UART8-CTS	UART8 Clear to Send	I
UART8-RTS	UART8 Request to Send	O
UART8-DCD	UART8 Data Carrier Detect	I
UART8-DSR	UART8 Data Set Ready	I
UART8-DTR	UART8 Data Terminal Ready	O
UART8-RI	UART8 Ring Information	I
UART8-RX	UART8 Data Receiver	I
UART8-TX	UART8 Data Transmitter	O
UART9		
UART9-CTS	UART9 Clear to Send	I
UART9-RTS	UART9 Request to Send	O
UART9-RX	UART9 Data Receiver	I
UART9-TX	UART9 Data Transmitter	O

17.10.4 Functional Description

17.10.4.1 Typical Application

The following figure shows the I/O signals of UART controllers. To connect external signals, you need to configure GPIO controller to enable UART model.

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Figure 17-45 I/O Signals of UART Controllers (2/4-wire)

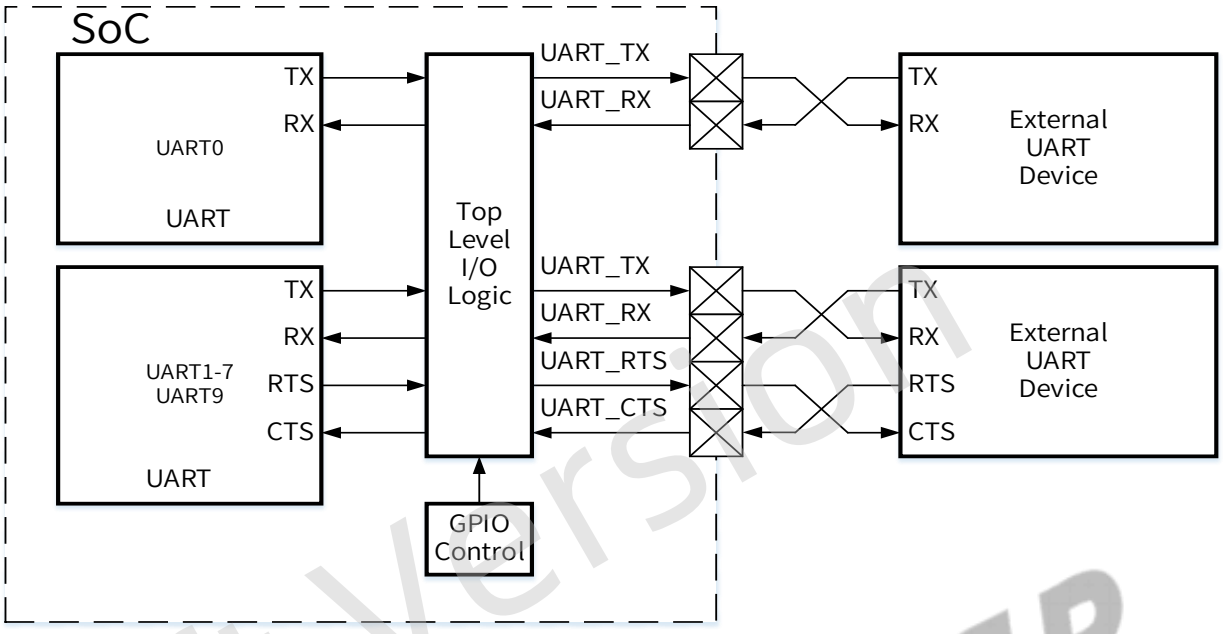
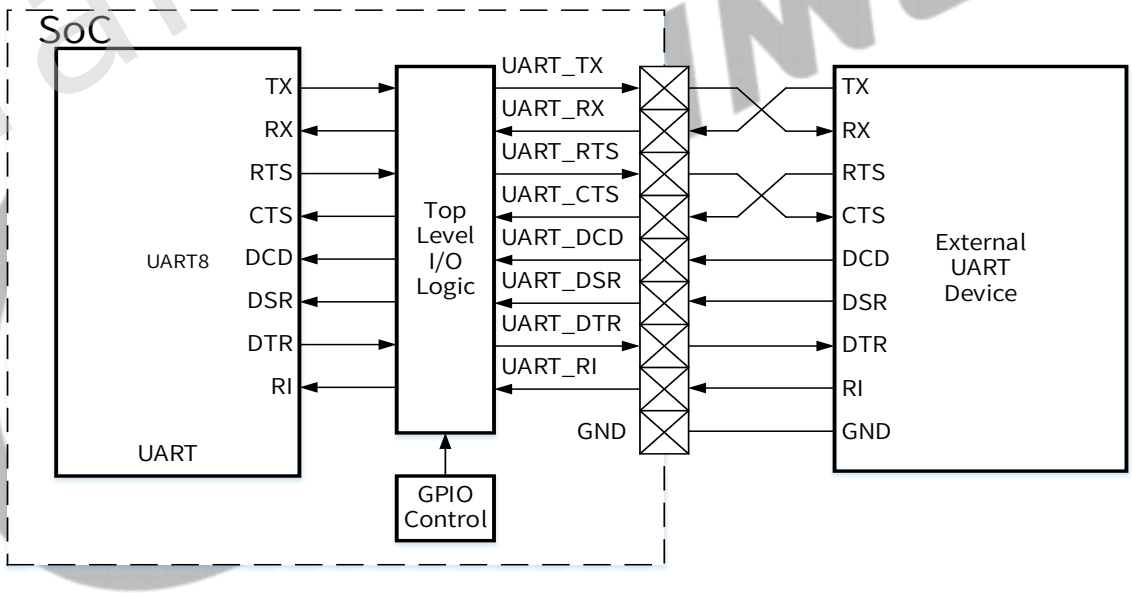


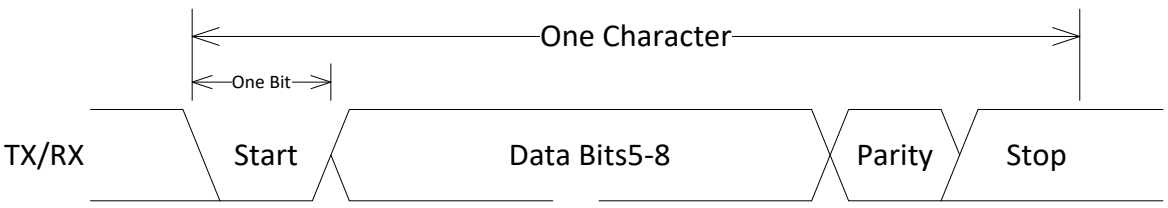
Figure 17-46 I/O Signals of UART Controllers (9-wire)



Serial Data Format

The following figure shows the UART serial data format. The start bit, data bit, parity bit, and stop bit can be configured.

Figure 17-47 UART Serial Data Format



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Start bit: It is the start flag of a data frame. In accordance with the UART protocol, the low level of the TXD signal indicates the start of a data frame. When the UART transmits data, the level needs to hold high.

Data bit: The data bit width can be configured as 5-bit, 6-bit, 7-bit, and 8-bit based on the different applications. If the RS-485 mode is enabled, the data bit width is 8-bit.

Parity bit: It is a 1-bit error correction signal. Parity bit includes odd parity, even parity. The UART can enable and disable the parity bit by setting the UART_LCR register. If RS-485 mode is enabled, the parity bit must be enabled.

Stop bit: It is the stop bit of a data frame. It can be set to 1-bit, 1.5-bit, and 2-bit by the UART_LCR register. The high level of the TXD signal indicates the end of a data frame.

RTS/CTS Auto Flow Control

The following figure shows the typical application diagram for RTS/CTS auto flow control. Figure 17-49 shows the data format of the RTS/CTS auto flow control.

Figure 17-48 Application Diagram for RTS/CTS Auto flow Control

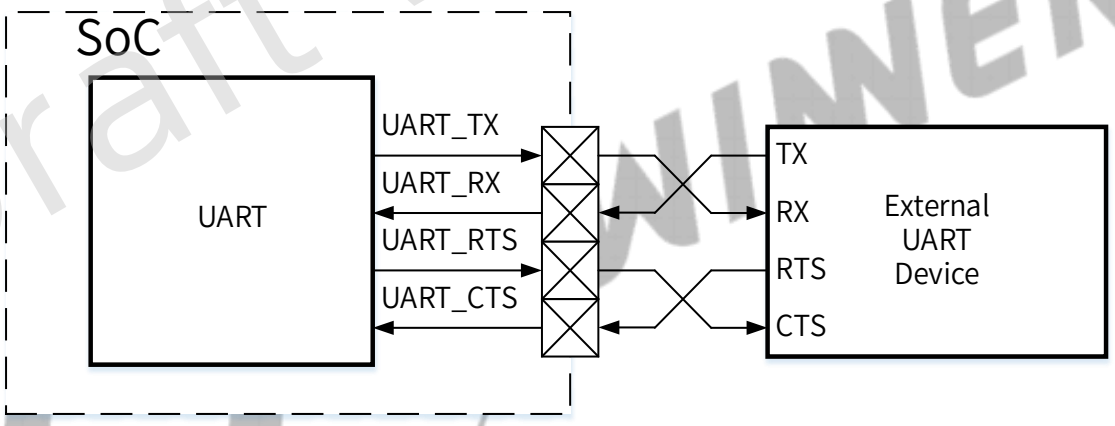
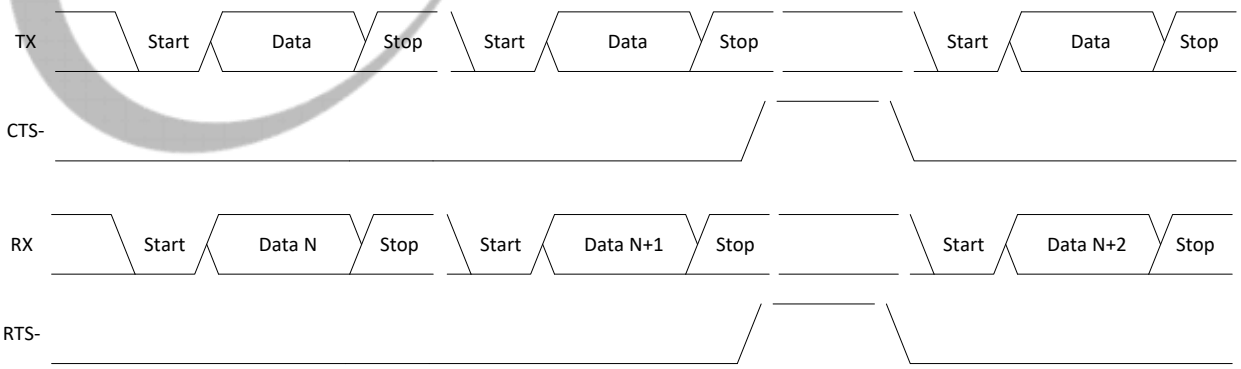


Figure 17-49 RTS/CTS Auto Flow Control Data Format



Serial IrDA

The following figure shows the application diagram for the IrDA transceiver. Figure 17-51 shows the data format of the serial IrDA.

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Figure 17-50 Application Diagram for IrDA Transceiver

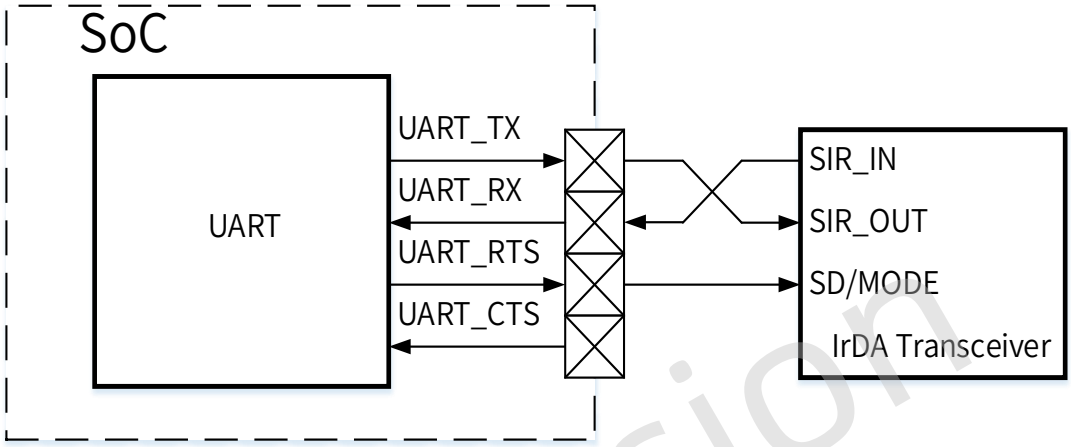
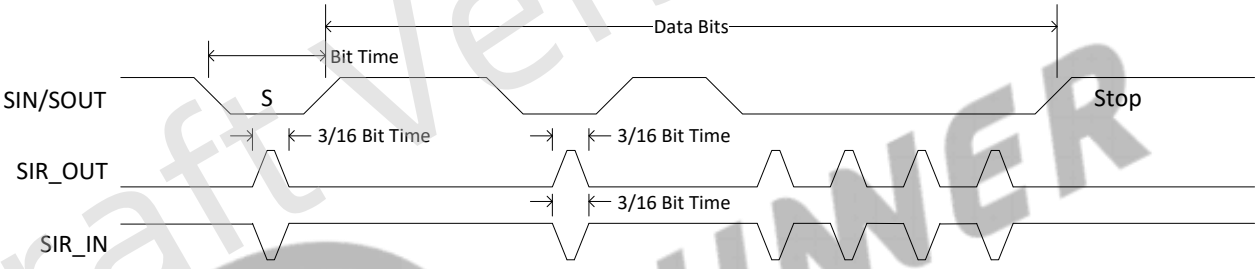


Figure 17-51 Serial IrDA Data Format



RS-485

Figure 17-52 shows the application diagram for the RS-485 transceiver. Figure 17-53 shows the data format of the RS-485.

Figure 17-52 Application Diagram for RS-485 Transceiver

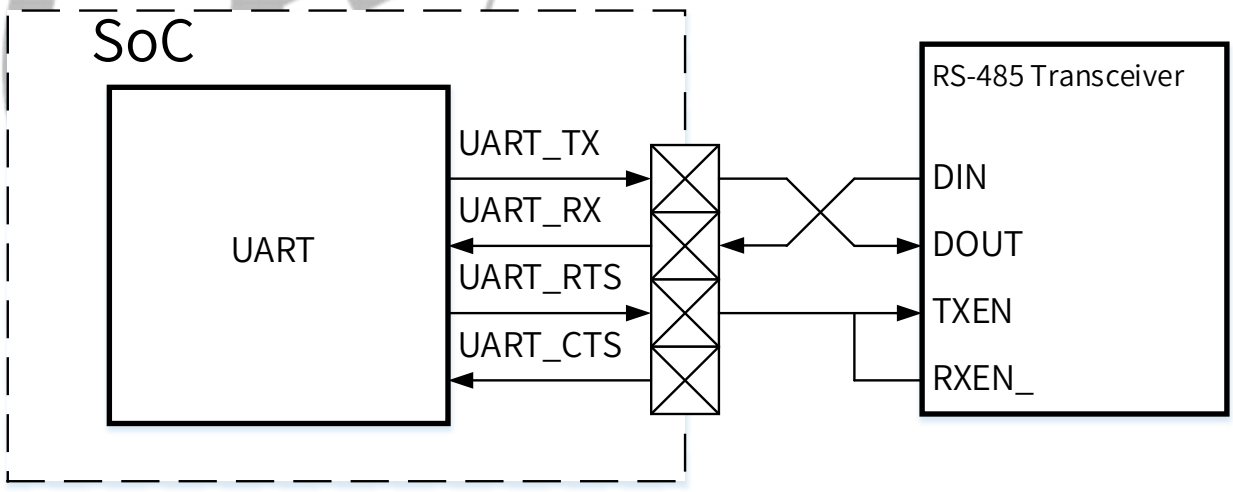


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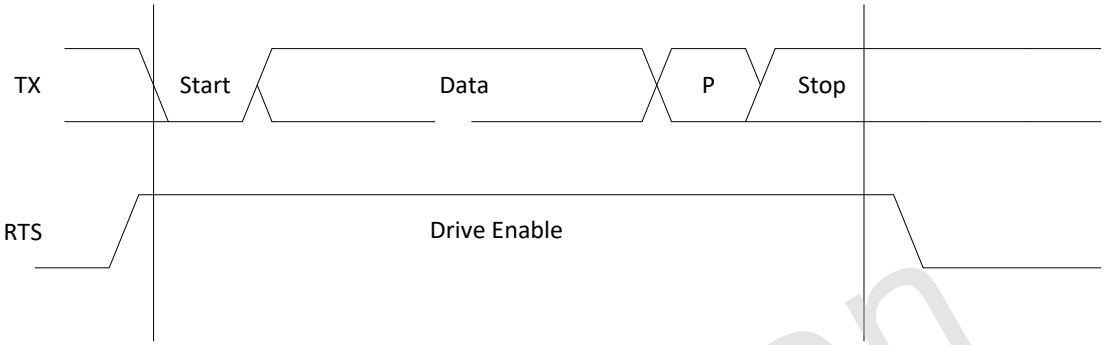
Figure 17-53 RS-485 Data Format



17.10.4.2 Baud and Error Rates

The baud rate is calculated as follows: Baud rate = APB_UART_CLK / (16 * divisor). APB_UART_CLK can be configured in section 4.1.7.27 0x0538 APB_UART Clock Register (Default Value: 0x0000_0000).

The divisor is frequency divider of UART, which has 16-bit. The low 8-bit is in the UART_DLL register, and the high 8-bit is in the UART_DLH register.

The following figures show the errors of baud rate in different clock sources.

Table 17-53 Errors of Baud Rate in UART Mode

Clock Source	Divisor	Baud Rate	Over Sampling	Error(%)
24000000	5000	300	16	0
24000000	2500	600	16	0
24000000	1250	1200	16	0
24000000	625	2400	16	0
24000000	313	4800	16	-0.16
24000000	156	9600	16	0.16
24000000	78	19200	16	0.16
24000000	39	38400	16	0.16
24000000	26	57600	16	0.16
24000000	13	115200	16	0.16
48000000	13	230400	16	0.16
60000000	1	3750000	16	0
75000000	5	921600	16	1.725
48000000	3	1000000	16	0
24000000	1	1500000	16	0
48000000	1	3000000	16	0
80000000	1	5000000	16	0
160000000	1	10000000	16	0

Table 17-54 Errors of Baud Rate in IrDA Mode

Clock source	Divisor	Baud rate	Encoding	Error(%)

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Clock source	Divisor	Baud rate	Encoding	Error(%)
24000000	5000	300	3/16	0
24000000	2500	600	3/16	0
24000000	1250	1200	3/16	0
24000000	625	2400	3/16	0
24000000	313	4800	3/16	-0.16
24000000	156	9600	3/16	0.16
24000000	78	19200	3/16	0.16
24000000	39	38400	3/16	0.16
24000000	26	57600	3/16	0.16
24000000	13	115200	3/16	0.16

Table 17-55 Errors of Baud Rate in RS485 Mode

Clock source	Divisor	Baud rate	Encoding	Error(%)
24000000	5000	300	16	0
24000000	2500	600	16	0
24000000	1250	1200	16	0
24000000	625	2400	16	0
24000000	313	4800	16	-0.16
24000000	156	9600	16	0.16
24000000	78	19200	16	0.16
24000000	39	38400	16	0.16
24000000	26	57600	16	0.16
24000000	13	115200	16	0.16
48000000	13	230400	16	0.16
60000000	1	3750000	16	0
75000000	5	921600	16	1.725
48000000	3	1000000	16	0
24000000	1	1500000	16	0
48000000	1	3000000	16	0
80000000	1	5000000	16	0
160000000	1	10000000	16	0

Table 17-56 Errors of Baud Rate in 19.2M and 26M Clock Source

Clock source	Divisor	Baud rate	Encoding	Error(%)
19200000	125	9600	16	0
26000000	14	115200	16	0.75%

17.10.4.3 DLAB Definition

The DLAB control bit (UART_LCR[7]) is the access control bit of the divisor Latch low/high register.

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If DLAB is 0, then the 0x00 offset address is the UART_RBR/UART_THR (RX/TX FIFO) register, and the 0x04 offset address is the UART_IER register.

If DLAB is 1, then the 0x00 offset address is the UART_DLL register, and the 0x04 offset address is the UART_DLH register.

When the UART initializes, the divisor needs to be set. Write 1 to DLAB to configure UART_DLL and UART_DLH, and then write 0 to DLAB to configure UART_RBR/UART_THR.

17.10.4.4 CHCFG_AT_BUSY Definition

The function of the CHCFG_AT_BUSY (UART_HALT [1]) and CHANGE_UPDATE (UART_HALT[2]) are as follows.

CHCFG_AT_BUSY: If this bit is enabled and UART is busy, the software can also set the UART controller, such as the UART_LCR, UART_DLH, UART_DLL register.

CHANGE_UPDATE: If CHCFG_AT_BUSY is enabled, and CHANGE_UPDATE is written to 1, the configuration of the UART controller can be updated. After the update completes, the bit is cleared to 0 automatically.

To set the divisor, write 1 to CHCFG_AT_BUSY, and Write 1 to DLAB to configure UART_DLL and UART_DLH, then Write 1 to CHANGE_UPDATE.

17.10.4.5 UART_USR Definition

UART_USR is the BUSY flag, used for indicating the UART controller is busy or not.

When TX is sending data, or RX is receiving data, or TX FIFO/RX FIFO is not null, the UART_USR will be asserted. It indicates the UART controller is busy.

17.10.5 Programming Guidelines

The following takes the UART module in the CPUX domain as an example.

17.10.5.1 Initialization

Step 1 System Initialization

- Configure APB_UART_CLK_REG in the Clock Controller Unit (CCU) to set the APB_UART clock.
- Set UARTx_BGR_REG[UARTx_GATING] to 1 to enable the module clock, and set UARTx_BGR_REG[UARTx_RST] to 1 to de-assert the module.

Step 2 UART Controller Initialization

- IO configuration: Configure GPIO multiplex as UART function, and set UART pins to internal pull-up mode (For detail, see the description in 17.9 GPIO).
- Baud-rate configuration:

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- Set UART baud-rate (refer to section 17.10.4.2 Baud and Error Rates);
- Write UART_FCR[FIFOE] to 1 to enable TX/RX FIFO;
- Write UART_HALT[HALT_TX] to 1 to disable TX transfer;
- Set UART_LCR[DLAB] to 1, remain default configuration for other bits; set 0x00 offset address to the UART_DLL register, set 0x04 offset address to the UART_DLH register;
- Write the high 8-bit of divisor to the UART_DLH register, and write the low 8-bit of divisor to the UART_DLL register;
- Set UART_LCR[DLAB] to 0, remain default configuration for other bits; set 0x00 offset address to the UART_RBR/UART_THR register, set 0x04 offset address to the UART_IER register;
- Set UART_HALT[HALT_TX] to 0 to enable TX transfer.

Step 3 Controller Parameter Configuration

- Set data width, stop bits, and even/odd parity type by writing the UART_LCR register.
- Reset, enable FIFO and set FIFO trigger condition by writing the UART_FCR register.
- Set the flow control parameter by writing the UART_MCR register.

Step 4 Interrupt Configuration

- Configure UART interrupt vector number to request UART interrupt (Refer to Section 11.1 GIC for interrupt vector number).
- In DMA mode, write UART_IER to 0 to disable interrupt; write UART_HSK[Handshake configuration] to 0xE5 to set DMA handshake mode; write UART_FCR[DMAM] to 1 to set DMA transmission/reception mode; set DMA parameter and request DMA interrupt according to DMA configuration process.
- In Interrupt mode, configure UART_IER to enable the corresponding interrupt according to requirements: such as transmit (TX) interrupt, receive (RX) interrupt, receive line status interrupt, RS48 interrupt, etc. (Here TX/RX interrupt is usually used).

17.10.5.2 Transmission in DMA Mode

Step 1 Initialize UART model. Refer to section 17.10.5.1 Initialization for initialization steps.

Step 2 Configure UART baud rate by setting UART_DLL and UART_DLH.

Step 3 Configure UART data format by setting UART_LCR.

Step 4 Initialize UART FIFO by setting UART_FCR and UART_FCC.

Step 5 Set PTE and DMA_PTE_RX by setting UART_HALT.

Step 6 Configure DMAC for UART.

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Step 7 Wait for DMA done by checking DMAC status.

17.10.5.3 Transmission in Interrupt Mode

Step 1 Initialize UART model. Refer to section 17.10.5.1 for initialization steps.

Step 2 Configure UART baud rate by setting UART_DLL and UART_DLH.

Step 3 Configure UART data format by setting UART_LCR.

Step 4 Initialize UART FIFO by setting UART_FCR and UART_FCC.

Step 5 Configure UART interrupt by UART_IER and UART_IIR.

Step 6 Wait for UART Interrupt service done.

17.10.5.4 Transmission in RS485 Mode

Select RS-485 Mode

Step 1 Initialize UART model. Refer to section 17.10.5.1 for initialization steps.

Step 2 Configure UART baud rate by setting UART_DLL and UART_DLH.

Step 3 Configure UART data format by setting UART_LCR.

Step 4 Initialize UART FIFO by setting UART_FCR and UART_FCC.

Step 5 Select RS485 Mode by setting UART_MCR [7:6].

Configure RS485 Receive Mode

Step 1 Configure UART as RS485 mode.

Step 2 Write 1 to UART_LCR [3].

Step 3 Configure UART RS485 Receive data format by setting UART_485_CTL [1:0].

----------If select ADD mode ----------

Step 4 Configure AAD broadcast mode by setting UART_485_CTL [8].

In broadcast mode, UART will receive all data.

In non-broadcast mode, UART will not receive data if it does not receive a matching address.

Step 5 Configure the matching address by setting UART_RS485_ADDR_MATCH.

Configure RS485 Half-Duplex Mode

Step 1 Configure UART as RS485 mode.

Step 2 Write 1 to UART_485_CTL [4] to select half-duplex mode.

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Step 3 Write 1 to UART_MCR [5] to enable auto flow control mode.

Step 4 Write 1 to the UART_FCR [0] to enable FIFOs.

The processing logic of RS485 half-duplex hardware auto flow control is as follows.

When there is data in the TXFIFO or TX shift register, the data will be sent first, and the RTS signal will be pulled high.

When there is no data in the TXFIFO or TX shift register, RTS is pulled low, indicating data reception. In this condition, you can also extend the duration that RTS is held high by configuring the RTSN_RST_DLY_TIME register. After the duration ends, the RTS will be pulled low.

NOTE

In RS485 half-duplex mode, only one UART can send data at a time.

17.10.5.5 FIFO Test Mode

Step 1 Configure FIFO test mode by setting FIFO_TM.

Step 2 Write data to the TXFIFO by write UART_THR and read data from the TXFIFO by read READ_TX_FIFO.

Step 3 Write data to the RXFIFO by write WRITE_RX_FIFO and read data from the RXFIFO by read READ_RX_FIFO.

17.10.6 Register List

Module Name	Base Address
UART0	0x02600000
UART1	0x02601000
UART2	0x02602000
UART3	0x02603000
UART4	0x02604000
UART5	0x02605000
UART6	0x02606000
UART7	0x02607000
UART8	0x02608000
UART9	0x02609000

Register Name	Offset	Description
UART_RBR	0x0000	UART Receive Buffer Register
UART_THR	0x0000	UART Transmit Holding Register
UART_DLL	0x0000	UART Divisor Latch Low Register
UART_DLH	0x0004	UART Divisor Latch High Register

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Register Name	Offset	Description
UART_IER	0x0004	UART Interrupt Enable Register
UART_IIR	0x0008	UART Interrupt Identity Register
UART_FCR	0x0008	UART FIFO Control Register
UART_LCR	0x000C	UART Line Control Register
UART_MCR	0x0010	UART Modem Control Register
UART_LSR	0x0014	UART Line Status Register
UART_MSR	0x0018	UART Modem Status Register
UART_SCH	0x001C	UART Scratch Register
UART_USR	0x007C	UART Status Register
UART_TFL	0x0080	UART Transmit FIFO Level Register
UART_RFL	0x0084	UART Receive FIFO Level Register
UART_HSK	0x0088	UART DMA Handshake Configuration Register
UART_DMA_REQ_EN	0x008C	UART DMA Request Enable Register
UART_FCR_1	0x00A0	UART FIFO Control Register1
UART_HALT	0x00A4	UART Halt TX Register
UART_DBG_DLL	0x00B0	UART Debug DLL Register
UART_DBG_DLH	0x00B4	UART Debug DLH Register
UART_485_CTL	0x00C0	UART RS485 Control Configure and Status Register
UART_RS485_ADDR_MATCH	0x00C4	UART RS485 Address Match Register
BUS_IDLE_CHECK	0x00C8	UART RS485 Bus Idle Check Register
TX_DELAY	0x00CC	UART TX Delay Register
FIFO_TM	0x00D4	UART FIFO Test Mode Register
READ_TX_FIFO	0x00D8	UART Read TX FIFO Register
WRITE_RX_FIFO	0x00DC	UART Write RX FIFO Register
READ_RX_FIFO	0x00E0	UART Read RX FIFO Register
RTSN_RST_DLY_TIME	0x00E4	UART RS485 RTSN Pull Down Delay Time Register
UART_FCC	0x00F0	UART FIFO Clock Control Register

17.10.7 Register Description

17.10.7.1 0x0000 UART Receive Buffer Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: UART_RBR
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/

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Offset: 0x0000			Register Name: UART_RBR
Bit	Read/Write	Default/Hex	Description
8:0	R	0x0	RBR Receiver Buffer Register Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LCR) is set. If in FIFO mode and FIFOs are enabled (FCR[0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an overrun error occurs.

17.10.7.2 0x0000 UART Transmit Holding Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: UART_THR
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:0	W	0x0	THR Transmit Holding Register Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. Any attempt to write data when the FIFO is full results in the write data being lost.

17.10.7.3 0x0000 UART Divisor Latch Low Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: UART_DLL
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x0000			Register Name: UART_DLL
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x0	DLL Divisor Latch Low Lower 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLL is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

17.10.7.4 0x0004 UART Divisor Latch High Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: UART_DLH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	DLH Divisor Latch High Upper 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

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17.10.7.5 0x0004 UART Interrupt Enable Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: UART_IER
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	PTIME Programmable THRE Interrupt Mode Enable This is used to enable/disable the generation of THRE Interrupt. 0: Disable 1: Enable
6:5	/	/	/
4	R/W	0x0	RS485_INT_EN RS485 Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	EDSSI Enable Modem Status Interrupt This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. 0: Disable 1: Enable
2	R/W	0x0	ELSI Enable Receiver Line Status Interrupt This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. 0: Disable 1: Enable
1	R/W	0x0	ETBEI Enable Transmit Holding Register Empty Interrupt This is used to enable/disable the generation of Transmitter Holding Register Empty Interrupt. This is the third highest priority interrupt. 0: Disable 1: Enable

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Offset: 0x0004			Register Name: UART_IER
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	ERBFI Enable Received Data Available Interrupt This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts. 0: Disable 1: Enable

17.10.7.6 0x0008 UART Interrupt Identity Register (Default Value: 0x0000_0001)

Offset: 0x0008			Register Name: UART_IIR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R	0x0	FEFLAG FIFOs Enable Flag This is used to indicate whether the FIFOs are enabled or disabled. 00: Disable 11: Enable
5:4	/	/	/
3:0	R	0x1	IID Interrupt ID This indicates the highest priority pending interrupt which can be one of the following types: 0000: Modem status 0001: No interrupt pending 0010: THR empty 0011: RS485 Interrupt (pending by UART_485_CTL[6:5]) 0100: Received data available 0110: Receiver line status 0111: Busy detect 1100: Character timeout Bit 3 indicates an interrupt can only occur when the FIFOs are enabled and used to distinguish a Character Timeout condition interrupt.

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Interrupt ID	Priority Level	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	Receiver line status	Overrun/parity/ framing errors or break interrupt	Reading the line status register
0011	Second	RS485 Interrupt	In RS485 mode, receives address data and match setting address	Writes 1 to addr flag to reset
0100	Third	Received data available	Receiver data available (non-FIFO mode or FIFOs disabled) or RCVR FIFO trigger level reached (FIFO mode and FIFOs enabled)	Reading the receiver buffer register (non-FIFO mode or FIFOs disabled) or the FIFO drops below the trigger level (FIFO mode and FIFOs enabled)
1100	Fourth	Character timeout indication	No characters in or out of the RCVR FIFO during the last 4 character times and there is at least 1character in it during This time	Reading the receiver buffer register
0010	Fifth	Transmit holding register empty	Transmitter holding register empty (Program THRE Mode disabled) or XMIT FIFO at or below threshold (Program THRE Mode enabled)	Reading the IIR register (if source of interrupt); or, writing into THR (FIFOs or THRE Mode not selected or disabled) or XMIT FIFO above threshold (FIFOs and THRE Mode selected and enabled).
0000	Sixth	Modem status	Clear to send or data set ready or ring indicator or data carrier detect. Note that if auto flow control mode is enabled, a change in CTS (that is, DCTS set) does not cause an interrupt.	Reading the Modem status Register
0111	Seventh	Busy detect indication	UART_16550_COMPATIBLE = NO and master has tried to write to the Line Control Register while the UART is busy (USR[0] is set to one).	Reading the UART status register

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17.10.7.7 0x0008 UART FIFO Control Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: UART_FCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	W	0x0	RT RCVR Trigger This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation. 00: 1 character in the FIFO 01: FIFO ¼ full 10: FIFO ½ full 11: FIFO-2 less than full
5:4	W	0x0	TFT TX Empty Trigger This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation. 00: FIFO empty 01: 2 characters in the FIFO 10: FIFO ¼ full 11: FIFO ½ full

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Offset: 0x0008			Register Name: UART_FCR
Bit	Read/Write	Default/Hex	Description
3	W	0x0	DMAM DMA Mode 0: Mode 0 In this mode, when PTE is high and TX FIFO is enable, the TX DMA request will be set when TFL is less than or equal to FIFO Trigger Level(otherwise it will be cleared). When PTE is high and TX FIFO is disabled, the TX DMA request will be set only if THR is empty. If PTE is low, the TX DMA request will be set only if the TX FIFO(TX FIFO Enabled) or THR(TX FIFO Disabled) is empty. When dma_pte_rx is high and RX FIFO is enabled, the rx drq will be set only if RFL is equal to or more than FIFO Trigger Level, otherwise it will be cleared. 1: Mode 1 In this mode, TX FIFO should be enable. If the PTE is high, the TX DMA request will be set when TFL is less than or equal to FIFO Trigger Level; If PTE is low, the TX DMA request will be set when TX FIFO is empty. Once the request is set, it is cleared only when TX FIFO is full. If RFL is equal to or more than FIFO Trigger Level or there is a character timeout , the rx drq will be set; Once the rx drq is set, it is cleared only when RX FIFO(RX FIFO enabled) or RBR(RX FIFO disabled) is empty.
2	W	0x0	XFIFOR XMIT FIFO Reset This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request. It is 'self-clearing'. It is not necessary to clear this bit.
1	W	0x0	RFIFOR RCVR FIFO Reset This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request. It is 'self-clearing'. It is not necessary to clear this bit.

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Offset: 0x0008			Register Name: UART_FCR
Bit	Read/Write	Default/Hex	Description
0	W	0x0	FIFOE Enable FIFOs This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.

17.10.7.8 0x000C UART Line Control Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: UART_LCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	DLAB Divisor Latch Access Bit It is writable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable reading and writing of the Divisor Latch register (DLL and DLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers. 0: Select RX Buffer Register (RBR) / TX Holding Register(THR) and Interrupt Enable Register (IER) 1: Select Divisor Latch LS Register (DLL) and Divisor Latch MS Register (DLM)
6	R/W	0x0	BC Break Control Bit This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If SIR_MODE = Enabled and active (MCR[6] set to one) the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to the receiver and the sir_out_n line is forced low.

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Offset: 0x000C			Register Name: UART_LCR
Bit	Read/Write	Default/Hex	Description
5:4	R/W	0x0	EPS Even Parity Select It is writable only when UART is not busy (USR[0] is zero) and always writable readable. This is used to select between even and odd parity, when parity is enabled (PEN set to one). Setting the LCR[5] is unset to reverse the LCR[4]. 00: Odd Parity 01: Even Parity
3	R/W	0x0	PEN Parity Enable It is writable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 0: Parity disabled 1: Parity enabled Note: When use RS485 mode, PEN(LCR[3]) must set to 1.
2	R/W	0x0	STOP Number of stop bits It is writable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to zero, one stop bit is transmitted in the serial data. If set to one and the data bits are set to 5 (LCR[1:0] set to zero) one and a half stop bits is transmitted. Otherwise, two stop bits are transmitted. Note that regardless of the number of stop bits selected, the receiver checks only the first stop bit. 0: 1 stop bit 1: 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit

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Offset: 0x000C			Register Name: UART_LCR
Bit	Read/Write	Default/Hex	Description
1:0	R/W	0x0	DLS Data Length Select It is writable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of data bits per character that the peripheral transmits and receives. The number of bit that may be selected areas follows: 00: 5 bits 01: 6 bits 10: 7 bits 11: 8 bits Note: When use RS485 mode, DLS must be set to 11.

17.10.7.9 0x0010 UART Modem Control Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: UART_MCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R/W	0x0	UART_FUNCTION Select IrDA or RS485 0: UART Mode 1: IrDA SIR Mode 2: RS485 Mode 3: Reversed
5	R/W	0x0	AFCE Auto Flow Control Enable When FIFOs are enabled and the Auto Flow Control Enable (AFCE) bit is set, Auto Flow Control features are enabled. 0: Auto Flow Control Mode is disabled 1: Auto Flow Control Mode is enabled

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Offset: 0x0010			Register Name: UART_MCR
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	LOOP Loop Back Mode 0: Normal Mode 1: Loop Back Mode This is used to put the UART into a diagnostic mode for test purposes. If operating in UART mode (SIR_MODE != Enabled or not active, MCR[6] set to zero), data on the sout line is held high, while serial data output is looped back to the sin line, internally. In this mode all the interrupts are fully functional. Also, in loopback mode, the modem control inputs (dsr_n, cts_n, ri_n, dcd_n) are disconnected and the modem control outputs (dtr_n, rts_n, out1_n, out2_n) are looped back to the inputs, internally. If operating in infrared mode (SIR_MODE == Enabled AND active, MCR[6] set to one), data on the sir_out_n line is held low, while serial data output is inverted and looped back to the sir_in line.
3:2	/	/	/
1	R/W	0x0	RTS Request to Send This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. When Auto RTS Flow Control is not enabled (MCR[5] set to zero), the rts_n signal is set low by programming MCR[1] (RTS) to a high. In Auto Flow Control, AFCE_MODE == Enabled and active (MCR[5] set to one) and FIFOs enable (FCR[0] set to one), the rts_n output is controlled in the same way, but is also gated with the receiver FIFO threshold trigger (rts_n is inactive high when above the threshold). The rts_n signal is de-asserted when MCR[1] is set low. 0: rts_n de-asserted (logic 1) 1: rts_n asserted (logic 0) Note that in Loopback mode (MCR[4] set to one), the rts_n output is held inactive high while the value of this location is internally looped back to an input.

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Offset: 0x0010			Register Name: UART_MCR
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	DTR Data Terminal Ready This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n. 0: dtr_n de-asserted (logic 1) 1: dtr_n asserted (logic 0) The Data Terminal Ready output is used to inform the modem or data set that the UART is ready to establish communications. Note that in Loopback mode (MCR[4] set to one), the dtr_n output is held inactive high while the value of this location is internally looped back to an input.

17.10.7.10 0x0014 UART Line Status Register (Default Value: 0x0000_0060)

Offset: 0x0014			Register Name: UART_LSR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R	0x0	FIFOERR RX Data Error in FIFO When FIFOs are disabled, this bit is always 0. When FIFOs are enabled, this bit is set to 1 when there is at least one PE, FE, or BI in the RX FIFO. It is cleared by a read from the LSR register provided there are no subsequent errors in the FIFO.
6	R	0x1	TEMT Transmitter Empty If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register and the TX Shift Register are empty. If the FIFOs are enabled, this bit is set whenever the TX FIFO and the TX Shift Register are empty. In both cases, this bit is cleared when a byte is written to the TX data channel.

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Offset: 0x0014			Register Name: UART_LSR
Bit	Read/Write	Default/Hex	Description
5	R	0x1	THRE TX Holding Register Empty If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register is empty and ready to accept new data and it is cleared when the CPU writes to the TX Holding Register. If the FIFOs are enabled, this bit is set to "1" whenever the TX FIFO is empty and it is cleared when at least one byte is written to the TX FIFO.
4	R	0x0	BI Break Interrupt This is used to indicate the detection of a break sequence on the serial input data. If in UART mode (SIR_MODE == Disabled), it is set whenever the serial input, sir_in, is held in a logic '0' state for longer than the sum of *start time* + *data bits* + *parity* + *stop bits*. If in infrared mode (SIR_MODE == Enabled), it is set whenever the serial input, sir_in, is continuously pulsed to logic '0' for longer than the sum of *start time* + *data bits* + *parity* + *stop bits*. A break condition on serial input causes one and only one character, consisting of all zeros, to be received by the UART. In the FIFO mode, the character associated with the break condition is carried through the FIFO and is revealed when the character is at the top of the FIFO. Reading the LSR clears the BI bit. In the non-FIFO mode, the BI indication occurs immediately and persists until the LSR is read.

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Offset: 0x0014			Register Name: UART_LSR
Bit	Read/Write	Default/Hex	Description
3	RC	0x0	FE Framing Error This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data. In the FIFO mode, since the framing error is associated with a character received, it is revealed when the character with the framing error is at the top of the FIFO. When a framing error occurs, the UART tries to resynchronize. It does this by assuming that the error was due to the start bit of the next character and then continues receiving the other bit i.e. data, and/or parity and stop. It should be noted that the Framing Error (FE) bit (LSR[3]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]). 0: No framing error 1: Framing error Reading the LSR clears the FE bit.
2	RC	0x0	PE Parity Error This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set. In the FIFO mode, since the parity error is associated with a character received, it is revealed when the character with the parity error arrives at the top of the FIFO. It should be noted that the Parity Error (PE) bit (LSR[2]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]). 0: No parity error 1: Parity error Reading the LSR or FIFO clears the PE bit.

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Offset: 0x0014			Register Name: UART_LSR
Bit	Read/Write	Default/Hex	Description
1	RC	0x0	OE Overrun Error This occurs if a new data character was received before the previous data was read. In the non-FIFO mode, the OE bit is set when a new character arrives in the receiver before the previous character was read from the RBR. When this happens, the data in the RBR is overwritten. In the FIFO mode, an overrun error occurs when the FIFO is full and a new character arrives at the receiver. The data in the FIFO is retained and the data in the receive shift register is lost. 0: No overrun error 1: Overrun error Reading the LSR clears the OE bit.
0	R	0x0	DR Data Ready This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. 0: No data ready 1: Data ready This bit is cleared when the RBR is read in non-FIFO mode, or when the receiver FIFO is empty, in FIFO mode.

17.10.7.11 0x0018 UART Modem Status Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: UART_MSR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R	0x0	DCD Line State of Data Carrier Detect This is used to indicate the current state of the modem control line dcd_n. This bit is the complement of dcd_n. When the Data Carrier Detect input (dcd_n) is asserted it is an indication that the carrier has been detected by the modem or data set. 0: dcd_n input is de-asserted (logic 1) 1: dcd_n input is asserted (logic 0)

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Offset: 0x0018			Register Name: UART_MSR
Bit	Read/Write	Default/Hex	Description
6	R	0x0	RI Line State of Ring Indicator This is used to indicate the current state of the modem control line ri_n. This bit is the complement of ri_n. When the Ring Indicator input (ri_n) is asserted it is an indication that a telephone ringing signal has been received by the modem or data set. 0: ri_n input is de-asserted (logic 1) 1: ri_n input is asserted (logic 0)
5	R	0x0	DSR Line State of Data Set Ready This is used to indicate the current state of the modem control line dsr_n. This bit is the complement of dsr_n. When the Data Set Ready input (dsr_n) is asserted it is an indication that the modem or data set is ready to establish communications with UART. 0: dsr_n input is de-asserted (logic 1) 1: dsr_n input is asserted (logic 0) In Loopback Mode (MCR[4] set to one), DSR is the same as MCR[0] (DTR).
4	R	0x0	CTS Line State of Clear To Send This is used to indicate the current state of the modem control line cts_n. This bit is the complement of cts_n. When the Clear to Send input (cts_n) is asserted it is an indication that the modem or data set is ready to exchange data with UART. 0: cts_n input is de-asserted (logic 1) 1: cts_n input is asserted (logic 0) In Loopback Mode (MCR[4] = 1), CTS is the same as MCR[1] (RTS).

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Offset: 0x0018			Register Name: UART_MSR
Bit	Read/Write	Default/Hex	Description
3	RC	0x0	DDCD Delta Data Carrier Detect This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read. 0: No change on dcd_n since last read of MSR 1: Change on dcd_n since last read of MSR Reading the MSR clears the DDCD bit. Note: Ff the DDCD bit is not set and the dcd_n signal is asserted (low) and a reset occurs (software or otherwise), then the DDCD bit is set when the reset is removed if the dcd_n signal remains asserted.
2	RC	0x0	TERI Trailing Edge Ring Indicator This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read. 0: No change on ri_n since last read of MSR 1: Change on ri_n since last read of MSR Reading the MSR clears the TERI bit.
1	RC	0x0	DDSR Delta Data Set Ready This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read. 0: No change on dsr_n since last read of MSR 1: Change on dsr_n since last read of MSR Reading the MSR clears the DDSR bit. In Loopback Mode (MCR[4] = 1), DDSR reflects changes on MCR[0] (DTR). Note: If the DDSR bit is not set and the dsr_n signal is asserted (low) and a reset occurs (software or otherwise), then the DDSR bit is set when the reset is removed if the dsr_n signal remains asserted.

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Offset: 0x0018			Register Name: UART_MSR
Bit	Read/Write	Default/Hex	Description
0	RC	0x0	DCTS Delta Clear to Send This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read. 0: No change on ctsdsr_n since last read of MSR 1: Change on ctsdsr_n since last read of MSR Reading the MSR clears the DCTS bit. In Loopback Mode (MCR[4] = 1), DCTS reflects changes on MCR[1] (RTS). Note: If the DCTS bit is not set and the cts_n signal is asserted (low) and a reset occurs (software or otherwise), then the DCTS bit is set when the reset is removed if the cts_n signal remains asserted.

17.10.7.12 0x001C UART Scratch Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: UART_SCH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:1	R/W	0x0	SCRATCH_REG Scratch Register This register is for programmers to use as a temporary storage space. It has no defined purpose in the UART.
0	R/W	0x0	RS485_AUTO_RTS_EN RS485_AUTO_RTS_ENABLE 1: Enable 0: Disable

17.10.7.13 0x007C UART Status Register (Default Value: 0x0000_0006)

Offset: 0x007C			Register Name: UART_USR
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/

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Offset: 0x007C			Register Name: UART_USR
Bit	Read/Write	Default/Hex	Description
4	R	0x0	RFF Receive FIFO Full This is used to indicate that the receive FIFO is completely full. 0: Receive FIFO not full 1: Receive FIFO Full This bit is cleared when the RX FIFO is no longer full.
3	R	0x0	RFNE Receive FIFO Not Empty This is used to indicate that the receive FIFO contains one or more entries. 0: Receive FIFO is empty 1: Receive FIFO is not empty This bit is cleared when the RX FIFO is empty.
2	R	0x1	TFE Transmit FIFO Empty This is used to indicate that the transmit FIFO is completely empty. 0: Transmit FIFO is not empty 1: Transmit FIFO is empty This bit is cleared when the TX FIFO is no longer empty.
1	R	0x1	TFNF Transmit FIFO Not Full This is used to indicate that the transmit FIFO in not full. 0: Transmit FIFO is full 1: Transmit FIFO is not full This bit is cleared when the TX FIFO is full.
0	R	0x0	BUSY UART Busy Bit 0: Idle or inactive 1: Busy

17.10.7.14 0x0080 UART Transmit FIFO Level Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: UART_TFL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/

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Offset: 0x0080			Register Name: UART_TFL
Bit	Read/Write	Default/Hex	Description
8:0	R	0x0	TFL Transmit FIFO Level This is indicates the number of data entries in the transmit FIFO.

17.10.7.15 0x0084 UART Receive FIFO Level Register (Default Value: 0x0000_0000)

Offset: 0x0084			Register Name: UART_RFL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:0	R	0x0	RFL Receive FIFO Level This indicates the number of data entries in the receive FIFO.

17.10.7.16 0x0088 UART DMA Handshake Configuration Register (Default Value: 0x0000_00A5)

Offset: 0x0088			Register Name: UART_HSK
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0xA5	Handshake configuration 0xA5: DMA wait cycle mode 0xE5: DMA handshake mode

17.10.7.17 0x008C UART DMA Request Enable Register (Default Value: 0x0000_0003)

Offset: 0x008C			Register Name: UART_DMA_REQ_EN
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	DMA Timeout Enable 0: Disable 1: Enable
1	R/W	0x1	DMA TX REQ Enable 0: Disable 1: Enable
0	R/W	0x1	DMA RX REQ Enable 0: Disable 1: Enable

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17.10.7.18 0x00A0 UART FIFO Control Register1 (Default Value: 0x0001_0000)

Offset: 0x00A0			Register Name: UART_FCR_1
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28	R/W	0x0	TRIG_SEL 0: Use RT/TFT 1: Use RT_NEW/TFT_NEW The bit width of RT_NEW and TFT_NEW is 12 bits. When the FIFO depth is greater than 4095, RT_NEW and TFT_NEW cannot completely express all thresholds.
27:16	R/W	0x1	RT_NEW RCVR Trigger This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation.
15:12	/	/	/
11:0	R/W	0x0	TFT_NEW TX Empty Trigger This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation.

17.10.7.19 0x00A4 UART Halt TX Register (Default Value: 0x0000_0000)

Offset: 0x00A4			Register Name: UART_HALT
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	DMA_PTE_RX The sending of RX_DRQ. In DMA1 mode, when RFL is more than or equal to trig or receive timeout, send DRQ. In DMA0 mode, when DMA_PTE_RX = 1 and FIFO on, if RFL is more than or equal to trig, send DRQ, else DRQ is cleared. In other cases, once the receive data is valid, send DRQ.

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Offset: 0x00A4			Register Name: UART_HALT
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PTE The sending of TX_REQ. In DMA1 mode (FIFO on), if PTE is set 1, when TFL is less than or equal to trig, send the DMA request. If PTE is set 0, when FIFO is empty, send the DMA request. The DMA request will stop when FIFO is full. In DMA0 mode, if PTE is set 1 and FIFO on, when TFL is less than or equal to trig, send DMA request. If PTE is set 1 and FIFO off, when THR is empty, send DMA request. If PTE is set 0, when FIFO(FIFO Enable) or THR(FIFO Enable) is empty, send DMA request. Otherwise, DMA request is cleared.
5	R/W	0x0	SIR_RX_INVERT SIR Receiver Pulse Polarity Invert 0: Not invert receiver signal 1: Invert receiver signal
4	R/W	0x0	SIR_TX_INVERT SIR Transmit Pulse Polarity Invert 0: Not invert transmit pulse 1: Invert transmit pulse
3	/	/	/
2	R/WAC	0x0	CHANGE_UPDATE After the user using HALT[1] to change the baudrate or LCR configuration, write 1 to update the configuration and waiting this bit self clear to 0 to finish update process. Write 0 to this bit has no effect. 1: Update trigger, Self clear to 0 when finish update.
1	R/W	0x0	CHCFG_AT_BUSY This is an enable bit for the user to change LCR register configuration and baudrate register (DLH and DLL) when the UART is busy (USB[0] is 1). 1: Enable change when busy
0	R/W	0x0	HALT_TX Halt TX This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled. 0 : Halt TX disabled 1 : Halt TX enabled Note: If FIFOs are not enabled, the setting has no effect on operation.

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17.10.7.20 0x00B0 UART Debug DLL Register (Default Value: 0x0000_0000)

Offset: 0x00B0			Register Name: UART_DBG_DLL
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	DEBUG DLL

17.10.7.21 0x00B4 UART Debug DLH Register (Default Value: 0x0000_0000)

Offset: 0x00B4			Register Name: UART_DBG_DLH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	DEBUG DLH

17.10.7.22 0x00C0 UART RS485 Control Configure and Status Register (Default Value: 0x0000_0000)

Offset: 0x00C0			Register Name: UART_485_CTL
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	AAD_BROADCAST_MODE AAD Mode Receive address detect broadcast mode 0: AAD broadcast mode 1: Not aad broadcast mode(when use this mode need set UART_485_CTL[1:0] is 01 then set UART_485_CTL[8] is 1)
7	R/W	0x0	ADD_BIT_POLARITY Address bit polarity 0: The address bit is 1 representative the data is address 1: The address bit is 0 representative the data is address
6	R/W1C	0x0	AAD_ADDR_DETECTED AAD Mode Receive address detected 0: Receive address is different with set 1: Receive address is same with set
5	R/W1C	0x0	ADDRESS_DETECTED RS485 Address detect 0: No address detected 1: Address detected

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Offset: 0x00C0			Register Name: UART_485_CTL
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	DUPLEX RS485 Transfer mode 0: Full duplex 1: Half duplex
3:2	/	/	/
1:0	R/W	0x0	RCM RS485 Receive mode This is used to choose the RS485 Receive mode 00: RS_9BITM 01: AAD In RS-9bit mode, addresses and data are not distinguished. All data and addresses received by RX are written into RX FIFO. In AAD mode, addresses and data are distinguished. The data and addresses received by RX are filtered according to the AAD configuration.

17.10.7.23 0x00C4 UART RS485 Address Match Register (Default Value: 0x0000_0000)

Offset: 0x00C4			Register Name: UART_RS485_ADDR_MATCH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	AAD_RECEIVE _ADDR AAD Mode Receive address

17.10.7.24 0x00C8 UART RS485 Bus Idle Check Register (Default Value: 0x0000_0000)

Offset: 0x00C8			Register Name: BUS_IDLE_CHECK
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	BUS_IDLE_CHK_EN 1: Enable bus idle check function 0: Disable bus idle check function
6	R	0x0	BUS_STATUS The flag of bus status 1: Busy 0: Idle
5:0	R	0x0	ADJ_TIME How long the bus is idle. The unit is 8*16*Tbit

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17.10.7.25 0x00CC UART TX Delay Register (Default Value: 0x0000_0000)

Offset: 0x00CC			Register Name: TX_DELAY
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	DLY The delay time between the last stop bit and the next start bit. The unit is 16*Tbit. It is use to control the space between tow bytes in TX.

17.10.7.26 0x00D4 UART FIFO Test Mode Register (Default Value: 0x0000_0000)

Offset: 0x00D4			Register Name: FIFO_TM
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	FTM FIFO Test Mode 0: Normal mode 1: FIFO test mode

17.10.7.27 0x00D8 UART Read TX FIFO Register (Default Value: 0x0000_0000)

Offset: 0x00D8			Register Name: READ_TX_FIFO
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:0	R	0x0	R_TXFIFO Read TX FIFO In FIFO test mode, reading the address could read the data from the txfifo.

17.10.7.28 0x00DC UART Write RX FIFO Register (Default Value: 0x0000_0000)

Offset: 0x00DC			Register Name: WRITE_RX_FIFO
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	W	0x0	W_RXFIFO Write RX FIFO In FIFO test mode, writing the address could write the data to the rxfifo.

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17.10.7.29 0x00E0 UART Read RX FIFO Register (Default Value: 0x0000_0000)

Offset: 0x00E0			Register Name: READ_RX_FIFO
Bit	Read/Write	Default/Hex	Description
31:11	/	/	/
10:0	R	0x0	R_RXFIFO Read RX FIFO In FIFO test mode, read the address could read the data from the rxfifo.

17.10.7.30 0x00E4 UART RS485 RTSN Pull Down Delay Time Register (Default Value: 0x0000_0000)

Offset: 0x00E4			Register Name: RTSN_RST_DLY_TIME
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	RTSN_RST_DLYT_TIME_ENABLE Rts_n reset delay time enable 0: Not enable 1: Enable
7:0	R/W	0x0	RTSN_RST_DLYT Rts_n Reset Delay Time Rts_n delay number of Tbit to reset in RS485 auto flow control.

17.10.7.31 0x00F0 UART FIFO Clock Control Register (Default Value: 0x0000_0003)

Offset: 0x00F0			Register Name: UART_FCC
Bit	Read/Write	Default/Hex	Description
31:8	R	0x0	FIFO Depth Indicates the depth of TX/RX FIFO
7:3	/	/	/
2	R/W	0x0	RXFIFO Clock Mode 0: Sync mode, R/W clocks use APB clock 1: Async mode, write clock uses APB clock, read clock uses AHB clock
1	R/W	0x1	TX FIFO Clock Enable 0: Clock disable 1: Clock enable
0	R/W	0x1	RX FIFO Clock Enable 0: Clock disable 1: Clock enable

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17.11 SPI

17.11.1 Overview

The Serial Peripheral Interface (SPI) is a full-duplex, synchronous, four-wire serial communication interface between a CPU and SPI-compliant external devices. It can work in master mode and slave mode.

The SPI has the following features:

- Up to 3 SPI controllers: SPI0, SPI2, SPI3
- Support multiple SPI modes
  - Master mode and slave mode for standard SPI
  - Master mode for Dual-Output/Dual-Input SPI and Dual I/O SPI
  - Master mode for Quad-Output/Quad-Input SPI
  - Master mode for 3-wire SPI, with programmable serial data frame length of 1 bit to 32 bits
  - Camera slave mode for Single-Output/Single-Input SPI (only SPI2 and SPI3)
  - Camera slave mode for Dual-Input SPI (only SPI2 and SPI3)
  - Camera slave mode for Quad-Input SPI (only SPI2 and SPI3)
  - Ordinary slave mode for Single-Output/Single-Input SPI
- Master mode supports the maximum clock frequency: 100 MHz
- Camera mode supports the maximum clock frequency: 50 MHz
- Support TX/RX DMA Slave interface
- Support Data Sample Mode for master mode and both two slave modes
  - Mode0, Mode1, Mode2 and Mode3
- Supports control signal configuration for master mode and both two slave modes
  - Two chip selects for SPI0 and three chip selects for SPI2, SPI3 to support multiple peripherals
  - Polarity and phase of the Chip Select (SPI_CS) and SPI Clock (SPI_CLK) are configurable

NOTE

This chapter only describes SPI0, SPI2, and SPI3. For detailed information of SPI1 (supports SPI mode and DBI mode), please refer to Section 17.12 SPI_DBI.

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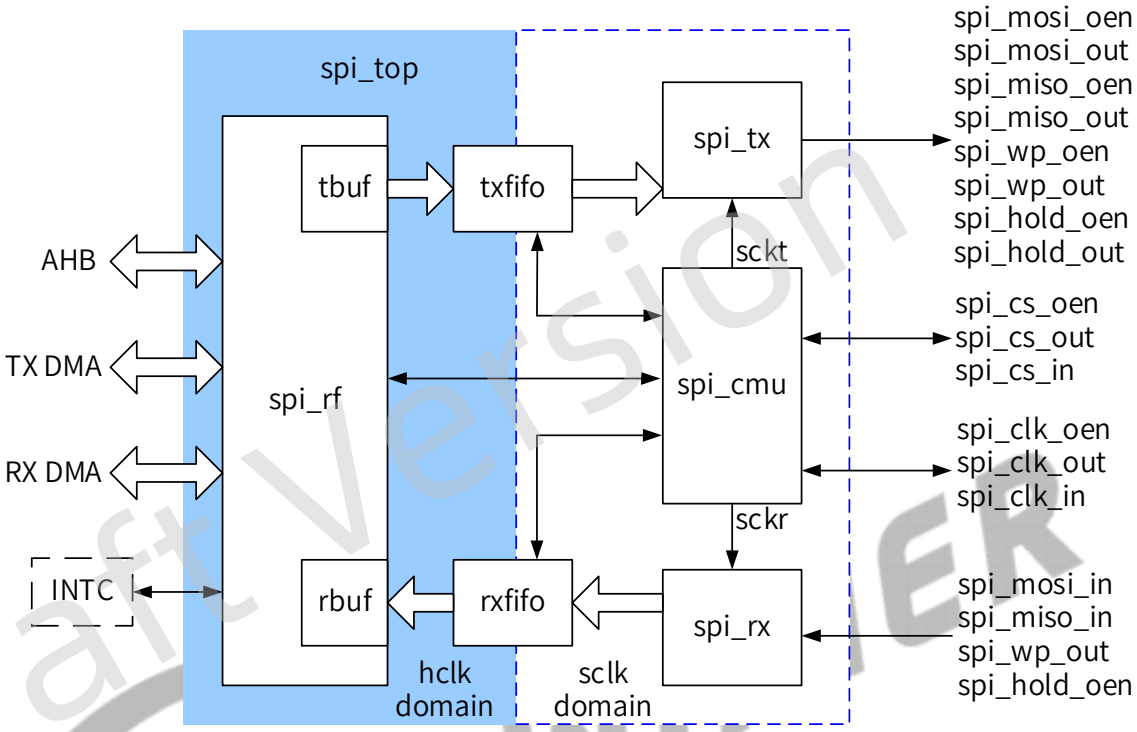
17.11.2 Block Diagram

The following figure shows a block diagram of the SPI.

Figure 17-54 SPI Block Diagram



SPI contains the following sub-blocks:

Table 17-57 SPI Sub-blocks

Sub-block	Description
spi_rf	Responsible for implementing the internal register, interrupt, and DMA Request.
spi_tbuf	The data length transmitted from AHB to TXFIFO is converted into 8 bits, then the data is written into the RXFIFO.
spi_rbuf	The block is used to convert the RXFIFO data into the reading data length of AHB.
txfifo, rxfifo	The data transmitted from the SPI to the external serial device is written into the TXFIFO; the data received from the external serial device into SPI is pushed into the RXFIFO.
spi_cmu	Responsible for implementing SPI bus clock, chip select, internal sample, and the generation of transfer clock.
spi_tx	Responsible for implementing SPI data transfer, the interface of the internal TXFIFO, and status register.
spi_rx	Responsible for implementing SPI data receive, the interface of the internal RXFIFO, and status register.

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17.11.3 External Signals

The following table describes the external signals of SPI. The MOSI and MISO are bidirectional I/O, when SPI is as a master device, the CLK and CS are the output pin; when SPI is as a slave device, the CLK and CS are the input pin. When using SPI, the corresponding PADs are selected as SPI function via Section 17.9 GPIO.

Table 17-58 SPI External Signals

Signal Name	Description	Type
SPI0-CS0	SPI0 Chip Select 0 (Active Low)	I/O
SPI0-CS1	SPI0 Chip Select 1 (Active Low)	I/O
SPI0-CLK	SPI0 Clock	I/O
SPI0-MOSI	SPI0 Master Data Out, Slave Data In	I/O
SPI0-MISO	SPI0 Master Data In, Slave Data Out	I/O
SPI0-WP	SPI0 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI0-HOLD	SPI0 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI2-CS0	SPI2 Chip Select 0 (Active Low)	I/O
SPI2-CS1	SPI2 Chip Select 1 (Active Low)	I/O
SPI2-CS2	SPI2 Chip Select 2 (Active Low)	I/O
SPI2-CLK	SPI2 Clock	I/O
SPI2-MOSI	SPI2 Master Data Out, Slave Data In	I/O
SPI2-MISO	SPI2 Master Data In, Slave Data Out	I/O
SPI2-WP	SPI2 Write Protection (Active Low)/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI2-HOLD	SPI2 Hold Signal/ Serial Data Input and Output for Quad Input or Quad Output	I/O
SPI3-CS0	SPI3 Chip Select 0 (Active Low)	I/O
SPI3-CS1	SPI3 Chip Select 1 (Active Low)	I/O
SPI3-CS2	SPI3 Chip Select 2 (Active Low)	I/O
SPI3-CLK	SPI3 Clock	I/O
SPI3-MOSI	SPI3 Master Data Out, Slave Data In	I/O
SPI3-MISO	SPI3 Master Data In, Slave Data Out	I/O

17.11.4 Functional Description

17.11.4.1 SPI Master and Slave Mode

The SPI controller can be configured to a master or slave device. The master mode is selected by setting the MODE bit (SPI_GCR [1]); the slave mode is selected by clearing the MODE bit.

In master mode, the SPI_CLK is generated and transmitted to the external device, and the data from the TX FIFO is transmitted on the MOSI pin, the data from the slave is received on the MISO pin and sent to RX FIFO. The Chip Select (SPI_CS) is an active low signal, and it must be set low

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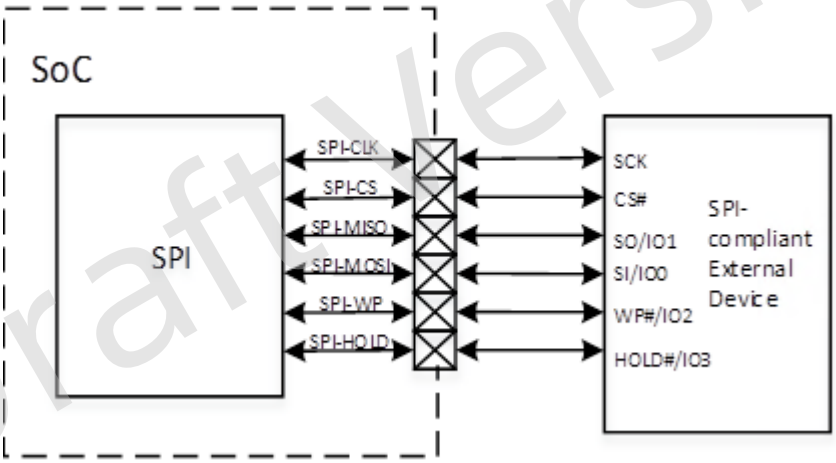
before the data are transmitted or received. The SPI_CS can be selected the auto control mode or the software manual control mode. When using auto control, the SS_OWNER (SPI_TCR [6]) must be cleared (default value is 0); when using manual control, the SS_OWNER must be set. And the level of SPI_CS is controlled by SS_LEVEL (SPI_TCR [7]).

In slave mode, after the software selects the MODE bit (SPI_GCR [1]) to '0', it waits for master initiate a transaction. When the master asserts SPI_CS, and SPI_CLK is transmitted to the slave, the slave data is transmitted from TX FIFO on the MISO pin, and the data from the MOSI pin is received in RX FIFO.

The following figure shows the application block diagram when the SPI master device is connected to a slave device.

Figure 17-55 SPI Application Block Diagram



17.11.4.2 SPI Transmit Format

The SPI supports 4 different formats for data transfer. The software can select one of the four modes in which the SPI works by setting the bit1 (Polarity) and bit0 (Phase) of SPI_TCR (Offset: 0x0008). The SPI controller master uses the SPI_CLK signal to transfer data in and out of the shift register. Data is clocked using any one of four programmable clock phase and polarity combinations.

The CPOL (SPI_TCR [1]) defines the polarity of the clock signal (SPI_CLK). The SPI_CLK is a high level when CPOL is '1' and it is a low level when CPOL is '0'. The CPHA (SPI_TCR [0]) decides whether the leading edge of SPI_CLK is used to setup or sample data. The leading edge is used to setup data when CPHA is '1', and sample data when CPHA is '0'. The following table lists the four modes.

Table 17-59 SPI Transmit Format

Mode	Polarity (CPOL)	Phase (CPHA)	Leading Edge	Trailing Edge
Mode0	0	0	Sample on the rising edge	Setup on the falling edge
Mode1	0	1	Setup on the rising edge	Sample on the falling edge
Mode2	1	0	Sample on the falling edge	Setup on the rising edge

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Mode	Polarity (CPOL)	Phase (CPHA)	Leading Edge	Trailing Edge
Mode3	1	1	Setup on the falling edge	Sample on the rising edge

The following figures describe four waveforms for SPI_CLK.

Figure 17-56 SPI Phase 0 Timing Diagram

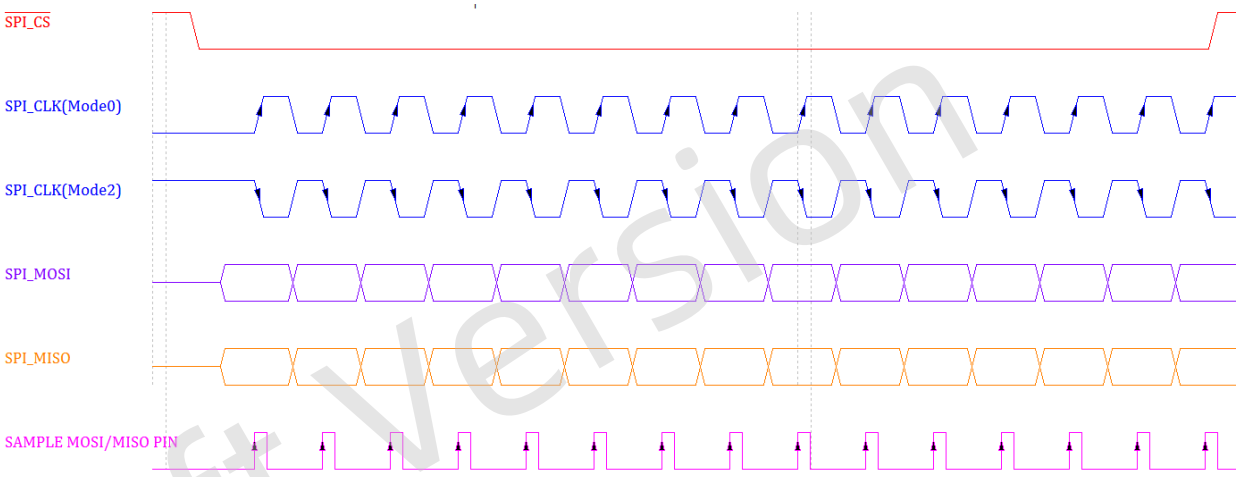
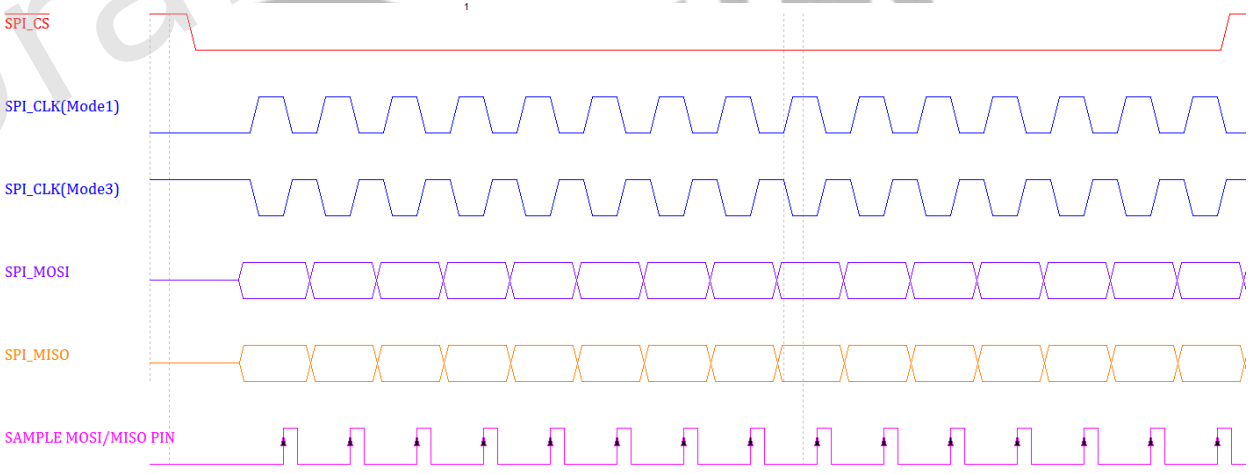


Figure 17-57 SPI Phase 1 Timing Diagram



17.11.4.3 SPI 3-Wire Mode

The SPI 3-wire mode is only valid when the SPI controller work in master mode, and is selected when the Work Mode Select bit (SPI_BATCR [1:0]) is equal to 0x2. And in the 3-wire mode, the input data and the output data use the same single data line. The following figure describes the 3-wire mode.

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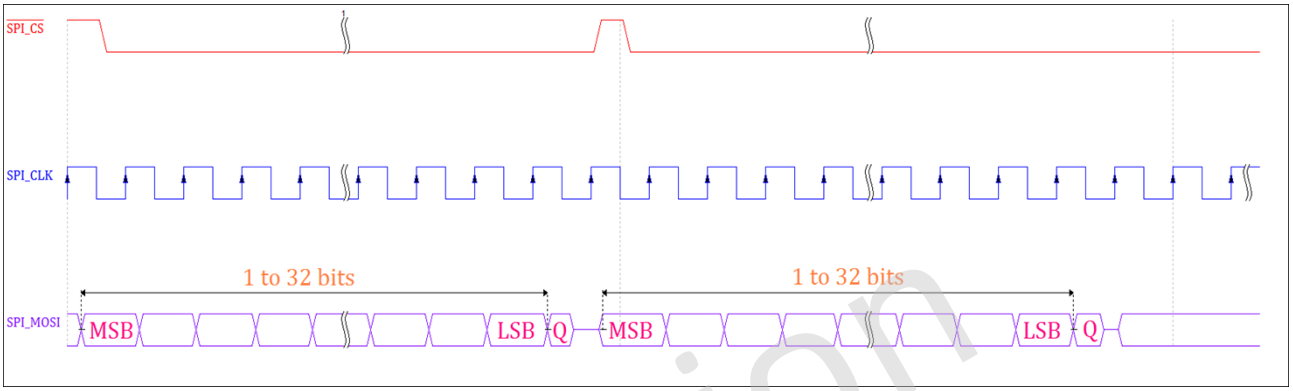
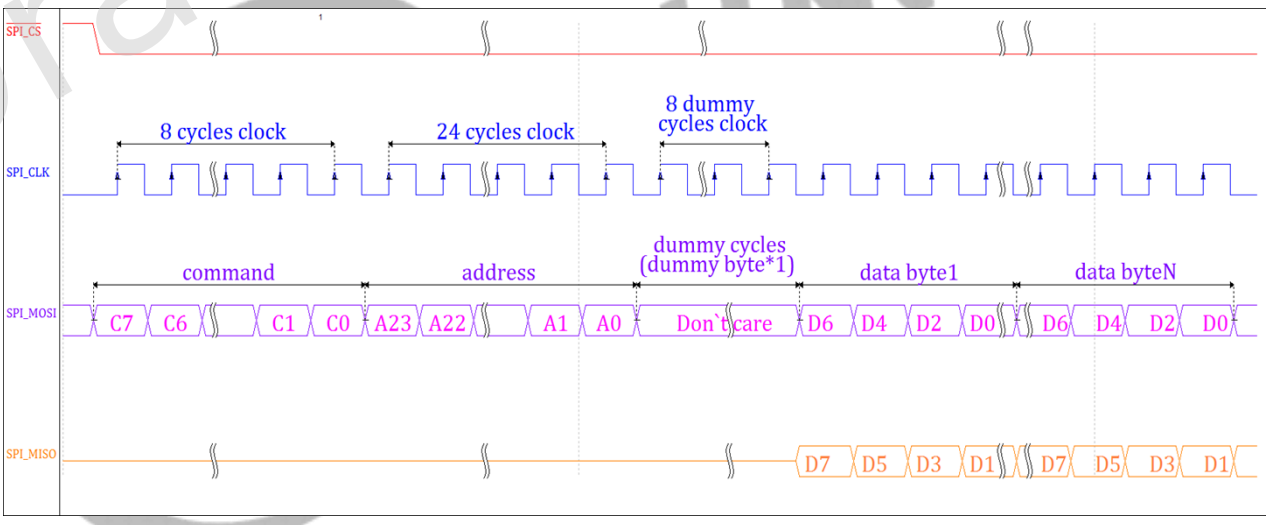
Figure 17-58 SPI 3-Wire Mode



17.11.4.4 SPI Dual Mode

The dual read mode (SPI x2) is selected when the DRM is set in SPI_BCC (Offset: 0x0038) [28]. Using the dual mode allows data to be transferred to or from the device at double the rate of standard single mode, the data can be read at fast speed using two data bits (MOSI and MISO) at a time. The following figure describes the dual-input/dual-output SPI and the dual I/O SPI.

Figure 17-59 SPI Dual-Input/Dual-Output Mode



In the dual-input/dual-output SPI mode, the command, address, and the dummy bytes output in a unit of a single bit in serial mode through the SPI_MOSI line, only the data bytes are output (write) and input (read) in a unit of dual bits through the SPI_MOSI and SPI_MISO.

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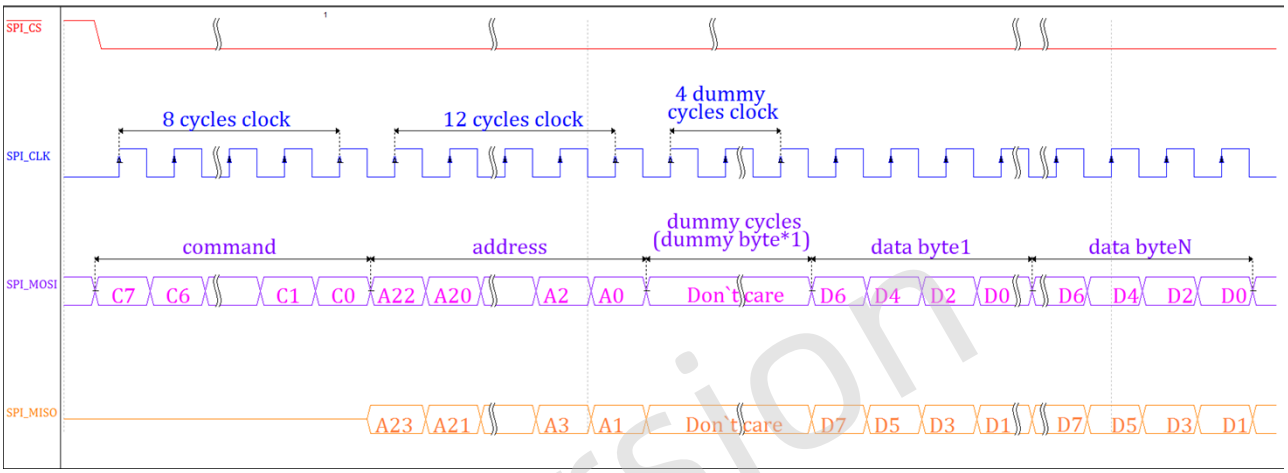
Figure 17-60 SPI Dual I/O Mode

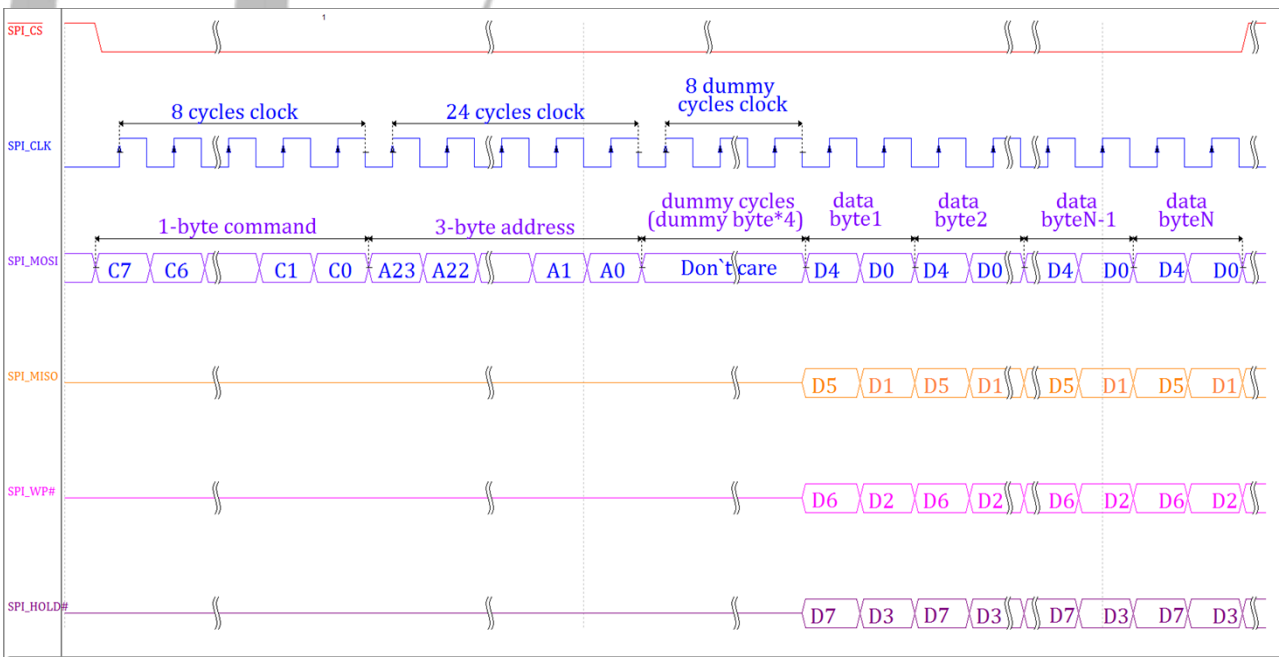


In the dual I/O SPI mode, only the command bytes output in a unit of a single bit in serial mode through the SPI_MOSI line. The address bytes and the dummy bytes output in a unit of dual bits through the SPI_MOSI and SPI_MISO. And the data bytes output (write) and input (read) in a unit of dual bits through the SPI_MOSI and SPI_MISO.

17.11.4.5 SPI Quad Mode

The quad read mode (SPI x4) is selected when the Quad_EN is set in SPI_BCC (Offset: 0x0038) [29]. Using the quad mode allows data to be transferred to or from the device at 4 times the rate of standard single mode, the data can be read at fast speed using four data bits (MOSI, MISO, IO2 (WP#) and IO3 (HOLD#)) at the same time. The following figure describes the quad-input/quad-output SPI.

Figure 17-61 SPI Quad-Input/Quad-Output Mode



In the quad-input/quad-output SPI mode, the command, address, and the dummy bytes output in a unit of a single bit in serial mode through SPI_MOSI line. Only the data bytes output (write)

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and input (read) in a unit of quad bits through the SPI_MOSI, SPI_MISO, SPI_WP#, and SPI_HOLD#.

17.11.4.6 SPI Error Conditions

If any error conditions occur, the hardware will set the corresponding status bits in the SPI_ISR and stop the transfer. For the SPI controller, the following error scenarios can happen.

- TX_FIFO Underrun

The TX_FIFO underrun happens when the CPU/DMA reads data from the empty TX FIFO. In the case, the SPI controller will end the transaction and flag the error bit along with the TF_UDF bit in the SPI_ISR. The SPI controller will generate an interrupt if interrupts are enabled. The software has to clear the error bit and the TF_UDF bit. To start a new transaction, the software has to reset the FIFO by writing to the SRST (soft reset) bit in the SPI_GCR.

- TX_FIFO Overflow

The TX_FIFO overflow happens when the CPU/DMA writes data into the full TX FIFO. In the case, the SPI controller will end the transaction and flag the error bit along with the TF_OVF bit in the SPI_ISR. The SPI controller will generate an interrupt if interrupts are enabled. The software has to clear the error bit and the TF_OVF bit. To start a new transaction, the software has to reset the FIFO by writing to the SRST (soft reset) bit in the SPI_GCR.

- RX_FIFO Underrun

The RX_FIFO underrun happens when the CPU/DMA reads data from RX FIFO when it is empty. In the case, the SPI controller will end the transaction and flag the error bit along with the RF_UDF bit in the SPI_ISR. The SPI controller will generate an interrupt if interrupts are enabled. The software has to clear the error bit and the RF_UDF bit. To start a new transaction, the software has to reset the FIFO by writing to the SRST (soft reset) bit in the SPI_GCR.

- RX_FIFO Overflow

The RX_FIFO overflow happens when the CPU/DMA writes data into the RX FIFO when it is full. In the case, the SPI controller will end the transaction and flag the error bit along with the RF_OVF bit in the SPI_ISR. The SPI controller will generate an interrupt if interrupts are enabled. The software has to clear the error bit and the RF_OVF bit. To start a new transaction, the software has to reset the FIFO by writing to the SRST (soft reset) bit in the SPI_GCR.

17.11.4.7 SPI Clock Frequency

The SPI clock source frequency can be configured according to CCU, which is assumed to be frequency A MHz. 2 stage division coefficient N & M should be set.

The SPI working frequency is $\frac{A}{(N+1)\times(M+1)}$ MHz.

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17.11.5 Programming Guidelines

17.11.5.1 Writing/Reading Data Process

The SPI transfers serial data between the processor and peripheral devices. The CPU mode and DMA mode are the two main operational modes for SPI. For each SPI, the data is simultaneously transmitted (shifted out serially) and received (shifted in serially). The SPI has 2 channels, including the TX channel and RX channel. The TX channel has the path from TX FIFO to peripheral devices. The RX channel has the path from peripheral devices to RX FIFO.

Write Data: The CPU or DMA must write data on the SPI_TXD (Offset: 0x0200), the data on the register are automatically moved to TX FIFO.

Read Data: To read data from RX FIFO, the CPU or DMA must access the SPI_RXD (Offset: 0x0300) and the data are automatically sent to the SPI_RXD (Offset: 0x0300).

In CPU or DMA mode, the SPI sends a completed interrupt (SPI_ISR[TC]) to the processor after each transmission is complete.

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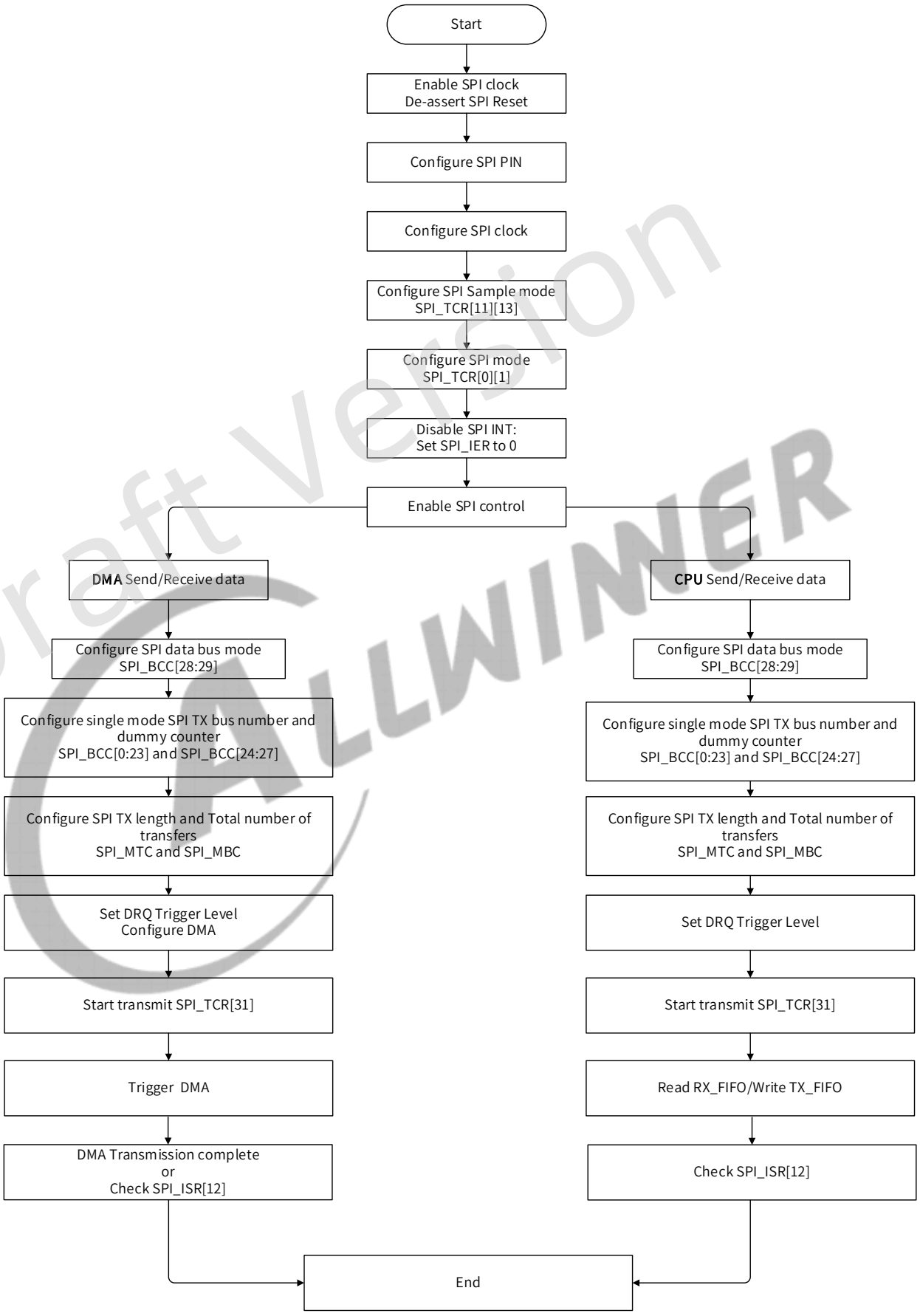
Figure 17-62 Data Transfer in Query Mode(including CPU and DMA mode)



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17.11.5.2 Master Transmit and Receive Burst

In the SPI Master mode, the transmit and receive bursts (byte in unit) are configured before the SPI transfers serial data between the processor and external device. The transmit burst write in MWTC (bit[23:0]) of SPI_MTC. The transmit burst in single mode before automatically sending dummy burst write in STC (bit[23:0]) of SPI_BCC. For dummy data, SPI controller can automatically sent before receive by writing DBC (bit[27:24]) in SPI_BCC. If users don't use SPI controller to send dummy data automatically, then the dummy bursts are used as the transmit counters to write together in MWTC (bit[23:0]) of SPI_MTC. In Master mode, the total burst numbers write in MBC (bit[23:0]) of SPI Master Burst Counter Register. When all transmit burst and receive burst are transferred, SPI controller will send a completed interrupt, at the same time, SPI controller will clear DBC, MWTC and MBC.

17.11.5.3 SPI Run Clock and Sample Mode

The SPI Controller runs at 24KHz-100MHz at its interface to external SPI devices. The internal SPI Clock should run at the same frequency as the outgoing clock in master mode. The SPI Clock is selected different clock sources, and SPI must configure different work mode. There are three work mode: normal sample mode, delay half cycle sample mode, delay one cycle sample mode. Delay half cycle sample mode is the default mode of SPI controller. When SPI runs at 48MHz or below 48MHz, SPI can work at normal sample mode or delay half cycle sample mode. When SPI runs over 48MHz, set SDC in SPI Transfer Control Register to '1' to make the internal read sample point with a delay of half cycle of SPI_CLK, which is used in high-speed read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave device. The different configuration of new/old SPI sample mode shows in the following tables.

Table 17-60 SPI Old Sample Mode

SPI Sample Mode	SDM(bit13)	SDC(bit11)	Run Clock
normal sample	1	0	<=24MHz
delay half cycle sample	0	0	<=40MHz
delay one cycle sample	0	1	>=80MHz

NOTE

The remaining spectrum is not recommended. Because when the output delay of SPI (refer to the datasheet of the manufactures for the specific delay time) is the same with the half-cycle time of SPI working clock, the variable edge of the output data for the device bumps into the clock sampling edge of the controller, so setting 1 cycle of sampling delay would cause stability problem.

Table 17-61 SPI New Sample Mode

SPI Sample Mode	SDM(bit13)	SDC(bit11)	SDC1(bit15)
normal sample	1	0	0
delay half cycle sample	0	0	0

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SPI Sample Mode	SDM(bit13)	SDC(bit11)	SDC1(bit15)
delay one cycle sample	0	1	0
delay 1.5 cycle sample	1	1	0
delay 2 cycle sample	1	0	1
delay 2.5 cycle sample	0	0	1
delay 3 cycle sample	0	1	1

17.11.5.4 SPI Camera Configuration

Vsync Signal Feature

Step 1 Configure SPI_GCR [10:8] to select SPI slave RX mode and SINGLE/DUAL/QUAD mode.

Step 2 Configure SPI_TCR [17:16] to enable Vsync feature and select Vsync valid edge.

Step 3 Configure other SPI feature such as SPI Mode and SPI clock polarity.

Step 4 Configure SPI_GCR [5] to 1.

Step 5 Configure SPI_SVCNR according to camera feature.

Step 6 Configure SPI_GCR [0] to start SPI feature. Then SPI will start receiving data starting from every valid Vsync input signal and ends until enough data is received.

Framehead Detect Feature

Step 1 Configure SPI_GCR [10:8] to select SPI slave RX mode and SINGLE/DUAL/QUAD mode.

Step 2 Configure SPI_TCR [18] to enable Framehead Detect feature.

Step 3 Configure other SPI feature such as SPI Mode, SPI clock polarity, etc..

Step 4 Configure SPI_GCR [5] to 1.

Step 5 Configure SPI_SVCNR [31:24], SPI_SFHNR, SPI_SFHLR, SPI_SFHHR according to camera feature.

Step 6 Configure SPI_GCR [0] to start SPI feature. Then SPI will start receiving data since framehead is detected.

Idle Wait Feature

Step 1 Configure SPI_GCR [10:8] to select SPI slave RX mode and SINGLE/DUAL/QUAD mode.

Step 2 Configure SPI_TCR [19] to enable Idle Wait feature.

Step 3 Configure other SPI feature such as SPI Mode, SPI clock polarity, etc..

Step 4 Configure SPI_GCR [5] to 1.

Step 5 Configure SPI_SVCNR [31:24], SPI_SIWVER, SPI_SIWVRR according to camera feature.

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Step 6 Configure SPI_GCR [0] to start SPI feature. Then SPI will detect SPI_CLK input signal. When the input clock signal is found stable for a period of time longer than (SIWVE+SIWVR), then it will start receiving data.

17.11.6 Register List

Module Name	Base Address
SPI0	0x04025000
SPI2	0x04027000
SPI3	0x04028000

NOTE

Only SPI_TXD and SPI_RXD support byte/half word/word access. Other registers only supports word access.

Register Name	Offset	Description
SPI_GCR	0x0004	SPI Global Control Register
SPI_TCR	0x0008	SPI Transfer Control Register
SPI_IER	0x0010	SPI Interrupt Control Register
SPI_ISR	0x0014	SPI Interrupt Status Register
SPI_FCR	0x0018	SPI FIFO Control Register
SPI_FSR	0x001C	SPI FIFO Status Register
SPI_WCR	0x0020	SPI Wait Clock Counter Register
SPI_MBC	0x0030	SPI Master Burst Counter Register
SPI_MTC	0x0034	SPI Master Transmit Counter Register
SPI_BCC	0x0038	SPI Master Burst Control Counter Register
SPI_BATCR	0x0040	SPI Bit-Aligned Transfer Configure Register
SPI_BA_CCR	0x0044	SPI Bit-Aligned CLOCK Configuration Register
SPI_TBR	0x0048	SPI TX Bit Register
SPI_RBR	0x004C	SPI RX Bit Register
SPI_NDMA_MODE_CTL	0x0088	SPI Normal DMA Mode Control Register
SPI_SVCNR	0x0160	SPI Slave Vsync Cycle Number Register
SPI_SFHNR	0x0164	SPI Slave Frame Head Number Register
SPI_SFHLR	0x0168	SPI Slave Frame Head Low Register
SPI_SFHHR	0x016C	SPI Slave Frame Head High Register
SPI_SFHLRR	0x0170	SPI Slave Frame Head Low Receive Register
SPI_SFHHRR	0x0174	SPI Slave Frame Head High Receive Register
SPI_SIWVER	0x0178	SPI Slave Idle Wait Value Enable Register
SPI_SIWVRR	0x017C	SPI Slave Idle Wait Value Reset Register
SPI_TXD	0x0200	SPI TX Data Register
SPI_RXD	0x0300	SPI RX Data Register
SPI_BSR	0x0400	SPI BUF Status Register

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17.11.7 Register Description

17.11.7.1 0x0004 SPI Global Control Register (Default Value: 0x0000_0080)

Offset: 0x0004			Register Name: SPI_GCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	SRST Soft reset Write '1' to this bit will clear the SPI controller, and auto clear to '0' when reset operation completes Write '0' has no effect.
30:11	/	/	/
10:9	R/W	0x0	SLV_MODE Slave Mode Wire Mode 0: Single 1: Dual 2: Quad
8	R/W	0x0	SLV_DIR Slave Mode Direction 0: Slave RX 1: Reserved
7	R/W	0x1	TP_EN Transmit Pause Enable In master mode, it is used to control transmit state machine to stop smart burst sending when RX FIFO is full. 1: Stop transmit data when RXFIFO is full 0: Normal operation. Ignore RXFIFO status Note: When XCH=1, do not write to this bit.
6	/	/	/
5	R/W	0x0	SLAVE_MODE_SEL Slave Mode Sample Timing Mode Select 0: Old mode of Sample Timing 1: New mode of Sample Timing Note: Cannot be written when slave is transferring. For Dual/Quad RX, old mode is not valid.
4:3	/	/	/
2	R/W	0x0	MODE_SEL Sample Timing Mode Select 0: Old mode of Sample Timing 1: New mode of Sample Timing Note: When XCH=1, do not write to this bit.

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Offset: 0x0004			Register Name: SPI_GCR
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	MODE SPI Function Mode Select 0: Slave Mode 1: Master Mode Note: When XCH=1, do not write to this bit.
0	R/W	0x0	EN SPI Module Enable Control 0: Disable 1: Enable Note: After transforming from bit_mode to byte_mode, it must Enable the SPI Module again.

17.11.7.2 0x0008 SPI Transfer Control Register (Default Value: 0x0000_0087)

Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	XCH Exchange Burst In master mode it is used to start SPI burst 0: Idle 1: Initiates exchange. Write "1" to this bit will start the SPI burst, and will auto clear after finishing the bursts transfer specified by BC. Write "1" to SRST will also clear this bit. Write '0' to this bit has no effect. Note: When XCH=1, do not write to this bit.
30:20	/	/	/
19	R/W	0x0	SIWE Slave Idle Wait Enable 0: Disable 1: Enable In slave mode, VIE and SFHE should be disabled when SIWE is enabled.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	SFHE Slave Mode frame head auto distinguish feature enable 0: Disable 1: Enable In slave mode, VIE and SIWE should be disabled when SFHE is enabled. When frame head auto distinguishing, the frame head must be written in the way of MSB first.
17	R/W	0x0	VIE Vsync Input Enable 0: Disable 1: Enable In slave mode, SPI_CS input signal can be configured into SPI_CS or Vsync.
16	R/W	0x0	VIS Vsync Input Select 0: Vsync negative edge 1: Vsync positive edge
15	R/W	0x0	SDC1 Master Sample Data Control register1 Set this bit to '1' to make the internal read sample point with a delay of half cycle of SPI_CLK. It is used in high speed read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave. 0: Normal operation, do not delay internal read sample point 1: Delay internal read sample point Note: When XCH=1, do not write to this bit.
14	R/W	0x0	SDDM Sending Data Delay Mode 0: Normal sending 1: Delay sending Set the bit to"1" to make the data that should be sent with a delay of half cycle of SPI_CLK in dual I/O mode for SPI mode 0.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
13	R/W	0x0	SDM Sample Data Mode 1: Normal Sample Mode 0: Delay Sample Mode In Master Normal Sample Mode, SPI master samples the data at the correct edge for each SPI mode; In Master Delay Sample Mode, SPI master samples data at the edge that is half cycle delayed by the correct edge defined in respective SPI mode. In Slave Mode, it should be configured as the opposite value of CPOL.
12	R/W	0x0	FBS First Transmit Bit Select 0: MSB first 1: LSB first Note: When XCH=1, do not write to this bit.
11	R/W	0x0	SDC Master Sample Data Control Set this bit to '1' to make the internal read sample point with a delay of half cycle of SPI_CLK. It is used in high speed read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave. 0: Normal operation, do not delay internal read sample point 1: Delay internal read sample point Note: When XCH=1, do not write to this bit.
10	R/W	0x0	RPSM Rapids mode select Select Rapids mode for high speed write. 0: Normal write mode 1: Rapids write mode Note: When XCH=1, do not write to this bit.
9	R/W	0x0	DDB Dummy Burst Type 0: The bit value of dummy SPI burst is zero 1: The bit value of dummy SPI burst is one Note: When XCH=1, do not write to this bit.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	DHB Discard Hash Burst In master mode it controls whether discarding unused SPI bursts 0: Receiving all SPI bursts in BC period 1: Discard unused SPI bursts, only fetching the SPI bursts during dummy burst period. The bursts number is specified by TC. Note: When XCH=1, do not write to this bit.
7	R/W	0x1	SS_LEVEL When control CS signal manually (SPI_CTRL_REG.SS_CTRL==1), set this bit to '1' or '0' to control the level of CS signal. 0: Set CS to low 1: Set CS to high Note: When XCH=1, do not write to this bit.
6	R/W	0x0	SS_OWNER CS Output Owner Select Usually, controller sends CS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of CS signal. 0: SPI controller 1: Software Note: When XCH=1, do not write to this bit.
5:4	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPI_CS0 will be asserted 01: SPI_CS1 will be asserted 10: SPI_CS2 will be asserted 11: SPI_CS3 will be asserted (not used) Note: When XCH=1, do not write to this bit.
3	R/W	0x0	SSCTL In master mode, this bit selects the output wave form for the SPI_CSx signal. Only valid when SS_OWNER = 0. 0: SPI_CSx remains asserted between SPI bursts 1: Negate SPI_CSx between SPI bursts Note: When XCH=1, do not write to this bit.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
2	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: When XCH=1, do not write to this bit.
1	R/W	0x1	CPOL SPI Clock Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: When XCH=1, do not write to this bit.
0	R/W	0x1	CPHA SPI Clock/Data Phase Control 0: Phase 0 (Leading edge for sample data) 1: Phase 1 (Leading edge for setup data) Note: When XCH=1, do not write to this bit.

17.11.7.3 0x0010 SPI Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: SPI_IER
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W	0x0	SLV_VSYNC_ERR_EN Slave Vsync Error Enable 0: Disable 1: Enable
13	R/W	0x0	SS_INT_EN SSI Interrupt Enable Chip Select Signal (CSx) from valid state to invalid state 0: Disable 1: Enable
12	R/W	0x0	TC_INT_EN Transfer Completed Interrupt Enable 0: Disable 1: Enable
11	R/W	0x0	TF_UDR_INT_EN TXFIFO under run Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0010			Register Name: SPI_IER
Bit	Read/Write	Default/Hex	Description
10	R/W	0x0	TF_OVF_INT_EN TX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
9	R/W	0x0	RF_UDR_INT_EN RXFIFO under run Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	RF_OVF_INT_EN RX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
7	/	/	/
6	R/W	0x0	TF_FUL_INT_EN TX FIFO Full Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	TX_EMP_INT_EN TX FIFO Empty Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	TX_ERQ_INT_EN TX FIFO Empty Request Interrupt Enable 0: Disable 1: Enable
3	/	/	/
2	R/W	0x0	RF_FUL_INT_EN RX Full Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	RX_EMP_INT_EN RX Empty Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	RF_RDY_INT_EN RX Ready Request Interrupt Enable 0: Disable 1: Enable

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17.11.7.4 0x0014 SPI Interrupt Status Register (Default Value: 0x0000_0032)

Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1C	0x0	SVEI Slave Vsync Error Interrupt When in Slave Mode and Vsync is valid, it indicates that there is no enough input clock cycles between 2 Vsync edge. Writing 1 to this bit clears it.
13	R/W1C	0x0	SSI CS Invalid Interrupt When SSI is 1, it indicates that SS has changed from valid state to invalid state. Writing 1 to this bit clears it.
12	R/W1C	0x0	TC Transfer Completed In master mode, it indicates that all bursts specified by BC has been exchanged. In other condition, when set, this bit indicates that all the data in TXFIFO has been loaded in the Shift register, and the Shift register has shifted out all the bits. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed
11	R/W1C	0x0	TF_UDF TXFIFO underrun This bit is set when if the TXFIFO is underrun. Writing 1 to this bit clears it. 0: TXFIFO is not underrun 1: TXFIFO is underrun
10	R/W1C	0x0	TF_OVF TXFIFO Overflow This bit is set when if the TXFIFO is overflow. Writing 1 to this bit clears it. 0: TXFIFO is not overflow 1: TXFIFO is overflowed
9	R/W1C	0x0	RX_UDF RXFIFO Underrun When set, this bit indicates that RXFIFO has underrun. Writing 1 to this bit clears it.

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Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x0	RX_OVF RXFIFO Overflow When set, this bit indicates that RXFIFO has overflowed. Writing 1 to this bit clears it. 0: RXFIFO is available. 1: RXFIFO has overflowed.
7	/	/	/
6	R/W1C	0x0	TX_FULL TXFIFO Full This bit is set when if the TXFIFO is full. Writing 1 to this bit clears it. 0: TXFIFO is not Full 1: TXFIFO is Full
5	R/W1C	0x1	TX_EMP TXFIFO Empty This bit is set if the TXFIFO is empty. Writing 1 to this bit clears it. 0: TXFIFO contains one or more words. 1: TXFIFO is empty
4	R/W1C	0x1	TX_READY TXFIFO Ready 0: TX_WL > TX_TRIG_LEVEL 1: TX_WL <= TX_TRIG_LEVEL This bit is set any time if TX_WL <= TX_TRIG_LEVEL. Writing “1” to this bit clears it. Where TX_WL is the water level of RXFIFO
3	/	/	/
2	R/W1C	0x0	RX_FULL RX Full This bit is set when both the RXFIFO and the RX buffer are full. Writing 1 to this bit clears it. 0: Not Full 1: Full
1	R/W1C	0x1	RX_EMP RX Empty This bit is set when both the RXFIFO and the RX buffer are empty. Writing 1 to this bit clears it. 0: Not empty 1: Empty

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Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	RX_RDY RX Ready 0: RX_WL <= RX_TRIG_LEVEL 1: RX_WL > RX_TRIG_LEVEL This bit is set any time if RX_WL > RX_TRIG_LEVEL. Writing "1" to this bit clears it. Where RX_WL is the total water level of both the RXFIFO and the RX buffer.

17.11.7.5 0x0018 SPI FIFO Control Register (Default Value: 0x0040_0001)

Offset: 0x0018			Register Name: SPI_FCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	TX_FIFO_RST TX FIFO Reset Write '1' to this bit will reset the control portion of the TX FIFO and auto clear to '0' when completing reset operation, write to '0' has no effect.
30	R/W	0x0	TF_TEST_ENB TX Test Mode Enable 0: Disable 1: Enable In normal mode, TX FIFO can only be read by SPI controller, write '1' to this bit will switch TX FIFO read and write function to AHB bus. This bit is used to test the TX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time.
29:25	/	/	/
24	R/W	0x0	TF_DRQ_EN TX FIFO DMA Request Enable 0: Disable 1: Enable
23:16	R/W	0x40	TX_TRIG_LEVEL TX FIFO Empty Request Trigger Level
15	R/WAC	0x0	RF_RST RXFIFO Reset Write '1' to this bit will reset the control portion of the receiver FIFO, and auto clear to '0' when completing reset operation, write '0' to this bit has no effect.

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Offset: 0x0018			Register Name: SPI_FCR
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	RF_TEST RX Test Mode Enable 0: Disable 1: Enable In normal mode, RX FIFO can only be written by SPI controller, write '1' to this bit will switch RX FIFO read and write function to AHB bus. This bit is used to test the RX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time.
13:9	/	/	/
8	R/W	0x0	RF_DRQ_EN RX FIFO DMA Request Enable 0: Disable 1: Enable
7:0	R/W	0x1	RX_TRIG_LEVEL RX FIFO Ready Request Trigger Level CAUTION: Whether in CPU mode or in DMA mode, trigger level should be set properly that actual data amount reaches trigger level.

17.11.7.6 0x001C SPI FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: SPI_FSR
Bit	Read/Write	Default/Hex	Description
31	R	0x0	TB_WR TX FIFO Write Buffer Write Enable
30:24	/	/	/
23:16	R	0x0	TF_CNT TX FIFO Counter These bits indicate the number of words in TX FIFO 0: 0 byte in TX FIFO 1: 1 byte in TX FIFO ... 64 64 bytes in TX FIFO other: Reserved
15	R	0x0	RB_WR RX FIFO Read Buffer Write Enable
14:8	/	/	/

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Offset: 0x001C			Register Name: SPI_FSR
Bit	Read/Write	Default/Hex	Description
7:0	R	0x0	RF_CNT RX FIFO Counter These bits indicate the number of words in RX FIFO + RX BUFFER 0: 0 byte in RX FIFO & RX BUFFER 1: 1 byte in RX FIFO & RX BUFFER ... 64: 64 bytes in RX FIFO & RX BUFFER ... 192:192bytes in RX FIFO & RX BUFFER other: Reserved

17.11.7.7 0x0020 SPI Wait Clock Counter Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: SPI_WCR
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:16	R/W	0x0	SWC Dual mode direction switch waits clock counter (for master mode only). 0: No wait states inserted n: n SPI_CLK wait states inserted Note: These bits control the number of wait states to be inserted before start dual data transfer in dual SPI mode. The SPI module counts SPI_CLK by SWC for delaying next word data transfer. Note: When XCH=1, do not write to this bit.
15:0	R/W	0x0	WCC Wait Clock Counter (In Master mode) These bits control the number of wait states to be inserted in data transfers. The SPI module counts SPI_CLK by WCC for delaying next word data transfer. 0: No wait states inserted N: N SPI_CLK wait states inserted Note: When XCH=1, do not write to this bit.

17.11.7.8 0x0030 SPI Master Burst Counter Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: SPI_MBC
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0030			Register Name: SPI_MBC
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	MBC Master Burst Counter In master mode, this field specifies the total burst number. 0: 0 burst 1: 1 burst ... N: N bursts Note: • When XCH=1, do not write to this bit. • Total transfer data, include the TXD, RXD and dummy burst.

17.11.7.9 0x0034 SPI Master Transmit Counter Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: SPI_MTC
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	MWTC Master Write Transmit Counter In master mode, this field specifies the burst number that should be sent to TXFIFO before automatically sending dummy burst. For saving bus bandwidth, the dummy burst (all zero bits or all one bits) is sent by SPI Controller automatically. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.

17.11.7.10 0x0038 SPI Master Burst Control Counter Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: SPI_BCC
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/

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Offset: 0x0038			Register Name: SPI_BCC
Bit	Read/Write	Default/Hex	Description
29	R/W	0x0	Quad_EN Master Quad_Mode_EN 0: Quad mode disable 1: Quad mode enable Note: • When XCH=1, do not write to this bit. • Quad mode includes Quad-Input and Quad-Output.
28	R/W	0x0	DRM Master Dual Mode RX Enable 0: RX use single-bit mode 1: RX use dual mode Note: • When XCH=1, do not write to this bit. • It is only valid when Quad_Mode_EN=0.
27:24	R/W	0x0	DBC Master Dummy Burst Counter In master mode, this field specifies the burst number that should be sent before receive in dual SPI mode. The data is don't care by the device. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.
23:0	R/W	0x0	STC Master Single Mode Transmit Counter In master mode, this field specifies the burst number that should be sent in single mode before automatically sending dummy burst. This is the first transmit counter in all bursts. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.

17.11.7.11 0x0040 SPI Bit-Aligned Transfer Configure Register (Default Value: 0x0000_00A0)

Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description

Draft Version

Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	TCE Transfer Control Enable In master mode, it is used to start t1o transfer the serial bit's frame, it is only valid when Work Mode Select==0x2/0x3. 0: Idle 1: Initiates transfer. Write "1" to this bit will start to transfer serial bits' frame (the value comes from the SPI TX Bit Register or SPI RX Bit Register), and will auto clear after the bursts transfer completely. Write '0' to this bit has no effect.
30	R/W	0x0	MSMS Master Sample Standard 0: Standard Sample Mode 1: Delay Sample Mode In Standard Sample Mode, SPI master samples the data at the standard rising edge of CLK for each SPI mode; In Delay Sample Mode, SPI master samples data at the edge that is half cycle delayed by the standard rising edge of CLK defined in respective SPI mode.
29:26	/	/	/
25	R/W1C	0x0	TBC Transfer Bits Completed When set, this bit indicates that the last bit of the serial data frame in SPI TX Bit Register (or SPI RX Bit Register) has been transferred completely. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed Note: It is only valid when Work Mode Select is 0x2/0x3.
24	R/W	0x0	TBC_INT_EN Transfer Bits Completed Interrupt Enable 0: Disable 1: Enable Note: It is only valid when Work Mode Select is 0x2/0x3.
23:22	/	/	/

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Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
21:16	R/W	0x00	RX_FEM_LEN Configure the length of serial data frame(burst) of RX 000000: 0bit 000001: 1bit ... 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select==0x2/0x3. and cannot be written when TCE=1.
15:14	/	/	/
13:8	R/W	0x00	TX_FRM_LEN Configure the length of serial data frame(burst) of TX 000000: 0bit 000001: 1bit ... 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select==0x2/0x3, and cannot be written when TCE=1.
7	R/W	0x1	SS_LEVEL When control CS signal manually, set this bit to '1' or '0' to control the level of CS signal. 0: Set CS to low 1: Set CS to high Note: It is only valid when Work Mode Select==0x2/0x3, and only work in Mode0, cannot be written when TCE=1.

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Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	SS_OWNER CS Output Owner Select Usually, controller sends CS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of CS signal. 0: SPI controller 1: Software Note: It is only valid when Work Mode Select==0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
5	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: It is only valid when Work Mode Select==0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
4	/	/	/
3:2	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPI_CS0 will be asserted 01: SPI_CS1 will be asserted 10: SPI_CS2 will be asserted 11: SPI_CS3 will be asserted (not used) Note: It is only valid when Work Mode Select==0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
1:0	R/W	0x0	WMS Work Mode Select 00: Data frame is byte aligned in Standard SPI, Dual-Output/Dual Input SPI, Dual IO SPI and Quad-Output/Quad-Input SPI. 01: Reserve 10: Data frame is bit aligned in 3-Wire SPI 11: Data frame is bit aligned in Standard SPI

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17.11.7.12 0x0044 SPI Bit-Aligned CLOCK Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: SPI_BA_CCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	CDR_N Clock Divide Rate (Master Mode Only) The SPI_CLK is determined according to the following equation: SPI_CLK = Source_CLK / (2*(CDR_N + 1)). This register is only valid when Work Mode Select==0x2/0x3.

17.11.7.13 0x0048 SPI TX Bit Register (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: SPI_TBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VTB The Value of the Transmit Bits This register is used to store the value of the transmitted serial data frame. Note: In the process of transmission, the LSB is transmitted first. This register is only valid when Work Mode Select==0x2/0x3.

17.11.7.14 0x004C SPI RX Bit Register (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: SPI_RBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VRB The Value of the Receive Bits This register is used to store the value of the received serial data frame. Note: In the process of transmission, the LSB is transmitted first. This register is only valid when Work Mode Select==0x2/0x3.

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17.11.7.15 0x0088 SPI Normal DMA Mode Control Register (Default Value: 0x0000_00E5)

Offset: 0x0088			Register Name: SPI_NDMA_MODE_CTL
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R/W	0x3	SPI_ACT_M SPI NDMA Active Mode 00: DMA_active is low 01: DMA_active is high 10: DMA_active is controlled by dma_request(DRQ) 11: DMA_active is controlled by controller
5	R/W	0x1	SPI_ACK_M SPI NDMA Acknowledge Mode 0: Active fall do not care ACK 1: Active fall must after detect ACK is high
4:0	R/W	0x05	SPI_DMA_WAIT The counts of hold cycles from DMA last signal high to dma_active high

17.11.7.16 0x0160 SPI Slave Vsync Cycle Number Register (Default Value: 0x0000_0000)

Offset: 0x0160			Register Name: SPI_SVCNR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	SFR Slave Frequency Relation 0: AHB frequency is the same as input SPI_CLK 1: AHB frequency is twice as much as input SPI_CLK ... N: AHB frequency is (N+1) times as much as input SPI_CLK
23:0	R/W	0x0	SVCN Slave Vsync Cycle Number In slave mode, when VIE is enabled, this number indicates valid cycles numbers after one valid Vsync edge. 0: 0 cycle 1: 1 cycle ... N: N cycles

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17.11.7.17 0x0164 SPI Slave Frame Head Number Register (Default Value: 0x0000_0000)

Offset: 0x0164			Register Name: SPI_SFHNR
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2:0	R/W	0x0	SFHN Slave Frame Head Number In slave mode, when SLV_FRAME_HEAD_EN is enabled, SFHN is used to configure the frame head length in byte. 0: 1 byte 1: 2 byte ... 7: 8 bytes

17.11.7.18 0x0168 SPI Slave Frame Head Low Register (Default Value: 0x0000_0000)

Offset: 0x0168			Register Name: SPI_SFHLR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SFHL Slave Frame Head Low Low 4 bytes of the wanted frame head. When not all bytes are used, lower bytes are valid.

17.11.7.19 0x016C SPI Slave Frame Head High Register (Default Value: 0x0000_0000)

Offset: 0x016C			Register Name: SPI_SFHHR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SFHH Slave Frame Head High High 4 bytes of the wanted frame head. When not all bytes are used, lower bytes are valid.

17.11.7.20 0x0170 SPI Slave Frame Head Low Receive Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: SPI_SFHLRR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SFHLR Slave Frame Head Low Receive Low 4 bytes of the received frame head. When not all bytes are used, lower bytes are valid.

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17.11.7.21 0x0174 SPI Slave Frame Head High Receive Register (Default Value: 0x0000_0000)

Offset: 0x0174			Register Name: SPI_SFHHRR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SFHHR Slave Frame Head High Receive High 4 bytes of the received frame head. When not all bytes are used, lower bytes are valid.

17.11.7.22 0x0178 SPI Slave Idle Wait Value Enable Register (Default Value: 0x0000_0000)

Offset: 0x0178			Register Name: SPI_SIWVER
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SIWVE Slave Idle Wait Value Enable Wait time in AHB cycle. The SPI slave controller will wait until enough IDLE time is detected before receiving data.

17.11.7.23 0x017C SPI Slave Idle Wait Value Reset Register (Default Value: 0x0000_0000)

Offset: 0x017C			Register Name: SPI_SIWVRR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R/W	0x0	SIWVR Slave Idle Wait Value Reset Wait time in AHB cycle. The SPI slave controller will further wait until enough IDLE time is detected after enable the clock is enabled. Note: Total wait IDLE time is SIWVE + SIWVR. And SIWVR time must be larger than 2 SPI clk.

17.11.7.24 0x0200 SPI TX Data Register (Default Value: 0x0000_0000)

Offset: 0x0200			Register Name: SPI_TXD
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0200			Register Name: SPI_TXD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TDATA Transmit Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are rooms in TXFIFO, one burst data is written to TXFIFO and the depth is increased by 1. In half-word accessing method, two SPI burst data are written and the TXFIFO depth is increase by 2. In word accessing method, four SPI burst data are written and the TXFIFO depth is increased by 4. Note: This address is write-only if TF_TEST is '0', and if TF_TEST is set to '1', this address is readable and writable to test the TX FIFO through the AHB bus.

17.11.7.25 0x0300 SPI RX Data Register (Default Value: 0x0000_0000)

Offset: 0x0300			Register Name: SPI_RXD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RDATA Receive Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are data in RXFIFO, the top word is returned and the RXFIFO depth is decreased by 1. In half-word accessing method, two SPI bursts are returned and the RXFIFO depth is decrease by 2. In word accessing method, the four SPI bursts are returned and the RXFIFO depth is decreased by 4. Note: This address is read-only if RF_TEST is '0', and if RF_TEST is set to '1', this address is readable and writable to test the RX FIFO through the AHB bus.

17.11.7.26 0x0400 SPI BUF Status Register (Default Value: 0x0000_0000)

Offset: 0x0400			Register Name: SPI_BSR
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/

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Offset: 0x0400			Register Name: SPI_BSR
Bit	Read/Write	Default/Hex	Description
26:24	R	0x0	TX_SHIFT_BUFFER_CNT TX SHIFT Buffer Counter These bits indicate the number of bytes in TX Shift Buffer, which is after TX FIFO, and maximum depth is 4 level with 8 bits.
23:16	R	0x0	TB_CNT TX FIFO Write Buffer Counter These bits indicate the number of bytes in TX FIFO Write Buffer
15:8	/	/	/
7:0	R	0x0	RB_CNT RX FIFO Read Buffer Counter These bits indicate the number of bytes in RX FIFO Read Buffer

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17.12 SPI_DBI

17.12.1 Overview

The SoC provides a 3/4 line SPI display bus interface (SPI_DBI) for video data transmission. It supports DBI mode or SPI mode. The DBI mode can be configured with 3-wire/4-wire transmission and compatible with multiple video data formats at the same time. The integrated SPI mode is used for low-cost display schemes.

The SPI mode has the following features:

- Supports multiple SPI mode
  - Master mode and Slave mode Standard SPI (4wire)
  - Master mode Dual-Output/Dual-Input SPI/ Dual I/O SPI
  - Master mode Quad-Output/Quad-Input SPI
  - Master mode 3wire SPI, with programmable serial data frame length: 1bit to 32bits
  - Ordinary slave mode for Single-Output/Single-Input SPI
- Master mode supports the Maximum clock frequency: 100MHz
- Supports TX/RX DMA Slave interface
- Supports Data Sample Mode: Mode0, Mode1, Mode2 and Mode3
- Supports Control signal configuration
  - Three chip selects to support multiple peripherals
  - Polarity and phase of the Chip Select (SPI_CS) and SPI Clock (SPI_CLK) are configurable

The DBI mode has the following features:

- Supports DBI Type C 3 Line/4 Line Interface Mode
- Supports 2 Data Lane Interface Mode
- Support data source from CPU or DMA
- Supports RGB111/444/565/666/888 video format
- Supports maximum resolution of RGB666 240 x 320@30Hz with single data lane
- Supports maximum resolution of RGB888 240 x 320@60Hz or 320 x 480@30Hz with dual data lane
- Supports tearing effect
- Supports software flexible control video frame rate

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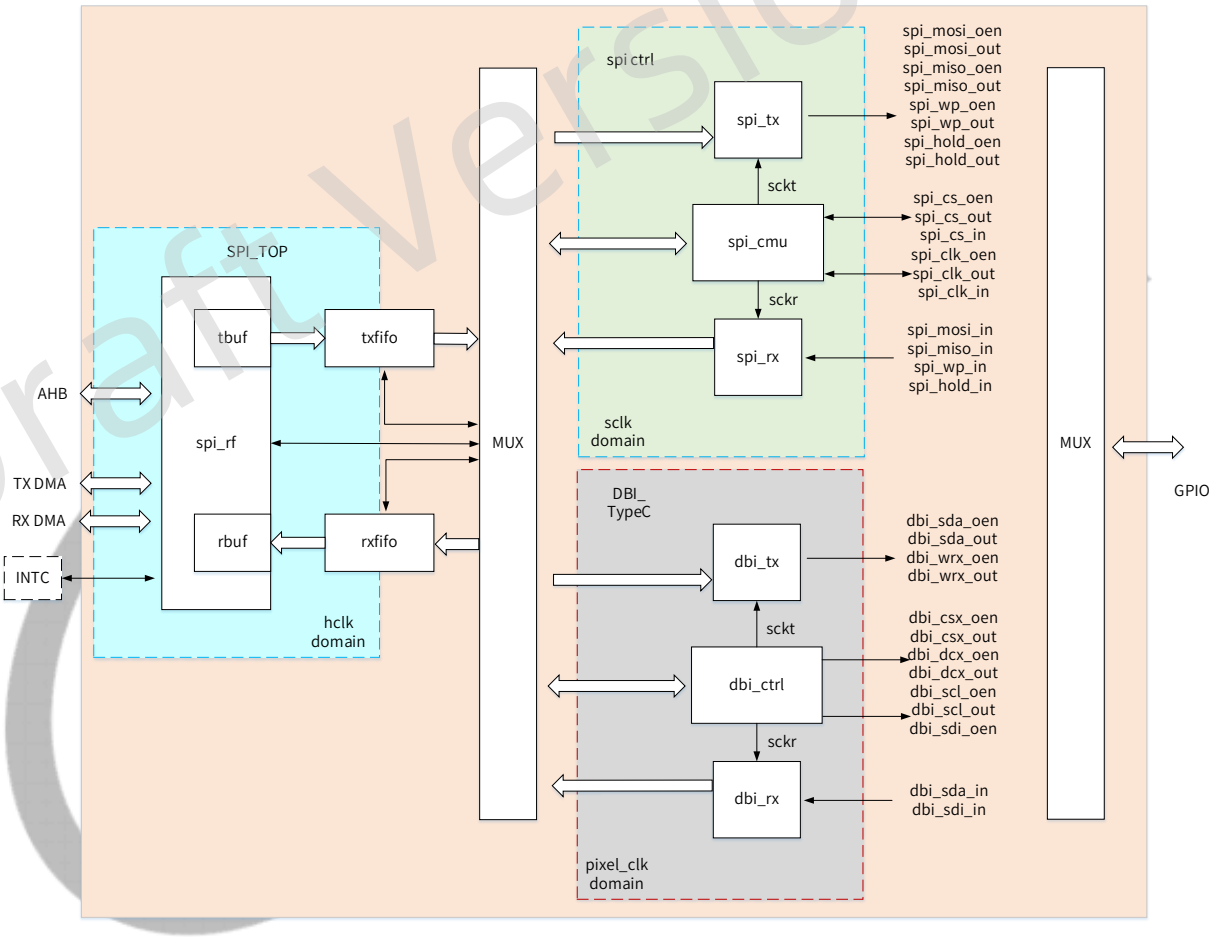
NOTE

This chapter only describes SPI1 (SPI mode and DBI mode). For detailed information of SPI0, SPI2, and SPI3, please refer to Section 17.11 SPI.

17.12.2 Block Diagram

The following figure shows a block diagram of the SPI_DBI.

Figure 17-63 SPI_DBI Block Diagram



SPI_DBI contains the following sub-blocks:

Table 17-62 SPI_DBI Sub-blocks

Sub-block	Description
spi_rf	Responsible for implementing the internal register, interrupt, and DMA Request.
spi_tbuf	The data length transmitted from AHB to TXFIFO is converted into 8 bits, then the data is written into the RXFIFO.
spi_rbuf	The block is used to convert the RXFIFO data into the reading data length of AHB.

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Sub-block	Description
txfifo, rxfifo	The data transmitted from the SPI to the external serial device is written into the TXFIFO; the data received from the external serial device into SPI is pushed into the RXFIFO.
spi_cmu	Responsible for implementing SPI bus clock, chip select, internal sample, and the generation of transfer clock.
spi_tx	Responsible for implementing SPI data transfer, the interface of the internal TXFIFO, and status register.
spi_rx	Responsible for implementing SPI data receive, the interface of the internal RXFIFO, and status register.
dbi_ctrl	Responsible for implementing DBI bus clock, chip select, data command select, RGB format reshape.
dbi_tx	Responsible for implementing DBI data transfer, the interface of the internal TXFIFO, and status register.
dbi_rx	Responsible for implementing DBI data receive, the interface of the internal RXFIFO, and status register.

17.12.3 External Signals

The following table describes the external signals of SPI_DBI. When using SPI_DBI, the corresponding PADs are selected as SPI_DBI function via Section 17.9 GPIO .

Table 17-63 GPIO multiplexing of SPI1 and DBI

DBI	SPI1
DBI-CSX	SPI1-CS0
DBI-SCLK	SPI1-CLK
DBI-SDO	SPI1-MOSI
DBI-SDI/ DBI-TE/ DBI-DCX	SPI1-MISO
DBI-DCX/DBI-WRX	SPI1-HOLD
DBI-TE	SPI1-WP

Table 17-64 SPI_DBI External Signals

Signal Name	Description	Type
SPI Mode		
SPI1-CS [2:0]	SPI1 Chip Select Signal, Low Active	I/O
SPI1-CLK	SPI1 Clock Signal Provides serial interface timing.	I/O
SPI1-MOSI	SPI1 Master Data Out, Slave Data In	I/O
SPI1-MISO	SPI1 Master Data In, Slave Data Out	I/O
SPI1-WP	SPI1 Write Protect, Low Active Protects the memory area against all program or erase instructions. It also can be used for serial data input and output for SPI Quad Input or Quad Output mode.	I/O

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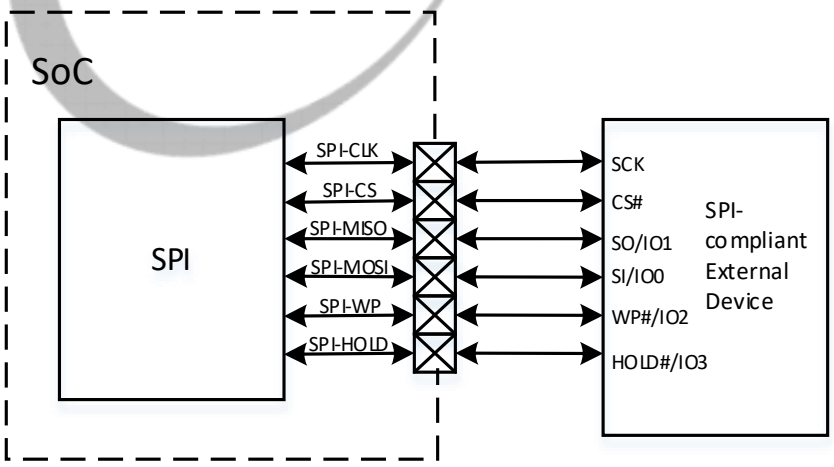
Signal Name	Description	Type
SPI1-HOLD	SPI1 Hold Signal Pauses any serial communication with the device without deselecting or resetting it. It also can be used for serial data input and output for SPI Quad Input or Quad Output mode.	I/O
DBI Mode		
DBI-CSX	Chip Select Signal, Low Active	I/O
DBI-SCLK	Serial Clock Signal	I/O
DBI-SDO	Data Output Signal	I/O
DBI-SDI	Data Input Signal The data is sampled on the rising edge and the falling edge	I/O
DBI-TE	Tearing Effect Input It is used to capture the external TE signal edge. The rising and falling edge is configurable.	I/O
DBI-DCX	DCX pin is the select output signal of data and command. DCX = 0: Register command; DCX = 1: Data or parameter.	I/O
DBI-WRX	When DBI operates in dual data lane format, the RGB666 format 2 can use WRX to transfer data	I/O

17.12.4 Functional Description

17.12.4.1 Typical Application

The following figure shows the application block diagram when the SPI master device is connected to a slave device.

Figure 17-64 SPI Application Block Diagram



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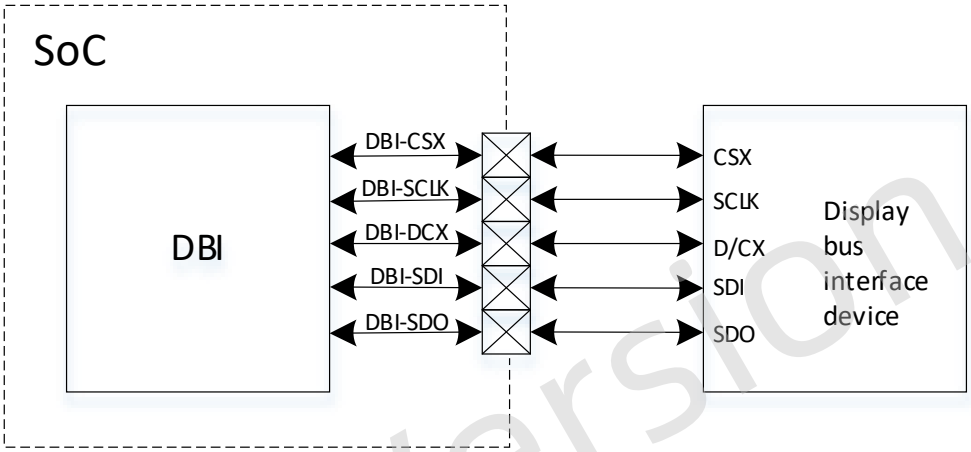
The following figure shows the application block diagram when the DBI master device is connected to a display bus interface device.

Figure 17-65 DBI Application Block Diagram



NOTE

For the function descriptions in SPI mode, refer to the section SPI.

17.12.4.2 DBI 3-Line Interface Write and Read Timing

DBI 3-Line Interface Write Timing

The 3-line DBI Interface I contains CSX, SDA, and SCL, where SDA shares this port for bidirectional port data input and output.

The 3-line DBI Interface II contains CSX, SDA, SCL, and SDI; Data input and output ports are independent of each other.

As there is no Data/Command data line in the 3-line display bus mode, whether Data or Command is currently being transmitted cannot be confirmed. Thus, an extra bit is added to the data-stream before MSB to indicate whether Data or Command is currently being transmitted. (0: Command, 1: Data)

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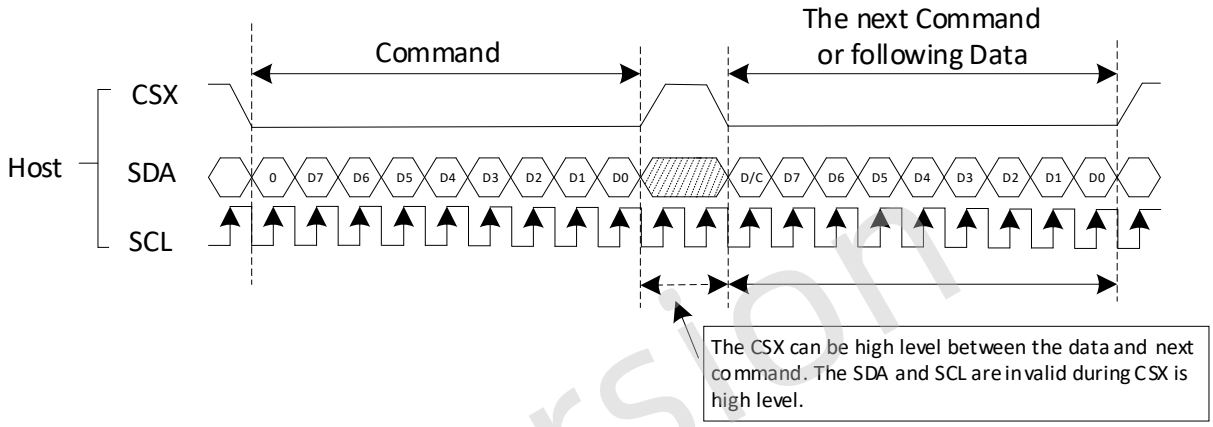
The following figure shows the write operation format of 3-line DBI Interface I and Interface II.

Figure 17-66 DBI 3-Line Display Bus Serial Interface Write Operation Format

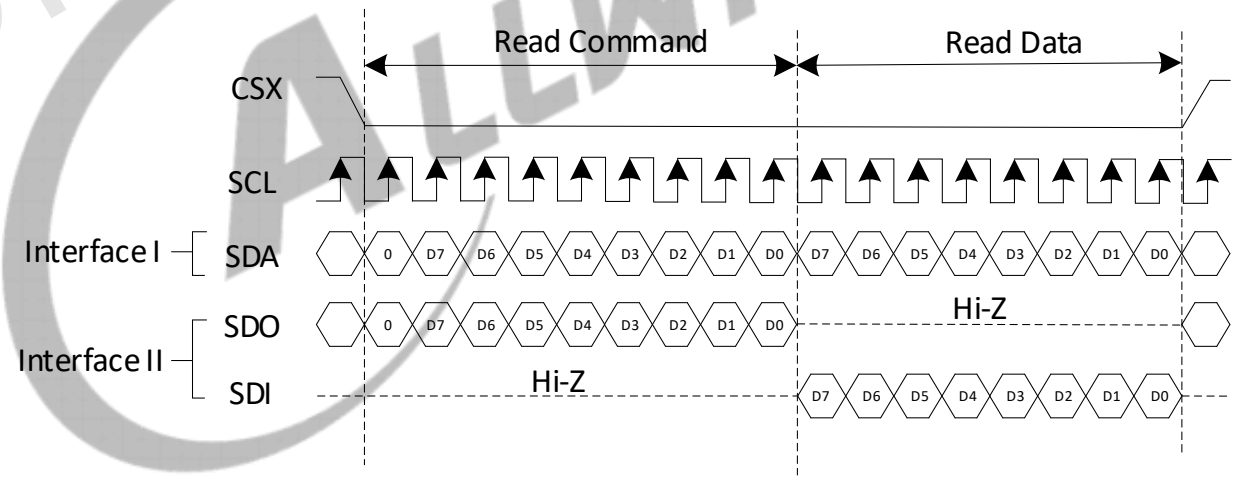


DBI 3-Line Interface Read Timing

The 3-line DBI Interface I uses the SDA port as bidirectional data input and output port. There are only three cases of data reading volume, 8bits/24bits/32bits, and the first data sampled is high.

The following figure shows the 8 bits read operation format of 3-line DBI Interface I and Interface II. After the read command is transmitted, the data is read immediately with no dummy period.

Figure 17-67 DBI 3-Line Display Bus Serial Interface 8-bit Read Operation Format



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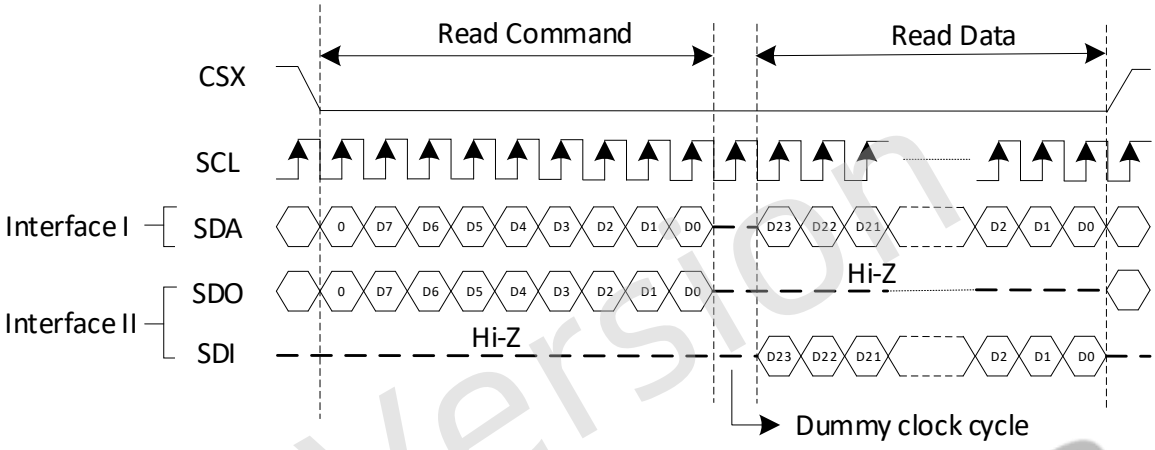
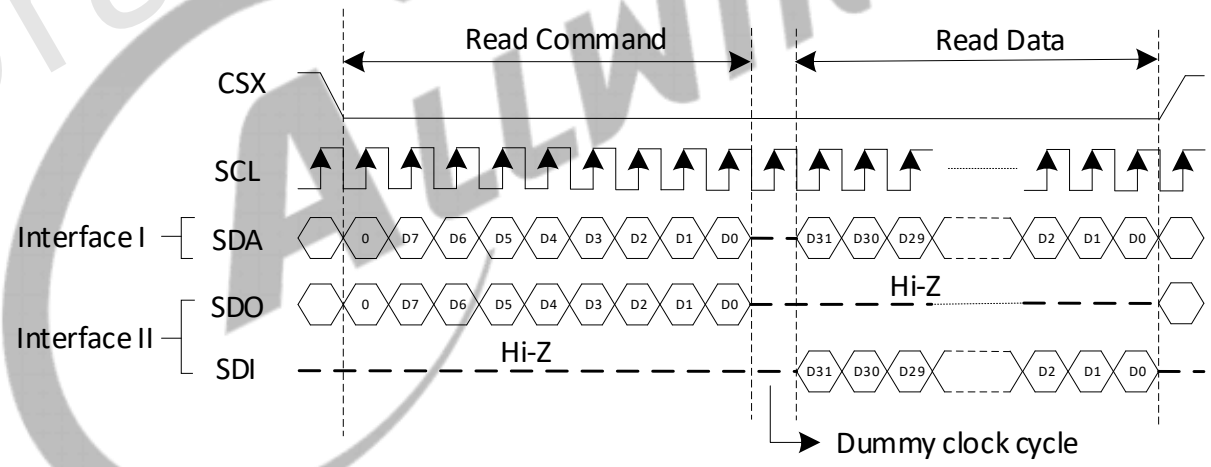
The following figure shows the 24 bits read operation format of 3-line DBI Interface I and Interface II. When the read command is transmitted, the data will be read after the dummy clock cycle.

Figure 17-68 DBI 3-Line Display Bus Serial Interface 24-bit Read Operation Format



The following figure shows the 32 bits read operation format of 3-line DBI Interface I and Interface II. When the read command is transmitted, the data will be read after the dummy clock cycle.

Figure 17-69 DBI 3-Line Display Bus Serial Interface 32-bit Read Operation Format



17.12.4.3 DBI 4-Line Interface Write and Read Timing

DBI 4-Line Interface Write Timing

The 4-line DBI Interface I contains CSX, D/CX, SDA, and SCL, where SDA shares this port for bidirectional port data input and output.

The 4-line DBI Interface II contains CSX, D/CX, SDA, SCL, and SDI; Data input and output ports are independent of each other.

Since the 4-line display bus mode has a Data/Command data line indicating whether Data or Command is currently being transmitted (0: Command, 1: Data). So there is no need to add an extra bit to data-stream before MSB like the 3-line DBI.

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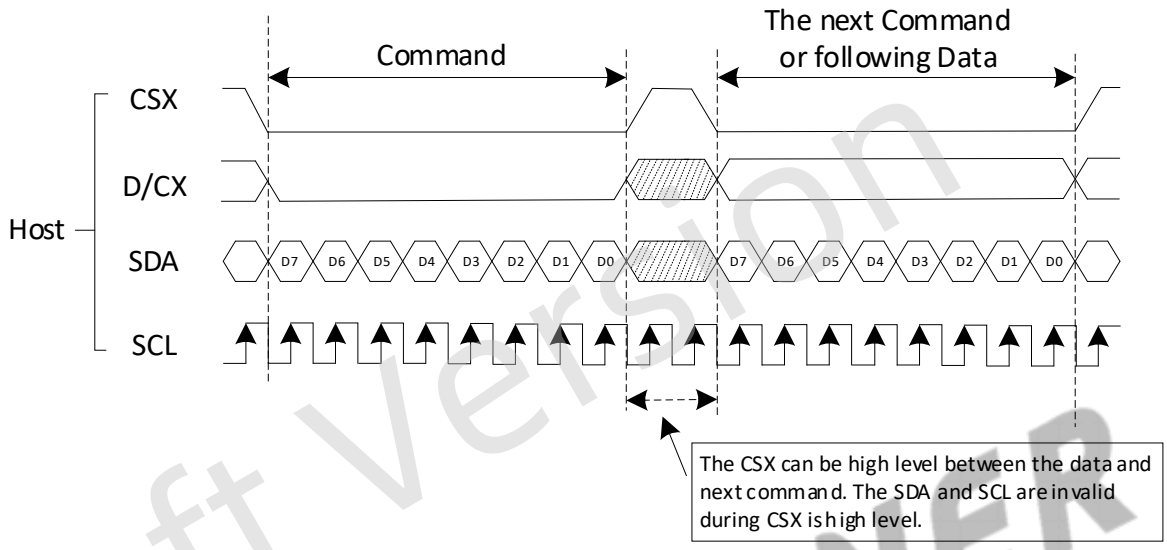
There is a Data/Command data line in the 4-line display bus mode to indicate whether Data or Command is currently being transmitted. (0: Command, 1: Data) Thus, no extra bit is needed.

The following figure shows the writing operation format of 4-line DBI Interface I and Interface II.

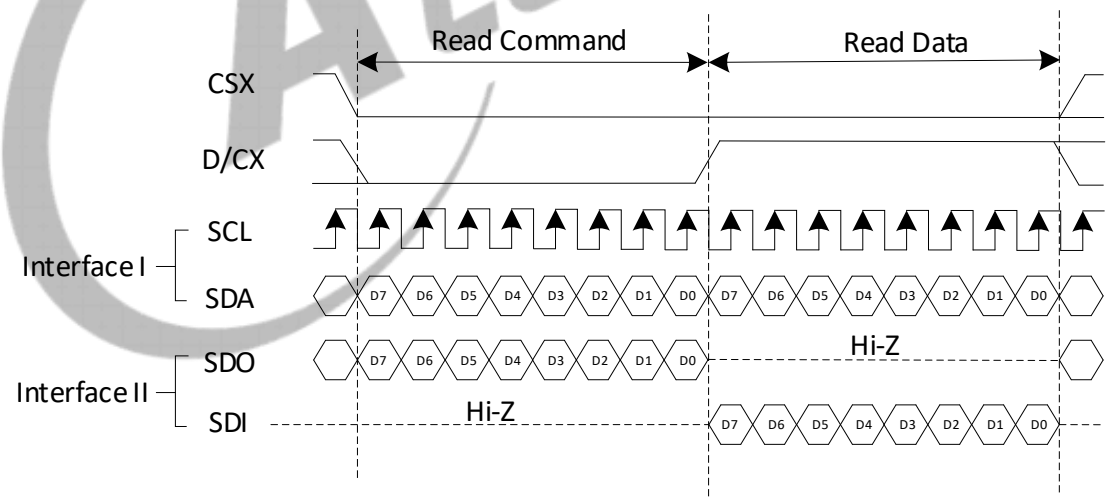
Figure 17-70 DBI 4-Line Display Bus Serial Interface Write Operation Format



DBI 4-Line Interface Read Timing

The following figure shows the 8 bits read operation format of 4-line DBI Interface I and Interface II.

Figure 17-71 DBI 4-Line Display Bus Serial Interface 8-bit Read Operation Format



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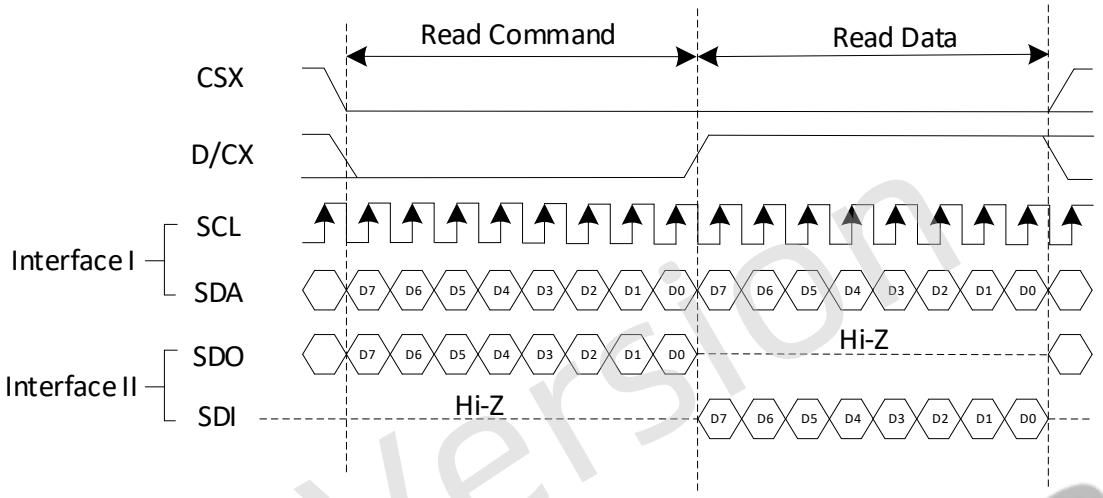
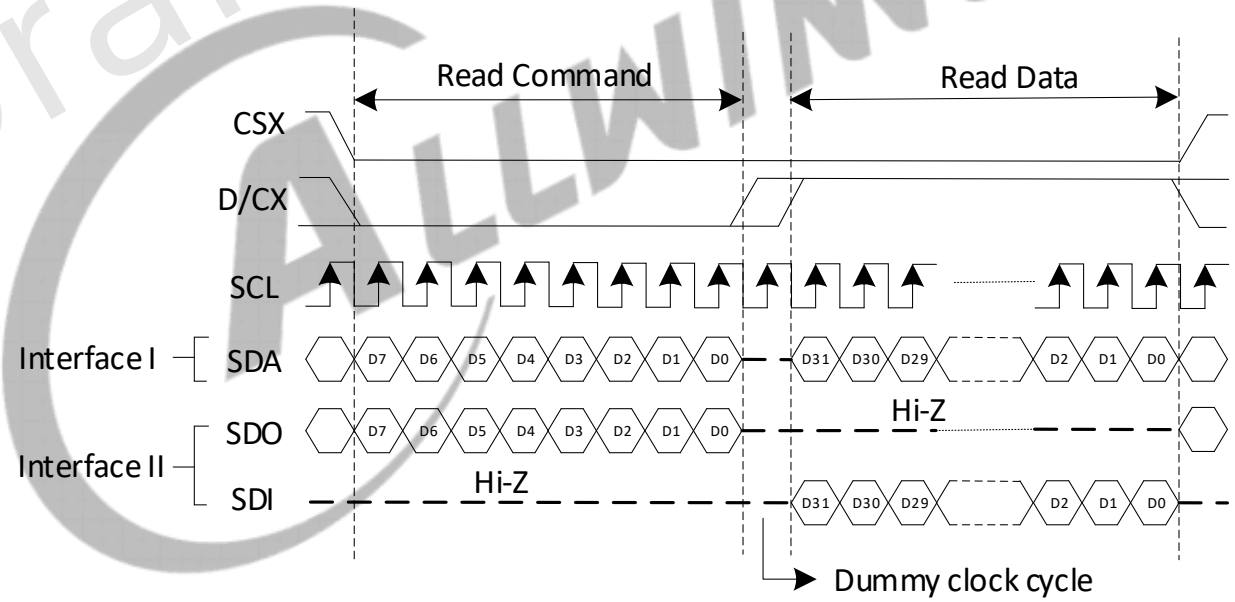
The following figure shows the 24 bits read operation format of 4-line DBI Interface I and Interface II.

Figure 17-72 DBI 4-Line Display Bus Serial Interface 24-bit Read Operation Format



The following figure shows the 32 bits read operation format of 4-line DBI Interface I and Interface II.

Figure 17-73 DBI 4-Line Display Bus Serial Interface 32-bit Read Operation Format



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17.12.4.4 DBI 3-Line Interface Transmit Video Format

Figure 17-74 RGB111 3-Line Interface Transmit Video Format

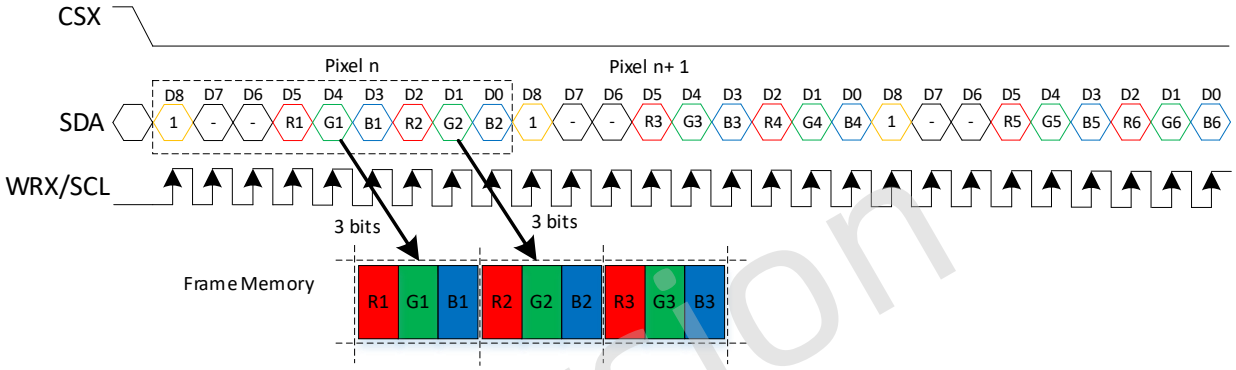
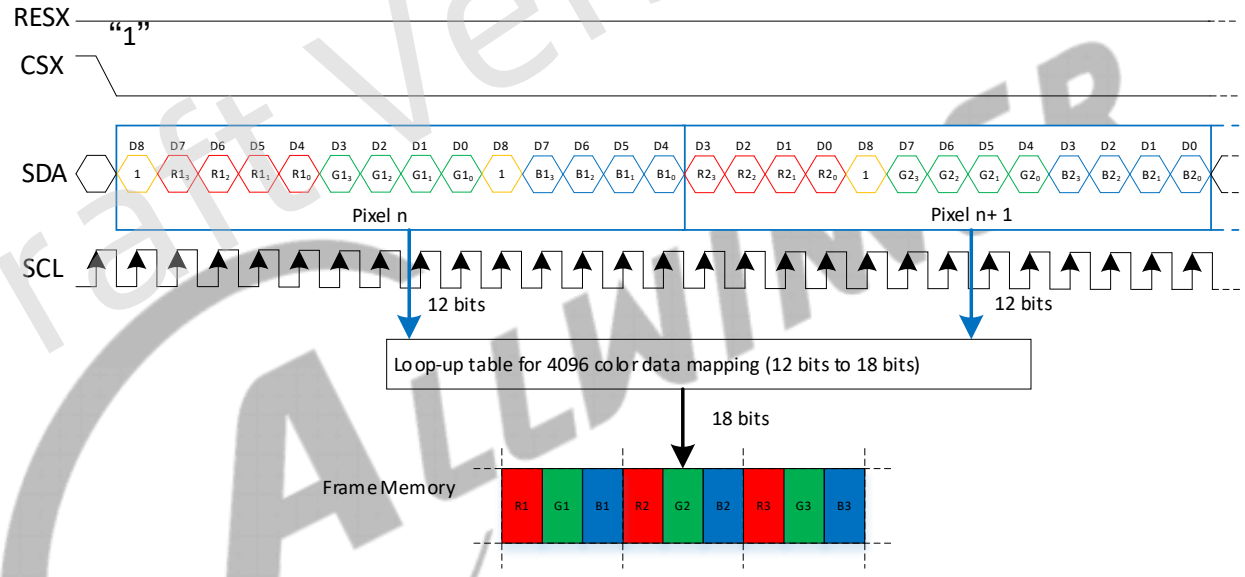


Figure 17-75 RGB444 3-Line Interface Transmit Video Format



Note 1. Pixel data with 12-bit color depth information
Note 2. The most significant bits are: Rx3, Gx3 and Bx3
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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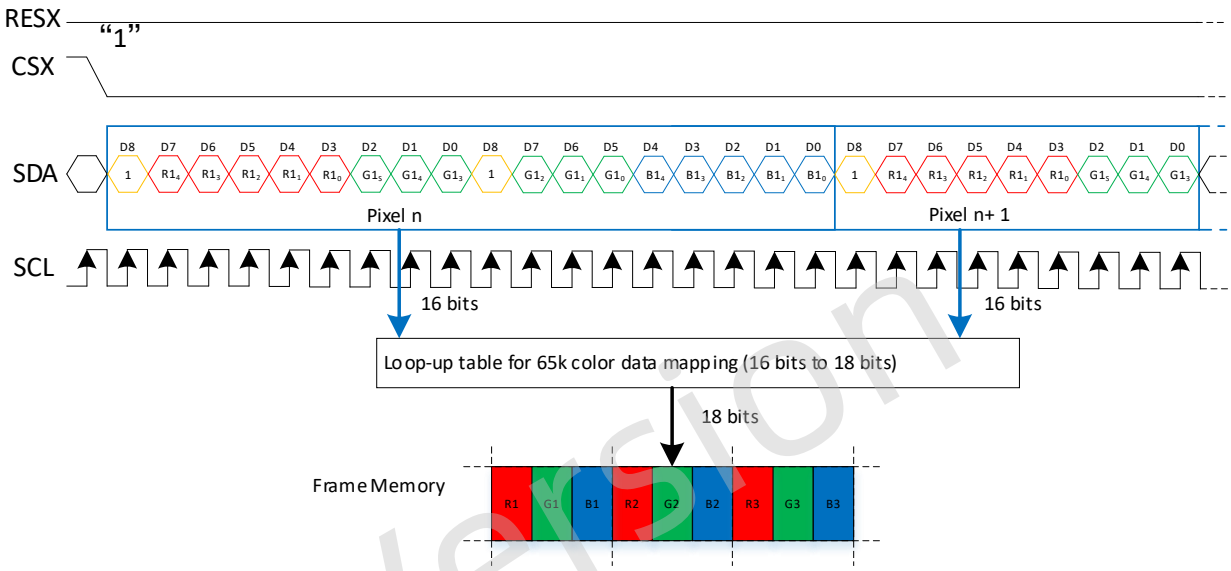
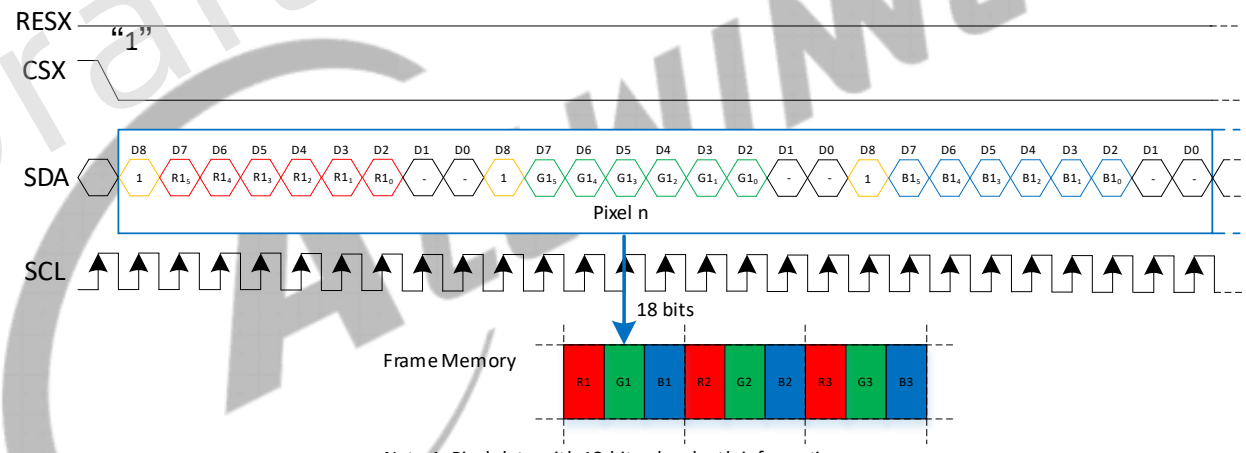
Figure 17-76 RGB565 3-Line Interface Transmit Video Format



Note 1. Pixel data with 16-bit color depth information
Note 2. The most significant bits are: Rx4, Gx5 and Bx4
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

Figure 17-77 RGB666 3-Line Interface Transmit Video Format



Note 1. Pixel data with 18-bit color depth information
Note 2. The most significant bits are: Rx5, Gx5 and Bx5
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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17.12.4.5 DBI 4-Line Interface Transmit Video Format

Figure 17-78 RGB111 4-Line Interface Transmit Video Format

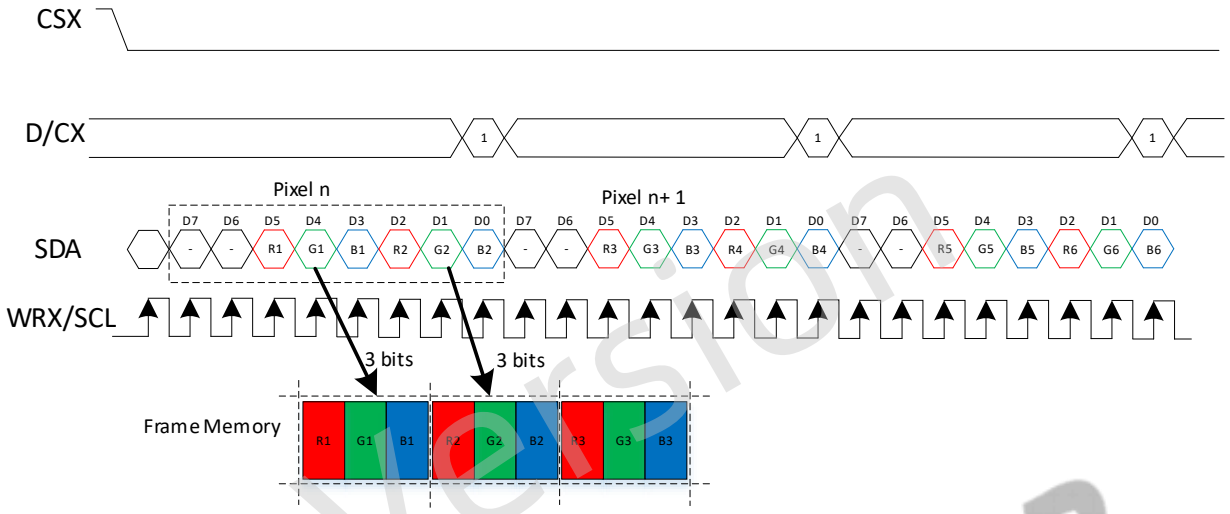
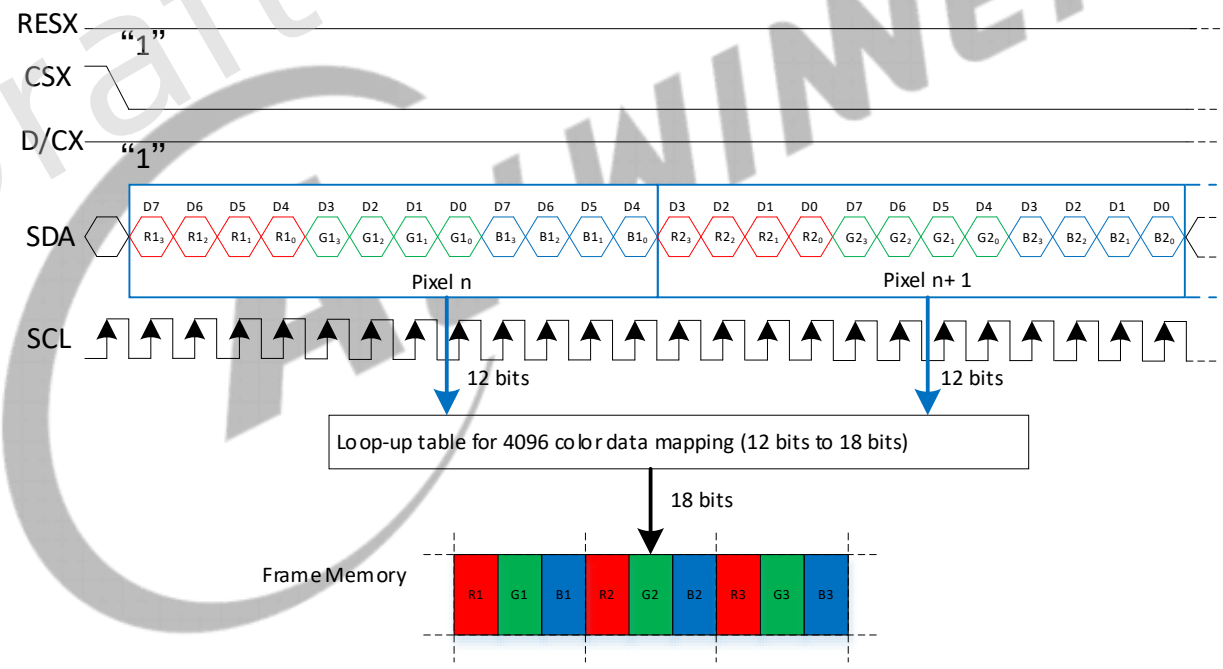


Figure 17-79 RGB444 4-Line Interface Transmit Video Format



Note 1. Pixel data with 12-bit color depth information
Note 2. The most significant bits are: Rx3, Gx3 and Bx3
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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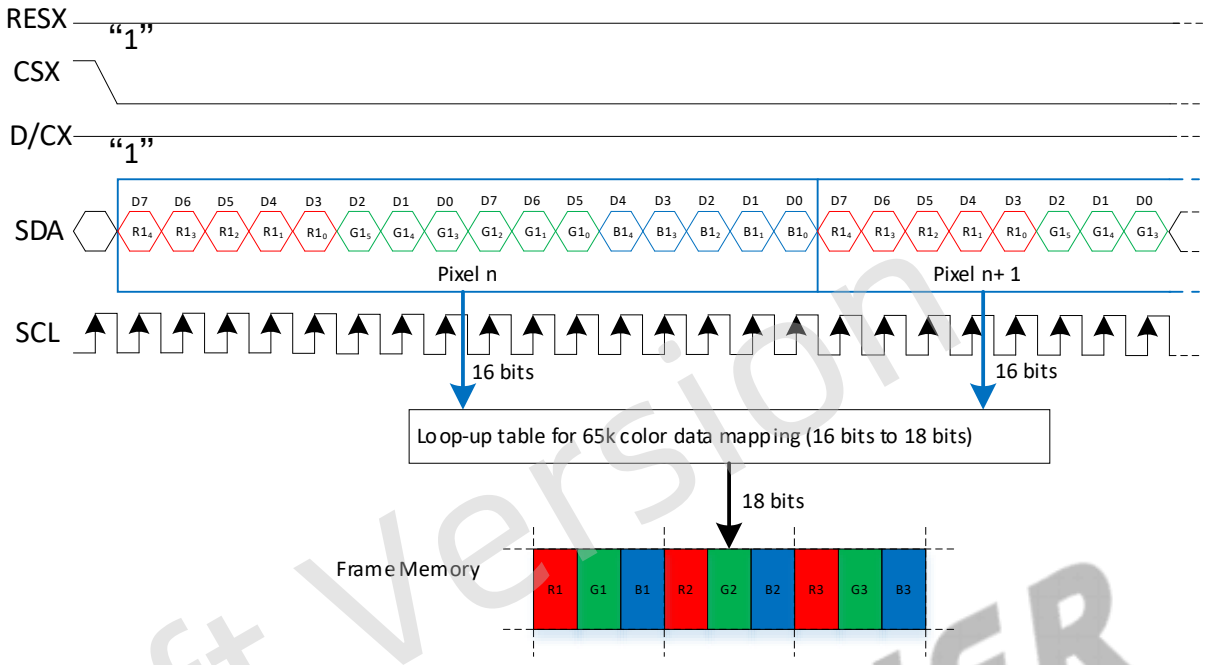
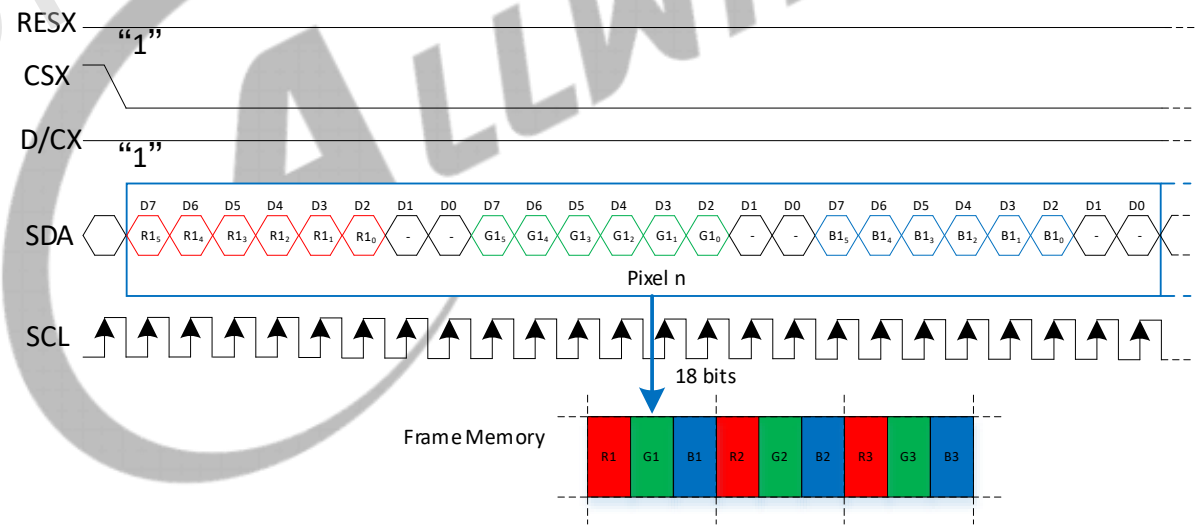
Figure 17-80 RGB565 4-Line Interface Transmit Video Format



Note 1. Pixel data with 16-bit color depth information
Note 2. The most significant bits are: Rx4, Gx5 and Bx4
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

Figure 17-81 RGB666 4-Line Interface Transmit Video Format



Note 1. Pixel data with 18-bit color depth information
Note 2. The most significant bits are: Rx5, Gx5 and Bx5
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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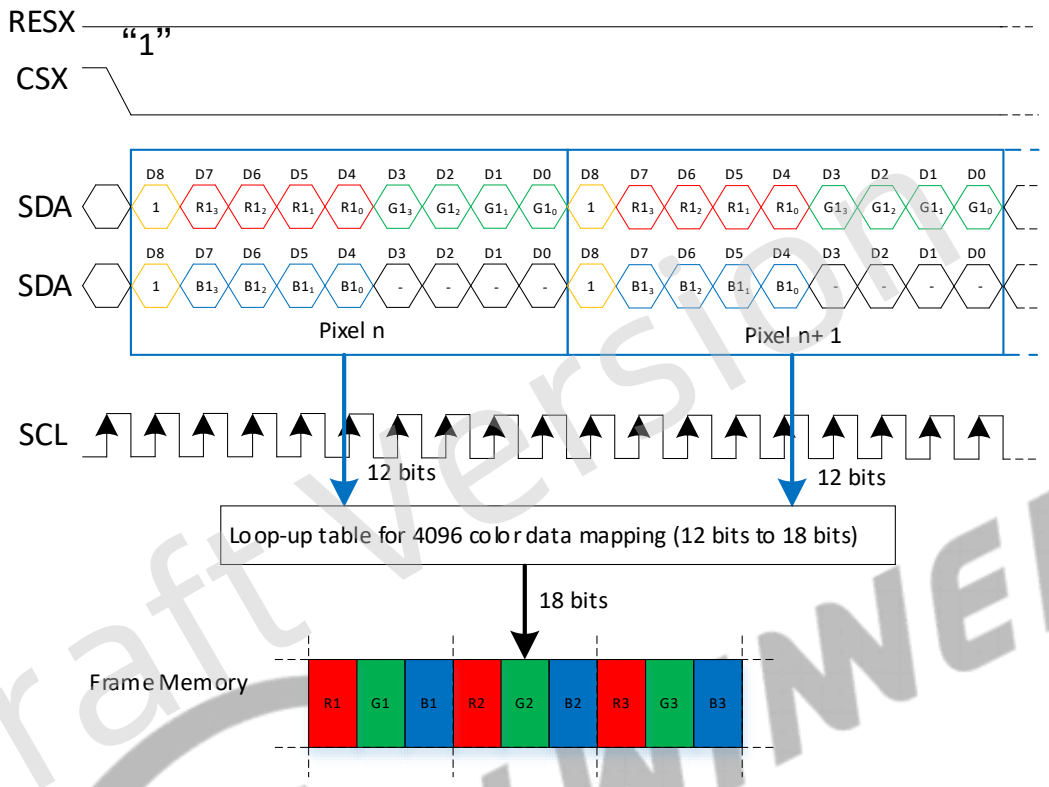
17.12.4.6 DBI 2 Data Lane Interface Transmit Video Format

Figure 17-82 RGB444 2 Data Lane Interface Transmit Video Format



Note 1. Pixel data with 12-bit color information
Note 2. The most significant bits are: Rx3, Gx3 and Bx3
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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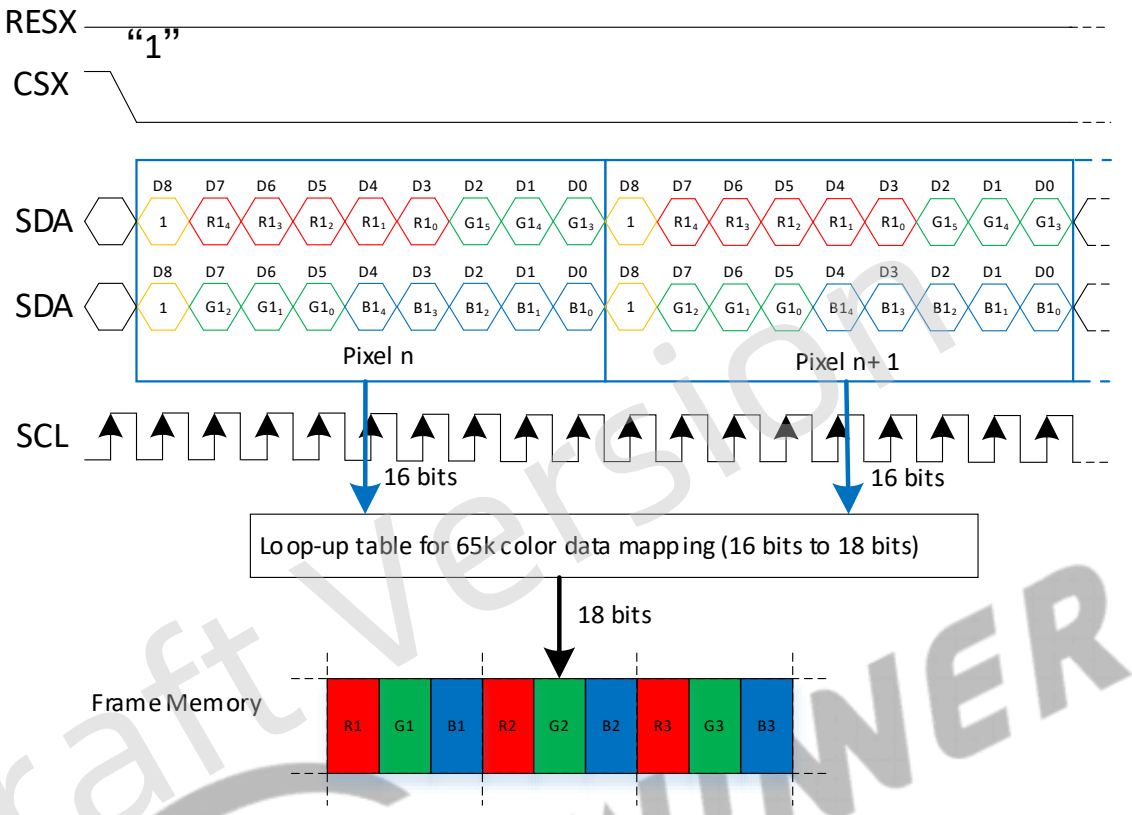
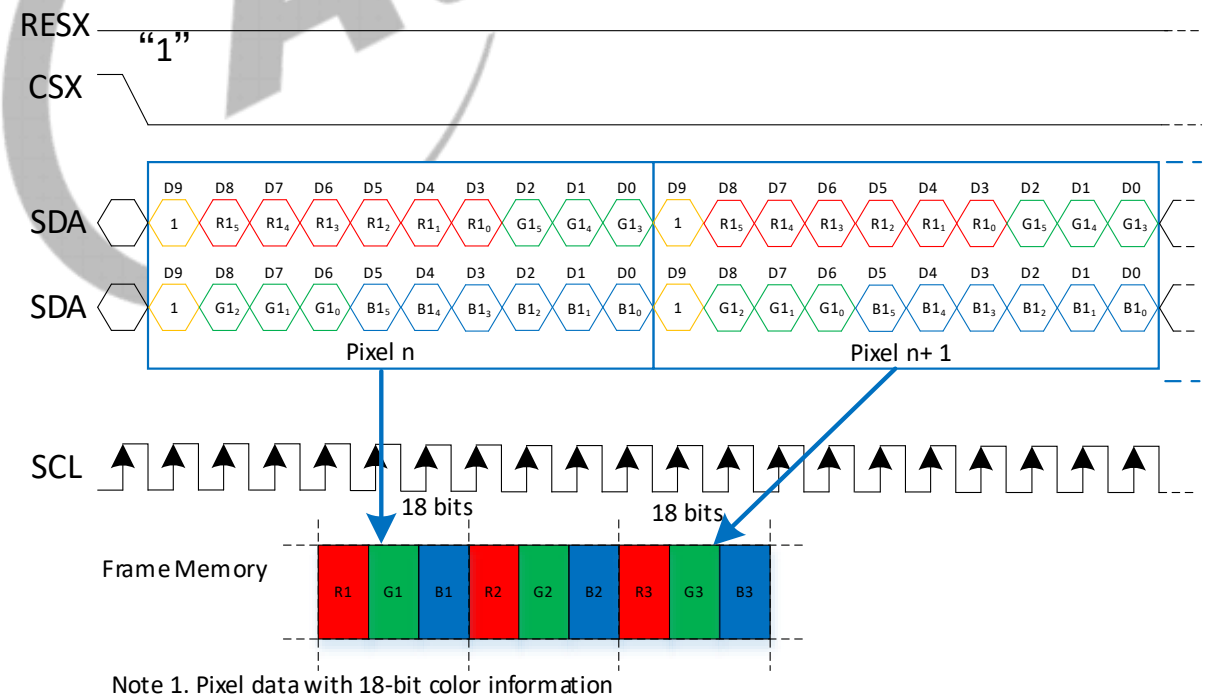
Figure 17-83 RGB565 2 Data Lane Interface Transmit Video Format



Note 1. Pixel data with 16-bit color information
Note 2. The most significant bits are: Rx4, Gx5 and Bx4
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

Figure 17-84 RGB666 2 Data Lane Interface Transmit Video Format 0



Note 1. Pixel data with 18-bit color information
Note 2. The most significant bits are: Rx5, Gx5 and Bx5
Note 3. The least significant bits are: Rx0, Gx0 and Bx0

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Figure 17-85 RGB666 2 Data Lane Interface Transmit Video Format 1 (ilitek)

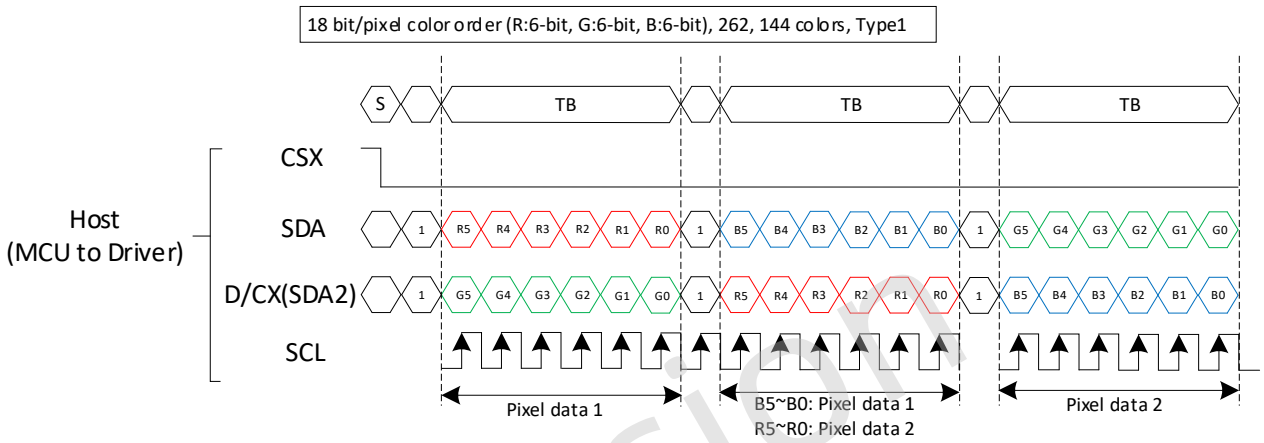


Figure 17-86 RGB666 2 Data Lane Interface Transmit Video Format 2 (New vision)

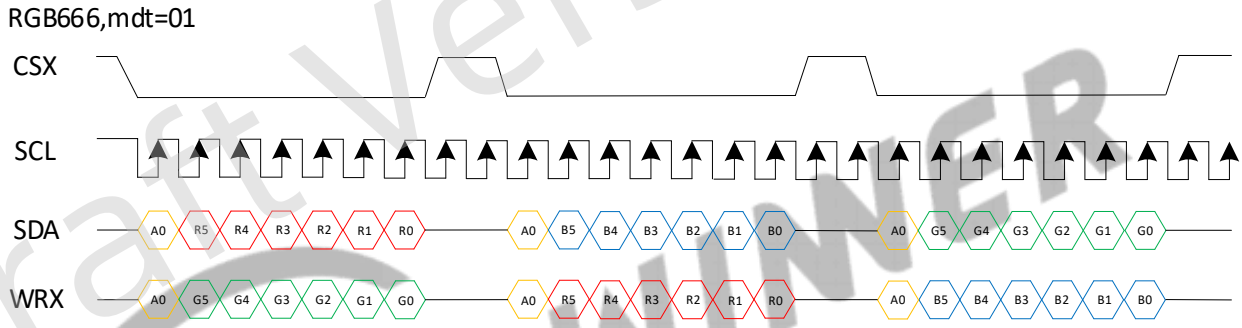
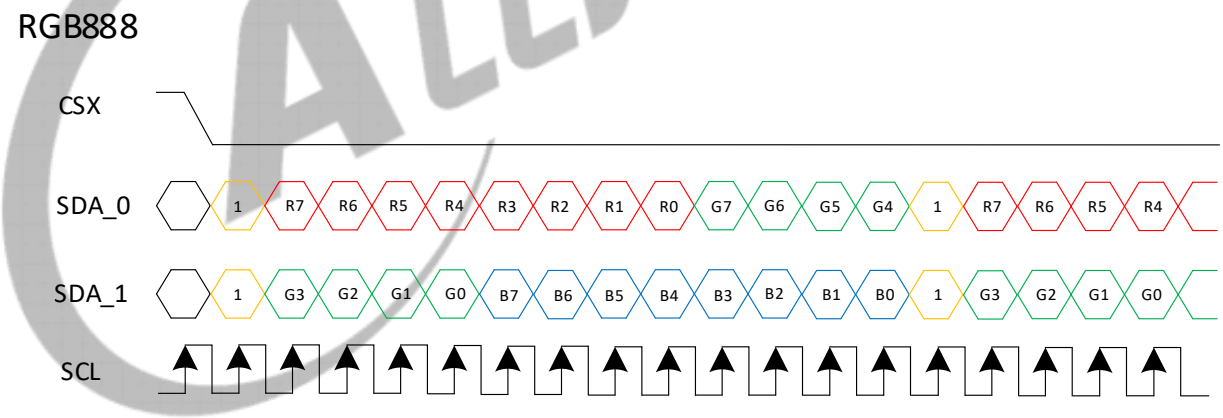


Figure 17-87 RGB888 2 Data Lane Interface Transmit Video Format



Note 1. Pixel data with 24-bit color information
Note 2. The most significant bits are: R7, G7 and B7
Note 3. The least significant bits are: R0, G0 and B0

17.12.4.7 SPI_DBI Clock Frequency

The equation of SPI_DBI clock frequency is

f_{SPI_CLK} = frame_rate * one_pixel_clock_cycle * one_frame_pixel_value

The meaning of each sub-item is as follows:

- frame_rate: The frame rate at the unit of Hz.

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- one_pixel_clock_cycle: The clock period cycle of transferring a pixel. The period cycle differs from different transferring format.
  - 2-data lane-RGB444: 9 cycles
  - 2-data lane-RGB565: 9 cycles
  - 2-data lane-RGB666-Format0: 10 cycles
  - 2-data lane-RGB888: 13 cycles
- one_frame_pixel_value: The number of pixel in one frame.
  - 240*320
  - 320*480

For example, RGB data is transferred with 2 data line and RGB666 mode, and the resolution and frame rate of the display are 240*320 and 60Hz, then the SPI clock frequency is 60*10*240*320=46.08MHz.

The following tables show the SCLK frequency in different RGB modes.

Table 17-65 2 data lane sclk frequency

2 data lane					
RGB	Frame	30Hz		60Hz	
	Resolution	240*320	320*480	240*320	320*480
RGB111		/	/	/	/
RGB444		20.736	41.472	41.472	82.944
RGB565		20.736	41.472	41.472	82.944
RGB666		23.04	46.08	46.08	92.16
RGB888		29.952	59.904	59.904	/

Table 17-66 3 line sclk frequency

3 line					
RGB	Frame	30Hz		60Hz	
	Resolution	240*320	320*480	240*320	320*480
RGB111		10.368	20.736	20.736	41.472
RGB444		31.104	62.208	62.208	/
RGB565		41.472	82.944	82.944	/
RGB666		62.208	/	/	/
RGB888		/	/	/	/

Table 17-67 4 line sclk frequency

4 line					
RGB	Frame	30Hz		60Hz	
	Resolution	240*320	320*480	240*320	320*480
RGB111		9.216	18.432	18.432	36.864
RGB444		27.648	55.296	55.296	/
RGB565		36.864	73.728	73.728	/
RGB666		55.296	/	/	/

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4 line					
RGB	Frame	30Hz		60Hz	
	Resolution	240*320	320*480	240*320	320*480
RGB888		/	/	/	/

17.12.5 Programming Guidelines

NOTE

The following are the programming guidelines in DBI mode. For those in SPI mode, see the Section 17.11.5 Programming Guidelines.

17.12.5.1 Write Command Transfer

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI_EN_MODE_SEL (bit[30:29]) of DBI_CTL_1 (0x0104) to 0 to select the trigger mode of DBI.

Step 3 Configure the DBI_CTL_0 (0x0100).

a) Set DBI_CTL_0[Command Type] (bit31) to 0 to configure the writing command.

b) Set DBI_CTL_0[Write Command Dummy Cycles] (bit[30:20]) to configure the number of dummy cycles between commands.

c) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

d) Set DBI_CTL_0[Transmit Mode] (bit15) to 0 to select the command path.

e) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to 0 to transmit the command.

f) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

g) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Set DBI_CTL_1[DCX_DATA] (bit22) to 0 to send the command.

Step 5 DMA Path: Configure the SPI_FCR register (0x0018).

a) Set SPI_FCR[TF_DRQ_EN] (bit24) to 1 to enable TXFIFO DMA.

b) Set SPI_FCR[TX_TRIG_LEVEL] (bit[23:16]) to 255. It indicates the controller requests data from DMA if the remaining space of TX FIFO is greater than 255.

CPU Path: Write the command to be sent to the 0x200 address.

Step 6 Set SPI_GCR[DBI_EN] (bit4) to 1 to start transmitting the command.

Step 7 Wait until the TX FIFO underrun interrupt (SPI_ISR[TF_UDF]) is 1. It indicates that the command written to the TX FIFO is transmitted completely.

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17.12.5.2 Parameter Transfer

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI_EN_MODE_SEL (bit[30:29]) of DBI_CTL_1 (0x0104) to 0 to select the trigger mode of DBI.

Step 3 Configure the DBI_CTL_0 register (0x0100).

a) Set DBI_CTL_0[Command Type] (bit31) to 0 to configure the writing command.

b) Set DBI_CTL_0[Write Command Dummy Cycles] (bit[30:20]) to configure the number of dummy cycles between commands.

c) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

d) Set DBI_CTL_0[Transmit Mode] (bit15) to 0 to select the command path.

e) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to 0 to transmit the command.

f) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

g) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Set DBI_CTL_1[DCX_DATA] (bit22) to 1 to send the parameter.

Step 5 DMA Path: Configure the SPI_FCR register (0x0018).

a) Set SPI_FCR[TF_DRQ_EN] (bit24) to 1 to enable TXFIFO DMA.

b) Set SPI_FCR[TX_TRIG_LEVEL] (bit[23:16]) to 255. It indicates the controller requests data from DMA if the remaining space of TX FIFO is greater than 255.

CPU Path: Write the command to be sent to the 0x200 address.

Step 6 Set SPI_GCR[DBI_EN] (bit4) to 1 to start transmitting the command.

Step 7 Wait until the TX FIFO underrun interrupt (SPI_ISR[TF_UDF]) is 1. It indicates that the command written to the TX FIFO is transmitted completely.

17.12.5.3 Video Transfer

Before the video transmission configuration, confirm whether the data is from CPU path or DMA path.

a) If the data is from the CPU path, the controller writes the command to be sent to the 0x0200 address by the AHB bus.

b) If the data is from the DMA path, configure DBI_CTL_2 [DBI_FIFO_DRQ_EN] (bit15) to 1 and DBI_CTL_2 [TX_TRIG_LEVEL] (bit[14:8]) to 64. The controller will request data from DMA if the remaining space of TX FIFO is greater than 64. The empty & full interrupts and the status bit of DBI FIFO can be configured in the DBI_INT and DBI_DEBUG_0.

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Software Trigger Mode

The software enables DBI_en_trigger when the edge interrupt of TE is detected.

After transmitting a frame of image, the controller automatically clears the line_cnt, pixel_cnt and stops data transmission.

The operation process is as follows.

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI EN MODE SEL (bit[30:29]) of DBI_CTL_1 (0x0104) to 01 to select the software trigger mode.

Step 3 Configure the DBI_CTL_0 register (0x0100).

a) Set DBI_CTL_0[Command Type] (bit31) to 0 to set the writing command.

b) Set DBI_CTL_0[Write Command Dummy Cycles] (bit[30:20]) to configure the number of dummy cycles between commands.

c) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

d) Set DBI_CTL_0[Transmit Mode] (bit15) to 1 to select the image path.

e) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to select RGB111/444/565/666/888.

f) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

g) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Set DBI_CTL_1[DCX_DATA] (bit22) to 0 to send the image data.

Step 5 Configure DBI_Video_Size (0x110) according to the sent image size.

Step 6 Configure DBI_CTL_2 (0x0108) to set the TE-related parameter.

Step 7 Detect the TE interrupt of the DBI_INT (0x0120) register.

Step 8 Configure DBI_CTL_1[DBI_soft_trigger] to 1.

Timer Trigger Mode

The software configures timer_en to enable timer counting, and when the counter reaches the specified value, the DBI_EN automatically will be enabled to start data transmission.

After transmitting a frame of image, the controller will automatically clear the line_cnt, pixel_cnt, and stop data transmission.

The timer starts counting again. When the counter reaches the specified value, the controller automatically enables DBI_EN again until the software turns off the timer_en.

The operation process is as follows.

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI EN MODE SEL (bit30:29) of DBI_CTL_1 (0x0104) to 10 to select the timer trigger mode.

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Step 3 Configure the DBI_CTL_0 register (0x0100).

a) Set DBI_CTL_0[Command Type] (bit31) to 0 to set the writing command.

b) Set DBI_CTL_0[Write Command Dummy Cycles] (bit[30:20]) to configure the number of dummy cycles between commands.

c) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

d) Set DBI_CTL_0[Transmit Mode] (bit15) to 1 to select the image path.

e) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to select RGB111/444/565/666/888.

f) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

g) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Set DBI_CTL_1[DCX_DATA] (bit22) to 0 to send the image data.

Step 5 Configure DBI_Video_Size (0x110) to transmit the image size.

Step 6 Configure the related parameter of DBI_Timer (0x10C).

TE Trigger Mode

When the edge changes of the TE are detected (The rising and falling edges are optional), the DBI_EN can be automatically enabled to start data transmission.

After transmitting a frame of image, the controller will automatically clear the line_cnt, pixel_cnt, and stop transmitting data.

When the edge changes of the TE are detected (The rising and falling edges are optional), the DBI_EN can be automatically enabled to restart data transmission until the software shuts down TE_EN or the screen no longer sends TE signals.

The operation process is as follows.

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI EN MODE SEL (bit30:29) of DBI_CTL_1 (0x0104) to 11 to select the TE Configure the DBI_CTL_0 register (0x0100).

Step 3 Set DBI_CTL_0[Command Type] (bit31) to 0 to set the writing command.

a) Set DBI_CTL_0[Write Command Dummy Cycles] (bit[30:20]) to configure the number of dummy cycles between commands.

b) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

c) Set DBI_CTL_0[Transmit Mode] (bit15) to 1 to select the image path.

d) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to select RGB111/444/565/666/888.

e) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

f) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Configure DBI_CTL_1[DCX_DATA] (bit22) to 0 to send the image data.

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Step 5 Configure DBI_Video_Size (0x0110) to transmit the image size.

Step 6 Configure DBI_CTL_2 (0x0108) to set the TE-related parameter.

17.12.5.4 Read Command & Read Data Transfer

Step 1 Set the SPI_DBI_MODE_SEL (bit3) of SPI_GCR (0x0004) to 1 to select DBI mode.

Step 2 Set the DBI EN MODE SEL (bit[30:29]) of DBI_CTL_1 (0x0104) to 0.

Step 3 Configure the DBI_CTL_0 register (0x0100).

a) Set DBI_CTL_0[Command Type] (bit31) to 0 to set the reading command.

b) Set DBI_CTL_0[Output Data Sequence] (bit19) to select the MSB or LSB.

c) Set DBI_CTL_0[Transmit Mode] (bit15) to 0 to select the command path.

d) Set DBI_CTL_0[Output Data Format] (bit[14:12]) to 0.

e) Set DBI_CTL_0[DBI interface Select] (bit[10:8]) to select the DBI interface type.

f) The remaining values of the DBI_CTL_0 register remain the default value.

Step 4 Configure the DBI_CTL_1 register (0x0104).

a) Configure DBI_CTL_1[DCX_DATA] (bit22) to 0 to send the command.

b) Configure DBI_CTL_1[Read_MSB_First] (bit20) to select whether the first bit of the read data is the highest or lowest bit of data.

c) Configure DBI_CTL_1[Read Data Number of Bytes] to set the byte number to be read.

d) Configure DBI_CTL_1[Read Command Dummy Cycles] to set the dummy cycle between the read command and the read data. When the dummy cycle is complete, the data starts to be sampled.

Step 5 DMA Path: Configure the SPI_FCR register (0x0018).

a) Set SPI_FCR[RF_DRQ_EN] (bit8) to 1 to enable RXFIFO DMA.

b) Set SPI_FCR[RX_TRIG_LEVEL] (bit[7:0]) to 32, which indicates the controller requests receiving data from DMA if the data of the RX FIFO is greater than 64.

CPU Path: Read data in RX FIFO from the 0x0300 address.

Step 6 Set SPI_GCR[DBI_EN] (bit4) to 1 to start transmitting command.

Step 7 Wait until DBI_INT[RD_DONE_INT] is 1. It indicates that the data is read completely.

17.12.6 Register List

Module Name	Base Address
SPI1	0x04026000

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NOTE

Only SPI_TXD and SPI_RXD support byte/half word/word access. Other registers only supports word access.

Register Name	Offset	Description
SPI_GCR	0x0004	SPI Global Control Register
SPI_TCR	0x0008	SPI Transfer Control Register
SPI_IER	0x0010	SPI Interrupt Control Register
SPI_ISR	0x0014	SPI Interrupt Status Register
SPI_FCR	0x0018	SPI FIFO Control Register
SPI_FSR	0x001C	SPI FIFO Status Register
SPI_WCR	0x0020	SPI Wait Clock Counter Register
SPI_MBC	0x0030	SPI Master Burst Counter Register
SPI_MTC	0x0034	SPI Master Transmit Counter Register
SPI_BCC	0x0038	SPI Master Burst Control Counter Register
SPI_BATCR	0x0040	SPI Bit-Aligned Transfer Configure Register
SPI_BA_CCR	0x0044	SPI Bit-Aligned CLOCK Configuration Register
SPI_TBR	0x0048	SPI TX Bit Register
SPI_RBR	0x004C	SPI RX Bit Register
SPI_NDMA_MODE_CTL	0x0088	SPI Normal DMA Mode Control Register
DBI_CTL_0	0x0100	DBI Control Register 0
DBI_CTL_1	0x0104	DBI Control Register 1
DBI_CTL_2	0x0108	DBI Control Register 2
DBI_TIMER	0x010C	DBI Timer Control Register
DBI_VIDEO_SIZE	0x0110	DBI Video Size Register
DBI_INT	0x0120	DBI Interrupt Register
DBI_DEBUG_0	0x0124	DBI Debug Register 0
DBI_DEBUG_1	0x0128	DBI Debug Register 1
SPI_SVCNR	0x0160	SPI Slave Vsync Cycle Number Register
SPI_SFHNR	0x0164	SPI Slave Frame Head Number Register
SPI_SFHLR	0x0168	SPI Slave Frame Head Low Register
SPI_SFHHR	0x016C	SPI Slave Frame Head High Register
SPI_SFHLRR	0x0170	SPI Slave Frame Head Low Receive Register
SPI_SFHHRR	0x0174	SPI Slave Frame Head High Receive Register
SPI_SIWVER	0x0178	SPI Slave Idle Wait Value Enable Register
SPI_SIWVRR	0x017C	SPI Slave Idle Wait Value Reset Register
SPI_TXD	0x0200	SPI TX Data Register
SPI_RXD	0x0300	SPI RX Data Register
SPI_BSR	0x0400	SPI BUF Status Register

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17.12.7 Register Description

17.12.7.1 0x0004 SPI Global Control Register (Default Value: 0x0000_0080)

Offset: 0x0004			Register Name: SPI_GCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	SRST Soft reset Write '1' to this bit will clear the SPI controller, and auto clear to '0' when reset operation completes Write '0' has no effect.
30:11	/	/	/
10:9	R/W	0x0	SLV_MODE Slave Mode Wire Mode 0: Single 1: Dual 2: Quad
8	R/W	0x0	SLV_DIR Slave Mode Direction 0: Slave RX 1: Reserved
7	R/W	0x1	TP_EN Transmit Pause Enable In master mode, it is used to control transmit state machine to stop smart burst sending when RX FIFO is full. 1: Stop transmitting data when RXFIFO is full. 0: Normal operation, ignore RXFIFO status. Note: When XCH=1, do not write to this bit.
6	/	/	/
5	R/W	0x0	SLAVE_MODE_SEL Slave Mode Sample Timing Mode Select 0: Old mode of Sample Timing 1: New mode of Sample Timing Note: Cannot be written when slave is transferring For Dual/Quad RX, old mode is not valid.
4	R/W	0x0	DBI_EN DBI Module Enable Control 0: Disable 1: Enable

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Offset: 0x0004			Register Name: SPI_GCR
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	SPI_DBI_MODE_SEL SPI_DBI Working Mode Select 0: SPI MODE 1: DBI MODE
2	R/W	0x0	MODE_SELEC Sample Timing Mode Select 0: Old mode of Sample Timing 1: New mode of Sample Timing Note: When XCH=1, do not write to this bit.
1	R/W	0x0	MODE SPI Function Mode Select 0: Slave Mode 1: Master Mode Note: When XCH=1, do not write to this bit.
0	R/W	0x0	EN SPI Module Enable Control 0: Disable 1: Enable Note: After transforming from bit_mode to byte_mode, the SPI Module must be enabled again.

17.12.7.2 0x0008 SPI Transfer Control Register (Default Value: 0x0000_0087)

Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	XCH Exchange Burst In master mode it is used to start SPI burst 0: Idle 1: Initiates exchange. Write “1” to this bit will start the SPI burst, and will auto clear after finishing the bursts transfer specified by BC. Write “1” to SRST will also clear this bit. Write ‘0’ to this bit has no effect. Note: When XCH=1, do not write to this bit.
30:20	/	/	/

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
19	R/W	0x0	SIWE Slave Idle Wait Enable 0: Disable 1: Enable In slave mode, VIE and SFHE should be disabled when SIWE is enabled.
18	R/W	0x0	SFHE Slave Mode frame head auto distinguish feature enable 0: Disable 1: Enable In slave mode, VIE and SIWE should be disabled when SFHE is enabled. When frame head auto distinguishing, the frame head must be written in the way of MSB first.
17	R/W	0x0	VIE Vsync Input Enable 0: Disable 1: Enable In slave mode, SPI_CS input signal can be configured into SPI_CS or Vsync.
16	R/W	0x0	VIS Vsync Input Select 0: Vsync negative edge 1: Vsync positive edge
15	R/W	0x0	SDC1 Master Sample Data Control register1 Set this bit to '1' to make the internal read sample point with a delay of half cycle of SPI_CLK. It is used in high speed read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave. 0: Normal operation, do not delay internal read sample point 1: Delay internal read sample point Note: When XCH=1, do not write to this bit.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	SDDM Sending Data Delay Mode 0: Normal sending 1: Delay sending Set the bit to"1" to make the data that should be sent with a delay of half cycle of SPI_CLK in dual IO mode for SPI mode 0.
13	R/W	0x0	SDM Master Sample Data Mode 1: Normal Sample Mode 0: Delay Sample Mode In Normal Sample Mode, SPI master samples the data at the correct edge for each SPI mode; In Delay Sample Mode, SPI master samples data at the edge that is half cycle delayed by the correct edge defined in respective SPI mode.
12	R/W	0x0	FBS First Transmit Bit Select 0: MSB first 1: LSB first Note: When XCH=1, do not write to this bit.
11	R/W	0x0	SDC Master Sample Data Control Set this bit to '1' to make the internal read sample point with a delay of half cycle of SPI_CLK. It is used in high speed read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave. 0: Normal operation, do not delay internal read sample point 1: Delay internal read sample point Note: When XCH=1, do not write to this bit.
10	R/W	0x0	RPSM Rapids mode select Select Rapids mode for high speed write. 0: Normal write mode 1: Rapids write mode Note: When XCH=1, do not write to this bit.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	DDB Dummy Burst Type 0: The bit value of dummy SPI burst is zero 1: The bit value of dummy SPI burst is one Note: When XCH=1, do not write to this bit.
8	R/W	0x0	DHB Discard Hash Burst In master mode it controls whether discarding unused SPI bursts 0: Receiving all SPI bursts in BC period 1: Discard unused SPI bursts, only fetching the SPI bursts during dummy burst period. The bursts number is specified by TC. Note: When XCH=1, do not write to this bit.
7	R/W	0x1	SS_LEVEL When control CS signal manually (SPI_CTRL_REG.SS_CTRL==1), set this bit to '1' or '0' to control the level of CS signal. 0: Set CS to low 1: Set CS to high Note: When XCH=1, do not write to this bit.
6	R/W	0x0	SS_OWNER CS Output Owner Select Usually, controller sends CS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of CS signal. 0: SPI controller 1: Software Note: When XCH=1, do not write to this bit.
5:4	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPI_CS0 will be asserted 01: SPI_CS1 will be asserted 10: SPI_CS2 will be asserted 11: SPI_CS3 will be asserted (not used) Note: When XCH=1, do not write to this bit.

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Offset: 0x0008			Register Name: SPI_TCR
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	SSCTL In master mode, this bit selects the output wave form for the SPI_CSx signal. Only valid when SS_OWNER = 0. 0: SPI_CSx remains asserted between SPI bursts 1: Negate SPI_CSx between SPI bursts Note: When XCH=1, do not write to this bit.
2	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: When XCH=1, do not write to this bit.
1	R/W	0x1	CPOL SPI Clock Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: When XCH=1, do not write to this bit.
0	R/W	0x1	CPHA SPI Clock/Data Phase Control 0: Phase 0 (Leading edge for sample data) 1: Phase 1 (Leading edge for setup data) Note: When XCH=1, do not write to this bit.

17.12.7.3 0x0010 SPI Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: SPI_IER
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W	0x0	SLV_VSYNC_ERR_EN Slave Vsync Error Enable 0: Disable 1: Enable
13	R/W	0x0	SS_INT_EN SSI Interrupt Enable Chip Select Signal (CSx) from valid state to invalid state 0: Disable 1: Enable

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Offset: 0x0010			Register Name: SPI_IER
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	TC_INT_EN Transfer Completed Interrupt Enable 0: Disable 1: Enable
11	R/W	0x0	TF_UDR_INT_EN TXFIFO under run Interrupt Enable 0: Disable 1: Enable
10	R/W	0x0	TF_OVF_INT_EN TX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
9	R/W	0x0	RF_UDR_INT_EN RXFIFO under run Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	RF_OVF_INT_EN RX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
7	/	/	/
6	R/W	0x0	TF_FUL_INT_EN TX FIFO Full Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	TX_EMP_INT_EN TX FIFO Empty Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	TX_ERQ_INT_EN TX FIFO Empty Request Interrupt Enable 0: Disable 1: Enable
3	/	/	/
2	R/W	0x0	RF_FUL_INT_EN RX Full Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	RX_EMP_INT_EN RX Empty Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0010			Register Name: SPI_IER
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	RF_RDY_INT_EN RX Ready Request Interrupt Enable 0: Disable 1: Enable

17.12.7.4 0x0014 SPI Interrupt Status Register (Default Value: 0x0000_0032)

Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1C	0x0	SVEI Slave Vsync Error Interrupt When in Slave Mode and Vsync is valid, it indicates that there is no enough input clock cycles between 2 Vsync edge. Writing 1 to this bit clears it.
13	R/W1C	0x0	SSI CS Invalid Interrupt When SSI is 1, it indicates that CS has changed from valid state to invalid state. Writing 1 to this bit clears it.
12	R/W1C	0x0	TC Transfer Completed In master mode, it indicates that all bursts specified by BC has been exchanged. In other condition, When set, this bit indicates that all the data in TXFIFO has been loaded in the Shift register, and the Shift register has shifted out all the bits. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed
11	R/W1C	0x0	TF_UDF TXFIFO under run This bit is set when if the TXFIFO is underrun. Writing 1 to this bit clears it. 0: TXFIFO is not underrun 1: TXFIFO is underrun

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Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
10	R/W1C	0x0	TF_OVF TXFIFO Overflow This bit is set when if the TXFIFO is overflow. Writing 1 to this bit clears it. 0: TXFIFO is not overflow 1: TXFIFO is overflowed
9	R/W1C	0x0	RX_UDF RXFIFO Underrun When set, this bit indicates that RXFIFO has underrun. Writing 1 to this bit clears it.
8	R/W1C	0x0	RX_OVF RXFIFO Overflow When set, this bit indicates that RXFIFO has overflowed. Writing 1 to this bit clears it. 0: RXFIFO is available. 1: RXFIFO has overflowed.
7	/	/	/
6	R/W1C	0x0	TX_FULL TXFIFO Full This bit is set when if the TXFIFO is full . Writing 1 to this bit clears it. 0: TXFIFO is not full 1: TXFIFO is Full
5	R/W1C	0x1	TX_EMP TXFIFO Empty This bit is set if the TXFIFO is empty. Writing 1 to this bit clears it. 0: TXFIFO contains one or more words. 1: TXFIFO is empty
4	R/W1C	0x1	TX_READY TXFIFO Ready 0: TX_WL > TX_TRIG_LEVEL 1: TX_WL <= TX_TRIG_LEVEL This bit is set any time if TX_WL <= TX_TRIG_LEVEL. Writing “1” to this bit clears it. Where TX_WL is the water level of RXFIFO
3	/	/	/

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Offset: 0x0014			Register Name: SPI_ISR
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x0	RX_FULL RX Full This bit is set when the RX FIFO and RX buffer are full. Writing 1 to this bit clears it. 0: Not Full 1: Full
1	R/W1C	0x1	RX_EMP RX Empty This bit is set when the RX FIFO and RX buffer are empty. Writing 1 to this bit clears it. 0: Not empty 1: Empty
0	R/W1C	0x0	RX_RDY RX Ready 0: RX_WL <= RX_TRIG_LEVEL 1: RX_WL > RX_TRIG_LEVEL This bit is set any time if RX_WL > RX_TRIG_LEVEL. Writing "1" to this bit clears it. Where RX_WL is the water level of RX FIFO and RX buffer.

17.12.7.5 0x0018 SPI FIFO Control Register (Default Value: 0x0040_0001)

Offset: 0x0018			Register Name: SPI_FCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	TX_FIFO_RST TX FIFO Reset Write '1' to this bit will reset the control portion of the TX FIFO and auto clear to '0' when completing reset operation, write to '0' has no effect.
30	R/W	0x0	TF_TEST_ENB TX Test Mode Enable 0: Disable 1: Enable In normal mode, TX FIFO can only be read by SPI controller, write '1' to this bit will switch TX FIFO read and write function to AHB bus. This bit is used to test the TX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time.
29:25	/	/	/

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Offset: 0x0018			Register Name: SPI_FCR
Bit	Read/Write	Default/Hex	Description
24	R/W	0x0	TF_DRQ_EN TX FIFO DMA Request Enable 0: Disable 1: Enable
23:16	R/W	0x40	TX_TRIG_LEVEL TX FIFO Empty Request Trigger Level
15	R/WAC	0x0	RF_RST RXFIFO Reset Write '1' to this bit will reset the control portion of the receiver FIFO, and auto clear to '0' when completing reset operation, write '0' to this bit has no effect.
14	R/W	0x0	RF_TEST RX Test Mode Enable 0: Disable 1: Enable In normal mode, RX FIFO can only be written by SPI controller, write '1' to this bit will switch RX FIFO read and write function to AHB bus. This bit is used to test the RX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time.
13:9	/	/	/
8	R/W	0x0	RF_DRQ_EN RX FIFO DMA Request Enable 0: Disable 1: Enable
7:0	R/W	0x1	RX_TRIG_LEVEL RX FIFO Ready Request Trigger Level

17.12.7.6 0x001C SPI FIFO Status Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: SPI_FSR
Bit	Read/Write	Default/Hex	Description
31	R	0x0	TB_WR TX FIFO Write Buffer Write Enable
30:24	/	/	/

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Offset: 0x001C			Register Name: SPI_FSR
Bit	Read/Write	Default/Hex	Description
23:16	R	0x0	TF_CNT TX FIFO Counter These bits indicate the number of words in TX FIFO 0: 0 byte in TX FIFO 1: 1 byte in TX FIFO ... 64 64 bytes in TX FIFO other: Reserved
15	R	0x0	RB_WR RX FIFO Read Buffer Write Enable
14:8	/	/	/
7:0	R	0x0	RF_CNT RX FIFO Counter These bits indicate the number of words in RX FIFO 0: 0 byte in RX FIFO 1: 1 byte in RX FIFO ... 64: 64 bytes in RX FIFO other: Reserved

17.12.7.7 0x0020 SPI Wait Clock Counter Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: SPI_WCR
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:16	R/W	0x0	SWC Dual mode direction switch wait clock counter (for master mode only). 0: No wait states inserted n: n SPI_CLK wait states inserted These bits control the number of wait states to be inserted before start dual data transfer in dual SPI mode. The SPI module counts SPI_CLK by SWC for delaying next word data transfer. Note: When XCH=1, do not write to this bit.

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Offset: 0x0020			Register Name: SPI_WCR
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	WCC Wait Clock Counter (In Master mode) These bits control the number of wait states to be inserted in data transfers. The SPI module counts SPI_CLK by WCC for delaying next word data transfer. 0: No wait states inserted N: N SPI_CLK wait states inserted Note: When XCH=1, do not write to this bit.

17.12.7.8 0x0030 SPI Master Burst Counter Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: SPI_MBC
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:0	R/W	0x0	MBC Master Burst Counter In master mode, this field specifies the total burst number. 0: 0 burst 1: 1 burst ... N: N bursts Note: • When XCH=1, do not write to this bit. • Total transfer data include the TXD, RXD and dummy burst.

17.12.7.9 0x0034 SPI Master Transmit Counter Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: SPI_MTC
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

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Offset: 0x0034			Register Name: SPI_MTC
Bit	Read/Write	Default/Hex	Description
23:0	R/W	0x0	MWTC Master Write Transmit Counter In master mode, this field specifies the burst number that should be sent to TXFIFO before automatically sending dummy burst. For saving bus bandwidth, the dummy burst (all zero bits or all one bits) is sent by SPI Controller automatically. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.

17.12.7.10 0x0038 SPI Master Burst Control Counter Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: SPI_BCC
Bit	Read/Write	Default/Hex	Description
31: 30	/	/	/
29	R/W	0x0	Quad_EN Quad_Mode_EN 0: Quad mode disable 1: Quad mode enable Note: • When XCH=1, do not write to this bit. • Quad mode includes Quad-Input and Quad-Output.
28	R/W	0x0	DRM Master Dual Mode RX Enable 0: RX use single-bit mode 1: RX use dual mode Note: • When XCH=1, do not write to this bit. • It is only valid when Quad_Mode_EN=0.

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Offset: 0x0038			Register Name: SPI_BCC
Bit	Read/Write	Default/Hex	Description
27:24	R/W	0x0	DBC Master Dummy Burst Counter In master mode, this field specifies the burst number that should be sent before receive in dual SPI mode. The data is don't care by the device. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.
23:0	R/W	0x0	STC Master Single Mode Transmit Counter In master mode, this field specifies the burst number that should be sent in single mode before automatically sending dummy burst. This is the first transmit counter in all bursts. 0: 0 burst 1: 1 burst ... N: N bursts Note: When XCH=1, do not write to this bit.

17.12.7.11 0x0040 SPI Bit-Aligned Transfer Configure Register (Default Value: 0x0000_00A0)

Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	TCE Transfer Control Enable In master mode, it is used to start t1o transfer the serial bits frame, it is only valid when Work Mode Select==0x2/0x3. 0: Idle 1: Initiates transfer. Write "1" to this bit will start to transfer serial bits frame (the value comes from the SPI TX Bit Register or SPI RX Bit Register), and will auto clear after the bursts transfer completely. Write '0' to this bit has no effect.

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Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
30	R/W	0x0	MSMS Master Sample Standard 0: Standard Sample Mode 1: Delay Sample Mode In Standard Sample Mode, SPI master samples the data at the standard rising edge of CLK for each SPI mode. In Delay Sample Mode, SPI master samples data at the edge that is half cycle delayed by the standard rising edge of CLK defined in respective SPI mode.
29:26	/	/	/
25	R/W1C	0x0	TBC Transfer Bits Completed When set, this bit indicates that the last bit of the serial data frame in SPI TX Bit Register (or SPI RX Bit Register) has been transferred completely. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed Note: It is only valid when Work Mode Select=0x2/0x3.
24	R/W	0x0	TBC_INT_EN Transfer Bits Completed Interrupt Enable 0: Disable 1: Enable Note: It is only valid when Work Mode Select=0x2/0x3.
23:22	/	/	/
21:16	R/W	0x00	RX_FEM_LEN Configure the length of serial data frame(burst) of RX 000000: 0bit 000001: 1bit ... 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select=0x2/0x3, and cannot be written when TCE=1.
15:14	/	/	/

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Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
13:8	R/W	0x00	TX_FEM_LEN Configure the length of serial data frame(burst) of TX 000000: 0bit 000001: 1bit ... 100000: 32bits Other values: Reserved Note: It is only valid when Work Mode Select=0x2/0x3, and cannot be written when TCE=1.
7	R/W	0x1	SS_LEVEL When control CS signal manually, set this bit to '1' or '0' to control the level of CS signal. 0: Set CS to low 1: Set CS to high Note: It is only valid when Work Mode Select=0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
6	R/W	0x0	SS_OWNER CS Output Owner Select Usually, controller sends CS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of CS signal. 0: SPI controller 1: Software Note: It is only valid when Work Mode Select=0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
5	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: It is only valid when Work Mode Select=0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
4	/	/	/

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Offset: 0x0040			Register Name: SPI_BATCR
Bit	Read/Write	Default/Hex	Description
3:2	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPI_CS0 will be asserted 01: SPI_CS1 will be asserted 10: SPI_CS2 will be asserted 11: SPI_CS3 will be asserted (not used) Note: It is only valid when Work Mode Select =0x2/0x3, and only work in Mode0, cannot be written when TCE=1.
1:0	R/W	0x0	WMS Work Mode Select 00: Data frame is byte aligned in Standard SPI, Dual-Output/Dual Input SPI, Dual IO SPI and Quad-Output/Quad-Input SPI. 01: Reserved 10: Data frame is bit aligned in 3-Wire SPI 11: Data frame is bit aligned in Standard SPI

17.12.7.12 0x0044 SPI Bit-Aligned CLOCK Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: SPI_BA_CCR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	CDR_N Clock Divide Rate (Master Mode Only) The SPI_CLK is determined according to the following equation: SPI_CLK = Source_CLK / (2*(CDR_N + 1)). Note: This register is only valid when Work Mode Select=0x2/0x3.

17.12.7.13 0x0048 SPI TX Bit Register (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: SPI_TBR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0048			Register Name: SPI_TBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VTB The Value of the Transmit Bits This register is used to store the value of the transmitted serial data frame. In the process of transmission, the LSB is transmitted first. Note: This register is only valid when Work Mode Select=0x2/0x3.

17.12.7.14 0x004C SPI RX Bit Register (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: SPI_RBR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	VRB The Value of the Receive Bits This register is used to store the value of the received serial data frame. In the process of transmission, the LSB is transmitted first. Note: This register is only valid when Work Mode Select=0x2/0x3.

17.12.7.15 0x0088 SPI Normal DMA Mode Control Register (Default Value: 0x0000_00E5)

Offset: 0x0088			Register Name: SPI_NDMA_MODE_CTL
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R/W	0x3	SPI_ACT_M SPI NDMA Active Mode 00: DMA_ACTIVE is low 01: DMA_ACTIVE is high 10: DMA_ACTIVE is controlled by DMA_REQUEST(DRQ) 11: DMA_ACTIVE is controlled by controller
5	R/W	0x1	SPI_ACK_M SPI NDMA Acknowledge Mode 0: Active fall do not care ack 1: Active fall must after detect ack is high
4:0	R/W	0x05	SPI_DMA_WAIT The counts of hold cycles from DMA last signal high to DMA_ACTIVE high.

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17.12.7.16 0x0100 DBI Control Register 0 (Default Value: 0x0010_0000)

Offset: 0x0100			Register Name: DBI_CTL_0
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	CMDT Command Type 0: Write Command 1: Read Command
30:20	R/W	0x1	WCDC Write Command Dummy Cycles Controls dummy cycles between two write commands Range 1-255 Default Condition: There is a dbi_clk cycle between each command or parameter.
19	R/W	0x0	DAT_SEQ Output Data Sequence 0: MSB First 1: LSB First
18:16	R/W	0x0	RGB_SEQ Output RGB Sequence 000: RGB 001: RBG 010: GRB 011: GBR 100: BRG 101: BGR 110-111: Reserved
15	R/W	0x0	TRAN_MOD Transmit Mode 0: Command / Parameter 1: Video
14:12	R/W	0x0	DAT_FMT Output Data Format 000: RGB111 001: RGB444 010: RGB565 011: RGB666 100: RGB888 (only for 2 Data Lane Interface) 101-111: Reserved
11	/	/	/

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Offset: 0x0100			Register Name: DBI_CTL_0
Bit	Read/Write	Default/Hex	Description
10:8	R/W	0x0	DBI_INTF DBI Interface 000: 3 Line Interface I 001: 3 Line Interface II 010: 4 Line Interface I 011: 4 Line Interface II 100: 2 Data Lane Interface
7:4	R/W	0x0	RGB_FMT RGB_Source_Format When Video Source_Type is RGB32 0000: RGB 0001: RBG 0010: GRB 0011: GBR 0100: BRG 0110: BGR Others: Reserved When Video Source_Type is RGB16 0: RGB 0001-0100: Reserved 0101: BGR 0110: GRBG_0: {G[5:3] R[4:0] B[4:0] G[2:0]} 0111: GBRG_0: {G[5:3] B[4:0] R[4:0] G[2:0]} 1000: GRBG_1: {G[2:0] R[4:0] B[4:0] G[5:3]} 1001: GBRG_1: {G[2:0] B[4:0] R[4:0] G[5:3]} Others: Reserved
3	R/W	0x0	DUM_VAL Dummy Cycle Value Output Value During Dummy Cycle
2	R/W	0x0	RGB_BO RGB Bit Order 0: Remain the sequence of RGB data 1: Swap the higher bit and the lower bit for each component of DRAM RGB
1	R/W	0x0	ELEMENT_A_POS Element A Position Only for RGB32 Data Format 0: A component is in the bit [31:24] of data source 1: A component is in the bit [7:0] of data source

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Offset: 0x0100			Register Name: DBI_CTL_0
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	VI_SRC_TYPE RGB Source Type 0: RGB32 1: RGB16

17.12.7.17 0x0104 DBI Control Register 1 (Default Value: 0x0000_0001)

Offset: 0x0104			Register Name: DBI_CTL_1
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	DBI_SOFT_TRG It is only available for software trigger mode. Write '1' to this bit will start DBI TX module and auto clear to '0' when completing start operation, write to '0' has no effect.
30:29	R/W	0x0	DBI_EN_MOD_SEL DBI Enable Mode Select 0: Always on mode 1: Software trigger mode 2: Timer trigger mode 3: Te trigger mode
28	/	/	/
27:26	R/W	0x0	RGB666_FMT 2 Data Lane RGB666 Format 00: Normal Format 01: Special Format for ILITEK 10: Special Format for New Vision
25	R/W	0x0	DBI_RXCLK_INV DBI rx clock inverse 0: Sample data by using the positive edge of the output clock 1: Sample data by using the negative edge of the output clock
24	R/W	0x0	DBI_CLKO_MOD DBI output clock mode 0: DBI clock always on(DCX Setup/hold equals one clock cycle) 1: DBI clock auto gating(DCX Setup/hold equals to a half clock cycle)

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Offset: 0x0104			Register Name: DBI_CTL_1
Bit	Read/Write	Default/Hex	Description
23	R/W	0x0	DBI_CLKO_INV DBI clock output inverse When the bit24 (DBI output clock mode) is 0. 0: The falling edge releases the CSX signal, and the falling edge releases data 1: The rising edge releases the CSX signal, and the rising edge releases data When the bit24 (DBI output clock mode) is 1. 0: The rising edge releases the CSX signal, and the falling edge releases data 1: The falling edge releases the CSX signal, and the rising edge releases data
22	R/W	0x0	DCX_DATA DCX Data Value 0: DCX Value equal to 0 1: DCX Value equal to 1
21	R/W	0x0	RGB 16 Data Source Select RGB 16 Data Source Select 0: Pixel1 is stored in the higher bit of address, and Pixel0 is stored in the lower bit of address 1: Pixel0 is stored in the higher bit of address, and Pixel1 is stored in the lower bit of address
20	R/W	0x0	RDAT_LSB Bit Order of Read Data 0: A reading data is the higher bit 1: A reading data is the lower bit
19:16	/	/	/
15:8	R/W	0x0	RCDC Read Command Dummy Cycles The dummy cycle between the read command and read data Reading 1-byte (8 bits) data has not dummy cycle.
7:0	R/W	0x1	RDBN Read Data Number of Bytes Sample Bytes data based on configuration.

17.12.7.18 0x0108 DBI Control Register 2 (Default Value: 0x0000_4000)

Offset: 0x0108			Register Name: DBI_CTL_2
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0108			Register Name: DBI_CTL_2
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	DBI_FIFO_DRQ_EN DBI FIFO DMA Request Enable 0: Disable 1: Enable
14:8	R/W	0x40	DBI_TRIG_LEVEL DBI FIFO Empty Request Trigger Level
7	/	/	/
6	R/W	0x0	DBI_SDI_OUT_SEL DBI SDI PIN Output Select The signal is used with the DBI SDI PIN Function Sel bit. 0: Output WRX (When DBI DCX PIN Function Sel = 0, the SDI pin outputs data) 1: Output DCX
5	R/W	0x0	DBI_DCX_SEL DBI DCX PIN Function Sel 0: DBI DCX Function 1: WRX(2 Data Lane Interface)
4:3	R/W	0x0	DBI_SDI_SEL DBI SDI PIN Function sel 0: DBI_SDI (Interface II) , WRX(2 Data Lane Interface) 1: DBI_TE 2: DBI_DCX 3: Reserved
2	R/W	0x0	TE_DBC_SEL TE debounce function select 0: Debounce 1: No-debounce
1	R/W	0x0	TE_TRIG_SEL TE edge trigger select 0: TE rising edge 1: TE falling edge
0	R/W	0x0	TE_EN TE enable 0: TE Disable 1: TE Enable

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17.12.7.19 0x010C DBI Timer Control Register (Default Value: 0x0000_0000)

Offset: 0x010C			Register Name: DBI_TIMER
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DBI_TM_EN DBI Timer enable 0: Enable 1: Disable
30:0	R/W	0x0	DBI_TM_VAL DBI Timer Value It is used to set the time at which the interrupt of the DBI Timer is triggered. When the Timer_EN is 1, the timer starts to count (the counting clock is CLK). When the counter reaches the target value to trigger the Timer_INT of DBI, the data will start to be sent in serially. Note: Counting will stop when the data is being sent serially. The interval between two data transfers set by DBI Timer Value is called frame blanking.

17.12.7.20 0x0110 DBI Video Size Register (Default Value: 0x01E0_0140)

Offset: 0x0110			Register Name: DBI_VIDEO_SIZE
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:16	R/W	0x1e0	V_SIZE It is used to generate the Frame int.
15:11	/	/	/
10:0	R/W	0x140	H_SIZE It is used to generate the Line int.

17.12.7.21 0x0120 DBI Interrupt Register (Default Value: 0x0000_4000)

Offset: 0x0120			Register Name: DBI_INT
Bit	Read/Write	Default/Hex	Description
31:15	/	/	/
14	R/W1C	0x1	DBI_FIFO_EMPTY_INT DBI FIFO Empty Interrupt Status 0: DBI_FIFO is not empty 1: DBI_FIFO is empty

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Offset: 0x0120			Register Name: DBI_INT
Bit	Read/Write	Default/Hex	Description
13	R/W1C	0x0	DBI_FIFO_FULL_INT DBI FIFO Full Interrupt Status 0: DBI_FIFO is not full. 1: DBI_FIFO is full.
12	R/W1C	0x0	TIMER_INT It indicates that timer has been count clk cycles to the value of DBI_Timer Register[30:0] . Writing 1 to this bit clears it. 0: Timer hasn't been achieved objective 1: Timer has been achieved objective
11	R/W1C	0x0	RD_DONE_INT It indicates that the number of byte setting in DBI_Control Register 1 [19:8] has been read. Writing 1 to this bit clears it. 0: All of data hasn't been read. 1: All of data has been read.
10	R/W1C	0x0	TE_INT It indicates that TE signal has been changed. Writing 1 to this bit clears it. 0: TE signal isn't changed. 1: TE signal has been changed.
9	R/W1C	0x0	FRAM_DONE_INT It indicates that a frame video data has been send. Writing 1 to this bit clears it. 0: A frame of video has not been send. 1: A frame of video has been send.
8	R/W1C	0x0	LINE_DONE_INT It indicates that a line of video data has been send. Writing 1 to this bit clears it. 0: A line of video data has not been send 1: A line of video data has been send
7	/	/	/
6	R/W	0x0	DBI_FIFO_EMPTY_INT_EN DBI FIFO Empty Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	DBI_FIFO_FULL_INT_EN DBI FIFO Full Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0120			Register Name: DBI_INT
Bit	Read/Write	Default/Hex	Description
4	R/W	0x0	TIMER_INT_EN Timer Interrupt Enable 0: Disable 1: Enable
3	R/W	0x0	RD_DONE_INT_EN Read Done Interrupt Enable 0: Disable 1: Enable
2	R/W	0x0	TE_INT_EN TE Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	FRAM_DONE_INT_EN Frame Done Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	LINE_DONE_INT_EN Line Done Interrupt Enable 0: Disable 1: Enable

17.12.7.22 0x0124 DBI Debug Register 0 (Default Value: 0x007F_0000)

Offset: 0x0124			Register Name: DBI_DEBUG_0
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22:16	R	0x7F	DBI_FIFO_AVAIL DBI FIFO ROOM VALID 0-127 Words
15:13	/	/	/
12	R	0x0	TE_VAL TE input value 0: TE not Trigger 1: TE Trigger
11:8	R	0x0	DBI_RXCS FSM for DBI Receive RX_BS0 - RX_BS6 , Gray - Code
7:4	R	0x0	SH_CS FSM for shifter 0000-0011: SH0-SH11

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Offset: 0x0124			Register Name: DBI_DEBUG_0
Bit	Read/Write	Default/Hex	Description
3:2	R	0x0	DBI_TXCS FSM for DBI Transmit 00: IDLE 01: SHIF 10: DUMY 11: READ
1:0	R	0x0	MEM_CS FSM for DBI Memory 00: IDLE_FRM 01: FRM_POS 10: FRM_RDY

17.12.7.23 0x0128 DBI Debug Register 1 (Default Value: 0x0000_0000)

Offset: 0x0128			Register Name: DBI_DEBUG_1
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:16	R	0x0	LCNT Line counter The number of pixel lines that are currently sent
15:12	/	/	/
11:0	R	0x0	CCNT Component counter The number of RGB components that are currently sent The field is equal to pixel_cnt *3.

17.12.7.24 0x0160 SPI Slave Vsync Cycle Number Register (Default Value: 0x0000_0000)

Offset: 0x0160			Register Name: SPI_SVCNR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	SFR Slave Frequency Relation 0: AHB frequency is the same as input SPI_CLK 1: AHB frequency is twice as much as input SPI_CLK ... N: AHB frequency is (N+1) times as much as input SPI_CLK

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Offset: 0x0160			Register Name: SPI_SVCNR
Bit	Read/Write	Default/Hex	Description
23:0	R/W	0x0	SVCN Slave Vsync Cycle Number In slave mode, when VIE is enabled, this number indicates valid cycles numbers after one valid Vsync edge. 0: 0 cycle 1: 1 cycle ... N: N cycles

17.12.7.25 0x0164 SPI Slave Frame Head Number Register (Default Value: 0x0000_0000)

Offset: 0x0164			Register Name: SPI_SFHNR
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2:0	R/W	0x0	SFHN Slave Frame Head Number In slave mode, when SLV_FRAME_HEAD_EN is enabled, SFHN is used to configure the frame head length in byte. 0: 1 byte 1: 2 byte ... 7: 8 bytes

17.12.7.26 0x0168 SPI Slave Frame Head Low Register (Default Value: 0x0000_0000)

Offset: 0x0168			Register Name: SPI_SFHLR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SFHL Slave Frame Head Low Low 4 bytes of the wanted frame head. When not all bytes are used, lower bytes are valid.

17.12.7.27 0x016C SPI Slave Frame Head High Register (Default Value: 0x0000_0000)

Offset: 0x016C			Register Name: SPI_SFHHR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x016C			Register Name: SPI_SFHHR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SFHH Slave Frame Head High High 4 bytes of the wanted frame head. When not all bytes are used, lower bytes are valid.

17.12.7.28 0x0170 SPI Slave Frame Head Low Receive Register (Default Value: 0x0000_0000)

Offset: 0x0170			Register Name: SPI_SFHLRR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SFHLR Slave Frame Head Low Receive Low 4 bytes of the received frame head. When not all bytes are used, lower bytes are valid.

17.12.7.29 0x0174 SPI Slave Frame Head High Receive Register (Default Value: 0x0000_0000)

Offset: 0x0174			Register Name: SPI_SFHHRR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SFHHR Slave Frame Head High Receive High 4 bytes of the received frame head. When not all bytes are used, lower bytes are valid.

17.12.7.30 0x0178 SPI Slave Idle Wait Value Enable Register (Default Value: 0x0000_0000)

Offset: 0x0178			Register Name: SPI_SIWVER
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	SIWVE Slave Idle Wait Value Enable Wait time in AHB cycle. The SPI slave controller will wait until enough IDLE time is detected before receiving data.

17.12.7.31 0x017C SPI Slave Idle Wait Value Reset Register (Default Value: 0x0000_0000)

Offset: 0x017C			Register Name: SPI_SIWVRR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/

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Offset: 0x017C			Register Name: SPI_SIWVRR
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	SIWVR Slave Idle Wait Value Reset Wait time in AHB cycle. The SPI slave controller will further wait until enough IDLE time is detected after enable the clock is enabled. Note: Total wait IDLE time is SIWVE + SIWVR. And SIWVR time must be larger than 2 SPI clk.

17.12.7.32 0x0200 SPI TX Data Register (Default Value: 0x0000_0000)

Offset: 0x0200			Register Name: SPI_TXD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TDATA Transmit Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are rooms in TXFIFO, one burst data is written to TXFIFO and the depth is increased by 1. In half-word accessing method, two SPI burst data are written and the TXFIFO depth is increased by 2. In word accessing method, four SPI burst data are written and the TXFIFO depth is increased by 4. Note: This address is writing-only if TF_TEST is '0', and if TF_TEST is set to '1', this address is readable and writable to test the TX FIFO through the AHB bus.

17.12.7.33 0x0300 SPI RX Data Register (Default Value: 0x0000_0000)

Offset: 0x0300			Register Name: SPI_RXD
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0300			Register Name: SPI_RXD
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	RDATA Receive Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are data in RXFIFO, the top word is returned and the RXFIFO depth is decreased by 1. In half-word accessing method, two SPI bursts are returned and the RXFIFO depth is decrease by 2. In word accessing method, the four SPI bursts are returned and the RXFIFO depth is decreased by 4. Note: This address is read-only if RF_TEST is '0', and if RF_TEST is set to '1', this address is readable and writable to test the RX FIFO through the AHB bus.

17.12.7.34 0x0400 SPI BUF Status Register (Default Value: 0x0000_0000)

Offset: 0x0400			Register Name: SPI_BSR
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:24	R	0x0	TX_SHIFT_BUFFER_CNT TX SHIFT Buffer Counter These bits indicate the number of bytes in TX Shift Buffer, which is after TX FIFO, and maximum depth is 4 level with 8 bits.
23:16	R	0x0	TB_CNT TX FIFO Write Buffer Counter These bits indicate the number of bytes in TX FIFO Write Buffer
15:8	/	/	/
7:0	R	0x0	RB_CNT RX FIFO Read Buffer Counter These bits indicate the number of bytes in RX FIFO Read Buffer

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17.13 Two Wire Interface (TWI)

17.13.1 Overview

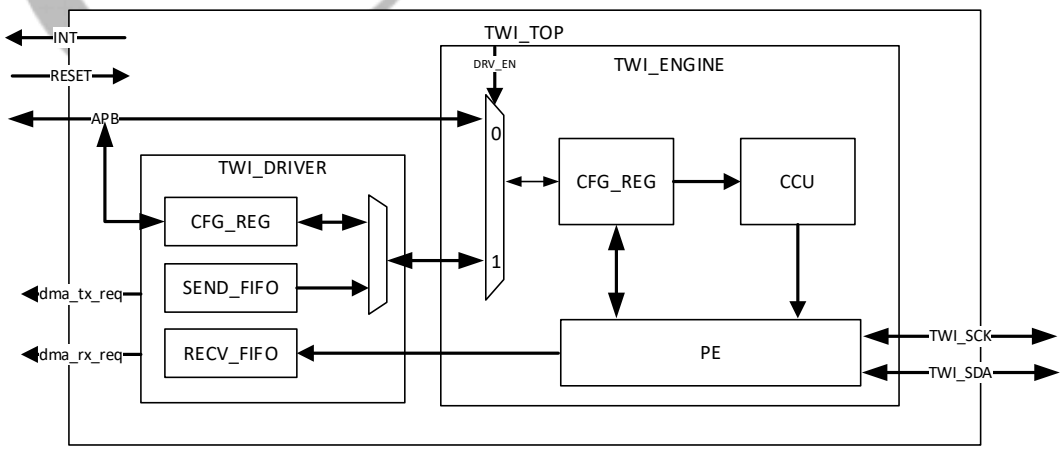
This Two Wire Interface (TWI) Controller is used to connect SoC on the 2-wire bus. It can support all the standard 2-wire bus transfer. Master mode and Slave mode are all supported. The TWI has the following features:

- Up to 6 TWI controllers
- Compliant with I2C bus standard
- Support master mode or slave mode
- Supports 7-bit standard address and 10-bit extended address.
- Supports standard mode (100 kbit/s) and high-speed mode (400 kbit/s).
- Supports general call and start byte.
- Master mode features:
  - Supports the bus arbitration in the case of multiple master devices.
  - Supports clock synchronization and bit and byte waiting.
  - Supports packet transmission and DMA
- Salve mode features:
  - Interrupt on address detection

17.13.2 Block Diagram

The following figure shows the block diagram of TWI.

Figure 17-88 TWI Block Diagram



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TWI contains the following sub-blocks:

Table 17-68 TWI Sub-blocks

Sub-block	Description
RESET	Module reset signal
INT	Module output interrupt signal
CFG_REG	Module configuration register in TWI
PE	Packet encoding/decoding
CCU	Module clock controller unit
SEND_FIFO	The register address bytes and the written data bytes are buffered in SEND_FIFO
RECV_FIFO	The read data bytes are buffered in RECV_FIFO

The controller includes TWI engine and TWI driver. Each time the TWI engine sends a START signal, a STOP signal, or a BYTE data, or a corresponding ACK, the TWI engine will generate an interrupt, and wait for the CPU to process and clear the interrupt before the next START, STOP, or BYTE, ACK transmission can be performed. Therefore, when a device communication is completed, many interrupts will be generated, and the CPU needs to wait for the previous interrupt before it can configure the next one. The TWI driver defines each communication with the device as a packet transmission. The CPU can directly configure the slave address, register address and data transmission for one or more package transmissions without waiting for interruption, then start the TWI driver, and the TWI driver can control the TWI engine to complete a pre-configured communication, and report an interrupt to the CPU after completion.

17.13.3 External Signals

The following table describes the external signals of the TWI. The TWIn-SCK and TWIn-SDA are bidirectional I/O, when the TWI is configured as a master device, the TWIn-SCK is an output pin; when the TWI is configured as a slave device, the TWIn-SCK is an input pin.

Table 17-69 TWI External Signals

Signal Name	Description	Type
TWI0-SCK	TWI0 Serial Clock Signal	I/O
TWI0-SDA	TWI0 Serial Data Signal	I/O
TWI1-SCK	TWI1 Serial Clock Signal	I/O
TWI1-SDA	TWI1 Serial Data Signal	I/O
TWI2-SCK	TWI2 Serial Clock Signal	I/O
TWI2-SDA	TWI2 Serial Data Signal	I/O
TWI3-SCK	TWI3 Serial Clock Signal	I/O
TWI3-SDA	TWI3 Serial Data Signal	I/O
TWI4-SCK	TWI4 Serial Clock Signal	I/O
TWI4-SDA	TWI4 Serial Data Signal	I/O
TWI5-SCK	TWI5 Serial Clock Signal	I/O
TWI5-SDA	TWI5 Serial Data Signal	I/O

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17.13.4 Functional Description

17.13.4.1 TWI Engine Master and Slave Mode in TWI Engine

There are four operation modes on the TWI bus which dictates the communications method. They are Master Transmit, Master Receive, Slave Transmit and Slave Receive. In general, CPU host controls TWI Engine by writing commands and data to its registers. TWI Engine transmits an interrupt to CPU when each time a byte transfer is done or a START/STOP conditions is detected. The CPU host can also poll the status register for current status if the interrupt mechanism is not disabled by the CPU host.

When the CPU host wants to start a bus transfer, it initiates a bus START to enter the master mode by setting IM_STA bit in the 2WIRE_CNTR register to high (before it must be low). The TWI Engine will assert INT line and INT_FLAG to indicate a completion for the START condition and each consequent byte transfer. At each interrupt, the micro-processor needs to check the 2WIRE_STAT register for current status. A transfer has to be concluded with STOP condition by setting M_STP bit high.

In Slave Mode, the TWI Engine also constantly samples the bus and look for its own slave address during addressing cycles. Once a match is found, it is addressed and interrupt the CPU host with the corresponding status. Upon request, the CPU host should read the status, read/write 2WIRE_DATA data register, and set the 2WIRE_CNTR control register. After each byte transfer, a slave device always halts the operation of remote master by holding the next low pulse on SCL line until the microprocessor responds to the status of previous byte transfer or START condition.

The following figures describes the write/reading timing in 7-bit standard addressing mode.

Figure 17-89 Write Timing in 7-bit Standard Addressing Mode

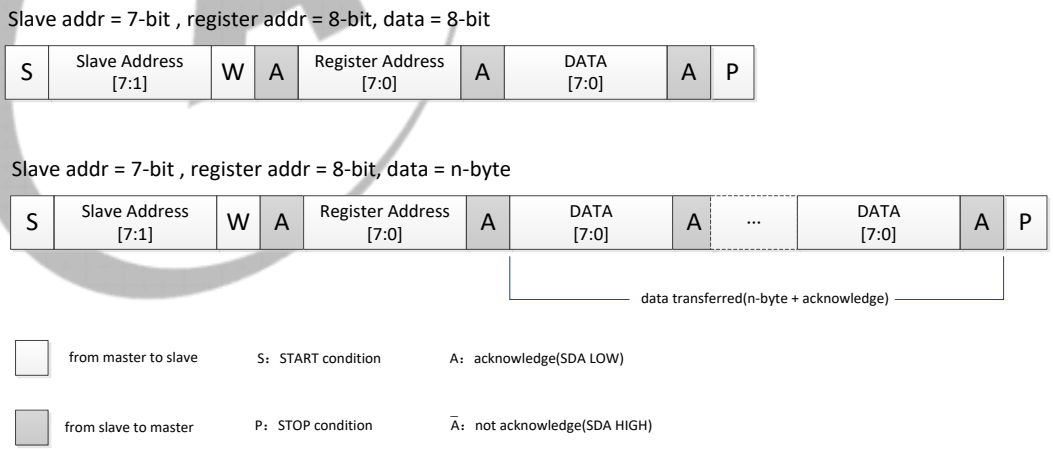

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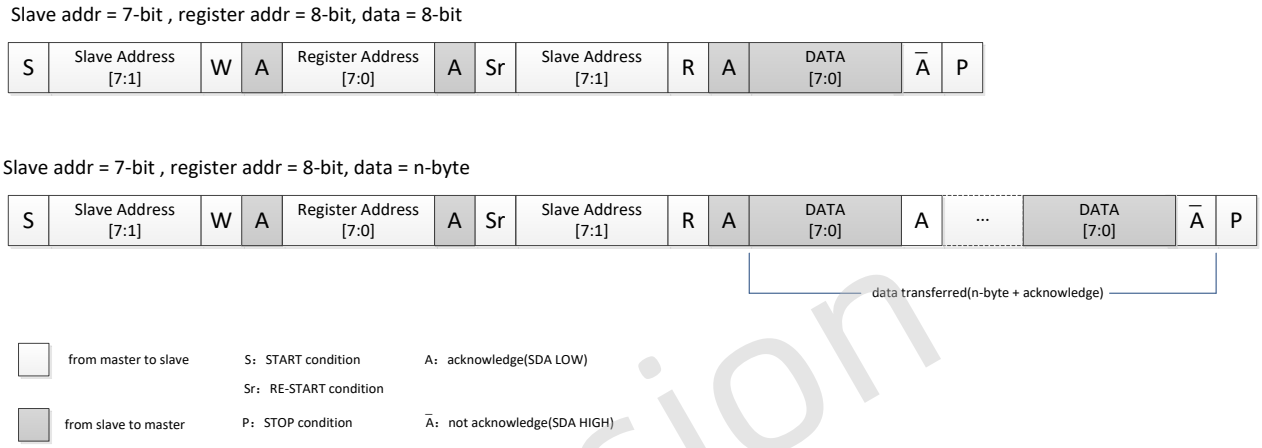
Figure 17-90 Read Timing in 7-bit Standard Addressing Mode



The following figures describe the write/read timing in 10-bit extended address mode.

Figure 17-91 Write Timing in 10-bit Extended Addressing Mode

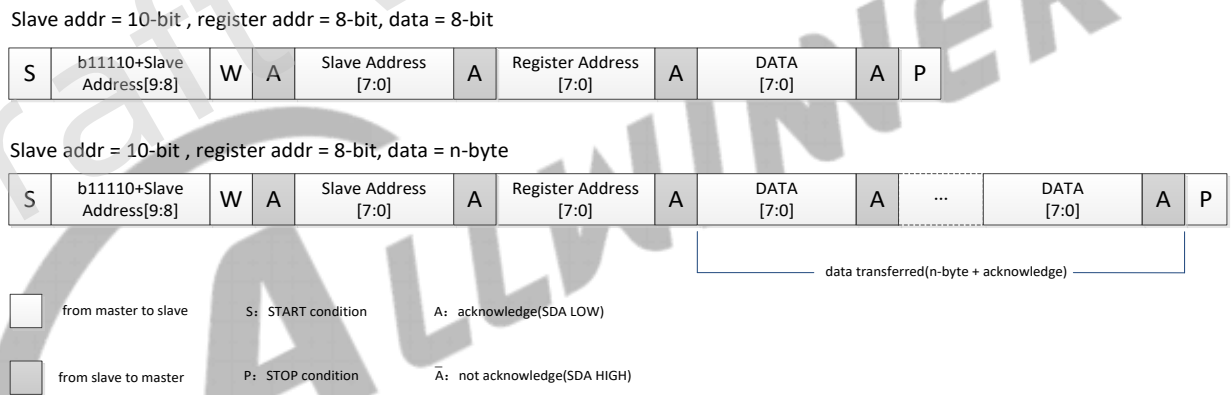
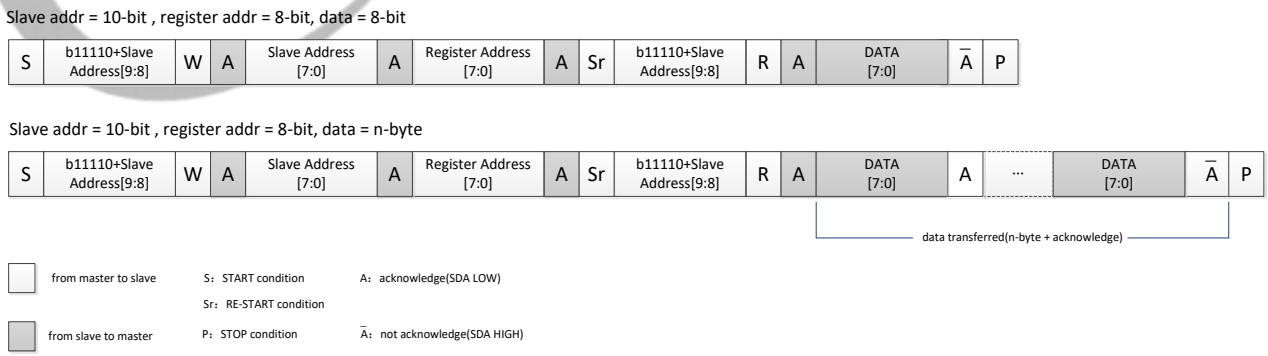


Figure 17-92 Read Timing in 10-bit Extended Addressing Mode



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17.13.4.2 Write/Read Packet Transmission in TWI Driver

The TWI driver is only supported for master mode. When the TWI works in master mode, the TWI driver drives the TWI engine for one or more packet transmission instead of the CPU host. Packet transmission is defined in the following figures. The register address bytes and the written data bytes are buffered in SEND_FIFO, the read data bytes are buffered in RECV_FIFO.

Figure 17-93 TWI Driver Write Packet Transmission

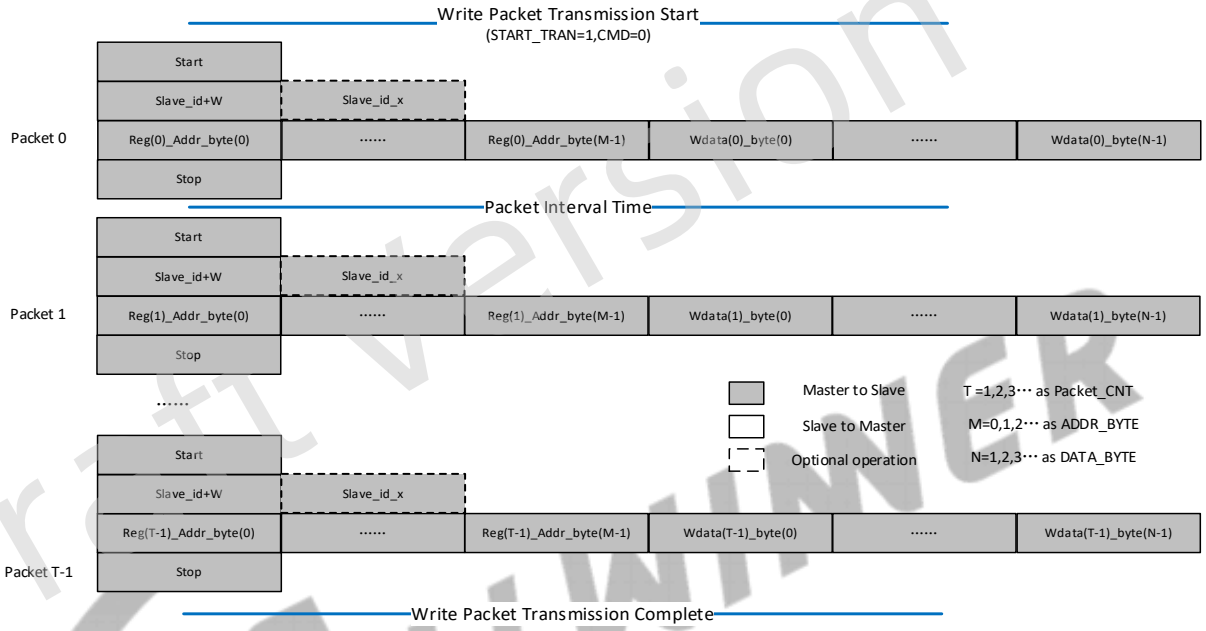

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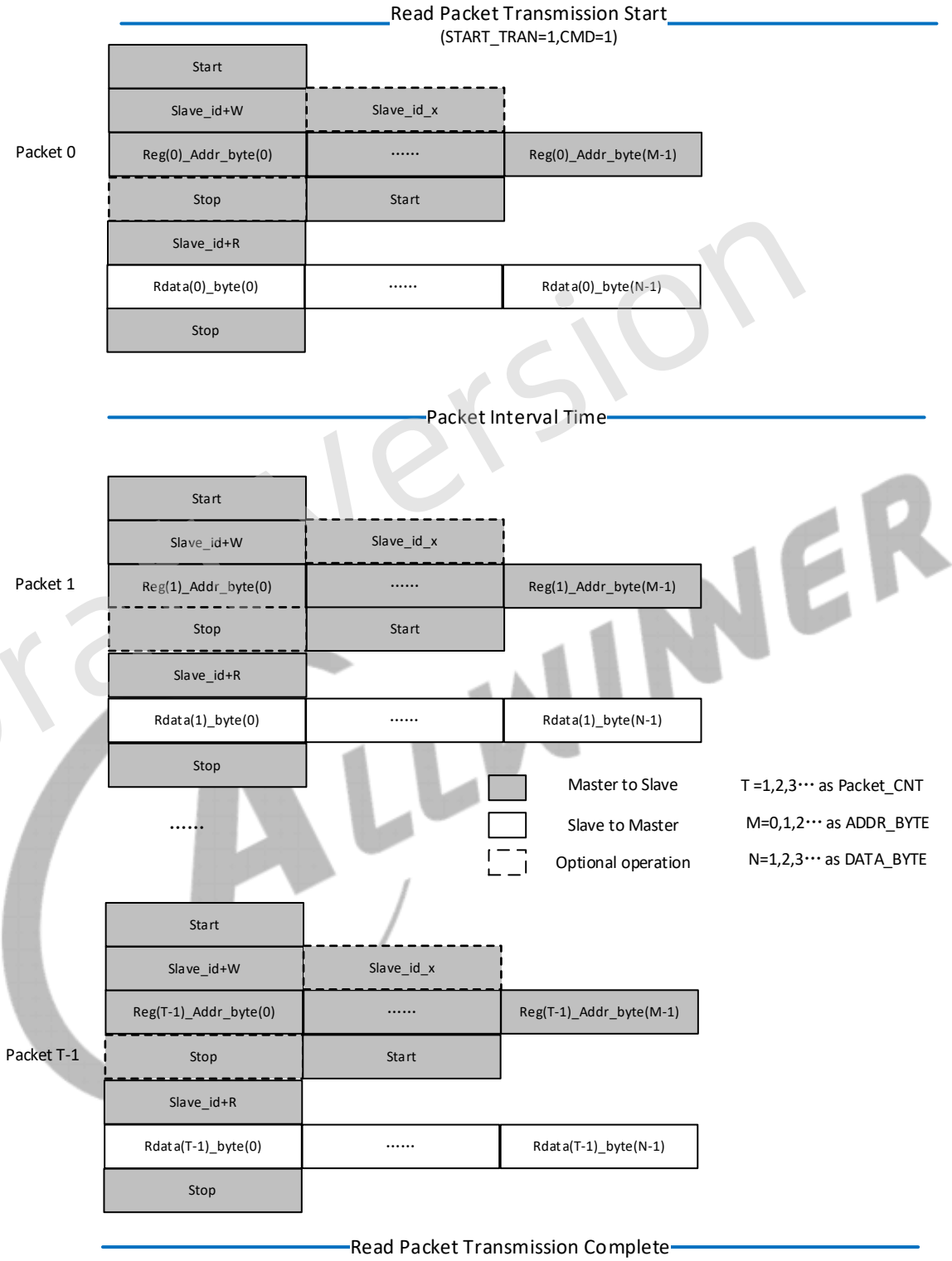
Figure 17-94 TWI Driver Read Packet Transmission



17.13.4.3 Programming State Diagram

Figure 17-95 shows the TWI programming state diagram. For the value between two states, see the TWI_STAT register.

M_SEND_S: master sends START signal;

M_SEND_ADDR: master sends slave address;

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M_SEND_XADD: master sends slave extended address;

M_SEND_SR: master repeated start;

M_SEND_DATA: master sends data;

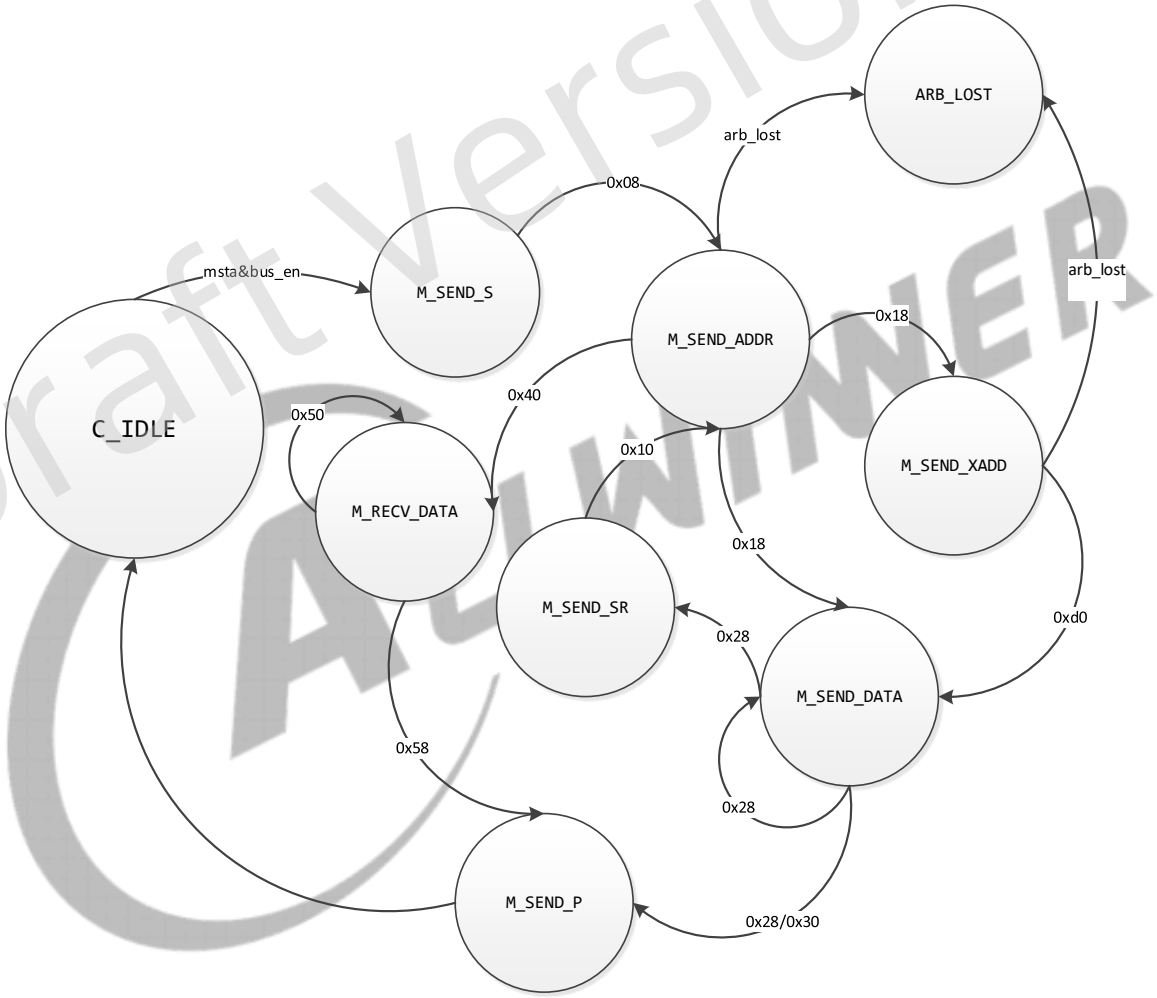
M_SEND_P: master sends STOP signal;

M_RECV_DATA: master receives data;

ARB_LOST: Arbitration lost;

C_IDLE: Idle.

Figure 17-95 TWI Programming State Diagram



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17.13.5 Programming Guidelines

The TWI controller operates in an 8-bit data format. The data on the TWI_SDA line is always 8 bits long. At first, the TWI controller sends a start condition. When in the addressing formats of 7-bit, the TWI sends out an 8-bit message which includes 7 MSB slave address and 1 LSB read/write flag. The least significant of the salve address indicates the direction of transmission. When the TWI works in 10-bit slave address mode, the operation will be divided into two steps, for details on the operation, refer to register description in Section 17.13.7.1 and 17.13.7.2.

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17.13.5.1 TWI Engine Initialization

Follow the steps below to initialize the TWI engine.

Step 1 Configure the corresponding GPIO multiplex function as TWI mode.

Step 2 For TWIn, set TWIn_GAR_REG[TWIn_APB_CLK_EN] in CCU module to 0 to close TWIn clock.

Step 3 For TWIn, set TWIn_GAR_REG [TWIn_RST_N] in CCU module to 0, then set to 1 to reset TWIn.

Step 4 For TWIn, set TWIn_GAR_REG [TWIn_APB_CLK_EN] in CCU module to 1 to open TWIn clock.

Step 5 Configure TWI_CCR[CLK_M] and TWI_CCR[CLK_N] to get the needed rate (The clock source of TWI is from APB1).

Step 6 Configure TWI_CNTR[BUS_EN] and TWI_CNTR[A_ACK]. When using interrupt mode, set TWI_CNTR[INT_EN] to 1, and register the system interrupt through GIC module. In slave mode, configure TWI_ADDR and TWI_XADDR registers to finish TWI initialization configuration.

17.13.5.2 Writing Data Operation in TWI Engine

Follow the steps below to write data to the device.

Step 1 Clear TWI_EFR register, and configure TWI_CNTR[M_STA] to 1 to transmit the START signal.

Step 2 After the START signal is transmitted, the first interrupt is triggered, then write device ID to TWI_DATA (For a 10-bit device ID, firstly write the first byte ID, secondly write the second byte ID in the next interrupt).

Step 3 The Interrupt is triggered again after device ID transmission completes, write device data address to be read to TWI_DATA (For a 16-bit address, firstly write the first-byte address, secondly write the second-byte address).

Step 4 Interrupt is triggered after data address transmission completes, write data to be transmitted to TWI_DATA (For consecutive write data operation, every byte transmission completion triggers interrupt, during interrupt write the next byte data to TWI_DATA).

Step 5 After transmission completes, write TWI_CNTR[M_STP] to 1 to transmit the STOP signal and end this write-operation.

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17.13.5.3 Reading Data Operation in TWI Engine

Follow the steps below to read data from the device.

Step 1 Clear TWI_EFR register, and set TWI_CNTR[A_ACK] to 1, and configure TWI_CNTR[M_STA] to 1 to transmit the START signal.

Step 2 After the START signal is transmitted, the first interrupt is triggered, then write device ID to TWI_DATA (For a 10-bit device ID, firstly write the first-byte ID, secondly write the second-byte ID in the next interrupt).

Step 3 The Interrupt is triggered again after device ID transmission completes, write device data address to be read to TWI_DATA (For a 16-bit address, firstly write the first-byte address, secondly write the second-byte address).

Step 4 The Interrupt is triggered after data address transmission completes, write TWI_CNTR[M_STA] to 1 to transmit new START signal, and after interrupt triggers, write device ID to TWI_DATA to start read-operation.

Step 5 After device address transmission completes, each receive completion will trigger an interrupt, in turn, read TWI_DATA to get data. When receiving the previous interrupt of the last byte data, clear [A_ACK] to stop acknowledge signal of the last byte.

Step 6 Write TWI_CNTR[M_STP] to 1 to transmit the STOP signal and end this read-operation.

17.13.5.4 TWI Driver Initialization

Follow the steps below to initialize the TWI driver.

Step 1 Configure corresponding GPIO multiplex function as TWI mode.

Step 2 For TWIn, set TWIn_GAR_REG[TWIn_APB_CLK_EN] in CCU module to 0 to close TWIn clock.

Step 3 For TWIn, set TWIn_GAR_REG [TWIn_RST_N] in CCU module to 0, then set to 1 to reset TWIn.

Step 4 For TWIn, set TWIn_GAR_REG [TWIn_APB_CLK_EN] in CCU module to 1 to open TWIn clock.

Step 5 Set TWI_DRV_CTRL [TWI_DRV_EN] to 1 to enable the TWI driver.

Step 6 Configure TWI_DRV_BUS_CTRL[CLK_M] and TWI_DRV_BUS_CTRL[CLK_N] to get the needed rate (The clock source of TWI is from APB1).

Step 7 Set TWI_DRV_CTRL[RESTART_MODE] to 0 and [READ_TRAN_MODE] to 1, and set TWI_DRV_INT_CTRL[TRAN_COM_INT_EN] to 1.

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Step 8 When using DMA for data transmission, set TWI_DRV_DMA_CFG[DMA_RX_EN] and TWI_DRV_DMA_CFG[DMA_TX_EN] to 1, and configure TWI_DRV_DMA_CFG[RX_TRIG] and TWI_DRV_DMA_CFG[TX_TRIG] to set the thresholds of RXFIFO and TXFIFO.

17.13.5.5 Writing Packet Transmission for TWI Driver

Follow the steps below to write package to the device.

Step 1 Configure TWI_DRV_SLV[SLV_ID] to set the device ID, and configure TWI_DRV_SLV[CMD] to 0 to set the write operation.

Step 2 Configure TWI_DRV_FMT[ADDR_BYTE] according to the address width of the device register, and TWI_DRV_FMT[DATA_BYTE] according to the written data count in a packet.

Step 3 Configure TWI_DRV_CFG[PACKET_CNT] to set the written packet number.

Step 4 Configure DMA channel, including TWI TXFIFO, device register address, and the written data.

Step 5 Set [START_TRAN] to 1 to start TWI Driver transmission.

Step 6 When TWI driver transmission completes, the interrupt is triggered, it indicates that the write packet transmission ends.

17.13.5.6 Reading Packet Transmission for TWI Driver

To read package from the device, perform the following steps:

Step 1 Configure TWI_DRV_SLV[SLV_ID] to set the device ID, and configure TWI_DRV_SLV[CMD] to 1 to set the read operation.

Step 2 Configure TWI_DRV_FMT[ADDR_BYTE] according to the address width of the device register, and TWI_DRV_FMT[DATA_BYTE] according to the read data count in a packet.

Step 3 Configure TWI_DRV_CFG[PACKET_CNT] to set the read packet number.

Step 4 Configure DMA channel, including TWI TXFIFO, TWI RXFIFO, device register address and the read data.

Step 5 Set [START_TRAN] to 1 to start TWI Driver transmission.

Step 6 When TWI driver transmission completes, the interrupt is triggered, it indicates that the read packet transmission ends.

17.13.5.7 Slave Mode Configuration

Step 1 Configure TWI_CNTR as 0xC4 to enable A_ACK, BUS_EN, and INT_EN.

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Step 2 Set the TWI_ADDR (bit[7:1]) and TWI_XADDR (bit[7:0]) based on the device address bit width.

17.13.6 Register List

Module Name	Base Address	Comments
TWI0	0x02510000	
TWI1	0x02511000	TWI1 register is the same with TWI0.
TWI2	0x02512000	TWI2 register is the same with TWI0.
TWI3	0x02513000	TWI3 register is the same with TWI0.
TWI4	0x02514000	TWI4 register is the same with TWI0.
TWI5	0x02515000	TWI5 register is the same with TWI0.

Register Name	Offset	Description
TWI_ADDR	0x0000	TWI Slave Address Register
TWI_XADDR	0x0004	TWI Extend Address Register
TWI_DATA	0x0008	TWI Data Register
TWI_CNTR	0x000C	TWI Control Register
TWI_STAT	0x0010	TWI Status Register
TWI_CCR	0x0014	TWI Clock Register
TWI_SRST	0x0018	TWI Soft Reset Register
TWI_EFR	0x001C	TWI Enhance Feature Register
TWI_LCR	0x0020	TWI Line Control Register
TWI_DRV_CTRL	0x0200	TWI_DRV Control Register
TWI_DRV_CFG	0x0204	TWI_DRV Transmission Configuration Register
TWI_DRV_SLV	0x0208	TWI_DRV Slave ID Register
TWI_DRV_FMT	0x020C	TWI_DRV Packet Format Register
TWI_DRV_BUS_CTRL	0x0210	TWI_DRV Bus Control Register
TWI_DRV_INT_CTRL	0x0214	TWI_DRV Interrupt Control Register
TWI_DRV_DMA_CFG	0x0218	TWI_DRV DMA Configure Register
TWI_DRV_FIFO_CON	0x021C	TWI_DRV FIFO Content Register
TWI_DRV_SEND_FIFO_ACC	0x0300	TWI_DRV Send Data FIFO Access Register
TWI_DRV_RECV_FIFO_ACC	0x0304	TWI_DRV Receive Data FIFO Access Register

17.13.7 Register Description

17.13.7.1 0x0000 TWI Slave Address Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: TWI_ADDR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x0000			Register Name: TWI_ADDR
Bit	Read/Write	Default/Hex	Description
7:1	R/W	0x0	SLA Slave Address For 7-bit addressing, the bit [7:1] indicates: SLA6, SLA5, SLA4, SLA3, SLA2, SLA1, SLA0 For 10-bit addressing, the bit [7:1] indicates: 1, 1, 1, 1, 0, SLAX[9:8]
0	R/W	0x0	GCE General Call Address Enable 0: Disable 1: Enable

NOTE

For 7-bit addressing:

SLA6 (bSLA0 is the 7-bit address of the TWI in slave mode. When the TWI receives this address after a START condition, it will generate an interrupt and enter slave mode. (SLA6 corresponds to the first bit received from the TWI bus.) If GCE is set to 1, the TWI will also recognize the general call address (00h).

For 10-bit addressing:

When the address received starts with b 11110, the TWI recognizes b 11110 as the first part of a 10-bit address and if the next two bits match ADDR [2:1] (For example, SLAX9 and SLAX8 for the extended address of the device), it sends an ACK. (The device does not generate an interrupt at this point.) If the next byte of the address matches the XADDR register (SLAX7 (bSLAX0), the TWI generates an interrupt and goes into slave mode.

17.13.7.2 0x0004 TWI Extend Address Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: TWI_XADDR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	SLAX Extend Slave Address SLAX[7:0]

17.13.7.3 0x0008 TWI Data Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: TWI_DATA
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x0008			Register Name: TWI_DATA
Bit	Read/Write	Default/Hex	Description
7:0	R/W	0x0	TWI_DATA Data byte transmitted or received

17.13.7.4 0x000C TWI Control Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: TWI_CNTR
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	INT_EN Interrupt Enable 0: The interrupt line always low 1: The interrupt line will go high when INT_FLAG is set.
6	R/W	0x0	BUS_EN TWI Bus Enable 0: The TWI bus SDA/SCL is ignored and the TWI controller will not respond to any address on the bus. 1: The TWI will respond to call to its slave address (b) and to the general call address if the GCE bit in the ADDR register is set. Note: In master operation mode, this bit should be set to '1'.
5	R/WAC	0x0	M_STA Master Mode Start When the M_STA is set to '1', the TWI controller enters master mode and will transmit a START condition on the bus when the bus is free. If the M_STA bit is set to '1' when the TWI controller is already in master mode and one or more bytes have been transmitted, then a repeated START condition will be sent. If the M_STA bit is set to '1' when the TWI is accessed in slave mode, the TWI will complete the data transfer in slave mode then enter master mode when the bus has been released. The M_STA bit is cleared automatically after a START condition has been sent. Writing a '0' to this bit has no effect.

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Offset: 0x000C			Register Name: TWI_CNTR
Bit	Read/Write	Default/Hex	Description
4	R/WAC	0x0	M_STP Master Mode Stop If the M_STP is set to '1' in master mode, a STOP condition is transmitted on the TWI bus. If the M_STP bit is set to '1' in slave mode, the TWI will indicate if a STOP condition has been received, but no STOP condition will be transmitted on the TWI bus. If both M_STA and M_STP bits are set, the TWI will first transmit the STOP condition (if in master mode), then transmit the START condition. The M_STP bit is cleared automatically. Writing a 0 to this bit has no effect.
3	R/W1C	0x0	INT_FLAG Interrupt Flag The INT_FLAG is automatically set to '1' when any of the 28 (out of the possible 29) states is entered (see 'STAT Register' below). The state that does not set INT_FLAG is state F8h. If the INT_EN bit is set, the interrupt line goes high when INT_FLAG is set to '1'. If the TWI is operating in slave mode, the data transfer is suspended when INT_FLAG is set and the low period of the TWI bus clock line (SCL) is stretched until '1' is written to INT_FLAG. The TWI clock line is then released and the interrupt line goes low.

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Offset: 0x000C			Register Name: TWI_CNTR
Bit	Read/Write	Default/Hex	Description
2	R/W	0x0	A_ACK Assert Acknowledge When A_ACK is set to '1', an Acknowledge (low level on SDA) will be sent during the acknowledge clock pulse on the TWI bus if: (1). Either the whole of a matching 7-bit slave address or the first or the second byte of a matching 10-bit slave address has been received. (2). The general call address has been received and the GCE bit in the ADDR register is set to '1'. (3). A data byte has been received in master or slave mode. When A_ACK is '0', a Not Acknowledge (high level on SDA) will be sent when a data byte is received in master or slave mode. If A_ACK is cleared to '0' in slave transmitter mode, the byte in the DATA register is assumed to be the last byte . After this byte has been transmitted, the TWI will enter state C8h then return to the idle state (status code F8h) when INT_FLAG is cleared. The TWI will not respond as a slave unless A_ACK is set.
1	/	/	/
0	R/W	0x0	CLK_COUNT_MODE 0: scl clock high period count on oscl 1: scl clock high period count on iscl

17.13.7.5 0x0010 TWI Status Register (Default Value: 0x0000_00F8)

Offset: 0x0010			Register Name: TWI_STAT
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0xF8	STA Status Information Byte Code Status 0x00: Bus error 0x08: START condition transmitted 0x10: Repeated START condition transmitted 0x18: Address + Write bit transmitted, ACK received 0x20: Address + Write bit transmitted, ACK not received

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Offset: 0x0010			Register Name: TWI_STAT
Bit	Read/Write	Default/Hex	Description
			0x28: Data byte transmitted in master mode, ACK received 0x30: Data byte transmitted in master mode, ACK not received 0x38: Arbitration lost in address or data byte 0x40: Address + Read bit transmitted, ACK received 0x48: Address + Read bit transmitted, ACK not received 0x50: Data byte received in master mode, ACK transmitted 0x58: Data byte received in master mode, not ACK transmitted 0x60: Slave address + Write bit received, ACK transmitted 0x68: Arbitration lost in the address as master, slave address + Write bit received, ACK transmitted 0x70: General Call address received, ACK transmitted 0x78: Arbitration lost in the address as master, General Call address received, ACK transmitted 0x80: Data byte received after slave address received, ACK transmitted 0x88: Data byte received after slave address received, not ACK transmitted 0x90: Data byte received after General Call received, ACK transmitted 0x98: Data byte received after General Call received, not ACK transmitted 0xA0: STOP or repeated START condition received in slave mode 0xA8: Slave address + Read bit received, ACK transmitted 0xB0: Arbitration lost in the address as master, slave address + Read bit received, ACK transmitted 0xB8: Data byte transmitted in slave mode, ACK received 0xC0: Data byte transmitted in slave mode, ACK not received 0xC8: The Last byte transmitted in slave mode, ACK received 0xD0: Second Address byte + Write bit transmitted, ACK received 0xD8: Second Address byte + Write bit transmitted,

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Offset: 0x0010			Register Name: TWI_STAT
Bit	Read/Write	Default/Hex	Description
			ACK not received 0xF8: No relevant status information, INT_FLAG=0 0xF9: Data is being transmitted normally. Others: Reserved

17.13.7.6 0x0014 TWI Clock Register (Default Value: 0x0000_0080)

Offset: 0x0014			Register Name: TWI_CCR
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	CLK_DUTY_30_EN Enabling duty cycle 30% of Clock as Master 0: Bit [7] takes effect 1: 30% Configure the duty cycle as 30% when the rate is 400 kbit/s
7	R/W	0x1	CLK_DUTY Setting duty cycle of clock as master 0: 50% 1: 40% Configure the duty cycle as 50% when the rate is 100 kbit/s
6:3	R/W	0x0	CLK_M Clock M setting

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Offset: 0x0014			Register Name: TWI_CCR
Bit	Read/Write	Default/Hex	Description
2:0	R/W	0x0	CLK_N The TWI bus is sampled by the TWI at the frequency defined by F0: Fsamp = F0 = Fin/2^CLK_N The TWI SCL output frequency, in master mode, is F1/10: F1 = F0/ (CLK_M + 1) Fscl = F1/10 = Fin/ (2^CLK_N * (CLK_M + 1) *10) Specially, Fscl = F1/11 when CLK_M=0 and CLK_DUTY=40% due to the delay of SCL sample debounce. For Example: Fin = 24 MHz (APB clock input) For 400 kHz full speed 2-wire, CLK_N = 1, CLK_M = 2 F0 = 24 MHz/2^1 = 12 MHz, F1 = F0/ (10*(2+1)) = 0.4 MHz For 100 kHz standard speed 2-wire, CLK_N = 1, CLK_M = 11 F0 = 24 MHz/2^1 = 12 MHz, F1 = F0/ (10*(11+1)) = 0.1 MHz In the slave mode, the following is the recommended configuration: When the APB is 100 MHz and TWI is 100K, CLK_N is 4. When the APB is 100 MHz and TWI is 400K, CLK_N is 2. When the APB is 24 MHz and TWI is 100K, CLK_N is 2. When the APB is 24 MHz and TWI is 400K, CLK_N is 1.

17.13.7.7 0x0018 TWI Soft Reset Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: TWI_SRST
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/WAC	0x0	SOFT_RST Soft Reset Write '1' to this bit to reset the TWI and clear to '0' when completing Soft Reset operation.

17.13.7.8 0x001C TWI Enhance Feature Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: TWI_EFR
Bit	Read/Write	Default/Hex	Description

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Offset: 0x001C			Register Name: TWI_EFR
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1:0	R/W	0x0	DBN Data Byte Number Follow Read Command Control 00: No data byte can be written after the read command 01: Only 1-byte data can be written after the read command 10: 2-bytes data can be written after the read command 11: 3-bytes data can be written after the read command

17.13.7.9 0x0020 TWI Line Control Register (Default Value: 0x0000_003A)

Offset: 0x0020			Register Name: TWI_LCR
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R	0x1	SCL_STATE Current State of TWI_SCL 0: Low 1: High
4	R	0x1	SDA_STATE Current State of TWI_SDA 0: Low 1: High
3	R/W	0x1	SCL_CTL TWI_SCL Line State Control Bit When the line control mode is enabled (bit [2] is set), this bit decides the output level of TWI_SCL. 0: Output low level 1: Output high level
2	R/W	0x0	SCL_CTL_EN TWI_SCL Line State Control Enable When this bit is set, the state of TWI_SCL is controlled by the value of bit [3]. 0: Disable TWI_SCL line control mode 1: Enable TWI_SCL line control mode

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Offset: 0x0020			Register Name: TWI_LCR
Bit	Read/Write	Default/Hex	Description
1	R/W	0x1	SDA_CTL TWI_SDA Line State Control Bit When the line control mode is enabled (bit [0] is set), this bit decides the output level of TWI_SDA. 0: Output low level 1: Output high level
0	R/W	0x0	SDA_CTL_EN TWI_SDA Line State Control Enable When this bit is set, the state of TWI_SDA is controlled by the value of bit [1]. 0: Disable TWI_SDA line control mode 1: Enable TWI_SDA line control mode

17.13.7.10 0x0200 TWI_DRV Control Register (Default Value: 0x00F8_0040)

Offset: 0x0200			Register Name: TWI_DRV_CTRL
Bit	Read/Write	Default/Hex	Description
31	R/WAC	0x0	START_TRAN Start transmission 0: Transmission idle 1: Start transmission Automatically cleared to '0' when finished. If the slave is not responding for the expected status over the time defined by TIMEOUT, the current transmission will stop. All setting formats and data will be loaded from registers and FIFO when the transmission starts.
30	/	/	/
29	R/W	0x0	RESTART_MODE Restart mode 0: RESTART 1: STOP+START Define the TWI_DRV action after sending the register address.
28	R/W	0x0	READ_TRAN_MODE Read transition mode 0: Send slave_id+W 1: Not send slave_id+W Setting this bit to 1 if reading from a slave in which the register width is equal to 0.

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Offset: 0x0200			Register Name: TWI_DRV_CTRL
Bit	Read/Write	Default/Hex	Description
27:24	R	0x0	TRAN_RESULT Transition result 000: OK 001: FAIL Other: Reserved
23:16	R	0xF8	TWI_STA TWI status 0x00: Bus error 0x08: START condition transmitted 0x10: Repeated START condition transmitted 0x18: Address + Write bit transmitted, ACK received 0x20: Address + Write bit transmitted, ACK not received 0x28: Data byte transmitted in master mode, ACK received 0x30: Data byte transmitted in master mode, ACK not received 0x38: Arbitration lost in address or data byte 0x40: Address + Read bit transmitted, ACK received 0x48: Address + Read bit transmitted, ACK not received 0x50: Data byte received in master mode, ACK received 0x58: Data byte received in master mode, ACK not received 0x01: Timeout when sending the 9th SCL clock When TWI_STA=0x01 (Timeout), the code status can be checked in the TWI_STAT register. Other: Reserved
15:2	R/W	0x10	TIMEOUT_N Timeout number When sending the 9th clock, assert fail signal when the slave device does not respond after N*F_{SCL} cycles. And the software must do a reset to the TWI_DRV module and send a stop condition to slave.
1	R/W	0x0	SOFT_RESET Software reset 0: Normal 1: Reset

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Offset: 0x0200			Register Name: TWI_DRV_CTRL
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	TWI_DRV_EN TWI driver enable 0: Module disable 1: Module enable (only use in TWI Master Mode)

17.13.7.11 0x0204 TWI_DRV Transmission Configuration Register (Default Value: 0x0000_0001)

Offset: 0x0204			Register Name: TWI_DRV_CFG
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PKT_INTERVAL Define the interval between each packet for PKT_INTERVAL F_{SCL} cycles.
15:0	R/W	0x1	PACKET_CNT The FIFO data is transmitted as PACKET_CNT packets in current format.

17.13.7.12 0x0208 TWI_DRV Slave ID Register (Default Value: 0x0000_0000)

Offset: 0x0208			Register Name: TWI_DRV_SLV
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:9	R/W	0x0	SLV_ID Slave device ID • 7-bit addressing: SLA6, SLA5, SLA4, SLA3, SLA2, SLA1, SLA0 • 10-bit addressing: 1, 1, 1, 1, 0, SLAX[9:8]
8	R/W	0x0	CMD R/W operation to slave device 0: Write 1: Read
7:0	R/W	0x0	SLV_ID_X SLAX [7:0] The low 8 bits for slave device ID with 10-bit addressing.

17.13.7.13 0x020C TWI_DRV Packet Format Register (Default Value: 0x0001_0001)

Offset: 0x020C			Register Name: TWI_DRV_FMT
Bit	Read/Write	Default/Hex	Description

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Offset: 0x020C			Register Name: TWI_DRV_FMT
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x1	ADDR_BYTE How many bytes be sent as slave device reg address 0-255
15:0	R/W	0x1	DATA_BYTE How many bytes be sent/received as data 1-65535

17.13.7.14 0x0210 TWI_DRV Bus Control Register (Default Value: 0x0000_80C0)

Offset: 0x0210			Register Name: TWI_DRV_BUS_CTRL
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	CLK_DUTY_30_EN Enabling duty cycle 30% of Clock as Master 0: Bit [15] takes effect 1: 30% The duty cycle is configured as 30% when the rate is 400 kbit/s.
16	W	0x0	CLK_COUNT_MODE Clock count mode 0: scl clock high period count on oscl 1: scl clock high period count on iscl
15	R/W	0x1	CLK_DUTY Setting duty cycle of clock as Master 0: 50% 1: 40% The duty cycle is configured as 50% when the rate is 100 kbit/s.
14:12	R/W	0x0	CLK_N TWI_DRV bus sampling clock F0=24MHz/2^CLK_N
11:8	R/W	0x0	CLK_M TWI_DRV output SCL frequency is F_{SCL}=F1/10=(F0/(CLK_M+1))/10 Specially, Foscl = F1/11 when CLK_M=0 and CLK_DUTY=40% due to the delay of SCL sample debounce.
7	R	0x1	SCL_STA SCL current status
6	R	0x1	SDA_STA SDA current status

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Offset: 0x0210			Register Name: TWI_DRV_BUS_CTRL
Bit	Read/Write	Default/Hex	Description
5:4	/	/	/
3	R/W	0x0	SCL_MOV SCL manual output value
2	R/W	0x0	SDA_MOV SDA manual output value
1	R/W	0x0	SCL_MOE SCL manual output enable
0	R/W	0x0	SDA_MOE SDA manual output enable

17.13.7.15 0x0214 TWI_DRV Interrupt Control Register (Default Value: 0x0000_0054)

Offset: 0x0214			Register Name: TWI_DRV_INT_CTRL
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W	0x0	RF_OVF_INT_EN RECV FIFO Overflow Interrupt Enable 0: Disable 1: Enable
26	R/W	0x0	RF_UDF_INT_EN RECV_FIFO under run Interrupt Enable 0: Disable 1: Enable
25	R/W	0x0	SF_OVF_INT_EN SEND FIFO Overflow Interrupt Enable 0: Disable 1: Enable
24	R/W	0x0	SF_UDF_INT_EN SEND_FIFO under run Interrupt Enable 0: Disable 1: Enable
23	R/W	0x0	RF_FULL_INT_EN RECV_FIFO Full Interrupt Enable 0: Disable 1: Enable
22	R/W	0x0	RF_EMP_INT_EN RECV_FIFO Empty Interrupt Enable 0: Disable 1: Enable

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Offset: 0x0214			Register Name: TWI_DRV_INT_CTRL
Bit	Read/Write	Default/Hex	Description
21	R/W	0x0	SF_FULL_INT_EN SEND_FIFO Full Interrupt Enable 0: Disable 1: Enable
20	R/W	0x0	SF_EMP_INT_EN SEND_FIFO Empty Interrupt Enable 0: Disable 1: Enable
19	R/W	0x0	RX_REQ_INT_EN If set, an interrupt is sent when RX_REQ_PD sets.
18	R/W	0x0	TX_REQ_INT_EN If set, an interrupt is sent when TX_REQ_PD sets.
17	R/W	0x0	TRAN_ERR_INT_EN If set, an interrupt is sent when TRAN_ERR_PD sets.
16	R/W	0x0	TRAN_COM_INT_EN If set, an interrupt is sent when TRAN_COM_PD sets.
15:12	/	/	/
11	R/W1C	0x0	RF_OVF RECV_FIFO Overflow This bit is set when if the RECV_FIFO is overflow. Writing 1 to this bit clears it. 0: RECV_FIFO is not overflow 1: RECV_FIFO is overflowed
10	R/W1C	0x0	RF_UDF RECV_FIFO Underrun When set, this bit indicates that RECV_FIFO has underrun. Writing 1 to this bit clears it.
9	R/W1C	0x0	SF_OVF SEND_FIFO Overflow This bit is set when if the SEND_FIFO is overflow. Writing 1 to this bit clears it. 0: SEND_FIFO is not overflow 1: SEND_FIFO is overflowed
8	R/W1C	0x0	SF_UDF SEND_FIFO Underrun When set, this bit indicates that SEND_FIFO has underrun. Writing 1 to this bit clears it.

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Offset: 0x0214			Register Name: TWI_DRV_INT_CTRL
Bit	Read/Write	Default/Hex	Description
7	R/W1C	0x0	RF_FULL RECV_FIFO Full This bit is set when if the RECV_FIFO is full. Writing 1 to this bit clears it. 0: RECV_FIFO is not Full 1: RECV _FIFO is Full
6	R/W1C	0x1	RF_EMP RECV_FIFO Empty This bit is set if the RECV_FIFO is empty. Writing 1 to this bit clears it. 0: RECV_FIFO contains one or more words. 1: RECV_FIFO is empty.
5	R/W1C	0x0	SF_FULL SEND_FIFO Full This bit is set when if the SEND_FIFO is full. Writing 1 to this bit clears it. 0: SEND_FIFO is not Full 1: SEND_FIFO is Full
4	R/W1C	0x1	SF_EMP SEND_FIFO Empty This bit is set if the SEND_FIFO is empty. Writing 1 to this bit clears it. 0: SEND_FIFO contains one or more words. 1: SEND_FIFO is empty.
3	R/W1C	0x0	RX_REQ_PD Set when the data byte number in RECV_FIFO reaches RX_TRIG.
2	R/W1C	0x1	TX_REQ_PD Set when there is no less than DMA_TX_TRIG empty byte number in SEND_FIFO.
1	R/W1C	0x0	TRAN_ERR_PD Packet transmission failure pending
0	R/W1C	0x0	TRAN_COM_PD Packet transmission completion pending

17.13.7.16 0x0218 TWI_DRV DMA Configure Register (Default Value: 0x0010_0010)

Offset: 0x0218			Register Name: TWI_DRV_DMA_CFG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/

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Offset: 0x0218			Register Name: TWI_DRV_DMA_CFG
Bit	Read/Write	Default/Hex	Description
24	R/W	0x0	DMA_RX_EN DMA RX Enable
23:22	/	/	/
21:16	R/W	0x10	RX_TRIG RX trigger When DMA_RX_EN is set, send DMA RX Req when the data byte number in RECV_FIFO reaches RX_TRIG, or the read transmission is completed, the data of RECV_FIFO does not reach RX_TRIG but as long as the RECV_FIFO is not empty.
15:9	/	/	/
8	R/W	0x0	DMA_TX_EN DMA TX Enable
7:6	/	/	/
5:0	R/W	0x10	TX_TRIG TX trigger When DMA_TX_EN is set, send DMA TX Req when the space of SEND_FIFO (FIFO Level (b)data volume) reaches TX_TRIG.

17.13.7.17 0x021C TWI_DRV FIFO Content Register (Default Value: 0x0000_0000)

Offset: 0x021C			Register Name: TWI_DRV_FIFO_CON
Bit	Read/Write	Default/Hex	Description
31:23	/	/	/
22	R/WAC	0x0	RECV_FIFO_CLEAR Set this bit to clear RECV_FIFO pointer, and this bit is cleared automatically.
21:16	R	0x0	RECV_FIFO_CONTENT The number of data in RECV_FIFO
15:7	/	/	/
6	R/WAC	0x0	SEND_FIFO_CLEAR Set this bit to clear SEND_FIFO pointer, and this bit is cleared automatically.
5:0	R	0x0	SEND_FIFO_CONTENT The number of data in SEND_FIFO

17.13.7.18 0x0300 TWI_DRV Send Data FIFO Access Register (Default Value: 0x0000_0000)

Offset: 0x0300	Register Name: TWI_DRV_SEND_FIFO_ACC

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Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	W	0x0	SEND_DATA_FIFO Address of a 32x8 SEND_FIFO, which stores reg address and data sending to the slave device.

17.13.7.19 0x0304 TWI_DRV Receive Data FIFO Access Register (Default Value: 0x0000_0000)

Offset: 0x0304			Register Name: TWI_DRV_RECV_FIFO_ACC
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	RECV_DATA_FIFO Address of a 32x8 RECV_FIFO, which stores data received from the slave device.

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17.14 PWM

17.14.1 Overview

The Pulse Width Modulation (PWM) module can output the configurable PWM waveforms and measure the external input waveforms. This device has 3 PWM controllers (PWMCS0, PWMCS1, and PWM2). This chapter only describes PWM2. For detailed information about PWMCS (compatible with PWM function), see section 17.15 PWMCS.

The PWM2 has the following features:

- Maximum 14 independent PWM channels
  - Supports PWM output in cycle mode
  - Supports PWM output in pulse mode, and the pulse number is configurable
  - Output frequency range:
    - 0 to 24 MHz if the clock source is HOSC
    - 0 to 100 MHz if the clock source is APB0 clock
  - Various duty-cycle: 0% to 100%
  - Minimum resolution: 1/65536
- Maximum 7 complementary pairs output
  - Supports dead-zone generator, and the dead-time is configurable
  - The pairing methods for each controller are as follows.

PWM Controller	PWM Pairs	PWM Channels
PWM2	PWM01 pair	PWM2-0+PWM2-1
	PWM23 pair	PWM2-2+PWM2-3
	PWM45 pair	PWM2-4+PWM2-5
	PWM67 pair	PWM2-6+PWM2-7
	PWM89 pair	PWM2-8+PWM2-9
	PWMAB pair	PWM2-10+PWM2-11
	PWMCD pair	PWM2-12+PWM2-13

- Maximum 4 groups of PWM channel output for controlling stepper motors
  - Supports any plural channels to form a group, and output the same duty-cycle pulse
  - In the group mode, the relative phase of the output waveform for each channel is configurable
- Maximum 14 channels of input capture
  - Supports detection at the rising and falling edges
  - Supports pulse-width measurement
- Maximum 14 channels of input waveform counting

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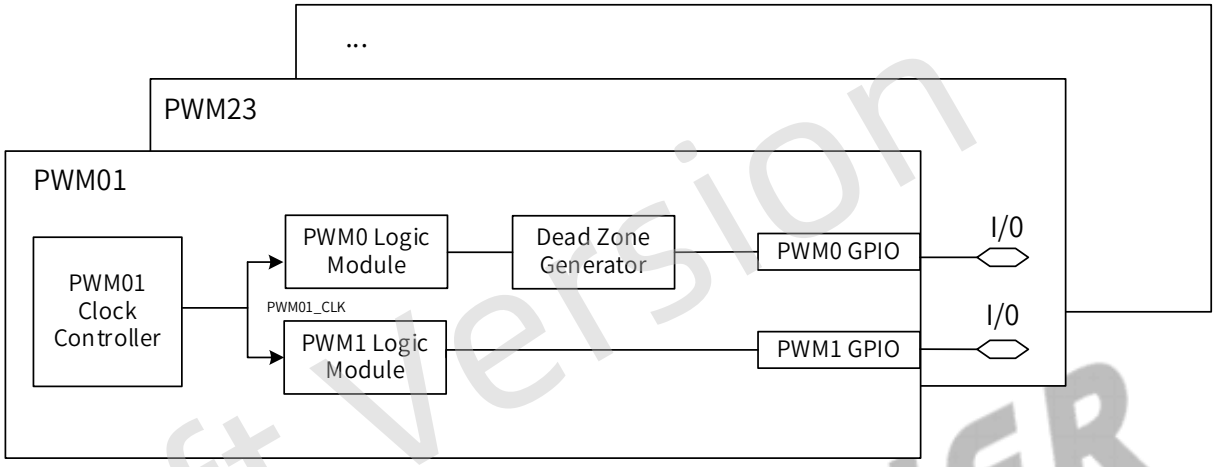
17.14.2 Block Diagram

The PWM includes multi PWM channels. Each channel can generate different PWM waveform by the independent counter and duty-ratio configuration register. Each PWM pair shares one group of clock and dead-zone generator to generate PWM waveform.

Figure 17-96 PWM Block Diagram



Each PWM pair consists of 1 clock module, 2 timer logic module, and 1 programmable dead-zone generator.

17.14.3 External Signals

The following table describes the external signals of the PWM.

Table 17-70 PWM External Signals

Signal	Description	Type
PWM2		
PWM2-0	PWM2 Wave Output /Capture Wave Input 0	I/O
PWM2-1	PWM2 Wave Output /Capture Wave Input 1	I/O
PWM2-2	PWM2 Wave Output /Capture Wave Input 2	I/O
PWM2-3	PWM2 Wave Output /Capture Wave Input 3	I/O
PWM2-4	PWM2 Wave Output /Capture Wave Input 4	I/O
PWM2-5	PWM2 Wave Output /Capture Wave Input 5	I/O
PWM2-6	PWM2 Wave Output /Capture Wave Input 6	I/O
PWM2-7	PWM2 Wave Output /Capture Wave Input 7	I/O
PWM2-8	PWM2 Wave Output /Capture Wave Input 8	I/O
PWM2-9	PWM2 Wave Output /Capture Wave Input 9	I/O
PWM2-10	PWM2 Wave Output /Capture Wave Input 10	I/O
PWM2-11	PWM2 Wave Output /Capture Wave Input 11	I/O
PWM2-12	PWM2 Wave Output /Capture Wave Input 12	I/O
PWM2-13	PWM2 Wave Output /Capture Wave Input 13	I/O

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17.14.4 Functional Description

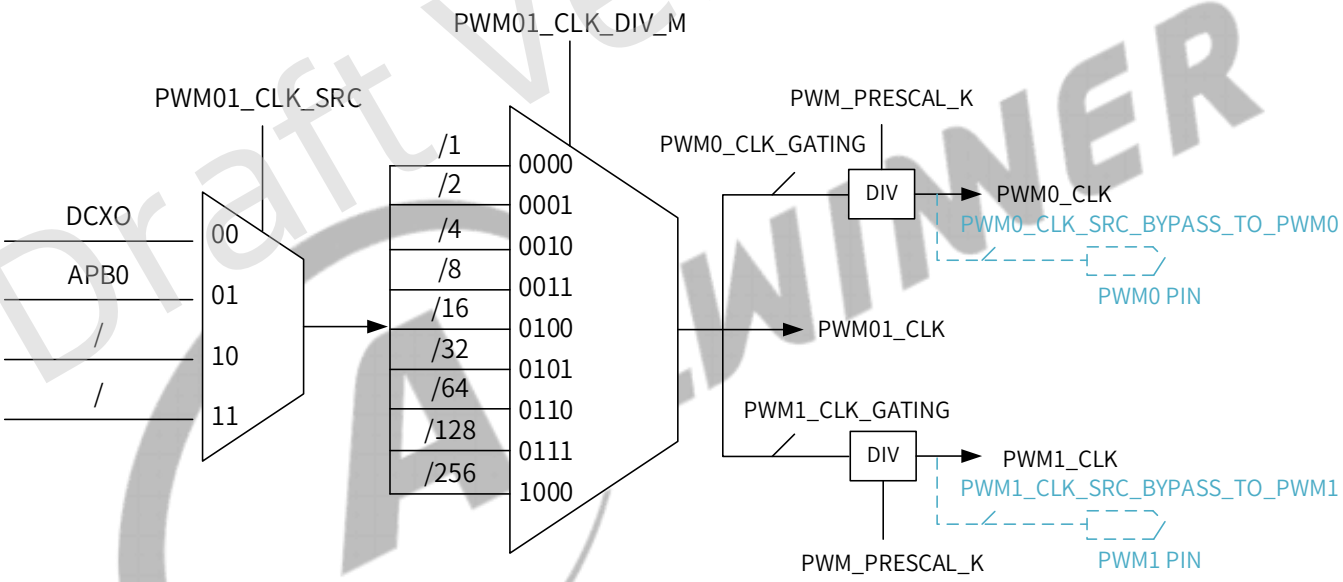
17.14.4.1 PWM Clock Controller

The clock controller of each PWM pair, for instance PWM01, includes clock source select (PWM01_CLK_SRC), 1-256 scaler (PWM01_CLK_DIV_M). Each PWM channel has the secondary frequency division (PWM_PRESCAL_K), clock source bypass (PWMx_CLK_BYPASS) and clock switch (PWMx_CLK_GATING).

The clock sources of PWM include DCXO and APB0. The DCXO comes from the external high-frequency oscillator; the APB0 is usually the 100MHz APB0 bus clock.

The clock bypass is to select the clock for PWM output. The following figure shows the PWM01 Clock Source Bypass.

Figure 17-97 PWM01 Clock Source Bypass



17.14.4.2 PWM Output

Take PWM01 as an example. Figure 17-98 indicates the PWM01 output logic diagram. The logic diagrams of other PWM pairs are the same as that of PWM01.

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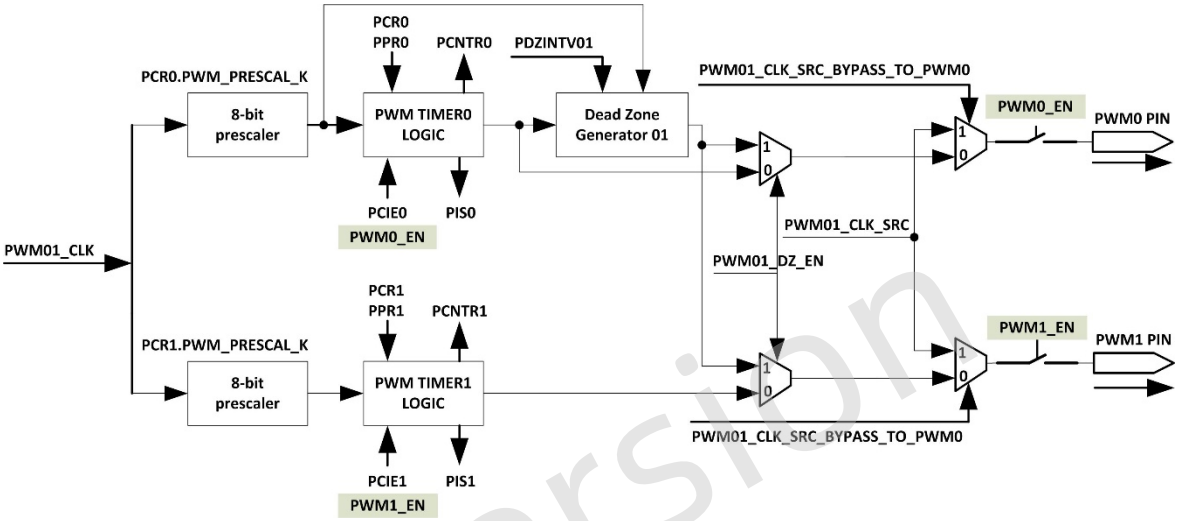
Figure 17-98 PWM01 Output Logic Module Diagram



The timer logic module of PWM consists of one 16-bit up-counter (PCNTR) and three 16-bit parameters (PWM_ENTIRE_CYCLE, PWM_ACT_CYCLE, PWM_COUNTER_START). The PWM_ENTIRE_CYCLE is used to control the PWM cycle, the PWM_ACT_CYCLE is used to control the duty cycle, the PWM_COUNTER_START is used to control the output phase (multi-channel synchronization work requirements).

The PWM_ENTIRE_CYCLE and the PWM_ACT_CYCLE support the cache load. After PWM output is enabled, the register values of the PWM_ENTIRE_CYCLE and the PWM_ACT_CYCLE can be changed anytime, and the changed value caches into the cache register. When the PCNTR counter outputs a period of PWM waveform, the value of the cache register can be updated for the PCNTR control. The purpose of the cache load is to avoid the unstable PWM output waveform with glitches when updating the values of the PWM_ENTIRE_CYCLE and PWM_ACT_CYCLE. The PWM_CNTSTART doesn t support cache load, and only can be configured before enabling PWM output.

The PWM supports waveform output in cycle mode and pulse mode.

Cycle mode: The PWM outputs the set PWM waveform continually; that is, the output waveform is a continuous PWM square wave.

Pulse mode: After setting the PWM_PUL_NUM parameter, the PWM outputs (PWM_PULNUM+1) periods of PWM waveform; that is, the waveform with several pulses are output.

NOTE

As the data width of PWM_PUL_CNT is 16 bits, it output maximum 65536 PWM waveforms.

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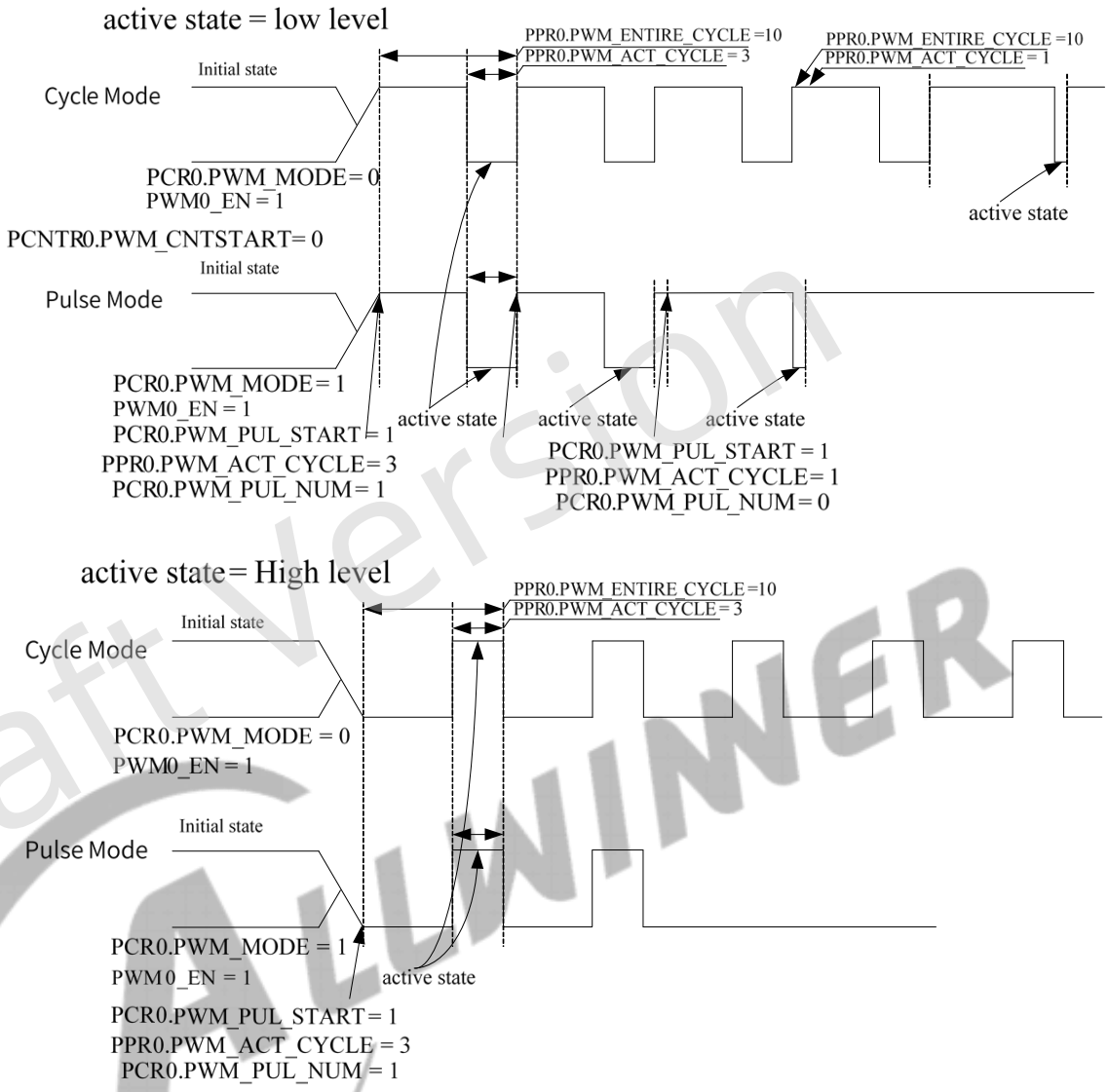
The following figure shows active state in cycle mode and pulse mode.

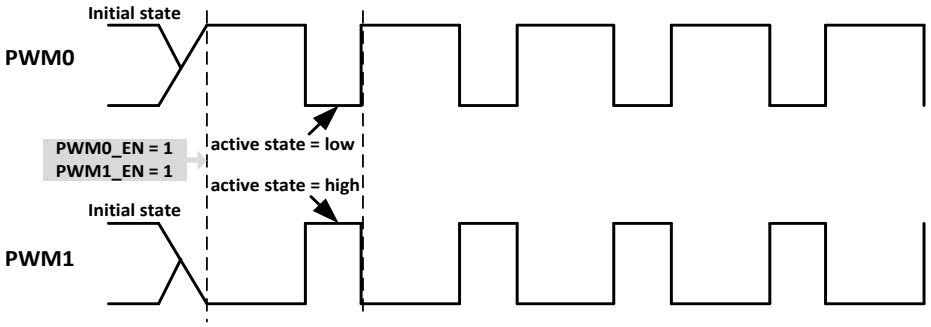


Every PWM pair supports two kinds of waveform output, including complementary pair Waveform output and PWM pair output with dead-time.

Complementary Pair Waveform Output

The following figure shows the complementary pair waveform output of PWM01.

Figure 17-99 PWM01 Complementary Pair Output



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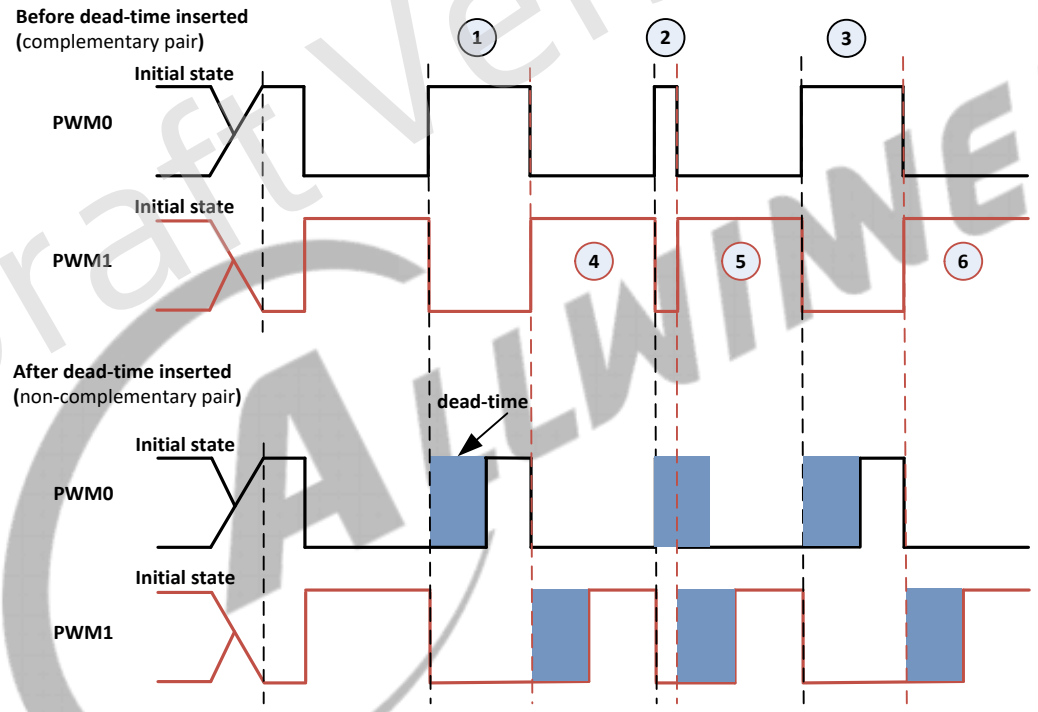
The complementary pair output needs to satisfy the following conditions:

- PWM0 and PWM1 share the same clock divider, frequency, duty-cycle, and phase
- PWM0 and PWM1 have an opposite active state
- Enable the clock gating of PWM0 and PWM1 at the same time
- Enable the waveform output of PWM0 and PWM1 at the same time

Waveform Output with Dead-time Insertion

Every PWM pair has a programmable dead-time generator. When the dead-time function of the PWM pair is enabled, the PWM01 waveform output is decided by PWM timer logic and DeadZone Generator. The following figure shows the waveform output with dead-time insertion.

Figure 17-100 Output Waveform with Dead-time Insertion



The principle of inserting dead-time is to insert dead-time at every rising edge. If the high level period in mark⊣ is less than dead-time, then dead-time will override the high level period. Therefore, the setting of dead-time needs to consider the period and the duty-cycle of the output waveform. The dead-time formula is defined as follows:

Dead time = (PWM01_CLK / PWM0_PRESCALE_K)$^{-1}$ * PDZINTV01

17.14.4.3 PWM Group Mode

The PWM group mode is used to satisfy the PWM group signals of stepper motors in different phases. Take PWM Group0 as an example. The PGR0.CS selects the PWM channel working in the same group; the same parameters of PWM_ENTIRE_CYCLE, PWM_ACT_CYCLE are set through the same clock configuration; the different PWM_COUNTER_START can output PWM group signals with the same duty-cycle and the different phase. PCNTR counting starts from 0 by default, and

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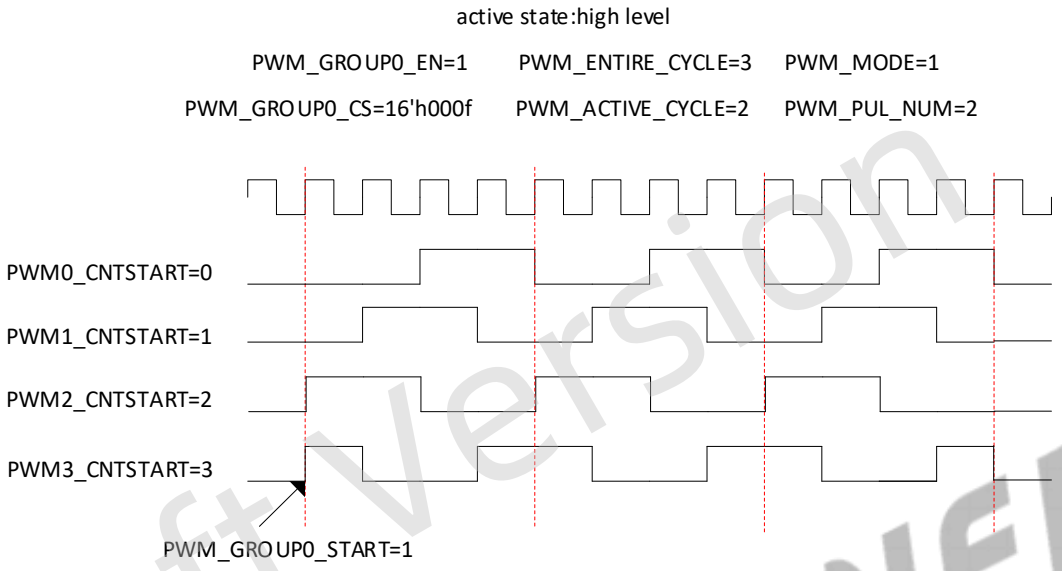
the phase of output waveform can be controlled through PWM_CNTSTART configuration. If the stepper motor uses the PWM group mode, it is recommended to divide 0, 1, 2, 3 into a group, and 4, 5, 6, 7 another group.

Figure 17-101 Group 0-3 PWM Signal Output



17.14.4.4 Waveform Capture

Each PWM pair is also used for capturing external clock. The following figure shows the PWM01 capture logic module diagram.

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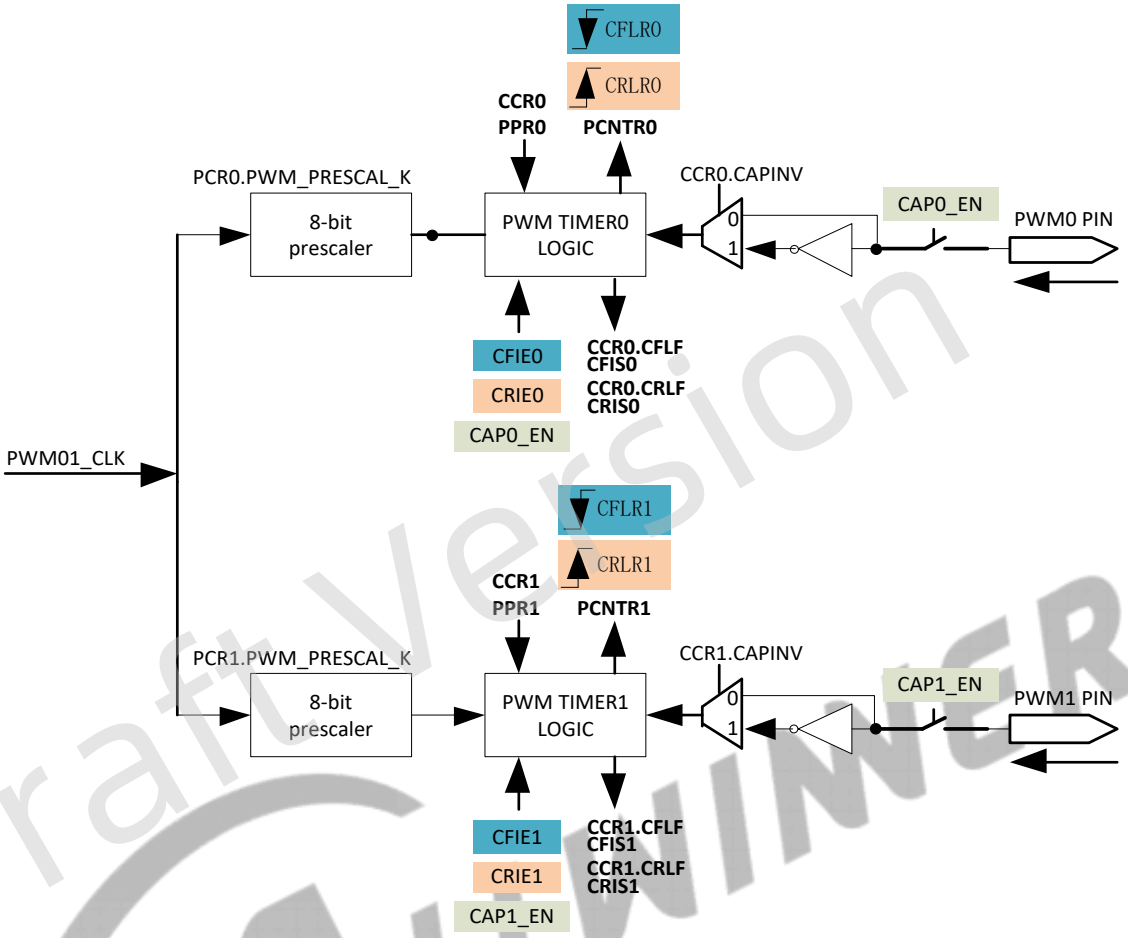
Figure 17-102 PWM01 Capture Logic Module Diagram



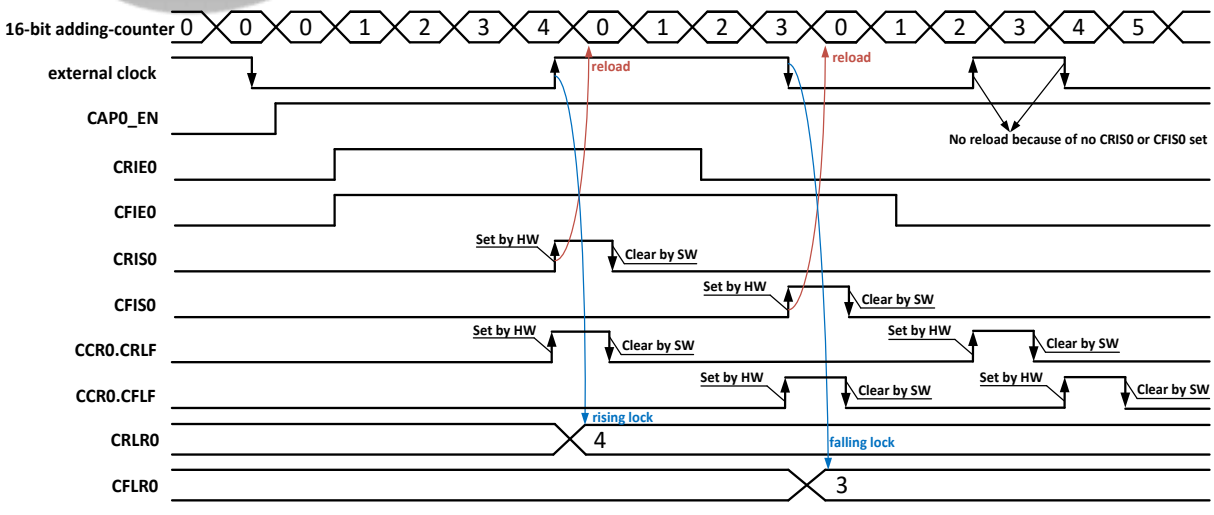
Take the PWM0 channel as an example. The PWM0 channel has one CFLR 0 and one CRLR 0 for capturing the values of up-counters on the falling edge and rising edge. You can calculate the period of the external clock by CFLR 0 and CRLR 0.

$T_{low\text{-}level}$ = (PWM01_CLK / PWM0_PRESCALE_K)$^{-1}$ * (CRLR0+1)

$T_{high\text{-}level}$ = (PWM01_CLK / PWM0_PRESCALE_K)$^{-1}$ * (CFLR0+1)

$T_{period} = T_{high\text{-}level} + T_{low\text{-}level}$

Figure 17-103 PWM0 Channel Capture Timing



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When the capture input function of the PWM0 channel is enabled, the PCNTR of the PWM0 channel starts to work.

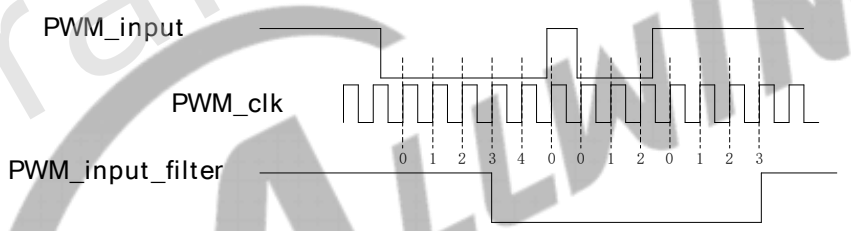
When the timer logic module of PWM0 captures a rising edge, the current value of PCNTR is locked to CRLR0, and CCR0[CRLF] is set to 1. If CRIE0 is 1, and CRIS0 is set to 1, the PWM0 channel will send interrupt requests, and the up-counter will be reloaded to 0 and continue to count. If CRIE0 is 0, CRIS0 will not be set to 1 when the timer logic module of PWM0 captures a rising edge; thus, the up-counter will not be reloaded to 0.

When the timer logic module of PWM0 captures a falling edge, the current value of PCNTR is locked to CFLR0, and CCR0[CFLF] is set to 1. If CFIE0 is 1, and CFIS0 is set to 1, the PWM0 channel will send interrupt requests, and the up-counter will be reloaded to 0 and continue to count. If CFIE0 is 0, CFIS0 will not be set to 1 when the timer logic module of PWM0 captures a falling edge; thus, the up-counter will not be reloaded to 0.

17.14.4.5 Waveform Capture Filtering

During waveform capture, you could choose whether to filter the input waveform to avoid the impact of input glitches on waveform capture.

Figure 17-104 Waveform Capture Filtering



17.14.5 Programming Guidelines

The following working mode takes PWM01 as an example, other PWM pairs and PWM01 are consistent.

17.14.5.1 Configuring Clock

Step 1 PWM gating: When using PWM, write 1 to PCGR [PWMx_CLK_GATING].

Step 2 PWM clock source select: Set PCCR01 [PWM01_CLK_SRC] to select HOSC or APB0 clock.

Step 3 PWM clock divider: Set PCCR01 [PWM01_CLK_DIV_M] to select different frequency division coefficient (1/2/4/8/16/32/64/128/256).

Step 4 PWM clock bypass: Set PCGR [PWM_CLK_SRC_BYPASS_TO_PWM] to 1, output the PWM clock after the secondary frequency division to the corresponding PWM output pin.

Step 5 PWM internal clock configuration: Set PCR [PWM_PRESCAL_K] to select any frequency division coefficient from 1 to 256.

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NOTE

The channels in the complementary output and group mode needs the same clock configuration, simultaneously opened clock gating, PWM parameter configuration, and enabled PWM output at the same time to ensure the synchronization among channels. Besides, as a PWM pair shares the same first-level clock division and gating, their two channels cannot be allocated to different groups in the group mode. If the allocation still exists, the same first-level clock division and the simultaneously opened gating is required. The total module needs to be reset when the group mode regroups.

17.14.5.2 Configuring PWM

Step 1 PWM mode: Set PCR [PWM_MODE] to select cycle mode or pulse mode, if pulse mode, PCR[PWM_PUL_NUM] needs to be configured.

Step 2 PWM active level: Set PCR [PWM_ACT_STA] to select a low level or high level.

Step 3 PWM duty-cycle: Configure PPR [PWM_ENTIRE_CYCLE] and PPR [PWM_ACT_CYCLE] after clock gating is opened.

Step 4 PWM starting/stopping phase: Configure PCNTR [PWM_COUNTER_START] after the clock gating is enabled and before the PWM is enabled. You can verify whether the configuration was successful by reading back PCNTR [PWM_COUNTER_STATUS].

Step 5 Enable PWM: Configure PER to select the corresponding PWM enable bit; when selecting pulse mode, PCR [PWM_PUL_START] needs to be enabled.

17.14.5.3 Configuring Deadzone

Step 1 Set initial value: set PDZCR01 [PDZINTV01].

Step 2 Enable Deadzone: set PDZCR01 [PWM01_DZ_EN].

17.14.5.4 Configuring Capture Input

Step 1 Enable capture: Configure CER to enable the corresponding channel.

Step 2 Capture mode: Configure CCR [CRLF] and CCR [CFLF] to select rising edge capture or falling edge capture, configure CCR [CAPINV] to select whether the input signal does reverse processing.

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17.14.5.5 Number of Captured Input Signals

Enable Input Counting and Input Capture Simultaneously

To enable input counting and input capture simultaneously, follow the steps as below.

Step 1 To configure the capture method, confirm whether to capture in the rising edge or falling edge through CRLF and CFLF in the CCR. Confirm whether to invert the input signal through CAPINV in the CCR.

Step 2 Set the CCSR to 1 in the CCR.

Step 3 Confirm whether to count in the falling edge or rising edge through CCCRFR in the CCCR .

Step 4 Confirm whether to stop or restart counting when reaching the count value through CCCSRR in the CCCR.

Step 5 Set the number of CCCNR and enable the corresponding channels of CCCIER. When the input signal reach the set value of CCCNR, an interrupt will be generated.

Step 6 Set CFIEn or CRIEn to 1 in the CIER to generate the capture channel pending.

Step 7 Set CAPn_EN to 1 in the CER to enable corresponding capture functions.

Enable Input Counting Separately

To enable input counting separately, follow the steps as below.

Step 1 Confirm whether to invert the input signal through CAPINV in CCR.

Step 2 Set CCSR to 0 in CCR.

Step 3 Confirm whether to count in the falling edge or rising edge through CCCRFR in the CCCR.

Step 4 Confirm whether to stop or restart counting when reaching the count value through CCCSRR in the CCCR.

Step 5 Set the number of CCCNR and enable the corresponding channels of CCCIER. When the input signal reach the set value of CCCNR, an interrupt will be generated.

Step 6 Set CAP_CNTn_EN to 1 to enable the counting of corresponding channels of CCCER.

Step 7 Set CAP_CNTn_EN to 0 to disable the transfer of corresponding channels of CCCER.

17.14.6 Register List

Module Name	Base Address	The Value of N
PWM2	0x020C0000	0-13

Register Name	Offset	Description

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Register Name	Offset	Description
PIER	0x0000	PWM IRQ Enable Register
PISR	0x0004	PWM IRQ Status Register
CIER	0x0010	Capture IRQ Enable Register
CISR	0x0014	Capture IRQ Status Register
CCCIER	0x0018	Capture Cycle Counter IRQ Enable Register
CCCISR	0x001C	Capture Cycle Counter IRQ Status Register
PCCR01	0x0020	PWM01 Clock Configuration Register
PCCR23	0x0024	PWM23 Clock Configuration Register
PCCR45	0x0028	PWM45 Clock Configuration Register
PCCR67	0x002C	PWM67 Clock Configuration Register
PCCR89	0x0030	PWM89 Clock Configuration Register
PCCRAB	0x0034	PWMAB Clock Configuration Register
PCCRCD	0x0038	PWMCD Clock Configuration Register
PCGR	0x0040	PWM Clock Gating Register
PDZCR01	0x0060	PWM01 Dead Zone Control Register
PDZCR23	0x0064	PWM23 Dead Zone Control Register
PDZCR45	0x0068	PWM45 Dead Zone Control Register
PDZCR67	0x006C	PWM67 Dead Zone Control Register
PDZCR89	0x0070	PWM89 Dead Zone Control Register
PDZCRAB	0x0074	PWMAB Dead Zone Control Register
PDZCRCD	0x0078	PWMCD Dead Zone Control Register
PER	0x0080	PWM Enable Register
PGR0	0x0090	PWM Group0 Register
PGR1	0x0094	PWM Group1 Register
PGR2	0x0098	PWM Group2 Register
PGR3	0x009C	PWM Group3 Register
CCCER	0x00A0	Capture Cycle Counter Enable Register
CER	0x00C0	Capture Enable Register
PCR	0x0100+0x0000+N*0x0040	PWM Control Register
PPR	0x0100+0x0004+N*0x0040	PWM Period Register
PCNTR	0x0100+0x0008+N*0x0040	PWM Counter Register
PPCNTR	0x0100+0x000C+N*0x0040	PWM Pulse Counter Register
CCR	0x0100+0x0010+N*0x0040	Capture Control Register
CRLR	0x0100+0x0014+N*0x0040	Capture Rise Lock Register
CFLR	0x0100+0x0018+N*0x0040	Capture Fall Lock Register
CCCSR	0x0100+0x001C+N*0x0040	Capture Cycle Counter Status Register
CCCNR	0x0100+0x0020+N*0x0040	Capture Cycle Counter Number Register
CCCR	0x0100+0x0024+N*0x0040	Capture Cycle Counter Register
CFG	0x0100+0x0028+N*0x0040	Capture Filter Register

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17.14.7 Register Description

17.14.7.1 0x0000 PWM IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x0000			Register Name: PIER
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19	R/W	0x0	PGIE3 PWM group 3 Interrupt Enable 0: Disable 1: Enable
18	R/W	0x0	PGIE2 PWM group 2 Interrupt Enable 0: Disable 1: Enable
17	R/W	0x0	PGIE1 PWM group 1 Interrupt Enable 0: Disable 1: Enable
16	R/W	0x0	PGIE0 PWM group 0 Interrupt Enable 0: Disable 1: Enable
15:14	/	/	/
13	R/W	0x0	PCIE13. PWM channel 13 Interrupt Enable. 0: PWM channel 13 Interrupt Disable; 1: PWM channel 13 Interrupt Enable.
12	R/W	0x0	PCIE12. PWM channel 12 Interrupt Enable. 0: PWM channel 12 Interrupt Disable; 1: PWM channel 12 Interrupt Enable.
11	R/W	0x0	PCIE11. PWM channel 11 Interrupt Enable. 0: PWM channel 11 Interrupt Disable; 1: PWM channel 11 Interrupt Enable.
10	R/W	0x0	PCIE10. PWM channel 10 Interrupt Enable. 0: PWM channel 10 Interrupt Disable; 1: PWM channel 10 Interrupt Enable.
9	R/W	0x0	PCIE9. PWM channel 9 Interrupt Enable. 0: PWM channel 9 Interrupt Disable; 1: PWM channel 9 Interrupt Enable.

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Offset:0x0000			Register Name: PIER
Bit	Read/Write	Default/Hex	Description
8	R/W	0x0	PCIE8. PWM channel 8 Interrupt Enable. 0: PWM channel 8 Interrupt Disable; 1: PWM channel 8 Interrupt Enable.
7	R/W	0x0	PCIE7. PWM channel 7 Interrupt Enable. 0: PWM channel 7 Interrupt Disable; 1: PWM channel 7 Interrupt Enable.
6	R/W	0x0	PCIE6. PWM channel 6 Interrupt Enable. 0: PWM channel 6 Interrupt Disable; 1: PWM channel 6 Interrupt Enable.
5	R/W	0x0	PCIE5. PWM channel 5 Interrupt Enable. 0: PWM channel 5 Interrupt Disable; 1: PWM channel 5 Interrupt Enable.
4	R/W	0x0	PCIE4. PWM channel 4 Interrupt Enable. 0: PWM channel 4 Interrupt Disable; 1: PWM channel 4 Interrupt Enable.
3	R/W	0x0	PCIE3. PWM channel 3 Interrupt Enable. 0: PWM channel 3 Interrupt Disable; 1: PWM channel 3 Interrupt Enable.
2	R/W	0x0	PCIE2. PWM channel 2 Interrupt Enable. 0: PWM channel 2 Interrupt Disable; 1: PWM channel 2 Interrupt Enable.
1	R/W	0x0	PCIE1. PWM channel 1 Interrupt Enable. 0: PWM channel 1 Interrupt Disable; 1: PWM channel 1 Interrupt Enable.
0	R/W	0x0	PCIE0. PWM channel 0 Interrupt Enable. 0: PWM channel 0 Interrupt Disable; 1: PWM channel 0 Interrupt Enable.

17.14.7.2 0x0004 PWM IRQ Status Register (Default Value: 0x0000_0000)

Offset:0x0004			Register Name: PISR
Bit	Read/Write	Default/Hex	Description

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Offset:0x0004			Register Name: PISR
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19	R/W1C	0x0	PGIS3 PWM group 3 Interrupt Status
18	R/W1C	0x0	PGIS2 PWM group 2 Interrupt Status
17	R/W1C	0x0	PGIS1 PWM group 1 Interrupt Status
16	R/W1C	0x0	PGIS0 PWM group 0 Interrupt Status
15:14	/	/	/
13	R/W1C	0x0	PIS13. PWM channel 13 Interrupt Status. When PWM channel 13 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 13 interrupt is not pending. 1: PWM channel 13 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 13 interrupt status.
12	R/W1C	0x0	PIS12. PWM channel 12 Interrupt Status. When PWM channel 12 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 12 interrupt is not pending. 1: PWM channel 12 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 8 interrupt status.
11	R/W1C	0x0	PIS11. PWM channel 11 Interrupt Status. When PWM channel 11 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 11 interrupt is not pending. 1: PWM channel 11 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 11 interrupt status.
10	R/W1C	0x0	PIS10. PWM channel 10 Interrupt Status. When PWM channel 10 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 10 interrupt is not pending. 1: PWM channel 10 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 10 interrupt status.

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Offset:0x0004			Register Name: PISR
Bit	Read/Write	Default/Hex	Description
9	R/W1C	0x0	PIS9. PWM channel 9 Interrupt Status. When PWM channel 9 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 9 interrupt is not pending. 1: PWM channel 9 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 9 interrupt status.
8	R/W1C	0x0	PIS8. PWM channel 8 Interrupt Status. When PWM channel 8 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 8 interrupt is not pending. 1: PWM channel 8 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 8 interrupt status.
7	R/W1C	0x0	PIS7. PWM channel 7 Interrupt Status. When PWM channel 7 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 7 interrupt is not pending. 1: PWM channel 7 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 7 interrupt status.
6	R/W1C	0x0	PIS6. PWM channel 6 Interrupt Status. When PWM channel 6 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 6 interrupt is not pending. 1: PWM channel 6 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 6 interrupt status.
5	R/W1C	0x0	PIS5. PWM channel 5 Interrupt Status. When PWM channel 5 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 5 interrupt is not pending. 1: PWM channel 5 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 5 interrupt status.

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Offset:0x0004			Register Name: PISR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	PIS4. PWM channel 4 Interrupt Status. When PWM channel 4 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 4 interrupt is not pending. 1: PWM channel 4 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 4 interrupt status.
3	R/W1C	0x0	PIS3. PWM channel 3 Interrupt Status. When PWM channel 3 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 3 interrupt is not pending. 1: PWM channel 3 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 3 interrupt status.
2	R/W1C	0x0	PIS2. PWM channel 2 Interrupt Status. When PWM channel 2 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 2 interrupt is not pending. 1: PWM channel 2 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 2 interrupt status.
1	R/W1C	0x0	PIS1. PWM channel 1 Interrupt Status. When PWM channel 1 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 1 interrupt is not pending. 1: PWM channel 1 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 1 interrupt status.
0	R/W1C	0x0	PIS0. PWM channel 0 Interrupt Status. When PWM channel 0 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 0 interrupt is not pending. 1: PWM channel 0 interrupt is pending. Writes 0: No effect. 1: Clear PWM channel 0 interrupt status.

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17.14.7.3 0x0010 Capture IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x0010			Register Name: CIER
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W	0x0	CFIE13. If this enable bit is set 1, when capture channel 13 captures falling edge, it generates a capture channel 13 pending. 0: Capture channel 13 fall lock Interrupt disable; 1: Capture channel 13 fall lock Interrupt enable.
26	R/W	0x0	CRIE13. If this enable bit is set 1, when capture channel 13 captures rising edge, it generates a capture channel 13 pending. 0: Capture channel 13 rise lock Interrupt disable; 1: Capture channel 13 rise lock Interrupt enable.
25	R/W	0x0	CFIE12. If this enable bit is set 1, when capture channel 12 captures falling edge, it generates a capture channel 12 pending. 0: Capture channel 12 fall lock Interrupt disable; 1: Capture channel 12 fall lock Interrupt enable.
24	R/W	0x0	CRIE12. If this enable bit is set 1, when capture channel 12 captures rising edge, it generates a capture channel 12 pending. 0: Capture channel 12 rise lock Interrupt disable; 1: Capture channel 12 rise lock Interrupt enable.
23	R/W	0x0	CFIE11. If this enable bit is set 1, when capture channel 11 captures falling edge, it generates a capture channel 11 pending. 0: Capture channel 11 fall lock Interrupt disable; 1: Capture channel 11 fall lock Interrupt enable.
22	R/W	0x0	CRIE11. If this enable bit is set 1, when capture channel 11 captures rising edge, it generates a capture channel 11 pending. 0: Capture channel 11 rise lock Interrupt disable; 1: Capture channel 11 rise lock Interrupt enable.

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Offset:0x0010			Register Name: CIER
Bit	Read/Write	Default/Hex	Description
21	R/W	0x0	CFIE10. If this enable bit is set 1, when capture channel 10 captures falling edge, it generates a capture channel 10 pending. 0: Capture channel 10 fall lock Interrupt disable; 1: Capture channel 10 fall lock Interrupt enable.
20	R/W	0x0	CRIE10. If this enable bit is set 1, when capture channel 10 captures rising edge, it generates a capture channel 10 pending. 0: Capture channel 10 rise lock Interrupt disable; 1: Capture channel 10 rise lock Interrupt enable.
19	R/W	0x0	CFIE9. If this enable bit is set 1, when capture channel 9 captures falling edge, it generates a capture channel 9 pending. 0: Capture channel 9 fall lock Interrupt disable; 1: Capture channel 9 fall lock Interrupt enable.
18	R/W	0x0	CRIE9. If this enable bit is set 1, when capture channel 9 captures rising edge, it generates a capture channel 9 pending. 0: Capture channel 9 rise lock Interrupt disable; 1: Capture channel 9 rise lock Interrupt enable.
17	R/W	0x0	CFIE8. If this enable bit is set 1, when capture channel 8 captures falling edge, it generates a capture channel 8 pending. 0: Capture channel 8 fall lock Interrupt disable; 1: Capture channel 8 fall lock Interrupt enable.
16	R/W	0x0	CRIE8. If this enable bit is set 1, when capture channel 8 captures rising edge, it generates a capture channel 8 pending. 0: Capture channel 8 rise lock Interrupt disable; 1: Capture channel 8 rise lock Interrupt enable.
15	R/W	0x0	CFIE7. If this enable bit is set 1, when capture channel 7 captures falling edge, it generates a capture channel 7 pending. 0: Capture channel 7 fall lock Interrupt disable; 1: Capture channel 7 fall lock Interrupt enable.

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Offset:0x0010			Register Name: CIER
Bit	Read/Write	Default/Hex	Description
14	R/W	0x0	CRIE7. If this enable bit is set 1, when capture channel 7 captures rising edge, it generates a capture channel 7 pending. 0: Capture channel 7 rise lock Interrupt disable; 1: Capture channel 7 rise lock Interrupt enable.
13	R/W	0x0	CFIE6. If this enable bit is set 1, when capture channel 6 captures falling edge, it generates a capture channel 6 pending. 0: Capture channel 6 fall lock Interrupt disable; 1: Capture channel 6 fall lock Interrupt enable.
12	R/W	0x0	CRIE6. If this enable bit is set 1, when capture channel 5 captures rising edge, it generates a capture channel 5 pending. 0: Capture channel 5 rise lock Interrupt disable; 1: Capture channel 5 rise lock Interrupt enable.
11	R/W	0x0	CFIE5. If this enable bit is set 1, when capture channel 5 captures falling edge, it generates a capture channel 5 pending. 0: Capture channel 5 fall lock Interrupt disable; 1: Capture channel 5 fall lock Interrupt enable.
10	R/W	0x0	CRIE5. If this enable bit is set 1, when capture channel 5 captures rising edge, it generates a capture channel 5 pending. 0: Capture channel 5 rise lock Interrupt disable; 1: Capture channel 5 rise lock Interrupt enable.
9	R/W	0x0	CFIE4. If this enable bit is set 1, when capture channel 4 captures falling edge, it generates a capture channel 4 pending. 0: Capture channel 4 fall lock Interrupt disable; 1: Capture channel 4 fall lock Interrupt enable.
8	R/W	0x0	CRIE4. If this enable bit is set 1, when capture channel 4 captures rising edge, it generates a capture channel 4 pending. 0: Capture channel 4 rise lock Interrupt disable; 1: Capture channel 4 rise lock Interrupt enable.

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Offset:0x0010			Register Name: CIER
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	CFIE3. If this enable bit is set 1, when capture channel 3 captures falling edge, it generates a capture channel 3 pending. 0: Capture channel 3 fall lock Interrupt disable; 1: Capture channel 3 fall lock Interrupt enable.
6	R/W	0x0	CRIE3. If this enable bit is set 1, when capture channel 3 captures rising edge, it generates a capture channel 3 pending. 0: Capture channel 3 rise lock Interrupt disable; 1: Capture channel 3 rise lock Interrupt enable.
5	R/W	0x0	CFIE2. If this enable bit is set 1, when capture channel 2 captures falling edge, it generates a capture channel 2 pending. 0: Capture channel 2 fall lock Interrupt disable; 1: Capture channel 2 fall lock Interrupt enable.
4	R/W	0x0	CRIE2. If this enable bit is set 1, when capture channel 2 captures rising edge, it generates a capture channel 2 pending. 0: Capture channel 2 rise lock Interrupt disable; 1: Capture channel 2 rise lock Interrupt enable.
3	R/W	0x0	CFIE1. If this enable bit is set 1, when capture channel 1 captures falling edge, it generates a capture channel 1 pending. 0: Capture channel 1 fall lock Interrupt disable; 1: Capture channel 1 fall lock Interrupt enable.
2	R/W	0x0	CRIE1. If this enable bit is set 1, when capture channel 1 captures rising edge, it generates a capture channel 1 pending. 0: Capture channel 1 rise lock Interrupt disable; 1: Capture channel 1 rise lock Interrupt enable.
1	R/W	0x0	CFIE0. If this enable bit is set 1, when capture channel 0 captures falling edge, it generates a capture channel 0 pending. 0: Capture channel 0 fall lock Interrupt disable; 1: Capture channel 0 fall lock Interrupt enable.

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Offset:0x0010			Register Name: CIER
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CRIE0. If this enable bit is set 1, when capture channel 0 captures rising edge, it generates a capture channel 0 pending. 0: Capture channel 0 rise lock Interrupt disable; 1: Capture channel 0 rise lock Interrupt enable.

17.14.7.4 0x0014 Capture IRQ Status Register (Default Value: 0x0000_0000)

Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27	R/W1C	0x0	CRIS13. Capture channel 13 rising lock interrupt status. When capture channel 13 captures rising edge, if capture channel 13 rise lock interrupt (CRIE13) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 13 interrupt is not pending. 1: Capture channel 13 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 13 interrupt status.
26	R/W1C	0x0	CFIS13. Capture channel 13 falling lock interrupt status. When capture channel 13 captures falling edge, if capture channel 13 fall lock interrupt (CFIE13) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 13 interrupt is not pending. 1: Capture channel 13 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 13 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
25	R/W1C	0x0	CRIS12. Capture channel 12 rising lock interrupt status. When capture channel 12 captures rising edge, if capture channel 12 rise lock interrupt (CRIE12) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 12 interrupt is not pending. 1: Capture channel 12 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 12 interrupt status.
24	R/W1C	0x0	CFIS12. Capture channel 12 falling lock interrupt status. When capture channel 12 captures falling edge, if capture channel 12fall lock interrupt (CFIE12) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 12 interrupt is not pending. 1: Capture channel 12 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 12 interrupt status.
23	R/W1C	0x0	CRIS11. Capture channel 11 rising lock interrupt status. When capture channel 11 captures rising edge, if capture channel 11 rise lock interrupt (CRIE11) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 11 interrupt is not pending. 1: Capture channel 11 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 11 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
22	R/W1C	0x0	CFIS11. Capture channel 11 falling lock interrupt status. When capture channel 11 captures falling edge, if capture channel 11 fall lock interrupt (CFIE11) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 11 interrupt is not pending. 1: Capture channel 11 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 11 interrupt status.
21	R/W1C	0x0	CFIS10. Capture channel 10 falling lock interrupt status. When capture channel 10 captures falling edge, if capture channel 10 fall lock interrupt (CFIE10) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 10 interrupt is not pending. 1: Capture channel 10 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 10 interrupt status.
20	R/W1C	0x0	CRIS10. Capture channel 10 rising lock interrupt status. When capture channel 10 captures rising edge, if capture channel 10 rise lock interrupt (CRIE10) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 10 interrupt is not pending. 1: Capture channel 10 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 10 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
19	R/W1C	0x0	CFIS9. Capture channel 9 falling lock interrupt status. When capture channel 9 captures falling edge, if capture channel 9 fall lock interrupt (CFIE9) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 9 interrupt is not pending. 1: Capture channel 9 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 9 interrupt status.
18	R/W1C	0x0	CRIS9. Capture channel 9 rising lock interrupt status. When capture channel 9 captures rising edge, if capture channel 9 rise lock interrupt (CRIE9) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 9 interrupt is not pending. 1: Capture channel 9 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 9 interrupt status.
17	R/W1C	0x0	CFIS8. Capture channel 8 falling lock interrupt status. When capture channel 8 captures falling edge, if capture channel 8 fall lock interrupt (CFIE8) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 8 interrupt is not pending. 1: Capture channel 8 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 8 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
16	R/W1C	0x0	CRIS8. Capture channel 8 rising lock interrupt status. When capture channel 8 captures rising edge, if capture channel 8 rise lock interrupt (CRIE8) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 8 interrupt is not pending. 1: Capture channel 8 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 8 interrupt status.
15	R/W1C	0x0	CFIS7. Capture channel 7 falling lock interrupt status. When capture channel 7 captures falling edge, if capture channel 7 fall lock interrupt (CFIE7) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 7 interrupt is not pending. 1: Capture channel 7 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 7 interrupt status.
14	R/W1C	0x0	CRIS7. Capture channel 7 rising lock interrupt status. When capture channel 7 captures rising edge, if capture channel 7 rise lock interrupt (CRIE7) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 7 interrupt is not pending. 1: Capture channel 7 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 7 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
13	R/W1C	0x0	CFIS6. Capture channel 6 falling lock interrupt status. When capture channel 6 captures falling edge, if capture channel 6 fall lock interrupt (CFIE6) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 6 interrupt is not pending. 1: Capture channel 6 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 6 interrupt status.
12	R/W1C	0x0	CRIS6. Capture channel 6 rising lock interrupt status. When capture channel 6 captures rising edge, if capture channel 6 rise lock interrupt (CRIE6) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 6 interrupt is not pending. 1: Capture channel 6 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 6 interrupt status.
11	R/W1C	0x0	CFIS5. Capture channel 5 falling lock interrupt status. When capture channel 5 captures falling edge, if capture channel 5 fall lock interrupt (CFIE5) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 5 interrupt is not pending. 1: Capture channel 5 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 5 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
10	R/W1C	0x0	CRIS5. Capture channel 5 rising lock interrupt status. When capture channel 5 captures rising edge, if capture channel 5 rise lock interrupt (CRIE5) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 5 interrupt is not pending. 1: Capture channel 5 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 5 interrupt status.
9	R/W1C	0x0	CFIS4. Capture channel 4 falling lock interrupt status. When capture channel 4 captures falling edge, if capture channel 4 fall lock interrupt (CFIE4) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 4 interrupt is not pending. 1: Capture channel 4 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 4 interrupt status.
8	R/W1C	0x0	CRIS4. Capture channel 4 rising lock interrupt status. When capture channel 4 captures rising edge, if capture channel 4 rise lock interrupt (CRIE4) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 4 interrupt is not pending. 1: Capture channel 4 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 4 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
7	R/W1C	0x0	CFIS3. Capture channel 3 falling lock interrupt status. When capture channel 3 captures falling edge, if capture channel 3 fall lock interrupt (CFIE3) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 3 interrupt is not pending. 1: Capture channel 3 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 3 interrupt status.
6	R/W1C	0x0	CRIS3. Capture channel 3 rising lock interrupt status. When capture channel 3 captures rising edge, if capture channel 3 rise lock interrupt (CRIE3) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 3 interrupt is not pending. 1: Capture channel 3 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 3 interrupt status.
5	R/W1C	0x0	CFIS2. Capture channel 2 falling lock interrupt status. When capture channel 2 captures falling edge, if capture channel 2 fall lock interrupt (CFIE2) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 2 interrupt is not pending. 1: Capture channel 2 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 2 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	CRIS2. Capture channel 2 rising lock interrupt status. When capture channel 2 captures rising edge, if capture channel 2 rise lock interrupt (CRIE2) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 2 interrupt is not pending. 1: Capture channel 2 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 2 interrupt status.
3	R/W1C	0x0	CFIS1. Capture channel 1 falling lock interrupt status. When capture channel 1 captures falling edge, if capture channel 1 fall lock interrupt (CFIE1) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 1 interrupt is not pending. 1: Capture channel 1 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 1 interrupt status.
2	R/W1C	0x0	CRIS1. Capture channel 1 rising lock interrupt status. When capture channel 1 captures rising edge, if capture channel 1 rise lock interrupt (CRIE1) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 1 interrupt is not pending. 1: Capture channel 1 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 1 interrupt status.

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Offset:0x0014			Register Name: CISR
Bit	Read/Write	Default/Hex	Description
1	R/W1C	0x0	CFIS0. Capture channel 0 falling lock interrupt status. When capture channel 0 captures falling edge, if capture channel 0 fall lock interrupt (CFIE0) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 0 interrupt is not pending. 1: Capture channel 0 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 0 interrupt status.
0	R/W1C	0x0	CRIS0. Capture channel 0 rising lock interrupt status. When capture channel 0 captures rising edge, if capture channel 0 rise lock interrupt (CRIE0) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture channel 0 interrupt is not pending. 1: Capture channel 0 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 0 interrupt status.

17.14.7.5 0x0018 Capture Cycle Counter IRQ Enable Register (Default Value: 0x0000_0000)

Offset:0x0018			Register Name: CCCIER
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13	R/W	0x00	CCIE13. Capture counter channel 13 Interrupt Enable. 0: Capture counter channel 13 Interrupt disable; 1: Capture counter channel 13 Interrupt enable.
12	R/W	0x00	CCIE12. Capture counter channel 12 Interrupt Enable. 0: Capture counter channel 12 Interrupt disable; 1: Capture counter channel 12 Interrupt enable.
11	R/W	0x00	CCIE11. Capture counter channel 11 Interrupt Enable. 0: Capture counter channel 11 Interrupt disable; 1: Capture counter channel 11 Interrupt enable.

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Offset:0x0018			Register Name: CCCIER
Bit	Read/Write	Default/Hex	Description
10	R/W	0x00	CCIE10. Capture counter channel 10 Interrupt Enable. 0: Capture counter channel 10 Interrupt disable; 1: Capture counter channel 10 Interrupt enable.
9	R/W	0x00	CCIE9. Capture counter channel 9 Interrupt Enable. 0: Capture counter channel 9 Interrupt disable; 1: Capture counter channel 9 Interrupt enable.
8	R/W	0x00	CCIE8. Capture counter channel 8 Interrupt Enable. 0: Capture counter channel 8 Interrupt disable; 1: Capture counter channel 8 Interrupt enable.
7	R/W	0x00	CCIE7. Capture counter channel 7 Interrupt Enable. 0: Capture counter channel 7 Interrupt disable; 1: Capture counter channel 7 Interrupt enable.
6	R/W	0x00	CCIE6. Capture counter channel 6 Interrupt Enable. 0: Capture counter channel 6 Interrupt disable; 1: Capture counter channel 6 Interrupt enable.
5	R/W	0x00	CCIE5. Capture counter channel 5 Interrupt Enable. 0: Capture counter channel 5 Interrupt disable; 1: Capture counter channel 5 Interrupt enable.
4	R/W	0x00	CCIE4. Capture counter channel 4 Interrupt Enable. 0: Capture counter channel 4 Interrupt disable; 1: Capture counter channel 4 Interrupt enable.
3	R/W	0x00	CCIE3. Capture counter channel 3 Interrupt Enable. 0: Capture counter channel 3 Interrupt disable; 1: Capture counter channel 3 Interrupt enable.
2	R/W	0x00	CCIE2. Capture counter channel 2 Interrupt Enable. 0: Capture counter channel 2 Interrupt disable; 1: Capture counter channel 2 Interrupt enable.
1	R/W	0x00	CCIE1. Capture counter channel 1 Interrupt Enable. 0: Capture counter channel 1 Interrupt disable; 1: Capture counter channel 1 Interrupt enable.

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Offset:0x0018			Register Name: CCCIER
Bit	Read/Write	Default/Hex	Description
0	R/W	0x00	CCIE0. Capture counter channel 0 Interrupt Enable. 0: Capture counter channel 0 Interrupt disable; 1: Capture counter channel 0 Interrupt enable.

17.14.7.6 0x001C Capture Cycle Counter IRQ Status Register (Default Value: 0x0000_0000)

Offset:0x001C			Register Name: CCCISR
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13	R/W1C	0x00	CCIS13. If capture counter 13 is enabled(CCIE13) and channel 13 capture counter up to channel 13 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 13 interrupt is not pending. 1: Capture counter channel 13 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 13 interrupt status.
12	R/W1C	0x00	CCIS12. If capture counter12 is enabled(CCIE8) and channel 12 capture counter up to channel12 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 12 interrupt is not pending. 1: Capture counter channel 12 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 12 interrupt status.

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Offset:0x001C			Register Name: CCCISR
Bit	Read/Write	Default/Hex	Description
11	R/W1C	0x00	CCIS11. If capture counter 11 is enabled(CCIE11) and channel 11 capture counter up to channel 11 setting value (CCNR), this bit is set 11 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 11 interrupt is not pending. 1: Capture counter channel 11 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 11 interrupt status.
10	R/W1C	0x00	CCIS10. If capture counter 10 is enabled(CCIE10) and channel 10 capture counter up to channel 10 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 10 interrupt is not pending. 1: Capture counter channel 10 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 10 interrupt status.
9	R/W1C	0x00	CCIS9. If capture counter 9 is enabled(CCIE9) and channel 9 capture counter up to channel 9 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 9 interrupt is not pending. 1: Capture counter channel 9 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 9 interrupt status.

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Offset:0x001C			Register Name: CCCISR
Bit	Read/Write	Default/Hex	Description
8	R/W1C	0x00	CCIS8. If capture counter 8 is enabled(CCIE8) and channel 9 capture counter up to channel 8 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 8 interrupt is not pending. 1: Capture counter channel 8 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 8 interrupt status.
7	R/W1C	0x00	CCIS7. If capture counter 7 is enabled(CCIE7) and channel 7 capture counter up to channel 7 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 7 interrupt is not pending. 1: Capture counter channel 7 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 7 interrupt status.
6	R/W1C	0x00	CCIS6. If capture counter 6 is enabled(CCIE6) and channel 6 capture counter up to channel 6 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 6 interrupt is not pending. 1: Capture counter channel 6 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 6 interrupt status.

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Offset:0x001C			Register Name: CCCISR
Bit	Read/Write	Default/Hex	Description
5	R/W1C	0x00	CCIS5. If capture counter 5 is enabled(CCIE5) and channel 5 capture counter up to channel 5 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 5 interrupt is not pending. 1: Capture counter channel 5 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 5 interrupt status.
4	R/W1C	0x00	CCIS4. If capture counter 4 is enabled(CCIE4) and channel 4 capture counter up to channel 4 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 4 interrupt is not pending. 1: Capture counter channel 4 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 4 interrupt status.
3	R/W1C	0x00	CCIS3. If capture counter 3 is enabled(CCIE3) and channel 3 capture counter up to channel 3 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 3 interrupt is not pending. 1: Capture counter channel 3 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 3 interrupt status.

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Offset:0x001C			Register Name: CCCISR
Bit	Read/Write	Default/Hex	Description
2	R/W1C	0x00	CCIS2. If capture counter 2 is enabled(CCIE2) and channel 2 capture counter up to channel 2 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 2 interrupt is not pending. 1: Capture counter channel 2 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 2 interrupt status.
1	R/W1C	0x00	CCIS1. If capture counter 1 is enabled(CCIE1) and channel 1 capture counter up to channel 1 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 1 interrupt is not pending. 1: Capture counter channel 1 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 1 interrupt status.
0	R/W1C	0x00	CCIS0. If capture counter 0 is enabled(CCIE0) and channel 0 capture counter up to channel 0 setting value (CCNR), this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: Capture counter channel 0 interrupt is not pending. 1: Capture counter channel 0 interrupt is pending. Writes 0: No effect. 1: Clear capture channel 0 interrupt status.

17.14.7.7 0x0020 PWM01 Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: PCCR01
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/

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Offset: 0x0020			Register Name: PCCR01
Bit	Read/Write	Default/Hex	Description
8:7	R/W	0x00	PWM01_CLK_SRC. Select PWM01 clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/
3:0	R/W	0x0000	PWM01_CLK_DIV_M. PWM01 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.8 0x0024 PWM23 Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: PCCR23
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWM23_CLK_SRC_SEL. Select PWM23 clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/

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Offset: 0x0024			Register Name: PCCR23
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0000	PWM23_CLK_DIV_M. PWM23 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.9 0x0028 PWM45 Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: PCCR45
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWM45_CLK_SRC_SEL. Select PWM45 clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/
3:0	R/W	0x0000	PWM45_CLK_DIV_M. PWM45 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.10 0x002C PWM67 Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: PCCR67
Bit	Read/Write	Default/Hex	Description

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Offset: 0x002C			Register Name: PCCR67
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWM67_CLK_SRC_SEL. Select PWM67 clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/
3:0	R/W	0x0000	PWM67_CLK_DIV_M. PWM67 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.11 0x0030 PWM89 Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: PCCR89
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWM89_CLK_SRC_SEL. Select PWM89 clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/

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Offset: 0x0030			Register Name: PCCR89
Bit	Read/Write	Default/Hex	Description
3:0	R/W	0x0000	PWM89_CLK_DIV_M. PWM67 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.12 0x0034 PWMAB Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: PCCRAB
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWMAB_CLK_SRC_SEL. Select PWMAB clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/
3:0	R/W	0x0000	PWMAB_CLK_DIV_M. PWMAB clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.13 0x0038 PWMCD Clock Configuration Register (Default Value: 0x0000_0000)

Offset: 0x0038			Register Name: PCCRCD
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0038			Register Name: PCCRCD
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x00	PWMCD_CLK_SRC_SEL. Select PWMCD clock source. 00: OSC24M 01: APB0 Others: /
6:4	/	/	/
3:0	R/W	0x0000	PWMCD_CLK_DIV_M. PWMCD clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

17.14.7.14 0x0040 PWM Clock Gating Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: PCGR
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R/W	0x0	PWM13_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM13 output. 0: Not bypass 1: Bypass
28	R/W	0x0	PWM12_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM12 output. 0: Not bypass 1: Bypass
27	R/W	0x0	PWM11_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM11 output. 0: Not bypass 1: Bypass

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Offset: 0x0040			Register Name: PCGR
Bit	Read/Write	Default/Hex	Description
26	R/W	0x0	PWM10_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM10 output. 0: Not bypass 1: Bypass
25	R/W	0x0	PWM9_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM9 output. 0: Not bypass 1: Bypass
24	R/W	0x0	PWM8_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM8 output. 0: Not bypass 1: Bypass
23	R/W	0x0	PWM7_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM7 output. 0: Not bypass 1: Bypass
22	R/W	0x0	PWM6_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM6 output. 0: Not bypass 1: Bypass
21	R/W	0x0	PWM5_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM5 output. 0: Not bypass 1: Bypass
20	R/W	0x0	PWM4_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM4 output. 0: Not bypass 1: Bypass
19	R/W	0x0	PWM3_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM3 output. 0: Not bypass 1: Bypass

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Offset: 0x0040			Register Name: PCGR
Bit	Read/Write	Default/Hex	Description
18	R/W	0x0	PWM2_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM2 output. 0: Not bypass 1: Bypass
17	R/W	0x0	PWM1_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM1 output. 0: Not bypass 1: Bypass
16	R/W	0x0	PWM0_CLK_BYPASS. Bypass clock source (after pre-scale) to PWM0 output. 0: Not bypass 1: Bypass
15:14	/	/	/
13	R/W	0x0	PWM13_CLK_GATING. Gating clock for PWM13. 0: Mask 1: Pass
12	R/W	0x0	PWM12_CLK_GATING. Gating clock for PWM12. 0: Mask 1: Pass
11	R/W	0x0	PWM11_CLK_GATING. Gating clock for PWM11. 0: Mask 1: Pass
10	R/W	0x0	PWM10_CLK_GATING. Gating clock for PWM10. 0: Mask 1: Pass
9	R/W	0x0	PWM9_CLK_GATING. Gating clock for PWM9. 0: Mask 1: Pass
8	R/W	0x0	PWM8_CLK_GATING. Gating clock for PWM8. 0: Mask 1: Pass

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Offset: 0x0040			Register Name: PCGR
Bit	Read/Write	Default/Hex	Description
7	R/W	0x0	PWM7_CLK_GATING. Gating clock for PWM7. 0: Mask 1: Pass
6	R/W	0x0	PWM6_CLK_GATING. Gating clock for PWM6. 0: Mask 1: Pass
5	R/W	0x0	PWM5_CLK_GATING. Gating clock for PWM5. 0: Mask 1: Pass
4	R/W	0x0	PWM4_CLK_GATING. Gating clock for PWM4. 0: Mask 1: Pass
3	R/W	0x0	PWM3_CLK_GATING. Gating clock for PWM3. 0: Mask 1: Pass
2	R/W	0x0	PWM2_CLK_GATING. Gating clock for PWM2. 0: Mask 1: Pass
1	R/W	0x0	PWM1_CLK_GATING. Gating clock for PWM1. 0: Mask 1: Pass
0	R/W	0x0	PWM0_CLK_GATING. Gating clock for PWM0. 0: Mask 1: Pass

17.14.7.15 0x0060 PWM01 Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0060			Register Name: PDZCR01
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PDZINTV01. PWM01 Dead Zone interval value.
7:1	/	/	/

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Offset: 0x0060			Register Name: PDZCR01
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PWM01_DZ_EN. PWM01 Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.16 0x0064 PWM23 Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0064			Register Name: PDZCR23
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM23_DZ_INTV. PWM23 Dead Zone interval value.
7:1	/	/	/
0	R/W	0x0	PWM23_DZ_EN. PWM23 Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.17 0x0068 PWM45 Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0068			Register Name: PDZCR45
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM45_DZ_INTV. PWM45 Dead Zone interval value.
7:1	/	/	/
0	R/W	0x0	PWM45_DZ_EN. PWM45 Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.18 0x006C PWM67 Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x006C			Register Name: PDZCR67
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM67_DZ_INTV. PWM67 Dead Zone interval value.
7:1	/	/	/

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Offset: 0x006C			Register Name: PDZCR67
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PWM67_DZ_EN. PWM67 Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.19 0x0070 PWM89 Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0070			Register Name: PDZCR89
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM89_DZ_INTV. PWM89 Dead Zone interval value.
7:1	/	/	/
0	R/W	0x0	PWM89_DZ_EN. PWM89 Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.20 0x0074 PWMAB Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0074			Register Name: PDZCRAB
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWMAB_DZ_INTV. PWMAB Dead Zone interval value.
7:1	/	/	/
0	R/W	0x0	PWMAB_DZ_EN. PWMAB Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.21 0x0078 PWMCD Dead Zone Control Register (Default Value: 0x0000_0000)

Offset: 0x0078			Register Name: PDZCRCD
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWMCD_DZ_INTV. PWMCD Dead Zone interval value.
7:1	/	/	/

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Offset: 0x0078			Register Name: PDZCRCD
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	PWMCD_DZ_EN. PWMCD Dead Zone enable. 0: Dead Zone disable 1: Dead Zone enable.

17.14.7.22 0x0080 PWM Enable Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: PER
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13	R/W	0x0	PWM13_EN. PWM Channel 13 Enable 0: PWM disable 1: PWM enable.
12	R/W	0x0	PWM12_EN. PWM Channel 12 Enable 0: PWM disable 1: PWM enable.
11	R/W	0x0	PWM11_EN. PWM Channel 11 Enable 0: PWM disable 1: PWM enable.
10	R/W	0x0	PWM10_EN. PWM Channel 10 Enable 0: PWM disable 1: PWM enable.
9	R/W	0x0	PWM9_EN. PWM Channel 9 Enable 0: PWM disable 1: PWM enable.
8	R/W	0x0	PWM8_EN. PWM Channel 8 Enable 0: PWM disable 1: PWM enable.
7	R/W	0x0	PWM7_EN. PWM Channel 7 Enable 0: PWM disable 1: PWM enable.

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Offset: 0x0080			Register Name: PER
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	PWM6_EN. PWM Channel 6 Enable 0: PWM disable 1: PWM enable.
5	R/W	0x0	PWM5_EN. PWM Channel 5 Enable 0: PWM disable 1: PWM enable.
4	R/W	0x0	PWM4_EN. PWM Channel 4 Enable 0: PWM disable 1: PWM enable.
3	R/W	0x0	PWM3_EN. PWM Channel 3 Enable 0: PWM disable 1: PWM enable.
2	R/W	0x0	PWM2_EN. PWM Channel 2 Enable 0: PWM disable 1: PWM enable.
1	R/W	0x0	PWM1_EN. PWM Channel 1 Enable 0: PWM disable 1: PWM enable.
0	R/W	0x0	PWM0_EN. PWM Channel 0 Enable 0: PWM disable 1: PWM enable.

17.14.7.23 0x0090 PWM Group0 Register (Default Value: 0x0000_0000)

Offset: 0x0090			Register Name: PGR0
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/WAC	0x0	PWMG0_START PWM channels selected in PWMG0_CS start to output PWM waveform at the same time.
16	R/W	0x0	PWMG0_EN PWM Group0 Enable.

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Offset: 0x0090			Register Name: PGR0
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	PWMG0_CS If bit[i] is set, PWM i is selected as one channel of PWM Group0.

17.14.7.24 0x0094 PWM Group1 Register (Default Value: 0x0000_0000)

Offset: 0x0094			Register Name: PGR1
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/WAC	0x0	PWMG1_START PWM channels selected in PWMG1_CS start to output PWM waveform at the same time.
16	R/W	0x0	PWMG1_EN PWM Group1 Enable.
15:0	R/W	0x0	PWMG1_CS If bit[i] is set, PWM i is selected as one channel of PWM Group1.

17.14.7.25 0x0098 PWM Group2 Register (Default Value: 0x0000_0000)

Offset: 0x0098			Register Name: PGR2
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/WAC	0x0	PWMG2_START PWM channels selected in PWMG2_CS start to output PWM waveform at the same time.
16	R/W	0x0	PWMG2_EN PWM Group2 Enable.
15:0	R/W	0x0	PWMG2_CS If bit[i] is set, PWM i is selected as one channel of PWM Group2.

17.14.7.26 0x009C PWM Group3 Register (Default Value: 0x0000_0000)

Offset: 0x009C			Register Name: PGR3
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/WAC	0x0	PWMG3_START PWM channels selected in PWMG3_CS start to output PWM waveform at the same time.

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Offset: 0x009C			Register Name: PGR3
Bit	Read/Write	Default/Hex	Description
16	R/W	0x0	PWMG3_EN PWM Group2 Enable.
15:0	R/W	0x0	PWMG3_CS If bit[i] is set, PWM i is selected as one channel of PWM Group3.

17.14.7.27 0x00A0 Capture Cycle Counter Enable Register (Default Value: 0x0000_0000)

Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29	R	0x0	ATUO_CLEAR_FLAG_13 When choosing the CAP_CNT13_EN to enable capture counter this bit is used to judge whether the channel13 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT13_EN to 1 until ATUO_CLEAR_FLAG_13=0.
28	R	0x0	ATUO_CLEAR_FLAG_12 When choosing the CAP_CNT12_EN to enable capture counter this bit is used to judge whether the channel12 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT12_EN to 1 until ATUO_CLEAR_FLAG_12=0.
27	R	0x0	ATUO_CLEAR_FLAG_11 When choosing the CAP_CNT11_EN to enable capture counter this bit is used to judge whether the channel11 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT11_EN to 1 until ATUO_CLEAR_FLAG_11=0.

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Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
26	R	0x0	ATUO_CLEAR_FLAG_10 When choosing the CAP_CNT10_EN to enable capture counter this bit is used to judge whether the channel6 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT10_EN to 1 until ATUO_CLEAR_FLAG_10=0.
25	R	0x0	ATUO_CLEAR_FLAG_9 When choosing the CAP_CNT9_EN to enable capture counter this bit is used to judge whether the channel9 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT9_EN to 1 until ATUO_CLEAR_FLAG_9=0.
24	R	0x0	ATUO_CLEAR_FLAG_8 When choosing the CAP_CNT1_EN to enable capture counter this bit is used to judge whether the channel8 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT8_EN to 1 until ATUO_CLEAR_FLAG_8=0.
23	R	0x0	ATUO_CLEAR_FLAG_7 When choosing the CAP_CNT7_EN to enable capture counter this bit is used to judge whether the channel7 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT7_EN to 1 until ATUO_CLEAR_FLAG_7=0.
22	R	0x0	ATUO_CLEAR_FLAG_6 When choosing the CAP_CNT6_EN to enable capture counter this bit is used to judge whether the channel6 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT6_EN to 1 until ATUO_CLEAR_FLAG_6=0.

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Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
21	R	0x0	ATUO_CLEAR_FLAG_5 When choosing the CAP_CNT5_EN to enable capture counter this bit is used to judge whether the channel5 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT5_EN to 1 until ATUO_CLEAR_FLAG_5=0.
20	R	0x0	ATUO_CLEAR_FLAG_4 When choosing the CAP_CNT1_EN to enable capture counter this bit is used to judge whether the channel4 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT4_EN to 1 until ATUO_CLEAR_FLAG_4=0.
19	R	0x0	ATUO_CLEAR_FLAG_3 When choosing the CAP_CNT3_EN to enable capture counter this bit is used to judge whether the channel3 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT3_EN to 1 until ATUO_CLEAR_FLAG_3=0.
18	R	0x0	ATUO_CLEAR_FLAG_2 When choosing the CAP_CNT2_EN to enable capture counter this bit is used to judge whether the channel2 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT2_EN to 1 until ATUO_CLEAR_FLAG_2=0.
17	R	0x0	ATUO_CLEAR_FLAG_1 When choosing the CAP_CNT1_EN to enable capture counter this bit is used to judge whether the channel1 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT1_EN to 1 until ATUO_CLEAR_FLAG_1=0.

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Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
16	R	0x0	ATUO_CLEAR_FLAG_0 When choosing the CAP_CNT0_EN to enable capture counter this bit is used to judge whether the channel0 capture counter is working or not. This bit=1 means the channel is processing, this bit=0 means the channel is disable. The user should not set CAP_CNT0_EN to 1 until ATUO_CLEAR_FLAG_0=0.
15:14	/	/	
13	R/W	0x0	CAP_CNT13_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT13_EN=1 will enabling capture counter. NOTE: CAP_CNT13_EN and CAP13_EN cannot be enabled together.
12	R/W	0x0	CAP_CNT12_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT12_EN=1 will enabling capture counter. NOTE: CAP_CNT12_EN and CAP12_EN cannot be enabled together.
11	R/W	0x0	CAP_CNT11_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT11_EN=1 will enabling capture counter. NOTE: CAP_CNT11_EN and CAP11_EN cannot be enabled together.
10	R/W	0x0	CAP_CNT10_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT10_EN=1 will enabling capture counter. NOTE: CAP_CNT10_EN and CAP10_EN cannot be enabled together.

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Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
9	R/W	0x0	CAP_CNT9_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT9_EN=1 will enabling capture counter. NOTE: CAP_CNT9_EN and CAP9_EN cannot be enabled together.
8	R/W	0x0	CAP_CNT8_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT8_EN=1 will enabling capture counter. NOTE: CAP_CNT8_EN and CAP8_EN cannot be enabled together.
7	R/W	0x0	CAP_CNT7_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT7_EN=1 will enabling capture counter. NOTE: CAP_CNT7_EN and CAP7_EN cannot be enabled together.
6	R/W	0x0	CAP_CNT6_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT6_EN=1 will enabling capture counter. NOTE: CAP_CNT6_EN and CAP6_EN cannot be enabled together.
5	R/W	0x0	CAP_CNT5_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT5_EN=1 will enabling capture counter. NOTE: CAP_CNT5_EN and CAP5_EN cannot be enabled together.
4	R/W	0x0	CAP_CNT4_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT4_EN=1 will enabling capture counter. NOTE: CAP_CNT4_EN and CAP4_EN cannot be enabled together.

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Offset: 0x00A0			Register Name: CCCER
Bit	Read/Write	Default/Hex	Description
3	R/W	0x0	CAP_CNT3_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT3_EN=1 will enabling capture counter. NOTE: CAP_CNT3_EN and CAP3_EN cannot be enabled together.
2	R/W	0x0	CAP_CNT2_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT2_EN=1 will enabling capture counter. NOTE: CAP_CNT2_EN and CAP2_EN cannot be enabled together.
1	R/W	0x0	CAP_CNT1_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT1_EN=1 will enabling capture counter. NOTE: CAP_CNT1_EN and CAP1_EN cannot be enabled together.
0	R/W	0x0	CAP_CNT0_EN When relative PWM channel Capture Control Register bit5=0, setting CAP_CNT0_EN=1 will enabling capture counter. NOTE: CAP_CNT0_EN and CAP0_EN cannot be enabled together.

17.14.7.28 0x00C0 Capture Enable Register (Default Value: 0x0000_0000)

Offset: 0x00C0			Register Name: CER
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13	R/W	0x0	CAP13_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 9 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.

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Offset: 0x00C0			Register Name: CER
Bit	Read/Write	Default/Hex	Description
12	R/W	0x0	CAP12_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 8 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
11	R/W	0x0	CAP11_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 7 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
10	R/W	0x0	CAP10_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 6 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
9	R/W	0x0	CAP9_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 9 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
8	R/W	0x0	CAP8_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 8 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
7	R/W	0x0	CAP7_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 7 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.

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Offset: 0x00C0			Register Name: CER
Bit	Read/Write	Default/Hex	Description
6	R/W	0x0	CAP6_EN. When enable capture function, the 16-bit up-counter starts working and capture channel 6 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
5	R/W	0x0	CAP5_EN. When enable capture function, the 16-bit up-counter starts working and capture channel5 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
4	R/W	0x0	CAP4_EN. When enable capture function, the 16-bit up-counter starts working and capture channel4 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
3	R/W	0x0	CAP3_EN. When enable capture function, the 16-bit up-counter starts working and capture channel3 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
2	R/W	0x0	CAP2_EN. When enable capture function, the 16-bit up-counter starts working and capture channel2 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.
1	R/W	0x0	CAP1_EN. When enable capture function, the 16-bit up-counter starts working and capture channel1 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.

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Offset: 0x00C0			Register Name: CER
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	CAP0_EN. When enable capture function, the 16-bit up-counter starts working and capture channel is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable.

17.14.7.29 0x00C4 ADC Notification Register (Default Value: 0x0000_0000)

Offset: 0x00C4			Register Name: ANR
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x0	CHANNEL_SEL. 0x0: channel 0 0x1: channel 1 ..... 0x13: channel 13 Other: No used This register is used to select which channel to notify the ADC that the pulse output is complete.

17.14.7.30 0x0100+0x0000+N*0x0040 PWM Control Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0000+N*0x0040			Register Name: PCR
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PWM_PUL_NUM In pulse mode, PWM output pulse for PWM_CYCLE_NUM+1 times and then stop.
15:13	/	/	/
12	R/W	0x0	ADC_WORK In pulse mode, if ADC_WORK=1, the PWM will inform ADC to work when PWM output pulse up to PWM_CYCLE_NUM+1 times. 0: Disable to inform ADC to work in pulse mode 1: Enable to inform ADC to work in pulse mode
11	R	0x0	PWM_PERIOD_RDY. PWM period register ready. 0: PWM period register is ready to write 1: PWM period register is busy.

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Offset: 0x0100+0x0000+N*0x0040			Register Name: PCR
Bit	Read/Write	Default/Hex	Description
10	R/WAC	0x0	PWM_PUL_START. PWM pulse output start. 0: No effect 1: Output pulse for PWM_CYCLE_NUM+1. After finishing configuration for outputting pulse, set this bit once and then PWM would output waveform. After the waveform is finished, the bit will be cleared automatically.
9	R/W	0x0	PWM_MODE. PWM output mode select. 0: Cycle mode 1: Pulse mode.
8	R/W	0x0	PWM_ACT_STA. PWM active state. 0: Low Level 1: High Level.
7:0	R/W	0x0	PWM_PRESCAL_K. PWM pre-scale K, actual pre-scale is (K+1). K = 0, actual pre-scale: 1 K = 1, actual pre-scale: 2 K = 2, actual pre-scale: 3 K = 3, actual pre-scale: 4 ..... K = 255, actual pre-scale: 256.

17.14.7.31 0x0100+0x0004+N*0x0040 PWM Period Register (Default Value: 0x0000_0000)

Offset:0x0100+0x0004+N*0x0040			Register Name: PPR
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PWM_ENTIRE_CYCLE. Number of the entire cycles in the PWM clock. 0 = 1 cycle 1 = 2 cycles ..... N = N+1 cycles If the register need to be modified dynamically, the PCLK should be faster than the PWM CLK .

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Offset:0x0100+0x0004+N*0x0040			Register Name: PPR
Bit	Read/Write	Default/Hex	Description
15:0	R/W	0x0	PWM_ACT_CYCLE. Number of the active cycles in the PWM clock. 0 = 0 cycle 1 = 1 cycles ..... N = N cycles

17.14.7.32 0x0100+0x0008+N*0x0040 PWM Counter Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0008+N*0x0040			Register Name: PCNTR
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	PWM_COUNTER_START PWM counter value is set for phase control.
15:0	R	0x0	PWM_COUNTER_STATUS On PWM output or capture input, reading this register could get the current value of the PWM 16bit up-counter.

17.14.7.33 0x0100+0x000C+N*0x0040 PWM Pulse Counter Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x000C+N*0x0040			Register Name: PPCNTR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	PWM_PUL_COUNTER_STATUS On PWM output , reading this register could get the current value of the PWM pulse counter.

17.14.7.34 0x0100+0x0010+N*0x0040 Capture Control Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0010+N*0x0040			Register Name: CCR
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	CCSR This bit is used to configure starting manner of capture counter. 0: Staring capture counter when CAP_CNT*_EN=1 1: Starting capture counter when CAP*_EN=1 NOTE: This bit needs to be configured before CAP_CNT*_EN=1 or CAP*_EN=1

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Offset: 0x0100+0x0010+N*0x0040			Register Name: CCR
Bit	Read/Write	Default/Hex	Description
4	R/W1C	0x0	CRLF. When capture channel captures rising edge, the 16-bit up-counter's current value is latched to CRLR and then this bit is set 1 by hardware. Write 1 to clear this bit.
3	R/W1C	0x0	CFLF. When capture channel captures falling edge, the 16-bit up-counter's current value is latched to CFLR and then this bit is set 1 by hardware. Write 1 to clear this bit.
2	R/W	0x0	CRTE. Rising edge capture trigger enable
1	R/W	0x0	CFTE. Falling edge capture trigger enable
0	R/W	0x0	CAPINV. Inversing the signal inputted form capture channel before capture channel's 16bit counter. 0: Not inverse 1: Inverse

17.14.7.35 0x0100+0x0014+N*0x0040 Capture Rise Lock Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0014+N*0x0040			Register Name: CRLR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	CRLR. When capture channel captures rising edge, the 16-bit up-counter's current value is latched to this register.

17.14.7.36 0x0100+0x0018+N*0x0040 Capture Fall Lock Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0018+N*0x0040			Register Name: CFLR
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	CFLR. When capture channel captures falling edge, the 16-bit up-counter's current value is latched to this register.

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17.14.7.37 0x0100+0x001C+N*0x0040 Capture Cycle Counter Status Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x001C+N*0x0040			Register Name: CCCSR
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	CCCSR. When capture channel captures the cycles of input signal, the 32-bit up-counter's current value can be observed by this register.

17.14.7.38 0x0100+0x0020+N*0x0040 Capture Cycle Counter Number Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0020+N*0x0040			Register Name: CCCNR
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	CCCNR. This number is used to compare with capture counter number, and if capture counter up to the number, it will generate capture counter interrupt if capture counter interrupt is enable.

17.14.7.39 0x0100+0x0024+N*0x0040 Capture Cycle Counter Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0024+N*0x0040			Register Name: CCCR
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R	0x0	CCCNR_RDY. Capture Cycle Counter Number Register ready. 0: Capture Cycle Counter Number Register is ready to write. 1: Capture Cycle Counter Number Register is busy.
1	R/W	0x0	CCCSRR This number is used to choose the behavior after counter value up to CCCNR. 0: Stop counting and set CAP_CNT*_EN=0 if PWM channel Capture Control Register bit5=0 1: Restart counting
0	R/W	0x0	CCCRFR. This number is used to choose capture cycle counter using rising edge or falling edge to count. 0: Rising edge 1: Falling edge

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17.14.7.40 0x0100+0x0028+N*0x0040 Capture Filter Register (Default Value: 0x0000_0000)

Offset: 0x0100+0x0028+N*0x0040			Register Name: CFG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	CFG_EN 0: Do not filter the input wave 1: Filter the input wave
4:0	R/W	0x0	FILTER_LVL 0x0: 0 0x1: 1 ..... 0x1F: 31 This bit is used to set how many cycles of input waveform should be kept to be regarded as valid. NOTE: The configuration should be ready before PWM capture enable. After PWM capture is enabled, no modification is permitted.

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17.15 PWMCS

17.15.1 Overview

Pulse Width Modulation Control System (PWM) supports configurable PWM waveform output (single-end, A/B phase, pulse+direction, CW/CCW) and supports waveform capture to detect input pulse signals.

This device has two PWMCS controllers, and each PWMCS controller has following features:

Waveform Output

- Up to 8-channel independent output
- Up to 4-channel group mode output.
  - Phase configuration range for each channel is 0 to 2^{23}-1.
- Up to 4-channel complementary pairs mode output
  - Supports dead-zone generator.
  - The dead-zone time is configurable, ranging between 0 and 2^{8}-1.
- Up to 4-channel IO pairs mode output
  - Supports pulse+direction, AB phase, CW/CCW
  - IO pairs support four groups: PWMCS_CH_01, PWMCS_CH_23, PWMCS_CH_45, and PWMCS_CH_67
- Supports periodic output (not support in group mode and IO pair mode)
- Supports pulse output
  - The number of pulse is configurable, ranging from 0 to 2^{23}-1.
  - Supports continuous output of pulses with different duty cycles.
- Supports outputting high/low level and high resistance
- Configurable initial phase and polarity
- Configurable pulse output suspension state
- Supports pulse output delay, ranging from 0 to 2^{23}-1
- Configurable output waveform duty cycle (0-100%)
- Supports input filtering with configurable filtering time in the range of 0 to 2^{32}-1
- 0-4 MHz output frequency. Supports maximum and minimum frequency limit

Input Waveform Capture

- Up to 8-channel independent capture
  - Input frequency ranging from 0 to 4 MHz
  - Input waveform rising/falling edge detection

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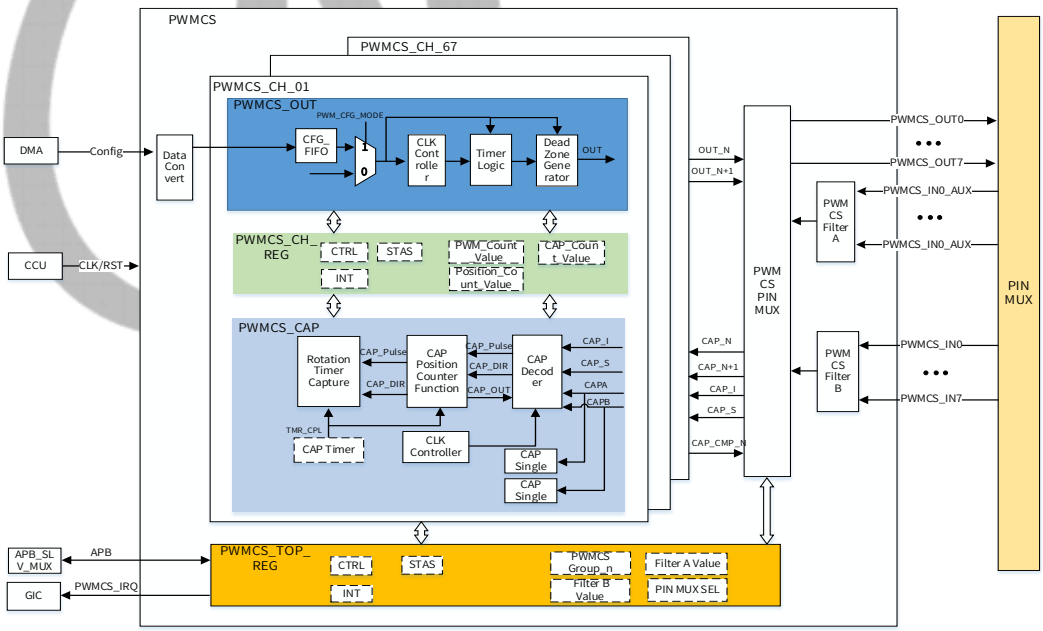
- Continuous capture mode and single capture mode
- Input waveform periodic counting, ranging from 0 to 2^{23}-1.
- Capture of high/low logic level, ranging from 0 to 2^{23}-1.

- Up to 4-channel IO pairs mode capture
  - Supports pulse+direction, AB phase, CW/CCW
  - Incremental/decremental counting, ranging from 0 to 2^{32}-1.
  - Status output and latching of position/time counters
  - Orthogonal decoding counter (single-fold frequency, double-fold frequency, and quadruple-fold frequency)
  - Position measurement with fixed interval and interval measurement with fixed position
  - IO pairs support four groups: PWMCS_CH_01, PWMCS_CH_23, PWMCS_CH_45, and PWMCS_CH_67
- Supports INDEX, STROB, software initiation and capture position counter
- Supports input level detection. The time of detection is configurable, ranging from 0 to 2^{32}-1
- Supports input filtering with configurable filtering time in the range of 0 to 2^{32}-1

17.15.2 Block Diagram

The following figure shows a block diagram of the PWMCS.

Figure 17-105 PWMCS Block Diagram



PWMCS mainly consists of the top block and channel block.

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Table 17-71 PWMCS Block Description

Blocks	Sub-Blocks	Description
Top blocks	Data Convert	It is responsible for data concatenation written by DMA. It can select the CFG FIFO corresponding to each channel according to register configuration.
	PWMCS_TOP_REG	Top registers, including group registers and I/O multiplex configuration registers.
	PWMCS_PIN_MUX	It implements signal mapping between external signals from GPIO pins and internal PWM-IN/OUT signals.
	PWMCS_FILTER	For input filtering and level detection.
Channel blocks	PWMCS_OUT	It implements single-end, A/B phase, pulse+direction, CW/CCW pulse output. • **CFG FIFO**: parameters configuration FIFO, such as configuration for output number of pulse, frequency, and duty cycle. • **CLK Control**: clock configuration, including clock source selection, two-level division ratio selection, and clock gating configuration. • **Timer Logic**: it contains control logics for output number of pulse, duty cycle, phase, and so on. • **Dead Zone Generator**: it inserts dead-time into output waveform, and the dead-time is configurable.
	PWMCS_CH_REG	It impeachments register control input, status output, and interrupt input.
	PWMCS_CAP	It implements single-end, A/B phase, pulse+direction, CW/CCW pulse capture. • **CLK Control**: clock configuration, including clock source selection, two-level division ratio selection, and clock gating. • **CAP Decoder**: used to decode the capture to get information such as CAP_Pulse and CAP_DIR. CAP_Pulse and CAP_DIR are the counting pulse signal and counting direction signal for CAP position counter. • **CAP Position Counter**: it counts based on analytic signals from CAP Decoder. • **Rotation Timer Capture**: rotation speed calculation based on the captured position and moving interval. • **CAP Timer**: used to generate signals with intervals of unit time. • **CAP Single**: used to capture single-end duty cycle and edges.

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17.15.3 External Signals

The following table describes the external signals of the PWMCS.

Table 17-72 PWMCS External Signals

Signal	Description	Type
PWMCS-O0	PWMCS Waveform Output 0	O
PWMCS-O1	PWMCS Waveform Output 1	O
PWMCS-O2	PWMCS Waveform Output 2	O
PWMCS-O3	PWMCS Waveform Output 3	O
PWMCS-O4	PWMCS Waveform Output 4	O
PWMCS-O5	PWMCS Waveform Output 5	O
PWMCS-O6	PWMCS Waveform Output 6	O
PWMCS-O7	PWMCS Waveform Output 7	O
PWMCS-O8	PWMCS Waveform Output 8	O
PWMCS-O9	PWMCS Waveform Output 9	O
PWMCS-OA	PWMCS Waveform Output 10	O
PWMCS-OB	PWMCS Waveform Output 11	O
PWMCS-OC	PWMCS Waveform Output 12	O
PWMCS-OD	PWMCS Waveform Output 13	O
PWMCS-OE	PWMCS Waveform Output 14	O
PWMCS-OF	PWMCS Waveform Output 15	O
PWMCS-I0	PWMCS Capture Waveform Input 0	I
PWMCS-I1	PWMCS Capture Waveform Input 1	I
PWMCS-I2	PWMCS Capture Waveform Input 2	I
PWMCS-I3	PWMCS Capture Waveform Input 3	I
PWMCS-I4	PWMCS Capture Waveform Input 4	I
PWMCS-I5	PWMCS Capture Waveform Input 5	I
PWMCS-I6	PWMCS Capture Waveform Input 6	I
PWMCS-I7	PWMCS Capture Waveform Input 7	I
PWMCS-I8	PWMCS Capture Waveform Input 8	I
PWMCS-I9	PWMCS Capture Waveform Input 9	I
PWMCS-IA	PWMCS Capture Waveform Input 10	I
PWMCS-IB	PWMCS Capture Waveform Input 11	I
PWMCS-IC	PWMCS Capture Waveform Input 12	I
PWMCS-ID	PWMCS Capture Waveform Input 13	I
PWMCS-IE	PWMCS Capture Waveform Input 14	I
PWMCS-IF	PWMCS Capture Waveform Input 15	I
PWM0-0	PWM0 Waveform Output /Capture Waveform Input 0	I/O
PWM0-1	PWM0 Waveform Output /Capture Waveform Input 1	I/O
PWM0-2	PWM0 Waveform Output /Capture Waveform Input 2	I/O
PWM0-3	PWM0 Waveform Output /Capture Waveform Input 3	I/O
PWM0-4	PWM0 Waveform Output /Capture Waveform Input 4	I/O
PWM0-5	PWM0 Waveform Output /Capture Waveform Input 5	I/O

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Signal	Description	Type
PWM0-6	PWM0 Waveform Output /Capture Waveform Input 6	I/O
PWM0-7	PWM0 Waveform Output /Capture Waveform Input 7	I/O
PWM1-0	PWM1 Waveform Output /Capture Waveform Input 0	I/O
PWM1-1	PWM1 Waveform Output /Capture Waveform Input 1	I/O
PWM1-2	PWM1 Waveform Output /Capture Waveform Input 2	I/O
PWM1-3	PWM1 Waveform Output /Capture Waveform Input 3	I/O
PWM1-4	PWM1 Waveform Output /Capture Waveform Input 4	I/O
PWM1-5	PWM1 Waveform Output /Capture Waveform Input 5	I/O
PWM1-6	PWM1 Waveform Output /Capture Waveform Input 6	I/O
PWM1-7	PWM1 Waveform Output /Capture Waveform Input 7	I/O

The following table shows the multiplexing relationship of internal and external signals.

Table 17-73 PWMCS Signal Multiplexing

	External Signals		Internal Signals
	PWM Function	PWMCS Function	
PWMCS0	PWM0-0	PWMCS-O0	PWMCS-DO0
	PWM0-1	PWMCS-O1	PWMCS-DO1
	PWM0-2	PWMCS-O2	PWMCS-DO2
	PWM0-3	PWMCS-O3	PWMCS-DO3
	PWM0-4	PWMCS-O4	PWMCS-DO4
	PWM0-5	PWMCS-O5	PWMCS-DO5
	PWM0-6	PWMCS-O6	PWMCS-DO6
	PWM0-7	PWMCS-O7	PWMCS-DO7
	PWM0-0	PWMCS-I0	PWMCS-DI0
	PWM0-1	PWMCS-I1	PWMCS-DI1
	PWM0-2	PWMCS-I2	PWMCS-DI2
	PWM0-3	PWMCS-I3	PWMCS-DI3
	PWM0-4	PWMCS-I4	PWMCS-DI4
	PWM0-5	PWMCS-I5	PWMCS-DI5
	PWM0-6	PWMCS-I6	PWMCS-DI6
	PWM0-7	PWMCS-I7	PWMCS-DI7
PWMCS1	PWM1-0	PWMCS-O8	PWMCS-DO0
	PWM1-1	PWMCS-O9	PWMCS-DO1
	PWM1-2	PWMCS-OA	PWMCS-DO2
	PWM1-3	PWMCS-OB	PWMCS-DO3
	PWM1-4	PWMCS-OC	PWMCS-DO4
	PWM1-5	PWMCS-OD	PWMCS-DO5
	PWM1-6	PWMCS-OE	PWMCS-DO6
	PWM1-7	PWMCS-OF	PWMCS-DO7
	PWM1-0	PWMCS-I8	PWMCS-DI0
	PWM1-1	PWMCS-I9	PWMCS-DI1
	PWM1-2	PWMCS-IA	PWMCS-DI2

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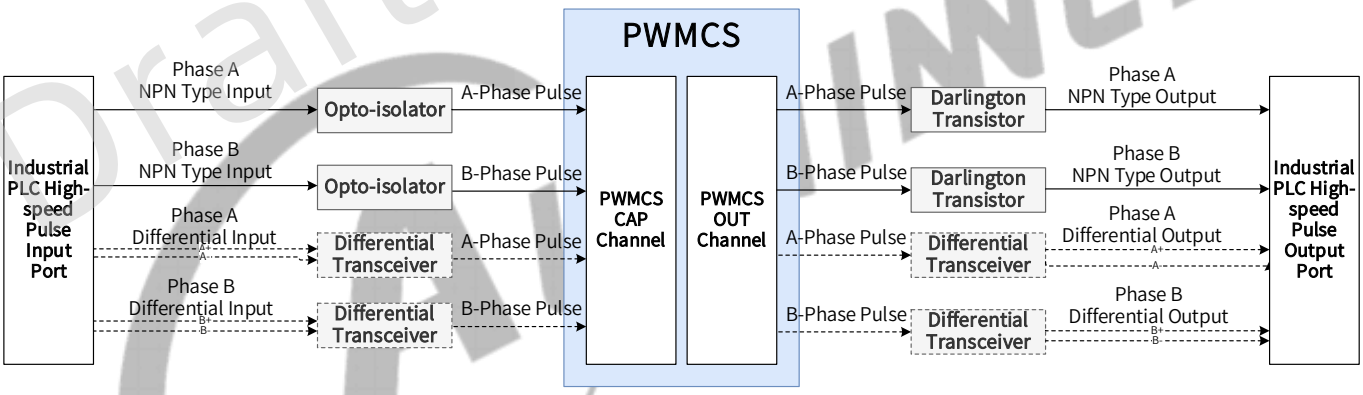
	External Signals		Internal Signals
	PWM Function	PWMCS Function	
	PWM1-3	PWMCS-IB	PWMCS-DI3
	PWM1-4	PWMCS-IC	PWMCS-DI4
	PWM1-5	PWMCS-ID	PWMCS-DI5
	PWM1-6	PWMCS-IE	PWMCS-DI6
	PWM1-7	PWMCS-IF	PWMCS-DI7

17.15.4 Functional Description

17.15.4.1 Typical Application

The pulse output function of the PWMCS module is mainly used in industry to control pulse-type AC servo drives rather than directly drive motors. Typical application is shown in the figure below.

Figure 17-106 PWMCS Typical Application



17.15.4.2 A/B Phase Output

The following figure shows the diagram of A/B phase mode output, taking PWMCS_O0/1 as an example.

When the position of PWMCS_O0 waveform is 90° before PWMCS_O1, it means the external motor rotates forward.

When the position of PWMCS_O1 waveform is 90° before PWMCS_O0, it means the external motor rotates backward.

AB phase switching can be realized by adjusting pulse phase value through the delay of output channel.

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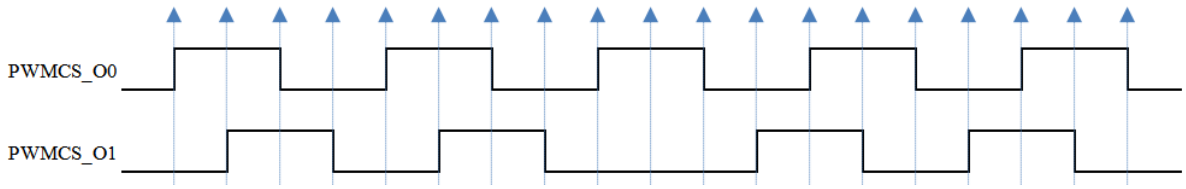
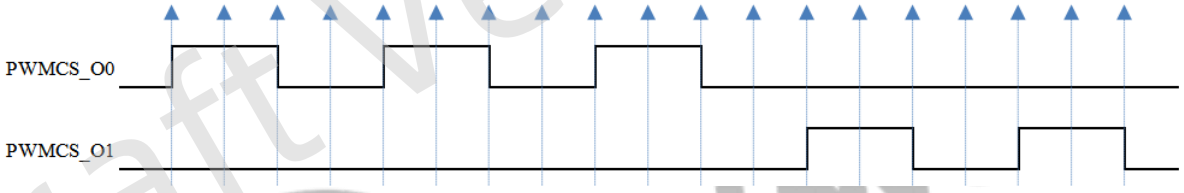
Figure 17-107 A/B Phase Output



17.15.4.3 CW/CCW Output

The figure below is a schematic diagram of the CW/CCW mode output, taking PWMCS_O0/1 as an example. When PWMCS_O0 has a pulse output, it means the external motor rotates forward, and when PWMCS_O1 has a pulse output, it means the external motor rotates backward. The output interval value can be changed through channel delay.

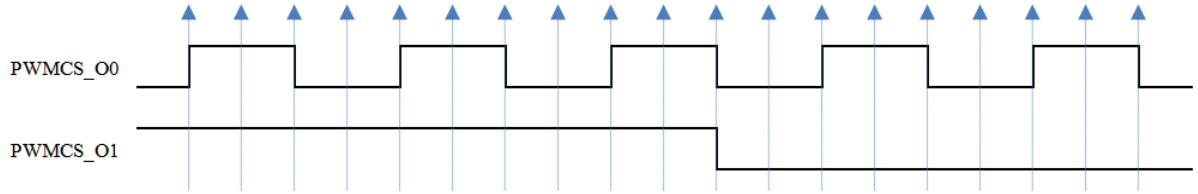
Figure 17-108 CW/CCW Output



17.15.4.4 Pulse + Direction Mode Output

The figure below is a schematic diagram of the pulse + direction mode output, taking PWMCS_O0/1 as an example. PWMCS_O0 represents pulse, PWMCS_O1 represents direction, 1 represents forward rotation, and 0 represents reverse rotation. The output pulse and direction interval values can be changed through channel delay.

Figure 17-109 Pulse + Direction Mode Output



17.15.4.5 A/B Phase Capture

In industry, the pulse capture function is mainly used to connect incremental encoders or grating rulers as feedback for motion control. PWMCS can capture pulse types commonly used in industry, such as orthogonal AB pulses and orthogonal ABZ pulses.

The figure below is a schematic diagram of decoding in A/B phase mode (quadruple-fold frequency), taking PWMCS-I0/1 as an example. CAP_Pulse is generated on both the edges of PWMCS_I0 and PWMCS_I1. A/B phase can be configured as quadruple-fold frequency (all edges), double-fold frequency (edges of PWMCS_I1), single-fold frequency (the rising edge of PWMCS_I1

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represents forward rotation, and the falling edge of PWMCS_I1 represents reverse rotation). Meanwhile, the CAP_Dir, representing motor's rotation direction, is decoded according to the A/B phase decoding rules. The position counter increases during forward rotation and decreases during reverse rotation.

Figure 17-110 A/B Phase Capture (Quadruple-fold Frequency)

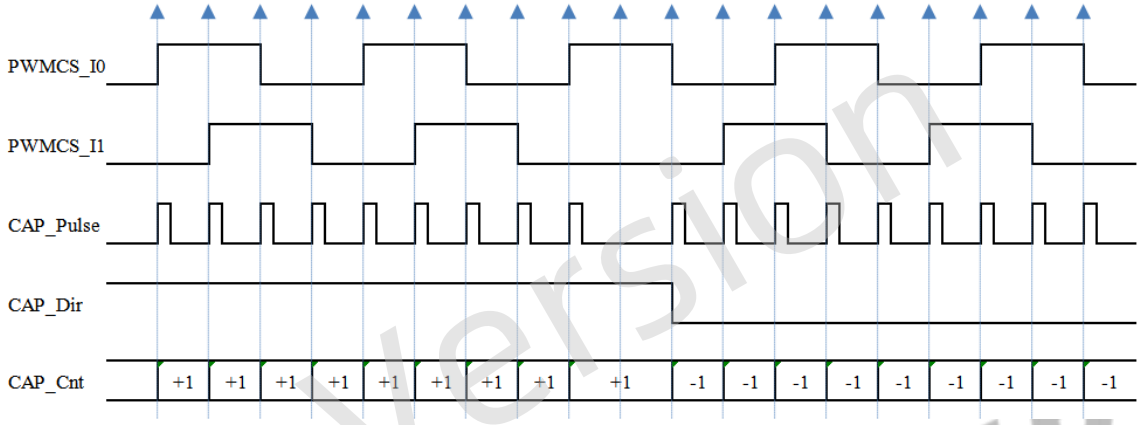
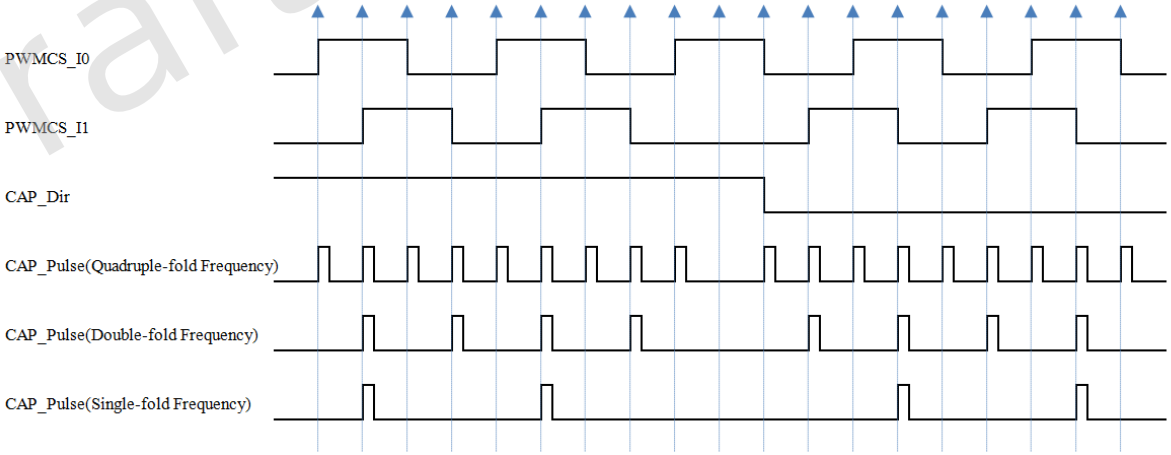


Figure 17-111 A/B Phase Capture (All Frequencies)



17.15.4.6 CW/CCW Capture

The following figure is a decoding diagram of CW/CCW mode, taking PWMCS_I0/1 as an example. When PWMCS_I0 has an input pulse and PWMCS_I1 has no input pulse, it means forward rotation; when PWMCS_I0 has no input pulse and PWMCS_I1 has an input pulse, it means reverse rotation. CAP_Pulse is generated on the edge of PWMCS_I0/ PWMCS_I1 (configurable as rising edge or falling edge). The position counter increases during forward rotation and decreases during reverse rotation.

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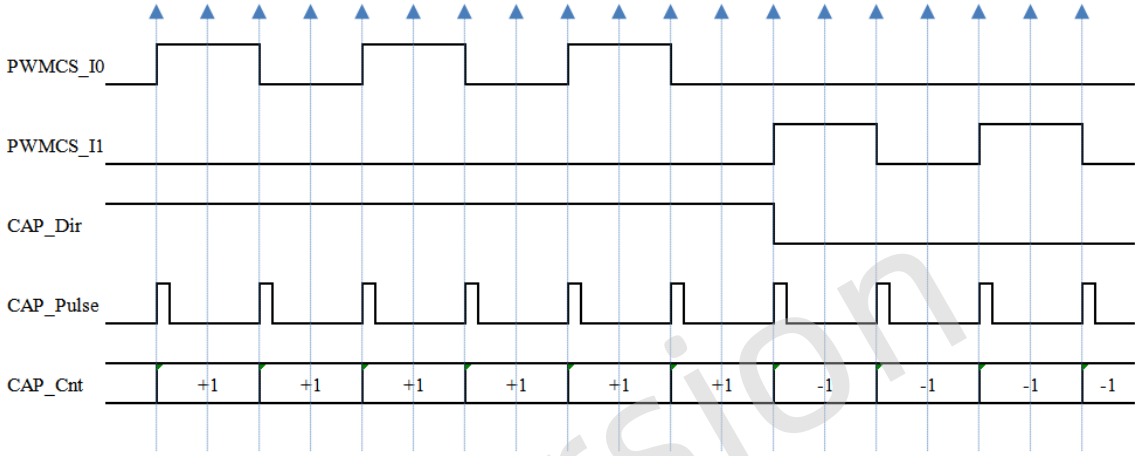
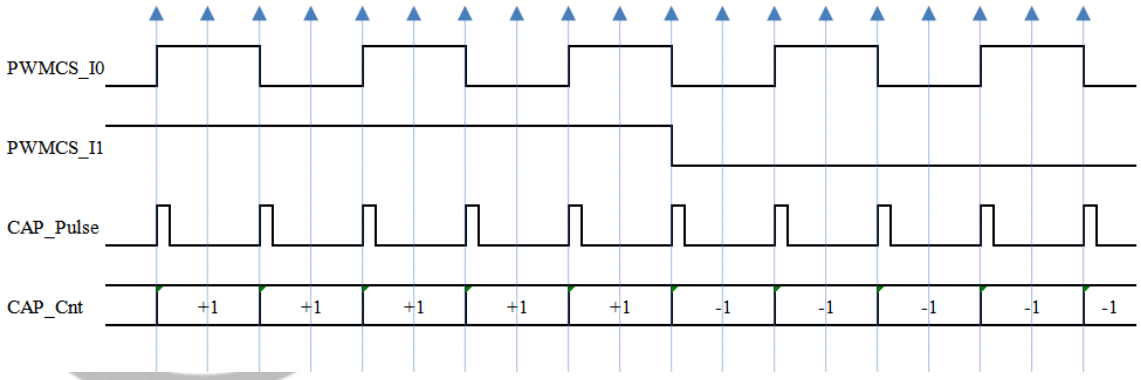
Figure 17-112 CW/CCW Decoding



17.15.4.7 Pulse + Direction Mode Capture

The figure below is a schematic diagram of the pulse + direction mode capture, taking PWMCS_I0/1 as an example. PWMCS_I0 represents input pulse, PWMCS_I1 represents input direction. CAP_Pulse is generated on the edge of PWMCS_I0 (configurable as rising edge or falling edge). The position counter increases during forward rotation and decreases during reverse rotation

Figure 17-113 Pulse + Direction Mode Capture



17.15.4.8 Filtering

The input signal from the external pins can be filtered through PWMCS_FILTER. The figure below is a schematic diagram of glitch filtering. When the width of input signals is less than the register configuration value, the filter module will consider it a glitch and will not transmit it to the internal PWMCS. In the following figure, the first red arrow points to a low-level glitch, and the second red arrow points to a high-level glitch. Both of the two glitches will not be transmitted to the subsequent modules.

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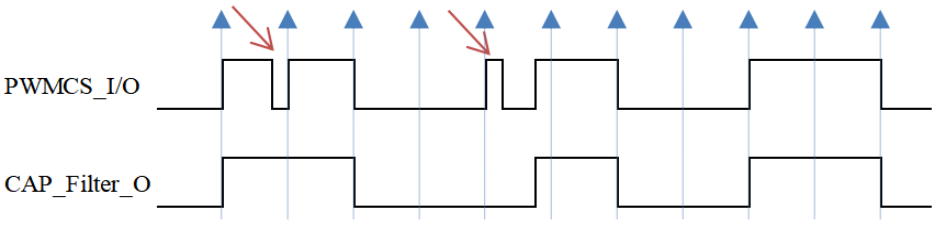
Figure 17-114 CAP Filtering



17.15.4.9 Duty Cycle Pulse Waveforms Output

Register Configuration Mode

In this mode, continuous output of pulses with different duty cycles (only single-end) can be implemented by writing the next command parameter into registers in advance.

FIFO Configuration Mode

PWMCS can continuously output different duty cycle pulse waveforms through internal FIFO according to the written commands, and supports a maximum of 8 command parameter sets to be written at a time. Figure 17-115 shows the command format and writing order. CPU/DMA can write command parameters through PCS_CFG_DATA register. One single command needs to be written to PCS_CFG_DATA in four segments (see Table 17-74).

Figure 17-115 FIFO Command Format

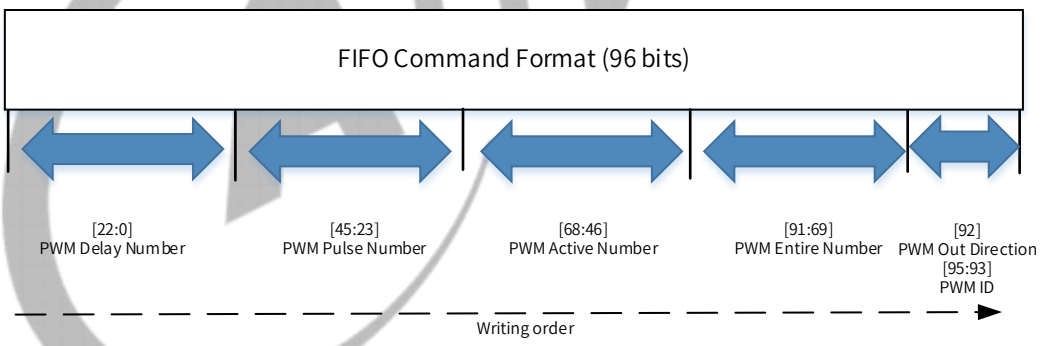


Table 17-74 FIFO Command Write Step

Step	Segments
1	Bit[22:0] PWM Delay Number
2	Bit [45:23] PWM Pulse Number
3	Bit [68:46] PWM Active Number
4	Bit [91:69] PWM Entire Number+bit[92] PWM Out Direction + Bit [95:93] PWMID

When PWMCS continuously outputs different duty cycle pulse waveforms, the FIFO of output channels share the same read enable.

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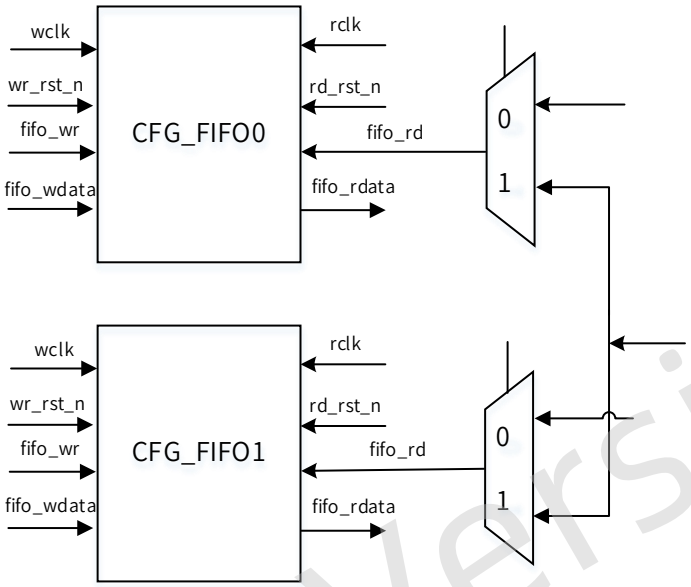
Figure 17-116 Internal FIFO Diagram



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17.16 LEDC

17.16.1 Overview

The LEDC is used to control the external LED lamp.

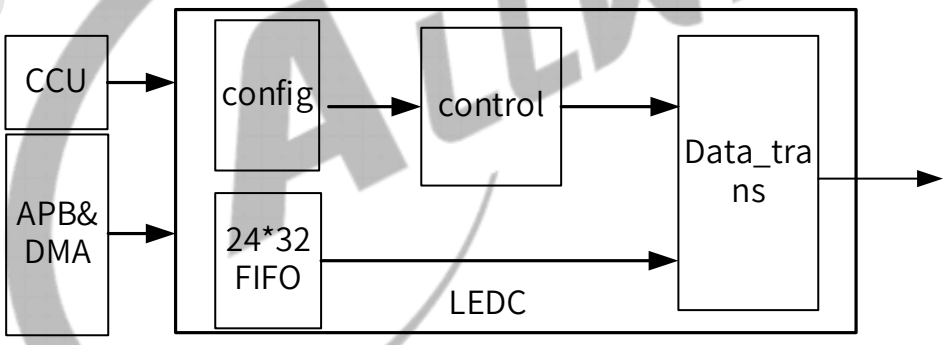
The LEDC has the following features:

- Configurable LED output high-/low-level width
- Configurable LED reset time
- LEDC data supports DMA configuration mode and CPU configuration mode
- Maximum 1024 LEDs serial connection
- Configurable interval time for packets and frame data
- Configurable RGB display mode
- The default level of non-data output is configurable

17.16.2 Block Diagram

The following figure shows a block diagram of the LEDC.

Figure 17-117 LEDC Block Diagram



LEDC contains the following sub-blocks:

Table 17-75 LEDC Sub-blocks

Sub-block	Description
Config	Register configuration.
Control	It controls the timing and status of LEDC.
FIFO	The size is up to 32 layer*24bit.
Data_trans	The input data is transferred to 0 and 1.

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17.16.3 External Signals

The following table describes the external signals of the LEDC.

Table 17-76 LEDC External Signals

Signal Name	Description	Type
LEDC	Intelligent Control LED Signal Output	O

17.16.4 Functional Description

17.16.4.1 LED Data Input

The RGB mode of LEDC data is configurable. By default, the data is sent in GRB order, and the higher bit is transmitted first. The following tables and figure show LED input data structure and code, and data input time and mode.

Figure 17-118 LED Input Data Structure

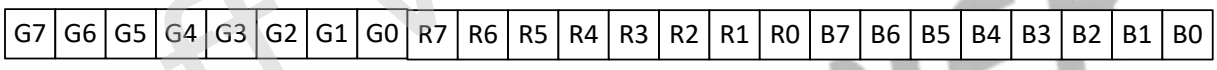
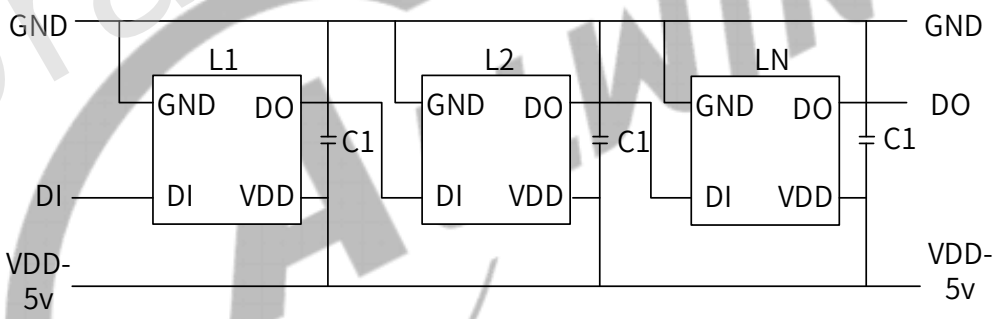


Figure 17-119 LED Typical Circuit



C1 is the filter capacitor of LED lights, which is usually set as 100NF.

Figure 17-120 LED Input Data Code

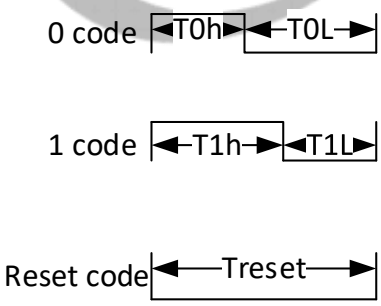


Table 17-77 LED Data Input Time

T0H	0 code, high-level time	220 ns to 380 ns
T0L	0 code, low-level time	580 ns to 1.6 us
T1H	1 code, high-level time	580 ns to 1.6 us
T1L	1 code, low-level time	220 ns to 420 ns

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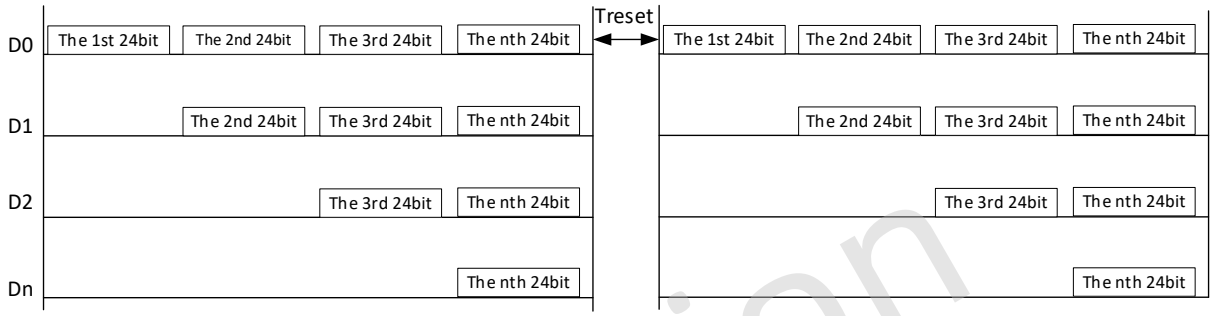
RESET	Frame unit, low-level time	> 280 us

Figure 17-121 LED Data Transfer Mode



17.16.4.2 LEDC Data Transfer

The LEDC supports DMA data transfer mode and CPU data transfer mode through APB bus.

- Data transfer in DMA mode

When the valid space of internal FIFO is greater than the setting FIFO space threshold, the LEDC will send the DMA_REQ to require DMA to transfer data from DRAM to LEDC. The maximum data transfer size in DMA mode is 16 words. (The internal FIFO level is 32.) The DMA data transfer mode is set by LEDC_DMA_EN.

- Data transfer in CPU mode

When the valid space of internal FIFO is greater than the setting FIFO space threshold, the LEDC will send LEDC_CPUREQ_INT to require CPU to transfer data to LEDC. The transfer data size in CPU mode is controlled by software. The internal FIFO destination address is 0x02085014. The data width is 32-bit. (The lower 24-bit is valid.)

17.16.4.3 LEDC Interrupt

Interrupt Name	Description
FIFO_OVERFLOW_INT	FIFO overflow interrupt. The amount of written data exceeds the maximum storage space of LED FIFO. The LEDC is in the data loss state. At this time, the anomaly should be handled by software as follows. a) Query the LED_FIFO_DATA_REG to determine which data has been stored in the internal FIFO of LEDC. Resend the overflow data to LEDC. b) Soft_reset the LEDC to refresh all the data.

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Interrupt Name	Description
WAITDATA_TIMEOUT_INT	Wait Data Timeout Interrupt When internal FIFO of LEDC cannot get data resulting from some anomaly, the timeout interrupt will be enabled after the led_wait_data_time. At this time, the LEDC is in WAIT_DATA state, and the LEDC will output the level configured by the LED_PAUSE_LEVEL. During the wait_data, the LEDC will continue to transfer data. In this condition, if LEDC waiting time exceeds the LED reset time, it is as if the LEDC has sent a reset operation. The LED will enter refreshing state by mistake, and the data hasn t been completely sent.
FIFO_CPUREQ_INT	FIFO Request CPU Data Interrupt When FIFO data amount is less than the threshold, the interrupt will be reported to the CPU.
LEDC_TRANS_FINISH_INT	Data Transfer Finish Interrupt It indicates that the configured data of total_data_length finishes transfer.

LEDC interrupt usage scenario:

- CPU mode

The software can enable GLOBAL_INT_EN, FIFO_CPUREQ_INT_EN, WAITDATA_TIMEOUT_INT_EN, FIFO_OVERFLOW_INT_EN, LEDC_TRANS_FINISH_INT_EN, and cooperate with LEDC_FIFO_TRIG_LEVEL to use. When FIFO_CPUREQ_INT is set to 1, the software can configure data of LEDC_FIFO_TRIG_LEVEL to LEDC.

- DMA mode

The software can enable GLOBAL_INT_EN, WAITDATA_TIMEOUT_INT_EN, FIFO_OVERFLOW_INT_EN, LEDC_TRANS_FINISH_INT_EN, and cooperate with LEDC_FIFO_TRIG_LEVEL to use. After receiving LEDC DMA_REQ, DMA can transfer data of LEDC_FIFO_TRIG_LEVEL to LEDC.

17.16.5 Programming Guidelines

17.16.5.1 Normal Configuration

Step 1 Configure LEDC_CLK and bus pclk as described in the Section 4.1 Clock Controller Unit (CCU).

Step 2 Configure the LEDC data to be written.

Step 3 Configure 0-code, 1-code, reset time, LEDC wait time, and the number of connected LED and the threshold of DMA transferred data by setting LED_T01_TIMING_CTRL_REG, LEDC_DATA_FINISH_CNT_REG, LED_RESET_TIMING_CTRL_REG, LEDC_WAIT_TIME0_CTRL_REG, LEDC_DMA_CTRL_REG, LEDC_INT_CTRL_REG.

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Step 4 Enable LEDC_EN through LEDC_CTRL_REG. The LEDC will start outputting data.

Step 5 When the LEDC interrupt is pulled up, it indicates the configured data has been transferred. Then, LED_EN will be set to 0, and the read/write point of LEDC FIFO is cleared.

Step 6 Repeat the first four steps to start a new round of configuration, and enable LEDC_EN. The LEDC will restart data transfer.

17.16.5.2 Abnormal Processing

WAITDATA_TIMEOUT

Step 1 When a WAITDATA_TIMEOUT_INT occurs, it indicates the internal FIFO data request of LEDC doesn t get a response. In this condition, if the default output level is low, the external LED device will mistakenly believe it is a reset operation, resulting in refreshing LED data by mistake.

Step 2 Write 1 to LEDC_SOFT_RESET to reset the LEDC by software. Afterwards, LEDC_EN will be automatically disabled. All the status registers will go back to the idle status. LEDC FIFO read &write point are cleared to 0. LEDC The interrupts are cleared.

Step 3 Write 1 to RESET_LED_EN. The LEDC will send the reset order to ensure the external LED lights stay in the right status.

Step 4 Read the reset_led_en. 1 indicates that LEDC hasn t sent the reset order. 0 indicates that LEDC has sent the reset order.

Step 5 When the LEDC reset operation finishes, the LEDC will restart the configuration of data and registers. The data transfer starts again.

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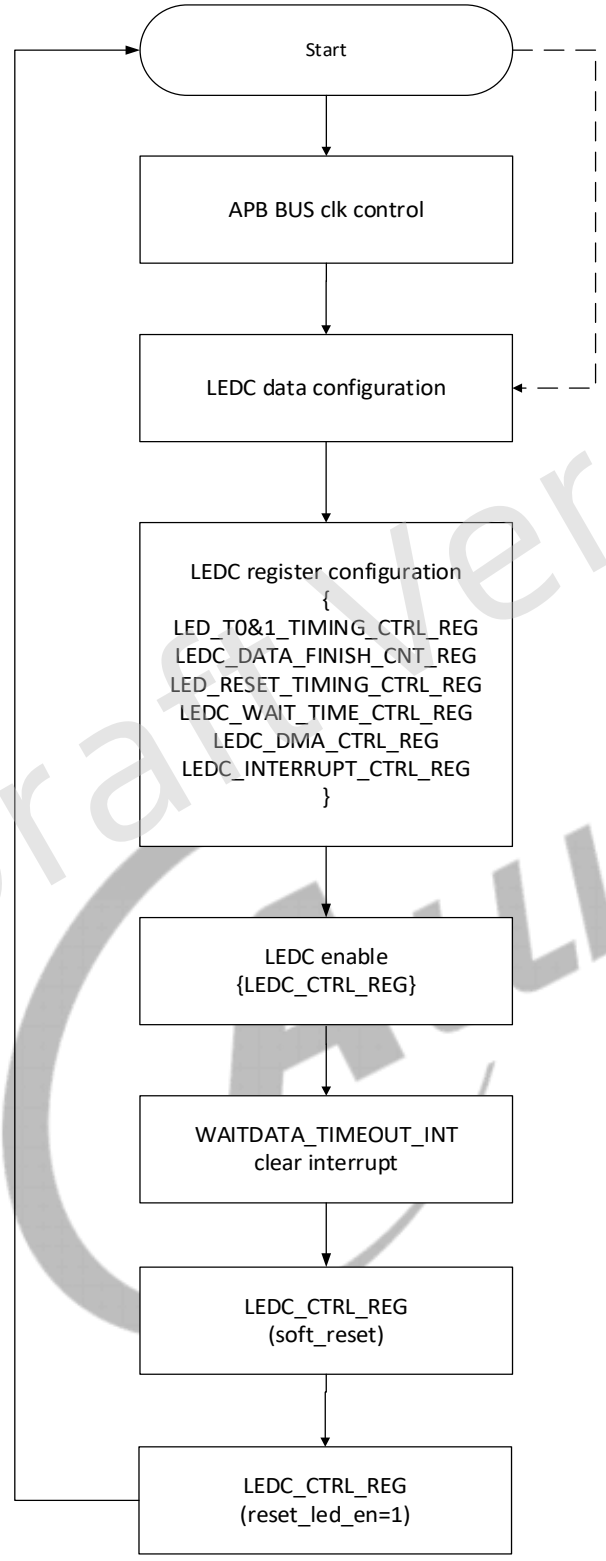
Figure 17-122 LEDC Timeout Abnormal Processing Flow



FIFO_OVERFLOW

Step 1 When FIFO_OVERFLOW_INT occurs, it indicates the data configured by software exceeds the LEDC FIFO space, at this time the redundant data will be lost.

Step 2 Read the data in LEDC_FIFO_DATA_REG to confirm the lost data.

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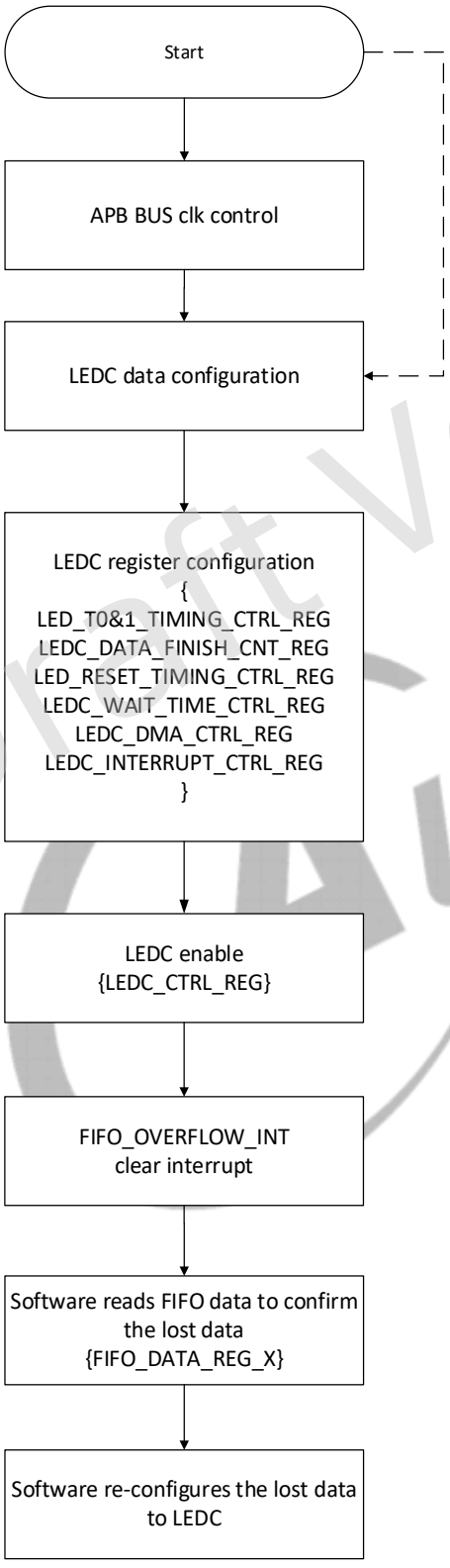
Step 3 Re-configure the lost data to the LEDC.

Step 4 Refer to the WAITDATA_TIMEOUT processing flow above if soft_reset is needed.

Figure 17-123 FIFO Overflow Abnormal Processing Flow



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17.16.6 Register List

Module Name	Base Address
LEDC	0x02085000

Register Name	Offset	Description
LEDC_CTRL_REG	0x0000	LEDC Control Register
LED_T01_TIMING_CTRL_REG	0x0004	LED T0 & T1 Timing Control Register
LEDC_DATA_FINISH_CNT_REG	0x0008	LEDC Data Finish Counter Register
LED_RESET_TIMING_CTRL_REG	0x000C	LED Reset Timing Control Register
LEDC_WAIT_TIME0_CTRL_REG	0x0010	LEDC Wait Time0 Control Register
LEDC_DATA_REG	0x0014	LEDC Data Register
LEDC_DMA_CTRL_REG	0x0018	LEDC DMA Control Register
LEDC_INT_CTRL_REG	0x001C	LEDC Interrupt Control Register
LEDC_INT_STS_REG	0x0020	LEDC Interrupt Status Register
LEDC_WAIT_TIME1_CTRL_REG	0x0028	LEDC Wait Time1 Control Register
LEDC_FIFO_DATA_REG	0x0030+0x04*N (N=0(b31)	LEDC FIFO Data Register

17.16.7 Register Description

17.16.7.1 0x0000 LEDC Control Register (Default Value: 0x0000_003C)

Offset: 0x0000			Register Name: LEDC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30	R/W	0x0	LED_PAUSE_LEVEL The output level in the status of LED_WAIT_TIME0, LED_WAIT_TIME1, and LED_WAIT_DATA when LED is working. 1: High level 0: Low level
29	R/W	0x0	LED_IDLE_LEVEL The default output level when LED doesn't work. 1: High level 0: Low level

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Offset: 0x0000			Register Name: LEDC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
28:16	R/W	0x0	TOTAL_DATA_LENGTH Total Length of Data to be transferred (range: 0 to 8K, unit: 32-bit, only low 24-bit is valid) It is recommended to be set to an integer multiple of (LED_NUM+1). If TOTAL_DATA_LENGTH is larger than (LED_NUM+1), but non-integer multiple, the last frame of data will transfer data less than (LED_NUM+1).
15:11	/	/	/
10	R/WAC	0x0	RESET_LED_EN LED Reset Enable **Write Operation:** Write 1 by software. The CPU will trigger LEDC to send a reset order to LED. Note: a) Only when LEDC is in IDLE status, the reset will be performed. After the reset finishes, the state machine returns to the IDLE status. Then, the LEDC stays in the IDLE status through SOFT_RESET. b) Read this bit to check if the reset is complete. **Read operation:** 0: LEDC has sent the LED reset order. 1: LEDC doesn't send the LED reset order.
9	/	/	/
8:6	R/W	0x0	LED_RGB_MODE 000 GRB (bypass) 001 GBR 010 RGB 011 RBG 100 BGR 101 BRG • The data is configured and sent to LEDC by software according to GRB (MSB) mode. Then, the LEDC arrays data and outputs them to the external LED based on the LED_RGB_MODE configuration. • The configuration in other software input modes is as follows.

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Offset: 0x0000			Register Name: LEDC_CTRL_REG
Bit	Read/Write	Default/Hex	Description

Software Input Mode	Configuration	LEDC Output Mode
GRB	000	GRB
	001	GBR
	010	RGB
	011	RBG
	100	BGR
	101	BRG
GBR	000	GBR
	001	GRB
	010	BGR
	011	BRG
	100	RGB
	101	RBG
RGB	000	RGB
	001	RBG
	010	GRB
	011	GBR
	100	BRG
	101	BGR
RBG	000	RBG
	001	RGB
	010	BRG
	011	BGR
	100	GRB
	101	GBR
BGR	000	BGR
	001	BRG
	010	GBR
	011	GRB
	100	RBG
	101	RGB
BRG	000	BRG
	001	BGR
	010	RBG
	011	RGB
	100	GBR
	101	GRB

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Offset: 0x0000			Register Name: LEDC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
5	R/W	0x1	LED_MSB_TOP Adjust sequence of the arrayed GRB data 0: LSB 1: MSB (by default)
4	R/W	0x1	LED_MSB_G MSB control for Green data 0: LSB 1: MSB (by default)
3	R/W	0x1	LED_MSB_R MSB control for Red data 0: LSB 1: MSB (by default)
2	R/W	0x1	LED_MSB_B MSB control for Blue data 0: LSB 1: MSB (by default)
1	R/W1C	0x0	LEDC_SOFT_RESET LEDC Soft Reset Write 1 to clear it. In this condition, the LEDC will be reset by software. Consequently, all the internal state register and state machine returns to the idle state. LEDC FIFO read &write point is cleared to 0. The interrupts of LEDC are cleared. 1. LEDC_CTRL_REG (LEDC_EN is cleared to 0). 2. LEDC_T01_TIMING_CTRL_REG remains unchanged. 3. LEDC_DATA_FINISH_CNT_REG (LEDC_DATA_FINISH_CNT is cleared to 0). 4. LED_RESET_TIMING_CTRL_REG remains unchanged. 5. LEDC_WAIT_TIME_CTRL_REG remains unchanged. 6. LEDC_DMA_CTRL_REG remains unchanged. 7. LEDC_INT_CTRL_REG remains unchanged. 8. LEDC_INT_STS_REG is cleared to 0. 9. LEDC_CLK_GATING_REG remains unchanged. 10. LEDC_FIFO_DATA_REG remains unchanged.

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Offset: 0x0000			Register Name: LEDC_CTRL_REG
Bit	Read/Write	Default/Hex	Description
0	R/W	0x0	LEDC_EN LEDC Enable 0: Disable 1: Enable LEDC_EN indicates that the LEDC can start working. When LEDC finishes data transmission or LEDC_SOFT_RESET, LEDC_EN will be auto cleared to 0 by hardware.

17.16.7.2 0x0004 LED T0 & T1 Timing Control Register (Default Value: 0x0286_01D3)

Offset: 0x0004			Register Name: LED_T01_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26:21	R/W	0x14	T1H_TIME LED T1H time Unit: Cycle (24 MHz), T1H_TIME=42 ns*(N+1) The default value is 882 ns; the range is 80 ns(b2)560 ns. N: 1(b3)F. When is 0, t1h_time = 3F
20:16	R/W	0x6	T1L_TIME LED T1L time Unit: Cycle (24 MHz), t1l_time =42 ns*(N+1) The default value is 294 ns; the range is 80 ns(b1)280 ns. N: 1(b1)F. When is 0, t1l_time = 1F
15:11	/	/	/
10:6	R/W	0x7	T0H_TIME LED T0h time Unit: cycle (24 MHz), t0h_time =42 ns*(N+1) The default value is 336 ns; the range is 80 ns(b1)280 ns. N: 1(b1)F. When is 0, t0h_time = 1F
5:0	R/W	0x13	T0L_TIME LED T0l time Unit: cycle (24 MHz), t0L_time =42 ns*(N+1) The default value is 840 ns; the range is 80 ns(b2)560 ns. N: 1(b3)F. When is 0, t0L_time = 3F

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17.16.7.3 0x0008 LEDC Data Finish Counter Register (Default Value: 0x1D4C_0000)

Offset: 0x0008			Register Name: LEDC_DATA_FINISH_CNT_REG
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x1D4C	LED_WAIT_DATA_TIME The value is the time that internal FIFO in LEDC waits for data. When the time is out, the LEDC will send the wait_data_timeout_int. (This is an abnormal scene, LEDC needs to be soft_reset.) The value is about 300 us by default. The adjustable range is 80 ns(b)55 us. led_wait_data_time=42ns*(N+1). N ranges from 1 to 1FFF. When the field is 0, LEDC_WAIT_DATA_TIME=1FFF
15:13	/	/	/
12:0	R	0x0	LED_DATA_FINISH_CNT The value is the total number (0(b)8k) of LED data that have been sent.

17.16.7.4 0x000C LED Reset Timing Control Register (Default Value: 0x1D4C_0000)

Offset: 0x000C			Register Name: LED_RESET_TIMING_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x1D4C	TR_TIME Reset time control of LED light Unit: cycle (24 MHz), tr_time=42 ns*(N+1) The default value is 300 us. The adjust range is 80 ns(b)327 us. N ranges from 1 to 1FFF If the value is 0, TR_TIME=1FFF.
15:10	/	/	/
9:0	R/W	0x0	LED_NUM The value is the number of LED lights, up to 1024. The default value is 0, indicating that 1 LED light is connected. The range is from 0 to 1023.

17.16.7.5 0x0010 LEDC Wait Time 0 Control Register (Default Value: 0x0000_00FF)

Offset: 0x0010			Register Name: LEDC_WAIT_TIME0_CTRL_REG
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0010			Register Name: LEDC_WAIT_TIME0_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	WAIT_TIM0_EN WAIT_TIME0 enable When it is 1, the waiting time will be automatically inserted among LED package data. 0: Disable 1: Enable
7:0	R/W	0xFF	TOTAL_WAIT_TIME0 Waiting time between 2 LED data. The LEDC output level is controlled by the LED_PAUSE_LEVEL The adjust range is 80 ns(b)10 us. wait_time0=42 ns*(N+1) Unit: cycle (24 MHz) N: 1(b)FF

17.16.7.6 0x0014 LEDC Data Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: LEDC_DATA_REG
Bit	Read/Write	Default/Hex	Description
31:0	W	0x0	LEDC DATA LED display data (the lower 24-bit is valid)

17.16.7.7 0x0018 LEDC DMA Control Register (Default Value: 0x0000_002F)

Offset: 0x0018			Register Name: LEDC_DMA_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x1	LEDC_DMA_EN LEDC DMA Request Enable 0: Disable the request of DMA transfer data 1: Enable the request of DMA transfer data
4:0	R/W	0x0F	LEDC_FIFO_TRIG_LEVEL The remaining space of internal FIFO in LEDC The internal FIFO in LEDC is 24*32. When the remaining space of internal FIFO in LEDC is larger or equal to LEDFIFO_TRIG_LEVEL, the DMA or the CPU request will be generated. The value is 15 by default, which ranges from 1 to 31 and is recommended to be 7 or 15. If the value is 0, LEDFIFO_TRIG_LEVEL is F.

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17.16.7.8 0x001C LEDC Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: LEDC_INT_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	GLOBAL_INT_EN Global Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	FIFO_OVERFLOW_INT_EN FIFO Overflow Interrupt Enable When the data written by the software is larger than the internal FIFO level of LEDC, the LEDC is in the data loss state. 0: Disable 1: Enable
3	R/W	0x0	WAITDATA_TIMEOUT_INT_EN The internal FIFO in LEDC cannot get data in some abnormal scene. After led_wait_data_time, the interrupt will be enabled. 0: Disable 1: Enable
2	/	/	/
1	R/W	0x0	FIFO_CPUREQ_INT_EN FIFO request CPU Data Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	LED_TRANS_FINISH_INT_EN Data Transmission Finish Interrupt Enable 0: Disable 1: Enable

17.16.7.9 0x0020 LEDC Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: LEDC_INT_STS_REG
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R	0x0	FIFO_EMPTY FIFO Empty Status Flag
16	R	0x0	FIFO_FULL FIFO Full Status Flag

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Offset: 0x0020			Register Name: LEDC_INT_STS_REG
Bit	Read/Write	Default/Hex	Description
15:10	R	0x0	FIFO_WLW FIFO Internal Valid Data Depth It indicates the space FIFO has been occupied.
9:5	/	/	/
4	R/W1C	0x0	FIFO_OVERFLOW_INT FIFO Overflow Interrupt The amount of written data exceeds the maximum storage space of LED FIFO. The LEDC is in the data loss state. At this time, the anomaly should be handled by software as follows. 1. Query the LED_FIFO_DATA_REG to determine which data has been stored in the internal FIFO of LEDC. Resend the overflow data to LEDC. 2. Soft_reset the LEDC to refresh all the data. 0: FIFO doesn't overflow 1: FIFO overflows Write 1 to clear the interrupts.
3	R/W1C	0x0	WAITDATA_TIMEOUT_INT When internal FIFO of LEDC cannot get data resulting from some anomaly, the timeout interrupt will be enabled after the led_wait_data_time. At this time, the LEDC is in WAIT_DATA state, and the LEDC will output the level configured by the LED_PAUSE_LEVEL. During the wait_data, the LEDC will continue to transfer data. In this condition, if LEDC waiting time exceeds the LED reset time, it is as if the LEDC has sent a reset operation. The LED will enter refreshing state by mistake, and the data hasn't been completely sent. 0: LEDC not timeout 1: LEDC timeout Write 1 to clear the interrupts.
2	/	/	/
1	R/W1C	0x0	FIFO_CPUREQ_INT FIFO Request CPU Data Interrupt When FIFO data amount is less than the threshold, the interrupt will be reported to the CPU. 0: FIFO doesn't request that CPU transfers data 1: FIFO requests that CPU transfers data Write 1 to clear the interrupts.

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Offset: 0x0020			Register Name: LEDC_INT_STS_REG
Bit	Read/Write	Default/Hex	Description
0	R/W1C	0x0	LED_TRANS_FINISH_INT Data Transfer Finish Interrupt It indicates that the configured data of total_data_length finishes transfer. 0: Data is not transferred completely 1: Data is transferred completely Write 1 to clear the interrupts.

17.16.7.10 0x0028 LEDC Wait Time 1 Control Register (Default Value: 0x01FF_FFFF)

Offset: 0x0028			Register Name: LEDC_WAIT_TIME1_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	WAIT_TIM1_EN WAIT_TIME1 enable When it is 1, the waiting time will be automatically inserted among the LED frame data. 0: Disable 1: Enable
30:0	R/W	0x01FFFFFF	TOTAL_WAIT_TIME1 Waiting Time between Two Frames of Data. The LEDC output is controlled by the LED_PAUSE_LEVEL. The adjust range is 80 ns(b)85 s. wait_time1=42 ns*(N+1) Unit: cycle (24 MHz) N: 0x80(b)0x7FFFFFFF If it is 0, TOTAL_WAIT_TIME1 is 0x7FFFFFFF

17.16.7.11 0x0030+N*0x04(N=0(b)31) LEDC FIFO Data Register (Default Value: 0x0000_0000)

Offset: 0x0030+N*0x04 (N=0(b)31)			Register Name: LEDC_FIFO_DATA_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	LEDC_FIFO_DATA_X Internal FIFO data of LEDC The lower 24-bit is valid.

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17.17 IR Receiver (IR_RX)

17.17.1 Overview

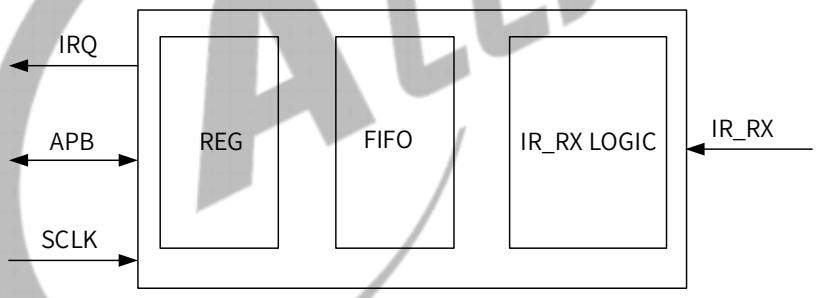
The Consumer Infrared (IR) receiver captures pulse from the IR Receiver module and uses the Run-Length Code (RLC) to encode the pulse.

The CIR receiver has the following features:

- Four IR_RX interfaces in CPUX domain
- Supports CIR for remote control
- 64x8 bits FIFO for data buffer
- Sample clock up to 1 MHz
- Programmable FIFO thresholds
- RLC encoding
- NEC format infra data
- Noise filtering

17.17.2 Block Diagram

Figure 17-124 IR Receiver Block Diagram



The IR receiver samples the input signal on the programmable frequency and records these samples into RX FIFO when one IR signal is found on the air. The IR receiver uses Run-Length Code (RLC) to encode pulse width. The encoded data is buffered in 64 levels and 8-bit width RX FIFO; the MSB bit is used to record the polarity of the receiving IR signal, the rest 7 bits are used for the length of RLC. The maximum length of the RLC is 128. If the duration of one level (high or low level) is more than 128, another byte is used.

17.17.3 External Signals

The following table describes the external signals of IR_RX.

Table 17-78 CIR Receiver External Signals

Signal Name	Description	Type
IR0-RX	Consumer Infrared Receiver	I

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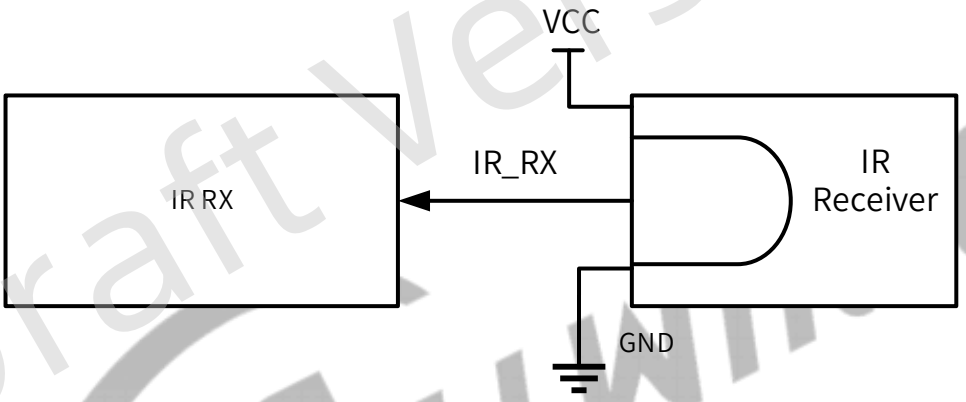
Signal Name	Description	Type
IR1-RX	Consumer Infrared Receiver	I
IR2-RX	Consumer Infrared Receiver	I
IR3-RX	Consumer Infrared Receiver	I

17.17.4 Functional Description

17.17.4.1 Typical Application

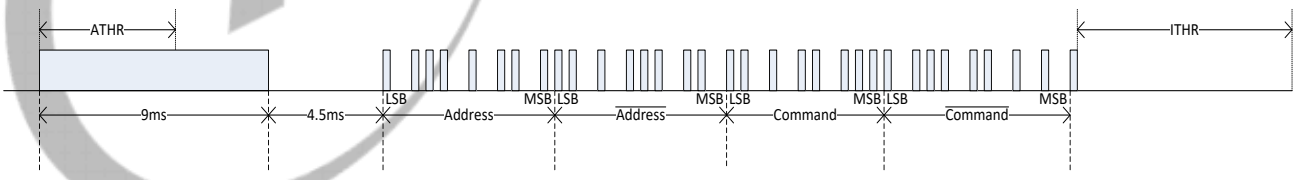
The following figure shows IRRX controller IO signals. To connect with external devices, you could configure GPIO controller to enable IRRX model.

Figure 17-125 CIR Receiver Application Diagram



17.17.4.2 NEC Protocol

Figure 17-126 A Typical Pulse Train of the NEC



The CIR receiver module is a timer with a capture function.

When IR_RX signals satisfy the Active Threshold (ATHR), the CIR receiver starts capturing. The signals whose pulse width less than Noise Threshold (NTHR) will be ignored. When the signals satisfy Idle Threshold (ITHR), the capture process will be stopped and the Receiver Packet End interrupt will be generated, then the Receiver Packet End Flag will be asserted.

In a capture process, every effective pulse is buffered to FIFO in bytes according to the form of the Run-Length Code. The MSB bit of a byte indicates the polarity of pulse, and the rest 7 bits indicates the pulse width by taking Sample Clock as a basic unit. This is the code form of the RLC-Byte. When the level changes or the pulse width counter overflows, the RLC-Byte is buffered to FIFO. Afterwards, the IR_RX module receives the infrared signals transmitted by the infrared remote control, the software decodes the signals.

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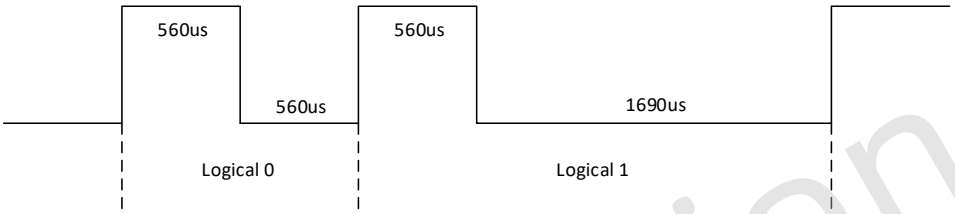
17.17.4.3 Operating Mode

Sample Clock

Figure 17-127 Logical '0' and Logical '1' of NEC Protocol



For NEC protocol, a logical "1" takes 2.25 ms (560 us+1680 us) to transmit, while a logical "0" is only half of that, being 1.12 ms (560 us+560 us).

For example, if the sample clock is 31.25 kHz, and a sample cycle is 32 us, then 18 sample cycles are 560 us. So the RLC of 560 us low level is 0x12 (b'00010010), the RLC of 560 us high level is 0x92 (b'10010010). Then a logical "1" takes code 0x12 (b'00010010) and code 0xb5 (b'10110101) to transmit, a logical "0" takes code 0x12 and code 0x92 to transmit.

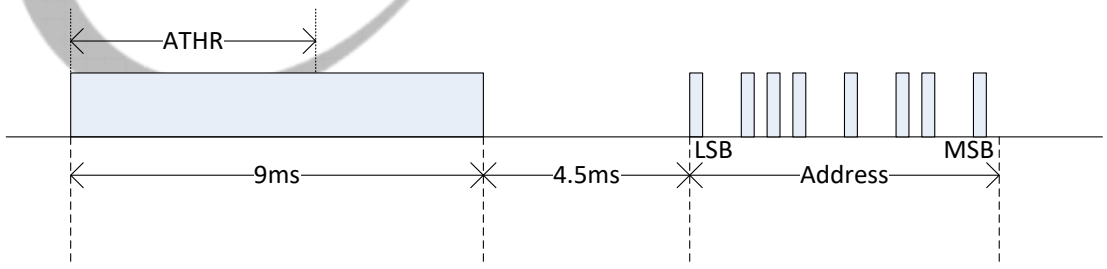
NOTE

Input signals can be inverted by setting the RPPI of the CIR_RXCTL register to 1.

ATHR

When the CIR receiver is in the idle state, if the electrical level of the IR_RX signal changes (positive jump or negative jump), and the duration reaches this threshold, then the CIR receiver takes the starting of the signal as a lead code, and the CIR receiver turns into an active state and starts to capture IR_RX signals.

Figure 17-128 ATHR Definition



ITHR

If the electrical level of IR_RX signals has no change, and the duration reaches this threshold, then the CIR receiver enters into Idle state and ends this capture.

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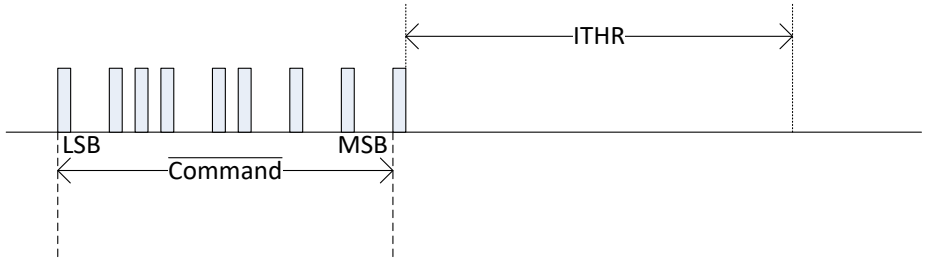
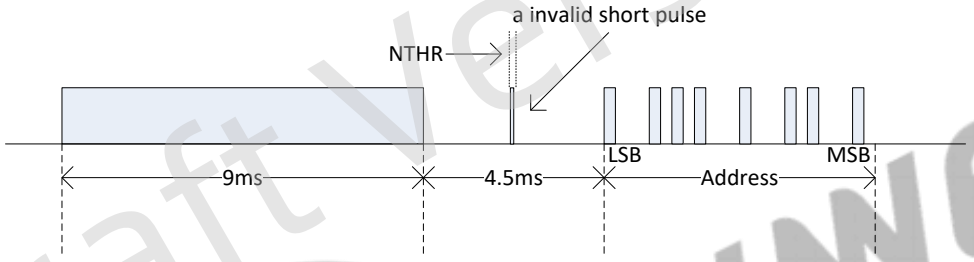
Figure 17-129 ITHR Definition



NTHR

In the capture process, the pulse is ignored if the pulse width is less than the Noise Threshold.

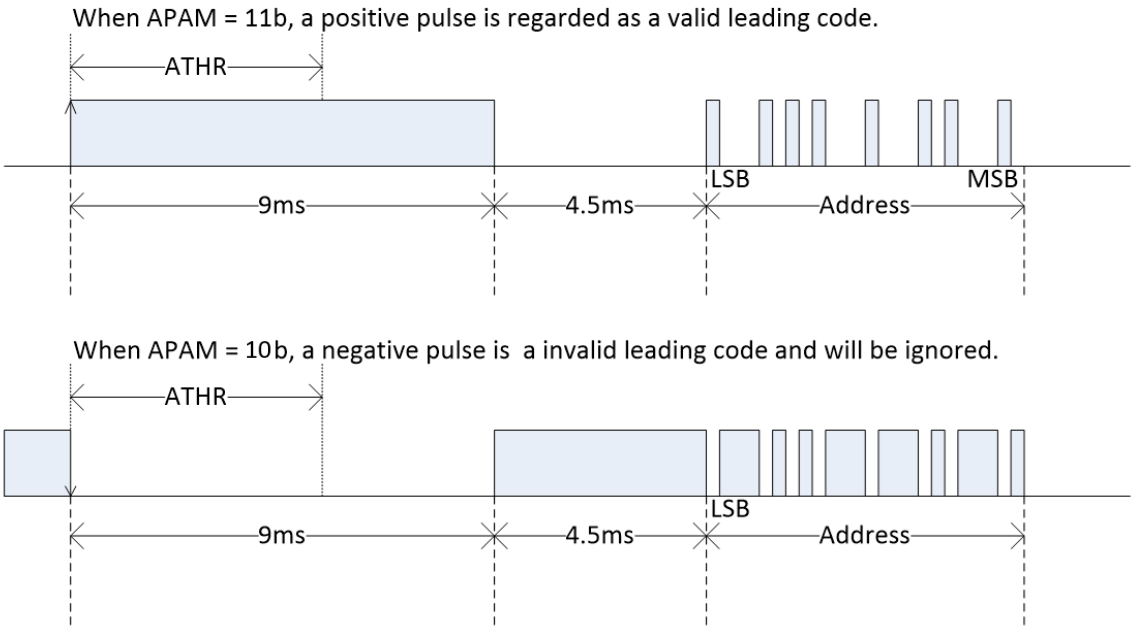
Figure 17-130 NTHR Definition



APAM

The APAM is used to fit the type of lead code. If a pulse does not fit the type of lead code, it is not regarded as a lead code even if the pulse width reaches ATHR.

Figure 17-131 APAM Definition



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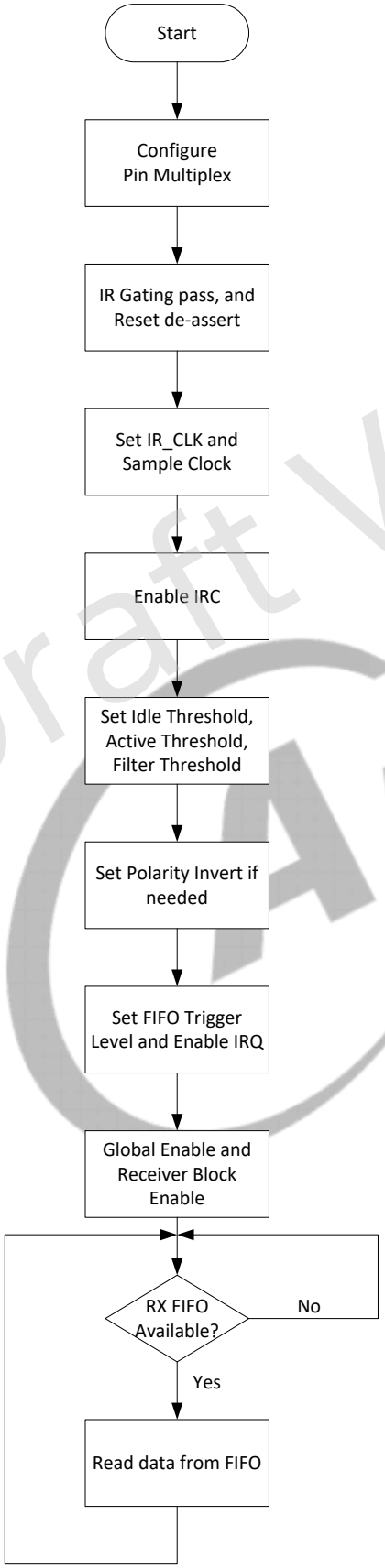
17.17.5 Operation Guidelines

Figure 17-132 CIR Receiver Process



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17.17.6 Register List

Module Name	Base Address
IRRX0	0x02081000
IRRX1	0x02082000
IRRX2	0x02083000
IRRX3	0x02084000

Register Name	Offset	Description
CIR_CTL	0x0000	CIR Receiver Control Register
CIR_RXCTL	0x0010	CIR Receiver Configure Register
CIR_RXFIFO	0x0020	CIR Receiver FIFO Register
CIR_RXINT	0x002C	CIR Receiver Interrupt Control Register
CIR_RXSTA	0x0030	CIR Receiver Status Register
CIR_CONFIG	0x0034	CIR Receiver Configure Register
CIR_STORE0	0x0080	CIR Receiver Store Register0
CIR_STORE1	0x0084	CIR Receiver Store Register1
CIR_STORE2	0x0088	CIR Receiver Store Register2
CIR_STORE3	0x008C	CIR Receiver Store Register3

17.17.7 Register Description

17.17.7.1 0x0000 CIR Receiver Control Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: CIR_CTL
Bit	Read/Write	Default/HEX	Description
31:8	/	/	/
7:6	R/W	0x0	APAM Active Pulse Accept Mode 00, 01: Both positive and negative pulses are valid as a leading code. 10: Only negative pulse is valid as a leading code. 11: Only positive pulse is valid as a leading code.
5:4	R/W	0x0	CEN CIR ENABLE 00-10: Reserved 11: CIR mode enable
3:2	/	/	/
1	R/W	0x0	RXEN Receiver Block Enable 0: Disable 1: Enable

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Offset: 0x0000			Register Name: CIR_CTL
Bit	Read/Write	Default/HEX	Description
0	R/W	0x0	GEN Global Enable A disable on this bit overrides any other block or channel enables and flushes all FIFOs. 0: Disable 1: Enable

17.17.7.2 0x0010 CIR Receiver Configure Register (Default Value: 0x0000_0004)

Offset: 0x0010			Register Name: CIR_RXCTL
Bit	Read/Write	Default/HEX	Description
31:3	/	/	/
2	R/W	0x1	RPPI Receiver Pulse Polarity Invert. 0: Do not invert receiver signal 1: Invert receiver signal
1:0	/	/	/

17.17.7.3 0x0020 CIR Receiver FIFO Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: CIR_RXFIFO
Bit	Read/Write	Default/HEX	Description
31:8	/	/	/
7:0	R	0x0	RBF Receiver Byte FIFO

17.17.7.4 0x002C CIR Receiver Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: CIR_RXINT
Bit	Read/Write	Default/HEX	Description
31:14	/	/	/
13:8	R/W	0x0	RAL RX FIFO Available Received Byte Level for interrupt and DMA request TRIGGER_LEVEL = RAL + 1

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Offset: 0x002C			Register Name: CIR_RXINT
Bit	Read/Write	Default/HEX	Description
5	R/W	0x0	DRQ_EN RX FIFO DMA Enable 0: Disable 1: Enable When set to '1', the Receiver FIFO DRQ is asserted if reaching RAL. The DRQ is de-asserted when condition fails.
4	R/W	0x0	RAI_EN RX FIFO Available Interrupt Enable 0: Disable 1: Enable When set to '1', the Receiver FIFO IRQ is asserted if reaching RAL. The IRQ is de-asserted when condition fails.
3:2	/	/	/
1	R/W	0x0	RPEI_EN Receiver Packet End Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	ROI_EN Receiver FIFO Overrun Interrupt Enable 0: Disable 1: Enable

17.17.7.5 0x0030 CIR Receiver Status Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: CIR_RXSTA
Bit	Read/Write	Default/HEX	Description
31:15	/	/	/
14:8	R	0x0	RAC RX FIFO Available Counter 0: No available data in RX FIFO 1: 1-byte available data in RX FIFO 2: 2-byte available data in RX FIFO (p) 64: 64 byte available data in RX FIFO
7	R	0x0	STAT Status of CIR 0x0: Idle 0x1: Busy
6:5	/	/	/

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Offset: 0x0030			Register Name: CIR_RXSTA
Bit	Read/Write	Default/HEX	Description
4	R/W1C	0x0	RA RX FIFO Available 0: RX FIFO not available according its level 1: RX FIFO available according its level This bit is cleared by writing a '1'.
3:2	/	/	/
1	R/W1C	0x0	RPE Receiver Packet End Flag 0: STO was not detected. In CIR mode, one CIR symbol is receiving or not detected. 1: STO field or packet abort symbol (7 b0000,000 and 8 b0000,0000 for MIR and FIR) is detected. In CIR mode, one CIR symbol is received. This bit is cleared by writing a '1'.
0	R/W1C	0x0	ROI Receiver FIFO Overrun 0: Receiver FIFO not overrun 1: Receiver FIFO overrun This bit is cleared by writing a '1'.

17.17.7.6 0x0034 CIR Receiver Configure Register (Default Value: 0x0000_1828)

Offset: 0x0034			Register Name: CIR_CONFIG
Bit	Read/Write	Default/HEX	Description
31:25	/	/	/
24	R/W	0x0	SCS2 Bit2 of Sample Clock Select for CIR This bit is defined by SCS bits below.
23	R/W	0x0	ATHC Active Threshold Control for CIR 0x0: ATHR in Unit of (Sample Clock) 0x1: ATHR in Unit of (128*Sample Clocks)
22:16	R/W	0x0	ATHR Active Threshold for CIR These bits control the duration of CIR from Idle to Active State. The duration can be calculated by ((ATHR + 1) *(ATHC? Sample Clock: 128*Sample Clock)).

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Offset: 0x0034			Register Name: CIR_CONFIG
Bit	Read/Write	Default/HEX	Description
15:8	R/W	0x18	ITHR Idle Threshold for CIR The Receiver uses it to decide whether the CIR command has been received. If there is no CIR signal on the air, the receiver is staying in IDLE status. One active pulse will bring the receiver from IDLE status to Receiving status. After the CIR is end, the inputting signal will keep the specified level (high or low level) for a long time. The receiver can use this idle signal duration to decide that it has received the CIR command. The corresponding flag is asserted. If the corresponding interrupt is enabled, the interrupt line is asserted to CPU. When the duration of signal keeps one status (high or low level) for the specified duration ((ITHR + 1)*128 sample_clk), this means that the previous CIR command has been finished.
7:2	R/W	0xa	NTHR Noise Threshold for CIR When the duration of signal pulse (high or low level) is less than NTHR, the pulse is taken as noise and should be discarded by hardware. 0: All samples are recorded into RX FIFO 1: If the signal is only one sample duration, it is taken as noise and discarded. 2: If the signal is less than (<=) two sample duration, it is taken as noise and discarded. (p) 61: If the signal is less than (<=) sixty-one sample duration, it is taken as noise and discarded.

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Offset: 0x0034			Register Name: CIR_CONFIG
Bit	Read/Write	Default/HEX	Description
1:0	R/W	0x0	SCS Sample Clock Select for CIR SCS2 / SCS[1] / SCS[0] / Sample Clock 0 / 0 / 0 / IR_CLK/64 0 / 0 / 1 / IR_CLK /128 0 / 1 / 0 / IR_CLK /256 0 / 1 / 1 / IR_CLK /512 1 / 0 / 0 / IR_CLK 1 / 0 / 1 / Reserved 1 / 1 / 0 / Reserved 1 / 1 / 1 / Reserved

17.17.7.7 0x0080 CIR Receiver Store Register0 (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: CIR_STORE0
Bit	Read/Write	Default/HEX	Description
31:0	R/W	0x0	SV0 Receiver Store Value 0

17.17.7.8 0x0084 CIR Receiver Store Register1 (Default Value: 0x0000_0000)

Offset: 0x0084			Register Name: CIR_STORE1
Bit	Read/Write	Default/HEX	Description
31:0	R/W	0x0	SV1 Receiver Store Value 1

17.17.7.9 0x0088 CIR Receiver Store Register2 (Default Value: 0x0000_0000)

Offset: 0x0088			Register Name: CIR_STORE2
Bit	Read/Write	Default/HEX	Description
31:0	R/W	0x0	SV2 Receiver Store Value 2

17.17.7.10 0x008C CIR Receiver Store Register3 (Default Value: 0x0000_0000)

Offset: 0x008C			Register Name: CIR_STORE3
Bit	Read/Write	Default/HEX	Description

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Offset: 0x008C			Register Name: CIR_STORE3
Bit	Read/Write	Default/HEX	Description
31:0	R/W	0x0	SV3 Receiver Store Value 3

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17.18 IR Transmitter (IR_TX)

17.18.1 Overview

The CIR transmitter (IR_TX) transfer arbitrary waves which can be modulated with configurable carrier waves such as 38 kHz. IR_TX only uses lower 8 bits of the 32-bit registers, and stores a 16-bit number in 2 registers, where one register contains the higher 8 bits while the other contains the lower 8 bits.

The IR_TX has the following features:

- One IR_TX interface in CPUX domain
- Arbitrary wave generator
- Configurable carrier frequency
- Configurable carrier duty cycle
- 128 bytes FIFO for data buffer
- Supports Interrupts and DMA
- Supports DMA shake and wait mode
- RLC encoding
- NEC format infra data
- Single/cyclic transmitting mode

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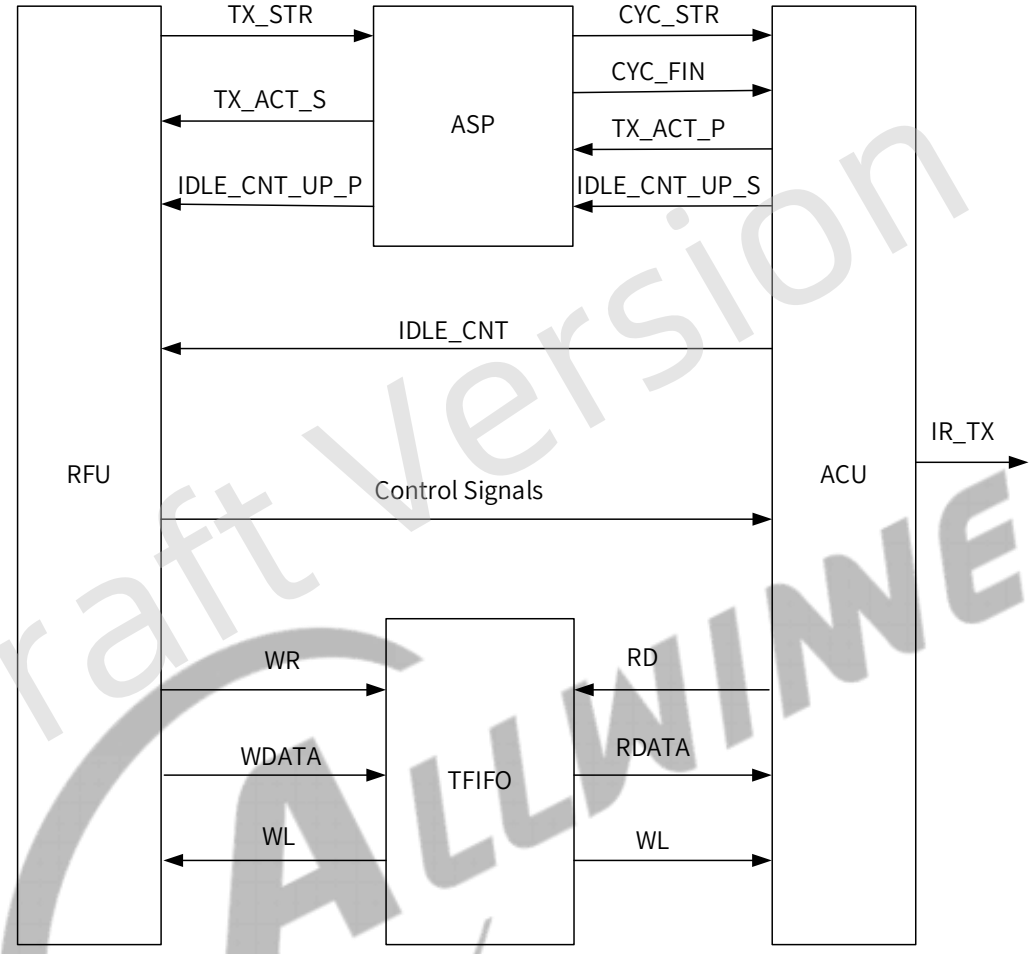
17.18.2 Block Diagram

The following figure shows a block diagram of the IR_TX.

Figure 17-133 IR_TX Block Diagram



17.18.3 External Signals

The following table describes the external signals of IR_TX.

Table 17-79 IR_TX External Signals

Signal Name	Description	Type
IR-TX	Consumer Infrared Receiver	O

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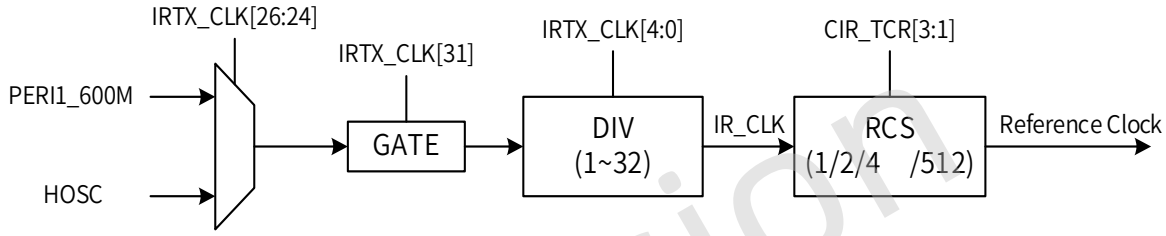
17.18.4 Functional Description

17.18.4.1 Clock Sources

Figure 17-134 IR_TX Clock Description



17.18.4.2 Function Implementation

The IR_TX is used to generate a waveform of arbitrary length, arbitrary shape, and no high-speed requirement, and it can change the data into the high-/low-level sequence of a certain length. Every transmitting data is in bytes, the Bit [7] of a byte means whether the level of a transmitting wave is high or low, the Bit [6:0] is the length of this wave. If the current transmitting frequency-division is 1, 0x88 is a high level of 8 cycles, 0x08 is a low level of 8 cycles. If the current transmitting frequency-division is 4, 0x88 is a high level of 32 cycles, 0x08 is a low level of 32 cycles.

The IR_TX has two transmission modes: non-cycle transmission and cycle transmission.

The non-cycle transmission is to transmit all the data in TX_FIFO until the FIFO is empty.

The cycle transmission is to transmit all the data in TX_FIFO, after the transmission completion, wait for a certain time to recover the data in TX_FIFO and then send it until a stop signal is detected. The data recovery in FIFO is implemented by clearing the read pointer.

Timing Diagram

The IR remote control contains many protocols designed by different manufacturers. Here take NEC protocol as an example, the IR_TX module uses a variable pulse-width modulation technique to encompass the various formats of infrared encoding for remote-control applications. A message is started by a 9 ms AGC burst, which is used to set the gain of the earlier CIR receivers. This AGC burst is then followed by a 4.5 ms space, which is then followed by the address and command.

Bit definition: the logical “1” takes 2.25 ms to transmit, while a logical “0” is only 1.12 ms.

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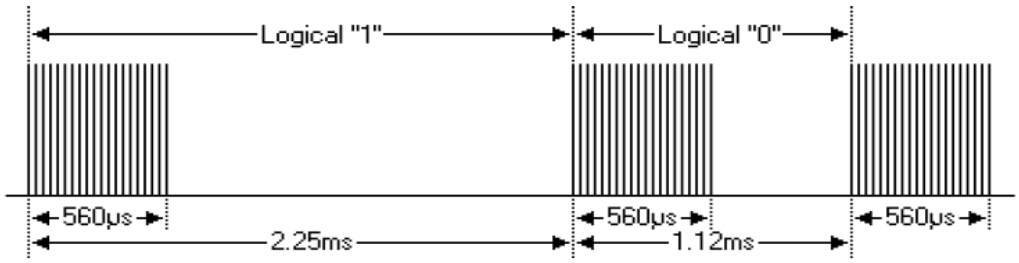
Figure 17-135 Definitions of Logical “1” and Logical “0”



Timing for a message:

Figure 17-136 CIR Message Timing Diagram

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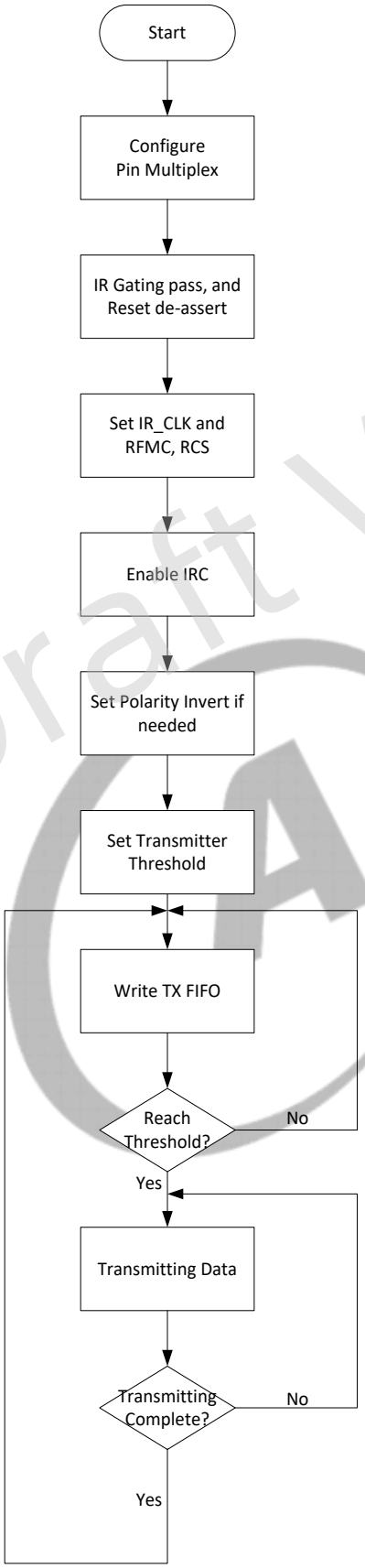
17.18.5 Operation Guidelines

Figure 17-137 CIR Transmitter Process



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17.18.6 Register List

Module Name	Base Address
IRTX	0x02080000

Register Name	Offset	Description
CIR_TGLR	0x0000	CIR Transmitter Global Register
CIR_TMCR	0x0004	CIR Transmitter Modulation Control Register
CIR_TCR	0x0008	CIR Transmitter Control Register
CIR_IDC_H	0x000C	CIR Transmitter Idle Duration Counter Register
CIR_IDC_L	0x0010	CIR Transmitter Idle Duration Counter Register
CIR_TICR_H	0x0014	CIR Transmitter Idle Counter Register
CIR_TICR_L	0x0018	CIR Transmitter Idle Counter Register
CIR_TEL	0x0020	CIR Transmitter FIFO Empty Level Register
CIR_TXINT	0x0024	CIR Transmitter Interrupt Control Register
CIR_TAC	0x0028	CIR Transmitter FIFO Available Counter Register
CIR_TXSTA	0x002C	CIR Transmitter Status Register
CIR_TXT	0x0030	CIR Transmitter Threshold Register
CIR_DMA	0x0034	CIR Transmitter DMA Control Register
CIR_TXFIFO	0x0080	CIR Transmitter FIFO Data Register

17.18.7 Register Description

17.18.7.1 0x0000 CIR Transmitter Global Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: CIR_TGLR
Bit	Read/Write	Default	Description
31:8	/	/	/
7	R/W	0x0	IMS Internal Modulation Select 0: The transmitting signal is not modulated 1: The transmitting signal is modulated internally
6:5	R/W	0x0	DRMC Duty ratio of modulated carrier is high level /low level. 00: Low level is the one time of high level 01: Low level is the two times of high level 10: Low level is the three times of high level 11: Reserved
4:3	/	/	/

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Offset: 0x0000			Register Name: CIR_TGLR
Bit	Read/Write	Default	Description
2	R/W	0x0	TPPI Transmit Pulse Polarity Invert 0: Not invert transmit pulse 1: Invert transmit pulse
1	R/W	0x0	TR Transmit Reset When this bit is set, the transmitting is reset. The FIFO will be flush, the TIC field and CSS field will be clean during Transmit Reset. This field will automatically clean when the Transmit Reset is finished, and the CIR transmitter will state Idle .
0	R/W	0x0	TXEN Transmit Block Enable 0: Disable the CIR Transmitter 1: Enable the CIR Transmitter

17.18.7.2 0x0004 CIR Transmitter Modulation Control Register (Default Value:0x0000_009E)

Offset: 0x0004			Register Name: CIR_TMCR
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R/W	0x9E	RFMC Reference Frequency of Modulated Carrier Reference Frequency of modulated carrier based on a division of a fixed functional clock(FCLK). The range of the modulated carrier is usually 30KHZ to 60KHz.The most consumer electronics is 38KHz. The default modulated carrier is 38KHz when FCLK is 12MHz. RFMC= FCLK/((N+1)*(DRMC+2)).

17.18.7.3 0x0008 CIR Transmitter Control Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: CIR_TCR
Bit	Read/Write	Default	Description
31:8	/	/	/

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Offset: 0x0008			Register Name: CIR_TCR
Bit	Read/Write	Default	Description
7	R/W	0x0	CSS Cyclical Pulse Start/Stop Control Start to transmit when set to '1', 0: Stop when cleared to '0'. From start to stop, all data in FIFO must be transmitted. 1: Start.
6:4	/	/	/
3:1	R/W	0x0	RCS Reference Clock Select for CIR Transmit 000: CIR Transmit reference clock is ir_clk 001: CIR Transmit reference clock is ir_clk/2 010: CIR Transmit reference clock is ir_clk/4 011: CIR Transmit reference clock is ir_clk/8 100: CIR Transmit reference clock is ir_clk/64 101: CIR Transmit reference clock is ir_clk/128 110: CIR Transmit reference clock is ir_clk/256 111: CIR Transmit reference clock is ir_clk/512
0	R/W	0x0	TTS Type of the transmission signal 0: The transmitting wave is single non-cyclical pulse. 1: The transmitting wave is cyclical short-pulse.

17.18.7.4 0x000C CIR Transmitter Idle Duration Counter Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: CIR_IDC_H
Bit	Read/Write	Default	Description
31:8	/	/	/
3:0	R/W	0x0	IDC_H Idle Duration Counter threshold (High 4 bit) Idle Duration = 128*IDC*Ts (IDC = 0-4095)

17.18.7.5 0x0010 CIR Transmitter Idle Duration Counter Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: CIR_IDC_L
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R/W	0x0	IDC_L Idle Duration Counter threshold (Low 8 bit) Idle Duration = 128*IDC*Ts (IDC = 0-4095)

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17.18.7.6 0x0014 CIR Transmitter Idle Counter Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: CIR_TICR_H
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R	0x0	TIC_H Transmit Idle Counter_H (High 8 bit) Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active. When this counter reaches the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero.

17.18.7.7 0x0018 CIR Transmitter Idle Counter Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: CIR_TICR_L
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R	0x0	TIC_L Transmit Idle Counter_L (Low 8 bit) Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active. When this counter reaches the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero.

17.18.7.8 0x0020 CIR Transmitter FIFO Empty Level Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: CIR_TEL
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R/W	0x0	TEL TX FIFO empty Level for DRQ and IRQ. TRIGGER_LEVEL = TEL + 1

17.18.7.9 0x0024 CIR Transmitter Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: CIR_TXINT
Bit	Read/Write	Default	Description
31:3	/	/	/

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Offset: 0x0024			Register Name: CIR_TXINT
Bit	Read/Write	Default	Description
2	R/W	0x0	DRQ_EN TX FIFO DMA Enable 0: Disable 1: Enable When set to '1', the TX FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less than the RAL. The DRQ is de-asserted when condition fails.
1	R/W	0x0	TAI_EN TX FIFO Available Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	TPEI_EN Transmit Packet End Interrupt Enable for Cyclical Pulse 0: Disable 1: Enable TUI_EN Transmitter FIFO under run Interrupt Enable for Non-Cyclical Pulse 0: Disable 1: Enable

17.18.7.10 0x0028 CIR Transmitter FIFO Available Counter Register (Default Value: 0x0000_0080)

Offset: 0x0028			Register Name: CIR_TAC
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R	0x80	TAC TX FIFO Available Space Counter 0x00: No available space in TX FIFO 0x01: 1-byte available space in TX FIFO 0x02: 2-byte available space in TX FIFO (p) 0x80: 128 byte available space in TX FIFO

17.18.7.11 0x002C CIR Transmitter Status Register (Default Value: 0x0000_0002)

Offset: 0x002C	Register Name: CIR_TXSTA

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Bit	Read/Write	Default	Description
31:4	/	/	/
3	R	0x0	STCT Status of CIR Transmitter 0: Idle 1: Active This bit will automatically set when the controller begins transmit the data in the FIFO. The “1” will last when the data in the FIFO. It will automatically be cleaned to “0” when all data in the FIFO is transmitted. The bit is for debug. Output Level of Idle state determined by level of the last data output.
2	R	0x0	DRQ DMA Request Flag When set to ‘1’, the TX FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less than the RAL. The DRQ is de-asserted when condition fails. This bit is for debug.
1	R/W	0x1	TAI TX FIFO Available Interrupt Flag 0: TX FIFO not available by its level 1: TX FIFO available by its level This bit can be cleared by software writing ‘1’.
0	R/W	0x0	TPE Transmitter Packet End Flag for Cyclical Pulse 0: Transmissions of address, control and data fields not completed 1: Transmissions of address, control and data fields completed TUR Transmitter FIFO Under Run Flag for Non-Cyclical Pulse 0: No transmitter FIFO under run 1: Transmitter FIFO under run This bit is cleared by writing a ‘1’.

17.18.7.12 0x0030 CIR Transmitter Threshold Register (Default Value: 0x0000_0000)

Offset: 0x0030	Register Name: CIR_TXT

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Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R/W	0x0	NCTT Non-cyclical Pulse Transmit Threshold The controller will trigger transmitting the data in the FIFO when the data byte number has reaches the Transmit Threshold set in this field.

17.18.7.13 0x0034 CIR Transmitter DMA Control Register (Default Value: 0x0000_00A5)

Offset: 0x0034			Register Name: CIR_DMA_CTL
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	R/W	0xA5	DMA Handshake configuration 0xA5: DMA wait cycle mode 0xEA: DMA handshake mode

17.18.7.14 0x0080 CIR Transmitter FIFO Data Register (Default Value: 0x0000_0000)

Offset: 0x0080			Register Name: CIR_TXFIFO
Bit	Read/Write	Default	Description
31:8	/	/	/
7:0	W	0x0	Transmit Byte FIFO When the transmitting is trigger, the data in the FIFO will be transmitted until the data number has been transmitted finished.

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18 Security

18.1 Crypto Engine (CE)

18.1.1 Overview

The Crypto Engine (CE) module is one encryption/decryption algorithms accelerator. It supports kinds of symmetric, asymmetric, HASH, and RBG algorithms. There are two software interfaces for secure and non-secure world each. Algorithm control information is written in memory by task descriptor, then CE automatically reads it when executing request. It supports parallel requests from 4 channels each world and supports 4 different types of algorithms simultaneously. This module also has an internal DMA controller to transfer data between CE and memory. It supports parallel running for symmetric, HASH, asymmetric algorithms.

The CE has the following features:

- Symmetrical algorithm:
  - AES symmetrical algorithm
    - Key size: 128/192/256 bits
    - CFB mode includes: CFB1, CFB8, CFB64, and CFB128
    - CTR mode includes: CTR16, CTR32, CTR64, and CTR128
    - Supports ECB, CBC, CTS, OFB, CBC-MAC, GCM, and XTS modes
  - Support XTS-AES symmetrical algorithm: key size: 256/512 bits
  - DES symmetrical algorithm
    - CTR mode, includes: CTR16, CTR32, and CTR64
    - Supports ECB, CBC, and CBC-MAC mode
  - Supports 3DES, SM4 symmetrical algorithm
- Hash algorithms
  - Support MD5, SHA1, SHA224, SHA256, SHA384, SHA512, and SM3
  - Support HMAC-SHA1, HMAC-SHA256
  - Support multi-package[1] mode for these ones
  - Support hardware padding

[1] If not last package, input should aligned with computation block, namely 512bits or 1024bit

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- Random bit generator algorithms
  - Support PRNG, 175 bits seed width, and output with multiple of 5 words
  - Support TRNG, post-process by hardware with SHA256, output with multiple of 8 words
  - Support TRNG single/double entropy source mode
  - Support TRNG healthy test
  - Support Instantiate/Reseed/Generate/Uninstantiate 4 process
  - Support prediction resistance requests
  - Support 8 separate suits of Internal State
  - Maxim 2^32 BYTE length of Entropy input, Nonce, Personalization, Additional input. And length is multiple of word
- Public key algorithms
  - Supports RSA public key algorithms: 512/1024/2048/3072/4096-bit width
  - Supports ECC public key algorithms: 160/224/256/384/521-bit width
  - Supports SM2 algorithms
- Security Strategy and System Feature
  - Symmetric, asymmetric, HASH/RBG ctrl logics are separate, can handle task simultaneously. Symmetric logic can select instantiate 2 suits at implementation time.
  - Support task chain mode for each request. Task or task chain are executed at request order.
  - multi- scatter group(sg) are supported for both input and output data
  - Support secure and non-secure interfaces respectively, each world issues task request through its own interface, don't know each other's existence.
  - Each world has 4 channels for software request, each channel has an interrupt control and status bit, and channels are independent with each other.
  - Supports byte-aligned address for all configurations

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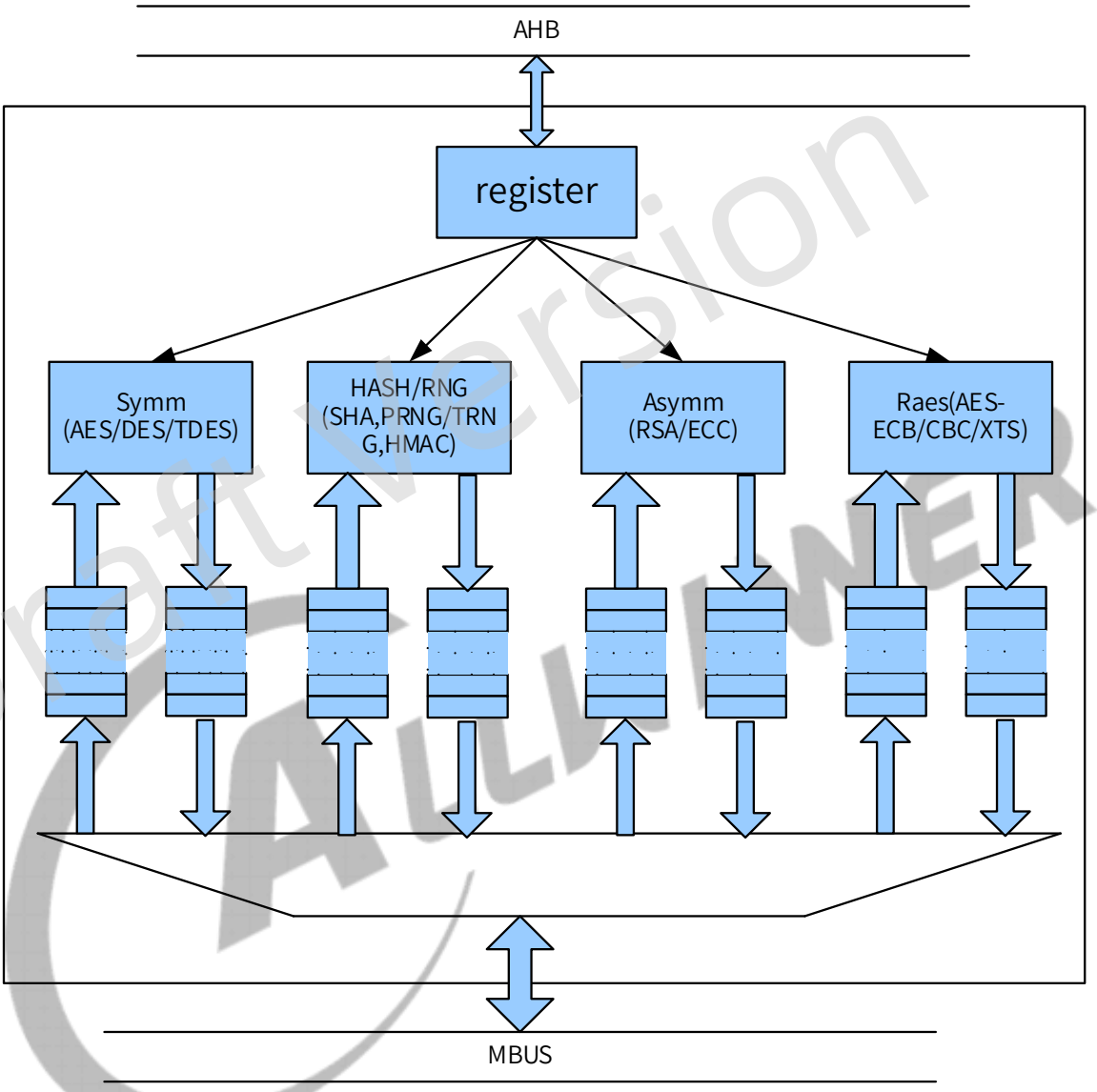
18.1.2 Block Diagram

The following figure shows a block diagram of CE.

Figure 18-1 CE Block Diagram



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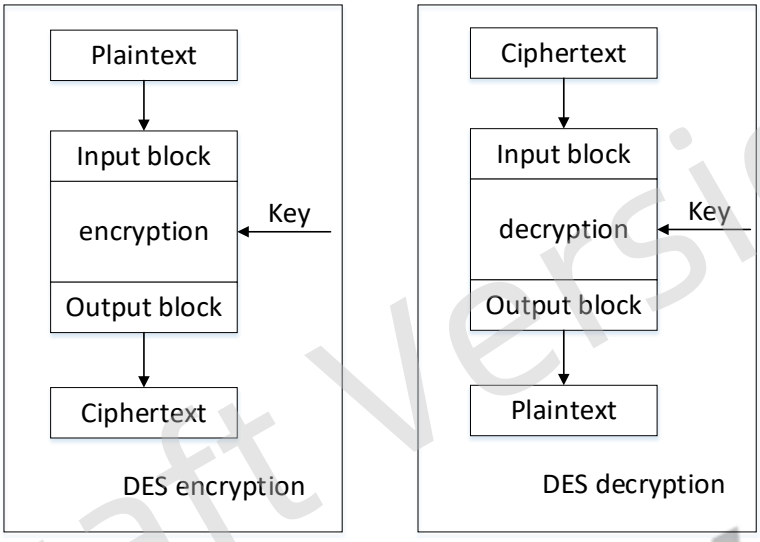
18.1.3 Functional Description

18.1.3.1 DES Algorithm

The following figure shows the DES encryption and decryption operation.

Figure 18-2 DES Encryption and Decryption



18.1.3.2 3DES Algorithm

The 3DES algorithm supports both 3-key and 2-key operations. A 2-key operation can be regarded as a simplified 3-key operation. To be specific, key 3 is represented by key 1 in a 2-key operation. The following figure shows the 3DES encryption and decryption operation of a 3-key operation and a 2-key operation.

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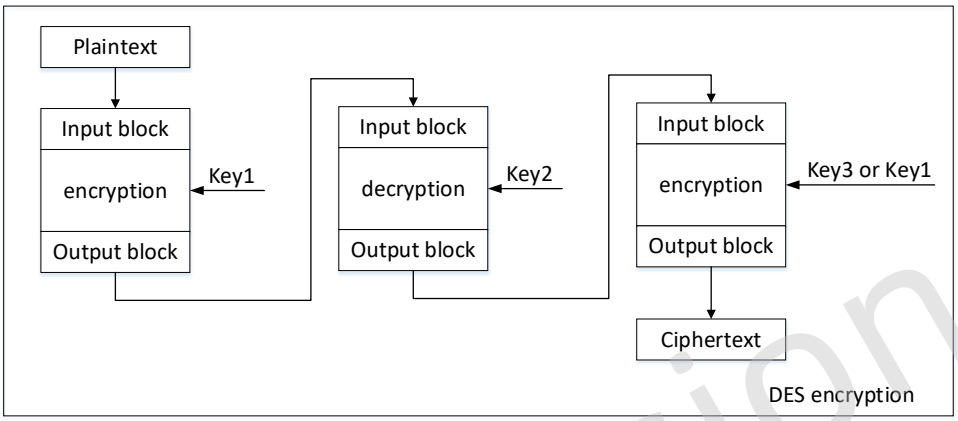
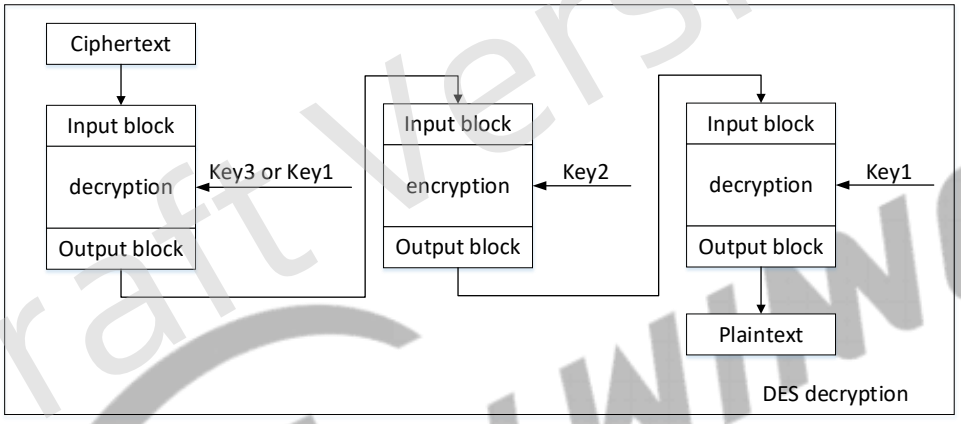
Figure 18-3 3DES Encryption and Decryption of a 3-key Operation and a 2-key Operation





18.1.3.3 ECB Mode

The ECB mode is a confidentiality mode that features, for a given key, the assignment of a fixed ciphertext block to each plaintext block, analogous to the assignment of code words in a codebook.

In ECB mode, encryption and decryption algorithms are directly applied to the block data. The operation of each block is independent, so the plaintext encryption and ciphertext decryption can be performed concurrently.

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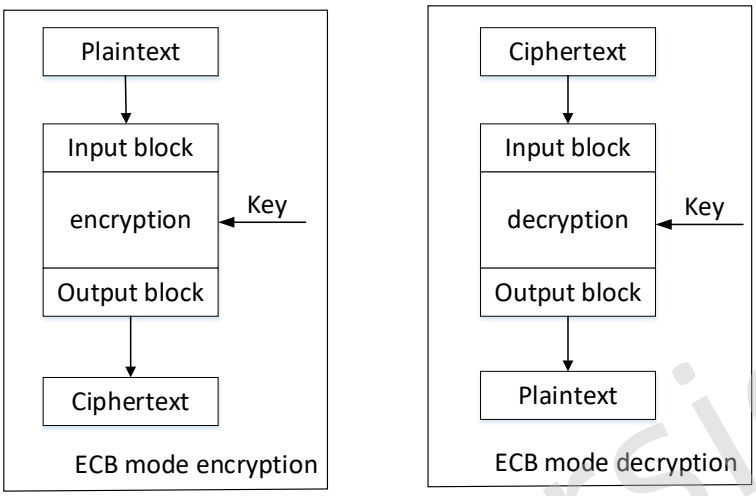
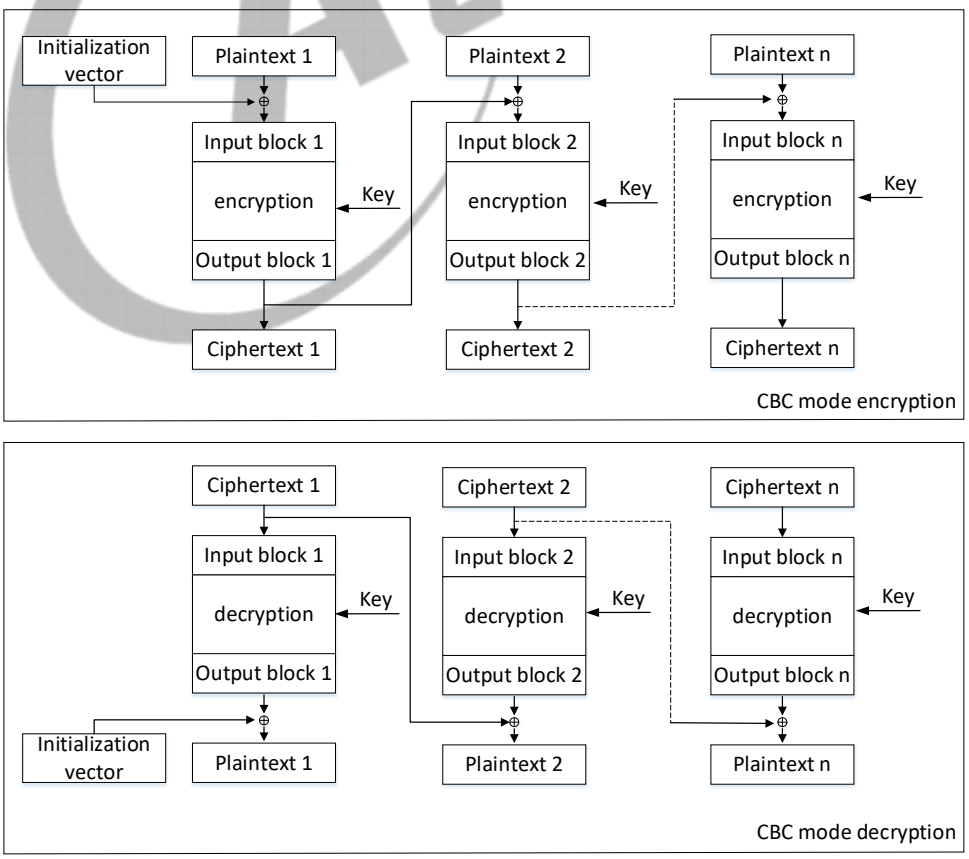
Figure 18-4 ECB Mode Encryption and Decryption



18.1.3.4 CBC Mode

The CBC mode is a confidentiality mode whose encryption process features the combining of the plaintext blocks with the previous ciphertext blocks. The CBC mode requires an initialization vector (IV) to combine with the first plaintext block. The encryption process of each plaintext block is related to the block processing result of the previous ciphertext blocks, so encryption operations cannot be concurrently performed in CBC mode. The decryption operation is independent of output plain text of the previous block, so decryption operations can be performed concurrently.

Figure 18-5 CBC Mode Encryption and Decryption



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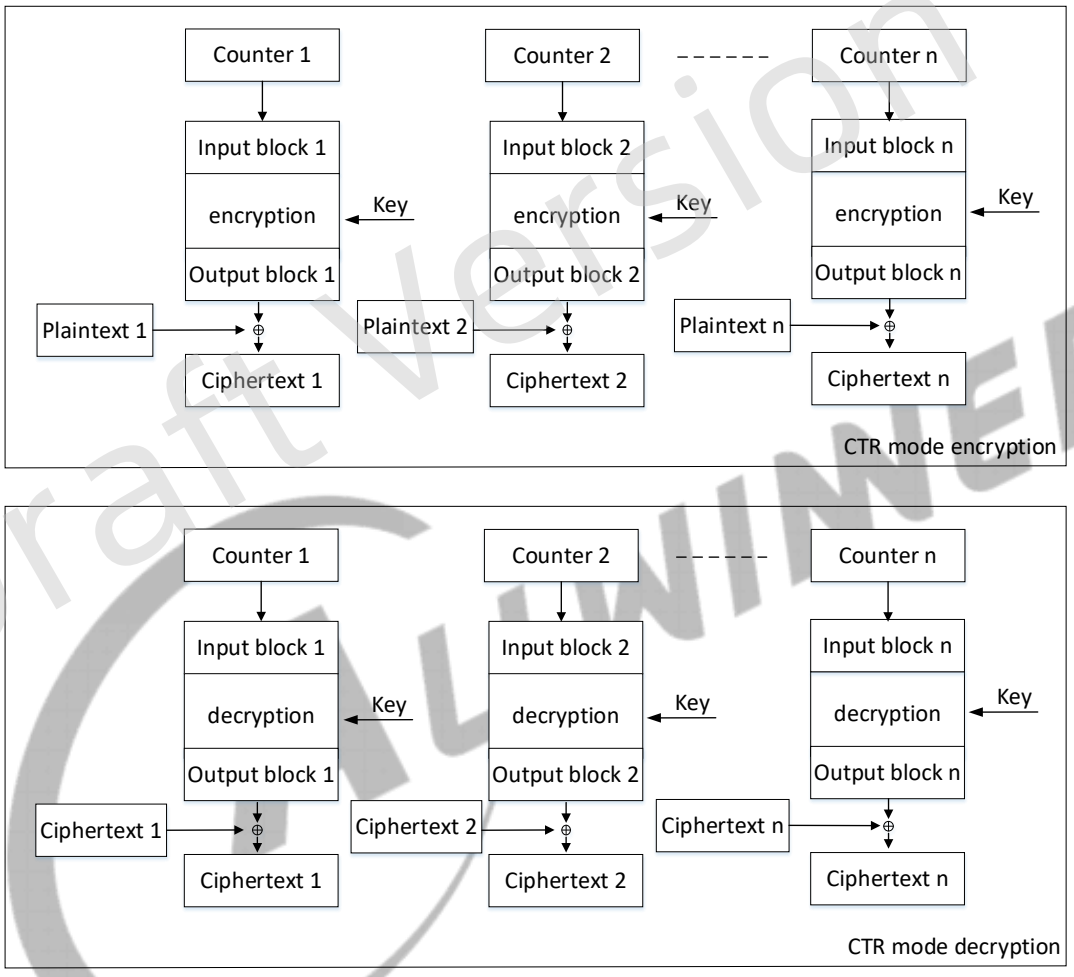
18.1.3.5 CTR Mode

The CTR mode is a confidentiality mode that features the application of the forward cipher to a set of input blocks, called counters, to produce a sequence of output blocks that are exclusive-ORed with the plaintext to produce the ciphertext, and vice versa. All of the counters must be distinct.

Figure 18-6 CTR Mode Encryption and Decryption



18.1.3.6 CFB Mode

The CFB mode is a confidentiality mode that features the feedback of successive ciphertext segments into the input blocks of the forward cipher to generate output blocks that are exclusive-ORed with the plaintext to produce the ciphertext, and vice versa. The CFB mode requires an IV as the initial input block, and the forward cipher operation is applied to the IV to produce the first output block. The first ciphertext segment is produced by exclusive-ORing the first plaintext segment with the s most significant bits of the first output block. The value of s is 1 bit, 8 bits, 64 bits, or 128 bits.

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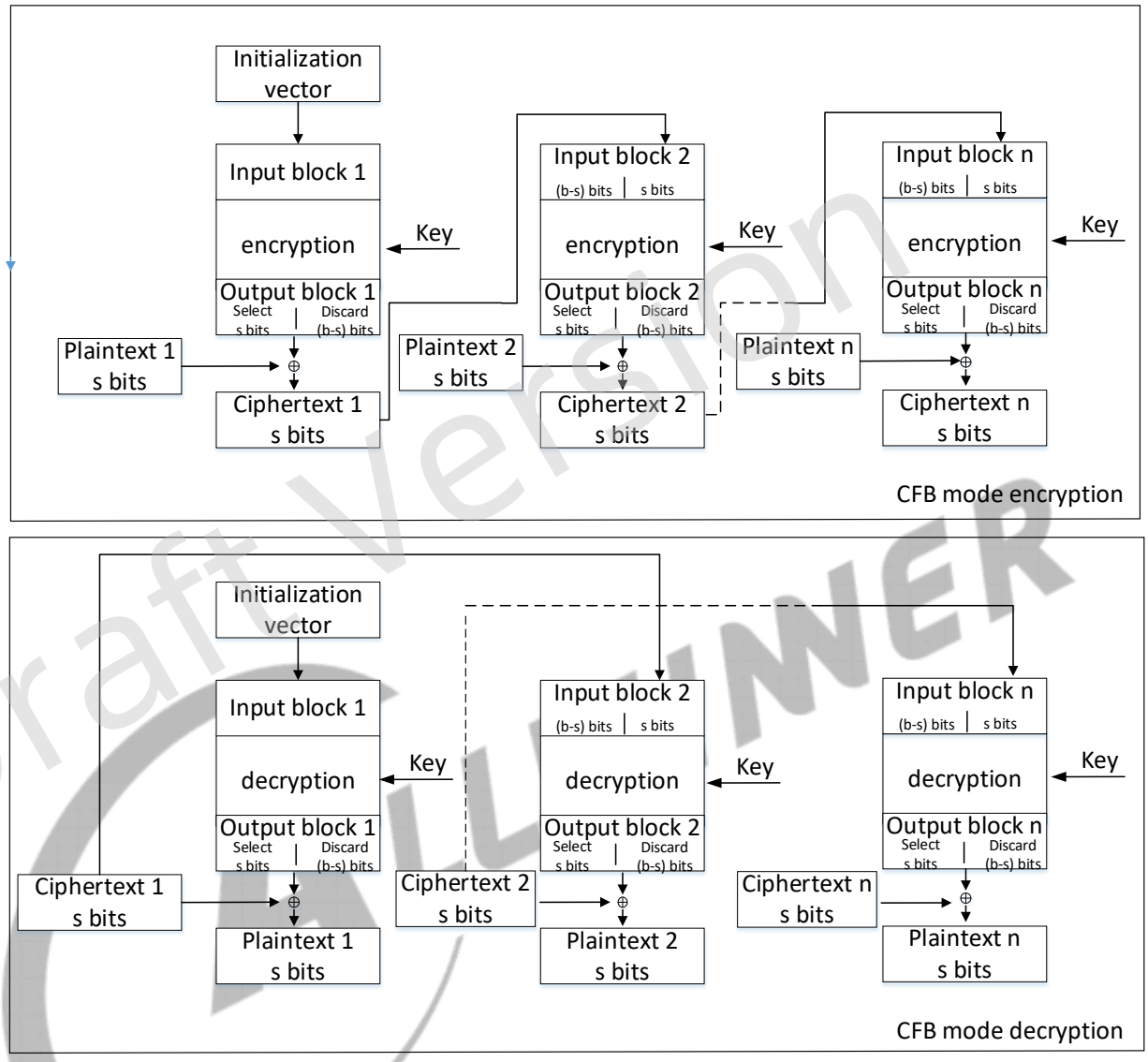
The following figure shows the s-bit CFB mode of the AES algorithms.

Figure 18-7 CFB Mode Encryption and Decryption



18.1.3.7 OFB Mode

The OFB mode is a confidentiality mode that features the iteration of the forward cipher on an IV to generate a sequence of output blocks that are exclusive-ORed with the plaintext to produce the ciphertext, and vice versa. If a same key is used, different IVs must be used to ensure operation security.

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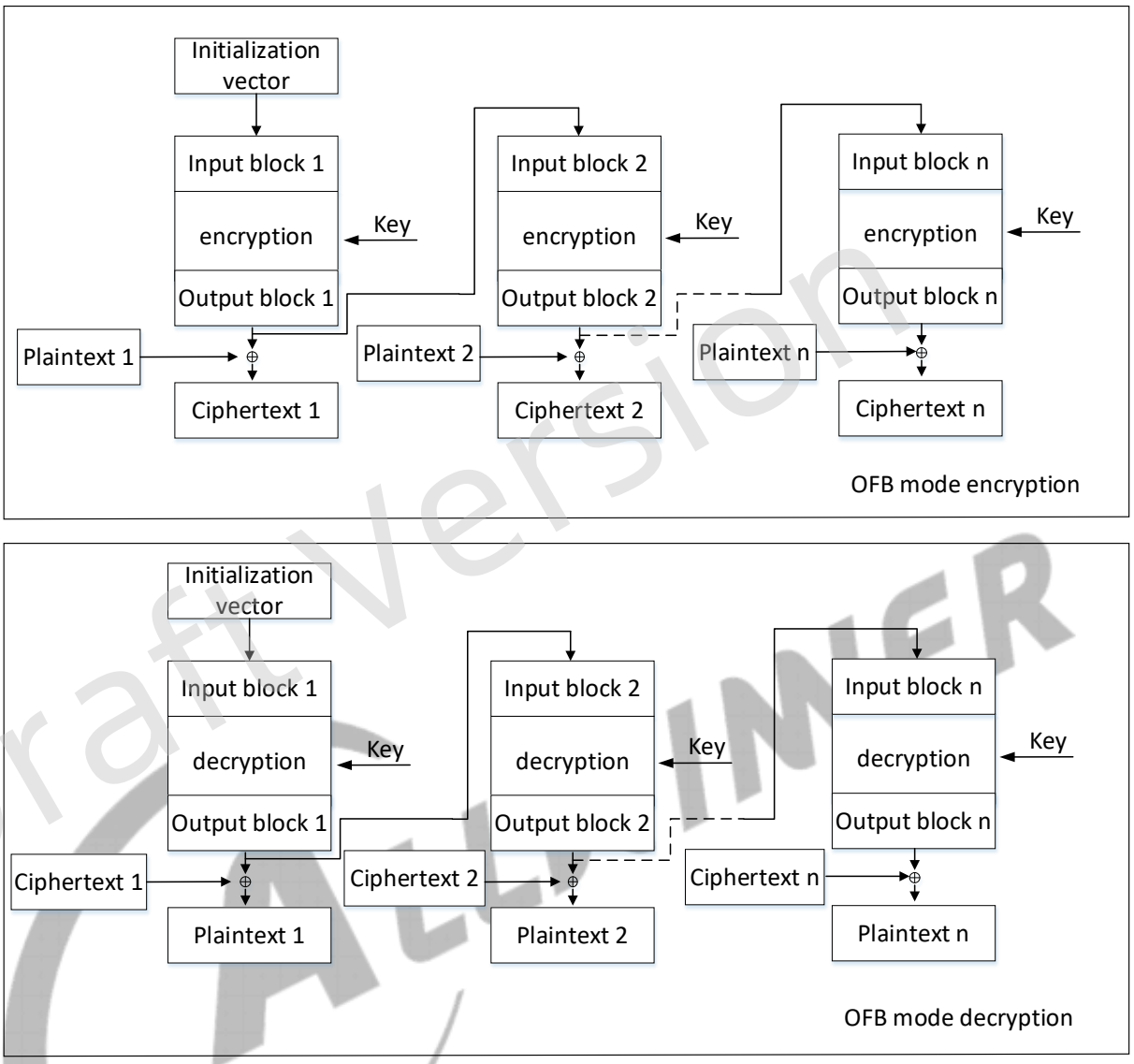
Figure 18-8 OFB Mode Encryption and Decryption



18.1.3.8 CTS Mode

The CTS mode is a confidentiality mode that accepts any plaintext input whose bit length is greater than or equal to the block size but not necessarily a multiple of the block size. Below are the diagrams for CTS encryption and decryption.

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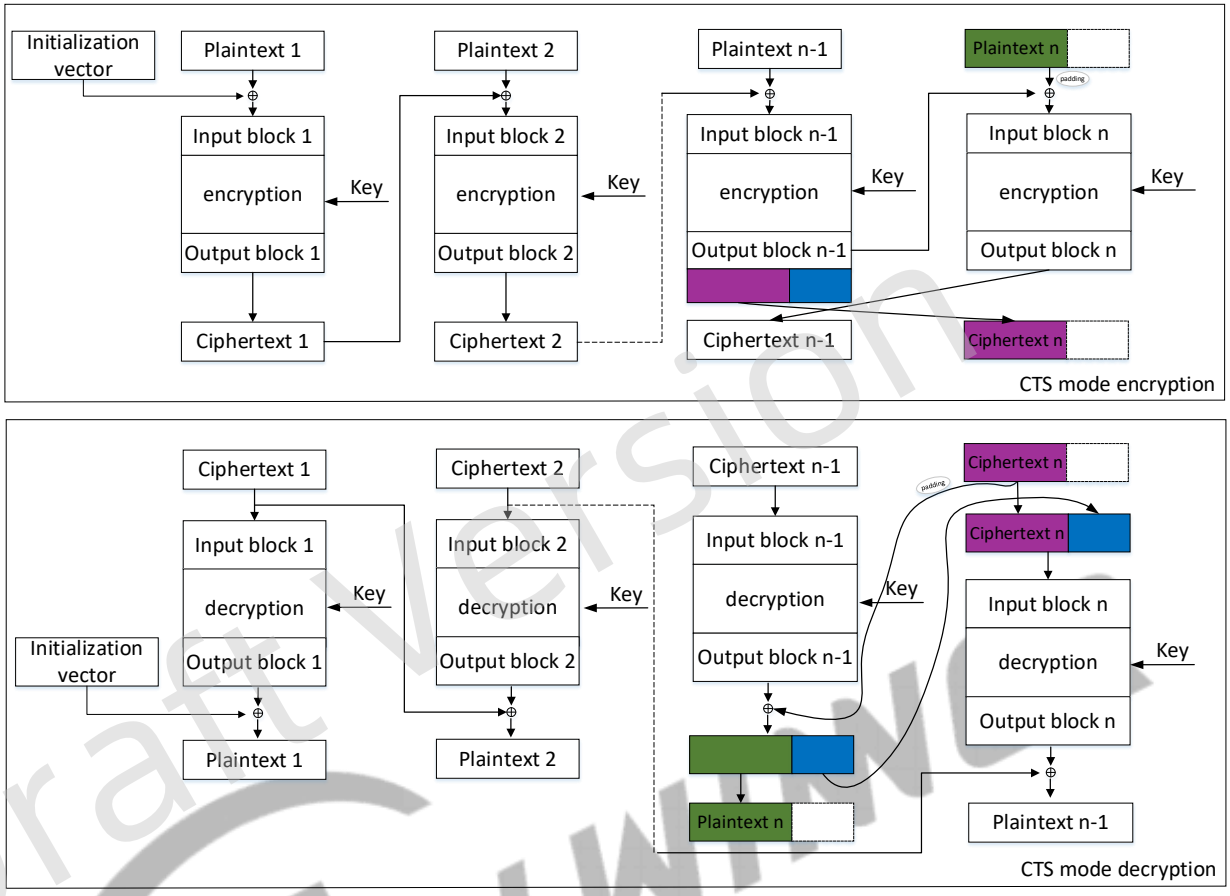
Figure 18-9 CTS Mode Encryption and Decryption



18.1.3.9 HASH Algorithm

The hash algorithms support MD5, SHA1, SHA224, SHA256, SHA384, SHA512, HMAC-SHA1, and HMAC-SHA256. All algorithms are iterative, one-way hash functions that can process a message to produce a condensed representation called a message digest. When a message is received, the message digest can be used to verify whether the data has changed, that is, to verify its integrity.

The hash algorithm of the CE supports block-aligned total length of the input data (padded by software), that is, a multiple of 64 bytes. The message length after padding by software is used as the configured data length for the hash algorithm.

18.1.3.10 RSA Algorithm

The RSA is a public key encryption/decryption algorithm implemented through the modular exponentiation operation.

The ciphertext is obtained as follows: C = ME mod N. The plaintext is obtained as follows: M = CD mod N.

M indicates the plaintext, C indicates the ciphertext, (N, E) indicates the public key, and (N, D) indicates the private key.

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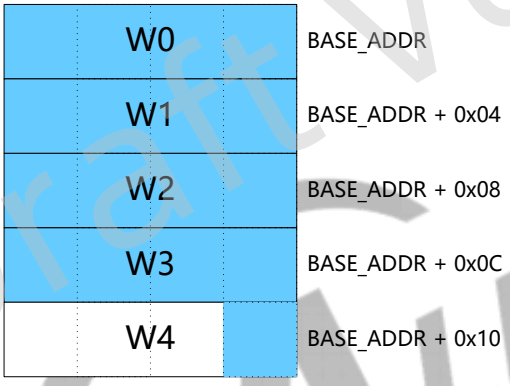
18.1.3.11 Storing Message

In the application, a message may not be stored contiguously in the memory, but divided into multiple segments. Or a piece of continuously stored messages can be artificially split into multiple pieces as needs. Then each segment corresponds to a set of the source address and source length in the descriptor. Multiple segments correspond to groups 0-7 source address/source length in sequence.

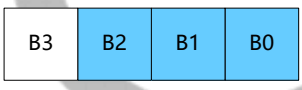
Each task supports up to 8 message segments, and the data volume of each message segment supports up to 4 GWord (AES-CTS is 1 GByte). The total amount of all segments in a task (that is a package) supports up to 4 GWord (AES-CTS is 1 GByte). If a message is divided into multiple packages, all others are required to be whole words; when the last package of AES-CTS is less than one word, 0 needs to be padded, and those less than one word are counted as one word. The following figure shows the address order structure.

Figure 18-10 Word Address of Message



Byte order: low byte first, high byte last. When the data is less than one word, the low byte is filled first. The following figure shows the byte order structure (blue means it is filled by the message).

Figure 18-11 Byte Order



Bit order: high bit first, low bit last. When the data is less than one Byte, the high bit is filled first. The following figure shows the bit order structure.

Figure 18-12 Bit Order

18.1.3.12 Storing Key

The length of KEY must be an integer multiple of word.

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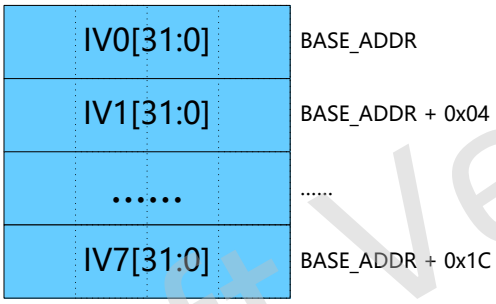
18.1.3.13 Storing IV

For different algorithms, the length of IV is different. But they are integer multiples of word. To keep the byte order of IV and HASH digest output consistent, the byte order of IV is different from that of the message. For the multi-packet operation, the first address of the digest output result of the previous HASH can be directly configured to the first address of the next IV, and the software does not need to do any processing on the digest.

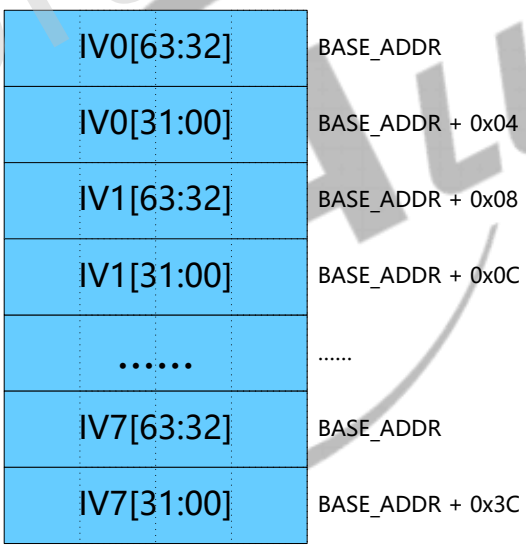
The following figure shows the storage method of 32-bit IV value.

Figure 18-13 The Storage Method of 32-bit IV



The following figure shows the storage method of 64-bit IV value.

Figure 18-14 The Storage Method of 64-bit IV



18.1.3.14 Task Descriptor of Hash Algorithms and RBG Algorithms

The task descriptor is data written by software to a contiguous space in memory. The data describes the various properties of a task, such as algorithm type, mode, subcommand, key address, data source address, the data size read from data source, abstract destination address, the written destination data size, and the information of other tasks. First, we configure the task descriptor by software; then we operate the registers of CE to start this task. After the task starts, CE will read task descriptor based on the address of the task descriptor configured in register, and perform the task one time based on the described properties.

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In applications, the “NEXT TASK ADDR” field can be configured as the starting address of the next task descriptor, to concatenate multi task descriptors into a task chain. After starting the first task, CE will perform every task in order until the “NEXT TASK ADDR” field is invalid (that is 0).

The HASH/RBG algorithms and Symmetrical/Asymmetrical algorithms use the different descriptor structure, separately.

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Figure 18-15 Task Chaining of Hash Algorithms and Random Bit Generator Algorithms

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The detail structures are as follows.

No.	Descriptor	Name	Width	Description
0	CTRL	CHN	[1:0]	Channel ID
		IVE	[8]	IV mode enable, active high
		LPKG	[12]	Last Package flag, Active High
		DLAV	[13]	Data length valid For last package, the bit needs be configured. For non-last package, the bit needs not be configured. (Please configure it as 0 in PRNG/TRNG) 1: DLA means the WORD address where data total length (by bits) is saved. 0: DLA means the value of message total length (by bits).
		IE	[16]	Interrupt enable for current task, active high
1	CMD	HASH SEL	[3:0]	Hash algorithms select 0: MD5 1: SHA1 2: SHA224 3: SHA256 4: SHA384 5: SHA512 6: SM3 Other: Reserved
		HME	[4]	HMAC mode enable, active high
		RGB SEL	[11:8]	RGB algorithms select 0: No RGB use 1: PRNG 2: TRNG Other: Reserved
		SUB CMD	[31:16]	Sub-command in a specific algorithm When using PRNG, sub_cmd[15] means PRNG seed reload; sub_cmd[14:0] means PRNG linearly shifted seed
2	DLA	DLA	[31:0]	Data length OR its address. For last package, the field needs be configured. For non-last package, the field needs not be configured. (Not used in PRNG/TRNG) When DLAV=1, here is the byte address where data total length (by bits) is saved. When DLAV=0, here is the value of message total length (by bits)

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No.	Descriptor	Name	Width	Description
3	DLA	DLA	[7:0]	When DLAV=1, here is the byte address bit[39:32] where data total length (by bits) is saved.
	KA	KA	[31:8]	KEY Address: The byte address bit [23:0]. where HMAC KEY or PRNG KEY is saved.
4	KA	KA	[15:0]	KEY Address: The byte address bit [39:24]. where HMAC KEY or PRNG KEY is saved.
	IVA	IVA	[31:16]	IV Address: The byte address bit[15:0] where IV is saved.
5	IVA	IVA	[23:0]	IV Address: The byte address bit [39:16] where IV is saved.
	Reversed	Reversed	[31:24]	/
6+5*x	SGx_W0	SGx_WORD0	[31:0]	Source Data Address x: The byte address bit [31:0] where Source Datax is saved.
7+5*x	SGx_W1	SGx_WORD1	[7:0]	Source Data Address x: The byte address bit [39:32] where Source Datax is saved.
			[31:8]	Output Data Address x: The byte address bit [23:0] where Output Datax to be saved.
8+5*x	SGx_W2	SGx_WORD2	[15:0]	Output Data Address x: The byte address bit [39:24] where Output Datax to be saved.
	Reversed	Reversed	[31:16]	/
9+5*x	SGx_W3	SGx_WORD3	[31:0]	Source Data length x: The Length (by bytes) of Source Datax.
10+5*x	SGx_W4	SGx_WORD4	[31:0]	Output Data length x: The Length (by bytes) of output Datax.
46	NSA	NSA	[31:0]	Next SG Address: The byte address bit [31:0]. where the descriptor of the next 8 sg in a task is saved. If this is the only one group sg or the last group of a task, NSA must be 32'h0.

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No.	Descriptor	Name	Width	Description
47	NSA	NSA	[7:0]	Next SG Address: The byte address bit [39:32]. where the descriptor of the next 8 sg in a task is saved. If this is the only one group sg or the last group of a task, NSA must be 8'h0.
	NTA	NTA	[31:8]	Next task Address: The byte address bit[23:0] where the descriptor of the next task in a task-chain is saved. If this is the only task or the last task of a task-chain, NTA must be 24'h0.
48	NTA	NTA	[7:0]	Next task Address: The byte address bit[39:24] where the descriptor of the next task in a task-chain is saved. If this is the only task or the last task of a task-chain, NTA must be 16'h0.
	Reversed	Reversed	[31:8]	/
49	Reversed	Reversed	/	/
50	Reversed	Reversed	/	/
51	Reversed	Reversed	/	/

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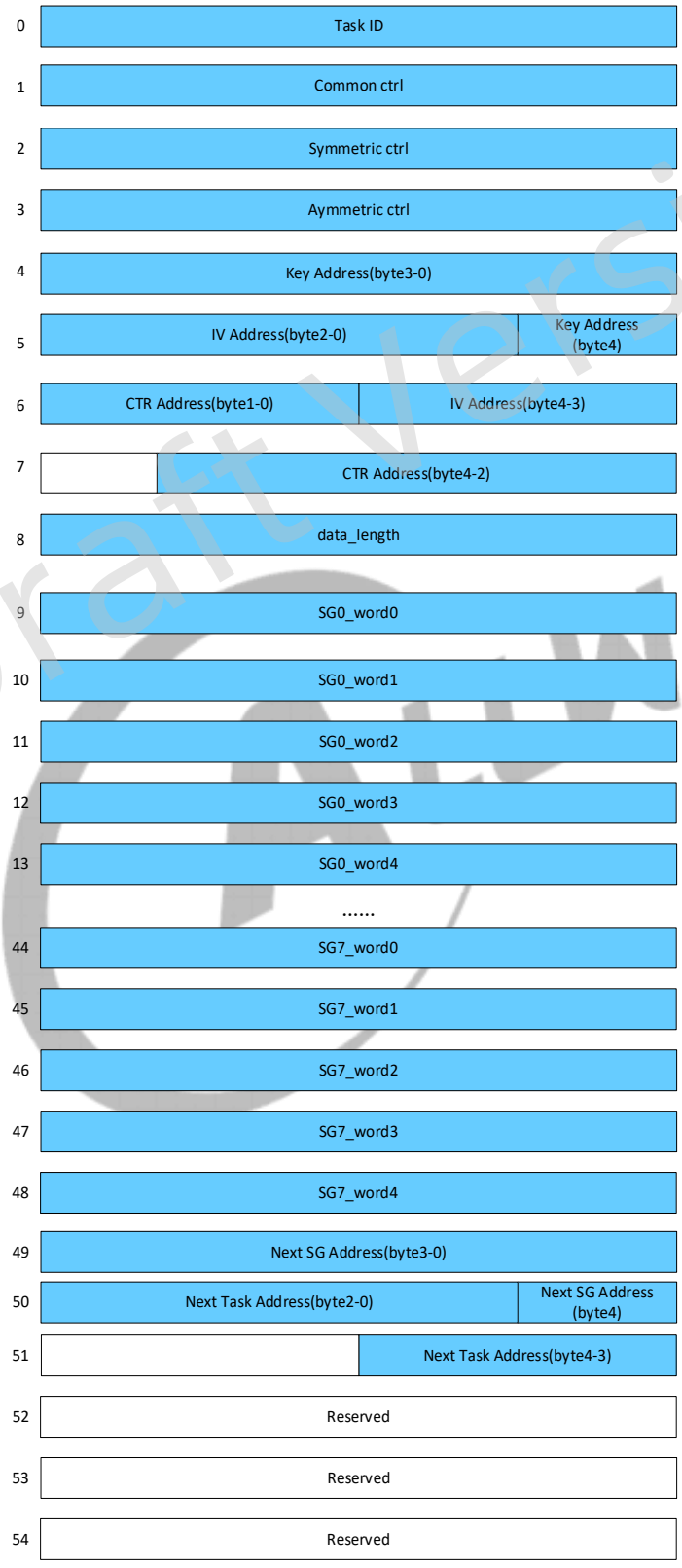
18.1.3.15 Other Algorithms Task Descriptor

Software make request through task descriptor, providing algorithm type, mode, key address, source/destination sg address and size, etc. The task descriptor is as follows.

Figure 18-16 Task Chaining of Other Algorithms



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Channel id supports 0-3 for each world.

- Common ctrl

Bit	Read/Write	Default	Description
31	R/W	0	interrupt enable for current task 0: Disable interrupt 1: Enable interrupt
30:25	/	/	/
24:17	R/W	0	cbc_mac_len the outcome bit length of CBC-MAC when in CBC-MAC mode. The part also be used as gcm/ocb mode tag_len.
16:9	/	/	/
8	R/W	0	OP DIR Algorithm Operation Direction 0: Encryption 1: Decryption
7	/	/	/
6:0	R/W	0	Algorithm type 0x0: AES 0x1: DES 0x2: Triple DES (3DES) 0x3: SM4 others: Reserved 0x20: RSA 0x21: ECC 0x22: SM2 others: Reserved 0x30: RAES Others: Reserved

- Symmetric ctrl

Bit	Read/Write	Default	Description
31:27	/	/	/
26	R/W	0	no_modk 0: Have module key derivation function 1: No module key derivation function
25:24	/	/	/

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Bit	Read/Write	Default	Description
23:20	R/W	0	KEY Select key select for AES/SM4/TDES (TDES only configured as 0/8-15) 0: Select input CE_KEYx (Normal Mode) 1: Select {SSK} 2: Select {HUK} 3-7: Reserved 8-15: Select internal Key n (n from 0 to 7)
19:18	R/W	0	cfb_width For AES-CFB width 0: CFB1 1: CFB8 2: CFB64 3: CFB128
17	/	/	/
16	R/W	0	AES CTS last package flag When set to '1', it means this is the last package for AES-CTS mode (the size of the last package >128bit). The part also be used as gcm/ocb mode gcm_last/ocb_last .
15:14	/	/	/
13	R/W	0	xts_last 0: not last block for XTS 1: last block for XTS
12	R/W	0	xts_first 0: not first block for XTS 1: first block for XTS
11:8	R/W	0	AES/DES/3DES/RAES modes. DES/3DES only supports ECB/CBC/CTR. RAES only supports ECB/CBC/XTS operation mode for symmetric 0: Electronic Code Book (ECB) mode 1: Cipher Block Chaining (CBC) mode 2: Counter (CTR) mode 3: CipherText Stealing (CTS) mode 4: Output feedback (OFB)mode 5: Cipher feedback (CFB)mode 6: CBC-MAC mode 7: OCB mode 8: GCM mode 9: XTS mode Other: Reserved
7:6	/	/	/

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Bit	Read/Write	Default	Description
5:4	R/W	0	gcm_iv_mode [1:0] gcm_iv_mode [0]: Value 1 show the last req for iv calculate gcm_iv_mode [1]: 0: No GHASH calculate mode 1: GHASH calculate mode gcm_iv_mode [1:0]: 00: IDLE state, this calculate do not have the process from iv to J0. 01: By iv padding generating J0. On the mode, iv padding is 96 bits, so iv_length will be 96bits. 10: By GHASH calculate for iv generating J0, and this is not the last req for iv calculate. 11: By GHASH calculate for iv generating J0, and this is the last req for iv calculate
3:2	R/W	0	CTR Width Counter Width for CTR Mode 0: 16-bits Counter 1: 32-bits Counter, gcm mode always use this setting without software 2: 64-bits Counter 3: 128-bits Counter
1:0	R/W	0	AES Key Size 0: 128-bits 1: 192-bits 2: 256-bits 3: Reserved

- Asymmetric ctrl

Bit	Read/Write	Default	Description
31:21	/	/	/

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Bit	Read/Write	Default	Description
20:16	R/W	0	PKC algorithm mode. For modular computation: 00000: Modular exponent(RSA) 00001: Modular add 00010: Modular minus 00011: Modular multiplication others: Reserved For ECC: 00000: Point add 00001: Point double 00010: Point multiplication 00011: Point verification 00100: Encryption 00101: Decryption 00110: Sign 00111: Sign verify others: Reserved For SM2: 00000: Encryption 00001: Decryption 00010: Sign 00011: Sign verify 00100: Key exchange
15:8	/	/	/
7:0	R/W	0	Asymmetric algorithms operation width field. It indicates how much width this request apply, as words.

key addr field is address for each algorithm's key, also for extension feature micro codes address. (By byte)

iv addr field is tweak value address for XTS.

ctr addr is address for next block's IV. (By byte)

src/dst sgX addr field indicate 40bits address for source and destination data. (By byte)

src/dst sgX size field indicates size for each sg respectively (by byte)

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For SG, the detail as flow:

byte3	byte2	byte1	byte0	
SRC_ADDR0 [B3]	SRC_ADDR0 [B2]	SRC_ADDR0 [B1]	SRC_ADDR0 [B0]	SG_WORD0
DST_ADDR0 [B2]	DST_ADDR0 [B1]	DST_ADDR0 [B0]	SRC_ADDR0 [B4]	SG_WORD1
		DST_ADDR0 [B4]	DST_ADDR0 [B3]	SG_WORD2
SRC_SIZE0 [B3]	SRC_SIZE0 [B2]	SRC_SIZE0 [B1]	SRC_SIZE0 [B0]	SG_WORD3
DST_SIZE0 [B3]	DST_SIZE0 [B2]	DST_SIZE0 [B1]	DST_SIZE0 [B0]	SG_WORD4

1 group SG has 8 sg, each sg has 5 words, the ADDR is 40 bits and byte-addr; the SIZE is 32bits and byte-unit. We will support unlimited SG number, but the 1860 just use for test. This can have many group SG in a task, using the next_sg_addr to create the new SG information in the task.

Next sg field should be set to 0 when no next group sg, else set to next sg's descriptor.

next task field should be set to 0 when no next task, else set to next task's descriptor.

18.1.3.16 PKC Microcode

PKC module supports RSA, ECC, SM2, asymmetric algorithms in the form of microcode. It implements basic modular add, minus, multiplication, point add, point double, and logic computing, etc. Complete RSA/ECC/SM2 encryption, decryption, sign, verify are implemented with this microcode.

Asymmetric algorithms RSA/ECC/SM2 are implemented as microcode in PKC module. The encryption, decryption, sign, verify operations of asymmetric algorithms are composed with certain fixed microcode with hardware.

18.1.3.17 PKC Configuration

Before starting PKC, task description must be configured. Parameters to PKC are assigned to source sg, outcome is put to destination sg.

For RSA, parameters should be at the order of key, modulus, plaintext.

For ECC point add P2 = P0 + P1, parameters should be at the order of p, P0x, P0y, P1x, P1y. Output is at the order of P2x, P2y.

For ECC point double P2 = 2*P0, parameters should be at the order of p, a, P0x, P0y. Output is at the order of P2x, P2y.

For ECC point multiplication P2 = k*P0, parameters should be at the order of p, k, a, P0x, P0y. Output is at the order of P2x, P2y.

For ECC point verification, parameters should be at the order of p, a, P0x, P0y, b. Output is 1 or 0.

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For ECC encryption, parameters should be at the order of random k, p, a, Gx, Gy, Qx, Qy, m. Output is at the order of Rx, Ry, c.

For ECC decryption, parameters should be at the order of random k, p, a, Rx, Ry, c. Output is m.

For ECC signature, parameters should be at the order of random k, p, a, Gx, Gy, n, d, e. Output is at the order of r, s.

For ECC signature verification, parameters should be at the order of n, s, e, r, p, a, Gx, Gy, Qx, Qy, n, r. Output is 1 or 0.

18.1.3.18 Error Check

After CE reads the task descriptor, CE can monitor error during algorithm operation. When the error is monitored, CE will do the following operations:

The task will pause immediately

Generates interrupt

The corresponding channel of the task status register is Fail

The corresponding channel bit of error status register can be read error number

The error number has the following types.

Code	Name	Description	Algorithms Type
0x01	algorithm not support	The algorithm type is not supported.	All
0x21	key ladder configuration error	/	KL
0x31	data length error	Input size or output size configuration size error.	All

18.1.4 Register List

Module Name	Base Address	Comments
CE_NS	0x03040000	Non-Security CE
CE_S	0x03040800	Security CE

18.1.4.1 CE_NS Register List

Module Name	Base Address	Comments
CE_NS	0x03040000	Non-Security CE

Register Name	Offset	Description
CE_TDA0_NS	0x0000	Non-Security CE Task Descriptor Address0 Register
CE_TDA1_NS	0x0004	Non-Security CE Task Descriptor Address1 Register

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Register Name	Offset	Description
CE_ICR_NS	0x0008	Non-Security CE Interrupt Control Register
CE_ISR_NS	0x000C	Non-Security CE Interrupt Status Register
CE_TLR_NS	0x0010	Non-Security CE Task Load Register
CE_TSR_NS	0x0014	Non-Security CE Task Status Register
CE_ESR_NS	0x0018	Non-Security CE Error Status Register

18.1.4.2 CE_S Register List

Module Name	Base Address	Comments
CE_S	0x03040800	Security CE

Register Name	Offset	Description
CE_TDA0_S	0x0000	Security CE Task Descriptor Address0 Register
CE_TDA1_S	0x0004	Security CE Task Descriptor Address1 Register
CE_ICR_S	0x0008	Security CE Interrupt Control Register
CE_ISR_S	0x000C	Security CE Interrupt Status Register
CE_TLR_S	0x0010	Security CE Task Load Register
CE_TSR_S	0x0014	Security CE Task Status Register
CE_ESR_S	0x0018	Security CE Error Status Register
CE_SCSA0_S	0x0020	Security CE Symmetric Algorithm DMA Current Source Address0 Register
CE_SCSA1_S	0x0024	Security CE Symmetric Algorithm DMA Current Source Address1 Register
CE_SCDA0_S	0x0028	Security CE Symmetric Algorithm DMA Current Destination Address0 Register
CE_SCDA1_S	0x002C	Security CE Symmetric Algorithm DMA Current Destination Address1 Register
CE_HCSA0_S	0x0030	Security CE HASH Algorithm DMA Current Source Address0 Register
CE_HCSA1_S	0x0034	Security CE HASH Algorithm DMA Current Source Address1 Register
CE_HCDA0_S	0x0038	Security CE HASH Algorithm DMA Current Destination Address0 Register
CE_HCDA1_S	0x003C	Security CE HASH Algorithm DMA Current Destination Address1 Register
CE_ACSA0_S	0x0040	Security CE Asymmetric Algorithm DMA Current Source Address0 Register
CE_ACSA1_S	0x0044	Security CE Asymmetric Algorithm DMA Current Source Address1 Register
CE_ACDA0_S	0x0048	Security CE Asymmetric Algorithm DMA Current Destination Address0 Register

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Register Name	Offset	Description
CE_ACDA1_S	0x004C	Security CE Asymmetric Algorithm DMA Current Destination Address1 Register
CE_XCSA0_S	0x0050	Security CE XTS Algorithm DMA Current Source Address0 Register
CE_XCSA1_S	0x0054	Security CE XTS Algorithm DMA Current Source Address1 Register
CE_XCDA0_S	0x0058	Security CE XTS Algorithm DMA Current Destination Address0 Register
CE_XCDA1_S	0x005C	Security CE XTS Algorithm DMA Current Destination Address1 Register
CE_RCTC_S	0x0090	Security CE Repetition Count Test C Register
CE_APTC_S	0x0094	Security CE Adaptive Proportion Test C Register
CE_APTWP_S	0x0098	Security CE Adaptive Proportion Test W Period Register
CE_HTCR_S	0x009C	Security CE Healthy Test Control Register

18.1.5 CE_NS Register Description

18.1.5.1 0x0000 Non-Security CE Task Descriptor Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: CE_TDA0_NS
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TASK_DES_ADDR0 Task Descriptor Address is bit[31:0] (byte addr)

18.1.5.2 0x0004 Non-Security CE Task Descriptor Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: CE_TDA1_NS
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	TASK_DES_ADDR1 Task Descriptor Address is bit[39:32] (byte addr)

18.1.5.3 0x0008 Non-Security CE Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: CE_ICR_NS
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0008			Register Name: CE_ICR_NS
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	DBL_ENT_SRC_EN Double Entropy Source Enable 1: Double entropy source enable 0: Single entropy source enable (double entropy source disable)
30	R/W	0x0	LOW_POWER_EN CE Low Power Enable 1: Low power function enable 0: Low power function disable
29:4	/	/	/
3	R/W	0x0	CH3_TASK_IE Channel 3 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
2	R/W	0x0	CH2_TASK_IE Channel 2 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
1	R/W	0x0	CH1_TASK_IE Channel 1 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
0	R/W	0x0	CH0_TASK_IE Channel 0 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable

18.1.5.4 0x000C Non-Security CE Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: CE_ISR_NS
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:6	R/W1C	0x0	CH3_TASK_S Channel 3 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.

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Offset: 0x000C			Register Name: CE_ISR_NS
Bit	Read/Write	Default/Hex	Description
5:4	R/W1C	0x0	CH2_TASK_S Channel 2 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.
3:2	R/W1C	0x0	CH1_TASK_S Channel 1 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.
1:0	R/W1C	0x0	CH0_TASK_S Channel 0 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.

18.1.5.5 0x0010 Non-Security CE Task Load Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: CE_TLR_NS
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	CE_RAES_TLR Task Load For Channel RAES type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect
2	R/W	0x0	CE_ASYM_TLR Task Load For Channel ASYM type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect

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Offset: 0x0010			Register Name: CE_TLR_NS
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	CE_HASH_RBG_TLR Task Load For Channel HASH/RBG type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect
0	R/W	0x0	CE_SYMM_TLR Task Load For Channel SYMM type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect

18.1.5.6 0x0014 Non-Security CE Task Status Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: CE_TSR_NS
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	XTS_CH_INDEX indicate which channel in run for XTS. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3
23:16	R	0x0	ASYMM_CH_INDEX indicate which channel in run for asymmetric. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3
15:8	R	0x0	DIG_CH_INDEX indicate which channel in run for digest. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3

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Offset: 0x0014			Register Name: CE_TSR_NS
Bit	Read/Write	Default/Hex	Description
7:0	R	0x0	SYMM_CH_INDEX indicate which channel in run for symmetric. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3

18.1.5.7 0x0018 Non-Security CE Error Status Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: CE_ESR_NS
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	CH3_ERR Error code for task channel 3 0x00: No error 0x01: Algorithm not support 0x31: Data length error other: Reserved
23:16	R	0x0	CH2_ERR Error code for task channel 2 0x00: No error 0x01: Algorithm not support 0x31: Data length error other: Reserved
15:8	R	0x0	CH1_ERR Error code for task channel 1 0x00: No error 0x01: Algorithm not support 0x31: Data length error other: Reserved
7:0	R	0x0	CH0_ERR Error code for task channel 0 0x00: No error 0x01: Algorithm not support 0x31: Data length error other: Reserved

18.1.6 CE_S Register Description

18.1.6.1 0x0000 Security CE Task Descriptor Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0000	Register Name: CE_TDA0_S

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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	TASK_ DESCRIPTOR _ADDR0 Task Descriptor Address is bit[31:0] (byte addr)

18.1.6.2 0x0004 Security CE Task Descriptor Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: CE_TDA1_S
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	TASK_ DESCRIPTOR _ADDR1 Task Descriptor Address is bit[39:32] (byte addr)

18.1.6.3 0x0008 Security CE Interrupt Control Register (Default Value: 0x0000_0000)

Offset: 0x0008			Register Name: CE_ICR_S
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	CH3_TASK_IE Channel 3 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
2	R/W	0x0	CH2_TASK_IE Channel 2 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
1	R/W	0x0	CH1_TASK_IE Channel 1 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable
0	R/W	0x0	CH0_TASK_IE Channel 0 Task Interrupt Enable 0: Interrupt disable 1: Interrupt enable

18.1.6.4 0x000C Security CE Interrupt Status Register (Default Value: 0x0000_0000)

Offset: 0x000C			Register Name: CE_ISR_S
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/

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Offset: 0x000C			Register Name: CE_ISR_S
Bit	Read/Write	Default/Hex	Description
7:6	R/W1C	0x0	CH3_TASK_S Channel 3 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.
5:4	R/W1C	0x0	CH2_TASK_S Channel 2 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.
3:2	R/W1C	0x0	CH1_TASK_S Channel 1 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.
1:0	R/W1C	0x0	CH0_TASK_S Channel 0 task Status: Read 0x1: Task completed successfully Read 0x2: Task fail and stop Write 0x3 to clear it.

18.1.6.5 0x0010 Security CE Task Load Register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: CE_TLR_S
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	CE_RAES_TLR Task Load For Channel RAES type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect
2	R/W	0x0	CE_ASYM_TLR Task Load For Channel ASYM type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect

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Offset: 0x0010			Register Name: CE_TLR_S
Bit	Read/Write	Default/Hex	Description
1	R/W	0x0	CE_HASH_RBG_TLR Task Load For Channel HASH/RBG type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect
0	R/W	0x0	CE_SYMM_TLR Task Load For Channel SYMM type Read as 1: CE is busy to load another task, or the task queue is full. Write this bit is not allowed now. Read as 0: CE is ready to load another task Write 1: Load task and start to run. Write 0: No effect

18.1.6.6 0x0014 Security CE Task Status Register (Default Value: 0x0000_0000)

Offset: 0x0014			Register Name: CE_TSR_S
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	XTS_CH_INDEX indicate which channel in run for XTS. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3
23:16	R	0x0	ASYMM_CH_INDEX indicate which channel in run for asymmetric. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3
15:8	R	0x0	DIG_CH_INDEX indicate which channel in run for digest. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3

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Offset: 0x0014			Register Name: CE_TSR_S
Bit	Read/Write	Default/Hex	Description
7:0	R	0x0	SYMM_CH_INDEX indicate which channel in run for symmetric. 0: Task channel0 1: Task channel1 2: Task channel2 3: Task channel3

18.1.6.7 0x0018 Security CE Error Status Register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: CE_ESR_S
Bit	Read/Write	Default/Hex	Description
31:24	R	0x0	CH3_ERR Error code for task channel 3 0x00: No error 0x01: Algorithm not support 0x31: Data length error 0x43: TRNG rep_cnt (repetition count) test error 0x4c: TRNG adap_pro (adaptive proportion) test error 0x4f: TRNG rep_cnt and adap_pro tests occur both errors other: Reserved
23:16	R	0x0	CH2_ERR Error code for task channel 2 0x00: No error 0x01: Algorithm not support 0x31: Data length error 0x43: TRNG rep_cnt (repetition count) test error 0x4c: TRNG adap_pro (adaptive proportion) test error 0x4f: TRNG rep_cnt and adap_pro tests occur both errors other: Reserved
15:8	R	0x0	CH1_ERR Error code for task channel 1 0x00: No error 0x01: Algorithm not support 0x31: Data length error 0x43: TRNG rep_cnt (repetition count) test error 0x4c: TRNG adap_pro (adaptive proportion) test error 0x4f: TRNG rep_cnt and adap_pro tests occur both errors other: Reserved

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Offset: 0x0018			Register Name: CE_ESR_S
Bit	Read/Write	Default/Hex	Description
7:0	R	0x0	CH0_ERR Error code for task channel 0 0x00: No error 0x01: Algorithm not support 0x31: Data length error 0x43: TRNG rep_cnt (repetition count) test error 0x4c: TRNG adap_pro (adaptive proportion) test error 0x4f: TRNG rep_cnt and adap_pro tests occur both errors other: Reserved

18.1.6.8 0x0020 Security CE Symmetric Algorithm DMA Current Source Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: CE_SCSA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SYMM_SRCD_DMA0 Symmetric algorithm current source address DMA reads. Bit[31:0], byte addr.

18.1.6.9 0x0024 Security CE Symmetric Algorithm DMA Current Source Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0024			Register Name: CE_SCSA1_S
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	SYMM_SRCD_DMA1 Symmetric algorithm current source address DMA reads. Bit[39:32], byte addr.

18.1.6.10 0x0028 Security CE Symmetric Algorithm DMA Current Destination Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0028			Register Name: CE_SCDA0_S
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0028			Register Name: CE_SCDA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SYMM_DESD_DMA0 Symmetric algorithm current destination address DMA writes. Bit[31:0], byte addr.

18.1.6.11 0x002C Security CE Symmetric Algorithm DMA Current Destination Address1 Register (Default Value: 0x0000_0000)

Offset: 0x002C			Register Name: CE_SCDA1_S
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	SYMM_DESD_DMA1 Symmetric algorithm current destination address DMA writes. Bit[39:32], byte addr.

18.1.6.12 0x0030 Security CE HASH Algorithm DMA Current Source Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: CE_HCSA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HASH_SRCD_DMA0 HASH algorithm current source address DMA reads. Bit[31:0], byte addr.

18.1.6.13 0x0034 Security CE HASH Algorithm DMA Current Source Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0034			Register Name: CE_HCSA1_S
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R	0x0	HASH_SRCD_DMA1 HASH algorithm current source address DMA reads. Bit[39:32], byte addr.

18.1.6.14 0x0038 Security CE HASH Algorithm DMA Current Destination Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0038	Register Name: CE_HCDA0_S

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Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HASH_DESD_DMA0 HASH algorithm current destination address DMA writes. Bit[31:0], byte addr.

18.1.6.15 0x003C Security CE HASH Algorithm DMA Current Destination Address1 Register (Default Value: 0x0000_0000)

Offset: 0x003C			Register Name: CE_HCDA1_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HASH_DESD_DMA1 HASH algorithm current destination address DMA writes. Bit[39:32], byte addr.

18.1.6.16 0x0040 Security CE Asymmetric Algorithm DMA Current Source Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0040			Register Name: CE_ACSA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HASH_SRCD_DMA0 Asymmetric algorithm current source address DMA reads. Bit[31:0], byte addr.

18.1.6.17 0x0044 Security CE Asymmetric Algorithm DMA Current Source Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0044			Register Name: CE_ACSA1_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	HASH_SRCD_DMA1 Asymmetric algorithm current source address DMA reads. Bit[39:32], byte addr.

18.1.6.18 0x0048 Security CE Asymmetric Algorithm DMA Current Destination Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0048			Register Name: CE_ACDA0_S
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0048			Register Name: CE_ACDA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	ASYMM_DESD_DMA0 Asymmetric algorithm current destination address DMA writes. Bit[31:0], byte addr.

18.1.6.19 0x004C Security CE Asymmetric Algorithm DMA Current Destination Address1 Register (Default Value: 0x0000_0000)

Offset: 0x004C			Register Name: CE_ACDA1_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	ASYMM_DESD_DMA1 Asymmetric algorithm current destination address DMA writes. Bit[39:32], byte addr.

18.1.6.20 0x0050 Security CE XTS Algorithm DMA Current Source Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0050			Register Name: CE_XCSA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	XTS_SRCD_DMA0 XTS algorithm current source address DMA reads. Bit[31:0], byte addr.

18.1.6.21 0x0054 Security CE XTS Algorithm DMA Current Source Address1 Register (Default Value: 0x0000_0000)

Offset: 0x0054			Register Name: CE_XCSA1_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	XTS_SRCD_DMA1 XTS algorithm current source address DMA reads. Bit[39:32], byte addr.

18.1.6.22 0x0058 Security CE XTS Algorithm DMA Current Destination Address0 Register (Default Value: 0x0000_0000)

Offset: 0x0058			Register Name: CE_XCDA0_S
Bit	Read/Write	Default/Hex	Description

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Offset: 0x0058			Register Name: CE_XCDA0_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	XTS_DESD_DMA0 XTS algorithm current destination address DMA writes. Bit[31:0], byte addr.

18.1.6.23 0x005C Security CE XTS Algorithm DMA Current Destination Address1 Register (Default Value: 0x0000_0000)

Offset: 0x005C			Register Name: CE_XCDA1_S
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	XTS_DESD_DMA1 XTS algorithm current destination address DMA writes. Bit[39:32], byte addr.

18.1.6.24 0x0090 Security CE Repetition Count Test C Register (Default Value 0x0000_0000)

Offset: 0x0090			Register Name: CE_RCTC_S
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	REP_CNT_TEST_C Repetition Count Test, C value is bit[31:0] (by binary) The C value is calculated as follows: $C = 1 + [\frac{-\log_2 \alpha}{H}]$, $\alpha=10^{-20}$-10^{-40} H is sampling value and the unit is bit. Here is an example: $\alpha=10^{-20}$ and H = 160.

18.1.6.25 0x0094 Security CE Adaptive Proportion Test C Register (Default Value 0x0000_0000)

Offset: 0x0094			Register Name: CE_APTC_S
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADAP_PRO_TEST_C Adaptive Proportion Test, C value is bit [31:0] (by binary) The C value is calculated as follows: C=1+CRITBINOM (W, power (2, (-H)), 1-α) $\alpha=10^{-20}$-10^{-40} H is sampling value and the unit is bit. W is selected according to needs. Here is an example: $\alpha=10^{-20}$ and H = 160.

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18.1.6.26 0x0098 Security CE Adaptive Proportion Test W Period Register (Default Value 0x0000_0000)

Offset: 0x0098			Register Name: CE_APTWP_S
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADAP_PRO_TEST_W_Period Adaptive Proportion Test, window W period between two tests is bit [31:0] (by binary) 0 no wait start next test directly >0 wait window W period time start next test

18.1.6.27 0x009C Security CE Healthy Test Control Register (Default Value 0x0000_0000)

Offset: 0x009C			Register Name: CE_HTCR_S
Bit	Read/Write	Default/Hex	Description
31:18	/	/	/
17	R/W	0x0	ADAP_PRO_TEST_EN Adaptive Proportion Test Enable 1: Adap_pro healthy test enable 0: Adap_pro healthy test disable
16	R/W	0x0	REP_CNT_TEST_EN Repetition Count Test Enable 1: Rep_cnt healthy test enable 0: Rep_cnt healthy test disable
15:11	/	/	/
10:0	R/W	0x0	ADAP_PRO_TEST_W Adaptive Proportion Test, window size W value is bit [10:0] 1024: Noise source is binary 512: Noise source is not binary

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